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The AME2020 atomic mass evaluation ^{**}

(I). Evaluation of input data, and adjustment procedures

W.J. Huang (黄文嘉)^{1,2,3,4} Meng Wang (王猛)^{1,5,*} F.G. Kondev⁶ G. Audi (欧乔治)³ S. Naimi⁷

¹ Institute of Modern Physics, Chinese Academy of Sciences, Lanzhou 730000, China

² Advanced Energy Science and Technology Guangdong Laboratory, Huizhou 516003, China

³ Université Paris-Saclay, CNRS/IN2P3, IJCLab, 91405 Orsay, France

⁴ Max-Planck-Institut für Kernphysik, Saupfercheckweg 1, 69117 Heidelberg, Germany

⁵ School of Nuclear Science and Technology, University of Chinese Academy of Sciences, Beijing 100049, China

⁶ Physics Division, Argonne National Laboratory, Lemont, IL 60439, USA

⁷ RIKEN Nishina Center, Wako, Saitama 351-0198, Japan

Abstract This is the first of two articles (Part I and Part II) that presents the results of the new atomic mass evaluation, AME2020. It includes complete information on the experimental input data that were used to derive the tables of recommended values which are given in Part II. This article describes the evaluation philosophy and procedures that were implemented in the selection of specific nuclear reaction, decay and mass-spectrometric data which were used in a least-squares fit adjustment in order to determine the recommended mass values and their uncertainties. All input data, including both the accepted and rejected ones, are tabulated and compared with the adjusted values obtained from the least-squares fit analysis. Differences with the previous AME2016 evaluation are discussed and specific examples are presented for several nuclides that may be of interest to AME users.

Keywords: atomic mass evaluation, mass spectrometry, nuclear decays, nuclear reactions, least-squares adjustment, trends from the mass surface (TMS), atomic-mass tables

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1 Introduction

The Atomic Mass Evaluation (AME) was created in 1950's to produce a reliable atomic mass table from the available experimental data. A number of AME mass tables have been produced and published since then [1], serving the need of the broader research community to have reliable and comprehensive information about the atomic masses. The last complete evaluation of experimental atomic mass data, AME2016, was published in March, 2017 [2, 3]. Since then the experimental knowledge of atomic masses has continuously expanded and a large amount of new data has been published in the scientific literature.

This paper is the first of two AME2020 articles (Part I and Part II). It contains the details on the evaluation procedures and the main AME2020 evaluation table, Table I, where all accepted and rejected experimental data are presented and compared with the adjusted values deduced using a least-squares fit analysis. In Part II (the next article of this issue) [4], the recommended mass values and their uncertainties are given, followed by ta-

bles and graphs of derived quantities and detailed bibliographical information. As in previous AME publications, the AME2020 is accompanied by the NUBASE2020 evaluation [5], which provides information about the basic properties of the known ground and excited isomeric states that are used in the AME.

There is no strict literature cut-off date for the data used in the present AME2020 evaluation: all data, available to the authors until the end of October 2020 were included. The final mass-adjustment calculations were performed on February 1st, 2021. The present publication contains updated information presented in the previous AME tables, including data that were not used in the final adjustment due to specific reasons, e.g. data that have too large uncertainties.

Remark: In the following text, several data of general interest are discussed. References to articles that can be found in Table I are avoided. When it is necessary to provide specific references, those are given using the NSR key-numbers style [6] (e.g. [2020Fu05]) and are listed at the end of Part II, under "References used in

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* E-mail: wangm@impcas.ac.cn



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the AME2020 and the NUBASE2020 evaluations”.

As in the previous AME versions, all uncertainties are one-standard deviation (1σ).

2 Units and energy recalibration

The unit used in the AME is the ‘unified atomic mass unit’ u , which is defined as one twelfth of the mass of one free atom of carbon-12, at rest and in its atomic and nuclear ground states, $1 u = m(^{12}\text{C})/12$. The ‘unified mass unit’ stands for the unification of the physical and chemical units in 1960 following the proposal of Kohman, Mattauch and Wapstra [7]. It is worth mentioning that the chemical mass-spectrometry community (e.g. bio-chemistry, polymer chemistry) widely uses the dalton unit (symbol Da, named after John Dalton*). The dalton is an alternative name for the same unit and allows to express the number of nucleons in a molecule.

Until May 20th, 2019, the SI (International System of Units) unit of amount of substance, the mole, was defined by fixing the value of the molar mass of ^{12}C to be exactly 0.012 kg/mol. At that time the kilogram was defined as exactly the mass of the international prototype of the kilogram, an artifact made of platinum-iridium. In the current SI, the base units are defined by exact numerical values for the Planck constant h , elementary charge e , Boltzmann constant k , and Avogadro constant N_A [8]. The kilogram is now defined by taking the fixed numerical value of the Planck constant, h , to be $6.626\,070\,15 \times 10^{-34} \text{ J s}$, which is equal to $\text{kg m}^2 \text{ s}^{-1}$, where the meter and the second are defined by the fixed values of the speed of light in vacuum and frequency of the ground-state cesium hyperfine splitting [9].

With the new SI definition, the mass of an atom can be expressed in unit ‘kg’ with higher precision. Recently, the mass of ^{87}Rb atom was determined with a precision of 142 ppt in unit ‘kg’ [10]. This result is not included in AME as it will be used to determine the fine-structure constant by combining with the atomic mass of ^{87}Rb . In the present evaluation, the atomic mass of ^{87}Rb is known with a precision of 69 ppt in unit ‘u’.

In general, the atomic mass of a particular nuclide can be obtained as an inertial mass from the movement of an ionized atom in an electro-magnetic field, with some other masses used as references. Thus, the mass is derived from the ratio of masses and it is expressed in units of ‘u’. ^{12}C serves as the ultimate, absolute reference in the mass determination. In other cases, the atomic mass can be determined from an energy relation between the mass of the nuclide of interest and that of one or more well-known masses associated with a nuclear decay or reaction measurement. This energy re-

lation is then expressed in units of electron volts, eV. In AME1993 through AME2016, the unit of the *maintained* volt V_{90} was used, which was defined in 1990 by $2e/h = 483597.9$ (exact) GHz/ V_{90} . The γ -ray energies determined in wavelength measurements can be expressed in eV_{90} without any loss in precision, since the conversion factor is an exact quantity. These energies serve as standards in other energy measurements, so the reaction and decay energies used in AME can be more accurately expressed in *maintained* volts, as discussed by Cohen and Wapstra [11]. At that time, the precision of the conversion factor between mass units and *maintained* volt (V_{90}) was also higher than that between the former and *international* volt.

Since in the most recent CODATA evaluation [8] the Plank constant, h and the elementary charge, e are exact quantities, the *international* volt is used in AME2020. The relation between *maintained* and *international* volts is given as $V_{90} = 1.000\,000\,106\,66$ (exact) V, a difference of 106.66 ppb.

Here we discuss the influence of this new definition by using the reaction energy for $^1\text{H}(n,\gamma)^2\text{H}$ as an example. In an experiment at the Institut Laue-Langevin (ILL) [1999Ke05], the wavelength of the emitted γ ray was determined by using a high-resolution, silicon crystal spectrometer. In AME2003, the recommended value was $2224.5660(4) \text{ keV}_{90}$, based on the work of Kessler *et al.* [1999Ke05]. This result had the highest precision for energy measurement in the input data of AME, with a relative uncertainty of 180 ppb. In the later work by the same group [2006De21], the value was determined as $2224.55610(44) \text{ keV}$, using new evaluation on the lattice spacing of the crystal. The latter is used as an adjusted parameter in the CODATA evaluation, but it is not expressed explicitly. Using the same value for the wavelength as in [2006De21] and the length-energy conversion coefficient, we derived $2224.55600(44) \text{ keV}_{90}$ as an input to AME2016. In the present work, we recalibrated the result and obtained $2224.55624(44) \text{ keV}$ as input value.

In Table A, the relations between several constants of interest, obtained from the CODATA evaluation [8] are presented. The ratio of mass units to electronvolts is also given. In addition, values for the masses of the proton, neutron and alpha particle, as derived from the present evaluation, are also given, together with the mass difference between the neutron and the hydrogen atom.

The recommended $2e/h$ values and mass-energy conversion factors are given in Table B.

Some more historical points are worth mentioning.

In 1986, Taylor and Cohen [12] showed that the empirical ratio between the *maintained* and *international* volts, which had of course been selected to be nearly

*John Dalton, 1766-1844, who first speculated that elements combine in proportions following simple laws, and was the first to create a table of (very approximate) atomic weights.

Table A. Constants used in this work or resulting from the present evaluation.

1 u	=	$m(^{12}\text{C})/12$	=	atomic mass unit				
1 u	=	1 660 539.0666	$\pm$	0.0005	$\times 10^{-33}$ kg	0.3	ppb	<i>a</i>
1 u	=	931 494.10242	$\pm$	0.00028	keV	0.3	ppb	<i>a</i>
1 MeV	=	1 073 544.10233	$\pm$	0.00032	nu	0.3	ppb	<i>a</i>
m_e	=	548 579.909065	$\pm$	0.000016	nu	0.03	ppb	<i>a</i>
	=	510 998.95000	$\pm$	0.00015	eV	0.3	ppb	<i>a</i>
m_p	=	1 007 276 466.588	$\pm$	0.014	nu	0.014	ppb	<i>b</i>
M_α	=	4 001 506 179.13	$\pm$	0.16	nu	0.04	ppb	<i>c</i>
$m_n - m_H$	=	839 884.01	$\pm$	0.47	nu	560	ppb	<i>c</i>
	=	782 347.00	$\pm$	0.44	eV	560	ppb	<i>c</i>

a) derived from the work of the latest CODATA [8].

b) derived from this work combined with m_e and the ionization energy for ^1H and ^4He from [8].

c) this work.

equal to 1, had changed by as much as 7 ppm. For this reason, a new value to define the *maintained* volt V_{90} was introduced in 1990 [13]. Since older high-precision, reaction-energy measurements were essentially expressed in keV_{86} , the difference in voltage definition had to be taken into account. In 1990, A.H. Wapstra [14] discussed the need for recalibration of several γ rays produced in (n,γ) and (p,γ) reactions, which are often used as calibration standards and have precisions better than 8 ppm. This work has been updated in AME2003 to evaluate the influence of new calibrators, as well as of the fundamental constants for γ -ray and particle energies used in (n,γ) , (p,γ) and (p,n) reactions [15]. In doing this, the calibration work of Helmer and van der Leun [16], based on the fundamental constant values at that time, was used. For each of the data concerned, the changes were relatively minor.

For α -particle energies, Rytz has taken this change into account, when updating his earlier evaluation of α -particle energies [17]. The calibration for proton energies has also been undertaken in AME2003 [15]. These recalibrated data are used in the present work and marked by “Z” behind the reference key-number in Table I. When it was not possible to do so, for example when this position was used to indicate that a remark was added, the same “Z” symbol was added to the uncertainty value mentioned in the remark.

The adoption of the *international* unit eV in the present evaluation, instead of eV_{90} used in earlier AME, does not influence the input data except for the four results from [2006De21], which have been recalibrated as discussed above. For all the other energy measurements, the uncertainties are far larger than the difference of the two units.

Until the end of the last century, the relative precision of $M - A$ expressed in keV was for several nuclides

worse than the same quantity expressed in mass units. Due to the increased precision of fundamental constants, now the relative precision of $M - A$ expressed in keV is as good as the same quantity expressed in mass units. In the earlier mass tables (e.g. AME1993 [18]), the binding energies, $ZM_H + NM_n - M$, were presented. The main reason for this was that the uncertainty of this quantity (in keV_{90}) was larger than that of the mass excess, $M - A$. However, due to the increased precision of the neutron mass, this is no longer important. Since AME2003, we give instead the binding energy per nucleon for educational reasons, connected to the Aston Curve and the maximum stability around the ‘iron-peak’ which is of importance in astrophysics.

3 Input data

In general, two different methods are used in atomic mass measurements: the mass-spectrometric one (often called a “direct method”), where the inertial mass is determined from the trajectory or cyclotron frequency of an ion in a magnetic field, or from its time of flight; and the so-called “indirect method” where the reaction energy, i.e. the difference between two or more masses, is determined using a specific nuclear reaction or a decay process. Regardless of which method is used, it is the relation between atomic masses that is determined experimentally. In the AME data treatment, we try our best to include only the primary experimental information, rather than the absolute mass values reported in the literature. In this way, the masses can be recalibrated automatically for any future changes, and the original correlation information can be properly preserved.

In the present work all available experimental data related to atomic masses (both energy and mass-spectrometric data) are considered. The input data are

Table B. Definition of Volt unit, and resulting mass-energy conversion constants.

$2e/h$				u		
1983	483594.21	(1.34)	GHz/V	931501.2	(2.6)	keV
1983	483594	(exact)	GHz/V ₈₆	931501.6	(0.3)	keV ₈₆
1986	483597.67	(0.14)	GHz/V	931494.32	(0.28)	keV
1990	483597.9	(exact)	GHz/V ₉₀	931493.86	(0.07)	keV ₉₀
1999	483597.9	(exact)	GHz/V ₉₀	931494.009	(0.007)	keV ₉₀
2010	483597.9	(exact)	GHz/V ₉₀	931494.0023	(0.0007)	keV ₉₀
2012	483597.9	(exact)	GHz/V ₉₀	931494.0038	(0.0004)	keV ₉₀
2018	483597.8484	(exact)	GHz/V	931494.10242	(0.00028)	keV

extracted from the available literature, compiled in an appropriate format and then carefully evaluated. The data are compared with other results and with values obtained from the trends from the mass surface (TMS) in the region, and the calibrations are checked, if necessary. As a consequence, some of the input data are accepted directly or after appropriate corrections and used in the subsequent mass adjustment procedure. There are also data that are not used due to large uncertainties or other reasons, as explained in section 4.5. All input data are given in Table I, and these used in the mass adjustment are also presented graphically in Fig. 1.

3.1 Connection diagram

Fig. 1 shows a connection diagram of $(N - Z)$ versus $(N + Z)$, where each dot represents a nuclide and each line connecting the dots an experimentally established relation. It is straightforward to show many mass measurements results that way, including these from mass-doublet spectroscopy or decay energy measurements. For example, mass measurements using Penning-trap almost always give a ratio of masses of two ions, as the inverse proportion to their cyclotron frequencies in the trap. Thus, in most case those measurements can be represented as a connection of two masses. For a relation of several atomic masses determined by reaction energy measurements, usually one mass difference between two nuclides is the most important one and it is then presented in the diagrams. Most lines in Fig. 1 link two nuclides, with exceptions for five mass-spectrometric “triplet” measurements of Rb and Cs isotopes from [1982Au01, 1986Au02], which can be best represented as a linear combination of three isotopes with non-integer coefficients [1982Au01].

However, not all of the experimental results can be clearly represented in Fig. 1. For example, direct experimental results from ‘Smith-type’, ‘Schottky’, ‘Isochronous’, ‘Time-of-flight’ and some ‘Multi-reflection Time-of-flight’ mass spectrometers, are calibrated in a

more complex way, and are thus published by their authors as absolute mass values. They are then shown as open squares in Fig. 1, and presented in Table I as a difference ${}^A\text{El}-u$, where ${}^A\text{El}$ represents the nuclide with the mass number A and element name El.

Atomic masses are exacted from the network in Fig. 1. The absolute mass values in the unit ‘u’ can be determined experimentally for nuclides that are eventually connected to ${}^{12}\text{C}$, the anchor point of the network. As can be seen from Fig. 1, the accepted data may allow determination of the mass of a particular nuclide using several different routes; such a nuclide is called *primary*. Their mass values in the table are then derived by a least squares method. In the other cases, the mass of a nuclide can be derived from only one connection to another nuclide; it is called a *secondary* nuclide. This classification is of importance for our calculation procedure.

The diagrams in Fig. 1 also show many cases where the relation between two atomic masses is experimentally known, but none of them are connected to ${}^{12}\text{C}$. Since it is our policy to include all available experimental results, in such cases we have produced estimated mass values that are based on TMS in the neighborhood. Using this data representation, vacancies also occur, and these are filled with estimated values produced by the same TMS procedure. Estimates of unknown masses are further discussed in section 5.2.

Excited isomeric states may be involved in mass measurements and the related information is included in this work. Although some isomeric states are indicated in Fig. 1, the diagram itself is two-dimensional and cannot represent the experimental results involving isomers completely. Instead, all relevant information is given in Table I.

3.2 Representation of the input data

All input data are given as linear equations in Table I. Penning-trap mass spectrometers provide cyclotron frequencies of charged ions in the trap, which are always

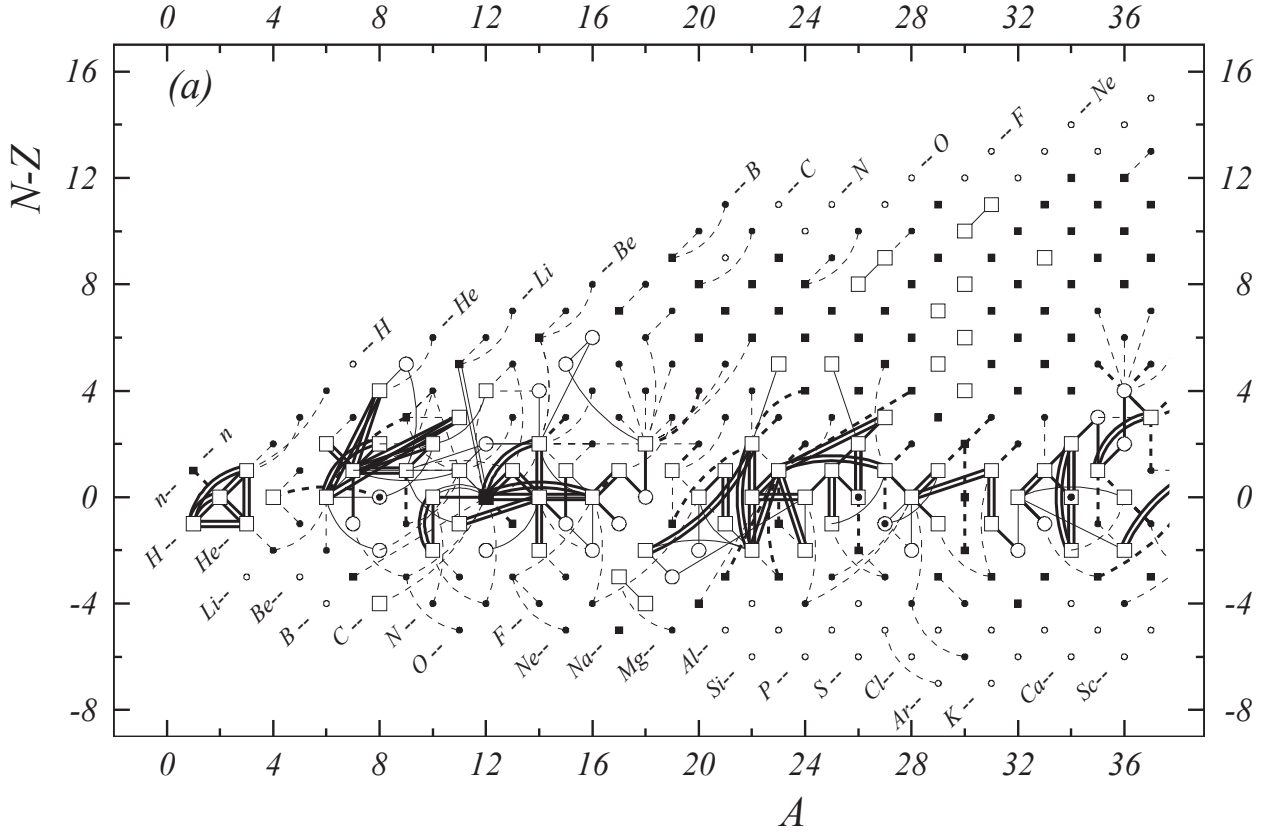


Figure 1. (a) - (j). Diagram of connections for input data.

For *primary data* (those checked by other data):

- absolute mass-doublet nuclide (or nuclide connected by a unique secondary relative mass-doublet to a remote reference nuclide);
- other primary nuclide;
- ◻ ⊙ primary nuclide with relevant isomer;
- // mass-spectrometric connection;
- other primary reaction connection.

Primary connections are drawn with two different thicknesses. Thicker lines represent the highest precision data in the given mass region

(limits: 1 keV for $A < 36$,
2 keV for $A = 36$ to 165 and
3 keV for $A > 165$).

For *secondary data* (cases where masses are known from one type of data and are therefore not checked by a different connection):

- secondary experimental nuclide determined from mass spectrometry;
- secondary experimental nuclide determined by a reaction or a decay;
- nuclide for which mass is estimated from trends from the Mass Surface (TMS);
- - - - connection to a secondary nuclide. Note that an experimental connection may exist between two estimated TMS nuclides when neither of them is connected to the network of primaries.

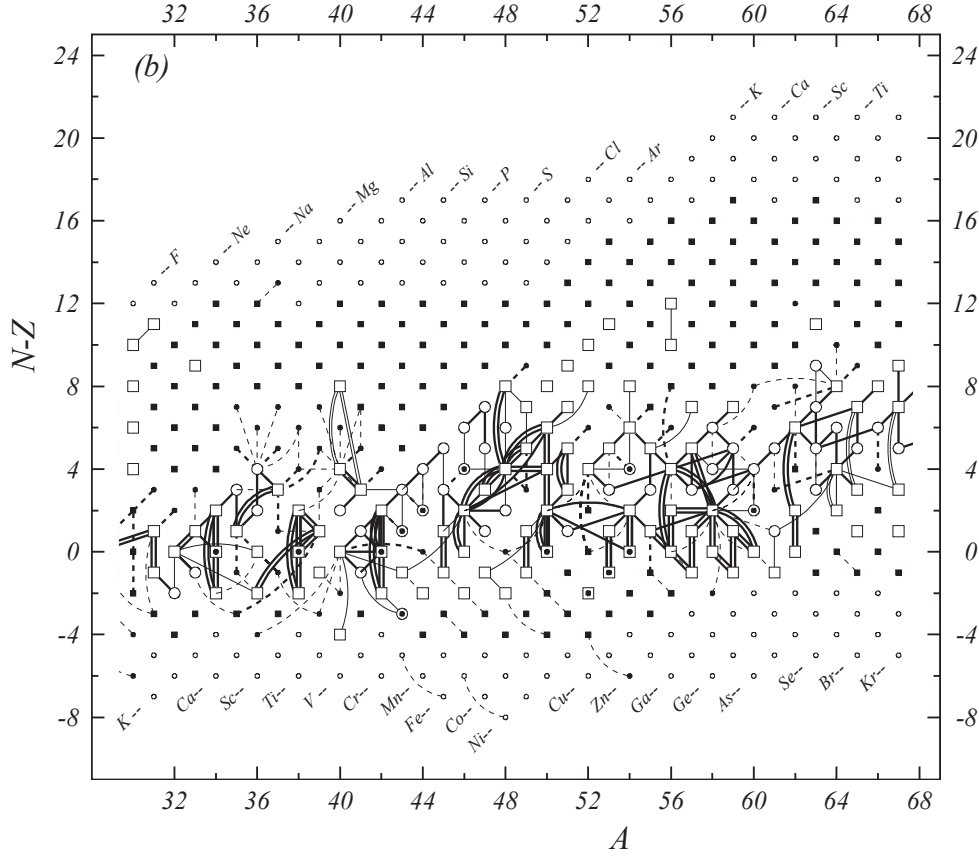


Figure 1 (b). Diagram of connections for input data — continued.

compared to that of a well known reference nuclide in order to determine the ratio of two masses. The mass ratios are converted to a linear combination of masses of electrically neutral atoms without any loss of accuracy. The procedure for converting frequency ratios to linear equations can be found in Appendix A. Some authors publish their results directly as masses, but this is not a recommended practice for high-precision mass measurements since the mass values could be modified due to change of the reference mass. An energy measurement of nuclear reaction or decay can be easily represented with a linear equation, too.

When it is too difficult to express a primary experimental result in a simple equation, the absolute mass value is given in Table I as a difference $^A\text{El}-u$ and it is usually accompanied with the relevant comment to explain the original information. Such comment is not applied for results from time-of-flight mass spectrometry, where the mass of interest is obtained using complex calibrations with many reference masses and reported as an absolute mass value in the original literature.

Values in the equations listed in Table I and the corresponding uncertainties are given in ' μu ' for results

from mass spectrometry and in 'keV' for energy measurements.

3.3 New input data

A total of 777 pieces of new experimental data are included in AME2020 in comparison to AME2016. Among them, 477 data are from mass spectrometry measurements, and 300 data from energy measurements.

Updated values for the masses of π^+ and π^- are included from the latest Particle Data Group publication [19]. These two mesons are present since they are involved in some atomic mass experiments, although they are not considered as "nuclides" themselves.

3.3.1 Most precise results

The most precise input data are provided by Penning traps, where the cyclotron frequencies of the ion of interest and a reference ion are measured in a uniform magnetic field. Three traps, namely LIONTRAP at Mainz, PENTATRAP at Heidelberg and FSU-MIT-TRAP at Florida, are dedicated for mass measurements of stable or long-lived nuclides with ultra-high accuracy, and provide 15 results with precisions of 0.04 nu ($10^{-9} u$) for

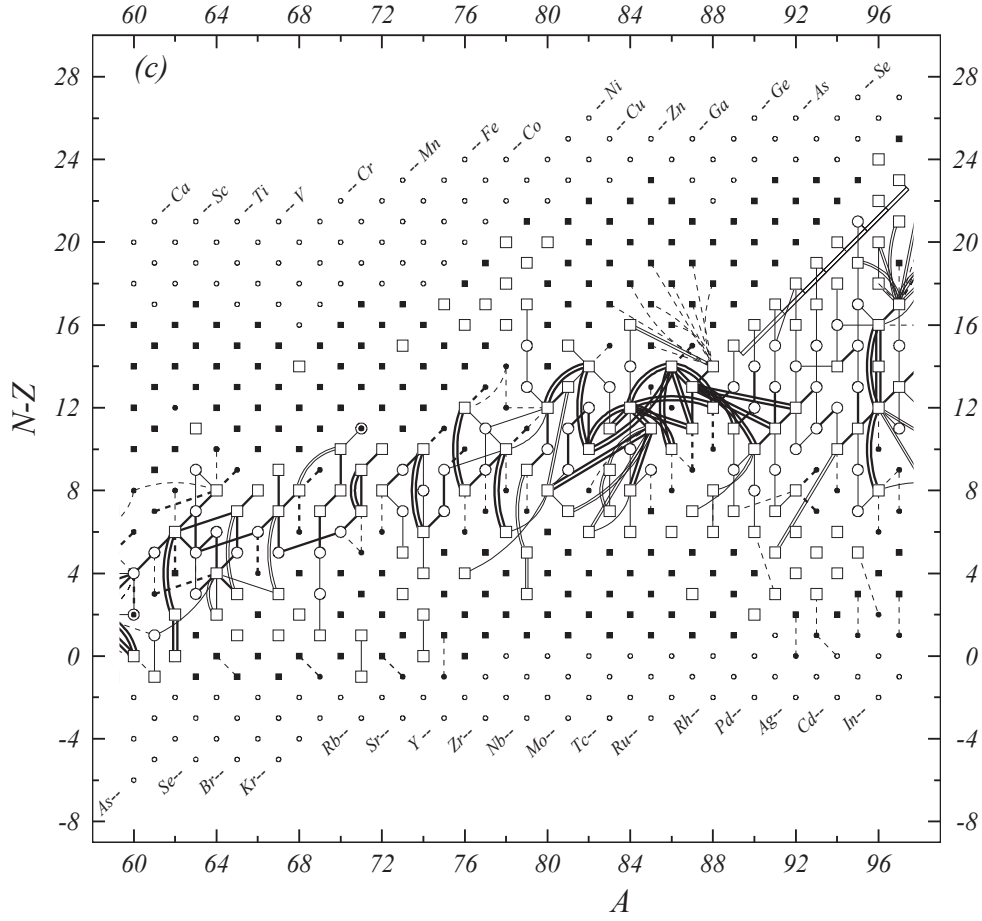


Figure 1 (c). Diagram of connections for input data —continued.

light ions [2020Fi05], and 1.5–2.8 nu for xenon isotopes [2020Ri04]. These results provide reliable reference standards for other mass measurements.

For measurements with such high precisions, not only the electronic binding energy (single or multiple ionization), but also the molecular binding energy (dissociation energy) should be considered in the data treatment. As described for example in Ref. [2018Sm03], even the rotational energy of the molecule has to be taken into account. For most molecules used in the experiments, the binding energy represents typically a correction of a few parts in 10^{10} and its uncertainty only limits the accuracy of the neutral atomic mass to a few parts in 10^{12} . For measurements with precisions not better than 10^{-9} (100 eV/100 u), the molecular binding energy could be neglected without loss of accuracy. In cases where the precision is better than 10^{-10} (10 eV/100 u), e.g. at MIT [1995Di08] and FSU [2015My03], it is necessary to take into account the molecular binding energy. More details on the binding energy corrections can be found in Appendix A.

3.3.2 Mass spectrometry away from β -stability

The domain of nuclides with experimentally known masses has extended impressively over the last few years, thanks to the developments of radioactive nuclear beam facilities and novel mass spectrometers. Around 20 years ago, masses of short-lived nuclides were mainly known from Q_β end-point measurements, while in the present evaluation, mass spectrometry dominates. It can be concluded that the shape of the atomic mass surface, and hence understanding of nuclear interactions, has been changed significantly over the last 10–20 years.

In the past, several types of mass spectrometry have provided some important results for radioactive nuclides, but are not used any more. Among them, the classical double-focussing mass spectrometry and the ‘Smith-type’ Radio-frequency mass spectrometry MISTRAL at CERN, Cyclotron time-of-flight mass spectrometry at SARA-Grenoble and at CSS2-GANIL, were described in AME2016 and will not be discussed here. The Schottky mass spectrometry at ESR-GSI, which measures the revolution frequency of cooled nuclides in a storage ring,

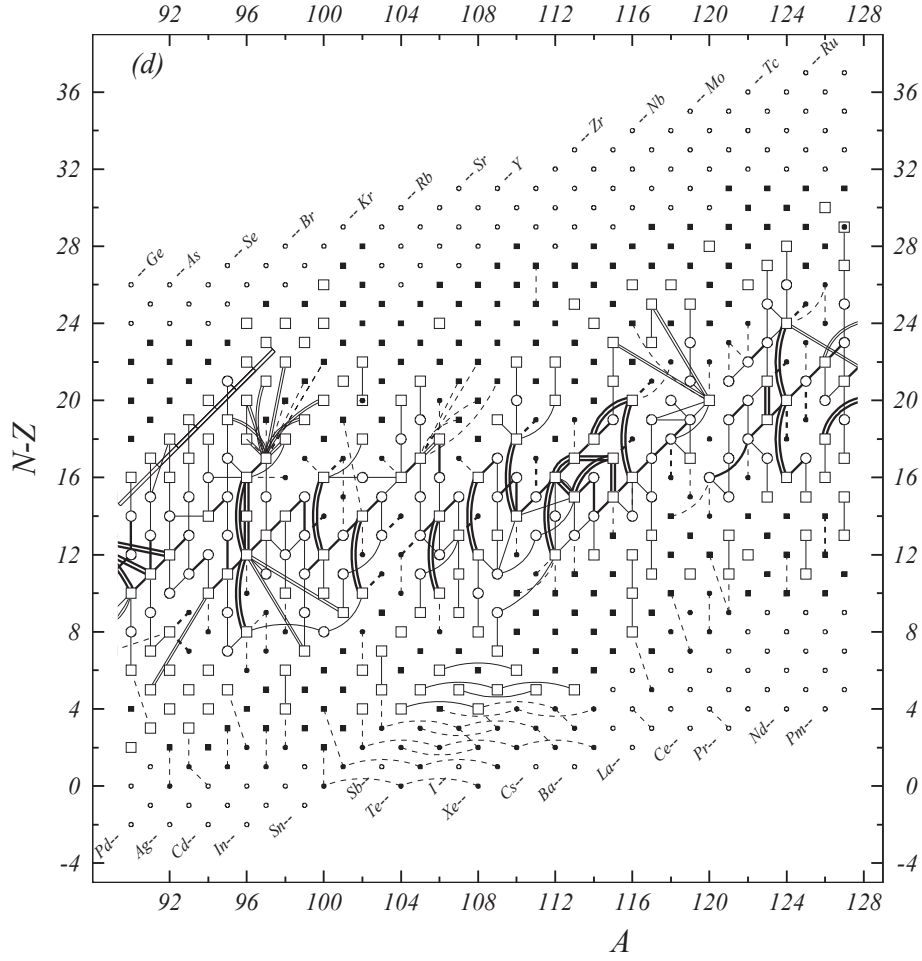


Figure 1 (d). Diagram of connections for input data —continued.

has not provided new data either.

All new mass-spectrometry results included in AME2020 are obtained by using one of the four methods, namely Penning traps and Multi-Reflection Time-of-Flight spectrometers (MR-TOF) for low-energy radioactive nuclides, the Magnetic-Rigidity time-of-flight ($B\rho$ -ToF) mass spectrometry and storage-ring based isochronous-mass-spectrometry (IMS) at high-energy facilities.

Penning trap spectrometers Nowadays, seven Penning traps are being operated at major facilities for radioactive nuclide research around the world: ISOLTRAP-Cern, CPT-Argonne, JYFLTRAP-Jyväskylä, LEBIT-East-Lansing, SHIPTRAP-Darmstadt, TITAN-Vancouver, and TRIGA-TRAP-Mainz. Conventionally the Time-of-Flight Ion-Cyclotron-Resonance (ToF-ICR) method is used to determine the cyclotron frequencies of ions in the trap. High resolving power (10^6) and accuracies (to 10^{-8}) are routinely achieved for nuclides located quite far from the

line of β -stability. Recently, a novel method of Phase-Imaging Ion-Cyclotron-Resonance (PI-ICR) was firstly developed at SHIPTRAP using a position-sensitive detector [2013El01]. This method is more efficient and more precise than the conventional ToF-ICR. In AME2016 only SHIPTRAP reported results using PI-ICR, while in the current work ISOLTRAP, CPT and JYFLTRAP also reported such results. Thanks to the high resolving power of PI-ICR, many low-lying isomeric states can be clearly identified. For example, the isomeric state in ^{104}Nb with an excitation energy of 9.8(2.3) keV was measured by CPT [2018Or.A].

Multi-Reflection Time-of-Flight spectrometers

The Multi-Reflection Time-of-Flight spectrometer (MR-TOF) was initially developed for beam purification for ISOLTRAP and then used as an independent mass spectrometer. This method started to provide mass results of short-lived nuclides in 2013 and results from four MR-TOF's, operated at ISOLDE-CERN, RIBF-RIKEN, GSI and

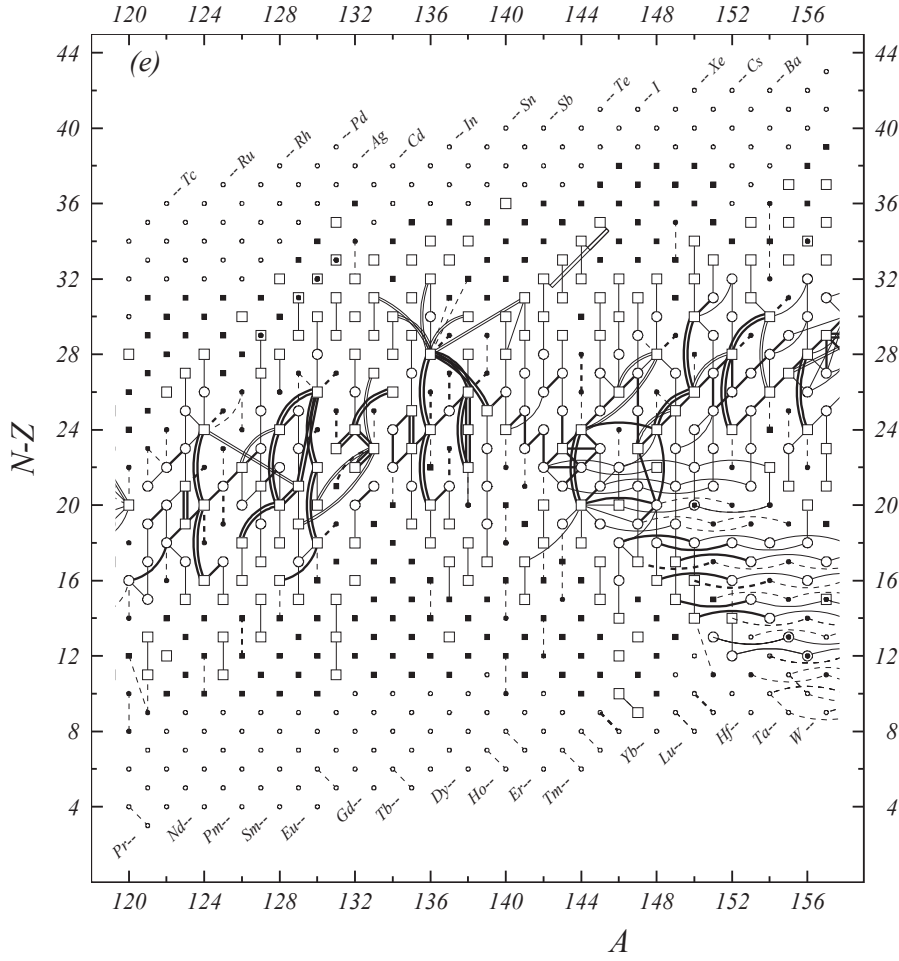


Figure 1 (e). Diagram of connections for input data —continued.

TRIUMF, are included in AME2020.

The MR-TOF mass measurements are based on time-of-flight, aiming at extending the flight path by reflecting ions back and forth in a static electric field. With this method, a relative mass precision of 10^{-7} is routinely achieved with a typical resolution of 10^{-5} . MR-TOF spectrometer is in principle more sensitive than Penning trap and it can be used in measurements of nuclides with very short half-lives and/or low production rates. As demonstrated at ISOLDE-CERN, it can always extend the measured region by one more nuclide further away from stability compared with PENNING TRAP [2017We16, 2018Mo14, 2020Ma09].

For mass determination with MR-TOF, the general relation between mass-to-charge ratio (m/q) and time-of-flight t is [2013Wi06]:

$$t = \alpha \sqrt{m/q} + \beta, \quad (1)$$

where α and β are constants related to the experimental set-up and are the same for the ion of interest and for

the reference ions.

At RIKEN, GSI and TRIUMF, the β parameter can be determined independently, thus only one reference nuclide is needed for mass calibration. In this way, the ratio of time-of-flight between the ion of interest and the reference ion is used to extract the linear equation, as it is the case for Penning trap measurement. However, at ISOLDE-CERN, two reference masses are used to determine the mass of interest, and the so-called C_{tof} method is used. When using this method, we express the linear equation in term of absolute mass.

Magnetic-Rigidity Time-of-Flight spectrometers

Projectile fragmentation and in-flight fission reactions are powerful tools to produce exotic nuclides far from the stability. Masses of almost undecelerated fragment products, produced in bombardment of thin targets with beams of heavy ions, can be determined from a combination of magnetic deflection and time-of-flight measurements. Nuclides in an extended region in A/Z and Z

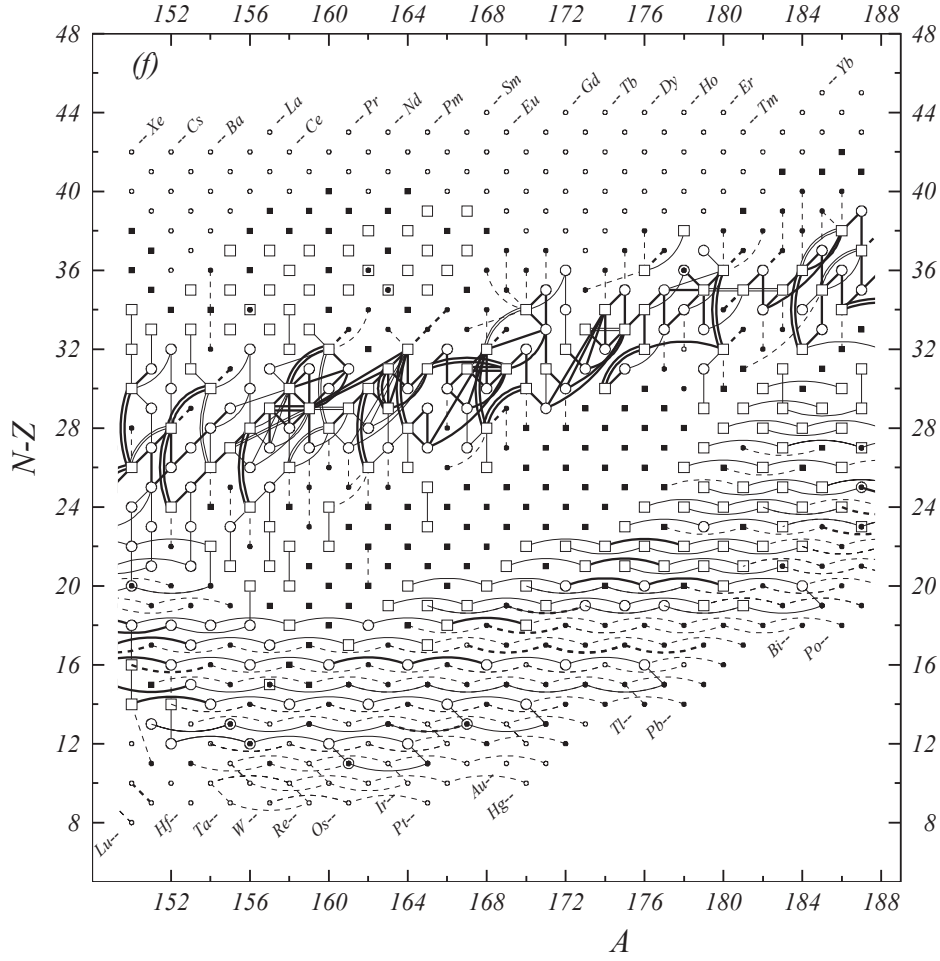


Figure 1 (f). Diagram of connections for input data —continued.

are analyzed simultaneously. Each individual ion, even very short-lived ($1\mu\text{s}$) one, can be identified and its mass measured. The limited resolving power of $m/\delta m \sim 10^4$ and cross-contaminations can cause significant shifts in masses. Using this method, mass values with precisions of 3-50 ppm have been obtained for a large number of neutron-rich nuclides up to $A \sim 70$.

One difficulty in such experiments is the presence of isomeric states with half-lives comparable or longer than the time of flight (about $1\mu\text{s}$) which may affect the mass of the ground state. In the recent work at RIKEN [2020Mi13], the isomeric ratios are also measured using γ -ray detectors at the experimental counting station, providing corrections for isomeric contamination. The most critical part in these experiments is the calibration, since it is frequently from an empirically determined function, which, in several cases, had to be extrapolated rather far from the calibrating masses. The MSU group re-evaluated their old B ρ -ToF experimental data [2015Me01, 2015Me08, 2016Me07], by incorporat-

ing recently published high precision Penning trap mass data [2018Le03, 2018Mo14] as additional calibration nuclides. This allowed to reduce the systematic uncertainty in previously measured data and to expand the number of nuclides for which masses were obtained [2020Me06].

Isochronous Mass Spectrometry If the flight path length of the spectrometer can be extended, then the resolving power can be improved. This is the basic idea of the storage-ring based spectrometer, which has been proven to be a powerful tool for direct mass measurement of short-lived nuclides. In an Isochronous Mass Spectrometry (IMS) experiment, the revolution times of the ions stored in the storage ring are measured, from which the nuclear masses can be extracted. Since no beam cooling is needed, the IMS method is suitable to mass measurements of nuclei with half-lives as short as several microseconds. The first set-up of this type was operated at GSI-ESR at Darmstadt, but there are no new results from that group in recent years. The Cooler Stor-

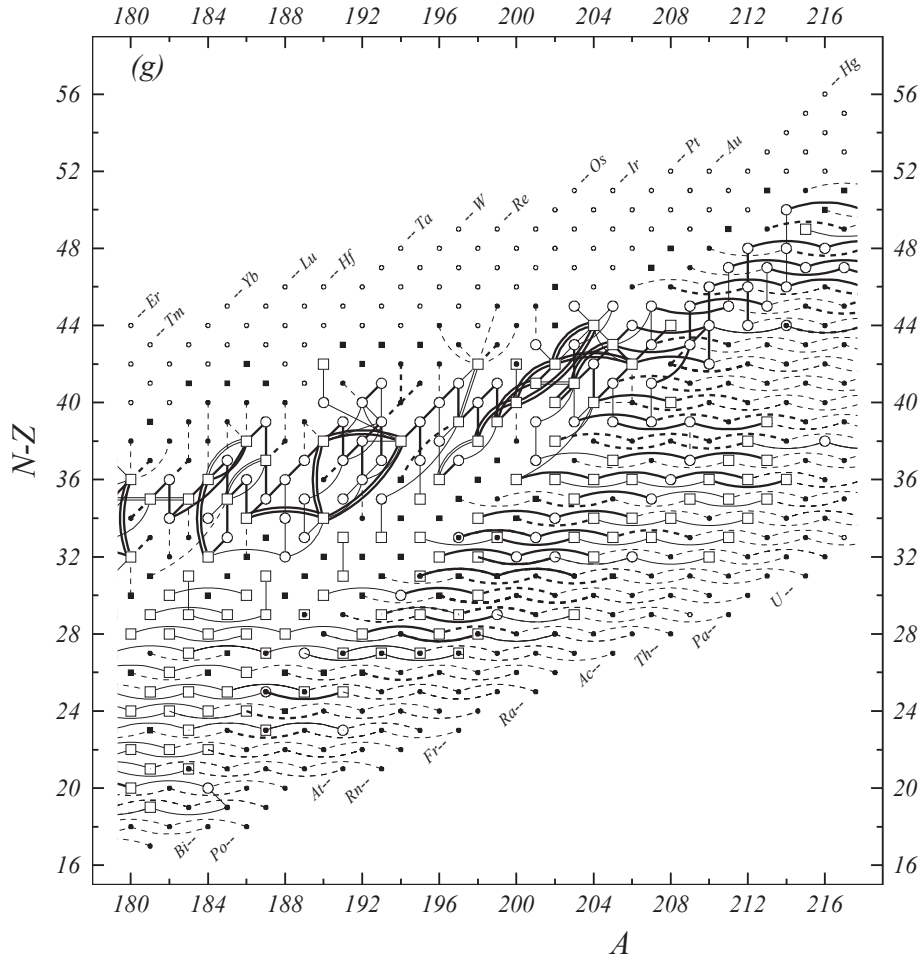


Figure 1 (g). Diagram of connections for input data —continued.

age Ring for experiment (CSRe) at the IMP-LANZHOU is the second spectrometer for IMS mass measurements. In these experiments, the drifts of the revolution times caused by instabilities of the magnetic fields are carefully corrected, and as a consequence the appropriate statistical uncertainty is assigned to each ion by evaluating the revolution times. In this way no apparent systematic uncertainty is observed, and a relative precision of 1×10^{-7} can be routinely achieved. A new isomeric state in ^{101}In has been discovered recently, demonstrating the excellent resolving power [2019Xu13]. This result was later confirmed by MR-TOF at GSI [2020Ho03].

3.3.3 Energy measurement

Although mass spectrometry dominates the field of high-precision mass measurements, decay energy measurements play an indispensable role in discovering new nuclides and in determining their masses.

Presently, many nuclides beyond the driplines can be accessed in the light-mass region. They can decay

via direct proton or neutron emission. The half-lives of some unbound nuclides are too short to acquire their outer electrons (which takes around 10^{-14} s), and to form atoms. However, we still convert their masses to “atomic masses” so we can treat them consistently with other nuclides. It is challenging to experimentally study these unbound nuclides far from stability where only a very few events can be observed. Frequently, theoretical calculations are required to extract their properties from the experimental data. For example, light unbound nuclides ^{20}B and ^{21}B [2018Le18], and ^{31}K [2019Ko18] have been discovered using complete kinematic measurement of all decay products, and their masses determined with the invariant mass spectrum. Another example is the mass of ^{73}Rb which was determined in β -delayed proton measurement [2020Ho17].

Of the 300 newly included energy results in AME2020, 237 are α decay energy measurements. Fusion reactions are frequently used for the synthesis of new neutron-deficient nuclides in the medium and heavy mass

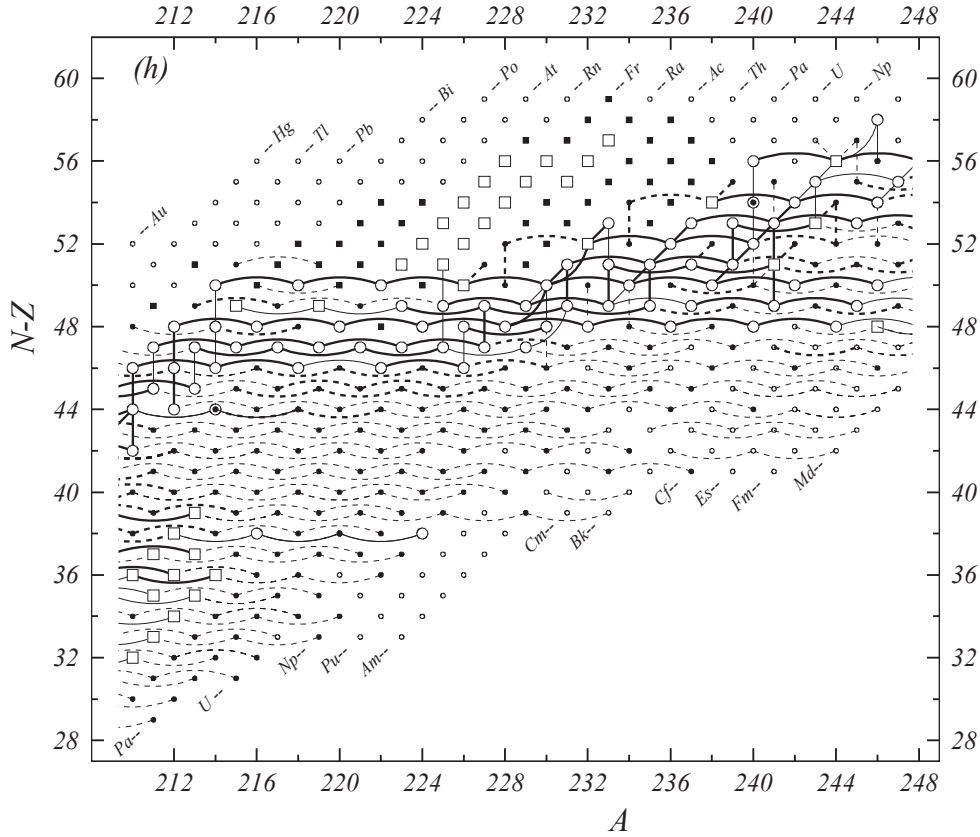


Figure 1 (h). Diagram of connections for input data — continued.

regions. Several new neutron-deficient uranium and neptunium isotopes were discovered using this approach in Lanzhou. The nuclides were identified by the characteristic α decays along the decay chains, and their masses are deduced from the measured α -decay energies [2019Zh23, 2020Ma09].

3.4 Corrections of the input data

In most cases, values given by authors in the original publication are accepted by the AME evaluators, but there are exceptions. One example is the performed recalibration due to the change in the definition of volt, as discussed in Section 2. For more complex cases, a remark is added in Table I at the end of the concerned A-group.

Changes in the uncertainties of the α -decay energies are also introduced, when it is not clear if the final state is the ground state or an excited one. For these cases, an extra 50 keV is added in quadrature to the original uncertainty.

3.4.1 Isomeric mixtures

The resolving power of mass spectrometers improved significantly in recent years and many isomers can be clearly identified and their masses unambiguously mea-

sured. However, there are cases where a mixture of isomers is actually measured. Since the mass of the ground state is the primary aim of the present evaluation, it can be derived only in cases where information on the excitation energies and production rates of the isomers is available. In AME, we use a special treatment of isomer mixtures (see Appendix B) and information about such changes is given as comments to the input data in Table I.

The mass M_0 of the ground state can be calculated if both the excitation energy E_1 of the upper isomer and its relative production rates are known, as discussed in [2020Ca08]. However, such detailed information is usually lacking in the literature. In such cases, if E_1 is known but not the production ratio, one must assume equal probabilities for all possible relative intensities. The estimated mass of the ground state is then $M_0 = M_{exp} - E_1/2$, and the uncertainty is $0.29E_1$ (see Appendix B). For cases where two or more excited isomers are involved, the correction method is discussed in Appendix B.

A further complication arises when E_1 is not known. In such a case, we use an estimate value for E_1 , flagged with '#', which is incorporated in the NUBASE evaluation. When estimating the E_1 values, we first consider

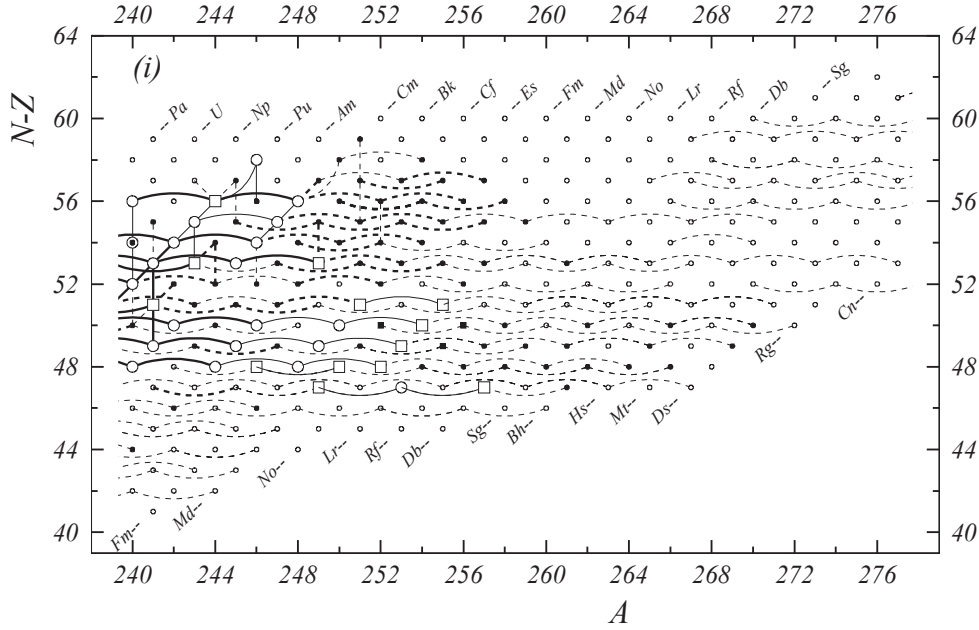


Figure 1 (i). Diagram of connections for input data —continued.

the available experimental data possibly giving lower limits: e.g. if it is known that one of two isomers decays to the other; or if γ rays of known energy occur in such decays. If not, we try to interpolate between E_1 values in neighboring nuclides that can be expected to have the same spin and configuration assignments. If such a comparison does not yield useful results, theoretical predictions were sometimes accepted, including upper limits for transition energies following from the measured half-lives. Values estimated this way are provided with somewhat generous uncertainties.

In general, an isomer can affect the mass of the ground state if its lifetime is relatively long (hundreds of milliseconds or longer). However, half-life values given in NUBASE are those for neutral atoms. For bare nuclides, where all electrons are fully stripped from the atom, the lifetimes of such isomers can be considerably longer, since the decay by conversion electrons is switched off. One example is the half-life measurement of the fully-stripped $^{94m}\text{Ru}^{44+}$ by means of IMS at CSRe [2017Ze02].

3.4.2 Particle energy vs. decay energy

Sometimes, conflicting meanings of particle and decay energies are given in the literature. For example, energy values are given by some authors as the particle energy E , while others quote it as decay energy Q , which is the sum of kinetic energies of the emitted particle and the recoiling daughter nuclide. In general, the α particle carries about 97-98% of its Q value and the recoiling nuclide accounts for about 2-3%. The relation between

the Q_α and E_α values is discussed in Appendix C.

Some authors derive Q_α value by not only correcting for the recoil energy, but also for the effect of screening by the atomic electrons (see Appendix C). In our adjustment procedure, the latter corrections have been removed.

Recalibration of decay energies in implantation experiments In experiments where α - and proton-emitting nuclei are implanted in a silicon detector, both the α particle and the recoiling daughter nuclide deposit energies in the detector. In such cases, α particles and protons with energies of a few MeV have almost 100% detection efficiency, which is not the case for the heavy recoiling nuclides, where only part of the recoiling energy contributes to the signal.

This effect has been discussed in Ref. [2012Ho12] and the partial recoil energy of the daughter nuclide has been taken into account in the energy calibration. However, not all authors make such corrections to their published data. We have developed a procedure [20] to calculate the detection efficiency for heavy nuclides in silicon detectors based on the Lindhard's theory [21], which has been experimentally proven to be reliable [22, 23].

The correction has been applied only to a few experimental results in the present work and a remark is added in Table I. After discussions with the authors of the original publication, the α -decay energy of ^{255m}Lr from Ref. [2008Ha31] has been corrected and the difference turns out to be 7 keV. Another example is the proton-decay energy of ^{53m}Co from [2015Sh16], as dis-

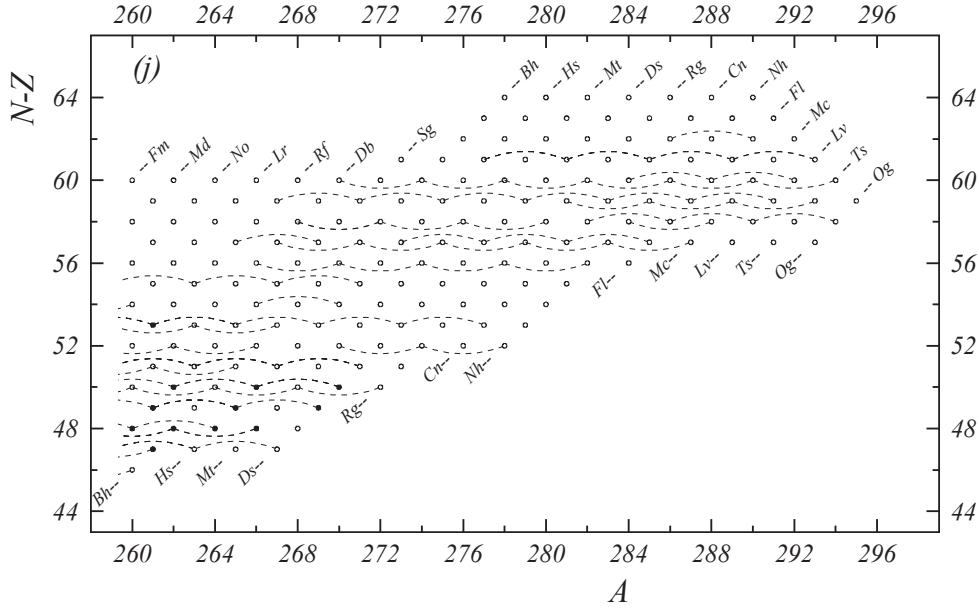


Figure 1 (j). Diagram of connections for input data —continued.

cussed in Ref. [20].

4 Determination of the adjusted mass values

The atomic mass evaluation is unique when compared to the other evaluations of data, in the sense that all mass determinations are relative measurements, not absolute ones. Each experimental datum defines a relation involving two or more nuclides. The ensemble of these relations generates a highly entangled network, as represented in Fig. 1. All the input data are given as linear equations in Table I.

To determine the mass values, we use a least-squares method weighed according to the uncertainty with which each piece of data is known. In the first step, we pre-average the same and parallel data to reduce the size of the system of equations, and then separate primary data from secondary ones. The primary data are used in the following least-squares calculations. At last, we recommend the atomic masses (Part II, Table I), the nuclear reaction and separation energies (Part II, Table III), the adjusted values for the input data (Table I), the *influences* of data on the primary nuclides (Table I), the *influences* received by each primary nuclide (Part II, Table II), and display information on the inversion errors, the correlations coefficients (Part II, Table B), the χ^2 values, the distribution of the v_i (see below), among others.

4.1 Compacting the set of data

4.1.1 Pre-averaging

Two or more measurements of the same physical quantities can be replaced without loss of information by their average value and precision, reducing thus the system of equations to be treated. By extending this procedure, we consider *parallel* data: reaction data occur that give values essentially for the mass difference of the same two nuclides, except in rare cases where the precision is comparable to that in the masses of the reaction particles. Example: $^{14}\text{C}(^7\text{Li}, ^7\text{Be})^{14}\text{B}$ and $^{14}\text{C}(^{14}\text{C}, ^{14}\text{N})^{14}\text{B}$; or $^{22}\text{Ne}(t, ^3\text{He})^{22}\text{F}$ and $^{22}\text{Ne}(^7\text{Li}, ^7\text{Be})^{22}\text{F}$. For this kind of case, one of the two links is replaced by an equivalent value for the other, and the pre-averaging procedure works as for the same quantities.

If the Q data to be pre-averaged are strongly conflicting, i.e. if the consistency factor (or Birge ratio) $\chi_n = \sqrt{\chi^2/(Q-1)}$ resulting in the calculation of the pre-average is greater than 2.5, the (internal) precision σ_i in the average is multiplied by the Birge ratio ($\sigma_e = \sigma_i \chi_n$). The quantity σ_e is often called the ‘external error’. Unweighted average is also used in some cases, where the corresponding remarks are given in Table I. The pre-averaged data with largest discrepancies are listed in Table C. Most of these have already been displayed in the similar table in AME2016 [2]. Only one new item, namely $^{12}\text{O}(2p)^{10}\text{C}$, is added in the current table due to the new input data. Six items shown in AME2016 are now removed and instead listed in Table D, where the resolution of the discrepancies is given.

Table C. Worst pre-averagings. n is the number of data in the pre-average.

Item	n	χ_n	σ_e	Item	n	χ_n	σ_e
$^{186}\text{W}(n,\gamma)^{187}\text{W}$	2	2.44	0.11	$^{177}\text{Pt}(\alpha)^{173}\text{Os}$	2	2.06	6.06
$^{144}\text{Ce}(\beta^-)^{144}\text{Pr}$	2	2.44	2.18	$^{244}\text{Cf}(\alpha)^{240}\text{Cm}$	2	2.03	3.97
$^{220}\text{Fr}(\alpha)^{216}\text{At}$	2	2.34	4.66	$^{15}\text{N}(p,n)^{15}\text{O}$	2	2.03	1.28
$^{75}\text{As}(n,\gamma)^{76}\text{As}$	2	2.32	0.17	$^{58}\text{Fe}(t,p)^{60}\text{Fe}$	4	2.03	7.38
$^{110}\text{In}(\beta^+)^{110}\text{Cd}$	3	2.29	28.4	$^{204}\text{Tl}(\beta^-)^{204}\text{Pb}$	2	2.03	0.39
$^{40}\text{Cl}(\beta^-)^{40}\text{Ar}$	2	2.21	76.1	$^{106}\text{Ag}(\varepsilon)^{106}\text{Pd}$	2	1.98	6.63
$^{153}\text{Gd}(n,\gamma)^{154}\text{Gd}$	2	2.16	0.39	$^{78}\text{Se}(n,\gamma)^{79}\text{Se}$	3	1.96	0.28
$^{113}\text{Cs}(p)^{112}\text{Xe}$	3	2.10	5.03	$^{46}\text{Ca}(n,\gamma)^{47}\text{Ca}$	2	1.94	0.56
$^{27}\text{P}^i(2p)^{25}\text{Al}$	2	2.08	74.7	$^{181}\text{Pb}(\alpha)^{177}\text{Hg}$	3	1.93	15.2
$^{12}\text{O}(2p)^{10}\text{C}$	2	2.08	26.8	$^{145}\text{Sm}(\varepsilon)^{145}\text{Pm}$	2	1.92	7.92
$^{204}\text{Rn}-^{208}\text{Pb}_{0.981}$	2	2.06	18.2	$^{234}\text{Th}(\beta^-)^{234}\text{Pa}^m$	3	1.90	2.10

In the present evaluation, we have replaced 2980 data by 1189 averages. As much as 21.8% of which have values of χ_n (Birge ratio) beyond unity, 1.34% beyond two and none beyond 3, giving an overall very satisfactory distribution for our treatment. As a matter of fact, in a complex system like the one here, many values of χ_n beyond 1 or 2 are expected to exist, and if the uncertainties were multiplied by χ_n in all these cases, the χ^2 -test on the total adjustment would have been invalidated. This explains the choice we made here of a rather high threshold ($\chi_n^0 = 2.5$), compared e.g. to $\chi_n^0 = 1$ used in a different context by the Particle Data Group [19], for departing from the rule of ‘internal error’ of the weighted average.

Besides the computer-automated pre-averaging, we found it convenient, in some β^+ -decay cases, to combine results stemming from various capture ratios in an average. These cases are $^{109}\text{Cd}(\varepsilon)^{109}\text{Ag}$ (average of 3 data), $^{139}\text{Ce}(\varepsilon)^{139}\text{La}$ (average of 10) and $^{195}\text{Au}(\varepsilon)^{195}\text{Pt}$ (5 results), and they are detailed in Table I. Four more cases (^{147}Tb , ^{152}Ho , ^{166}Yb and ^{207}Bi) occur in our list, but they carry no weight and are labeled with ‘U’ in Table I.

4.1.2 Used policies in treating parallel data

In averaging β^- (α^-) decay energies derived from branches observed in the same experiment, to or from different levels in the decay of a given nuclide, the uncertainty we use for further evaluation is not the one resulting from the weighted average adjustment, but instead we use the smallest experimental one. In this way, we avoid decreasing artificially the part of the uncertainty that is not due to statistics. In some cases, however, when it is obvious that the uncertainty is dominated by weak statistics, we do not follow the above rule (e.g. $^{23}\text{Al}^i(p)^{22}\text{Mg}$ of [1997BI04]).

Some quantities have been reported more than once by the same group. If the results are obtained by the

same method in different experiments and are published in regular refereed journals, usually only the most recent one is used in the calculation. There are two reasons for this policy. The first is that one might expect that the authors, who believe their two results are of the same quality, would have averaged them in their latest publication. The second is that if we accept and average the two results, we would have no control on the part of the uncertainty that is not due to statistics. Our policy is different when the newer result is published in a secondary reference (not refereed abstract, preprint, private communication, conference, thesis or annual report). In such cases, the older result is used in the calculations, except when the newer one is an update of the previous value. In some case, when the systematic uncertainties are discussed explicitly, all the independent results are used.

4.1.3 Insignificant data

Another feature to increase the meaning of the final χ^2 in the least-squares procedure, is to not use data with weights at least a factor 10 smaller than the combination of *all* other data giving the same result. Such data were labeled with ‘U’ in the list of input data; comparison with the output values allows to check our judgment. Until the AME2003 evaluation, our policy was not to print data labeled ‘U’ if they already appeared in one of our previous tables, reducing thus the size of the table of data to be printed. This policy has been changed since AME2012 [24, 25], and we try as much as possible to give all relevant data, also including insignificant ones. The reason for this is that conflicts might appear amongst recent results, and access to older ones might shed some light on evaluating the new data.

Table D. Worst pre-averagings shown in AME2016 and removed in the current work, together with the solutions.

Item	n	χ_n	σ_e	solution
$^{146}\text{Ba}(\beta^-)^{146}\text{La}$	2	2.24	107.4	new data available for ^{146}Ba
$^{219}\text{U}(\alpha)^{215}\text{Th}$	2	2.18	38.5	new data available
$^{36}\text{S}(^{11}\text{B}, ^{13}\text{N})^{34}\text{Si}$	3	2.14	32.4	new data available for ^{34}Si
$^{223}\text{Pa}(\alpha)^{219}\text{Ac}$	2	2.09	10.0	uncertainty increased for ambiguous transition
$^{278}\text{Mt}(\alpha)^{274}\text{Bh}$	3	1.98	43.6	uncertainty increased for ambiguous transition
$^{167}\text{Os}(\alpha)^{163}\text{W}$	4	1.98	3.50	uncertainty increased for ambiguous transition

4.2 Separating secondary data

In Section 3, while examining the diagrams of connections (Fig. 1), we noticed that, whereas the masses of *secondary* nuclides can be determined uniquely from the chain of secondary connections going down to a *primary* nuclide, only the latter see the complex entanglement that necessitated the use of the least-squares method.

In terms of equations and parameters, we consider that if, in a collection of equations to be treated with the least-squares method, a parameter occurs in only one equation, removing this equation and this parameter will not affect the result of the fit for all other data. Thus, we can redefine more precisely what was called *secondary* in Section 3: the parameter above is a *secondary* parameter (or mass) and its related equation is a *secondary* equation. After the reduced set has been solved, then the *secondary* equation can be used to determine the final value and uncertainty for that particular *secondary* parameter. The equations and parameters remaining after taking out all secondaries are called *primary*.

Therefore, only the system of *primary* data is overdetermined, and thus will be improved in the adjustment, so that each *primary* nuclide will benefit from all the available information. *Secondary* data will remain unchanged; they do not contribute to χ^2 . The diagrams in Fig. 1 show, that many *secondary* data exist. Thus, taking them out simplifies considerably the system. More importantly, if a better value is found for a *secondary* datum, the mass of the *secondary* nuclide can easily be improved. The procedure is more complicated for new *primary* data.

We define DEGREES for *secondary* nuclides and *secondary* data. They reflect their distances along the chains connecting them to the network of primaries. The first secondary nuclide connected to a primary one will be a nuclide of degree 2; and the connecting datum will be a datum of degree 2 as well. Degree 1 is for primary nuclides and data. Degrees for secondary nuclides and data range from 2 to 17. It is the nuclide ^{291}Lv that has the highest degree number 17. Its mass is determined through a long α chain, although many of them are just

estimates. In Table I, the degree of data is indicated in column ‘Dg’. In the table of atomic masses (Part II, Table I), each *secondary* nuclide is marked with a label in column ‘Orig.’ indicating from which other nuclide its mass value is determined. The word *primary* used for these nuclides and for the data connecting them does not mean that they are more important than the others, but only that they are subject to the least-squares treatment above. The labels *primary* and *secondary* are not intrinsic properties of data or nuclides. They may change from primary to secondary or vice versa when other information becomes available.

To summarize, separating secondary nuclides and secondary data from primaries significantly reduces the size of the system that will be treated by the least-squares method, and allows better insight into the relations between data and masses. After treatment of the primary data alone, the adjusted masses for primary nuclides can be easily combined with the secondary data to yield masses of secondary nuclides.

4.3 Least-squares method

Each piece of data has a value $q_i \pm dq_i$ with the accuracy dq_i (one standard deviation) and makes a relation between two, three or four masses with unknown values m_μ . An overdetermined system of Q data to M masses ($Q > M$) can be represented by a system of Q linear equations with M parameters:

$$\sum_{\mu=1}^M k_i^\mu m_\mu = q_i \pm dq_i, \quad (2)$$

e.g. for a nuclear reaction $A(a,b)B$ requiring an energy q_i to occur, the energy balance is written:

$$m_A + m_a - m_b - m_B = q_i \pm dq_i. \quad (3)$$

Thus, $k_i^A = +1$, $k_i^a = +1$, $k_i^b = -1$ and $k_i^B = -1$.

In matrix notation, $\mathbf{K}$ being the (Q, M) matrix of coefficients, Eq. 2 is written: $\mathbf{K}|m\rangle = |q\rangle$. Elements of matrix $\mathbf{K}$ are almost all null: e.g. for $A(a,b)B$, Eq. 3 yields a line of $\mathbf{K}$ with only four non-zero elements.

We define the diagonal weight matrix $\mathbf{W}$ by its elements $w_i^i = 1/(dq_i dq_i)$. The solution of the least-squares method leads to a very simple construction:

$${}^t\mathbf{KWK}|m\rangle = {}^t\mathbf{KW}|q\rangle. \quad (4)$$

The NORMAL matrix $\mathbf{A} = {}^t\mathbf{KWK}$ is a square matrix of order M , positive-definite, symmetric and regular and hence invertible [26]. Thus the vector $|\bar{m}\rangle$ for the adjusted masses is:

$$|\bar{m}\rangle = \mathbf{A}^{-1} {}^t\mathbf{KW}|q\rangle \quad \text{or} \quad |\bar{m}\rangle = \mathbf{R}|q\rangle. \quad (5)$$

The rectangular (M, Q) matrix $\mathbf{R}$ is called the RESPONSE matrix.

The diagonal elements of $\mathbf{A}^{-1}$ are the squared errors on the adjusted masses, and the non-diagonal ones $(a^{-1})_{\mu\nu}^{-1}$ are the coefficients for the correlations between masses m_μ and m_ν . Values for correlation coefficients for the most precise nuclides are given in Table B of Part II. Starting from AME2012 [24, 25], we now give on the website of the AMDC [27] the full list of correlation coefficients, allowing any user to perform exact calculation of any combination of masses.

One of the most powerful tools in the least-squares calculation described above is the flow-of-information matrix, discovered in 1984 [28]. This matrix allows to trace back the contribution of each individual piece of datum to each of the parameters (here the atomic masses). The AME uses this method since 1993.

The flow-of-information matrix $\mathbf{F}$ is defined as follows: $\mathbf{K}$, the matrix of coefficients, is a rectangular (Q, M) matrix. The transpose of the response matrix ${}^t\mathbf{R}$ is also a (Q, M) rectangular one. The (i, μ) element of $\mathbf{F}$ is defined as the product of the corresponding elements of ${}^t\mathbf{R}$ and of $\mathbf{K}$. In Ref. [28], it is demonstrated that such an element represents the “influence” of datum i on parameter (mass) m_μ . A column of $\mathbf{F}$ thus represents all the contributions brought by all data to a given mass m_μ , and a line of $\mathbf{F}$ represents all the influences given by a single piece of data. The sum of influences along a line is the “significance” of that datum. It has also been proven [28] that the influences and significances have all the expected properties, namely that the sum of all the influences on a given mass (along a column) is unity, that the significance of a datum is always less than unity and that it always decreases when new data are added. The significance defined in this way is exactly the quantity obtained by squaring the ratio of the uncertainty on the adjusted value over that of the input one, which was the recipe used before the discovery of the $\mathbf{F}$ matrix to calculate the relative importance of data.

A simple interpretation of influences and significances can be obtained in calculating, from the adjusted masses and Eq. 2, the adjusted data:

$$|\bar{q}\rangle = \mathbf{KR}|q\rangle. \quad (6)$$

The i^{th} diagonal element of $\mathbf{KR}$ represents then the contribution of datum i to the determination of $\bar{q}_i$ (same datum): this quantity is exactly what is called above the *significance* of datum i . This i^{th} diagonal element of $\mathbf{KR}$ is the sum of the products of line i of $\mathbf{K}$ and column i of $\mathbf{R}$. The individual terms in this sum are precisely the *influences* defined above.

The flow-of-information matrix $\mathbf{F}$, provides thus insight on how the information from datum i flows into each of the masses m_μ .

The flow-of-information matrix cannot be given in full in a printed table. It can be observed along lines, displaying for each datum, the nuclides influenced by this datum and the values of these *influences*. It can be observed also along columns to display for each primary mass all contributing data with their *influence* on that mass.

The first display is partly given in the table of input data (Table I) in column ‘Signf.’ for the *significance* of primary data and ‘Main infl.’ for the largest *influence*. Since in the large majority of cases only two nuclides are concerned in each piece of data, the second largest *influence* could easily be deduced. It is therefore not felt necessary to give a table of all *influences* for each primary datum.

The second display is given in Part II, Table II for the up to three most important data with their *influence* in the determination of each primary mass.

4.4 Consistency of data

The system of equations being largely over-determined ($Q \gg M$) offers the evaluator several interesting possibilities to examine and judge the data. One might for example examine all data for which the adjusted values deviate significantly from the input ones. This helps to locate erroneous pieces of information. One could also examine a group of data in one experiment and check if the uncertainties assigned to them in the experimental paper were not underestimated.

If the precisions dq_i assigned to the data q_i were indeed all accurate, the normalized deviations v_i between adjusted $\bar{q}_i$ (Eq. 6) and input q_i data, $v_i = (\bar{q}_i - q_i)/dq_i$, would be distributed as a Gaussian function of standard deviation $\sigma = 1$, and would make χ^2 :

$$\chi^2 = \sum_{i=1}^Q \left(\frac{\bar{q}_i - q_i}{dq_i} \right)^2 \quad \text{or} \quad \chi^2 = \sum_{i=1}^Q v_i^2 \quad (7)$$

equal to $Q - M$, the number of degrees of freedom, with a standard deviation of $\sqrt{2(Q - M)}$.

One can define as above the NORMALIZED CHI, χ_n (or ‘consistency factor’ or ‘Birge ratio’): $\chi_n = \sqrt{\chi^2/(Q - M)}$ for which the expected value is $1 \pm 1/\sqrt{2(Q - M)}$.

Another quantity of interest for the evaluator is the PARTIAL CONSISTENCY FACTOR, χ_n^p , defined for a (homogeneous) group of p data as:

$$\chi_n^p = \sqrt{\frac{Q}{Q-M} \frac{1}{p} \sum_{i=1}^p v_i^2}. \quad (8)$$

The χ_n^p reduces to χ_n if the sum is taken over all the input data.

One can consider groups of data related to a given laboratory and with a given method of measurement and examine the χ_n^p of each of them. Since various methods are used in mass determination, possible systematic errors associated with a specific method, that are beyond control in experiments and not accounted for in the reported uncertainties, can be observed by the mass evaluators when considering the mass adjustment as a whole. There are presently 278 groups of data in Table I (among which 185 have at least one measurement used in determining the masses), identified in column ‘Lab’. A high value of χ_n^p might be a warning on the validity of the considered group of data within the reported uncertainties. We used such analyses to locate questionable groups of data. Each group of mass-spectrometric data is assigned a factor F according to its partial consistency factor χ_n^p , due to the fact that its statistical uncertainties and its internal systematic error alone do not reflect the real experimental situation. From comparison to all other data and more specially to combination of reaction and decay energy measurements, we have assigned factors F of 1.5, 2.5 or 4.0 to the different labs. Example: the group of data H25 has been assigned $F=2.5$, this means that the total uncertainty assigned to ^{155}Gd ^{35}Cl – ^{153}Eu ^{37}Cl is $2.5 \times 2.4 \mu\text{u}$. The weight of this piece of data is then very low, compared to $0.79 \mu\text{u}$ derived from all other data. This is why it is labeled “U”.

As in our previous works, we have estimated the average accuracy for 185 groups of data used in the evaluation that were related to a given laboratory and a specific method of measurement, by calculating their partial consistency factors χ_n^p . As much as 98 groups have χ_n^p larger than unity, and 2 groups larger than 2.

4.5 Treatment of undependable data

The important interdependence of most data, as illustrated by the connection diagrams (Figs. 1a–1j), allows local and global consistency tests. These can indicate that something may be wrong with the input values. We follow the policy of checking all significant data that differ by more than two (sometimes 1.5) standard deviations from the adjusted values. Fairly often, examination of the experimental results shows that a correction is necessary. Possible reasons could be that a particular decay has been assigned to a wrong final level or that a

reported decay energy belongs to an isomer, rather than to a ground state, or even that the mass number assigned to a decay was incorrect. In such cases, the values are corrected and remarks are added below the corresponding A -group of data in Table I, in order to explain the reasons for the corrections.

It also happens that a careful examination of a particular paper would lead to serious doubts about the validity of the results within the reported precision, but could not permit making a specific correction. Doubts can also be expressed by the authors themselves. The results are given in Table I and compared with the adjusted values. They are labeled ‘F’, and not used in the final adjustment, but always followed by a comment to explain the reason for this label. The reader may observe that in several cases the difference between the experimental and adjusted values is small compared to the experimental uncertainty: this does not disprove the correctness of the label ‘F’ assignment.

It happens quite often that two (or more) pieces of data are discrepant, leading to important contribution to the χ^2 . A detailed examination of the papers may not allow correction or rejection, indicating that at least the results in one of them could not be trusted within the given uncertainties. Then, based on past experience, we use in the calculations the value that seems to be the most reliable, while the other is labeled ‘B’, if published in a regular refereed journal, or ‘C’ otherwise.

In some cases, detailed analysis of strongly conflicting data could not lead to reasons to assume that one of them is more dependable than the others or could not lead to a rejection of a particular data entry. Also, bad agreement with other data is not the only reason to doubt the correctness of the reported data. As in previous AME, and as explained above (see Section 4), we made use of the property of regularity of the surface of masses in making a choice, as well as in further checks on the other data.

When one mass is only determined by one single experimental result, but the result is strongly doubted, the policy defined for ‘irregular masses’ with ‘D’-label assignment may apply and discussed in section 5.1.

We do not accept experimental results if information on other quantities (e.g. half-lives), derived in the same experiment and for the same nuclide, were in strong contradiction with well established values.

Data with labels ‘F’, ‘B’, ‘C’ or ‘D’ are not used in the calculations. The least-squares calculations are then done in iterations. This recipe is unavoidably subjective, but has proven to be efficient through the agreement of these estimates with newly measured masses in a great majority of cases.

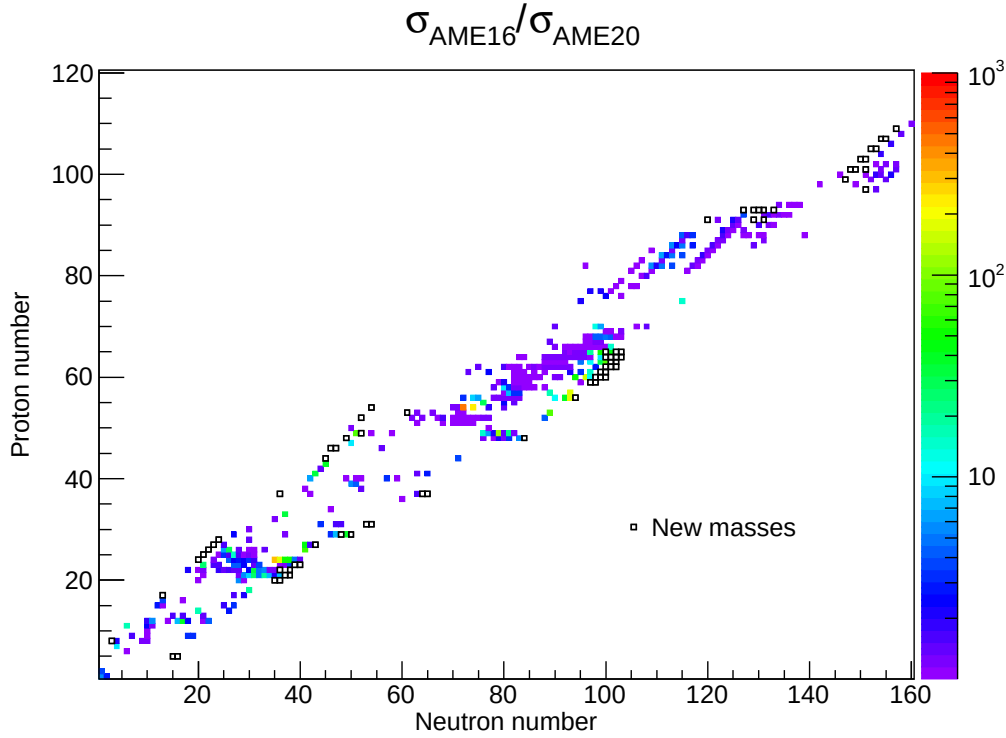


Figure 2. Ratios of mass uncertainties in AME2016 and AME2020 for nuclides whose mass precisions have been improved. The new experimental masses in AME2020 are shown as the black squares.

4.6 Results of the calculation

In this evaluation, we have 13812 experimental data of which 6023 are labeled ‘U’ (see above), 997 are labeled ‘O’ (old result from same group) and 951 are not accepted and labeled ‘B’, ‘C’, ‘D’ or ‘F’ (respectively 533, 161, 81 and 176 items). In the calculation we have thus 5841 valid input data, compressed to 4050 in the pre-averaging procedure. Separating secondary data, leaves a system of 2201 primary data, representing 1135 primary reactions and decays, and 1066 primary mass-spectrometric measurements. These primary data are used to determine 1304 primary masses (^{12}C not included). Thus, we have to solve a system of 2201 equations with 1304 parameters. Theoretically, the expectation value for χ^2 should be 897 ± 42 (and the theoretical $\chi_n = 1 \pm 0.024$). The total χ^2 of the adjustment is actually 938 ($\chi_n = 1.023$), thus showing that the ensemble of evaluated data was of good quality, and that the adopted criteria of selection and rejection of input data were adequate.

In the present AME, there are 2550 masses for ground state and 406 masses for excited isomers derived from experimental results. Among the 2550 experimental ground state masses, 122 nuclides have a precision better than 0.1 keV, 402 better than 1 keV, 1532 better than 10 keV, 2407 better than 100 keV (respectively 107, 366,

1499 and 2340 in AME2016). There are 143 nuclides known with uncertainties larger than or equal to 100 keV (158 in AME2016).

The mass precision has been improved for 427 nuclides in comparison with AME2016, as illustrated in Fig. 2, where the ratios of their mass uncertainties in AME2016 and AME2020 are displayed in a color code. Meanwhile, masses of 74 nuclides, which were unknown experimentally in AME2016, are measured and provided in the current mass table, as shown with black squares in Fig. 2.

5 Mass Estimation

We recommend the estimated mass values for some nuclides whose mass is not determined experimentally as well as for nuclides where the experimental value is seriously questioned. The values are estimated based on the trends from the mass surface (TMS), with the assumption that the mass surface is generally smooth. An interactive graphical program is used [29] that allows a simultaneous observation of four graphs, either for the particle energies or for the differences compared with a mass model, as a function of any of the variables N , Z , A , $N-Z$ or $N-2Z$, while drawing iso-lines (lines connecting nuclides having same parameter) of any of these quantities. The mass of a nuclide can be modified or created in

Table E. Experimental data replaced by estimated values in AME2016, and the adjusted values in AME2020. The unit for the type AEI-u is μu , while for others is keV.

Item	Experimental data		In AME2016		In AME2020	
$^{70}\text{Co-u}$	-50370	320	-50060#	320#	-49947	12
$^{101}\text{Rb}(\beta^-)^{101}\text{Sr}$	11810	110	12480#	200#	12757	22
$^{150}\text{Ba-u}$	-55309	371	-53570#	320#	-53559	6
$^{220}\text{Pa}(\alpha)^{216}\text{Ac}$	9829	50	9650#	50#	9704	11

any view and we can determine how much freedom is left in setting a value for this mass. At the same time, interdependence through secondary connections are taken into account. In cases where two tendencies may alternate, following the parity of the proton or of the neutron numbers, one of the parities may be deselected.

The replaced values for data yielding the ‘irregular masses’ as well as the ‘estimated unknown masses’ (see below) are thus derived by observing the continuity property in several views of the mass surface, with all the consequences due to connections to masses in the same chain. The estimated values are also compared with the predictions of available mass models. All values not derived from experimental data alone, have been clearly marked with the hashmark ‘#’ symbol in all tables, here and in Part II. Since publication of AME1983 [30], the uncertainties are also given for the estimated data derived from TMS. These uncertainties are obtained by considering several graphs of TMS with a guess on how much the estimated mass may change without the extrapolated surface looking too much distorted.

5.1 Irregular mass values

As mentioned in section 4.5, we label some questioned results as ‘D’ and do not use them in mass calculation. When a single mass deviates significantly from regularity with no similar pattern for nuclides with same N or with same Z values, then the correctness of the data determining this mass may be questioned. Following the policy defined in AME1995 [31], in order to avoid strongly fluctuating plots, we replace seriously *irregular* masses by values derived from trends from the mass surface (TMS), when they are derived from a unique mass relation, without verification by a different experimental method. In most of the cases only one measurement is reported for such unique mass relation. But sometimes there are two or more measurements that are treated the same way, since use of the same method and the same type of relation may well lead to the same systematic uncertainty.

In the last evaluation AME2016, there were 33 such physical quantities, as listed in Table C of Ref.[2]. In that list, 4 items have been superseded by new experimental data in this work. The new results available justify our

earlier treatment, as shown in Table E. The rest 29 irregular experimental values are treated similarly as in AME2016. Meanwhile, we use the estimated values in a larger extent in the present evaluation, by adding 48 more items in such a list in the present evaluation. Thus there are 77 such irregular experimental values replaced by estimated ones, as shown in Table F. Such data, as well as other local irregularities that can be observed in the figures in Part II could be considered as incentive to remeasure the masses of the involved nuclides, preferably by different methods, in order to remove any doubt and possibly point out true irregularities due to physical properties.

Because of the treatment, masses of 22 nuclides that were deemed to be known experimentally in AME2016, are now replaced with the estimated values. 17 nuclides among them, namely, ^{29}Cl , ^{38}Al , ^{41}Si , ^{43}P , ^{45}S , ^{59}V , ^{68}Fe , ^{103}Sn , ^{105}Y , ^{106}Zr , ^{107}Zr , ^{115}Tc , ^{138}Sb , ^{184}Bi , ^{188}Ta , ^{189}W and ^{228}Np , are displayed in Table F. Masses of 5 nuclides that are not listed in Table F, namely, ^{104}Sb , ^{107}Te , ^{108}I , ^{111}Xe , ^{112}Cs , are also experimentally unknown because they are connected to ^{12}C through ^{103}Sn , whose mass is now an estimated value. Other irregular experimental data do not change the number of known masses, because they involve only nuclides with unknown mass, i.e., none of them is connected to ^{12}C eventually. The item of $^{130}\text{Eu}(\text{p})^{129}\text{Sm}$ shown in Table F is such an example. Six experimental results for 2-proton separation energies are also listed in Table F as it seems that they distort the mass surface in neighboring regions. The mass surface looks more smooth by using the estimated values. It is noted that the experimental proton-decay energies of five proton emitters, namely ^{131}Eu , ^{135}Tb , ^{140}Ho , ^{146}Tm and ^{150}Lu , are about 150-300 keV smaller than expected values from TMS. It is likely that this is due to the effect of the Thomas-Ehrmann shift, which makes proton-emitting nuclides more bound than suggested by TMS [32]. Nuclides at the proton dripline show different wave functions than its neighboring particle-bound nuclides and the assumption of regularity might not be applicable to these cases. Thus, we decided not to replace these experimental results.

As shown in Table F, the Q_α value of ^{284}Cn from [2017Ka66] is replaced by the estimated value which is lower by 200 keV. The relevant results in [2017Ka66] can

Table F. Experimental data that we recommend replaced by values derived from TMS in AME2020. The unit for the type $^4\text{El-u}$ is μu , while for others is keV.

Item	Reference	Experimental value		Recommended value	
$^{28}\text{Cl(p)}^{27}\text{S}$	18Mu18	1600	80	3190#	300#
$^{29}\text{Cl(p)}^{28}\text{S}$	18Xu04	1800	100	2260#	100#
$^{29}\text{Ar(2p)}^{27}\text{S}$	18Mu18	5500	100	5900#	180#
$^{30}\text{Ar(2p)}^{28}\text{S}$	18Xu04	2420	80	3420#	80#
$^{31}\text{K-u}$	19Ko18	32783	275	35320#	322#
$^{38}\text{Al-u}$	07Ju03	17402	268	17681#	161#
$^{39}\text{Al-u}$	07Ju03	21653	676	23070#	322#
$^{41}\text{Si-u}$	07Ju03	13011	397	13698#	322#
$^{42}\text{Si-u}$	07Ju03	16275	623	17896#	322#
$^{43}\text{P-u}$	07Ju03	5024	397	5411#	322#
$^{44}\text{P-u}$	07Ju03	10070	966	11927#	429#
$^{45}\text{S-u}$	07Ju03	-4283	741	-3586#	322#
$^{45}\text{Fe(2p)}^{43}\text{Cr}$	18Xu04	1154	16	1800#	200#
$^{47}\text{Cl-u}$	07Ju03	-9576	1074	-10285#	215#
$^{48}\text{Ni(2p)}^{46}\text{Fe}$	14Po05	1290	40	2390#	300#
$^{49}\text{Ar-u}$	20Me06	-17499	1180	-18454#	429#
$^{54}\text{Zn(2p)}^{52}\text{Ni}$	05Bl15	1480	20	2280#	200#
$^{57}\text{Ca-u}$	18Mi08	-7912	1063	-7171#	429#
$^{59}\text{Ti-u}$	20Me06	-27075	290	-27783#	322#
$^{59}\text{Ti-u}$	20Mi13	-27053	258	-27783#	322#
$^{60}\text{Sc-u}$	20Mi13	-3811	1116	-4885#	537#
$^{61}\text{Ti-u}$	20Mi13	-17370	483	-17005#	322#
$^{62}\text{Ti-u}$	20Mi13	-14278	483	-12700#	429#
$^{64}\text{V-u}$	20Mi13	-17917	548	-17520#	429#
$^{65}\text{Cr-u}$	20Me06	-29286	837	-30392#	215#
$^{67}\text{Mn-u}$	20Me06	-36458	354	-36050#	215#
$^{67}\text{Kr(2p)}^{65}\text{Se}$	16Go26	1480	20	2280#	200#
$^{68}\text{Mn-u}$	20Me06	-29748	1406	-31047#	322#
$^{68}\text{Fe-u}$	20Me06	-47622	344	-47408#	215#
$^{69}\text{Fe-u}$	20Me06	-43232	429	-42082#	215#
$^{70}\text{Fe-u}$	20Me06	-40483	526	-39603#	322#
$^{74}\text{Ni-u}$	11Es06	-52830	1060	-52282#	215#
$^{80}\text{Zr-u}$	98Is06	-59600	1600	-58358#	322#
$^{90}\text{Rh}(\epsilon)^{90}\text{Ru}$	19Pa16	13410	1330	13250#	200#
$^{91}\text{Pd}(\epsilon)^{91}\text{Rh}$	18Pa20	11800	2200	12400#	300#
$^{94}\text{Br-u}$	16Kn03	-50242	429	-51047#	215#
$^{94}\text{Ag}(\epsilon)^{94}\text{Pd}$	19Pa16	13400	650	13700#	400#
$^{95}\text{Cd}(\epsilon)^{95}\text{Ag}$	18Pa20	10200	1700	12750#	400#
$^{96}\text{Cd}(\epsilon)^{96}\text{Ag}$	19Pa16	8480	460	8940#	400#
$^{98}\text{In}(\epsilon)^{98}\text{Cd}$	19Pa16	12930	400	13730#	300#
$^{99}\text{Sn}(\epsilon)^{99}\text{In}$	18Pa20	14700	3600	13400#	500#
$^{103}\text{Sn}(\beta^-)^{103}\text{In}$	05Ka34	7660	70	7540#	100#

Table F. Experimental data that we recommend replaced by values derived from TMS in AME2020. The unit for the type ^AEl-u is μu , while for others is keV. Continued

Item	Reference	Experimental value	Recommended value
¹⁰⁵ Y-u	16Kn03	-55041	574 -54452# 429#
¹⁰⁶ Zr-u	16Kn03	-62856	186 -63070# 215#
¹⁰⁷ Zr-u	16Kn03	-58379	482 -57993# 322#
¹⁰⁷ Zr-u	16Kn03	-58379	482 -57993# 322#
¹¹³ Mo-u	16Kn03	-57317	337 -56522# 322#
¹¹⁵ Tc-u	16Kn03	-60462	339 -59900# 210#
¹¹⁶ Cs($\epsilon\alpha$) ¹¹² Te	77Bo28	12300	400 13100# 100#
¹¹⁶ Cs($\epsilon\alpha$) ¹¹² Te	76Jo.A	12400	900 13100# 100#
¹¹⁸ Ru-u	16Kn03	-61879	196 -61192# 215#
¹¹⁸ In(β^-) ¹¹⁸ Sn	64Ka10	4270	100 4525# 50#
¹²⁰ In(β^-) ¹²⁰ Sn	64Ka10	5280	200 5420# 50#
¹²⁰ In(β^-) ¹²⁰ Sn	78Al18	5340	170 5420# 50#
¹²⁴ Pd-u	16Kn03	-64617	399 -62695# 322#
¹²⁶ Ag-u	16Kn03	-65926	329 -65186# 215#
¹²⁹ Nd(ϵp) ¹²⁸ Ce	78Bo.A	5300	300 5870# 200#
¹³⁰ Eu(p) ¹²⁹ Sm	04Da04	1028.0	15.0 1528# 200#
¹³⁵ Pm(β^+) ¹³⁵ Nd	95Ve08	6040	150 6386# 50#
¹³⁸ Sb-u	16Kn03	-58208	457 -58562# 322#
¹⁴² Dy(β^+) ¹⁴² Tb	91Fi03	7100	200 6440# 200#
¹⁴³ I-u	16Kn03	-53849	495 -54461# 215#
¹⁵⁴ Ce-u	16Kn03	-56404	619 -56060# 215#
¹⁶⁶ Eu(β^-) ¹⁶⁶ Gd	14Ha38	7322	300 7422# 100#
¹⁸⁴ Bi(α) ¹⁸⁰ Tl	03An27	8024.8	50 8220# 100#
¹⁸⁸ Ta-u	13Sh30	-36084	59 -36329# 215#
¹⁸⁹ W(β^-) ¹⁸⁹ Re	65Ka07	2500	200 2170# 200#
²²⁸ Np(α) ²²⁴ Pa	03Ni10	7308.5	36. 7538# 100#
²³⁵ Cm(α) ²³¹ Pu	20Kh.A	7131.6	20.0 7280# 100#
²³⁶ Bk(α) ²³² Am	20Po07	7447	14 7797# 200#
²⁶⁸ Hs(α) ²⁶⁴ Sg	10Ni14	9622.9	16. 9760# 100#
²⁷⁰ Db(α) ²⁶⁶ Lr	14Kh04	8019.0	30.5 8310# 200#
²⁷⁰ Mt(α) ²⁶⁶ Bh	12Mo25	10414.5	71. 10180# 100#
²⁸¹ Rg(α) ²⁷⁷ Mt	17Og01	9410	50 9904# 400#
²⁸² Mt(α) ²⁷⁸ Bh	16Ho09	8960.	180. 8660# 200#
²⁸⁴ Cn(α) ²⁸⁰ Ds	17Ka66	9869.2	192.7 9670# 150#
²⁸⁶ Rg(α) ²⁸² Mt	16Ho09	8793.	45. 8630# 100#
²⁹⁰ Nh(α) ²⁸⁶ Rg	16Ho09	9846.	45. 9320# 100#

be alternatively interpreted. Recently, the decay from ²⁸⁴Cn has been firmly established and the decay energy is reported to be 9.46(1) MeV [33]. This result supports the general trend of a smaller value, but is even smaller by 210 keV than the estimated value. It is too late to include the new results but we mention it nonetheless.

We replace the irregular experimental values by the ones we recommend in all tables and graphs in AME2020. Data which are not used following this policy, can be easily located in Table I where they are flagged 'D' and always accompanied by a comment explaining in which direction the value has been changed and by what amount.

5.2 Estimates for unknown masses

Estimates for unknown masses are also made with use of trends from the mass surface, by demanding that all graphs should be as smooth as possible, except where they are expected to show the effects of shell closures or nuclear deformations effects. Therefore, we warn the user of our tables that the present extrapolations, based on trends in known masses, will be wrong if unsuspected new regions of deformation or (semi-) magic numbers occur.

In addition to the rather severe constraints imposed by the requirement of simultaneous REGULARITY of all graphs, all available experimental information is considered in the estimates. Knowledge of stability or instability against particle emission, or limits on proton or α emission, contribute to upper or lower limits on the separation energies. For example, the proton separation energy S_p of ^{64}As is estimated to be from its half-life, and S_p of ^{97}In is estimated based on the experimental information from [2019Pa20].

For proton-rich nuclides with $N < Z$, mass estimates can be obtained from the charge symmetry. This feature gives a relation between masses of isobars around the one with $N = Z$.

Another often good estimate can be obtained from the observation that masses of nuclidic states belonging to an isobaric multiplet are represented quite accurately by a quadratic equation of the charge number Z (or of the third component of the isospin, $T_3 = \frac{1}{2}(N - Z)$): the Isobaric Multiplet Mass Equation (IMME) [34]. Use of this relation is attractive since it uses experimental information of the isobaric analogues states. Up to the AME1983, the IMME was used for deriving mass values for nuclides where little experimental information was available. This policy was questioned with respect to the correctness in stating as ‘experimental’ a quantity that was derived by combination with a calculation. Since AME1993, it was decided not to present any IMME-derived mass values in our evaluation, but rather use the IMME as a guideline when estimating masses of unknown nuclides. We continue this policy here, and do not replace experimental values by one estimated from IMME, even if more precise. Typical examples are ^{28}S and ^{40}Ti , for which the IMME predicts masses with precisions of respectively 24 keV and 22 keV, whereas the experimental masses are known both with 160 keV precision, from double charge-exchange reactions.

Our estimates ensure that every nuclide for which there is any experimental Q -value is connected to the main group of primary nuclides. In addition, we seek continuity of the mass surface. Therefore an estimated value is included for any nuclide if it is between two experimentally studied nuclides on a line defined by either $Z = \text{constant}$ (isotopes), $N = \text{constant}$ (isotones),

$N - Z = \text{constant}$ (isodiaspheres), or, in a few cases $N + Z = \text{constant}$ (isobars). It would have been desirable to give also estimates for all unknown nuclides that are within reach of the present accelerator and mass separator technologies. Unfortunately, such an ensemble is practically not easy to define. Instead, we estimate mass values for all nuclides for which at least one piece of experimental information is available (e.g. identification or half-life measurement or proof of instability towards proton or neutron emission). This way, the ensemble of experimental masses and estimated ones has the same contour as in the NUBASE2020 evaluation.

We add 911 data estimated from TMS, some of which are essential for linking unconnected experimental data to the network of experimentally known masses (see Figs. 1a–1j). Thus a total of 1008 masses for the ground state and 101 masses for excited isomers are deduced due to the estimated values. Combining the experimental and the estimated mass values, there are in total 3557 ground state masses in the atomic mass table (Part II, Table I).

6 Special cases

Special cases have been discussed in the AME series to highlight the issues raised in the evaluation and to call for more efforts to solve them. Some of the special cases discussed in AME2016 have been solved so we removed them from the current list. Meanwhile some new cases have been added.

6.1 Evaluation in the lightest mass region

The precise knowledge of the atomic masses of lightest nuclei is important for several fundamental tests in physics. Nowadays, Penning traps provide the most precise values for the masses of light nuclei. Local adjustments in AME2016 for the lightest masses, i.e., hydrogen ^1H , deuterium D , tritium ^3H , and helium-3 ^3He , gave rise to a consistency factor $\chi_n = 3.4$, which indicates underestimation of systematic uncertainties in some experiments. Given the limited information at that time, the results from the FSU group were adopted, while the ones from [2015Za13] were not used.

Since AME2016, significant effort has been put into mass measurements of the lightest nuclei, which allows us to examine the sources of inconsistency. The FSU group remeasured the frequency ratio of $^3\text{He}^+$ to HD . The new result agrees with the previous one in [2015My03] and the precision is improved by a factor of two, validating our choice of their results in AME2016. Later on, three more frequency ratios of H_3^+/HD^+ [2018Sm03], H_2^+/D^+ [2020Fi05], and $\text{H}_3^+/^3\text{He}^+$ [2018Sm03] were measured, which gave precise values of the mass differences for $\text{H}_2 - \text{D}$ and $\text{H}_3 - ^3\text{He}$. Note that H_2 and H_3 are the

hydrogen molecules, while in the linear equations they stand for 2 and 3 times of the hydrogen mass since the dissociation energies of the molecules have been corrected. The upper limit derived for the mass difference from [2018Sm03] is assumed to correspond to the ground state of H_3 . The new mass difference of $\text{H}_2\text{--D}$ obtained in this experiment differs from the recommended values in AME2016 by 1.9σ , and the new mass difference of $\text{H}_3\text{--}^3\text{He}$ agrees with the recommended one in AME2016 within 1σ .

LIONTRAP at Mainz uses ^{12}C as reference ions and obtains the “absolute” masses for the ions of interest. In [2017He14], the frequency ratio of H^+/C^{6+} is measured and the mass of H is deduced. This value is three times more precise than our previous adjusted value in AME2016 but differs by 3.4σ . In [2020Ra.1], the frequency ratio of $^2\text{H}^+/\text{C}^{6+}$ is measured and we can derive the mass of deuterium to be $2.014\,101\,777\,842(17)$ u (the most precise value (8 ppt)). This value differs from the recommended value in AME2016 by 2.2σ but is 7 times more precise. One should note that the experimental data from the Washington group played an important role in mass determination for H and D in AME2016, meaning there exists strong conflict between the ones from LIONTRAP and those from Washington. For a crosscheck, LIONTRAP also measured the frequency ratio of HD/C^{4+} [2020Ra.1], which results in a sum of the individual masses of H and D. Subtracting the summed masses by their reported H mass gives an independent mass of D, which agrees with the deuterium mass determined directly in the same experiment.

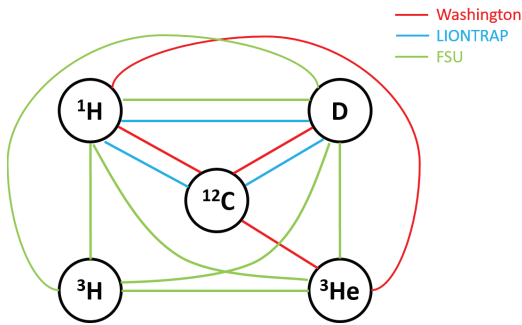


Figure 3. The connection diagram in the lightest mass region.

Fig. 3 shows the connection diagram in the lightest mass region. Note that hydrogen also connects to ^{16}O by frequency ratio $^{16}\text{O}^{8+}/\text{H}^+$ [2019He19]. Given the large inconsistency found from the Washington group, and the consistent adjustment between the FSU trap and LIONTRAP, the results from the Washington group are not used. Based on the statistical analysis, we also assigned

a factor $F = 2.5$ for the Washington group. The new assignment also affects other nuclides that are measured by the Washington group. As a result, the uncertainty for ^4He increases from 63 pu in AME2016 to 158 pu in AME2020, and ^{16}O from 173 pu to 319 pu.

6.2 The mass of ^{52}Co

The mass of ^{52}Co was measured firstly with the CSR-IMS in Lanzhou [2016Xu10], and the value was adopted in AME2016 as the only available experimental result. After the publication of AME2016, the JYFLTRAP group published their result for ^{52}Co [2017Ne05]. The two results deviate with each other by 2.8σ . In both measurements, an isomeric state of ^{52}Co was observed and can be separated fairly well from the ground state. The resolving powers of these two measurements are comparable. From the spectra in the two papers, it is difficult to identify possible problems. We keep both results in the present evaluation and appeal for new studies to solve this issue.

6.3 The Q -value for $^{99}\text{Rh}(\beta^+)^{99}\text{Ru}$

This case has been discussed in AME2012 and AME2016. The Q -value for $^{99}\text{Rh}(\beta^+)^{99}\text{Ru}$ was adjusted to be 2044(7) keV in AME2016, where the Q_{β^+} was mainly determined from the two β -decay results in Ref. [1952Sc11] and [1959To25]. A higher Q -value of 2170(30) keV was reported in the later work [1974An23], but was not used in AME due to its large uncertainty. However, it was found that the Q -value should be larger than 2059.34 keV because this state was populated in beta-decay experiments.

Both ^{99}Rh and ^{99}Ru are primary nuclides in our evaluation. The mass of ^{99}Ru is mainly determined by (n,γ) reaction to ^{100}Ru , which in turn is determined from Penning trap measurement. The uncertainty of the adjusted ^{99}Rh mass is 6.7 keV. The mass of ^{99}Rh can also be determined through the Q_{β^+} value of ^{99}Pd , which in turn is measured with a Penning trap with an uncertainty of 5.4 keV. Consequently, even if we do not use any of the experimental results from the beta-spectrum measurements of ^{99}Rh , we can still determine the Q -value of $^{99}\text{Rh}(\beta^+)^{99}\text{Ru}$ to be 2032(21) keV. If we increase the Q -value of $^{99}\text{Rh}(\beta^+)^{99}\text{Ru}$, then strong tension will be built around this region.

In AME2020, we discard the result from [1952Sc11]. The unweighted average of results from [1959To25] and [1974An23] is then used in the evaluation. The adjusted value is 2041(19) keV, still marginally smaller than the required decay energy, as discussed above.

6.4 The ^{102}Pd double-electron capture energy

In the AME2012 evaluation, we found large discrepancies between data coming from combinations of the

very precise (n, γ) reactions with β^+ , β^- and ε decay energies versus direct Penning-trap data [2011Go23]. We finally decided at that time to provisionally not use the new Penning trap result and called for more measurements in order to clarify this issue. In the AME2016 evaluation, we were aware of the experimental results for ^{102}Pd and ^{103}Pd from the ESR mass measurements through a private communication [2014Ya.A]. Their results support the SHIPTRAP datum, although with relatively large uncertainties. We therefore decided to discard two pieces of data: $^{102}\text{Rh}(\beta^-)^{102}\text{Pd}$ and $^{103}\text{Pd}(\varepsilon)^{103}\text{Rh}$, and to use the SHIPTRAP result. The local consistency was then restored.

The treatment is now further supported by the recent results from JYFLTRAP [2019Ne04]. The mass difference of ^{102}Pd and ^{102}Ru was measured using both the Time-of-Flight and the Phase-Imaging Ion Cyclotron Resonance techniques. The obtained result is around 8 times more precise and in excellent agreement with the value obtained at SHIPTRAP, verifying the reliability of the Penning trap mass spectrometry. This issue is thus considered to be resolved.

7 General information

The full content of the present issue is accessible online at the AMDC website [27]. In addition to several graphs representing the mass surface, beyond the main ones given in Part II, are available on the AMDC website.

Tables of masses (Part II, Table I) and nuclear reaction and separation energies (Part II, Table III) are available in ASCII format to simplify their input to computer programs. The headers of these files give information on the formats. The first file, named **mass.mas20**, contains the table of masses. The next two files correspond to the table of reaction and separation energies, in two parts of 6 entries each, as in Part II, Table III: **rct1.mas20** for S_{2n} , S_{2p} , Q_α , $Q_{2\beta}$, $Q_{\varepsilon p}$ and $Q_{\beta n}$ (odd pages in this issue); and **rct2.mas20** for S_n , S_p , $Q_{4\beta}$, $Q_{d,\alpha}$, $Q_{p,\alpha}$ and $Q_{n,\alpha}$ (facing even pages).

8 Acknowledgments

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work prior to publication. Continuing interest, discussions, suggestions and encouragements from D. Lunney are highly appreciated.

Appendix A Converting frequency ratios to linear equations

Primary data from Penning-trap measurement are typically given in the form of a frequency ratio, which is converted to a linear equation of atomic masses in the AME. The code of a Fortran program for converting frequency ratios to linear equations was given in AME2016 [2]. Here we just describe the principle of this method.

In the following, quantities with the subscript r describe the characteristics of the reference ion in the Penning trap. Equivalent quantities, with no subscript, describe characteristics of the ion being measured.

In AME2012 [24], the frequency ratio was described as:

$$R = \frac{f_r}{f} = \frac{\mathcal{M} - m_e q + B}{\mathcal{M}_r - m_e q_r + B_r} \frac{q_r}{q}, \quad (9)$$

This function is only valid for atoms. When molecules are involved, an extra term for the molecular binding energy should be included. Therefore in AME2016 [2], the equation was rewritten as:

$$R = \frac{f_r}{f} = \frac{\mathcal{M} - D - m_e q + B}{\mathcal{M}_r - D_r - m_e q_r + B_r} \frac{q_r}{q}, \quad (10)$$

where q is the charged state of the given ion, D is the molecular binding energy (dissociation energy), B is the electron binding energy, m_e is the mass of the electron and $\mathcal{M}$ the total atomic mass. All masses and energies are in atomic mass units u and so, $u=1$. This expression can be written in terms of the mass excess M and atomic mass number A :

$$A + M - D - m_e q + B = R \frac{q}{q_r} (A_r + M_r - D_r - m_e q_r + B_r) \quad (11)$$

or, alternatively:

$$\begin{aligned} M - R \frac{q}{q_r} M_r = m_e q (1 - R) &+ A_r \left(\frac{q}{q_r} R - \frac{A}{A_r} \right) \\ &+ R \frac{q}{q_r} (B_r - D_r) - (B - D). \end{aligned}$$

The general aim is to establish some quantity y and its associated precision dy . We define C to be a truncated, three-digit decimal approximation of the ratio A to A_r , and then we can write:

$$y = M - C M_r \quad (12)$$

and so

$$y = y_1 + y_2 + y_3 + y_4, \quad (13)$$

where

$$y_1 = M_r \left(R \frac{q}{q_r} - C \right), \quad (14)$$

$$y_2 = m_e q (1 - R), \quad (15)$$

$$y_3 = A_r \left(\frac{q}{q_r} R - \frac{A}{A_r} \right), \quad (16)$$

and

$$y_4 = R \frac{q}{q_r} (B_r - D_r) - (B - D). \quad (17)$$

To fix relative orders of magnitude, M_r is generally smaller than 0.1 u, $R - C$ is a few 10^{-4} , $(1 - R)$ is usually smaller than unity (and typically 0.2 for a 20% mass change), $R - \frac{A}{A_r}$ varies from 1 to 100×10^{-6} , $(B_r - D_r)$ is generally smaller than 0.1 μ u, and A_r is typically 100 u for atomic mass $A = 100$. The four terms y_1 , y_2 , y_3 , and y_4 take values of the order of 10 μ u, 100 μ u, 10 to 10000 μ u, and 0.1 μ u, respectively.

The associated precision dy is written:

$$dy = dy_1 + dy_2 + dy_3 + dy_4, \quad (18)$$

where

$$dy_1 = \frac{q}{q_r} M_r dR + \left(R \frac{q}{q_r} - C \right) dM_r \simeq dR \times 10^5 \mu\text{u}, \quad (19)$$

$$dy_2 = m_e q dR \simeq dR \times 10^3 \mu\text{u}, \quad (20)$$

$$dy_3 = \frac{q}{q_r} A_r dR \simeq dR \times 10^8 \mu\text{u}, \quad (21)$$

and

$$dy_4 = \frac{q}{q_r} (B_r - D_r) dR + R \frac{q}{q_r} (dB_r - dD_r) - (dB - dD) \simeq dR \times 10^{-1} \mu\text{u}. \quad (22)$$

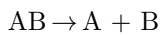
Consequently, only the 3rd term contributes significantly to the precision of the measurement, and so we write: $dy = dy_3$.

If the two frequencies are measured with a typical precision of 10^{-7} for ions at $A = 100$, then the precision on the frequency ratio R is 1.4×10^{-7} and the precision on the mass is approximatively 14 μ u.

Next we will illustrate how to calculate the molecular binding energy for the most precise Penning trap experiment.

A.1 Bond Dissociation Energy

The bond dissociation energy D is a quantity which signifies the strength of a chemical bond. The bond dissociation energy for a bond A–B, which is broken through reaction:



is defined as the standard enthalpy change at a specified temperature[35]:

$$D^\circ(\text{AB}) = \Delta H_f^\circ(\text{A}) + \Delta H_f^\circ(\text{B}) - \Delta H_f^\circ(\text{AB}), \quad (23)$$

where ΔH_f° is the standard heat of formation and its value is available for a large number of atoms and compounds on NIST Chemistry Webbook [36]. All D° values refer to the gaseous state at temperature either 0 K or 298 K. When data on the standard heat of formation is absent at 0 K, $D^\circ(\text{AB})$ will be converted to $D_{298}^\circ(\text{AB})$ by the approximation:

$$D_{298}^\circ(\text{AB}) \approx D^\circ(\text{AB}) + 3.72 \text{ kJ/mol}. \quad (24)$$

Unlike diatomic molecules which involve only one bond, polyatomic molecules have several bonds and their D 's are the sum of all the single bonds. For example, if one wants to know the dissociation energy of CH_4 , one needs to calculate the dissociation energy of the four bonds $\text{CH}_3\text{--H}$, $\text{CH}_2\text{--H}$, CH--H , and C--H , respectively:

$$\begin{aligned} D^\circ(\text{CH}_3\text{--H}) &= \Delta H_f^\circ(\text{CH}_3) + \Delta H_f^\circ(\text{H}) - \Delta H_f^\circ(\text{CH}_4), \\ D^\circ(\text{CH}_2\text{--H}) &= \Delta H_f^\circ(\text{CH}_2) + \Delta H_f^\circ(\text{H}) - \Delta H_f^\circ(\text{CH}_3), \\ D^\circ(\text{CH--H}) &= \Delta H_f^\circ(\text{CH}) + \Delta H_f^\circ(\text{H}) - \Delta H_f^\circ(\text{CH}_2), \\ D^\circ(\text{C--H}) &= \Delta H_f^\circ(\text{C}) + \Delta H_f^\circ(\text{H}) - \Delta H_f^\circ(\text{CH}). \end{aligned} \quad (25)$$

Summing over all four bonds above, we can obtain:

$$\begin{aligned} D^\circ(\text{CH}_4) &= D(\text{CH}_3\text{--H}) + D(\text{CH}_2\text{--H}) \\ &\quad + D(\text{CH--H}) + D(\text{C--H}) \\ &= \Delta H_f^\circ(\text{C}) + 4 \times \Delta H_f^\circ(\text{H}) - \Delta H_f^\circ(\text{CH}_4). \end{aligned}$$

The dissociation energy can be generalized for a polyatomic molecule which has the form $\text{A}_n\text{B}_k\text{C}_i$:

$$\begin{aligned} D^\circ(\text{A}_n\text{B}_k\text{C}_i) &= n\Delta H_f^\circ(\text{A}) + k\Delta H_f^\circ(\text{B}) + i\Delta H_f^\circ(\text{C}) \\ &\quad - \Delta H_f^\circ(\text{A}_n\text{B}_k\text{C}_i). \end{aligned} \quad (26)$$

The ionization energies for molecules are available on NIST Atomic Spectra Database [37]. We now illustrate this treatment with one example.

A.1.1 $^{13}\text{C}_2\text{H}_2^+$ and $^{14}\text{N}_2^+$ mass doublet

In the work of [2004Ra33], a cyclotron frequency ratio $R = 0.999421460888(7)$ of relative precision 7×10^{-12} has been obtained for ions $^{13}\text{C}_2\text{H}_2^+$ and $^{14}\text{N}_2^+$. We first calculate $D^\circ(\text{C}_2\text{H}_2)$ by applying Eq. 26:

$$\begin{aligned} D^\circ(\text{C}_2\text{H}_2) &= 2 \times \Delta H_f^\circ(\text{C}) + 2 \times \Delta H_f^\circ(\text{H}) - \Delta H_f^\circ(\text{C}_2\text{H}_2) \\ &= 2 \times 716.68 + 2 \times 218.998 - 227.40 \\ &= 1641.956 \text{ kJ/mol} \\ &= 17.018 \text{ eV}. \end{aligned}$$

$D^\circ(\text{N}_2) = 944.9 \text{ kJ/mol} = 9.793 \text{ eV}$ is obtained from Ref. [38], since $\Delta H_f^\circ(\text{N}_2)$ is not on the list of [36]. Combining the ionization energy $B(\text{C}_2\text{H}_2) = 11.400 \text{ eV}$ and $B(\text{N}_2) = 15.581 \text{ eV}$, we obtain the mass difference:

$$^{13}\text{C} + \text{H} - ^{14}\text{N} = 8105.862995(98) \mu\text{u}.$$

In the original paper [2004Ra33], this equation is given as:

$$^{13}\text{C} + \text{H} - ^{14}\text{N} = 8105.86288(10) \mu\text{u},$$

which differs by 12(10) nu from our calculation. The difference is because we used the updated molecular binding energy from Ref. [36, 38]. In this case, the molecular binding energy is 170 times larger than the uncertainty, and even the updates of the molecular energies have significant impact on the deduced values.

Appendix B Mixtures of isomers or of isobars in mass spectrometry

In cases where two or more unresolved lines may combine into a single one in an observed spectrum, while one cannot decide which ones are present and in which proportion, a special procedure has to be used.

The first goal is to determine what is the most probable value M_{exp} that will be observed in the measurement, and what is the uncertainty σ of this prediction. We assume that all the lines may contribute and that all contributions have equal probabilities. The measured mass reflects the mixing. We call M_0 the mass of the lowest line, and $M_1, M_2, M_3, \dots$ the masses of the other lines. For a given composition of the mixture, the resulting mass m is given by

$$m = (1 - \sum_{i=1}^n x_i) M_0 + \sum_{i=1}^n x_i M_i, \quad \text{with} \begin{cases} 0 \leq x_i \leq 1 \\ \sum_{i=1}^n x_i \leq 1 \end{cases} \quad (27)$$

in which the relative unknown contributions $x_1, x_2, x_3, \dots$ have each a uniform distribution of probability within the allowed range.

If $P(m)$ is the normalized probability of measuring the value m , then :

$$\overline{M} = \int P(m) m dm \quad (28)$$

$$\text{and} \quad \sigma^2 = \int P(m) (m - \overline{M})^2 dm. \quad (29)$$

It is thus assumed that the experimentally measured mass will be $M_{exp} = \overline{M}$, and that σ , which reflects the uncertainty on the composition of the mixture, will have to be quadratically added to the experimental uncertainties.

B.1 Case of 2 spectral lines

In the case of two lines, one simply gets

$$m = (1 - x_1) M_0 + x_1 M_1 \quad \text{with} \quad 0 \leq x_1 \leq 1. \quad (30)$$

The relation between m and x_1 is biunivocal so that

$$P(m) = \begin{cases} 1/(M_1 - M_0) & \text{if } M_0 \leq m \leq M_1, \\ 0 & \text{elsewhere} \end{cases} \quad (31)$$

i.e. a rectangular distribution, and one obtains:

$$M_{exp} = \frac{1}{2}(M_0 + M_1), \quad (32)$$

$$\sigma = \frac{\sqrt{3}}{6}(M_1 - M_0) = 0.290 (M_1 - M_0).$$

Therefore, the ground state mass $M_0 \pm \sigma_0$ can be determined from the measured mass $M_{exp} \pm \sigma_{exp}$ and the knowledge of the excitation energies $E_1 \pm \sigma_1$ with following equations:

$$M_0 = M_{exp} - \frac{1}{2} E_1,$$

$$\sigma^2 = \frac{1}{12} E_1^2 \quad \text{or} \quad \sigma = 0.29 E_1,$$

$$\sigma_0^2 = \sigma_{exp}^2 + \left(\frac{1}{2} \sigma_1\right)^2 + \sigma^2.$$

B.2 Case of 3 spectral lines

In the case of three spectral lines, we derive from Eq. 27:

$$m = (1 - x_1 - x_2) M_0 + x_1 M_1 + x_2 M_2, \quad (33)$$

$$\text{with} \begin{cases} 0 \leq x_1 \leq 1 \\ 0 \leq x_2 \leq 1 \\ 0 \leq x_1 + x_2 \leq 1. \end{cases} \quad (34)$$

After normalization, one gets:

$$P(m) = \frac{2k}{M_2 - M_0}, \quad (35)$$

$$\text{with} \begin{cases} k = (m - M_0)/(M_1 - M_0) & \text{if } M_0 \leq m \leq M_1 \\ k = (M_2 - m)/(M_2 - M_1) & \text{if } M_1 \leq m \leq M_2 \end{cases} \quad (36)$$

and finally:

$$M_{exp} = \frac{1}{3}(M_0 + M_1 + M_2) \quad (37)$$

$$\sigma = \frac{\sqrt{2}}{6} \sqrt{M_0^2 + M_1^2 + M_2^2 - M_0 M_1 - M_1 M_2 - M_2 M_0}.$$

The ground state mass $M_0 \pm \sigma_0$ can be determined from the measured mass $M_{exp} \pm \sigma_{exp}$ and the knowledge of the excitation energies $E_1 \pm \sigma_1$ and $E_2 \pm \sigma_2$:

$$M_0 = M_{exp} - \frac{1}{3}(E_1 + E_2),$$

$$\sigma^2 = \frac{1}{18}(E_1^2 + E_2^2 - E_1 E_2)$$

$$\text{or} \quad \sigma = 0.236 \sqrt{E_1^2 + E_2^2 - E_1 E_2},$$

$$\sigma_0^2 = \sigma_{exp}^2 + \left(\frac{1}{3} \sigma_1\right)^2 + \left(\frac{1}{3} \sigma_2\right)^2 + \sigma^2.$$

If the levels are regularly spaced, *i.e.* $E_2 = 2E_1$,

$$\sigma = \frac{\sqrt{6}}{12} E_2 = 0.204 E_2,$$

while for a value of E_1 very near 0 or E_2 ,

$$\sigma = \frac{\sqrt{2}}{6} E_2 = 0.236 E_2.$$

B.3 Case of more than 3 spectral lines

For more than 3 lines, one may easily infer $M_{exp} = \sum_{i=0}^n M_i / (n+1)$, but the determination of σ requires the knowledge of $P(m)$. As the exact calculation of $P(m)$ becomes rather difficult, it is more simple to do simulations.

However, care must be taken that the values of the x_i 's are explored with an exact equality of chance to occur. For each set of x_i 's, m is calculated, and the histogram $N_j(m_j)$ of its distribution is built. Calling $nbin$ the number of bins of the histogram, one gets :

$$P(m_j) = \frac{N_j}{\sum_{j=1}^{nbin} N_j}, \quad (38)$$

and

$$M_{exp} = \sum_{j=1}^{nbin} P(m_j) m_j,$$

$$\sigma^2 = \sum_{j=1}^{nbin} P(m_j) (m_j - M_{exp})^2.$$

For three excited isomers, the ground state mass $M_0 \pm \sigma_0$ can be determined from the measured mass $M_{exp} \pm \sigma_{exp}$ and the knowledge of the excitation energies $E_1 \pm \sigma_1$, $E_2 \pm \sigma_2$ and $E_3 \pm \sigma_3$ as:

$$M_0 = M_{exp} - \frac{1}{4}(E_1 + E_2 + E_3),$$

$$\sigma_0^2 = \sigma_{exp}^2 + \left(\frac{1}{4}\sigma_1\right)^2 + \left(\frac{1}{4}\sigma_2\right)^2 + \left(\frac{1}{4}\sigma_3\right)^2 + \sigma^2.$$

An example of a Fortran program based on the CERN library was given for the cases of two, three and four lines in AME2016 [2].

Appendix C Definition of decay energies

Conventionally, the total energy released in α decay, Q_α , is defined as the difference between the atomic masses of the mother and daughter nuclides:

$$Q_\alpha = M_{\text{mother}} - M_{\text{daughter}} - M_{\text{He}}. \quad (39)$$

This value equals the sum of the observed energy of the α particle and that of the recoiling nuclide (with only a minor correction for the fact that the cortège of atomic electrons in the latter may be in an excited state). Unfortunately, some authors in the literature quote Q_α as a value 'corrected for screening', which essentially means that they take the masses of bare nuclides for the M values in the above equation.

A similar bad habit has been observed for some published proton-decay energies. We very strongly object to this custom; at the very least, the symbol Q should not be used for the difference in nuclear masses.

High precision α -decay energies

The most precise α -decay energies are those measured absolutely using magnetic spectrographs, as summarized in Ref. [1991Ry01] and references therein. All α -energy standards are determined using this method. The relation between the α -particle energy E_α and the decay energy, Q_α is:

$$Q_\alpha = (M - m_\alpha) - \sqrt{(M - m_\alpha)^2 - 2 \times M \times E_\alpha} + 78.6 \text{ eV}. \quad (40)$$

where M is the mass of the parent nuclide, m_α is the mass of the doubly ionized α particle and 78.6 eV is the binding energy of two electrons in helium. This formula had been discussed in AME1977 and has been used since then. The electron binding energy of helium has just been taken into account since AME2016.

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Table I. Input data compared with adjusted values**EXPLANATION OF TABLE**

The ordering is in groups according to highest occurring relevant mass number.

Item	K^m , Cs^m , Cs^n , In^p , Tl^q : higher isomers, see NUBASE.	In nuclear reactions: ε = electron capture.	In mass-doublet equation: $H = {}^1H$, $N = {}^{14}N$, $D = {}^2H$, $O = {}^{16}O$, $C = {}^{12}C$, u = absolute mass-doublet.	In mass-triplet equation: Rb^x , Rb^y : different mixtures of isomers or contaminants.
Input value	Mass doublet: value and its standard precision in μu . Triplet: value and its standard precision in keV. Reaction: value and its standard precision in keV. The value is the combination of mass excesses $\Delta(M - A)$ given under ‘item’. It is the author’s experimental result and the author’s stated uncertainty, except in a few cases for which comments are given and for some α -reactions: if the α -decay does not clearly feed the ground state, then an extra 50 keV is added quadratically to the original uncertainty. If more than one group report such energies, another 50 keV is added to the averaged precision in the adjustment.			
Adjusted value	Output of calculation. For secondary data ($Dg = 2-18$) the adjusted value is the same as the input value and is not repeated. Also, the adjusted value is only given once for a group of results for the same reaction or doublet. Values and precisions were rounded off, but not to more than tens of keV. # Value and precision derived not from purely experimental data, but at least partly from trends of the mass surface (TMS). * No mass value has been calculated for one of the masses involved.			
v_i	Normalized deviation between input and adjusted value, given as their difference divided by the input precision (see Section 4.4).			
Dg	1 Primary data (see Section 3.1, p. 030002-4). 2–18 Secondary data of different degrees (see Section 4.2, p. 030002-16). B Well-documented data, or data from regular reviewed journals, which disagree with other well-documented values. C Data from incomplete reports, at variance with other data. o Data included in or superseded by later work of same group. D Data not checked by another method and at large variance with TMS, replaced by an estimated value (see Section 5.1, p. 030002-19). F Study of paper raises doubts about validity of data within the reported precision. R Item replaced for computational reasons by an equivalent one giving same result. U Data with much less weight than that of a combination of other data. – Data that will be averaged.			
Signf.	<i>Significance</i> ($\times 100$) of primary data only (see Section 4.3); the significance of secondary data is always 100%.			
Main infl.	Largest <i>influence</i> ($\times 100$) and nucleus to which the data contributes the most (see Section 4.3).			
Lab	Identifies the group which measured the corresponding item. Example of Lab key: MA8 Penning Trap data of Mainz-Isolde group. The numbers refer to different experimental conditions.			
F	Multiplying factor for mass spectrometric data (see Section 4.4). The standard precision given in the ‘Input value’ column has been multiplied by this factor before being used in the least-squares adjustment.			

Reference	Reference keys: (in order to reduce the width of the Table, the two digits for the centuries are omitted; at the end of this volume however, the full reference key-number is given: 2003Ba49 and not 03Ba49).
20Ma09	Results derived from regular journal. These keys are copied from Nuclear Data Sheets. Where not yet available, the style 20Ra.1 has been used.
20Ba.A	Result from abstract, preprint, private communication, conference, thesis or annual report.
Ens169	References to energies of excited states, when of interest, are mentioned in remarks in the Qfile. Their reference-keys refer to the “Evaluated Nuclear Structure Data Files” (ENSDF) (the electronic version of the Nuclear Data Sheets NDS), the reference-keys are indicated Ens169 in which ‘16’ indicates the year (here 2016) and ‘9’ the month (Oct, Nov, Dec indicated a b c) of the released ENSDF file.
Nub211	When the excited energy is derived or estimated in NUBASE2020, it is indicated with ‘Nubase’ (see previous item).
Averag	Average of data from the following lines.
AHW	(or FGK, GAu, HWJ, MMC, Sar, WgM) : comment written by one of the evaluators.
*	A remark on the corresponding item is given below the block of data corresponding to the same (highest) A.
Y	recalibrations of 65Ry01 for charged particle recalibrations, and recalculated triplets for isomeric mixtures.
Z	recalibrations of 91Ry01 for α particles, 90Wa22 for γ in (n, γ) and (p, γ) reactions and 91Wa.A for protons and γ in (p, γ) reactions (see Section 2).

Remarks. For data indicated with a star in the reference column, remarks have been added. They are collected in groups at the end of each block of data in which the highest occurring relevant mass number is the same. They give:

- Information explaining how the values in column ‘Input value’ have been derived for papers not mentioning e.g. the mass differences as derived from measured ratios of frequencies, or the reaction energies, or values for transitions to excited states in the final nuclei (for which better values of the excitation energies are now known).
- Reasons for changing values (e.g. recalibrations) or precisions as given by the authors or for rejecting them (i.e. for labelling them B, C or F).
- Value suggested by TMS and recommended in this evaluation as the best estimate (see Section 5, p. 030002-19).

Special notation in remarks:

E_{β^-} , Q_{β^-}	β^- endpoint energy, β^- decay energy
E_{β^+} , Q_{β^+}	β^+ endpoint energy, β^+ decay energy
E_p , Q_p	proton energy in the laboratory, proton-decay energy
a_s	scattering length
T	threshold for given reaction
ε	electron capture; $\beta^+ = \varepsilon + e^+$ (see NUBASE2020, p. 030001-18)
p^+ , pK, pL	fraction β^+ , $\varepsilon(K)$ or $\varepsilon(L)$ in transition to mentioned states
L/K, L/M	$\varepsilon(L)/\varepsilon(K)$, $\varepsilon(L)/\varepsilon(M)$
IBE	internal bremsstrahlung endpoint
M-A, D_M	mass excess (in keV), mass difference (in μu)
TMS	Trends from Mass Surface
‘Z’ (after uncertainty)	recalibrated (see above, under ‘Reference’)

Table I. Comparison of input data and adjusted values (Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
π^+	140081.39	0.18	140081.39	0.18	0.0	1	100	100 π^+			20PaDG *
$\pi^+(2\beta^+)\pi^-$	1021.998	0.001	1021.9980	0.0010	-0.0	1	100	100 π^-			20PaDG *
$*\pi^+$	By convention! This is $M=139570.39(0.18)$ keV + $m(e^-)$										WgM20a**
$*\pi^+(2\beta^+)\pi^-$	By convention = $2m(e^-)$.										WgM20a**
$*\pi^+(2\beta^+)\pi^-$	π^+, π^- should have same mass, with relative difference determined as $2(5) \times 10^{-4}$										20PaDG **
$H_{12}-C$	93902.7	0.4	93900.38277	0.00017	-2.3	U			M17	2.5	66Be10
	93900.66	0.48			-0.2	U			A2	2.5	70St25
	93900.32	0.12			0.3	U			B07	1.5	71Sm01
	93900.391	0.012			-0.3	U			WA1	2.5	95Va38
	93900.3804	0.0084			0.3	U			MI1	1.0	95Di08
	93900.3865	0.0017			-0.9	U			WA1	2.5	01Va33
	93900.3860	0.0042			-0.8	U			ST2	1.0	02Be64
	93900.38272	0.00040			0.1	o			MZ3	1.0	17He14
	93900.38292	0.00036			-0.4	1	22	22 1H	MZ3	1.0	19He19
$n(\beta^-)^1H$	782	13	782.3470	0.0004	0.0	U					51Ro50
D_6-C	84610.56	0.12	84610.66706	0.00009	0.4	U			A2	2.5	70St25
	84610.62	0.09			0.3	U			B07	1.5	71Sm01
	84611.60	0.34			-1.1	U			J5	2.5	72Ka57 *
	84611.47	0.40			-0.8	U			J6	2.5	76Ka50
	84610.644	0.005			1.8	C			WA1	2.5	92Va.A
	84610.584	0.078			0.4	U			OH1	2.5	93Ma.A
	84610.662	0.007			0.3	o			WA1	2.5	93Va.C
	84610.6616	0.0067			0.3	o			WA1	2.5	95Va38
	84610.6710	0.0054			-0.7	U			MI1	1.0	95Di08
	84610.6656	0.0036			0.4	U			MI1	1.0	95Di08
	84610.66897	0.00086			-0.9	U			WA1	2.5	06Va22
	84610.66834	0.00024			-2.1	F			WA1	2.5	15Za13 *
	84610.66705	0.00010			0.1	1	82	82 2H	MZ3	1.0	20Ra.1 *
	1547.77	0.28	1548.28595	0.00003	0.7	U			C1	2.5	64Mo.A
	1548.22	0.05			0.5	o			M19	2.5	67Jo18
	1548.08	0.08			1.0	o			J2	2.5	69Na21
	1548.286	0.004			-0.0	o			B07	1.5	71Sm01
	1548.222	0.063			0.4	o			J5	2.5	72Ka57
	1548.176	0.133			0.3	o			J5	2.5	72Ka57
	1548.298	0.008			-1.0	U			B08	1.5	75Sm02
	1548.301	0.005			-2.0	U			B08	1.5	75Sm02
H_2-D	1548.190	0.023			1.7	U			J6	2.5	76Ka50
	1548.28	0.05			0.0	U			M25	2.5	78Ha14
	1548.302	0.012			-0.5	U			OH1	2.5	93Go37
	1548.2836	0.0018			1.3	U			MI1	1.0	95Di08
	1548.28649	0.00035			-1.5	U			ST2	1.0	08So20
	1548.28588	0.00006			1.2	-			FS1	1.0	18Sm03 *
	1548.28597	0.00004			-0.5	-			FS1	1.0	20Fi05 *
	ave. 1548.28594	0.00003			0.3	1	67	58 1H			average
	2224.564	0.017	2224.5662	0.0004	0.1	U			BNL		80Gr02
	2224.5	0.12			0.6	U			MMn		80Is02
	2224.561	0.009			0.6	U			Utr		82Va13 Z
	2224.549	0.009			1.9	U					82Vy10 Z
	2224.560	0.009			0.7	U					83Ad05 Z
	2224.5756	0.008			-1.2	U			NBS		86Gr01 *
	2224.5727	0.0500			-0.1	U			PTc		97Ro26 *
	2224.5660	0.0004			0.6	o			NBS		99Ke05 *
	2224.58	0.05			-0.3	U			Bdn		06Fi.A *
	2224.56624	0.00044			-0.0	1	100	100 1n	NBS		06De21 *
$*D_6-C$	For all 72Ka57 doublets, see also reference										72Og03 **
$*D_6-C$	F: the other result from same paper is not trusted within error bar										GAu **
$*D_6-C$	From freq. ratio $D^+/(^{12}C^{6+})=1.0070527379117(85)$										20Ra.1 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference
*H ₂ –D	Uncorrected frequency ratio H ₃ ⁺ /(HD ⁺)=0.999487821168(19); Reconstructed										HWJ201 **
*	from upper limit for the mass difference between ions										HWJ201 **
*H ₂ –D	From frequency ratio H ₂ ⁺ /(² H ⁺)=0.999231660004(19)										HWJ201 **
* ¹ H(n,γ) ² H	Original 2224.5890(0.0022) revised in reference; error increased by evaluator										90Wa22 **
* ¹ H(n,γ) ² H	Original error 0.0005 increased for calibration										GAu **
* ¹ H(n,γ) ² H	More precisely, H+n–D=2388170.07(0.42) nu corrected to 2388169.95(0.42) nu										99Ke05 **
* ¹ H(n,γ) ² H	All errors in reference increased by 20 ppm for calibration										06Fi.A **
* ¹ H(n,γ) ² H	Original 2224.56610(0.00044) recalibrated with 2018 Codata										WgM211**
³ H ₄ –C	64197.0690	0.0062	64197.1253	0.0003	3.6	B			WA1	2.5	93Va04 *
	64197.1136	0.0116			1.0	o			ST2	1.0	02Bf02
	64197.1148	0.0100			1.0	U			ST2	1.0	06Na49
H D–C	21926.80973	0.00006	21926.80974	0.00002	0.2	1	16	8 ² H	MZ3	1.0	20Ra.1
³ He ₄ –C	64117.2399	0.0039	64117.28787	0.00024	4.9	B			WA1	2.5	93Va04
	64117.252	0.030			0.5	U			WA1	2.5	93Va04 *
	64117.294	0.011			–0.6	o			ST2	1.0	01Fr18
	64117.2868	0.0100			0.1	o			ST2	1.0	06Na49 *
	64117.28668	0.00017			2.8	B			WA1	2.5	15Za13
H ₃ – ³ He	7445.858	0.012	7445.77373	0.00006	–2.8	U			MZ1	2.5	91Ha31 *
	7445.7737	0.0001			0.3	1	33	26 ³ He	FS1	1.0	18Sm03 *
D ₂ –H ³ H	4329.257	0.003	4329.24247	0.00008	–3.2	B			B08	1.5	75Sm02
³ H–H D	–5877.2	0.7	–5877.52842	0.00008	–0.2	U			C1	2.5	64Mo.A
	–5877.52837	0.00014			–0.4	1	30	30 ³ H	FS1	1.0	15My03
	–5896.84	0.42	–5897.48777	0.00005	–0.6	U			C1	2.5	64Mo.A
	–5897.512	0.005			3.2	B			B08	1.5	75Sm02
	–5897.495	0.006			0.8	U			B09	1.5	81Sm02
³ He–H D	–5897.48771	0.00014			–0.5	o			FS1	1.0	15My03
	–5897.48780	0.00007			0.4	1	60	62 ³ He	FS1	1.0	17Ha33 *
	19.83	0.18	19.95935	0.00006	0.3	U			C1	2.5	64Mo.A
	19.951	0.004			0.8	o				2.5	84Ni16 *
	19.967	0.003			–1.0	o				2.5	84Li24
³ H– ³ He	19.967	0.002			–1.5	U				2.5	85Li02
	19.948	0.003			1.5	U				2.5	85Ta.A
	19.9570	0.0013			1.8	U			ST2	1.0	06Na49
	19.95934	0.00007			0.2	1	82	70 ³ H	FS1	1.0	15My03
	6257.6	0.3	6257.2301	0.0004	–1.2	U					69Pr06
	6256.96	0.25			1.1	U			ILn		79Br25 Z
	4029	12	4032.66383	0.00008	0.3	U			CIT		49To23 Y
² H(d,p) ³ H	4034	6			–0.2	U			MIT		64Sp12
	4033.7	1.7			–0.6	U			NDm		67Od01
	3260	9	3268.9088	0.0004	1.0	U			CIT		49To23 Y
² H(d,n) ³ He	3269	11			–0.0	U			Wis		56Do41 Y
	18.645	0.016	18.59202	0.00006	–3.3	B					72Be11 *
³ H(β [–]) ³ He	18.619	0.040			–0.7	U					73Pi01 *
	18.607	0.013			–1.2	o					76Tr07
	18.614	0.013			–1.7	U					81Lu07 *
	18.562	0.020			1.5	U					83De47 *
	18.590	0.008			0.3	U					85Si07 *
	18.604	0.006			–2.0	o					85Bo34
	18.603	0.010			–1.1	o					86Fr09 *
	18.600	0.004			–2.0	U					87Bo07
	18.598	0.015			–0.4	o					87Bu.A
	18.603	0.004			–2.7	B					88Ka32
	18.589	0.003			1.0	o					89St05
	18.595	0.006			–0.5	o					91Bu12
	18.592	0.003			0.0	U					91Ka41 *
	18.591	0.002			0.5	U					91Ro07 *
	18.595	0.006			–0.5	U					92Bu13 *
	18.589	0.003			1.0	o					92Ot.A
	18.593	0.003			–0.3	U					92Ho09 *
	18.591	0.003			0.3	U					93We03

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference
${}^3\text{H}(\beta^-){}^3\text{He}$	18.597	0.014	18.59202	0.00006	-0.4	U					95Hi14
	18.5895	0.0025			1.0	U					95St26
${}^3\text{H}(\text{p},\text{n}){}^3\text{He}$	-764.08	0.15	-763.7550	0.0004	2.1	o			Zur		61Ry05
	-764.39	0.37			1.7	U			NRL		64Bo10
	-763.82	0.08			0.8	U			Zur		64Sa12
${}^3\text{H}_4\text{-C}$	Item preliminarily disregarded										AHW **
${}^3\text{He}_4\text{-C}$	Original changed after discussion with authors										AHW **
${}^3\text{He}_4\text{-C}$	Use instead the most precise difference between ${}^3\text{H}$ and ${}^3\text{He}$ (see below)										GAu **
${}^3\text{H}_3\text{-}{}^3\text{He}$	From ${}^3\text{He}^+/\text{H}_2^+ = 1.496441095(6) + 3\text{eV}$ ionization										AHW **
${}^3\text{H}_3\text{-}{}^3\text{He}$	Uncorrected frequency ratio $\text{H}_3^+/({}^3\text{He}^+) = 0.997536905902(14)$; Reconstructed										HWJ201 **
*	from upper limit for the mass difference between ions										HWJ201 **
${}^3\text{He-H D}$	Other data are related to a metastable state in H_3 , not used										HWJ183 **
${}^3\text{H-H}{}^3\text{He}$	Atom mass difference=ion mass difference $18.573 + 0.011\text{ keV}$										AHW **
*	required correction cannot be estimated										85Au07 **
${}^3\text{H}(\beta^-){}^3\text{He}$	For corrections to 72Be11 see reference										82Di01 **
${}^3\text{H}(\beta^-){}^3\text{He}$	For corrections to 73Pi01 and 81Lu07 see reference										85Au07 **
${}^3\text{H}(\beta^-){}^3\text{He}$	Error for 83De47 increased, see reference										85Au07 **
${}^3\text{H}(\beta^-){}^3\text{He}$	Original value 18580(7) corrected in reference										89Re04 **
${}^3\text{H}(\beta^-){}^3\text{He}$	As calculated from their data in reference										89Re04 **
${}^3\text{H}(\beta^-){}^3\text{He}$	$E_{\beta^-} = 18.5721(0.0030)$, SFS and recoil as in reference										88Ka32 **
${}^3\text{H}(\beta^-){}^3\text{He}$	$E_{\beta^-} = 18.5705(0.0020)$, SFS and recoil as in reference										89St05 **
${}^3\text{H}(\beta^-){}^3\text{He}$	$E_{\beta^-} = 18.556(0.006)$, corrections as in reference										91Bu12 **
${}^3\text{H}(\beta^-){}^3\text{He}$	$E_{\beta^-} = 18.5733(0.0002+\text{syst})$, SFS and recoil as in reference										88Ka32 **
${}^4\text{He}_3\text{-C}$	7809.706	0.009	7809.7624	0.0005	2.5	B			WA1	2.5	92Va.A
	7809.7493	0.0030			1.7	B			WA1	2.5	95Va38
	7809.7704	0.0039			-2.1	U			ST2	1.0	01Fr18
	7809.7620	0.0003			0.5	o			WA1	2.5	01Va.A
	7809.7467	0.0066			1.0	U			MZ2	2.5	01Br27
	7809.76246	0.00019			-0.1	o			WA1	2.5	04Va14
	7809.76239	0.00019			-0.0	1	100	100 ${}^4\text{He}$	WA1	2.5	06Va22
${}^4\text{He-H}_4$	-28696.8747	0.0026	-28696.87346	0.00017	0.5	o			ST2	1.0	06Na49
	-28696.8750	0.0026			0.6	U			ST2	1.0	06Na13
$\text{D}_2\text{-}{}^4\text{He}$	25600.315	0.014	25600.30156	0.00016	-0.6	U			B08	1.5	75Sm02
	25600.331	0.005			-2.4	U			MZ1	2.5	90Ge12 *
	25600.328	0.005			-2.1	U			MZ1	2.5	92Ke06 *
	25600.308	0.005			-0.3	U			BL1	4.0	01He36
$\text{H } {}^3\text{H-}{}^4\text{He}$	21271.075	0.012	21271.05909	0.00018	-0.9	U			B08	1.5	75Sm02
${}^4\text{H}(\gamma,\text{n}){}^3\text{H}$	5200	1700	1600	100	-2.1	U					62Ar05
	2900	500			-2.6	U					69Mi10 *
	8000	3000			-2.1	U					79Me13 *
	2700	600			-1.8	U					81Se11
	2600	200			-5.0	B					85Fr01 *
	3500	500			-3.8	B					86Be35 *
	2600	400			-2.5	U					86Mi14 *
	3000	200			-7.0	B					87Go25 *
	3800	300			-7.3	B					90Am04 *
	3100	300			-5.0	B					91Bl05 *
	2300	300			-2.3	U					95Al31 *
	2670	310			-3.5	B					03Me11
	1600	100			2						09Gu17 *
${}^3\text{He}(\text{d},\text{p}){}^4\text{He}$	18380	10	18353.05484	0.00016	-2.7	U			Mex		64Ma.B
	18382	15			-1.9	U			Mex		64Ma.B
	18350.1	3.9			0.8	U			NDm		67Od01
${}^4\text{Li}(\text{p}){}^3\text{He}$	3300	300	3100	210	-0.7	2					87Br.B
${}^4\text{D}_2\text{-}{}^4\text{He}$	Error has to be confirmed										GAu **
${}^4\text{H}(\gamma,\text{n}){}^3\text{H}$	From ${}^7\text{Li}(\pi^-, \text{t}){}^4\text{H}$ reaction										69Mi10 **
${}^4\text{H}(\gamma,\text{n}){}^3\text{H}$	From ${}^7\text{Li}(\pi^-, \text{t}){}^4\text{H}$ reaction										AHW **
${}^4\text{H}(\gamma,\text{n}){}^3\text{H}$	From ${}^7\text{Li}({}^3\text{He}, {}^3\text{He}){}^4\text{H}$ reaction										85Fr01 **
${}^4\text{H}(\gamma,\text{n}){}^3\text{H}$	From ${}^9\text{Be}({}^{11}\text{B}, {}^{16}\text{O}){}^4\text{H}$ reaction										86Be35 **
${}^4\text{H}(\gamma,\text{n}){}^3\text{H}$	From ${}^7\text{Li}(\text{n}, \alpha){}^4\text{H}$ reaction										86Mi14 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^4\text{H}(\gamma, n)^3\text{H}$	From $^9\text{Be}(\pi^-, dt)^4\text{H}$, same data in reference										91Go19 **
$^4\text{H}(\gamma, n)^3\text{H}$	From $^7\text{Li}(\pi^-, t)^4\text{H}$										90Am04 **
$^4\text{H}(\gamma, n)^3\text{H}$	From $^2\text{D}(t, n)^4\text{H}$										91B105 **
$^4\text{H}(\gamma, n)^3\text{H}$	From $^6\text{Li}(^6\text{Li}, ^8\text{B})$										95Al31 **
$^4\text{H}(\gamma, n)^3\text{H}$	Fit with 3 resonances 1.6(0.1), 3.4(0.1), 6.0(0.1) MeV										09Gu17 **
$^5\text{H}(\gamma, 2n)^3\text{H}$	1800	800	1800	90	0.0	U					68Yo06
	11000	1500			-6.1	F					81Se.A *
	7400	700			-8.0	F					87Go25 *
	5200	400			-8.5	F					95Al31 *
	1700	300			0.3	U					01Ko52 *
	1800	100			0.0	2					03Go11 *
	1800	200			0.0	2					04St18
$^4\text{He}(n, \gamma)^5\text{He}$	-890	50	-735	20	3.1	B					66La04 *
	-735	20				3					09Ak03
$^4\text{He}(p, \gamma)^5\text{Li}$	-1965	50				3					65Ma56 *
$^5\text{H}(\gamma, 2n)^3\text{H}$	From $^6\text{Li}(\pi^-, p)^5\text{H}$. F : private communication by author										AHW **
$^5\text{H}(\gamma, 2n)^3\text{H}$	From $^9\text{Be}(\pi^-, pt)^5\text{H}$, same data in reference										91Go19 **
$^5\text{H}(\gamma, 2n)^3\text{H}$	F : probably higher state										01Ko52 **
$^5\text{H}(\gamma, 2n)^3\text{H}$	From $^7\text{Li}(^6\text{Li}, ^8\text{B})$										95Al31 **
$^5\text{H}(\gamma, 2n)^3\text{H}$	F : probably higher state										01Ko52 **
$^5\text{H}(\gamma, 2n)^3\text{H}$	From $p(^6\text{He}, ^2\text{He})$										01Ko52 **
$^5\text{H}(\gamma, 2n)^3\text{H}$	From $t(t, p)$										03Go11 **
$^4\text{He}(n, \gamma)^5\text{He}$	Average of many reactions leading to ^5He										AHW **
$^4\text{He}(p, \gamma)^5\text{Li}$	Average of many reactions leading to ^5Li										AHW **
$^6\text{Li}_2-\text{C}$	30246.152	0.119	30245.7750	0.0030	-0.8	F			BL1	4.0	98He.B *
	30245.575	0.034			1.5	U			BL1	4.0	01He36
	30245.7748	0.0031			0.0	1	100	100 ^6Li	FS1	1.0	10Mo30
$^6\text{Li}-\text{H}_6$	-31827.302	0.040	-31827.3040	0.0016	-0.0	U			ST2	1.0	06Na13
$^6\text{Li}-\text{D}_3$	-27182.500	0.040	-27182.4461	0.0016	0.3	U			BL1	4.0	01He36
$^4\text{He}-^6\text{Li}_{.667}$	-7483.694	0.046	-7483.7117	0.0010	-0.4	U			TT1	1.0	09Br10
$^6\text{He}-^7\text{Li}_{.857}$	5170.947	0.057	5170.95	0.06	-0.0	1	100	100 ^6He	TT1	1.0	12Br03
$^6\text{H}(\gamma, 3n)^3\text{H}$	2700	400	2710	250	0.0	2					84Al08 *
	2600	500			0.2	2					86Be35 *
	2800	500			-0.2	2					92Al.A *
	2850	900			-0.2	2					08Ca22 *
$^6\text{Li}(n, \alpha)^3\text{H}$	4794	6	4783.4717	0.0015	-1.8	U			Win		67De15
$^6\text{Li}(p, \alpha)^3\text{He}$	4017	12	4019.7168	0.0015	0.2	U			CIT		49To16 Y
	4021	5			-0.3	U			Wis		51Wi26 Y
	4023	2			-1.6	U			Bir		53Co02 Y
	4025	6			-0.9	o			MIT		64Sp12
	4018.2	1.1			1.4	U			MIT		81Ro02
$^6\text{Li}(d, \alpha)^4\text{He}$	22396	12	22372.7716	0.0015	-1.9	U			Bir		53Co02 Y
	22376	14			-0.2	U			Ric		53Ph28 Y
	22403	12			-2.5	U			Mex		64Ma.B
$^6\text{Li}(p, t)^4\text{Li}$	-18700	300	-18900	210	-0.7	R			Brk		65Ce02
$^6\text{He}(\beta^-)^6\text{Li}$	3509.8	3.8	3505.21	0.05	-1.2	U					63Jo04
$^6\text{Li}(p, n)^6\text{Be}$	-5074	13	-5071	5	0.3	2			CIT		67Ho01
$^6\text{Li}(^3\text{He}, t)^6\text{Be}$	-4306	6	-4307	5	-0.1	2			CIT		66Wh01
$^6\text{Li}_2-\text{C}$	F : leak during the measurement										98He.B **
$^6\text{H}(\gamma, 3n)^3\text{H}$	From $^7\text{Li}(^7\text{Li}, ^8\text{B})^6\text{H}$										84Al08 **
$^6\text{H}(\gamma, 3n)^3\text{H}$	From $^9\text{Be}(^{11}\text{B}, ^{14}\text{O})^6\text{H}$										86Be35 **
*	^6H not observed in $^6\text{Li}(\pi^-, \pi^+)$										87Se.A **
$^6\text{H}(\gamma, 3n)^3\text{H}$	From $^7\text{Li}(^7\text{Li}, ^8\text{B})^6\text{H}$										92Al.A **
$^6\text{H}(\gamma, 3n)^3\text{H}$	Symmetrized from 2910(+850-950) keV										08Ca22 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
${}^7\text{Li}-\text{H}_7$	-38771.7889	0.0045	-38771.789	0.004	-0.0	1	100	100 ${}^7\text{Li}$	ST2	1.0	06Na13 *
${}^7\text{B}-\text{u}$	29712	27				2			1.0	1.0	11Ch32 *
${}^7\text{Li}-{}^6\text{Li}_{1.167}$	-1644.991	0.079	-1644.976	0.005	0.2	o			TT1	1.0	09Br10
${}^6\text{Li}-{}^7\text{Li}_{.857}$	1407.954	0.013	1407.944	0.004	-0.7	U			TT1	1.0	09Br.A
${}^3\text{He}(\alpha,\gamma){}^7\text{Be}$	1586.3	0.6	1587.14	0.07	1.4	U					82Kr05
${}^7\text{Li}(\text{p},\alpha){}^4\text{He}$	17364	11	17346.244	0.004	-1.6	U			CIT		51Wh05 Y
	17352	9			-0.6	U			Bir		53Co02 Y
	17345	13			0.1	U			Ric		53Fa18 Y
	17373	6			-4.5	C			Mex		64Ma.B
	17357	14			-0.8	U			MIT		64Sp12
${}^7\text{H}(\gamma,2\text{n}){}^5\text{H}$	-1100	340	100#	1000#	3.5	F					08Ca22 *
${}^7\text{He}(\gamma,\text{n}){}^6\text{He}$	450	20	410	8	-2.0	3			MSU		01Ch31
	430	20			-1.0	3					02Me07
	360	50			1.0	U					06Sk03
	400	10			1.0	3					08De29
	388	20			1.1	3					09Ak03
	380	28			1.1	U					16Re14
${}^7\text{Li}(\text{t},\alpha){}^6\text{He}$	9788	30	9839.90	0.05	1.7	U			ChR		54Al35 Y
${}^7\text{Li}(\text{d},{}^3\text{He}){}^6\text{He}-{}^{19}\text{F}(){}^{18}\text{O}$	-1981.09	0.42	-1980.36	0.05	1.7	U			MSU		78Ro01 *
${}^6\text{Li}(\text{n},\gamma){}^7\text{Li}$	7250.0	0.5	7251.094	0.004	2.2	U			Utr		68Sp01
	7250.3	0.9			0.9	U					72Op01
	7250.98	0.09			1.3	U			Ptn		85Ko47 *
	7249.94	0.15			7.7	C			Bdn		06Fi.A
${}^6\text{Li}(\text{d},\text{p}){}^7\text{Li}$	5028	2	5026.528	0.004	-0.7	U			Bir		53Co02 Y
	5035	5			-1.7	U			Mex		61Ja23
	5024	7			0.4	U			MIT		64Sp12
${}^6\text{Li}(\text{t},\text{d}){}^7\text{Li}$	986	7	993.864	0.004	1.1	U			ChR		54Al35
${}^7\text{Li}({}^3\text{He},\alpha){}^6\text{Li}$	13322	10	13326.527	0.004	0.5	U			Mex		64Ma.B
${}^6\text{Li}(\text{n},\gamma){}^7\text{Li}^i$	-3947	50	-4000	30	-1.0	1	39	39 ${}^7\text{Li}^i$			69Pr04 *
${}^6\text{Li}({}^3\text{He},\text{d}){}^7\text{Be}$	136	3	113.38	0.07	-7.5	C			Mex		64Ma.B
${}^7\text{Li}(\text{t},{}^3\text{He}){}^7\text{He}$	-11184	30	-11147	8	1.2	U			LAI		69St02
${}^7\text{Be}(\varepsilon){}^7\text{Li}$	866	7	861.89	0.07	-0.6	U					72Pe05
${}^7\text{Li}(\text{p},\text{n}){}^7\text{Be}$	-1644.04	0.22	-1644.24	0.07	-0.9	U			Zur		61Ry05 Z
	-1643.68	0.26			-2.2	U			Wis		63Ga09 Y
	-1644.30	0.10			0.6	-			Mar		70Ro07 *
	-1644.18	0.10			-0.6	-			Auc		85Wh03 *
ave.	-1644.24	0.07			0.0	1	100	100 ${}^7\text{Be}$			average
${}^7\text{Li}(\pi^+,\pi^-){}^7\text{B}$	-11870	100	-11747	25	1.2	U					81Se.A
${}^7\text{Be}^i(\text{IT}){}^7\text{Be}$	11000	50	10980	30	-0.4	3					67Ha08
	10970	40			0.3	3					67Mc14
${}^7\text{Li}-\text{H}_7$	$D_M=7016003.4256(45)-7\times 1007825.03207(10)$ using Ame2003										06Na13 **
${}^7\text{B}-\text{u}$	Represents ${}^7\text{B} \rightarrow 3\text{p} + {}^4\text{He}$, yielding $M-A=27677(25)$ keV										GAu **
${}^7\text{H}(\gamma,2\text{n}){}^5\text{H}$	From ${}^7\text{H}(\gamma,4\text{n}){}^3\text{H}=704(323)$, and ${}^5\text{H}(\gamma,2\text{n}){}^3\text{H}=1800(100)$ keV										08Ca22 **
${}^7\text{H}(\gamma,2\text{n}){}^5\text{H}$	F : not confirmed in later work of reference with higher statistics										10Ni10 **
${}^7\text{Li}(\text{d},{}^3\text{He}){}^6\text{He}-{}^{19}\text{F}(){}^{18}\text{O}$	$Q-Q=0.98(0.41)$ to 2^+ level at $1982.07(0.09)$ keV in ${}^{18}\text{O}$										Ens967 **
${}^6\text{Li}(\text{n},\gamma){}^7\text{Li}$	Original 7251.02 recalibrated using ${}^{35}\text{Cl}(\text{n},\gamma)$ of reference										82Kr12 **
${}^6\text{Li}(\text{n},\gamma){}^7\text{Li}$	Typo 7250.02 in Ame1986 recalib. 7249.97 in Ame1993, 7249.98 Ame2003										GAu **
${}^6\text{Li}(\text{n},\gamma){}^7\text{Li}^i$	IT=11200(50); Q rebuilt with Ame1965										MMC128**
${}^7\text{Li}(\text{p},\text{n}){}^7\text{Be}$	T=1880.64(0.09,Z); error in Q increased										AHW **
${}^7\text{Li}(\text{p},\text{n}){}^7\text{Be}$	T=1880.43(0.02,Z); error in Q increased										AHW **
${}^8\text{Li}-\text{u}$	22488.2	4.0	22486.24	0.05	-0.5	U			RI1	1.0	13It01
${}^8\text{C}-\text{u}$	37606	32	37643	20	1.2	1	37	37 ${}^8\text{C}$	1.0	1.0	11Ch32 *
${}^8\text{He}-{}^6\text{Li}_{1.333}$	13776.88	0.72	13775.58	0.10	-1.8	o			TT1	1.0	08Ry03
	13775.50	0.19			0.4	1	25	25 ${}^8\text{He}$	TT1	1.0	08Br.D

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^8\text{Li}-^6\text{Li}_{1.333}$	2327.426	0.034	2327.44	0.05	0.3	o			TT1	1.0	08Sm.A
	2327.42	0.11			0.2	o			TT1	1.0	08Sm03
	2327.42	0.11			0.2	1	21	21 ^8Li	TT1	1.0	09Br10
$^8\text{He}-^7\text{Li}_{1.143}$	15642.49	0.11	15642.46	0.10	-0.2	1	75	75 ^8He	TT1	1.0	12Br03
$^4\text{He}(^{18}\text{O},^{14}\text{O})^8\text{He}$	-37967	25	-37975.36	0.09	-0.3	U			MIT		75Ja10
$^4\text{He}(^{26}\text{Mg},^{22}\text{Mg})^8\text{He}$	-44962	30	-44999.32	0.18	-1.2	U			Brk		74Ce05
$^4\text{He}(^{64}\text{Ni},^{60}\text{Ni})^8\text{He}$	-31818	15	-31810.6	0.4	0.5	U			Pri		75Ko18
	-31796	8			-1.8	U			Tex		77Tr07
	91.88	0.05	91.84	0.04	-0.8	3			Zur		68Be02 *
$^8\text{Be}(\alpha)^4\text{He}$	91.80	0.05			0.8	3					92Wu09 *
	92.2	0.4			-0.9	U					16Re14
	790	11	801.92	0.05	1.1	U			ChR		54Al35
$^6\text{Li}(^3\text{He},\text{p})^8\text{Be}$	16824	12	16787.46	0.04	-3.0	C			Mex		64Ma.B
$^6\text{Li}(\text{d},\gamma)^8\text{Be}^j$	-5216.5	3.0	-5213.4	2.0	1.0	1	43	43 $^8\text{Be}^j$			76No07 *
$^6\text{Li}(^3\text{He},\text{n})^8\text{B}$	-1974.8	1.0	-1974.8	1.0	0.0	1	100	100 ^8B	Nvl		58Du78 Y
$^7\text{Li}(\text{n},\gamma)^8\text{Li}$	2032.78	0.15	2032.62	0.05	-1.1	-					74Ju.A *
	2032.77	0.18			-0.8	-			ORn		91Ly01 Z
	2032.57	0.06			0.8	-			Bdn		06Fi.A
$^7\text{Li}(\text{d},\text{p})^8\text{Li}$	-192	1	-191.95	0.05	0.1	U			Wis		51Wi26 Y
	-188	7			-0.6	U			MIT		64Sp12
	ave.	2032.61	0.05	2032.62	0.05	0.1	1	79	79 ^8Li		average
$^7\text{Li}(^3\text{He},\text{d})^8\text{Be}$	11795	13	11760.93	0.04	-2.6	U			Mex		64Ma.B
$^8\text{C}-\text{u}$	Represents $^8\text{C} \rightarrow 4\text{p} + ^4\text{He}$, yielding $M - A = 35030(30)$ keV										GAu **
$^8\text{Be}(\alpha)^4\text{He}$	For atomic binding energy correction see reference										67St30 **
$^6\text{Li}(\text{d},\gamma)^8\text{Be}^j$	$E_d = 6962.8(3.0)$ keV										76No07 **
$^7\text{Li}(\text{n},\gamma)^8\text{Li}$	PrvCom to reference										74Aj01 **
$^9\text{Li}-^6\text{Li}_{1.500}$	4105.867	0.092	4105.86	0.20	-0.1	o			TT1	1.0	08Sm.A
	4105.86	0.20				2			TT1	1.0	08Sm03
	-8397.39	0.10	-8397.35	0.08	0.4	1	67	67 ^9Be	TT1	1.0	09Ri03
$^9\text{Be}(\text{p},\alpha)^6\text{Li}$	2117	7	2125.63	0.08	1.2	U			CIT		49To16 Y
	2130	10			-0.4	U			Chi		51Ca37 Y
	2125	4			0.2	U			Wis		51Wi26 Y
	2126	2			-0.2	U			Bir		53Co02 Y
	2144	6			-3.1	B			MIT		64Sp12
	2125.4	1.8			0.1	U			NDm		67Od01
	-2125.6	1.2	-2125.63	0.08	-0.0	U			NDm		65Br28
$^6\text{Li}(\alpha,\text{p})^9\text{Be}$	-3974	12	-3976.0	0.9	-0.2	U			Tal		63Me08
$^7\text{Li}(\alpha,\text{n})^9\text{B}$	-2397	20	-2386.96	0.19	0.5	U					64Mi04
$^7\text{Li}(\text{t},\text{p})^9\text{Li}$	-2385.7	3.0			-0.4	U			MSU		75Ka18
$^9\text{Be}(^{15}\text{N},^{17}\text{F})^7\text{He}$	-16610	30	-16575	8	1.2	U			Ber		99Bo26
$^9\text{Be}(\text{d},\alpha)^7\text{Li}$	7162	10	7152.15	0.08	-1.0	U			CIT		51Wh05 Y
	7153	3			-0.3	U			Bir		53Co02 Y
	7162	4			-2.5	U			Mex		64Ma.B
	7157	8			-0.6	U			MIT		64Sp12
	11215	15	11200.90	0.08	-0.9	U			Mex		64Ma.B
$^7\text{Li}(^3\text{He},\text{p})^9\text{Be}$	-22499	50	-22450	30	1.0	o			Brk		65De08
$^9\text{Be}(\text{p},^3\text{He})^7\text{Li}^i$	-22479	40			0.8	1	61	61 $^7\text{Li}^i$	Brk		67Mc14
$^7\text{Be}(^3\text{He},\text{n})^9\text{C}$	-6287	5	-6282.1	2.1	1.0	3			CIT		67Ba.A Z
	-6275.2	3.5			-2.0	3			CIT		71Mo01 Z
	100	60	1250	50	19.2	B			MSU		01Ch31 *
$^9\text{He}(\gamma,\text{n})^8\text{He}$	1270	100			-0.2	-			Ber		99Bo26
	2000	200			-3.7	B					07Go24
	1330	80			-0.9	-					10Jo06 *
	ave.	1310	60		-0.8	1	56	56 ^9He			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^9\text{Be}(\gamma, n)^8\text{Be}$	-1665	1	-1664.54	0.08	0.5	U			Wis	50Mo56	Y
$^9\text{Be}(p, d)^8\text{Be}$	557	3	560.03	0.08	1.0	U			CIT	49To16	Y
	558	5			0.4	U			Chi	51Ca37	Y
	557.5	1.			2.5	U			Wis	51Wi26	Y
	560	2			0.0	U			Bir	53Co02	Y
	562	4			-0.5	U			MIT	64Sp12	
	559.0	1.1			0.9	U			Zur	66Re02	
	559.6	0.6			0.7	U			NDm	67Od01	Z
$^9\text{Be}(d, t)^8\text{Be}$	4602	13	4592.69	0.08	-0.7	U			MIT	64Sp12	
	4591.7	3.1			0.3	U			NDm	67Od01	
$^9\text{Be}(^3\text{He}, \alpha)^8\text{Be}$	18931	13	18913.08	0.08	-1.4	U			Mex	64Ma.B	
$^9\text{Be}(\pi^-, \pi^+)^9\text{He}$	-30472	100	-30610	50	-1.4	-				87Se05	
$^9\text{Be}(^{13}\text{C}, ^{13}\text{O})^9\text{He}$	-50200	600	-49580	50	1.0	o			Ber	88Bo20	
	-49470	80			-1.3	o			Ber	91Bo.B	
$^9\text{Be}(^{14}\text{C}, ^{14}\text{O})^9\text{He}$	-34580	100	-34580	50	0.0	-			Ber	95Bo.B	
$^9\text{Be}(\pi^-, \pi^+)^9\text{He}$	ave. -30540	70	-30610	50	-0.9	1	44	44 ^9He		average	
$^9\text{Be}(p, n)^9\text{B}$	-1850.4	1.0				2			Wis	50R159	Z
	-1852	3	-1850.4	0.9	0.5	U			Ric	55Ma84	Z
$^9\text{He}(\gamma, n)^8\text{He}$	From scattering length $a_s = -10$ fm; questioned in reference									10Jo06	**
$^9\text{He}(\gamma, n)^8\text{He}$	Scattering length $a_s = -3.17(66)$ fm									10Jo06	**
$^{10}\text{B } ^{37}\text{Cl}-\text{C } ^{35}\text{Cl}$	9987.21	0.56	9986.74	0.07	-0.3	U			H38	2.5	84El05
$^{10}\text{Be}-^7\text{Li}_{1.429}$	-9334.16	0.13	-9334.22	0.09	-0.4	1	44	44 ^{10}Be	TT1	1.0	09Ri03
$^{10}\text{B}-u$	12936.862	0.016	12936.862	0.016	0.0	1	100	100 ^{10}B	MS1	1.0	16Gu02
$^{10}\text{C}-^{10}\text{B}$	3916.413	0.090	3916.36	0.07	-0.6	1	67	67 ^{10}C	JY1	1.0	11Er02
$^7\text{Li}(t, \gamma)^{10}\text{Be}^i$	-3929.8	21.0				2					73Ab10
$^{10}\text{B}(n, \alpha)^7\text{Li}$	2801	4	2789.908	0.015	-2.8	U					67De15
$^7\text{Li}(\alpha, n)^{10}\text{B}$	-2787	4	-2789.908	0.015	-0.7	U			Ric	57Bi84	Y
$^{10}\text{B}(p, \alpha)^7\text{Be}$	1147	5	1145.67	0.07	-0.3	U			CIT	49Ch35	Y
	1146	6			-0.1	U			CIT	51Br10	Y
	1146	2			-0.2	U			Wis	52Cr30	Y
	1153	4			-1.8	U			MIT	64Sp12	
$^{10}\text{B}(^3\text{He}, ^6\text{He})^7\text{B}$	-18550	100	-18287	25	2.6	U			Brk	67Mc14	
$^{10}\text{He}(\gamma, 2n)^8\text{He}$	1200	300	1440	90	0.8	U					94Ko16
	1420	100			0.2	2					10Jo06
	2100	200			-3.3	B					12Si07
	1600	250			-0.6	2					12Ko43
	1400	300			0.1	U					15Ma54
$^{10}\text{Be}(p, ^3\text{He})^8\text{Li}^i$	-26802.3	5.4				2			MSU	75Ro01	*
$^{10}\text{Be}(p, t)^8\text{Be}^j$	-27487.0	2.6	-27489.3	2.0	-0.9	1	57	57 $^8\text{Be}^j$	MSU	75Ro01	*
$^{10}\text{B}(d, \alpha)^8\text{Be}$	17829	10	17819.75	0.04	-0.9	U			Bir	54El10	Y
	17830	6			-1.7	U			Mex	64Ma.B	
	17818.6	4.1			0.3	U			NDm	67Od01	
$^{10}\text{B}(p, ^3\text{He})^8\text{Be}$	-535.5	2.5	-533.31	0.04	0.9	U			Wis	52Cr30	Y
$^{10}\text{Li}(\gamma, n)^9\text{Li}$	150	150	26	13	-0.8	U					90Am05
	25	15			0.1	3					95Zi03
	30	24			-0.1	3					08Ak03
$^{10}\text{Li}^m(\gamma, n)^9\text{Li}$	240	60	220	40	-0.3	3					97Bo10
	210	50			0.2	3					97Zi04
$^9\text{Be}(^9\text{Be}, ^8\text{B})^{10}\text{Li}^n$	-33770	260	-33750	40	0.1	U			Brk	75Wi26	*
$^9\text{Be}(^{13}\text{C}, ^{12}\text{N})^{10}\text{Li}^n$	-36370	50	-36390	40	-0.4	2			Ber	93Bo03	*
$^{10}\text{Be}(d, ^3\text{He})^9\text{Li}$	-14142.8	2.5	-14142.91	0.20	-0.0	U			MSU	75Ka18	
$^9\text{Be}(n, \gamma)^{10}\text{Be}$	6812.33	0.06	6812.28	0.05	-0.8	-			MMn	86Ke14	Z
	6812.10	0.14			1.3	-			Bdn	06Fi.A	
$^9\text{Be}(d, p)^{10}\text{Be}$	4583	8	4587.72	0.05	0.6	U			Ric	51Ki55	Y
	4595	4			-1.8	U			Mex	64Ma.B	
	4590	8			-0.3	U			MIT	64Sp12	
$^9\text{Be}(n, \gamma)^{10}\text{Be}$	ave. 6812.29	0.06	6812.28	0.05	-0.2	1	88	56 ^{10}Be		average	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^9\text{Be}(^3\text{He},\text{d})^{10}\text{B}$	1123	5	1093.34	0.08	-5.9	C			Mex		64Ma.B
$^{10}\text{B}(\text{d},\text{t})^9\text{B}$	-2189	10	-2180.0	0.9	0.9	U			MIT		64Sp12
$^{10}\text{B}(^3\text{He},\alpha)^9\text{B}$	12130	15	12140.4	0.9	0.7	U			Ric		60Sp08
	12171	15			-2.0	U			Mex		64Ma.B
$^{10}\text{Be}(^{14}\text{C},^{14}\text{O})^{10}\text{He}$	-41190	70	-41580	90	-5.5	B			Ber		94Os04
$^{10}\text{Be}(\beta^-)^{10}\text{B}$	560	5	556.88	0.08	-0.6	U					50Hu27
	555	5			0.4	U					52Fe16
$^{10}\text{C}(\beta^+)^{10}\text{B}$	3604	16	3648.06	0.07	2.8	U					63Ba52
$^{10}\text{B}(\text{p},\text{n})^{10}\text{C}$	-4433.7	1.5	-4430.41	0.07	2.2	U			Har		75Fr.A
	-4430.17	0.34			-0.7	o			Auc		84Ba12 *
	-4430.17	0.09			-2.7	o			Auc		89Ba28 *
	-4430.30	0.12			-0.9	1	33	33 ^{10}C	Auc		98Ba83 *
$^{10}\text{B}(^3\text{He},\text{t})^{10}\text{C}$	-3667	10	-3666.65	0.07	0.0	U			Brk		68Br23
$^{10}\text{B}(^{14}\text{N},^{14}\text{B})^{10}\text{N}$	-47550	400				2					02Le16
$^{10}\text{Be}(\text{p},^3\text{He})^8\text{Li}^i$	Original value -26804.1(5.4) recalibrated										AHW **
$^{10}\text{Be}(\text{p},\text{t})^8\text{Be}^j$	Original -27487.6(2.6) recalibrated										GAu **
$^{10}\text{Li}(\gamma,\text{n})^9\text{Li}$	From $^{11}\text{B}(\pi^-, \text{p})^{10}\text{Li}$										GAu **
$^{10}\text{Li}(\gamma,\text{n})^9\text{Li}$	Resonance less than 50 above the one neutron threshold, but										95Zi03 **
*	could also be final state interaction; then ^{10}Li would be 200 higher										97Bo10 **
$^{10}\text{Li}(\gamma,\text{n})^9\text{Li}$	Deduced from s-state scattering length $a_s = -22.4(4.8)$ fm										08Ak03 **
$^{10}\text{Li}^m(\gamma,\text{n})^9\text{Li}$	From $^{10}\text{Be}(^{12}\text{C},^{12}\text{N})^{10}\text{Li}^m$ (1^+ level)										GAu **
$^{10}\text{Li}^m(\gamma,\text{n})^9\text{Li}$	Theoretical work: 1^+ level above 1^- ground state										02Ga12 **
$^9\text{Be}(^9\text{Be},^8\text{B})^{10}\text{Li}^n$	$Q = -34060(250)$ to 2^+ level 290(80) above 1^+ level										93Bo03 **
*	revised with Breit-Wigner line shape. Probably 2^+ level										97Bo10 **
$^9\text{Be}(^{13}\text{C},^{12}\text{N})^{10}\text{Li}^n$	Revised with Breit-Wigner line shape (probably 2^+ level)										97Bo10 **
$^{10}\text{B}(\text{p},\text{n})^{10}\text{C}$	$T=4876.90(0.37)$; withdrawn by author										89Ba28 **
$^{10}\text{B}(\text{p},\text{n})^{10}\text{C}$	$T=4876.88(0.10, Z)$; original $T=4876.95(0.10)$ keV										MMC126**
$^{10}\text{B}(\text{p},\text{n})^{10}\text{C}$	Average of two datasets; withdrawn by author										98Ba83 **
$^{10}\text{B}(\text{p},\text{n})^{10}\text{C}$	$T=4877.03(0.13)$; this is the second 89Ba28 dataset, recalibrated by author										98Ba83 **
$^{11}\text{B } ^{37}\text{Cl}-^{13}\text{C } ^{35}\text{Cl}$	2998.15	1.30	3000.21	0.07	0.6	U			H38	2.5	84El05
$^{11}\text{Li}-^6\text{Li}_{1.833}$	16003.5	1.2	16003.3	0.7	-0.1	o			TT1	1.0	08Sm.A
	16003.33	0.66				2			TT1	1.0	08Sm03
$^{11}\text{Be}-^6\text{Li}_{1.833}$	-6059.27	0.28	-6059.17	0.26	0.4	1	83	83 ^{11}Be	TT1	1.0	08Br.C
$^{11}\text{Be}-^7\text{Li}_{1.571}$	-3479.83	0.62	-3480.31	0.26	-0.8	1	17	17 ^{11}Be	TT1	1.0	09Ri03
$^{11}\text{C}-^{14}\text{N}_{.786}$	9016.430	0.064	9016.43	0.06	0.0	1	100	100 ^{11}C	MS1	1.0	16Gu02
$^{11}\text{Li}-\text{u}$	43780	130	43723.6	0.7	-0.3	U			TO2	1.5	88Wo09
	43805	28			-2.9	U			P40	1.0	03Ba.A
	43715.4	5.0			1.6	o			P40	1.0	04Ba.A
	43714.5	5.1			1.8	U			P40	1.0	09Ga24
$^{11}\text{Be}-\text{u}$	21654.0	3.6	21661.08	0.26	2.0	o			P40	1.0	04Ba.A
	21653.5	3.5			2.2	U			P40	1.0	09Ga24
	21658.5	3.8			0.7	U			P40	1.0	09Ga24 *
$^{11}\text{B}-\text{u}$	9305.167	0.013	9305.167	0.013	-0.0	1	100	100 ^{11}B	MS1	1.0	16Gu02
$^9\text{Li}-^{11}\text{Li}_{.491} \ ^6\text{Li}_{.600}$	-3949	175	-3494.8	0.4	1.0	U			P12	2.5	75Th08
$^9\text{Li}-^{11}\text{Li}_{.409} \ ^7\text{Li}_{.643}$	-1250	86	-1288.2	0.3	-0.3	U			P11	1.5	75Th08 *
	-1223	195			-0.3	U			P13	1.0	75Th08
$^9\text{Li}-^{11}\text{Li}_{.273} \ ^8\text{Li}_{.750}$	-1928	31	-1873.26	0.25	0.7	U			P12	2.5	75Th08
	-1923	31			1.6	U			P13	1.0	75Th08
$^7\text{Li}(\alpha, \gamma)^{11}\text{B}^i$	-3885.6	20.0	-3896	9	-0.5	1	21	21 $^{11}\text{B}^i$			66Cu02 *
$^{11}\text{B}(\text{p}, \alpha)^8\text{Be}$	8583	15	8590.09	0.04	0.5	U			CIT		51Li26 Y
	8589	4			0.3	U			Bir		53Co02 Y
	8597	6			-1.2	U			Mex		61Ja23
	8575	11			1.4	U			MIT		64Sp12
$^{11}\text{B}(^3\text{He}, ^6\text{He})^8\text{B}^i$	-27539	8				2			MSU		75Ro01 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{11}\text{Be}^i(\gamma, d)^9\text{Li}$	3245	20				3					97Te07 *
$^9\text{Be}(t, p)^{11}\text{Be}$	-1164	15	-1167.88	0.25	-0.3	U			Ald		62Pu01
$^{11}\text{B}^j(2p)^9\text{Li}$	2700	80				3					12Ch40
$^{11}\text{B}(d, \alpha)^9\text{Be}$	8029	4	8030.06	0.08	0.3	U			Bir		54El10 Y
	8035	9			-0.5	U			Mex		64Ma.B
	8024	7			0.9	U			MIT		64Sp12
	8029.7	2.8			0.1	U			NDm		67Od01
$^9\text{Be}(^3\text{He}, p)^{11}\text{B}$	10344	13	10322.99	0.08	-1.6	U			Mex		64Ma.B
	10322.1	2.3			0.4	U			NDm		67Od01
$^{11}\text{B}(p, ^3\text{He})^9\text{Be}^i$	-24713.3	1.7				2			MSU		74Ka15 *
$^9\text{Be}(^3\text{He}, p)^{11}\text{B}^i$	-2240	12	-2237	9	0.2	-					63Gr.A *
	-2240.6	20.			0.2	-			MSU		82Zw02 *
ave.	-2240	10			0.3	1	79	79 $^{11}\text{B}^i$			average
$^{11}\text{B}(p, t)^9\text{B}^i$	-26064.1	2.3				2			MSU		74Ka15 *
$^9\text{Be}(^3\text{He}, n)^{11}\text{C}^i$	-4612	50	-4600	40	0.2	1	50	50 $^{11}\text{C}^i$			71Wa21 *
$^{11}\text{O}(2p)^9\text{C}$	4250	60				4					20We08
$^{10}\text{Be}(d, p)^{11}\text{Be}$	-1721	7	-1722.93	0.25	-0.3	U			CIT		70Go11
$^{11}\text{B}(^7\text{Li}, ^8\text{B})^{10}\text{Li}$	-32431	80	-32399	13	0.4	U			MSU		94Yo01 *
$^{11}\text{B}(^7\text{Li}, ^8\text{B})^{10}\text{Li}^n$	-32908	62	-32870	40	0.5	R			MSU		94Yo01
$^{10}\text{Be}(p, \gamma)^{11}\text{B}^i$	-1322	30	-1332	9	-0.3	U					70Go04 *
$^{10}\text{B}(n, \gamma)^{11}\text{B}$	11454.1	0.2	11454.221	0.019	0.6	U			Ptn		86Ko19 Z
	11454.15	0.27			0.3	U			Bdn		06Fi.A
$^{10}\text{B}(d, p)^{11}\text{B}$	9227	5	9229.654	0.019	0.5	U			Bir		54El10 Y
	9234	6			-0.7	U			Mex		64Ma.B
	9244	11			-1.3	U			MIT		64Sp12
	9232.9	2.			-1.6	U			NDm		66Br18
$^{11}\text{B}(^3\text{He}, \alpha)^{10}\text{B}$	9101	20	9123.400	0.019	1.1	U			Man		60Ta12
$^{10}\text{B}(^3\text{He}, d)^{11}\text{C}$	3174	15	3196.71	0.06	1.5	U			Man		60Fo01
	3226	10			-2.9	C			Mex		64Ma.B
$^{11}\text{N}(p)^{10}\text{C}$	1973	180	1378	5	-3.3	B			MSU		74Be20 *
	1300	40			1.9	o			Lis		96Ax01
	1450	400			-0.2	U			MSU		98Az01 *
	1630	50			-5.0	B			Spe		00OI01 *
	1350	120			0.2	U			Lis		00Ma62 *
	1310	50			1.4	U			INS		03Gu06
	1540	20			-8.1	B					06Ca05
	1378	5				2					19We11
$^{11}\text{B}(\pi^-, \pi^+)^{11}\text{Li}$	-33120	50	-33082.5	0.6	0.7	U					91Ko.B
$^{11}\text{B}(^{14}\text{C}, ^{14}\text{O})^{11}\text{Li}$	-37120	35	-37048.4	0.6	2.0	U			MSU		93Yo07
$^{11}\text{B}^i(\text{IT})^{11}\text{B}$	12510	50	12560	9	1.0	U					71Wa21 *
$^{11}\text{C}(\beta^+)^{11}\text{B}$	1982.8	2.6	1981.69	0.06	-0.4	U					75Be28
$^{11}\text{B}(p, n)^{11}\text{C}$	-2759.7	3.	-2764.04	0.06	-1.4	U			Wis		50Ri59 Z
	-2763.2	1.4			-0.6	U			Ric		61Be13 Z
$^{11}\text{B}(^3\text{He}, t)^{11}\text{C}$	-2002.1	1.2	-2000.28	0.06	1.5	U			Str		65Go05 Z
$^{11}\text{B}(^3\text{He}, t)^{11}\text{C}^i$	-14151	50	-14160	40	-0.2	1	50	50 $^{11}\text{C}^i$			71Wa21 *
* $^{11}\text{Be}-u$	Result from the "cooling" experiment										09Ga24 **
* $^9\text{Li}-^{11}\text{Li}_{409} \ ^7\text{Li}_{643}$	Symmetric double-doublet 6-9-8-11 included										GAu **
* $^7\text{Li}(\alpha, \gamma)^{11}\text{B}^i$	IT=12550(30); Q rebuilt with Ame1964										GAu **
* $^{11}\text{B}(^3\text{He}, ^6\text{He})^8\text{B}^i$	IT=10619(9); rebuilt $Q = -27538.9(8.2)$ keV										GAu **
*	IT=10614(20) keV by a different method										14Br15 **
* $^{11}\text{Be}^i(\gamma, d)^9\text{Li}$	$Q(d)=Q(p+n)+2224.5660(0.0004)$ $Q(p+n)=1020(20)$ keV										MMC162**
* $^{11}\text{B}(p, ^3\text{He})^9\text{Be}^i$	IT=14392.2(1.8); rebuilt $Q = -24715.2(1.7)$; recalibrated +1.87 keV										MMC121**
* $^9\text{Be}(^3\text{He}, p)^{11}\text{B}^i$	IT=12565(12); Q rebuilt with Ame1961										MMC121**
* $^9\text{Be}(^3\text{He}, p)^{11}\text{B}^i$	IT=12563(20); Q rebuilt with Ame1977										MMC121**
* $^{11}\text{B}(p, t)^9\text{B}^i$	IT=14655.4(2.5); rebuilt $Q = -26064.3$; recalibrated +0.16 keV										MMC121**
* $^9\text{Be}(^3\text{He}, n)^{11}\text{C}^i$	IT=12170(40); Q rebuilt; possibly not pure T=3/2										MMC121**
* $^{11}\text{B}(^7\text{Li}, ^8\text{B})^{10}\text{Li}$	Original (>-32471) re-evaluated										GAu **
*	existence of this level not completely certain										94Yo01 **
* $^{10}\text{Be}(p, \gamma)^{11}\text{B}^i$	IT=12550(30); Q rebuilt with Ame1964										MMC121**
* $^{11}\text{N}(p)^{10}\text{C}$	From $^{14}\text{N}(^3\text{He}, ^6\text{He})^{11}\text{N}$ $Q = -25010(100)$ to 250(150) level										90Aj01 **
* $^{11}\text{N}(p)^{10}\text{C}$	From $^9\text{Be}(^{12}\text{N}, ^{10}\text{Be})^{11}\text{N}$										98Az01 **
* $^{11}\text{N}(p)^{10}\text{C}$	From $^{10}\text{B}(^{14}\text{N}, ^{13}\text{B})^{11}\text{N}$										00OI01 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{11}\text{N}(\text{p})^{10}\text{C}$	From scattering ^{10}C on H; precisely, 1270(+180–50) keV										00Ma62 **
$^{11}\text{B}^i(\text{IT})^{11}\text{B}$	From $^{11}\text{B}(\text{}^3\text{He}, \text{}^3\text{He}')$										AHW **
$^{11}\text{B}(\text{}^3\text{He}, \text{t})^{11}\text{C}^i$	IT=12150(50); Q rebuilt; possibly not pure T=3/2										MMC121**
$^{12}\text{Be}-\text{u}$	26911.3	14.2	26922.1	2.0	0.8	U			P40	1.0	09Ga24
$^{12}\text{Be}-\text{C}$	26922.4	2.3			−0.1	1	79	$^{79}\text{}^{12}\text{Be}$	TT1	1.0	10Et01
$\text{C}_{14}-^{12}\text{C}_{12}$	1.2	4.9	0.00000	0.00013	−0.2	U			TG1	1.5	09Ke.A
$\text{C}_{14}-^{12}\text{C}_{15}$	2.1	2.7			−0.5	U			TG1	1.5	11Ke03
$\text{C}_{15}-^{12}\text{C}_{14}$	−4.0	4.7			0.6	U			TG1	1.5	11Ke03
	−0.5	6.2			0.1	U			TG1	1.5	11Ke03
$\text{C}_{15}-^{12}\text{C}_{16}$	−1.6	2.1			0.5	U			TG1	1.5	10Ke09
	4.7	4.2			−0.7	U			TG1	1.5	11Ke03
	3.7	4.1			−0.6	U			TG1	1.5	11Ke03
$\text{C}_{16}-^{12}\text{C}_{12}$	−0.3	6.1			0.0	U			TG1	1.5	09Ke.A
	2.6	5.0			−0.3	U			TG1	1.5	09Ke.A
$\text{C}_{16}-^{12}\text{C}_{15}$	−1.3	3.9			0.2	U			TG1	1.5	11Ke03
	−0.7	1.3			0.4	U			TG1	1.5	12Sm07
$\text{C}_{21}-^{12}\text{C}_{22}$	1.6	1.3			−0.8	U			TG1	1.5	14Ei01
$\text{C}_{23}-^{12}\text{C}_{22}$	1.2	1.7			−0.5	U			TG1	1.5	14Ei01
$\text{C}_{23}-^{12}\text{C}_{21}$	2.2	1.8			−0.8	U			TG1	1.5	14Ei01
$^7\text{Li}(\text{}^7\text{Li}, 2\text{p})^{12}\text{Be}$	−9710	100	−9841.5	1.9	−1.3	U			LAI		71Ho26
$^{12}\text{C}(\alpha, \text{}^8\text{He})^8\text{C}$	−64520	200	−64249	18	1.4	U					74Ro17
	−64278	26			1.1	−			Tex		76Tr01
ave.	−64270	23			0.9	1	63	$^{63}\text{}^8\text{C}$			average
$^9\text{Be}(\text{}^7\text{Li}, \alpha)^{12}\text{B}^i$	−2308.4	50.	−2258	19	1.0	1	14	$^{14}\text{}^{12}\text{B}^i$	Phi		75Aj03 *
$^{12}\text{C}(\text{}^3\text{He}, \text{}^6\text{He})^9\text{C}$	−31578	8	−31571.8	2.1	0.8	U			MSU		71Tr03
	−31575.6	3.2			1.2	R			MSU		79Ka.A
$^{10}\text{Be}(\text{t}, \text{p})^{12}\text{Be}$	−4809	15	−4809.4	1.9	−0.0	U			Brk		78Al29
	−4808.3	4.2			−0.3	1	21	$^{21}\text{}^{12}\text{Be}$	Phi		94Fo08
$^{10}\text{B}(\text{t}, \text{p})^{12}\text{B}$	6346	6	6342.1	1.3	−0.7	U			Man		60Ja17
$^{10}\text{B}(\alpha, \text{d})^{12}\text{C}$	1340.3	0.8	1339.804	0.015	−0.6	U			Wis		56Do41 Z
	1340.6	1.5			−0.5	U			NDm		65Br28
$^{12}\text{C}(\text{d}, \alpha)^{10}\text{B}$	−1340.1	1.2	−1339.804	0.015	0.2	U			NDm		65Br28
$^{10}\text{B}(\text{}^3\text{He}, \text{p})^{12}\text{C}$	19694.5	3.6	19692.858	0.015	−0.5	U			NDm		67Od01
	19692.86	0.44			−0.0	U			Mun		83Ch08 *
$^{10}\text{B}(\text{}^3\text{He}, \text{p})^{12}\text{C}^i$	4585	6	4585	3	−0.1	1	31	$^{31}\text{}^{12}\text{C}^i$			62Br10
$^{10}\text{B}(\text{}^3\text{He}, \text{n})^{12}\text{N}$	1570	25	1572.4	1.0	0.1	U			CIT		64Fi02
	1561	9			1.3	U			CIT		64Ka08
	1568	20			0.2	U			LAI		66Za01
	1574	7			−0.2	U			Har		68Ad03
$^{12}\text{N}^i(2\text{p})^{10}\text{B}$	2905	29	2952	4	1.6	U					12Ja11 *
$^{12}\text{O}(2\text{p})^{10}\text{C}$	1770	20	1737	12	−1.7	2					95Kr03
	1638	24			4.1	B					12Ja11
	1688	29			1.7	o					19We03
	1718	15			1.2	2					19We11
$^{12}\text{Li}(\gamma, \text{n})^{11}\text{Li}$	120	15	210	30	6.0	B					08Ak03 *
	210	30				3					13Ko03
$^{11}\text{B}(\text{d}, \text{p})^{12}\text{B}$	1141	4	1145.1	1.3	1.0	1	11	$^{11}\text{}^{12}\text{B}$	Mex		61Ja23
	1137	5			1.6	U			MIT		64Sp12
$^{11}\text{B}(\text{}^3\text{He}, \text{d})^{12}\text{C}$	10436	17	10463.204	0.012	1.6	U			Man		60Fo01
	10469.7	5.7			−1.1	U			NDm		67Od01
$^{11}\text{B}(\text{d}, \text{n})^{12}\text{C}^i$	−1376.2	4.0	−1376	3	0.0	1	69	$^{69}\text{}^{12}\text{C}^i$			55Ma76 *
$^{12}\text{N}(\beta^+)^{12}\text{C}$	17406	15	17338.1	1.0	−4.5	B					63Gi04

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{12}\text{C}(\text{p},\text{n})^{12}\text{N}$	-18119.9	4.4	-18120.4	1.0	-0.1	U			Yal	69Ov01	Z
$^{12}\text{C}(\pi^+, \pi^-)^{12}\text{O}$	-31037	48	-30991	12	1.0	B				80Bu15	
	-31014	24			0.9	B				92Iv.A	
$^{12}\text{N}^i(\text{IT})^{12}\text{N}$	12242	4				3				19We11	
$^9\text{Be}(^7\text{Li}, \alpha)^{12}\text{B}^i$	IT=12770(50) using $Q_{gs}=10461.6(1.6)\text{keV}$									75Aj03	**
*	energy and resolution arguments for T=1, not an IAS									08Ch28	**
$^{10}\text{B}(^3\text{He}, \text{p})^{12}\text{C}$	Original $Q=15305.45(0.3)$ revised by authors to $15253.95(31)\text{keV}$									83Vo.A	**
*	to 2^+ level at $4438.91(0.31)\text{keV}$									Ens006	**
$^{12}\text{N}^i(2\text{p})^{10}\text{B}$	$Q_{2p}=1165(29)$ to $^{10}\text{B}^i$ at $1740.05(0.04)$									Nub211	**
$^{12}\text{Li}(\gamma, \text{n})^{11}\text{Li}$	E_{res} derived in reference from scattering length $a_s=-13.7(1.6)\text{fm}$									10Ha04	**
$^{11}\text{B}(\text{d}, \text{n})^{12}\text{C}^i$	$E_{res}=1627(4)$ $Q=-1376(4)$; recalibrated $Q=-1376.16(4.00)\text{keV}$									MMC121**	
C H— ^{13}C	4500	36	4470.19656	0.00025	-0.6	U			R08	1.5	69De19
	4470.185	0.008			1.0	U			B08	1.5	75Sm02
	4470.10	0.05			0.8	U			M25	2.5	78Ha14
C D— ^{13}C H	2921.923	0.008	2921.91061	0.00025	-1.0	U			B08	1.5	75Sm02
	2921.87	0.05			0.3	U			M25	2.5	78Ha14
	2921.9086	0.0012			1.7	U			MI1	1.0	95Di08
	2921.9074	0.0015			2.1	U			MI1	1.0	95Di08
$^{13}\text{C}-\text{u}$	3354.8404	0.0041	3354.83534	0.00025	-0.5	U			WA1	2.5	95Va38
$^9\text{Be}(\alpha, \gamma)^{13}\text{C}^i$	-4458.4	2.0	-4460.4	1.1	-1.0	2					73Ad02
	-4461.4	1.4			0.7	2					78Hi06
$^{10}\text{B}(\alpha, \text{p})^{13}\text{C}$	4068	12	4061.546	0.015	-0.5	U			MIT		64Sp12
	4063.4	2.4			-0.8	U			NDm		67Od01
$^{13}\text{Li}(\gamma, 2\text{n})^{11}\text{Li}$	1470	310	110	70	-4.4	B					08Ak03
	1470	350			-3.9	B					10Jo07
	110	70				3					13Ko03
$^{11}\text{B}(\text{t}, \text{p})^{13}\text{B}$	-233	4	-233.4	1.0	-0.1	U			Man		60Mu07
	-233.4	1.0				2			Str		83An15
$^{13}\text{C}(\text{d}, \alpha)^{11}\text{B}$	5169	6	5168.108	0.012	-0.1	U			CIT		51Li29
	5166	5			0.4	U			Ric		53Ph28
	5165	10			0.3	U			MIT		64Sp12
	5166.6	2.5			0.6	U			NDm		70Br23
$^{11}\text{B}(^3\text{He}, \text{p})^{13}\text{C}$	13221	10	13184.947	0.012	-3.6	C			Mex		64Ma.B
	13185.4	4.0			-0.1	U			NDm		67Od01
$^{11}\text{B}(^3\text{He}, \text{n})^{13}\text{N}$	10183	11	10182.13	0.27	-0.1	U					71Hs03
$^{13}\text{Be}(\gamma, \text{n})^{12}\text{Be}$	100	70	510	10	5.9	B					01Th01
	60	10			45.0	B					08Ch07
	510	10				2					10Ko17
	400	30			3.7	B					14Ra07
$^{12}\text{C}(\text{n}, \gamma)^{13}\text{C}$	4946.47	0.17	4946.3087	0.0005	-0.9	U					67Pr10
	4946.03	0.15			1.9	U			Utr		68Sp01
	4946.51	0.31			-0.6	U			ILn		79Br25
	4946.321	0.024			-0.5	U					80Wa24
	4946.337	0.031			-0.9	U			Utr		81Va.B
	4946.31	0.10			-0.0	U			Bdn		06Fi.A
$^{12}\text{C}(\text{d}, \text{p})^{13}\text{C}$	2727	6	2721.74250	0.00024	-0.9	o			Ric		51KI55
	2722	4			-0.1	U			Ric		53Fa18
	2720	2			0.9	U			Bir		54El10
	2725	5			-0.7	U			Mex		61Ja23
	2722	4			-0.1	U			MIT		64Sp12
	2722.3	0.6			-0.9	o			NDm		67Od01
	2721.9	0.8			-0.2	U			NDm		74Jo14
	2721.80	0.50			-0.1	U			Rez		90Pi05
$^{13}\text{C}(\text{p}, \text{d})^{12}\text{C}$	-2722	7	-2721.74250	0.00024	0.0	o			MIT		64Sp12

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{13}\text{C}(\text{d,t})^{12}\text{C}$	1311	3	1310.92133	0.00024	−0.0	U			CIT	51Li29	Y	
	1311	6			−0.0	U			Mex	64Ma.B		
	1311	6			−0.0	U			MIT	64Sp12		
	1310.9	0.7			0.0	U			NDm	67Od01		
$^{12}\text{C}(\text{p},\gamma)^{13}\text{N}$	1943.24	0.32	1943.49	0.27	0.8	2				77Fr20	Z	
	1944.1	0.5			−1.2	2				77He26	Z	
$^{12}\text{C}(\text{d,n})^{13}\text{N}$	−280.5	3.	−281.08	0.27	−0.2	U			Ric	49Bo67	Y	
$^{12}\text{C}(\text{p},\gamma)^{13}\text{N}^i$	−13121.62	0.18				2				73Hu07		
$^{13}\text{C}(^{14}\text{C},^{14}\text{O})^{13}\text{Be}^p$	−37020	50				2			Ber	92Os04		
$^{13}\text{N}(\beta^+)^{13}\text{C}$	2222.3	3.8	2220.47	0.27	−0.5	U				54Ki23		
$^{13}\text{C}(\text{p,n})^{13}\text{N}$	−3002.3	1.0	−3002.82	0.27	−0.5	o			Ric	61Be13	Z	
	−3004.1	1.5			0.9	U			NRL	64Bo10	Z	
	−3002.4	1.0			−0.4	U			Ric	66Bo20		
$^{13}\text{O}(\beta^+)^{13}\text{N}$	17500	200	17770	10	1.3	U				65Mc09		
$^{13}\text{Li}(\gamma,2\text{n})^{11}\text{Li}$	Corresponds to excited state										13Ko03	**
$^{13}\text{Li}(\gamma,2\text{n})^{11}\text{Li}$	Symmetrized from 120(+60−80)										GAu	**
$^{13}\text{Be}(\gamma,\text{n})^{12}\text{Be}$	From scattering length $a_s < -10$ fm; questioned in reference										10Ko17	**
$^{13}\text{Be}(\gamma,\text{n})^{12}\text{Be}$	From scattering length $a_s = -20$ fm; questioned in reference										10Ko17	**
*	$a_s = -3.4(0.6)$ fm in 10Ko17; $a_s = -3.2(+.9-1.1)$ fm in 07Si24										07Si24	**
$^{12}\text{C}(\text{n},\gamma)^{13}\text{C}$	$Q(\gamma)=1261.844(0.006,\text{Z})$ to $3684.477(0.023,\text{Z})$ level										81Va.B	**
$^{12}\text{C}(\text{n},\gamma)^{13}\text{C}$	$Q(\gamma)=1261.844(0.006,\text{Z})$ to $3684.493(0.030,\text{Z})$ level										81Va.B	**
$^{12}\text{C}(\text{d,p})^{13}\text{C}$	Estimated systematic error 0.5 added to statistical error 0.038 keV										AHW	**
$^{14}\text{Be}-\text{u}$	42660	150	42890	140	1.0	2			TO2	1.5	88Wo09	
$\text{C D}_2-^{14}\text{C H}_2$	9311.498	0.006	9311.503	0.004	0.6	1	20	20 ^{14}C	B08	1.5	75Sm02	
$\text{C H}_2-\text{N}$	12576.22	0.10	12576.05954	0.00024	−0.6	U			J2	2.5	69Na21	
	12576.086	0.009			−2.0	U			B07	1.5	71Sm01	
	12576.0598	0.0008			−0.3	U			MI1	1.0	95Di08	
$\text{C D}-\text{N}$	11027.815	0.018	11027.77359	0.00024	−1.5	o			B07	1.5	71Sm01	
	11027.773	0.007			0.1	U			B08	1.5	75Sm02	
$\text{C H}_4-\text{N D}$	14124.17	0.14	14124.34550	0.00024	0.5	U			J6	2.5	76Ka50	
$\text{C D}_2-\text{N H}_2$	9479.68	0.13	9479.48764	0.00025	−0.6	U			J6	2.5	76Ka50	
$^{14}\text{N}-\text{u}$	3074.014	0.019	3074.00425	0.00024	−0.2	U			OH1	2.5	93Ma.A	
	3074.0056	0.0018			−0.3	U			WA1	2.5	95Va38	
$^{13}\text{C H}-^{14}\text{N}$	8105.86299	0.00010	8105.86298	0.00010	−0.1	1	96	78 ^{13}C	MI3	1.0	04Ra33	
$^{14}\text{C H}_2-\text{N D}$	1716.269	0.003	1716.270	0.004	0.3	1	80	80 ^{14}C	B08	1.5	75Sm02	
$^{14}\text{O}-^{14}\text{N}$	5522.702	0.027	5522.702	0.027	−0.0	1	100	100 ^{14}O	MS1	1.0	15Va08	
$^{14}\text{N}(^3\text{He},^9\text{Li})^8\text{C}$	−42214	50	−42225	18	−0.2	R			MSU		76Ro04	
$^{11}\text{B}(\alpha,\text{p})^{14}\text{C}$	789	17	783.760	0.013	−0.3	U			MIT		64Sp12	
$^{14}\text{B}^i(\gamma,\text{d})^{12}\text{Be}$	2515	20				2					01Ta23	
$^{14}\text{C}(^{18}\text{O},^{20}\text{Ne})^{12}\text{Be}$	−15770	50	−15798.8	1.9	−0.6	U			ChR		74Ba15	
$^{14}\text{C}(\text{d},\alpha)^{12}\text{B}$	361.8	1.4	361.3	1.3	−0.4	1	89	89 ^{12}B	Wis		56Do41	
$^{14}\text{C}(\text{p},^3\text{He})^{12}\text{B}^i$	−30702.73	19.96	−30711	19	−0.4	1	86	86 $^{12}\text{B}^i$			71Ne.A	
$^{14}\text{C}(\text{p,t})^{12}\text{C}^j$	−32235.9	2.4				2			MSU		78Ro08	
$^{14}\text{N}(\text{d},\alpha)^{12}\text{C}$	13579	6	13574.22386	0.00027	−0.8	U			Mex		64Ma.B	
	13588	6			−2.3	U			MIT		64Sp12	
$^{12}\text{C}(^3\text{He},\text{p})^{14}\text{N}$	4779.0	1.4	4778.83098	0.00023	−0.1	U			CIT		62Ba26	
	4806	9			−3.0	C			Mex		64Ma.B	
	4776.3	1.5			1.7	U			NDm		67Od01	
$^{14}\text{N}(\text{p,t})^{12}\text{N}$	−22135.5	1.0	−22135.5	1.0	0.0	1	100	100 ^{12}N	MSU		75No.A	
$^{12}\text{C}(^3\text{He},\text{n})^{14}\text{O}$	−1146.86	0.72	−1147.880	0.025	−1.4	U			Nvl		61Bu04	
	−1148.61	0.56			1.3	U			CIT		62Ba26	
	−1149.01	0.48			2.4	U			Mar		70Ro07	
$^{14}\text{C}(^7\text{Li},^8\text{B})^{13}\text{Be}$	−39990	500	−38654	10	2.7	U			Dbn		83Al20	
$^{14}\text{C}(^{11}\text{B},^{12}\text{N})^{13}\text{Be}$	−39600	90	−39310	10	3.2	B			Dbn		98Be28	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{13}\text{C}(\text{n},\gamma)^{14}\text{C}$	8177	2	8176.434	0.004	-0.3	U					67Th05
	8176.61	0.24			-0.7	U			Bdn		06Fi.A
$^{13}\text{C}(\text{d},\text{p})^{14}\text{C}$	5946	4	5951.868	0.004	1.5	U			CIT		51Li29 Y
	5952	10			-0.0	U			Nob		54Ah47 Y
	5951	10			0.1	U			Mex		64Ma.B
	5951	8			0.1	U			MIT		64Sp12
	5951.85	0.54			0.0	U			Rez		90Pi05 *
$^{13}\text{C}(\text{p},\gamma)^{14}\text{N}$	7551.0	0.8	7550.56356	0.00009	-0.5	U					56Ma87 Z
	7551.1	0.5			-1.1	U					63Bo07 *
$^{13}\text{C}(^3\text{He},\text{d})^{14}\text{N}$	2048	14	2057.08848	0.00010	0.6	U			MIT		64Sp12
$^{14}\text{N}(^3\text{He},\alpha)^{13}\text{N}$	10015	10	10024.24	0.27	0.9	U			Ric		59Yo25
$^{14}\text{F}(\text{p})^{13}\text{O}$	1560	40				3					10Go16
$^{14}\text{C}(\pi^-, \pi^+)^{14}\text{Be}$	-38100	170	-37960	130	0.8	R					84Gi09 *
$^{14}\text{C}(^{14}\text{C}, ^{14}\text{O})^{14}\text{Be}^p$	-43440	60				2			Ber		95Bo10
$^{14}\text{C}(^7\text{Li}, ^7\text{Be})^{14}\text{B}$	-21499	30	-21506	21	-0.2	-			ChR		73Ba34
$^{14}\text{C}(^{14}\text{C}, ^{14}\text{N})^{14}\text{B}$	-20494	30	-20487	21	0.2	-			Ors		81Na.A
$^{14}\text{C}(^7\text{Li}, ^7\text{Be})^{14}\text{B}$	ave. -21506	21	-21506	21	0.0	1	100	100 ^{14}B			average
$^{14}\text{C}(\beta^-)^{14}\text{N}$	155.2	0.5	156.476	0.004	2.6	U					54Ki23
	155.74	0.08			9.2	F					91Su09 *
	155.95	0.22			2.4	U					95Wi20
	156.27	0.14			1.5	U					00Ku25
$^{14}\text{C}(\text{p},\text{n})^{14}\text{N}$	-626.15	0.3	-625.871	0.004	0.9	U			Wis		56Sa06
	-625.88	0.09			0.1	U			Zur		73Hi.A
$^{14}\text{N}(\text{p},\text{n})^{14}\text{O}$	-5930.7	2.8	-5926.711	0.025	1.4	U			Ric		65Ku02
	-5927.6	1.5			0.6	U			Har		73Cl12
	-5925.6	0.4			-2.8	F			Auc		77Wh01 *
	-5925.41	0.08			-16.3	F			Auc		81Wh03 *
	-5925.41	0.11			-11.8	F			Auc		98Ba83 *
	-5926.68	0.17			-0.2	U			Auc		03To03
$^{14}\text{N}(^3\text{He},\text{t})^{14}\text{O}$	-5161.3	0.8	-5162.956	0.025	-2.1	F			Mun		77Vo02 *
* $^{13}\text{C H}-^{14}\text{N}$	Original 8105.86288(10) μu recalculated for molecular and ionization										HWJ153 **
* $^{14}\text{B}^i(\gamma,\text{d})^{12}\text{Be}$	$Q(\text{d})=Q(\text{p}+\text{n})+2224.5660(0.0004)$ $Q(\text{p}+\text{n})=290(20)$ keV										MMC162**
* $^{14}\text{C}(\text{p}, ^3\text{He})^{12}\text{B}^i$	$IT=12710(20)$; Q rebuilt with Ame1964										MMC129**
*	energy and resolution arguments for $T=1$, not an IAS										08Ch28 **
* $^{14}\text{C}(\text{p},\text{t})^{12}\text{C}^j$	$IT=27595.0(2.4)$; Q rebuilt										MMC121**
* $^{12}\text{C}(^3\text{He},\text{n})^{14}\text{O}$	Originals $T=1436.2(0.9,\text{Bu})$ $1437.5(0.7,\text{Ba})$ $1437.9(0.6,\text{Ro})$ respectively, recalibrated										MMC126**
* $^{13}\text{C}(\text{d},\text{p})^{14}\text{C}$	Estimated systematic error 0.5 added to statistical error 0.20										AHW **
* $^{13}\text{C}(\text{p},\gamma)^{14}\text{N}$	$E_p=1747.06(0.53)$ to 2^+ level at 9172.25 keV										Ens01a **
* $^{14}\text{C}(\pi^-, \pi^+)^{14}\text{Be}$	Original error 160 increased with 60 calibration uncertainty										GAu **
* $^{14}\text{C}(\beta^-)^{14}\text{N}$	F : find 17 keV neutrino. See also reference										91No07 **
* $^{14}\text{N}(\text{p},\text{n})^{14}\text{O}$	F : withdrawn by author										81Wh03 **
* $^{14}\text{N}(\text{p},\text{n})^{14}\text{O}$	Authors recalibrated 77Wh01 for atomic effects										81Wh03 **
* $^{14}\text{N}(\text{p},\text{n})^{14}\text{O}$	F : withdrawn by author										98Ba83 **
* $^{14}\text{N}(\text{p},\text{n})^{14}\text{O}$	Original $T=6353.08(0.07)$ recalibrated to $T=6352.99(0.12)$ by author										98Ba83 **
* $^{14}\text{N}(\text{p},\text{n})^{14}\text{O}$	F : withdrawn by author										03To03 **
* $^{14}\text{N}(^3\text{He},\text{t})^{14}\text{O}$	F : rejected in reference of same group										09Fa15 **
$\text{C D H}-^{15}\text{N}$	21817.9119	0.0008	21817.9115	0.0006	-0.5	1	61	61 ^{15}N	MI1	1.0	95Di08
$\text{C H}_3-^{15}\text{N}$	23366.1979	0.0017	23366.1974	0.0006	-0.3	1	13	13 ^{15}N	MI1	1.0	95Di08
$^{15}\text{F}-\text{u}$	17477	86	17785	15	1.4	F				2.5	01Ze.A *
$^{14}\text{N D}-^{15}\text{N H}$	9242.29	0.11	9241.8519	0.0007	-1.6	U			C5	2.5	71Ke01
	9241.780	0.008			6.0	B			B08	1.5	75Sm02
$^{15}\text{N}(\text{p},\alpha)^{12}\text{C}$	4966	6	4965.4933	0.0006	-0.1	U			CIT		51Li26 Y
	4962	4			0.9	U			Bir		53Co02 Y
	4954	8			1.4	U			Mex		64Ma.B
	4965	7			0.1	U			MIT		64Sp12

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{12}\text{C}(\alpha, n)^{15}\text{O}$	-8503	12	-8502.0	0.5	0.1	U			Tal		63Ne05
$^{15}\text{N}(\text{d}, \alpha)^{13}\text{C}$	7675	9	7687.2358	0.0006	1.4	U			Mex		64Ma.B
	7689	6			-0.3	U			MIT		64Sp12
$^{15}\text{Ne}(2\text{p})^{13}\text{O}$	2522	66				3					14Wa09
$^{15}\text{Be}(\gamma, n)^{14}\text{Be}$	1800	100				3					13Sn02
$^{14}\text{C}(\text{d}, \text{p})^{15}\text{C}$	-1006.5	0.8				2			Wis		56Do41 Y
$^{14}\text{C}(\text{p}, \gamma)^{15}\text{N}^i$	-1407.8	3.5				2					59Fe99 *
$^{14}\text{N}(\text{n}, \gamma)^{15}\text{N}$	10833.2	0.6	10833.2968	0.0008	0.2	U					68Gr14
	10833.1	0.7			0.3	U					74Sp04
	10833.5	0.05			-4.1	B			MMn		80Is02
	10833.314	0.012			-1.4	U					97Ju02
	10833.2339	0.0500			1.3	U			PTc		97Ro26 *
	10833.32	0.22			-0.1	U			Bdn		06Fi.A
	10833.3	0.5			-0.0	U					06Be33
$^{14}\text{N}(\text{d}, \text{p})^{15}\text{N}$	8629	11	8608.7306	0.0006	-1.8	U			CIT		52Mi54 Y
	8614	6			-0.9	U			Mex		64Ma.B
	8623	3			-4.8	B			MIT		64Sp12
	8608.83	0.50			-0.2	U			Rez		90Pi05 *
$^{14}\text{N}(\text{p}, \gamma)^{15}\text{O}$	7297.1	0.9	7296.8	0.5	-0.4	1	30	30 ^{15}O	CIT		72Ne05
$^{14}\text{N}(^3\text{He}, \text{d})^{15}\text{O}$	1803	10	1803.3	0.5	0.0	U			Ric		59Yo25 Y
	1802	15			0.1	U			Man		60Fo01
$^{15}\text{F}(\text{p})^{14}\text{O}$	1410	150	1270	14	-0.9	o					03Le26
	1510	110			-2.2	U					03Pe23
	1490	130			-1.7	U					04Go15 *
	1560	130			-2.2	U					04Le12
	1230	50			0.8	U					05Gu25
	1270	14				3					16De15
$^{15}\text{C}(\beta^-)^{15}\text{N}$	9810	30	9771.7	0.8	-1.3	U					59Al06
$^{15}\text{O}(\beta^+)^{15}\text{N}$	2745	5	2754.2	0.5	1.8	U					57Ki22
$^{15}\text{N}(\text{p}, \text{n})^{15}\text{O}$	-3541.7	0.9	-3536.5	0.5	5.7	B					58Jo28 Y
	-3535.1	1.0			-1.5	-			CIT		72Je02 Z
	-3537.6	0.8			1.4	-					72Sh08 Z
ave.	-3536.6	0.6			0.2	1	70	70 ^{15}O			average
$^{15}\text{F}-\text{u}$	F : results distrusted (see also ^{18}Na and ^{19}Mg)										GAu **
$^{14}\text{C}(\text{p}, \gamma)^{15}\text{N}^i$	From a parametrized fit										MMC122**
$^{14}\text{N}(\text{n}, \gamma)^{15}\text{N}$	Original error 0.0005 increased for calibration										GAu **
$^{14}\text{N}(\text{d}, \text{p})^{15}\text{N}$	Estimated systematic error 0.5 added to statistical error 0.061 keV										AHW **
$^{15}\text{F}(\text{p})^{14}\text{O}$	Symmetrized from 1450(+160-100) keV										04Go15 **
$^{16}\text{O}-\text{H}_{16}$	-130285.89116	0.00069	-130285.891	10.000	40.	1 1		261 9 ^{16}O	MZ3	1.0	19He19 *
$\text{C H}_2 \text{ D}-\text{O}$	34837.406	0.033	34837.2224	0.0003	-3.7	B			B07	1.5	71Sm01
	34837.202	0.020			0.7	U			B08	1.5	75Sm02
$\text{C D}_2-\text{O}$	33289.129	0.033	33288.9364	0.0003	-3.9	B			B07	1.5	71Sm01
	33289.061	0.038			-2.2	U			B07	1.5	71Sm01
	33288.940	0.019			-0.1	U			B08	1.5	75Sm02
C_4-O_3	15256.131	0.018	15256.1422	0.0010	0.2	o			WA1	2.5	92Va.A
	15256.086	0.081			0.3	U			OH1	2.5	93Ma.A
	15256.121	0.009			0.9	o			WA1	2.5	95Va38
	15256.1425	0.0008			-0.1	o			WA1	2.5	01Va33
	15256.1415	0.0005			0.6	o			WA1	2.5	03Va.A
	15256.14129	0.00054			0.7	1	50	50 ^{16}O	WA1	2.5	06Va22
$\text{C H}_4-\text{O}$	36387.55	0.8	36385.5083	0.0003	-1.0	U			J1	2.5	68Ma45
	36386.01	0.24			-0.8	U			J2	2.5	69Na21
	36385.644	0.036			-2.5	U			B07	1.5	71Sm01
	36385.5062	0.0013			1.6	U			MI1	1.0	95Di08
	36385.5073	0.0019			0.5	U			MI1	1.0	95Di08
	36385.5060	0.0022			1.1	U			MI1	1.0	95Di08

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{16}\text{O}-\text{u}$	-5085.362	0.027	-5085.3807	0.0003	-0.3	U			OH1	2.5	93Ma.A
	-5085.3798	0.0011			-0.3	U				2.5	16Ho.A
$^{14}\text{C H}_2-\text{O}$	23977.413	0.014	23977.433	0.004	1.0	U			B08	1.5	75Sm02
$\text{N D}-\text{O}$	22261.160	0.013	22261.16284	0.00025	0.1	U			B08	1.5	75Sm02
$\text{N}_2-\text{C O}$	11233.57	0.20	11233.3892	0.0004	-0.4	U			J2	2.5	69Na21
	11233.543	0.025			-4.1	B			B07	1.5	71Sm01
	11233.43	0.21			-0.1	U			J6	2.5	76Ka50
	11259	27			-0.4	U			CR1	2.5	89Sh10
	11233.3909	0.0022			-0.8	U			MI1	1.0	95Di08
	11233.38932	0.00042			-0.2	1	82	78 ^{14}N	MI1	1.0	04Th17
$^{16}\text{O}(\alpha, ^8\text{He})^{12}\text{O}$	-66020	120	-65935	12	0.7	U			Brk		78Ke06
$^{16}\text{O}(\text{p}, \alpha)^{13}\text{N}$	-5211	10	-5218.43	0.27	-0.7	U			MIT		64Sp12
$^{16}\text{O}(^3\text{He}, ^6\text{He})^{13}\text{O}$	-30516	14	-30513	10	0.2	2			Brk		70Me11 *
	-30511	13			-0.2	2			MSU		71Tr03 *
$^{16}\text{Be}(\gamma, 2\text{n})^{14}\text{Be}$	1350	100				3					12Sp01
$^{14}\text{C}(^{14}\text{C}, ^{12}\text{N})^{16}\text{B}$	-48380	60	-48410	25	-0.5	o			Ber		95Bo10
	-48378	60			-0.5	1	17	17 ^{16}B	Ber		00Ka21
$^{14}\text{C}(\text{t}, \text{p})^{16}\text{C}$	-3015	8	-3013	4	0.2	2			MSU		77Fo09
	-3013	4			-0.1	2			LAI		78Se04
$^{14}\text{C}(^3\text{He}, \text{p})^{16}\text{N}$	4983	4	4978.2	2.3	-1.2	R			BNL		66Ga08
$^{14}\text{C}(^3\text{He}, \text{p})^{16}\text{N}^i$	-4951	7				2					68He03 *
$^{14}\text{N}(\text{t}, \text{p})^{16}\text{N}$	4853	10	4840.3	2.3	-1.3	U			Ald		66He10
$^{14}\text{C}(^3\text{He}, \text{n})^{16}\text{O}^j$	-8100	8	-8104	4	-0.5	1	23	23 $^{16}\text{O}^j$			70Ad01 *
$^{16}\text{O}(\text{d}, \alpha)^{14}\text{N}$	3110.	3.5	3110.38802	0.00028	0.1	U			Wis		52Cr30 Y
	3119	5			-1.7	U			Ric		53Fa18 Y
	3110	6			0.1	U			Mex		64Ma.B
	3113	6			-0.4	U			MIT		64Sp12
$^{14}\text{N}(^3\text{He}, \text{p})^{16}\text{O}^i$	2444	6	2447	4	0.5	1	54	54 $^{16}\text{O}^i$			64Br08 *
$^{14}\text{N}(\text{d}, \gamma)^{16}\text{O}^j$	-1986.3	4.4	-1985	4	0.3	1	77	77 $^{16}\text{O}^j$			72Ne10
$^{14}\text{N}(^3\text{He}, \text{n})^{16}\text{F}$	-963	40	-952	5	0.3	U			LAI		65Za01
	-970	15			1.2	R			Har		68Ad03
$^{16}\text{Ne}(2\text{p})^{14}\text{O}$	1350	80	1401	20	0.6	U					08Mu13
$^{16}\text{B}(\gamma, \text{n})^{15}\text{B}$	85	15	83	15	-0.1	1	95	83 ^{16}B			09Le02
$^{15}\text{N}(\text{d}, \text{p})^{16}\text{N}$	286	12	264.3	2.3	-1.8	U			CIT		55Pa50 Y
	269	10			-0.5	U			Pit		57Wa01 Y
	259	6			0.9	2			Mex		64Ma.B
	267	8			-0.3	2			MIT		64Sp12
	270	10			-0.6	U			Pen		66He10
$^{15}\text{N}(\text{p}, \gamma)^{16}\text{O}^i$	-665.3	6.6	-669	4	-0.5	1	46	46 $^{16}\text{O}^i$			57Ha99
$^{16}\text{O}(^3\text{He}, \alpha)^{15}\text{O}$	4920	10	4913.7	0.5	-0.6	U			Ald		59Hi68 Y
	4907	7			1.0	U			Ric		59Yo25 Y
$^{16}\text{F}(\text{p})^{15}\text{O}$	527	7	531	5	0.5	1	59	58 ^{16}F			18Ch25
$^{16}\text{N}(\beta^-)^{16}\text{O}$	10400	20	10420.9	2.3	1.0	U					59Al06
$^{16}\text{O}(^3\text{He}, \text{t})^{16}\text{F}$	-15430	10	-15431	5	-0.1	-			KVI		80Ja.A
ave.	-15436	8			0.6	1	42	42 ^{16}F			average
$^{16}\text{O}(\pi^+, \pi^-)^{16}\text{Ne}$	-27766	45	-27702	20	1.4	2					80Bu15
* $^{16}\text{O}(^3\text{He}, ^6\text{He})^{13}\text{O}$	$M-A$ increased by 7 for more recent calibrator $M-A(^9\text{C})=28913(2)$										AHW **
* $^{16}\text{O}(^3\text{He}, ^6\text{He})^{13}\text{O}$	Recalibrated using their $^{12}\text{C}(^3\text{He}, ^6\text{He})$ result										AHW **
* $^{14}\text{C}(^3\text{He}, \text{p})^{16}\text{N}^i$	$IT=9928(7)$, Q rebuilt with Ame1965										MMC121**
* $^{14}\text{C}(^3\text{He}, \text{n})^{16}\text{O}^j$	$IT=22717(8)$, Q rebuilt with Ame1964										MMC121**
* $^{14}\text{N}(^3\text{He}, \text{p})^{16}\text{O}^i$	$IT=12798(6)$, Q rebuilt										MMC121**
$^{17}\text{O}_2-^{28}\text{Si D}_3$	-20968.3557	0.0014	-20968.3560	0.0013	-0.2	1	84	83 ^{17}O	FS1	1.0	10Mo29
$^{17}\text{B}-\text{u}$	45970	860	46930	220	0.7	U			GA1	1.5	87Gi05
	46830	180			0.4	2			TO2	1.5	88Wo09
	47127	250			-0.5	2			GA3	1.5	91Or01

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{17}\text{Ne}-^{22}\text{Ne}_{.773}$	24373.27	0.38	24373.3	0.4	0.0	1	100	100 ^{17}Ne	MA8	1.0	08Ge07
$^{17}\text{Na}-u$	37273	64				2			1.0	1.0	17Br07 *
$^{17}\text{O}-^{16}\text{O H}$	-3607.8961	0.0016	-3607.8952	0.0007	0.6	1	19	17 ^{17}O	FS1	1.0	10Mo29
$^{17}\text{O}(n,\alpha)^{14}\text{C}$	1817.2	3.5	1817.745	0.004	0.2	U					01Wa50
$^{14}\text{C}(\alpha,n)^{17}\text{O}$	-1819.07	2.0	-1817.745	0.004	0.7	U			Wis		56Sa06 Y
$^{17}\text{O}(p,\alpha)^{14}\text{N}$	1200	17	1191.8742	0.0007	-0.5	U			MIT		64Sp12
$^{17}\text{O}(d,\alpha)^{15}\text{N}$	9818	12	9800.6047	0.0009	-1.4	U			Nob		54Pa39 Y
$^{16}\text{O}(n,\gamma)^{17}\text{O}$	4143.24	0.23	4143.0801	0.0008	-0.7	U					77Mc05 Z
	4143.06	0.13			0.2	U			Bdn		06Fi.A
$^{16}\text{O}(d,p)^{17}\text{O}$	1915	8	1918.5139	0.0006	0.4	U			Ric		51KI55 Y
	1918	4			0.1	U			MIT		57Br82
	1918	3			0.2	U			Mex		61Ja23
	1920	3			-0.5	U			MIT		64Sp12
	1918.74	0.5			-0.5	U			Rez		90Pi05 *
$^{16}\text{O}(n,\gamma)^{17}\text{O}^i$	-6935.70	0.17				2					81Hi01 *
$^{16}\text{O}(p,\gamma)^{17}\text{F}$	600.35	0.28	600.27	0.25	-0.3	-			CIT		75Ro05
$^{16}\text{O}(d,n)^{17}\text{F}$	-1626	4	-1624.30	0.25	0.4	U			Ric		51Bo49 Y
	-1624.6	0.5			0.6	-			Nvl		60Bo21 Z
$^{16}\text{O}(p,\gamma)^{17}\text{F}$	ave. 600.27	0.25	600.27	0.25	-0.0	1	100	100 ^{17}F			average
$^{16}\text{O}(p,\gamma)^{17}\text{F}^i$	-10592.8	1.9				2					76Hi09
$^{16}\text{O}(^3\text{He},2n)^{17}\text{Ne}$	-22420	190	-22448.9	0.4	-0.2	U			BNL		67Es02
$^{17}\text{F}(\beta^+)^{17}\text{O}$	2770	6	2760.47	0.25	-1.6	U					54Wo23
* $^{17}\text{Na}-u$	Authors says this is an upper limit $M-A<34720(60)$ keV										17Br07 **
* $^{17}\text{Na}-u$	Represents $^{17}\text{Na}\rightarrow 3p+^{14}\text{O}$, yielding $M-A<34720(60)$ keV										GAu **
* $^{16}\text{O}(d,p)^{17}\text{O}$	Estimated systematic error 0.5 added to statistical error 0.062 keV										AHW **
* $^{16}\text{O}(n,\gamma)^{17}\text{O}^i$	Original $Q=-6934.41(0.17)$ does not match original $T=7373.31(0.18)$										MMC129**
$\text{C D}_3-^{18}\text{O}$	43145.72216	0.00088	43145.7214	0.0007	-0.9	-			FS1	1.0	09Re15 *
	43145.72116	0.00136			0.2	-			FS1	1.0	09Re15 *
ave.	43145.7219	0.0007			-0.6	1	87	87 ^{18}O			average
$\text{C}_3-^{18}\text{O}_2$	1680.7695	0.0038	1680.7757	0.0014	1.6	1	13	13 ^{18}O	FS1	1.0	09Re15
$^{18}\text{F}-u$	943	85	937.3	0.5	-0.0	U				2.5	92Ge08
$^{18}\text{Na}-u$	25969	54	26880	100	6.7	F				2.5	01Ze.A *
	26882	183			-0.0	1	30	30 ^{18}Na	1.0	1.0	04Ze05 *
$^{18}\text{Ne}-^{22}\text{Ne}_{.818}$	12755.68	0.39	12755.7	0.4	-0.0	1	100	100 ^{18}Ne	MA8	1.0	04BI20
$^{14}\text{C}(^7\text{Li},^3\text{He})^{18}\text{N}$	-10170	60	-10117	19	0.9	U			Str		80Kr.A
$^{18}\text{O}(^{48}\text{Ca},^{51}\text{V})^{15}\text{B}$	-21760	50	-21762	21	-0.0	-			Hei		78Bh02
	-21768	25			0.2	-			Can		83Ho08
ave.	-21766	22			0.2	1	88	88 ^{15}B			average
$^{18}\text{O}(p,\alpha)^{15}\text{N}$	3954	9	3979.8008	0.0009	2.9	U			Nob		54Mi60 Y
	3964	10			1.6	U			Mex		64Ma.B
$^{18}\text{O}(d,\alpha)^{16}\text{N}$	4235	7	4244.1	2.3	1.3	R			CIT		55Pa50 Z
	4219	20			1.3	U			Mex		64Ma.B
	4249	15			-0.3	U			Phi		66He10
	4244	4			0.0	R			MIT		67Sp09 Z
$^{16}\text{O}(^3\text{He},p)^{18}\text{F}$	2033	5	2032.1	0.5	-0.2	U			Ric		59Yo25
	2055	5			-4.6	C			Mex		64Ma.B
$^{16}\text{O}(^3\text{He},n)^{18}\text{Ne}$	-3205	13	-3194.7	0.4	0.8	U			Nvl		61Du02 Y
	-3198	6			0.5	U			Ald		61To03 Y
	-3194.0	1.5			-0.5	U					94Ma14
$^{18}\text{B}(\gamma,n)^{17}\text{B}$	5	5				3					10Sp02 *
$^{18}\text{O}(^{48}\text{Ca},^{49}\text{Ti})^{17}\text{C}$	-17465	35	-17476	17	-0.3	2			Hei		77No08
	-17479	20			0.2	2			Can		82Fi10
$^{18}\text{O}(^{207}\text{Pb},^{208}\text{Po})^{17}\text{C}$	-26870	220	-26797	17	0.3	U			ChR		79Ba31
$^{18}\text{O}(t,\alpha)^{17}\text{N}$	3872	15				2			LAl		60Ja13

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{17}\text{O}(\text{n},\gamma)^{18}\text{O}$	8043.5	1.0	8045.3702	0.0010	1.9	U			Bdn		06Fi.A
$^{17}\text{O}(\text{d},\text{p})^{18}\text{O}$	5820	10	5820.8040	0.0009	0.1	U			Nob		54Ah37 Y
	5820	10			0.1	U			Man		65Mo16
$^{17}\text{O}(\text{p},\gamma)^{18}\text{F}$	5603	3	5607.1	0.5	1.4	U			Str		73Se03
	5606.2	0.6			1.5	1	60	60 ^{18}F	CIT		75Ro05 Z
$^{18}\text{Na}(\text{p})^{17}\text{Ne}$	1270	170	1250	90	-0.1	-					04Ze05
	1230	150			0.1	-					12Mu05
ave.	1250	110			0.0	1	70	70 ^{18}Na			average
$^{18}\text{O}(\pi^-, \pi^+)^{18}\text{C}$	-26712	150	-26720	30	-0.1	U					78Se07
$^{18}\text{O}(^{48}\text{Ca}, ^{48}\text{Ti})^{18}\text{C}$	-21434	30				2			Can		82Fi10
	-21331	300	-21430	30	-0.3	U			Ors		82Na04
$^{18}\text{N}(\beta^-)^{18}\text{O}$	13860	400	13896	19	0.1	U					64Ch19
$^{18}\text{O}(\text{d}, 2\text{p})^{18}\text{N}$	-15270	100	-15338	19	-0.7	U			Brk		78De.A
$^{18}\text{O}(\text{t}, ^3\text{He})^{18}\text{N}$	-13917	60	-13877	19	0.7	U			LAL		69St07 *
$^{18}\text{O}(^7\text{Li}, ^7\text{Be})^{18}\text{N}$	-14761	20	-14758	19	0.2	2			Can		83Pu01
$^{18}\text{O}(^{14}\text{C}, ^{14}\text{N})^{18}\text{N}$	-13720	50	-13740	19	-0.4	2			Ors		80Na14
$^{18}\text{F}(\beta^+)^{18}\text{O}$	1657	2	1655.9	0.5	-0.5	U					64Ho28
$^{18}\text{O}(\text{p}, \text{n})^{18}\text{F}$	-2451	4	-2438.3	0.5	3.2	B			Wis		50Ri59 Y
	-2436.97	0.73			-1.8	1	40	40 ^{18}F	Nvl		64Bo13 Z
	-2440.2	2.8			0.7	U					67Pr04
$^{18}\text{Ne}(\beta^+)^{18}\text{F}$	4438	9	4444.5	0.6	0.7	U					63Fr10
*C D ₃ - ^{18}O	Respectively CD ₃ ⁺ / $^{18}\text{O}^+$, C ₂ D ₅ ⁺ / $^{18}\text{O}^{2+}$ considered independent										GAu **
* $^{18}\text{Na}-\text{u}$	F : results distrusted (see also ^{15}F and ^{19}Mg)										GAu **
* $^{18}\text{Na}-\text{u}$	Other interpretation : $D_M=25969(172) \mu\text{u}$, $M-A=24190(160) \text{keV}$										04Ze05 **
* $^{18}\text{B}(\gamma, \text{n})^{17}\text{B}$	Decay energy <10 keV, derived from scattering length $a_s < -50 \text{fm}$										10Sp02 **
* $^{18}\text{O}(\text{t}, ^3\text{He})^{18}\text{N}$	From $Q=14038(30)$, reinterpreted as mainly to (2^-) level at 114.90 keV										Ens967 **
$^{28}\text{Si H}_3-\text{C } ^{19}\text{F}$	1998.4687	0.0022	1998.4680	0.0009	-0.3	1	18	14 ^{19}F	FS1	1.0	09Re15
$^{19}\text{B}-\text{u}$	64166	376				2			GA8	1.5	12Ga45
$^{19}\text{C}-\text{u}$	34680	260	34800	110	0.3	o			TO1	1.5	86Vi09
	35370	450			-0.8	U			GA1	1.5	87Gi05
	35180	130			-2.0	o			TO2	1.5	88Wo09
	35506	253			-1.9	U			GA3	1.5	91Or01
C D ₄ -H ^{19}F	50178.88	0.05	50178.9174	0.0009	0.5	U			B08	1.5	75Sm02
$^{19}\text{Mg}-\text{u}$	35470	270	34180	60	-1.9	F				2.5	01Ze.A *
$^{13}\text{C D}_3-^{19}\text{F}$	47257.00669	0.00091	47257.0068	0.0008	0.1	1	86	86 ^{19}F	FS1	1.0	09Re15
$^{19}\text{Ne}-^{22}\text{Ne}_{.864}$	9323.92	0.33	9324.17	0.17	0.8	2			MA8	1.0	04BI20
	9324.26	0.20			-0.5	2			MA8	1.0	08Ge07
$^{19}\text{F}(\text{p}, \alpha)^{16}\text{O}$	8115	10	8113.6122	0.0008	-0.1	U			CIT		50Ch53 Y
	8115	10			-0.1	U			CIT		57Yo04 Y
	8122	9			-0.9	U			MIT		64Sp12
$^{17}\text{O}(\text{t}, \text{p})^{19}\text{O}$	3524	7	3519.2	2.6	-0.7	R			Man		65Mo19
$^{19}\text{F}(\text{d}, \alpha)^{17}\text{O}$	10060	12	10032.1261	0.0010	-2.3	U			MIT		64Sp12
$^{19}\text{Mg}(2\text{p})^{17}\text{Ne}$	750	50	760	60	0.2	o					07Mu15
	760	60				2					12Mu05
	980	170			-1.3	U					18Xu04 *
$^{18}\text{C}(\text{n}, \gamma)^{19}\text{C}$	530	120	580	90	0.4	3					99Na27 *
	650	150			-0.5	3					01Ma08 *
$^{18}\text{O}(^{18}\text{O}, ^{17}\text{F})^{19}\text{N}$	-19374	50	-19374	16	0.0	2			Ors		81Na.A
	-19334	35			-1.1	2			Can		89Ca25
$^{18}\text{O}(^{48}\text{Ca}, ^{47}\text{Sc})^{19}\text{N}$	-16540	20	-16527	17	0.6	2			Can		83Ho08
$^{18}\text{O}(^{208}\text{Pb}, ^{207}\text{Bi})^{19}\text{N}$	-18440	150	-18333	17	0.7	U			ChR		79Ba31
$^{18}\text{O}(\text{d}, \text{p})^{19}\text{O}$	1727	8	1731.1	2.6	0.5	o			Nob		54Mi89 Y
	1732	8			-0.1	2			CIT		54Th30
	1731	5			0.0	2			Nob		57Ah19 Y
	1733	6			-0.3	2			Mex		64Ma.B
	1727	5			0.8	2			MIT		64Sp12 Z
	1734	10			-0.3	U			Man		65Mo16

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{19}\text{F}(^3\text{He},\alpha)^{18}\text{F}$	10166	15	10145.7	0.5	-1.4	U			Ald		59Hi67 Y
$^{19}\text{Na}(p)^{18}\text{Ne}$	160	110	323	11	1.5	U					04Ze05
	328	22			-0.2	1	23	$23\ ^{19}\text{Na}$			10Mu12
$^{19}\text{O}(\beta^-)^{19}\text{F}$	4800	12	4820.3	2.6	1.7	U					59Al06
$^{19}\text{Ne}(\beta^+)^{19}\text{F}$	3262	10	3239.50	0.16	-2.3	U					60Wa04
$^{19}\text{F}(p,n)^{19}\text{Ne}$	-4021.3	4.7	-4021.85	0.16	-0.1	U			Ric		55Ma84
	-4019.6	1.4			-1.6	U			Ric		61Be13 Z
	-4021.1	1.0			-0.7	U			Zur		61Ry04 Z
	-4020.7	0.8			-1.4	U					66Ma60
	-4019.6	0.7			-3.2	B					69Ov01 Z
$^{19}\text{F}(^3\text{He},t)^{19}\text{Ne}^i$	-10759	9				2					98Ut02 *
* $^{19}\text{Mg}-u$	F : results distrusted (see also ^{15}F and ^{18}Na)									GAu	**
* $^{19}\text{Mg}(2p)^{17}\text{Ne}$	Symmetrized from 870(+240-70); Confirmed previous result in 12Mu05									HWJ192	**
* $^{18}\text{C}(n,\gamma)^{19}\text{C}$	From Coulomb dissociation cross sections and angular distribution									99Na27	**
* $^{18}\text{C}(n,\gamma)^{19}\text{C}$	From momentum distribution following one neutron removal									01Ma08	**
* $^{19}\text{F}(^3\text{He},t)^{19}\text{Ne}^i$	rebuilt from $E_x=7500(9)$; with references from Ame1995; recalibrated +1keV									MMC141	**
$^{20}\text{C}-u$	39940	1210	40260	250	0.2	U			GA1	1.5	87Gi05
	40360	240			-0.3	2			TO2	1.5	88Wo09
	40165	491			0.1	2			GA3	1.5	91Or01
	40420	550			-0.2	2			GA5	1.5	99Sa.A
	40108	290			0.4	2			GA8	1.5	12Ga45
$^{20}\text{N}-u$	23230	280	23370	80	0.3	U			TO1	1.5	86Vi09
	23210	150			0.7	2			GA1	1.5	87Gi05
	23380	130			-0.1	2			TO2	1.5	88Wo09
	23397	69			-0.3	2			GA3	1.5	91Or01
$\text{C D}_4-^{20}\text{Ne}$	63966.9329	0.0026	63966.9361	0.0016	1.2	1	40	$40\ ^{20}\text{Ne}$	MI1	1.0	95Di08
$^{20}\text{Ne}-u$	-7559.309	0.090	-7559.8247	0.0016	-2.3	U			OH1	2.5	93Ma.A
	-7559.814	0.014			-0.8	U			ST2	1.0	02Bf02
$\text{O D}_2-^{20}\text{Ne}$	30677.497	0.067	30677.9997	0.0017	3.0	B			OH1	2.5	93Go38
$^{20}\text{Mg}-^{23}\text{Na}_{870}$	27663.8	2.0				2			TT1	1.0	14Ga20
$^{20}\text{Ne}-^{22}\text{Ne}_{909}$	270.89	0.43	271.107	0.017	0.5	U			MA8	1.0	04Bi20
	271.16	0.20			-0.3	U			MA8	1.0	08Ge07
$^{20}\text{Ne}(^3\text{He},^8\text{Li})^{15}\text{F}$	-29960	200	-29623	14	1.7	U			MSU		78Be26
	-29730	180			0.6	U			Brk		78Ke06
$^{20}\text{Ne}^i(\alpha)^{16}\text{O}$	5548.8	6.3	5542.6	2.0	-1.0	U					73To08
$^{20}\text{Ne}(\alpha,^8\text{He})^{16}\text{Ne}$	-60150	80	-60213	20	-0.8	U			Brk		78Ke06
	-60197	23			-0.7	R			Tex		83Wo01
$^{20}\text{Ne}(^3\text{He},^6\text{He})^{17}\text{Ne}$	-26188	50	-26203.3	0.4	-0.3	U			Brk		70Me11 *
	-26158	32			-1.4	F					98Gu10 *
$^{18}\text{O}(^{48}\text{Ca},^{46}\text{Sc})^{20}\text{N}$	-25873	60	-25010	80	14.3	B			Can		89Or03 *
$^{18}\text{O}(t,p)^{20}\text{O}$	3086	15	3081.9	0.9	-0.3	U			LAl		60Ja13
	3076	10			0.6	U			Ald		62Hi06
	3082.4	1.9			-0.3	2			Str		82An12
	3081.7	1.0			0.2	2			Str		85An17
$^{18}\text{O}(^3\text{He},p)^{20}\text{F}$	6875.2	1.5	6876.895	0.030	1.1	U			NDm		70Ro06
$^{18}\text{O}(^3\text{He},p)^{20}\text{F}^i$	356.0	3.0				2					76Mi01 *
$^{20}\text{Ne}(d,\alpha)^{18}\text{F}$	2795	9	2795.8	0.5	0.1	U			Nob		54Mi61 Y
	2766	20			1.5	U			Mex		64Ma.B
	2790	10			0.6	U					75Bo59
$^{20}\text{B}(\gamma,n)^{19}\text{B}$	1560	150				3					18Le18
$^{19}\text{F}(n,\gamma)^{20}\text{F}$	6601.1	0.3	6601.336	0.030	0.8	U			Utr		68Sp01
	6601.29	0.14			0.3	2			ILn		83Hu12 Z
	6601.32	0.05			0.3	2			MMn		87Ke09 Z
	6601.35	0.04			-0.3	2			ORn		96Ra04
	6601.34	0.13			-0.0	2			Bdn		06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{19}\text{F}(\text{d,p})^{20}\text{F}$	4377	7	4376.770	0.030	-0.0	U			MIT		64Sp12
	4377.7	0.9			-1.0	U			NDm		70Ro06
$^{19}\text{F}(\text{p},\gamma)^{20}\text{Ne}^j$	-3889.4	2.7				2					67B119
$^{20}\text{Ne}(^3\text{He},\alpha)^{19}\text{Ne}$	3750	13	3712.32	0.16	-2.9	U			MIT		64Sp12
$^{20}\text{Na}^i(\text{p})^{19}\text{Ne}$	4381	50	4307.9	1.2	-1.5	U			Brk		79Mo02
	4332	16			-1.5	U			MSU		92Go10
	4326	30			-0.6	U			Lis		95Pi03
$^{20}\text{F}(\beta^-)^{20}\text{Ne}$	7053	13	7024.469	0.030	-2.2	U					54Wo23 *
	7050	15			-1.7	U					59Al06 *
	7032	6			-1.3	U					76Ge08 *
	7026.9	1.8			-1.4	U					87Va20 *
	7019.8	1.7			2.7	U					89He11 *
$^{20}\text{Ne}^i(\text{IT})^{20}\text{Ne}$	10274	3	10272.5	2.0	-0.5	2					76In06
	10271.2	2.7			0.5	2					77Fi08
$^{20}\text{Na}(\beta^+)^{20}\text{Ne}$	13906	40	13892.4	1.1	-0.3	U					67Su05
$^{20}\text{Ne}(\text{p,n})^{20}\text{Na}$	-14672.1	7.	-14674.8	1.1	-0.4	U					71Wi07 Z
$^{20}\text{Ne}(^3\text{He},\text{t})^{20}\text{Na}-^{36}\text{Ar}()^{36}\text{K}$	-1078.06	1.06	-1078.1	1.1	-0.0	1	100	100 ^{20}Na	Mun		10Wr01
$^{20}\text{Na}^i(\text{IT})^{20}\text{Na}$	6498.4	0.5				2					15GI03
$^{20}\text{Ne}(^3\text{He},^6\text{He})^{17}\text{Ne}$	Original $M-A=16479(50)$ but revised calibrator $M(^9\text{C})=28910.2(2.1)$										AHW **
$^{20}\text{Ne}(^3\text{He},^6\text{He})^{17}\text{Ne}$	F : calibrated with $^{24}\text{Mg}(^3\text{He},^6\text{He})$ from reference for excitation energies										92Ku02 **
*	no details given. No correction possible										GAu **
$^{18}\text{O}(^{48}\text{Ca},^{46}\text{Sc})^{20}\text{N}$	Probably to excited levels in ^{20}N and ^{46}Sc										GAu **
$^{18}\text{O}(^3\text{He},\text{p})^{20}\text{F}^i$	IT=6519.4(3.0), Q rebuilt with Ame1971										MMC122**
$^{20}\text{F}(\beta^-)^{20}\text{Ne}$	$E_{\beta^-}=5419(13)$ 5416(15) 5398(6) 5392.3(1.8) 5386.1(1.7) respectively,										GAu **
*	to 2^+ level at 1633.674 keV										Ens992 **
$^{21}\text{N}-\text{u}$	26580	200	27090	140	1.7	U			TO1	1.5	86Vi09
	27060	190			0.1	2			GA1	1.5	87Gi05
	26930	210			0.5	2			TO2	1.5	88Wo09
	27162	131			-0.4	2			GA3	1.5	91Or01
$^{21}\text{Na}-^{39}\text{K}_{538}$	17180.51	0.29	17180.37	0.05	-0.5	o			MA8	1.0	04Mu26
	17180.51	0.29			-0.3	U			Ma8	1.5	08Mu05 *
$\text{H}_3\ ^{18}\text{O}-^{21}\text{Ne}$	28787.76	0.25	28788.02	0.04	1.1	U			CP1	1.0	04Sa53 *
$^{21}\text{Na}-^{23}\text{Na}_{913}$	6995.25	0.32	6995.10	0.05	-0.5	o			MA8	1.0	04Mu26
	6995.25	0.32			-0.3	U			Ma8	1.5	08Mu05
$^{21}\text{Mg}-^{23}\text{Na}_{913}$	21046.41	0.81				2			TT1	1.0	14Ga20
$^{21}\text{Ne}-^{22}\text{Ne}_{955}$	2073.82	0.40	2073.90	0.05	0.2	U			MA8	1.0	04Bi20
	2074.04	0.26			-0.5	U			MA8	1.0	08Ge07
$^{21}\text{Na}-^{21}\text{Ne}$	3808.017	0.097	3807.774	0.019	-2.5	U			MS1	1.0	15Ei01
	3807.774	0.019			-0.0	1	100	100 ^{21}Na	MA8	1.0	19Ka30 *
$^{21}\text{Na}-^{20}\text{Na}$	-9732	50	-9699.8	1.2	0.3	U			CR1	2.5	89Sh10
$^{18}\text{O}(^{18}\text{O},^{15}\text{O})^{21}\text{O}$	-12574	70	-12483	12	1.3	U			Ors		78Na02
	-12499	20			0.8	2			Can		89Ca25
$^{18}\text{O}(^{64}\text{Ni},^{61}\text{Ni})^{21}\text{O}$	-11713	15	-11722	12	-0.6	2			Dar		85Wo01
$^{18}\text{O}(^{208}\text{Pb},^{205}\text{Pb})^{21}\text{O}$	-6860	75	-6823	12	0.5	U			ChR		79Ba31
$^{21}\text{B}(\gamma,2\text{n})^{19}\text{B}$	2470	190				3					18Le18
$^{19}\text{F}(\text{t,p})^{21}\text{F}$	6221.0	1.8				2			Str		84An17
$^{19}\text{F}(^3\text{He},\text{p})^{21}\text{Ne}$	11911	15	11886.58	0.04	-1.6	U			Ald		59Hi75 Y
$^{20}\text{Ne}(\text{n},\gamma)^{21}\text{Ne}$	6760.8	1.5	6761.16	0.04	0.2	U					70Se14
	6761.16	0.04			0.1	-			MMn		86Pr05 Z
	6761.19	0.14			-0.2	-			Bdn		06Fi.A
$^{20}\text{Ne}(\text{d,p})^{21}\text{Ne}$	4531	9	4536.60	0.04	0.6	U			Nob		55Ah41 Y
	4532	6			0.8	U			Mex		64Ma.B
	4534	7			0.4	U			MIT		64Sp12
$^{20}\text{Ne}(\text{n},\gamma)^{21}\text{Ne}$	ave.	6761.16	6761.16	0.04	0.0	1	100	100 ^{21}Ne			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{20}\text{Ne}(p,\gamma)^{21}\text{Na}$	2431.2	0.7	2431.90	0.04	1.0	U					69B103 Z
$^{20}\text{Ne}(p,\gamma)^{21}\text{Na}^i$	-6547.9	14.3	-6543	4	0.3	U					81Fe05
$^{21}\text{Na}^i(p)^{20}\text{Ne}$	6543	4				2					73Se08 *
$^{21}\text{O}(\beta^-)^{21}\text{F}$	8150	175	8110	12	-0.2	U					81Al07
$^{21}\text{Na}(\beta^+)^{21}\text{Ne}$	3522	30	3546.919	0.018	0.8	U					52Sc15
	3532	20			0.7	U					60Wa04
$^{21}\text{Na}-^{39}\text{K}_{538}$	CF=1.5 for prelim. results; not trusted within given uncertainties										GAu **
$^*\text{H}_3\ ^{18}\text{O}-^{21}\text{Ne}$	$D_M=28787.78(0.25)$ corrected -0.02 keV for molecular and ionization										04Sa53 **
$^{21}\text{Na}-^{21}\text{Ne}$	Electron binding energies not considered by authors										HWJ201 **
$^{21}\text{Na}^i(p)^{20}\text{Ne}$	$Q_p=6548(4), 4904(4)$ to ground state and 2^+ level at 1633.674 keV										Ens992 **
$^{22}\text{C}-u$	57585	408	57550	250	-0.1	U					GA8 1.5 12Ga45
$^{22}\text{N}-u$	32990	790	34100	220	0.9	U					GA1 1.5 87Gi05
	34340	250			-0.6	2					TO2 1.5 88Wo09
	34683	389			-1.0	2					GA3 1.5 91Or01
	34240	320			-0.3	2					GA5 1.5 99Sa.A
	33398	279			1.7	2					GA8 1.5 12Ga45
$^{22}\text{O}-u$	9842	81	9970	60	1.0	R					GA3 1.5 91Or01
$^{22}\text{Ne}-u$	-8614.885	0.019	-8614.886	0.019	-0.1	1	100	100	^{22}Ne	ST2	1.0 02Bf02
$^{22}\text{Na}-^{39}\text{K}_{564}$	14907.33	0.30	14907.09	0.14	-0.8	o				MA8	1.0 04Mu26
	14907.33	0.30			-0.5	1	10	10	^{22}Na	Ma8	1.5 08Mu05
$^{22}\text{Mg}-^{39}\text{K}_{564}$	20040.33	0.35	20040.14	0.17	-0.5	o				MA8	1.0 04Mu26
	20040.33	0.35			-0.4	1	11	11	^{22}Mg	Ma8	1.5 08Mu05
$\text{O H}-^{22}\text{Ne}_{.773}$	9398.87	0.19	9398.958	0.015	0.5	U				MA8	1.0 08Ge07
$^{22}\text{Na}-^{24}\text{Mg}_{.917}$	8153.64	0.31	8154.32	0.14	2.2	o				MA8	1.0 04Mu26
	8153.64	0.31			1.5	1	9	9	^{22}Na	Ma8	1.5 08Mu05
$^{22}\text{Na}-^{23}\text{Na}_{.957}$	4228.11	0.29	4228.35	0.14	0.8	o				MA8	1.0 04Mu26
	4228.11	0.29			0.5	-				Ma8	1.5 08Mu05
	4228.35	0.25			-0.0	-				TT1	1.0 17Re10
ave.	4228.29	0.22			0.3	1	43	43	^{22}Na		average
$^{22}\text{Mg}-^{23}\text{Na}_{.957}$	9361.33	0.25	9361.39	0.17	0.3	1	47	47	^{22}Mg	TT1	1.0 17Re10
$^{22}\text{Na}-^{22}\text{Ne}$	3052.75	0.33	3052.43	0.14	-1.0	1	19	18	^{22}Na	CP1	1.0 04Sa53 *
$^{22}\text{Mg}-^{22}\text{Ne}$	8185.79	0.73	8185.48	0.17	-0.4	1	6	5	^{22}Mg	CP1	1.0 04Sa53 *
$^{22}\text{Mg}-^{22}\text{Na}$	5132.99	0.34	5133.05	0.18	0.2	o				MA8	1.0 04Mu26
	5132.99	0.34			0.1	-				Ma8	1.5 08Mu05
	5133.05	0.26			-0.0	-				TT1	1.0 17Re10
ave.	5133.04	0.23			0.1	1	57	37	^{22}Mg		average
$^{22}\text{Ne}-^{20}\text{Ne}$	-1056.415	0.290	-1055.062	0.019	1.9	U				OH1	2.5 93Go38
$^{18}\text{O}(^{18}\text{O}, ^{14}\text{O})^{22}\text{O}$	-19060	100	-18860	60	2.0	2				Can	76Hi10
$^{18}\text{O}(^{208}\text{Pb}, ^{204}\text{Pb})^{22}\text{O}$	-6710	180	-6700	60	0.0	2				ChR	79Ba31
$^{22}\text{Mg}^i(\alpha)^{18}\text{Ne}$	5885	40	5902	6	0.4	U				Bor	97B103 *
	5904	8			-0.3	1	60	60	$^{22}\text{Mg}^i$	Bor	06Ac04 *
$^{19}\text{F}(\alpha, p)^{22}\text{Ne}$	1674	11	1673.215	0.018	-0.1	U				MIT	64Sp12
$^{19}\text{F}(\alpha, n)^{22}\text{Na}$	-1958	10	-1952.46	0.13	0.6	U				Duk	60Wi07 Y
$^{22}\text{C}(\gamma, 2n)^{20}\text{C}$	-200	120	-35	20	1.4	o					11Ya25 *
	-110	60			1.3	U					12Fo04 *
	-35	20				3					13Mo12 *
$^{20}\text{Ne}(^3\text{He}, n)^{22}\text{Mg}$	197	25	217.95	0.16	0.8	U				Har	68Ad03
	209	11			0.8	U				CIT	70Mc06
$^{22}\text{Mg}^i(2p)^{20}\text{Ne}$	6098	13	6108	6	0.8	1	23	23	$^{22}\text{Mg}^i$		06Ac04 *
$^{22}\text{Ne}(t, \alpha)^{21}\text{F}$	4545	10	4547.8	1.8	0.3	U				LAI	61Si03 Y
$^{21}\text{Ne}(n, \gamma)^{22}\text{Ne}$	10364.4	0.3	10364.26	0.04	-0.5	U				MMn	86Pr05 Z
	10363.9	0.5			0.7	U				Bdn	06Fi.A
$^{21}\text{Ne}(d, p)^{22}\text{Ne}$	8152	11	8139.69	0.04	-1.1	U				CIT	52Mi54 Y
$^{21}\text{Ne}(p, \gamma)^{22}\text{Na}$	6739.0	0.7	6738.59	0.14	-0.6	U					70An06 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{22}\text{Mg}^i(\text{p})^{21}\text{Na}$	8547	15	8540	6	-0.4	1	17	17 $^{22}\text{Mg}^i$	Brk		82Ca16 *
$^{22}\text{F}(\beta^-)^{22}\text{Ne}$	11000	150	10818	12	-1.2	U					73Gu05
	10950	120			-1.1	U			ANB		74Da02
$^{22}\text{Ne}(\text{t}, ^3\text{He})^{22}\text{F}$	-10788	33	-10799	12	-0.3	2					69St07 *
	-10794	18			-0.3	2			Dar		88Cl04 *
$^{22}\text{Ne}(^7\text{Li}, ^7\text{Be})^{22}\text{F}$	-11691	20	-11680	12	0.6	2			Can		89Or04 *
$^{22}\text{Na}(\beta^+)^{22}\text{Ne}$	2842.2	0.5	2843.32	0.13	2.2	U					68Be35 *
	2840.4	1.5			1.9	U					68We02 *
	2841.5	1.0			1.8	U					72Gi17 *
$^{22}\text{Na}-^{22}\text{Ne}$	$D_M=3052.79(0.33) \mu\text{u}$, $M-A=-5181.06(0.31) \text{keV}$; corrected -0.04 keV for										04Sa53 **
*	ion-ion interaction										04Sa53 **
$^{22}\text{Mg}-^{22}\text{Ne}$	$D_M=8185.84(0.73) \mu\text{u}$, $M-A=-399.65(0.68) \text{keV}$; corrected -0.05 keV for										04Sa53 **
*	ion-ion interaction										04Sa53 **
$^{22}\text{Mg}^i(\alpha)^{18}\text{Ne}$	$E_\alpha=3270(40)$ to 2^+ level at 1887.3 keV										Ens967 **
$^{22}\text{Mg}^i(\alpha)^{18}\text{Ne}$	$Q_\alpha=4017(8)$ to 2^+ level at 1887.3 keV										Ens967 **
$^{22}\text{C}(\gamma, 2n)^{20}\text{C}$	From upper limit $S_{2n}<400$										GAu **
$^{22}\text{C}(\gamma, 2n)^{20}\text{C}$	From upper limit $S_{2n}<220$										GAu **
$^{22}\text{C}(\gamma, 2n)^{20}\text{C}$	The two items are estimates derived from the experimental result of reference										10Ta04 **
$^{22}\text{C}(\gamma, 2n)^{20}\text{C}$	From upper limit $S_{2n}<70$										GAu **
$^{22}\text{Mg}^i(2p)^{20}\text{Ne}$	Original $Q_{2p}=4464(8)$ to 1633.674(0.015) level										06Ac04 **
$^{22}\text{Mg}^i(2p)^{20}\text{Ne}$	Estimated systematic 10 keV uncertainty added										16Ma.A **
$^{21}\text{Ne}(\text{p}, \gamma)^{22}\text{Na}$	$T=701.8(0.5)$ to 1^+ level at 7408.6(0.5) keV										Ens157 **
$^{21}\text{Ne}(\text{p}, \gamma)^{22}\text{Na}$	Reanalysis using E(exc) for lower levels of reference										90En08 **
$^{22}\text{Mg}^i(\text{p})^{21}\text{Na}$	$E_p=8149(21)$, 7839(15) to $3/2^+$ ground state, $5/2^+$ level at 331.90 keV										Ens04c **
$^{22}\text{Ne}(\text{t}, ^3\text{He})^{22}\text{F}$	Original value -10834(30) re-calculated from Q to										GAu **
*	(2^+) level at 709.1, 1^+ at 1627.1 and 1^+ at 2571.7 keV										Ens157 **
$^{22}\text{Ne}(\text{t}, ^3\text{He})^{22}\text{F}$	Original value -10836(12) re-calculated										GAu **
$^{22}\text{Ne}(^7\text{Li}, ^7\text{Be})^{22}\text{F}$	$Q=-12400(20)$ to (2^+) level at 709.1 keV										Ens157 **
$^{22}\text{Na}(\beta^+)^{22}\text{Ne}$	$E_{\beta^+}=545.7(0.5) 543.9(1.5) 545(1)$ respectively, to 2^+ level at 1274.537 keV										Ens157 **
$^{23}\text{N}-\text{u}$	37110	2000	39420	450	0.8	o			GA5	1.5	99Sa.A
	39378	923			0.0	U			GA7	1.5	07Ju03
	39421	301				2			GA8	1.5	12Ga45
$^{23}\text{O}-\text{u}$	15700	320	15700	130	-0.0	o			TO1	1.5	86Vi09
	15860	320			-0.3	o			GA1	1.5	87Gi05
	15700	150			-0.0	2			TO2	1.5	88Wo09
	15621	186			0.3	o			GA3	1.5	91Or01
	15695	107			0.0	2			GA7	1.5	07Ju03
$^{23}\text{F}-\text{u}$	3530	210	3530	40	-0.0	U			TO1	1.5	86Vi09
	3553	43			-0.4	-			GT1	1.5	04Ma.A
	3503	48			0.5	-			LZ1	1.0	15Xu14
	ave. 3520	40			0.2	1	86	86 ^{23}F			average
$^{23}\text{Na}-\text{u}$	-10230.721	0.0037	-10230.7180	0.0019	0.8	-			MI2	1.0	99Br47
	-10230.716	0.0048			-0.4	-			MI2	1.0	99Br47
	-10230.7172	0.0026			-0.3	-			FS1	1.0	10Mo30
	ave. -10230.7181	0.0019			0.0	1	100	100 ^{23}Na			average
$^{23}\text{Ne}-^{22}\text{Ne}_{1.045}$	3469.59	0.37	3469.46	0.11	-0.3	U			MA8	1.0	04Bl20
$^{23}\text{Mg}-^{23}\text{Na}$	4354.80	0.83	4354.49	0.03	-0.4	U			JY1	1.0	09Sa38
	4354.66	0.17			-1.0	U			TT1	1.0	14Sc09
	4354.487	0.034				2			MA8	1.0	19Ka30
$^{23}\text{Al}-^{23}\text{Na}$	17475.07	0.37				2			JY1	1.0	09Sa38
$^{23}\text{Na}(\text{p}, \alpha)^{20}\text{Ne}$	2377	3	2376.1338	0.0024	-0.3	U			Wis		53Do04 Y
	2373	8			0.4	U			MIT		64Sp12
$^{23}\text{Na}(\text{d}, \alpha)^{21}\text{Ne}$	6911	9	6912.73	0.04	0.2	U			Mex		64Ma.B
	6909	10			0.4	U			MIT		64Sp12

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{23}\text{Al}^i(2p)^{21}\text{Na}$	6000	64	6080	40	1.2	1	44	44 $^{23}\text{Al}^i$			18Wa05 *
$^{22}\text{Ne}(^{18}\text{O}, ^{17}\text{F})^{23}\text{F}$	-14080	90	-14040	30	0.4	1	14	14 ^{23}F	Can		89Or04
$^{22}\text{Ne}(n, \gamma)^{23}\text{Ne}$	5200.2	2.0	5200.65	0.10	0.2	U					70Se14
	5200.65	0.12			-0.0	2			MMn		86Pr05 Z
	5200.64	0.20			0.0	2			Bdn		06Fi.A
$^{22}\text{Ne}(d, p)^{23}\text{Ne}$	2967	8	2976.08	0.10	1.1	U			Nob		54Ah20 Y
	2971	9			0.6	o			MIT		60Fr04
	2974	6			0.3	U			Mex		64Ma.B
	2968	7			1.2	U			MIT		64Sp12
$^{22}\text{Ne}(p, \gamma)^{23}\text{Na}$	8794.0	1.5	8794.109	0.018	0.1	U					71Pi08 Z
	8794.26	0.17			-0.9	U					89Ba42 Z
$^{22}\text{Ne}(p, \gamma)^{23}\text{Na}^j$	-10796.3	2.0				2					85Ev01 *
$^{23}\text{Al}^i(p)^{22}\text{Mg}$	11644	57	11580	40	-1.1	1	56	56 $^{23}\text{Al}^i$	Bor		97Bi04 *
$^{23}\text{F}(\beta^-)^{23}\text{Ne}$	8510	170	8440	30	-0.4	U					74Go17
$^{23}\text{Ne}(\beta^-)^{23}\text{Na}$	4383	8	4375.81	0.10	-0.9	U					63Ca06
$^{23}\text{Mg}(\beta^+)^{23}\text{Na}$	4121	12	4056.18	0.03	-5.4	B					63Fr10
$^{23}\text{Na}(p, n)^{23}\text{Mg}$	-4832	10	-4838.53	0.03	-0.7	U			Oak		55Ki28 Z
	-4836.5	6.			-0.3	U			Ric		58Bi41 Y
	-4848.0	7.			1.4	U			ChR		58Go77 Y
	-4835.8	2.5			-1.1	U			Har		62Fr09 Z
	-4843.2	5.1			0.9	U			Tkm		63Ok01 Z
* $^{23}\text{Al}^i(2p)^{21}\text{Na}$	Also $Q_{2p}=5857(66)$ keV to $5/2^+$ level at 331.9 keV										18Wa05 **
* $^{22}\text{Ne}(p, \gamma)^{23}\text{Na}^j$	Original $E_{res}=-10793.6(2.0)$ is a typo										16Ma.A **
* $^{23}\text{Al}^i(p)^{22}\text{Mg}$	$Q_p=11620(100)$, $10410(70)$ to ground state and 2^+ level at 1247.02 keV										Ens157 **
*	also $Q_{2p}=6180(100)$, $5860(100)$ to ground state and 331.90 level in ^{21}Na										97Bi04 **
$^{24}\text{O}-u$	20080	1070	19860	180	-0.1	o			GA1	1.5	87Gi05
	20000	500			-0.2	U			TO2	1.5	88Wo09
	20659	442			-1.2	o			GA3	1.5	91Or01
	20460	340			-1.2	o			GA5	1.5	99Sa.A
	19861	118				2			GA7	1.5	07Ju03
$^{24}\text{F}-u$	8070	170	8100	100	0.1	U			TO1	1.5	86Vi09
	8450	240			-1.0	U			GA1	1.5	87Gi05
	8135	86			-0.3	2			GA3	1.5	91Or01
	8030	120			0.4	2			TO4	1.5	91Zh24
$^{24}\text{Mg}-\text{H}_{24}$	-202759.080	0.014	-202759.076	0.014	0.3	1	98	98 ^{24}Mg	ST2	1.0	03Be02
$^{24}\text{Ne}-^{22}\text{Ne}_{1.091}$	3009.49	0.55				2			MA8	1.0	04Bi20
$^{24}\text{Mg}-^{23}\text{Na}_{1.043}$	-4287.23	0.32	-4287.672	0.014	-0.9	U			MA8	1.5	08Mu05
$^{24}\text{Al}-^{23}\text{Na}_{1.043}$	10618.18	0.25	10618.24	0.24	0.2	1	96	96 ^{24}Al	TT1	1.0	15Ch58
$^{24}\text{Mg}(p, ^6\text{He})^{19}\text{Na}$	-37213	70	-37166	11	0.7	U			Brk		69Ce01
$^{24}\text{Mg}(^3\text{He}, ^8\text{Li})^{19}\text{Na}$	-32876	12	-32878	11	-0.1	1	77	77 ^{19}Na	MSU		75Be38
$^{24}\text{Mg}(\alpha, ^8\text{He})^{20}\text{Mg}$	-60900	210	-60596.0	1.9	1.4	U					74Ro17
	-60677	27			3.0	B			Tex		76Tr03
$^{24}\text{Mg}(^3\text{He}, ^6\text{He})^{21}\text{Mg}$	-27488	40	-27498.3	0.8	-0.3	U			Brk		70Me11
	-27512	18			0.8	U			MSU		71Tr03
$^{22}\text{Ne}(t, p)^{24}\text{Ne}$	5587	10	5587.8	0.5	0.1	U			LA1		61Si03 Z
$^{24}\text{Mg}(d, \alpha)^{22}\text{Na}$	1955	12	1958.62	0.13	0.3	U			MIT		64Sp12
$^{24}\text{Mg}(p, t)^{22}\text{Mg}$	-21194	3	-21194.43	0.16	-0.1	U			MSU		74Ha02
	-21198.3	1.5			2.6	U			MSU		74No07
	-21193.9	1.0			-0.5	U			Yal		05Pa31
$^{23}\text{Na}(n, \gamma)^{24}\text{Na}$	6959.50	0.12	6959.365	0.016	-1.1	o			BNn		74Gr37 Z
	6959.42	0.07			-0.8	2			BNn		80Gr12 *
	6959.67	0.14			-2.2	U			ILn		83Hu11 Z
	6959.38	0.08			-0.2	U			Ptn		83Ti02
	6959.44	0.05			-1.5	2			ORn		04To03
	6959.59	0.14			-1.6	o			Bdn		06Fi.A
	6959.352	0.018			0.7	2			Bdn		14Fi01

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{23}\text{Na}(\text{d,p})^{24}\text{Na}$	4735	7	4734.799	0.016	-0.0	U			CIT		52Mi54 Y	
	4736	5			-0.2	U			Mex		64Ma.B	
	4736	7			-0.2	U			MIT		64Sp12	
$^{23}\text{Na}(\text{p},\gamma)^{24}\text{Mg}$	11692.95	0.17	11692.696	0.013	-1.5	U			Wis		67Mo17 Z	
	11691.2	1.1			1.4	U					72Me09	
	11692.43	0.31			0.9	U					85Uh01 Z	
	-14307.5	1.5	-14306.66	0.03	0.6	U			MSU		74No07	
$^{24}\text{Mg}(\text{p,d})^{23}\text{Mg}$	4051	15	4046.40	0.03	-0.3	U			Man		59Ba13 Y	
$^{24}\text{Mg}({}^7\text{Li}, {}^8\text{He})^{23}\text{Al}$	-37397	27	-37384.2	0.4	0.5	U					01Ca37	
$^{24}\text{Al}(\text{p})^{23}\text{Mg}$	4086	9	4085	3	-0.1	3			Brk		79Ay01	
	4084.5	3.5			0.1	3			MSU		80Le18	
	4093	20			-0.4	U			Bor		98Cz01	
$^{24}\text{Ne}(\beta^-)^{24}\text{Na}$	2449	50	2466.3	0.5	0.3	U					56Dr11	
$^{24}\text{Na}(\beta^-)^{24}\text{Mg}$	5511.8	2.	5515.677	0.021	1.9	U					61De25 *	
	5515.8	2.			-0.1	U					64Le09 *	
	5516.8	2.			-0.6	U					65Be24 *	
	5511.5	1.0			4.2	B					69Bo48 *	
	5511.8	2.			1.9	U					72Gi17 *	
	5512.5	1.2			2.6	U					76Ge06 *	
	13880	50	13884.77	0.23	0.1	U					68Ar03	
$^{24}\text{Mg}(\text{p,n})^{24}\text{Al}$	-14659.0	2.8	-14667.11	0.23	-2.9	B			Yal		69Ov01 Z	
$^{24}\text{Mg}({}^3\text{He,t})^{24}\text{Al}$	-13880	60	-13903.36	0.23	-0.4	U			Brk		66Ma18	
$^{24}\text{Mg}({}^3\text{He,t})^{24}\text{Al}-^{36}\text{Ar}({})^{36}\text{K}$	-1071.48	1.05	-1070.4	0.4	1.0	1	13	9	^{36}K	Mun	10Wr01	
$^{24}\text{Mg}(\pi^+, \pi^-)^{24}\text{Si}$	-23594	52	-23657	19	-1.2	2					80Bu15	
$^{23}\text{Na}(\text{n},\gamma)^{24}\text{Na}$	Original value (Z) increased by 0.037 for better recoil correction										AHW	**
$^{24}\text{Na}(\beta^-)^{24}\text{Mg}$	$E_{\beta^-}=1389(2) \ 1393(2) \ 1394(2) \ 1388.7(1.0) \ 1389(2) \ 1389.7(1.2)$ respectively,										GAu	**
*	to 4^+ level at 4122.889 keV										Ens07a	**
$^{25}\text{F}-\text{u}$	12010	220	12170	100	0.5	o			TO1	1.5	86Vi09	
	12010	290			0.4	o			GA1	1.5	87Gi05	
	12210	150			-0.2	2			TO2	1.5	88Wo09	
	12120	151			0.2	o			GA3	1.5	91Or01	
	11990	130			0.9	2			TO4	1.5	91Zh24	
	12249	97			-0.6	2			GA7	1.5	07Ju03	
	-2293	32	-2190	30	3.4	F			P40	1.0	01Lu20 *	
$^{25}\text{Ne}-\text{u}$	-2166	41			-0.5	1	58	58	^{25}Ne	LZ1	1.0	
	4591.342	0.048	4591.34	0.05	0.0	1	100	100	^{25}Al	JY1	1.0	
$^{25}\text{Al}-^{25}\text{Mg}$	-3151	8	-3147.34	0.14	0.5	U			MIT		59Br74 Y	
$^{25}\text{Mg}(\text{p},\alpha)^{22}\text{Na}$	7488.8	1.2				2			Str		84An17	
$^{23}\text{Na}(\text{t,p})^{25}\text{Na}$	7026	13	7047.88	0.05	1.7	U			MIT		64Sp12	
$^{25}\text{Mg}(\text{d},\alpha)^{23}\text{Na}$	7048	10			-0.0	U					67Ha17	
$^{25}\text{O}(\gamma,\text{n})^{24}\text{O}$	776	15	757	8	-1.2	3					08Ho03 *	
	740	40			0.4	U					13Ca18 *	
	749	10			0.8	3					16Ko11	
	7330.5	9.99	7330.52	0.05	0.0	U					69Ha.A	
$^{24}\text{Mg}(\text{n},\gamma)^{25}\text{Mg}$	7330.5	0.3			0.1	U			MMn		80Is02 Z	
	7330.78	0.14			-1.8	U			ILn		82Hu02 Z	
	7330.4	0.2			0.6	U			MMn		85Ke.A	
	7330.64	0.08			-1.5	-			MMn		90Pr02 Z	
	7330.69	0.05			-3.4	B			ORn		92Wa06	
	7330.53	0.15			-0.1	-			Bdn		06Fi.A	
	5098	12	5105.96	0.05	0.7	U			Har		61Hi11 Y	
$^{24}\text{Mg}(\text{d,p})^{25}\text{Mg}$	5112	12			-0.5	U			Mex		61Ja23	
	5102	7			0.6	U			MIT		64Sp12	
	ave.	7330.62	0.07	7330.52	0.05	-1.3	1	45	43	^{25}Mg	average	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{24}\text{Mg}(\text{p},\gamma)^{25}\text{Al}$	2271.6	1.1	2271.37	0.06	-0.2	U					71Ev01 Z
	2271.7	0.7			-0.5	U					72Pi07 Z
	2271.4	0.8			-0.0	U					85Uh01 Z
$^{24}\text{Mg}({}^3\text{He},\text{d})^{25}\text{Al}$	-3218.0	4.5	-3222.11	0.06	-0.9	U			NDm		73Br27
$^{24}\text{Mg}(\text{p},\gamma)^{25}\text{Al}^i$	-5629.3	5.8	-5629.7	1.8	-0.1	U					68Te01 *
$^{25}\text{Ne}(\beta^-)^{25}\text{Na}$	7380	300	7322	29	-0.2	U					73Go11
$^{25}\text{Na}(\beta^-)^{25}\text{Mg}$	3650	250	3835.0	1.2	0.7	U					54Na18
	4000	200			-0.8	U					55Ma63
$^{25}\text{Al}(\beta^+)^{25}\text{Mg}$	4292	30	4276.81	0.04	-0.5	U					60Wa04
$^{25}\text{Mg}(\text{p},\text{n})^{25}\text{Al}$	-5058	6	-5059.16	0.04	-0.2	U			Har		69Fr08
$^{25}\text{Al}^i(\text{IT})^{25}\text{Al}$	7901	2	7901.1	1.8	0.1	1	85	85^{25}Al^i			77Ro03
$^{25}\text{Ne}-\text{u}$	F : rejected by authors: "unreliable double peak"										06Ga04 **
$^{25}\text{O}(\gamma,\text{n})^{24}\text{O}$	Symmetrized from 770(+20-10)										08Ho03 **
$^{25}\text{O}(\gamma,\text{n})^{24}\text{O}$	Symmetrized from 725(+54-29)										13Ca18 **
$^{24}\text{Mg}(\text{p},\gamma)^{25}\text{Al}^i$	IT=7916(6), Q rebuilt with Ame1964, error estimated by evaluator										GAu **
$^{26}\text{F}-\text{u}$	19800	1000	20050	110	0.2	o					TO1 1.5 86Vi09
	20940	640			-0.9	o					GA1 1.5 87Gi05
	19820	210			0.7	-					TO2 1.5 88Wo09
	19544	300			1.1	o					GA3 1.5 91Or01
	19490	210			1.8	U					TO4 1.5 91Zh24
	20054	86			-0.0	-					GA7 1.5 07Ju03
	ave.	20020	120		0.2	1	93	93^{26}F			average
$^{26}\text{Ne}-\text{u}$	448	90	516	20	0.5	2					GA3 1.5 91Or01
	461	33			1.7	o					P40 1.0 01Lu20
	518	20			-0.1	2					P40 1.0 06Ga04
$^{26}\text{Na}-\text{u}$	-7367	7	-7365	4	0.2	o					P40 1.0 01Lu17
	-7365	4			-0.1	2					P40 1.0 06Ga04 *
	-7368	11			0.2	2					P40 1.0 06Ga04
$^{26}\text{Mg}-\text{H}_{26}$	-220857.848	0.034	-220857.86	0.03	-0.3	1	85	85^{26}Mg	ST2	1.0	03Be02
$^{26}\text{Al}-^{23}\text{Na}_{1.130}$	-1547.46	0.24	-1547.41	0.07	0.2	U			MA8	1.0	08Ge08
$^{26}\text{Si}-^{23}\text{Na}_{1.13}$	3895.1	2.1	3894.53	0.12	-0.3	U			MS1	1.0	10Kw02
$^{26}\text{Al}-^{25}\text{Mg}_{1.040}$	1621.46	0.48	1621.43	0.06	-0.1	U			JY1	1.0	06Er08
$^{26}\text{Al}^m-^{25}\text{Mg}_{1.040}$	1867.09	0.53	1866.53	0.06	-1.1	U			JY1	1.0	06Er08
$^{26}\text{Al}-^{26}\text{Mg}$	4299.14	0.17	4298.90	0.07	-1.4	1	16	15^{26}Al	JY1	1.0	06Er08
$^{26}\text{Al}^m-^{26}\text{Mg}$	4544.09	0.17	4544.00	0.07	-0.5	1	16	15^{26}Al^m	JY1	1.0	06Er08
$^{26}\text{Al}^m-^{26}\text{Al}$	245.09	0.17	245.096	0.014	0.0	U			JY1	1.0	06Er08
	244.91	0.14			1.3	U			JY1	1.0	09Er02
	245.114	0.049			-0.4	U			JY1	1.0	09Er07
$^{26}\text{Si}-^{26}\text{Al}$	5441.97	0.14	5441.94	0.09	-0.2	2			JY1	1.0	09Er02
	5441.92	0.12			0.2	2			JY1	1.0	09Er02 *
$^{25}\text{Na}-^{26}\text{Na}_{.721} \text{ } ^{22}\text{Na}_{.284}$	-2881	33	-2939.7	2.8	-1.8	U			P13	1.0	75Th08
	-2921	22			-0.8	U			P13	1.0	75Th08
$^{26}\text{Al}(\text{n},\alpha)^{23}\text{Na}$	2966.5	2.5	2966.12	0.07	-0.2	U					01Wa50
$^{23}\text{Na}(\alpha,\text{n})^{26}\text{Al}$	-2968	4	-2966.12	0.07	0.5	U			Duk		60Wi07 Y
$^{26}\text{O}(\gamma,2\text{n})^{24}\text{O}$	90	110	18	5	-0.7	U					12Lu07 *
	18	5				3					16Ko11
$^{24}\text{Mg}(\text{t},\text{p})^{26}\text{Mg}$	9940	12	9941.81	0.03	0.2	U			Har		61Hi11 Y
$^{24}\text{Mg}({}^3\text{He},\text{p})^{26}\text{Al}$	5932	15	5918.81	0.07	-0.9	U			Ald		59Hi66 Y
	5922	8			-0.4	U			Phi		72Be51
$^{24}\text{Mg}({}^3\text{He},\text{n})^{26}\text{Si}$	85	18	67.33	0.11	-1.0	U			CIT		67Mi02
	75	30			-0.3	U					67Mc03
	95	15			-1.8	U			Har		68Ad03
	65	30			0.1	U			Ber		68Ha09
$^{26}\text{Mg}({}^7\text{Li}, {}^8\text{B})^{25}\text{Ne}$	-22050	100	-22194	29	-1.4	-			Brk		73Wi06
$^{26}\text{Mg}({}^{13}\text{C}, {}^{14}\text{O})^{25}\text{Ne}$	-19067	50	-19062	29	0.1	-			Can		85Wo04
$^{26}\text{Mg}({}^7\text{Li}, {}^8\text{B})^{25}\text{Ne}$	ave.	-22170	40	-22194	29	-0.5	1	42	42^{25}Ne		average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{26}\text{Mg}(\text{d}, ^3\text{He})^{25}\text{Na}$	−8653	10	−8652.2	1.2	0.1	U			MSU		73Be14
$^{26}\text{Mg}(\text{t}, \alpha)^{25}\text{Na}$	5664	12	5668.2	1.2	0.3	U			Ald		62Hi01
$^{25}\text{Mg}(\text{n}, \gamma)^{26}\text{Mg}$	11092.9	0.3	11093.08	0.04	0.6	U			MMn		80Is02
	11091.84	0.44			2.8	U			ILn		82Hu02 Z
	11093.10	0.06			−0.3	1	54	46 ^{25}Mg	MMn		90Pr02 Z
	11093.17	0.06			−1.5	o			ORn		91Ki04 Z
	11093.23	0.05			−3.0	B			ORn		92Wa06 Z
	11093.16	0.22			−0.4	U			Bdn		06Fi.A
$^{25}\text{Mg}(\text{d}, \text{p})^{26}\text{Mg}$	8865	12	8868.51	0.04	0.3	U			Ald		61Hi11 Y
	8876	12			−0.6	U			Mex		61Ja23
	8889	12			−1.7	U			MIT		64Sp12
$^{25}\text{Mg}(\text{p}, \gamma)^{26}\text{Al}$	6305.0	1.2	6306.33	0.06	1.1	U					74De37
	6304.9	1.1			1.3	U					79El11
	6306.39	0.11			−0.6	−					85Be17 Z
	6306.38	0.08			−0.6	−			Utr		91Ki04 Z
ave.	6306.38	0.06			−0.8	1	75	64 ^{26}Al			average
$^{26}\text{Si}^i(\text{p})^{25}\text{Al}$	7563	15	7501	4	−4.1	B			Brk		83Ca06
	7544	15			−2.9	B			Brk		83Ho23 *
$^{26}\text{Mg}(\pi^-, \pi^+)^{26}\text{Ne}$	−17676	72	−17718	18	−0.6	U					80Na12
$^{26}\text{Na}(\beta^-)^{26}\text{Mg}$	9210	200	9354	4	0.7	U					73Al13
$^{26}\text{Mg}(\text{t}, ^3\text{He})^{26}\text{Na}$	−9292	20	−9335	4	−2.2	U			LAl		74Fi01
$^{26}\text{Mg}(^7\text{Li}, ^7\text{Be})^{26}\text{Na}$	−10182	40	−10216	4	−0.8	U			ChR		72Ba35 *
$^{26}\text{Al}(\beta^+)^{26}\text{Mg}$	3991	8	4004.40	0.06	1.7	U					58Fe16 *
$^{26}\text{Mg}(\text{p}, \text{n})^{26}\text{Al}$	−4786.7	10.	−4786.75	0.06	−0.0	U			Oak		55Ki28 *
	−4787.04	0.48			0.6	U			Utr		69De27
	−4786.1	1.6			−0.4	U			Har		69Fr08 *
	−4785.66	0.22			−5.0	C			Auc		84Ba.B *
	−4786.57	0.05			−3.6	C			Auc		92Ba.A *
	−4786.25	0.12			−4.2	B			Auc		94Br11 *
$^{26}\text{Mg}(^3\text{He}, \text{t})^{26}\text{Al}$	−4023.0	0.6	−4023.00	0.06	0.0	F			Mun		77Vo02 *
$^{26}\text{Mg}(^3\text{He}, \text{t})^{26}\text{Al} - ^{27}\text{Al}()^{27}\text{Si}$	808.2	2.0	807.95	0.12	−0.1	U			ChR		74Ha35
$^{26}\text{Mg}(^3\text{He}, \text{t})^{26}\text{Al} - ^{14}\text{N}()^{14}\text{O}$	1139.43	0.13	1139.96	0.07	4.1	B			ChR		87Ko34 *
$^{26}\text{Al}^m(\text{IT})^{26}\text{Al}$	228.305	0.013	228.306	0.013	0.0	1	99	85 $^{26}\text{Al}^m$			Ens164
$^{26}\text{Si}(\beta^+)^{26}\text{Al}$	5079	13	5069.14	0.08	−0.8	U					63Fr10
$^{26}\text{Si}^i(\text{IT})^{26}\text{Si}$	13015	4				3					04Th09
* $^{26}\text{Na}-\text{u}$	Result from the “Thermo” experiment. Next item from “Rilis”										06Ga04 **
* $^{26}\text{Si}-^{26}\text{Al}$	$D_M=5196.82(0.12) \mu\text{u}$ for $^{26}\text{Al}^m$ at 228.306(0.013); $M-A=-7141.05(0.13) \text{ keV}$										Nub211 **
* $^{26}\text{O}(\gamma, 2\text{n})^{24}\text{O}$	Symmetrized from 150(+50−150) keV										12Lu07 **
* $^{26}\text{O}(\gamma, 2\text{n})^{24}\text{O}$	less than 40 keV										13Ca18 **
* $^{26}\text{Si}^i(\text{p})^{25}\text{Al}$	$E_p=3699(15)$ to 3695.5 level; different from preceeding data										Ens098 **
* $^{26}\text{Mg}(^7\text{Li}, ^7\text{Be})^{26}\text{Na}$	$Q=-10222(30)$ corrected for contribution of unresolved 82.2 level										Ens164 **
* $^{26}\text{Al}(\beta^+)^{26}\text{Mg}$	$E_{\beta^+}=1160(8)$ to 2^+ level at 1808.74 level										Ens164 **
* $^{26}\text{Mg}(\text{p}, \text{n})^{26}\text{Al}$	$T=5191(10, Z)$ to $^{26}\text{Al}^m$ at 228.306 keV										Nub211 **
* $^{26}\text{Mg}(\text{p}, \text{n})^{26}\text{Al}$	$T=5209.3(1.6, Z)$ to $^{26}\text{Al}^m$ at 228.306 keV										Nub211 **
* $^{26}\text{Mg}(\text{p}, \text{n})^{26}\text{Al}$	$T=5208.86(0.23)$ to $^{26}\text{Al}^m$ at 228.306 keV										Nub211 **
* $^{26}\text{Mg}(\text{p}, \text{n})^{26}\text{Al}$	$T=5209.71(0.05)$ to $^{26}\text{Al}^m$ at 228.306 keV										Nub211 **
* $^{26}\text{Mg}(\text{p}, \text{n})^{26}\text{Al}$	$T=5209.46(0.12)$ to $^{26}\text{Al}^m$ at 228.306 keV										Nub211 **
* $^{26}\text{Mg}(^3\text{He}, \text{t})^{26}\text{Al}$	$Q=-4251.3(0.6, Z)$ to $^{26}\text{Al}^m$ at 228.306 keV										Nub211 **
* $^{26}\text{Mg}(^3\text{He}, \text{t})^{26}\text{Al}$	F : rejected in reference of same group										09Fa15 **
* $^{26}\text{Mg}(^3\text{He}, \text{t})^{26}\text{Al} - ^{14}\text{N}()^{14}\text{O}$	$Q(\text{to } 1057.740(0.023) \text{ level}) - ^{14}\text{N}()^{14}\text{O}=81.69(0.13)$										82Al19 **
$^{27}\text{F}-\text{u}$	27500	700	26980	130	−0.5	U			TO2	1.5	88Wo09
	26005	770			0.8	U			GA3	1.5	91Or01
	27100	900			−0.1	U			TO4	1.5	91Zh24
	26900	580			0.1	o			GA5	1.5	99Sa.A
	26441	204			1.8	B			GA7	1.5	07Ju03
	27322	279			−0.8	1	10	10 ^{27}F	GA8	1.5	12Ga45

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{27}\text{Ne}-u$	6010	640	7570	100	1.6	F			TO1	1.5	86Vi09 *
	7470	300			0.2	U			GA1	1.5	87Gi05
	7567	172			0.0	o			GA3	1.5	91Or01
	7670	130			-0.5	2			TO4	1.5	91Zh24
	7536	75			0.3	2			GA7	1.5	07Ju03
$^{27}\text{Na}-u$	-5922	11	-5924	4	-0.1	o			P40	1.0	01Lu17 *
	-5922	4			-0.4	o			P40	1.0	06Ga04 *
$^{27}\text{P}-u$	-735	45	-708	10	0.6	U			LZ1	1.0	18Fu11
$^{27}\text{Mg}-^{23}\text{Na}_{1.174}$	-3648.27	0.15	-3648.49	0.05	-1.5	1	11	11 ^{27}Mg	TT1	1.0	17Br14
$^{27}\text{Al}-^{23}\text{Na}_{1.174}$	-6450.79	0.25	-6450.73	0.05	0.2	U			MA8	1.0	08Ge08
	-6450.754	0.054			0.5	1	89	89 ^{27}Al	TT1	1.0	16Kw.A
$^{27}\text{Na}-^{27}\text{Al}$	12538	4				2			P40	1.0	01Lu17
$^{24}\text{Na}-^{27}\text{Na}_{.356}$ $^{22}\text{Na}_{.655}$	-3006	38	-3059.8	1.3	-0.9	U			P10	1.5	75Th08
$^{26}\text{Na}-^{27}\text{Na}_{.770}$ $^{22}\text{Na}_{.236}$	-1437	86	-1389	5	0.6	U			P13	1.0	75Th08
$^{26}\text{Na}-^{27}\text{Na}_{.481}$ $^{25}\text{Na}_{.520}$	676	66	659	4	-0.2	U			P10	1.5	75Th08
	734	86			-0.6	U			P11	1.5	75Th08
$^{23}\text{Na}(\alpha, \gamma)^{27}\text{Al}$	10090.0	1.3	10091.93	0.05	1.5	U			Utr		78Ma23
$^{27}\text{Al}(\text{p}, \alpha)^{24}\text{Mg}$	1601.7	0.7	1600.77	0.05	-1.3	o			Zur		63Ry04
	1598.4	1.0			2.4	U			NDm		65Br28
	1601.3	0.5			-1.1	U			Zur		67St30 Z
	1600.06	0.21			3.4	B			Utr		78Ma23 Z
$^{24}\text{Mg}(\alpha, \text{p})^{27}\text{Al}$	-1598.9	1.0	-1600.77	0.05	-1.9	U			NDm		65Br28
$^{25}\text{Mg}(\text{t}, \text{p})^{27}\text{Mg}$	9055	11	9054.65	0.06	-0.0	U			Tal		61Hi11 Y
$^{27}\text{Al}(\text{d}, \alpha)^{25}\text{Mg}$	6699	12	6706.73	0.07	0.6	U			Ald		61Hi11 Y
	6691	11			1.4	U			Tal		62Sh01 Y
	6700	10			0.7	U			MIT		64Sp12
$^{27}\text{Al}(\text{p}, \text{t})^{25}\text{Al}^i$	-23843.4	4.7	-23842.8	1.8	0.1	1	15	15 $^{25}\text{Al}^i$	MSU		73Be14 *
$^{27}\text{P}^i(2\text{p})^{25}\text{Al}$	6410	45	6350	30	-1.4	2			Lis		91Bo32
	6270	50			1.5	2					01Ca60 *
$^{27}\text{F}(\gamma, \text{n})^{26}\text{F}$	-1620	60	-1610	60	0.1	1	98	90 ^{27}F			20Re06
$^{26}\text{Mg}(^{18}\text{O}, ^{17}\text{F})^{27}\text{Na}$	-13295	55	-13431	4	-2.5	F			Mun		78Pa12 *
	-13433	60			0.0	U			Can		85Fi08
$^{26}\text{Mg}(\text{n}, \gamma)^{27}\text{Mg}$	6443.35	0.55	6443.37	0.04	0.0	U			ILn		82Hu02
	6443.56	0.25			-0.8	o			MMn		85Ke.A
	6443.26	0.08			1.4	-			MMn		90Pr02 Z
	6443.44	0.05			-1.4	-			ORn		92Wa06 Z
	6443.35	0.13			0.1	-			Bdn		06Fi.A
$^{26}\text{Mg}(\text{d}, \text{p})^{27}\text{Mg}$	4214	12	4218.80	0.04	0.4	U			Ald		61Hi11 Y
	4215	10			0.4	U			Mex		61Ja23
	4211	6			1.3	U			MIT		64Sp12
$^{26}\text{Mg}(\text{n}, \gamma)^{27}\text{Mg}$	ave. 6443.39	0.04	6443.37	0.04	-0.4	1	93	89 ^{27}Mg			average
$^{26}\text{Mg}(\text{p}, \gamma)^{27}\text{Al}$	8270.8	0.5	8271.29	0.06	1.0	U			Utr		59An33 *
	8271.2	0.5			0.2	U					63Va24 Z
	8271.3	0.5			-0.0	U			Utr		78Ma24 *
$^{27}\text{Al}(\text{t}, \alpha)^{26}\text{Mg}$	11541	12	11542.58	0.06	0.1	U			Ald		61Hi11
$^{27}\text{Al}(^3\text{He}, \alpha)^{26}\text{Al}$	7523	15	7519.58	0.08	-0.2	U			Ald		59Hi66
	7519	15			0.0	U			Man		60Ta12
$^{26}\text{Al}(\text{p}, \gamma)^{27}\text{Si}$	7464.9	0.9	7463.34	0.13	-1.7	U					84Bu09 Z
$^{27}\text{P}(\text{p})^{26}\text{Si}$	-788	31	-807	9	-0.6	U			Dbp		17Ja05 *
	-807	9				3					19Su14 *
$^{27}\text{Na}(\beta^-)^{27}\text{Mg}$	8930	150	9069	4	0.9	U					73Al13
$^{27}\text{Mg}(\beta^-)^{27}\text{Al}$	2600	11	2610.27	0.07	0.9	U					54Da22
$^{27}\text{Si}(\beta^+)^{27}\text{Al}$	4872	20	4812.36	0.10	-3.0	B					60Wa04
$^{27}\text{Al}(\text{p}, \text{n})^{27}\text{Si}$	-5573	10	-5594.71	0.10	-2.2	U			Oak		55Ki28 Z
	-5597.5	6.0			0.5	U			Tkm		63Ok01
	-5593.6	4.3			-0.3	U			Ric		65Ku02
	-5585.1	2.3			-4.2	B			Ric		66Bo20 Z
	-5592.0	1.0			-2.7	U			Yal		69Ov01 Z
	-5594.1	3.2			-0.2	U			Har		76Fr13
	-5593.8	0.26			-3.5	F			Auc		77Na24 *
	-5594.27	0.11			-4.0	F			Auc		85Wh03 *
	-5594.72	0.10				2			Auc		94Br37 Z

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{27}\text{Si}^i(\text{IT})^{27}\text{Si}$	6638	1	6625.0	2.3	-13.0	F					16Mc09 *
* $^{27}\text{Ne}-\text{u}$	F : contaminated by ^{27}Na										91Zh24 **
* $^{27}\text{Na}-\text{u}$	Not independent of $^{27}\text{Na}-^{27}\text{Al}$ from isobaric method, do not use										01Lu17 **
* $^{27}\text{Al}(\text{p},\text{t})^{25}\text{Al}^i$	IT=7904(5), rebuilt $Q=-23847.9(4.7)$; recalib +4.5keV										GAu **
* $^{27}\text{P}^i(2\text{p})^{25}\text{Al}$	And $E_{2p}=5315(60)$ to $3/2^+$ level at 944.9 keV										Ens098 **
* $^{26}\text{Mg}(^{18}\text{O},^{17}\text{F})^{27}\text{Na}$	F : shape of peak raises doubt on centroid determination										GAu **
* $^{26}\text{Mg}(\text{p},\gamma)^{27}\text{Al}$	$E_p=338.65(0.12)$ to $8596.8(0.5)$ level										78Ma24 **
* $^{26}\text{Mg}(\text{p},\gamma)^{27}\text{Al}$	$E_p=338.21(0.30)$ to $8596.8(0.5)$ level										78Ma24 **
* $^{26}\text{Mg}(\text{p},\gamma)^{27}\text{Al}$	$E_p=809.90(0.05,Z)$ to $9050.7(0.5,Z)$ level										78Ma24 **
* $^{27}\text{P}(\text{p})^{26}\text{Si}$	$Q_p=332(30)$ keV from $3/2^+$ level at 1120(8) keV from 08Ga10										17Ja05 **
* $^{27}\text{P}(\text{p})^{26}\text{Si}$	$Q_p=318(8)$ keV from $3/2^+$ level at 1125(2) keV										19Su14 **
* $^{27}\text{Al}(\text{p},\text{n})^{27}\text{Si}$	F : rejected by same group "measurement contains error"										94Br37 **
* $^{27}\text{Si}^i(\text{IT})^{27}\text{Si}$	F : Very weak peak, evaluator thinks this is not the IAS of ^{27}P										GAu **
$^{28}\text{Ne}-\text{u}$	11490	430	12130	140	1.0	o			TO1	1.5	86Vi09
	12270	560			-0.2	o			GA1	1.5	87Gi05
	11958	238			0.5	o			GA3	1.5	91Or01
	12160	140			-0.1	2			TO4	1.5	91Zh24
	12110	118			0.1	2			GA7	1.5	07Ju03
$^{28}\text{Na}-\text{u}$	-1220	190	-1061	11	0.6	o			TO1	1.5	86Vi09
	-1097	96			0.2	U			GA3	1.5	91Or01
	-1062	14			0.1	o			P40	1.0	01Lu17
	-1061	11				2			P40	1.0	06Ga04
$^{28}\text{Si}-\text{u}$	-23073.43	0.30	-23073.4656	0.0006	-0.1	U			ST1	1.0	93Je06
	-23073.4676	0.0020			1.0	U			MII	1.0	95Di08
	-23073.00	0.27			-0.7	U			OH1	2.5	94Go.A
	-23073.466	0.008			0.1	U			ST2	1.0	02Be64 *
$\text{C}_2\text{D}_2-^{28}\text{Si}$	51277.0224	0.0024	51277.0213	0.0006	-0.5	U			MII	1.0	95Di08
$^{15}\text{N}_2-^{28}\text{Si}\text{H}_2$	7641.2007	0.0024	7641.1983	0.0013	-1.0	1	30	^{26}N	MII	1.0	95Di08
$\text{C}_2\text{H}_4-^{28}\text{Si}$	54373.59360	0.00079	54373.5932	0.0005	-0.5	1	48	^{48}Si	FS1	1.0	08Re16
$^{13}\text{C}_2\text{H}_2-^{28}\text{Si}$	45433.19986	0.00071	45433.2000	0.0005	0.3	1	53	^{32}Si	FS1	1.0	08Re16
$^{28}\text{Si}_2\text{ }^{16}\text{O}-^{35}\text{Cl }^{37}\text{Cl}$	14013.07	0.70	14012.42	0.07	-0.6	U			H46	1.5	93Nx02
$^{28}\text{Mg}-^{23}\text{Na}_{1.217}$	-3673.79	0.28				2			TT1	1.0	17Br14
$^{25}\text{Na}-^{28}\text{Na}_{.446}\text{ }^{22}\text{Na}_{.568}$	-5869	75	-5974	5	-0.9	U			P10	1.5	75Th08 *
$^{26}\text{Na}-^{28}\text{Na}_{.619}\text{ }^{22}\text{Na}_{.394}$	-4229	613	-4208	7	0.0	U			P11	1.5	75Th08
	-4205	128			-0.0	U			P12	2.5	75Th08
	-4203	87			-0.1	U			P13	1.0	75Th08
$^{28}\text{Si}(\text{p},^6\text{He})^{23}\text{Al}$	-38569	80	-38544.0	0.3	0.3	U			Brk		69Ce01
$^{28}\text{Si}(^3\text{He},^8\text{Li})^{23}\text{Al}$	-34274	25	-34255.5	0.3	0.7	U			MSU		75Be38
$^{28}\text{Si}(\alpha,^8\text{He})^{24}\text{Si}$	-61433	21	-61423	19	0.5	R			Tex		80Tr04
$^{28}\text{Si}(\text{p},\alpha)^{25}\text{Al}$	-7709.3	2.6	-7712.77	0.06	-1.3	U			NDm		73Br27
$^{28}\text{Si}(^3\text{He},^6\text{He})^{25}\text{Si}$	-27976	50	-27981	10	-0.1	U			Brk		70Me11
	-27981	10				2			MSU		72Be12
$^{26}\text{Mg}(\text{t},\text{p})^{28}\text{Mg}$	6474	12	6466.24	0.26	-0.6	U			Har		61Hi11 Y
$^{26}\text{Mg}(^3\text{He},\text{p})^{28}\text{Al}$	8285	5	8278.42	0.06	-1.3	U			Phi		74Be07
$^{28}\text{Si}(\text{d},\alpha)^{26}\text{Al}$	1429	4	1428.15	0.07	-0.2	U			MIT		64Sp12
$^{28}\text{Si}(\text{p},\text{t})^{26}\text{Si}$	-22009	3	-22012.63	0.11	-1.2	U			MSU		74Ha02
	-22014.1	1.0			1.5	U			Yal		05Pa31
$^{28}\text{F}(\gamma,\text{n})^{27}\text{F}$	220	50	199	6	-0.4	U			MSU		12Ch02
	199	6				2					20Re06
$^{27}\text{Al}(\text{n},\gamma)^{28}\text{Al}$	7725.02	0.20	7725.174	0.011	0.8	U			BNn		78St25 Z
	7725.07	0.30			0.3	U			ILn		79Br25 Z
	7725.13	0.3			0.1	U			MMn		80Is02 Z
	7725.02	0.10			1.5	U					81Su.A Z
	7725.14	0.09			0.4	o			ILn		82Sc14 Z
	7725.17	0.15			0.0	U			Bdn		06Fi.A
	7725.174	0.011				2			ILn		18Rz01

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{27}\text{Al}(\text{d,p})^{28}\text{Al}$	5511	5	5500.608	0.011	-2.1	U			Mex		61Ja23
	5503	10			-0.2	U			MIT		64Sp12
$^{27}\text{Al}(\text{p},\gamma)^{28}\text{Si}$	11584.89	0.30	11584.90	0.05	0.0	U			Utr		78Ma23 Z
	11585.09	0.14			-1.3	1	11	11	^{27}Al		28Sir-0
$^{27}\text{Al}(^3\text{He,d})^{28}\text{Si}$	6049	18	6091.43	0.05	2.4	U					60Fo01
$^{27}\text{Al}(\text{p},\gamma)^{28}\text{Si}^r$	-956.15	0.03	-956.139	0.025	0.3	2			Utr		78Ma23 Z
	-956.025	0.020			-5.7	B			Auc		94Br37 Z
	-956.13	0.05			-0.4	2					98Wa.A Z
$^{28}\text{Si}(^3\text{He},\alpha)^{27}\text{Si}$	3407	15	3398.01	0.11	-0.6	U			Ald		59Hi68 Y
$^{28}\text{Si}(^3\text{He},\alpha)^{27}\text{Si}^i$	-3225.5	2.6	-3227.0	2.3	-0.6	1	79	79	$^{27}\text{Si}^i$		86Sc21 *
$^{28}\text{Si}(^7\text{Li},^8\text{He})^{27}\text{P}$	-37513	40	-37536	9	-0.6	U					01Ca37
$^{28}\text{P}^i(\text{p})^{27}\text{Si}$	3835	20				3			Lis		89Po10
$^{28}\text{Cl}(\text{p})^{27}\text{S}$	1600	80	3490#	300#	23.6	D					18Mu18 *
$^{28}\text{Mg}(\beta^-)^{28}\text{Al}$	1791	10	1830.77	0.27	4.0	B					53Ma23 *
	1831.8	2.0			-0.5	U					54OI03 *
$^{28}\text{Al}(\beta^-)^{28}\text{Si}$	4644	10	4642.08	0.05	-0.2	U					52Mo22 *
	4657	14			-1.1	U					54OI03 *
$^{28}\text{Si}^r(\text{IT})^{28}\text{Si}$	12541.23	0.14	12541.04	0.05	-1.3	R			Utr		90En02 Z
$^{28}\text{P}(\beta^+)^{28}\text{Si}$	14290	40	14344.9	1.1	1.4	U					68Ar03 *
$^{28}\text{Si}(\text{p,n})^{28}\text{P}$	-15118.3	4.1	-15127.3	1.1	-2.2	U			Yal		69Ov01 Z
	-15112.5	5.8			-2.5	U			BNL		71Go18 Z
$^{28}\text{Si}(^3\text{He,t})^{28}\text{P}$	-14380	60	-14363.5	1.1	0.3	U			Brk		66Ma18
$^{28}\text{Si}(^3\text{He,t})^{28}\text{P}-^{36}\text{Ar}(\text{O})^{36}\text{K}$	-1530.58	1.10	-1530.6	1.1	-0.0	1	100	100	^{28}P	Mun	10Wr01
$^{28}\text{Si}(\pi^+, \pi^-)^{28}\text{S}$	-24544	160				2					82Mo12 *
$^{28}\text{Si}-\text{u}$	Unc. was erroneously 0.0008 in Ame2012, was correct in Ame2003										GAu **
$^{25}\text{Na}-^{28}\text{Na}_{446}-^{22}\text{Na}_{568}$	Symmetric double-doublet 22-24 26-28 included										GAu **
$^{28}\text{Si}(^3\text{He},\alpha)^{27}\text{Si}^i$	IT=6626(3), Q rebuilt with Ame1977										GAu **
$^{28}\text{Cl}(\text{p})^{27}\text{S}$	Trends from Mass Surface TMS suggest ^{28}Cl 1890 keV less bound										GAu **
$^{28}\text{Mg}(\beta^-)^{28}\text{Al}$	$E_{\beta^-}=418(10)$ 459(2) respectively, to 1^+ level at 1372.917 keV										Ens13a **
$^{28}\text{Al}(\beta^-)^{28}\text{Si}$	$E_{\beta^-}=2865(10)$ 2878(14) respectively, to 2^+ level at 1779.030 keV										Ens13a **
$^{28}\text{P}(\beta^+)^{28}\text{Si}$	$E_{\beta^+}=11490(40)$ to 2^+ level at 1779.030 keV										Ens13a **
$^{28}\text{Si}(\pi^+, \pi^-)^{28}\text{S}$	Original -24603(160) recalibrated to $^{16}\text{O}(\pi^+, \pi^-)^{16}\text{Ne}$ $Q=-27704(20)$ keV										GAu **
$^{29}\text{F}-\text{u}$	43103	376				2			GA8	1.5	12Ga45
$^{29}\text{Ne}-\text{u}$	19433	551	19750	160	0.4	o			GA3	1.5	91Or01
	19300	400			0.8	U			TO4	1.5	91Zh24
	19400	410			0.6	o			GA5	1.5	00Sa21
	19753	107				2			GA7	1.5	07Ju03
$^{29}\text{Na}-\text{u}$	2820	230	2877	8	0.2	U			TO1	1.5	86Vi09
	2838	143			0.2	U			GA3	1.5	91Or01
	2861	14			1.1	o			P40	1.0	01Lu17
	2866	13			0.9	1	37	37	^{29}Na	1.0	06Ga04
$^{29}\text{Na}-^{39}\text{K}_{744}$	29885.9	9.9	29879	8	-0.6	1	63	63	^{29}Na	1.0	13Ch49
$^{29}\text{Mg}-\text{u}$	-11375	19	-11392.8	0.4	-0.9	-			P40	1.0	06Ga04 *
	-11388	16			-0.3	-			P40	1.0	06Ga04
	ave. -11383	12			-0.8	1	0	0	^{29}Mg		average
$^{29}\text{Al}-\text{O}_{1.812}$	-10328.8	1.0	-10332.1	0.4	-3.3	C			TT1	1.0	12Ch.A
	-10333.5	1.7			0.8	U			TT1	1.0	16Kw.A
$^{29}\text{P}^{40}\text{Ar}-\text{u}$	-55816.80	0.50	-55816.5	0.4	0.6	1	59	59	^{29}P	1.0	15Ei01
$^{29}\text{S}-\text{u}$	-3322	14				2			LZ1	1.0	18Fu11
$^{29}\text{Mg}-^{23}\text{Na}_{1.261}$	1508.09	0.37	1508.1	0.4	0.0	1	100	100	^{29}Mg	1.0	17Br14
$^{29}\text{Al}-^{23}\text{Na}_{1.261}$	-6645.90	0.37				2			TT1	1.0	17Ga20
$^{29}\text{Si}-^{28}\text{Si H}$	-8256.90198	0.00024	-8256.90198	0.00024	-0.0	1	100	100	^{29}Si	1.0	05Ra34
$^{26}\text{Na}-^{29}\text{Na}_{512}-^{22}\text{Na}_{506}$	-5763	91	-5611	5	1.1	U			P10	1.5	75Th08
	-6379	293			1.7	U			P11	1.5	75Th08
	-5252	277			-0.5	U			P12	2.5	75Th08
	-5576	66			-0.5	U			P13	1.0	75Th08

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{27}\text{Na}-^{29}\text{Na}_{466} \text{ } ^{25}\text{Na}_{540}$	-1708	124	-1713	5	-0.0	U			P10	1.5	75Th08
$^{18}\text{O}(^{13}\text{C},2\text{p})^{29}\text{Mg}$	-1456	50	-1623.4	0.3	-3.3	B					81Pa17
$^{26}\text{Mg}(^{11}\text{B},^8\text{B})^{29}\text{Mg}$	-19720	50	-19856.0	1.1	-2.7	U			Brk		74Sc26
$^{26}\text{Mg}(^{18}\text{O},^{15}\text{O})^{29}\text{Mg}$	-9207	55	-9240.6	0.6	-0.6	U			Mun		78Pa12
	-9250	45			0.2	U			Can		85Fi08
$^{26}\text{Mg}(\alpha,\text{p})^{29}\text{Al}$	-2880	40	-2870.8	0.3	0.2	U			Yal		57Gr47 Y
	-2874	10			0.3	U			ANL		68Be13
$^{29}\text{Si}(\text{n},\alpha)^{26}\text{Mg}$	-21	21	-34.136	0.029	-0.6	U			Ham		62An05
$^{27}\text{Al}(\text{t},\text{p})^{29}\text{Al}$	8679.5	1.2	8671.7	0.3	-6.5	B			Str		84An17
$^{29}\text{Si}(\text{d},\alpha)^{27}\text{Al}$	6000	11	6012.59	0.05	1.1	U			MIT		64Sp12
$^{29}\text{Si}(\text{p},\text{t})^{27}\text{Si}^i$	-23802	5	-23796.4	2.3	1.1	1	21	$^{21}\text{ } ^{27}\text{Si}^i$	MSU		77Be13
$^{27}\text{Al}(^3\text{He},\text{n})^{29}\text{P}$	6616	30	6615.9	0.4	-0.0	U			Oak		72Gr39
$^{29}\text{Ar}(2\text{p})^{27}\text{S}$	5500	180	5900#	180#	2.2	D					18Mu18 *
$^{28}\text{Si}(\text{n},\gamma)^{29}\text{Si}$	8473.6	0.3	8473.6025	0.0005	0.0	o			MMn		80Is02 Z
	8473.61	0.04			-0.2	U			MMn		90Is02 Z
	8473.55	0.04			1.3	U			ORn		92Ra19 Z
	8473.5509	0.0500			1.0	o			PTc		97Ro26 *
	8473.54	0.17			0.4	U			Bdn		06Fi.A
	8473.551	0.030			1.7	U			PTc		01Pa52 *
	8473.5957	0.0050			1.4	U			NBS		06De21
$^{28}\text{Si}(\text{d},\text{p})^{29}\text{Si}$	6252	10	6249.03627	0.00023	-0.3	U			Mex		64Ma.B
	6252	10			-0.3	U			MIT		64Sp12
	6249.35	0.5			-0.6	U			Rez		90Pi05 *
$^{28}\text{Si}(\text{p},\gamma)^{29}\text{P}$	2747.1	1.7	2749.0	0.4	1.1	-					73Ba35 Z
	2748.8	0.6			0.4	-					74By01 Z
$^{28}\text{Si}(\text{d},\text{n})^{29}\text{P}$	560	30	524.5	0.4	-1.2	U			Ald		60Ma21
$^{28}\text{Si}(^3\text{He},\text{d})^{29}\text{P}$	-2733	12	-2744.5	0.4	-1.0	U			Ald		60Hi03 Y
$^{28}\text{Si}(\text{p},\gamma)^{29}\text{P}$	ave. 2748.6	0.6	2749.0	0.4	0.7	1	40	$^{40}\text{ } ^{29}\text{P}$			average
$^{28}\text{Si}(\text{p},\gamma)^{29}\text{P}^i$	-5630	10	-5632.8	2.5	-0.3	U			ANL		66Yo01
	-5631.9	5.0			-0.2	1	24	$^{24}\text{ } ^{29}\text{P}^i$			68Te01 *
$^{29}\text{Cl}(\text{p})^{28}\text{S}$	1800	100	2660#	100#	8.6	D					15Mu13
	1800	100			8.6	D					18Xu04 *
$^{29}\text{Mg}(\beta^-)^{29}\text{Al}$	7624	400	7595.4	0.5	-0.1	U					73Go34
$^{29}\text{Al}(\beta^-)^{29}\text{Si}$	3850	100	3687.3	0.3	-1.6	U					54Na14 *
$^{29}\text{P}(\beta^+)^{29}\text{Si}$	4967	20	4942.2	0.4	-1.2	U					55Ro05
$^{29}\text{P}^i(\text{IT})^{29}\text{P}$	8382.1	2.8	8381.8	2.4	-0.1	1	76	$^{76}\text{ } ^{29}\text{P}^i$			72Ba26
$^{29}\text{Mg}-\text{u}$	Result from the "Plasma" experiment. Next item from "Rilis"										06Ga04 **
$^{29}\text{Ar}(2\text{p})^{27}\text{S}$	Trends from Mass Surface TMS suggest ^{29}Ar 400 keV less bound										GAu **
$^{28}\text{Si}(\text{n},\gamma)^{29}\text{Si}$	Original error 0.0005 increased for calibration										GAu **
$^{28}\text{Si}(\text{n},\gamma)^{29}\text{Si}$	Original error 0.005 increased for calibration										GAu **
$^{28}\text{Si}(\text{d},\text{p})^{29}\text{Si}$	Estimated systematic error 0.5 added to statistical error 0.037 keV										AHW **
$^{28}\text{Si}(\text{p},\gamma)^{29}\text{P}^i$	IT=8376(6), Q rebuilt with Ame1964, error estimated by evaluator										GAu **
$^{29}\text{Cl}(\text{p})^{28}\text{S}$	Trends from Mass Surface TMS suggest ^{29}Cl 860 keV less bound										GAu **
$^{29}\text{Al}(\beta^-)^{29}\text{Si}$	$E_{\beta^-}=1550(100)$ to $5/2^+$ level at 2028.16 and $3/2^+$ at 2425.97 keV										Ens125 **
$^{30}\text{Ne}-\text{u}$	23872	884	24990	270	0.8	o			GA3	1.5	91Or01
	25660	850			-0.5	o			GA5	1.5	00Sa21
	24734	301			0.6	-			GA7	1.5	07Ju03
	25024	301			-0.1	-			GA8	1.5	12Ga45
	ave. 24880	320			0.4	1	73	$^{73}\text{ } ^{30}\text{Ne}$			average
$^{30}\text{Na}-\text{u}$	7620	540	9098	5	1.8	F			TO1	1.5	86Vi09 *
	9200	370			-0.2	U			GA1	1.5	87Gi05
	9126	218			-0.1	U			GA3	1.5	91Or01
	9330	130			-1.2	U			TO4	1.5	91Zh24
	8976	27			4.5	B			P40	1.0	01Lu17
	8990	25			4.3	B			P40	1.0	06Ga04

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{30}\text{Na}-\text{O}_{1.876}$	18638.9	5.6	18638	5	-0.1	1	82	82 ^{30}Na	TT1	1.0	13Ch49
$^{30}\text{Na}-^{39}\text{K}_{.769}$	37004	12	37008	5	0.3	1	18	18 ^{30}Na	TT1	1.0	13Ch49
$^{30}\text{Mg}-\text{O}_{1.876}$	3.0	3.7	5.6	1.4	0.7	1	14	14 ^{30}Mg	TT1	1.0	13Ch49
$^{30}\text{Mg}-\text{u}$	-9700	230	-9534.5	1.4	0.5	o			TO1	1.5	86Vi09
	-9597	98			0.4	U			GA3	1.5	91Or01
	-9490	110			-0.3	U			TO4	1.5	91Zh24
	-9546	14			0.8	U			P40	1.0	06Ga04
$^{30}\text{Mg}-^{39}\text{K}_{.769}$	18375.6	1.5	18375.2	1.4	-0.3	1	86	86 ^{30}Mg	TT1	1.0	20Le06
$^{30}\text{Al}-\text{O}_{1.875}$	-7495.1	2.8	-7495.7	2.1	-0.2	1	55	55 ^{30}Al	TT1	1.0	20Le06
$^{30}\text{Al}-^{39}\text{K}_{.769}$	10878.1	3.1	10878.9	2.1	0.3	1	45	45 ^{30}Al	TT1	1.0	16Kw.A
$^{30}\text{P}-^{30}\text{Si}$	4543.353	0.066				3			JY1	1.0	16Ca22
$^{30}\text{S}-^{30}\text{P}$	6593.28	0.21				4			JY1	1.0	11So11
$^{26}\text{Na}-^{30}\text{Na}_{.433} \text{ } ^{22}\text{Na}_{.591}$	-7454	287	-7468	4	-0.0	U			P10	1.5	75Th08
	-8060	641			0.6	U			P11	1.5	75Th08
	-7045	225			-0.8	U			P12	2.5	75Th08
	-7515	117			0.4	U			P13	1.0	75Th08
$^{27}\text{Na}-^{30}\text{Na}_{.360} \text{ } ^{25}\text{Na}_{.648}$	-2750	213	-2505	4	0.8	U			P10	1.5	75Th08
$^{26}\text{Mg}(^{18}\text{O}, ^{14}\text{O})^{30}\text{Mg}$	-16234	55	-16123.8	1.3	2.0	U			Mun		78Pa12 *
$^{30}\text{Si}(\text{n}, \alpha)^{27}\text{Mg}$	-4193	21	-4199.97	0.05	-0.3	U			Ham		62An05
$^{30}\text{Si}(\text{p}, \alpha)^{27}\text{Al}$	-2368	10	-2372.04	0.05	-0.4	U			MIT		64Sp12
$^{27}\text{Al}(\alpha, \text{p})^{30}\text{Si}$	2375	8	2372.04	0.05	-0.4	U			Man		59Ba13 Y
$^{30}\text{Si}(\text{d}, \alpha)^{28}\text{Al}$	3123	10	3128.56	0.05	0.6	U			MIT		64Sp12
$^{28}\text{Si}(^3\text{He}, \text{n})^{30}\text{S}$	-573	15	-573.64	0.21	-0.0	U			CIT		67Mi02
$^{30}\text{Ar}(2\text{p})^{28}\text{S}$	2280	130	3420#	80#	8.8	B					15Mu13 *
	2420	80			12.5	D					18Xu04 *
$^{29}\text{Si}(\text{n}, \gamma)^{30}\text{Si}$	10609.6	0.3	10609.199	0.022	-1.3	o			MMn		80Is02 Z
	10609.21	0.04			-0.3	2			MMn		90Is02 Z
	10609.24	0.05			-0.8	2			ORn		92Ra19 Z
	10609.1776	0.0500			0.4	o			PTc		97Ro26 *
	10609.178	0.030			0.7	2			PTc		01Pa52 *
	10609.23	0.21			-0.1	U			Bdn		06Fi.A
$^{29}\text{Si}(\text{d}, \text{p})^{30}\text{Si}$	8413	10	8384.633	0.022	-2.8	U			Mex		61Ja23
	8396	13			-0.9	U			MIT		64Sp12
	8384.92	0.53			-0.5	U			Rez		90Pi05 *
$^{29}\text{Si}(\text{p}, \gamma)^{30}\text{P}$	5594.5	0.4	5594.75	0.07	0.6	U					85Re02
	5594.5	0.5			0.5	U					96Wa33
$^{30}\text{Cl}(\text{p})^{29}\text{S}$	480	20				3					18Mu18
$^{30}\text{Na}(\beta^-)^{30}\text{Mg}$	17167	330	17356	5	0.6	U					83De04 *
$^{30}\text{Mg}(\beta^-)^{30}\text{Al}$	6690	240	6982.7	2.3	1.2	U					83De04 *
$^{30}\text{Al}(\beta^-)^{30}\text{Si}$	8550	250	8568.8	1.9	0.1	U					61Ro12 *
$^{30}\text{Si}(\text{t}, ^3\text{He})^{30}\text{Al}$	-8520	40	-8550.3	1.9	-0.8	U					69Aj03
	-8545	15			-0.4	U					87Pe06
$^{30}\text{P}(\beta^+)^{30}\text{Si}$	4262	40	4232.11	0.06	-0.7	U					56Gr07
	4267	25			-1.4	U					63Fr10
$^{30}\text{Si}(\text{p}, \text{n})^{30}\text{P}$	-5012.1	5.	-5014.45	0.06	-0.5	U			Har		75Fr.A Z
$^{30}\text{S}(\beta^+)^{30}\text{P}$	6118	22	6141.60	0.20	1.1	U					63Fr10 *
$^{30}\text{Na}-\text{u}$	F : contaminated by ^{30}Mg										91Zh24 **
$^{26}\text{Mg}(^{18}\text{O}, ^{14}\text{O})^{30}\text{Mg}$	Tentative, say authors; four counts only										AHW **
$^{30}\text{Ar}(2\text{p})^{28}\text{S}$	Symmetrized from 2250(+150-100)										GAU **
$^{30}\text{Ar}(2\text{p})^{28}\text{S}$	Symmetrized from 2450(+50-100), supersedes 15Mu13										HWJ192 **
$^{30}\text{Ar}(2\text{p})^{28}\text{S}$	Trends from Mass Surface TMS suggest ^{30}Ar 1000 keV less bound										GAU **
$^{29}\text{Si}(\text{n}, \gamma)^{30}\text{Si}$	Original error 0.0005 increased for calibration										GAU **
$^{29}\text{Si}(\text{n}, \gamma)^{30}\text{Si}$	Original error 0.005 increased for calibration										GAU **
$^{29}\text{Si}(\text{d}, \text{p})^{30}\text{Si}$	Estimated systematic error 0.5 added to statistical error 0.16 keV										AHW **
$^{30}\text{Na}(\beta^-)^{30}\text{Mg}$	Calculated from 3 values used for calibration										GAU **
$^{30}\text{Mg}(\beta^-)^{30}\text{Al}$	Calculated from value used for calibration										GAU **
$^{30}\text{Al}(\beta^-)^{30}\text{Si}$	$E_{\beta^-}=5050(250)$ to 2^+ level at 3498.49 keV										Ens109 **
$^{30}\text{S}(\beta^+)^{30}\text{P}$	$E_{\beta^+}=4422(22)$ to 0^+ level at 677.01 keV										Ens109 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{31}\text{Ne}-u$	33087	1739	33470	290	0.1	U					07Ju03
	33752	333			-0.6	1	33	$^{33}\text{ }^{31}\text{Ne}$	GA7	1.5	12Ga45
$^{31}\text{Na}-u$	13440	1000	13147	15	-0.2	o			GA1	1.5	87Gi05
	13559	327			-0.8	o			GA3	1.5	91Or01
	13610	210			-1.5	U			TO4	1.5	91Zh24
	13441	118			-1.7	U			GA7	1.5	07Ju03
$^{31}\text{Mg}-\text{O}_{1.938}$	6503.7	3.3				2			TT1	1.0	13Ch49
$^{31}\text{Mg}-u$	-3830	220	-3352	3	1.4	o			TO1	1.5	86Vi09
	-3520	180			0.6	o			GA1	1.5	87Gi05
	-3458	149			0.5	U			GA3	1.5	91Or01
	-3370	120			0.1	U			TO4	1.5	91Zh24
	-3425	18			4.1	B			P40	1.0	06Ga04
$\text{O}_2-^{31}\text{P H}$	8242.20819	0.00086	8242.2089	0.0007	0.9	1	64	$^{48}\text{ }^{31}\text{P}$	FS1	1.0	08Re16
$^{31}\text{K}-u$	32783	275	36780#	320#	14.5	D			1.0	1.0	19Ko18 *
$^{31}\text{Na}-^{39}\text{K}_{.795}$	42017	25	42000	15	-0.7	o			TT1	1.0	13Ch49
	42000	15				2			TT1	1.0	17Ga20
$^{31}\text{Al}-^{39}\text{K}_{.795}$	12803.1	2.4				2			TT1	1.0	16Kw.A
$^{31}\text{P}-^{28}\text{Si H}_3$	-26639.6290	0.0056	-26639.6324	0.0007	-0.6	U			FS1	1.0	06Re19
	-26639.63324	0.00089			0.9	1	62	$^{52}\text{ }^{31}\text{P}$	FS1	1.0	08Re16
$^{31}\text{S}-^{31}\text{P}$	5794.98	0.25	5795.00	0.25	0.1	1	97	$^{97}\text{ }^{31}\text{S}$	JY1	1.0	10Ka30
$^{31}\text{Cl}-^{31}\text{P}$	18686.1	3.7				2			JY1	1.0	16Ka15
O_2-^{31}P	16067.228	0.096	16067.2408	0.0007	0.1	U			MS1	1.0	09Kw02 *
$^{26}\text{Na}-^{31}\text{Na}_{.373}\text{ }^{22}\text{Na}_{.657}$	-7457	286	-8024	6	-0.8	U			P12	2.5	75Th08
$^{18}\text{O}(^{15}\text{N},2\text{p})^{31}\text{Al}$	-170	90	-308.6	2.2	-1.5	U					81Pa11
$^{27}\text{Al}(\alpha,\gamma)^{31}\text{P}$	9667.4	1.3	9668.60	0.05	0.9	U			Utr		78Ma23
$^{31}\text{P}(\text{p},\alpha)^{28}\text{Si}$	1912	5	1916.3079	0.0007	0.9	U			Bar		56Va14 Y
	1919	4			-0.7	U			VUn		64Sm03
	1911	10			0.5	U			MIT		64Sp12
	1915.8	0.2			2.5	U			Zur		67Si30
$^{28}\text{Si}(\alpha,\text{n})^{31}\text{S}$	-8135	44	-8096.67	0.23	0.9	U			Tal		63Ne05
$^{31}\text{P}(\text{d},\alpha)^{29}\text{Si}$	8166	11	8165.3441	0.0007	-0.1	U			MIT		64Sp12
$^{31}\text{Cl}'(2\text{p})^{29}\text{P}$	7700	100	7631	3	-0.7	U					90Bo24 *
	7610	60			0.3	U			Lis		91Bo32
	7643	50			-0.2	U			Lis		92Ba01 *
	7627	15			0.3	o					98Ax02
	7631	3				2					00Fy01 *
$^{30}\text{Ne}(\text{n},\gamma)^{31}\text{Ne}$	190	130	170	130	-0.2	1	95	$^{67}\text{ }^{31}\text{Ne}$			14Na10 *
$^{30}\text{Si}(^{18}\text{O},^{17}\text{F})^{31}\text{Al}$	-12200	25	-12216.8	2.2	-0.7	U					88Wo02
	-12237	35			0.6	U			Ber		89Bo.A
$^{30}\text{Si}(\text{n},\gamma)^{31}\text{Si}$	6589.1	0.7	6587.39	0.04	-2.4	U					70Be48
	6587.5	0.8			-0.1	U					70Sp02
	6588.4	0.3			-3.4	B					72Dz13
	6587.32	0.20			0.4	U			MMn		90Is02 Z
	6587.39	0.05			0.1	3			ORn		92Ra19 Z
	6587.3970	0.0500			-0.1	o			PTc		97Ro26 *
	6587.39	0.14			0.0	U			Bdn		06Fi.A
	6587.397	0.057			-0.1	3			PTc		01Pa52
$^{30}\text{Si}(\text{d},\text{p})^{31}\text{Si}$	4368	7	4362.83	0.04	-0.7	U			MIT		64Sp12
	4364.18	0.55			-2.5	U			Rez		90Pi05 *
$^{30}\text{Si}(\text{p},\gamma)^{31}\text{P}$	7297.4	1.2	7296.553	0.022	-0.7	U					68Wo01
$^{31}\text{Cl}'(\text{p})^{30}\text{S}$	12033	10	12026	3	-0.7	o					98Ax02 *
	12033	14			-0.5	U					00Fy01 *
$^{31}\text{Mg}(\beta^-)^{31}\text{Al}$	10150	700	11829	4	2.4	U					83De04
$^{31}\text{Al}(\beta^-)^{31}\text{Si}$	7940	100	7998.3	2.2	0.6	U					73Go22
$^{31}\text{Si}(\beta^-)^{31}\text{P}$	1471	8	1491.51	0.04	2.6	U					52Mo12
	1486	12			0.5	U					52Wa12

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{31}\text{S}(\beta^+)^{31}\text{P}$	5412	30	5398.01	0.23	-0.5	U					60Wa04
$^{31}\text{P}(\text{p,n})^{31}\text{S}$	-6212.3	20.	-6180.36	0.23	1.6	o			ChR		58Go77 Y
	-6250	20			3.5	B			ChR		59Br06 Y
$^{31}\text{K}-\text{u}$	from three-proton separation energy -4600(200) keV										HWJ201 **
$^{31}\text{K}-\text{u}$	Trends from Mass Surface TMS suggest ^{31}K 3720 keV less bound										GAu **
$^{31}\text{O}_2-^{31}\text{P}$	For original doublet $^{31}\text{P}-\text{O}_{1.938}$, $D_M=-16382.522(0.096) \mu\text{u}$										GAu **
$^{31}\text{Cl}'(2\text{p})^{29}\text{P}$	Large error in E_{cm} due to sequential decay kinematics										MMC122**
$^{31}\text{Cl}'(2\text{p})^{29}\text{P}$	reference also finds 3p emission at 4870 keV										92Ba01 **
$^{31}\text{Cl}'(2\text{p})^{29}\text{P}$	$Q_{2p}=7620(5)$, 6245(2), 5679(3), 5223(5) keV										00Fy01 **
*	to ground state and levels $3/2^+$ at 1383.55, $5/2^+$ at 1953.91, $3/2^+$ 2422.7 keV										Ens125 **
$^{30}\text{Ne}(\text{n},\gamma)^{31}\text{Ne}$	Symmetrized from 150(+160-100) keV										14Na10 **
$^{30}\text{Si}(\text{n},\gamma)^{31}\text{Si}$	Original error 0.0005 increased for calibration										GAu **
$^{30}\text{Si}(\text{d,p})^{31}\text{Si}$	Estimated systematic error 0.5 added to statistical error 0.23 keV										AHW **
$^{31}\text{Cl}'(\text{p})^{30}\text{S}$	Average of 3 branches										AHW **
$^{31}\text{Cl}'(\text{p})^{30}\text{S}$	$E_p=11654(28)$, 9493(20), 8347(15), 8092(14) keV										00Fy01 **
$^{32}\text{Na}-\text{u}$	19720	636	20010	40	0.3	o			GA3	1.5	91Or01
	19900	1100			0.1	U			TO4	1.5	91Zh24
	20980	500			-1.3	o			GA5	1.5	00Sa21
	20193	129			-0.9	U			GA7	1.5	07Ju03
	9280.9	3.5				2			TT1	1.0	13Ch49
$^{32}\text{Mg}-\text{O}_2$	-800	260	-890	4	-0.2	o			TO1	1.5	86Vi09
$^{32}\text{Mg}-\text{u}$	-890	270			0.0	U			GA1	1.5	87Gi05
	-924	214			0.1	U			GA3	1.5	91Or01
	-820	130			-0.4	U			TO4	1.5	91Zh24
	-1142	113			2.2	o			P40	1.0	01Lu20
	-966	38			2.0	U			P40	1.0	06Ga04 *
	-983	22			4.2	B			P40	1.0	06Ga04
	-1744	13	-1745	8	-0.1	o			TT1	1.0	12Ch.A
	-1744.9	7.7				2			TT1	1.0	16Kw.A
$^{32}\text{Al}-\text{O}_2$	-12160	220	-11916	8	0.7	U			TO1	1.5	86Vi09
	-11870	200			-0.2	U			GA1	1.5	87Gi05
	-11877	104			-0.2	U			GA3	1.5	91Or01
$^{32}\text{Al}-\text{u}$	-67319.35	0.32				2			MS1	1.0	09Kw02 *
$^{32}\text{Si O}_2-\text{C}_5 \text{H}_4$	17754.2	1.0	17758.0650	0.0014	1.5	U			J1	2.5	68Ma45
O_2-^{32}S	90531.3	1.4	90529.0816	0.0014	-0.6	U			J1	2.5	68Ma45
$\text{C}_2 \text{H}_8-^{32}\text{S}$	-17758.0663	0.0020	-17758.0650	0.0014	0.7	1	50	45 ^{32}S	FS1	1.0	05Sh38
$^{32}\text{S}-\text{O}_2$	-84335.9367	0.0019	-84335.9378	0.0014	-0.6	1	55	55 ^{32}S	FS1	1.0	05Sh38
$^{32}\text{S}-\text{C}_2 \text{D}_4$	-34156.50	0.57	-34157.0204	0.0016	-0.9	U			MS1	1.0	09Kw02 *
$^{32}\text{S}-\text{H C F}$	49808	40				2			TT1	1.0	17Ga20
$^{32}\text{Na}-^{39}\text{K}_{821}$	27434.8	1.9				2			MA8	1.0	03Bl17
$^{32}\text{Ar}-^{39}\text{K}_{821}$	25483.43	0.34	25484.0423	0.0030	1.8	U			MS1	1.0	09Kw02 *
$\text{C F}_3-^{32}\text{S O}_2 \text{H}$	33310.02	0.59	33309.0741	0.0030	-1.6	U			MS1	1.0	09Kw02 *
$^{26}\text{Na}-^{32}\text{Na}_{325} \text{ } ^{22}\text{Na}_{709}$	-8569	354	-9245	13	-0.8	U			P12	2.5	75Th08
$^{32}\text{S}(^3\text{He}, ^8\text{Li})^{27}\text{P}$	-31277	35	-31371	9	-2.7	B			MSU		77Be13
$^{32}\text{S}(\text{p},\alpha)^{29}\text{P}$	-4171	20	-4198.6	0.4	-1.4	U			Tky		64Ej05
$^{32}\text{S}(^3\text{He}, ^6\text{He})^{29}\text{S}$	-25520	50	-25582	13	-1.2	U			MSU		73Be09
$^{30}\text{Si}(\text{t,p})^{32}\text{Si}$	7307	1	7305.57	0.30	-1.4	U			Str		80An.A
$^{32}\text{S}(\text{d},\alpha)^{30}\text{P}$	4892	10	4896.13	0.07	0.4	U			MIT		64Sp12
$^{32}\text{S}(\text{p,t})^{30}\text{S}$	-19614	3	-19617.12	0.21	-1.0	U			MSU		74Ha02
$^{31}\text{Si}(\text{n},\gamma)^{32}\text{Si}$	9203.2180	0.0500	9200.0	0.3	-65.0	B			PTc		97Ro26 *
	9203.22	0.76			-4.3	B			PTc		01Pa51
	7935.73	0.16	7935.65	0.04	-0.5	U			MMn		85Ke12 Z
	7935.65	0.04				2			ILn		89Mi16 Z
$^{31}\text{P}(\text{n},\gamma)^{32}\text{P}$	7935.60	0.16			0.3	U			Bdn		06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{31}\text{P}(\text{d,p})^{32}\text{P}$	5712	8	5711.08	0.04	−0.1	U			MIT		64Sp12
$^{31}\text{P}(\text{p},\gamma)^{32}\text{S}$	8864.9	0.9	8863.9638	0.0014	−1.0	U					72Co13
	8862.7	3.			0.4	U					73Ve06 *
	8865.6	1.0			−1.6	U					73Ve08 Z
	8865.1	0.9			−1.3	U					74Vi02
$^{31}\text{P}(^3\text{He,d})^{32}\text{S}$	3356	13	3370.4887	0.0014	1.1	U			MIT		68Gr17
$^{32}\text{S}(\text{p,d})^{31}\text{S}$	−12817.8	1.5	−12819.76	0.23	−1.3	U			MSU		73Mo23
$^{32}\text{S}(^3\text{He},\alpha)^{31}\text{S}$	5415	15	5533.30	0.23	7.9	B					66Gr26
	5515	15			1.2	U			MIT		67Sp09
	5486	20			2.4	U			Ors		67Ro17
	5538	6			−0.8	U			CIT		70Mo08
$^{32}\text{Cl}(\text{p})^{31}\text{S}$	−1583.5	3.1	−1581.1	0.5	0.8	U					85Bj01 *
	−1581.9	2.1			0.4	U					93Sc16 *
	−1581.3	0.6			0.3	1	79	$^{76}\text{^{32}Cl}$			08Ga.A *
	−1581.2	2.1			0.0	U					20Ar04 *
$^{32}\text{Na}(\beta^-)^{32}\text{Mg}$	18300	1400	19470	40	0.8	U					83De04
$^{32}\text{Si}(\beta^-)^{32}\text{P}$	213	7	227.2	0.3	2.0	U					64Br09
	221.4	1.2			4.8	B					84Po09
$^{32}\text{P}(\beta^-)^{32}\text{S}$	1707.6	0.7	1710.66	0.04	4.4	B					61Ni02
	1710.1	0.7			0.8	U					68Fi04
$^{32}\text{Cl}(\beta^+)^{32}\text{S}$	12720	30	12680.8	0.6	−1.3	U					68Ar03 *
$^{32}\text{S}(\text{p,n})^{32}\text{Cl}$	−13470	14	−13463.2	0.6	0.5	U			Yal		69Ov01 Z
	−13470	9			0.8	U			BNL		71Go18 Z
$^{32}\text{S}(^3\text{He,t})^{32}\text{Cl}$	−12699	15	−12699.4	0.6	−0.0	U					89Je07
$^{32}\text{S}(^3\text{He,t})^{32}\text{Cl}-^{36}\text{Ar}^{36}\text{K}$	133.01	1.10	133.5	0.6	0.5	1	30	$^{24}\text{^{32}Cl}$	Mun		10Wr01
$^{32}\text{S}(\pi^+,\pi^-)^{32}\text{Ar}$	−22813	50	−22793.2	1.8	0.4	U					80Bu15
$^{32}\text{Mg}-\text{u}$	Result from the “Plasma” experiment. Next item from “Rilis”										06Ga04 **
$^{32}\text{Si O}_2-\text{C}_5\text{H}_4$	For original doublet $^{32}\text{Si O}_2\text{H}_3-\text{C}_5\text{H}_7$										GAu **
$^{32}\text{S}-\text{H C F}$	For original doublet $^{32}\text{S O}_2\text{H}-\text{H}_2\text{C O}_2\text{F}$										GAu **
$^{32}\text{C F}_3-^{32}\text{S O}_2\text{H}$	For original doublet $^{32}\text{S O}_2\text{H}-(\text{C F}_3)_{0.942}$, $D_M=-25761.27(0.34)\mu\text{u}$										GAu **
$^{32}\text{C F}_3-^{32}\text{S O}_2$	For original doublet $^{32}\text{S O}_2-(\text{C F}_3)_{0.928}$, $D_M=-33654.93(0.59)\mu\text{u}$										GAu **
$^{31}\text{Si}(\text{n},\gamma)^{32}\text{Si}$	Original error 0.0005 increased for calibration										GAu **
$^{31}\text{P}(\text{p},\gamma)^{32}\text{S}$	$T=3289(3) Q=3185.3(3.)$ to $^{32}\text{S}^j$ at $12047.96(0.28)\text{keV}$										Nub211 **
$^{32}\text{Cl}(\text{p})^{31}\text{S}$	$E_p=3353.5(3.0) Q_p=3462.8(3.1)$ from $^{32}\text{Cl}^i 5046.3(0.3) T=2$ level										Nub211 **
$^{32}\text{Cl}(\text{p})^{31}\text{S}$	$E_p=3348.5(2.0) Q_p=3457.6(2.1)$ from $^{32}\text{Cl}^i 5046.3(0.3) T=2$ level										Nub211 **
*	corrected to $3464.4(2.1)\text{keV}$										02Py02 **
$^{32}\text{Cl}(\text{p})^{31}\text{S}$	$Q_p=3465.0(0.4)$ from $^{32}\text{Cl}^i 5046.3(0.3) T=2$ level										Nub211 **
*	this Q_p is quoted in reference as “A.Garcia et al. (in preparation)”										08Bh08 **
$^{32}\text{Cl}(\text{p})^{31}\text{S}$	$E_p=3356(2) Q_p=3465.1(2.1)$ from $^{32}\text{Cl}^i 5046.3(0.3) T=2$ level										Nub211 **
$^{32}\text{Cl}(\beta^+)^{32}\text{S}$	$E_{\beta^+}=9470(30)$ to 2^+ level at 2230.57keV										Ens119 **
$^{33}\text{Na}-\text{u}$	27386	1601	25530	480	−0.8	o			GA3	1.5	91Or01
	26370	1160			−0.5	o			GA5	1.5	00Sa21
	25142	376			0.7	o			GA7	1.5	07Ju03
	25529	322				2			GA8	1.5	12Ga45
$^{33}\text{Mg}-\text{O}_{2.062}$	15813.3	3.1	15813.9	2.9	0.2	1	85	$^{85}\text{^{33}Mg}$	TT1	1.0	13Ch49
$^{33}\text{Mg}-\text{u}$	5460	900	5327.9	2.9	−0.1	o			GA1	1.5	87Gi05
	5203	318			0.3	U			GA3	1.5	91Or01
	5710	180			−1.4	U			TO4	1.5	91Zh24
	5311	24			0.7	U			P40	1.0	06Ga04
$^{33}\text{Al}-\text{u}$	−9490	250	−9122	8	1.0	o			TO1	1.5	86Vi09
	−9250	160			0.5	o			GA1	1.5	87Gi05
	−9167	142			0.2	U			GA3	1.5	91Or01
	−9020	120			−0.6	U			TO4	1.5	91Zh24
	−9125	64			0.0	o			GT1	1.5	04Ma.A
	−8957	100			−0.7	o			GT2	2.5	08Kn.A
	−8915	128			−0.6	U			GT2	2.5	08Su19
	−9209	62			1.4	U			LZ1	1.0	15Xu14

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{33}\text{Si O}_2 - ^{13}\text{C C}_4 \text{ H}_4$	-66848.76	0.75				2			MS1	1.0	09Kw02 *
$^{33}\text{Cl}-\text{u}$	-22536.9	7.5	-22548.0	0.4	-1.5	U			LZ1	1.0	11Tu09
$^{33}\text{Mg}-^{39}\text{K}_{.846}$	36035.7	7.4	36032.2	2.9	-0.5	1	15	15 ^{33}Mg	MA8	1.0	19As04
$^{33}\text{Al}-^{39}\text{K}_{.846}$	21582.0	7.5				2			TT1	1.0	16Kw.A
$^{33}\text{Ar}-^{39}\text{K}_{.846}$	20629.86	0.43				2			MA8	1.0	03Bl17
$^{33}\text{Ar}-^{36}\text{Ar}_{.917}$	19689.2	4.5	19686.7	0.4	-0.6	U			MA6	1.0	01He29
$^{33}\text{S}-^{32}\text{S H}$	-8437.29682	0.00030	-8437.2968	0.0003	0.0	1	100	100 ^{33}S	MI3	1.0	05Ra34
$^{30}\text{Si}(\alpha, \text{p})^{33}\text{P}$	-2965	10	-2959.7	1.1	0.5	U			ANL		68Be13
$^{33}\text{S}(\text{n}, \alpha)^{30}\text{Si}$	3497.6	5.	3493.506	0.022	-0.8	U			ILL		81Wa31
	3496.9	5.0			-0.7	U					01Wa50
$^{31}\text{P}(\text{He}, \text{p})^{33}\text{S}$	9787	15	9787.5617	0.0015	0.0	U					71Gr04
$^{32}\text{S}(\text{n}, \gamma)^{33}\text{S}$	8641.5	0.3	8641.6392	0.0005	0.5	o			MMn		80Is02 Z
	8641.82	0.10			-1.8	U			ORn		83Ra04 Z
	8641.60	0.03			1.3	U			MMn		85Ke08 Z
	8641.81	0.17			-1.0	U			Bdn		06Fi.A
	8641.6398	0.0033			-0.2	U			NBS		06De21
$^{32}\text{S}(\text{d}, \text{p})^{33}\text{S}$	6420	6	6417.07299	0.00028	-0.5	U			MIT		64Sp12
$^{32}\text{S}(\text{p}, \gamma)^{33}\text{Cl}$	2276.4	0.9	2276.8	0.4	0.4	-					59Ku79
	2276.8	0.5			-0.1	-					76Al01
$^{32}\text{S}(\text{d}, \text{n})^{33}\text{Cl}$	62	9	52.2	0.4	-1.1	U					72El03
$^{32}\text{S}(\text{He}, \text{d})^{33}\text{Cl}$	-3218	15	-3216.7	0.4	0.1	U					66Gr26
	-3217	5			0.1	U			CIT		70Mo08
$^{32}\text{S}(\text{p}, \gamma)^{33}\text{Cl}$	ave. 2276.7	0.4	2276.8	0.4	0.2	1	80	80 ^{33}Cl			average
$^{32}\text{S}(\text{p}, \gamma)^{33}\text{Cl}^i$	-3267.0	1.0	-3271.7	0.5	-4.7	B					70Ab15
	-3271.4	2.0			-0.1	U					82Wi.A
	-3271.5	0.8			-0.3	1	37	37 $^{33}\text{Cl}^i$			02Py01
$^{33}\text{Cl}^i(\text{p})^{32}\text{S}$	3266.8	1.0	3271.7	0.5	4.9	B					87Bo21
$^{33}\text{Si}(\beta^-)^{33}\text{P}$	5768	50	5823.0	1.3	1.1	U					73Go33
$^{33}\text{P}(\beta^-)^{33}\text{S}$	249	2	248.5	1.1	-0.2	2					54Ni06
	248.3	1.3			0.2	2					84Po09
$^{33}\text{Cl}(\beta^+)^{33}\text{S}$	5532	50	5582.5	0.4	1.0	U					60Wa04
$^{33}\text{Cl}^i(\text{IT})^{33}\text{Cl}$	5548.5	0.4	5548.4	0.4	-0.1	1	83	63 $^{33}\text{Cl}^i$			06Tr10
* $^{33}\text{Si O}_2 - ^{13}\text{C C}_4 \text{ H}_4$	For original doublet $^{33}\text{Si O}_2 \text{ H}_3 - ^{13}\text{C C}_4 \text{ H}_7$										GAu **
$^{34}\text{Na}-\text{u}$	34010	429				2			GA8	1.5	12Ga45
$^{34}\text{Mg}-\text{O}_{2.126}$	19747	31	19747	7	-0.0	U			TT1	1.0	13Ch49
$^{34}\text{Mg}-\text{u}$	8855	476	8935	7	0.1	o			GA3	1.5	91Or01
	9190	350			-0.5	U			TO4	1.5	91Zh24
	9900	350			-1.8	U			GA5	1.5	00Sa21
	9190	97			-1.7	U			GA7	1.5	07Ju03
$^{34}\text{Al}-\text{u}$	-3760	430	-3218.1	2.3	0.8	o			TO1	1.5	86Vi09 *
	-3400	250			0.5	o			GA1	1.5	87Gi05 *
	-3262	218			0.1	o			GA3	1.5	91Or01 *
	-2940	120			-1.5	U			TO4	1.5	91Zh24 *
	-3199	97			-0.1	U			GT1	1.5	04Ma.A *
	-3328	86			0.9	U			GA7	1.5	07Ju03 *
$^{34}\text{Mg}-^{39}\text{K}_{.872}$	40583.4	7.4				2			MA8	1.0	19As04
$^{34}\text{Al}-^{39}\text{K}_{.872}$	28438.0	7.8	28429.9	2.3	-1.0	o			TT1	1.0	16Kw.A
	28427.0	3.3			0.9	2			TT1	1.0	17Ga20
	28432.4	3.1			-0.8	2			MA8	1.0	19As04 *
$^{34}\text{Si}-^{39}\text{K}_{.872}$	10185.99	0.86				2			MA8	1.0	19As04
$^{34}\text{Ar}-^{39}\text{K}_{.872}$	11919.02	0.36	11918.04	0.08	-2.7	U			MA8	1.0	02He23
$^{34}\text{Ar}-^{36}\text{Ar}_{.944}$	10907.4	3.8	10907.51	0.09	0.0	U			MA6	1.0	01He29
$^{34}\text{Cl}-^{34}\text{S}$	5895.548	0.058	5895.48	0.04	-1.2	1	49	31 ^{34}Cl	JY1	1.0	09Er07
$^{34}\text{Cl}^m-^{34}\text{S}$	6052.575	0.068	6052.60	0.04	0.4	1	41	31 $^{34}\text{Cl}^m$	JY1	1.0	09Er07

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{34}\text{S}-^{34}\text{Ar}$	-12403.19	0.20	-12403.08	0.08	0.5	1	14	13 ^{34}Ar	JY1	1.0	11Er02
$^{34}\text{Cl}^m-^{34}\text{Cl}$	157.05	0.11	157.124	0.029	0.7	U			JY1	1.0	09Er07
	157.30	0.27			-0.7	U			JY1	1.0	11Er02
$^{34}\text{Ar}-^{34}\text{Cl}$	6507.627	0.092	6507.60	0.07	-0.3	1	54	52 ^{34}Ar	JY1	1.0	11Er02
$^{34}\text{Cl}^m-^{34}\text{Ar}$	-6350.41	0.11	-6350.48	0.07	-0.6	1	39	35 ^{34}Ar	JY1	1.0	11Er02
$\text{C}_4\text{H}_3-^{34}\text{P O}$	54914.59	0.87				2			MS1	1.0	09Kw02 *
$^{30}\text{Si}(^7\text{Li},^3\text{He})^{34}\text{P}$	100	40	91.6	0.8	-0.2	U					77Pe17
$^{31}\text{P}(\alpha, p)^{34}\text{S}$	629.9	2.9	627.09	0.04	-1.0	U			Har		73Ry01
$^{31}\text{P}(\alpha, n)^{34}\text{Cl}$	-5632	10	-5646.86	0.05	-1.5	U			Tal		70Um01
	-5641.5	3.7			-1.4	U			Har		73Ry01
$^{34}\text{S}(\text{d}, \alpha)^{32}\text{P}$	5096	10	5083.99	0.06	-1.2	U					78Ba30
$^{32}\text{S}(^3\text{He}, n)^{34}\text{Ar}$	-759	15	-777.34	0.08	-1.2	U			CIT		67Mi02
$^{34}\text{S}(^{13}\text{C}, ^{14}\text{O})^{33}\text{Si}$	-14243	75	-14300.1	0.7	-0.8	U			Can		86Fi06
$^{33}\text{S}(n, \gamma)^{34}\text{S}$	11417.12	0.10	11417.15	0.04	0.3	-			ORn		83Ra04 Z
	11417.22	0.23			-0.3	-			Bdn		06Fi.A
$^{33}\text{S}(\text{d}, p)^{34}\text{S}$	9202	10	9192.58	0.04	-0.9	U			MIT		64Sp12
	9195	6			-0.4	U			Utr		71Va21
$^{33}\text{S}(n, \gamma)^{34}\text{S}$	ave. 11417.14	0.09	11417.15	0.04	0.1	1	24	24 ^{34}S			average
$^{33}\text{S}(p, \gamma)^{34}\text{Cl}$	5142.42	0.20	5143.20	0.05	3.9	B			Oak		83Ra04 *
	5142.4	0.3			2.7	U			Utr		83Wa27 Z
	5143.29	0.07			-1.3	1	48	48 ^{34}Cl	Auc		94Li20
$^{34}\text{Si}(\beta^-)^{34}\text{P}$	4700	300	4557.0	1.1	-0.5	U					77Na05
$^{34}\text{P}(\beta^-)^{34}\text{S}$	5383	45	5383.0	0.8	-0.0	U			ANB		73Go33
$^{34}\text{S}(\text{t}, ^3\text{He})^{34}\text{P}$	-5368	20	-5364.4	0.8	0.2	U			LAL		77Aj01
$^{34}\text{S}(^7\text{Li}, ^7\text{Be})^{34}\text{P}$	-6224	40	-6244.9	0.8	-0.5	U			Can		85Dr06
$^{34}\text{Cl}(\beta^+)^{34}\text{S}$	5522	30	5491.60	0.04	-1.0	U					56Gr07
$^{34}\text{S}(p, n)^{34}\text{Cl}$	-6252	10	-6273.95	0.04	-2.2	U			Tal		70Um01
	-6271.9	1.9			-1.1	U			Har		75Fr.A
	-6274.27	0.56			0.6	U			Auc		77Ba16
	-6273.11	0.25			-3.4	F			Auc		92Ba.A *
$^{34}\text{S}(^3\text{He}, \text{t})^{34}\text{Cl}$	-5510.8	0.4	-5510.20	0.04	1.5	F			Mun		77Vo02 *
$^{34}\text{S}(^3\text{He}, \text{t})^{34}\text{Cl}-^{27}\text{Al}(^27\text{Si})$	-678.7	2.3	-679.25	0.10	-0.2	U			ChR		74Ha35
$^{34}\text{Cl}^m(\text{IT})^{34}\text{Cl}$	146.36	0.03	146.360	0.027	-0.0	1	84	65 $^{34}\text{Cl}^m$			Ens126
* $^{34}\text{Al}-\text{u}$	Possible isomeric mixture 26(1) ms, E=46.5 keV										12Ro25 **
* $^{34}\text{Al}-^{39}\text{K}_{872}$	Authors stated only ground state										19As04 **
* $\text{C}_4\text{H}_3-^{34}\text{P O}$	For original doublet $^{34}\text{P H}_2\text{O}-\text{C}_4\text{H}_5$										GAu **
* $^{33}\text{S}(p, \gamma)^{34}\text{Cl}$	$E_p=974.76(0.15, Z)$ to $6088.20(0.10, Z)$ level										83Ra04 **
* $^{34}\text{S}(p, n)^{34}\text{Cl}$	F : disturbed by resonance; at least 0.5 keV uncertain										94Li20 **
* $^{34}\text{S}(^3\text{He}, \text{t})^{34}\text{Cl}$	F : rejected in reference of same group										09Fa15 **
$^{35}\text{Mg}-\text{u}$	18669	1721	16790	290	-0.7	o			GA3	1.5	91Or01
	18830	1070			-1.3	o			GA5	1.5	00Sa21
	16790	193				2			GA7	1.5	07Ju03
$^{35}\text{Al}-\text{u}$	-340	460	-240	8	0.1	o			GA1	1.5	87Gi05
	-296	298			0.1	o			GA3	1.5	91Or01
	80	190			-1.1	U			TO4	1.5	91Zh24
	-236	75			-0.0	U			GA7	1.5	07Ju03
$\text{C}_3-^{35}\text{Cl H}$	23320.8	0.3	23322.27	0.04	2.0	U			M17	2.5	66Be10
	23322.239	0.034			0.7	1	56	56 ^{35}Cl	B07	1.5	71Sm01
	23321.83	0.63			0.3	U			J5	2.5	72Ka57
	23322.328	0.325			-0.1	U			J5	2.5	72Ka57
$\text{C}_5\text{H}_{10}-^{35}\text{Cl}_2$	140549.37	2.98	140544.93	0.08	-0.6	U			C2	2.5	65De09
	140545.01	0.13			-0.4	1	15	15 ^{35}Cl	B07	1.5	71Sm01
$\text{C}_4\text{H}_6\text{O}-^{35}\text{Cl}_2$	104153.75	3.45	104159.42	0.08	0.7	U			C2	2.5	65De09
$\text{C}_2\text{D}_6-^{35}\text{Cl H}$	107934.90	0.54	107932.94	0.04	-1.5	U			J5	2.5	72Ka57
	107933.422	0.538			-0.4	U			J5	2.5	72Ka57

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_3 H-D^{35}Cl$	24871.92	0.75	24870.56	0.04	-0.7	U			C2	2.5	65De09
$C_8 H_9-^{35}Cl_3$	163867.25	0.90	163867.20	0.11	-0.0	U			A2	2.5	70St25
$^{35}Ar-u$	-24747.3	4.3	-24742.3	0.7	1.2	U			LZ1	1.0	11Tu09
$^{35}Al-^{39}K_{.897}$	32315.1	7.9				2			TT1	1.0	17Ga20
$^{35}K-^{39}K_{.897}$	20560.69	0.55				2			MA8	1.0	07Ya08
$^{35}Cl(p,\alpha)^{32}S$	1862	5	1866.06	0.04	0.8	U			Bar		57Va03 Y
	1865	8			0.1	U			MIT		64Sp12
$^{32}S(\alpha,p)^{35}Cl$	-1862	17	-1866.06	0.04	-0.2	U			MIT		64Sp12
$^{32}S(\alpha,n)^{35}Ar$	-8751	18	-8614.7	0.7	7.6	B			Tal		63Ne05
$^{35}Cl(d,\alpha)^{33}S$	8285	10	8283.13	0.04	-0.2	U			MIT		64Sp12
$^{33}S(^3He,n)^{35}Ar$	3335	16	3321.3	0.7	-0.9	U					75Da14
$^{35}K^i(2p)^{33}Cl$	4311	40				2					85Ay01
$^{34}S(^{18}O,^{17}F)^{35}P$	-7796	40	-7808.4	1.9	-0.3	U			Can		88Or01
$^{34}S(n,\gamma)^{35}S$	6986.00	0.10	6985.84	0.04	-1.6	-			ORn		83Ra04 Z
	6985.84	0.05			-0.0	-			MMn		85Ke08 Z
	6986.09	0.14			-1.8	U			Bdn		06Fi.A
$^{34}S(d,p)^{35}S$	4762	10	4761.27	0.04	-0.1	U			MIT		64Sp12
	4757	5			0.9	U			Utr		71Va18
$^{34}S(n,\gamma)^{35}S$	ave. 6985.87	0.04	6985.84	0.04	-0.8	1	75	46 ^{34}S			average
$^{34}S(p,\gamma)^{35}Cl$	6367.4	1.6	6370.81	0.04	2.1	U					72Hu10
	6370.7	0.4			0.3	U					76Sp08 Z
	6370.70	0.20			0.6	U			Oak		83Ra04 *
$^{35}Cl(\gamma,n)^{34}Cl$	-12660	40	-12644.76	0.05	0.4	U					61Sa11
$^{35}P(\beta^-)^{35}S$	3909	75	3988.4	1.9	1.1	U					72Go31
$^{35}S(\beta^-)^{35}Cl$	167.4	0.2	167.322	0.026	-0.4	U					57Co62
	166.80	0.15			3.5	B					85Al11
	167.288	0.100			0.3	U					85Ap01 *
	166.93	0.2			2.0	o					85Ma59
	167.4	0.1			-0.8	U					85Oh06 *
	166.7	0.2			3.1	B					89Si04 *
	167.56	0.03			-7.9	B					92Ch27 *
	167.35	0.10			-0.3	U					93Ab11 *
	167.23	0.10			0.9	U					93Be21 *
	167.27	0.10			0.5	U					93Mo01 *
	167.334	0.027			-0.5	1	91	71 ^{35}S			00Ho13
$^{35}Cl(n,p)^{35}S$	612	4	615.025	0.026	0.8	U			BNL		68Sc01
$^{35}Ar(\beta^+)^{35}Cl$	5980	40	5966.2	0.7	-0.3	U					56Ki29
	5950	50			0.3	U					60Wa04
$^{35}Cl(p,n)^{35}Ar$	-6747.2	1.6	-6748.6	0.7	-0.9	2			Har		75Fr.A Z
	-6747.9	1.0			-0.7	2			Auc		77Wh03 Z
	-6750.4	1.2			1.5	2			Mtr		78Az01 *
$^{34}S(p,\gamma)^{35}Cl$	$E_p=1264.97(0.13,Z)$ to $7598.91(0.15,Z)$ level										83Ra04 **
$^{35}S(\beta^-)^{35}Cl$	Original error (0.030) increased to 0.100										AHW **
$^{35}Cl(p,n)^{35}Ar$	Original T=6942.2(2.2) recalibrated 6945.5(1.2)										GAu **
$^{36}Mg-u$	24930	1610	21880	740	-1.3	o			GA5	1.5	00Sa21
	21879	494				2			GA7	1.5	07Ju03
$^{36}Al-u$	6187	421	6390	160	0.3	o			GA3	1.5	91Or01
	6500	400			-0.2	U			TO4	1.5	91Zh24
	6140	310			0.5	o			GA5	1.5	00Sa21
	6388	107				2			GA7	1.5	07Ju03
$^{36}Si-u$	-13850	640	-13350	80	0.5	U			TO1	1.5	86Vi09
	-13490	320			0.3	o			GA1	1.5	87Gi05
	-13578	191			0.8	o			GA3	1.5	91Or01
	-13110	150			-1.1	2			TO4	1.5	91Zh24
	-13376	75			0.2	2			GT1	1.5	04Ma.A
	-13280	118			-0.4	2			GA7	1.5	07Ju03
	-13484	163			0.8	2			LZ1	1.0	15Xu14

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{36}\text{Ar}-u$	-32454.895	0.015	-32454.894	0.029	0.1	o			ST2	1.0	02Bf02
	-32454.895	0.029			0.0	1	100	100 ^{36}Ar	ST2	1.0	03Fr08
$^{36}\text{K}-^{39}\text{K}_{923}$	14800.99	0.38	14800.8	0.3	-0.5	1	84	84 ^{36}K	MA8	1.0	07Ya08
$^{36}\text{Ar}(^3\text{He}, ^8\text{Li})^{31}\text{Cl}$	-29180	50	-29212	3	-0.6	U			MSU		77Be13
$^{36}\text{S}(^{48}\text{Ca}, ^{52}\text{V})^{32}\text{Al}$	-12651	370	-12347	7	0.8	o			Dar		87Ch.A
$^{36}\text{S}(^{48}\text{Ca}, ^{51}\text{V})^{33}\text{Al}$	-14150	140	-14189	7	-0.3	U			Dar		86Wo07
$^{36}\text{S}(^{14}\text{C}, ^{17}\text{O})^{33}\text{Si}$	-6380	20	-6321.2	0.7	2.9	U			Mun		84Ma49
$^{36}\text{S}(^{11}\text{B}, ^{14}\text{N})^{33}\text{Si}$	-4311	30	-4345.5	0.7	-1.2	U			Can		85Fi03
$^{36}\text{Ar}(^3\text{He}, ^6\text{He})^{33}\text{Ar}$	-23512	30	-23508.1	0.4	0.1	U			MSU		74Na07
$^{36}\text{S}(^{11}\text{B}, ^{13}\text{N})^{34}\text{Si}$	-7327	25	-7350.2	0.9	-0.9	U			Can		85Fi03
$^{36}\text{S}(^{14}\text{C}, ^{16}\text{O})^{34}\text{Si}$	-2989	20	-2915.6	0.8	3.7	B			Mun		84Ma49
$^{36}\text{S}(^{64}\text{Ni}, ^{66}\text{Zn})^{34}\text{Si}$	-8890	41	-8872.3	1.1	0.4	o			Dar		85Wo07 *
	-8903	33			0.9	U			Dar		86Sm05 *
$^{36}\text{S}(d, \alpha)^{34}\text{P}$	4604.4	5.	4595.4	0.8	-1.8	U					82So.A *
$^{36}\text{Ar}(p, t)^{34}\text{Ar}$	-19513	3	-19514.09	0.08	-0.4	U			MSU		74Ha02
$^{36}\text{Ar}(p, t)^{34}\text{Ar}^i$	-27473	50	-27448	5	0.5	U					69Br21 *
	-27448	5				2					72Pa02 *
$^{36}\text{S}(^{14}\text{C}, ^{15}\text{O})^{35}\text{Si}$	-16184	50	-16110	40	1.5	2			Mun		84Ma49
$^{36}\text{S}(^{13}\text{C}, ^{14}\text{O})^{35}\text{Si}$	-21122	60	-21160	40	-0.6	2			Can		86Fi06
$^{36}\text{S}(^{64}\text{Ni}, ^{65}\text{Zn})^{35}\text{Si}$	-17250	100	-17460	40	-2.1	2			Dar		86Sm05 *
$^{36}\text{S}(d, ^3\text{He})^{35}\text{P}$	-7607	5	-7601.8	1.9	1.0	2			BNL		84Th08
	-7601	2			-0.4	2			Hei		85Kh04
$^{36}\text{S}(^{14}\text{C}, ^{15}\text{N})^{35}\text{P}$	-2927	10	-2887.9	1.9	3.9	B			Mun		84Ma49 *
$^{36}\text{S}(^6\text{Li}, ^7\text{Be})^{35}\text{P}$	-7521	17	-7488.4	1.9	1.9	U			Can		85Dr06
$^{36}\text{S}(^{64}\text{Ni}, ^{65}\text{Cu})^{35}\text{P}$	-5659	34	-5641.7	2.0	0.5	U			Dar		85Wo.A
$^{35}\text{Cl}(n, \gamma)^{36}\text{Cl}$	8579.73	0.20	8579.794	0.005	0.3	U			BNn		78St25 Z
	8579.7	0.3			0.3	o			MMn		80Is02 Z
	8579.81	0.20			-0.1	U			MMn		81Ke02 Z
	8579.66	0.10			1.3	U					81Su.A Z
	8579.61	0.09			2.0	U			ILn		82Kr12 Z
	8579.67	0.17			0.7	U			Bdn		06Fi.A
	8579.7945	0.0048			-0.0	1	100	99 ^{36}Cl	NBS		06De21
$^{35}\text{Cl}(d, p)^{36}\text{Cl}$	6360	8	6355.228	0.005	-0.6	U			MIT		64Sp12
$^{35}\text{Cl}(p, \gamma)^{36}\text{Ar}$	8506.1	0.5	8506.98	0.04	1.8	U					72Ho40 Z
$^{35}\text{Cl}(p, \gamma)^{36}\text{Ar}^i$	-2346.8	1.5	-2345.2	1.2	1.1	2					76Hu01
	-2342.5	1.9			-1.4	2					76Ma40
$^{36}\text{Ar}(d, t)^{35}\text{Ar}$	-9007	10	-8998.3	0.7	0.9	U			Yal		70Wh04
$^{36}\text{K}^i(p)^{35}\text{Ar}$	2592	21	2623.8	2.3	1.5	U			Brk		81Ay01
	2623.8	2.3				3					95Ga16
$^{36}\text{S}(^7\text{Li}, ^7\text{Be})^{36}\text{P}$	-11277	27	-11275	13	0.1	2			Can		85Dr06
$^{36}\text{S}(^{14}\text{C}, ^{14}\text{N})^{36}\text{P}$	-10256	15	-10257	13	-0.0	2			Mun		84Ma49
$^{36}\text{Cl}(\beta^+)^{36}\text{S}$	1137	18	1142.13	0.19	0.3	U					68Pi03
$^{36}\text{Cl}(\epsilon)^{36}\text{S}$	1180	15			-2.5	U					64Li10
	1160	18			-1.0	U					65Be19
$^{36}\text{S}(p, n)^{36}\text{Cl}$	-1924.64	0.31	-1924.48	0.19	0.5	1	37	36 ^{36}S			01Wa50
$^{36}\text{Cl}(\beta^-)^{36}\text{Ar}$	708.7	0.6	709.53	0.04	1.4	U					67Sp06
$^{36}\text{Ar}(p, n)^{36}\text{K}$	-13588.3	8.	-13596.7	0.3	-1.1	U			BNL		71Go18 Z
	-13618	23			0.9	U					71Ja09
$^{36}\text{Ar}(^3\text{He}, t)^{36}\text{K}$	-12930	40	-12833.0	0.3	2.4	U			Duk		70Dz04
$^{36}\text{S}(^{64}\text{Ni}, ^{66}\text{Zn})^{34}\text{Si}$	Calibrated with $^{36}\text{S}(^{64}\text{Ni}, ^{62}\text{Ni}) M - A = -26862(12)$ now $-26861(7)$										AHW **
$^{36}\text{S}(d, \alpha)^{34}\text{P}$	Original error 1.2 judged too small										GAU **
$^{36}\text{Ar}(p, t)^{34}\text{Ar}^i$	IT=7950(50); Q rebuilt, estimated with 72Pa02 $Q = -19523$ for ground state										MMC12a**
$^{36}\text{Ar}(p, t)^{34}\text{Ar}^i$	IT=7925(5); Q rebuilt with author's $Q = -19523$ for ground state										MMC128**
$^{36}\text{S}(^{64}\text{Ni}, ^{65}\text{Zn})^{35}\text{Si}$	$M - A = -14482(59)$ for average of ground state and 54, 114, 207 levels										86Sm05 **
$^{36}\text{S}(^{14}\text{C}, ^{15}\text{N})^{35}\text{P}$	Original report -2693 is a typo										GAU **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{37}\text{Al}-\text{u}$	10310	579	10530	190	0.3	o			GA3	1.5	91Or01
	10900	450			-0.5	o			GA5	1.5	00Sa21
	10531	129				2			GA7	1.5	07Ju03
$^{37}\text{Si}-\text{u}$	-7550	1410	-7050	120	0.2	o			GA1	1.5	87Gi05
	-7310	305			0.6	o			GA3	1.5	91Or01
	-6930	150			-0.6	2			TO4	1.5	91Zh24
	-7107	97			0.4	2			GA7	1.5	07Ju03
	-20740	430	-20390	40	0.5	U			TO1	1.5	86Vi09
$^{37}\text{P}-\text{u}$	-19910	190			-1.7	o			GA1	1.5	87Gi05
	-20442	200			0.2	U			GA3	1.5	91Or01
$\text{C}_3\text{H}-^{37}\text{Cl}$	41924.73	1.09	41922.46	0.06	-0.8	U			C2	2.5	65De09
	41922.2	0.2			0.5	U			M17	2.5	66Be10
	41922.176	0.305			0.4	U			J5	2.5	72Ka57
$\text{C}_2\text{D}_8-^{37}\text{ClH}_3$	123436.51	0.12	123436.55	0.06	0.2	U			B07	1.5	71Sm01
$\text{C}_3\text{H}_6\text{O}_2-^{37}\text{Cl}_2$	104974.24	0.08	104974.28	0.11	0.4	1	86	$^{86}\text{ }^{37}\text{Cl}$	B07	1.5	71Sm01
$\text{C}_3\text{H}_5\text{D}_2-^{37}\text{Cl}$	45020.96	1.14	45019.03	0.06	-0.7	U			C2	2.5	65De09
$\text{C}_8\text{H}_{15}-^{37}\text{Cl}_3$	219665.80	0.90	219667.76	0.17	0.9	U			A2	2.5	70St25
$\text{C}_3\text{H}_3\text{D}-^{37}\text{Cl}$	43473.27	1.33	43470.74	0.06	-0.8	U			C2	2.5	65De09
$^{37}\text{K}-\text{u}$	-26632.5	6.4	-26624.11	0.10	1.3	U			LZ1	1.0	11Tu09
$\text{H}_3\text{O}-^{37}\text{Ca}_{514}$	25638.22	0.35				2			MS1	1.0	07Ri08 *
$\text{D}_2\text{ }^{35}\text{Cl}-\text{H}_2\text{ }^{37}\text{Cl}$	15505.41	0.71	15503.61	0.07	-1.0	U			C2	2.5	65De09
	15503.80	0.09			-0.8	U			H31	2.5	77So02
$\text{C}_5\text{H}_{12}-^{35}\text{Cl }^{37}\text{Cl}$	159145.17	0.12	159145.12	0.07	-0.3	1	14	$^{9 }^{37}\text{Cl}$	B07	1.5	71Sm01
$\text{H}_2\text{ }^{35}\text{Cl}-^{37}\text{Cl}$	18600.0	0.4	18600.18	0.07	0.2	U			M17	2.5	66Be10
$\text{D }^{35}\text{Cl}-^{37}\text{Cl}$	17052.95	1.02	17051.90	0.07	-0.4	U			C2	2.5	65De09
	17051.816	0.185			0.2	U			J5	2.5	72Ka57
$^{37}\text{K}-^{39}\text{K}_{949}$	7817.98	0.33	7818.44	0.10	1.4	U			MA8	1.0	07Ya08
$^{37}\text{Cl}(\text{p},\alpha)^{34}\text{S}$	3030	6	3034.19	0.07	0.7	U			Bar		57Va03 Y
	3029	8			0.6	U			MIT		64Sp12
$^{37}\text{Ar}(\text{n},\alpha)^{34}\text{S}$	4630	7	4630.41	0.21	0.1	U			ILL		78As06
$^{34}\text{S}(\alpha,\text{n})^{37}\text{Ar}$	-4625	90	-4630.41	0.21	-0.1	U			Tal		63Ne05
$^{37}\text{Cl}(\text{d},\alpha)^{35}\text{S}$	7791	12	7795.47	0.07	0.4	U			MIT		64Sp12
$^{37}\text{Cl}(\text{p},^3\text{He})^{35}\text{S}^i$	-19713	10				2					75Gu15
$^{35}\text{Cl}(^3\text{He},\text{p})^{37}\text{Ar}$	9582	15	9576.39	0.21	-0.4	U			MIT		67Sp09
$^{36}\text{Mg}(\text{n},\gamma)^{37}\text{Mg}$	240	110				3					14Ko14 *
$^{36}\text{S}(^{18}\text{O},^{17}\text{F})^{37}\text{P}$	-14410	40	-14400	40	0.2	2			Can		88Or.A *
$^{36}\text{S}(^{48}\text{Ca},^{47}\text{Sc})^{37}\text{P}$	-11490	120	-11560	40	-0.6	2			Dar		88Fi04 *
$^{36}\text{S}(\text{n},\gamma)^{37}\text{S}$	4303.52	0.12	4303.60	0.06	0.7	2			ORn		84Ra09 Z
	4303.61	0.09			-0.1	2			Bdn		06Fi.A
$^{36}\text{S}(\text{d},\text{p})^{37}\text{S}$	2079.12	0.13	2079.04	0.06	-0.6	2					84Pi03
$^{36}\text{S}(^{14}\text{C},^{13}\text{C})^{37}\text{S}$	-3874	7	-3872.83	0.06	0.2	U			Mun		84Ma49
$^{37}\text{Cl}(^{13}\text{C},^{14}\text{O})^{36}\text{P}$	-16433	50	-16393	13	0.8	U			Can		88Or01
$^{36}\text{S}(\text{p},\gamma)^{37}\text{Cl}$	8386.47	0.23	8386.38	0.19	-0.4	1	65	$^{64 }^{36}\text{S}$	Utr		84No05 Z
	-1835.5	0.3				2					84No05
$^{36}\text{Ar}(\text{n},\gamma)^{37}\text{Ar}$	8791.1	1.0	8787.45	0.21	-3.6	B					68Wi25 Z
	8788.8	1.2			-1.1	U					70Ha56 Z
	8789.9	0.9			-2.7	U			Bdn		06Fi.A
$^{36}\text{Ar}(\text{p},\gamma)^{37}\text{K}$	1857.3	1.0	1857.63	0.09	0.3	U					64Ar17
	1857.63	0.09				2			Utr		88De03 Z
$^{36}\text{Ar}(\text{d},\text{n})^{37}\text{K}$	-320	100	-366.94	0.09	-0.5	U			Yal		61Ya01
$^{36}\text{Ar}(\text{p},\gamma)^{37}\text{K}^i$	-3192.6	0.8				2					88De03
$^{37}\text{S}(\beta^-)^{37}\text{Cl}$	4750	40	4865.13	0.20	2.9	U					67Wi14
$^{37}\text{Cl}(\text{t},^3\text{He})^{37}\text{S}$	-4854	30	-4846.53	0.20	0.2	U			LAI		70Aj01
$^{37}\text{Ar}(\epsilon)^{37}\text{Cl}$	818	15	813.87	0.20	-0.3	U					53An01
$^{37}\text{Cl}(\text{p},\text{n})^{37}\text{Ar}$	-1595.5	4.0	-1596.22	0.20	-0.2	U			Wis		50Ri59 Y
	-1595.4	1.0			-0.8	U			MIT		52Sc09 Z
	-1596.9	2.4			0.3	U			Oak		64Jo11
	-1596.8	1.0			0.6	U			Duk		66Pa18 Z
	-1596.22	0.20				2			PTB		98Bo30
	-1596.3	1.0			0.1	U					01Wa50

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{37}\text{K}(\beta^+)^{37}\text{Ar}$	6120	70	6147.48	0.23	0.4	U					58Su60
	6170	70			-0.3	U					60Wa04
$^*\text{H}_3\text{O}-^{37}\text{Ca}_{514}$	Error in Table II : $M-A=13135.7(1.4)$ corrected to $-13136.06(0.64)$ keV										07Ri08 **
$^{36}\text{Mg}(\text{n},\gamma)^{37}\text{Mg}$	Symmetrized from 220(+120-90) keV										14Ko14 **
$^{36}\text{S}^{18}\text{O},^{17}\text{F}^{37}\text{P}$	And $Q=-13650(40)$, $M-A=-19750(40)$ if other peak is ground state one										88Or.A **
$^{36}\text{S}^{48}\text{Ca},^{47}\text{Sc}^{37}\text{P}$	And $Q=-11569(80)$, $M-A=-18980(80)$ if other peak due to ^{47}Sc 807.89 level										88Fi04 **
$^{38}\text{Al}-\text{u}$	15240	1500	17680#	160#	1.1	o			GA4	1.5	00Sa21
	17980	920			-0.2	o			GA5	1.5	00Sa21
	17402	268			0.7	D			GA7	1.5	07Ju03 *
$^{38}\text{Si}-\text{u}$	-4510	180	-4480	110	0.1	o			GA4	1.5	00Sa21
	-4020	290			-1.1	U			TO4	1.5	91Zh24
	-4100	320			-0.8	o			GA5	1.5	00Sa21
	-4477	75				2			GA7	1.5	07Ju03
$^{38}\text{P}-\text{u}$	-14420	620	-15700	80	-1.4	U			GA1	1.5	87Gi05
	-15910	140			1.0	2			GA4	1.5	00Sa21
	-15530	150			-0.7	2			TO4	1.5	91Zh24
	-16110	310			0.9	U			GA5	1.5	00Sa21
	-15717	75			0.2	o			GT1	1.5	04Ma.A
	-15660	100			-0.1	2			GT2	2.5	08Kn.A
	-15688	97			-0.1	2			LZ1	1.0	15Xu14
$^{38}\text{Ca}-\text{H}_6\text{O}_2$	-60460.24	0.30	-60460.21	0.21	0.1	o			MS1	1.0	06Bo11
	-60460.24	0.30			0.1	1	48	48 ^{38}Ca	MS1	1.0	07Ri08
$^{38}\text{Ar}-^{39}\text{K}_{974}$	-1917.88	0.37	-1918.01	0.21	-0.4	1	32	32 ^{38}Ar	MA8	1.0	02He23
$^{38}\text{K}-^{39}\text{K}_{974}$	4430.88	0.44	4431.00	0.21	0.3	1	23	23 ^{38}K	MA8	1.0	07Ya08
$^{38}\text{Ca}^{19}\text{F}-^{39}\text{K}_{1.462}$	27783.80	0.63	27783.51	0.21	-0.5	U			MA8	1.0	07Ge07
$^{38}\text{K}-^{38}\text{Ar}$	6348.974	0.068	6349.01	0.05	0.6	1	50	27 ^{38}K	JY1	1.0	09Er07
$^{38}\text{K}^m-^{38}\text{Ar}$	6488.743	0.049	6488.73	0.04	-0.2	1	72	45 $^{38}\text{K}^m$	JY1	1.0	09Er07
$^{38}\text{Ar}-^{38}\text{Ca}$	-13587.18	0.12	-13587.12	0.07	0.5	1	32	17 ^{38}Ar	JY1	1.0	11Er02
$^{38}\text{K}^m-^{38}\text{K}$	139.698	0.065	139.72	0.05	0.3	-			JY1	1.0	09Er07
	139.78	0.14			-0.4	-			JY1	1.0	11Er02
ave.	139.71	0.06			0.1	1	60	34 $^{38}\text{K}^m$			average
$^{38}\text{Ca}-^{38}\text{K}$	7238.04	0.10	7238.11	0.07	0.7	1	45	25 ^{38}K	JY1	1.0	11Er02
$^{38}\text{K}^m-^{38}\text{Ca}$	-7098.43	0.11	-7098.39	0.07	0.4	1	37	21 $^{38}\text{K}^m$	JY1	1.0	11Er02
$^{24}\text{Mg}(^{16}\text{O},2\text{n})^{38}\text{Ca}$	-12727	30	-12754.71	0.19	-0.9	U					72Zi02 *
$^{35}\text{Cl}(\alpha,\text{p})^{38}\text{Ar}$	837.2	2.4	837.24	0.20	0.0	U			Har		75Sq01
$^{35}\text{Cl}(\alpha,\text{n})^{38}\text{K}$	-5862.1	1.5	-5859.17	0.20	2.0	U			Mun		76Sh24 Z
	-5858.7	2.9			-0.2	U			Har		75Sq01 *
$^{36}\text{S}(\text{t,p})^{38}\text{S}$	3838	30	3858	7	0.7	U					85Da15
$^{36}\text{S}(^{14}\text{C},^{12}\text{C})^{38}\text{S}$	-781	10	-783	7	-0.2	R			Mun		84Ma49
$^{36}\text{Ar}(^3\text{He},\text{n})^{38}\text{Ca}$	-1365	21	-1313.14	0.20	2.5	U			CIT		69Sh04
$^{37}\text{Cl}(\text{n},\gamma)^{38}\text{Cl}$	6107.84	0.30	6107.88	0.08	0.1	U					73Sp06 Z
	6107.95	0.10			-0.7	2			MMn		81Ke02 Z
	6107.73	0.15			1.0	2			Bdn		06Fi.A
$^{37}\text{Cl}(\text{d,p})^{38}\text{Cl}$	3885	8	3883.32	0.08	-0.2	U			MIT		64Sp12
	3883.28	0.50			0.1	U			Rez		90Pi05 *
$^{37}\text{Cl}(\text{p},\gamma)^{38}\text{Ar}$	10243.0	1.0	10242.25	0.20	-0.8	U					68En01 Z
$^{38}\text{S}(\beta^-)^{38}\text{Cl}$	2947	20	2937	7	-0.5	3					71En01
	2936	12			0.1	3					72Vi11
$^{38}\text{Cl}(\beta^-)^{38}\text{Ar}$	4913	5	4916.71	0.22	0.7	U					68Va06
$^{38}\text{K}(\beta^+)^{38}\text{Ar}$	5870	30	5914.07	0.04	1.5	U					56Gr07 *
	5790	50			2.5	U					67Va27 *
$^{38}\text{Ar}(\text{p},\text{n})^{38}\text{K}$	-6695.5	4.	-6696.41	0.04	-0.2	U			Har		75Sq01
	-6695.65	0.70			-1.1	U					78Ja06 Z
$^{38}\text{Ar}(\text{p},\text{n})^{38}\text{K}^m$	-6826.73	0.12	-6826.56	0.04	1.4	U			Auc		98Ha36 Z
$^{38}\text{Al}-\text{u}$	Trends from Mass Surface TMS suggest ^{38}Al 260 keV less bound										GAu **
$^{24}\text{Mg}(^{16}\text{O},2\text{n})^{38}\text{Ca}$	$E(^{16}\text{O})=24880(30)$ to 2^+ level at 2213.13(0.10) keV										Ens082 **
$^{35}\text{Cl}(\alpha,\text{n})^{38}\text{K}$	$Q=-5989.1(2.9,\text{Z})$ to $^{38}\text{K}^m$ at 130.15(0.04) keV										Nub211 **
$^{37}\text{Cl}(\text{d,p})^{38}\text{Cl}$	Estimated systematic error 0.5 added to statistical error 0.064 keV										AHW **
$^{38}\text{K}(\beta^+)^{38}\text{Ar}$	$E_{\beta^+}=2680(30)$ 2600(50) respectively, to 2^+ level at 2167.64 keV										Ens082 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{39}\text{Al}-u$	22970	1580	23070#	320#	0.0	o			GA5	1.5	00Sa21
	21653	676			1.4	D			GA7	1.5	07Ju03
$^{39}\text{Si}-u$	1900	540	2490	150	0.7	o			GA4	1.5	00Sa21
	2210	490			0.4	o			GA5	1.5	00Sa21
	2491	97				2			GA7	1.5	07Ju03
$^{39}\text{P}-u$	-13890	140	-13710	120	0.8	2			GA4	1.5	00Sa21
	-13580	160			-0.6	2			TO4	1.5	91Zh24
	-13870	280			0.4	2			GA5	1.5	00Sa21
	-13602	140			-0.5	2			GT1	1.5	04Ma.A
$^{39}\text{K}-^{23}\text{Na}_{1.696}$	-18942.88	0.58	-18942.218	0.006	0.8	U			Ma8	1.5	08Mu05
$^{39}\text{Ca}-u$	-29278.8	6.4	-29289.2	0.6	-1.6	U			LZ1	1.0	11Tu09
$^{39}\text{K}-^{36}\text{Ar}_{1.083}$	-1144.65	0.44	-1144.87	0.03	-0.5	U			MA8	1.0	02He23
	-1144.83	0.40			-0.1	U			MA8	1.0	03B117
$^{39}\text{K}-^{37}\text{K}_{1.054}$	-8231.29	0.53	-8231.70	0.11	-0.5	U			Ma8	1.5	08Mu05
$^{39}\text{Ca } ^{19}\text{F}-^{39}\text{K}_{1.487}$	23082.43	0.64	23082.4	0.6	-0.0	1	100	100 ^{39}Ca	MA8	1.0	08Ge08
$^{39}\text{K}-^{40}\text{Ar}$	1323.3631	0.0043	1323.363	0.004	-0.1	1	100	100 ^{39}K	FS1	1.0	10Mo30
$^{39}\text{K}(\text{p},\alpha)^{36}\text{Ar}$	1287	7	1288.402	0.027	0.2	U			MIT		64Sp12
$^{37}\text{Cl}(\text{t},\text{p})^{39}\text{Cl}$	5701.9	2.5	5699.5	1.7	-1.0	2			Str		84An03
$^{39}\text{K}(\text{p},^3\text{He})^{37}\text{Ar}^i$	-15493.4	6.				2			MSU		73Be23
$^{39}\text{K}(\text{p},\text{t})^{37}\text{K}^i$	-21713.1	3.	-21718.1	0.8	-1.7	U			MSU		73Be23
$^{39}\text{Sc}^i(2\text{p})^{37}\text{K}$	4969	120	5170	40	1.7	U			Lis		90De43
	4877	40			7.3	B			Brk		92Mo15
	5146	40			0.6	o			Bor		01Gi01
	5170	40				3			Bor		07Do17
$^{38}\text{Ar}(\text{p},\gamma)^{39}\text{K}$	6380.9	1.1	6381.34	0.19	0.4	U					70Ma31
	6382.2	0.8			-1.1	U					84Ha27
$^{39}\text{K}(\text{p},\text{d})^{38}\text{K}$	-10851	2	-10853.19	0.20	-1.1	U			MSU		74Wi17
$^{39}\text{K}(^3\text{He},\alpha)^{38}\text{K}$	7498	15	7499.87	0.20	0.1	U			Roc		66B104
	7483	10			1.7	U			Roc		72Fe06
$^{39}\text{Cl}(\beta^-)^{39}\text{Ar}$	3440	20	3442	5	0.1	U					56Pe38
$^{39}\text{Ar}(\beta^-)^{39}\text{K}$	565	5				2					50Br66
$^{39}\text{Ca}(\beta^+)^{39}\text{K}$	6512	25	6524.5	0.6	0.5	U					58Ki40
$^{39}\text{K}(\text{p},\text{n})^{39}\text{Ca}$	-7302.5	6.	-7306.8	0.6	-0.7	U			Tal		70Ke08
	-7314.9	1.8			4.5	B					78Ra15
$^{39}\text{Al}-u$	Trends from Mass Surface TMS suggest ^{39}Al 1320 keV less bound										GAu
$^{39}\text{K}(\text{p},^3\text{He})^{37}\text{Ar}^i$	$M-A=-25954(6)$; rebuilt $Q=-15493.8(6.)$ with Ame1971; recalibration +0.35										MMC123**
$^{39}\text{K}(\text{p},\text{t})^{37}\text{K}^i$	$M-A=-19753(3)$; Q rebuilt with Ame1971										MMC123**
$^{39}\text{Sc}^i(2\text{p})^{37}\text{K}$	$E_{2p}=3600(120)$ to $1/2^+$ level at 1370.85 keV										Ens123
$^{39}\text{Sc}^i(2\text{p})^{37}\text{K}$	Other possibility $^{39}\text{Sc}^i(\alpha)^{35}\text{K}=3600(120)$ keV										90De43
$^{39}\text{Sc}^i(2\text{p})^{37}\text{K}$	$E_{2p}=4750(40)$ p+p at 90 degrees; deduced $Q=E_{2p}[1+\text{Mp}/\text{M}(^{37}\text{K})]$										MMC123**
$^{39}\text{Sc}^i(2\text{p})^{37}\text{K}$	$E_{2p}=4880(40)$ + recoil 266 keV; data reanalysed and included in 07Do17										MMC135**
$^{39}\text{Sc}^i(2\text{p})^{37}\text{K}$	IAS identification not sure										MMC135**
$^{40}\text{Si}-u$	5290	1010	6080	130	0.5	o			GA4	1.5	00Sa21
	6180	740			-0.1	o			GA5	1.5	00Sa21
	5829	247			0.7	2			GA7	1.5	07Ju03
	6120	140			-0.3	2			RT1	1.0	18Mi08
$^{40}\text{P}-u$	-8800	200	-8740	90	0.2	o			GA4	1.5	00Sa21
	-8950	210			0.7	2			TO4	1.5	91Zh24
	-8200	320			-1.1	o			GA5	1.5	00Sa21
	-8621	129			-0.6	2			GA7	1.5	07Ju03
	-8749	107			0.1	2			RT1	1.0	18Mi08
$^{40}\text{S}-u$	-24440	190	-24517	4	-0.3	o			GA4	1.5	00Sa21
	-24530	250			0.0	U			TO4	1.5	91Zh24
	-24910	340			0.8	o			GA5	1.5	00Sa21
	-24627	129			0.6	U			GA7	1.5	07Ju03

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_3 H_4 - {}^{40}Ar$	68917.0053	0.0035	68917.0056	0.0023	0.1	1	45	45 ${}^{40}Ar$	MI1	1.0	95Di08
$C_2 D_8 - {}^{40}Ar$	150431.1045	0.0040	150431.1007	0.0023	-0.9	1	34	34 ${}^{40}Ar$	MI1	1.0	95Di08
${}^{20}Ne_2 - {}^{40}Ar$	22497.2245	0.0042	22497.2285	0.0030	0.9	-	-	-	MI1	1.0	95Di08
	22497.2280	0.0060			0.1	-	-	-	MI1	1.0	95Di08
ave.	22497.2260	0.0030			0.8	1	74	60 ${}^{20}Ne$			average
${}^{40}Ar-u$	-37616.878	0.040	-37616.8780	0.0023	0.0	U			ST2	1.0	02Bf02
${}^{40}Ca-H_{40}$	-350410.425	0.022	-350410.425	0.022	-0.0	1	100	100 ${}^{40}Ca$	ST2	1.0	06Na18
${}^{40}Ti-u$	-9689	81	-9650	70	0.4	1	82	82 ${}^{40}Ti$	LZ1	1.0	20Fu05
${}^{40}S O - {}^{41}K_{1.366}$	22541	16	22544	4	0.2	U			MS1	1.0	09Ri12
${}^{40}S - {}^{41}K_{976}$	12752.0	9.4	12741	4	-1.2	1	21	21 ${}^{40}S$	MS1	1.0	09Ri12
${}^{40}S - {}^{40}Ar$	13096.6	4.8	13099	4	0.6	1	79	79 ${}^{40}S$	MS1	1.0	09Ri12
${}^{40}Ca - {}^{40}Ar$	208.2	0.5	207.729	0.022	-0.4	U			J3	2.5	68Fu11
${}^{40}Ca({}^3He, {}^8Li){}^{35}K$	-29693	20	-29688.1	0.5	0.2	U			MSU		76Be08
${}^{40}Ca(\alpha, {}^8He){}^{36}Ca$	-57580	40				2			Tex		77Tr03
${}^{40}Ar(n, \alpha){}^{37}S$	-2500	50	-2497.07	0.20	0.1	U					55Be78
	-2490	50			-0.1	U			Ric		64Da11
${}^{40}K(n, \alpha){}^{37}Cl$	3866	7	3872.46	0.08	0.9	U			BNL		68Sc01
${}^{40}Ca(p, \alpha){}^{37}K$	-5179	9	-5182.15	0.10	-0.3	U			CIT		66Mc13
${}^{40}Ca({}^3He, {}^6He){}^{37}Ca$	-24270	50	-24371.2	0.6	-2.0	U			Brk		68Bu02
	-24368	25			-0.1	U			MSU		73Be23 *
${}^{40}Ar(p, {}^3He){}^{38}Cl^i$	-21092	24				2			Brk		70Ha10 *
${}^{40}Ar(p, t){}^{38}Ar^j$	-26765	31				2			Brk		70Ha10 *
${}^{40}Ca(d, \alpha){}^{38}K$	4655	10	4665.16	0.20	1.0	U			MIT		64Sp12
${}^{40}Ca(p, t){}^{38}Ca$	-20459	25	-20448.74	0.20	0.4	U					66Ha32
	-20428	11			-1.9	U			MSU		72Pa02
	-20452	5			0.7	U			MSU		74Se05
${}^{40}Ar({}^{18}O, {}^{19}Ne){}^{39}S$	-14504	200	-14410	50	0.5	U			Can		84Ho.B
${}^{40}Ar({}^{13}C, {}^{14}O){}^{39}S$	-16760	50				2			Can		89Dr03
${}^{40}Ar(t, \alpha){}^{39}Cl$	7256	40	7285.2	1.7	0.7	U			LAI		61Ja07
${}^{40}Ar(d, {}^3He){}^{39}Cl - {}^{36}Ar({}^{35}Cl)$	-4024.13	2.42	-4021.7	1.7	1.0	R			Hei		93Ma50
${}^{40}Ar({}^3He, \alpha){}^{39}Ar^i$	1604	19	1627	7	1.2	2					67Gr01 *
	1631.3	8.0			-0.5	2					72Wi07 *
${}^{39}K(n, \gamma){}^{40}K$	7799.50	0.08	7799.62	0.06	1.5	-			ILn		84Vo01 Z
	7799.56	0.16			0.4	-			Bdn		06Fi.A
${}^{39}K(d, p){}^{40}K$	5579	10	5575.05	0.06	-0.4	U			MIT		64Sp12
${}^{39}K(n, \gamma){}^{40}K$	7799.51	0.07	7799.62	0.06	1.5	1	61	61 ${}^{40}K$			average
${}^{39}K(p, \gamma){}^{40}Ca$	8329.5	0.9	8328.178	0.021	-1.5	U					68Do12
	8329.6	0.9			-1.6	U					68Li12 *
	8328.24	0.09			-0.7	U			Utr		90Ki07 Z
${}^{39}K({}^3He, d){}^{40}Ca$	2845	8	2834.703	0.021	-1.3	U			Oak		67Se10
${}^{40}Ca({}^3He, \alpha){}^{39}Ca$	4950	20	4942.6	0.6	-0.4	U			Ald		66Hi06
	4919	15			1.6	U			MIT		71Ra35
${}^{40}Ca({}^7Li, {}^8He){}^{39}Sc$	-37400	40	-37376	24	0.6	2			MSU		88Mo18
${}^{40}Ca({}^{14}N, {}^{15}C){}^{39}Sc$	-27670	30	-27683	24	-0.4	2			Can		88Wo07
${}^{40}Sc^i(p){}^{39}Ca$	3840	120	3830	6	-0.1	o			Lis		90De43
	3820	30			0.3	U					90Zh.A *
	3827.7	10.			0.2	2			GSI		97Li25 *
	3830.8	7.			-0.1	2			Lis		98Bh12
	3841	20			-0.6	U			Bor		07Do17
${}^{40}Cl(\beta^-){}^{40}Ar$	7320	80	7480	30	2.0	2					89Mi03
${}^{40}Ar({}^7Li, {}^7Be){}^{40}Cl$	-8375	35	-8340	30	0.9	2					84Fi02
${}^{40}K(\epsilon){}^{40}Ar$	1504	7	1504.40	0.06	0.1	U					67Mc10 *
	1497	8			0.9	U					68Az01 *
${}^{40}K(n, p){}^{40}Ar$	2270	5	2286.75	0.06	3.4	B			BNL		68Sc01
	2286.7	1.0			0.1	U			ILL		81We12
${}^{40}Ar(p, n){}^{40}K$	-2286.3	1.0	-2286.75	0.06	-0.5	U			Duk		66Pa18 Z
	-2286.3	1.0			-0.5	U					01Wa50

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{40}\text{K}(\beta^-)^{40}\text{Ca}$	1325	15	1310.91	0.06	-0.9	U					52Fe16
	1350	20			-2.0	U					59Ke26
$^{40}\text{Sc}(\beta^+)^{40}\text{Ca}$	14330	40	14323.0	2.8	-0.2	U					68Ar03 *
$^{40}\text{Ca}(\text{p,n})^{40}\text{Sc}$	-15105.4	2.9				2			Yal		69Ov01 Z
$^{40}\text{Ca}(^3\text{He,t})^{40}\text{Sc}$	-14490	60	-14341.6	2.8	2.5	U			Bld		65Ri06
$^{40}\text{Ca}(\pi^+, \pi^-)^{40}\text{Ti}$	-24974	160	-24830	70	0.9	1	18	18	^{40}Ti		82Mo12 *
$^{40}\text{Ca}(^3\text{He}, ^6\text{He})^{37}\text{Ca}$	Average of 2 values with small calibration correction										AHW **
$^{40}\text{Ar}(\text{p}, ^3\text{He})^{38}\text{Cl}^i$	IT=8216(25); rebuilt $Q = -21093.65(23.68)$; recalibrated for $^{10}\text{B} + 1.5$ keV										MMC123**
$^{40}\text{Ar}(\text{p,t})^{38}\text{Ar}^j$	IT=18784(30); Q rebuilt with $^{10}\text{C} = 15702.5(1.8)$ from reference										68Br23 **
$^{40}\text{Ar}(^3\text{He}, \alpha)^{39}\text{Ar}^i$	IT=9089(20); Q rebuilt with Ame1961										MMC123**
$^{40}\text{Ar}(^3\text{He}, \alpha)^{39}\text{Ar}^j$	IT=9075(10); Q rebuilt with Ame1964										MMC123**
$^{39}\text{K}(\text{p}, \gamma)^{40}\text{Ca}$	$E_{\text{res}} = 1345.4(0.5)$ to 2^- level at 9641.1(0.8) keV										Ens048 **
$^{40}\text{Sc}^i(\text{p})^{39}\text{Ca}$	Uncertainty not given, estimated from graph: stat(9keV), calib(11)										GAu **
$^{40}\text{Sc}^j(\text{p})^{39}\text{Ca}$	$E_p = 3731(10)$; also $E_p = 1330(20)$ $Q_p = 1364.4(20)$ keV to 2468.5 level										Ens062 **
$^{40}\text{Sc}^i(\text{p})^{39}\text{Ca}$	IT=4370(10) in original paper										MMC123**
$^{40}\text{K}(\epsilon)^{40}\text{Ar}$	LMK=0.34(0.08) 0.47(0.16) respectively, to 2^+ level at 1460.851, recalculated Q										Ens048 **
$^{40}\text{Sc}(\beta^+)^{40}\text{Ca}$	$E_{\beta^+} = 9580(40)$ to 3^- level at 3736.69 keV, and other E_{β^+}										Ens048 **
$^{40}\text{Ca}(\pi^+, \pi^-)^{40}\text{Ti}$	Recalibrated to $^{16}\text{O}(\pi^+, \pi^-) Q = -27704(20)$ keV										GAu **
$^{41}\text{Si}-u$	14560	1980	14170#	320#	-0.1	o			GA5	1.5	00Sa21
	13011	397			1.9	D			GA7	1.5	07Ju03 *
$^{41}\text{P}-u$	-5930	300	-5350	130	1.3	o			GA4	1.5	00Sa21
	-5200	500			-0.2	U			TO4	1.5	91Zh24
	-5290	420			-0.1	o			GA5	1.5	00Sa21
	-5346	86				2			GA7	1.5	07Ju03
$^{41}\text{S}-u$	-20500	150	-20407	4	0.4	U			GA4	1.5	00Sa21
	-19970	230			-1.3	U			TO4	1.5	91Zh24
	-20430	330			0.0	U			GA5	1.5	00Sa21
	-20494	75			0.8	U			GT1	1.5	04Ma.A
$^{41}\text{S}-\text{C}_2 \text{H O}$	-23146.2	4.4				2			MS1	1.0	09Ri12 *
$^{41}\text{Cl}-u$	-29620	190	-29320	70	1.1	2			TO3	1.5	90Tu01
	-29500	270			0.5	2			TO4	1.5	91Zh24
$^{41}\text{Sc}-u$	-30741	12	-30748.84	0.08	-0.7	U			LZ1	1.0	11Tu09
$^{41}\text{Ti}-u$	-16200	390	-16850	30	-1.1	U			GT1	1.5	04St05
	-16852	30				2			LZ1	1.0	12Zh34 *
$^{41}\text{K}-^{39}\text{K}_{1.051}$	-30.05	0.32	-30.259	0.006	-0.7	U			MA8	1.0	02He23
	-29.5	2.4			-0.3	U			MA8	1.0	09Na.A
$^{41}\text{K}-^{40}\text{Ar H}$	-8382.9005	0.0061	-8382.8980	0.0030	0.4	-			FS1	1.0	10Mo30
$^{41}\text{K}-^{40}\text{Ar}$	-557.8652	0.0039	-557.8660	0.0030	-0.2	-			FS1	1.0	10Mo30
$^{41}\text{K}-^{40}\text{Ar H}$	ave. -8382.8980	0.0030	-8382.8980	0.0030	0.1	1	100	100	^{41}K		average
$^{41}\text{K}(\text{p}, \alpha)^{38}\text{Ar}$	4002	20	4019.33	0.20	0.9	U			ChR		60Cl02
	4018	10			0.1	U			MIT		64Sp12
$^{41}\text{K}(\text{d}, \alpha)^{39}\text{Ar}$	8397	15	8393	5	-0.2	U			MIT		67Sp09
$^{39}\text{K}(^3\text{He}, \text{p})^{41}\text{Ca}$	8920	20	8972.96	0.14	2.6	U			MIT		67Sp09
$^{40}\text{Ar}(^{18}\text{O}, ^{17}\text{F})^{41}\text{Cl}$	-10530	83	-10470	70	0.8	R			Can		84Ho.B
$^{40}\text{Ar}(\text{n}, \gamma)^{41}\text{Ar}$	6098.4	0.7	6098.9	0.3	0.8	2					70Ha56 Z
	6099.1	0.4			-0.4	2			Bdn		06Fi.A
$^{40}\text{Ar}(\text{d}, \text{p})^{41}\text{Ar}$	3878	6	3874.4	0.3	-0.6	U			MIT		64Sp12
$^{40}\text{Ar}(\text{p}, \gamma)^{41}\text{K}$	7807.8	0.3	7808.6200	0.0030	2.7	U					89Sm06 Z
$^{40}\text{Ar}(^3\text{He}, \text{d})^{41}\text{K}^i$	-6034	15				2					75Me10 *
$^{40}\text{K}(\text{n}, \gamma)^{41}\text{K}$	10095.19	0.10	10095.37	0.06	1.8	-			ILn		84Kr05 Z
	10095.25	0.20			0.6	-			Bdn		06Fi.A
	ave. 10095.20	0.09			1.9	1	39	39	^{40}K		average
$^{40}\text{Ca}(\text{n}, \gamma)^{41}\text{Ca}$	8363.0	0.5	8362.82	0.14	-0.4	-					69Ar.A Z
	8362.5	0.5			0.6	-					70Cr04 Z
	8362.72	0.3			0.3	-			MMn		80Is02 Z
	8362.86	0.17			-0.2	-			Bdn		06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{40}\text{Ca}(\text{d,p})^{41}\text{Ca}$		6134	4	6138.26	0.14	1.1	U		MIT		68Be36
$^{40}\text{Ca}(\text{n},\gamma)^{41}\text{Ca}$	ave.	8362.81	0.14	8362.82	0.14	0.1	1	100	100 ^{41}Ca		average
$^{40}\text{Ca}(\text{p},\gamma)^{41}\text{Sc}$		1085.7	1.4	1084.93	0.07	-0.6	U				73Al11
		1085.09	0.09			-1.8	1	69	69 ^{41}Sc		87Zi02 *
$^{40}\text{Ca}(\text{d,n})^{41}\text{Sc}$		-1145	15	-1139.64	0.07	0.4	U				61Ma08 Y
$^{40}\text{Ca}(\text{p},\gamma)^{41}\text{Sc}^r$		-1796.4	1.5	-1797.42	0.08	-0.7	U				77Ko10
$^{40}\text{Ca}(\text{p},\gamma)^{41}\text{Sc}^i$		-4851.4	4.9	-4854	3	-0.5	2				76Fo01
$^{41}\text{Sc}^i(\text{p})^{40}\text{Ca}$		4855.6	5.	4854	3	-0.4	2		Jyp		97Ho12
		4855.6	8.			-0.2	2		Lis		98Bh12
		4857	16			-0.2	U		Bor		07Do17
$^{41}\text{Cl}(\beta^-)^{41}\text{Ar}$		5670	150	5760	70	0.6	R				74Gu10
$^{41}\text{Ar}(\beta^-)^{41}\text{K}$		2492.0	1.1	2492.0	0.3	0.0	U				64Pa03 *
$^{41}\text{K}(\text{p,n})^{41}\text{Ca}$		-1209.6	1.5	-1203.99	0.14	3.7	B		Oak		64Jo11 Z
		-1203.8	0.5			-0.4	U		Can		70Kn03 Z
$^{41}\text{Sc}(\beta^+)^{41}\text{Ca}$		6630	100	6495.55	0.16	-1.3	U				62Cr04
$^{41}\text{Sc}^r(\text{IT})^{41}\text{Sc}$		2882.6	0.3	2882.35	0.05	-0.8	U				77Ko10
		2882.39	0.10			-0.4	-		Utr		87Zi02 Z
		2882.26	0.06			1.4	-		Utr		89Ki11 Z
	ave.	2882.29	0.05			1.0	1	90	58 $^{41}\text{Sc}^r$		average
$^{41}\text{Si}-\text{u}$	Trends from Mass Surface TMS suggest ^{41}Si 1080 keV less bound										GAu **
$^{41}\text{S}-\text{C}_2\text{H O}$	For original doublet $^{41}\text{S C H}-\text{C}_3\text{H}_2\text{O}$										09Ri12 **
$^{41}\text{Ti}-\text{u}$	Same result in reference										13Ya03 **
$^{40}\text{Ar}(\text{He},\text{d})^{41}\text{K}^i$	IT=8349(15); Q rebuilt with Ame1971										MMC123**
$^{40}\text{Ca}(\text{p},\gamma)^{41}\text{Sc}$	$E_p=647.25(0.05,\text{Z})$ to $1716.43(0.08,\text{Z})$ level										87Zi02 **
$^{41}\text{Ar}(\beta^-)^{41}\text{K}$	$E_{\beta^-}=1198.3(1.1)$ to $7/2^-$ level at 1293.609 keV										Ens163 **
$^{42}\text{Si}-\text{u}$		20860	3990	18080#	320#	-0.5	o		GA5	1.5	99Sa.A
		16275	623			1.9	D		GA7	1.5	07Ju03 *
$^{42}\text{P}-\text{u}$		260	740	1170	100	0.8	o		GA4	1.5	00Sa21
		1550	630			-0.4	o		GA5	1.5	00Sa21
		1084	225			0.3	2		GA7	1.5	07Ju03
		1181	107			-0.1	2		RT1	1.0	18Mi08
$^{42}\text{S}-\text{u}$		-18940	150	-18935	3	0.0	U		GA4	1.5	00Sa21
		-18510	350			-0.8	U		TO4	1.5	91Zh24
		-19390	350			0.9	U		GA5	1.5	00Sa21
		-18934.9	3.0				2		MS1	1.0	09Ri12 *
$^{42}\text{Cl}-\text{u}$		-27000	190	-26660	60	1.2	U		TO3	1.5	90Tu01
		-26870	190			0.7	U		TO4	1.5	91Zh24
		-26658	64				2		LZ1	1.0	15Xu14
$^{42}\text{Ti}-\text{u}$		-26973	25	-26950.63	0.29	0.9	U		LZ1	1.0	17Zh12
$^{42}\text{Ar}-^{36}\text{Ar}_{1.167}$		920.6	6.2				2		MA6	1.0	01He29
$^{42}\text{S}-^{41}\text{K}_{1.024}$		20151.8	9.5	20156	3	0.4	U		MS1	1.0	09Ri12
$^{42}\text{Sc}-^{42}\text{Ca}$		6898.74	0.22	6898.91	0.05	0.8	o		JY1	1.0	06Er08
		6898.960	0.064			-0.8	1	66	56 ^{42}Sc	1.0	17Er01 *
$^{42}\text{Sc}^m-^{42}\text{Ca}$		7560.68	0.23	7561.08	0.06	1.7	o		JY1	1.0	06Er08
		7561.120	0.081			-0.5	1	62	59 $^{42}\text{Sc}^m$	1.0	17Er01 *
$^{42}\text{Ti}-^{42}\text{Ca}$		14431.69	0.71	14431.59	0.24	-0.1	1	12	12 ^{42}Ti	1.0	09Ku19
$^{42}\text{Sc}^m-^{42}\text{Sc}$		661.97	0.24	662.17	0.07	0.8	U		JY1	1.0	06Er08
		662.50	0.42			-0.8	U		JY1	1.0	09Ku19
		662.109	0.088			0.7	1	55	39 $^{42}\text{Sc}^m$	1.0	17Er01 *
$^{42}\text{Ti}-^{42}\text{Sc}$		7532.92	0.34	7532.68	0.24	-0.7	1	50	49 ^{42}Ti	1.0	09Ku19
$^{42}\text{Ti}-^{42}\text{Sc}^m$		6870.19	0.38	6870.51	0.24	0.8	1	40	39 ^{42}Ti	1.0	09Ku19
$^{28}\text{Si}(^{16}\text{O},2\text{n})^{42}\text{Ti}$		-17250	13	-17268.08	0.27	-1.4	U				72Zi02
$^{42}\text{Ca}(\text{p},\alpha)^{39}\text{K}$		118	7	123.96	0.15	0.9	U		MIT		64Sp12
$^{39}\text{K}(\alpha,\text{n})^{42}\text{Sc}$		-7160	60	-7332.59	0.15	-2.9	U		Yal		61Sm05
		-7455	30			4.1	B		Tal		65Ne02

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{40}\text{Ar}(t,p)^{42}\text{Ar}$	7043	40	7044	6	0.0	U			LAI		61Ja07
$^{40}\text{Ca}(^3\text{He},p)^{42}\text{Sc}$	4966	20	4916.85	0.15	-2.5	U			MIT		64Sp12
	4905	5			2.4	U			ANL		74Ha55
$^{40}\text{Ca}(^3\text{He},n)^{42}\text{Ti}$	-2865	6	-2882.15	0.27	-2.9	U			CIT		67Mi02
$^{41}\text{K}(n,\gamma)^{42}\text{K}$	7533.78	0.15	7533.80	0.11	0.1	2			ILn		85Kr06 Z
	7533.82	0.15			-0.1	2			Bdn		06Fi.A
$^{41}\text{K}(d,p)^{42}\text{K}$	5314	12	5309.23	0.11	-0.4	U			MIT		64Sp12
$^{41}\text{K}(p,\gamma)^{42}\text{Ca}$	10275.5	3.4	10276.72	0.15	0.4	U					71Vi14
$^{41}\text{Ca}(n,\gamma)^{42}\text{Ca}$	11480.63	0.06	11480.70	0.06	1.2	1	86	^{86}Ca	ORn		89Ki11 Z
$^{42}\text{Ca}(^3\text{He},\alpha)^{41}\text{Ca}$	9102	15	9096.92	0.06	-0.3	U			MIT		71Ra35
$^{41}\text{Ca}(p,\gamma)^{42}\text{Sc}^r - ^{40}\text{Ca}()^{41}\text{Sc}^r$	-6.67	0.05	-6.72	0.05	-1.0	1	90	$^{49}\text{Sc}^r$	Utr		89Ki11 *
$^{42}\text{Cl}(\beta^-)^{42}\text{Ar}$	9760	220	9590	60	-0.8	U					89Mi03
$^{42}\text{K}(\beta^-)^{42}\text{Ca}$	3519.	3.5	3525.26	0.18	1.8	U					68Va06
	3524	6			0.2	U					75Ra09
$^{42}\text{Sc}(\beta^+)^{42}\text{Ca}$	6342	100	6426.29	0.05	0.8	U					61Ja22
	6486	100			-0.6	U					63Ro10 *
$^{42}\text{Ca}(p,n)^{42}\text{Sc}$	-7213.7	2.3	-7208.64	0.05	2.2	U			Har		75Fr.A
$^{42}\text{Ca}(^3\text{He},t)^{42}\text{Sc}$	-6442.3	0.4	-6444.88	0.05	-6.5	F			Mun		77Vo02 *
$^{42}\text{Ca}(^3\text{He},t)^{42}\text{Sc} - ^{27}\text{Al}()^{27}\text{Si}$	-1611.7	2.6	-1613.93	0.11	-0.9	U			ChR		74Ha35
$^{42}\text{Ca}(^3\text{He},t)^{42}\text{Sc} - ^{26}\text{Mg}()^{26}\text{Al}$	-2417.8	3.5	-2421.89	0.08	-1.2	U			ChR		74Ha35
	-2421.83	0.23			-0.2	1	11	^{7}Al	ChR		87Ko34 *
$^{42}\text{Sc}^m(\text{IT})^{42}\text{Sc}$	616.28	0.06	616.81	0.06	8.8	B					Ens167
$^{42}\text{Sc}^r(\text{IT})^{42}\text{Sc}$	6076.33	0.08	6076.20	0.07	-1.6	1	75	$^{51}\text{Sc}^r$	Utr		89Ki11 Z
$^{42}\text{Si}-u$	Trends from Mass Surface TMS suggest ^{42}Si 1680 keV less bound										GAu **
$^{42}\text{S}-u$	For original doublet $^{42}\text{S}-(\text{C}_2\text{H O})_{1.024}$, $D_M = -21740.2(3.1)\mu\text{u}$										09Ri12 **
$^{42}\text{Sc}-^{42}\text{Ca}$	Frequency ratio not correct for unc., values taken from paper										HWJ17b **
$^{42}\text{Sc}^m - ^{42}\text{Ca}$	Frequency ratio not correct for unc., values taken from paper										HWJ17b **
$^{42}\text{Sc}^m - ^{42}\text{Sc}$	Frequency ratio not correct for unc., values taken from paper										HWJ17b **
$^{41}\text{Ca}(p,\gamma)^{42}\text{Sc}^r - ^{40}\text{Ca}()^{41}\text{Sc}^r$	Calculated from resonance energy difference = 5.73(0.05) keV										GAu **
$^{42}\text{Sc}(\beta^+)^{42}\text{Ca}$	$E_{\beta^+} = 2870(100)$ from $^{42}\text{Sc}^m$ at 616.28 to 6^+ level at 3189.26 keV										Ens167 **
$^{42}\text{Ca}(^3\text{He},t)^{42}\text{Sc}$	F : rejected in reference of same group										09Fa15 **
$^{42}\text{Ca}(^3\text{He},t)^{42}\text{Sc} - ^{26}\text{Mg}()^{26}\text{Al}$	$Q = -2193.52(0.23)$ to $^{26}\text{Al}^m$ at 228.306 keV										Nub211 **
$^{43}\text{P}-u$	4220	1620	5410#	320#	0.5	U			GA4	1.5	00Sa21
	6190	1040			-0.5	o			GA5	1.5	00Sa21
	5024	397			0.6	D			GA7	1.5	07Ju03 *
$^{43}\text{S}-u$	-12810	250	-13092	5	-0.8	o			GA4	1.5	00Sa21
	-13400	900			0.2	U			TO4	1.5	91Zh24
	-12900	460			-0.3	o			GA5	1.5	00Sa21
	-12958	107			-0.8	U			GA7	1.5	07Ju03
	-13087	22			-0.2	2			MS1	1.0	09Ri12 *
	-13092.7	5.5			0.1	2			MS1	1.0	09Ri12 *
	-13022	236			-0.2	U			GA8	1.5	12Ga45
$^{43}\text{Cl}-u$	-26090	300	-25940	70	0.3	o			GA4	1.5	00Sa21
	-25740	200			-0.7	o			TO3	1.5	90Tu01
	-25970	350			0.1	U			TO4	1.5	91Zh24
	-26010	330			0.1	o			GA5	1.5	00Sa21
	-25905	86			-0.2	o			GT1	1.5	04Ma.A
	-25894	140			-0.2	2			GA7	1.5	07Ju03
	-26361	100			1.7	C			GT2	2.5	08Kn.A
	-25941	70			0.1	2			LZ1	1.0	15Xu14
$^{43}\text{Ti}-u$	-31461	10	-31472	6	-1.1	1	38	^{38}Ti	LZ1	1.0	18Zh29 *
$^{43}\text{V}-u$	-19234	46				2			LZ1	1.0	13Ya03
$^{43}\text{Ar}-^{36}\text{Ar}_{1.194}$	4387.2	5.7				2			MA6	1.0	01He29
$^{43}\text{K}-^{39}\text{K}_{1.103}$	766.45	0.44				2			MA8	1.0	07Ya08

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{43}\text{Ca}(\text{p},\alpha)^{40}\text{K}$	-14	8	-9.32	0.23	0.6	U			MIT		64Sp12
$^{40}\text{Ca}(\alpha,\text{p})^{43}\text{Sc}$	-3470	30	-3522.3	1.9	-1.7	U					61Ma03
$^{40}\text{Ca}(\alpha,\text{n})^{43}\text{Ti}$	-11169.9	10.	-11177	6	-0.8	1	40	40 ^{43}Ti	Tal		67Al08
$^{41}\text{K}(^3\text{He},\text{p})^{43}\text{Ca}^i$	2452	30	2497	14	1.5	1	23	23 $^{43}\text{Ca}^i$	MIT		68Do02
$^{43}\text{V}^i(2\text{p})^{41}\text{Sc}$	4320	50	4359	15	0.8	U			Lis		92Bo37
	4503	22			-6.5	B			Bor		01Gi01 *
	4348	16			0.7	1	89	89 $^{43}\text{V}^i$	Bor		07Do17
$^{42}\text{Ca}(\text{n},\gamma)^{43}\text{Ca}$	7933.1	0.5	7932.90	0.17	-0.4	-					69Ar.A Z
	7933.1	0.5			-0.4	-			Ptn		69Gr08 Z
	7933.1	0.4			-0.5	-					71Bi.A
	7932.73	0.23			0.7	-			Bdn		06Fi.A
$^{42}\text{Ca}(\text{d},\text{p})^{43}\text{Ca}$	5716	10	5708.33	0.17	-0.8	U			MIT		64Sp12
	5707	12			0.1	U			MIT		66Do02
$^{43}\text{Ca}(\text{d},\text{t})^{42}\text{Ca}$	-1672	10	-1675.67	0.17	-0.4	U			Ald		64Bj02
$^{42}\text{Ca}(\text{n},\gamma)^{43}\text{Ca}$	ave. 7932.89	0.17	7932.90	0.17	0.0	1	99	99 ^{43}Ca			average
$^{42}\text{Ca}(\text{p},\gamma)^{43}\text{Sc}$	4935	5	4929.8	1.9	-1.0	2					65Br31
	4929	2			0.4	2					69Wa19
$^{42}\text{Ca}(^3\text{He},\text{d})^{43}\text{Sc}^i$	-4808	8	-4795	3	1.6	1	17	17 $^{43}\text{Sc}^i$			66Sc17 *
$^{43}\text{V}^i(\text{p})^{42}\text{Ti}$	8200	45	8110	15	-2.0	1	11	11 $^{43}\text{V}^i$	Bor		01Gi01 *
$^{43}\text{K}(\beta^-)^{43}\text{Ca}$	1817	20	1833.5	0.5	0.8	U					54Li24 *
	1815	10			1.8	U					59Be72 *
$^{43}\text{Sc}(\beta^+)^{43}\text{Ca}$	2200	20	2220.7	1.9	1.0	U					52Ha44
	2220	10			0.1	U					54Li42
$^{43}\text{Ca}(\text{p},\text{n})^{43}\text{Sc}$	-3005	10	-3003.1	1.9	0.2	U			Har		60Mc12 Y
	-2998	10			-0.5	U					67Mc07
$^{43}\text{Ca}(^3\text{He},\text{t})^{43}\text{Sc}^i$	-6467	8	-6471	3	-0.5	-					71Al19 *
	-6469	4			-0.5	-					71Be29 *
ave.	-6469	4			-0.7	1	83	83 $^{43}\text{Sc}^i$			average
$^{43}\text{P}-\text{u}$	Trends from Mass Surface TMS suggest ^{43}P 360 keV less bound										GAu **
$^{43}\text{S}-\text{u}$	For original doublet $^{43}\text{S}-(\text{C}_3\text{H}_5\text{O})_{0.754}$, $D_M=-38753(22)\mu\text{u}$										09Ri12 **
$^{43}\text{S}-\text{u}$	For original doublet $^{43}\text{S C H}-(\text{C}_3\text{H}_5\text{O})_{0.982}$, $D_M=-38694.8(5.5)\mu\text{u}$										09Ri12 **
$^{43}\text{Ti}-\text{u}$	could be mixed with 313(1) state, without correction										18Zh29 **
$^{43}\text{V}^i(2\text{p})^{41}\text{Sc}$	$E_{2p}=4292(22) + \text{recoil}=211$; data reanalysed and included in 07Do17										MMC135**
$^{42}\text{Ca}(^3\text{He},\text{d})^{43}\text{Sc}^i$	$IT=4238(8)$; Q rebuilt with Ame1961										MMC123**
$^{43}\text{V}^i(\text{p})^{42}\text{Ti}$	$Q_p=4590(45)$ followed by γ 's 1938+1554 keV + recoil=118 keV										MMC135**
$^{43}\text{K}(\beta^-)^{43}\text{Ca}$	$E_{\beta^-}=827(20)$ 825(10) respectively, to $3/2^+$ level at 990.257 keV										Ens156 **
$^{43}\text{Ca}(^3\text{He},\text{t})^{43}\text{Sc}^i$	$IT=4226(8)$; Q rebuilt with Ame1965										MMC123**
$^{43}\text{Ca}(^3\text{He},\text{t})^{43}\text{Sc}^i$	$CDE=7238(4)$ $Q=-6474(4)$; recalibration +6 keV for $^{42}\text{Ca}(\text{p},\text{n})^{42}\text{Sc}$ from Ame1961										MMC123**
$^{44}\text{P}-\text{u}$	10070	966	11930#	430#	1.3	D			GA7	1.5	07Ju03 *
$^{44}\text{S}-\text{u}$	-10510	580	-9881	6	0.7	o			GA4	1.5	00Sa21
	-8960	620			-1.0	o			GA5	1.5	00Sa21
	-9769	150			-0.5	U			GA7	1.5	07Ju03
	-10027	301			0.3	U			GA8	1.5	12Ga45
$^{44}\text{S}-\text{C}_2\text{H}_4\text{O}$	-36095.9	5.6				2			MS1	1.0	09Ri12 *
$^{44}\text{Cl}-\text{u}$	-21700	130	-21990	90	-1.5	2			GA4	1.5	00Sa21
	-21500	500			-0.6	U			TO3	1.5	90Tu01
	-21450	270			-1.3	U			TO4	1.5	91Zh24
	-22150	370			0.3	2			GA5	1.5	00Sa21
	-22115	161			0.5	2			GT1	1.5	04Ma.A
	-22051	118			0.6	2			RT1	1.0	18Mi08
$^{44}\text{Ca}-\text{u}$	-44515.3	10.0	-44518.5	0.3	-0.3	U			MR1	1.0	15Wi.A
$^{44}\text{Sc}-\text{u}$	-40480	410	-40597.2	1.9	-0.2	U			TO6	1.5	98Ba.A *
$^{44}\text{V}-\text{u}$	-25890	120	-25559	8	1.8	B			GT1	1.5	04St05 *
	-25579	21			1.0	1	14	14 ^{44}V	LZ1	1.0	18Zh29

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{44}\text{V}^m\text{-u}$	-25272	20	-25268	6	0.2	U			LZ1	1.0	18Zh29
$^{44}\text{Cr-u}$	-14409	55				2			LZ1	1.0	20Fu05
$^{44}\text{Ar-}^{39}\text{K}_{1.128}$	5862.9	1.7				2			MA8	1.0	03B117
$^{44}\text{K-}^{39}\text{K}_{1.128}$	2526.07	0.45				2			MA8	1.0	07Ya08
	2529.2	2.2	2526.1	0.5	-1.4	o			TT1	1.0	10La.A
	2529.1	1.7			-1.8	U			TT1	1.0	12La05
$^{44}\text{V-}^{32}\text{S}$	2373.0	8.4	2370	8	-0.4	1	86	^{86}V	MS1	1.0	20Pu02 *
$^{44}\text{V}^m\text{-}^{32}\text{S}$	2660.8	5.9				2			MS1	1.0	20Pu02 *
$^{40}\text{Ca}(\alpha,\gamma)^{44}\text{Ti}$	5127.1	0.7				2					82Di05
$^{44}\text{Ca}(\text{p},\alpha)^{41}\text{K}$	-1058	10	-1045.1	0.3	1.3	U			MIT		64Sp12
$^{41}\text{K}(\alpha,\text{n})^{44}\text{Sc}$	-3420	60	-3389.9	1.8	0.5	U			Yal		61Sm05
$^{44}\text{Ca}(\text{d},\alpha)^{42}\text{K}$	4273	20	4264.1	0.3	-0.4	U					77Pa24
$^{42}\text{Ca}(\text{t},\text{p})^{44}\text{Ca}$	10593	15	10582.28	0.29	-0.7	U			Ald		67Bj06
$^{42}\text{Ca}(\text{}^3\text{He},\text{p})^{44}\text{Sc}$	6920	20	6911.0	1.7	-0.5	U			Hei		70Sc22
$^{43}\text{Ca}(\text{n},\gamma)^{44}\text{Ca}$	11130.6	0.5	11131.18	0.23	1.2	-					69Ar.A Z
	11130.1	0.7			1.5	-					72Wh02 Z
	11131.54	0.29			-1.3	-			Bdn		06Fi.A
$^{43}\text{Ca}(\text{d},\text{p})^{44}\text{Ca}$	8922	14	8906.61	0.23	-1.1	U			MIT		64Sp12
	8920	10			-1.3	U			Kop		67Bj02
$^{44}\text{Ca}(\text{}^3\text{He},\alpha)^{43}\text{Ca}$	9452	15	9446.45	0.23	-0.4	U			MIT		71Ra35
$^{43}\text{Ca}(\text{n},\gamma)^{44}\text{Ca}$	ave. 11131.17	0.24	11131.18	0.23	0.0	1	99	^{99}Ca			average
$^{44}\text{Ca}(\text{p},\text{d})^{43}\text{Ca}^i$	-16880	30	-16901	14	-0.7	-					72Ma23 *
$^{44}\text{Ca}(\text{d},\text{t})^{43}\text{Ca}^i$	-12858.7	19.7	-12869	14	-0.5	-					76Do05 *
$^{44}\text{Ca}(\text{p},\text{d})^{43}\text{Ca}^i$	ave. -16888	16	-16901	14	-0.8	1	77	$^{77}\text{Ca}^i$			average
$^{43}\text{Ca}(\text{p},\gamma)^{44}\text{Sc}$	6694	2	6696.1	1.7	1.1	2					71Po.A
$^{43}\text{Ca}(\text{}^3\text{He},\text{d})^{44}\text{Sc}^i$	-1583	5	-1575.1	2.5	1.6	1	24	$^{24}\text{Sc}^i$			68Sc15
$^{44}\text{V}^i(\text{p})^{43}\text{Ti}$	950	50	908	11	-0.8	U			Lis		92Bo37
	908	11				2			Bor		07Do17
	917	53			-0.2	U					14Po05
$^{44}\text{K}(\beta^-)^{44}\text{Ca}$	5580	80	5687.2	0.5	1.3	U					70Le05
$^{44}\text{Ca}(\text{t},\text{}^3\text{He})^{44}\text{K}$	-5660	40	-5668.6	0.5	-0.2	U			LAI		70Aj01
$^{44}\text{Sc}(\beta^+)^{44}\text{Ca}$	3642	5	3652.7	1.8	2.1	R					50Br52 *
	3650	5			0.5	R					55Bi23 *
$^{44}\text{Ca}(\text{p},\text{n})^{44}\text{Sc}$	-4410	15	-4435.0	1.8	-1.7	U			Har		60Mc12 Y
	-4447	10			1.2	U					67Mc07
$^{44}\text{Ca}(\text{}^3\text{He},\text{t})^{44}\text{Sc}^i$	-6444	4	-6449.0	2.5	-1.3	-					71Be29 *
	-6449	4			-0.0	-					72Ma50 *
ave.	-6446.5	2.8			-0.9	1	76	$^{76}\text{Sc}^i$			average
$^{44}\text{P-u}$	Trends from Mass Surface TMS suggest ^{44}P 1730 keV less bound										GAu **
$^{44}\text{S-C}_2\text{H}_4\text{O}$	For original doublet $^{44}\text{S C H-C}_3\text{H}_5\text{O}$										09Ri12 **
$^{44}\text{Sc-u}$	$M-A=-37570(370)$ keV for mixture gs+p at 271.240 keV										Nub211 **
$^{44}\text{V-u}$	$M-A=-23980(80)$ keV for mixture gs+m at 270#100 keV										Nub211 **
$^{44}\text{V-u}$	Authors have unduely increased the lower error to 380 keV										GAu **
$^{44}\text{V-}^{32}\text{S}$	From freq. ratio $^{44}\text{VO}^+/(^{32}\text{SCO})^+=0.99996043(14)$										HWJ207 **
$^{44}\text{V}^m\text{-}^{32}\text{S}$	From freq. ratio $^{44}\text{VmO}^+/(^{32}\text{SCO})^+=0.999955631(98)$										HWJ207 **
$^{44}\text{Ca}(\text{p},\text{d})^{43}\text{Ca}^i$	IT=7970(30); Q rebuilt with Ame1965										MMC123**
$^{44}\text{Ca}(\text{d},\text{t})^{43}\text{Ca}^i$	IT=7980(20); Q rebuilt with Ame1971										MMC123**
$^{43}\text{Ca}(\text{}^3\text{He},\text{d})^{44}\text{Sc}^i$	IT=2796(5); Q rebuilt with Ame1965										MMC123**
$^{44}\text{Sc}(\beta^+)^{44}\text{Ca}$	$E_{\beta^+}=1463(5)$ 1471(5) respectively, to 2^+ level at 1157.019 keV										Ens119 **
$^{44}\text{Ca}(\text{}^3\text{He},\text{t})^{44}\text{Sc}^i$	CDE=7214(4) $Q=-6450(4)$; recalibration +6 keV for $^{42}\text{Ca}(\text{p},\text{n})^{42}\text{Sc}$ from Ame1961										MMC123**
$^{44}\text{Ca}(\text{}^3\text{He},\text{t})^{44}\text{Sc}^i$	IT=2781(5); Q rebuilt with Ame1971										MMC123**
$^{45}\text{S-u}$	-3610	2460	-3590#	320#	0.0	o			GA4	1.5	00Sa21
	-3330	2880			-0.1	o			GA5	1.5	00Sa21
	-4283	741			0.6	D			GA7	1.5	07Ju03 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{45}\text{Cl}-u$	-19690	140	-19610	150	0.4	o			GA4	1.5	00Sa21
	-20300	700			0.7	U			TO3	1.5	90Tu01
	-19850	460			0.4	o			GA5	1.5	00Sa21
	-19710	107			0.7	2			GA7	1.5	07Ju03
	-19098	236			-1.4	2			GA8	1.5	12Ga45
$^{45}\text{V}-u$	-34225.7	9.7	-34231.5	0.9	-0.6	U			LZ1	1.0	11Tu09
	-34230	11			-0.1	U			LZ1	1.0	18Zh29
$^{45}\text{Cr}-u$	-20390	540	-20950	40	-0.7	U			GT1	1.5	04St05 *
	-20950	38				2			LZ1	1.0	12Zh34 *
$^{45}\text{Ar}-^{39}\text{K}_{1.154}$	9922.45	0.55				2			MA8	1.0	03Bl17
$^{45}\text{K}-^{39}\text{K}_{1.154}$	2574.21	0.56				2			MA8	1.0	07Ya08
$^{45}\text{V}-^{45}\text{Ti}$	7647.74	0.23	7647.74	0.23	-0.0	1	100	100 ^{45}V	JY1	1.0	14Ka22
$^{45}\text{Sc}(p,\alpha)^{42}\text{Ca}$	2343	8	2339.0	0.7	-0.5	U			MIT		64Sp12
$^{45}\text{Sc}(d,\alpha)^{43}\text{Ca}$	8028	12	8047.4	0.7	1.6	U			MIT		64Sp12
	8059	12			-1.0	U			Kop		67Ha.A
$^{43}\text{Ca}(^3\text{He},p)^{45}\text{Sc}$	10310	20	10305.7	0.7	-0.2	U			Hei		70Sc22
$^{45}\text{Fe}(2p)^{43}\text{Cr}$	1140	40	1800#	200#	16.5	o					02Gi09
	1100	100			7.0	U					02Pf02
	1154	16			40.4	D					05Do20 *
$^{44}\text{Ca}(n,\gamma)^{45}\text{Ca}$	7414.8	1.0	7414.82	0.17	0.0	U					69Ar.A Z
	7414.83	0.3			-0.0	-			MMn		80Is02 Z
	7414.79	0.21			0.1	-			Bdn		06Fi.A
$^{44}\text{Ca}(d,p)^{45}\text{Ca}$	5184	4	5190.25	0.17	1.6	U			MIT		68Be36
$^{44}\text{Ca}(n,\gamma)^{45}\text{Ca}$	ave. 7414.80	0.17	7414.82	0.17	0.1	1	99	97 ^{45}Ca			average
$^{44}\text{Ca}(p,\gamma)^{45}\text{Sc}$	6887.8	1.2	6892.6	0.7	4.0	B					74Sc02 Z
$^{45}\text{Sc}(^3\text{He},\alpha)^{44}\text{Sc}$	9249	15	9250.0	1.9	0.1	U			MIT		71Ra09
$^{45}\text{Sc}(d,t)^{44}\text{Sc}^i$	-7846	10	-7848.1	2.6	-0.2	U					71Oh01 *
$^{45}\text{V}^i(p)^{44}\text{Ti}$	3190	50	3170	9	-0.4	U					74Ja10 *
	3170	9				3			Bor		07Do17 *
$^{45}\text{K}(\beta^-)^{45}\text{Ca}$	4180	200	4196.6	0.6	0.1	U					64Mo18
$^{45}\text{Ca}(\beta^-)^{45}\text{Sc}$	258	2	260.1	0.7	1.0	1	14	11 ^{45}Sc			65Fr12
$^{45}\text{Ti}(\beta^+)^{45}\text{Sc}$	2066	5	2062.1	0.5	-0.8	U					66Po04
$^{45}\text{Sc}(p,n)^{45}\text{Ti}$	-2844.2	4.	-2844.4	0.5	-0.1	U			Ric		55Br16 Y
	-2843.6	4.0			-0.2	U			Can		70Kn03
	-2844.4	0.5			-0.0	1	100	100 ^{45}Ti	PTB		85Sc16 Z
$^{45}\text{Sc}(^3\text{He},t)^{45}\text{Ti}^i$	-6801	4	-6800	3	0.3	1	61	60 $^{45}\text{Ti}^i$			71Be29 *
* $^{45}\text{S}-u$	Trends from Mass Surface TMS suggest ^{45}S 650 keV less bound										GAu **
* $^{45}\text{Cr}-u$	$M-A=-18940(500)$ keV for mixture gs+m at 107(1) keV										Nub211 **
* $^{45}\text{Cr}-u$	Same result in reference										13Ya03 **
* $^{45}\text{Fe}(2p)^{43}\text{Cr}$	Trends from Mass Surface TMS suggest ^{45}Fe 650 keV less bound										GAu **
* $^{45}\text{Sc}(d,t)^{44}\text{Sc}^i$	IT=2784(10) combined with $Q=-5062$; Q rebuilt										MMC123**
* $^{45}\text{V}^i(p)^{44}\text{Ti}$	$Q_p=2060(50)$ 2087(9) respectively, to 2^+ level at 1083.06 keV										Ens119 **
* $^{45}\text{Sc}(^3\text{He},t)^{45}\text{Ti}^i$	CDE=7571(4) $Q=-6807(4)$; recalibration +6 keV for $^{42}\text{Ca}(p,n)^{42}\text{Sc}$ from Ame1961										MMC123**
$^{46}\text{Cl}-u$	-16000	860	-14750	100	1.0	o			GA4	1.5	00Sa21
	-14940	1730			0.1	o			GA5	1.5	00Sa21
	-14826	172			0.3	2			GA7	1.5	07Ju03
	-15040	301			0.7	2			GA8	1.5	12Ga45
	-14708	118			-0.3	2			RT1	1.0	18Mi08
$^{46}\text{Ar}-u$	-32013	107	-31960.8	2.5	0.3	U			GT1	1.5	04Ma.A
$^{46}\text{Sc}-u$	-44650	230	-44833.0	0.7	-0.5	U			TO6	1.5	98Ba.A *
$\text{C}_2\text{H}_8\text{N}-^{46}\text{Ti}$	113071	7	113047.90	0.10	-0.8	U			R09	4.0	72De11
$\text{C}^{13}\text{CH}_5\text{O}-^{46}\text{Ti}$	84799	13	84768.26	0.10	-0.6	U			R09	4.0	72De11
$\text{CH}_4\text{NO}-^{46}\text{Ti}$	76672	8	76662.39	0.10	-0.3	U			R09	4.0	72De11
$\text{C}_5\text{H}_2-^{46}\text{TiO}$	68145	15	68109.09	0.10	-0.6	U			R09	4.0	72De11

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference		
C H ₂ O ₂ — ⁴⁶ Ti	52881	14	52852.95	0.10	−0.5	U			R09	4.0	72De11		
¹³ C H O ₂ — ⁴⁶ Ti	48423	9	48382.75	0.10	−1.1	U			R09	4.0	72De11		
⁴⁶ Ti— ²² Ne _{2.091}	−29358.74	0.48	−29359.92	0.10	−2.4	U			CP1	1.0	05Sa44		
⁴⁶ V— ²² Ne _{2.091}	−21787.10	0.58	−21788.88	0.15	−3.1	U			CP1	1.0	05Sa44		
⁴⁶ Cr—u	−31638	15	−31639	12	−0.1	1	67	67	⁴⁶ Cr	LZ1	1.0	17Zh12	
⁴⁶ Mn—u	−13331	93				2			LZ1	1.0	20Fu05	*	
⁴⁶ Ar— ³⁹ K _{1.179}	10827.5	1.2	10829.3	2.5	1.5	o			MA8	1.0	15Mo.A		
	10829.3	2.5				2			MA8	1.0	20Mo25		
⁴⁶ K— ³⁹ K _{1.179}	4771.64	0.78				2			MA8	1.0	07Ya08		
⁴⁶ Ti— ⁴⁸ Ti ₉₅₈	2499.165	0.062	2499.19	0.06	0.4	1	94	94	⁴⁶ Ti	MS1	1.0	17Ka53	
⁴⁶ V— ⁴⁶ Ti	7571.67	0.41	7571.03	0.10	−1.6	U			CP1	1.0	05Sa44		
	7571.41	0.33			−1.1	o			JY1	1.0	06Er08		
	7571.10	0.11			−0.6	1	81	87	⁴⁶ V	JY1	1.0	11Er02	
³² S(¹⁶ O,2n) ⁴⁶ Cr	−17421.6	20.	−17424	11	−0.1	1	33	33	⁴⁶ Cr			72Zi02	
⁴⁶ Ti(p,α) ⁴³ Sc	−3065	14	−3076.1	1.9	−0.8	U			MIT			64Sp12	*
	−3083	10			0.7	U			Tal			65Pi01	
⁴⁶ Ti(³ He, ⁶ He) ⁴³ Ti	−17470	12	−17474	6	−0.3	1	23	23	⁴³ Ti	MSU		77Mu03	*
⁴⁴ Ca(t,p) ⁴⁶ Ca	9339	20	9331.7	2.3	−0.4	U			Kop			67Bj06	
⁴⁴ Ca(³ He,p) ⁴⁶ Sc	7940	20	7935.2	0.7	−0.2	U			Hei			70Sc22	
⁴⁶ Ti(d,α) ⁴⁴ Sc	4400	12	4398.6	1.8	−0.1	U			Kop			67Ha.A	
⁴⁶ Ti(p,t) ⁴⁴ Ti	−14235	10	−14240.5	0.7	−0.6	U			Oak			72Ra05	
⁴⁶ Ca(t,α) ⁴⁵ K	5998	10	6000.9	2.3	0.3	U			Ald			68Sa09	
⁴⁶ Ca(d,t) ⁴⁵ Ca	−4144	10	−4141.5	2.3	0.3	U			Ald			67Bj05	
⁴⁶ Ca(³ He,α) ⁴⁵ Ca	10194	10	10178.9	2.3	−1.5	U			MIT			71Ra35	
⁴⁵ Sc(n,γ) ⁴⁶ Sc	8760.61	0.3	8760.64	0.10	0.1	2			BNn			80Li07	Z
	8760.58	0.14			0.4	2			Utr			82Ti02	Z
	8760.75	0.18			−0.6	2			Bdn			06Fi.A	
⁴⁵ Sc(d,p) ⁴⁶ Sc	6541	8	6536.07	0.10	−0.6	U			MIT			64Sp12	
	6543	8			−0.9	U			Kop			67Ha.A	
⁴⁵ Sc(p,γ) ⁴⁶ Ti	10344.7	0.7	10344.9	0.7	0.3	1	88	88	⁴⁵ Sc			71Gu.A	
⁴⁶ Ti(p,d) ⁴⁵ Ti ⁱ	−15682	5	−15684	3	−0.4	1	40	40	⁴⁵ Ti ⁱ			78Ko27	
⁴⁶ Cr ⁱ (p) ⁴⁵ V	4350	50	4269	13	−1.6	U			Lis			92Bo37	
	4269	13				2			Bor			07Do17	*
⁴⁶ Mn ⁱ (p) ⁴⁵ Cr	3520	100	4840	30	13.2	B			Lis			92Bo37	
	4840	33				3			Bor			07Do17	*
⁴⁶ K(β [−]) ⁴⁶ Ca	7650	300	7725.7	2.3	0.3	U						66Pa20	
⁴⁶ Ca(³ He,t) ⁴⁶ Sc ⁱ	−6407	4	−6410	3	−0.8	1	72	63	⁴⁶ Sc ⁱ			71Be29	*
⁴⁶ Sc(β [−]) ⁴⁶ Ti	2367	3	2366.6	0.7	−0.1	U						53Yo03	*
	2364	6			0.4	U						56Wo09	*
⁴⁶ Ti(p,n) ⁴⁶ V	−7844	9	−7834.72	0.09	1.0	U			Tal			63Ja12	
	−7835.8	1.8			0.6	U			Har			76Sq01	Z
⁴⁶ Ti(³ He,t) ⁴⁶ V	−7069.0	0.6	−7070.96	0.09	−3.3	F			Mun			77Vo02	*
⁴⁶ Ti(³ He,t) ⁴⁶ V— ²⁷ Al(²⁷ Si)	−2230.8	2.7	−2240.01	0.13	−3.4	B			ChR			74Ha35	
⁴⁶ Ti(³ He,t) ⁴⁶ V— ⁴⁷ Ti(⁴⁷ V)	−4121.62	0.19	−4121.83	0.11	−1.1	1	35	14	⁴⁷ V	Mun		09Fa15	*
⁴⁶ Ti(³ He,t) ⁴⁶ V— ⁴⁸ Ti(⁴⁸ V ⁱ)	−18.57	0.20	−18.58	0.20	−0.0	1	100	100	⁴⁸ V ⁱ	Mun		09Fa15	
⁴⁶ Ti(³ He,t) ⁴⁶ V— ⁵⁰ Ti(⁵⁰ V ⁱ)	−31.21	0.25	−31.21	0.25	0.0	1	100	100	⁵⁰ V ⁱ	Mun		09Fa.A	
* ⁴⁶ Sc—u	$M - A = -41520(210)$ keV for mixture gs+n at 142.528 keV										Nub211	**	
* ⁴⁶ Ti(p,α) ⁴³ Sc	$Q = -3217$ probably to ⁴³ Sc ^m at 151.79 keV										Nub211	**	
* ⁴⁶ Ti(³ He, ⁶ He) ⁴³ Ti	Averaged with reference ; Q reduced by 3 for recalibration ²⁷ Al(³ He, ⁶ He)										75Mu09	**	
* ⁴⁶ Cr ⁱ (p) ⁴⁵ V	$Q_p = 4254(15)$ 3494(25) 3003(13) to ground state, (5/2 ⁺) level at 797.2, (7/2 ⁺) at 1272.2 keV										MMC128**		
*											Ens082	**	
* ⁴⁶ Mn ⁱ (p) ⁴⁵ Cr	$Q_p = 4239(33)$ to (5/2 ⁺) state at 493.6 + 107(1)										11Ho02	**	
* ⁴⁶ Ca(³ He,t) ⁴⁶ Sc ⁱ	CDE=7177(4) $Q = -6413(4)$; recalibration +6 keV for ⁴² Ca(p,n) ⁴² Sc from Ame1961										MMC123**		
* ⁴⁶ Sc(β [−]) ⁴⁶ Ti	$E_{\beta^-} = 357(3)$ to 4 ⁺ level at 2009.846 keV										Ens00b	**	
* ⁴⁶ Sc(β [−]) ⁴⁶ Ti	$E_{\beta^-} = 1475(6)$ to 2 ⁺ level at 889.286 keV										Ens00b	**	
* ⁴⁶ Ti(³ He,t) ⁴⁶ V	F : rejected in reference of same group										09Fa15	**	
* ⁴⁶ Ti(³ He,t) ⁴⁶ V— ⁴⁷ Ti(⁴⁷ V)	$Q - Q = 28.73(0.16)$ keV to ⁴⁷ V ⁱ IAS at 4150.35(0.11) keV										Ens075	**	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{47}\text{Cl}-u$	-9576	1074	-10290#	220#	-0.4	D			GA7	1.5	07Ju03 *
$^{47}\text{Ar}-u$	-25400	600	-27232.9	1.3	-2.0	U			TO3	1.5	90Tu01
	-26570	1360			-0.3	U			GA5	1.5	00Sa21
	-26903	204			-1.1	U			GA8	1.5	12Ga45
$^{47}\text{Sc}-u$	-47630	230	-47597.6	2.1	0.1	U			TO6	1.5	98Ba.A *
$\text{C } ^{35}\text{Cl}-^{47}\text{Ti}$	17085.94	0.82	17095.20	0.09	4.5	B			H32	2.5	79Ko10
$\text{C } ^{13}\text{C H}_8 \text{ N}-^{47}\text{Ti}$	117329	14	117271.60	0.09	-1.0	U			R09	4.0	72De11
$\text{C}_2 \text{ H}_7 \text{ O}-^{47}\text{Ti}$	98012	7	97932.35	0.09	-2.8	B			R09	4.0	72De11
$\text{C}_5 \text{ H}_3 -^{47}\text{Ti O}$	76869	10	76802.98	0.09	-1.7	U			R09	4.0	72De11
$\text{C H}_3 \text{ O}_2 -^{47}\text{Ti}$	61608	10	61546.84	0.09	-1.5	U			R09	4.0	72De11
$^{47}\text{Cr}-u$	-37103.8	8.6	-37105	6	-0.1	-			LZ1	1.0	11Tu09
	-37107	11			0.2	-			LZ1	1.0	18Zh29
	ave. -37105	7			0.0	1	68	68 ^{47}Cr			average
$^{47}\text{Mn}-u$	-24226	34				2			LZ1	1.0	13Ya03
$^{47}\text{Ar}-^{39}\text{K}_{1.205}$	16501.8	1.2	16500.8	1.3	-0.8	o			MA8	1.0	15Mo.A
	16500.8	1.3				2			MA8	1.0	20Mo25
$^{47}\text{K}-^{39}\text{K}_{1.205}$	5398.5	2.7	5395.3	1.5	-1.2	o			TT1	1.0	10La.A
	5395.3	1.5				2			TT1	1.0	12La05
$^{46}\text{Ti } ^{13}\text{C}-^{47}\text{Ti C}$	4218.03	0.94	4223.70	0.07	2.4	U			H32	2.5	79Ko10
$^{47}\text{Ti}-^{48}\text{Ti}_{979}$	2723.600	0.058	2723.57	0.03	-0.6	1	35	37 ^{47}Ti	MS1	1.0	17Ka53
$^{47}\text{Ti}-^{46}\text{Ti}$	-929	41	-868.87	0.07	0.4	U			R09	4.0	72De11
$^{47}\text{Ti}(d,\alpha)^{45}\text{Sc}$	6830	12	6845.5	0.7	1.3	U			Kop		67Ha.A
$^{46}\text{Ar}(d,p)^{47}\text{Ar}$	1327	80	1442.8	2.6	1.4	U					06Ga28
$^{46}\text{Ca}(n,\gamma)^{47}\text{Ca}$	7277.4	0.6	7276.37	0.27	-1.7	-					70Cr04
	7276.1	0.3			0.9	-			Bdn		06Fi.A
$^{46}\text{Ca}(d,p)^{47}\text{Ca}$	5055	8	5051.81	0.27	-0.4	U			Kop		67Ha.A
	5044	4			2.0	U			MIT		68Be36
$^{46}\text{Ca}(n,\gamma)^{47}\text{Ca}$	ave. 7276.36	0.27	7276.37	0.27	0.1	1	100	90 ^{46}Ca			average
$^{46}\text{Ti}(n,\gamma)^{47}\text{Ti}$	8875.1	3.0	8880.66	0.06	1.9	U					69Te01
	8880.5	0.3			0.5	U			Bdn		06Fi.A
$^{46}\text{Ti}(d,p)^{47}\text{Ti}$	6658	6	6656.09	0.06	-0.3	U			MIT		67Ba32 *
	6659	8			-0.4	U			Kop		67Ba32 *
	6654.3	1.7			1.1	U			NDm		76Jo01
$^{46}\text{Ti}(d,p)^{47}\text{Ti}-^{48}\text{Ti}()^{49}\text{Ti}$	738.15	0.25	738.28	0.07	0.5	U			Mun		09Fa15
$^{46}\text{Ti}(p,\gamma)^{47}\text{V}$	5167.80	0.07	5167.77	0.07	-0.4	1	91	86 ^{47}V	Utr		86De13 *
$^{46}\text{Ti}(^3\text{He},d)^{47}\text{V}$	-317	15	-325.70	0.07	-0.6	U			MIT		67Do03
$^{47}\text{Mn}^i(p)^{46}\text{Cr}$	6867	20	6992	13	6.2	B			Bor		01Gi01 *
	6992	13				2			Bor		07Do17 *
$^{47}\text{K}(\beta^-)^{47}\text{Ca}$	6700	300	6632.7	2.6	-0.2	U					64Ku02 *
$^{47}\text{Ca}(\beta^-)^{47}\text{Sc}$	1984.6	5.	1992.2	1.2	1.5	U					67Hs03 *
	1992.3	5.			-0.0	U					68Fi04 *
	1991.9	1.2			0.2	1	97	91 ^{47}Ca			87Ju04
$^{47}\text{Sc}(\beta^-)^{47}\text{Ti}$	600	2	600.8	1.9	0.4	1	93	93 ^{47}Sc			56Gr12
$^{47}\text{V}(\beta^+)^{47}\text{Ti}$	2912	10	2930.54	0.09	1.9	U					54Da31
$^{47}\text{Ti}(p,n)^{47}\text{V}$	-3706	13	-3712.89	0.09	-0.5	U			Har		60Mc12 Y
$^{*47}\text{Cl}-u$	Trends from Mass Surface TMS suggest ^{47}Cl 660 keV more bound										GAu **
$^{*47}\text{Sc}-u$	$M-A=-44320(210)$ keV for mixture gs+m at 766.83 keV and										Nub211 **
$^{*47}\text{Cr}-u$	assuming ratio $R=0.07(3)$, from half-life=272 ns and TOF=1 μs										GAu **
$^{*46}\text{Ti}(d,p)^{47}\text{Ti}$	independent from 2011Tu09										WgM194**
$^{*46}\text{Ti}(p,\gamma)^{47}\text{V}$	All 67Ba32 results decreased 0.2% for recalibration										AHW **
$^{*47}\text{Mn}^i(p)^{46}\text{Cr}$	$E_p=985.94(0.05,Z)$ to $1/2^+$ level at 6132.60(0.09) keV										Ens075 **
$^{*47}\text{K}(\beta^-)^{47}\text{Ca}$	$Q_p=5975(25)$ 4880(20) to 2^+ level at 892.16 and (4^+) at 1987.1 keV;										Ens102 **
$^{*47}\text{Ca}(\beta^-)^{47}\text{Sc}$	data reanalysed and included in 07Do17										07Do17 **
	$Q_p=6104(24)$ 5000(15) to 2^+ level at 892.16 and (4^+) at 1987.1 keV										Ens102 **
	also tentatively $Q_p=3973(20)$ to (3^-) at 3196.5 keV, not used										MMC128**
	$E_{\beta^-}=4100(300)$ to 2578.33 $3/2^+$ and 2599.53 $1/2^+$ levels										Ens075 **
	Original values increased by 4(4) for shape factor										87Ju04 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{48}\text{Ar}-u$	-23920	330	-23999	18	-0.2	o			MT1	1.0	15Me01
	-23972	129			-0.2	U			RT1	1.0	18Mi08
	-24037	279			0.1	U			MT1	1.0	20Me06
	-23999	18				2			MR1	1.0	20Mo25
$^{48}\text{K}-^{39}\text{K}_{1.231}$	10017.7	2.5	10018.5	0.8	0.3	o			TT1	1.0	10La.A
	10018.50	0.83				2			TT1	1.0	12La05
$^{48}\text{Ca}-^{39}\text{K}_{1.231}$	-2799.93	0.22	-2800.031	0.020	-0.5	U			MS1	1.0	12Re17
$^{48}\text{Ca}-\text{N}^{18}\text{O O}$	-44625.15	0.29	-44625.582	0.019	-1.5	U			TT1	1.0	14Kw04
$^{48}\text{Ca}-\text{C}_4$	-47477.348	0.019	-47477.346	0.019	0.1	1	99	99 ^{48}Ca	SH2	1.0	16Ko45
$^{13}\text{C}^{35}\text{Cl}-^{48}\text{Ti}$	24261.73	0.75	24266.85	0.09	2.7	U			H32	2.5	79Ko10
$\text{C}_5\text{H}_4-^{48}\text{Ti O}$	88492	24	88444.83	0.08	-0.5	U			R09	4.0	72De11
	88494	27			-0.5	U			R09	4.0	72De11
$\text{C}_4\text{H}_2\text{N}-^{48}\text{Ti O}$	75935	17	75868.77	0.08	-1.0	U			R09	4.0	72De11
$\text{C}_4-^{48}\text{Ti}$	52109	19	52059.32	0.08	-0.7	U			R09	4.0	72De11
$^{48}\text{Ti O}-^{85}\text{Rb}_{.753}$	9277.7	1.2	9277.63	0.08	-0.1	U			MA8	1.0	12Na15
$^{48}\text{Ti}-\text{N}^{18}\text{O O}$	-49207.30	0.23	-49207.56	0.08	-1.1	1	12	12 ^{48}Ti	TT1	1.0	14Kw04
$^{48}\text{Mn}-u$	-31480	120	-31451	7	0.2	U			GT1	1.5	04St05
	-31454	10			0.3	1	52	52 ^{48}Mn	LZ1	1.0	17Zh12
$^{48}\text{Fe}-u$	-19333	99				2			LZ1	1.0	20Fu05
$^{48}\text{Ca}-^{40}\text{Ca}_{1.200}$	-2586.23	0.23	-2586.37	0.03	-0.6	U			MS1	1.0	12Re17
$^{48}\text{Ca}-^{41}\text{K}_{1.171}$	-2774.65	0.22	-2774.721	0.019	-0.3	U			MS1	1.0	12Re17
$^{48}\text{Ti O}-^{55}\text{Mn}_{1.164}$	14972.6	1.2	14973.2	0.3	0.5	1	7	7 ^{55}Mn	MA8	1.0	12Na15
$^{46}\text{Ti}^{37}\text{Cl}-^{48}\text{Ti}^{35}\text{Cl}$	1726.8	1.1	1735.56	0.09	2.0	U			H18	4.0	64Ba03
	1730.29	0.87			2.4	U			H32	2.5	79Ko10
$^{48}\text{Ti}-^{48}\text{Ca}$	-4582.018	0.086	-4581.98	0.08	0.5	1	82	81 ^{48}Ti	MS1	1.0	13Bu12
	-4581.86	0.22			-0.5	U			TT1	1.0	14Kw04
$^{48}\text{Ti}-^{47}\text{Ti}$	-3791	48	-3816.81	0.03	-0.1	U			R09	4.0	72De11
$^{48}\text{Ca}(^3\text{He}, ^{11}\text{C})^{40}\text{S}$	-17416	35	-17105	4	8.9	F			Pri		79Ko.B *
$^{48}\text{Ca}(^3\text{He}, ^8\text{B})^{43}\text{Cl}$	-29070	60	-28060	60	16.9	F			MSU		76Ka24 *
$^{48}\text{Ca}(\alpha, ^9\text{Be})^{43}\text{Ar}$	-21165.5	70.2	-21139	5	0.4	U			Brk		74Je01
$^{48}\text{Ca}(^3\text{He}, ^7\text{Be})^{44}\text{Ar}$	-12362	20	-12389.4	1.6	-1.4	U			MSU		76Cr03 *
$^{48}\text{Ca}(\alpha, ^7\text{Be})^{45}\text{Ar}$	-27840.4	60.2	-27798.1	0.5	0.7	U			Brk		74Je01
$^{48}\text{Ti}(p, \alpha)^{45}\text{Sc}$	-2560	5	-2556.6	0.7	0.7	U			ANL		64Yn03
	-2545	15			-0.8	U			Tal		65PI01
$^{48}\text{Ca}(^6\text{Li}, ^8\text{B})^{46}\text{Ar}$	-23324.8	70.2	-23288.3	2.5	0.5	U			Brk		74Je01
$^{48}\text{Ca}(^{14}\text{C}, ^{16}\text{O})^{46}\text{Ar}$	-6739	50	-6696.7	2.3	0.8	U			Mun		80Ma40
$^{48}\text{Ca}(d, \alpha)^{46}\text{K}$	1915	15	1899.9	0.7	-1.0	U			ANL		65Ma07
$^{46}\text{Ca}(t, p)^{48}\text{Ca}$	8752	20	8746.1	2.2	-0.3	U			Ald		67Bj06
$^{48}\text{Ti}(d, \alpha)^{46}\text{Sc}$	3967	12	3979.5	0.7	1.0	U			Kop		67Ha.A
$^{48}\text{Ti}(p, ^3\text{He})^{46}\text{Sc}^i$	-19394	6	-19387	4	1.2	1	37	37 $^{46}\text{Sc}^i$			78Ko27
$^{48}\text{Ti}(p, t)^{46}\text{Ti}^i$	-21192	7				2					78Ko27
$^{48}\text{Ti}(p, t)^{46}\text{Ti}^j$	-26177	6				2					78Ko27
$^{46}\text{Ti}(^3\text{He}, n)^{48}\text{Cr}$	5550	18	5553	7	0.2	R			CIT		67Mi02
$^{48}\text{Ni}(2p)^{46}\text{Fe}$	1400	100	2390#	300#	9.9	D					05Gi15 *
	1280	120			9.2	o					11Po09
	1280	60			18.5	o					12Po03 *
	1290	40			27.5	D					14Po05 *
$^{48}\text{Ca}(^{14}\text{C}, ^{15}\text{O})^{47}\text{Ar}$	-18142	100	-18693.3	1.3	-5.5	B			MSU		85Be50
$^{48}\text{Ca}(d, ^3\text{He})^{47}\text{K}$	-10304	12	-10308.4	1.4	-0.4	U			ANL		66Ne01
$^{48}\text{Ca}(t, \alpha)^{47}\text{K}$	4006	15	4012.0	1.4	0.4	U			LAI		66Wi11
	4001	10			1.1	U			Ald		68Sa09
$^{48}\text{Ca}(d, t)^{47}\text{Ca}$	-3699	10	-3694.3	2.2	0.5	U			ANL		66Er02
$^{48}\text{Ca}(^3\text{He}, \alpha)^{47}\text{Ca}$	10630	12	10626.1	2.2	-0.3	U			ANL		66Er02
	10642	10			-1.6	U			MIT		71Ra35
$^{47}\text{Ti}(n, \gamma)^{48}\text{Ti}$	11626.39	0.3	11626.66	0.03	0.9	U			MMn		80Is02 Z
	11626.65	0.04			0.2	1	63	61 ^{47}Ti	Ptn		84Ru06 Z
	11626.66	0.23			-0.0	U			Bdn		06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{47}\text{Ti}(\text{d,p})^{48}\text{Ti}$	9401	8	9402.09	0.03	0.1	U			Kop		67Ba32
	9403	6			-0.2	U			MIT		67Ba32
$^{47}\text{Ti}(^3\text{He,d})^{48}\text{V}$	1337	15	1335.9	1.0	-0.1	U			MIT		68Do06
$^{47}\text{Ti}(^3\text{He,d})^{48}\text{V}^i$	-1706	20	-1682.96	0.22	1.2	U					68Do06
$^{48}\text{Mn}^i(\text{p})^{47}\text{Cr}$	979.6	32.	1014	6	1.1	o			Bor		95B105
	979.6	33.			1.0	o			Bor		96Fa09 *
	1013	12			0.1	-			Bor		07Do17 *
	1018	10			-0.4	-					16Or03
ave.	1016	8			-0.2	1	67	49 $^{48}\text{Mn}^i$			average
$^{48}\text{K}(\beta^-)^{48}\text{Ca}$	12000	500	11940.4	0.8	-0.1	U					75Mu08
$^{48}\text{Ca}(^7\text{Li},^7\text{Be})^{48}\text{K}$	-12959	27	-12802.3	0.8	5.8	B			Can		78We14
$^{48}\text{Ca}(^{14}\text{C},^{14}\text{N})^{48}\text{K}$	-11910	50	-11783.9	0.8	2.5	U			Mun		80Ma40
$^{48}\text{Ca}(\text{p,n})^{48}\text{Sc}$	-534	15	-503	5	2.1	U					67Mc07 Z
	-506	7			0.4	1	50	50 ^{48}Sc			68Mc10
$^{48}\text{Sc}(\beta^-)^{48}\text{Ti}$	3986	7	3989	5	0.4	1	50	50 ^{48}Sc			57Va08 *
$^{48}\text{V}(\beta^+)^{48}\text{Ti}$	4008	5	4014.9	1.0	1.4	U					53Ma64 *
	4013.6	3.			0.4	1	10	10 ^{48}V			67Ko01 *
	4014	7			0.1	U					74Me15 *
$^{48}\text{Ti}(\text{p,n})^{48}\text{V}$	-4803	10	-4797.3	1.0	0.6	U			Tal		62Ne08 Y
$^{48}\text{Ti}(^3\text{He,t})^{48}\text{V}^i$	-7048	4	-7052.39	0.22	-1.1	U					71Be29 *
$^{48}\text{V}(\text{IT})^{48}\text{V}$	3018.7	1.0	3018.8	0.9	0.1	1	90	90 ^{48}V			Ens067
$^{48}\text{Mn}^i(\text{IT})^{48}\text{Mn}$	3036.7	0.9	3036.7	0.9	0.0	1	100	51 $^{48}\text{Mn}^i$			07Do17 *
$^{48}\text{Ca}(^3\text{He},^{11}\text{C})^{40}\text{S}$	F : possible ^{40}Ca contamination; mismatch in cross-sections										GAu **
$^{48}\text{Ca}(^3\text{He},^8\text{B})^{43}\text{Cl}$	F : poor spectrum. Authors say: possibly not to ground state										76Ka24 **
$^{48}\text{Ca}(^3\text{He},^7\text{Be})^{44}\text{Ar}$	$M - A = -32270(20)$ $Q = -12791(20)$ for ^7Be 429 keV level										GAu **
$^{48}\text{Ni}(2\text{p})^{46}\text{Fe}$	From only 1 event, Si detector										05Gi15 **
$^{48}\text{Ni}(2\text{p})^{46}\text{Fe}$	From 4 events, gaseous detector										12Po03 **
$^{48}\text{Ni}(2\text{p})^{46}\text{Fe}$	Trends from Mass Surface TMS suggest ^{48}Ni 1100 keV less bound										GAu **
$^{48}\text{Mn}^i(\text{p})^{47}\text{Cr}$	Unexpectedly low intensity 3.6(1.1)%										96Fa09 **
$^{48}\text{Mn}^i(\text{p})^{47}\text{Cr}$	Measured intensity 1.8(0.3)%										07Do17 **
$^{48}\text{Sc}(\beta^-)^{48}\text{Ti}$	$E_{\beta^-} = 654(7)$ to 6^+ level at 3333.196 keV										Ens067 **
$^{48}\text{V}(\beta^+)^{48}\text{Ti}$	$E_{\beta^+} = 692(5)$ 698(3) 698(7) respectively, to 4^+ level at 2295.654 keV										Ens067 **
$^{48}\text{Ti}(^3\text{He,t})^{48}\text{V}^i$	CDE=7818(4) $Q = -7054(4)$; recalibration +6 keV for $^{42}\text{Ca}(\text{p,n})^{42}\text{Sc}$ from Ame1961										MMC123**
$^{48}\text{Mn}^i(\text{IT})^{48}\text{Mn}$	γ cascade 2633.5(0.5)+89.9(0.6)+313.3(0.4)										GAu **
$^{49}\text{Ar}-\text{u}$	-19110	1180	-18320#	430#	0.7	o			MT1	1.0	15Me01 *
	-17499	1180			-0.7	D			MT1	1.0	20Me06 *
$^{49}\text{K}-\text{u}$	-31981	225	-31789.2	0.9	0.6	U			GT1	1.5	04Ma.A
$^{49}\text{K}-^{39}\text{K}_{1.256}$	13794.9	2.8	13795.4	0.9	0.2	o			TT1	1.0	10La.A
	13795.41	0.86				2			TT1	1.0	12La05
$^{49}\text{Ca}-^{39}\text{K}_{1.256}$	1247.1	2.9	1247.28	0.19	0.1	o			TT1	1.0	10La.A
	1247.1	1.2			0.2	U			TT1	1.0	12La05
$^{49}\text{Sc}-\text{u}$	-49989.3	4.5	-49986.8	2.4	0.5	1	29	29 ^{49}Sc	MR1	1.0	20Ku19
$\text{C H}_2 ^{35}\text{Cl}-^{49}\text{Ti}$	36637	13	36638.37	0.09	0.0	U			R09	4.0	72De11
$\text{C}_4 \text{H}-^{49}\text{Ti}$	59967	10	59960.64	0.08	-0.2	U			R09	4.0	72De11
$\text{C}_5 \text{H}_5-^{49}\text{Ti O}$	96348	19	96346.15	0.08	-0.0	U			R09	4.0	72De11
$\text{C H}_5 ^{32}\text{S}-^{49}\text{Ti}$	63365	14	63331.94	0.08	-0.6	U			R09	4.0	72De11
$^{49}\text{Mn}-\text{u}$	-40410	12	-40386.6	2.4	1.9	U			LZ1	1.0	11Tu09
	-40373	15			-0.9	U			LZ1	1.0	18Zh29
$^{49}\text{Fe}-\text{u}$	-26571	26				2			LZ1	1.0	12Zh34 *
$^{47}\text{Ti } ^{37}\text{Cl}-^{49}\text{Ti } ^{35}\text{Cl}$	946.4	1.1	942.98	0.08	-0.8	U			H18	4.0	64Ba03
	944.46	0.35			-1.7	U			H32	2.5	79Ko10
$^{48}\text{Ti } ^{13}\text{C}-^{49}\text{Ti C}$	3432.64	0.80	3431.122	0.025	-0.8	U			H32	2.5	79Ko10
$^{48}\text{Ti H}-^{49}\text{Ti}$	7876	7	7901.319	0.025	0.9	U			R09	4.0	72De11
	7874	27			0.3	U			R09	4.0	72De11

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{49}\text{Ti}-^{48}\text{Ti}_{1.021}$	1016.967	0.053	1016.961	0.025	-0.1	1	23	20	^{49}Ti	MS1	1.0	17Ka53
$^{49}\text{Mn}-^{49}\text{Cr}$	8279.63	0.25	8279.63	0.25	-0.0	1	100	100	^{49}Mn	JY1	1.0	14Ka22
$^{49}\text{Ti}-^{48}\text{Ti}$	-43	36	-76.287	0.025	-0.2	U				R09	4.0	72De11
$^{49}\text{Ti}(\text{d},\alpha)^{47}\text{Sc}$	6476	12	6483.6	1.9	0.6	U				Kop		67Ha.A
$^{48}\text{Ca}(\text{n},\gamma)^{49}\text{Ca}$	5146.6	0.7	5146.45	0.18	-0.2	2						69Ar.A Z
	5146.38	0.30			0.2	2						70Cr04 Z
	5146.48	0.23			-0.1	2				Bdn		06Fi.A
$^{48}\text{Ca}(\text{d},\text{p})^{49}\text{Ca}$	2917	7	2921.89	0.18	0.7	U				ANL		66Er02
	2917	4			1.2	U				MIT		68Be36
$^{48}\text{Ca}(\text{p},\gamma)^{49}\text{Sc}$	9628.7	3.6	9626.6	2.3	-0.6	-						68Vi01 Z
$^{48}\text{Ca}(\text{d},\text{n})^{49}\text{Sc}$	7404	7	7402.0	2.3	-0.3	-						68Gr09
$^{48}\text{Ca}(\text{}^3\text{He},\text{d})^{49}\text{Sc}$	4150	12	4133.1	2.3	-1.4	U				ANL		66Er02
$^{48}\text{Ca}(\text{p},\gamma)^{49}\text{Sc}$	ave. 9629	3	9626.6	2.3	-0.7	1	50	50	^{49}Sc			average
$^{48}\text{Ti}(\text{n},\gamma)^{49}\text{Ti}$	8142.22	0.3	8142.379	0.023	0.5	U				MMn		80Is02 Z
	8142.39	0.03			-0.4	1	61	68	^{49}Ti	Ptn		83Ru08 Z
	8142.35	0.16			0.2	U				Bdn		06Fi.A
$^{48}\text{Ti}(\text{d},\text{p})^{49}\text{Ti}$	5907	8	5917.812	0.023	1.4	U				Kop		67Ba32
	5918	6			-0.0	U				MIT		67Ba32
	5918.6	1.7			-0.5	U				NDm		76Jo01
$^{48}\text{Ti}(\text{p},\gamma)^{49}\text{V}$	6756.8	1.5	6758.2	0.8	0.9	R						72Ki06
$^{49}\text{Mn}^i(\text{p})^{48}\text{Cr}$	2712.2	50.	2729	16	0.3	U						70Ce02 *
	2730	29			-0.0	o				Bor		96Fa09 *
	2729	16				3				Bor		07Do17 *
$^{49}\text{K}(\beta^-)^{49}\text{Ca}$	10970	70	11688.5	0.8	10.3	B						86Mi08
$^{49}\text{Ca}(\beta^-)^{49}\text{Sc}$	5200	100	5262.4	2.3	0.6	U						56Ma27
	4970	50			5.8	B						56Ok02
$^{49}\text{Sc}(\beta^-)^{49}\text{Ti}$	2010	5	2001.6	2.3	-1.7	1	21	21	^{49}Sc			61Re06
	1983	7			2.7	B						69Fl02
$^{49}\text{V}(\epsilon)^{49}\text{Ti}$	626	10	601.9	0.8	-2.4	U						56Ha59
$^{49}\text{Ti}(\text{p},\text{n})^{49}\text{V}$	-1383	9	-1384.2	0.8	-0.1	U				Har		60Mc12 Z
	-1383.6	1.0			-0.6	2				Oak		64Jo11 Z
$^{49}\text{Ti}(\text{}^3\text{He},\text{t})^{49}\text{Vi}$	-7052	4				2						71Be29 *
$^{49}\text{Cr}(\beta^+)^{49}\text{V}$	2590	20	2629.8	2.3	2.0	U						53Cr18 *
$^{*49}\text{Ar}-\text{u}$	Trends from Mass Surface TMS suggest ^{49}Ar 760 keV more bound											GAu **
$^{*49}\text{Fe}-\text{u}$	Same result in reference											13Ya03 **
$^{*49}\text{Mn}^i(\text{p})^{48}\text{Cr}$	$Q_p=1960(50)$ 1978(29) 1977(16) respectively, to 2^+ level at 752.19(0.11) keV											Ens067 **
$^{*49}\text{Ti}(\text{}^3\text{He},\text{t})^{49}\text{Vi}$	CDE=7822(4) $Q=-7058(4)$; recalibration +6 keV for $^{42}\text{Ca}(\text{p},\text{n})^{42}\text{Sc}$ from Ame1961											MMC123**
$^{*49}\text{Cr}(\beta^+)^{49}\text{V}$	$E_{\beta^+}=1540(10)$ 1390(20) to $(7/2^-)$ ground state + $(5/2^-)$ 90.6392 and $3/2^-$ at 152.9282											Ens089 **
$^{50}\text{K}-\text{u}$	-26100	800	-27620	8	-1.3	U				TO3	1.5	90Tu01
$^{50}\text{K}-^{39}\text{K}_{1.282}$	18899	11	18908	8	0.8	o				TT1	1.0	10La.A
	18908.3	8.3				2				TT1	1.0	12La05
$^{50}\text{Ca}-^{39}\text{K}_{1.282}$	4027.0	4.0	4027.5	1.7	0.1	o				TT1	1.0	10La.A
	4027.5	1.7				2				TT1	1.0	12La05
$^{50}\text{Sc}-\text{u}$	-47940	250	-47812.6	2.7	0.3	U				TO6	1.5	98Ba.A *
	-47822.6	9.8			1.0	U				MR1	1.0	20Ku19
$\text{C H}_3 \text{}^{35}\text{Cl}-^{50}\text{Ti}$	47550	23	47542.17	0.10	-0.1	U				R09	4.0	72De11
$\text{C}_4 \text{H}_2-^{50}\text{Ti}$	70860	8	70864.44	0.09	0.1	U				R09	4.0	72De11
$\text{C}_5 \text{H}_6-^{50}\text{Ti O}$	107253	18	107249.95	0.09	-0.0	U				R09	4.0	72De11
$\text{C}_3 \text{}^{13}\text{C H}-^{50}\text{Ti}$	66401	21	66394.25	0.09	-0.1	U				R09	4.0	72De11
$\text{C}_3 \text{N}-^{50}\text{Ti}$	58279	43	58288.38	0.09	0.1	U				R09	4.0	72De11
$\text{C}_4 \text{H}_2-^{50}\text{V}$	68485	14	68493.38	0.10	0.1	U				R09	4.0	72De11
$\text{C}_3 \text{N}-^{50}\text{V}$	55903	23	55917.32	0.10	0.2	U				R09	4.0	72De11
$\text{C H}_3 \text{}^{35}\text{Cl}-^{50}\text{V}$	45158	17	45171.11	0.11	0.2	U				R09	4.0	72De11
$\text{C}_4 \text{H}_2-^{50}\text{Cr}$	69608	8	69607.85	0.10	-0.0	U				R09	4.0	72De11

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
C ₃ N- ⁵⁰ Cr	57051	7	57031.80	0.10	-0.7	U			R09	4.0	72De11
C H ₃ ³⁵ Cl- ⁵⁰ Cr	46290	14	46285.58	0.11	-0.1	U			R09	4.0	72De11
⁵⁰ Fe-u	-37012	9				2			LZ1	1.0	17Zh12
⁵⁰ Co-u	-18883	135				2			LZ1	1.0	20Fu05 *
⁵⁰ Ti- ⁴⁸ Ti _{1.042}	-968.526	0.048	-968.56	0.03	-0.7	1	50	45 ⁵⁰ Ti	MS1	1.0	17Ka53
⁵⁰ V- ⁴⁸ Ti _{1.042}	1402.62	0.11	1402.50	0.06	-1.1	1	27	26 ⁵⁰ V	MS1	1.0	17Ka53
⁵⁰ Cr- ⁴⁸ Ti _{1.042}	287.979	0.072	288.03	0.06	0.7	1	65	64 ⁵⁰ Cr	MS1	1.0	17Ka53
⁴⁹ Ti ¹³ C- ⁵⁰ Ti C	6440.47	0.88	6433.60	0.03	-3.1	B			H32	2.5	79Ko10
⁵⁰ V- ⁵⁰ Ti	2371.14	0.12	2371.06	0.06	-0.7	1	27	21 ⁵⁰ V	MS1	1.0	17Ka53
⁵⁰ V- ⁵⁰ Cr	1114.414	0.075	1114.47	0.06	0.8	1	62	32 ⁵⁰ Cr	MS1	1.0	17Ka53
⁵⁰ Mn- ⁵⁰ Cr	8195.91	0.10	8195.95	0.07	0.4	1	52	52 ⁵⁰ Mn	JY1	1.0	08Er04
⁵⁰ Mn ^m - ⁵⁰ Cr	8437.852	0.065	8437.83	0.06	-0.3	1	81	81 ⁵⁰ Mn ^m	JY1	1.0	08Er04
⁵⁰ Mn ^m - ⁵⁰ Mn	241.840	0.100	241.88	0.07	0.4	1	55	37 ⁵⁰ Mn	JY1	1.0	08Er04
⁵⁰ Ti- ⁴⁹ Ti	-3075	38	-3078.77	0.03	-0.0	U			R09	4.0	72De11
⁵⁰ Sc O- ¹⁹ F ³⁵ Cl	-20153.8	2.7	-20153.8	2.7	-0.0	1	100	100 ⁵⁰ Sc	MS1	1.0	20Le.A
⁵⁰ Cr(p, ⁶ He) ⁴⁵ V	-28686	17	-28678.0	0.9	0.5	U			MSU		75Mu09 *
⁵⁰ Ti(p, α) ⁴⁷ Sc	-2231	15	-2231.0	1.9	0.0	U			Tal		65PI01
⁵⁰ V(p, α) ⁴⁷ Ti	572	23	578.43	0.06	0.3	U			MIT		67Sp09
⁵⁰ Cr(p, α) ⁴⁷ V	-3387	10	-3390.24	0.10	-0.3	U			Ald		66Br06
⁵⁰ Cr(³ He, ⁶ He) ⁴⁷ Cr	-18365	14	-18359	5	0.4	1	14	14 ⁴⁷ Cr	MSU		77Mu03 *
⁴⁸ Ca(t, p) ⁵⁰ Ca	3012	15	3025.2	1.6	0.9	U			Ald		66Hi01
	3020	10			0.5	U			LAI		66Wi11
⁴⁸ Ca(³ He, p) ⁵⁰ Sc	7965	15	7954.5	2.5	-0.7	U			ANL		69Oh01
⁵⁰ V(d, α) ⁴⁸ Ti	9982	15	9980.52	0.05	-0.1	U			MIT		66Do06
	9988	20			-0.4	U			Kop		67Ha.A
⁵⁰ Cr(d, α) ⁴⁸ V	4928	12	4927.4	1.0	-0.0	U			Kop		67Ha.A
	4923	15			0.3	U			MIT		68Do03
⁵⁰ Cr(d, α) ⁴⁸ V ⁱ	1880	11	1908.60	0.23	2.6	U			MIT		68Do03 *
⁵⁰ Cr(p, t) ⁴⁸ Cr	-15100	8	-15101	7	-0.1	2			Oak		71Do18
	-15100	30			-0.0	U			Bld		72Sh27
⁵⁰ Cr(p, t) ⁴⁸ Cr ^j	-23861	15				2			MSU		75Mo26 *
⁵⁰ Co ⁱ (2p) ⁴⁸ Mn	1972	13				2			Bor		07Do17
⁵⁰ Ti(t, α) ⁴⁹ Sc	7644	25	7655.5	2.3	0.5	U			LAI		66Wi11
⁴⁹ Ti(n, γ) ⁵⁰ Ti	10939.6	0.3	10939.17	0.03	-1.4	U			MMn		80Is02 Z
	10939.19	0.04			-0.4	1	60	49 ⁵⁰ Ti	Ptn		84Ru06 Z
	10939.20	0.22			-0.1	U			Bdn		06Fi.A
⁴⁹ Ti(d, p) ⁵⁰ Ti	8723	8	8714.61	0.03	-1.0	U			Kop		67Ba32
	8721	6			-1.1	U			MIT		67Ba32
⁵⁰ Cr(p, d) ⁴⁹ Cr	-10790	30	-10775.8	2.2	0.5	U			Pri		67Wh03
⁵⁰ Cr(d, t) ⁴⁹ Cr	-6743.1	2.2	-6743.1	2.2	0.0	1	100	100 ⁴⁹ Cr	NDm		76Jo01
⁵⁰ Fe ⁱ (p) ⁴⁹ Mn	4389	41	4332	10	-1.4	o			Bor		96Fa09 *
	4332	10				2			Bor		07Do17 *
⁵⁰ K(β^-) ⁵⁰ Ca	14050	300	13861	8	-0.6	U					86Mi08
⁵⁰ Sc(β^-) ⁵⁰ Ti	6500	200	6894.7	2.5	2.0	U					63Ch03
	6260	100			6.3	B					69Wa24
⁵⁰ V(n, p) ⁵⁰ Ti	2979	15	2990.97	0.06	0.8	U			ILL		81Wa31
	2984	10			0.7	U			ILL		94Wa17
⁵⁰ Ti(p, n) ⁵⁰ V	-2991	10	-2990.97	0.06	0.0	U			Har		60Mc12 Y
⁵⁰ Ti(³ He, t) ⁵⁰ V ⁱ	-7032	4	-7039.75	0.27	-1.9	U					71Be29 *
⁵⁰ Cr(p, n) ⁵⁰ Mn	-8416.1	1.9	-8416.82	0.07	-0.4	U			Har		75Fr.A
⁵⁰ Cr(³ He, t) ⁵⁰ Mn	-7650.5	0.4	-7653.07	0.07	-6.4	F			Mun		77Vo02 *
⁵⁰ Cr(³ He, t) ⁵⁰ Mn- ²⁷ Al(²⁷ Si	-2820.0	2.8	-2822.12	0.12	-0.8	U			ChR		74Ha35
⁵⁰ Cr(³ He, t) ⁵⁰ Mn- ⁴² Ca(⁴² Sc	-1207.6	2.3	-1208.19	0.08	-0.3	U			ChR		74Ha35
⁵⁰ Cr(³ He, t) ⁵⁰ Mn- ⁵⁴ Fe(⁵⁴ Co	610.09	0.17	610.07	0.10	-0.1	1	35	23 ⁵⁴ Co	ChR		87Ko34 *
* ⁵⁰ Sc-u	$M-A=-44530(220)$ keV for mixture gs+m at 256.895 keV										Nub211 **
* ⁵⁰ Co-u	Authors suggested ground state										20Fu05 **
* ⁵⁰ Cr(p, ⁶ He) ⁴⁵ V	Original Q increase by 1 for recalibration										AHW **
* ⁵⁰ Cr(³ He, ⁶ He) ⁴⁷ Cr	Original Q reduced by 3, see ⁴⁶ Ti(³ He, ⁶ He)										AHW **
* ⁵⁰ Cr(d, α) ⁴⁸ V ⁱ	IT=3043(9); rebuilt from their $Q_{gs}=4923(15)$ keV										MMC124**
* ⁵⁰ Cr(p, t) ⁴⁸ Cr ^j	Strongest of two fragments given as IT=8760(15); Q rebuilt with Ame1971										75Mo26 **
* ⁵⁰ Fe ⁱ (p) ⁴⁹ Mn	$E_p=2790(41)$ to $11/2^{(-)}$ level at 1541.3125 keV										Ens089 **
* ⁵⁰ Fe ⁱ (p) ⁴⁹ Mn	$Q_p=2770(12)$ 41.1%, 1874(16) 1.0% to $11/2^{(-)}$ level at 1541.3125, and										Ens089 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
* * ⁵⁰ Ti(³ He,t) ⁵⁰ Vi * ⁵⁰ Cr(³ He,t) ⁵⁰ Mn * ⁵⁰ Cr(³ He,t) ⁵⁰ Mn– ⁵⁴ Fe(⁵⁴ Co)	13/2 ⁽⁻⁾ at 2481.3 keV										Ens089 **
	CDE=7802(4) $Q=-7038(4)$; recalibration +6 keV for ⁴² Ca(p,n) ⁴² Sc from Ame1961										MMC123**
	F : rejected in reference of same group										09Fa15 **
	$Q-Q=40.90(0.16)$ to $650.99(0.06)$ level in ⁵⁰ Mn										Ens10c **
⁵¹ Ca– ³⁹ K _{1.308}	8467.58	0.56				2			MA8	1.0	13Wi06
⁵¹ Ca–u	–38800	350	–39004.3	0.6	–0.4	U			TO3	1.5	90Tu01
	–38900	400			–0.2	U			TO5	1.5	94Se12
	–39249	183			0.9	U			GT1	1.5	04Ma.A
⁵¹ Sc H– ¹⁹ F ³³ S	–8468.2	2.7	–8468.2	2.7	–0.0	1	100	100 ⁵¹ Sc	MS1	1.0	20Le.A
⁵¹ Ti– ³⁹ K _{1.308}	–5917.0	2.3	–5918.6	0.5	–0.7	1	5	5 ⁵¹ Ti	TT1	1.0	18Re11
⁵¹ Ti–u	–53378.8	16.1	–53390.5	0.5	–0.7	U			TR1	1.0	18Le03
	–53389.0	2.3			–0.7	o			TT1	1.0	18Le03 *
⁵¹ V– ³⁹ K _{1.308}	–8571.23	0.73	–8570.42	0.10	1.1	–			TT1	1.0	14Ma21
	–8570.68	0.99			0.3	–			TT1	1.0	14Ma21
	–8570.9	2.0			0.2	U			TT1	1.0	18Re11
ave.	–8571.0	0.6			1.1	1	3	3 ⁵¹ V			average
C ₄ H ₃ – ⁵¹ V	79526	9	79517.43	0.10	–0.2	U			R09	4.0	72De11
C ₅ H ₇ – ⁵¹ V O	115921	13	115902.94	0.10	–0.3	U			R09	4.0	72De11
C ₄ H ₅ N– ⁵¹ V O	103334	13	103326.88	0.10	–0.1	U			R09	4.0	72De11
C ₃ H N– ⁵¹ V	66943	7	66941.37	0.10	–0.1	U			R09	4.0	72De11
⁵¹ V–u	–56042.8	1.9	–56042.34	0.10	0.2	o			TT1	1.0	18Le03
⁵¹ Cr– ³⁹ K _{1.308}	–7763.66	0.78	–7762.69	0.18	1.2	–			TT1	1.0	14Ma21
	–7764.0	1.2			1.1	–			TT1	1.0	14Ma21
ave.	–7763.8	0.7			1.6	1	7	7 ⁵¹ Cr			average
⁵¹ Fe–u	–43148	12	–43144.9	1.5	0.3	U			LZ1	1.0	11Tu09
	–43154	15			0.6	U			LZ1	1.0	18Zh29
C H ₂ ³⁷ Cl– ⁵¹ Fe	24697.5	1.5				2			MS1	1.0	18On01
⁵¹ Co–u	–29353	52				2			LZ1	1.0	14Sh14
⁵¹ Ca– ⁵⁸ Ni ₈₇₉	17823	24	17830.4	0.6	0.3	U			TT1	1.0	12Ga29
⁴⁷ Ti ³⁷ Cl ₂ – ⁵¹ V ³⁵ Cl ₂	1906.	1.8	1899.59	0.16	–0.9	U			H18	4.0	64Ba03
⁵¹ V– ⁴⁸ Ti _{1.063}	–703.292	0.082	–703.28	0.06	0.2	1	61	58 ⁵¹ V	MS1	1.0	17Ka53
⁴⁹ Ti ³⁷ Cl– ⁵¹ V ³⁵ Cl	956.7	0.7	956.61	0.10	–0.0	U			H18	4.0	64Ba03
⁵¹ K– ⁵¹ V	31871	14				2			TT1	1.0	12Ga29
⁵¹ Cr– ⁵¹ V	807.46	0.71	807.72	0.16	0.4	–			TT1	1.0	14Ma21
	806.3	1.2			1.2	–			TT1	1.0	14Ma21
ave.	807.2	0.6			0.9	1	7	6 ⁵¹ Cr			average
⁴⁸ Ca(¹⁴ C, ¹¹ C) ⁵¹ Ca	–15900	150	–15522.1	0.5	2.5	U			Mun		80Ma40 *
	–16886	100			13.6	B			MSU		85Be50
⁴⁸ Ca(¹⁸ O, ¹⁵ O) ⁵¹ Ca	–12040	120	–11531.0	0.7	4.2	B			Hei		85Br03 *
	–13900	40			59.2	B			Can		88Ca21
⁴⁸ Ca(α ,p) ⁵¹ Sc	–5860	20	–5838.6	2.5	1.1	U			ANL		66Er02
⁵¹ V(p, α) ⁴⁸ Ti	1162	10	1153.90	0.06	–0.8	U			MIT		64Sp12
⁵¹ V(d, α) ⁴⁹ Ti	7066	12	7071.71	0.06	0.5	U			Kop		67Ha.A
⁵⁰ Ti(n, γ) ⁵¹ Ti	6372.3	1.2	6372.4	0.5	0.1	–					71Ar39 Z
	6372.6	0.6			–0.3	–			Bdn		06Fi.A
⁵⁰ Ti(d,p) ⁵¹ Ti	4143	6	4147.8	0.5	0.8	U			MIT		67Ba32
	4148	8			–0.0	U			Kop		67Ba32
	4147.7	1.2			0.1	–			NDm		76Jo01
⁵⁰ Ti(n, γ) ⁵¹ Ti	6372.5	0.5	6372.4	0.5	–0.2	1	95	95 ⁵¹ Ti			average
⁵⁰ Ti(p, γ) ⁵¹ V	8063.3	2.0	8060.21	0.07	–1.5	U					70K105 Z
	8063.6	2.0			–1.7	U					70Ma36 Z
⁵⁰ Ti(³ He,d) ⁵¹ V	2555	15	2566.73	0.07	0.8	U			MIT		67Ob04
⁵⁰ V(n, γ) ⁵¹ V	11051.18	0.10	11051.18	0.06	0.0	–			MMn		78Ro03 Z
	11051.05	0.17			0.8	–			ILn		91Mi08 Z
	11051.14	0.22			0.2	–			Bdn		06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{51}\text{V}(\gamma, n)^{50}\text{V}$	-11040	60	-11051.18	0.06	-0.2	U			Phi		60Ge01
$^{50}\text{V}(\text{d}, \text{p})^{51}\text{V}$	8840	15	8826.62	0.06	-0.9	U			MIT		67De02
	8828	20			-0.1	U			Kop		67Ha.A
$^{51}\text{V}(\text{p}, \text{d})^{50}\text{V}$	-8815	20	-8826.62	0.06	-0.6	U			Oak		65Ba29
$^{50}\text{V}(\text{n}, \gamma)^{51}\text{V}$	ave. 11051.15	0.08	11051.18	0.06	0.5	1	58	35 ^{51}V			average
$^{50}\text{V}(^3\text{He}, \text{d})^{51}\text{Cr}$	4031	12	4022.97	0.15	-0.7	U			MIT		69Do01
$^{50}\text{Cr}(\text{n}, \gamma)^{51}\text{Cr}$	9261.71	0.30	9260.67	0.15	-3.5	B			MMn		80Is02 Z
	9260.63	0.20			0.2	1	55	51 ^{51}Cr	Bdn		06Fi.A
$^{50}\text{Cr}(\text{d}, \text{p})^{51}\text{Cr}$	7049	8	7036.10	0.15	-1.6	U			Kop		67Ha.A
	7041	6			-0.8	U			MIT		68Ro09
$^{50}\text{Cr}(\text{p}, \gamma)^{51}\text{Mn}$	5270.8	0.3	5270.83	0.29	0.1	1	94	93 ^{51}Mn			72Fo25 Z
$^{50}\text{Cr}(^3\text{He}, \text{d})^{51}\text{Mn}$	-206	15	-222.64	0.29	-1.1	U			MIT		67Sp09
$^{50}\text{Cr}(\text{p}, \gamma)^{51}\text{Mn}^i$	819	2	820.2	1.5	0.6	2					72Fo25
$^{50}\text{Cr}(^3\text{He}, \text{d})^{51}\text{Mn}^i$	-4652	20	-4673.3	1.5	-1.1	U			MIT		67Ra14
	-4671.7	2.3			-0.7	2					79Pa14 *
$^{51}\text{Co}^i(\text{p})^{50}\text{Fe}$	6513	16				3			Bor		07Do17 *
$^{51}\text{Ti}(\beta^-)^{51}\text{V}$	2440	30	2470.1	0.5	1.0	U					55Bu01
	2450	30			0.7	U					55Ma01
$^{51}\text{Cr}(\epsilon)^{51}\text{V}$	756	5	752.39	0.15	-0.7	U					55Bi29
$^{51}\text{V}(\text{p}, \text{n})^{51}\text{Cr}$	-1533.5	2.0	-1534.74	0.15	-0.6	U			Nvl		59Go68 Z
	-1533.3	1.8			-0.8	U			Oak		64Jo11 Z
	-1533.7	1.5			-0.7	U			Can		70Kn03 Z
	-1534.93	0.24			0.8	1	39	35 ^{51}Cr	PTB		89Sc24 Z
$^{51}\text{V}(^3\text{He}, \text{t})^{51}\text{Cr}^i$	-7384	5				2					71Be29
$^{51}\text{Mn}(\beta^+)^{51}\text{Cr}$	3232	20	3207.5	0.3	-1.2	U					66Gi02
$^{51}\text{Ti}-\text{u}$	Frequency ratios given in 18Re11										HWJ192 **
$^{48}\text{Ca}(^{14}\text{C}, ^{11}\text{C})^{51}\text{Ca}$	May be a ^{40}Ca contamination. There is a -16900(150) peak										85Be50 **
$^{48}\text{Ca}(^{18}\text{O}, ^{15}\text{O})^{51}\text{Ca}$	Proposed 970(90) level reinterpreted as ground state in reference										85Be50 **
$^{48}\text{Ca}(^{18}\text{O}, ^{15}\text{O})^{51}\text{Ca}$	Weak $M-A=-36120(120)$ level disregarded										AHW **
$^{50}\text{Cr}(^3\text{He}, \text{d})^{51}\text{Mn}^i$	IT=4449(3); Q rebuilt with Ame1977										MMC124**
$^{51}\text{Co}^i(\text{p})^{50}\text{Fe}$	$Q_p=4662(16)$ to (4^+) level at 1851.5 keV										Ens10c **
$^{51}\text{Co}^i(\text{p})^{50}\text{Fe}$	$Q_p=6153$ was a misprint in Ame2012										XuX16b **
$^{51}\text{V}(^3\text{He}, \text{t})^{51}\text{Cr}^i$	CDE=8145(5) $Q=-7881(5)$; recalibration -3 keV for $^{50}\text{Cr}(\text{p}, \text{n})^{50}\text{Mn}$ from Ame1961										MMC123**
$^{52}\text{K}-\text{u}$	-18398	36				2			MR1	1.0	15Ro10
$^{52}\text{Ca}-^{39}\text{K}_{1.333}$	11592.90	0.72				2			MA8	1.0	13Wi06
$^{52}\text{Ca}-\text{u}$	-34900	500	-36786.4	0.7	-2.5	U			TO3	1.5	90Tu01
	-36792	11			0.5	U			MR1	1.0	13Wi06
$^{52}\text{Sc}-^{33}\text{S }^{19}\text{F}$	-13365.9	3.3	-13366	3	-0.0	1	100	100 ^{52}Sc	MS1	1.0	20Le.A
$^{52}\text{Sc}-\text{u}$	-43500	230	-43504	3	-0.0	U			TO3	1.5	90Tu01
	-43350	250			-0.4	U			TO5	1.5	94Se12
	-43110	240			-1.1	U			TO6	1.5	98Ba.A
	-43438	97			-0.7	o			LZ1	1.0	15Xu14
	-43260	560			-0.4	o			MT1	1.0	15Me08
	-43505	70			0.0	U			LZ1	1.0	19Xu09
	-43607	247			0.4	U			MT1	1.0	20Me06
$^{52}\text{Ti}-^{39}\text{K}_{1.333}$	-4738.7	3.2	-4737.2	2.9	0.5	1	85	85 ^{52}Ti	TT1	1.0	18Re11
$^{52}\text{Ti}-\text{u}$	-53104	17	-53116.5	2.9	-0.7	U			TR1	1.0	18Le03
	-53118.0	3.2			0.5	o			TT1	1.0	18Le03
$^{52}\text{V}-\text{u}$	-55198	28	-55226.36	0.17	-1.0	U			TR1	1.0	18Re11
$^{52}\text{Cr}-^{39}\text{K}_{1.333}$	-11116.29	0.48	-11116.03	0.12	0.5	U			MA8	1.0	19Hu15
	-11118.0	2.1			0.9	U			TT1	1.0	18Re11
$\text{C}_4 \text{H}_4-^{52}\text{Cr}$	90826	9	90795.41	0.12	-0.8	U			R09	4.0	72De11
$\text{C}_3 ^{13}\text{C} \text{H}_3-^{52}\text{Cr}$	86373	18	86325.22	0.12	-0.7	U			R09	4.0	72De11
$\text{C}_3 \text{H}_2 \text{N}-^{52}\text{Cr}$	78253	6	78219.35	0.12	-1.4	U			R09	4.0	72De11

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{52}\text{Cr}-^{85}\text{Rb}_{.612}$	-5510.46	0.57	-5510.61	0.12	-0.3	U			MA8	1.0	19Hu15
$^{52}\text{Cr}-u$	-59497.2	2.1	-59495.29	0.12	0.9	o			TT1	1.0	18Le03
$^{52}\text{Co}-u$	-36888	9	-36870	6	2.0	1	40	40 ^{52}Co	LZ1	1.0	16Xu10
$^{52}\text{Co}^m-u$	-36473	11	-36466	8	0.7	1	54	54 $^{52}\text{Co}^m$	LZ1	1.0	16Xu10
$^{52}\text{Ni}-u$	-24219	89				2			LZ1	1.0	20Fu05
$^{52}\text{Ca}-^{58}\text{Ni}_{.897}$	21220	110	21212.2	0.8	-0.1	U			TT1	1.0	12Ga29
$^{52}\text{Cr}-^{48}\text{Ti}_{1.083}$	-3114.916	0.096	-3115.04	0.08	-1.3	1	78	78 ^{52}Cr	MS1	1.0	17Ka53
$^{52}\text{Ca}-^{52}\text{Cr}$	22740	82	22708.9	0.7	-0.4	U			TT1	1.0	12Ga29
$^{52}\text{Mn}-^{52}\text{Cr}$	5054.376	0.068				2			JY1	1.0	17Ne05
$^{52}\text{Fe}-^{52}\text{Cr}$	7608.65	0.15				2			JY1	1.0	17Ne05
$^{52}\text{Fe}^m-^{52}\text{Cr}$	15081.23	0.29				2			JY1	1.0	17Ne05
$^{52}\text{Co}-^{52}\text{Cr}$	22637.5	7.3	22626	6	-1.6	1	60	60 ^{52}Co	JY1	1.0	17Ne05
$^{52}\text{Co}^m-^{52}\text{Cr}$	23038	12	23029	8	-0.7	1	46	46 $^{52}\text{Co}^m$	JY1	1.0	17Ne05
$^{52}\text{Cr}-^{50}\text{Cr}$	-5566	41	-5537.50	0.10	0.2	U			R09	4.0	72De11
$^{52}\text{Cr}(p,\alpha)^{49}\text{V}$	-2596	10	-2593.3	0.8	0.3	U			Ald		66Br06
$^{50}\text{Ti}(t,p)^{52}\text{Ti}$	5698	10	5706.7	2.7	0.9	-			LAl		66Wi11
	5700	10			0.7	-			LAl		71Ca19
ave.	5699	7			1.1	1	15	15 ^{52}Ti			average
$^{50}\text{Ti}(^3\text{He},p)^{52}\text{V}$	7653	15	7653.41	0.14	0.0	U			Phi		75Ca07
$^{52}\text{Cr}(d,\alpha)^{50}\text{V}$	4517	12	4514.54	0.10	-0.2	U			Kop		67Ha.A
$^{52}\text{Cr}(p,^3\text{He})^{50}\text{V}^i$	-18645	6	-18651.05	0.28	-1.0	U					78Ko27
$^{52}\text{Cr}(p,t)^{50}\text{Cr}^j$	-21244	7				2					78Ko27
$^{52}\text{Cr}(p,t)^{50}\text{Cr}^j$	-26041	6				2					78Ko27
$^{51}\text{V}(n,\gamma)^{52}\text{V}$	7311.2	0.5	7311.24	0.13	0.1	2					84De15
	7311.18	0.26			0.2	2			ILn		91Mi08 Z
	7311.27	0.15			-0.2	2			Bdn		06Fi.A
$^{51}\text{V}(d,p)^{52}\text{V}$	5098	9	5086.68	0.13	-1.3	U			MIT		64Sp12
	5086	8			0.1	U			Kop		67Ha.A
$^{51}\text{V}(p,\gamma)^{52}\text{Cr}$	10500.7	2.8	10505.37	0.10	1.7	U					74Ro44 Z
$^{52}\text{Co}^i(p)^{51}\text{Fe}$	1367	60	1481	8	1.9	o			Bor		94Fa06
	1349	10			13.2	B			Bor		07Do17
	1352	10			12.9	B					16Or03
$^{52}\text{Ca}(\beta^-)^{52}\text{Sc}$	5700	200	6257	3	2.8	B					85Hu03
$^{52}\text{Sc}(\beta^-)^{52}\text{Ti}$	8020	250	8954	4	3.7	B					85Hu03
$^{52}\text{Ti}(\beta^-)^{52}\text{V}$	1940	200	1965.3	2.8	0.1	U					67Mo11 *
$^{52}\text{V}(\beta^-)^{52}\text{Cr}$	3904	30	3976.48	0.16	2.4	U					65Ko09 *
	3854	30			4.1	B					67Va27 *
$^{52}\text{Mn}(\beta^+)^{52}\text{Cr}$	4710.9	4.	4708.12	0.06	-0.7	U					58Ko57 *
	4707.9	6.			0.0	U					60Ka20 *
$^{52}\text{Cr}(p,n)^{52}\text{Mn}$	-5479	10	-5490.47	0.06	-1.1	U			Ric		66Ri09
$^{52}\text{Cr}(^3\text{He},t)^{52}\text{Mn}^i$	-7653	5				2					71Be29 *
$^{52}\text{Fe}(\beta^+)^{52}\text{Mn}$	2372	10	2379.29	0.15	0.7	U					56Ar33 *
	2229	130			1.2	U					79Ge02 *
	2510	100			-1.3	U					95Ir01
$^{52}\text{Fe}^m(\text{IT})^{52}\text{Fe}$	6958.0	0.4	6960.7	0.3	6.7	B					Ens157
$^{52}\text{Co}^i(\text{IT})^{52}\text{Co}^m$	2548	2				2					16Or03
$^{52}\text{Ti}(\beta^-)^{52}\text{V}$	$E_{\beta^-}=1800(200)$ to 1^+ level at 141.610 keV										Ens159 **
$^{52}\text{V}(\beta^-)^{52}\text{Cr}$	$E_{\beta^-}=2470(30)$ 2420(30) respectively, to 2^+ level at 1434.091 keV										Ens159 **
$^{52}\text{Mn}(\beta^+)^{52}\text{Cr}$	$E_{\beta^+}=575(4)$ and $572(6)$ respectively, to 6^+ level at 3113.858 keV										Ens159 **
$^{52}\text{Cr}(^3\text{He},t)^{52}\text{Mn}^i$	CDE=8414(5) $Q=-7650(5)$; recalibration -3 keV for $^{50}\text{Cr}(p,n)^{50}\text{Mn}$ from Ame1961										MMC123**
$^{52}\text{Fe}(\beta^+)^{52}\text{Mn}$	$E_{\beta^+}=804(10)$ to 1^+ level at 546.438 keV										Ens159 **
$^{52}\text{Fe}(\beta^+)^{52}\text{Mn}$	$E_{\beta^+}=5350(130)$ from $^{52}\text{Fe}^m$ 12^+ at 6958.0 to 11^+ level at 3837.2 keV										Ens159 **
$^{53}\text{K}-u$	-13200	120				2			MR1	1.0	15Ro10
$^{53}\text{Ca}-u$	-31549	47				2			MR1	1.0	13Wi06
$^{53}\text{Sc}-^{34}\text{S }^{19}\text{F}$	-7891	19	-7891	19	-0.0	1	100	100 ^{53}Sc	MS1	1.0	20Le.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{53}\text{Sc}-\text{u}$	-41440	260	-41621	19	-0.5	o			TO3	1.5	90Tu01
	-41830	280			0.5	U			TO5	1.5	94Se12
	-41100	400			-0.9	U			TO6	1.5	98Ba.A
	-41694	118			0.4	U			GT1	1.5	04Ma.A
	-40910	290			-2.5	B			MT1	1.0	11Es06
	-41804	123			1.5	o			LZ1	1.0	15Xu14
	-40980	610			-1.1	o			MT1	1.0	15Me08
	-41772	86			1.8	U			LZ1	1.0	19Xu09
	-41224	225			-1.8	U			MT1	1.0	20Me06
$^{53}\text{Ti}-^{39}\text{K}_{1.359}$	-1006.4	3.1				2			TT1	1.0	18Re11
$^{53}\text{Ti}-\text{u}$	-50325	19	-50329	3	-0.2	U			TR1	1.0	18Le03
	-50329.3	3.1			0.0	o			TT1	1.0	18Le03
$^{53}\text{V}-\text{u}$	-55664	20	-55665	3	-0.1	U			TR1	1.0	18Re11
$^{53}\text{Cr}-^{39}\text{K}_{1.359}$	-10031.7	2.0	-10030.81	0.12	0.4	U			TT1	1.0	18Re11
$\text{C}_4\text{H}_5-^{53}\text{Cr}$	98529	8	98478.86	0.12	-1.6	U			R09	4.0	72De11
$\text{C}_3\text{H}_3\text{N}-^{53}\text{Cr}$	85958	10	85902.80	0.12	-1.4	U			R09	4.0	72De11
$\text{C}_2^{13}\text{CH}_2\text{N}-^{53}\text{Cr}$	81507	27	81432.60	0.12	-0.7	U			R09	4.0	72De11
$\text{C}_3\text{HO}-^{53}\text{Cr}$	62152	14	62093.35	0.12	-1.0	U			R09	4.0	72De11
$^{53}\text{Cr}-^{85}\text{Rb}_{.624}$	-4311.80	0.58	-4310.49	0.12	2.3	U			MA8	1.0	19Hu15
$^{53}\text{Cr}-\text{u}$	-59354.5	2.0	-59353.70	0.12	0.4	o			TT1	1.0	18Le03
$^{53}\text{Co}-\text{u}$	-45783	18	-45796.7	1.9	-0.8	U			LZ1	1.0	11Tu09
$^{53}\text{Ni}-\text{u}$	-31810	27				2			LZ1	1.0	12Zh34 *
$^{53}\text{Cr}-^{48}\text{Ti}_{1.104}$	-1880.44	0.12	-1880.21	0.09	2.0	1	56	55 ^{53}Cr	MS1	1.0	17Ka53
$^{53}\text{Co}-^{53}\text{Fe}$	8897.67	0.49	8897.6	0.5	-0.0	1	94	94 ^{53}Co	JY1	1.0	10Ka26
$^{53}\text{Co}^m-^{53}\text{Fe}$	12305.2	1.3	12305.3	1.0	0.1	1	60	60 $^{53}\text{Co}^m$	JY1	1.0	10Ka26
$^{53}\text{Co}^m-^{53}\text{Co}$	3407.9	1.5	3407.7	1.0	-0.1	1	46	40 $^{53}\text{Co}^m$	JY1	1.0	10Ka26
$^{53}\text{Cr}-^{52}\text{Cr}$	115	46	141.59	0.10	0.1	U			R09	4.0	72De11
$^{51}\text{V}(\text{t,p})^{53}\text{V}$	7325	25	7309	3	-0.6	U			Ald		67Hi02
$^{53}\text{Cr}(\text{d},\alpha)^{51}\text{V}$	7635	12	7626.29	0.10	-0.7	U			Kop		67Ha.A
$^{52}\text{Cr}(\text{n},\gamma)^{53}\text{Cr}$	7939.52	0.3	7939.43	0.10	-0.3	-			MMn		80Is02 Z
	7939.01	0.2			2.1	-			BNn		80Ko01 Z
	7939.10	0.28			1.2	-			Bdn		06Fi.A
	5725	6	5714.86	0.10	-1.7	U			MIT		64Sp12
$^{52}\text{Cr}(\text{d,p})^{53}\text{Cr}$	5719	8			-0.5	U			Kop		67Ha.A
$^{52}\text{Cr}(\text{n},\gamma)^{53}\text{Cr}$	ave.	7939.15	0.14	7939.43	0.10	1.9	1	46	25 ^{53}Cr		average
$^{52}\text{Cr}(\text{p},\gamma)^{53}\text{Mn}$	6559.1	1.1	6559.8	0.3	0.6	U					70Ma25 Z
	6559.72	0.36			0.3	1	85	84 ^{53}Mn			79Sw01 Z
$^{52}\text{Cr}(\text{He},\text{d})^{53}\text{Mn}$	1070	15	1066.3	0.3	-0.2	U			MIT		67Ob04
$^{53}\text{Co}^m(\text{p})^{52}\text{Fe}$	1559.7	40.	1558.0	1.9	-0.0	o					70Ja22
	1600.5	30.			-1.4	U					70Ce04
	1590	30			-1.1	U					72Ce01
	1590	30			-1.1	U					76Vi02
	1552.3	8.0			0.7	U					15Sh16 *
$^{53}\text{Co}^i(\text{p})^{52}\text{Fe}$	2789.5	50.	2708.7	2.6	-1.6	U					76Vi02 *
	2778.5	18.			-3.9	B			Bor		07Do17 *
$^{53}\text{Ti}(\beta^-)^{53}\text{V}$	5020	100	4970	4	-0.5	U			ANB		77Pa01
$^{53}\text{V}(\beta^-)^{53}\text{Cr}$	3536	50	3436	3	-2.0	U					56Sc.A *
$^{53}\text{Cr}(\text{p,n})^{53}\text{Mn}$	-1379	8	-1379.6	0.3	-0.1	U			MIT		52Lo06 Y
	-1381.1	1.6			0.9	U			Oak		64Jo11 Z
$^{53}\text{Cr}(\text{He},\text{t})^{53}\text{Mn}^i$	-7589	4				2					71Be29 *
$^{53}\text{Fe}(\beta^+)^{53}\text{Mn}$	3860	100	3742.9	1.7	-1.2	U					59Ju40
	3820	100			-0.8	U					75BI01
$^{53}\text{Co}^i(\text{IT})^{53}\text{Co}$	4325	2				2					16Su10
$^{53}\text{Ni}-\text{u}$	Same result in reference										13Ya03 **
$^{53}\text{Co}^m(\text{p})^{52}\text{Fe}$	Original $Q=1558(8)$ corrected for recoil										16Hu.A **
$^{53}\text{Co}^i(\text{p})^{52}\text{Fe}$	$Q_p=1940(50)$ 1929(18) respectively, to 2^+ level at 849.45 keV										Ens159 **
$^{53}\text{V}(\beta^-)^{53}\text{Cr}$	$E_{\beta^-}=2530(50)$ to $5/2^-$ level at 1006.27 keV										Ens09a **
$^{53}\text{Cr}(\text{He},\text{t})^{53}\text{Mn}^i$	CDE=8350(4) $Q=-7586(4)$; recalibration -3 keV for $^{50}\text{Cr}(\text{p,n})^{50}\text{Mn}$ from Ame1961										MMC123**

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{54}\text{Ca}-u$	-27011	52				2			MR1	1.0	13Wi06
$^{54}\text{Sc}-u$	-36060	500	-36971	15	-1.2	o			TO3	1.5	90Tu01 *
	-37060	500			0.1	o			TO5	1.5	94Se12 *
	-36960	400			-0.0	U			TO6	1.5	98Ba.A *
	-37059	225			0.3	U			GT1	1.5	04Ma.A
	-36070	390			-2.3	U			MT1	1.0	11Es06 *
	-37202	585			0.4	o			LZ1	1.0	15Xu14 *
	-36230	680			-1.1	o			MT1	1.0	15Me08
	-37021	386			0.1	U			LZ1	1.0	19Xu09
	-36554	258			-1.6	U			MT1	1.0	20Me06
$^{54}\text{Ti}-u$	-48820	230	-49108	17	-0.8	U			TO3	1.5	90Tu01
	-49130	250			0.1	U			TO5	1.5	94Se12
	-48820	280			-0.7	U			TO6	1.5	98Ba.A
	-48998	118			-0.9	o			LZ1	1.0	15Xu14
	-49108	17				2			TR1	1.0	18Le03
	-49050	107			-0.5	U			LZ1	1.0	19Xu09
$^{54}\text{V}-u$	-53574	18	-53568	12	0.3	1	44	44 ^{54}V	TR1	1.0	18Re11
$^{54}\text{Cr}-^{39}\text{K}_{1.385}$	-10849.6	5.1	-10856.12	0.14	-1.3	U			TT1	1.0	18Re11
$\text{C}_4\text{H}_6-^{54}\text{Cr}$	108018	17	108072.83	0.14	0.8	U			R09	4.0	72De11
$\text{C}_3\text{ }^{13}\text{C}\text{H}_5-^{54}\text{Cr}$	103569	15	103602.64	0.14	0.6	U			R09	4.0	72De11
$\text{C}_3\text{H}_4\text{N}-^{54}\text{Cr}$	95445	13	95496.77	0.14	1.0	U			R09	4.0	72De11
$\text{C}_2\text{ }^{13}\text{C}\text{H}_3\text{N}-^{54}\text{Cr}$	90960	24	91026.58	0.14	0.7	U			R09	4.0	72De11
$\text{C}_2\text{N O}-^{54}\text{Cr}$	59057	26	59111.26	0.14	0.5	U			R09	4.0	72De11
$^{54}\text{Cr}-^{85}\text{Rb}_{.635}$	-5110.22	0.69	-5109.12	0.14	1.6	U			MA8	1.0	19Hu15
$^{54}\text{Cr}-u$	-61116.1	4.9	-61122.64	0.14	-1.3	o			TT1	1.0	18Le03
$^{13}\text{C }^{37}\text{Cl}_3-^{54}\text{Fe }^{35}\text{Cl}_2$	23744.46	1.26	23749.0	0.4	0.9	U			H39	4.0	84Ha20
$\text{C}_4\text{H}_6-^{54}\text{Fe}$	107368	11	107342.0	0.4	-0.6	U			R09	4.0	72De11
$\text{C}_3\text{H}_4\text{N}-^{54}\text{Fe}$	94791	8	94765.9	0.4	-0.8	U			R09	4.0	72De11
$\text{C}_2\text{N O}-^{54}\text{Fe}$	58411	8	58380.4	0.4	-1.0	U			R09	4.0	72De11
$\text{C}_3\text{ }^{13}\text{C}\text{H}_5-^{54}\text{Fe}$	102908	48	102871.8	0.4	-0.2	U			R09	4.0	72De11
$^{54}\text{Ni}-u$	-42167	5				2			LZ1	1.0	17Zh12
$^{54}\text{Cr}-^{48}\text{Ti}_{1.125}$	-2555.97	0.16	-2555.90	0.11	0.4	1	50	49 ^{54}Cr	MS1	1.0	17Ka53
$^{54}\text{Sc}-^{54}\text{Cr}$	24152	15				2			TR1	1.0	20Le.A
$^{54}\text{Co}-^{54}\text{Fe}$	8850.94	0.14	8850.89	0.10	-0.4	1	47	47 ^{54}Co	JY1	1.0	08Er04
$^{54}\text{Co}^m-^{54}\text{Fe}$	9062.960	0.092	9062.99	0.08	0.3	1	81	81 $^{54}\text{Co}^m$	JY1	1.0	08Er04
$^{54}\text{Co}^m-^{54}\text{Co}$	212.18	0.15	212.10	0.10	-0.5	1	49	30 ^{54}Co	JY1	1.0	08Er04
$^{54}\text{Cr}-^{53}\text{Cr}$	-1662	48	-1768.94	0.11	-0.6	U			R09	4.0	72De11
$^{54}\text{Fe}(p,^6\text{He})^{49}\text{Mn}$	-28943	24	-28937.8	2.2	0.2	U			MSU		75Mu09 *
$^{54}\text{Fe}(\alpha,^8\text{He})^{50}\text{Fe}$	-50950	60	-50963	8	-0.2	U			Tex		77Tr05
$^{54}\text{Cr}(p,\alpha)^{51}\text{V}$	130	30	131.78	0.12	0.1	U			Kop		64Ve02
$^{54}\text{Fe}(p,\alpha)^{51}\text{Mn}$	-3145	9	-3147.3	0.4	-0.3	U			Ald		66Br05
	-3146.9	1.1			-0.4	1	15	9 ^{54}Fe	NDm		74Jo14
$^{54}\text{Fe}(p,\alpha)^{51}\text{Mn}^i$	-7606.6	5.0	-7598.0	1.5	1.7	U					79Ta22 *
$^{54}\text{Fe}(^3\text{He},^6\text{He})^{51}\text{Fe}$	-18694	15	-18726.3	1.4	-2.2	U			MSU		77Mu03 *
$^{54}\text{Cr}(d,\alpha)^{52}\text{V}$	5225	12	5218.46	0.17	-0.5	U			Kop		67Ha.A
$^{52}\text{Cr}(t,p)^{54}\text{Cr}$	9171	10	9176.71	0.12	0.6	U			LAL		71Ca19
$^{52}\text{Cr}(^3\text{He},p)^{54}\text{Mn}$	7785	15	7781.0	1.0	-0.3	U			MIT		69Ly06
	7788	9			-0.8	U			Phi		72Be07
$^{52}\text{Cr}(^3\text{He},p)^{54}\text{Mn}^i$	1633.6	3.9	1634.6	2.8	0.3	1	51	51 $^{54}\text{Mn}^i$			72Be07 *
$^{52}\text{Cr}(^3\text{He},n)^{54}\text{Fe}^j$	-7173	20				2					75Bo14
$^{54}\text{Fe}(d,\alpha)^{52}\text{Mn}$	5169	12	5167.6	0.4	-0.1	U			Kop		67Ha.A
	5159	15			0.6	U			MIT		67Sp09
	5163.3	2.2			1.9	U			NDm		76Jo01
$^{54}\text{Fe}(p,t)^{52}\text{Fe}$	-15584	8	-15583.4	0.4	0.1	U					78Ko27 *
$^{54}\text{Fe}(p,t)^{52}\text{Fe}^j$	-24139	7	-24140	6	-0.1	2					78Ko27
	-24141.3	11.0			0.1	2					78De18 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{54}\text{Zn}(2p)^{52}\text{Ni}$	1480	20	2280#	200#	40.0	D					05Gi15
	1480	20			40.0	D					05Bi15 *
	1280	210			4.8	U					11As08
$^{54}\text{Cr}(d,^3\text{He})^{53}\text{V}$	-6879.2	3.1				2			NDm		79Br.B
$^{53}\text{Cr}(n,\gamma)^{54}\text{Cr}$	9719.30	0.16	9719.08	0.10	-1.4	-					68Wh03 Z
	9718.3	0.4			1.9	-					72Lo26 Z
	9718.91	0.27			0.6	-			MMn		80Is02 Z
	9719.7	0.5			-1.2	-			SAn		89Ho15 Z
	9720.00	0.20			-4.6	C			Bdn		06Fi.A
$^{53}\text{Cr}(d,p)^{54}\text{Cr}$	7480	12	7494.51	0.10	1.2	U			MIT		64Sp12
	7514	10			-1.9	U			Kop		67Ha.A
$^{53}\text{Cr}(n,\gamma)^{54}\text{Cr}$	ave.	9719.14	0.13	9719.08	0.10	-0.5	1	65	45 ^{54}Cr		average
$^{53}\text{Cr}(p,\gamma)^{54}\text{Mn}$		7559.6	1.0			2					75We10 Z
$^{53}\text{Cr}(^3\text{He},d)^{54}\text{Mn}$	2080	12	2066.1	1.0	-1.2	U			MIT		69Ly06
$^{54}\text{Fe}(d,t)^{53}\text{Fe}$	-7121.5	2.1	-7121.2	1.6	0.1	-			NDm		74Jo14
$^{54}\text{Fe}(^3\text{He},\alpha)^{53}\text{Fe}$	7197	20	7199.2	1.6	0.1	U			MIT		68Tr01
	7199.6	2.6			-0.2	-			NDm		74Jo14
$^{54}\text{Fe}(d,t)^{53}\text{Fe}$	ave.	-7121.2	1.6	-7121.2	1.6	-0.0	1	100	100 ^{53}Fe		average
$^{54}\text{Ti}(\beta^-)^{54}\text{V}$	4280	160	4154	19	-0.8	U					96Do23
$^{54}\text{V}(\beta^-)^{54}\text{Cr}$	7000	100	7037	11	0.4	U					70Wa14
$^{54}\text{Cr}(t,^3\text{He})^{54}\text{V}$	-7023	15	-7019	11	0.3	1	56	56 ^{54}V	LAI		77Fi03
$^{54}\text{Mn}(\epsilon)^{54}\text{Cr}$	1359	8	1377.1	1.0	2.3	U					72Ko47 *
	1379	8			-0.2	U					00Hi08 *
$^{54}\text{Cr}(p,n)^{54}\text{Mn}$	-2160	5	-2159.5	1.0	0.1	U			MIT		52Lo06 Z
$^{54}\text{Cr}(^3\text{He},t)^{54}\text{Mn}^i$	-7541	4	-7542.1	2.8	-0.3	1	49	49 $^{54}\text{Mn}^i$			71Be29 *
$^{54}\text{Co}(\beta^+)^{54}\text{Fe}$	8023	110	8244.55	0.09	2.0	U					59Su.A *
	8459	41			-5.2	C					60Mi.A
$^{54}\text{Fe}(p,n)^{54}\text{Co}$	-9031.1	2.5	-9026.89	0.09	1.7	U			Yal		69Ov01 *
	-9023.7	1.8			-1.8	U			Har		74Ho21 Z
$^{54}\text{Fe}(^3\text{He},t)^{54}\text{Co}$	-8261.2	1.0	-8263.14	0.09	-1.9	F			Mun		77Vo02 *
$^{54}\text{Fe}(^3\text{He},t)^{54}\text{Co}-^{27}\text{Al}()^{27}\text{Si}$	-3432.5	3.0	-3432.19	0.13	0.1	U			ChR		74Ha35
$^{54}\text{Fe}(^3\text{He},t)^{54}\text{Co}-^{42}\text{Ca}()^{42}\text{Sc}$	-1817.24	0.18	-1818.26	0.10	-5.7	B			ChR		87Ko34
* $^{54}\text{Sc}-u$	Original -36000(500) μu or $M-A=-33500(470)$ keV										GAu **
* $^{54}\text{Sc}-u$	Original -37000(500) μu or $M-A=-34470(470)$ keV										GAu **
* $^{54}\text{Sc}-u$	$M-A=-34370(370)$ keV for mixture gs+m at 110.5 keV										Nub211 **
* $^{54}\text{Sc}-u$	$M-A=-33540(360)$ keV for mixture gs+m at 110.5 keV										Nub211 **
* $^{54}\text{Sc}-u$	No isomeric correction needed										HWJ151 **
* $^{54}\text{Fe}(p,^6\text{He})^{49}\text{Mn}$	Q increased 1 for recalibration										AHW **
* $^{54}\text{Fe}(p,\alpha)^{51}\text{Mn}^i$	IT=4459(5); Q rebuilt with Ame1977										MMC124**
* $^{54}\text{Fe}(^3\text{He},^6\text{He})^{51}\text{Fe}$	Averaged with reference See $^{46}\text{Ti}(^3\text{He},^6\text{He})$										75Mu09 **
* $^{52}\text{Cr}(^3\text{He},p)^{54}\text{Mn}^i$	IT=6151(5); Q rebuilt with Ame1971										MMC124**
* $^{54}\text{Fe}(p,t)^{52}\text{Fe}$	$Q=-21239(8)$ to $^{52}\text{Fe}^i$ at 5654.5										Nub211 **
* $^{54}\text{Fe}(p,t)^{52}\text{Fe}^j$	IT=8561(5); Q rebuilt with Ame1977										MMC124**
* $^{54}\text{Zn}(2p)^{52}\text{Ni}$	Trends from Mass Surface TMS suggest ^{54}Zn 800 keV less bound										GAu **
* $^{54}\text{Mn}(\epsilon)^{54}\text{Cr}$	IBE=518(8) to 2^+ level at 834.855 keV, B(K)=5.99										Ens148 **
* $^{54}\text{Mn}(\epsilon)^{54}\text{Cr}$	IBE=544(8) to 2^+ level at 834.855 keV										Ens148 **
* $^{54}\text{Cr}(^3\text{He},t)^{54}\text{Mn}^i$	CDE=8302(4) $Q=-7538(4)$; recalibration -3 keV for $^{50}\text{Cr}(p,n)^{50}\text{Mn}$ from Ame1961										MMC123**
* $^{54}\text{Co}(\beta^+)^{54}\text{Fe}$	$E_{\beta^+}=4250(110)$ from $^{54}\text{Co}^m$ at 197.57 to 2949.2 6^+ level										Nub211 **
* $^{54}\text{Fe}(p,n)^{54}\text{Co}$	Uncorrected for resonance. Orig T=9204.1(1.8) corrected in reference										76Fr13 **
* $^{54}\text{Fe}(^3\text{He},t)^{54}\text{Co}$	Original value -8260.2(0.6) recalibrated										AHW **
* $^{54}\text{Fe}(^3\text{He},t)^{54}\text{Co}$	F : rejected in reference of same group										09Fa15 **
$^{55}\text{Ca}-u$	-20022	172				2			RT1	1.0	18Mi08
$^{55}\text{Sc}-u$	-30600	1100	-33110	70	-1.5	U			TO3	1.5	90Tu01
	-32100	600			-1.1	U			TO6	1.5	98Ba.A
	-32460	640			-1.0	o			MT1	1.0	11Es06
	-32760	620			-0.6	o			MT1	1.0	15Me08
	-33376	236			1.1	U			MT1	1.0	20Me06
	-32861	204			-1.2	U			RT1	1.0	20Mi13

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{55}\text{Ti}-\text{u}$	-44650	280	-44910	30	-0.6	U			TO3	1.5	90Tu01
	-44880	260			-0.1	U			TO5	1.5	94Se12
	-44360	350			-1.0	U			TO6	1.5	98Ba.A
	-44909	31				2			TR1	1.0	18Le03
$^{55}\text{V}-\text{u}$	-52738	29				2			TR1	1.0	18Re11
$^{55}\text{Cr}-^{85}\text{Rb}_{.647}$	-2093.4	1.9	-2091.32	0.25	1.1	U			MA8	1.0	19Hu15
$\text{C}_4\text{H}_7-^{55}\text{Mn}$	116757	8	116732.18	0.28	-0.8	U			R09	4.0	72De11
$\text{C}_3^{13}\text{C H}_6-^{55}\text{Mn}$	112281	25	112261.99	0.28	-0.2	U			R09	4.0	72De11
$\text{C}_3\text{H}_5\text{N}-^{55}\text{Mn}$	104202	10	104156.12	0.28	-1.1	U			R09	4.0	72De11
$\text{C}_2\text{H}_3\text{N}_2-^{55}\text{Mn}$	91618	28	91580.06	0.28	-0.3	U			R09	4.0	72De11
$\text{C}_3\text{H}_3\text{O}-^{55}\text{Mn}$	80372	10	80346.67	0.28	-0.6	U			R09	4.0	72De11
$^{55}\text{Mn}-^{85}\text{Rb}_{.647}$	-4884.41	0.84	-4884.92	0.28	-0.6	o			MA8	1.0	12Na15
	-4883.81	0.80			-1.4	1	12	$^{12}\text{ }^{55}\text{Mn}$	MA8	1.0	19Hu15
$^{55}\text{Ni}-\text{u}$	-48678	18	-48670.2	0.8	0.4	U			LZ1	1.0	11Tu09
$^{55}\text{Cu}-\text{u}$	-33962	167				2			LZ1	1.0	13Ya03
$^{55}\text{Sc}-^{55}\text{Cr}$	26053	67				3			TR1	1.0	20Le.A
$^{55}\text{Ni}-^{55}\text{Co}$	9333.43	0.62				2			JY1	1.0	10Ka26
$^{55}\text{Mn}(\text{p},\alpha)^{52}\text{Cr}$	2570	8	2571.02	0.27	0.1	U			MIT		64Sp12
	2600	10			-2.9	U			ANL		67Ka11
	8283	8	8285.88	0.27	0.4	U			MIT		64Sp12
$^{55}\text{Mn}(\text{d},\alpha)^{53}\text{Cr}$	8277	15			0.6	U			Kop		67Ha.A
	$^{54}\text{Cr}(\text{n},\gamma)^{55}\text{Cr}$	6246.2	0.4	6246.26	0.19	0.2	2				72Wh05 Z
		6246.28	0.21		-0.1	2			Bdn		06Fi.A
$^{54}\text{Cr}(\text{d},\text{p})^{55}\text{Cr}$	4027	8	4021.70	0.19	-0.7	U			MIT		64Sp12
	4035	8			-1.7	U			Kop		67Ha.A
	4022.1	1.2			-0.3	U			NDm		74Jo14
$^{54}\text{Cr}(\text{p},\gamma)^{55}\text{Mn}$	8067.2	0.4	8066.13	0.26	-2.7	1	43	$^{37}\text{ }^{55}\text{Mn}$			78We12
$^{54}\text{Cr}(^3\text{He},\text{d})^{55}\text{Mn}$	2568	18	2572.66	0.26	0.3	U			MIT		69Ra02
$^{55}\text{Mn}(\gamma,\text{n})^{54}\text{Mn}$	-10192	20	-10225.6	1.0	-1.7	U			Phi		60Ge01
$^{54}\text{Fe}(\text{n},\gamma)^{55}\text{Fe}$	9297.91	0.3	9298.12	0.19	0.7	-			MMn		80Is02 Z
	9298.53	0.27			-1.5	-			Bdn		06Fi.A
$^{54}\text{Fe}(\text{d},\text{p})^{55}\text{Fe}$	7084	8	7073.56	0.19	-1.3	U			MIT		64Sp12
	7083	10			-0.9	U			Kop		67Ha.A
	7072.3	1.7			0.7	U			NDm		74Jo14
$^{54}\text{Fe}(\text{n},\gamma)^{55}\text{Fe}$	ave.	9298.25	0.20	9298.12	0.19	-0.6	1	89	$^{73}\text{ }^{54}\text{Fe}$		average
$^{54}\text{Fe}(\text{p},\gamma)^{55}\text{Co}$	5064.0	0.7	5064.35	0.30	0.5	-					77Er02 Z
	5063.9	0.4			1.1	-					80Ha36 Z
$^{54}\text{Fe}(^3\text{He},\text{d})^{55}\text{Co}$	-428	15	-429.12	0.30	-0.1	U			MIT		67Ob04
	-426.9	2.2			-1.0	U			NDm		74Jo14
$^{54}\text{Fe}(\text{p},\gamma)^{55}\text{Co}$	ave.	5063.9	0.3	5064.35	0.30	1.2	1	74	$^{56}\text{ }^{55}\text{Co}$		average
$^{55}\text{Ti}(\beta^-)^{55}\text{V}$	7440	200	7290	40	-0.7	U					96Do23
$^{55}\text{V}(\beta^-)^{55}\text{Cr}$	5956	100	5985	27	0.3	U			ANB		77Na17
$^{55}\text{Cr}(\beta^-)^{55}\text{Mn}$	2500	40	2602.2	0.3	2.6	U					63Me06
	2494	25			4.3	B					65Ko09
$^{55}\text{Fe}(\epsilon)^{55}\text{Mn}$	224.5	4.	231.12	0.18	1.7	U					65Be19
	224.5	3.			2.2	U					69Ka13
	231.4	0.4			-0.7	-					89Zl.A
	230.7	1.9			0.2	U					90Is06
	231.0	1.0			0.1	U					93Wi05 *
	231.37	0.30			-0.8	-					95Da14 *
	231.0	0.3			0.4	-					95Sy01 *
$^{55}\text{Mn}(\text{p},\text{n})^{55}\text{Fe}$	232.36	0.64			-1.9	U					01Ke14
	-1015.7	2.	-1013.47	0.18	1.1	U			Nvl		59Go68 Z
	-1014.6	0.8			1.4	U			Oak		64Jo11 Z
$^{55}\text{Fe}(\epsilon)^{55}\text{Mn}$	ave.	231.23	0.19	231.12	0.18	-0.6	1	91	$^{84}\text{ }^{55}\text{Fe}$		average
$^{55}\text{Mn}(^3\text{He},\text{t})^{55}\text{Fe}^i$	-7883	6				2					71Be29 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{55}\text{Co}(\beta^+)^{55}\text{Fe}$	3466	2	3451.4	0.3	-7.3	B					66Fi06 *
$^{55}\text{Fe}(\epsilon)^{55}\text{Mn}$	Error estimated by evaluator										AHW **
$^{55}\text{Fe}(\epsilon)^{55}\text{Mn}$	Original error 0.10 increased by evaluator										GAu **
$^{55}\text{Fe}(\epsilon)^{55}\text{Mn}$	Original statistical error 0.10 increased by evaluator										GAu **
$^{55}\text{Mn}(^3\text{He},t)^{55}\text{Fe}^i$	CDE=8654(6) $Q = -7890(6)$; recalibration +7 keV for $^{54}\text{Fe}(p,n)^{54}\text{Co}$ from Ame1961										MMC123**
$^{55}\text{Co}(\beta^+)^{55}\text{Fe}$	$E_{\beta^+}=1513(2)$ to $5/2^-$ level at 931.29 keV										Ens097 **
$^{56}\text{Ca}-u$	-14504	268				2			RT1	1.0	18Mi08
$^{56}\text{Sc}-u$	-26680	630	-27390	280	-1.1	o			MT1	1.0	15Me08 *
	-27520	350			0.4	2			MT1	1.0	20Me06 *
	-27171	461			-0.5	2			RT1	1.0	20Mi13 *
$^{56}\text{Ti}-u$	-41300	350	-42320	110	-1.9	-			TO3	1.5	90Tu01
	-42010	300			-0.7	-			TO5	1.5	94Se12
	-41770	270			-1.4	-			TO6	1.5	98Ba.A
	-42319	129			-0.0	-			GT1	1.5	04Ma.A
	-42700	290			1.3	o			LZ1	1.0	15Xu14
	-42738	203			2.0	-			LZ1	1.0	19Xu09
	-42384	258			0.2	-			MT1	1.0	20Me06
	ave. -42350	110			0.3	1	93	93 ^{56}Ti			average
$^{56}\text{V}-u$	-49470	250	-49580	190	-0.3	-			TO3	1.5	90Tu01
	-49640	260			0.2	-			TO5	1.5	94Se12
	-49310	250			-0.7	-			TO6	1.5	98Ba.A
	ave. -49470	220			-0.5	1	74	74 ^{56}V			average
$^{56}\text{Cr}-^{85}\text{Rb}_{.659}$	-1216.3	2.0	-1220.5	0.6	-2.1	U			MA8	1.0	05Gu27
$^{56}\text{Mn}-^{85}\text{Rb}_{.659}$	-2965.1	1.5	-2966.6	0.3	-1.0	U			MA8	1.0	05Gu37
$^{56}\text{Mn}-^{39}\text{K}_{1.436}$	-8979.0	2.7	-8979.7	0.3	-0.3	U			MA8	1.0	09Na.A
$\text{C}_4\text{H}_8-^{56}\text{Fe}$	127754	10	127664.72	0.29	-2.2	U			R09	4.0	72De11
$\text{C}_3^{13}\text{CH}_7-^{56}\text{Fe}$	123300	47	123194.52	0.29	-0.6	U			R09	4.0	72De11
$\text{C}_3\text{H}_6\text{N}-^{56}\text{Fe}$	115171	13	115088.66	0.29	-1.6	U			R09	4.0	72De11
$\text{C}_3\text{H}_4\text{O}-^{56}\text{Fe}$	91381	15	91279.21	0.29	-1.7	U			R09	4.0	72De11
$\text{C}_2\text{H}_2\text{NO}-^{56}\text{Fe}$	78790	24	78703.15	0.29	-0.9	U			R09	4.0	72De11
$\text{C}_2\text{O}_2-^{56}\text{Fe}$	54990	9	54893.70	0.29	-2.7	B			R09	4.0	72De11
$^{56}\text{Fe}-^{85}\text{Rb}_{.659}$	-6933.40	0.60	-6933.90	0.29	-0.8	1	23	23 ^{56}Fe	MA8	1.0	19Hu15
$^{56}\text{Cu}-u$	-41485	16	-41471	7	0.9	2			LZ1	1.0	18Zh29
	-41467.5	7.6			-0.4	2			MS1	1.0	18Va01 *
$^{56}\text{Fe}-^{58}\text{Ni}_{.966}$	-2604.70	0.47	-2604.50	0.26	0.4	1	31	26 ^{58}Ni	JY1	1.0	10Ka26
$^{56}\text{Co}-^{58}\text{Ni}_{.966}$	2297.85	0.55	2298.0	0.4	0.3	1	58	51 ^{56}Co	JY1	1.0	10Ka26
$^{56}\text{Ni}-^{55}\text{Co}_{1.018}$	1176.23	0.48	1175.4	0.4	-1.7	1	60	33 ^{55}Co	JY1	1.0	10Ka26
$^{56}\text{Cr}-^{56}\text{Fe}$	5713.44	0.55				2			MA8	1.0	19Hu15
$^{56}\text{Ni}-^{56}\text{Fe}$	7192.00	0.52	7192.2	0.3	0.4	1	42	40 ^{56}Ni	JY1	1.0	10Ka26
$^{56}\text{Ni}-^{56}\text{Co}$	2289.61	0.49	2289.7	0.4	0.2	1	67	49 ^{56}Co	JY1	1.0	10Ka26
$^{56}\text{Fe}-^{54}\text{Fe}$	-4755	47	-4672.65	0.30	0.4	U			R09	4.0	72De11
$^{56}\text{Fe}(p,\alpha)^{53}\text{Mn}$	-1060	9	-1052.8	0.4	0.8	U			MIT		64Sp12
	-1056	9			0.4	U			Ald		66Br05
	-1052.3	0.8			-0.6	1	25	16 ^{53}Mn	NDm		74Jo14
$^{54}\text{Cr}(t,p)^{56}\text{Cr}$	5995	30	6010.6	0.6	0.5	U			Ald		68Ch20
	6024	10			-1.3	U			LAl		71Ca19
$^{56}\text{Fe}(d,\alpha)^{54}\text{Mn}$	5662	12	5661.9	1.0	-0.0	U			Kop		67Ha.A
	5673	30			-0.4	U					67Hj01
$^{54}\text{Fe}(^3\text{He},p)^{56}\text{Co}$	7410	10	7428.2	0.5	1.8	U			CIT		67Mi02
	7408	15			1.3	U			MIT		68Be10
$^{54}\text{Fe}(^3\text{He},n)^{56}\text{Ni}$	4513	14	4512.9	0.4	-0.0	U			CIT		67Mi02
$^{55}\text{Mn}(n,\gamma)^{56}\text{Mn}$	7270.53	0.3	7270.44	0.13	-0.3	2			MMn		80Is02 Z
	7270.42	0.15			0.1	2			Bdn		06Fi.A
$^{55}\text{Mn}(d,p)^{56}\text{Mn}$	5052	5	5045.88	0.13	-1.2	U			MIT		64Sp12
	5053	8			-0.9	U			Kop		67Ha.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{55}\text{Mn}(p,\gamma)^{56}\text{Fe}$	10189	7	10183.59	0.16	-0.8	U					69Fr22
	10193.7	4.5			-2.2	U					70Sa19 *
	10195.7	3.6			-3.4	B					74Pe15 *
	10183.80	0.17			-1.2	1	85	49 ^{56}Fe	Utr		92Gu03 Z
$^{56}\text{Fe}(d,t)^{55}\text{Fe}$	-4938.3	1.3	-4939.83	0.23	-1.2	U			NDm		74Jo14
$^{56}\text{Ni}(p)^{55}\text{Co}$	-7148.5	30.	-7166.6	0.3	-0.6	U					08Jo04 *
$^{56}\text{Cu}^i(p)^{55}\text{Ni}$	2929	31	2948	10	0.6	U			Bor		07Do17 *
	2948	10			3						14Or04 *
$^{56}\text{Ti}(\beta^-)^{56}\text{V}$	7030	330	6760	190	-0.8	1	33	26 ^{56}V			96Do23
$^{56}\text{Cr}(\beta^-)^{56}\text{Mn}$	1610	150	1626.5	0.6	0.1	U					60Dr03
$^{56}\text{Mn}(\beta^-)^{56}\text{Fe}$	3685	5	3695.50	0.21	2.1	U					62Ho14 *
$^{56}\text{Co}(\beta^+)^{56}\text{Fe}$	4566.0	2.0	4566.6	0.4	0.3	U					65Pe18 *
$^{56}\text{Fe}(p,n)^{56}\text{Co}$	-5351	10	-5349.0	0.4	0.2	U			Tal		62Ne08 Y
$^{56}\text{Fe}(^3\text{He},t)^{56}\text{Co}^i$	-8178	9			2						71Be29 *
* $^{56}\text{Sc}-u$	$M-A=-24850(590)$ keV for mixture gs+m at 0#(100#) keV										15Me08 **
* $^{56}\text{Sc}-u$	$M-A=-25380(260)$ keV for mixture gs+m at 0#(100#) keV and 775.0(0.1) keV										Nub211 **
* $^{56}\text{Sc}-u$	Isomer mixture corrected by 19(3) for 774.9(0.3) keV, 0(390) for										20Mi13 **
* $^{56}\text{Sc}-u$	0#(100#) keV										20Mi13 **
* $^{56}\text{Cu}-u$	represents frequency ratio $^{56}\text{Cu}^+/(C_4H_7)^+=1.01641577(12)$										HWJ192 **
* $^{55}\text{Mn}(p,\gamma)^{56}\text{Fe}$	$E_p=1537(2)$ to $11703(4)$ level										70Sa19 **
* $^{55}\text{Mn}(p,\gamma)^{56}\text{Fe}$	$E_p=1537(2)$ to $11705(3)$ level										74Pe15 **
* $^{56}\text{Ni}(p)^{55}\text{Co}$	$E_p=2540(30)$ from 9735 level										08Jo04 **
* $^{56}\text{Cu}^i(p)^{55}\text{Ni}$	Strongest fragment (strength 2.7); weaker fragment 85(70) keV lower (strength 1.3)										14Or04 **
* $^{56}\text{Mn}(\beta^-)^{56}\text{Fe}$	$E_{\beta^-}=2838(5)$ to 2^+ level at 846.7778 keV										Ens115 **
* $^{56}\text{Co}(\beta^+)^{56}\text{Fe}$	$E_{\beta^+}=1459(3)$ to 4^+ level at 2085.1045 keV										Ens115 **
* $^{56}\text{Fe}(^3\text{He},t)^{56}\text{Co}^i$	Strongest fragment given as CDE=8950(6); $Q=-8186(6)$ rebuilt with Ame1965										71Be29 **
*	recalibration +7 keV for $^{54}\text{Fe}(p,n)^{54}\text{Co}$ from Ame1961										MMC123**
$^{57}\text{Ca}-u$	-7912	1063	-7040#	430#	0.8	D			RT1	1.0	18Mi08 *
$^{57}\text{Sc}-u$	-22540	1400	-22950	190	-0.3	o			MT1	1.0	15Me08
	-21664	944			-1.4	U			MT1	1.0	20Me06
	-22952	193				2			RT1	1.0	20Mi13
$^{57}\text{Ti}-u$	-35700	1000	-36930	220	-0.8	o			TO3	1.5	90Tu01
	-36200	400			-1.2	2			TO6	1.5	98Ba.A
	-37102	408			0.3	2			GT1	1.5	04Ma.A
	-36280	370			-1.8	o			MT1	1.0	11Es06
	-37037	258			0.4	2			MT1	1.0	20Me06
$^{57}\text{V}-u$	-47300	400	-47700	90	-0.7	U			TO3	1.5	90Tu01
	-47640	270			-0.2	U			TO5	1.5	94Se12
	-47320	250			-1.0	U			TO6	1.5	98Ba.A
	-47703	91				2			LZ1	1.0	15Xu14
	-47934	279			0.8	U			MT1	1.0	20Me06
$^{57}\text{Cr}-u$	-56240	250	-56387.9	2.0	-0.4	U			TO3	1.5	90Tu01
	-56300	260			-0.2	U			TO5	1.5	94Se12
	-56170	270			-0.5	U			TO6	1.5	98Ba.A
$^{57}\text{Cr}-^{85}\text{Rb}_{.671}$	2802.1	2.0	2801.2	2.0	-0.5	o			MA8	1.0	05Gu27
	2801.2	2.0				2			MA8	1.0	19Hu15
$^{57}\text{Mn}-^{85}\text{Rb}_{.671}$	-2525.1	2.3	-2525.0	1.6	0.1	1	49	49 ^{57}Mn	MA8	1.0	05Gu37
$^{57}\text{Mn}-^{39}\text{K}_{1.462}$	-8650.7	2.8	-8652.9	1.6	-0.8	1	33	33 ^{57}Mn	MA8	1.0	12Na15
$C_7\text{H}_8-^{57}\text{Fe}\ ^{35}\text{Cl}$	158378.5	3.5	158355.61	0.29	-2.6	U			M18	2.5	68Hu05
$C_4\text{H}_9-^{57}\text{Fe}$	135085	11	135033.34	0.29	-1.2	U			R09	4.0	72De11
$C_3\text{H}_7\text{N}-^{57}\text{Fe}$	122500	10	122457.28	0.29	-1.1	U			R09	4.0	72De11
$C_3\text{H}_5\text{O}-^{57}\text{Fe}$	98684	8	98647.83	0.29	-1.1	U			R09	4.0	72De11
$C_2\text{H}_3\text{NO}-^{57}\text{Fe}$	86104	17	86071.77	0.29	-0.5	U			R09	4.0	72De11
$^{57}\text{Ni}-^{85}\text{Rb}_{.671}$	-1019.8	2.7	-1019.5	0.6	0.1	U			MA8	1.0	07Gu09

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{57}\text{Cu}-u$	-50772	43	-50788.3	0.5	-0.4	U			LZ1	1.0	11Tu09
$^{56}\text{Fe } ^{13}\text{C}-^{57}\text{Fe C}$	2897.67	0.47	2898.422	0.018	0.6	U			H30	2.5	77Ba10
	2897.68	0.40			0.7	U			H30	2.5	77Ba10
$^{56}\text{Fe H}-^{57}\text{Fe}$	7325	7	7368.618	0.018	1.6	U			R09	4.0	72De11
$^{57}\text{Fe}-^{56}\text{Fe}_{1.018}$	1627.95	0.46	1627.574	0.019	-0.8	U			JY1	1.0	10Ka26
$^{57}\text{Fe}-^{58}\text{Ni}_{.983}$	-1048.75	0.46	-1048.89	0.27	-0.3	1	33	$^{29} ^{58}\text{Ni}$	JY1	1.0	10Ka26
$^{57}\text{Ni}-^{58}\text{Ni}_{.983}$	3350.77	0.72	3350.6	0.5	-0.3	1	55	$^{50} ^{57}\text{Ni}$	JY1	1.0	10Ka26
$^{57}\text{Cu}-^{56}\text{Ni}_{1.018}$	8126.29	0.55	8125.6	0.4	-1.2	1	63	$^{48} ^{57}\text{Cu}$	JY1	1.0	10Ka26
$^{57}\text{Cu}-^{57}\text{Fe}$	13817.80	0.86	13819.7	0.5	2.3	1	29	$^{29} ^{57}\text{Cu}$	JY1	1.0	10Ka26
$^{57}\text{Cu}-^{57}\text{Ni}$	9420.42	0.55	9420.3	0.5	-0.2	1	74	$^{50} ^{57}\text{Ni}$	JY1	1.0	10Ka26
$^{56}\text{Fe } ^{37}\text{Cl}-^{57}\text{Fe } ^{35}\text{Cl}$	-3413.7	4.3	-3406.53	0.07	0.7	U			M18	2.5	68Hu05
$^{57}\text{Fe}-^{56}\text{Fe}$	456.6	1.4	456.414	0.018	-0.1	U			M18	2.5	68Hu05
	453.2	2.1			0.6	U			M18	2.5	68Hu05
	491	39			-0.2	U			R09	4.0	72De11
$^{54}\text{Cr}(\alpha, p)^{57}\text{Mn}$	-4308	8	-4313.2	1.5	-0.6	U			NDm		76Ma03
	-4302	8			-1.4	U			Can		78An10
$^{57}\text{Fe}(p, \alpha)^{54}\text{Mn}$	237	9	240.3	1.0	0.4	U			MIT		64Sp12
$^{54}\text{Fe}(\alpha, p)^{57}\text{Co}$	-1770.3	1.8	-1773.0	0.5	-1.5	U			NDm		74Jo14
$^{55}\text{Mn}(t, p)^{57}\text{Mn}$	7438.2	3.6	7434.6	1.5	-1.0	1	18	$^{17} ^{57}\text{Mn}$	NDm		77Ma12
$^{57}\text{Fe}(d, \alpha)^{55}\text{Mn}$	8246	15	8241.33	0.16	-0.3	U			Kop		67Ha.A
$^{56}\text{Fe}(n, \gamma)^{57}\text{Fe}$	7645.9	0.5	7646.172	0.017	0.5	U			Utr		68Sp01
	7646.10	0.17			0.4	o			BNn		76Al16 Z
	7645.96	0.20			1.1	U			BNn		78St25 Z
	7646.13	0.21			0.2	U			MMn		80Is02 Z
	7645.93	0.15			1.6	U			Ptn		80Ve05 Z
	7646.0956	0.0500			1.5	-			PTc		97Ro26 *
	7646.08	0.09			1.0	U					02Bo11
	7646.10	0.15			0.5	o			Bdn		06Fi.A
	7646.183	0.018			-0.6	-			Bdn		17Fi03
$^{56}\text{Fe}(d, p)^{57}\text{Fe}$	5425	8	5421.605	0.017	-0.4	U			MIT		64Sp12
	5425	8			-0.4	U			Kop		67Ha.A
	5419.8	1.3			1.4	U			NDm		74Jo14
$^{56}\text{Fe}(n, \gamma)^{57}\text{Fe}$	ave. 7646.173	0.017	7646.172	0.017	-0.1	1	100	$^{88} ^{57}\text{Fe}$			average
$^{56}\text{Fe}(p, \gamma)^{57}\text{Co}$	6027.7	1.0	6027.5	0.4	-0.2	-					70Ob02 Z
	6029.3	1.5			-1.2	-					71Le21 Z
$^{56}\text{Fe}(^3\text{He}, d)^{57}\text{Co}$	538	20	534.0	0.4	-0.2	U			LAI		65B113
$^{56}\text{Fe}(p, \gamma)^{57}\text{Co}$	ave. 6028.2	0.8	6027.5	0.4	-0.9	1	29	$^{29} ^{57}\text{Co}$			average
$^{56}\text{Fe}(p, \gamma)^{57}\text{Co}^i$	-1226.4	0.5	-1225.9	0.3	0.9	2					70Ob02 *
	-1225.6	0.4			-0.6	2					71Le21 *
$^{57}\text{Cu}^i(p)^{56}\text{Ni}$	4650	50	4609	25	-0.8	2					76Vi02
	4568	10			4.1	C					98Jo.A
	4595	29			0.5	2					02Jo09
$^{57}\text{Ti}(\beta^-)^{57}\text{V}$	11020	950	10030	220	-1.0	U					96Do23
$^{57}\text{Cr}(\beta^-)^{57}\text{Mn}$	5100	100	4961.3	2.4	-1.4	U			ANB		78Da04
$^{57}\text{Mn}(\beta^-)^{57}\text{Fe}$	2690	50	2695.7	1.5	0.1	U					63Va37
$^{57}\text{Co}(\epsilon)^{57}\text{Fe}$	810	30	836.4	0.4	0.9	U					71La02 *
$^{57}\text{Fe}(p, n)^{57}\text{Co}$	-1619.4	2.0	-1618.7	0.4	0.3	-			Oak		64Jo11 Z
	-1618.2	2.0			-0.3	-			Can		70Kn03
	ave. -1618.8	1.4			0.0	1	10	$^{10} ^{57}\text{Co}$			average
$^{57}\text{Fe}(^3\text{He}, t)^{57}\text{Co}^i$	-8122	7	-8108.3	0.3	2.0	U					71Be29 *
$^{57}\text{Ni}(\beta^+)^{57}\text{Co}$	3245	10	3261.7	0.6	1.7	U					50Fr10 *
	3235	10			2.7	U					51Ca28 *
	3246	10			1.6	U					58Ko60 *
$^{57}\text{Cu}(\beta^+)^{57}\text{Ni}$	8742	130	8774.9	0.4	0.3	U					84Sh28
$^{57}\text{Ca}-u$	Trends from Mass Surface TMS suggest ^{57}Ca 810 keV less bound										GAu **
$^{56}\text{Fe}(n, \gamma)^{57}\text{Fe}$	Original error 0.0005 increased for calibration										GAu **
$^{56}\text{Fe}(p, \gamma)^{57}\text{Co}^i$	T=1247.9 recalibrated to T=1248.5(0.6) keV										AHW **
$^{56}\text{Fe}(p, \gamma)^{57}\text{Co}^i$	T=1247.1 recalibrated to T=1247.7(0.4) keV										AHW **
$^{57}\text{Co}(\epsilon)^{57}\text{Fe}$	IBE=674(30) to $5/2^-$ level at 136.4743 keV										Ens98c **
$^{57}\text{Fe}(^3\text{He}, t)^{57}\text{Co}^i$	CDE=8893(7) $Q=-8129(7)$; recalibration +7 keV for $^{54}\text{Fe}(p, n)^{54}\text{Co}$ from Ame1961										MMC123**
$^{57}\text{Ni}(\beta^+)^{57}\text{Co}$	$E_{\beta^+}=845(10)$ 835(10) 849(10) respectively, to $13/2^-$ level at 1377.663 keV										Ens98c **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{58}\text{Sc}-\text{u}$	-16618	204				2			RT1	1.0	20Mi13 *
$^{58}\text{Ti}-\text{u}$	-33162	268	-33190	200	-0.1	2			MT1	1.0	20Me06
	-33226	290			0.1	2			RT1	1.0	20Mi13
$^{58}\text{V}-\text{u}$	-43210	280	-43400	100	-0.5	U			TO3	1.5	90Tu01
	-43350	280			-0.1	U			TO5	1.5	94Se12
	-42700	400			-1.2	U			TO6	1.5	98Ba.A
	-43328	107			-0.5	2			GT1	1.5	04Ma.A
	-43457	134			0.4	2			LZ1	1.0	15Xu14
	-42641	247			-3.1	B			MT1	1.0	20Me06
$^{58}\text{Cr}-\text{u}$	-55680	230	-55815	3	-0.4	U			TO3	1.5	90Tu01
	-55750	260			-0.2	U			TO5	1.5	94Se12
	-55490	270			-0.8	U			TO6	1.5	98Ba.A
$^{58}\text{Cr}-^{85}\text{Rb}_{.682}$	4343.9	3.2				2			MA8	1.0	18Mo14
$^{58}\text{Mn}-^{39}\text{K}_{1.487}$	-5964.9	2.9				2			MA8	1.0	12Na15 *
$\text{C}_3\text{H}_8\text{N}-^{58}\text{Fe}$	132382	12	132400.7	0.3	0.4	U			R09	4.0	72De11
$\text{C}_3\text{H}_6\text{O}-^{58}\text{Fe}$	108576	13	108591.2	0.3	0.3	U			R09	4.0	72De11
$\text{C}_2\text{H}_4\text{NO}-^{58}\text{Fe}$	95999	13	96015.2	0.3	0.3	U			R09	4.0	72De11
$\text{C}_3\text{H}_6\text{O}-^{58}\text{Ni}$	106491	8	106523.2	0.4	1.0	U			R10	4.0	74De22
$\text{C}_3^{13}\text{CH}_9-^{58}\text{Ni}$	138424	14	138438.5	0.4	0.3	U			R10	4.0	74De22
$\text{C}_3\text{H}_8\text{N}-^{58}\text{Ni}$	130302	25	130332.6	0.4	0.3	U			R10	4.0	74De22
$\text{C}_2\text{H}_4\text{ON}-^{58}\text{Ni}$	93926	10	93947.1	0.4	0.5	U			R10	4.0	74De22
	93928	15			0.3	U			R10	4.0	74De22
$\text{C}_3\text{H}_6\text{O}-^{58}\text{Ni}$	106504	14	106523.2	0.4	0.3	U			R10	4.0	74De22
$^{58}\text{Ni}-^{58}\text{Fe}$	2059	32	2068.1	0.3	0.1	U			R09	4.0	72De11
$^{58}\text{Cu}-^{58}\text{Ni}$	9190.61	0.50	9190.6	0.5	0.0	1	90	90 ^{58}Cu	JY1	1.0	10Ka26
$^{58}\text{Ni}(\text{p}, ^6\text{He})^{53}\text{Co}$	-27889	18	-27872.6	1.7	0.9	U			MSU		75Mu09 *
$^{58}\text{Ni}(\alpha, ^8\text{He})^{54}\text{Ni}$	-50190	50	-50135	5	1.1	U			Tex		77Tr05
$^{58}\text{Fe}(\text{p}, \alpha)^{55}\text{Mn}$	420	9	421.33	0.24	0.1	U			MIT		64Sp12
$^{58}\text{Ni}(\text{p}, \alpha)^{55}\text{Co}$	-1341.0	2.9	-1334.8	0.4	2.1	U			BNL		73Go19
	-1335.1	0.9			0.3	1	18	11 ^{55}Co	NDm		74Jo14
$^{58}\text{Ni}(^3\text{He}, ^6\text{He})^{55}\text{Ni}$	-17556	11	-17553.8	0.7	0.2	U			MSU		77Mu03 *
$^{58}\text{Fe}(\text{d}, \alpha)^{56}\text{Mn}$	5470	12	5467.20	0.27	-0.2	U			Kop		67Ha.A
$^{56}\text{Fe}(^3\text{He}, \text{p})^{58}\text{Co}$	6853	15	6882.4	1.1	2.0	U			MIT		72Ly01
$^{58}\text{Ni}(\text{d}, \alpha)^{56}\text{Co}$	6522	12	6522.5	0.4	0.0	U			Kop		67Ha.A
	6506	10			1.6	U			MIT		68Be10
$^{58}\text{Ni}(\text{p}, \text{t})^{56}\text{Ni}$	-13987	18	-13982.1	0.3	0.3	U			Bld		65Ho07
$^{58}\text{Ni}(\text{p}, \text{t})^{56}\text{Ni}^j$	-23926	4				2					84Ka07 *
$^{57}\text{Fe}(\text{n}, \gamma)^{58}\text{Fe}$	10044.60	0.3	10044.57	0.18	-0.1	-			MMn		80Is02 Z
	10044.65	0.24			-0.3	-			Bdn		06Fi.A
$^{57}\text{Fe}(\text{d}, \text{p})^{58}\text{Fe}$	7815	8	7820.01	0.18	0.6	U			MIT		64Sp12
	7824	12			-0.3	U			Kop		67Ha.A
$^{57}\text{Fe}(\text{n}, \gamma)^{58}\text{Fe}$	ave. 10044.63	0.19	10044.57	0.18	-0.3	1	93	86 ^{58}Fe			average
$^{57}\text{Fe}(\text{p}, \gamma)^{58}\text{Co}$	6952	3	6954.2	1.1	0.7	1	14	14 ^{58}Co			70Er03
$^{58}\text{Ni}(\text{p}, \text{d})^{57}\text{Ni}$	-9971.2	7.	-9991.7	0.5	-2.9	U					79Ik04 *
$^{58}\text{Ni}(^3\text{He}, \alpha)^{57}\text{Ni}$	8360.3	4.	8361.4	0.5	0.3	U			MSU		76Na23
	8384.8	15.			-1.6	U					79Fo09 *
$^{58}\text{Ni}(^7\text{Li}, ^8\text{He})^{57}\text{Cu}$	-29564	50	-29622.4	0.5	-1.2	U			MSU		85Sh03
	-29613	17			-0.6	U			Tex		86Ga19
$^{58}\text{Ni}(^{14}\text{N}, ^{15}\text{C})^{57}\text{Cu}$	-19900	40	-19929.6	0.9	-0.7	U			Ber		87St04
$^{58}\text{Mn}(\beta^-)^{58}\text{Fe}$	5890	100	6327.7	2.7	4.4	B					69Wa10 *
	5958	100			3.7	B					71Dy01 *
$^{58}\text{Fe}(\text{t}, ^3\text{He})^{58}\text{Mn}$	-6318	15	-6309.1	2.7	0.6	U			LAI		77Fi03 *
$^{58}\text{Co}(\beta^+)^{58}\text{Fe}$	2305	6	2308.0	1.1	0.5	U					52Ch31 *
	2307	4			0.2	U					63Rh02 *
$^{58}\text{Fe}(^3\text{He}, \text{t})^{58}\text{Co}^i$	-8079	8				2					71Be29 *
$^{58}\text{Ni}(\text{p}, \text{n})^{58}\text{Cu}$	-9351	5	-9343.4	0.4	1.5	U			Mar		64Ma.A
	-9352.6	3.4			2.7	U			Ric		66Bo20 Z
	-9346	10			0.3	U			Ric		66Ri09
	-9346.6	1.7			1.9	U			Yal		69Ov01 Z
	-9347.8	4.0			1.1	U			Har		76Fr13

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{58}\text{Ni}(\pi^+, \pi^-)^{58}\text{Zn}$	-16908	50				2					86Se04
* $^{58}\text{Sc}-u$	Isomer mixture was corrected.										20Mi13 **
* $^{58}\text{Mn}-^{39}\text{K}_{1.487}$	$D_M = -5887.8(2.9) \mu\text{u}$ for $^{58}\text{Mn}^m$ at 71.77 keV; $M - A = -55755.6(2.7) \text{ keV}$										Nub211 **
* $^{58}\text{Ni}(p, ^6\text{He})^{53}\text{Co}$	Q increased 1 for recalibration										AHW **
* $^{58}\text{Ni}(^3\text{He}, ^6\text{He})^{55}\text{Ni}$	Averaged with reference See $^{46}\text{Ti}(^3\text{He}, ^6\text{He})$										75Mu09 **
* $^{58}\text{Ni}(p, t)^{56}\text{Ni}^j$	Strongest of three fragments IT=9943(4); Q rebuilt with Ame1977										MMC129**
* $^{58}\text{Ni}(p, d)^{57}\text{Ni}$	$Q = -15210(7)$ for $^{57}\text{Ni}^i$ at 5238.8(0.7) keV, strongest fragment IT=5230(7); rebuilt with $Q_{gs} = -9975 \text{ keV}$, average of 73Ed01 and 65Sh06										Nub211 **
* $^{58}\text{Ni}(^3\text{He}, \alpha)^{57}\text{Ni}$	IT=5235(15); $Q = 3146(15)$ for $^{57}\text{Ni}^i$ at 5238.8(0.7) rebuilt with Ame1977										73Ed01 **
* $^{58}\text{Mn}(\beta^-)^{58}\text{Fe}$	$Q_{\beta^-} = 6100(300)$; and 5930(100) from $^{58}\text{Mn}^m$ at 71.77 keV										MMC129**
* $^{58}\text{Mn}(\beta^-)^{58}\text{Fe}$	$Q_{\beta^-} = 6030(100)$ from $^{58}\text{Mn}^m$ at 71.77 keV										Nub211 **
* $^{58}\text{Fe}(t, ^3\text{He})^{58}\text{Mn}$	And $Q = -6318(15) - 77(8)$ to $^{58}\text{Mn}^m$ at 71.77 keV										Nub211 **
* $^{58}\text{Co}(\beta^+)^{58}\text{Fe}$	$E_{\beta^+} = 472(6) - 474(4)$ respectively, to 2^+ level at 810.7662 keV										Ens104 **
* $^{58}\text{Fe}(^3\text{He}, t)^{58}\text{Co}^i$	Strongest of two fragments IT=5759(8); Q rebuilt with Ame1964										71Be29 **
*	recalibration +7 keV for $^{54}\text{Fe}(p, n)^{54}\text{Co}$ from Ame1961										MMC123**
$^{59}\text{Sc}-u$	-11626	268				2			RT1	1.0	20Mi13
$^{59}\text{Ti}-u$	-27075	290	-27780#	320#	-2.4	D			MT1	1.0	20Me06
	-27053	258			-2.8	D			RT1	1.0	20Mi13 *
$^{59}\text{V}-u$	-38500	400	-40380	150	-3.1	B			TO3	1.5	90Tu01
	-40700	350			0.6	2			TO5	1.5	94Se12
	-39900	400			-0.8	2			TO6	1.5	98Ba.A
	-40677	129			1.6	2			GT1	1.5	04Ma.A
	-39764	279			-2.2	2			MT1	1.0	20Me06
$^{59}\text{Cr}-u$	-51490	290	-51654.6	0.7	-0.4	U			TO3	1.5	90Tu01 *
	-51640	310			-0.0	U			TO5	1.5	94Se12 *
	-51100	310			-1.2	U			TO6	1.5	98Ba.A *
	-52380	500			1.5	U			MT1	1.0	16Me07 *
	-51672	21			0.8	U			MR1	1.0	18Mo14
$^{59}\text{Cr}-^{85}\text{Rb}_{.694}$	9563.35	0.72				2			MA8	1.0	18Mo14
$^{59}\text{Mn}-^{39}\text{K}_{1.513}$	-4696.8	2.5				2			MA8	1.0	12Na15
$^{59}\text{Fe}-^{85}\text{Rb}_{.694}$	-3907.7	1.1	-3908.6	0.4	-0.8	1	10	10 ^{59}Fe	MA8	1.0	19Hu15
$\text{C}_3 \text{H}_7 \text{O}-^{59}\text{Co}$	116467	12	116496.3	0.4	0.6	U			R10	4.0	74De22
$\text{C}_2^{13}\text{C} \text{H}_6 \text{O}-^{59}\text{Co}$	112011	25	112026.1	0.4	0.2	U			R10	4.0	74De22
$\text{C}_2 \text{H}_5 \text{O N}-^{59}\text{Co}$	103901	6	103920.3	0.4	0.8	U			R10	4.0	74De22
$^{59}\text{Co}-^{39}\text{K}_{1.513}$	-11892.4	1.5	-11894.4	0.4	-1.3	U			MA8	1.0	19Hu15
$^{59}\text{Zn}-u$	-50698	29	-50688.1	0.8	0.3	U			LZ1	1.0	11Tu09
$^{58}\text{Ni H}-^{59}\text{Co}$	9970	15	9973.16	0.22	0.1	U			R10	4.0	74De22
$^{59}\text{Zn}-^{58}\text{Cu}_{1.017}$	5722.4	1.3	5722.6	0.8	0.1	1	36	27 ^{59}Zn	JY1	1.0	10Ka26
$^{59}\text{Zn}-^{59}\text{Cu}$	9815.22	0.72	9815.2	0.6	-0.1	1	81	73 ^{59}Zn	JY1	1.0	10Ka26
$^{59}\text{Co}-^{58}\text{Ni}$	-2182	35	-2148.13	0.22	0.2	U			R10	4.0	74De22
$^{59}\text{Co}(p, \alpha)^{56}\text{Fe}$	3245	8	3241.4	0.3	-0.5	U			MIT		64Sp12
	3243	9			-0.2	U			Ald		66Br05
	3240.4	1.4			0.7	U			NDm		74Jo14
$^{59}\text{Co}(d, \alpha)^{57}\text{Fe}$	8667	15	8663.0	0.3	-0.3	U			Kop		67Ha.A
	8659.3	3.2			1.2	U			NDm		74Jo14
$^{59}\text{Ni}(p, t)^{57}\text{Ni}$	-12738.2	3.3	-12733.7	0.5	1.4	U			MSU		76Na23
	-12738.4	5.0			0.9	U					78Na11 *
$^{58}\text{Fe}(n, \gamma)^{59}\text{Fe}$	6581.15	0.30	6581.00	0.11	-0.5	-			Ptn		73Sp06 Z
	6580.94	0.20			0.3	-			Ptn		80Ve05 Z
	6581.02	0.14			-0.1	-			Bdn		06Fi.A
$^{58}\text{Fe}(d, p)^{59}\text{Fe}$	4357	8	4356.44	0.11	-0.1	U			MIT		64Sp12
	4369	8			-1.6	U			Kop		67Ha.A
$^{58}\text{Fe}(n, \gamma)^{59}\text{Fe}$	ave.	6581.01	0.11	6581.00	0.11	-0.1	1	99	90 ^{59}Fe		average
$^{58}\text{Fe}(p, \gamma)^{59}\text{Co}$		7359.7	2.0	7363.5	0.4	1.9	U				74Ke14 Z

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{58}\text{Fe}(^3\text{He},\text{d})^{59}\text{Co}$	1871	20	1870.1	0.4	-0.0	U			LAI		65BI13
$^{58}\text{Fe}(\text{p},\gamma)^{59}\text{Co}-^{56}\text{Fe}()$	1336.5	0.7	1336.1	0.4	-0.6	1	41	28 ^{57}Co			75Br29
$^{59}\text{Co}(\gamma,\text{n})^{58}\text{Co}$	-10441	26	-10453.9	1.1	-0.5	U			Phi		60Ge01
$^{59}\text{Co}(\text{d},\text{t})^{58}\text{Co}$	-4196.0	1.4	-4196.6	1.1	-0.5	1	62	61 ^{58}Co	NDm		74Jo14
$^{58}\text{Ni}(\text{n},\gamma)^{59}\text{Ni}$	8999.37	0.30	8999.28	0.05	-0.3	U					75Wi06 Z
	8999.38	0.20			-0.5	U			MMn		77Is01 Z
	8999.10	0.23			0.8	U			ILn		93Ha05 Z
	8999.28	0.05			-0.0	1	99	74 ^{59}Ni	ORn		04Ra23
	8999.15	0.18			0.7	U			Bdn		06Fi.A
$^{58}\text{Ni}(\text{d},\text{p})^{59}\text{Ni}$	6797	10	6774.71	0.05	-2.2	U			Kop		67Ha.A
	6785	5			-2.1	U			MIT		70An25
	6773.5	1.7			0.7	U			NDm		74Jo14
$^{58}\text{Ni}(\text{p},\gamma)^{59}\text{Cu}$	3418.5	0.5	3418.6	0.4	0.1	1	62	62 ^{59}Cu			63Bo07 Z
	3419	2			-0.2	U					70Fo09
	3416.7	2.0			0.9	U					75KI06 Z
$^{58}\text{Ni}(\text{p},\pi^-)^{59}\text{Zn}$	-144735	40	-144783.6	0.7	-1.2	U					83Sh31
$^{59}\text{Mn}(\beta^-)^{59}\text{Fe}$	5200	100	5139.6	2.4	-0.6	U			ANB		77Pa18
$^{59}\text{Fe}(\beta^-)^{59}\text{Co}$	1570	4	1564.9	0.4	-1.3	U					52Me53 *
	1563	3			0.6	U					63Wo01 *
$^{59}\text{Ni}(\epsilon)^{59}\text{Co}$	1074.5	1.3	1073.00	0.19	-1.2	U					76Be02 *
$^{59}\text{Co}(\text{p},\text{n})^{59}\text{Ni}$	-1855.8	2.0	-1855.35	0.19	0.2	U			MIT		51Mc48 Z
	-1854.3	4.0			-0.3	U					57Bu37 Z
	-1861	5			1.1	U			Ric		57Ch30 Z
	-1855.8	1.6			0.3	U			Oak		64Jo11 Z
	-1855.33	0.20			-0.1	1	95	90 ^{59}Co	PTB		98Bo30
$^{59}\text{Co}(^3\text{He},\text{t})^{59}\text{Ni}^i$	-8436	8	-8433.5	2.1	0.3	U					71Be29 *
$^{59}\text{Zn}(\beta^+)^{59}\text{Cu}$	9120	100	9142.8	0.6	0.2	U					81Ar13
$^{59}\text{Ti}-\text{u}$	Isomer mixture was corrected										20Mi13 **
$^{59}\text{Ti}-\text{u}$	Trends from Mass Surface TMS suggest ^{59}Ti 670 keV more bound										GAu **
$^{59}\text{Cr}-\text{u}$	Original -51220(240) μu or $M-A=-47710(230)$ keV										GAu **
$^{59}\text{Cr}-\text{u}$	Original -51370(270) μu or $M-A=-47850(250)$ keV										GAu **
$^{59}\text{Cr}-\text{u}$	$M-A=-47350(250)$ keV for mixture gs+m at 502.7 keV										Nub211 **
$^{59}\text{Cr}-\text{u}$	$M-A=-48540(440)$ keV for mixture gs+m at 502.7 keV										Nub211 **
$^{59}\text{Ni}(\text{p},\text{t})^{57}\text{Ni}$	Strongest of three IAS fragments, $Q=-17977.2(5.0)$ for $^{57}\text{Ni}^i$ at 5238.8										Nub211 **
$^{59}\text{Fe}(\beta^-)^{59}\text{Co}$	$E_{\beta^-}=-475(3)$ to $3/2^-$ level at 1099.256 keV										Ens024 **
$^{59}\text{Fe}(\beta^-)^{59}\text{Co}$	$E_{\beta^-}=-462(3)$, 273(3) to $3/2^-$ levels at 1099.256, 1291.605 keV										Ens024 **
$^{59}\text{Ni}(\epsilon)^{59}\text{Co}$	Authors add B(K)=8.3 of Ni, changed in 7.7 of Co										AHW **
$^{59}\text{Co}(^3\text{He},\text{t})^{59}\text{Ni}^i$	Strongest fragment $Q=-8441(8)$; recalibration +5 keV for $^{58}\text{Ni}(\text{p},\text{n})^{58}\text{Cu}^i$ from Ame1961										MMC129**
$^{60}\text{Sc}-\text{u}$	-3811	1116	-4890#	540#	-1.0	D			RT1	1.0	20Mi13 *
$^{60}\text{Ti}-\text{u}$	-23725	258				2			RT1	1.0	20Mi13
$^{60}\text{V}-\text{u}$	-33860	700	-35520	200	-1.6	U			TO3	1.5	90Tu01 *
	-35560	600			0.0	2			TO5	1.5	94Se12 *
	-35180	520			-0.4	2			TO6	1.5	98Ba.A *
	-35889	215			1.1	2			GT1	1.5	04Ma.A
	-35510	430			-0.0	o			MT1	1.0	11Es06 *
	-35300	270			-0.8	2			MT1	1.0	20Me06 *
$^{60}\text{Cr}-\text{u}$	-49680	240	-50358.3	1.2	-1.9	U			TO3	1.5	90Tu01
	-50270	280			-0.2	U			TO5	1.5	94Se12
	-49910	280			-1.1	U			TO6	1.5	98Ba.A
	-50929	494			1.2	U			MT1	1.0	16Me07
	-50368	20			0.5	U			MR1	1.0	18Mo14
$^{60}\text{Cr}-^{85}\text{Rb}_{.706}$	11918.1	1.2				2			MA8	1.0	18Mo14
$^{60}\text{Mn}-\text{u}$	-56550	240	-56863.4	2.5	-0.9	U			TO3	1.5	90Tu01 *
	-56810	290			-0.1	U			TO5	1.5	94Se12 *
	-56530	280			-0.8	U			TO6	1.5	98Ba.A *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{60}\text{Mn}-^{39}\text{K}_{1.538}$	-1044.0	2.5				2			MA8	1.0	12Na15 *
$^{60}\text{Co}-\text{u}$	-66380	280	-66184.5	0.4	0.5	U			TO6	1.5	98Ba.A *
$\text{C}_3 \text{H}_8 \text{O}-^{60}\text{Ni}$	126796	14	126729.7	0.4	-1.2	U			R10	4.0	74De22
$\text{C}_2 \text{H}_6 \text{O N}-^{60}\text{Ni}$	114231	10	114153.7	0.4	-1.9	U			R10	4.0	74De22
$\text{C}_2^{13}\text{C H}_7 \text{O}-^{60}\text{Ni}$	122315	10	122259.5	0.4	-1.4	U			R10	4.0	74De22
$\text{C H}_2 \text{N O}_2-^{60}\text{Ni}$	77843	16	77768.2	0.4	-1.2	U			R10	4.0	74De22
$\text{C}_5-^{60}\text{Ni}$	69275	14	69214.9	0.4	-1.1	U			R10	4.0	74De22
$^{60}\text{Ni}-^{85}\text{Rb}_{.706}$	-6937.8	1.6	-6938.4	0.4	-0.4	U			MA8	1.0	07Gu09
$^{60}\text{Zn}-^{58}\text{Ni}_{1.034}$	8698.02	0.55	8698.1	0.4	0.1	1	65	^{65}Zn	JY1	1.0	10Ka26
$^{60}\text{Zn}-^{59}\text{Cu}_{1.017}$	3373.19	0.55	3373.2	0.4	-0.1	1	65	^{35}Zn	JY1	1.0	10Ka26
$^{60}\text{Ni}-^{58}\text{Ni H}$	-12513	30	-12381.55	0.08	1.1	U			R10	4.0	74De22
$^{60}\text{Ni}-^{59}\text{Co}$	-2503	40	-2408.39	0.22	0.6	U			R10	4.0	74De22
$^{60}\text{Ni}-^{58}\text{Ni}$	-4624	25	-4556.52	0.08	0.7	U			R10	4.0	74De22
	-4627	45			0.4	U			R10	4.0	74De22
$^{60}\text{Ni H}-^{59}\text{Co}$	5310	40	5416.64	0.22	0.7	U			R10	4.0	74De22
$^{60}\text{Ni}(\text{p},\alpha)^{57}\text{Co}$	-263.6	0.7	-263.5	0.4	0.1	1	37	^{33}Co	NDm		74Jo14
$^{58}\text{Fe}(\text{t},\text{p})^{60}\text{Fe}$	6907	15	6919	3	0.8	2			LAI		71Ca19
	6947	10			-2.8	2			MSU		76St11
	6913	4			1.4	2			LAI		78No05
$^{60}\text{Ni}(\text{d},\alpha)^{58}\text{Co}$	6084.5	2.2	6084.9	1.1	0.2	1	25	^{25}Co	NDm		74Jo14
$^{58}\text{Ni}(\text{t},\text{p})^{60}\text{Ni}$	11905	10	11905.21	0.07	0.0	U			Ald		71Da16
$^{60}\text{Ni}(\text{p},\text{t})^{58}\text{Ni}^i$	-20735	40				2					74Ko08 *
$^{60}\text{Ni}(\text{p},\text{t})^{58}\text{Ni}^j$	-26444	7				2					84Ka07 *
$^{58}\text{Ni}(\text{He},\text{p})^{60}\text{Cu}$	5770	12	5758.6	1.6	-0.9	U			CIT		67Mi02
	5746	20			0.6	U			MIT		68Yo01
$^{58}\text{Ni}(\text{He},\text{p})^{60}\text{Cu}^i$	3210	10	3218	5	0.8	1	26	$^{26}\text{Cu}^i$	MIT		68Yo01
$^{58}\text{Ni}(\text{He},\text{n})^{60}\text{Zn}$	818	18	805.5	0.4	-0.7	U			CIT		67Mi02
	821	13			-1.2	U			Oak		72Gr39
$^{58}\text{Ni}(\text{He},\text{n})^{60}\text{Zn}^j$	-6562	24				2					74Ev02 *
$^{59}\text{Co}(\text{n},\gamma)^{60}\text{Co}$	7491.88	0.08	7491.92	0.07	0.5	2			BNn		84Ko29 Z
	7492.05	0.15			-0.9	2			Bdn		06Fi.A
$^{59}\text{Co}(\text{d},\text{p})^{60}\text{Co}$	5267	11	5267.35	0.07	0.0	U			MIT		64Sp12
	5272	8			-0.6	U			Kop		67Ha.A
$^{59}\text{Co}(\text{p},\gamma)^{60}\text{Ni}^i$	-1594	4				2					67Ar01
$^{59}\text{Ni}(\text{n},\gamma)^{60}\text{Ni}$	11387.6	0.4	11387.73	0.05	0.3	U					75Wi06 Z
	11387.73	0.05			-0.0	1	99	^{78}Ni	ORn		04Ra23
$^{60}\text{Ni}(\text{p},\text{d})^{59}\text{Ni}$	-9180	50	-9163.16	0.05	0.3	U			Pri		64Le10
$^{60}\text{Ni}(\text{d},\text{t})^{59}\text{Ni}$	-5130.2	2.1	-5130.50	0.05	-0.1	U			NDm		74Jo14
$^{60}\text{Ni}(\text{p},\text{d})^{59}\text{Ni}^i$	-16505.1	2.1				2					78Ik02 *
$^{60}\text{Mn}(\beta^-)^{60}\text{Fe}$	8234	86	8445	4	2.5	U			ANB		78No03 *
$^{60}\text{Co}(\beta^-)^{60}\text{Ni}$	2823.6	1.0	2822.81	0.21	-0.8	U					68Wo02 *
$^{60}\text{Cu}(\beta^+)^{60}\text{Ni}$	6250	40	6128.0	1.6	-3.1	B					54Nu26
$^{60}\text{Ni}(\text{p},\text{n})^{60}\text{Cu}$	-6912	20	-6910.3	1.6	0.1	U			ChR		58Go77
	-6909	10			-0.1	U			Ric		66Ri09
	-6910.3	1.6				2			Yal		69Ov01 Z
$^{60}\text{Ni}(\text{He},\text{t})^{60}\text{Cu}^i$	-8685	6	-8688	5	-0.5	1	74	$^{74}\text{Cu}^i$			71Be29 *
$^{60}\text{Zn}(\beta^+)^{60}\text{Cu}$	4166	64	4170.8	1.6	0.1	U					86Ka38
$^{60}\text{Sc}-\text{u}$	Trends from Mass Surface TMS suggest ^{60}Sc 1000 keV more bound										GAu **
$^{60}\text{V}-\text{u}$	Original -33800(700) μu or $M-A=-31500(650)$ keV										GAu **
$^{60}\text{V}-\text{u}$	Original -35500(600) μu or $M-A=-33070(560)$ keV										GAu **
$^{60}\text{V}-\text{u}$	$M-A=-32700(470)$ keV for mixture gs+m+n at 0#150 and 203.7(0.7) keV										Nub211 **
$^{60}\text{V}-\text{u}$	$M-A=-33010(390)$ keV for mixture gs+m+n at 0#150 and 203.7(0.7) keV										Nub211 **
$^{60}\text{V}-\text{u}$	$M-A=-32810(230)$ keV for mixture gs+m+n at 0#150 and 203.7(0.7) keV										Nub211 **
$^{60}\text{Mn}-\text{u}$	$M-A=-52540(230)$ keV for mixture gs+m at 271.90 keV										Nub211 **
$^{60}\text{Mn}-\text{u}$	$M-A=-52780(260)$ keV for mixture gs+m at 271.90 keV										Nub211 **
$^{60}\text{Mn}-\text{u}$	$M-A=-52520(250)$ keV for mixture gs+m at 271.90 keV										Nub211 **
$^{60}\text{Mn}-^{39}\text{K}_{1.538}$	$D_M=-752.1(2.5)$ μu for $^{60}\text{Mn}^m$ at 271.90 keV; $M-A=-52695.9(2.4)$ keV										Nub211 **
$^{60}\text{Co}-\text{u}$	$M-A=-61800(260)$ keV for mixture gs+m at 58.59 keV										Nub211 **
$^{60}\text{Ni}(\text{p},\text{t})^{58}\text{Ni}^i$	IT=8830(40); Q rebuilt with Ame1971										MMC124**
$^{60}\text{Ni}(\text{p},\text{t})^{58}\text{Ni}^j$	IT=14537(7); Q rebuilt with Ame1977										MMC124**
$^{58}\text{Ni}(\text{He},\text{n})^{60}\text{Zn}^j$	IT=7380(30); Q rebuilt with Ame1971										MMC124**
$^{60}\text{Ni}(\text{p},\text{d})^{59}\text{Ni}^i$	Strongest fragment IT=7341; Q rebuilt with Ame1977										MMC129**

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{60}\text{Mn}(\beta^-)^{60}\text{Fe}$	$E_{\beta^-}=5714(86)$ from $^{60}\text{Mn}^m$ at 271.90 to $(3^+, 4^+)$ 2792.68 level										Ens13c **
$^{60}\text{Co}(\beta^-)^{60}\text{Ni}$	$E_{\beta^-}=317.88(0.10)$ to 4^+ level at 2505.753 keV										Ens13c **
$^{60}\text{Ni}(^3\text{He}, t)^{60}\text{Cu}^i$	CDE=9454(6) $Q=-8690(6)$; recalibration +5 keV for $^{58}\text{Ni}(p, n)^{58}\text{Cu}^i$ from Ame1961										MMC123**
$^{61}\text{Ti}-u$	-17370	483	-17570#	320#	-0.4	D			RT1	1.0	20Mi13 *
$^{61}\text{V}-u$	-32750	960	-32400	250	0.4	o			MT1	1.0	11Es06
	-32614	301			0.7	2			MT1	1.0	20Me06
	-31884	462			-1.1	2			RT1	1.0	20Mi13
$^{61}\text{Cr}-u$	-44500	400	-45621.9	2.0	-1.9	U			TO3	1.5	90Tu01
	-45910	300			0.6	U			TO5	1.5	94Se12
	-45120	280			-1.2	U			TO6	1.5	98Ba.A
	-45679	107			0.4	U			GT1	1.5	04Ma.A
	-45634	176			0.1	U			LZ1	1.0	15Xu14
	-46248	548			1.1	U			MT1	1.0	16Me07
	-45629	22			0.3	U			MR1	1.0	18Mo14
$^{61}\text{Cr}-^{85}\text{Rb}_{.718}$	17713.1	2.0				2			MA8	1.0	18Mo14
$^{61}\text{Mn}-u$	-55160	300	-55547.5	2.5	-0.9	U			TO3	1.5	90Tu01
	-55540	280			-0.0	U			TO5	1.5	94Se12
	-55320	270			-0.6	U			TO6	1.5	98Ba.A
$^{61}\text{Mn}-^{39}\text{K}_{1.564}$	1215.6	2.5				2			MA8	1.0	12Na15
$^{61}\text{Fe}-^{39}\text{K}_{1.564}$	-6490.7	2.8				2			MA8	1.0	12Na15
$\text{C H}_3 \text{ N O}_2-^{61}\text{Ni}$	85373	14	85323.5	0.4	-0.9	U			R10	4.0	74De22
$\text{C}_5 \text{ H}-^{61}\text{Ni}$	76810	10	76770.2	0.4	-1.0	U			R10	4.0	74De22
$^{61}\text{Ga}-u$	-50654	59	-50600	40	0.9	1	48	48 ^{61}Ga	LZ1	1.0	11Tu09
$^{60}\text{Ni H}-^{61}\text{Ni}$	7539	14	7555.34	0.05	0.3	U			R10	4.0	74De22
$^{61}\text{Ni}-^{60}\text{Ni}$	339	60	269.69	0.05	-0.3	U			R10	4.0	74De22
$^{61}\text{Ni}-^{58}\text{Ni H}$	-12187	30	-12111.86	0.09	0.6	U			R10	4.0	74De22
$^{61}\text{Ni}-^{59}\text{Co}$	-2220	30	-2138.70	0.22	0.7	U			R10	4.0	74De22
$^{58}\text{Ni}(\alpha, n)^{61}\text{Zn}$	-9810	30	-9526	16	9.5	B			Oak		64St01
$^{58}\text{Ni}(^6\text{Li}, t)^{61}\text{Zn}$	-4736	23	-4743	16	-0.3	R			LAI		78Wo01
$^{59}\text{Co}(^3\text{He}, p)^{61}\text{Ni}$	9635	10	9634.44	0.21	-0.1	U			MIT		67Sp09
$^{60}\text{Ni}(n, \gamma)^{61}\text{Ni}$	7820.22	0.40	7820.10	0.05	-0.3	U					75W06 Z
	7819.96	0.20			0.7	U			MMn		77Is01 Z
	7820.02	0.20			0.4	U			ILn		93Ha05 Z
	7820.11	0.05			-0.1	-			ORn		04Ra23
	7820.06	0.16			0.3	-			Bdn		06Fi.A
$^{60}\text{Ni}(d, p)^{61}\text{Ni}$	5604	8	5595.54	0.05	-1.1	U			MIT		70An25
	5596.1	1.3			-0.4	U			NDm		74Jo14
$^{60}\text{Ni}(n, \gamma)^{61}\text{Ni}$	ave.	7820.11	0.05	7820.10	0.05	-0.0	1	100	82 ^{61}Ni		average
$^{61}\text{Ga}(p)^{60}\text{Zn}$	3110	30				2					87Ho.A
$^{61}\text{Fe}(\beta^-)^{61}\text{Co}$	3827	100	3977.7	2.7	1.5	U					67Eh02 *
	3887	100			0.9	U					67Gu06 *
$^{61}\text{Co}(\beta^-)^{61}\text{Ni}$	1290	40	1323.9	0.8	0.8	U					56Nu02
$^{61}\text{Cu}(\beta^+)^{61}\text{Ni}$	2227	5	2238.0	1.0	2.2	U					50Ow03
$^{61}\text{Ni}(p, n)^{61}\text{Cu}$	-3024.0	4.	-3020.3	1.0	0.9	U			Oak		64Jo11 Z
$^{61}\text{Ni}(^3\text{He}, t)^{61}\text{Cu}^i$	-8630	7				2					71Be29 *
$^{61}\text{Zn}(\beta^+)^{61}\text{Cu}$	5400	200	5635	16	1.2	U					59Cu86
$^{61}\text{Ga}(\beta^+)^{61}\text{Zn}$	9255	50	9210	40	-0.8	1	57	52 ^{61}Ga			02We07
$^{61}\text{Ti}-u$	Isomer mixture was corrected.										20Mi13 **
$^{61}\text{Ti}-u$	Trends from Mass Surface TMS suggest ^{61}Ti 190 keV more bound										GAu **
$^{61}\text{Fe}(\beta^-)^{61}\text{Co}$	$E_{\beta^-}=2800(100)$ 2860(100) respectively, to $3/2^-$ level at 1027.48 keV										Ens153 **
$^{61}\text{Ni}(^3\text{He}, t)^{61}\text{Cu}^i$	Strongest fragment IT=6380(7); Q rebuilt with Ame1964										MMC129**
*	recalibration +5 keV for $^{58}\text{Ni}(p, n)^{58}\text{Cu}^i$ from Ame1961										MMC123**

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{62}\text{Ti}-u$	-14278	483	-13100#	430#	2.4	D			RT1	1.0	20Mi13 *
$^{62}\text{V}-u$	-27204	451	-27070	280	0.3	2			MT1	1.0	20Me06
	-26978	365			-0.2	2			RT1	1.0	20Mi13
$^{62}\text{Cr}-u$	-42400	600	-43857	4	-1.6	U			TO3	1.5	90Tu01
	-44200	400			0.6	U			TO5	1.5	94Se12
	-43100	350			-1.4	U			TO6	1.5	98Ba.A
	-44026	118			1.0	U			GT1	1.5	04Ma.A
	-43897	526			0.1	U			MT1	1.0	16Me07
	-43845	19			-0.6	U			MR1	1.0	18Mo14
$^{62}\text{Cr}-^{85}\text{Rb}_{.729}$	20448.2	3.7				2			MA8	1.0	18Mo14
$^{62}\text{Mn}-u$	-51510	270	-52093	7	-1.4	U			TO3	1.5	90Tu01
	-52030	280			-0.1	U			TO5	1.5	94Se12
	-51180	280			-2.2	U			TO6	1.5	98Ba.A
$^{62}\text{Mn}^m-^{39}\text{K}_{1.590}$	5982.3	2.8				2			MA8	1.0	12Na15
$^{62}\text{Fe}-^{39}\text{K}_{1.590}$	-5501.5	3.0				2			MA8	1.0	12Na15
$\text{C}_5\text{H}_2-^{62}\text{Ni}$	87299	10	87305.3	0.5	0.2	U			R10	4.0	74De22
$\text{CH}_4\text{N O}_2-^{62}\text{Ni}$	95859	12	95858.6	0.5	-0.0	U			R10	4.0	74De22
$^{62}\text{Cu}-^{62}\text{Ni}$	4250.05	0.51				2			JY1	1.0	06Er03
$^{62}\text{Zn}-^{62}\text{Ni}$	5988.49	0.58	5988.6	0.5	0.2	1	68	^{68}Zn	JY1	1.0	06Er03
$^{62}\text{Ga}-^{62}\text{Ni}$	15845.06	0.71	15844.9	0.5	-0.2	1	52	^{52}Ga	JY1	1.0	06Er03
$^{62}\text{Ga}-^{62}\text{Zn}$	9856.21	0.45	9856.3	0.4	0.2	1	81	^{48}Ga	JY1	1.0	06Er03
$^{62}\text{Ni}-^{61}\text{Ni}$	-2669	15	-2710.1	0.3	-0.7	U			R10	4.0	74De22
$^{62}\text{Ni}-^{60}\text{Ni}$	-2333	30	-2440.4	0.3	-0.9	U			R10	4.0	74De22
$^{62}\text{Ni}(p,\alpha)^{59}\text{Co}$	342	10	347.5	0.4	0.5	U			MIT		64Sp12
	343.3	0.7			5.9	B			NDm		74Jo14
$^{59}\text{Co}(\alpha,p)^{62}\text{Ni}$	-346.5	2.3	-347.5	0.4	-0.4	U			NDm		74Jo14
$^{62}\text{Ni}(^{18}\text{O},^{20}\text{Ne})^{60}\text{Fe}$	911	20	926	3	0.7	U			Hei		84Ha31
$^{62}\text{Ni}(d,\alpha)^{60}\text{Co}$	5611.2	2.4	5614.8	0.4	1.5	U			NDm		74Jo14
$^{60}\text{Ni}(t,p)^{62}\text{Ni}$	9937	10	9934.0	0.3	-0.3	U			Ald		71Da16
$^{60}\text{Ni}(^3\text{He},p)^{62}\text{Cu}$	5938	25	5956.5	0.6	0.7	U			MIT		67Sp09
$^{60}\text{Ni}(^3\text{He},n)^{62}\text{Zn}$	3580	30	3554.7	0.5	-0.8	U			Oak		72Gr39
$^{62}\text{Ni}(^{14}\text{C},^{15}\text{O})^{61}\text{Fe}$	-7921	100	-7661.7	2.7	2.6	F			Ors		84De33 *
$^{62}\text{Ni}(t,\alpha)^{61}\text{Co}$	8689	20	8676.6	0.7	-0.6	U			LAI		66BI15
$^{61}\text{Ni}(n,\gamma)^{62}\text{Ni}$	10596.2	1.5	10595.7	0.3	-0.3	-					70Fa06
	10595.8	0.7			-0.1	-					75Wi06
	10595.6	0.4			0.3	-			Bdn		06Fi.A
$^{61}\text{Ni}(d,p)^{62}\text{Ni}$	8379	8	8371.2	0.3	-1.0	U			MIT		64Sp12
	8369	15			0.1	U			Ald		67Te02
$^{62}\text{Ni}(d,t)^{61}\text{Ni}$	-4340.6	1.3	-4338.5	0.3	1.6	-			NDm		74Jo14
$^{61}\text{Ni}(n,\gamma)^{62}\text{Ni}$	ave. 10595.8	0.3	10595.7	0.3	-0.3	1	87	^{69}Ni			average
$^{62}\text{Mn}^m(\text{IT})^{62}\text{Mn}$	343	6				3					15Ga38 *
$^{62}\text{Fe}(\beta^-)^{62}\text{Co}$	3000	200	2546	19	-2.3	U					75Fr16
$^{62}\text{Co}(\beta^-)^{62}\text{Ni}$	5195	30	5322	19	4.2	C					57Ga15 *
$^{62}\text{Ni}(t,^3\text{He})^{62}\text{Co}$	-5350	50	-5303	19	0.9	2					72Ba31
	-5296	20			-0.4	2			LAI		76Aj03
$^{62}\text{Cu}(\beta^+)^{62}\text{Ni}$	3932	10	3958.9	0.5	2.7	U					54Nu27
	3942	10			1.7	U					64Sa32
	3956	7			0.4	U					67An01
$^{62}\text{Ni}(p,n)^{62}\text{Cu}$	-4733	10	-4741.2	0.5	-0.8	U			Bar		61Ri02
	-4734.8	10.			-0.6	U			Ric		66Ri09
$^{62}\text{Ni}(^3\text{He},t)^{62}\text{Cu}^i$	-8591	6				2					71Be29 *
$^{62}\text{Zn}(\beta^+)^{62}\text{Cu}$	1682	10	1619.5	0.7	-6.3	B					50Ha65
	1697	10			-7.8	B					54Nu27
$^{62}\text{Ga}(\beta^+)^{62}\text{Zn}$	9171	26	9181.1	0.4	0.4	U			ANB		79Da04
$^{62}\text{Ti}-u$	Trends from Mass Surface TMS suggest ^{62}Ti 1200 keV less bound										GAu **
$^{62}\text{Ni}(^{14}\text{C},^{15}\text{O})^{61}\text{Fe}$	F : not unambiguously ground state transition										84De33 **
$^{62}\text{Mn}^m(\text{IT})^{62}\text{Mn}$	$E_x = 418(2) - 72(+8-3)$										GAu **
$^{62}\text{Co}(\beta^-)^{62}\text{Ni}$	$E_{\beta^-} = 5217(30)$ from $^{62}\text{Co}^m$ at 22(5) keV										Nub211 **
$^{62}\text{Ni}(^3\text{He},t)^{62}\text{Cu}^i$	CDE=9360(6) $Q = -8596(6)$; recalibration +5 keV for $^{58}\text{Ni}(p,n)^{58}\text{Cu}^i$ from Ame1961										MMC124**

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{63}\text{V}-\text{u}$	-23339	365				2			RT1	1.0	20Mi13
$^{63}\text{Cr}-\text{u}$	-38819	462	-38840	80	-0.0	U			GT1	1.5	04Ma.A
	-37870	700			-1.4	o			MT1	1.0	11Es06
	-38583	462			-0.6	U			MT1	1.0	16Me07
	-38839	78				2			MR1	1.0	18Mo14
	-49300	400	-50335	4	-1.7	U			TO3	1.5	90Tu01
$^{63}\text{Mn}-\text{u}$	-50190	300			-0.3	U			TO5	1.5	94Se12
	-49600	290			-1.7	U			TO6	1.5	98Ba.A
	-50500	107			1.0	o			GT1	1.5	04Ma.A
	-50829	102			1.9	U			GT2	2.5	08Kn.A
						2			MA8	1.0	12Na15
$^{63}\text{Mn}-^{39}\text{K}_{1.615}$	8278.7	4.0									
$^{63}\text{Fe}-\text{u}$	-59190	240	-59727	5	-1.5	U			TO3	1.5	90Tu01
	-59570	290			-0.4	U			TO5	1.5	94Se12
	-58990	300			-1.6	U			TO6	1.5	98Ba.A
$^{63}\text{Fe}-^{39}\text{K}_{1.615}$	-1114.5	6.1	-1113	5	0.2	1	57	57 ^{63}Fe	MA8	1.0	12Na15
$^{63}\text{Fe}-\text{H C}_2\text{ F}_2$	-64354	10	-64359	5	-0.5	o			MS1	1.0	08Bi05
	-64353	10			-0.6	1	21	21 ^{63}Fe	MS1	1.0	10Fe01
$^{63}\text{Fe}-\text{C }^{32}\text{S F}$	-30204	10	-30202	5	0.2	1	21	21 ^{63}Fe	MS1	1.0	10Fe01
$\text{C}_5\text{ H}_3-^{63}\text{Cu}$	93930	4	93878.0	0.5	-3.3	B			R10	4.0	74De22
$\text{C}_4\text{ H N}-^{63}\text{Cu}$	81347	10	81301.9	0.5	-1.1	U			R10	4.0	74De22
$\text{C}_4\text{ }^{13}\text{C H}_2-^{63}\text{Cu}$	89466	14	89407.8	0.5	-1.0	U			R10	4.0	74De22
$\text{C}_2\text{ H}_7\text{ O}_2-^{63}\text{Cu}$	115064	16	115007.3	0.5	-0.9	U			R10	4.0	74De22
$^{13}\text{C C H}_8\text{ O N}-^{63}\text{Cu}$	134404	18	134346.6	0.5	-0.8	U			R10	4.0	74De22
$^{47}\text{Ti O}-^{63}\text{Cu}$	17036	23	17075.0	0.5	0.4	U			R09	4.0	72De11
$^{63}\text{Cu H}_2\text{ O}-^{81}\text{Sr}$	16962	14	16950	3	-0.8	U			RI1	1.0	18Ki21
$^{63}\text{Ga}-^{85}\text{Rb}_{741}$	4658.0	1.4				2			MA8	1.0	07Gu09
$\text{C}_5\text{ H}_2-^{63}\text{Ga}_{984}$	75382.6	6.7	75384.6	1.4	0.3	U			MS1	1.0	07Sc24
$^{63}\text{Ge}-\text{u}$	-50372	40				2			LZ1	1.0	11Tu02
$^{63}\text{Cu}-^{62}\text{Ni}$	1193	35	1252.37	0.06	0.4	U			R10	4.0	74De22
$^{63}\text{Cu}-^{61}\text{Ni}$	-1449	30	-1457.7	0.3	-0.1	U			R10	4.0	74De22
$^{63}\text{Cu}(\text{p},\alpha)^{60}\text{Ni}$	3757	8	3757.4	0.3	0.1	U			MIT		64Sp12
	3780	10			-2.3	U			Min		67Jo03
	3754.9	1.5			1.7	U			NDm		76Jo01
$^{60}\text{Ni}(\alpha,\text{n})^{63}\text{Zn}$	-7970	40	-7906.2	1.6	1.6	U			Oak		64St01
	-7910	20			0.2	U					67Bi04
$^{63}\text{Cu}(\text{d},\alpha)^{61}\text{Ni}$	9376	30	9353.0	0.3	-0.8	U					67Hj01
$^{62}\text{Ni}(\text{n},\gamma)^{63}\text{Ni}$	6838.04	0.20	6837.77	0.06	-1.4	-			MMn		77Is01 Z
	6837.88	0.18			-0.6	-			ILn		92Ha21 Z
	6837.89	0.14			-0.9	-			Bdn		06Fi.A
	6837.75	0.18			0.1	-			JAn		12Os04
$^{62}\text{Ni}(\text{d},\text{p})^{63}\text{Ni}$	4620	6	4613.20	0.06	-1.1	U			MIT		70An25
	4614.0	1.1			-0.7	U			NDm		74Jo14
$^{62}\text{Ni}(\text{n},\gamma)^{63}\text{Ni}$	ave. 6837.88	0.09	6837.77	0.06	-1.3	1	48	34 ^{63}Ni			average
$^{62}\text{Ni}(\text{p},\gamma)^{63}\text{Cu}$	6119.2	1.5	6122.40	0.06	2.1	U					72Ki15
	6122.30	0.08			1.2	1	54	39 ^{63}Cu	Utr		86De14 Z
$^{63}\text{Cu}(\gamma,\text{n})^{62}\text{Cu}$	-10833	17	-10863.6	0.5	-1.8	U			Phi		60Ge01
$^{63}\text{Co}(\beta^-)^{63}\text{Ni}$	3590	50	3661	19	1.4	1	14	14 ^{63}Co			69Ki.A
$^{63}\text{Ni}(\beta^-)^{63}\text{Cu}$	65.87	0.15	66.977	0.015	7.4	B					66Hs01
	66.946	0.020			1.5	o					87He14
	66.945	0.004			7.9	F					92Ka29 *
	66.9459	0.0054			5.7	F					93Oh02 *
	66.980	0.015			-0.2	1	98	55 ^{63}Ni			99Ho09
$^{63}\text{Zn}(\beta^+)^{63}\text{Cu}$	3352	20	3366.4	1.5	0.7	U					61Cu02
	3390	30			-0.8	U					61Va08
$^{63}\text{Cu}(\text{p},\text{n})^{63}\text{Zn}$	-4146.5	4.	-4148.8	1.5	-0.6	-			Ric		55Br16
	-4139.5	8.			-1.2	U			Oak		55Ki28 Z
	-4150.1	4.4			0.3	-			Tkm		63Ok01
	ave. -4148.1	2.9			-0.2	1	28	27 ^{63}Zn			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{63}\text{Cu}(^3\text{He,t})^{63}\text{Zn}^i$	-8875	6				2					71Be29 *
$^{63}\text{Ga}(\beta^+)^{63}\text{Zn}$	5520	100	5666.3	2.0	1.5	U					72Fi.A
$^{63}\text{Ni}(\beta^-)^{63}\text{Cu}$	F : excitation of atomic electron not taken into account										99Ho09 **
$^{63}\text{Cu}(^3\text{He,t})^{63}\text{Zn}^i$	CDE=9644(6) $Q = -8880(6)$; recalibration +5 keV for $^{58}\text{Ni}(p,n)^{58}\text{Cu}^i$ from Ame1961										MMC124**
$^{64}\text{V}-u$	-17917	548	-17520#	430#	0.7	D			RT1	1.0	20Mi13 *
$^{64}\text{Cr}-u$	-35942	472	-36110	320	-0.4	o			MT1	1.0	16Me07
	-36114	322				2			MT1	1.0	20Me06
$^{64}\text{Mn}-u$	-45340	350	-46151	4	-1.5	U			TO3	1.5	90Tu01 *
	-46340	350			0.4	U			TO5	1.5	94Se12 *
	-45664	306			-1.1	U			TO6	1.5	98Ba.A *
	-46280	129			0.7	U			GT1	1.5	04Ma.A
$^{64}\text{Mn}-^{85}\text{Rb}_{.753}$	20271.7	3.8				2			MA8	1.0	12Na15
$^{64}\text{Fe}-u$	-58600	400	-59012	5	-0.7	U			TO3	1.5	90Tu01
	-59130	300			0.3	U			TO5	1.5	94Se12
	-58500	350			-1.0	U			TO6	1.5	98Ba.A
	-59012.2	5.3			-0.0	o			MS1	1.0	08BI05
$\text{H C}_2 \text{ F}_2-^{64}\text{Fe}_{.984}$	62699.4	5.3				2			MS1	1.0	10Fe01
$^{64}\text{Co}^m-u$	-64075.5	4.5	-64075	4	0.1	o			MS1	1.0	08BI05
$\text{H C}_2 \text{ F}_2-^{64}\text{Co}^m_{.984}$	67681.6	4.6	67681	4	-0.1	1	87	$^{64}\text{Co}^m$	MS1	1.0	10Fe01
$^{64}\text{Co}^m-^{32}\text{S O}_2$	-25974	12	-25976	4	-0.1	1	13	$^{64}\text{Co}^m$	MS1	1.0	10Fe01
$\text{C}_5 \text{ H}_4-^{64}\text{Ni}$	103278	10	103333.9	0.5	1.4	U			R10	4.0	74De22
$\text{C}_4 \text{ }^{13}\text{C H}_3-^{64}\text{Ni}$	98809	12	98863.7	0.5	1.1	U			R10	4.0	74De22
$\text{C}_4 \text{ H}_2 \text{ N}-^{64}\text{Ni}$	90703	16	90757.8	0.5	0.9	U			R10	4.0	74De22
$^{64}\text{Ni}-^{85}\text{Rb}_{.753}$	-5609.2	1.4	-5611.4	0.5	-1.6	1	13	^{64}Ni	MA8	1.0	07Gu09
$^{64}\text{Zn}-^{85}\text{Rb}_{.753}$	-4430.1	8.4	-4435.9	0.7	-0.7	U			MA8	1.0	07Ke09
$^{64}\text{Ga}-^{85}\text{Rb}_{.753}$	3261.3	2.5	3262.7	1.5	0.6	1	38	^{64}Ga	MA8	1.0	07Gu09
$\text{C}_5 \text{ H}_2-^{64}\text{Ga}_{.969}$	76851.5	2.6	76851.7	1.5	0.1	1	33	^{64}Ga	MS1	1.0	07Sc24
$^{64}\text{Ge}-u$	-57090	690	-58310	4	-1.2	U			GA6	1.5	02Li24
$\text{H }^{32}\text{S O}_2-^{64}\text{Ge H}$	20210.5	4.0				2			MS1	1.0	07Sc24
$^{64}\text{Ge}-^{85}\text{Rb}_{.753}$	8070	43	8112	4	1.0	U			MS1	1.0	19Sc11
$^{64}\text{Ga}-^{64}\text{Zn}$	7698.5	4.1	7698.6	1.6	0.0	1	15	^{64}Ga	CP1	1.0	07CI01
$^{64}\text{Ge}-^{64}\text{Zn}$	12517	33	12548	4	0.9	U			CP1	1.0	07CI01
$^{64}\text{Ni}-^{63}\text{Cu}$	-1523	30	-1630.89	0.22	-0.9	U			R10	4.0	74De22
$^{64}\text{Ga}-^{63}\text{Ga}$	-2730	150	-2453.8	2.1	0.7	U			CR1	2.5	89Sh10
$^{64}\text{Ni}-^{62}\text{Ni}$	-352	25	-378.53	0.23	-0.3	U			R10	4.0	74De22
$^{64}\text{Ni}(^3\text{He},^8\text{B})^{59}\text{Mn}$	-19610	30	-19564.1	2.6	1.5	U			MSU		76Ka24
$^{64}\text{Ni}(^3\text{He},^7\text{Be})^{60}\text{Fe}$	-6511	10	-6524	3	-1.3	R			MSU		76St11
$^{64}\text{Ni}(\alpha,^7\text{Be})^{61}\text{Fe}$	-21523	20	-21522.6	2.6	0.0	U			Tex		77Co08
$^{64}\text{Ni}(^{14}\text{C},^{17}\text{O})^{61}\text{Fe}$	-4609	100	-4349.9	2.6	2.6	U			Ors		84Be.A *
$^{64}\text{Ni}(p,\alpha)^{61}\text{Co}$	663.2	0.7				2			NDm		74Jo14
$^{64}\text{Zn}(p,\alpha)^{61}\text{Cu}$	830	15	844.1	0.7	0.9	U					67Br10
	830	10			1.4	U			Min		67Jo03
	844.1	0.7				2			NDm		76Jo01
$^{64}\text{Zn}(^3\text{He},^6\text{He})^{61}\text{Zn}$	-12331	23	-12316	16	0.7	-			MSU		79We02
ave.	-12320	16			0.3	1	95	^{61}Zn			average
$^{64}\text{Ni}(^{11}\text{B},^{13}\text{N})^{62}\text{Fe}$	-4930	70	-4898.8	2.8	0.4	U			Tex		77Co08
$^{64}\text{Ni}(^{14}\text{C},^{16}\text{O})^{62}\text{Fe}$	-501	40	-464.1	2.8	0.9	U			Ors		81Be40
$^{64}\text{Ni}(^{18}\text{O},^{20}\text{Ne})^{62}\text{Fe}$	-1915	50	-1961.9	2.8	-0.9	U			Can		76Hi14
	-1920	21			-2.0	U			Hei		77Bh03 *
	-1947	26			-0.6	U			Hei		84Ha31
$^{64}\text{Ni}(d,\alpha)^{62}\text{Co}$	5190	20	5036	19	-7.7	B					72Ba31
$^{62}\text{Ni}(t,p)^{64}\text{Ni}$	7999	20	8013.43	0.21	0.7	U			Ald		71Da16
$^{62}\text{Ni}(^3\text{He},p)^{64}\text{Cu}$	6299	25	6320.23	0.06	0.8	U			MIT		67Sp09

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{62}\text{Ni}(^3\text{He},n)^{64}\text{Zn}$	6118	12	6117.5	0.6	-0.0	U			Oak		72Gr39
$^{64}\text{Zn}(d,\alpha)^{62}\text{Cu}$	7508	15	7494.3	0.8	-0.9	U			MIT		67Sp09
$^{64}\text{Zn}(p,t)^{62}\text{Zn}$	-12493	10	-12496.8	0.8	-0.4	U			Bld		72Fa08
$^{64}\text{Ni}(^{14}\text{C},^{15}\text{O})^{63}\text{Fe}$	-11387	60	-11299	4	1.5	U			Ors		82De.A *
$^{64}\text{Ni}(^{34}\text{S},^{35}\text{Ar})^{63}\text{Fe}$	-17931	260	-18348	4	-1.6	U			Hei		83Wi.B
$^{64}\text{Ni}(t,\alpha)^{63}\text{Co}$	7266	20	7277	19	0.6	1	86	86 ^{63}Co	LAI		66Bl15
$^{63}\text{Ni}(n,\gamma)^{64}\text{Ni}$	9657.32	0.4	9657.46	0.20	0.4	-					75Wi06 Z
	9657.58	0.24			-0.5	-			ILn		92Ha21
	ave. 9657.51	0.21			-0.2	1	98	87 ^{64}Ni			average
$^{63}\text{Cu}(n,\gamma)^{64}\text{Cu}$	7916.07	0.12	7915.868	0.024	-1.7	U			BNn		83De28 Z
	7915.52	0.08			4.4	B					02Bo11
	7916.14	0.16			-1.7	U			Bdn		06Fi.A
	7915.867	0.024			0.0	1	100	91 ^{64}Cu			16Te05
$^{63}\text{Cu}(d,p)^{64}\text{Cu}$	5697	8	5691.302	0.024	-0.7	U			MIT		64Sp12
$^{64}\text{Zn}(n,d)^{63}\text{Cu}$	-5520	50	-5488.6	0.6	0.6	U					65Wa14
$^{64}\text{Zn}(d,t)^{63}\text{Zn}$	-5604.9	1.7	-5604.7	1.5	0.1	1	76	73 ^{63}Zn	NDm		76Jo01
$^{64}\text{Co}(\beta^-)^{64}\text{Ni}$	7000	500	7307	20	0.6	U					69Wa15
	7000	400			0.8	U					74Ra31
$^{64}\text{Ni}(t,^3\text{He})^{64}\text{Co}$	-7288	20				2			LAI		72Fi17
$^{64}\text{Cu}(\beta^+)^{64}\text{Ni}$	1673.4	1.0	1674.62	0.21	1.2	U					83Ch47
$^{64}\text{Ni}(p,n)^{64}\text{Cu}$	-2458	6	-2456.96	0.21	0.2	U					61Va19
	-2458.22	0.31			4.1	B			PTB		92Bo02 Z
$^{64}\text{Cu}(\beta^-)^{64}\text{Zn}$	577.8	1.0	579.6	0.6	1.8	1	42	32 ^{64}Zn			83Ch47
$^{64}\text{Ga}(\beta^+)^{64}\text{Zn}$	7072	30	7171.2	1.5	3.3	B					60Ja07
$^{64}\text{Zn}(p,n)^{64}\text{Ga}$	-7951	4	-7953.5	1.5	-0.6	1	14	12 ^{64}Ga	Tex		72Da.A
$^{64}\text{Zn}(^3\text{He},t)^{64}\text{Ga}$	-7206	8	-7189.8	1.5	2.0	U			MSU		74Ro16 *
$^{64}\text{Zn}(^3\text{He},t)^{64}\text{Ga}^i$	-9141	17	-9096.8	2.5	2.6	U			MIT		70Hi06
	-9110	6			2.2	1	17	17 $^{64}\text{Ga}^i$			71Be29 *
$^{64}\text{Ga}^i(\text{IT})^{64}\text{Ga}$	1905.1	2.3	1907.0	2.2	0.8	1	88	83 $^{64}\text{Ga}^i$			74Ro16
$^{64}\text{Ge}(\beta^+)^{64}\text{Ga}$	4410	250	4517	4	0.4	U					73Da01 *
* $^{64}\text{V}-u$	Trends from Mass Surface TMS suggest ^{64}V 370 keV less bound										GAu **
* $^{64}\text{Mn}-u$	Original -45270(350) μu or $M-A=-42170(330)$ keV										GAu **
* $^{64}\text{Mn}-u$	Original -46270(350) μu or $M-A=-43100(330)$ keV										GAu **
* $^{64}\text{Mn}-u$	$M-A=-42430(280)$ keV for mixture gs+m at 174.1(0.5) (4^+) keV										Nub211 **
* $^{64}\text{Ni}(^{14}\text{C},^{17}\text{O})^{61}\text{Fe}$	Cited in reference and confirmed in PrvCom sep 88										84De33 **
* $^{64}\text{Ni}(^{18}\text{O},^{20}\text{Ne})^{62}\text{Fe}$	$Q-Q(^{62}\text{Ni}(^{18}\text{O},^{20}\text{Ne}))=-2843(20)$, $Q(62)=923(4)$ keV										AHW **
* $^{64}\text{Ni}(^{14}\text{C},^{15}\text{O})^{63}\text{Fe}$	Original -11743(60) reinterpreted as ($3/2^-$) 356.2 level in ^{63}Fe										GAu **
* $^{64}\text{Zn}(^3\text{He},t)^{64}\text{Ga}$	$M-A=-58819(8)$; Q rebuilt with Ame1971										GAu **
* $^{64}\text{Zn}(^3\text{He},t)^{64}\text{Ga}^i$	CDE=9879(6) $Q=-9115(6)$; recalibration +5 keV for $^{58}\text{Ni}(p,n)^{58}\text{Cu}^i$ from Ame1961										MMC124**
* $^{64}\text{Ge}(\beta^+)^{64}\text{Ga}$	$E_{\beta^+}=2960(250)$ to (1^+) level at 427.03 keV										Ens073 **
$^{65}\text{Cr}-u$	-29286	837	-30390#	220#	-1.3	D			MT1	1.0	20Me06 *
$^{65}\text{Mn}-u$	-43900	600	-43980	4	-0.1	U			TO5	1.5	94Se12
	-43500	500			-0.6	U			TO6	1.5	98Ba.A
	-43790	330			-0.6	U			MT1	1.0	11Es06
$^{65}\text{Mn}-^{85}\text{Rb}_{.765}$	23500.6	4.0				2			MA8	1.0	12Na15
$^{65}\text{Fe}-u$	-54680	300	-54985	5	-0.7	U			TO3	1.5	90Tu01 *
	-55270	320			0.6	U			TO5	1.5	94Se12 *
	-54290	380			-1.2	U			TO6	1.5	98Ba.A *
	-54985.9	5.4			0.2	o			MS1	1.0	08Bl05 *
$\text{O}_2-^{65}\text{Fe}_{.492}$	16881.7	2.7				2			MS1	1.0	10Fe01 *
$^{65}\text{Co}-u$	-63537.9	2.3	-63537.9	2.2	-0.0	o			MS1	1.0	08Bl05
$\text{O}_2-^{65}\text{Co}_{.492}$	21089.9	1.1				2			MS1	1.0	10Fe01
$^{65}\text{Ni}-^{85}\text{Rb}_{.765}$	-2438.0	2.4	-2434.6	0.5	1.4	U			MA8	1.0	07Gu09
$\text{C}_5\text{H}_5-^{65}\text{Cu}$	111384	4	111335.7	0.7	-3.0	B			R10	4.0	74De22

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_4 H_3 N-^{65}Cu$	98800	8	98759.6	0.7	-1.3	U			R10	4.0	74De22
$C_4 ^{13}C H_4-^{65}Cu$	106921	18	106865.5	0.7	-0.8	U			R10	4.0	74De22
$^{49}Ti O-^{65}Cu$	15030	10	14989.5	0.7	-1.0	U			R09	4.0	72De11
$^{65}Cu-^{85}Rb_{.765}$	-4730.6	1.2	-4729.7	0.7	0.8	1	33	33 ^{65}Cu	MA8	1.0	07Gu09
$^{65}Ga-^{85}Rb_{.765}$	215.4	1.5	215.3	0.8	-0.1	1	32	32 ^{65}Ga	MA8	1.0	07Gu09
$^{65}Ge-u$	-60080	270	-60631.9	2.3	-1.4	U			GA6	1.5	02Li24
$C_5 H_2-^{65}Ge_{.939}$	72585.2	4.0	72583.4	2.2	-0.5	-			MS1	1.0	07Sc24 *
$C_5 H_5-^{65}Ge_{.985}$	98847.2	4.2	98847.5	2.3	0.1	-			MS1	1.0	07Sc24 *
$C_5 H_2-^{65}Ge_{.939}$	ave. 72584.2	2.9	72583.4	2.2	-0.3	1	57	57 ^{65}Ge			average
$^{65}Ge H-^{85}Rb_{.776}$	15634.4	6.2	15644.3	2.3	1.6	1	14	14 ^{65}Ge	MS1	1.0	07Sc24 *
$^{65}Ge O H-^{85}Rb_{.965}$	27237.1	4.3	27230.7	2.3	-1.5	1	29	29 ^{65}Ge	MS1	1.0	19Sc11
$^{65}As-u$	-50389	91				2			LZ1	1.0	11Tu02
$^{64}Zn H-^{65}Cu$	9171	16	9177.3	0.4	0.4	U			RI1	1.0	18Ki21
$^{65}Zn-^{65}Cu$	1459	13	1451.1	0.4	-0.6	U			RI1	1.0	18Ki21
$^{65}Ga-^{65}Cu$	4945.2	2.1	4944.9	0.7	-0.1	1	12	9 ^{65}Ga	RI1	1.0	18Ki21
$^{65}Ge-^{65}Cu$	11592	22	11578.7	2.4	-0.6	U			RI1	1.0	18Ki21
$^{65}Cu-^{64}Ni$	-275	40	-176.8	0.7	0.6	U			R10	4.0	74De22
$^{65}Cu-^{63}Cu$	-1784	10	-1807.6	0.7	-0.6	U			R10	4.0	74De22
$^{65}Cu(p,\alpha)^{62}Ni$	4345	8	4346.8	0.7	0.2	U			MIT		64Sp12
	4340	10			0.7	U			Min		67Jo03
	4344.6	1.8			1.2	1	13	10 ^{65}Cu	NDm		76Jo01
$^{62}Ni(\alpha,n)^{65}Zn$	-6510	40	-6480.8	0.6	0.7	U			Oak		64St01
$^{65}Cu(d,\alpha)^{63}Ni$	9012	40	8960.0	0.7	-1.3	U					67Hj01
$^{63}Cu(t,p)^{65}Cu$	9351	25	9344.6	0.7	-0.3	U			Ald		66Bj02
$^{64}Ni(n,\gamma)^{65}Ni$	6097.86	0.20	6098.08	0.14	1.1	2			MMn		77Is01 Z
	6098.28	0.19			-1.0	2			Bdn		06Fi.A
$^{64}Ni(d,p)^{65}Ni$	3876	6	3873.51	0.14	-0.4	U			MIT		64Sp12
	3867	15			0.4	U			Ald		67Te02
	3870	5			0.7	U			MIT		70An25
$^{65}Cu(t,\alpha)^{64}Ni$	12352	11	12360.3	0.7	0.8	U					72He23
$^{65}Cu(\gamma,n)^{64}Cu$	-9896	28	-9910.6	0.7	-0.5	U			Phi		60Ge01
$^{65}Cu(d,t)^{64}Cu$	-3650	60	-3653.3	0.7	-0.1	U			ANL		60Ze02
$^{64}Zn(n,\gamma)^{65}Zn$	7979.3	0.8	7979.32	0.17	0.0	U					71Ot01 Z
	7979.2	0.5			0.2	U					75De.A Z
	7979.28	0.17			0.3	1	98	54 ^{65}Zn	Bdn		06Fi.A
$^{64}Zn(d,p)^{65}Zn$	5758	10	5754.76	0.17	-0.3	U			ANL		67Vo05
$^{64}Zn(p,\gamma)^{65}Ga$	3942.0	1.0	3942.4	0.6	0.4	-					75We24 Z
	3943.0	1.0			-0.6	-					87Vi01
	ave. 3942.5	0.7			-0.1	1	76	59 ^{65}Ga			average
$^{65}Ge(\epsilon p)^{64}Zn$	2300	100	2236.8	2.3	-0.6	U			ChR		81Ha44
$^{65}As^i(p)^{64}Ge$	3603	30	3575	10	-0.9	U					93Ba12
	3564	20			0.5	o					11Ro47 *
	3575	10			3						14Ro14
$^{65}Ni(\beta^-)^{65}Cu$	2140	10	2137.9	0.7	-0.2	U					64Fr04
$^{65}Zn(\beta^+)^{65}Cu$	1347	2	1351.7	0.4	2.3	U					49Ma57
	1347	2			2.3	U					53Ba82
	1347	3			1.6	U					53Pe14
	1349	3			0.9	U					53Sa26
	1342	4			2.4	U					53Yu04
	1346	4			1.4	U					56Av28
$^{65}Cu(p,n)^{65}Zn$	-2131.4	1.5	-2134.0	0.4	-1.7	U					56Ma14 Z
	-2135.8	2.5			0.7	U					57Be44 Z
	-2135.3	1.8			0.7	U			Tkm		63Ok01
	-2135.6	1.7			0.9	U			Oak		64Jo11 Z
	-2134.6	0.8			0.7	-			Yal		69Ov01 Z
	-2133.55	0.43			-1.0	-			PTB		89Sc24
	ave. -2133.8	0.4			-0.6	1	89	46 ^{65}Zn			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{65}\text{Ga}(\beta^+)^{65}\text{Zn}$	3277	30	3254.5	0.6	-0.7	U					57Da07
$^{65}\text{Ge}(\beta^+)^{65}\text{Ga}$	5220	400	6179.3	2.3	2.4	U					58Po79
$^{65}\text{Cr}-\text{u}$	Trends from Mass Surface TMS suggest ^{65}Cr 780 keV more bound										GAu **
$^{65}\text{Fe}-\text{u}$	$M-A=-50740(250)$ keV for mixture gs+m at 393.7(0.2) keV										Nub211 **
$^{65}\text{Fe}-\text{u}$	$M-A=-51290(280)$ keV for mixture gs+m at 393.7(0.2) keV										Nub211 **
$^{65}\text{Fe}-\text{u}$	$M-A=-50370(330)$ keV for mixture gs+m at 393.7(0.2) keV and										Nub211 **
*	assuming ratio $R=0.13(6)$, from half-life=430 ns and TOF=1 μs										GAu **
$^{65}\text{Fe}-\text{u}$	$D_M=-54988.9(7.1)$ μu to ground state, $D_M=-54557.0(8.4)$ to $^{65}\text{Fe}^m$ at 393.7 keV										Nub211 **
$^*\text{O}_2-^{65}\text{Fe}_{.492}$	$D_M=16883.6(3.6)$ μu to ground state, $D_M=16671.2(4.2)$ to $^{65}\text{Fe}^m$ at 393.7 keV										Nub211 **
$^*\text{C}_5\text{H}_5-^{65}\text{Ge}_{.939}$	For original doublet $\text{C}_5\text{H}_5-(^{65}\text{GeH})_{0.939}$, $D_M=65237.5(4.0)$ μu										GAu **
$^*\text{C}_5\text{H}_5-^{65}\text{Ge}_{.985}$	For original doublet $\text{C}_5\text{H}_5-(^{65}\text{GeH})_{0.985}$, $D_M=91139.5(4.2)$ μu										GAu **
$^{65}\text{GeH}-^{85}\text{Rb}_{.776}$	Combining their 3 results yields a precision for ^{65}Ge of 2.4 keV, while										GAu **
*	their Table III gives 1.2 keV										GAu **
$^{65}\text{As}^i(\text{p})^{64}\text{Ge}$	And $E_p=2620(30)$ to 901.7 level										11Ro47 **
$^{66}\text{Mn}-\text{u}$	-39860	860	-39453	12	0.5	U			MT1	1.0	11Es06 *
$^{66}\text{Mn}-^{85}\text{Rb}_{.776}$	28998	12				2			MA8	1.0	12Na15
$^{66}\text{Fe}-\text{u}$	-52300	700	-53750	4	-1.4	U			TO3	1.5	90Tu01
	-54020	350			0.5	U			TO5	1.5	94Se12
	-52800	300			-2.1	U			TO6	1.5	98Ba.A
	-53935	150			0.8	U			GT1	1.5	04Ma.A
$^{66}\text{Fe}-^{28}\text{Si F}_2$	-27482.9	4.4				2			MS1	1.0	10Fe01
$^{66}\text{Co}-\text{u}$	-60470	300	-60557	15	-0.2	U			TO5	1.5	94Se12 *
	-59870	290			-1.6	U			TO6	1.5	98Ba.A *
$^{66}\text{Co}-\text{O C F}_2$	-52278	15	-52278	15	0.0	o			MS1	1.0	08Bl05
	-52278	15				2			MS1	1.0	10Fe01
$^{66}\text{Ni}-^{85}\text{Rb}_{.776}$	-2409.5	1.5				2			MA8	1.0	07Gu09
$^{66}\text{Cu}-^{85}\text{Rb}_{.776}$	-2680.6	2.2	-2680.0	0.7	0.3	1	10	10 ^{66}Cu	MA8	1.0	07Gu09
$\text{C}_5\text{H}_5-^{66}\text{Ge}_{.970}$	103278.9	2.5				2			MS1	1.0	07Se24 *
$^{66}\text{As}-\text{u}$	-55290	730	-55851	6	-0.5	U			GA6	1.5	02Li24
$^{66}\text{As}-^{85}\text{Rb}_{.776}$	12607	32	12600	6	-0.2	U			MS1	1.0	07Sc24
$^{66}\text{As O}-^{85}\text{Rb}_{.965}$	24186.3	6.1				2			MS1	1.0	19Sc11
$^{65}\text{GeH}-^{66}\text{Zn}$	21182	24	21159.5	2.5	-0.9	U			RII	1.0	18Ki21
$^{66}\text{Ga}-^{66}\text{Zn}$	5528	11	5556.1	0.9	2.6	F			RII	1.0	18Ki21 *
$^{66}\text{Ge}-^{66}\text{Zn}$	7824	14	7828.5	2.7	0.3	U			RII	1.0	18Ki21
$^{66}\text{Zn}(\text{p},\alpha)^{63}\text{Cu}$	1544.3	0.8	1544.7	0.7	0.5	1	74	66 ^{66}Zn	NDm		76Jo01
$^{63}\text{Cu}(\alpha,\text{n})^{66}\text{Ga}$	-7670	30	-7502.5	1.1	5.6	B			Oak		64St01
$^{64}\text{Ni}(\text{t},\text{p})^{66}\text{Ni}$	6559	25	6568.1	1.5	0.4	U			Ald		71Da16
$^{64}\text{Zn}(\text{t},\text{p})^{66}\text{Zn}$	10582	15	10556.1	0.9	-1.7	U			Ald		72Hu06
$^{65}\text{Cu}(\text{n},\gamma)^{66}\text{Cu}$	7065.80	0.12	7065.93	0.09	1.1	-			BNn		83De29 Z
	7066.13	0.15			-1.3	-			Bdn		06Fi.A
$^{65}\text{Cu}(\text{d},\text{p})^{66}\text{Cu}$	4837	8	4841.36	0.09	0.5	U			MIT		64Sp12
$^{65}\text{Cu}(\text{n},\gamma)^{66}\text{Cu}$	ave. 7065.93	0.09	7065.93	0.09	0.0	1	100	90 ^{66}Cu			average
$^{66}\text{Zn}(\text{d},\text{t})^{65}\text{Zn}$	-4770	60	-4801.3	0.9	-0.5	U			ANL		60Ze02
$^{66}\text{Co}(\beta^-)^{66}\text{Ni}$	9700	500	9598	14	-0.2	U					88Bo06
$^{66}\text{Ni}(\beta^-)^{66}\text{Cu}$	200	30	252.0	1.5	1.7	U					56Jo20
$^{66}\text{Cu}(\beta^-)^{66}\text{Zn}$	2650	30	2640.9	0.9	-0.3	U					51Fr19 *
	2650	30			-0.3	U					56Jo20
$^{66}\text{Ga}(\beta^+)^{66}\text{Zn}$	5175.0	3.0	5175.5	0.8	0.2	U					63Ca03 *
	5175.5	0.8				2					14Se12
$^{66}\text{Zn}(^3\text{He},\text{t})^{66}\text{Ga}^i$	-9044	6				2					71Be29 *
$^{66}\text{Ge}(\beta^+)^{66}\text{Ga}$	2490	50	2116.7	2.6	-7.5	F					69Ba31 *
	2420	30			-10.1	F					69Sa08 *
	2100	30			0.6	U					70De39 *
$^{66}\text{As}(\beta^+)^{66}\text{Ge}$	9550	50	9582	6	0.6	U			ANB		79Da.A
$^{66}\text{Mn}-\text{u}$	$M-A=-36900(790)$ keV for mixture gs+m at 464.5 keV (5^-)										Nub211 **
$^{66}\text{Co}-\text{u}$	Original $-60160(300)$ μu or $M-A=-56040(280)$ keV										GAu **
$^{66}\text{Co}-\text{u}$	$M-A=-55480(270)$ keV for mixture gs+m+n at 175.1 and 642(5) keV										Nub211 **
*	and assuming for first isomer a ratio $R=0.5(0.2)$ to ground state,										GAu **
*	from half-life=1.21 μs and TOF=1 μs										GAu **
$^*\text{C}_5\text{H}_5-^{66}\text{Ge}_{.970}$	For original doublet $\text{C}_5\text{H}_5-(^{66}\text{GeH})_{0.970}$, $D_M=95688.6(2.5)$ μu										GAu **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{66}\text{Ga}-^{66}\text{Zn}$	Compromised measurement due to the influence of neighboring ToF peaks										Sar201	**
$^{66}\text{Cu}(\beta^-)^{66}\text{Zn}$	$E_{\beta^-}=2630(30)$ 1640(30) to ground state and 2^+ level at 1039.2279 keV										Ens104	**
$^{66}\text{Ga}(\beta^+)^{66}\text{Zn}$	Original $E_{\beta^+}=4153(3)$ corrected for recoil										14Se12	**
$^{66}\text{Zn}(^3\text{He,t})^{66}\text{Ga}^i$	CDE=9813(6) $Q=-9049(6)$; recalibration +5 keV for $^{58}\text{Ni}(p,n)^{58}\text{Cu}^i$ from Ame1961										MMC124	**
$^{66}\text{Ge}(\beta^+)^{66}\text{Ga}$	$E_{\beta^+}=1440(50)$ to 43.9 level; F : probably distorted by annihilation pile up										AHW	**
$^{66}\text{Ge}(\beta^+)^{66}\text{Ga}$	$E_{\beta^+}=1370(30)$ to 43.9 level; F : probably distorted by annihilation pile up										AHW	**
$^{66}\text{Ge}(\beta^+)^{66}\text{Ga}$	$E_{\beta^+}=1028(30)$, 668(30), 558(50) to 43.812 1^+ , 381.859 1^+ , 536.618 1^+ level										Ens104	**
$^{67}\text{Mn}-u$	-36600	670	-36050#	220#	0.8	o			MT1	1.0	15Me.A	
	-36458	354			1.2	D			MT1	1.0	20Me06	*
$^{67}\text{Fe}-u$	-50190	500	-49070	4	1.5	U			TO5	1.5	94Se12	*
	-48450	380			-1.1	U			TO6	1.5	98Ba.A	*
	-49641	440			0.9	U			GT1	1.5	04Ma.A	
	-49580	300			1.7	o			MT1	1.0	11Es06	*
	-48510	460			-1.2	o			MT1	1.0	15Me.A	
	-49220	370			0.4	U			MT1	1.0	20Me06	*
	-49070.0	4.1				2			JY1	1.0	20Ca08	
$^{67}\text{Co}-u$	-59390	300	-59390	7	-0.0	U			TO5	1.5	94Se12	
	-58730	350			-1.3	U			TO6	1.5	98Ba.A	
$^{28}\text{Si F}_2-^{67}\text{Co}_{.985}$	32232.9	8.0	32232	7	-0.1	2			MS1	1.0	10Fe01	
	32231	13			0.1	2			MS1	1.0	10Fe01	*
$^{67}\text{Ni}-u$	-68370	430	-68431	3	-0.1	U			TO5	1.5	94Se12	*
	-68090	470			-0.5	U			TO6	1.5	98Ba.A	*
$^{67}\text{Ni}-^{85}\text{Rb}_{.788}$	1079.1	3.1				2			MA8	1.0	07Gu09	
$^{67}\text{Cu}-^{85}\text{Rb}_{.788}$	-2760.0	1.3	-2760.8	1.0	-0.6	1	54	54 ^{67}Cu	MA8	1.0	07Gu09	
$^{67}\text{As}-u$	-60500	260	-60748.9	0.5	-0.6	U			GA6	1.5	02Li24	
$^{67}\text{As}-^{86}\text{Kr}_{.779}$	8885.7	2.9	8885.4	0.5	-0.1	U			MS1	1.0	07Sc24	
	8808	30			2.6	U			MS1	1.0	07Sc24	
$^{67}\text{As}-^{85}\text{Rb}_{.788}$	8762.5	1.7	8760.8	0.5	-1.0	U			MS1	1.0	07Sc24	
	8760.87	0.54			-0.1	1	77	77 ^{67}As	MS1	1.0	19Sc11	
$^{67}\text{As O}-^{85}\text{Rb}_{.976}$	20258.7	1.0	20258.9	0.5	0.2	1	23	23 ^{67}As	MS1	1.0	19Sc11	
$^{67}\text{Se}-u$	-50006	72				2			LZ1	1.0	11Tu02	
$^{67}\text{Zn N}-^{66}\text{Zn }^{15}\text{N}$	4060.21	0.32	4058.89	0.25	-1.7	U			H30	2.5	77Ba10	*
$^{66}\text{Zn H}-^{67}\text{Zn}$	6741	15	6731.25	0.25	-0.7	U			RI1	1.0	18Ki21	
$^{67}\text{Ga}-^{67}\text{Zn}$	1112	11	1074.9	1.2	-3.4	F			RI1	1.0	18Ki21	*
$^{67}\text{Ge}-^{67}\text{Zn}$	5587.9	4.9	5590	5	0.3	1	87	87 ^{67}Ge	RI1	1.0	18Ki21	
$^{67}\text{As}-^{67}\text{Zn}$	12122	27	12123.7	0.9	0.1	U			RI1	1.0	18Ki21	
$^{64}\text{Zn}(\alpha,n)^{67}\text{Ge}$	-9240	60	-8977	4	4.4	B			Oak		64St01	
	-8987.5	12.			0.9	1	13	13 ^{67}Ge	ANL		78Mu05	
	-8993	5			3.3	B					79Al04	
$^{65}\text{Cu}(t,p)^{67}\text{Cu}$	7716	25	7716.7	1.1	0.0	U			Ald		66Bj02	
$^{65}\text{Cu}(^3\text{He,p})^{67}\text{Zn}$	8185	40	8258.9	0.9	1.8	U			MIT		74Is01	
$^{67}\text{Kr}(2p)^{65}\text{Se}$	1690	17	2890#	300#	70.6	D					16Go26	*
$^{66}\text{Zn}(n,\gamma)^{67}\text{Zn}$	7052.5	0.6	7052.47	0.23	-0.1	-					71Ot01	Z
	7052.5	0.5			-0.1	-					75De.A	Z
	7052.5	0.3			-0.1	-			Bdn		06Fi.A	
$^{66}\text{Zn}(d,p)^{67}\text{Zn}$	4827	10	4827.90	0.23	0.1	U			ANL		67Vo05	
	4820	5			1.6	U			MIT		74Is01	
$^{67}\text{Zn}(d,t)^{66}\text{Zn}$	-800	60	-795.24	0.23	0.1	U			ANL		60Ze02	
$^{66}\text{Zn}(n,\gamma)^{67}\text{Zn}$	ave. 7052.50	0.24	7052.47	0.23	-0.1	1	98	64 ^{67}Zn			average	
$^{67}\text{Ni}(\beta^-)^{67}\text{Cu}$	3830	90	3577	3	-2.8	U					75Re09	
$^{67}\text{Cu}(\beta^-)^{67}\text{Zn}$	577	8	560.8	0.8	-2.0	U					53Ea11	
	560.3	1.0			0.5	1	69	46 ^{67}Cu	ANB		15Ch57	
$^{67}\text{Zn}(p,n)^{67}\text{Ga}$	-1776	5	-1783.6	1.1	-1.5	U			Ric		57Ch30	Y
	-1783.3	1.4			-0.2	1	66	54 ^{67}Ga	Oak		64Jo11	Z

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{67}\text{Ge}(\beta^+)^{67}\text{Ga}$	4330	100	4205	4	-1.2	U					59Ri35 *
	4370	150			-1.1	U					69Ba07 *
$^{67}\text{As}(\beta^+)^{67}\text{Ge}$	6010	100	6086	4	0.8	U			ANB		80Mu12
$^{67}\text{Mn}-\text{u}$	Trends from Mass Surface TMS suggest ^{67}Mn 380 keV less bound										GAu **
$^{67}\text{Fe}-\text{u}$	Original -50000(500) μu or $M-A=-46570(470)$ keV										GAu **
$^{67}\text{Fe}-\text{u}$	$M-A=-44930(330)$ keV for mixture gs+m at 403(9) keV										Nub211 **
$^{67}\text{Fe}-\text{u}$	$M-A=-45980(250)$ keV for mixture gs+m at 403(9) keV										Nub211 **
$^{67}\text{Fe}-\text{u}$	$M-A=-45560(320)$ keV for mixture gs+m+n at 403(9) and 450#(100#) keV										Nub211 **
$^{28}\text{Si F}_2-^{67}\text{Co}_{.985}$	$M-A=-54829(12)$ for $^{67}\text{Co}^m$ at 491.55(0.11) keV										10Fe01 **
$^{67}\text{Ni}-\text{u}$	Original -67840(300) μu or $M-A=-63190(280)$ keV										GAu **
$^{67}\text{Ni}-\text{u}$	$M-A=-62930(330)$ keV for mixture gs+m at 1006.6(0.2) keV										Nub211 **
$^{67}\text{Zn N}-^{66}\text{Zn }^{15}\text{N}$	Original 4060.21(0.25); plus syst 0.20 μu estimated by evaluator										GAu **
$^{67}\text{Ga}-^{67}\text{Zn}$	Compromised measurement due to the influence of neighboring ToF peaks										Sar201 **
$^{67}\text{Kr}(2p)^{65}\text{Se}$	Trends from Mass Surface TMS suggest ^{67}Kr 1200 keV less bound										GAu **
$^{67}\text{Ge}(\beta^+)^{67}\text{Ga}$	$E_{\beta^+}=3140(100)$ 3180(100) respectively, to $1/2^-$ level at 166.98 keV										Ens05c **
$^{68}\text{Mn}-\text{u}$	-29748	1406	-31050#	320#	-0.9	D			MT1	1.0	20Me06 *
$^{68}\text{Fe}-\text{u}$	-46300	500	-47130#	210#	-1.1	2			TO6	1.5	98Ba.A
	-47330	460			0.4	o			MT1	1.0	11Es06
	-46830	460			-0.6	o			MT1	1.0	15Me.A
	-47622	344			1.4	D			MT1	1.0	20Me06 *
$^{68}\text{Co}-\text{u}$	-55640	350	-55441	4	0.4	o			TO5	1.5	94Se12
	-54750	300			-1.5	U			TO6	1.5	98Ba.A
	-55730	140			0.8	U			GT2	2.5	08Kn.A *
	-55760	250			1.3	U			MT1	1.0	11Es06 *
$\text{O }^{18}\text{O}-^{68}\text{Co}_{.5}$	21794.5	2.4	21794.5	2.1	0.0	1	74	74 ^{68}Co	MS1	1.0	18Iz01
$^{68}\text{Co}-^{34}\text{S}_{2.000}$	8825.2	8.2	8825	4	0.0	1	26	26 ^{68}Co	MS1	1.0	18Iz01
$^{68}\text{Ni}-\text{u}$	-68030	930	-68131	3	-0.1	U			TO5	1.5	94Se12 *
	-67530	930			-0.4	U			TO6	1.5	98Ba.A *
$^{68}\text{Ni}-^{85}\text{Rb}_{.800}$	2437.0	3.2				2			MA8	1.0	07Gu09
$^{68}\text{Cu}-\text{u}$	-70570	440	-70389.1	1.7	0.3	U			TO6	1.5	98Ba.A *
$^{68}\text{Cu}-^{85}\text{Rb}_{.800}$	179.1	1.7				2			MA8	1.0	07Gu09 *
$^{68}\text{Ga}-^{85}\text{Rb}_{.800}$	-1484	37	-1451.6	1.5	0.9	U			MA8	1.0	07Gu09
$^{68}\text{Ge}-\text{C}_5\text{H}_8$	-134496.7	8.6	-134504.9	2.0	-1.0	U			CP1	1.0	04Cl03
	-134506.3	2.8			0.5	2			CP1	1.0	04Cl03
	-134503.5	2.9			-0.5	2			CP1	1.0	04Cl03
$^{68}\text{As}-\text{u}$	-63221	107	-63225.9	2.0	-0.0	U			GT1	1.5	01Ha66
$^{68}\text{As}-\text{C}_5\text{H}_8$	-125839	13	-125826.1	2.0	1.0	U			CP1	1.0	04Cl03
	-125827.7	9.9			0.2	U			CP1	1.0	04Cl03
	-125827.1	2.9			0.3	-			CP1	1.0	04Cl03
	-125824.4	3.1			-0.6	-			CP1	1.0	04Cl03
ave.	-125825.8	2.1			-0.1	1	88	88 ^{68}As			average
$\text{C F}_3-^{68}\text{As}_{1.015}$	59385.8	5.7	59383.7	2.0	-0.4	1	12	12 ^{68}As	MS1	1.0	07Sc24
$^{68}\text{Se}-\text{u}$	-56197	86	-58174.8	0.5	-23.0	F			CS1	1.0	01La31 *
	-57560	1070			-0.4	U			GA6	1.5	02Li24
	-57900	300			-0.9	U			CS1	1.0	08Go23
$^{68}\text{Se}-\text{C}_5\text{H}_8$	-120801	31	-120775.0	0.5	0.8	U			CP1	1.0	04Cl03
$\text{C F}_3-^{68}\text{Se}_{1.015}$	54256.87	0.54				2			MS1	1.0	09Sa12
$^{68}\text{Zn }^{35}\text{Cl}-^{66}\text{Zn }^{37}\text{Cl}$	1757.9	1.0	1760.7	0.3	0.7	U			H18	4.0	64Ba03
$^{68}\text{As}-^{68}\text{Ge}$	8698.8	9.9	8678.8	2.8	-2.0	U			CP1	1.0	04Cl03
$^{68}\text{Se}-^{68}\text{Ge}$	13669	27	13729.9	2.1	2.3	U			CP1	1.0	04Cl03
$^{65}\text{Cu}(\alpha, n)^{68}\text{Ga}$	-5800	40	-5824.0	1.5	-0.6	U			Oak		64St01
$^{66}\text{Ni}(t, p)^{68}\text{Ni}-^{68}\text{Zn}()^{70}\text{Zn}$	-2110	21	-2100	4	0.5	U			Hei		77Bh03
$^{66}\text{Zn}(t, p)^{68}\text{Zn}$	8758	15	8768.8	0.3	0.7	U			Ald		72Hu06
$^{68}\text{Zn}(^{14}\text{C}, ^{15}\text{O})^{67}\text{Ni}$	-6052	150	-6100	3	-0.3	U			Ors		84De33
$^{67}\text{Zn}(n, \gamma)^{68}\text{Zn}$	10198.2	0.4	10198.10	0.19	-0.3	-					71Ot01 Z
	10198.06	0.22			0.2	-			Bdn		06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{68}\text{Zn}(\text{d,t})^{67}\text{Zn}$	ave.	-3930	60	-3940.87	0.19	-0.2	U	100	99 ^{68}Zn	ANL	60Ze02	
$^{67}\text{Zn}(\text{n},\gamma)^{68}\text{Zn}$		10198.09	0.19	10198.10	0.19	0.0	1				average	
$^{68}\text{Cu}(\beta^-)^{68}\text{Zn}$		4580	60	4440.1	1.8	-2.3	U				64Ba13	
		4590	50			-3.0	B				72Sw01	
$^{68}\text{Zn}(\text{t},^3\text{He})^{68}\text{Cu}$		-4410	20	-4421.5	1.8	-0.6	U		LAI		77Sh08	
$^{68}\text{Ga}(\beta^+)^{68}\text{Zn}$	2921.1	1.2				2		72Si03				
	2915	10	2921.1	1.2	0.6	U		85Bo58				
$^{68}\text{Zn}(\text{p,n})^{68}\text{Ga}$		-3693	6	-3703.4	1.2	-1.7	U		Ric		55Br16 Z	
		-3703	5			-0.1	U		Ric		57Ch30 Z	
		-3707	5			0.7	U		Oak		64Jo11 Z	
$^{68}\text{As}(\beta^+)^{68}\text{Ge}$		8100	100	8084.3	2.6	-0.2	U		ANB		77Pa13	
		8073	54			0.2	U				02Cl.A *	
$^{68}\text{Se}(\beta^+)^{68}\text{As}$		4710	200	4705.1	1.9	-0.0	U				04Wo16	
$^{*68}\text{Mn}-\text{u}$	Trends from Mass Surface TMS suggest ^{68}Mn 1210 keV more bound										GAu	**
$^{*68}\text{Fe}-\text{u}$	Trends from Mass Surface TMS suggest ^{68}Fe 400 keV less bound										GAu	**
$^{*68}\text{Co}-\text{u}$	$M-A=-51838(96)$ keV for mixture gs+m at 150#(150#) keV										Nub211	**
$^{*68}\text{Co}-\text{u}$	$M-A=-51860(210)$ keV for mixture gs+m at 150#(150#) keV										Nub211	**
$^{*68}\text{Ni}-\text{u}$	$M-A=-61950(280)$ keV for mixture gs+n at 2849.1 keV										Nub211	**
$^{*68}\text{Ni}-\text{u}$	$M-A=-61480(280)$ keV for mixture gs+n at 2849.1 keV										Nub211	**
$^{*68}\text{Cu}-\text{u}$	$M-A=-65380(350)$ keV for mixture gs+m at 721.26 keV										Nub211	**
$^{*68}\text{Cu}-^{85}\text{Rb}_{.800}$	This result was first published in reference										04Bl16	**
$^{*68}\text{Cu}-^{85}\text{Rb}_{.800}$	Also 948.6(1.6) μu for $^{68}\text{Cu}^m-^{85}\text{Rb}_{.800}$, yielding excit. of 716.7(2.2) keV										07Gu09	**
$^{*68}\text{Se}-\text{u}$	F : other results in same paper not trusted, see ^{80}Y and ^{80}Zr										GAu	**
$^{*68}\text{As}(\beta^+)^{68}\text{Ge}$	From mass difference 8667(64) μu										02Cl.A	**
$^{69}\text{Fe}-\text{u}$		-42240	640	-42080#	220#	0.2	o		MT1	1.0	15Me.A *	
		-43232	429			2.7	D			1.0	20Me06 *	
$^{69}\text{Co}-\text{u}$		-54800	400	-54090	90	1.2	o			TO5	1.5	94Se12
		-53050	300			-2.3	U			TO6	1.5	98Ba.A
		-54070	230			-0.1	U		MT1	1.0	11Es06	
		-54117	223			0.1	U		LZ1	1.0	15Xu14	
		-54091	92				2		JY1	1.0	20Ca08 *	
$^{69}\text{Co}^m-^{39}\text{K}_{1.769}$		10296	15	10297	14	0.1	1	93	93 $^{69}\text{Co}^m$	MS1	1.0	18Iz01 *
$^{69}\text{Co}^m-\text{u}$		-53895	54	-53906	14	-0.2	1	7	7 $^{69}\text{Co}^m$	JY1	1.0	20Ca08
$^{69}\text{Ni}-\text{u}$		-64600	400	-64390	4	0.4	U		TO5	1.5	94Se12 *	
		-64250	450			-0.2	U		TO6	1.5	98Ba.A *	
$^{69}\text{Ni}-^{85}\text{Rb}_{.812}$		7237.0	4.0				2		MA8	1.0	07Gu09	
$^{69}\text{Cu}-^{85}\text{Rb}_{.812}$		1056.0	1.5				2		MA8	1.0	07Gu09	
$^{69}\text{Zn}-\text{u}$		-73580	400	-73449.6	0.9	0.2	U		TO6	1.5	98Ba.A *	
$\text{C}_5\text{H}_9-^{69}\text{Ga}$		144852.7	2.4	144851.8	1.3	-0.2	U		M15	2.5	63Ri07	
$^{69}\text{Ga}-^{85}\text{Rb}_{.812}$		-2799.8	1.6	-2799.7	1.3	0.0	1	65	65 ^{69}Ga	MA8	1.0	07Gu09
$\text{C F}_3-^{69}\text{Se}$		55794.7	1.6	55794.6	1.6	-0.0	1	100	100 ^{69}Se	MS1	1.0	07Sc24
$^{69}\text{Ga}(\text{p},\alpha)^{66}\text{Zn}$		4440	10	4435.5	1.4	-0.5	U		ANL		67Ka11	
$^{66}\text{Zn}(\alpha,\text{n})^{69}\text{Ge}$		-7520	30	-7445.0	1.5	2.5	U		Oak		64St01	
$^{67}\text{Zn}(\text{t,p})^{69}\text{Zn}$		8168	20	8198.37	0.25	1.5	U		Ald		72Hu06	
$^{68}\text{Zn}(\text{n},\gamma)^{69}\text{Zn}$		6482.3	0.8	6482.07	0.16	-0.3	U				71Ot01 Z	
		6481.8	0.5			0.5	U				75De.A Z	
		6482.07	0.16				2		Bdn		06Fi.A	
$^{68}\text{Zn}(\text{d,p})^{69}\text{Zn}$		4259	10	4257.50	0.16	-0.1	U		ANL		67Vo05	
		4243	10			1.5	U		MIT		75Is04	
$^{68}\text{Zn}(^3\text{He,d})^{69}\text{Ga}$		1126	20	1116.2	1.4	-0.5	U				74Ri08	
$^{69}\text{Se}(\text{ep})^{68}\text{Ge}$		3390	50	3255.1	2.4	-2.7	U		ChR		76Ha29	
		3370	70			-1.6	U		ChR		77Ma24	
$^{69}\text{Br}(\text{p})^{68}\text{Se}$		789	37	640	40	-4.0	B		MSU		11Ro18 *	
		641	42				3		MSU		14De41 *	
$^{69}\text{Br}^i(\text{p})^{68}\text{Se}$		4131	50	4129	19	-0.0	3				97Xu01 *	
		4210.6	50.			-1.6	3		MSU		11Ro47 *	
		4113	22			0.7	3		MSU		14De41 *	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{69}\text{Cu}(\beta^-)^{69}\text{Zn}$	2480	70	2681.7	1.6	2.9	U					66Va12
$^{69}\text{Zn}(\beta^-)^{69}\text{Ga}$	897	5	909.9	1.4	2.6	U					53Du03
$^{69}\text{Ge}(\beta^+)^{69}\text{Ga}$	2225	15	2227.1	0.5	0.1	U					51Hu38 *
$^{69}\text{Ga}(\text{p,n})^{69}\text{Ge}$	-3008.8	3.2	-3009.5	0.5	-0.2	U			Tkm		63Ok01
	-3006.0	4.			-0.9	U			Oak		64Jo11 Z
	-3009.50	0.55			0.0	1	100	100 ^{69}Ge	PTB		92Bo.B Z
$^{69}\text{As}(\beta^+)^{69}\text{Ge}$	3972	50	3990	30	0.3	-					70Bo19
	4067	50			-1.6	-			ChR		77Ma24 *
	ave.	4020	40		-0.9	1	82	82 ^{69}As			average
$^{69}\text{Se}(\beta^+)^{69}\text{As}$	6817	75	6680	30	-1.9	1	18	18 ^{69}As	ChR		77Ma24 *
* $^{69}\text{Fe}-\text{u}$	Trends from Mass Surface TMS suggest ^{69}Fe 1070 keV less bound										GAu **
* $^{69}\text{Co}-\text{u}$	PrvCom with authors. Derived from the gs. and iso. production ratio										20Ca08 **
* $^{69}\text{Co}^m-^{39}\text{K}_{1.769}$	Re-assigned to the short-lived iso. state, suggested by 2020Ca08										20Ca08 **
* $^{69}\text{Co}^m-^{39}\text{K}_{1.769}$	PrvCom with authors. Derived from the gs. and iso. production ratio										20Ca08 **
* $^{69}\text{Ni}-\text{u}$	$M-A=-59940(330)$ keV for mixture gs+m+n at 321(2) and 2700.0 keV										Nub211 **
* $^{69}\text{Ni}-\text{u}$	$M-A=-59620(380)$ keV for mixture gs+m+n at 321(2) and 2700.0 keV										Nub211 **
*	and assuming for second isomer a ratio $R=0.13(0.06)$ to ground state,										GAu **
*	from half-life=439 ns and TOF=1 μs										GAu **
* $^{69}\text{Zn}-\text{u}$	$M-A=-68320(350)$ keV for mixture gs+m at 438.636 keV										Nub211 **
* $^{69}\text{Br}(\text{p})^{68}\text{Se}$	Symmetrized from $Q_p=785(+40-34)$ keV										GAu **
* $^{69}\text{Br}(\text{p})^{68}\text{Se}$	And $E_p=751(+132-82)$ in good agreement with previous item										14De41 **
* $^{69}\text{Br}^i(\text{p})^{68}\text{Se}$	Might be also a more intense peak around 3 MeV										GAu **
* $^{69}\text{Br}^i(\text{p})^{68}\text{Se}$	$E_p=2970(50)$ to (2^+) level at 1196.7 keV										GAu **
* $^{69}\text{Br}^i(\text{p})^{68}\text{Se}$	A weaker peak around 4 MeV cannot be excluded, could feed lower (2^+)										GAu **
* $^{69}\text{Br}^i(\text{p})^{68}\text{Se}$	Original $Q=2939(22)$ corrected to $Q_p=2916(22)$ to (2^+) level at 1196.7 keV										16Hu.A **
* $^{69}\text{Ge}(\beta^+)^{69}\text{Ga}$	$E_{\beta^+}=1215, 610$ to ground state $3/2^-$, 574.220 $5/2^-$ levels										Ens141 **
* $^{69}\text{As}(\beta^+)^{69}\text{Ge}$	$E_{\beta^+}=2812(50)$ to $3/2^-$ level at 232.70 keV										Ens141 **
* $^{69}\text{Se}(\beta^+)^{69}\text{As}$	$E_{\beta^+}=5006(75)$ to 789.46 $(1/2^-, 3/2^-)$ level, and others										Ens141 **
$^{70}\text{Fe}-\text{u}$	-40483	526	-39600#	320#	1.7	D			MT1	1.0	20Me06 *
$^{70}\text{Co}-\text{u}$	-49000	600	-49947	12	-1.1	U			TO6	1.5	98Ba.A
	-50370	320			1.3	U			MT1	1.0	11Es06 *
	-49946.6	11.8				2			JY1	1.0	20Ca08
$^{70}\text{Ni}-\text{u}$	-63980	350	-63568.7	2.3	0.8	U			TO5	1.5	94Se12 *
	-63020	350			-1.0	U			TO6	1.5	98Ba.A *
$^{70}\text{Cu}-^{85}\text{Rb}_{.824}$	5077.6	1.7	5077.3	1.2	-0.2	2			MA8	1.0	07Gu09 *
	5077.2	2.2			0.1	2			MA8	1.0	07Gu09 *
	5077.0	2.3			0.1	2			MA8	1.0	07Gu09 *
$^{70}\text{Ga}-^{85}\text{Rb}_{.824}$	-1293.0	2.3	-1292.8	1.3	0.1	1	31	31 ^{70}Ga	MA8	1.0	07Gu09
$\text{C}_5 \text{H}_{10}-^{70}\text{Ge}$	154001.3	2.2	154001.8	0.9	0.1	U			M15	2.5	63Ri07
$\text{C}_4 \text{H}_6 \text{O}-^{70}\text{Ge}$	117616.1	1.8	117616.3	0.9	0.0	U			M15	2.5	63Ri07
$^{70}\text{As}-^{85}\text{Rb}_{.824}$	3619.9	1.5				2			MA8	1.0	20Ku19
$^{70}\text{Se}-\text{u}$	-66890	490	-66484.5	1.7	0.6	o			GA6	1.5	98Ch20
	-66635	75			1.3	U			GT1	1.5	01Ha66
	-66520	140			0.2	U			GA6	1.5	02Li24
$^{70}\text{Se}-^{13}\text{C F}_3$	-65048.8	1.7				2			MS1	1.0	09Sa12
$^{70}\text{Se}-^{85}\text{Rb}_{.824}$	6209	18	6200.8	1.7	-0.5	U			MA8	1.0	11He10
$^{70}\text{Br}-^{13}\text{C F}_3$	-53772	16				2			MS1	1.0	09Sa12 *
$^{70}\text{Ni}-^{72}\text{Ge}_{.972}$	12173.6	2.3				2			JY1	1.0	07Ra27
$^{70}\text{Zn }^{35}\text{Cl}-^{68}\text{Zn }^{37}\text{Cl}$	3429.5	1.7	3425.1	2.2	-0.7	1	10	9 ^{70}Zn	H18	4.0	64Ba03
$^{70}\text{Zn}(^3\text{He}, ^8\text{B})^{65}\text{Co}$	-18385	13	-18370	3	1.2	U			Pri		78Ko24
$^{70}\text{Zn}(\alpha, ^7\text{Be})^{67}\text{Ni}$	-19155	36	-19166	3	-0.3	U			Tex		78Co.A
	-19164	22			-0.1	U			Pri		78Ko28
$^{70}\text{Zn}(^{14}\text{C}, ^{17}\text{O})^{67}\text{Ni}$	-1661	100	-1993	3	-3.3	B			Ors		88Gi04
$^{70}\text{Ge}(\text{p}, \alpha)^{67}\text{Ga}$	1180.9	1.5	1181.2	1.2	0.2	1	60	46 ^{67}Ga	NDm		76Jo01

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{70}\text{Ge}(^3\text{He}, ^6\text{He})^{67}\text{Ge}$	-10572	30	-10549	4	0.8	U			MSU		78Pa11
$^{70}\text{Zn}(^{14}\text{C}, ^{16}\text{O})^{68}\text{Ni}$	1727	30	1656	4	-2.4	U			Ors		88Gi04
$^{70}\text{Zn}(^{18}\text{O}, ^{20}\text{Ne})^{68}\text{Ni}$	172	26	158	4	-0.5	U			Hei		84Ha31
$^{68}\text{Zn}(\text{t}, \text{p})^{70}\text{Zn}$	7196	15	7218.4	2.0	1.5	U			Ald		72Hu06
$^{70}\text{Ge}(\text{p}, \text{t})^{68}\text{Ge}$	-11251	13	-11244.1	2.0	0.5	U			ChR		72Hs01
	-11242	7			-0.3	U			Ors		77Gu02
$^{70}\text{Zn}(^{14}\text{C}, ^{15}\text{O})^{69}\text{Ni}$	-8936	150	-9422	4	-3.2	B			Ors		84De33
$^{70}\text{Zn}(\text{d}, ^3\text{He})^{69}\text{Cu}$	-5605	10	-5624.0	2.4	-1.9	U			ANL		78Ze04
	-5622	13			-0.2	U			Hei		84Ha31
$^{70}\text{Zn}(\text{t}, \alpha)^{69}\text{Cu}$	8682	20	8696.4	2.4	0.7	U			LAI		81Aj02
$^{69}\text{Ga}(\text{n}, \gamma)^{70}\text{Ga}$	7654.0	1.0	7653.65	0.17	-0.4	U					71Ar12
	7651.6	1.0			2.0	F					71Ve03
	7653.65	0.17			-0.0	1	100	^{64}Ga	Bdn		06Fi.A
$^{69}\text{Ga}(\text{d}, \text{p})^{70}\text{Ga}$	5430	10	5429.08	0.17	-0.1	U			Kop		71Ar12
$^{70}\text{Ge}(\text{d}, ^3\text{He})^{69}\text{Ga}$	-3030	7	-3029.7	1.4	0.0	U			Ors		78Ro14
$^{70}\text{Cu}(\beta^-)^{70}\text{Zn}$	6310	110	6588.4	2.2	2.5	U					75Re09
	5928	110			6.0	B					75Re09
$^{70}\text{Zn}(\text{t}, ^3\text{He})^{70}\text{Cu}$	-6559	20	-6569.8	2.2	-0.5	U			LAI		77Sh08
	-6602	20			1.6	U			LAI		87Aj.A
$^{70}\text{Zn}(\text{p}, \text{n})^{70}\text{Ga}$	-1436.3	2.0	-1436.9	1.6	-0.3	-			Nvl		59Go68
	-1439.1	3.0			0.7	-			Oak		64Jo11
ave.	-1437.2	1.6			0.1	1	92	^{88}Zn			average
$^{70}\text{Ga}(\beta^-)^{70}\text{Ge}$	1650	10	1651.9	1.5	0.2	U					57Bu41
$^{70}\text{As}(\beta^+)^{70}\text{Ge}$	6220	50	6228.1	1.6	0.2	U					63Bo14
$^{70}\text{Se}(\beta^+)^{70}\text{As}$	2780	200	2404.1	2.1	-1.9	F					75La02
	2736	85			-3.9	B					01To06
$^{70}\text{Br}(\beta^+)^{70}\text{Se}$	9970	170	10504	15	3.1	C			ANB		79Da.A
	9898	80			7.6	B					04Ka38
* $^{70}\text{Fe}-\text{u}$	Trends from Mass Surface TMS suggest ^{70}Fe 820 keV less bound										GAu
* $^{70}\text{Ni}-\text{u}$	Original -63860(350) μu or $M-A=-59490(330)$ keV										GAu
* $^{70}\text{Ni}-\text{u}$	$M-A=-58590(330)$ keV for mixture gs+m at 2860.9(0.1) keV and										Nub211
*	assuming ratio $R=0.04(2)$, from half-life=232 ns and TOF=1 μs										GAu
* $^{70}\text{Cu}-^{85}\text{Rb}_{.824}$	The three results for ^{70}Cu were first published in reference										04Va07
* $^{70}\text{Cu}-^{85}\text{Rb}_{.824}$	$D_M=5185.7(2.2)$ μu for $^{70}\text{Cu}^m$ at 101.1(0.3) keV; $M-A=-62875.4(2.0)$ keV										Nub211
* $^{70}\text{Cu}-^{85}\text{Rb}_{.824}$	$D_M=5337.4(2.3)$ μu for $^{70}\text{Cu}^n$ at 242.6(0.5) keV; $M-A=-62734.1(2.2)$ keV										Nub211
* $^{70}\text{Br}-^{13}\text{C F}_3$	$D_M=-51311(16)$ μu for $^{70}\text{Br}^m$ at 2292.3(0.8) keV										Nub211
* $^{69}\text{Ga}(\text{n}, \gamma)^{70}\text{Ga}$	$F: E(\gamma)$ systematically lower than for other authors; Z recalibrated										AHW
* $^{70}\text{Cu}(\beta^-)^{70}\text{Zn}$	$E_{\beta^-}=4550(120), 3370(170)$ to 4^+ level at 1786.33, 5^- at 3037.61 keV										Ens04c
* $^{70}\text{Cu}(\beta^-)^{70}\text{Zn}$	$E_{\beta^-}=6170(110)$ from $^{70}\text{Cu}^n$ 1^+ at 242.6 keV										Nub211
* $^{70}\text{As}(\beta^+)^{70}\text{Ge}$	$E_{\beta^+}=2144(50)$ to 3^+ level at 3046.427, 4^+ at 3058.707 keV										Ens04c
* $^{70}\text{Se}(\beta^+)^{70}\text{As}$	$E_{\beta^+}=1500(200)$ to 1^+ level at 81.49, 1^+ at 234.70, 1^+ at 458.12 keV										Ens04c
* $^{70}\text{Se}(\beta^+)^{70}\text{As}$	F : author's half-life 20(2)m disagrees with Nubase 41.1(0.3)m										Nub211
* $^{70}\text{Br}(\beta^+)^{70}\text{Se}$	$Q_{\beta^+}=12190(80)$ from 2292.3 $^{70}\text{Br}^m$										04Ka38
$^{71}\text{Co}-\text{u}$	-47100	600	-47630	500	-0.6	2			TO6	1.5	98Ba.A
	-47870	600			0.4	2			MT1	1.0	11Es06
$^{71}\text{Ni}-\text{u}$	-60000	400	-59481.0	2.4	0.9	U			TO5	1.5	94Se12
	-58700	350			-1.5	U			TO6	1.5	98Ba.A
$^{71}\text{Cu}-^{85}\text{Rb}_{.835}$	6332.4	1.6				2			MA8	1.0	07Gu09
$^{71}\text{Zn}-\text{u}$	-72080	380	-72280.4	2.8	-0.4	U			TO6	1.5	98Ba.A
$^{71}\text{Zn}^m-^{85}\text{Rb}_{.835}$	1544.3	2.6	1544.4	2.5	0.0	1	95	$^{95}\text{Zn}^m$	MA8	1.0	08Ba54
$\text{C}_5\text{H}_{11}-^{71}\text{Ga}$	161370.2	3.2	161372.8	0.9	0.3	U			M15	2.5	63Ri07
$^{71}\text{Ga}-^{85}\text{Rb}_{.835}$	-1641.6	3.0	-1641.9	0.9	-0.1	-			MA8	1.0	07Gu09
	-1640.2	1.3			-1.3	-			MA8	1.0	07Ke09
ave.	-1640.4	1.2			-1.2	1	53	^{71}Ga			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{71}\text{Se}-u$	-68160	340	-67791	3	0.7	o			GA6	1.5	98Ch20
	-67687	75			-0.9	U			GT1	1.5	01Ha66
	-67830	120			0.2	U			GA6	1.5	02Li24
$^{71}\text{Se}-^{85}\text{Rb}_{.835}$	5865.0	3.0				2			MA8	1.0	11He10
$^{71}\text{Br}-u$	-61260	610	-60658	6	0.7	U			GA6	1.5	02Li24
$^{71}\text{Br H}_2-\text{C}_4 \text{ H}_9 \text{ O}$	-110347.7	5.8	-110348	6	0.0	1	100	100	^{71}Br	1.0	09Sa12
$^{71}\text{Kr}-u$	-49727	151	-49730	140	-0.0	1	84	84	^{71}Kr	1.0	11Tu02
$^{71}\text{Ni}-^{72}\text{Ge}_{.986}$	17352.2	2.4				2			JY1	1.0	07Ra27
$^{71}\text{Ga}-^{71}\text{Ge}$	-249.538	0.106	-249.57	0.10	-0.3	1	89	77	^{71}Ge	1.0	16Al30
$^{68}\text{Zn}(\alpha, n)^{71}\text{Ge}$	-5630	40	-5746.9	1.0	-2.9	U			Oak		64St01
$^{70}\text{Zn}(^{18}\text{O}, ^{17}\text{F})^{71}\text{Cu}$	-9529	35	-9588.1	2.4	-1.7	U			Ber		89Bo.A
$^{70}\text{Zn}(d, p)^{71}\text{Zn}$	3609	10	3611	3	0.2	1	10	7	^{71}Zn		67Vo05
$^{70}\text{Zn}(^3\text{He}, d)^{71}\text{Ga}$	2380	20	2369.9	2.1	-0.5	U					74Ri08
$^{71}\text{Ga}(\gamma, n)^{70}\text{Ga}$	-9240	60	-9300.3	1.4	-1.0	U			Phi		60Ge01
$^{71}\text{Ga}(d, t)^{70}\text{Ga}$	-3054	10	-3043.1	1.4	1.1	U			Kop		71Ar12
$^{70}\text{Ge}(n, \gamma)^{71}\text{Ge}$	7415.3	1.5	7415.94	0.11	0.4	U					70Or.A
	7415.1	2.			0.4	U					72Gr34
	7415.95	0.15			-0.1	-			MMn		91Is01
	7415.93	0.15			0.1	-			Bdn		06Fi.A
$^{70}\text{Ge}(d, p)^{71}\text{Ge}$	5182	10	5191.37	0.11	0.9	U			Kyu		73Ka03
$^{70}\text{Ge}(n, \gamma)^{71}\text{Ge}$	ave. 7415.94	0.11	7415.94	0.11	-0.0	1	100	86	^{70}Ge		average
$^{70}\text{Ge}(p, \gamma)^{71}\text{As}$	4619	5	4620	4	0.2	R					75Li14
$^{71}\text{Zn}^m(\text{IT})^{71}\text{Zn}$	157.7	1.3	157.7	1.3	-0.0	1	98	93	^{71}Zn		Ens10c
$^{71}\text{Zn}(\beta^-)^{71}\text{Ga}$	2610	50	2810.3	2.8	4.0	B					61Th01
	2786	50			0.5	U					61Th01
	2796	50			0.3	U					64So01
$^{71}\text{Ge}(\epsilon)^{71}\text{Ga}$	233.0	0.5	232.47	0.09	-1.1	U			Hei		84Ha.A
	229.3	1.0			3.2	F					91ZI01
	232.1	0.5			0.7	U					93Di03
	232.71	0.29			-0.8	1	10	9	^{71}Ge		95Le19
	233.5	1.2			-0.9	U			TT1		13Fr13
$^{71}\text{Ga}(p, n)^{71}\text{Ge}$	-1018.4	2.0	-1014.82	0.09	1.8	U			Oak		64Jo11
$^{71}\text{Ga}(^3\text{He}, t)^{71}\text{Ge}-^{65}\text{Cu}()^{65}\text{Zn}$	1122.0	0.9	1119.2	0.4	-3.1	B			Pri		84Ko10
$^{71}\text{As}(\beta^+)^{71}\text{Ge}$	1997	20	2013	4	0.8	U					53St31
	2010	10			0.3	2					54Th36
	2012	10			0.1	2					55Gr08
$^{71}\text{Se}(\beta^+)^{71}\text{As}$	4428	125	4747	5	2.5	U					73Sc17
	4762	35			-0.4	U					01To06
$^{71}\text{Kr}(\epsilon)^{71}\text{Br}$	10140	320	10180	130	0.1	1	16	16	^{71}Kr		97Oio1
$^{71}\text{Zn}-u$	$M-A=-67060(350)$ keV for mixture gs+m at 157.7 keV										Nub211
$^{71}\text{Zn}(\beta^-)^{71}\text{Ga}$	$E_{\beta^-}=1450(50)$ 1460(50) respectively, from $^{71}\text{Zn}^m$ at 157.7(1.3) to $9/2^+$ at 1493.74										Ens10c
$^{71}\text{Ge}(\epsilon)^{71}\text{Ga}$	F : sees 17 keV neutrino										AHW
$^{71}\text{Ge}(\epsilon)^{71}\text{Ga}$	Original error 0.1 increased for calibration uncertainty										GAu
$^{71}\text{As}(\beta^+)^{71}\text{Ge}$	$E_{\beta^+}=800(20)$ 813(10) 815(10) respectively, to $5/2^-$ level at 174.943 keV										Ens10c
$^{72}\text{Ni}-u$	-58700	500	-58214.1	2.4	0.6	U			TO5	1.5	94Se12
	-57400	400			-1.4	U			TO6	1.5	98Ba.A
$^{72}\text{Cu}-u$	-64250	510	-64179.7	1.5	0.1	U			TO6	1.5	98Ba.A
$^{72}\text{Cu}-^{85}\text{Rb}_{.847}$	10534.4	1.5				2			MA8	1.0	07Gu09
$^{72}\text{Zn}-^{85}\text{Rb}_{.847}$	1556.9	2.3				2			MA8	1.0	08Ba54
$^{72}\text{Ga}-^{85}\text{Rb}_{.847}$	1079.5	1.5	1081.5	0.9	1.4	1	34	34	^{72}Ga	1.0	07Gu09
$\text{C}_4 \text{ H}_8 \text{ O}-^{72}\text{Ge}$	135438.4	2.1	135439.05	0.08	0.1	U			M15	2.5	63Ri07
$^{72}\text{Ge}-u$	-77906	25	-77924.18	0.08	-0.7	U			MR1	1.0	15Wi.A
	-77927	22			0.1	U			MR1	1.0	15Wi.A
	-77898	20			-1.3	U			MR1	1.0	15Wi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{72}\text{Se}-^{85}\text{Rb}_{.847}$	1854.6	2.1				2			MA8	1.0	11He10
$^{72}\text{Br } ^{27}\text{Al}-^{85}\text{Rb}_{1.165}$	20892.1	7.2	20898.0	1.1	0.8	U			MA8	1.0	11He10
$^{72}\text{Br}-^{85}\text{Rb}_{.847}$	11308.7	1.1				2			MS1	1.0	15Va05 *
$^{72}\text{Kr}-^{85}\text{Rb}_{.847}$	16806.5	8.6				2			MA8	1.0	06Ro11
$^{70}\text{Ge H}_2-^{72}\text{Ge}$	17821.3	1.7	17822.8	0.9	0.3	U			M15	2.5	63Ri07
$^{72}\text{Ge } ^{35}\text{Cl}-^{70}\text{Ge } ^{37}\text{Cl}$	779.8	5.9	777.4	0.9	-0.2	U			H40	2.5	85El01
$^{72}\text{Ni}-^{72}\text{Ge}$	19710.1	2.4				2			JY1	1.0	07Ra27
$^{70}\text{Zn}(\text{t,p})^{72}\text{Zn}$	6231	20	6241.6	2.9	0.5	U			Ald		72Hu06
$^{71}\text{Ga}(\text{n},\gamma)^{72}\text{Ga}$	6521.1	1.0	6520.48	0.19	-0.6	U					70Li04 Z
	6519.8	1.0			0.7	F					71Ve03 *
	6520.44	0.19			0.2	1	99	66 ^{72}Ga	Bdn		06Fi.A
$^{72}\text{Ge}(\text{d},^3\text{He})^{71}\text{Ga}$	-4241	7	-4242.3	0.8	-0.2	U			Ors		78Ro14
$^{72}\text{Zn}(\beta^-)^{72}\text{Ga}$	422	20	442.8	2.3	1.0	U					63De11 *
	458	6			-2.5	B					63Th03 *
$^{72}\text{Ga}(\beta^-)^{72}\text{Ge}$	4000	20	3997.6	0.8	-0.1	U					55Jo09 *
	3984	10			1.4	U					60La04 *
$^{72}\text{As}(\beta^+)^{72}\text{Ge}$	4361	10	4356	4	-0.5	2					50Me55
	4345	10			1.1	2					68Vi05
$^{72}\text{Ge}(\text{p,n})^{72}\text{As}$	-5140	5	-5138	4	0.3	2			Kyu		76Ki12
$^{72}\text{Br}(\beta^+)^{72}\text{Se}$	8869	95	8806.4	2.2	-0.7	U					01To06
$^{72}\text{Kr}(\beta^+)^{72}\text{Br}$	5040	80	5121	8	1.0	U					73Sc17 *
* $^{72}\text{Cu}-\text{u}$	$M-A=-59710(470)$ keV for mixture gs+m at 270.3 keV										Nub211 **
* $^{72}\text{Br}-^{85}\text{Rb}_{.847}$	$D_M=11308.4(1.1) \mu\text{u}$, 11417.2(1.2) for ground state and $^{72}\text{Br}^m$ at 100.76(0.15) keV										Nub211 **
*	respectively $M-A=-59062.0(1.0)$ and $-58960.7(1.1)$ keV										15Va05 **
* $^{71}\text{Ga}(\text{n},\gamma)^{72}\text{Ga}$	$F: E(\gamma)$ systematically lower than for other authors; Z recalibrated										AHW **
* $^{72}\text{Zn}(\beta^-)^{72}\text{Ga}$	$E_{\beta^-}=260(20) 296(6)$ respectively, to 1^+ level at 161.53 keV										Ens102 **
* $^{72}\text{Ga}(\beta^-)^{72}\text{Ge}$	$E_{\beta^-}=3166(20) 3150(10)$ respectively, to 2^+ level at 834.01 keV										Ens102 **
* $^{72}\text{Kr}(\beta^+)^{72}\text{Br}$	$E_{\beta^+}=3794(180), 3626(105), 3682(80), 3364(155)$ to 162.67, 309.84 1^+ ,										73Sc17 **
*	415.05 1^+ and 576.74 1^+ levels										Ens102 **
$^{73}\text{Ni}-\text{u}$	-52500	500	-53793.3	2.6	-1.7	U			TO6	1.5	98Ba.A
$^{73}\text{Cu}-\text{u}$	-62740	350	-63325.6	2.1	-1.1	U			TO6	1.5	98Ba.A
$^{73}\text{Cu}-^{85}\text{Rb}_{.859}$	12447.9	4.2	12447.0	2.1	-0.2	1	25	25 ^{73}Cu	MA8	1.0	07Gu09
$^{73}\text{Zn}-\text{u}$	-70100	380	-70417.4	2.0	-0.6	U			TO6	1.5	98Ba.A *
$^{73}\text{Zn}-^{85}\text{Rb}_{.859}$	5355.2	2.0				2			MA8	1.0	08Ba54
$^{73}\text{Ga}-^{85}\text{Rb}_{.859}$	947.3	1.8				2			MA8	1.0	07Gu09
$\text{C}_4 \text{H}_9 \text{O}-^{73}\text{Ge}$	141878.4	2.1	141880.95	0.06	0.5	U			M15	2.5	63Ri07
$^{73}\text{Se}-^{85}\text{Rb}_{.859}$	2511	11	2528	8	1.5	1	52	52 ^{73}Se	MA8	1.0	11He10 *
$^{73}\text{Br}-\text{u}$	-68428	97	-68327	7	0.7	U			GT1	1.5	01Ha66
	-68315.3	19.4			-0.6	1	14	14 ^{73}Br	MR1	1.0	20Ku19
$^{73}\text{Br } ^{27}\text{Al}-^{85}\text{Rb}_{1.176}$	16945.3	7.8	16947	7	0.2	1	86	86 ^{73}Br	MA8	1.0	11He10
$^{73}\text{Kr}-^{85}\text{Rb}_{.859}$	15061.8	7.1	15062	7	0.0	o			MA8	1.0	04Ro32 *
	15062.8	9.7			-0.1	2			MA8	1.0	06Ro11
	15060.7	10.3			0.1	2			MA8	1.0	06Ro11
$^{73}\text{Ni}-^{72}\text{Ge}_{1.014}$	25221.8	2.6				2			JY1	1.0	07Ra27
$^{73}\text{Cu}-^{72}\text{Ge}_{1.014}$	15689.2	2.4	15689.5	2.1	0.1	1	75	75 ^{73}Cu	JY1	1.0	07Ra27
$^{72}\text{Ge H}-^{73}\text{Ge}$	6443.9	1.3	6441.90	0.05	-0.6	U			M15	2.5	63Ri07
$^{73}\text{Br}-^{72}\text{Br}$	-4610	330	-4921	7	-0.4	U			CR1	2.5	89Sh10 *
	-4709	166			-0.9	U			CR2	1.5	91Sh19 *
$^{72}\text{Ge}(\text{n},\gamma)^{73}\text{Ge}$	6783.4	0.9	6782.94	0.05	-0.5	U					72Gr34
	6780.9	2.			1.0	U					72Ha74
	6782.94	0.05			0.0	1	100	100 ^{72}Ge	MMn		91Is01 Z
	6783.12	0.15			-1.2	U			Bdn		06Fi.A
$^{72}\text{Ge}(\text{d,p})^{73}\text{Ge}$	4571	4	4558.37	0.05	-3.2	B			Ald		69He05
	4563	10			-0.5	U			Kop		72Ha74
	4541	10			1.7	U			Kyu		73Ka03

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{72}\text{Ge}(^3\text{He,d})^{73}\text{As}$	160	4	162	4	0.6	1	93	93 ^{73}As	Hei		76Sc13
$^{73}\text{Kr}(\epsilon\text{p})^{72}\text{Se}$	3700	150	4027	7	2.2	U			ChR		81Ha44
$^{73}\text{Rb}(\text{p})^{72}\text{Kr}$	640	40				3					20Ho17
$^{73}\text{Rb}'(\text{p})^{72}\text{Kr}$	3803	40	3843	18	1.0	3					93Ba61
	3853	20			-0.5	3					20Ho06
$^{73}\text{Ga}(\beta^-)^{73}\text{Ge}$	1554	40	1598.2	1.7	1.1	U					58Yt22 *
	1564	80			0.4	U					70Wa21 *
$^{73}\text{Ge}(\text{p,n})^{73}\text{As}$	-1121.6	15.	-1127	4	-0.4	U			Oak		64Jo11 *
$^{73}\text{Se}(\beta^+)^{73}\text{As}$	2740	10	2725	7	-1.5	1	55	48 ^{73}Se			56Ha10 *
$^{73}\text{Br}(\beta^+)^{73}\text{Se}$	4748	500	4582	10	-0.3	U					70Mu02 *
	4648	400			-0.2	U					74Ro11 *
	4688	140			-0.8	U					87He21 *
	4610	70			-0.4	U					01To06
$^{73}\text{Kr}(\beta^+)^{73}\text{Br}$	6790	350	7094	9	0.9	U					73Sc17 *
	6860	220			1.1	U					97Oi01
$^{73}\text{Zn}-\text{u}$	$M-A=-65200(350)$ keV for mixture gs+m at 195.5 keV										Nub211 **
$^{73}\text{Se}-^{85}\text{Rb}_{.859}$	$D_M=2524.6(7.3)$ μu for mixture gs+m at 25.71 keV; $M-A=-68230.0(6.8)$ keV										Nub211 **
$^{73}\text{Kr}-^{85}\text{Rb}_{.859}$	Combined results of next two items										GAu **
$^{73}\text{Br}-^{72}\text{Br}$	$D_M=-4660(330)$ μu corrected for ^{72}Br gs+m mixture at 100.76 keV										Nub211 **
$^{73}\text{Br}-^{72}\text{Br}$	From $^{72}\text{Br}/^{73}\text{Br}=0.98635312(227)$										AHW **
$^{73}\text{Ga}(\beta^-)^{73}\text{Ge}$	$E_{\beta^-}=1190(40)$ $1200(80)$ respectively, to $3/2^-$ level at 364.02 keV										Ens043 **
$^{73}\text{Ge}(\text{p,n})^{73}\text{As}$	$T=1205(15)$ to $5/2^-$ level at 67.039(0.008)keV										Ens043 **
$^{73}\text{Se}(\beta^+)^{73}\text{As}$	$E_{\beta^+}=1290(10)$ to $9/2^+$ level at 427.906 keV										Ens043 **
$^{73}\text{Br}(\beta^+)^{73}\text{Se}$	$E_{\beta^+}=3700(500)$ $3600(400)$ $3640(140)$ respectively, to $^{73}\text{Se}^m$ at 25.71 keV										Nub211 **
$^{73}\text{Kr}(\beta^+)^{73}\text{Br}$	$E_{\beta^+}=5589(350)$ to $3/2^-$ level at 178.08 keV										Ens043 **
$^{74}\text{Ni}-\text{u}$	-52830	1060	-52280#	220#	0.5	D			MT1	1.0	11Es06 *
$^{74}\text{Cu}-\text{u}$	-59400	400	-60125	7	-1.2	U			TO6	1.5	98Ba.A
$^{74}\text{Cu}-^{85}\text{Rb}_{.871}$	16706.0	6.6				2			MA8	1.0	07Gu09
$^{74}\text{Zn}-^{85}\text{Rb}_{.871}$	6238.4	2.7				2			MA8	1.0	08Ba54
$^{74}\text{Ga}-^{85}\text{Rb}_{.871}$	3776.9	22.6	3777	3	-0.0	U			MA8	1.0	07Ke09 *
	3776.9	4.0			-0.0	2			MA8	1.0	07Gu09
	3806.5	34.6			-0.9	U			MA8	1.0	07Ke09 *
	3776.8	5.4			0.0	2			TT1	1.0	15Ma30
$\text{C }^{32}\text{S}_2-^{74}\text{Ge H}_2$	7314.0	1.4	7314.522	0.014	0.1	U			M15	2.5	63Ri07
$^{74}\text{Ge}-^{84}\text{Kr}$	9680.0337	0.0128	9680.034	0.013	0.0	1	100	100 ^{74}Ge	FS1	1.0	10Mo03
$\text{C}_6\text{H}_2-^{74}\text{Se}$	93173.8	3.8	93174.130	0.016	0.0	U			M15	2.5	63Ri07
$^{74}\text{Se}-^{84}\text{Kr}$	10978.2066	0.0128	10978.207	0.015	0.0	o			FS1	1.0	10Mo03 *
$^{74}\text{Se}-^{85}\text{Rb}_{.871}$	-691.4	7.3	-692.927	0.016	-0.2	U			MA8	1.0	11He10
$^{74}\text{Br }^{27}\text{Al}-^{85}\text{Rb}_{1.188}$	16246.0	6.8	16242	6	-0.5	1	85	85 ^{74}Br	MA8	1.0	11He10 *
$^{74}\text{Kr}-^{85}\text{Rb}_{.871}$	9915.0	2.2	9915.2	2.2	0.1	o			MA8	1.0	04Ro32 *
	9916.8	2.6			-0.6	-			MA8	1.0	06Ro11
	9909.7	4.4			1.2	-			MA8	1.0	06Ro11
	ave.	9915.0			0.1	1	93	93 ^{74}Kr			average
$^{74}\text{Rb}-^{85}\text{Rb}_{.871}$	21109	19	21097	3	-0.6	U			MA8	1.0	07Ke09
	21097.9	4.3			-0.2	o			MA8	1.0	04Ke10 *
	21095.7	5.2			0.3	-			MA8	1.0	07Ke09
	21102.7	7.5			-0.8	-			MA8	1.0	07Ke09
	21096.4	6.5			0.1	-			TT1	1.0	15Ma30
	ave.	21097			-0.1	1	83	83 ^{74}Rb			average
$^{74}\text{Rb}-\text{u}$	-55765	125	-55734	3	0.2	U			P40	1.0	06Lu19
$^{74}\text{Ge }^{35}\text{Cl}-^{72}\text{Ge }^{37}\text{Cl}$	2047.5	1.1	2052.06	0.11	1.0	U			H18	4.0	64Ba03
	2047.74	0.71			2.4	U			H40	2.5	85El01
	2052.01	0.26			0.1	U			H44	1.5	91Hy01
$^{74}\text{Se }^{35}\text{Cl}-^{72}\text{Ge }^{37}\text{Cl}$	3347.9	4.7	3350.23	0.11	0.2	U			H40	2.5	85El01

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{73}\text{Ge H}-^{74}\text{Ge}$	10105.1	1.7	10106.23	0.06	0.3	U			M15	2.5	63Ri07
$^{74}\text{Se}-^{74}\text{Ge}$	1298.5	8.5	1298.173	0.008	-0.0	U			H40	2.5	85El01
	1298.7	3.7			-0.1	U			H40	2.5	85El01
	1298.096	0.053			1.5	U			JY1	1.0	10Ko15
	1298.1729	0.0080			0.0	1	100	100 ^{74}Se	FS1	1.0	10Mo03
$^{74}\text{Br}-^{73}\text{Br}$	-1244	410	-1763	10	-0.5	U			CR1	2.5	89Sh10 *
$^{74}\text{Se}(\text{p},\text{t})^{72}\text{Se}$	-11979	24	-12005.9	2.0	-1.1	U			Win		74De31 *
$^{74}\text{Ge}(^{14}\text{C}, ^{15}\text{O})^{73}\text{Zn}$	-8018	150	-7664.8	1.9	2.4	U			Ors		84De33
$^{74}\text{Ge}(\text{d}, ^3\text{He})^{73}\text{Ga}$	-5515	7	-5518.6	1.7	-0.5	U			Ors		78Ro14
	-5509	13			-0.7	U			Hei		84Ha31
$^{73}\text{Ge}(\text{n},\gamma)^{74}\text{Ge}$	10200.2	0.6	10196.24	0.06	-6.6	B					70Ha60
	10198	2			-0.9	U					74Ch18
	10195.90	0.15			2.2	-			ILn		85Ho.A Z
	10196.32	0.14			-0.6	-					89Bu.A
	10196.31	0.07			-1.0	-			MMn		91Is01 Z
	10196.06	0.20			0.9	-			Bdn		06Fi.A
ave.	10196.24	0.06			0.0	1	100	100 ^{73}Ge			average
$^{74}\text{Se}(\text{d}, ^3\text{He})^{73}\text{As}$	-3027	8	-3056	4	-3.6	B			Ors		83Ro08 *
$^{74}\text{Zn}(\beta^-)^{74}\text{Ga}$	2350	100	2293	4	-0.6	U					72Er05 *
$^{74}\text{Ga}(\beta^-)^{74}\text{Ge}$	5400	100	5372.8	3.0	-0.3	U					62Ei02 *
$^{74}\text{As}(\beta^+)^{74}\text{Ge}$	2558	4	2562.4	1.7	1.1	-					71Bo01 *
$^{74}\text{Ge}(\text{p},\text{n})^{74}\text{As}$	-3343.5	5.6	-3344.7	1.7	-0.2	-			Tkm		63Ok01
	-3348.3	5.			0.7	-			Oak		64Jo11 Z
	-3346	5			0.3	-					70Fi03 Z
	-3347	3			0.8	-			Kyu		73Ki11
$^{74}\text{As}(\beta^+)^{74}\text{Ge}$	ave.	2562.9	1.9	2562.4	1.7	-0.3	1	82	82 ^{74}As		average
$^{74}\text{As}(\beta^-)^{74}\text{Se}$	1351	4	1353.1	1.7	0.5	1	18	18 ^{74}As			71Bo01 *
$^{74}\text{Br}(\beta^+)^{74}\text{Se}$	6857	100	6925	6	0.7	U					69La15 *
$^{74}\text{Se}(\text{p},\text{n})^{74}\text{Br}$	-7689	15	-7707	6	-1.2	1	15	15 ^{74}Br			75Lu02 *
$^{74}\text{Kr}(\beta^+)^{74}\text{Br}$	3000	200	2956	6	-0.2	U					74Ro11
	3327	125			-3.0	B					75Sc07
$^{74}\text{Rb}(\beta^+)^{74}\text{Kr}$	10000	1500	10416	3	0.3	U					76Da.D
	10413.8	7.0			0.3	1	24	17 ^{74}Rb			03Pi08 *
* $^{74}\text{Ni}-\text{u}$	Trends from Mass Surface TMS suggest ^{74}Ni 510 keV less bound										GAu **
* $^{74}\text{Ga}-^{85}\text{Rb}_{.871}$	$D_M=3780.1(22.5) \mu\text{u}$ corrected $-3.0(1.7) \text{ keV}$ for gs+m mixture $R<0.1$ at 59.571 keV										Nub211 **
* $^{74}\text{Ga}-^{85}\text{Rb}_{.871}$	$D_M=3809.7(34.6) \mu\text{u}$ corrected $-3.0(1.7) \text{ keV}$ for gs+m mixture $R<0.1$ at 59.571 keV										Nub211 **
* $^{74}\text{Se}-^{84}\text{Kr}$	Not independent measurement, use $^{74}\text{Ge}-^{74}\text{Se}$ below										10Mo03 **
* $^{74}\text{Br}-^{85}\text{Rb}_{1.188}$	$D_M=16253.4(5.3) \mu\text{u}$ for mixture gs+m at 13.58 keV; $M-A=-82474.9(4.9) \text{ keV}$										Nub211 **
* $^{74}\text{Kr}-^{85}\text{Rb}_{.871}$	Combined results of next two items										GAu **
* $^{74}\text{Rb}-^{85}\text{Rb}_{.871}$	Combined results of next two items										GAu **
* $^{74}\text{Br}-^{73}\text{Br}$	$D_M=-1230(410) \mu\text{u}$ for $^{74}\text{Br}^m$ at 13.58 keV										Nub211 **
* $^{74}\text{Se}(\text{p},\text{t})^{72}\text{Se}$	Original error 12; added systematic error 21 keV										GAu **
* $^{74}\text{Se}(\text{d}, ^3\text{He})^{73}\text{As}$	$Q=-3033(8)$ for $Q(^{76}\text{Se}(\text{d}, ^3\text{He}))=-4020.7(2.0)$, now 4014.5 keV										AHW **
* $^{74}\text{Zn}(\beta^-)^{74}\text{Ga}$	$E_{\beta^-}=2100(100)$ to 1^+ level at 251.787 keV										Ens067 **
* $^{74}\text{Ga}(\beta^-)^{74}\text{Ge}$	$E_{\beta^-}=2450(100)$ to 3^- level at 2949.48 keV										Ens067 **
* $^{74}\text{As}(\beta^+)^{74}\text{Ge}$	Original error increased: authors report $E(2^+)=593.1(1.5)$ while $E(2^+)=595.850(0.006) \text{ keV}$; see also $^{84}\text{Rb}(\beta^+)$										AHW **
* $^{74}\text{As}(\beta^-)^{74}\text{Se}$	Original value 1350.1(0.7), error increased, see $^{84}\text{Rb}(\beta^+)$										Ens067 **
* $^{74}\text{Br}(\beta^+)^{74}\text{Se}$	$E_{\beta^+}=5200(100)$, 4500(100) to 634.76, 1363.21 levels										69La15 **
*	from $^{74}\text{Br}^m$ at 13.8(0.5) keV										93Do05 **
* $^{74}\text{Se}(\text{p},\text{n})^{74}\text{Br}$	$T=7868(15)$ to (2^-) level at 72.65 keV										Ens067 **
* $^{74}\text{Rb}(\beta^+)^{74}\text{Kr}$	Deduced from measured half-life and branching ratio										GAu **
* $^{74}\text{Rb}(\beta^+)^{74}\text{Kr}$	Original 10405(9) re-evaluated in reference										11To.A **
$^{75}\text{Cu}-\text{u}$	-58100	700	-58476.2	0.8	-0.4	U			TO6	1.5	98Ba.A
$^{75}\text{Cu}-^{85}\text{Rb}_{.882}$	19325.40	0.81	19325.3	0.8	-0.2	1	91	91 ^{75}Cu	MA8	1.0	17We16
$^{75}\text{Zn}-^{85}\text{Rb}_{.882}$	10641.7	2.1				2			MA8	1.0	08Ba54

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{75}\text{Ga}-^{85}\text{Rb}_{.882}$	4301.7	2.6	4305.9	0.7	1.6	U			MA8	1.0	07Gu09
	4305.94	0.72				2			MA8	1.0	19Hu15
$\text{C}_3\text{H}_7\text{O}_2-^{75}\text{As}$	123009.8	2.6	123009.9	0.9	0.0	U			M15	2.5	63Ri07
$^{75}\text{As}-^{85}\text{Rb}_{.882}$	-601.3	7.6	-604.0	0.9	-0.4	U			MA8	1.0	02Ke.A
$^{75}\text{Br }^{27}\text{Al}-^{85}\text{Rb}_{1.200}$	13201.3	4.6				2			MA8	1.0	11He10
$^{75}\text{Kr}-^{85}\text{Rb}_{.882}$	8747.2	8.7				2			MA8	1.0	06Ro11
$^{75}\text{Rb}-^{85}\text{Rb}_{.882}$	16371	8	16374.7	1.3	0.5	U			MA2	1.0	94Ot01
	16374.7	1.7			-0.0	2			MA8	1.0	07Ke09
	16368	21			0.3	U			MA8	1.0	07Ke09
	16374.6	1.9			0.0	2			TT1	1.0	15Ma30
$^{75}\text{Cu}-^{72}\text{Ge}_{1.042}$	22719.6	2.5	22720.8	0.8	0.5	1	10	9	^{75}Cu	1.0	07Ra27
$^{75}\text{As }^{35}\text{Cl}-^{73}\text{Ge }^{37}\text{Cl}$	1079.6	5.0	1085.7	1.0	0.5	U			H40	2.5	85El01
$^{74}\text{Ge}(\text{n},\gamma)^{75}\text{Ge}$	6505.9	1.1	6505.84	0.05	-0.1	U					72Gr34
	6505.5	2.			0.2	U					72Ha74
	6505.81	0.30			0.1	U					89Bu.A *
	6505.26	0.08			7.2	B			MMn		91Is01 Z
	6505.45	0.14			2.8	C			Bdn		06Fi.A
	6505.84	0.05				2					12Me04
$^{74}\text{Ge}(\text{d},\text{p})^{75}\text{Ge}$	4265	15	4281.27	0.05	1.1	U			MIT		67Sp09
	4282	10			-0.1	U			Kop		72Ha74
	4268	10			1.3	U			Kyu		73Ka03
$^{74}\text{Ge}(\text{p},\gamma)^{75}\text{As}$	6901.6	5.	6900.7	0.9	-0.2	U					74Wa08
$^{74}\text{Ge}(^3\text{He},\text{d})^{75}\text{As}$	1414	4	1407.2	0.9	-1.7	U			Hei		76Sc13
$^{75}\text{As}(\gamma,\text{n})^{74}\text{As}$	-10259	31	-10245.5	1.9	0.4	U			Phi		60Ge01
$^{74}\text{Se}(\text{n},\gamma)^{75}\text{Se}$	8027.84	0.30	8027.60	0.07	-0.8	U			BNn		81En07 Z
	8027.60	0.08			-0.0	-			ILn		84To11 Z
	8027.59	0.16			0.0	-			Bdn		06Fi.A
ave.	8027.60	0.07			-0.0	1	100	100	^{75}Se		average
$^{75}\text{Zn}(\beta^-)^{75}\text{Ga}$	6060	80	5901.7	2.1	-2.0	U			Stu		86Ek01
$^{75}\text{Ga}(\beta^-)^{75}\text{Ge}$	3300	200	3396.3	0.7	0.5	U					60Mo01
$^{75}\text{Ge}(\beta^-)^{75}\text{As}$	1188	20	1177.2	0.9	-0.5	U					55Sc09
$^{75}\text{As}(\text{p},\text{n})^{75}\text{Se}$	-1647.2	2.0	-1647.1	0.9	0.1	-			Nvl		59Go68 Z
	-1647.3	1.1			0.3	-			Oak		64Jo11 Z
	-1643	5			-0.8	U					70Fi03
ave.	-1647.3	1.0			0.3	1	85	85	^{75}As		average
$^{75}\text{Br}(\beta^+)^{75}\text{Se}$	3010	20	3062	4	2.6	U					52Fu04 *
	3030	50			0.6	U					61Ba43 *
	3050	20			0.6	U					69Ra24 *
$^{75}\text{Kr}(\beta^+)^{75}\text{Br}$	4400	200	4783	9	1.9	U					74Ro12 *
$^{75}\text{Sr}(\epsilon)^{75}\text{Rb}$	10600	220				3					03Hu01
$^{74}\text{Ge}(\text{n},\gamma)^{75}\text{Ge}$	Original error 0.03 keV increased										GAu **
$^{75}\text{Br}(\beta^+)^{75}\text{Se}$	$E_{\beta^+}=1700(20) 1720(50) 1740(20)$ respectively, to $3/2^-$ level at 286.5714 keV										Ens139 **
$^{75}\text{Kr}(\beta^+)^{75}\text{Br}$	$E_{\beta^+}=3200(200)$ to $132.46 (5/2)^+$, $154.61 (3/2)^+$ levels										Ens139 **
$^{76}\text{Cu}-^{85}\text{Rb}_{.894}$	24135.0	7.2	24128.9	1.0	-0.8	o			MA8	1.0	07Gu09
	24128.95	0.98				2			MA8	1.0	17We16
$^{76}\text{Zn}-^{85}\text{Rb}_{.894}$	11975.5	2.0	11974.9	1.6	-0.3	1	61	61	^{76}Zn	1.0	08Ba54
$^{76}\text{Zn}-^{88}\text{Rb}_{.864}$	9737.4	2.5	9738.3	1.6	0.4	1	39	39	^{76}Zn	1.0	08Ha23
$^{76}\text{Ga}-^{85}\text{Rb}_{.894}$	7687.6	2.1				2			MA8	1.0	07Gu09
$\text{C }^{32}\text{S}_2-^{76}\text{Ge}$	22741.6	1.5	22739.622	0.019	-0.5	U			M15	2.5	63Ri07
$^{76}\text{Ge}-\text{u}$	-78597.242	0.096	-78597.275	0.019	-0.3	U			ST2	1.0	01Do08
$\text{C}_6\text{H}_4-^{76}\text{Se}$	112100	8	112086.425	0.017	-0.7	U			M15	2.5	63Ri07
$^{76}\text{Se}-\text{u}$	-80786.205	0.081	-80786.297	0.017	-1.1	U			ST2	1.0	01Do08
$^{76}\text{Kr}-^{85}\text{Rb}_{.894}$	4774.3	4.7	4771	4	-0.8	1	84	84	^{76}Kr	1.0	06Ro11
$^{76}\text{Rb}-^{85}\text{Rb}_{.894}$	13931	8	13933.0	1.0	0.3	U			MA2	1.0	94Ot01
	13932.2	2.0			0.4	2			MA8	1.0	07Ke09
	13923	15			0.7	U			MA8	1.0	07Ke09
	13935.3	1.6			-1.4	2			MA8	1.0	07Ke09
	13931.0	1.7			1.2	2			TT1	1.0	15Ma30
	13925.2	4.5			1.7	U			TT1	1.0	15Ma30
	13933.5	3.4			-0.1	o			TT1	1.0	15Ma30 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{76}\text{Sr } ^{19}\text{F}-^{85}\text{Rb}_{1.118}$	38785	37				2			MA8	1.0	05Si34
$^{76}\text{Sr}-\text{u}$	-58813	107	-58240	40	5.4	F			CS1	1.0	01La31 *
$^{76}\text{Ge}-^{84}\text{Kr}$	9904.9983	0.0175	9904.998	0.019	-0.0	o			FS1	1.0	10Mo03 *
$^{76}\text{Se}-^{84}\text{Kr}$	7715.9762	0.0169	7715.976	0.017	-0.0	1	100	100 ^{76}Se	FS1	1.0	10Mo03
$^{74}\text{Ge } \text{H}_2-^{76}\text{Ge}$	15425.0	1.7	15425.100	0.023	0.0	U			M15	2.5	63Ri07
$^{76}\text{Ge } ^{35}\text{Cl}-^{74}\text{Ge } ^{37}\text{Cl}$	3175.7	1.5	3175.08	0.07	-0.1	U			H18	4.0	64Ba03
	3170.41	0.74			2.5	U			H40	2.5	85El01
	3174.61	0.41			0.8	U			H44	1.5	91Hy01
$^{76}\text{Se } ^{35}\text{Cl}-^{74}\text{Ge } ^{37}\text{Cl}$	986.30	0.65	986.06	0.07	-0.2	U			H44	1.5	91Hy01
$^{76}\text{Ge}-^{76}\text{Se}$	2190.92	0.59	2189.022	0.008	-1.3	U			H40	2.5	85El01
	2188.60	0.42			0.7	U			H44	1.5	91Hy01
	2188.963	0.054			1.1	U			ST2	1.0	01Do08
	2188.98	0.16			0.3	U			JY1	1.0	08Ra09
	2189.0221	0.008			-0.0	1	100	100 ^{76}Ge	FS1	1.0	10Mo03
$^{75}\text{Rb}-^{76}\text{Rb}_{.493} \text{ } ^{74}\text{Rb}_{.507}$	-1140	170	-1081.1	2.0	0.1	U			P20	2.5	82Au01
$^{76}\text{Ge}(^{14}\text{C}, ^{17}\text{O})^{73}\text{Zn}$	-3779	40	-3790.8	1.9	-0.3	U			Ors		84Be10 *
$^{76}\text{Ge}(^{14}\text{C}, ^{16}\text{O})^{74}\text{Zn}$	163	40	300.7	2.5	3.4	B			Ors		84Be10
$^{76}\text{Ge}(^{18}\text{O}, ^{20}\text{Ne})^{74}\text{Zn}$	-1219	21	-1197.1	2.5	1.0	U			Hei		84Ha31
$^{76}\text{Ge}(^{14}\text{C}, ^{15}\text{O})^{75}\text{Zn}$	-10354	150	-10489.7	2.0	-0.9	U			Ors		84De33
$^{76}\text{Ge}(\text{d}, ^3\text{He})^{75}\text{Ga}$	-6545	7	-6547.8	0.7	-0.4	U			Ors		78Ro14
	-6536	22			-0.5	U			Hei		84Ha31
$^{75}\text{As}(\text{n}, \gamma)^{76}\text{As}$	7329	2	7328.50	0.07	-0.3	U					68Jo11
	7328.421	0.075			1.0	2			ILn		90Ho10 Z
	7328.81	0.15			-2.1	2			Bdn		06Fi.A
$^{75}\text{As}(\text{d}, \text{p})^{76}\text{As}$	5105	5	5103.93	0.07	-0.2	U					76Mo32
$^{75}\text{Se}(\text{n}, \gamma)^{76}\text{Se}$	11154.15	0.30	11153.79	0.07	-1.2	U			ILn		83To20 Z
$^{76}\text{Zn}(\beta^-)^{76}\text{Ga}$	4160	80	3993.6	2.4	-2.1	U			Stu		86Ek01
$^{76}\text{Ga}(\beta^-)^{76}\text{Ge}$	6770	150	6916.3	2.0	1.0	o			Stu		77Al17
	7010	90			-1.0	U			Stu		86Ek01
$^{76}\text{Ge}(\text{p}, \text{n})^{76}\text{As}$	-1705	5	-1703.9	0.9	0.2	U					70Fi03
$^{76}\text{As}(\beta^-)^{76}\text{Se}$	2970	2	2960.6	0.9	-4.7	B					69Na11
$^{76}\text{Br}(\beta^+)^{76}\text{Se}$	5002	20	4963	9	-2.0	2					71Dz08
$^{76}\text{Br}(\text{n}, \text{p})^{76}\text{Se}$	5730	15	5745	9	1.0	2			ILL		78An14
$^{76}\text{Se}(\text{p}, \text{n})^{76}\text{Br}$	-5738.6	15.	-5745	9	-0.4	2					75Lu02
$^{76}\text{Rb}(\beta^+)^{76}\text{Kr}$	8793	570	8535	4	-0.5	U					75We23
	8063	44			10.7	F					82Mo10 *
	8094	162			2.7	F			BNL		83Li11 *
	8250	150			1.9	U			IRS		93Al03
* $^{76}\text{Rb}-^{85}\text{Rb}_{.894}$	no accuracy check was performed for this 12^+ charge state										15Ma30 **
* $^{76}\text{Sr}-\text{u}$	F : other results in same paper not trusted, see ^{80}Y										GAu **
* $^{76}\text{Ge}-^{84}\text{Kr}$	Not independent measurement, use $^{76}\text{Ge}-^{76}\text{Se}$ below										10Mo03 **
* $^{76}\text{Ge}(^{14}\text{C}, ^{17}\text{O})^{73}\text{Zn}$	$Q = -3974(40) \text{ MeV} - A = -65410(40) \text{ MeV}$ to $^{73}\text{Zn}^m$ at 195.5 keV										GAu **
* $^{76}\text{Rb}(\beta^+)^{76}\text{Kr}$	$E_{\beta^+} = 4558(30) \text{ keV}$ to level $(1^+, 2^+)$ at 2570.95 keV and corrections by authors										Ens953 **
* $^{76}\text{Rb}(\beta^+)^{76}\text{Kr}$	F : 29.6% feeding above 2570.95 level from reference										84Mo22 **
$^{77}\text{Cu}-\text{u}$	-51850	540	-52456.4	1.3	-1.1	U			OR1	1.0	06Ha62
$^{77}\text{Cu}-^{85}\text{Rb}_{.906}$	27462.1	1.3				2			MA8	1.0	17We16
$^{77}\text{Zn}-\text{u}$	-62790	780	-63112.8	2.1	-0.3	U			TO6	1.5	98Ba.A *
	-63380	260			0.4	U			GT2	2.5	08Kn.A *
$^{77}\text{Zn}-^{88}\text{Rb}_{.875}$	14485.7	4.5	14486.1	2.1	0.1	1	22	22 ^{77}Zn	JY1	1.0	08Ha23
$\text{C}_6 \text{H}_5-^{77}\text{Se}$	119211.9	4.2	119211.01	0.07	-0.1	U			M15	2.5	63Ri07
$^{77}\text{Rb}-^{39}\text{K}_{1.974}$	1912	27	2045.0	1.4	4.9	B			MA2	1.0	87Bo59
$^{77}\text{Zn}-^{85}\text{Rb}_{.906}$	16805.8	2.4	16805.7	2.1	-0.0	1	78	78 ^{77}Zn	MA8	1.0	08Ba54
$^{77}\text{Ga}-^{85}\text{Rb}_{.906}$	9072.8	2.6				2			MA8	1.0	07Gu09
	9070.0	4.6	9072.8	2.6	0.6	U			MA8	1.0	19Hu15

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{77}\text{Kr}-^{85}\text{Rb}_{.906}$	4588.5	2.1				2			MA8	1.0	06Ro11
$^{77}\text{Rb}-^{85}\text{Rb}_{.906}$	10327	8	10320.1	1.4	-0.9	U			MA2	1.0	94Ot01
	10320.1	1.4				2			MA8	1.0	07Ke09
$^{77}\text{Sr}-^{19}\text{F}-^{85}\text{Rb}_{1.129}$	35938.0	8.5				2			MA8	1.0	05Si34
$^{75}\text{Rb}-^{77}\text{Rb}_{.325}-^{74}\text{Rb}_{.676}$	-1340	380	-1053.6	2.4	0.3	U			P20	2.5	82Au01
$^{76}\text{Rb}-^{77}\text{Rb}_{.494}-^{75}\text{Rb}_{.507}$	525	30	557.1	1.3	0.4	U			P20	2.5	82Au01
$^{76}\text{Ge}(n,\gamma)^{77}\text{Ge}$	6072.5	1.0	6071.29	0.05	-1.2	U					72Gr34 Z
	6071.7	1.2			-0.3	U					72Ha74 Z
	6072.3	0.4			-2.5	U			Bdn		06Fi.A
	6071.29	0.05				2					12Me04
$^{76}\text{Ge}(d,p)^{77}\text{Ge}$	3839	10	3846.72	0.05	0.8	U			Kop		72Ha74
	3823	12			2.0	U			Kyu		73Ka03
$^{76}\text{Ge}(^3\text{He},d)^{77}\text{As}$	2497	3	2498.9	1.7	0.6	1	32	32 ^{77}As	Hei		76Sc13
$^{76}\text{Se}(n,\gamma)^{77}\text{Se}$	7418.87	0.20	7418.86	0.06	-0.1	-			BNn		81En07
	7418.85	0.07			0.1	-			ILn		85To10 Z
	7418.85	0.15			0.0	-			Bdn		06Fi.A
$^{76}\text{Se}(d,p)^{77}\text{Se}$	5192	10	5194.29	0.06	0.2	U			Ald		63Ma27
$^{76}\text{Se}(n,\gamma)^{77}\text{Se}$	ave. 7418.85	0.06	7418.86	0.06	0.1	1	99	99 ^{77}Se			average
$^{77}\text{Sr}(\epsilon p)^{76}\text{Kr}$	3850	200	3921	9	0.4	U			ChR		76Ha29
$^{77}\text{Zn}(\beta^-)^{77}\text{Ga}$	7270	120	7203	3	-0.6	U			Stu		86Ek01
$^{77}\text{Ga}(\beta^-)^{77}\text{Ge}$	5340	60	5220.5	2.4	-2.0	U			Stu		77Al17
	5690	300			-1.6	U			Stu		86Ek01
$^{77}\text{Ge}(\beta^-)^{77}\text{As}$	2670	100	2703.5	1.7	0.3	U					52Sm13 *
$^{77}\text{As}(\beta^-)^{77}\text{Se}$	700	7	683.2	1.7	-2.4	U					51Ca04
	679	4			1.0	1	18	18 ^{77}As			51Je01
$^{77}\text{Br}(\beta^+)^{77}\text{Se}$	1358	20	1364.7	2.8	0.3	U					51Ca28
$^{77}\text{Se}(p,n)^{77}\text{Br}$	-2147	4	-2147.0	2.8	-0.0	2			Oak		58Jo01
	-2147.0	4.			0.0	2			Tkm		63Ok01
$^{77}\text{Kr}(\beta^+)^{77}\text{Br}$	3012	30	3065	3	1.8	U					55Th01 *
	3027	40			1.0	U					73Ba22 *
	3300	100			-2.3	U					74Ro11 *
	2760	42			7.3	B					82Mo10 *
$^{77}\text{Rb}(\beta^+)^{77}\text{Kr}$	5180	390	5339.0	2.4	0.4	U					75We23
	5272	26			2.6	U					82Mo10
	5113	69			3.3	B			BNL		83Li11
	5320	70			0.3	U			IRS		93Al03
$^{77}\text{Sr}(\beta^+)^{77}\text{Rb}$	6986	227	7027	8	0.2	U			BNL		83Li11
* $^{77}\text{Zn}-u$	$M-A=-58100(700)$ keV for mixture gs+m at 772.440 keV										Nub211 **
* $^{77}\text{Zn}-u$	$M-A=-58648(95)$ keV for mixture gs+m at 772.440 keV ($1/2^-$)										Nub211 **
* $^{77}\text{Ge}(\beta^-)^{77}\text{As}$	$E_{\beta^-}=2196(100)$ to $9/2^+$ $^{77}\text{As}^m$ at 475.48 keV										Nub211 **
* $^{77}\text{Kr}(\beta^+)^{77}\text{Br}$	Error not in 55Th01, estimated by evaluator										AHW **
* $^{77}\text{Kr}(\beta^+)^{77}\text{Br}$	$E_{\beta^+}=1860(30)$ 1875(40) respectively, to $5/2^+$ level at 129.64 keV										Ens126 **
* $^{77}\text{Kr}(\beta^+)^{77}\text{Br}$	$E_{\beta^+}=2000(100)$ 1528(36) respectively, to $(3/2)^+$ level at 276.22 keV										Ens126 **
$^{78}\text{Cu}-u$	-47770	540	-48083	14	-0.6	U			OR1	1.0	06Ha62
	-48115.2	23.6			1.3	1	37	37 ^{78}Cu	MR1	1.0	17We16
$^{78}\text{Cu}-^{85}\text{Rb}_{.918}$	32912	18	32894	14	-1.0	1	63	63 ^{78}Cu	MA8	1.0	17We16
$^{78}\text{Zn}-^{88}\text{Rb}_{.886}$	16863.8	2.9	16863.6	2.1	-0.1	1	52	52 ^{78}Zn	JY1	1.0	08Ha23
$^{78}\text{Ga}-^{88}\text{Rb}_{.886}$	10184.3	3.3	10185.2	1.1	0.3	1	12	12 ^{78}Ga	JY1	1.0	08Ha23
$\text{C}_6\text{H}_6-^{78}\text{Se}$	129642.6	2.2	129640.95	0.19	-0.3	U			M15	2.5	63Ri07
$\text{C}_6\text{H}_6-^{78}\text{Kr}$	126548.3	3.6	126583.8	0.3	3.9	B			M15	2.5	63Ri07
	126554	17			1.2	U			R11	1.5	78Di09
	126560	7			2.3	U			R11	1.5	78Di09
$\text{C}_5\text{N H}_4-^{78}\text{Kr}$	113994	20	114007.8	0.3	0.5	U			R11	1.5	78Di09
$^{78}\text{Kr}-^{86}\text{Kr}_{.907}$	1441.2	1.0	1442.5	0.3	1.3	1	11	11 ^{78}Kr	MS1	1.0	06Ri15

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{78}\text{Zn}-^{85}\text{Rb}_{.918}$	19266.0	3.0	19266.2	2.1	0.1	1	48	48 ^{78}Zn	MA8	1.0	08Ba54
$^{78}\text{Ga}-^{85}\text{Rb}_{.918}$	12585.2	2.6	12587.9	1.1	1.0	U			MA8	1.0	07Gu09
	12588.0	1.2			-0.1	1	88	88 ^{78}Ga	MA8	1.0	19Hu15
$^{78}\text{Kr}-^{85}\text{Rb}_{.918}$	1342.3	1.4	1343.4	0.3	0.8	U			MA8	1.0	06Ro11
	1338.9	2.2			2.0	U			MA8	1.0	06Ro11
$^{78}\text{Rb}-^{85}\text{Rb}_{.918}$	9118	8	9119	3	0.1	2			MA2	1.0	94Ot01
	9121.3	7.5			-0.3	2			TT1	1.0	12Ga15 *
	9118.3	4.5			0.1	2			TT1	1.0	12Ga15 *
$^{78}\text{Sr}-^{85}\text{Rb}_{.918}$	13157	8				2			MA2	1.0	94Ot01
$^{78}\text{Se } ^{35}\text{Cl}_2-^{74}\text{Ge } ^{37}\text{Cl}_2$	2030.4	2.2	2031.72	0.24	0.4	U			H44	1.5	91Hy01
$^{78}\text{Se } ^{35}\text{Cl}-^{76}\text{Ge } ^{37}\text{Cl}$	-1147.60	0.92	-1143.36	0.20	1.8	U			H40	2.5	85El01
	-1143.57	0.72			0.2	U			H44	1.5	91Hy01
$^{78}\text{Se } ^{35}\text{Cl}-^{76}\text{Se } ^{37}\text{Cl}$	1042.03	1.35	1045.66	0.20	1.1	U			H40	2.5	85El01
	1044.58	0.45			1.6	U			H44	1.5	91Hy01
$^{76}\text{Se } \text{H}_2-^{78}\text{Kr}$	14440	25	14497.4	0.3	1.5	U			R11	1.5	78Di09
$^{77}\text{Se } \text{H}-^{78}\text{Kr}$	7367	26	7372.8	0.3	0.1	U			R11	1.5	78Di09
$^{78}\text{Kr}-^{78}\text{Se}$	3074	16	3057.10	0.28	-0.7	U			R11	1.5	78Di09
	3098	20			-1.4	U			R11	1.5	78Di09
	3057.18	0.29			-0.3	1	92	89 ^{78}Kr	MS1	1.0	13Bu17
$^{78}\text{Se } \text{H}-^{78}\text{Kr}$	4724	33	4767.93	0.28	0.9	U			R11	1.5	78Di09
$^{76}\text{Rb}-^{78}\text{Rb}_{.325}^{x} \text{ } ^{75}\text{Rb}_{.676}$	-130	40	-69	4	0.6	U			P20	2.5	82Au01
$^{77}\text{Rb}-^{78}\text{Rb}_{.494}^{x} \text{ } ^{76}\text{Rb}_{.507}$	-1192	19	-1138	6	1.1	U			P20	2.5	82Au01
$^{78}\text{Kr}(\alpha, ^8\text{He})^{74}\text{Kr}$	-41080	75	-41031.2	2.0	0.7	U			Tex		82Mo23 *
$^{78}\text{Se}(\text{p}, \alpha)^{75}\text{As}$	870.9	2.3	872.3	0.9	0.6	1	15	15 ^{75}As	NDm		82Zu04
$^{78}\text{Kr}(^3\text{He}, ^6\text{He})^{75}\text{Kr}$	-12581	14	-12516	8	4.7	B					87Mo06
$^{76}\text{Ge}(\text{t}, \text{p})^{78}\text{Ge}$	6310	5	6310	4	-0.0	2			LAl		78Ar12
	6310	5			-0.0	2			Phi		81St18
$^{78}\text{Se}(\text{p}, \text{t})^{76}\text{Se}$	-9433.7	4.3	-9434.83	0.18	-0.3	U			NDm		82Zu04
$^{78}\text{Kr}(\alpha, ^6\text{He})^{76}\text{Kr}$	-20351	10	-20332	4	1.9	o			Tex		82Mo23 *
$^{78}\text{Kr}(\text{p}, \text{t})^{76}\text{Kr}$	-12840	15	-12825	4	1.0	U			Tky		81Ma30
$^{78}\text{Se}(\text{d}, ^3\text{He})^{77}\text{As}$	-4904	4	-4905.1	1.7	-0.3	1	18	18 ^{77}As	Ors		83Ro08 *
$^{77}\text{Se}(\text{n}, \gamma)^{78}\text{Se}$	10497.7	0.3	10497.77	0.17	0.2	-			BNn		81En07 Z
	10497.75	0.21			0.1	-			Bdn		06Fi.A
$^{78}\text{Se}(\text{p}, \text{d})^{77}\text{Se}$	-8271.9	4.0	-8273.21	0.17	-0.3	U			NDm		82Zu04
$^{77}\text{Se}(\text{n}, \gamma)^{78}\text{Se}$	ave. 10497.73	0.17	10497.77	0.17	0.2	1	96	95 ^{78}Se			average
$^{78}\text{Kr}(\text{d}, \text{t})^{77}\text{Kr}$	-5804	7	-5822.9	2.0	-2.7	U					87Mo06
$^{78}\text{Zn}(\beta^-)^{78}\text{Ga}$	6440	140	6220.8	2.2	-1.6	o			Stu		86Ek01
	6364	90			-1.6	U			Stu		00Me.A
$^{78}\text{Ga}(\beta^-)^{78}\text{Ge}$	8140	160	8158	4	0.1	o			Stu		77Al17
	8200	80			-0.5	o			Stu		86Ek01
	8054	43			2.4	U			Stu		00Me.A
$^{78}\text{Ge}(\beta^-)^{78}\text{As}$	967	30	955	10	-0.4	R					65Fr04 *
	987	20			-1.6	R					65Kv01 *
$^{78}\text{As}(\beta^-)^{78}\text{Se}$	4270	100	4209	10	-0.6	U					70Mc01
	4310	100			-1.0	U					71Mo20 *
$^{78}\text{Br}(\beta^+)^{78}\text{Se}$	3542	50	3574	4	0.6	U			Bar		61Ri02
$^{78}\text{Se}(\text{p}, \text{n})^{78}\text{Br}$	-4344	10	-4356	4	-1.2	2			Bar		61Ri02
	-4370	10			1.4	2			LAl		61Sc11
	-4355.5	7.4			-0.1	2			Tkm		63Ok01 Z
	-4356	5			-0.0	2					70Fi03 Z
$^{78}\text{Rb}(\beta^+)^{78}\text{Kr}$	7085	370	7243	3	0.4	U					75We23 *
	7240	50			0.1	U					81Ba40
	7185	50			1.2	U			IRS		93Al03 *
$^{78}\text{Rb}^{\text{t}}(\text{IT})^{78}\text{Rb}$	74	12				3					82Au01 *
* $^{78}\text{Rb}-^{85}\text{Rb}_{.918}$	Correction for e^- binding=+97 eV is negligible										12Ga15 **
* $^{78}\text{Rb}-^{85}\text{Rb}_{.918}$	$D_M = 9237.6(4.5) \mu\text{u } M - A = -66824.9(4.2) \text{ keV}$ corrected for e^- binding=+97 eV										12Ga15 **
	from $^{78}\text{Rb}^n$ at 111.19(0.22) keV										Nub211 **
* $^{78}\text{Kr}(\alpha, ^8\text{He})^{74}\text{Kr}$	Original -41120(75) for 4 events included 1 background event										GAu **
* $^{78}\text{Kr}(\alpha, ^6\text{He})^{76}\text{Kr}$	Replaced by calibration free $^{80}\text{Kr}(\alpha, ^6\text{He})^{78}\text{Kr}-^{78}\text{Kr}()$										GAu **
* $^{78}\text{Se}(\text{d}, ^3\text{He})^{77}\text{As}$	Original value -4910(4) corrected, see $^{74}\text{Se}(\text{d}, ^3\text{He})$										AHW **
* $^{78}\text{Ge}(\beta^-)^{78}\text{As}$	$E_{\beta^-} = 690(30) \text{ } 710(20) \text{ respectively, to } 1^+ \text{ level at } 277.3 \text{ keV}$										Ens09a **
* $^{78}\text{As}(\beta^-)^{78}\text{Se}$	$E_{\beta^-} = 3000(100) \text{ to } 2^+ \text{ level at } 1308.644 \text{ keV}$										Ens09a **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{78}\text{Rb}(\beta^+)^{78}\text{Kr}$	E_{β^+} =3410(370) from $^{78}\text{Rb}^n$ $4^{(-)}$ at 111.19(0.22) to $(4)^-$ level at 2764.10 keV										Ens09a **
$^{78}\text{Rb}(\beta^+)^{78}\text{Kr}$	Q_{β^+} =7180(80); and 7300(50) from $^{78}\text{Rb}^n$ at 111.19(0.22) keV										Ens09a **
$^{78}\text{Rb}^{\text{v}}(\text{IT})^{78}\text{Rb}$	Corrected; using $^{78}\text{Rb}^n(\text{IT})$ =111.2 keV										GAu **
$^{79}\text{Cu}-\text{u}$	-46700	540	-45530	110	2.2	U			OR1	1.0	06Ha62
	-45526.9	112.7				2			MR1	1.0	17We16
$^{79}\text{Zn}-^{88}\text{Rb}_{.898}$	22278.1	2.9	22276.7	2.4	-0.5	1	68	68 ^{79}Zn	JY1	1.0	08Ha23
$^{79}\text{Ga}-^{88}\text{Rb}_{.898}$	12490.9	2.0	12490.2	1.3	-0.4	1	42	42 ^{79}Ga	JY1	1.0	08Ha23
$^{79}\text{Ga}-\text{u}$	-67064	129	-67148.4	1.3	-0.3	U			GT2	2.5	08Su19
$\text{C}_6\text{H}_7-^{79}\text{Br}$	136444.3	2.4	136437.6	1.1	-1.1	U			M15	2.5	63Ri07
	136444	15			-0.3	U			R11	1.5	78Di09
	136449	12			-0.6	U			R11	1.5	78Di09
$\text{C}_5\text{}^{13}\text{C}\text{H}_6-^{79}\text{Br}$	131976	16	131967.5	1.1	-0.4	U			R11	1.5	78Di09
	131974	17			-0.3	U			R11	1.5	78Di09
$\text{C}_5\text{}^{\text{N}}\text{H}_5-^{79}\text{Br}$	123870	7	123861.6	1.1	-0.8	U			R11	1.5	78Di09
	123871	14			-0.4	U			R11	1.5	78Di09
$\text{C}_5\text{}^{\text{O}}\text{H}_3-^{79}\text{Br}$	100061	15	100052.1	1.1	-0.4	U			R11	1.5	78Di09
	100057	20			-0.2	U			R11	1.5	78Di09
$\text{C}_4\text{}^{\text{N}}\text{O}\text{H}-^{79}\text{Br}$	87489	20	87476.1	1.1	-0.4	U			R11	1.5	78Di09
$^{79}\text{Kr}-\text{u}$	-79981	52	-79917	4	1.2	U			GS2	1.0	05Li24 *
$^{79}\text{Y}-\text{u}$	-62054	86				2			LZ1	1.0	18Xi04
$^{79}\text{Zn}-^{85}\text{Rb}_{.929}$	24582.4	4.2	24585.4	2.4	0.7	1	32	32 ^{79}Zn	MA8	1.0	08Ba54
$^{79}\text{Ga}-^{85}\text{Rb}_{.929}$	14798.4	1.7	14798.9	1.3	0.3	1	58	58 ^{79}Ga	MA8	1.0	19Hu15
$^{79}\text{Rb}-^{85}\text{Rb}_{.929}$	5934	8	5937.4	2.1	0.4	U			MA2	1.0	94Ot01
	5937.2	2.3			0.1	1	82	82 ^{79}Rb	MA8	1.0	07Ke09
$^{79}\text{Sr}-^{85}\text{Rb}_{.929}$	11655	9	11652	8	-0.3	1	78	78 ^{79}Sr	MA2	1.0	94Ot01
$^{77}\text{Se}\text{H}_2-^{79}\text{Br}$	17239	8	17226.6	1.1	-1.0	U			R11	1.5	78Di09
$^{78}\text{Se}\text{H}-^{79}\text{Br}$	6806	8	6796.7	1.1	-0.8	U			R11	1.5	78Di09
$^{79}\text{Br}-^{79}\text{Rb}$	-5652.7	5.0	-5652.5	2.3	0.0	1	20	17 ^{79}Rb	R11	1.0	18Ki21
$^{79}\text{Kr}-^{79}\text{Rb}$	-3870	41	-3907	4	-0.9	U			R11	1.0	18Ki21 *
$^{79}\text{Sr}-^{79}\text{Rb}$	5704	17	5715	8	0.6	1	23	22 ^{79}Sr	R11	1.0	18Ki21
$^{79}\text{Br}-^{78}\text{Kr}$	-2072	30	-2028.8	1.1	1.0	U			R11	1.5	78Di09
$^{77}\text{Rb}-^{79}\text{Rb}_{.487}$ $^{75}\text{Rb}_{.513}$	-1010	40	-996.4	1.7	0.1	U			P20	2.5	82Au01
$^{77}\text{Rb}-^{79}\text{Rb}_{.325}$ $^{76}\text{Rb}_{.675}$	-1060	40	-996.2	1.6	0.6	U			P20	2.5	82Au01
	-990	70			-0.0	U			P20	2.5	82Au01
$^{78}\text{Rb}^{\text{v}}-^{79}\text{Rb}_{.494}$ $^{77}\text{Rb}_{.506}$	940	40	919	12	-0.2	U			P20	2.5	82Au01
$^{78}\text{Se}(\text{n},\gamma)^{79}\text{Se}$	6962.6	0.3	6962.83	0.13	0.8	2					79Br.A Z
	6962.2	0.3			2.1	2			BNn		81En07 Z
	6963.11	0.17			-1.6	2			Bdn		06Fi.A
$^{78}\text{Se}(\text{d},\text{p})^{79}\text{Se}$	4756	6	4738.27	0.13	-3.0	B			MIT		64Sp12
$^{78}\text{Kr}(\text{d},\text{p})^{79}\text{Kr}$	5980	50	6111	3	2.6	U			Yal		56B110
$^{78}\text{Kr}(^3\text{He},\text{d})^{79}\text{Rb}$	-1585	10	-1580.0	2.0	0.5	U			Phi		87St11
$^{79}\text{Zn}(\beta^-)^{79}\text{Ga}$	8550	240	9116.1	2.5	2.4	U			Stu		86Ek01
$^{79}\text{Ga}(\beta^-)^{79}\text{Ge}$	6770	80	6980	40	2.6	o			Stu		77Al17
	7000	80			-0.3	o			Stu		86Ek01
	6979	40			-0.0	1	86	86 ^{79}Ge	Stu		00Me.A
$^{79}\text{Ge}(\beta^-)^{79}\text{As}$	4300	200	4110	40	-1.0	U					70Ka04
	4110	100			-0.0	1	14	14 ^{79}Ge	Stu		81Al20
$^{79}\text{As}(\beta^-)^{79}\text{Se}$	2230	50	2281	5	1.0	U					61Ku09 *
$^{79}\text{Se}(\beta^-)^{79}\text{Br}$	160	5	150.6	1.0	-1.9	U					49Pa.A
$^{79}\text{Kr}(\beta^+)^{79}\text{Br}$	1612	10	1626	3	1.4	2					52Be55
	1620	5			1.2	2					54Th39
	1635	5			-1.8	2					64Bo25
$^{79}\text{Rb}(\beta^+)^{79}\text{Kr}$	3530	50	3640	4	2.2	U					71Li02 *
	3720	90			-0.9	U					72Br31 *
	3650	70			-0.1	U			IRS		93Al03

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference		
$^{79}\text{Sr}(\beta^+)^{79}\text{Rb}$	5259	78	5323	8	0.8	U			BNL		81Li12		
	5059	67			3.9	B			Ors		82De36		
$^{79}\text{Y}(\beta^+)^{79}\text{Sr}$	7120	450	7680	80	1.2	U					92Mu12		
$^{79}\text{Kr}-\text{u}$	$M-A=-74437(30)$ keV for mixture gs+m at 129.77 7/2 ⁺ keV										Nub211	**	
$^{79}\text{Kr}-^{79}\text{Rb}$	$D_M=-3800(9)$ μu for ^{79}Kr gs+m mixture at 129.77 keV; $M-A=-74342.7(9.3)$ keV										Nub211	**	
$^{79}\text{As}(\beta^-)^{79}\text{Se}$	$E_{\beta^-}=1700(50)$ to 527.93 3/2 ⁻ level, and other E_{β^-}										Ens167	**	
$^{79}\text{Rb}(\beta^+)^{79}\text{Kr}$	$E_{\beta^+}=1825(50)$ 2010(90) respectively, to 3/2 ⁺ level at 688.17 keV										Ens167	**	
$^{80}\text{Ga}-\text{u}$	-63441	129	-63579	3	-0.4	U			GT2	2.5	08Su19		
	-63567	52			-0.2	U			TR1	1.0	20Re04		
$\text{C}_6\text{H}_8-^{80}\text{Se}$	146068.5	2.9	146078.5	1.0	1.4	U			M15	2.5	63Ri07		
$\text{C}_6\text{H}_8-^{80}\text{Kr}$	146225.7	4.6	146222.3	0.7	-0.3	U			M15	2.5	63Ri07		
	146235	18			-0.5	U			R11	1.5	78Di09		
	146215	16			0.3	U			R11	1.5	78Di09		
$\text{C}_5\text{O H}_4-^{80}\text{Kr}$	109834	20	109836.8	0.7	0.1	U			R11	1.5	78Di09		
$^{80}\text{Y}-\text{u}$	-65720	190	-65645	7	0.4	U			1.0	1.0	98Is06		
	-66664	86			11.8	F			CS1	1.0	01La31	*	
	-65600	200			-0.2	U			CS1	1.0	08Go23		
$^{80}\text{Y O}-^{96}\text{Mo}$	24594.6	6.7				2			JY1	1.0	06Ka48		
$^{80}\text{Zr}-\text{u}$	-59600	1600	-58790#	320#	0.5	D			1.0	1.0	98Is06	*	
	-59740	161			5.9	F			CS1	1.0	01La31	*	
$^{80}\text{Zn}-^{88}\text{Rb}_{.909}$	25165.2	7.3	25167.1	2.8	0.3	1	14	14	^{80}Zn	JY1	1.0	08Ha23	
$^{80}\text{Ga}-^{88}\text{Rb}_{.909}$	17034.9	3.1				2			JY1	1.0	08Ha23		
$^{80}\text{Ge}-^{88}\text{Rb}_{.909}$	5964.9	2.2				2			JY1	1.0	08Ha23		
$^{80}\text{Kr}-^{86}\text{Kr}_{.930}$	-488.9	1.1	-489.9	0.7	-0.9	1	46	46	^{80}Kr	MS1	1.0	06Ri15	
$^{80}\text{Zn}-^{85}\text{Rb}_{.941}$	27559.1	3.0	27558.8	2.8	-0.1	1	86	86	^{80}Zn	MA8	1.0	08Ba54	
$^{80}\text{Kr}-^{85}\text{Rb}_{.941}$	-614.5	1.7	-616.2	0.7	-1.0	1	19	19	^{80}Kr	MA8	1.0	06Ro11	
	-627.1	9.6			1.1	U			MA8	1.0	10Na13	*	
$^{80}\text{Rb}-^{85}\text{Rb}_{.941}$	5528	8	5522.3	2.0	-0.7	U			MA2	1.0	94Ot01		
	5522.3	2.0				2			MA8	1.0	07Ke09		
$^{80}\text{Sr}-^{85}\text{Rb}_{.941}$	7531	8	7523	4	-1.0	2			MA2	1.0	94Ot01		
	7513	14			0.7	U			MA8	1.0	05Si34		
	7521.3	4.2			0.5	2			SH1	1.0	11Ha08		
$^{80}\text{Se }^{35}\text{Cl}-^{78}\text{Se }^{37}\text{Cl}$	2164.8	1.4	2162.6	1.0	-0.4	U			H18	4.0	64Ba03		
	2160.8	9.2			0.1	U			H40	2.5	85El01		
$^{80}\text{As}-^{80}\text{Kr}$	6096.5	3.5				2			MS1	1.0	07Bo50		
$^{80}\text{Rb}-^{80}\text{Kr}$	6128	12	6138.5	2.1	0.9	U			RI1	1.0	18Ki21		
$^{80}\text{Sr}-^{80}\text{Kr}$	8110	16	8140	4	1.8	F			RI1	1.0	18Ki21	*	
$^{80}\text{Kr}-^{79}\text{Br}$	-1955	28	-1959.6	1.2	-0.1	U			R11	1.5	78Di09		
$^{80}\text{Kr}-^{78}\text{Kr}$	-4046	30	-3988.4	0.8	1.3	U			R11	1.5	78Di09		
$^{79}\text{Rb}-^{80}\text{Rb}_{.658}^{77}\text{Rb}_{.342}$	-1218	27	-1139.3	2.3	1.2	U			P20	2.5	82Au01		
$^{79}\text{Rb}-^{80}\text{Rb}_{.494}^{78}\text{Rb}_{.506}$	-1313	24	-1316	7	-0.1	U			P20	2.5	82Au01		
$^{80}\text{Se}(\text{p},\alpha)^{77}\text{As}$	1020.0	2.8	1020.9	1.8	0.3	1	40	32	^{77}As	NDm		82Zu04	
$^{80}\text{Kr}(\alpha,^6\text{He})^{77}\text{Kr}$	-10398	24	-10384.9	2.1	0.5	U						87Mo06	
$^{80}\text{Se}(\text{d},\alpha)^{78}\text{As}$	5755	12	5768	10	1.1	2			Phi			77Mo13	
$^{80}\text{Se}(\text{p},\text{t})^{78}\text{Se}$	-8395.1	3.0	-8394.4	1.0	0.2	-			NDm			82Zu04	
ave.	-8394.1	2.1			-0.2	1	20	20	^{80}Se			average	
$^{80}\text{Kr}(\alpha,^6\text{He})^{78}\text{Kr}-^{78}\text{Kr}(\gamma)^{76}\text{Kr}$	1432	10	1449	4	1.7	1	17	16	^{76}Kr			82Mo23	
$^{78}\text{Kr}(\alpha,^3\text{He},\text{n})^{80}\text{Sr}$	2990	30	2993	3	0.1	U						79A119	
$^{80}\text{Se}(\text{d},^3\text{He})^{79}\text{As}$	-5921	7	-5919	5	0.3	-			Ors			83Ro08	*
	-5921	13			0.2	-			Hei			83Wi14	
$^{80}\text{Se}(\text{t},\alpha)^{79}\text{As}$	8407	10	8401	5	-0.6	-			Phi			83Mo09	
$^{80}\text{Se}(\text{d},^3\text{He})^{79}\text{As}$	-5919	5	-5919	5	0.0	1	100	100	^{79}As			average	
$^{80}\text{Se}(\text{p},\text{d})^{79}\text{Se}$	-7687.6	3.0	-7688.8	1.0	-0.4	R			NDm			82Zu04	
$^{79}\text{Br}(\text{n},\gamma)^{80}\text{Br}$	7892.11	0.20	7892.28	0.13	0.8	-			ILn			78Do06	Z
	7892.41	0.18			-0.7	-			Bdn			06Fi.A	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference			
$^{79}\text{Br}(\text{d,p})^{80}\text{Br}$	ave.	5640	20	5667.71	0.13	1.4	U	100	$^{96}\text{ }^{79}\text{Br}$	Mtr	72Ch33			
$^{79}\text{Br}(\text{n},\gamma)^{80}\text{Br}$		7892.28	0.13	7892.28	0.13	-0.0	1				average			
$^{80}\text{Zn}(\beta^-)^{80}\text{Ga}$		7540	200	7575	4	0.2	U				86Ek01			
$^{80}\text{Ga}(\beta^-)^{80}\text{Ge}$		7150	150			2.8	U				Trs	86Gi07		
		10000	300	10312	4	1.0	o				Stu	81Al20		
		10380	120			-0.6	U				Stu	86Ek01		
$^{80}\text{Ge}(\beta^-)^{80}\text{As}$		2640	70	2679	4	0.6	U				Stu	77Al17		
		2630	20			2.5	U				Trs	86Gi07		
$^{80}\text{As}(\beta^-)^{80}\text{Se}$		6000	200	5545	3	-2.3	U					Trs	59Me68	
		5470	90			0.8	U						86Gi07	
$^{80}\text{Se}(\text{t},^3\text{He})^{80}\text{As}$	-5560	25	-5526	3	1.3	U		LAl	79Aj02					
$^{80}\text{Br}(\beta^+)^{80}\text{Se}$	1884	10	1870.5	0.3	-1.4	U	100	$^{96}\text{ }^{80}\text{Br}$	PTB	54Li19				
	1872	7			-0.2	U				69Ka06				
$^{80}\text{Se}(\text{p,n})^{80}\text{Br}$	-2655.2	2.8	-2652.8	0.3	0.9	U					Tkm	63Ok01		
	-2652.5	3.0			-0.1	U						Oak	64Jo11	Z
	-2653.2	5.			0.1	U							70Fi03	
	-2652.81	0.31			0.0	1							92Bo02	Z
$^{80}\text{Br}(\beta^-)^{80}\text{Kr}$	1970	30	2004.4	1.1	1.1	U						52Fu04		
	2040	20			-1.8	U							54Li19	
	1997	10			0.7	U							69Ka06	
	5120	500	5718.0	2.0	1.2	U							61Ho13	
$^{80}\text{Rb}(\beta^+)^{80}\text{Kr}$	5500	350			0.6	U		IRS	75We23	*				
	5650	100			0.7	U				93Al03				
	-6484.0	20.	-6500.3	2.0	-0.8	U				72Ja.A				
$^{80}\text{Kr}(\text{p,n})^{80}\text{Rb}$	6952	152	9163	7	14.5	F		BNL	Ors	81Li12	*			
$^{80}\text{Y}(\beta^+)^{80}\text{Sr}$	6934	242			9.2	F				82De36	*			
	6200	600			4.9	F				96Sh27	*			
* $^{80}\text{Y}-\text{u}$	F : below lower limit $M>-65890(90)\text{ }\mu\text{u}$ -61376(83) keV determined in reference										03Ba18	**		
* $^{80}\text{Zr}-\text{u}$	Trends from Mass Surface TMS suggest ^{80}Zr 770 keV less bound										GAu	**		
* $^{80}\text{Zr}-\text{u}$	F : other results in same paper not trusted, see ^{80}Y and ^{68}Se										GAu	**		
* $^{80}\text{Kr}-^{85}\text{Rb}_{.941}$	Only one measurement										GAu	**		
* $^{80}\text{Sr}-^{80}\text{Kr}$	Compromised measurement due to the influence of neighboring ToF peaks										Sar201	**		
* $^{80}\text{Se}(\text{d},^3\text{He})^{79}\text{As}$	Originally -5927(7), see $^{74}\text{Se}(\text{d},^3\text{He})$										AHW	**		
* $^{80}\text{Rb}(\beta^+)^{80}\text{Kr}$	$E_{\beta^+}=3860(350)$ to 2^+ level at 616.60 keV										Ens058	**		
* $^{80}\text{Y}(\beta^+)^{80}\text{Sr}$	F : below lower limit $Q_{\beta^-} > 8929(23)$ keV determined in reference										03Ba18	**		
$^{81}\text{Ga}-\text{u}$	-61853	33	-61866	4	-0.4	U			TR1	1.0	20Re04			
$^{81}\text{Ge}-\text{u}$	-71710	240	-71167.1	2.2	0.9	U			GT2	2.5	08Kn.A	*		
$\text{C}_6\text{H}_9-^{81}\text{Br}$	154135.3	3.8	154137.1	1.0	0.2	U			M15	2.5	63Ri07			
	154143	17			-0.2	U			R11	1.5	78Di09			
	154134	10			0.2	U			R11	1.5	78Di09			
	141561	10	141561.0	1.0	0.0	U			R11	1.5	78Di09			
$\text{C}_5\text{N H}_7-^{81}\text{Br}$	141553	18			0.3	U			R11	1.5	78Di09			
	117742	12	117751.6	1.0	0.5	U			R11	1.5	78Di09			
$\text{C}_4\text{O}_2\text{H}-^{81}\text{Br}$	81356	20	81366.1	1.0	0.3	U			R11	1.5	78Di09			
$\text{C}_4^{13}\text{C O H}_4-^{81}\text{Br}$	113275	14	113281.4	1.0	0.3	U			R11	1.5	78Di09			
$^{81}\text{Rb}-\text{u}$	-80958	41	-81006	5	-1.2	U			GS2	1.0	05Li24	*		
$^{81}\text{Y O}-^{97}\text{Mo}$	18352.0	5.8				2			JY1	1.0	06Ka48			
$^{81}\text{Zr}-\text{u}$	-61755	99				2			LZ1	1.0	18Xi04			
$^{81}\text{Ga}-^{88}\text{Rb}_{.920}$	19723.5	3.5				2			JY1	1.0	08Ha23			
$^{81}\text{Ge}-^{88}\text{Rb}_{.920}$	10422.6	2.2				2			JY1	1.0	08Ha23			
$^{81}\text{As}-^{88}\text{Rb}_{.920}$	3721.9	3.3	3721.9	2.8	0.0	1	74	$^{74}\text{ }^{81}\text{As}$	JY1	1.0	08Ha23			
$^{81}\text{Zn}-^{85}\text{Rb}_{.953}$	34467.0	5.4				2			MA8	1.0	08Ba54			
$^{81}\text{Rb}-^{85}\text{Rb}_{.953}$	3063	8	3058	5	-0.6	-			MA2	1.0	94Ot01			
	3055.4	9.2			0.3	-			SH1	1.0	11Ha08	*		
$^{81}\text{Sr}-^{85}\text{Rb}_{.953}$	ave.	3060	6			-0.2	1	76	$^{76}\text{ }^{81}\text{Rb}$		average			
	7278	8	7276	3	-0.3	2			MA2	1.0	94Ot01			
	7272	12			0.3	U			MA8	1.0	05Si34			
	7275.3	3.7			0.1	2			SH1	1.0	11Ha08			

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{81}\text{Se}-^{80}\text{Kr}_{1.013}$	2704.2	2.4	2702.2	1.2	-0.8	1	25	17 ^{81}Se	MS1	1.0	07Bo50 *
$^{80}\text{Se H}-^{81}\text{Br}$	8023	8	8058.6	1.5	3.0	B			R11	1.5	78Di09
$^{80}\text{Kr H}-^{81}\text{Br}$	7922	18	7914.8	1.3	-0.3	U			R11	1.5	78Di09
$^{81}\text{Br}-^{81}\text{Sr}$	-6899.5	4.9	-6923	4	-4.8	B			R11	1.0	18Ki21
$^{81}\text{Rb}-^{81}\text{Sr}$	-4193	27	-4217	6	-0.9	U			R11	1.0	18Ki21 *
$^{81}\text{Br}-\text{H }^{79}\text{Br}$	-9865	13	-9874.4	1.5	-0.3	U			M15	2.5	63Ri07
$^{81}\text{Br}-^{80}\text{Kr}$	-91	32	-89.7	1.3	0.0	U			R11	1.5	78Di09
$^{81}\text{Br}-^{79}\text{Br}$	-2020	32	-2049.4	1.5	-0.6	U			R11	1.5	78Di09
	-2014	35			-0.7	U			R11	1.5	78Di09
$^{79}\text{Rb}-^{81}\text{Rb}_{.325} \text{ } ^{78}\text{Rb}_{.675}^x$	-1130	30	-1148	9	-0.2	U			P20	2.5	82Au01 Y
$^{80}\text{Rb}-^{81}\text{Rb}_{.494} \text{ } ^{79}\text{Rb}_{.506}$	927	29	926	3	-0.0	U			P20	2.5	82Au01 Y
$^{80}\text{Se}(n,\gamma)^{81}\text{Se}$	6700.9	0.5	6700.8	0.3	-0.1	-			BNn		81En07 Z
	6700.9	0.5			-0.1	-			Bdn		06Fi.A
$^{80}\text{Se}(d,p)^{81}\text{Se}$	4490	6	4476.3	0.3	-2.3	U			MIT		64Sp12
	4477.5	3.0			-0.4	U			NDm		82Zu04
$^{80}\text{Se}(n,\gamma)^{81}\text{Se}$	ave. 6700.9	0.4	6700.8	0.3	-0.1	1	97	72 ^{81}Se			average
$^{81}\text{Br}(\gamma,n)^{80}\text{Br}$	-10130	35	-10159.3	1.4	-0.8	U			Phi		60Ge01
$^{80}\text{Kr}(d,p)^{81}\text{Kr}$	5660	15	5649.5	1.3	-0.7	U			Tex		75Ch11 *
	5646	4			0.9	1	10	7 ^{81}Kr	Oak		86Bu18
$^{80}\text{Kr}(^3\text{He},d)^{81}\text{Rb}$	-637	10	-641	5	-0.4	1	24	24 ^{81}Rb	Phi		87St11
$^{81}\text{Zr}(\text{ep})^{80}\text{Sr}$	4700	200	5500	90	4.0	B					99Hu05
$^{81}\text{Ga}(\beta^-)^{81}\text{Ge}$	8320	150	8664	4	2.3	U			Stu		81Al20
$^{81}\text{Ge}(\beta^-)^{81}\text{As}$	6230	120	6242	3	0.1	U			Stu		81Al20 *
$^{81}\text{As}(\beta^-)^{81}\text{Se}$	3800	200	3855.7	2.8	0.3	U					60Mo01
	3730	100			1.3	U			Stu		77Al17
$^{81}\text{Se}(\beta^-)^{81}\text{Br}$	1600	50	1588.0	1.4	-0.2	U					60Ku06
	1560	50			0.6	U					67Yt03
$^{81}\text{Kr}(\epsilon)^{81}\text{Br}$	280.7	0.5	280.9	0.5	0.3	1	89	84 ^{81}Kr			88Ax01 *
$^{81}\text{Br}(p,n)^{81}\text{Kr}$	-1062	4	-1063.2	0.5	-0.3	U					84Fi.A
$^{81}\text{Br}(^3\text{He},t)^{81}\text{Kr}$	-296	6	-299.4	0.5	-0.6	U					84Bu23
$^{81}\text{Br}(^3\text{He},t)^{81}\text{Kr}-^{51}\text{V}()^{51}\text{Cr}$	470.6	1.8	471.5	0.5	0.5	U			Pri		82Ko06 *
$^{81}\text{Rb}(\beta^+)^{81}\text{Kr}$	2260	30	2239	5	-0.7	U					75Va24 *
	2290	50			-1.0	U					77Li14
$^{81}\text{Sr}(\beta^+)^{81}\text{Rb}$	3990	30	3929	6	-2.0	U					73Br32 *
$^{81}\text{Y}(\beta^+)^{81}\text{Sr}$	5408	86	5815	6	4.7	B			BNL		81Li12
	5620	89			2.2	U			Ors		82De36
$^{81}\text{Zr}(\beta^+)^{81}\text{Y}$	7160	290	8190	90	3.5	B			Ors		82De36
$^{81}\text{Ge}-u$	$M-A=-66454(93)$ keV for mixture gs+m at 679.14 keV										Nub211 **
$^{81}\text{Rb}-u$	$M-A=-75369(29)$ keV for mixture gs+m at 86.31 keV										Nub211 **
$^{81}\text{Rb}-^{85}\text{Rb}_{.953}$	$D_M=3148.1(9.2)$ keV for $^{81}\text{Rb}^m$ at 86.31(0.07) keV; $M-A=-75373.1(8.6)$ keV										Nub211 **
$^{81}\text{Se}-^{80}\text{Kr}_{1.013}$	$D_M=2814.8(2.4)$ μ u for $^{81}\text{Se}^m$ at 103.00(0.06)keV; $M-A=-76283.2(2.4)$ keV										Nub211 **
$^{81}\text{Rb}-^{81}\text{Sr}$	$D_M=4147.0(0.9)$ μ u for mixture gs+m at 86.31 keV; $M-A=-75391.5$ keV										Nub211 **
$^{80}\text{Kr}(d,p)^{81}\text{Kr}$	Original value 5610(15) reinterpreted as going to 49.57 level										76Me08 **
$^{81}\text{Ge}(\beta^-)^{81}\text{As}$	$Q_{\beta^-}=6230(120)$; and 6930(280) from $^{81}\text{Ge}^m$ at 679.14 keV										Nub211 **
$^{81}\text{Kr}(\epsilon)^{81}\text{Br}$	$LM=0.42(0.05)$, $Q(\epsilon)=4.7(0.5)$ to $5/2^-$ level at 275.985 keV										Ens08a **
$^{81}\text{Br}(^3\text{He},t)^{81}\text{Kr}-^{51}\text{V}()^{51}\text{Cr}$	$Q-Q$ to 456.89(0.03) level=13.7(1.8) keV										GAu **
$^{81}\text{Rb}(\beta^+)^{81}\text{Kr}$	$E_{\beta^+}=1050(30)$ to $1/2^-$ level at 190.64 keV										Ens08a **
$^{81}\text{Sr}(\beta^+)^{81}\text{Rb}$	$E_{\beta^+}=2684(30)$ to 301.241 $3/2^-$ level, and other E_{β^+}										Ens08a **
$^{82}\text{Ga}-u$	-56812	268	-56823.5	2.6	-0.0	U			GT1	1.5	04Ma.A
	-56870	33			1.4	U			TR1	1.0	20Re04 *
$^{82}\text{Ge}-u$	-70400	129	-70226.0	2.4	0.5	U			GT2	2.5	08Su19
$\text{C}_6 \text{H}_{10}-^{82}\text{Se}$	161545.0	4.6	161550.8	0.5	0.5	U			M15	2.5	63Ri07
$\text{C}_6 \text{H}_{10}-^{82}\text{Kr}$	164769.8	3.4	164769.165	0.006	-0.1	U			M15	2.5	63Ri07
	164787	14			-0.8	U			R11	1.5	78Di09
	164784	16			-0.6	U			R11	1.5	78Di09

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_5 N H_8 - ^{82}Kr$	152200	25	152193.106	0.006	-0.2	U			R11	1.5	78Di09
$C_5 O H_6 - ^{82}Kr$	128396	20	128383.657	0.006	-0.4	U			R11	1.5	78Di09
$^{82}Rb-u$	-81775	39	-81791	3	-0.4	U			GS2	1.0	05Li24 *
$^{82}Sr-u$	-81604	63	-81600	6	0.1	U			GS2	1.0	05Li24
$^{82}Y O - ^{98}Mo$	16441.2	5.9				2			JY1	1.0	06Ka48
$^{82}Zr-u$	-68312	11	-68292.5	1.7	1.8	U			LZ1	1.0	18Xi04
$^{82}Ga - ^{88}Rb_{932}$	25830.4	2.6				2			JY1	1.0	08Ha23
$^{82}Ge - ^{88}Rb_{932}$	12427.9	2.4				2			JY1	1.0	08Ha23
$^{82}As - ^{88}Rb_{932}$	7392.6	4.0				2			JY1	1.0	08Ha23 *
$^{82}Kr - ^{86}Kr_{953}$	-1329.4	1.1	-1330.771	0.005	-1.2	U			MS1	1.0	06Ri15
	-1330.7684	0.0094			-0.3	1	26	$^{25} ^{82}Kr$	FS1	1.0	13Ho22
$^{82}Zn - ^{85}Rb_{965}$	39697.0	3.3				2			MA8	1.0	13Wo06
$^{82}Kr - ^{85}Rb_{965}$	-1394.9	2.6	-1395.944	0.006	-0.4	U			MA8	1.0	06Ro11
$^{82}Rb^m - ^{85}Rb_{965}$	3407	9	3406.0	2.8	-0.1	U			MA2	1.0	94Ot01
	3406.0	2.8				2			MA8	1.0	05Gu37
$^{82}Sr - ^{85}Rb_{965}$	3517	8	3523	6	0.7	1	65	$^{65} ^{82}Sr$	MA2	1.0	94Ot01
$^{82}Zr - ^{85}Rb_{965}$	16830.4	1.7				2			JY1	1.0	19Vi05
$^{82}Se ^{35}Cl - ^{80}Se ^{37}Cl$	3128.92	0.63	3127.9	1.0	-0.7	1	40	$^{36} ^{80}Se$	H40	2.5	85El01
$^{80}Se H_2 - ^{82}Kr$	18665	18	18690.7	1.0	1.0	U			R11	1.5	78Di09
	18671	19			0.7	U			R11	1.5	78Di09
$^{82}Kr - ^{84}Kr_{976}$	-140.6277	0.0051	-140.627	0.005	0.1	1	78	$^{75} ^{82}Kr$	FS1	1.0	13Ho22
$^{81}Br H - ^{82}Se$	7419	8	7413.7	1.2	-0.4	U			R11	1.5	78Di09
$^{81}Br H - ^{82}Kr$	10662	20	10632.1	1.0	-1.0	U			R11	1.5	78Di09
$^{82}Se - ^{82}Kr$	3222	16	3218.4	0.5	-0.2	U			R11	1.5	78Di09
	3218	22			0.0	U			R11	1.5	78Di09
	3216.1	1.6			0.9	U			H45	1.5	93Nx01
	3218.36	0.52			0.0	1	93	$^{93} ^{82}Se$	MS1	1.0	13Li01
$^{82}Kr - ^{78}Se H_3$	-27269	35	-27303.19	0.19	-0.7	U			R11	1.5	78Di09
$^{80}Se H_3 - ^{82}Kr$	26466	32	26515.7	1.0	1.0	U			R11	1.5	78Di09
$^{82}Kr - ^{81}Br$	-2805	32	-2807.0	1.0	-0.0	U			R11	1.5	78Di09
$^{82}Se H - ^{82}Kr$	11082	40	11043.4	0.5	-0.6	U			R11	1.5	78Di09
$^{79}Rb - ^{82}Rb_{241} ^{78}Rb_{760}^x$	-1536	29	-1627	10	-1.3	U			P20	2.5	82Au01 Y
$^{81}Rb - ^{82}Rb_{741} ^{78}Rb_{260}^x$	-1680	40	-1618	6	0.6	U			P20	2.5	82Au01 Y
$^{80}Rb - ^{82}Rb_{325} ^{79}Rb_{675}$	440	40	377.4	2.5	-0.6	U			P20	2.5	82Au01 Y
$^{82}Kr(^3He, ^6He) ^{79}Kr$	-8822	31	-8810	3	0.4	U					87Mo06
$^{82}Se(^{14}C, ^{16}O) ^{80}Ge$	-449	60	-301.7	2.1	2.5	U			Ors		83Be.C
$^{82}Se(^{18}O, ^{20}Ne) ^{80}Ge$	-2020	40	-1799.5	2.1	5.5	B			Hei		83Wi14 *
$^{82}Se(p, t) ^{80}Se$	-7496.1	3.0	-7495.2	0.9	0.3	1	10	$^{9} ^{80}Se$	NDm		82Zu04
$^{82}Se(d, ^3He) ^{81}As$	-6864	10	-6856.1	2.7	0.8	-			Ors		83Ro08 *
	-6861	18			0.3	U			Hei		83Wi14
	7467	6	7464.3	2.7	-0.4	-			Phi		82Mo04
$^{82}Se(t, \alpha) ^{81}As$	-6856	5	-6856.1	2.7	0.0	1	27	$^{26} ^{81}As$			average
$^{82}Se(d, ^3He) ^{81}As$	-7051.8	2.8	-7051.6	1.0	0.1	1	12	$^{11} ^{81}Se$	NDm		82Zu04
$^{81}Br(n, \gamma) ^{82}Br$	7592.80	0.20	7592.94	0.12	0.7	-			ILn		78Do06 Z
	7593.02	0.15			-0.5	-			Bdn		06Fi.A
$^{81}Br(d, p) ^{82}Br$	5400	20	5368.38	0.12	-1.6	U			Mtr		72Ch33
$^{81}Br(n, \gamma) ^{82}Br$	7592.94	0.12	7592.94	0.12	0.0	1	100	$^{94} ^{81}Br$			average
$^{82}Ge(\beta^-) ^{82}As$	4700	140	4690	4	-0.1	U			Stu		81Al20
$^{82}As(\beta^-) ^{82}Se$	7270	200	7488	4	1.1	U					70Va31 *
	6360	200			5.6	B					70Ka04 *
	7740	30			-8.4	C			Stu		00Me.A *
	7531	21			-2.0	U			Stu		04Ga44 *
$^{82}Se(t, ^3He) ^{82}As$	-7500	25	-7470	4	1.2	U			LAl		79Aj02
$^{82}Br(\beta^-) ^{82}Kr$	3092.9	1.0	3093.1	1.0	0.2	1	94	$^{94} ^{82}Br$			56Wa24 *
$^{82}Rb(\beta^+) ^{82}Kr$	4400	15	4404	3	0.3	U					69Be74 *
	4420	60			-0.3	U			IRS		93Al03 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{82}\text{Kr}(\text{p,n})^{82}\text{Rb}$	-5161	20	-5186	3	-1.3	U					72Ja.A
$^{82}\text{Rb}^m(\text{IT})^{82}\text{Rb}$	69.0	1.5				3					Ens035
$^{82}\text{Y}(\beta^+)^{82}\text{Sr}$	7868	185	7946	8	0.4	U			BNL		81Li12
	7793	123			1.2	U			Ors		82De36
$^{82}\text{Zr}(\beta^+)^{82}\text{Y}$	4000	500	4450	6	0.9	F			Ors		82De36 *
$^{82}\text{Ga}-\text{u}$	corr. for gs+m of ^{82}Rb as calibrant, -52939(23) keV assuming mainly iso.										20Re04 **
$^{82}\text{Rb}-\text{u}$	$M-A=-76138(30)$ keV for mixture gs+m at 69.0(1.5) keV										Nub211 **
$^{82}\text{As}-^{88}\text{Rb}_{.932}$	$D_M=7395.1(4.6)$ 7532.5(4.0) μu to ground state and $^{82}\text{As}^m$ at 131.6(0.5) keV										Nub211 **
$^{82}\text{Se}(^{18}\text{O}, ^{20}\text{Ne})^{80}\text{Ge}$	Recalibrated to $^{64}\text{Ni}(^{62}\text{Fe})=-1938(15)$ keV										AHW **
$^{82}\text{Se}(\text{d}, ^3\text{He})^{81}\text{As}$	Originally -6870(10), see $^{74}\text{Se}(\text{d}, ^3\text{He})$										AHW **
$^{82}\text{As}(\beta^-)^{82}\text{Se}$	$E_{\beta^-}=7200(200)$ to ground state (80%) and 654.75 2^+ level (10%)										Ens035 **
$^{82}\text{As}(\beta^-)^{82}\text{Se}$	$E_{\beta^-}=3600(200)$ from $^{82}\text{As}^m$ at 132.1 to 5^- level at 2893.70 and higher levels										Ens035 **
$^{82}\text{As}(\beta^-)^{82}\text{Se}$	and $E_{\beta^-}=7625(22)$ from $^{82}\text{As}^m$ at 132.1 keV										00Me.A **
$^{82}\text{As}(\beta^-)^{82}\text{Se}$	Average of 3 branches; and 7677(17) average of 2 branches from $^{82}\text{As}^m$ at 132.1										04Ga44 **
$^{82}\text{Br}(\beta^-)^{82}\text{Kr}$	$E_{\beta^-}=444(1)$ to 4^- level at 2648.360 keV										Ens035 **
$^{82}\text{Rb}(\beta^+)^{82}\text{Kr}$	$E_{\beta^+}=3350(60)$; and 800(15) from $^{82}\text{Rb}^m$ at 69.0(1.5) to 4^- level at 2648.360										Ens035 **
$^{82}\text{Rb}(\beta^+)^{82}\text{Kr}$	$Q_{\beta^+}=4360(100)$; and 4510(60) of $^{82}\text{Rb}^m$ at 69.0(1.5) keV										Nub211 **
$^{82}\text{Zr}(\beta^+)^{82}\text{Y}$	F : for 2.5(0.1) m activity, but Ensdf adopts 32(5) s										Ens035 **
$^{83}\text{Ga}-\text{u}$	-52881	27	-52879.7	2.8	0.0	U			TR1	1.0	20Re04
$^{83}\text{Ge}-\text{u}$	-65626	268	-65460.9	2.6	0.4	o			GT1	1.5	04Ma.A
	-65270	320			-0.6	U			OR1	1.0	06Ha62
	-65276	129			-0.6	U			GT2	2.5	08Su19
$^{83}\text{As}-\text{u}$	-74677	129	-74793	3	-0.4	U			GT2	2.5	08Su19
$\text{C}_6\text{H}_{11}-^{83}\text{Kr}$	171946.8	3.4	171948.834	0.010	0.2	U			M15	2.5	63Ri07
	171948	16			0.0	U			R11	1.5	78Di09
$\text{C}_5\text{N H}_9-^{83}\text{Kr}$	159344	25	159372.775	0.010	0.8	U			R11	1.5	78Di09
	159360	19			0.4	U			R11	1.5	78Di09
$\text{C}_5\text{O H}_7-^{83}\text{Kr}$	135543	25	135563.326	0.010	0.5	U			R11	1.5	78Di09
$^{83}\text{Y O}-^{98}\text{Mo}_{1.010}$	12941	20				2			JY1	1.0	06Ka48 *
$^{83}\text{Zr O}-^{98}\text{Mo}_{1.010}$	19697.9	6.9				2			JY1	1.0	06Ka48
$^{83}\text{Nb}-\text{u}$	-61850	174				2			LZ1	1.0	18Xi04
$^{83}\text{Ga}-^{88}\text{Rb}_{.943}$	30749.7	2.8				2			JY1	1.0	08Ha23
$^{83}\text{Ge}-^{88}\text{Rb}_{.943}$	18168.5	2.6				2			JY1	1.0	08Ha23
$^{83}\text{As}-^{88}\text{Rb}_{.943}$	8836.3	3.0				2			JY1	1.0	08Ha23
$^{80}\text{Se H}_3-^{83}\text{Kr}$	25825	25	25870.3	1.0	1.2	U			R11	1.5	78Di09
$^{83}\text{Kr}-^{86}\text{Kr}_{.965}$	386.6	1.1	387.261	0.009	0.6	U			MS1	1.0	06Ri15
$^{83}\text{Rb}-^{85}\text{Rb}_{.976}$	1207	8	1207.4	2.5	0.0	U			MA2	1.0	94Ot01
	1207.4	2.5			-0.0	1	100	100 ^{83}Rb	MA8	1.0	07Ke09
$^{82}\text{Se H}-^{83}\text{Kr}$	10380	18	10398.0	0.5	0.7	U			R11	1.5	78Di09
	10368	16			1.3	U			R11	1.5	78Di09
$^{82}\text{Kr H}-^{83}\text{Kr}$	7160	18	7179.669	0.010	0.7	U			R11	1.5	78Di09
$^{83}\text{Kr}-^{84}\text{Kr}_{.988}$	1566.7601	0.0089	1566.760	0.009	0.0	1	100	100 ^{83}Kr	FS1	1.0	13Ho22
$^{83}\text{Sr}-^{83}\text{Rb}$	2447	9	2440	7	-0.8	1	59	59 ^{83}Sr	MA2	1.0	94Ot01
$^{83}\text{Kr}-^{80}\text{Se H}_2$	-18022	36	-18045.3	1.0	-0.4	U			R11	1.5	78Di09
$^{85}\text{Rb}-^{83}\text{Kr H}$	-10211	45	-10161.812	0.010	0.7	U			R11	1.5	78Di09
$^{83}\text{Kr}-^{82}\text{Se}$	-2572	35	-2573.0	0.5	-0.0	U			R11	1.5	78Di09
$^{83}\text{Kr}-^{82}\text{Kr}$	648	12	645.363	0.010	-0.1	U			M15	2.5	63Ri07
$^{81}\text{Rb}-^{83}\text{Rb}_{.488}$ $^{79}\text{Rb}_{.513}$	-529	26	-548	5	-0.3	U			P20	2.5	82Au01 Y
$^{81}\text{Rb}-^{83}\text{Rb}_{.325}$ $^{80}\text{Rb}_{.675}$	-1054	27	-1040	5	0.2	U			P20	2.5	82Au01 Y
$^{82}\text{Rb}-^{83}\text{Rb}_{.659}$ $^{80}\text{Rb}_{.342}$	627	24	604	3	-0.4	U			P20	2.5	82Au01 Y
$^{82}\text{Rb}-^{83}\text{Rb}_{.494}$ $^{81}\text{Rb}_{.506}$	1098	23	1054	4	-0.8	U			P21	2.5	82Au01 Y
$^{82}\text{Ge}(\text{d,p})^{83}\text{Ge}$	1470	70	1408	3	-0.9	U			NDm		05Th03
$^{82}\text{Se}(\text{d,p})^{83}\text{Se}$	3593.4	3.0				2			NDm		78Mo12
$^{82}\text{Se}(^3\text{He,d})^{83}\text{Br}$	3207.4	5.6	3215	4	1.4	1	46	46 ^{83}Br	NDm		83Zu01

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{82}\text{Kr}(^3\text{He,d})^{83}\text{Rb}$	288	10	274.3	2.3	-1.4	U			Phi		87St11
$^{83}\text{Zr}(\varepsilon\text{p})^{82}\text{Sr}$	2750	100	2809	9	0.6	U					83Ha06
$^{83}\text{As}(\beta^-)^{83}\text{Se}$	5460	220	5671	4	1.0	U			Stu		77Al17
$^{83}\text{Se}(\beta^-)^{83}\text{Br}$	3610	40	3673	5	1.6	U					67Ma35
	3681	20			-0.4	U					68Sc10 *
$^{83}\text{Br}(\beta^-)^{83}\text{Kr}$	982	10	977	4	-0.5	-					51Du03 *
	967	15			0.7	U					63Pa09 *
	966	6			1.8	-					69Ph03 *
ave.	970	5			1.3	1	54	54	^{83}Br		average
$^{83}\text{Rb}(\varepsilon)^{83}\text{Kr}$	750	20	920.0	2.3	8.5	B					70Go45 *
$^{83}\text{Sr}(\beta^+)^{83}\text{Rb}$	2264	10	2273	6	0.9	1	41	41	^{83}Sr		68Et01 *
$^{83}\text{Y}(\beta^+)^{83}\text{Sr}$	4509	85	4592	20	1.0	U			BNL		81Li12 *
	4455	50			2.7	B			Ors		82De36 *
$^{83}\text{Zr}(\beta^+)^{83}\text{Y}$	5868	85	6294	20	5.0	B			Ors		82De36 *
$^{83}\text{Nb}(\beta^+)^{83}\text{Zr}$	7500	300	8300	160	2.7	B					88Ku14
$^{83}\text{Y O}-^{98}\text{Mo}_{1.010}$	$D_M=12973.8(5.9) \mu\text{u}$ for mixture gs+m at 62.04(0.10) keV; $M-A=-72172.9(5.8) \text{ keV}$										Nub211 **
$^{83}\text{Se}(\beta^-)^{83}\text{Br}$	$Q_{\beta^-}=3910(20)$ from $^{83}\text{Se}^m$ at 228.92 keV										Nub211 **
$^{83}\text{Br}(\beta^-)^{83}\text{Kr}$	$E_{\beta^-}=940(10) 925(15) 924(6)$ respectively, to $^{83}\text{Kr}^n$ at 41.5575 keV										Nub211 **
$^{83}\text{Rb}(\varepsilon)^{83}\text{Kr}$	$LK=0.132(0.002)$ to $5/2^-$ level at 561.9585, recalculated Q										Ens153 **
$^{83}\text{Sr}(\beta^+)^{83}\text{Rb}$	$E_{\beta^+}=1227(8) 24\%$ to ground state, 20% to $^{83}\text{Rb}^m$ at 42.0780, and other E_{β^+}										Nub211 **
$^{83}\text{Y}(\beta^+)^{83}\text{Sr}$	$E_{\beta^+}=2868(85)$ from $^{83}\text{Y}^m$ at 62.04 keV to $(3/2^-)$ level at 681.11 keV										Ens153 **
$^{83}\text{Y}(\beta^+)^{83}\text{Sr}$	$E_{\beta^+}=3353(50)$ to $9/2^+$ level at 35.47 keV										Ens153 **
*	and $E_{\beta^+}=2941(84)$ from $^{83}\text{Y}^m$ at 62.04 to $(3/2^-)$ level at 681.11 keV										Ens153 **
$^{83}\text{Zr}(\beta^+)^{83}\text{Y}$	$Q_{\beta^+}=5806(85)$ to $^{83}\text{Y}^m$ at 62.04 keV										Nub211 **
$^{83}\text{Zr}(\beta^+)^{83}\text{Y}$	Recalculated value 5802(50) of reference not accepted										87Ra06 **
$^{84}\text{Ga}-\text{u}$	-47337	32				2			TR1	1.0	20Re04
$^{84}\text{Ge}-\text{u}$	-62270	430	-62425	3	-0.4	U			OR1	1.0	06Ha62
$^{84}\text{As}-\text{u}$	-70530	320	-70697	3	-0.5	U			OR1	1.0	06Ha62 *
	-70710	128			0.0	U			GT2	2.5	08Su19
$\text{C}_6 \text{H}_{12}-^{84}\text{Kr}$	182399.4	2.5	182402.656	0.004	0.5	U			M15	2.5	63Ri07
	182392	6			1.2	U			R11	1.5	78Di09
$\text{C}_5 \text{N H}_{10}-^{84}\text{Kr}$	169819	18	169826.596	0.004	0.3	U			R11	1.5	78Di09
	169819	13			0.4	U			R11	1.5	78Di09
$\text{C}_5 \text{O H}_8-^{84}\text{Kr}$	146010	20	146017.147	0.004	0.2	U			R11	1.5	78Di09
$\text{C}_4 ^{13}\text{C O H}_7-^{84}\text{Kr}$	141543	18	141546.951	0.004	0.1	U			R11	1.5	78Di09
$^{84}\text{Kr}-\text{N}_6$	-106946.3154	0.0152	-106946.298	0.004	1.1	o			FS1	1.0	05Sh38 *
	-106946.2971	0.0086			-0.2	1	22	21	^{84}Kr	1.0	09Re03
$^{84}\text{Kr H}-^{85}\text{Rb}$	7515	18	7533.023	0.004	0.7	U			R11	1.5	78Di09
$\text{C}_6 \text{H}_{12}-^{84}\text{Sr}$	180470.8	2.6	180481.3	1.3	1.6	U			M15	2.5	63Ri07
$^{84}\text{Y O}-^{97}\text{Mo}_{1.031}$	12483.5	5.1	12482	5	-0.2	1	82	82	^{84}Y	1.0	08We10
$^{84}\text{Zr O}-^{98}\text{Mo}_{1.020}$	14728.6	5.9				2			JY1	1.0	06Ka48
$^{84}\text{Nb}-\text{u}$	-65721	13	-65694.3	0.4	2.1	U			LZ1	1.0	18Xi04
$^{84}\text{Ge}-^{88}\text{Rb}_{.955}$	22268.7	3.4				2			JY1	1.0	08Ha23
$^{84}\text{As}-^{88}\text{Rb}_{.955}$	13996.9	3.4				2			JY1	1.0	08Ha23
$^{84}\text{Se}-^{88}\text{Rb}_{.955}$	3160.4	2.1	3160.4	2.1	-0.0	1	100	100	^{84}Se	1.0	08Ha23
$^{82}\text{Se H}_2-^{84}\text{Kr}$	20834	16	20851.9	0.5	0.7	U			R11	1.5	78Di09
$^{84}\text{Kr}-^{86}\text{Kr}_{.977}$	-1168.3	1.0	-1168.8534	0.0022	-0.6	U			MS1	1.0	06Ri15
$^{83}\text{Kr H}-^{84}\text{Kr}$	10465	16	10453.821	0.009	-0.5	U			R11	1.5	78Di09
$^{84}\text{Kr}-^{85}\text{Rb}_{.988}$	-1349.4	1.5	-1350.534	0.004	-0.8	U			MA8	1.0	06De36
$^{84}\text{Rb}-^{85}\text{Rb}_{.988}$	1536	8	1527.0	2.4	-1.1	U			MA2	1.0	94Ot01
$^{84}\text{Sr}-^{85}\text{Rb}_{.988}$	570.9	1.5	570.9	1.3	-0.0	-			MA8	1.0	07Ke09
	572.1	4.3			-0.3	-			SH1	1.0	11Ha08
ave.	571.0	1.4			-0.1	1	89	89	^{84}Sr		average
$^{84}\text{Nb}-^{85}\text{Rb}_{.988}$	21457.45	0.43				2			JY1	1.0	19Vi05

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{84}\text{Kr}-^{80}\text{Se H}_3$	-28505	48	-28499.1	1.0	0.1	U			R11	1.5	78Di09
$^{84}\text{Kr}-^{83}\text{Kr}$	-2628	12	-2628.790	0.009	-0.0	U			M15	2.5	63Ri07
	-2646	30			0.4	U			R11	1.5	78Di09
$^{84}\text{Kr}-^{40}\text{Ar}_2$	-13268.5136	0.0171	-13268.517	0.006	-0.2	1	12	7 ^{40}Ar	FS1	1.0	05Sh38 *
$\text{C}_2 \text{O}_4-^{84}\text{Kr}$	68160.7359	0.0205	68160.750	0.004	0.7	o			FS1	1.0	05Sh38 *
	68160.7516	0.0131			-0.1	1	10	9 ^{84}Kr	FS1	1.0	09Re03
$^{82}\text{Se}(\text{t,p})^{84}\text{Se}$	6016	15	6014.7	2.0	-0.1	U			LAl		74Kn02
$^{84}\text{Sr}(\text{p,t})^{82}\text{Sr}$	-12310	10	-12300	6	1.0	1	36	35 ^{82}Sr	Oak		73Ba56
	-12295	24			-0.2	U			Win		74De31 *
$^{83}\text{Kr}(\text{n},\gamma)^{84}\text{Kr}$	10519.5	1.8	10520.020	0.008	0.3	U					72Ma42 Z
	10520.6	0.3			-1.9	U			Bdn		06Fi.A
$^{84}\text{Sr}(\text{d,t})^{83}\text{Sr}$	-5720	30	-5666	7	1.8	U					70Be24 *
$^{84}\text{As}(\beta^-)^{84}\text{Se}$	7195	200	10094	4	14.5	F			Trs		94Gi07 *
	9120	880			1.1	U					96WaZX
$^{84}\text{Se}(\beta^-)^{84}\text{Br}$	1818	50	1835	26	0.3	1	27	26 ^{84}Br			68Re12 *
	1808	100			0.3	U					70Ei02 *
$^{84}\text{Br}(\beta^-)^{84}\text{Kr}$	4650	30	4656	26	0.2	1	74	74 ^{84}Br			70Ha21 *
$^{84}\text{Br}^m(\beta^-)^{84}\text{Kr}$	4970	100				2					70Ha21 *
$^{84}\text{Rb}(\beta^+)^{84}\text{Kr}$	2679	3	2680.4	2.2	0.5	-					64La03
	2682	5			-0.3	-					71Bo01 *
$^{84}\text{Rb}(\text{n,p})^{84}\text{Kr}$	3450	10	3462.7	2.2	1.3	U			ILL		76An05
$^{84}\text{Rb}(\beta^+)^{84}\text{Kr}$	ave. 2679.8	2.6	2680.4	2.2	0.2	1	73	73 ^{84}Rb			average
$^{84}\text{Rb}(\beta^-)^{84}\text{Sr}$	892	4	890.6	2.3	-0.3	1	34	27 ^{84}Rb			71Bo01 *
$^{84}\text{Y}(\beta^+)^{84}\text{Sr}$	6950	30	6755	4	-6.5	F					70Va.A *
	6750	10			0.5	1	19	18 ^{84}Y			70Re.A *
	6423	135			2.5	U			BNL		81Li12 *
	6408	124			2.8	U			Ors		82De36 *
$^{84}\text{Nb}(\beta^+)^{84}\text{Zr}$	7200	300	10228	6	10.1	B					96Sh27
* $^{84}\text{As}-\text{u}$	Erroneously reported as -75700(300) keV in the publication										08Ha.A **
* $^{84}\text{Kr}-\text{N}_6$	Corrected in reference of same group										09Re03 **
* $^{84}\text{Kr}-^{40}\text{Ar}_2$	Corrected in reference of same group										09Re03 **
* $\text{C}_2 \text{O}_4-^{84}\text{Kr}$	Corrected in reference of same group										09Re03 **
* $^{84}\text{Sr}(\text{p,t})^{82}\text{Sr}$	Original error 12; added systematic error 21 keV										GAu **
* $^{84}\text{Sr}(\text{d,t})^{83}\text{Sr}$	$Q = -5755(30)$ to $9/2^+$ level at 35.47 keV										Ens153 **
* $^{84}\text{As}(\beta^-)^{84}\text{Se}$	F : observed $(\beta^- \text{n})$ decay implies $Q_{\beta^-} > 8681(15)$ keV										93Ru01 **
* $^{84}\text{Se}(\beta^-)^{84}\text{Br}$	$E_{\beta^-} = 1410(50)$ $1400(100)$ respectively, to 1^+ level at 408.2 keV										Ens09a **
* $^{84}\text{Br}(\beta^-)^{84}\text{Kr}$	$E_{\beta^-} = 4626(15), 3810(50), 2700(50)$ to ground state, $881.615\ 2^+$, $1897.784\ 0^+$										Ens09a **
* $^{84}\text{Br}^m(\beta^-)^{84}\text{Kr}$	$E_{\beta^-} = 2200(100)$ to 5^- level at 2770.94 keV										Ens09a **
* $^{84}\text{Rb}(\beta^+)^{84}\text{Kr}$	Original error increased: authors report $E(2^+) = 877.2(1.5)$ while										AHW **
*	$E(2^+) = 881.615(0.003)$ keV; see also $^{74}\text{As}(\beta^+)$										Ens09a **
* $^{84}\text{Rb}(\beta^-)^{84}\text{Sr}$	Originally $891.8(2.0)$, error increased, see $^{84}\text{Rb}(\beta^+)$										AHW **
* $^{84}\text{Y}(\beta^+)^{84}\text{Sr}$	F : possibly additioned with e^+e^-										AHW **
* $^{84}\text{Y}(\beta^+)^{84}\text{Sr}$	$E_{\beta^+} = 1641(10)$ and $2242(17)$ to levels at 4062 and 3511 keV										70Re.A **
* $^{84}\text{Y}(\beta^+)^{84}\text{Sr}$	$Q_{\beta^+} = 6409(170)$, and $6499(135)$ from $^{84}\text{Y}^m$ at 67 keV										00Do10 **
* $^{84}\text{Y}(\beta^+)^{84}\text{Sr}$	$Q_{\beta^+} = 6475(124)$ from $^{84}\text{Y}^m$ at 67 keV										00Do10 **
$^{85}\text{Ga}-\text{u}$	-42667	40				2			TR1	1.0	20Re04
$^{85}\text{Ge}-\text{u}$	-57220	540	-57030	4	0.4	U			OR1	1.0	06Ha62
$^{85}\text{As}-\text{u}$	-68095	225	-67836	3	0.8	o			GT1	1.5	04Ma.A
	-67887	129			0.2	U			GT2	2.5	08Su19
$\text{C}_6 \text{H}_{13}-^{85}\text{Rb}$	189927.6	3.9	189935.679	0.005	0.8	U			M15	2.5	63Ri07
	189930	15			0.3	U			R11	1.5	78Di09
$\text{C}_4 \text{N O H}_7-^{85}\text{Rb}$	140985	18	140974.111	0.005	-0.4	U			R11	1.5	78Di09
$^{85}\text{Rb}-^{39}\text{K}_{2.179}$	-9124.6	2.7	-9126.698	0.012	-0.8	U			MA8	1.0	09Na.A
$^{85}\text{Rb}-^{120}\text{Sn}_{.708}$	-18970.8	2.2	-18969.7	0.7	0.5	U			JY1	1.0	11Ha48
$^{85}\text{Y}-\text{u}$	-83559	31	-83567	20	-0.3	2			GS2	1.0	05Li24 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{85}\text{Zr O}-^{98}\text{Mo}_{1.031}$	13886.7	6.9				2			JY1	1.0	06Ka48
$^{85}\text{Nb O}-^{98}\text{Mo}_{1.031}$	21246	26	21289	4	1.7	U			JY1	1.0	06Ka48 *
$^{85}\text{Ge}-^{88}\text{Rb}_{.966}$	28638.8	4.0				2			JY1	1.0	08Ha23
$^{85}\text{As}-^{88}\text{Rb}_{.966}$	17832.8	3.3				2			JY1	1.0	08Ha23
$^{85}\text{Se}-^{88}\text{Rb}_{.966}$	7929.9	2.8				2			JY1	1.0	08Ha23
$^{85}\text{Br}-^{88}\text{Rb}_{.966}$	1314.9	3.3				2			JY1	1.0	07Ra23
$^{85}\text{Nb}-^{85}\text{Rb}$	17056.1	4.4				2			SH1	1.0	11Ha08 *
$^{85}\text{Mo}-^{85}\text{Rb}$	26471	17				2			SH1	1.0	11Ha08
$\text{C}_6 \text{H}_{14}-^{85}\text{Rb}$	197760.706	0.014	197760.711	0.005	0.3	U			MI2	1.0	99Br47
$^{85}\text{Rb}-\text{C}_6 \text{H}_{12}$	-182110.662	0.024	-182110.647	0.005	0.6	U			MI2	1.0	99Br47
$^{85}\text{Rb}-^{84}\text{Kr}$	300	32	292.009	0.004	-0.2	U			R11	1.5	78Di09
	292.0121	0.0064			-0.5	1	39	$^{34} \text{Rb}$	FS1	1.0	10Mo30
$^{83}\text{Rb}-^{85}\text{Rb}_{.488} \text{ } ^{81}\text{Rb}_{.512}$	-351	22	-339	3	0.2	U			P21	2.5	82Au01 Y
$^{84}\text{Kr}(\text{d,p})^{85}\text{Kr}$	4895	8	4887.7	2.0	-0.9	U			MIT		63Ho.A
$^{85}\text{Rb}(\gamma, \text{n})^{84}\text{Rb}$	-10650	80	-10479.7	2.2	2.1	U			Phi		60Ge01
$^{85}\text{Rb}(\text{p,d})^{84}\text{Rb}$	-8275	6	-8255.1	2.2	3.3	B			Bld		78Sh11
$^{84}\text{Sr}(\text{d,p})^{85}\text{Sr}$	6303	8	6300	3	-0.3	1	14	$^{12} \text{Sr}$			71Mo02
$^{85}\text{Mo}(\epsilon \text{p})^{84}\text{Zr}$	5100	200	6623	17	7.6	B					99Hu05
$^{85}\text{Se}(\beta^-)^{85}\text{Br}$	6185	90	6162	4	-0.3	o			Bwg		87Gr.A
	6182	23			-0.9	U			Bwg		92Gr.A
$^{85}\text{Br}(\beta^-)^{85}\text{Kr}$	2870	19	2905	4	1.8	U			Stu		79Al05
$^{85}\text{Kr}(\beta^-)^{85}\text{Rb}$	687	2				2					70Wo08
$^{85}\text{Sr}(\epsilon)^{85}\text{Rb}$	1007	30	1064.1	2.8	1.9	U					69Mc05
$^{85}\text{Rb}(\text{p,n})^{85}\text{Sr}$	-1890	30	-1846.4	2.8	1.5	U			BNL		58El44
$^{85}\text{Rb}(\text{}^3\text{He,t})^{85}\text{Sr}$	-1083	3	-1082.6	2.8	0.1	1	88	$^{88} \text{Sr}$	Pri		82Ko06
$^{85}\text{Y}(\beta^+)^{85}\text{Sr}$	3255	25	3261	19	0.2	R					63Do07 *
$^{85}\text{Zr}(\beta^+)^{85}\text{Y}$	4693	99	4667	20	-0.3	U			Ors		82De36
$^{85}\text{Nb}(\beta^+)^{85}\text{Zr}$	6000	200	6896	8	4.5	F					88Ku14 *
$^{85}\text{Y}-\text{u}$	$M-A=-77824(28)$ keV for mixture gs+m at 19.68 keV										Nub211 **
$^{85}\text{Nb O}-^{98}\text{Mo}_{1.031}$	$D_M=21292.2(6.9)$ μu for mixture gs+m at 150#80 keV; $M-A=-66274.8(6.6)$ keV										Nub211 **
$^{85}\text{Nb}-^{85}\text{Rb}$	Misprint in publication 1.000200869 (not 1.00200869)										GAu **
$^{85}\text{Y}(\beta^+)^{85}\text{Sr}$	$E_{\beta^+}=1540(20)$ to $3/2^-$ level at 743.25 keV										Ens148 **
*	and $E_{\beta^+}=2240(10)$ from $^{85}\text{Y}^m$ at 19.68 (conflicting $\rightarrow$ outer error used)										Nub211 **
$^{85}\text{Nb}(\beta^+)^{85}\text{Zr}$	F : see discussion of this result in reference										06Ka48 **
*	$Q_{\beta^+}=6100(200)$ in text p.268 and in Table 1										GAu **
$^{86}\text{Ge}-\text{u}$	-54750	540	-53030	470	3.2	B			OR1	1.0	06Ha62
	-53033	188				2			GT3	2.5	16Kn03
$^{86}\text{As}-\text{u}$	-63586	247	-63298	4	0.8	o			GT1	1.5	04Ma.A
	-63189	129			-0.3	U			GT2	2.5	08Su19
$^{86}\text{Se}-\text{u}$	-75702	128	-75688.3	2.7	0.0	U			GT2	2.5	08Su19
$\text{C}_6 \text{H}_{14}-^{86}\text{Kr}$	198936.7	2.7	198939.822	0.004	0.5	U			M15	2.5	63Ri07
	198933	15			0.3	U			R11	1.5	78Di09
$\text{C}_5 \text{N H}_{12}-^{86}\text{Kr}$	186366	20	186363.762	0.004	-0.1	U			R11	1.5	78Di09
$^{86}\text{Kr}-\text{u}$	-89389.271	0.110	-89389.375	0.004	-0.9	U			ST2	1.0	02Bf02
$^{86}\text{Kr}-^{120}\text{Sn}_{.717}$	-19269.6	2.2	-19268.6	0.7	0.5	U			JY1	1.0	11Ha48
$\text{C}_6 \text{H}_{14}-^{86}\text{Sr}$	200264.9	3.6	200289.722	0.006	2.8	U			M15	2.5	63Ri07
$^{86}\text{Y}-\text{u}$	-85019	75	-85114	15	-1.3	U			GS2	1.0	05Li24 *
$^{86}\text{Zr O}-^{98}\text{Mo}_{1.041}$	9692.8	6.9	9686	4	-0.9	1	31	$^{31} \text{Zr}$	JY1	1.0	06Ka48
$^{86}\text{Nb O}-^{98}\text{Mo}_{1.041}$	19171.0	5.9				2			JY1	1.0	06Ka48
$^{86}\text{As}-^{88}\text{Rb}_{.977}$	23346.2	3.7				2			JY1	1.0	08Ha23
$^{86}\text{Se}-^{88}\text{Rb}_{.977}$	10956.4	2.7				2			JY1	1.0	08Ha23
$^{86}\text{Br}-^{88}\text{Rb}_{.977}$	5450.1	3.3				2			JY1	1.0	07Ra23
$^{86}\text{Kr}-^{84}\text{Kr}_{1.024}$	1236.9544	0.0042	1236.9565	0.0023	0.5	1	30	$^{20} \text{Kr}$	FS1	1.0	12Ra34
$^{86}\text{Sr}-^{84}\text{Kr}_{1.024}$	-112.9463	0.0054	-112.944	0.004	0.5	1	59	$^{54} \text{Sr}$	FS1	1.0	12Ra34

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{86}\text{Kr}-^{85}\text{Rb}_{1.012}$	-120.3	3.6	-120.591	0.004	-0.1	U			MA8	1.0	06Ro11
	-119.1	1.6			-0.9	U			MA8	1.0	06De36
	-54	88			-0.8	U			MA9	1.0	10Na13 *
$^{86}\text{Rb}-^{85}\text{Rb}_{1.012}$	441	9	436.23	0.21	-0.5	U			MA2	1.0	94Ot01
$^{86}\text{Sr } ^{19}\text{F}-^{85}\text{Rb}_{1.235}$	16608	13	16603.564	0.006	-0.3	U			MA8	1.0	05Si34
$^{86}\text{Zr}-^{85}\text{Rb}_{1.012}$	5562.7	4.6	5566	4	0.6	1	69	$^{69}\text{ }^{86}\text{Zr}$	SH1	1.0	11Ha08
$^{86}\text{Mo}-^{85}\text{Rb}_{1.012}$	20443.6	4.0	20443	3	-0.2	2			SH1	1.0	11Ha08
	20441.7	5.1			0.2	2			JY1	1.0	19Vi05
$^{86}\text{Sr}-^{86}\text{Kr}$	-1349.8965	0.0060	-1349.900	0.004	-0.6	1	49	$^{46}\text{ }^{86}\text{Sr}$	FS1	1.0	12Ra34
$^{86}\text{Kr}-^{85}\text{Rb}$	-1206	42	-1179.111	0.004	0.4	U			R11	1.5	78Di09
	-1179.1083	0.0071			-0.4	-			FS1	1.0	10Mo30 *
	-1179.1109	0.0059			-0.1	-			FS1	1.0	10Mo30 *
ave.	-1179.110	0.005			-0.3	1	69	$^{66}\text{ }^{85}\text{Rb}$			average
$^{86}\text{Kr}-\text{N}_6$	-107833.3986	0.0074	-107833.401	0.004	-0.3	1	28	$^{27}\text{ }^{86}\text{Kr}$	FS1	1.0	09Re03
$\text{C}_2\text{O}_4-^{86}\text{Kr}$	69047.8337	0.0155	69047.852	0.004	1.2	o			FS1	1.0	05Sh38 *
	69047.8440	0.0113			0.7	1	12	$^{12}\text{ }^{86}\text{Kr}$	FS1	1.0	09Re03
$^{86}\text{Kr}-^{84}\text{Kr}$	-908	32	-887.1024	0.0023	0.4	U			R11	1.5	78Di09
	-887.1041	0.0125			0.1	o			FS1	1.0	05Sh38 *
	-887.1080	0.0069			0.8	-			FS1	1.0	09Re03 *
	-887.0954	0.0060			-1.2	-			FS1	1.0	10Mo30 *
ave.	-887.101	0.005			-0.3	1	25	$^{15}\text{ }^{84}\text{Kr}$			average
$^{86}\text{Sr}(\text{p},\text{t})^{84}\text{Sr}$	-11535	10	-11534.4	1.2	0.1	U			Oak		73Ba56
$^{85}\text{Rb}(\text{n},\gamma)^{86}\text{Rb}$	8651.1	1.0	8650.98	0.20	-0.1	U					69Da15 Z
	8651.3	1.5			-0.2	U					70Or.A
	8650.98	0.20				2			Bdn		06Fi.A
$^{85}\text{Rb}(\text{d},\text{p})^{86}\text{Rb}$	6433	10	6426.41	0.20	-0.7	U			Tal		69Da15
$^{86}\text{Se}(\beta^-)^{86}\text{Br}$	5095	100	5129	4	0.3	o			Bwg		87Gr.A
	5099	11			2.7	C			Bwg		92Gr.A
$^{86}\text{Br}(\beta^-)^{86}\text{Kr}$	7620	60	7633	3	0.2	U			Stu		79Al05
	7626	11			0.7	U			Bwg		92Gr.A
$^{86}\text{Rb}(\beta^-)^{86}\text{Sr}$	1774	5	1776.10	0.20	0.4	U					64Da16
	1770	3			2.0	U					66An10
	1779.2	2.5			-1.2	U					75Be21
	1775	3			0.4	U					75Ra09
$^{86}\text{Y}(\beta^+)^{86}\text{Sr}$	5220	20	5240	14	1.0	2					62Ya01 *
	5260	20			-1.0	2					65Va02 *
$^{86}\text{Nb}(\beta^+)^{86}\text{Zr}$	7978	80	8835	7	10.7	F					82De43 *
$^{86}\text{Mo}(\beta^+)^{86}\text{Nb}$	5019	430	5023	6	0.0	U					94Sh07 *
$^{86}\text{Y}-\text{u}$	$M-A=-79086(29)$ keV for mixture gs+m at 218.21 keV										Nub211 **
$^{86}\text{Kr}-^{85}\text{Rb}_{1.012}$	Typo in original paper, ratio should read 1.011 763 90(1 03)										GAu **
$^{86}\text{Kr}-^{85}\text{Rb}$	Different charge states : 3^+ and 2^+										10Mo30 **
$\text{C}_2\text{O}_4-^{86}\text{Kr}$	Corrected in reference of same group										09Re03 **
$^{86}\text{Kr}-^{84}\text{Kr}$	Corrected in reference of same group										09Re03 **
$^{86}\text{Kr}-^{84}\text{Kr}$	Different charge states in these two results: 3^+ and 2^+										10Mo30 **
$^{86}\text{Y}(\beta^+)^{86}\text{Sr}$	$E_{\beta^+}=2019(20)$ 1960(20) respectively, to 2229.81 4^+ level, and other E_{β^+}										Ens14c **
$^{86}\text{Nb}(\beta^+)^{86}\text{Zr}$	F : see discussion of this result in reference										06Ka48 **
$^{86}\text{Mo}(\beta^+)^{86}\text{Nb}$	$E_{\beta^+}=3900(400)$ to (1^+) level at 97.1 keV										94Sh07 **
$^{87}\text{Se}-\text{u}$	-71357	128	-71311.4	2.4	0.1	U			GT2	2.5	08Su19
$^{87}\text{Kr}-\text{u}$	-86622	30	-86645.24	0.26	-0.8	U			GS2	1.0	05Li24
$\text{C}_4\text{H}_7\text{O}_2-^{87}\text{Rb}$	135417.8	2.7	135423.933	0.006	0.9	U			M15	2.5	63Ri07
$\text{C}_5^{13}\text{C H}_{14}-^{87}\text{Rb}$	203767	15	203724.753	0.006	-1.9	U			R11	1.5	78Di09
$\text{C}_4\text{O N H}_9-^{87}\text{Rb}$	159277	15	159233.381	0.006	-1.9	U			R11	1.5	78Di09
$\text{C}_3^{13}\text{C O N H}_8-^{87}\text{Rb}$	154809	25	154763.185	0.006	-1.2	U			R11	1.5	78Di09
$\text{C}_4\text{H}_7\text{O}_2-^{87}\text{Sr}$	135722.2	3.5	135726.967	0.005	0.5	U			M15	2.5	63Ri07

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{87}\text{Y}-\text{u}$	-89153	30	-89123.9	1.2	1.0	U			GS2	1.0	03Li.A *
$^{87}\text{Zr}-\text{u}$	-85222	30	-85183	4	1.3	U			GS2	1.0	05Li24
$^{87}\text{Zr O}-^{97}\text{Mo}_{1.062}$	9543.3	5.2	9542	4	-0.2	1	73	73 ^{87}Zr	JY1	1.0	08We10
$^{87}\text{Nb}_{1.069}-\text{C}_7\text{H}_9$	-155224	30	-155205	8	0.6	U			CP1	1.0	11Fa10
$^{87}\text{Nb O}-^{98}\text{Mo}_{1.051}$	15027.9	7.3				2			JY1	1.0	06Ka48 *
$^{87}\text{Mo}_{1.069}-\text{C}_7\text{H}_9$	-147186.1	4.8	-147184	3	0.5	1	47	47 ^{87}Mo	CP1	1.0	11Fa10
$^{87}\text{Sr}-^{84}\text{Kr}_{1.036}$	565.8516	0.0059	565.850	0.004	-0.2	1	47	41 ^{87}Sr	FS1	1.0	12Ra34
$^{87}\text{Kr}-^{85}\text{Rb}_{1.024}$	3683.0	2.9	3682.07	0.26	-0.3	U			MA8	1.0	06De36
	3684.1	4.7			-0.4	U			MA8	1.0	10Na13
$^{87}\text{Rb}-^{85}\text{Rb}_{1.024}$	-490	9	-492.156	0.007	-0.2	U			MA2	1.0	94Ot01
	-493.0	2.7			0.3	U			MA8	1.0	06De36
	-492.33	0.80			0.2	U			MA8	1.0	07Ke09
	-492.4	1.4			0.2	U			MA8	1.0	09Na.A
	-492.04	0.87			-0.1	U			MA8	1.0	11He10
	-492.166	0.056			0.2	U			JY2	1.0	18Ne09 *
$^{87}\text{Sr}-^{85}\text{Rb}_{1.024}$	-780	9	-795.191	0.005	-1.7	U			MA2	1.0	94Ot01
$^{87}\text{Mo}-^{85}\text{Rb}_{1.024}$	18525.6	4.2	18524	3	-0.5	1	53	53 ^{87}Mo	SH1	1.0	11Ha08
$^{87}\text{Tc}-^{85}\text{Rb}_{1.024}$	28394.5	4.5				2			SH1	1.0	11Ha08 *
$^{87}\text{As}-^{88}\text{Rb}_{.989}$	28000.6	3.2				2			JY1	1.0	08Ha23
$^{87}\text{Se}-^{88}\text{Rb}_{.989}$	16397.5	2.4				2			JY1	1.0	08Ha23
$^{87}\text{Br}-^{88}\text{Rb}_{.989}$	8382.9	3.4				2			JY1	1.0	07Ra23
$^{86}\text{Kr H}-^{87}\text{Rb}$	9309	16	9255.127	0.006	-2.2	U			R11	1.5	78Di09
$^{87}\text{Sr}-^{86}\text{Kr}_{1.012}$	-660.4616	0.0050	-660.461	0.004	0.2	1	62	59 ^{87}Sr	FS1	1.0	12Ra34
$\text{C}_6\text{H}_{16}-^{87}\text{Rb}$	216019.966	0.023	216019.981	0.006	0.7	U			MI2	1.0	99Br47
$^{87}\text{Rb}-\text{C}_6\text{H}_{14}$	-200369.931	0.015	-200369.917	0.006	0.9	1	19	19 ^{87}Rb	MI2	1.0	99Br47
$^{87}\text{Rb}-^{86}\text{Kr}$	-1477	30	-1430.096	0.006	1.0	U			R11	1.5	78Di09
	-1430.0932	0.0059			-0.4	1	87	81 ^{87}Rb	FS1	1.0	10Mo30
$^{87}\text{Sr}-^{86}\text{Sr}$	-382	12	-383.230	0.006	-0.0	U			M15	2.5	63Ri07
$^{87}\text{Rb}-^{85}\text{Rb}$	-2620	35	-2609.207	0.007	0.2	U			R11	1.5	78Di09
$^{85}\text{Rb}-^{87}\text{Rb}_{.489}$ $^{83}\text{Rb}_{.512}$	-310	30	-314.8	1.2	-0.1	U			P21	2.5	82Au01
$^{84}\text{Rb}-^{87}\text{Rb}_{.241}$ $^{83}\text{Rb}_{.759}$	850	72	643.7	2.8	-1.1	U			P21	2.5	82Au01 *
$^{87}\text{Sr}(\text{p,t})^{85}\text{Sr}$	-11440	10	-11437.6	2.8	0.2	U			Oak		73Ba56
$^{87}\text{Br}(\beta^- \text{n})^{86}\text{Kr}$	1335	25	1303	3	-1.3	U					84Kr.B
$^{86}\text{Kr}(\text{n},\gamma)^{87}\text{Kr}$	5515.04	0.6	5515.17	0.25	0.2	2					77Je03 Z
	5515.20	0.27			-0.1	2			Bdn		06Fi.A
$^{86}\text{Kr}(\text{d,p})^{87}\text{Kr}$	3286	8	3290.61	0.25	0.6	U			MIT		63Ho.A
$^{87}\text{Rb}(\gamma,\text{n})^{86}\text{Rb}$	-9990	70	-9922.12	0.20	1.0	U			Phi		60Ge01
$^{87}\text{Rb}(\text{d,t})^{86}\text{Rb}$	-3659	15	-3664.89	0.20	-0.4	U			Tal		69Da15
$^{86}\text{Sr}(\text{n},\gamma)^{87}\text{Sr}$	8428.12	0.17	8428.295	0.005	1.0	U			ILn		86Wi16 Z
	8428.17	0.17			0.7	U			Bdn		06Fi.A
$^{86}\text{Sr}(\text{d,p})^{87}\text{Sr}$	6203	8	6203.728	0.005	0.1	U					71Mo02
$^{86}\text{Sr}(\text{p},\gamma)^{87}\text{Y}$	5785.4	3.3	5784.3	1.1	-0.3	R					71Um03
$^{86}\text{Sr}(\text{He,d})^{87}\text{Y}$	346	15	290.8	1.1	-3.7	B			ANL		71Ma11
$^{87}\text{Mo}(\text{ep})^{86}\text{Zr}$	3700	300	3795	5	0.3	U					83Ha06
$^{87}\text{Se}(\beta^-)^{87}\text{Br}$	7250	150	7466	4	1.4	o			Bwg		87Gr.A
	7275	35			5.4	B			Bwg		92Gr.A
$^{87}\text{Br}(\beta^-)^{87}\text{Kr}$	6830	120	6818	3	-0.1	U			Stu		79Al05
	6750	150			0.5	o			Bwg		87Gr.B
	6855	25			-1.5	U			Bwg		92Gr.A
$^{87}\text{Kr}(\beta^-)^{87}\text{Rb}$	3888	7	3888.27	0.25	0.0	U					73Wo01
$^{87}\text{Rb}(\beta^-)^{87}\text{Sr}$	272	3	282.275	0.006	3.4	B					59Fl40
	274	3			2.8	B					61Be41
$^{87}\text{Rb}(\text{He,t})^{87}\text{Sr}-^{81}\text{Br}(\text{He,t})^{81}\text{Kr}$	564.0	1.5	563.1	0.5	-0.6	1	10	9 ^{81}Kr	Pri		82Ko06
$^{87}\text{Y}(\beta^+)^{87}\text{Sr}$	2190	50	1861.7	1.1	-6.6	B					67Mi13 *
	1791	40			1.8	U					69Zo04 *
$^{87}\text{Sr}(\text{p,n})^{87}\text{Y}$	-2644.2	1.2	-2644.0	1.1	0.1	2					71Um03 Z

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{87}\text{Zr}(\beta^+)^{87}\text{Y}$	3663	40	3671	4	0.2	U					65Ba48 *
$^{87}\text{Nb}(\beta^+)^{87}\text{Zr}$	5165	60	5473	8	5.1	B					82De43 *
$^{87}\text{Mo}(\beta^+)^{87}\text{Nb}$	6382	308	6990	7	2.0	U					82De43 *
	6589	300			1.3	U					91Mi15 *
$^{87}\text{Y}-u$	$M-A=-82665(28)$ keV for $^{87}\text{Y}^m$ at 380.82 keV										Nub211 **
$^{87}\text{Nb}-^{98}\text{Mo}_{1.051}$	$D_M=15030.0(6.9)$ μu for mixture gs+m at 3.9(0.1) keV; $M-A=-73870.2(6.7)$ keV										Nub211 **
$^{87}\text{Rb}-^{85}\text{Rb}_{1.024}$	from $^{85}\text{Rb}^+ / ^{87}\text{Rb}^+ = 1.023523278677(654)$, alt.in reference 1.023523278629(662)										18Ne09 **
$^{87}\text{Tc}-^{85}\text{Rb}_{1.024}$	Most probably the high-spin isomer										11Ha08 **
$^{84}\text{Rb}-^{87}\text{Rb}_{.241}$ $^{83}\text{Rb}_{.759}$	$D_M=1080(40)$ keV corrected $-230(60)$ for mixture gs+m at 463.59 keV										Nub211 **
$^{87}\text{Y}(\beta^+)^{87}\text{Sr}$	$E_{\beta^+}=780(50)$ to $^{87}\text{Sr}^m$ at 388.82 keV										Nub211 **
$^{87}\text{Y}(\beta^+)^{87}\text{Sr}$	$E_{\beta^+}=1150(40)$ from $^{87}\text{Y}^m$ at 380.82 keV										Nub211 **
$^{87}\text{Zr}(\beta^+)^{87}\text{Y}$	$E_{\beta^+}=2260(40)$ to $^{87}\text{Y}^m$ at 380.82 keV										Nub211 **
$^{87}\text{Nb}(\beta^+)^{87}\text{Zr}$	$Q_{\beta^+}=5169(60)$ from $^{87}\text{Nb}^m$ at 3.9(0.1) keV										Nub211 **
$^{87}\text{Mo}(\beta^+)^{87}\text{Nb}$	$Q_{\beta^+}=6378(308)$ to $^{87}\text{Nb}^m$ at 3.9(0.1) keV										Nub211 **
$^{87}\text{Mo}(\beta^+)^{87}\text{Nb}$	$E_{\beta^+}=5300(300)$ to $(7/2)^+$ level at 266.9 keV										Ens15a **
$^{88}\text{Se}-u$	-68555	129	-68583	4	-0.1	U			GT2	2.5	08Su19
$^{88}\text{Br}-u$	-75832	100	-75917	3	-0.3	o			GT2	2.5	08Kn.A
	-75823	129			-0.3	U			GT2	2.5	08Su19
$\text{C}_4 \text{H}_8 \text{O}_2 - ^{88}\text{Sr}$	146789.1	4.7	146817.240	0.006	2.4	U			M15	2.5	63Ri07
$^{88}\text{Y}-u$	-90500	31	-90498.7	1.6	0.0	U			GS2	1.0	05Li24
$^{88}\text{Zr}-^{98}\text{Mo}_{1.061}$	5502.3	6.9	5502	6	-0.0	1	71	71 ^{88}Zr	JY1	1.0	06Ka48
$^{88}\text{Nb}-^{98}\text{Mo}_{1.061}$	13458	76	13510	60	0.7	1	67	67 ^{88}Nb	JY1	1.0	06Ka48 *
$^{88}\text{Sr}-^{84}\text{Kr}_{1.048}$	-1637.3606	0.0069	-1637.366	0.005	-0.8	1	46	42 ^{88}Sr	FS1	1.0	12Ra34
$^{88}\text{Kr}-^{85}\text{Rb}_{1.035}$	5745.5	2.8				2			MA8	1.0	06De36
$^{88}\text{Rb}-^{85}\text{Rb}_{1.035}$	2615	9	2613.21	0.17	-0.2	U			MA4	1.0	02Ra23
$^{88}\text{Sr}-^{85}\text{Rb}_{1.035}$	-3108	20	-3090.126	0.006	0.9	U			MA8	1.0	07Ke09
	-3088	11			-0.2	U			MA8	1.0	05Si34
$^{88}\text{Mo}-^{85}\text{Rb}_{1.035}$	13265.4	4.1				2			JY1	1.0	08We10
$^{88}\text{Tc}-^{85}\text{Rb}_{1.035}$	25045	23	25092	4	2.0	U			JY1	1.0	08We10 *
$^{88}\text{Sr}-^{86}\text{Kr}_{1.023}$	-2942.4188	0.0059	-2942.415	0.005	0.7	1	61	58 ^{88}Sr	FS1	1.0	12Ra34
$^{88}\text{Tc}-^{87}\text{Rb}_{1.011}$	25612.7	4.4				2			JY1	1.0	19Vi05
$^{88}\text{Se}-^{88}\text{Rb}$	20101.9	3.6				2			JY1	1.0	08Ha23
$^{88}\text{Br}-^{88}\text{Rb}$	12767.7	3.4				2			JY1	1.0	07Ra23
$^{88}\text{Tc}^m - ^{88}\text{Tc}$	75.6	3.5				3			JY1	1.0	19Vi05
$^{88}\text{Sr}-^{87}\text{Sr}$	-3260	12	-3265.241	0.006	-0.2	U			M15	2.5	63Ri07
$^{86}\text{Kr}(t,p)^{88}\text{Kr}$	4091	15	4086.5	2.6	-0.3	U			LAI		76Fi02
$^{88}\text{Sr}(p,t)^{86}\text{Sr}$	-11060	10	-11059.369	0.006	0.1	U			Oak		73Ba56
$^{87}\text{Rb}(n,\gamma)^{88}\text{Rb}$	6082.52	0.16	6082.52	0.16	0.0	1	99	99 ^{88}Rb	Bdn		06Fi.A
$^{87}\text{Rb}(d,p)^{88}\text{Rb}$	3858	15	3857.95	0.16	-0.0	U			Oak		71Ra17
	3837	20			1.0	U					71To05
$^{87}\text{Sr}(n,\gamma)^{88}\text{Sr}$	11112.63	0.22	11112.870	0.006	1.1	U			ILn		87Wi15 Z
	11112.64	0.22			1.0	U			Bdn		06Fi.A
$^{87}\text{Sr}(d,p)^{88}\text{Sr}$	8865	5	8888.304	0.005	4.7	B			MIT		68Co20
$^{88}\text{Se}(\beta^-)^{88}\text{Br}$	6854	31	6832	5	-0.7	U			Bwg		92Gr.A
$^{88}\text{Br}(\beta^-)^{88}\text{Kr}$	8970	130	8975	4	0.0	U			Stu		79Al05
	8880	200			0.5	o			Bwg		87Gr.B
	8960	36			0.4	U			Bwg		92Gr.A
$^{88}\text{Kr}(\beta^-)^{88}\text{Rb}$	2930	30	2917.7	2.6	-0.4	U			Trs		78Wo15
$^{88}\text{Rb}(\beta^-)^{88}\text{Sr}$	5310	60	5312.62	0.16	0.0	U			Trs		78Wo15
	5318	9			-0.6	U			Gsn		80De02 *
	5313	5			-0.1	U			Trs		82Br23
$^{88}\text{Y}(\beta^+)^{88}\text{Sr}$	3622.6	1.5				2					79An36 *
$^{88}\text{Sr}(p,n)^{88}\text{Y}$	-4402	6	-4404.9	1.5	-0.5	U			Bar		61Sh23
	-4406	10			0.1	U			Tal		62Ne08

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{88}\text{Nb}(\beta^+)^{88}\text{Zr}$	7550	100	7460	60	-0.9	1	34	33	^{88}Nb		84Ox01
$^{88}\text{Nb}^m(\beta^+)^{88}\text{Zr}$	7590	100				2					84Ox01
$^{88}\text{Tc}(\beta^+)^{88}\text{Mo}$	8600	1300	11016	6	1.9	U					96Od01
	7800	600			5.4	B					96Sh27
$^{88}\text{Nb O}-^{98}\text{Mo}_{1.061}$	$D_M=13527.5(6.9) \mu\text{u}$ for mixture gs+m at 130(120) keV; $M-A=-76150.9(6.7) \text{ keV}$										Nub211 **
$^{88}\text{Tc}-^{85}\text{Rb}_{1.035}$	Most probably the low-spin isomer										08We10 **
$^{88}\text{Tc}-^{85}\text{Rb}_{1.035}$	$D_M=25082.4(4.1) \mu\text{u}$ for mixture gs+m at 70(3) keV; $M-A=-61679.1(3.8) \text{ keV}$										Nub211 **
$^{88}\text{Rb}(\beta^-)^{88}\text{Sr}$	Original error 4 corrected in reference										94Ha.A **
$^{88}\text{Y}(\beta^+)^{88}\text{Sr}$	$E_{\beta^+}=764.6(1.5) \text{ to } 2^+$ level at 1836.090 keV										Ens141 **
$^{89}\text{Se}-\text{u}$	-63285	225	-63331	4	-0.1	o			GT1	1.5	04Ma.A
	-63291	129			-0.1	U			GT2	2.5	08Su19
$\text{C}_7 \text{ H}_5-^{89}\text{Y}$	133247.0	3.4	133287.0	0.4	4.7	B			M15	2.5	63Ri07
$^{89}\text{Nb}-\text{u}$	-86588	34	-86555	25	1.0	-			GS2	1.0	05Li24 *
ave.	-86582	29			0.9	1	78	78	^{89}Nb		average
$^{89}\text{Kr}-^{85}\text{Rb}_{1.047}$	10191.6	2.3				2			MA8	1.0	06De36
$^{89}\text{Rb}-^{85}\text{Rb}_{1.047}$	4628	9	4634	6	0.7	1	42	42	^{89}Rb	1.0	02Ra23
$^{89}\text{Y}-^{85}\text{Rb}_{1.047}$	-1806.32	0.52	-1805.7	0.4	1.2	1	49	49	^{89}Y	1.0	19Sa39
$^{89}\text{Mo}-^{85}\text{Rb}_{1.047}$	11824.3	4.2				2			JY1	1.0	08We10
$^{89}\text{Tc}-^{85}\text{Rb}_{1.047}$	20007	17	20005	4	-0.1	U			SH1	1.0	08We10
	20004.8	4.1				2			JY1	1.0	08We10
$^{89}\text{Ru}-^{85}\text{Rb}_{1.047}$	29694	26				2			JY1	1.0	19Vi05
$^{89}\text{Y}-^{87}\text{Rb}_{1.023}$	-1252.96	0.51	-1253.5	0.4	-1.1	1	51	51	^{89}Y	1.0	19Sa39
$^{89}\text{Se}-^{88}\text{Rb}_{1.011}$	26329.0	4.0				2			JY1	1.0	08Ha23
$^{89}\text{Br}-^{88}\text{Rb}_{1.011}$	16364.5	3.5				2			JY1	1.0	07Ra23
$^{88}\text{Rb}-^{89}\text{Rb}_{.494} \text{ } ^{87}\text{Rb}_{.506}$	545	23	563.4	2.7	0.3	U			P21	2.5	82Au01
$^{89}\text{Y}(\text{d},\alpha)^{87}\text{Sr}$	7889	15	7879.7	0.3	-0.6	U			Mtr		72Br13
$^{88}\text{Sr}(\text{n},\gamma)^{89}\text{Sr}$	6358.70	0.13	6358.72	0.09	0.1	-			ILn		89Wi05 Z
	6358.73	0.13			-0.1	-			Bdn		06Fi.A
$^{88}\text{Sr}(\text{d},\text{p})^{89}\text{Sr}$	4133	5	4134.15	0.09	0.2	U			MIT		67Sp09
$^{88}\text{Sr}(\text{n},\gamma)^{89}\text{Sr}$	6358.71	0.09	6358.72	0.09	0.0	1	100	100	^{89}Sr		average
$^{88}\text{Sr}(\text{p},\gamma)^{89}\text{Y}$	7078	4	7078.5	0.3	0.1	U					75Be.B Z
$^{89}\text{Y}(\gamma,\text{n})^{88}\text{Y}$	-11540	40	-11483.5	1.5	1.4	U			Phi		63Ge02
$^{89}\text{Br}(\beta^-)^{89}\text{Kr}$	8140	140	8262	4	0.9	U			Stu		81Ho17
	8120	120			1.2	o			Bwg		87Gr.B
	8155	30			3.6	C			Bwg		92Gr.A
$^{89}\text{Kr}(\beta^-)^{89}\text{Rb}$	5150	30	5177	6	0.9	U					67Ki01
	5191	60			-0.2	U			Trs		78Wo15 *
	5140	120			0.3	U			Stu		81Ho17 *
$^{89}\text{Rb}(\beta^-)^{89}\text{Sr}$	4486	12	4497	5	0.9	-					66Ki06
	4491	15			0.4	o			Gsn		80Bl.A
	4510	9			-1.5	-			Gsn		80De02 *
ave.	4501	7			-0.7	1	57	57	^{89}Rb		average
$^{89}\text{Sr}(\beta^-)^{89}\text{Y}$	1463	5	1502.2	0.4	7.8	B					49La06
	1488	4			3.5	B					70Wo05
$^{89}\text{Zr}(\beta^+)^{89}\text{Y}$	2841	10	2833.2	2.8	-0.8	U					51Hy24 *
	2832	10			0.1	U					53Sh48 *
	2828	7			0.7	-					60Ha26 *
$^{89}\text{Y}(\text{p},\text{n})^{89}\text{Zr}$	-3612.8	4.	-3615.6	2.8	-0.7	-			Tkm		63Ok01 Z
	-3619.4	6.			0.6	-			Oak		64Jo11 Z
$^{89}\text{Zr}(\beta^+)^{89}\text{Y}$	2832	3	2833.2	2.8	0.5	1	85	84	^{89}Zr		average
$^{89}\text{Nb}(\beta^+)^{89}\text{Zr}$	3870	100	4252	24	3.8	B					55Ma13
	4340	50			-1.8	1	23	22	^{89}Nb		74Vo08
$^{89}\text{Mo}(\beta^+)^{89}\text{Nb}$	5970	300	5611	24	-1.2	U					64Bu12

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{89}\text{Tc}(\beta^+)^{89}\text{Mo}$	7510	210	7620	5	0.5	U					91He04 *
$^{89}\text{Nb}-u$	$M-A=-80656(28)$ keV for mixture gs+m at 0#30 keV										Nub211 **
$^{89}\text{Kr}(\beta^-)^{89}\text{Rb}$	$E_{\beta^-}=4970(60)$ to 220.948 level										GAu **
$^{89}\text{Kr}(\beta^-)^{89}\text{Rb}$	Splitting Table 3a into 3 groups yields 4610(120) 4867(152) 5135(123) keV										GAu **
$^{89}\text{Rb}(\beta^-)^{89}\text{Sr}$	Original error 8 corrected in reference										94Ha.A **
$^{89}\text{Zr}(\beta^+)^{89}\text{Y}$	$E_{\beta^+}=910(10)$ 901(10) 897(7) respectively, to $^{89}\text{Y}^m$ at 908.97 keV										Nub211 **
$^{89}\text{Tc}(\beta^+)^{89}\text{Mo}$	$E_{\beta^+}=6370(210)$ to 118.8 level; no Fermi-Kurie plot										91He04 **
$^{90}\text{Se}-u$	-59904	236				2			GT1	1.5	04Ma.A
$^{90}\text{Rb}-u$	-85420	210	-85202	7	1.0	U			TG2	1.0	20Gr.1
$\text{C}_4\text{H}_{10}\text{O}_2-^{90}\text{Zr}$	163377	6	163380.80	0.13	0.3	U			M15	2.5	63Ri07
$^{90}\text{Zr}-u$	-95301.36	0.23	-95301.24	0.13	0.5	1	30	30 ^{90}Zr	MS1	1.0	15Gu09
$^{90}\text{Nb}-u$	-88872	50	-88741	4	2.6	U			GS2	1.0	05Li24 *
$^{90}\text{Mo}-\text{C}_7\text{H}_6$	-133018.9	4.7	-133019	4	-0.0	1	63	63 ^{90}Mo	CP1	1.0	11Fa10
$^{90}\text{Mo}_{1.033}-\text{C}_7\text{H}_9$	-159318	23	-159334	4	-0.7	U			CP1	1.0	11Fa10
$^{90}\text{Tc}-\text{C}_7\text{H}_6$	-122880.3	6.7	-122876.3	1.1	0.6	U			CP1	1.0	11Fa10
$^{90}\text{Tc}_{1.033}-\text{C}_7\text{H}_9$	-148835	22	-148856.9	1.1	-1.0	U			CP1	1.0	11Fa10
	-148854.2	8.5			-0.3	U			CP1	1.0	11Fa10
$^{90}\text{Ru}_{1.033}-\text{C}_7\text{H}_9$	-142382	11	-142380	4	0.2	1	14	14 ^{90}Ru	CP1	1.0	11Fa10
$^{90}\text{Kr}-^{85}\text{Rb}_{1.059}$	12942.6	2.0				2			MA8	1.0	06De36
$^{90}\text{Rb}-^{85}\text{Rb}_{1.059}$	8211	9	8212	7	0.1	1	59	59 ^{90}Rb	MA4	1.0	02Ra23 *
$^{90}\text{Tc}-^{85}\text{Rb}_{1.059}$	17489.2	8.0	17488.6	1.1	-0.1	U			SH1	1.0	08We10
	17489.8	4.2			-0.3	U			JY1	1.0	08We10
$^{90}\text{Ru}-^{85}\text{Rb}_{1.059}$	23775	11	23759	4	-1.5	-			SH1	1.0	08We10
	23756.6	4.7			0.5	-			JY1	1.0	08We10
ave.	23759	4			-0.1	1	86	86 ^{90}Ru			average
$^{90}\text{Tc}-^{86}\text{Kr}_{1.047}$	17664.6	1.1				2			JY1	1.0	12Ka12
$^{90}\text{Tc}^m-^{86}\text{Kr}_{1.047}$	17819.2	1.4				2			JY1	1.0	12Ka12
$^{90}\text{Zr}-^{87}\text{Rb}_{1.034}$	-1393.79	0.16	-1393.91	0.13	-0.7	1	62	62 ^{90}Zr	MS1	1.0	15Gu09
$^{90}\text{Br}-^{88}\text{Rb}_{1.023}$	22017.0	3.6				2			JY1	1.0	07Ra23
$^{89}\text{Rb}-^{90}\text{Rb}_{791}^{85}\text{Rb}_{209}$	-1826	24	-1817	12	0.1	U			P21	2.5	82Au01
$^{90}\text{Zr}(\alpha,^8\text{He})^{86}\text{Zr}$	-40136	30	-39988	4	4.9	B			INS		90Ka01
$^{90}\text{Zr}(^3\text{He},^6\text{He})^{87}\text{Zr}$	-12083	8	-12086	4	-0.4	1	27	27 ^{87}Zr	MSU		78Pa11
$^{90}\text{Zr}(p,t)^{88}\text{Zr}$	-12805	10	-12805	5	0.0	1	29	29 ^{88}Zr	Oak		71Ba43
$^{89}\text{Y}(n,\gamma)^{90}\text{Y}$	6857.1	1.0	6857.03	0.10	-0.1	U			ORn		81Ra07
	6857.26	0.30			-0.8	-					83De17
	6856.98	0.17			0.3	-			ILn		93Mi04 Z
	6857.01	0.14			0.1	-			Bdn		06Fi.A
$^{89}\text{Y}(d,p)^{90}\text{Y}$	4635	5	4632.46	0.10	-0.5	U					64Wa14
$^{89}\text{Y}(n,\gamma)^{90}\text{Y}$	6857.03	0.10	6857.03	0.10	0.0	1	100	100 ^{90}Y			average
$^{89}\text{Y}(p,\gamma)^{90}\text{Zr}$	8351	4	8350.3	0.4	-0.2	U					75Be.B
$^{90}\text{Zr}(\gamma,n)^{89}\text{Zr}$	-11940	50	-11965.9	2.8	-0.5	U			Phi		63Ge02
$^{90}\text{Zr}(p,d)^{89}\text{Zr}$	-9728	10	-9741.3	2.8	-1.3	U			Oak		71Ba43
$^{90}\text{Zr}(d,t)^{89}\text{Zr}$	-5719.2	7.1	-5708.7	2.8	1.5	1	15	15 ^{89}Zr	SPa		79Bo37
$^{90}\text{Zr}(^3\text{He},\alpha)^{89}\text{Zr}$	8580	50	8611.7	2.8	0.6	U			Phi		67Fo04
$^{90}\text{Br}(\beta^-)^{90}\text{Kr}$	9800	400	10959	4	2.9	U			Stu		81Ho17
	10280	110			6.2	C			Bwg		87Gr.B
	10350	75			8.1	C			Bwg		92Gr.A
$^{90}\text{Kr}(\beta^-)^{90}\text{Rb}$	4410	30	4406	7	-0.1	U					70Ma11
	4390	40			0.4	U			Trs		78Wo15
	4380	25			1.1	U			Bwg		87Gr.A
$^{90}\text{Rb}^*(\text{IT})^{90}\text{Rb}$	71	12				2					82Au01
$^{90}\text{Rb}(\beta^-)^{90}\text{Sr}$	6550	60	6585	6	0.6	U			Trs		78Wo15
	6560	150			0.2	U			Bwg		78St02
	6585	15			0.0	o			Gsn		80Bl.A
	6578	15			0.5	o			Gsn		80De02
	6587	10			-0.2	1	42	41 ^{90}Rb	Gsn		92Pr03
$^{90}\text{Sr}(\beta^-)^{90}\text{Y}$	546	2	546.0	1.4	-0.0	-					64Da16
	546	2			-0.0	-					83Ha35
ave.	546.0	1.4			-0.0	1	99	99 ^{90}Sr			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{90}\text{Y}(\beta^-)^{90}\text{Zr}$	2271	2	2275.6	0.4	2.3	B					61Ni02
	2284	5			-1.7	U					64Da16
	2273	5			0.5	U					64La13
	2280	5			-0.9	U					66Ri01
	2278	8			-0.3	U			Gsn		80Bl.A
	2279.5	2.9			-1.3	U					83Ha35
$^{90}\text{Nb}(\beta^+)^{90}\text{Zr}$	6111	4	6111	3	0.0	1	69	69 ^{90}Nb			68Pe01 *
$^{90}\text{Mo}(\beta^+)^{90}\text{Nb}$	2489	4	2489	3	0.0	1	69	37 ^{90}Mo			66Pe10 *
$^{90}\text{Tc}(\beta^+)^{90}\text{Mo}$	8920	410	9448	4	1.3	U					74Ia01 *
	9270	300			0.6	U					81Ox01 *
	8726	300			2.4	U					81Ox01 *
$^{90}\text{Rh}(\epsilon)^{90}\text{Ru}$	13410	1330	13250#	200#	-0.1	D					19Pa16 *
$^{90}\text{Nb}-u$	$M-A=-82721(29)$ keV for mixture gs+n at 124.67 keV										Nub211 **
$^{90}\text{Rb}-^{85}\text{Rb}_{1.059}$	$D_M=8326(9)$ μ u for $^{90}\text{Rb}^m$ at 106.90 keV; $M-A=-79260(9)$ keV										Nub211 **
$^{90}\text{Nb}(\beta^+)^{90}\text{Zr}$	$E_{\beta^+}=1500(4)$ to 6^+ level at 3589.419 keV										Ens981 **
$^{90}\text{Mo}(\beta^+)^{90}\text{Nb}$	$E_{\beta^+}=1085(4)$ to 1^+ level at 382.01 keV										Ens981 **
$^{90}\text{Tc}(\beta^+)^{90}\text{Mo}$	$E_{\beta^+}=7900(400)$ from $^{90}\text{Tc}^m$ at 144.1 to ground state (85%) and 947.97 (15%) level										74Ia01 **
$^{90}\text{Tc}(\beta^+)^{90}\text{Mo}$	$E_{\beta^+}=5300(300)$ to 2946.82 level										Ens981 **
$^{90}\text{Tc}(\beta^+)^{90}\text{Mo}$	$E_{\beta^+}=6900(300)$ from $^{90}\text{Tc}^m$ at 144.0(1.7) to 2^+ level a 947.97 keV										Ens981 **
$^{90}\text{Rh}(\epsilon)^{90}\text{Ru}$	Symmetrized from 13190(+1500-1160)										HWJ199 **
$^{90}\text{Rh}(\epsilon)^{90}\text{Ru}$	Trends from Mass Surface TMS suggest ^{90}Rh 160 keV more bound										GAu **
$^{91}\text{Se}-u$	-54300	186				2			GT3	2.5	16Kn03
$^{91}\text{Rb}-u$	-83532	21	-83463	8	1.3	U			Pb1	2.5	89Al33
	-83330	170			-0.8	U			TG2	1.0	20Gr.1
$\text{C}_7\text{H}_7-^{91}\text{Zr}$	149143.1	4.4	149135.02	0.10	-0.7	U			M15	2.5	63Ri07
$^{91}\text{Zr}-u$	-94359.85	0.25	-94359.79	0.10	0.2	1	17	17 ^{91}Zr	MS1	1.0	15Gu09
$^{91}\text{Nb}-u$	-93064	46	-93010	3	1.2	U			GS2	1.0	05Li24 *
$^{91}\text{Mo}-\text{C}_7\text{H}_7$	-143031.3	8.3	-143030	7	0.2	1	65	65 ^{91}Mo	CP1	1.0	11Fa10
$^{91}\text{Tc}-\text{C}_7\text{H}_7$	-136340.9	6.7	-136350.3	2.5	-1.4	-			CP1	1.0	11Fa10
	-136353.6	4.6			0.7	-			CP1	1.0	11Fa10
ave.	-136350	4			-0.2	1	45	45 ^{91}Tc			average
$^{91}\text{Ru}_{1.011}-\text{C}_7\text{H}_8$	-136730	620	-136664.6	2.4	0.1	U			CP1	1.0	11Fa10
$^{91}\text{Ru}-\text{C}_7\text{H}_7$	-128035.6	3.9	-128033.7	2.4	0.5	1	37	37 ^{91}Ru	CP1	1.0	11Fa10
$^{91}\text{Ru}_{1.022}-\text{C}_7\text{H}_9$	-145260	23	-145295.4	2.4	-1.5	U			CP1	1.0	11Fa10
$^{91}\text{Kr}-^{85}\text{Rb}_{1.071}$	18279.5	2.4				2			MA8	1.0	06De36
$^{91}\text{Rb}-^{85}\text{Rb}_{1.071}$	11003	10	11010	8	0.7	1	70	70 ^{91}Rb	MA4	1.0	02Ra23
$^{91}\text{Sr}-^{85}\text{Rb}_{1.071}$	4702	9	4669	6	-3.7	B			MA4	1.0	02Ra23
$^{91}\text{Tc}-^{85}\text{Rb}_{1.071}$	12898.3	5.4	12898.2	2.5	-0.0	1	22	22 ^{91}Tc	SH1	1.0	08We10
$^{91}\text{Ru}-^{85}\text{Rb}_{1.071}$	21223	11	21214.7	2.4	-0.8	-			SH1	1.0	08We10
	21215.5	4.2			-0.2	-			JY1	1.0	08We10
ave.	21216	4			-0.4	1	37	37 ^{91}Ru			average
$^{91}\text{Zr}-^{87}\text{Rb}_{1.046}$	637.32	0.19	637.37	0.10	0.3	1	29	29 ^{91}Zr	MS1	1.0	15Gu09
$^{91}\text{Br}-^{88}\text{Rb}_{1.034}$	26098.3	3.8				2			JY1	1.0	07Ra23
$^{91}\text{Tc}-^{94}\text{Mo}_{.968}$	10303.0	4.4	10304.1	2.5	0.2	1	33	33 ^{91}Tc	JY1	1.0	08We10
$^{91}\text{Ru}-^{94}\text{Mo}_{.968}$	18620.9	4.7	18620.6	2.4	-0.1	1	26	26 ^{91}Ru	JY1	1.0	08We10
$^{91}\text{Zr}-^{90}\text{Zr}$	942	12	941.45	0.16	-0.0	U			M15	2.5	63Ri07
$^{90}\text{Rb}^+-^{91}\text{Rb}_{.824}\text{ }^{85}\text{Rb}_{.176}$	-686	24	-771	15	-1.4	U			P21	2.5	82Au01
$^{91}\text{Zr}(\text{p},\text{t})^{89}\text{Zr}$	-10677	10	-10678.5	2.8	-0.1	U			Oak		71Ba43
$^{90}\text{Zr}(\text{n},\gamma)^{91}\text{Zr}$	7194.4	0.5	7194.36	0.15	-0.1	-					81Lo.A Z
	7192.7	0.8			2.1	-			Bdn		06Fi.A
$^{90}\text{Zr}(\text{d},\text{p})^{91}\text{Zr}$	4959	20	4969.80	0.15	0.5	U			Pit		64Co11
	4969	8			0.1	U			MIT		72Gr12
	4970.3	2.2			-0.2	U			SPa		79Bo37
$^{91}\text{Zr}(\text{p},\text{d})^{90}\text{Zr}$	-4977	10	-4969.80	0.15	0.7	U			Oak		71Ba43

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{91}\text{Zr}(\text{d,t})^{90}\text{Zr}$	−932	20	−937.13	0.15	−0.3	U			Pit		64Co11
	−940.3	3.7			0.9	U			SPa		79Bo37
$^{90}\text{Zr}(\text{n},\gamma)^{91}\text{Zr}$	ave. 7193.9	0.4	7194.36	0.15	1.0	1	12	7 ^{90}Zr			average
$^{90}\text{Zr}(\text{p},\gamma)^{91}\text{Nb}$	5167	5	5154.5	2.9	−2.5	B					71Ra08
	5167	4			−3.1	C					75Be.B Z
$^{90}\text{Zr}(^3\text{He,d})^{91}\text{Nb}$	−227	20	−339.0	2.9	−5.6	B			Hei		70Kn05
$^{90}\text{Zr}(\alpha,\text{t})^{91}\text{Nb}$	−14643	27	−14659.4	2.9	−0.6	U			Brk		71Zi03
$^{91}\text{Ru}^m(\varepsilon\text{p})^{90}\text{Mo}$	4300	500				2					83Ha06
$^{91}\text{Br}(\beta^-)^{91}\text{Kr}$	9790	100	9867	4	0.8	U			Bwg		89Gr03
	9805	50			1.2	U			Bwg		92Gr.A
$^{91}\text{Kr}(\beta^-)^{91}\text{Rb}$	6420	80	6771	8	4.4	B			Trs		78Wo15
	6450	80			4.0	B			Bwg		89Gr03
$^{91}\text{Rb}(\beta^-)^{91}\text{Sr}$	5830	45	5907	9	1.7	U			Trs		78Wo15 *
	5927	24			−0.8	o			Gsn		80De02 *
	5920	28			−0.5	−			McG		83Ia02 *
	5930	22			−1.0	−			Gsn		92Pr03 *
	ave. 5926	17			−1.1	1	26	18 ^{91}Rb			average
$^{91}\text{Sr}(\beta^-)^{91}\text{Y}$	2669	10	2699	5	3.0	B					53Am08 *
	2684	10			1.5	−					73Ha11 *
	2704	8			−0.6	−			Gsn		80De02 *
	2709	15			−0.6	−			McG		83Ia02
	ave. 2698	6			0.2	1	83	81 ^{91}Sr			average
$^{91}\text{Y}(\beta^-)^{91}\text{Zr}$	1545	5	1544.3	1.8	−0.1	−					64La13
	1544	2			0.1	−					75Ra08
	ave. 1544.1	1.9			0.1	1	98	98 ^{91}Y			average
$^{91}\text{Zr}(\text{p,n})^{91}\text{Nb}$	−2045	6	−2039.9	2.9	0.8	−			Oak		70Ki01
	−2038.8	3.4			−0.3	−			Kyu		71Ma47
	ave. −2040.3	3.0			0.1	1	98	98 ^{91}Nb			average
$^{91}\text{Mo}(\beta^+)^{91}\text{Nb}$	4460	30	4429	7	−1.0	−					56Sm96
	4435	23			−0.3	−					93Os06
	ave. 4444	18			−0.8	1	14	11 ^{91}Mo			average
$^{91}\text{Tc}(\beta^+)^{91}\text{Mo}$	6220	200	6222	7	0.0	U					74Ia01
$^{91}\text{Pd}(\varepsilon)^{91}\text{Rh}$	11800	2200	12400#	300#	0.3	D					18Pa20 *
$^{91}\text{Nb}-\text{u}$	$M-A=-86636(30)$ keV for mixture gs+m at 104.60 keV										Nub211 **
$^{91}\text{Rb}(\beta^-)^{91}\text{Sr}$	$E_{\beta^-}=5760(40)$ to ^{91}Sr ground state <8% and 93.628 keV $(3/2)^+$ level 25%										Ens13a **
$^{91}\text{Rb}(\beta^-)^{91}\text{Sr}$	Original error 8 corrected to 13 keV in reference										94Ha.A **
$^{91}\text{Rb}(\beta^-)^{91}\text{Sr}$	$E_{\beta^-}=5857$ to mixture ^{91}Sr ground state <8% and 93.628 $(3/2)^+$ level 25%										Ens13a **
$^{91}\text{Rb}(\beta^-)^{91}\text{Sr}$	$E_{\beta^-}=5850(20)$ and $E_{\beta^-}=5860(10)$ respectively, to ^{91}Sr ground state <8% and 93.628 level 25%										Ens13a **
$^{91}\text{Sr}(\beta^-)^{91}\text{Y}$	$E_{\beta^-}=2665(10)$, 2030(20), 1359(10), 1093(10) to										53Am08 **
*	ground state, $3/2^-$ level at 653.02 keV, $(5/2)^+$ at 1305.39, $(5/2)^-$ at 1545.90										Ens13a **
$^{91}\text{Sr}(\beta^-)^{91}\text{Y}$	Original error 4 increased: in disagreement with other results										AHW **
$^{91}\text{Sr}(\beta^-)^{91}\text{Y}$	Original error 3 corrected in reference										94Ha.A **
$^{91}\text{Pd}(\varepsilon)^{91}\text{Rh}$	Trends from Mass Surface TMS suggest ^{91}Pd 600 keV less bound										GAu **
$^{92}\text{Br}-\text{u}$	−60711	103	−60368	7	1.3	U			GT2	2.5	08Kn.A
$^{92}\text{Rb}-\text{u}$	−80323	32	−80272	7	0.6	U			Pb1	2.5	89Al33
$\text{C}_7\text{H}_8-^{92}\text{Zr}$	157569.4	3.8	157564.92	0.10	−0.5	U			M15	2.5	63Ri07
$^{92}\text{Zr}-\text{u}$	−94964.66	0.18	−94964.66	0.10	−0.0	1	32	32 ^{92}Zr	MS1	1.0	15Gu09
$^{92}\text{Nb}-\text{u}$	−92851	56	−92811.4	1.9	0.7	U			GS2	1.0	05Li24 *
$\text{C}_7\text{H}_8-^{92}\text{Mo}$	155790.0	3.2	155793.10	0.17	0.4	U			M15	2.5	63Ri07
$^{92}\text{Mo}_{1.011}-\text{C}_7\text{H}_9$	−164641.3	7.0	−164643.26	0.17	−0.3	U			CP1	1.0	11Fa10
$^{92}\text{Mo}-\text{u}$	−93193.14	0.47	−93192.85	0.17	0.6	1	13	13 ^{92}Mo	MS1	1.0	15Gu09
$^{92}\text{Tc}-\text{C}_7\text{H}_8$	−147328	13	−147330	3	−0.2	U			CP1	1.0	11Fa10
$^{92}\text{Tc}_{.989}-\text{C}_7\text{H}_7$	−138569	10	−138573	3	−0.4	−			CP1	1.0	11Fa10
$^{92}\text{Tc}_{1.011}-\text{C}_7\text{H}_9$	−156087.6	6.1	−156088	3	0.0	−			CP1	1.0	11Fa10
$^{92}\text{Tc}_{.989}-\text{C}_7\text{H}_7$	ave. −138572	5	−138573	3	−0.2	1	40	40 ^{92}Tc			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{92}\text{Ru}-\text{C}_7\text{H}_8$	-142352	18	-142365.9	2.9	-0.8	o			CP1	1.0	08Fa11
	-142352	18			-0.8	U			CP1	1.0	11Fa10
	-142377	10			1.1	o			CP1	1.0	08Fa11
	-142378	10			1.2	U			CP1	1.0	11Fa10
$^{92}\text{Ru}_{1.011}-\text{C}_7\text{H}_9$	-151074.5	5.6	-151068.3	2.9	1.1	o			CP1	1.0	08Fa11
	-151074.5	5.6			1.1	1	28	28 ^{92}Ru	CP1	1.0	11Fa10
$^{92}\text{Rh}_{1.011}-\text{C}_7\text{H}_9$	-138825	23	-138802	5	1.0	U			CP1	1.0	11Fa10
	-138818	17			1.0	U			CP1	1.0	11Fa10
$^{92}\text{Kr}-^{85}\text{Rb}_{1.082}$	21616.6	2.9				2			MA8	1.0	06De36
$^{92}\text{Rb}-^{85}\text{Rb}_{1.082}$	15176	9	15172	7	-0.4	1	53	53 ^{92}Rb	MA4	1.0	02Ra23
$^{92}\text{Sr}-^{85}\text{Rb}_{1.082}$	6482	9	6482	4	-0.0	-			MA4	1.0	02Ra23
	6484.0	4.3			-0.5	-			MA8	1.0	05Gu37
	ave. 6484	4			-0.5	1	90	90 ^{92}Sr			average
$^{92}\text{Mo}-^{85}\text{Rb}_{1.082}$	2251.43	0.85	2250.66	0.17	-0.9	U			JY1	1.0	12Ka13
$^{92}\text{Tc}-^{85}\text{Rb}_{1.082}$	10728	12	10713	3	-1.2	U			SH1	1.0	08We10
	10712.5	4.3			0.2	1	60	60 ^{92}Tc	JY1	1.0	08We10
$^{92}\text{Ru}-^{85}\text{Rb}_{1.082}$	15684.3	5.7	15677.9	2.9	-1.1	-			SH1	1.0	08We10
	15677.9	4.3			-0.0	-			JY1	1.0	08We10
	ave. 15680	3			-0.7	1	72	72 ^{92}Ru			average
$^{92}\text{Rh}-^{85}\text{Rb}_{1.082}$	27841	37	27811	5	-0.8	U			SH1	1.0	08We10
	27811.2	4.7				2			JY1	1.0	08We10
$^{92}\text{Zr}-^{87}\text{Rb}_{1.057}$	1031.51	0.21	1031.52	0.10	0.1	1	23	23 ^{92}Zr	MS1	1.0	15Gu09
$^{92}\text{Mo}-^{87}\text{Rb}_{1.057}$	2803.38	0.18	2803.34	0.17	-0.2	1	87	87 ^{92}Mo	MS1	1.0	15Gu09
$^{92}\text{Br}-^{88}\text{Rb}_{1.045}$	32306.8	7.2				2			JY1	1.0	07Ra23
$^{92}\text{Zr}-^{90}\text{Zr}-^{37}\text{Cl}$	3285	2	3286.70	0.17	0.2	U			H13	4.0	63Ba20
$^{92}\text{Zr}-^{91}\text{Zr}$	-603	12	-604.87	0.04	-0.1	U			M15	2.5	63Ri07
$^{88}\text{Rb}-^{92}\text{Rb}_{.410}-^{85}\text{Rb}_{.592}$	-3258	22	-3309.2	2.5	-0.9	U			P21	2.5	82Au01
$^{89}\text{Rb}-^{92}\text{Rb}_{.553}-^{85}\text{Rb}_{.449}$	-3457	24	-3470	6	-0.2	U			P21	2.5	82Au01
$^{91}\text{Rb}-^{92}\text{Rb}_{.848}-^{85}\text{Rb}_{.153}$	-1703	25	-1766	9	-1.0	U			P21	2.5	82Au01
$^{90}\text{Rb}^x-^{92}\text{Rb}_{.699}-^{85}\text{Rb}_{.303}$	-2059	24	-2132	14	-1.2	U			P21	2.5	82Au01
$^{90}\text{Rb}^x-^{92}\text{Rb}_{.326}-^{89}\text{Rb}_{.674}$	209	24	155	14	-0.9	U			P21	2.5	82Au01
$^{92}\text{Mo}(\alpha, ^8\text{He})^{88}\text{Mo}$	-43278	20	-43307	4	-1.4	U			INS		90Ka01
$^{92}\text{Mo}(\text{p}, \alpha)^{89}\text{Nb}$	-1306	50	-1319	24	-0.3	R			ANL		75Se.A
$^{92}\text{Mo}(\text{He}, ^6\text{He})^{89}\text{Mo}$	-14465	15	-14455	4	0.7	U			MSU		80Pa02
$^{92}\text{Zr}(\text{p}, \text{t})^{90}\text{Zr}$	-7372	14	-7347.32	0.15	1.8	U			Bld		66St15
	-7350	10			0.3	U			Oak		71Ba43
$^{92}\text{Mo}(\text{p}, \text{t})^{90}\text{Mo}$	-14330	30	-14297	3	1.1	U			VUn		76Ka08
$^{92}\text{Rb}(\beta^- \text{n})^{91}\text{Sr}$	785	15	808	8	1.5	1	26	15 ^{92}Rb			84Kr.B
$^{91}\text{Zr}(\text{n}, \gamma)^{92}\text{Zr}$	8634.91	0.20	8634.75	0.04	-0.8	U			ILn		79Br25 Z
	8634.64	0.15			0.7	U					81Su.A Z
	8635.00	0.24			-1.0	U			Bdn		06Fi.A
	8634.75	0.04			-0.0	1	96	50 ^{91}Zr	ILn		18Rz01
$^{91}\text{Zr}(\text{d}, \text{p})^{92}\text{Zr}$	6470	30	6410.18	0.04	-2.0	U					62Ma06
	6395	20			0.8	U			Pit		64Co11
	6410.9	4.3			-0.2	U			SPa		79Bo37
$^{92}\text{Zr}(\text{p}, \text{d})^{91}\text{Zr}$	-6410	11	-6410.18	0.04	-0.0	U			Bld		66St15
	-6410	10			-0.0	U			Oak		71Ba43
$^{92}\text{Zr}(\text{d}, \text{t})^{91}\text{Zr}$	-2363	25	-2377.52	0.04	-0.6	U			Pit		64Co11
$^{92}\text{Mo}(\text{p}, \text{d})^{91}\text{Mo}$	-10446	15	-10447	6	-0.0	-			Tex		73Ko03
	-10432	25			-0.6	-			Grn		73Mo03
	ave. -10442	13			-0.3	1	24	24 ^{91}Mo			average
$^{92}\text{Br}(\beta^-)^{92}\text{Kr}$	12155	100	12537	7	3.8	B			Bwg		89Gr03
	12220	55			5.8	C			Bwg		92Gr.A
$^{92}\text{Kr}(\beta^-)^{92}\text{Rb}$	6160	80	6003	7	-2.0	U			Trs		78Wo15
	6045	80			-0.5	o			Bwg		89Gr03
	5987	10			1.6	o			Bwg		92Gr.A
	5993	27			0.4	U			Bwg		92Gr06

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{92}\text{Rb}(\beta^-)^{92}\text{Sr}$		8080	160	8095	6	0.1	U		Trs		78Wo15
		8091	15			0.3	o		Gsn		80Bl.A
		8111	15			-1.1	o		Gsn		80De02
		8080	30			0.5	-		McG		83Ia02
		8095	25			-0.0	o		Bwg		87Gr.A
		8096	16			-0.1	-		Bwg		92Gr.A
		8107	15			-0.8	-		Gsn		92Pr03
$^{92}\text{Sr}(\beta^-)^{92}\text{Y}$	ave.	8099	10			-0.4	1	39	32 ^{92}Rb		average
		1929	50	1949	9	0.4	U				57He39 *
		1930	30			0.6	-		Trs		78Wo15 *
		1920	20			1.5	-		McG		83Ia02
$^{92}\text{Y}(\beta^-)^{92}\text{Zr}$	ave.	1923	17			1.6	1	32	29 ^{92}Y		average
		3640	20	3643	9	0.1	-				62Bu16
		3630	15			0.8	-		McG		83Ia02
$^{92}\text{Nb}(\beta^+)^{92}\text{Zr}$	ave.	3634	12			0.7	1	58	58 ^{92}Y		average
		2005	6	2005.7	1.8	0.1	U				59We30 *
$^{92}\text{Zr}(\text{p,n})^{92}\text{Nb}$		2008	6			-0.4	U				62Bu16 *
		-2790.7	2.3	-2788.1	1.8	1.1	-		Kyu		74Ku01
		-2792	5			0.8	-				75Ke12
	ave.	-2790.9	2.1			1.4	1	73	73 ^{92}Nb		average
$^{92}\text{Tc}(\beta^+)^{92}\text{Mo}$		7880	100	7883	3	0.0	U				64Va05
$^{92}\text{Mo}(\text{p,n})^{92}\text{Tc}$		-8672	50	-8665	3	0.1	U		Tal		66Mo06 *
$^{92}\text{Mo}(^3\text{He,t})^{92}\text{Tc}$		-7882	30	-7901	3	-0.6	U		ChR		73Ha02
$^{92}\text{Pd}(\epsilon)^{92}\text{Rh}$		8220	345				3				19Pa16 *
$^{92}\text{Nb}-\text{u}$	$M-A=-86422(34)$ keV for mixture gs+m at 135.5 keV										Nub211 **
$^{92}\text{Sr}(\beta^-)^{92}\text{Y}$	$E_{\beta^-}=545(50)$ 546(30) respectively, to 1^+ level at 1383.91 keV										Ens12a **
$^{92}\text{Nb}(\beta^+)^{92}\text{Zr}$	$p^+=56(6)\times 10^{-5}$, $60(6)\times 10^{-5}$ respectively, to 2^+ level at 934.51 keV										Ens12a **
*	recalculated $Q_{\beta^+}=2140(6)$ 2143(6) respectively, from $^{92}\text{Nb}^m$ at 135.5 keV										AHW **
$^{92}\text{Mo}(\text{p,n})^{92}\text{Tc}$	$T=9040(50)$ to (4^+) level at 270.09 keV										Ens12a **
$^{92}\text{Pd}(\epsilon)^{92}\text{Rh}$	$Q_{\epsilon}=8170(330)$ keV decays to the daughter excited state at 50#(100#) keV										HWJ202 **
$^{93}\text{Br}-\text{u}$		-56866	322	-56780	460	0.2	o		GT1	1.5	04Ma.A
		-56780	185				2		GT3	2.5	16Kn03
$^{93}\text{Rb}-\text{u}$		-78036	21	-77961	8	1.4	U		Pb1	2.5	89Al33
		-77868	100			-0.4	U		GT2	2.5	08Kn.A
		-77940	130			-0.2	U		TG2	1.0	20Gr.1
		-87010	420	-85976	8	2.5	U		TG2	1.0	20Gr.1
$\text{C}_7\text{H}_9-^{93}\text{Nb}$		164046.9	3.5	164052.1	1.6	0.6	U		M15	2.5	63Ri07
$^{93}\text{Mo}-\text{u}$		-93194	30	-93191.23	0.19	0.1	U		GS2	1.0	05Li24 *
$^{93}\text{Tc}-\text{u}$		-89729	31	-89754.9	1.1	-0.8	U		GS2	1.0	05Li24
$^{93}\text{Tc}-\text{C}_7\text{H}_9$		-160189.5	7.7	-160180.1	1.1	1.2	U		CP1	1.0	11Fa10
		-160170	22			-0.5	U		CP1	1.0	11Fa10
		-160189.4	8.5			1.1	U		CP1	1.0	11Fa10
		-151270	190	-151367.8	1.1	-0.5	U		CP1	1.0	11Fa10
$^{93}\text{Tc}_{989}-\text{C}_7\text{H}_8$		-153318.2	6.4	-153320.8	2.2	-0.4	-		CP1	1.0	11Fa10
$^{93}\text{Ru}-\text{C}_7\text{H}_9$		-153307	23			-0.6	U		CP1	1.0	11Fa10
		-153324.0	4.8			0.7	-		CP1	1.0	11Fa10
		-153321.9	3.5			0.3	-		CP1	1.0	11Fa10
		-82853	47	-82895.6	2.2	-0.9	U		GR1	1.0	19An10
$^{93}\text{Ru}-\text{u}$		-153321.9	2.6	-153320.8	2.2	0.4	1	73	73 ^{93}Ru		average
$^{93}\text{Rh}-\text{C}_7\text{H}_9$	ave.	-144485	25	-144512.5	2.8	-1.1	o				
		-144485	26			-1.1	U		CP1	1.0	08Fa11
		-144527.7	5.3			2.9	U		CP1	1.0	11Fa10
		-144527.7	5.2			2.9	U		CP1	1.0	08Fa11
		-144512.9	3.8			0.1	1	55	55 ^{93}Rh		11Fa10

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{93}\text{Kr}-^{85}\text{Rb}_{1.094}$	27649.2	2.7				2			MA8	1.0	06De36
$^{93}\text{Rb}-^{85}\text{Rb}_{1.094}$	18549	10	18541	8	-0.8	1	71	$^{71}\text{ }^{93}\text{Rb}$	MA4	1.0	02Ra23
$^{93}\text{Sr}-^{85}\text{Rb}_{1.094}$	10526	10	10526	8	0.0	1	66	$^{66}\text{ }^{93}\text{Sr}$	MA4	1.0	02Ra23
$^{93}\text{Ru}-^{85}\text{Rb}_{1.094}$	13609.4	4.3	13606.5	2.2	-0.7	1	27	$^{27}\text{ }^{93}\text{Ru}$	JY1	1.0	08We10
$^{93}\text{Rh}-^{85}\text{Rb}_{1.094}$	22428	12	22414.8	2.8	-1.1	-			SH1	1.0	08We10
	22413.5	4.5			0.3	-			JY1	1.0	08We10
ave.	22415	4			-0.1	1	45	$^{45}\text{ }^{93}\text{Rh}$			average
$^{91}\text{Rb}-^{93}\text{Rb}_{.489}\text{ }^{89}\text{Rb}_{.511}$	-471	9	-479	9	-0.4	1	15	$^{12}\text{ }^{91}\text{Rb}$	P31	2.5	86Au02
$^{91}\text{Rb}-^{93}\text{Rb}_{.326}\text{ }^{90}\text{Rb}_{.674}$	-656	23	-626	12	0.5	U			P21	2.5	82Au01
$^{92}\text{Rb}-^{93}\text{Rb}_{.495}\text{ }^{91}\text{Rb}_{.505}$	465	23	436	8	-0.5	U			P21	2.5	82Au01
$^{93}\text{Rb}(\beta^-)^{92}\text{Sr}$	2220	30	2176	9	-1.5	U					84Kr.B
$^{92}\text{Sr}(n,\gamma)^{93}\text{Sr}$	5230	6	5290	8	10.0	B					80Kr07
$^{92}\text{Zr}(n,\gamma)^{93}\text{Zr}$	6733.7	1.1	6734.3	0.4	0.6	-					72Gr23 Z
	6734.0	0.7			0.5	-					79Ke.D Z
	6735.3	0.7			-1.4	-			Bdn		06Fi.A
$^{92}\text{Zr}(d,p)^{93}\text{Zr}$	4493	20	4509.8	0.4	0.8	U			Pit		64Co11
$^{92}\text{Zr}(n,\gamma)^{93}\text{Zr}$	ave. 6734.5	0.5	6734.3	0.4	-0.4	1	98	$^{98}\text{ }^{93}\text{Zr}$			average
$^{93}\text{Nb}(\gamma,n)^{92}\text{Nb}$	-8780	60	-8830.9	2.0	-0.8	U			Phi		60Ge01
	-8825	3			-2.0	1	44	$^{27}\text{ }^{92}\text{Nb}$	McM		79Ba06
$^{93}\text{Nb}(d,t)^{92}\text{Nb}$	-2581	20	-2573.6	2.0	0.4	U			Pit		64Co11
	-2571	10			-0.3	U			Tal		64Sh04
$^{92}\text{Mo}(n,\gamma)^{93}\text{Mo}$	8067.4	1.5	8069.81	0.09	1.6	U					73Wa17
	8066	2			1.9	U					77Ri04
	8069.81	0.09				2			MMn		91Is02 Z
	8070.0	0.3			-0.6	U			Bdn		06Fi.A
$^{92}\text{Mo}(d,p)^{93}\text{Mo}$	5853	20	5845.24	0.09	-0.4	U			Pit		64Co11
$^{92}\text{Mo}(p,\gamma)^{93}\text{Tc}$	4081	5	4086.5	1.0	1.1	U					75Be.B
	4086.5	1.0				2					83Ay01
$^{92}\text{Mo}(^3\text{He},d)^{93}\text{Tc}$	-1411	4	-1407.0	1.0	1.0	U			Hei		83Wi.A
$^{93}\text{Kr}(\beta^-)^{93}\text{Rb}$	8700	500	8484	8	-0.4	U			Trs		78Wo15
	8600	100			-1.2	U			Bwg		87Gr.A
$^{93}\text{Rb}(\beta^-)^{93}\text{Sr}$	7560	120	7466	9	-0.8	U			Trs		78Wo15
	7488	15			-1.5	o			Gsn		80Bl.A
	7485	15			-1.3	o			Gsn		80De02
	7440	30			0.9	-			McG		83Ia02
	7455	35			0.3	-			Bwg		87Gr.A
	7456	15			0.7	-			Gsn		92Pr03
ave.	7453	13			1.0	1	50	$^{26}\text{ }^{93}\text{Rb}$			average
$^{93}\text{Sr}(\beta^-)^{93}\text{Y}$	4130	100	4141	12	0.1	U			Bwg		78St02
	4110	20			1.6	1	34	$^{24}\text{ }^{93}\text{Y}$	McG		83Ia02
$^{93}\text{Y}(\beta^-)^{93}\text{Zr}$	2890	20	2895	10	0.2	-					59Kn38
	2880	15			1.0	-			McG		83Ia02
ave.	2884	12			0.9	1	76	$^{76}\text{ }^{93}\text{Y}$			average
$^{93}\text{Zr}(\beta^-)^{93}\text{Nb}$	93.8	2.	90.8	1.5	-1.5	1	55	$^{53}\text{ }^{93}\text{Nb}$			53Gl.A *
$^{93}\text{Mo}(\epsilon)^{93}\text{Nb}$	158	15	405.8	1.5	16.5	B					64Ho08 *
$^{93}\text{Nb}(p,n)^{93}\text{Mo}$	-1188	10	-1188.1	1.5	-0.0	U					68Fi01
	-1190	5			0.4	U					75Ch05
$^{93}\text{Tc}(\beta^+)^{93}\text{Mo}$	3185.1	5.	3201.0	1.0	3.2	B					51Bo48 *
	3192.1	3.			3.0	B					74An24 *
$^{93}\text{Ru}(\beta^+)^{93}\text{Tc}$	6337	85	6389.4	2.3	0.6	U					83Ay01
$^{93}\text{Pd}(\epsilon)^{93}\text{Rh}$	10030	370				2					19Pa16
$^{*93}\text{Mo}-u$	$M-A=-84385(28)$ keV for $^{93}\text{Mo}^m$ at 2424.95 keV										Ens115 **
$^{*93}\text{Zr}(\beta^-)^{93}\text{Nb}$	$E_{\beta^-}=63(2)$ to $1/2^-$ level at 30.77 keV										Ens115 **
$^{*93}\text{Mo}(\epsilon)^{93}\text{Nb}$	$L/K=0.36(0.04)$ to $1/2^-$ level at 30.82 keV, recalculated Q										Ens115 **
$^{*93}\text{Tc}(\beta^+)^{93}\text{Mo}$	$E_{\beta^+}=800(5)$ 807(3) respectively, to $7/2^+$ level 1363.048 keV										Ens115 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{94}\text{Br}-\text{u}$	-50242	429	-51150#	220#	-0.9	D			GT3	2.5	16Kn03 *
$^{94}\text{Kr}-\text{u}$	-66238	247	-65860	13	1.0	U			GT1	1.5	04Ma.A
$^{94}\text{Kr}-^{85}\text{Rb}_{1.106}$	31701	13				2			MA8	1.0	06De36
	31665	24	31701	13	1.5	U			MA8	1.0	10Na13
	31649	97			0.5	U			MA9	1.0	10Na13 *
	-73602	54	-73605.2	2.2	-0.0	F			Pb1	2.5	89Al33 *
$^{94}\text{Rb}-\text{u}$ $^{94}\text{Rb}-^{85}\text{Rb}_{1.106}$	23958	10	23955.4	2.2	-0.3	U			MA4	1.0	02Ra23
	23955.6	2.6			-0.1	1	70	70 ^{94}Rb	TT1	1.0	12Si10 *
	12924	10	12916.2	1.8	-0.8	U			MA4	1.0	02Ra23
$^{94}\text{Sr}-^{85}\text{Rb}_{1.106}$	12916.0	1.8			0.1	1	98	98 ^{94}Sr	TT1	1.0	12Si10 *
$\text{C}_7\text{H}_{10}-^{94}\text{Zr}$	171929.4	3.9	171937.80	0.18	0.9	U			M15	2.5	63Ri07
$^{94}\text{Zr}-\text{u}$	-93687.34	0.20	-93687.48	0.18	-0.7	1	77	77 ^{94}Zr	MS1	1.0	15Gu09
$^{94}\text{Mo}-^{85}\text{Rb}_{1.106}$	2645.6	1.0	2644.13	0.15	-1.5	U			JY1	1.0	12Ka13
$\text{C}_7\text{H}_{10}-^{94}\text{Mo}$	173159.6	3.2	173166.73	0.15	0.9	U			M15	2.5	63Ri07
$^{94}\text{Mo}-\text{u}$	-94916.31	0.42	-94916.41	0.15	-0.2	1	13	13 ^{94}Mo	MS1	1.0	15Gu09
$^{94}\text{Tc}-\text{u}$	-90362	39	-90348	4	0.4	U			GS2	1.0	05Li24 *
$^{94}\text{Ru}-^{85}\text{Rb}_{1.106}$	8891	25	8903	3	0.5	U			SH1	1.0	08We10
	8907.1	4.5			-0.8	1	56	56 ^{94}Ru	JY1	1.0	08We10
	-166912.2	5.1	-166907	3	0.9	1	44	44 ^{94}Ru	CP1	1.0	11Fa10
$^{94}\text{Ru}-\text{u}$	-88618	28	-88657	3	-1.4	U			GR1	1.0	19An10
$^{94}\text{Rh}-^{85}\text{Rb}_{1.106}$	19291.2	4.6	19291	4	-0.0	1	62	62 ^{94}Rh	JY1	1.0	08We10
$^{94}\text{Rh}-\text{C}_7\text{H}_{10}$	-156520.2	5.9	-156520	4	0.1	1	38	38 ^{94}Rh	CP1	1.0	11Fa10
$^{94}\text{Rh}_{989}-\text{C}_7\text{H}_9$	-147834	30	-147834	4	0.0	U			CP1	1.0	11Fa10
$^{94}\text{Rh}-\text{u}$	-78370	140	-78270	4	0.7	U			GR1	1.0	19An10 *
$^{94}\text{Kr}-^{86}\text{Kr}_{1.093}$	31710	110	31843	13	1.2	U			MA9	1.0	10Na13
$^{94}\text{Zr}-^{87}\text{Rb}_{1.080}$	4397.13	0.37	4397.56	0.18	1.2	1	23	23 ^{94}Zr	MS1	1.0	15Gu09
$^{94}\text{Mo}-^{87}\text{Rb}_{1.080}$	3168.68	0.35	3168.62	0.15	-0.2	1	19	19 ^{94}Mo	MS1	1.0	15Gu09
$^{94}\text{Rb}-^{88}\text{Rb}_{1.068}$	21109.1	4.0	21109.8	2.2	0.2	1	30	30 ^{94}Rb	JY1	1.0	07Ra23
$^{94}\text{Zr}-^{35}\text{Cl}-^{92}\text{Zr}-^{37}\text{Cl}$	4235.0	2.	4227.31	0.21	-1.0	U			H13	4.0	63Ba20
$^{94}\text{Mo}-^{35}\text{Cl}-^{92}\text{Mo}-^{37}\text{Cl}$	1234.0	2.	1226.55	0.24	-0.9	U			H11	4.0	63Bi12
$^{94}\text{Pd}-^{94}\text{Mo}$	23952.7	4.6				2			JY1	1.0	08We10
$^{92}\text{Rb}-^{94}\text{Rb}_{.587}-^{89}\text{Rb}_{.413}$	-764	24	-779	7	-0.2	U			P21	2.5	82Au01 Y
$^{92}\text{Rb}-^{94}\text{Rb}_{.489}-^{90}\text{Rb}_{.511}$	-717	23	-726	9	-0.2	U			P21	2.5	82Au01 Y
$^{93}\text{Rb}-^{94}\text{Rb}_{.742}-^{90}\text{Rb}_{.258}$	-1296	25	-1288	9	0.1	U			P21	2.5	82Au01 Y
$^{93}\text{Rb}-^{94}\text{Rb}_{.495}-^{92}\text{Rb}_{.505}$	-840	40	-921	8	-0.8	F			P31	2.5	86Au02 *
$^{94}\text{Zr}(\text{d},\alpha)^{92}\text{Y}$	8278	25	8258	9	-0.8	1	13	13 ^{92}Y	Grn		74Gi09
$^{94}\text{Zr}(\text{p},\text{t})^{92}\text{Zr}$	-6466	12	-6471.15	0.19	-0.4	U			Bld		66St15
	-6470	10			-0.1	U			Oak		71Ba43
	3449	100	2520	370	-9.3	F					06Mu03 *
$^{94}\text{Ag}^n(2\text{p})^{92}\text{Rh}$	3580	80	3452	8	-1.6	U					84Kr.B
$^{94}\text{Rb}(\beta^- \text{n})^{93}\text{Sr}$	-5983	15	-5994.1	0.5	-0.7	U			Bld		66St15
$^{94}\text{Zr}(\text{p},\text{d})^{93}\text{Zr}$	-6000	10			0.6	U			Oak		71Ba43
	-1969	20	-1961.4	0.5	0.4	U			Pit		64Co11
	-1960.2	2.4			-0.5	U			SPa		79Bo37
$^{93}\text{Nb}(\text{n},\gamma)^{94}\text{Nb}$	7229.13	0.12	7227.54	0.08	-13.2	C					84Bo.C
	7227.51	0.09			0.3	-			MMn		88Ke09 Z
	7227.63	0.15			-0.6	-			Bdn		06Fi.A
	ave. 7227.54	0.08			-0.0	1	100	69 ^{94}Nb			average
$^{94}\text{Mo}(\text{d},\text{t})^{93}\text{Mo}$	-3441	20	-3421.09	0.23	1.0	U			Pit		64Co11
$^{94}\text{Ag}^n(\text{p})^{93}\text{Pd}$	5780	30	5790	17	0.3	3					05Mu15 *
	5794	20			-0.2	3					09Ce04 *
$^{94}\text{Rb}(\beta^-)^{94}\text{Sr}$	10304	20	10282.9	2.6	-1.1	o			Gsn		80Bl.A
	10322	100			-0.4	o			Gsn		80De02 *
	10353	140			-0.5	U			Trs		82Br23 *
	10335	45			-1.2	U			Bwg		82Pa24 *
	10312	20			-1.5	U			Gsn		92Pr03 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{94}\text{Sr}(\beta^-)^{94}\text{Y}$	3512	10	3506	6	-0.6	1	41	40 ^{94}Y	Gsn		80De02 *
$^{94}\text{Y}(\beta^-)^{94}\text{Zr}$	4920	9	4918	6	-0.2	1	50	50 ^{94}Y	Gsn		80De02 *
$^{94}\text{Nb}(\beta^-)^{94}\text{Mo}$	2043.3	6.	2045.0	1.5	0.3	-					66Sn02 *
	2046.3	3.			-0.4	-					68Ho10 *
	ave.	2045.7	2.7		-0.3	1	31	31 ^{94}Nb			average
	4261	5	4256	4	-1.1	2					64Ha29 *
$^{94}\text{Tc}(\beta^+)^{94}\text{Mo}$	4261	5	4256	4	-1.1	2					64Ha29 *
$^{94}\text{Mo}(\text{p,n})^{94}\text{Tc}$	-5027.8	7.	-5038	4	-1.5	2					73Mc04 *
$^{94}\text{Rh}(\beta^+)^{94}\text{Ru}$	9930	400	9676	5	-0.6	U					80Ox01 *
	9750	320			-0.2	U					06Ba55
$^{94}\text{Pd}(\beta^+)^{94}\text{Rh}$	6700	320	6805	5	0.3	U					06Ba55
$^{94}\text{Ag}(\epsilon)^{94}\text{Pd}$	13400	650	13700#	400#	0.5	D					19Pa16 *
$^{94}\text{Ag}^n(\beta^+)^{94}\text{Pd}$	17700	500	20200	370	5.0	B					04Mu30 *
$^{94}\text{Br}-\text{u}$	Trends from Mass Surface TMS suggest ^{94}Br 850 keV more bound										GAu **
$^{94}\text{Kr}-^{85}\text{Rb}_{1.106}$	Typo in original paper, ratio should read 1.00625564(114)										GAu **
$^{94}\text{Rb}-\text{u}$	F : possibly isomeric mixture										92Al.B **
$^{94}\text{Rb}-^{85}\text{Rb}_{1.106}$	$D_M=23956.4(2.6) \mu\text{u}$ $M-A=-68561.8(2.4)\text{keV}$ corrected for e^- binding=-775 eV										12Si10 **
$^{94}\text{Sr}-^{85}\text{Rb}_{1.106}$	$D_M=12916.7(1.8) \mu\text{u}$ $M-A=-78845.2(1.7)\text{keV}$ corrected for e^- binding=-625 eV										12Si10 **
$^{94}\text{Tc}-\text{u}$	$M-A=-84133(29) \text{keV}$ for mixture gs+m at 76(3) keV										Nub211 **
$^{94}\text{Rh}-\text{u}$	$M-A=-72848(24) \text{keV}$ for mixture gs+n at 300#(200#) keV										Nub211 **
$^{93}\text{Rb}-^{94}\text{Rb}_{.495} \quad ^{92}\text{Rb}_{.505}$	F : rejection based on line-shape analysis										86Au02 **
$^{94}\text{Ag}^n(2p)^{92}\text{Rh}$	$Q_{2p}=1900(100) \text{ to } (11^+) \text{ level at } 1548.6 \text{ keV}$										Ens12a **
$^{94}\text{Ag}^n(2p)^{92}\text{Rh}$	F : no evidence from He-jet experiment										09Ce04 **
$^{94}\text{Ag}^n(2p)^{92}\text{Rh}$	F : possibly misidentified ^{92}Rh γ rays										09Je05 **
$^{94}\text{Ag}^n(p)^{93}\text{Pd}$	$E_p=790(30), 1010(30) \text{ to } (33/2^+) \text{ at } 4995.6, (33/2^-, 35/2^-) \text{ at } 4752.7$										Ens115 **
$^{94}\text{Ag}^n(p)^{93}\text{Pd}$	$E_p=790(20) \text{ to level } (33/2^+) \text{ at } 4995.6 \text{ keV}$										Ens115 **
$^{94}\text{Rb}(\beta^-)^{94}\text{Sr}$	Original value 10304(30) corrected in reference										94Ha.A **
$^{94}\text{Rb}(\beta^-)^{94}\text{Sr}$	Original error 100 keV increased by 100 in reference for lower level feeding										94Ha.A **
$^{94}\text{Rb}(\beta^-)^{94}\text{Sr}$	As corrected in reference										87Gr.A **
$^{94}\text{Rb}(\beta^-)^{94}\text{Sr}$	$E_{\beta^-}=9475(20) \text{ to } 2^+ \text{ level at } 836.91 \text{ keV}$										Ens116 **
$^{94}\text{Sr}(\beta^-)^{94}\text{Y}$	Original error 6 corrected in reference										94Ha.A **
$^{94}\text{Y}(\beta^-)^{94}\text{Zr}$	Original error 5 corrected in reference										94Ha.A **
$^{94}\text{Nb}(\beta^-)^{94}\text{Mo}$	$E_{\beta^-}=470(6) \text{ to } 473(3) \text{ respectively, to level } 4^+ \text{ at } 1573.76 \text{ keV}$										Ens069 **
$^{94}\text{Tc}(\beta^+)^{94}\text{Mo}$	$E_{\beta^+}=816(5) \text{ to } 6^+ \text{ level at } 2423.45 \text{ keV}$										Ens069 **
$^{94}\text{Mo}(\text{p,n})^{94}\text{Tc}$	$T=5158(7) \text{ to } ^{94}\text{Tc}^m \text{ at } 76(3) \text{ keV}$										Nub211 **
$^{94}\text{Rh}(\beta^+)^{94}\text{Ru}$	$E_{\beta^+}=6400(400) \text{ to } (3,4,5) \text{ level at } 2503.2 \text{ keV}$										Ens069 **
$^{94}\text{Ag}(\epsilon)^{94}\text{Pd}$	Symmetried from 13350(+690-610)										HWJ199 **
$^{94}\text{Ag}(\epsilon)^{94}\text{Pd}$	Trends from Mass Surface TMS suggest ^{94}Ag 300 keV less bound										HWJ199 **
$^{94}\text{Ag}^n(\beta^+)^{94}\text{Pd}$	Q_{β^+} larger than 17.7 MeV, uncertainty not given										04Mu30 **
$^{95}\text{Kr}-\text{u}$	-60183	150	-60289	20	-0.5	U			GT1	1.5	04Ma.A
$^{95}\text{Kr}-^{85}\text{Rb}_{1.118}$	38330	20				2			MA8	1.0	06De36
$^{95}\text{Rb}-\text{u}$	-70618	86	-70736	22	-0.5	U			Pb1	2.5	89Al33
$^{95}\text{Sr}-^{85}\text{Rb}_{1.118}$	17987	10	17977	6	-1.0	1	39	39 ^{95}Sr	MA4	1.0	02Ra23
$^{95}\text{Mo}-^{85}\text{Rb}_{1.118}$	4457.6	1.0	4456.51	0.13	-1.1	U			JY1	1.0	12Ka13
$\text{C}_7 \text{H}_{11}-^{95}\text{Mo}$	180236.5	3.5	180237.91	0.13	0.2	U			M15	2.5	63Ri07
$^{95}\text{Mo}-\text{u}$	-94161.96	0.38	-94162.56	0.13	-1.6	1	12	12 ^{95}Mo	MS1	1.0	15Gu09
$^{95}\text{Tc}-\text{u}$	-92417	32	-92348	5	2.2	U			GS2	1.0	05Li24 *
$^{95}\text{Rh}-^{85}\text{Rb}_{1.118}$	14515.1	4.5	14517	4	0.4	1	86	86 ^{95}Rh	JY1	1.0	08We10
$^{95}\text{Rh}_{989}-\text{C}_7 \text{H}_{10}$	-161416	11	-161427	4	-1.0	1	14	14 ^{95}Rh	CP1	1.0	11Fa10
$^{95}\text{Mo}-^{87}\text{Rb}_{1.092}$	5012.33	0.44	5012.30	0.13	-0.1	U			MS1	1.0	15Gu09
$^{95}\text{Sr}-^{97}\text{Zr}_{979}$	6529	10	6525	6	-0.4	1	39	39 ^{95}Sr	JY1	1.0	06Ha03
$^{95}\text{Y}-^{97}\text{Zr}_{979}$	-32.4	6.7	-14	7	2.8	B			JY1	1.0	07Ha32
$^{95}\text{Pd}-^{94}\text{Mo}_{1.011}$	20848.9	4.7	20849	3	0.0	2			JY1	1.0	08We10
	20849.1	4.5			-0.0	2			JY1	1.0	08We10 *
$^{95}\text{Mo}-^{94}\text{Mo}$	757	12	753.85	0.10	-0.1	U			M15	2.5	63Ri07

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{93}\text{Rb}-^{95}\text{Rb}_{.653} \ ^{89}\text{Rb}_{.348}$	-1323	25	-1158	15	2.6	U			P21	2.5	82Au01
$^{93}\text{Rb}-^{95}\text{Rb}_{.587} \ ^{90}\text{Rb}_{.413}^x$	-1376	24	-1194	15	3.0	B			P21	2.5	82Au01
$^{94}\text{Rb}-^{95}\text{Rb}_{.792} \ ^{90}\text{Rb}_{.209}^x$	-16	28	195	16	3.0	B			P21	2.5	82Au01 Y
$^{92}\text{Rb}-^{95}\text{Rb}_{.242} \ ^{91}\text{Rb}_{.758}$	80	23	104	10	0.4	U			P21	2.5	82Au01
$^{93}\text{Rb}-^{95}\text{Rb}_{.489} \ ^{91}\text{Rb}_{.511}$	-654	12	-672	13	-0.6	F			P31	2.5	86Au02 *
$^{94}\text{Rb}-^{95}\text{Rb}_{.660} \ ^{92}\text{Rb}_{.341}$	433	15	422	14	-0.3	1	13	13 ^{95}Rb	P31	2.5	86Au02
	462	28			-0.6	U			P31	2.5	86Au02
$^{95}\text{Mo}(\text{n},\alpha)^{92}\text{Zr}$	6405	30	6393.55	0.16	-0.4	U			ILL		75Em04
$^{95}\text{Rb}(\beta-\text{n})^{94}\text{Sr}$	5120	100	4884	20	-2.4	U					84Kr.B
$^{94}\text{Zr}(\text{n},\gamma)^{95}\text{Zr}$	6461.6	1.0	6461.9	0.9	0.3	-					79Ke.D Z
	6357.8	0.3			347.1	F			Bdn		06Fi.A *
	6460.3	0.5			3.3	C			Bdn		08Fi.A *
$^{94}\text{Zr}(\text{d},\text{p})^{95}\text{Zr}$	4223	20	4237.4	0.9	0.7	U			Pit		64Co11
	4237.4	2.0			-0.0	-			SPa		79Bo37
$^{94}\text{Zr}(\text{n},\gamma)^{95}\text{Zr}$	ave.	6461.7	0.9	6461.9	0.9	0.3	1	92	91 ^{95}Zr		average
$^{94}\text{Mo}(\text{n},\gamma)^{95}\text{Mo}$	7367	2	7369.11	0.09	1.1	U					77Ri04
	7369.10	0.10			0.1	1	89	68 ^{94}Mo	MMn		91Is02 Z
	7368.4	0.5			1.4	U			Bdn		06Fi.A
$^{94}\text{Mo}(\text{d},\text{p})^{95}\text{Mo}$	5137	20	5144.54	0.09	0.4	U			Pit		64Co11
$^{95}\text{Pd}(\epsilon\text{p})^{94}\text{Ru}$	5116	300	5329	4	0.7	U					82Ku15 *
$^{95}\text{Rb}(\beta-)^{95}\text{Sr}$	9224	30	9227	20	0.1	o			Gsn		80Bl.A
	9276	100			-0.5	o			Gsn		80De02 *
	9300	30			-2.4	C			Gsn		84Bl.A
	9280	45			-1.2	-			Bwg		87Gr.A
	9272	35			-1.3	-			Gsn		92Pr03
$^{95}\text{Sr}(\beta-)^{95}\text{Y}$	ave.	9275	28		-1.7	1	53	51 ^{95}Rb			average
	6110	150	6091	7	-0.1	U					70Ma.A
	6060	100			0.3	U			Bwg		78St02
	6082	10			0.9	1	52	32 ^{95}Y	Gsn		84Bl.A
	6052	25			1.5	U					90Ma03
$^{95}\text{Y}(\beta-)^{95}\text{Zr}$	4445	9	4452	7	0.8	1	57	56 ^{95}Y	Gsn		80De02 *
$^{95}\text{Zr}(\beta-)^{95}\text{Nb}$	1125	8	1126.3	1.0	0.2	U					54Za05
	1119	5			1.5	U					55Dr43
	1122.7	3.			1.2	1	11	8 ^{95}Zr			74An22 *
$^{95}\text{Nb}(\beta-)^{95}\text{Mo}$	925.5	0.5	925.6	0.5	0.2	1	98	97 ^{95}Nb			63La06 *
$^{95}\text{Tc}(\beta+)^{95}\text{Mo}$	1683	10	1691	5	0.8	-					65Cr04 *
	1693	6			-0.4	-					74An05 *
$^{95}\text{Mo}(\text{p},\text{n})^{95}\text{Tc}$	-2440	30	-2473	5	-1.1	U					57Le27
	-2490	6			2.9	B			Oak		70Ki01
$^{95}\text{Tc}(\beta+)^{95}\text{Mo}$	ave.	1690	5	1691	5	0.0	1	97	97 ^{95}Tc		average
$^{95}\text{Ru}(\beta+)^{95}\text{Tc}$	2558	30	2564	11	0.2	1	12	10 ^{95}Ru			68Pi03 *
$^{95}\text{Rh}(\beta+)^{95}\text{Ru}$	5110	150	5117	10	0.0	U					75We03 *
$^{95}\text{Ag}(\epsilon)^{95}\text{Pd}$	10560	480	10060#	400#	-1.0	D					19Pa16 *
$^{95}\text{Cd}(\epsilon)^{95}\text{Ag}$	10200	1700	12850#	400#	1.6	D					18Pa20 *
$^{95}\text{Tc}-\text{u}$	$M-A=-86066(28)$ keV for mixture gs+m at 38.91 keV										Nub211 **
$^{95}\text{Pd}-^{94}\text{Mo}_{1.011}$	$D_M=22862.1(4.5)$ μu for $^{95}\text{Pd}^m$ at 1875.13 keV; $M-A=-68090.2(4.4)$ keV										Nub211 **
$^{93}\text{Rb}-^{95}\text{Rb}_{.489} \ ^{91}\text{Rb}_{.511}$	F : Rejected by authors										86Au02 **
$^{94}\text{Zr}(\text{n},\gamma)^{95}\text{Zr}$	F : value from 06Fi.A retracted										08Fi.A **
$^{94}\text{Zr}(\text{n},\gamma)^{95}\text{Zr}$	Weak evidence										08Fi.A **
$^{95}\text{Pd}(\epsilon\text{p})^{94}\text{Ru}$	$E_p=4300(300)$ from $^{95}\text{Pd}^m$ at 1875.13 to $^{94}\text{Ru}^m$ at 2644.1 keV										Nub211 **
*	same E_p ; both from figures										82No06 **
$^{95}\text{Rb}(\beta-)^{95}\text{Sr}$	$E_{\beta-}=8595(100)$ to $(3/2^+, 5/2^+)$ level at 680.70, corrected in reference										92Pr03 **
$^{95}\text{Y}(\beta-)^{95}\text{Zr}$	Original error 5 corrected in reference										94Ha.A **
*	$Q_{\beta-}=4417(10)$ given by same group, not used										84Bl.A **
$^{95}\text{Zr}(\beta-)^{95}\text{Nb}$	$E_{\beta-}=887(3)$ to $1/2^-$ level at 235.69 keV										Ens10a **
$^{95}\text{Nb}(\beta-)^{95}\text{Mo}$	$E_{\beta-}=159.7(0.5)$ to $7/2^+$ level at 765.803 keV										Ens10a **
$^{95}\text{Tc}(\beta+)^{95}\text{Mo}$	$E_{\beta+}=700(10)$ 710(6) respectively, from $^{95}\text{Tc}^m$ at 38.91 keV										Nub211 **
$^{95}\text{Ru}(\beta+)^{95}\text{Tc}$	$E_{\beta+}=1200(30)$ to $7/2^+$ level at 336.413 keV										Ens10a **
$^{95}\text{Rh}(\beta+)^{95}\text{Ru}$	$E_{\beta+}=3150(150)$ to $7/2^+$ level at 941.79 keV										Ens10a **
$^{95}\text{Ag}(\epsilon)^{95}\text{Pd}$	Trends from Mass Surface TMS suggest ^{95}Ag 500 keV more bound										GAu **
$^{95}\text{Cd}(\epsilon)^{95}\text{Ag}$	Trends from Mass Surface TMS suggest ^{95}Cd 2650 keV less bound										GAu **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{96}\text{Kr}-u$	-57002	61	-56986	21	0.3	1	12	12	^{96}Kr	TR1	1.0	20Sm.1
$^{96}\text{Kr}-^{85}\text{Rb}_{1.129}$	42606	22	42604	21	-0.1	1	88	88	^{96}Kr	MA8	1.0	10Na13
$^{96}\text{Rb}-u$	-65508	43	-65867	4	-3.3	F				Pb1	2.5	89Al33
$^{96}\text{Zr}-^{87}\text{Rb}_{1.103}$	8451.49	0.34	8451.50	0.12	0.0	1	13	13	^{96}Zr	MS1	1.0	15Gu09
$\text{C}_7\text{H}_{12}-^{96}\text{Zr}$	185628	6	185622.77	0.12	-0.3	U				M15	2.5	63Ri07
$^{96}\text{Zr}-u$	-91691	43	-91722.38	0.12	-0.7	U				JY0	1.0	04Ri12
	-91722.60	0.17			1.3	1	52	52	^{96}Zr	MS1	1.0	15Gu09
$^{96}\text{Mo}-^{85}\text{Rb}_{1.129}$	4265.7	1.1	4264.15	0.13	-1.4	U				JY1	1.0	12Ka13
$^{96}\text{Mo}-^{87}\text{Rb}_{1.103}$	4848.96	0.43	4848.65	0.13	-0.7	U				MS1	1.0	15Gu09
$\text{C}_7\text{H}_{12}-^{96}\text{Mo}$	189226.9	3.0	189225.61	0.13	-0.2	U				M15	2.5	63Ri07
$^{96}\text{Mo}-u$	-95324.94	0.47	-95325.23	0.13	-0.6	U				MS1	1.0	15Gu09
$^{96}\text{Tc}-u$	-92192	32	-92133	6	1.8	U				GS2	1.0	05Li24
$\text{C}_7\text{H}_{12}-^{96}\text{Ru}$	186304.6	3.8	186311.47	0.18	0.7	U				M16	2.5	63Da10
$^{96}\text{Pd}-u$	-81853	41	-81786	5	1.6	U				GR1	1.0	19An10
$^{96}\text{Rb}-^{88}\text{Rb}_{1.091}$	30887.7	3.6	30888	4	0.1	1	100	100	^{96}Rb	JY1	1.0	07Ra23
$^{96}\text{Zr}\text{ }^{35}\text{Cl}-^{94}\text{Zr}\text{ }^{37}\text{Cl}$	4929	3	4915.21	0.22	-1.1	U				H13	4.0	63Ba20
$^{96}\text{Mo}\text{ }^{35}\text{Cl}-^{94}\text{Mo}\text{ }^{37}\text{Cl}$	2539	2	2541.30	0.13	0.3	U				H11	4.0	63Bi12
$^{96}\text{Pd}-^{94}\text{Mo}_{1.021}$	15123.4	4.5				2				JY1	1.0	08We10
$^{96}\text{Sr}-^{97}\text{Zr}_{.990}$	9868	10	9865	9	-0.3	1	83	83	^{96}Sr	JY1	1.0	06Ha03
$^{96}\text{Y}-^{97}\text{Zr}_{.990}$	4053.7	6.8	4055	7	0.2	1	92	92	^{96}Y	JY1	1.0	07Ha32
$^{96}\text{Y}^m-^{97}\text{Zr}_{.990}$	5708.1	6.7				2				JY1	1.0	07Ha32
$^{96}\text{Mo}-^{97}\text{Mo}_{.990}$	-2280.5	5.8	-2281.96	0.17	-0.3	U				JY1	1.0	06Ka48
$^{96}\text{Zr}-^{96}\text{Nb}$	176.02	0.13	176.03	0.11	0.1	1	68	63	^{96}Nb	JY1	1.0	16Al03
$^{96}\text{Zr}-^{96}\text{Mo}$	3602.919	0.092	3602.85	0.08	-0.8	1	75	46	^{96}Mo	JY1	1.0	16Al03
$^{96}\text{Nb}-^{96}\text{Mo}$	3426.80	0.17	3426.82	0.11	0.1	1	46	37	^{96}Nb	JY1	1.0	16Al03
$^{96}\text{Ru}-^{96}\text{Mo}$	2914.14	0.13	2914.14	0.13	0.0	1	100	100	^{96}Ru	SH1	1.0	11El04
$^{96}\text{Mo}-^{95}\text{Mo}$	-1161	12	-1162.67	0.05	-0.1	U				M15	2.5	63Ri07
$^{93}\text{Rb}-^{96}\text{Rb}_{.554}\text{ }^{89}\text{Rb}_{.448}$	-2210	27	-2022	8	2.8	U				P21	2.5	82Au01
$^{95}\text{Rb}-^{96}\text{Rb}_{.848}\text{ }^{89}\text{Rb}_{.152}$	-1590	30	-1442	20	2.0	U				P21	2.5	82Au01
$^{94}\text{Rb}-^{96}\text{Rb}_{.699}\text{ }^{89}\text{Rb}_{.302}$	-1250	30	-999	4	3.3	B				P21	2.5	82Au01
$^{94}\text{Rb}-^{96}\text{Rb}_{.588}\text{ }^{91}\text{Rb}_{.413}$	-380	25	-378	4	0.0	U				P21	2.5	82Au01
$^{95}\text{Rb}-^{96}\text{Rb}_{.742}\text{ }^{92}\text{Rb}_{.258}$	-1116	27	-1074	20	0.6	U				P21	2.5	82Au01
	-1143	16			1.7	1	26	26	^{95}Rb	P31	2.5	86Au02
$^{96}\text{Zr}(\text{d},\alpha)^{94}\text{Y}$	7609	20	7623	6	0.7	1	10	10	^{94}Y	Grn		74Gi09
$^{96}\text{Zr}(\text{p},\text{t})^{94}\text{Zr}$	-5825	10	-5830.37	0.20	-0.5	U				Oak		71Ba43
$^{96}\text{Ru}(\text{p},\text{t})^{94}\text{Ru}$	-11165	10	-11158	3	0.7	U				Oak		71Ba01
$^{96}\text{Zr}(\text{t},\alpha)^{95}\text{Y}$	8294	20	8294	7	-0.0	1	11	11	^{95}Y	LAl		83FI06
$^{96}\text{Zr}(\text{p},\text{d})^{95}\text{Zr}$	-5440	20	-5625.7	0.9	-9.3	B				Bld		67St24
	-5630	10			0.4	U				Oak		71Ba43
$^{96}\text{Zr}(\text{d},\text{t})^{95}\text{Zr}$	-1603	20	-1593.0	0.9	0.5	U				Pit		64Co11
	-1595.8	2.8			1.0	U				SPa		79Bo37
$^{96}\text{Mo}(\text{t},\alpha)^{95}\text{Nb}$	10524	20	10516.3	0.5	-0.4	U				LAl		83FI06
$^{95}\text{Mo}(\text{n},\gamma)^{96}\text{Mo}$	9154.2	0.5	9154.34	0.05	0.3	U						70He27
	9154.32	0.05			0.3	1	96	67	^{95}Mo	MMn		91Is02
	9153.90	0.20			2.2	U				Bdn		06Fi.A
$^{96}\text{Mo}(\text{d},\text{t})^{95}\text{Mo}$	-2923	20	-2897.10	0.05	1.3	U				Pit		64Co11
$^{96}\text{Ru}(\text{p},\text{d})^{95}\text{Ru}$	-8470	10	-8469	10	0.1	1	90	90	^{95}Ru	Oak		71Ba01
$^{96}\text{Ag}(\text{ep})^{95}\text{Rh}$	6540	90				2						03Ba39
$^{96}\text{Rb}(\beta^-)^{96}\text{Sr}$	10800	220	11564	9	3.5	B						79Pe17
	11303	250			1.0	o				Gsn		80De02
	11547	100			0.2	U				Trs		82Br23
	11553	45			0.2	U				Gsn		84Bl.A
	11590	80			-0.3	U				Bwg		87Gr.A
	11709	40			-3.6	B				Gsn		92Pr03
$^{96}\text{Sr}(\beta^-)^{96}\text{Y}$	5332	30	5412	10	2.7	F						79Pe17
	5413	22			-0.1	-				Gsn		80De02
	5345	50			1.3	U				Bwg		87Gr.A
	5354	40			1.4	-						90Ma03
ave.	5399	19			0.6	1	25	17	^{96}Sr			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{96}\text{Y}(\beta^-)^{96}\text{Zr}$	7120	50	7109	6	-0.2	U			Gsn		80De02 *
	7030	70			1.1	U			Bwg		87Gr.A
	7067	30			1.4	U					90Ma03 *
$^{96}\text{Y}^m(\beta^-)^{96}\text{Zr}$	8030	150	8649	6	4.1	C			Bwg		87Gr.A
	8600	200			0.2	U					88St.A
	8237	21			19.6	C			Bwg		92Gr.A
$^{96}\text{Nb}(\beta^-)^{96}\text{Mo}$	3186.8	3.2	3192.06	0.11	1.6	U					68An03 *
$^{96}\text{Mo}(\text{p,n})^{96}\text{Tc}$	-3760	10	-3756	5	0.4	2					74Do09
	-3754	6			-0.3	2					78Ke10
$^{96}\text{Rh}(\beta^+)^{96}\text{Ru}$	6472	200	6393	10	-0.4	U					75Gu01 *
$^{96}\text{Ru}(\text{p,n})^{96}\text{Rh}$	-7175	10				2					70As08 Z
$^{96}\text{Pd}(\beta^+)^{96}\text{Rh}$	3450	150	3504	11	0.4	U					85Ry02 *
$^{96}\text{Cd}(\epsilon)^{96}\text{Ag}$	8480	460	8940#	400#	1.0	D					19Pa16 *
* $^{96}\text{Rb}-\text{u}$	F : possibly isomeric mixture										92Al.B **
* $^{96}\text{Tc}-\text{u}$	$M-A=-85860(28)$ keV for mixture gs+m at 34.23 keV										Nub211 **
* $^{96}\text{Ag}(\epsilon\text{p})^{95}\text{Rh}$	Original 6430(60) corrected by -110 keV for mixture of two β -decaying ^{96}Ag states, to two isomeric states in ^{95}Rh										03Ba39 **
*											03Ba39 **
* $^{96}\text{Rb}(\beta^-)^{96}\text{Sr}$	$E_{\beta-}=10894(40)$ to 2^+ level at 814.93 keV										Ens08a **
* $^{96}\text{Sr}(\beta^-)^{96}\text{Y}$	$E_{\beta-}=4400(30)$ to 1^+ level at 931.70 keV, and other $E_{\beta-}$										Ens08a **
* $^{96}\text{Sr}(\beta^-)^{96}\text{Y}$	F : all other results of reference are strongly conflicting										79Pe17 **
* $^{96}\text{Sr}(\beta^-)^{96}\text{Y}$	Original error 20 corrected in reference										94Ha.A **
*	$Q_{\beta-}=5362(10)$ given by same group, not used										84Bl.A **
* $^{96}\text{Y}(\beta^-)^{96}\text{Zr}$	$Q_{\beta-}=7079(15)$ given by same group, not used										84Bl.A **
* $^{96}\text{Y}(\beta^-)^{96}\text{Zr}$	$E_{\beta-}=5326(36)$ to 2^+ level at 1750.497 keV, and other $E_{\beta-}$										Ens08a **
* $^{96}\text{Nb}(\beta^-)^{96}\text{Mo}$	$E_{\beta-}=748.4(3.1)$ to 5^+ level at 2438.477 keV										Ens08a **
* $^{96}\text{Rh}(\beta^+)^{96}\text{Ru}$	$E_{\beta+}=3300(200)$ to 6^+ level at 2149.74 keV										Ens08a **
* $^{96}\text{Pd}(\beta^+)^{96}\text{Rh}$	$p^+=0.257(0.03)$ to 1^+ level at 1274.78 keV										Ens08a **
* $^{96}\text{Cd}(\epsilon)^{96}\text{Ag}$	Decays to the daughter excited state at 0#(100#) keV										19Pa16 **
* $^{96}\text{Cd}(\epsilon)^{96}\text{Ag}$	Trends from Mass Surface TMS suggest ^{96}Cd 460 keV less bound										HWJ199 **
$^{97}\text{Kr}-^{86}\text{Kr}_{1.128}$	49920	140				2			MA9	1.0	10Na13
$^{97}\text{Rb}-\text{u}$	-62512	64	-62822.9	2.1	-1.9	U			Pb1	2.5	89Al33
$^{97}\text{Rb}-^{88}\text{Rb}_{1.102}$	34908.0	5.7	34907.3	2.1	-0.1	1	13	13	^{97}Rb	1.0	07Ra23
$^{97}\text{Rb}-^{85}\text{Rb}_{1.141}$	37824.9	2.2	37825.0	2.1	0.1	1	87	87	^{97}Rb	1.0	12Si10 *
$^{97}\text{Sr}-^{85}\text{Rb}_{1.141}$	27022.9	3.9	27024	4	0.2	1	87	87	^{97}Sr	1.0	12Si10 *
$^{97}\text{Sr}-\text{u}$	-73599	99	-73624	4	-0.1	U			GT2	2.5	08Kn.A
$^{97}\text{Mo}-^{85}\text{Rb}_{1.141}$	6666.9	1.2	6664.82	0.18	-1.7	U			JY1	1.0	12Ka13
$^{97}\text{Mo}-^{87}\text{Rb}_{1.115}$	7280.86	0.39	7280.61	0.18	-0.6	1	21	21	^{97}Mo	1.0	15Gu09
$\text{C}_5\text{H}_5\text{O}_2-^{97}\text{Mo}$	122937.6	2.3	122937.49	0.18	-0.0	U			M15	2.5	63Ri07
$^{97}\text{Mo}-\text{u}$	-93982.95	0.36	-93983.10	0.18	-0.4	1	24	24	^{97}Mo	1.0	15Gu09
$^{97}\text{Ru}-\text{u}$	-92471	30	-92454.2	3.0	0.6	U			GS2	1.0	05Li24
$^{97}\text{Pd}-^{85}\text{Rb}_{1.141}$	17119.9	5.2				2			JY1	1.0	09El08
$^{97}\text{Pd}-\text{u}$	-83511	40	-83528	5	-0.4	U			GR1	1.0	19An10
$^{97}\text{Ag}-\text{u}$	-76118.6	12.9				2			GR1	1.0	20Ho03
$^{97}\text{Ag}^m-\text{u}$	-75455.1	38.6				2			GR1	1.0	20Ho03 *
$^{97}\text{Mo}-^{35}\text{Cl}-^{95}\text{Mo}-^{37}\text{Cl}$	3138	2	3129.59	0.19	-1.1	U			H11	4.0	63Bi12
$^{97}\text{Sr}-^{97}\text{Zr}$	15416	10	15412	4	-0.4	1	13	13	^{97}Sr	1.0	06Ha03
$^{97}\text{Y}-^{97}\text{Zr}$	7322.9	7.2				2			JY1	1.0	07Ha32 *
$^{97}\text{Mo}-^{96}\text{Mo}$	1346	12	1342.13	0.18	-0.1	U			M15	2.5	63Ri07
$^{94}\text{Rb}-^{97}\text{Rb}_{.485}-^{91}\text{Rb}_{.516}$	-21	25	-65	5	-0.7	U			P21	2.5	82Au01 Y
$^{96}\text{Rb}-^{97}\text{Rb}_{.792}-^{92}\text{Rb}_{.209}$	650	30	620	4	-0.4	U			P21	2.5	82Au01
$^{95}\text{Rb}-^{97}\text{Rb}_{.490}-^{93}\text{Rb}_{.511}$	-165	25	-107	21	0.9	1	11	10	^{95}Rb	2.5	82Au01
$^{96}\text{Rb}-^{97}\text{Rb}_{.742}-^{93}\text{Rb}_{.258}$	848	19	803	4	-1.0	U			P31	2.5	86Au02
$^{96}\text{Zr}(\text{n},\gamma)^{97}\text{Zr}$	5574	5	5569.15	0.04	-1.0	U					77Ba33
	5575.1	0.4			-14.9	C			Bdn		06Fi.A
	5569.15	0.04			0.0	1	100	100	^{97}Zr		18Rz01

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{96}\text{Zr}(\text{d,p})^{97}\text{Zr}$	3338	20	3344.58	0.04	0.3	U			Pit		64Co11
$^{96}\text{Mo}(\text{n},\gamma)^{97}\text{Mo}$	6821.1	1.0	6821.13	0.16	0.0	U					73De39
	6820	2			0.6	U					77Ri04
	6821.15	0.25			-0.1	-			MMn		91Is02
	6821.5	0.4			-0.9	-			Bdn		06Fi.A
$^{96}\text{Mo}(\text{d,p})^{97}\text{Mo}$	4582	20	4596.56	0.16	0.7	U			Pit		64Co11
$^{96}\text{Mo}(\text{n},\gamma)^{97}\text{Mo}$	ave. 6821.25	0.21	6821.13	0.16	-0.6	1	59	44 ^{97}Mo			average
$^{96}\text{Mo}(\text{}^3\text{He,d})^{97}\text{Tc}$	229	8	225	4	-0.5	-			ANL		74Co27
	220	8			0.6	-			Pit		74Co27
	ave. 225	6			0.1	1	53	53 ^{97}Tc			average
$^{96}\text{Ru}(\text{d,p})^{97}\text{Ru}$	5886	3	5886.9	2.8	0.3	2			Can		77Ho02
	5892	7			-0.7	2			ANL		77Me04
$^{97}\text{Rb}(\beta^-)^{97}\text{Sr}$	10020	50	10062	4	0.8	U			Gsn		80De02
	10450	30			-12.9	C			Gsn		84Bl.A
	10440	60			-6.3	C			Bwg		87Gr.A
	10462	40			-10.0	B			Gsn		92Pr03
$^{97}\text{Sr}(\beta^-)^{97}\text{Y}$	7452	40	7535	8	2.1	U			Gsn		84Bl.A
	7420	80			1.4	o			Bwg		87Gr.A
	7480	18			3.0	C			Bwg		92Gr.A
$^{97}\text{Y}(\beta^-)^{97}\text{Zr}$	6702	25	6821	7	4.8	C			Gsn		84Bl.A
	6640	70			2.6	o			Bwg		87Gr.A
	6689	13			10.2	C			Bwg		92Gr.A
$^{97}\text{Zr}(\beta^-)^{97}\text{Nb}$	2657.3	6.	2666	4	1.5	1	50	50 ^{97}Nb			74Ra.A
$^{97}\text{Nb}(\beta^-)^{97}\text{Mo}$	1933.1	6.	1942	4	1.5	1	50	50 ^{97}Nb			74Ra.A
$^{97}\text{Mo}(\text{p,n})^{97}\text{Tc}$	-1128	9	-1103	4	2.8	B			Oak		70Ki01
	-1102	6			-0.1	1	47	47 ^{97}Tc	ANL		74Co27
$^{97}\text{Ru}(\beta^+)^{97}\text{Tc}$	1150	100	1104	5	-0.5	U					70Ho01
$^{97}\text{Rh}(\beta^+)^{97}\text{Ru}$	3533	50	3520	40	-0.2	3					62Ba28
	3513	50			0.2	3					62Ch21
$^{97}\text{Pd}(\beta^+)^{97}\text{Rh}$	4790	300	4790	40	0.0	U					80Go11
$^{97}\text{Ag}(\beta^+)^{97}\text{Pd}$	6980	110	6902	13	-0.7	U					99Hu10
$^{97}\text{Cd}(\epsilon)^{97}\text{Ag}$	10170	420				3					19Pa16
$^{97}\text{In}(\epsilon)^{97}\text{Cd}$	10000	3000	13340#	580#	1.1	U					18Pa20
$^{97}\text{Rb}-^{85}\text{Rb}_{1.141}$	$D_M=37825.7(2.2) \mu\text{u}$ $M-A=-58518.5(2.1)\text{keV}$ corrected for e^- binding=-703eV										12Si10
$^{97}\text{Sr}-^{85}\text{Rb}_{1.141}$	$D_M=27023.5(3.9) \mu\text{u}$ $M-A=-68580.7(3.7)\text{keV}$ corrected for e^- binding=-553eV										12Si10
$^{97}\text{Ag}^m-\text{u}$	reconstructed from $E_x=618(38) \text{keV}$										20Ho03
$^{97}\text{Y}-^{97}\text{Zr}$	$D_M=8039.5(7.2) \mu\text{u}$ for $^{97}\text{Y}^m$ at 667.52(0.23) keV; $M-A=-75453.9(6.7) \text{keV}$										Nub211
$^{97}\text{Y}(\beta^-)^{97}\text{Zr}$	$E_{\beta^-}=6645(70)$; and 7280(150) from $^{97}\text{Y}^m$ at 667.52 keV										Nub211
$^{97}\text{Y}(\beta^-)^{97}\text{Zr}$	$E_{\beta^-}=6688(13)$; and 7361(26) from $^{97}\text{Y}^m$ at 667.52 keV										Nub211
$^{97}\text{Zr}(\beta^-)^{97}\text{Nb}$	$E_{\beta^-}=1914(2)$ to $1/2^-$ level at 743.35 keV; error increased										Ens104
$^{97}\text{Nb}(\beta^-)^{97}\text{Mo}$	$E_{\beta^-}=1275(2)$ to $7/2^+$ level at 658.13 keV; error increased										Ens104
$^{97}\text{Ru}(\beta^+)^{97}\text{Tc}$	$p^+ < 0.0001$ (see fig 1), decay to $7/2^+$ level at 970.03 keV										Ens104
$^{97}\text{Rh}(\beta^+)^{97}\text{Ru}$	$E_{\beta^+}=2090(50)$ 2070(50) respectively, to $7/2^+$ level at 421.54 keV										Ens104
$^{97}\text{Pd}(\beta^+)^{97}\text{Rh}$	$E_{\beta^+}=3500(300)$ to $7/2^+$ level at 265.36 keV										Ens104
$^{98}\text{Rb}-^{85}\text{Rb}_{1.153}$	43331	32	43339	17	0.2	1	29	29 ^{98}Rb	TT1	1.0	12Si10
$^{98}\text{Rb}-\text{u}$	-58359	29	-58368	17	-0.3	-			MA8	1.0	13Ma81
	-58370	29			0.1	-			TT1	1.0	16KI04
	ave. -58365	21			-0.2	1	71	71 ^{98}Rb			average
$^{98}\text{Rb}^m-\text{u}$	-58289	22				2			TT1	1.0	12Si10
$^{98}\text{Sr}-^{85}\text{Rb}_{1.153}$	30396.7	4.3	30399	3	0.6	-			TT1	1.0	12Si10
	30405.3	7.2			-0.9	-			TT1	1.0	16KI04
	ave. 30399	4			0.0	1	88	88 ^{98}Sr			average
$^{98}\text{Zr}-\text{u}$	-87247	43	-87260	9	-0.3	U			JY0	1.0	04Ri12
$^{98}\text{Mo}-^{85}\text{Rb}_{1.153}$	7104.1	5.7	7110.04	0.19	1.0	U			MA8	1.0	11He10
	7111.6	1.3			-1.2	U			JY1	1.0	12Ka13

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{98}\text{Mo}-^{87}\text{Rb}_{1.126}$	7665.96	0.64	7666.34	0.19	0.6	U			MS1	1.0	15Gu09
$\text{C}_5\text{H}_6\text{O}_2-^{98}\text{Mo}$	131375.4	2.8	131375.82	0.19	0.1	U			M15	2.5	63Ri07
$^{98}\text{Mo}-\text{u}$	-94596.21	0.53	-94596.39	0.19	-0.3	1	12	12 ^{98}Mo	MS1	1.0	15Gu09
$\text{C}_7\text{H}_{14}-^{98}\text{Ru}$	204263.5	2.9	204264	7	0.0	1	92	92 ^{98}Ru	M16	2.5	63Da10
$^{98}\text{Rh}-\text{u}$	-89300	37	-89292	13	0.2	U			GS2	1.0	05Li24 *
$^{98}\text{Pd}-^{85}\text{Rb}_{1.153}$	14404.5	5.1	14405	5	0.1	1	100	100 ^{98}Pd	JY1	1.0	09El08
$^{98}\text{Ag}-^{85}\text{Rb}_{1.153}$	23283	40	23270	40	-0.4	1	78	78 ^{98}Ag	MA8	1.0	11He10
$^{98}\text{Mo}-^{35}\text{Cl}-^{96}\text{Mo}-^{37}\text{Cl}$	3690	2	3678.96	0.20	-1.4	U			H11	4.0	63Bi12
$^{98}\text{Sr}-^{97}\text{Zr}_{1.010}$	18620	10	18619	3	-0.1	1	12	12 ^{98}Sr	JY1	1.0	06Ha03
$^{98}\text{Y}-^{97}\text{Zr}_{1.010}$	12321.4	8.5				2			JY1	1.0	07Ha32
$^{98}\text{Zr}-^{97}\text{Zr}_{1.010}$	2668	10	2667	9	-0.1	1	82	82 ^{98}Zr	JY1	1.0	06Ha03
$^{98}\text{Mo}-^{97}\text{Mo}_{1.010}$	327.9	5.8	326.54	0.07	-0.2	U			JY1	1.0	06Ka48
$^{98}\text{Y}^n-^{98}\text{Y}$	499.99	0.78				3			JY1	1.0	17Ur03
$^{98}\text{Mo}-^{97}\text{Mo}$	-614	12	-613.29	0.07	0.0	U			M15	2.5	63Ri07
$^{94}\text{Rb}-^{98}\text{Rb}_{.411}-^{91}\text{Rb}_{.590}$	-290	40	-347	8	-0.6	U			P21	2.5	82Au01 Y
$^{97}\text{Rb}-^{98}\text{Rb}_{.792}-^{93}\text{Rb}_{.209}$	-250	60	-281	13	-0.2	U			P21	2.5	82Au01
$^{96}\text{Rb}-^{98}\text{Rb}_{.490}-^{94}\text{Rb}_{.511}$	330	30	322	9	-0.1	U			P21	2.5	82Au01 Y
$^{97}\text{Rb}-^{98}\text{Rb}_{.660}-^{95}\text{Rb}_{.340}$	-300	50	-233	13	0.5	U			P21	2.5	82Au01
	-232	27			-0.0	U			P31	2.5	86Au02
$^{96}\text{Zr}(\text{t,p})^{98}\text{Zr}$	3508	20	3504	8	-0.2	1	18	18 ^{98}Zr	LAL		69B101
$^{96}\text{Zr}(\text{t,He,p})^{98}\text{Nb}$	5728	5				2			Phi		75Me13
$^{96}\text{Ru}(\text{t,}^{16}\text{O},^{14}\text{C})^{98}\text{Pd}$	-12529	20	-12516	5	0.6	U			BNL		82Th01
$^{98}\text{Mo}(\text{t},\alpha)^{97}\text{Nb}$	10019	20	10012	4	-0.4	U			LAL		83FI06
$^{97}\text{Mo}(\text{n},\gamma)^{98}\text{Mo}$	8642.4	0.5	8642.60	0.06	0.4	U					71He10
	8642.60	0.07			-0.0	-			MMn		91Is02 Z
	8642.57	0.18			0.2	-			Bdn		06Fi.A
$^{98}\text{Mo}(\text{d,t})^{97}\text{Mo}$	-2379	20	-2385.37	0.06	-0.3	U			Pit		64Co11
$^{97}\text{Mo}(\text{n},\gamma)^{98}\text{Mo}$	ave. 8642.60	0.07	8642.60	0.06	0.0	1	98	87 ^{98}Mo			average
$^{97}\text{Mo}(\text{t},\alpha)^{98}\text{Tc}$	680	8	683	3	0.4	-			ANL		74Co27
	686	10			-0.3	-			McM		76Ma16
	ave. 682	6			0.1	1	29	29 ^{98}Tc			average
$^{98}\text{Rb}(\beta^-)^{98}\text{Sr}$	11200	110	12053	16	7.8	B					79Pe17
	12343	150			-1.9	U			Trs		82Br23
	12519	60			-7.8	C			Gsn		84Bl.A
	12270	30			-7.2	C			McG		84Ia.A
	12440	75			-5.2	C			Bwg		87Gr.A
	12380	65			-5.0	B			Gsn		92Pr03
$^{98}\text{Rb}^m(\beta^-)^{98}\text{Sr}$	12710	120	12127	21	-4.9	C			Bwg		87Gr.A
$^{98}\text{Sr}(\beta^-)^{98}\text{Y}$	5730	40	5866	9	3.4	C					79Pe17
	5821	10			4.5	C			Gsn		84Bl.A
	5815	40			1.3	U			Bwg		87Gr.A
$^{98}\text{Y}(\beta^-)^{98}\text{Zr}$	8974	100	8993	12	0.2	U					79Pe17 *
	8780	30			7.1	C			Gsn		84Bl.A
	8840	55			2.8	C			Bwg		87Gr.A *
	8963	41			0.7	U					88Ma.A
	8830	17			9.6	C			Bwg		92Gr.A
	9062	27			-2.6	C			Bwg		92Gr.A *
$^{98}\text{Zr}(\beta^-)^{98}\text{Nb}$	2300	200	2243	10	-0.3	U					76He10
$^{98}\text{Nb}(\beta^-)^{98}\text{Mo}$	4300	200	4591	5	1.5	U					66Gu05
	4300	200			1.5	U					67Hu07
	4800	200			-1.0	U					76He10
	4580	100			0.1	U			Bwg		78St02
$^{98}\text{Mo}(\text{p,n})^{98}\text{Tc}$	-2458	10	-2466	3	-0.8	1	11	11 ^{98}Tc	ANL		74Co27
$^{98}\text{Tc}(\beta^-)^{98}\text{Ru}$	1795	22	1793	7	-0.1	1	11	8 ^{98}Ru			73Ok.A *
$^{98}\text{Rh}(\beta^+)^{98}\text{Ru}$	5120	100	5050	10	-0.7	U					72Ba37 *
	5151	50			-2.0	U					94Ba06 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{98}\text{Ru}(\text{p,n})^{98}\text{Rh}$	−5832	10				2					70As08 Z
$^{98}\text{Ag}(\beta^+)^{98}\text{Pd}$	8420	150	8250	30	−1.1	U					79Ve.A *
	8200	70			0.8	1	22	22 ^{98}Ag			00Hu17
$^{98}\text{Cd}(\beta^+)^{98}\text{Ag}$	5330	140	5430	40	0.7	U					92PI01
$^{98}\text{Cd}(\epsilon)^{98}\text{Ag}$	5430	40				2					01St.A
$^{98}\text{In}(\epsilon)^{98}\text{Cd}$	12930	400	13730#	300#	2.0	D					19Pa16 *
$^{98}\text{Rb}-^{85}\text{Rb}_{1.153}$	Original $D_M=43394.0(3.7) \mu\text{u}$ $M-A=-54317.7(3.4)$ keV corrected for gs+m mixture										16K104 **
$^{98}\text{Rb}-^{85}\text{Rb}_{1.153}$	Original $M-A=-54309.4(4.0)$ corrected for isom mixture $E_x=80(30)$ $R=0.65(0.15)$										16K104 **
$^{98}\text{Rb}-\text{u}$	Original $M-A=-54319.5(5.5)$ corrected for isom mixture $E_x=80(30)$ $R=0.65(0.15)$										16K104 **
$^{98}\text{Rb}^m-\text{u}$	Data re-analysis										16K104 **
$^{98}\text{Sr}-^{85}\text{Rb}_{1.153}$	$D_M=30397.3(4.3) \mu\text{u}$ $M-A=-66424.0(4.0)$ keV corrected for e^- binding= -529eV										12Si10 **
$^{98}\text{Sr}-^{85}\text{Rb}_{1.153}$	Former measured using 15^+ ion; latter with 10^+ ion										16K104 **
$^{98}\text{Rh}-\text{u}$	$M-A=-83154(30)$ keV for mixture gs+m at $56.3(1.0)$ keV										Nub211 **
$^{98}\text{Y}(\beta^-)^{98}\text{Zr}$	$E_{\beta^-}=4810(100)$ to 1^- level at 4164.60 keV										Ens035 **
$^{98}\text{Y}(\beta^-)^{98}\text{Zr}$	also $Q_{\beta^-}=9780(200)$ from $^{98}\text{Y}^m$ at 170.76 keV										87Gr.A **
$^{98}\text{Y}(\beta^-)^{98}\text{Zr}$	$Q_{\beta^-}=9233(27)$ from $^{98}\text{Y}^m$ at 170.76 keV										HWJ189 **
$^{98}\text{Tc}(\beta^-)^{98}\text{Ru}$	$E_{\beta^-}=397(22)$ to 4^+ level at 1397.82 keV										Ens035 **
$^{98}\text{Rh}(\beta^+)^{98}\text{Ru}$	$E_{\beta^+}=3450(100)$ to 2^+ level at 652.44 keV, and others										Ens035 **
$^{98}\text{Rh}(\beta^+)^{98}\text{Ru}$	$E_{\beta^+}=3476(50)$ to 2^+ level at 652.44 keV										Ens035 **
$^{98}\text{Ag}(\beta^+)^{98}\text{Pd}$	$Q_{\beta^+}=6880(150)$ to 4^+ level at 1541.40 keV										Ens035 **
$^{98}\text{In}(\epsilon)^{98}\text{Cd}$	Trends from Mass Surface TMS suggest ^{98}In 600 keV less bound										HWJ199 **
$^{99}\text{Rb}-^{85}\text{Rb}_{1.165}$	47885.1	4.8	47884	4	−0.2	2			MA8	1.0	13Ma81
	47880	10			0.4	2			TT1	1.0	16K104
$^{99}\text{Sr}-^{85}\text{Rb}_{1.165}$	35663	21	35649	5	−0.7	o			TT1	1.0	12Si10 *
	35644.5	7.0			0.6	1	53	53 ^{99}Sr	TT1	1.0	16K104
$^{99}\text{Zr}-\text{u}$	−83323	19	−83325	11	−0.1	1	35	35 ^{99}Zr	JY0	1.0	04Ri12
$\text{C}_7\text{H}_{15}-^{99}\text{Ru}$	211442.8	3.0	211445.2	0.4	0.3	U			M16	2.5	63Da10
$^{99}\text{Ag}-^{85}\text{Rb}_{1.165}$	20401.0	8.5	20411	7	1.1	2			SH1	1.0	07Ma92
	20427	11			−1.5	2			MA8	1.0	11He10
$^{99}\text{Cd}-^{85}\text{Rb}_{1.165}$	27690.8	1.7				2			MA8	1.0	09Br09
$^{99}\text{Pd}-^{96}\text{Mo}_{1.031}$	10052.8	5.5	10053	5	0.1	1	99	99 ^{99}Pd	JY1	1.0	09El08
$^{99}\text{Sr}-^{97}\text{Zr}_{1.021}$	23794.1	7.4	23790	5	−0.6	1	47	47 ^{99}Sr	JY1	1.0	06Ha03
$^{99}\text{Y}-^{97}\text{Zr}_{1.021}$	15066.8	7.1				2			JY1	1.0	07Ha32
$^{99}\text{Zr}-^{97}\text{Zr}_{1.021}$	7580	14	7581	11	0.1	1	65	65 ^{99}Zr	JY1	1.0	06Ha03
$^{99}\text{Ru}-^{98}\text{Ru}$	652	11	644	7	−0.3	U			M16	2.5	63Da10
$^{97}\text{Rb}-^{99}\text{Rb}_{.653}$ $^{93}\text{Rb}_{.348}$	100	100	135	4	0.1	U			P21	2.5	82Au01
$^{98}\text{Rb}-^{99}\text{Rb}_{.742}$ $^{95}\text{Rb}_{.258}$	690	180	562	17	−0.3	U			P21	2.5	82Au01
$^{97}\text{Rb}-^{99}\text{Rb}_{.490}$ $^{95}\text{Rb}_{.511}$	350	60	200	11	−1.0	U			P31	2.5	86Au02
$^{99}\text{Ru}(\text{n},\alpha)^{96}\text{Mo}$	6822	5	6815.9	0.4	−1.2	U					01Wa50
$^{96}\text{Ru}(^{16}\text{O},^{13}\text{C})^{99}\text{Pd}$	−11723	20	−11760	5	−1.8	U			BNL		82Th01
$^{98}\text{Mo}(\text{n},\gamma)^{99}\text{Mo}$	5927.7	1.	5925.44	0.15	−2.3	U					73De39
	5927	1			−1.6	U					74Er.A
	5924.6	0.6			1.4	U					76Ch02
	5923	2			1.2	U					77Ri04
	5925.42	0.15			0.2	1	100	99 ^{99}Mo	MMn		91Is02 Z
	5927.7	0.5			−4.5	C			Bdn		06Fi.A
$^{98}\text{Mo}(\text{d,p})^{99}\text{Mo}$	3687	20	3700.88	0.15	0.7	U			Pit		64Co11
$^{98}\text{Mo}(^3\text{He,d})^{99}\text{Tc}$	1010	20	1007.4	0.9	−0.1	U			McM		77Ch06
$^{99}\text{Tc}(\text{p,d})^{98}\text{Tc}$	−6740	5	−6742	3	−0.5	−					76Si06
	−6755	9			1.4	−			Bld		77Em02
ave.	−6744	4			0.3	1	59	57 ^{98}Tc			average
$^{99}\text{Rb}(\beta^-)^{99}\text{Sr}$	11340	120	11397	6	0.5	U			McG		84Ia.A
	10960	130			3.4	C			Bwg		87Gr.A
$^{99}\text{Sr}(\beta^-)^{99}\text{Y}$	8030	80	8125	8	1.2	U			McG		84Ia.A
	8360	75			−3.1	C			Bwg		87Gr.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{99}\text{Y}(\beta^-)^{99}\text{Zr}$	7605	60	6973	12	-10.5	C			Bwg		87Gr.A
	7568	14			-42.5	C			Bwg		92Gr.A
$^{99}\text{Zr}(\beta^-)^{99}\text{Nb}$	4550	35	4719	16	4.8	C			Bwg		87Gr.A
	4559	15			10.6	C			Bwg		92Gr.A
$^{99}\text{Nb}(\beta^-)^{99}\text{Mo}$	3740	200	3635	12	-0.5	U					70Ei02 *
$^{99}\text{Mo}(\beta^-)^{99}\text{Tc}$	1356.7	1.0	1357.8	0.9	1.1	1	79	78 ^{99}Tc			71Na01 *
$^{99}\text{Tc}(\beta^-)^{99}\text{Ru}$	292	3	297.5	0.9	1.8	U					51Ta05
	290	4			1.9	U					52Fe16
	293.5	2.0			2.0	1	22	20 ^{99}Tc			80Al02 *
$^{99}\text{Rh}(\beta^+)^{99}\text{Ru}$	2038	10	2041	19	0.3	C					52Sc11 *
	2053	10			-1.2	U					59To25 *
	2170	30			-4.3	U					74An23 *
	2111	58			-1.2	1	11	11 ^{99}Rh			averag *
$^{99}\text{Pd}(\beta^+)^{99}\text{Rh}$	3410	20	3402	19	-0.4	1	89	89 ^{99}Rh			69Ph01 *
$^{99}\text{Ag}(\beta^+)^{99}\text{Pd}$	5430	150	5470	8	0.3	U					81Hu03
$^{99}\text{Sn}(\epsilon)^{99}\text{In}$	14700	3600	13400#	500#	-0.4	D					18Pa20 *
Original $D_M=35661.6(4.4) \mu\text{u}$ $M-A=-62506.3(4.1)\text{keV}$ re-evaluated											
$^{99}\text{Sr}-^{85}\text{Rb}_{1.165}$	$E_{\beta^-}=3500(200)$ to $7/2^+$ level at 235.508 keV										
$^{99}\text{Nb}(\beta^-)^{99}\text{Mo}$	$E_{\beta^-}=1214(1)$ to $1/2^-$ level at 142.6832 keV										
$^{99}\text{Mo}(\beta^-)^{99}\text{Tc}$	$E_{\beta^+}=434.8(2.6)$, $346.7(2.0)$ from $^{99}\text{Tc}^m$ at 142.6832 to ground state, 89.57 level										
$^{99}\text{Tc}(\beta^-)^{99}\text{Ru}$	$E_{\beta^+}=740(10)$ from $^{99}\text{Rh}^m$ at 64.5 to 340.90 $7/2^+$ level										
$^{99}\text{Rh}(\beta^+)^{99}\text{Ru}$	$E_{\beta^+}=1030(10)$, $710(10)$, $590(10)$, $420(20)$ keV										
$^{99}\text{Rh}(\beta^+)^{99}\text{Ru}$	to ground state, $321.99\ 3/2^+$, $442.59\ 3/2^+$, $618.13\ 1/2^+$ levels										
*	Q_{β^+} larger than 2059.34 keV because that state populated via β decay										
$^{99}\text{Rh}(\beta^+)^{99}\text{Ru}$	Ref. proposed that $E_{\beta^+}=1030(10)$ keV decays to 89.57 $3/2^+$ level										
$^{99}\text{Rh}(\beta^+)^{99}\text{Ru}$	$E_{\beta^+}=1100(50)$, $680(30)$ and $540(30)$ to 89.57, 442.59 and 618.13 levels										
$^{99}\text{Rh}(\beta^+)^{99}\text{Ru}$	unweighted average $59\text{To}25=2053(10)$ $74\text{An}23=2170(30)$ keV										
$^{99}\text{Pd}(\beta^+)^{99}\text{Rh}$	$E_{\beta^+}=2180(20)$, $1930(20)$, $1510(20)$ keV										
*	to $200.7\ 7/2^+$, $464.4\ (5/2, 7/2)^+$, $874.5\ 5/2^+$ levels above $(1/2^-)$ ground state										
$^{99}\text{Sn}(\epsilon)^{99}\text{In}$	Trends from Mass Surface TMS suggest ^{99}Sn 1300 keV more bound										
$^{100}\text{Rb}-u$	-49685	19	-49668	14	0.9	1	55	55 ^{100}Rb	MA8	1.0	17De18 *
$^{100}\text{Rb}-^{85}\text{Rb}_{1.176}$	54087	21	54067	14	-1.0	1	45	45 ^{100}Rb	MA8	1.0	13Ma81
	54150	140			-0.6	U			TT1	1.0	16Kl04
$^{100}\text{Sr}-^{85}\text{Rb}_{1.176}$	39520	12	39519	7	-0.1	-			TT1	1.0	16Kl04
	39515	29			0.1	-			MA8	1.0	17De18
ave.	39519	11			-0.1	1	45	45 ^{100}Sr			average
$^{100}\text{Y}-u$	-72270	110	-72272	12	-0.0	o			GT2	2.5	08Kn.A *
	-72290	140			0.1	U			GT2	2.5	08Su19 *
$^{100}\text{Zr}-u$	-82016	18	-81990	9	1.5	1	24	24 ^{100}Zr	JY0	1.0	04Ri12
$^{100}\text{Mo}-^{85}\text{Rb}_{1.176}$	11216	27	11203.3	0.3	-0.5	U			MA8	1.0	11He10
	11205.5	1.4			-1.6	U			JY1	1.0	12Ka13
$^{100}\text{Mo}-^{87}\text{Rb}_{1.149}$	11820.25	0.57	11819.6	0.3	-1.2	1	32	32 ^{100}Mo	MS1	1.0	15Gu09
$\text{C}_7\ \text{H}_{16}-^{100}\text{Mo}$	217730.3	4.2	217732.5	0.3	0.2	U			M15	2.5	63Ri07
$^{100}\text{Mo}-u$	-92532.51	0.40	-92532.0	0.3	1.2	1	65	65 ^{100}Mo	MS1	1.0	15Gu09
$\text{C}_7\ \text{H}_{16}-^{100}\text{Ru}$	220983.8	3.7	220990.1	0.4	0.7	U			M16	2.5	63Da10
$^{100}\text{Rh}-u$	-91855	46	-91886	19	-0.7	1	18	18 ^{100}Rh	GS2	1.0	05Li24 *
$^{100}\text{Ag}-^{85}\text{Rb}_{1.176}$	19849.9	7.1	19851	5	0.1	2			JY1	1.0	09Ei08 *
	19851.8	8.2			-0.1	2			MA8	1.0	11He10 *
$^{100}\text{Ag}-u$	-83902	44	-83885	5	0.4	U			GR1	1.0	19An10 *
$^{100}\text{Cd}-u$	-79636	214	-79651.2	1.8	-0.1	U			CS1	1.0	96Ch32
$^{100}\text{Cd}-^{85}\text{Rb}_{1.176}$	24084.1	1.8				2			MA8	1.0	09Br09
$^{100}\text{In}-u$	-69405	322	-68898.1	2.4	1.6	U			CS1	1.0	96Ch32
$^{100}\text{In}-^{85}\text{Rb}_{1.176}$	34837.2	2.4				2			MA8	1.0	19Lu.A
$^{100}\text{Sn}-u$	-62020	1020	-61350	260	0.7	U			CS1	1.0	96Ch32

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{100}\text{Sr}-^{97}\text{Zr}_{1.031}$	27579	10	27580	7	0.1	1	55	55 ^{100}Sr	JY1	1.0	06Ha03
$^{100}\text{Y}-^{97}\text{Zr}_{1.031}$	19524	12				2			JY1	1.0	07Ha32
$^{100}\text{Y}^m-^{97}\text{Zr}_{1.031}$	19679	12				2			JY1	1.0	07Ha32
$^{100}\text{Zr}-^{97}\text{Zr}_{1.031}$	9815	10	9807	9	-0.8	1	76	76 ^{100}Zr	JY1	1.0	06Ha03
$^{100}\text{Nb}^m-^{97}\text{Zr}_{1.031}$	6472.6	2.1				2			JY1	1.0	07Ha32
$^{100}\text{Mo}-^{35}\text{Cl}-^{98}\text{Mo }^{37}\text{Cl}$	5019	2	5014.5	0.4	-0.6	U			H11	4.0	63Bi12
$^{100}\text{Nb}-^{100}\text{Nb}^m$	-335.7	8.3				3			JY1	1.0	07Ha32
$^{100}\text{Mo}-^{100}\text{Ru}$	3257.55	0.18	3257.52	0.18	-0.2	1	99	97 ^{100}Ru	JY1	1.0	08Ra09
$^{100}\text{Ru}-^{99}\text{Ru}$	-1718	11	-1719.825	0.028	-0.1	U			M16	2.5	63Da10
$^{96}\text{Ru}(^{16}\text{O}, ^{12}\text{C})^{100}\text{Pd}$	-5599	26	-5605	18	-0.2	1	46	46 ^{100}Pd	BNL		82Th01
$^{100}\text{Mo}(\text{d}, ^3\text{He})^{99}\text{Nb}$	-5639	15	-5653	12	-0.9	2			Tex		74Bi08
$^{100}\text{Mo}(\text{t}, \alpha)^{99}\text{Nb}$	8642	20	8667	12	1.3	2			LAI		83FI06
$^{100}\text{Mo}(\text{d}, \text{t})^{99}\text{Mo}$	-2038	20	-2037.0	0.4	0.0	U			Pit		64Co11
$^{99}\text{Tc}(\text{n}, \gamma)^{100}\text{Tc}$	6764.4	1.				2					79Pi08
	6765.20	0.04	6764.4	1.0	-20.0	C					04Fu.A
$^{99}\text{Ru}(\text{n}, \gamma)^{100}\text{Ru}$	9673.2	0.7	9673.324	0.026	0.2	U					74Ri03
	9672.65	0.06			11.2	B			ILn		88Co18 Z
	9673.39	0.05			-1.3	-			MMn		91Is02 Z
	9673.30	0.03			0.8	-			ILn		00Ge01
	9673.41	0.19			-0.5	U			Bdn		06Fi.A
ave.	9673.324	0.026			0.0	1	100	98 ^{99}Ru			average
$^{100}\text{Sr}(\beta^-)^{100}\text{Y}$	7460	140	7504	13	0.3	U			McG		84Ia.A *
	7075	100			4.3	C			Bwg		87Gr.A
$^{100}\text{Y}(\beta^-)^{100}\text{Zr}$	7920	100	9051	14	11.3	C			McG		84Ia.A *
	9310	70			-3.7	C			Bwg		87Gr.A
$^{100}\text{Zr}(\beta^-)^{100}\text{Nb}$	3335	25	3419	11	3.3	C			Bwg		87Gr.A
$^{100}\text{Nb}(\beta^-)^{100}\text{Mo}$	6245	25	6402	8	6.3	C			Bwg		87Gr.A
$^{100}\text{Nb}^m(\beta^-)^{100}\text{Mo}$	6745	75	6714.5	2.0	-0.4	U			Bwg		87Gr.A
$^{100}\text{Mo}(\text{t}, ^3\text{He})^{100}\text{Nb}^m$	-6690	30	-6695.9	2.0	-0.2	U			LAI		79Aj03
$^{100}\text{Rh}(\beta^+)^{100}\text{Ru}$	3630	20	3636	18	0.3	1	82	82 ^{100}Rh			53Ma64
$^{100}\text{Ag}(\beta^+)^{100}\text{Pd}$	7075	90	7075	18	-0.0	U					79Ve.A *
	7022	200			0.3	U					80Ha20 *
$^{100}\text{Cd}(\beta^+)^{100}\text{Ag}$	3890	70	3943	5	0.8	U					89Ry02
$^{100}\text{In}(\beta^+)^{100}\text{Cd}$	10900	930	10016.4	2.8	-1.0	U			Lvp		95Sz01 *
	10080	230			-0.3	U					02PI03
$^{100}\text{Sn}(\beta^+)^{100}\text{In}$	7390	660	7030	240	-0.5	o			GSI		97Su06 *
	7840	660			-1.2	o			GSI		02Fa13 *
	7030	240				3			GSI		12Hi07 *
	7690	160			-4.1	B					19Lu08
* $^{100}\text{Rb}-\text{u}$	MR-TOF data in the paper recalculated by evaluators										HWJ187 **
* $^{100}\text{Y}-\text{u}$	$M-A=-67245(93)$ keV for mixture gs+m at 144(16) keV										Nub211 **
* $^{100}\text{Y}-\text{u}$	$M-A=-67264(119)$ keV for mixture gs+m at 144(16) keV										Nub211 **
* $^{100}\text{Rh}-\text{u}$	$M-A=-85508(29)$ keV for mixture gs+m at 107.6 keV										Nub211 **
* $^{100}\text{Ag}-^{85}\text{Rb}_{1.176}$	$D_M=19858.2(5.2)$ μu for mixture gs+m at 15.52 keV; $M-A=-78131.0(4.9)$ keV										Nub211 **
* $^{100}\text{Ag}-^{85}\text{Rb}_{1.176}$	$D_M=19860.2(6.6)$ μu for mixture gs+m at 15.52 keV; $M-A=-78129.1(6.2)$ keV										Nub211 **
* $^{100}\text{Ag}-\text{u}$	$M-A=-78146(41)$ keV for mixture gs+m at 15.52 keV										Nub211 **
* $^{100}\text{Sr}(\beta^-)^{100}\text{Y}$	$E_{\beta^-}=7450(140)$ assumed by evaluator to 1^+ level at 10.70 keV										Ens082 **
* $^{100}\text{Y}(\beta^-)^{100}\text{Zr}$	Not unambiguously ground state transition										GAu **
* $^{100}\text{Ag}(\beta^+)^{100}\text{Pd}$	From 5^+ ground state to high spin level at 2920.4 keV										79Ve.A **
* $^{100}\text{Ag}(\beta^+)^{100}\text{Pd}$	$E_{\beta^+}=5350(200)$ from $^{100}\text{Ag}^m 2^+$ at 15.52 to 2^+ level at 665.50 keV										Ens082 **
* $^{100}\text{In}(\beta^+)^{100}\text{Cd}$	From lower and upper limits 9300-12500										GAu **
* $^{100}\text{Sn}(\beta^+)^{100}\text{In}$	$Q_{\beta^+}=7200(+800-500)$; also $E_{\beta^+}=3400(+700-300)$ keV to 2760(430) level										97Su06 **
* $^{100}\text{Sn}(\beta^+)^{100}\text{In}$	$E_{\beta^+}=3800(+700-300)$ to 2760(430) level										96Ki23 **
* $^{100}\text{Sn}(\beta^+)^{100}\text{In}$	$E_{\beta^+}=3290(200)$ to 1^+ level at 2721+x, with $x<80$ keV										12Hi07 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{101}\text{Rb}-u$	-45698	22				2			MA8	1.0	17De18
$^{101}\text{Sr}-^{85}\text{Rb}_{1.188}$	45410	22	45400	9	-0.5	2			MA8	1.0	17De18
	45398	10			0.2	2			TT1	1.0	16KI04
$^{101}\text{Zr}-u$	-78573	20	-78542	9	1.6	1	20	20 ^{101}Zr	JY0	1.0	04Ri12
$\text{C}_8\text{H}_5-^{101}\text{Ru}$	133549.5	2.2	133552.1	0.4	0.5	U			M16	2.5	63Da10
$^{101}\text{Rh}-u$	-93821	58	-93841	6	-0.3	U			GS2	1.0	05Li24 *
$^{101}\text{Pd}-u$	-91816	30	-91715	5	3.4	C			GS2	1.0	05Li24
$^{101}\text{Ag}-^{85}\text{Rb}_{1.188}$	17470.5	7.2	17478	5	1.0	2			SH1	1.0	07Ma92
	17485.6	7.5			-1.0	2			MA8	1.0	11He10
$^{101}\text{Cd}-^{85}\text{Rb}_{1.188}$	23367	11	23380.0	1.6	1.2	U			SH1	1.0	07Ma92
	23380.0	1.6				2			MA8	1.0	09Br09
$^{101}\text{In}-u$	-73591.4	15.4	-73586	13	0.4	2			LZ1	1.0	19Xu13
	-73575.4	21.5			-0.5	2			GR1	1.0	20Ho03
$^{101}\text{In}^m-u$	-72884.0	52.0	-72900	40	-0.3	2			LZ1	1.0	19Xu13
	-72922.6	57.3			0.4	2			GR1	1.0	20Ho03 *
$^{101}\text{Pd}-^{96}\text{Mo}_{1.052}$	8567.4	5.1	8567	5	-0.1	1	93	93 ^{101}Pd	JY1	1.0	09El08
$^{101}\text{Cd}-^{96}\text{Mo}_{1.052}$	18872.7	5.5	18868.4	1.6	-0.8	U			JY1	1.0	09El08
$^{101}\text{Y}-^{97}\text{Zr}_{1.041}$	22847.5	7.6				2			JY1	1.0	07Ha32
$^{101}\text{Zr}-^{97}\text{Zr}_{1.041}$	14153	10	14145	9	-0.8	1	80	80 ^{101}Zr	JY1	1.0	06Ha03
$^{101}\text{Nb}-^{102}\text{Ru}_{0.990}$	10009.6	4.0				2			JY1	1.0	07Ha32
$^{101}\text{Ru}-^{100}\text{Ru}$	1368	11	1362.63	0.25	-0.2	U			M16	2.5	63Da10
$^{100}\text{Mo}(n,\gamma)^{101}\text{Mo}$	5398.4	0.5	5398.24	0.07	-0.3	U			ILn		75Ka.A
	5399.6	0.7			-1.9	U			ORn		79We07 Z
	5398.23	0.08			0.1	2			ILn		90Se17 Z
	5398.27	0.13			-0.2	2			Bdn		06Fi.A
$^{100}\text{Mo}(d,p)^{101}\text{Mo}$	3161	6	3173.67	0.07	2.1	U					72Si25
$^{100}\text{Ru}(n,\gamma)^{101}\text{Ru}$	6802.0	0.7	6802.04	0.23	0.1	-					82Ba69
	6802.04	0.25			-0.0	-			Bdn		06Fi.A
$^{100}\text{Ru}(d,p)^{101}\text{Ru}$	4581	4	4577.47	0.23	-0.9	U					77Ho02
$^{100}\text{Ru}(n,\gamma)^{101}\text{Ru}$	ave. 6802.04	0.24	6802.04	0.23	0.0	1	99	98 ^{101}Ru			average
$^{101}\text{Sn}(\epsilon p)^{100}\text{Cd}$	6600	300				3					10St.A
$^{101}\text{Rb}(\beta^-)^{101}\text{Sr}$	11810	110	12757	22	8.6	B			Bwg		92Ba28
$^{101}\text{Sr}(\beta^-)^{101}\text{Y}$	9505	80	9730	11	2.8	B			Bwg		92Ba28
$^{101}\text{Y}(\beta^-)^{101}\text{Zr}$	8545	90	8106	11	-4.9	B			Bwg		92Ba28
$^{101}\text{Zr}(\beta^-)^{101}\text{Nb}$	5520	45	5731	9	4.7	B			Bwg		87Gr18
	5485	25			9.8	C			Bwg		92Gr.A
$^{101}\text{Nb}(\beta^-)^{101}\text{Mo}$	4575	30	4628	4	1.8	U			Bwg		87Gr.A
	4590	30			1.3	U			Bwg		87Gr18
	4569	18			3.3	C			Bwg		92Gr.A
$^{101}\text{Mo}(\beta^-)^{101}\text{Tc}$	2836	40	2825	24	-0.3	R					57Ok.A *
$^{101}\text{Tc}(\beta^-)^{101}\text{Ru}$	1620	30	1614	24	-0.2	2					71Ar23 *
$^{101}\text{Pd}(\beta^+)^{101}\text{Rh}$	1980	4	1980	4	0.1	1	95	88 ^{101}Rh			71Ib01 *
$^{101}\text{Ag}(\beta^+)^{101}\text{Pd}$	4100	200	4098	7	-0.0	U					72We.A
	4350	200			-1.3	U					78Ha11
	4180	150			-0.5	U					79Ve.A *
$^{101}\text{Cd}(\beta^+)^{101}\text{Ag}$	5530	130	5498	5	-0.2	U					70Be.A *
	5350	200			0.7	U					72We.A
* $^{101}\text{Rh}-u$	$M-A=-87315(29)$ keV for mixture gs+m at 157.32 keV										Nub211 **
* $^{101}\text{In}^m-u$	reconstructed from $E_x=608(57)$ keV										20Ho03 **
* $^{101}\text{Mo}(\beta^-)^{101}\text{Tc}$	$E_{\beta^-}=2230(40)$ to $(1/2^+, 3/2^+)$ level at 606.47 keV										Ens06a **
* $^{101}\text{Tc}(\beta^-)^{101}\text{Ru}$	$E_{\beta^-}=1320(30)$ to 306.858 $7/2^+$ and 1070(30) to 545.115 $7/2^+$ levels										Ens06a **
* $^{101}\text{Pd}(\beta^+)^{101}\text{Rh}$	$E_{\beta^+}=776(4)$ to $7/2^+$ level at 181.78 keV										Ens06a **
* $^{101}\text{Ag}(\beta^+)^{101}\text{Pd}$	$E_{\beta^+}=2895(150)$ to $7/2^+$ level at 261.0 keV, and others										Ens06a **
* $^{101}\text{Cd}(\beta^+)^{101}\text{Ag}$	Measured E_{β^+} may go to excited state										70Be.A **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{102}\text{Rb}-u$	-39992	89				2			MA8	1.0	17De18
$^{102}\text{Sr}-^{85}\text{Rb}_{1.200}$	49857	72				2			MA8	1.0	17De18
$^{102}\text{Y}-^{120}\text{Sn}_{.850}$	17456.3	4.3				2			JY1	1.0	11Ha48 *
$^{102}\text{Zr}-u$	-76780	43	-76846	9	-1.5	U			JY0	1.0	04Ri12
$\text{C}_8 \text{H}_6-^{102}\text{Ru}$	142604.8	3.2	142609.9	0.4	0.6	U			M16	2.5	63Da10
$\text{C}_8 \text{H}_6-^{102}\text{Pd}$	141324	19	141317.9	0.4	-0.1	U			M16	2.5	63Da10
	141346	18			-1.0	U			R12	1.5	83De51
$\text{C}_7 \text{N H}_4-^{102}\text{Pd}$	128775	19	128741.8	0.4	-1.2	U			R12	1.5	83De51
$^{102}\text{Pd}-u$	-94370.9	15.6	-94367.7	0.4	0.2	U			GS4	1.0	14Ya.A
$^{102}\text{Ag}-u$	-88315	30	-88295	9	0.7	U			GS2	1.0	05Li24 *
$^{102}\text{Cd}-^{85}\text{Rb}_{1.200}$	20320.9	7.3	20334.1	1.8	1.8	U			SH1	1.0	07Ma92
	20334.2	1.9			-0.0	1	88	88 ^{102}Cd	MA8	1.0	09Br09
$^{102}\text{In}-^{85}\text{Rb}_{1.200}$	29959	13	29958	5	-0.1	1	14	14 ^{102}In	SH1	1.0	07Ma92
$^{102}\text{Cd}-^{96}\text{Mo}_{1.063}$	15811.9	5.2	15812.5	1.8	0.1	1	12	12 ^{102}Cd	JY1	1.0	09El08
$^{102}\text{In}-^{96}\text{Mo}_{1.063}$	25436.5	5.3	25437	5	0.0	1	86	86 ^{102}In	JY1	1.0	09El08
$^{102}\text{Zr}-^{97}\text{Zr}_{1.052}$	16822.0	9.8	16820	9	-0.2	1	92	92 ^{102}Zr	JY1	1.0	06Ha03
$^{102}\text{Nb}-^{97}\text{Zr}_{1.052}$	11756.4	2.7	11756.5	2.7	0.0	1	99	99 ^{102}Nb	JY1	1.0	07Ha32
$^{102}\text{Mo}-^{97}\text{Zr}_{1.052}$	3961.0	9.8	3960	9	-0.1	1	83	83 ^{102}Mo	JY1	1.0	06Ha03
$^{102}\text{Nb}^m-^{102}\text{Nb}$	100.2	7.9	101	8	0.1	1	95	94 $^{102}\text{Nb}^m$	JY1	1.0	07Ha32
$^{102}\text{Pd}-^{102}\text{Ru}$	1291.76	0.39	1291.98	0.05	0.6	U			SH1	1.0	11Go23
	1291.980	0.048			-0.0	1	100	100 ^{102}Pd	JY1	1.0	19Ne08
$^{102}\text{Ru}-^{101}\text{Ru}$	-1233	11	-1232.77	0.05	0.0	U			M16	2.5	63Da10
$^{100}\text{Mo}(\text{t,p})^{102}\text{Mo}$	5034	20	5029	8	-0.3	1	17	17 ^{102}Mo	LAI		72Ca10
$^{100}\text{Mo}(\text{t,He,p})^{102}\text{Tc}$	6054	20	6022	9	-1.6	1	21	21 ^{102}Tc	Pri		82De03
$^{102}\text{Pd}(\text{p,t})^{100}\text{Pd}$	-10356	24	-10351	18	0.2	1	54	54 ^{100}Pd	Win		74De31 *
$^{101}\text{Ru}(\text{n},\gamma)^{102}\text{Ru}$	9220.4	0.9	9219.64	0.05	-0.8	U					74Ri03
	9219.64	0.05			0.0	1	100	98 ^{102}Ru	MMn		91Is02 Z
	9219.63	0.19			0.1	U			Bdn		06Fi.A
$^{102}\text{In}(\varepsilon\text{p})^{101}\text{Ag}$	3420	310	3351	7	-0.2	o			Lvp		91Re.A *
$^{102}\text{Sr}(\beta^-)^{102}\text{Y}$	8815	70	9010	70	2.8	B			Bwg		92Ba28
$^{102}\text{Y}(\beta^-)^{102}\text{Zr}$	9850	70	10409	10	8.0	B			Bwg		92Ba28
$^{102}\text{Zr}(\beta^-)^{102}\text{Nb}^m$	4605	30	4622	11	0.6	1	14	8 ^{102}Zr	Bwg		87Gr18
$^{102}\text{Nb}(\beta^-)^{102}\text{Mo}$	7300	50	7263	9	-0.7	o			Bwg		87Gr.A
	7335	40			-1.8	U			Bwg		87Gr18
$^{102}\text{Nb}^m(\beta^-)^{102}\text{Mo}$	7215	40	7357	11	3.5	C			Bwg		87Gr.A
	7210	35			4.2	B			Bwg		87Gr18
$^{102}\text{Tc}(\beta^-)^{102}\text{Ru}$	4420	100	4534	9	1.1	U					69B116
$^{102}\text{Rh}(\beta^+)^{102}\text{Ru}$	2317	10	2323	6	0.6	2					61Hi06
	2325	10			-0.2	2					63Bo17
$^{102}\text{Ru}(\text{p,n})^{102}\text{Rh}$	-3115	15	-3105	6	0.6	2					83Do11
$^{102}\text{Rh}(\beta^-)^{102}\text{Pd}$	1150	6	1120	6	-5.1	B					61Hi06
$^{102}\text{Ag}(\beta^+)^{102}\text{Pd}$	5800	200	5656	8	-0.7	U					67Ch05
	5966	100			-3.1	B					67Ch05 *
	4910	140			5.3	C					70Be.A *
	5350	200			1.5	U					72We.A
	5880	110			-2.0	U					79Ve.A
$^{102}\text{Cd}(\beta^+)^{102}\text{Ag}$	2554	57	2587	8	0.6	U					72We.A
	2587	8				2			GSI		91Ke08 *
$^{102}\text{In}(\beta^+)^{102}\text{Cd}$	9250	380	8965	5	-0.8	U			Lvp		95Sz01 *
	8970	150			-0.0	U			GSI		98Ka.A
	8910	170			0.3	U			GSI		03Gi06 *
$^{102}\text{Sn}(\beta^+)^{102}\text{In}$	5780	70	5760	100	-0.3	o			GSI		01St.A
	5760	100				2			GSI		02Fa13
* $^{102}\text{Y}-^{120}\text{Sn}_{.850}$	Associated with low-spin isomer										11Ha48 **
* $^{102}\text{Ag}-u$	$M-A=-82260(28)$ keV for mixture gs+m at 9.40 keV										Nub211 **
* $^{102}\text{Pd}(\text{p,t})^{100}\text{Pd}$	Original error 12; added systematic error 21 keV										GAu **
* $^{102}\text{In}(\varepsilon\text{p})^{101}\text{Ag}$	Estimated using proton spectrum from 1450 to 3200 keV										GAu **
* $^{102}\text{Ag}(\beta^+)^{102}\text{Pd}$	$E_{\beta^+}=3340(100), 3070(130)$ from $^{102}\text{Ag}^m$ at 9.40 to 1534.48, 2017.8 levels										Ens098 **
* $^{102}\text{Ag}(\beta^+)^{102}\text{Pd}$	$Q_{\beta^+}=4920(100)$ from $^{102}\text{Ag}^m$ at 9.40 keV										Nub211 **
* $^{102}\text{Cd}(\beta^+)^{102}\text{Ag}$	$E_{\beta^+}=1075(8)$ to 1^+ level at 490.44 keV										Ens098 **
* $^{102}\text{In}(\beta^+)^{102}\text{Cd}$	From 9900 keV upper and 8600 keV lower limits										GAu **
* $^{102}\text{In}(\beta^+)^{102}\text{Cd}$	Good agreement with authors' earlier measurement, average=8950(120) keV										03Gi06 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{103}\text{Y}-\text{u}$	-63060	183	-62756	12	1.1	o			GT1	1.5	04Ma.A
	-62803	106			0.2	U			GT2	2.5	08Kn.A
$^{103}\text{Y}-^{120}\text{Sn}_{858}$	21154	12				2			JY1	1.0	11Ha48
$^{103}\text{Zr}-\text{u}$	-72765	64	-72796	10	-0.5	U			JY0	1.0	04Ri12
$\text{C}_8\text{H}_7-^{103}\text{Rh}$	149263.5	3.3	149281.1	2.5	2.1	U			M16	2.5	63Da10
	149261	19			0.7	U			R12	1.5	83De51
$\text{C}_7\text{N H}_5-^{103}\text{Rh}$	136681	18	136705.1	2.5	0.9	U			R12	1.5	83De51
$^{103}\text{Pd}-\text{u}$	-93894.8	14.5	-93888.9	0.9	0.4	U			GS4	1.0	14Ya.A
$^{103}\text{Ag}-\text{u}$	-91091	52	-91039	4	1.0	U			GS2	1.0	05Li24 *
$^{103}\text{Ag}-^{85}\text{Rb}_{1.212}$	15875	14	15871	4	-0.3	U			SH1	1.0	07Ma92
	15871.4	4.4				2			MA8	1.0	11He10
$^{103}\text{Cd}-^{85}\text{Rb}_{1.212}$	20328	11	20327.8	1.9	-0.0	U			SH1	1.0	07Ma92
	20328.2	2.1			-0.2	1	86	86 ^{103}Cd	MA8	1.0	09Br09
$^{103}\text{In}-^{85}\text{Rb}_{1.212}$	26785	11	26790	10	0.4	1	77	77 ^{103}In	SH1	1.0	07Ma92
$^{103}\text{In}-\text{u}$	-80119.7	26.8	-80121	10	-0.1	1	13	13 ^{103}In	GR1	1.0	20Ho03
$^{103}\text{Cd}-^{96}\text{Mo}_{1.073}$	15699.2	5.2	15700.9	1.9	0.3	1	14	14 ^{103}Cd	JY1	1.0	09El08
$^{103}\text{Zr}-^{97}\text{Zr}_{1.062}$	21760.5	9.9				2			JY1	1.0	06Ha03
$^{103}\text{Mo}-^{97}\text{Zr}_{1.062}$	7648.4	9.9				2			JY1	1.0	06Ha03
$^{103}\text{Nb}-^{102}\text{Ru}_{1.010}$	16069.7	4.2				2			JY1	1.0	07Ha32
$^{103}\text{Cd}-^{102}\text{Cd}$	-1534	154	-1064.9	2.6	2.0	U			CR2	1.5	92Sh.A *
$^{103}\text{Rh}(\text{p},\text{t})^{101}\text{Rh}$	-8275	17	-8280	6	-0.3	1	13	12 ^{101}Rh	Pri		64Th05
$^{102}\text{Ru}(\text{n},\gamma)^{103}\text{Ru}$	6232.2	0.3	6232.05	0.15	-0.5	-					82Ba69 Z
	6232.00	0.17			0.3	-			Bdn		06Fi.A
$^{102}\text{Ru}(\text{d},\text{p})^{103}\text{Ru}$	4005	15	4007.48	0.15	0.2	U			ANL		71Fo01
$^{102}\text{Ru}(\text{n},\gamma)^{103}\text{Ru}$	ave. 6232.05	0.15	6232.05	0.15	0.0	1	100	99 ^{103}Ru			average
$^{103}\text{Rh}(\gamma,\text{n})^{102}\text{Rh}$	-9307	32	-9320	7	-0.4	U			Phi		60Ge01
$^{103}\text{Rh}(\text{p},\text{d})^{102}\text{Rh}$	-7144	16	-7095	7	3.1	B			Pri		64Th05
$^{102}\text{Pd}(\text{n},\gamma)^{103}\text{Pd}$	7624.6	1.5	7625.3	0.8	0.5	2					70Bo29
	7625.6	0.9			-0.3	2			Bdn		06Fi.A
$^{103}\text{Zr}(\beta^-)^{103}\text{Nb}$	6945	85	7220	10	3.2	B			Bwg		87Gr18
$^{103}\text{Nb}(\beta^-)^{103}\text{Mo}$	5535	35	5926	10	11.2	C			Bwg		87Gr.A
	5530	30			13.2	B			Bwg		87Gr18
$^{103}\text{Mo}(\beta^-)^{103}\text{Tc}$	3750	60	3650	13	-1.7	U			Bwg		87Gr18
$^{103}\text{Tc}(\beta^-)^{103}\text{Ru}$	2615	45	2663	10	1.1	U			Bwg		87Gr.A
$^{103}\text{Ru}(\beta^-)^{103}\text{Rh}$	764	4	764.5	2.3	0.1	-					58Ro09 *
	760	6			0.8	-					65Mu09 *
	762	5			0.5	-					70Pe04 *
	769	4			-1.1	-					82Oh04
	ave. 764.6	2.3			-0.0	1	98	98 ^{103}Rh			average
$^{103}\text{Pd}(\epsilon)^{103}\text{Rh}$	564	20	574.7	2.4	0.5	U					54Ri09 *
	543.0	0.8			39.7	B					86Be53
$^{103}\text{Ag}(\beta^+)^{103}\text{Pd}$	2580	100	2654	4	0.7	U					62Pa05 *
	2700	100			-0.5	U					66Ja12
	2320	50			6.7	B					69Ba02
	2622	34			0.9	U			Dlf		88Bo28 *
$^{103}\text{Cd}(\beta^+)^{103}\text{Ag}$	4200	130	4151	4	-0.4	U					70Be.A
	4250	200			-0.5	U					72We.A
	4131	23			0.9	U			Dlf		88Bo28 *
$^{103}\text{In}(\beta^+)^{103}\text{Cd}$	5380	200	6019	9	3.2	B			Brk		83Wo04
	6050	28			-1.1	1	11	10 ^{103}In	Dlf		88Bo28 *
	6040	60			-0.3	U					98Ka42
$^{103}\text{Sn}(\beta^+)^{103}\text{In}$	7500	600	7540#	100#	0.1	o			GSI		04Mu32
	7660	70			-1.7	D			GSI		05Ka34 *
* $^{103}\text{Ag}-\text{u}$	$M-A=-84784(29)$ keV for mixture gs+m at 134.45 keV										Nub211 **
* $^{103}\text{Cd}-^{102}\text{Cd}$	From $^{102}\text{Cd}/^{103}\text{Cd}=0.99029800(150)$										AHW **
* $^{103}\text{Ru}(\beta^-)^{103}\text{Rh}$	$E_{\beta^-}=227(4)$ to $5/2^+$ level at 536.840 keV, and other E_{β^-}										Ens09a **
* $^{103}\text{Ru}(\beta^-)^{103}\text{Rh}$	$E_{\beta^-}=112(6)$ to $5/2^+$ level at 650.064 keV, and other E_{β^-}										Ens09a **
* $^{103}\text{Ru}(\beta^-)^{103}\text{Rh}$	$E_{\beta^-}=225(5)$ to $5/2^+$ level at 536.840 keV, and other E_{β^-}										Ens09a **
* $^{103}\text{Pd}(\epsilon)^{103}\text{Rh}$	IBE=500(20) to $^{103}\text{Rh}^m 7/2^+$ at 39.753 keV										Nub211 **
* $^{103}\text{Ag}(\beta^+)^{103}\text{Pd}$	$E_{\beta^+}=1290(100)$ to $5/2^+$ level at 266.861 keV, and other E_{β^+}										Ens09a **
* $^{103}\text{Ag}(\beta^+)^{103}\text{Pd}$	$E_{\beta^+}=1601(27)$, plus systematic error 20 keV estimated by evaluator										GAu **
* $^{103}\text{Cd}(\beta^+)^{103}\text{Ag}$	$E_{\beta^+}=3109(11)$, plus systematic error 20 keV estimated by evaluator										GAu **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{103}\text{In}(\beta^+)^{103}\text{Cd}$	$E_{\beta^+}=4841(20)$, plus systematic error 20 keV estimated by evaluator										Ens09a **
$^{103}\text{Sn}(\beta^+)^{103}\text{In}$	Original 7640(70) recalibrated										05Ka34 **
$^{103}\text{Sn}(\beta^+)^{103}\text{In}$	Trends from Mass Surface TMS suggest ^{103}Sn 120 keV more bound										HWJ20a **
$^{104}\text{Zr}-u$	-70661	54	-70551	10	2.0	U			JY0	1.0	04Ri12
$^{104}\text{Nb}-u$	-77355	100	-77092.3	1.9	1.1	U			GT2	2.5	08Kn.A *
$^{104}\text{Nb}_{-52}\text{Cr}_{2.000}$	41898.3	1.9				2			CP2	1.0	18Or.A
$^{104}\text{Nb}^m_{-52}\text{Cr}_{2.000}$	41908.8	2.0				2			CP2	1.0	18Or.A
$\text{C}_8\text{H}_8-^{104}\text{Ru}$	157171.5	3.4	157174.9	2.7	0.4	1	10	^{104}Ru	M16	2.5	63Da10
$\text{C}_8\text{H}_8-^{104}\text{Pd}$	158612	10	158569.9	1.4	-1.7	U			M16	2.5	63Da10
	158599	12			-1.6	U			R12	1.5	83De51
$\text{C}_7\text{N H}_6-^{104}\text{Pd}$	146013	8	145993.8	1.4	-1.6	U			R12	1.5	83De51
$\text{C}_6^{13}\text{C N H}_5-^{104}\text{Pd}$	141552	20	141523.6	1.4	-0.9	U			R12	1.5	83De51
$^{104}\text{Pd}-u$	-95938	30	-95969.6	1.4	-1.1	U			GS2	1.0	05Li24
$^{104}\text{Ag}-u$	-91410	30	-91376	5	1.1	U			GS2	1.0	05Li24 *
$^{104}\text{Cd}-u$	-90147	30	-90143.8	1.8	0.1	U			GS2	1.0	05Li24
$^{104}\text{Cd}-^{85}\text{Rb}_{1.224}$	17813.7	5.5	17825.6	1.8	2.2	U			SH1	1.0	07Ma92
	17825.5	1.9			0.0	1	89	^{104}Cd	MA8	1.0	09Br09
$^{104}\text{In}-^{85}\text{Rb}_{1.224}$	26183.9	6.2				2			SH1	1.0	07Ma92
	26140.3	29.6	26184	6	1.5	U			JY1	1.0	09El08 *
$^{104}\text{Sn}-^{87}\text{Rb}_{1.195}$	31636.9	6.4	31634	6	-0.4	1	93	^{104}Sn	JY1	1.0	09El07
$^{104}\text{Cd}-^{96}\text{Mo}_{1.083}$	13094.2	5.5	13093.4	1.8	-0.1	1	11	^{104}Cd	JY1	1.0	09El08
$^{104}\text{Zr}-^{97}\text{Zr}_{1.072}$	24896	10				2			JY1	1.0	06Ha03
$^{104}\text{Nb}-^{97}\text{Zr}_{1.072}$	18352.8	2.9	18354.5	1.9	0.6	B			JY1	1.0	07Ha32 *
$^{104}\text{Mo}-^{97}\text{Zr}_{1.072}$	9194.0	9.7	9194	10	0.0	1	97	^{104}Mo	JY1	1.0	06Ha03
$^{104}\text{In}-^{103}\text{In}$	-1241	231	-1664	11	-1.2	U			CR2	1.5	91Sh19 *
$^{104}\text{Pd}-^{102}\text{Pd}$	-1617	30	-1601.9	1.5	0.3	U			R12	1.5	83De51
$^{104}\text{Te}(\alpha)^{100}\text{Sn}$	5096.4	208.0				4			Ara		18Au04
$^{104}\text{Ru}(\text{d},\alpha)^{102}\text{Tc}$	7180	10	7188	9	0.8	1	80	^{102}Tc	Pri		82De03
$^{102}\text{Ru}(\text{t},\text{p})^{104}\text{Ru}$	6648	30	6650.2	2.5	0.1	U			LAl		72Ca10
$^{104}\text{Ru}(\text{d},^3\text{He})^{103}\text{Tc}$	-5289	10	-5287	9	0.2	2			VUn		83De20
$^{104}\text{Ru}(\text{t},\alpha)^{103}\text{Tc}$	9048	30	9033	9	-0.5	2			LAl		81FI02
$^{104}\text{Ru}(\text{d},\text{t})^{103}\text{Ru}-^{148}\text{Gd}()^{147}\text{Gd}$	85	3	83.9	2.4	-0.4	1	65	^{104}Ru	Jul		86Ru04 *
$^{103}\text{Rh}(\text{n},\gamma)^{104}\text{Rh}$	6998.96	0.10	6998.96	0.08	-0.0	2			MMn		81Ke03 Z
	6998.95	0.14			0.0	2			Bdn		06Fi.A
$^{103}\text{Rh}(\text{d},\text{p})^{104}\text{Rh}$	4786	10	4774.39	0.08	-1.2	U			MIT		64Sp12
$^{104}\text{Pd}(\text{d},\text{t})^{103}\text{Pd}$	-3762	25	-3752.2	1.6	0.4	U			Pit		64Co11
$^{104}\text{Sb}(\text{p})^{103}\text{Sn}$	510	100	509	15	-0.0	U					94Pa12 *
$^{104}\text{Nb}(\beta^-)^{104}\text{Mo}$	8105	90	8533	9	4.8	B			Bwg		87Gr18
$^{104}\text{Nb}^m(\beta^-)^{104}\text{Mo}$	8320	80	8543	9	2.8	B			Bwg		87Gr18 *
$^{104}\text{Mo}(\beta^-)^{104}\text{Tc}$	4800	200	2155	24	-13.2	B					64Te02
	2155	40			0.0	-			Bwg		87Gr18
	2160	40			-0.1	-			Jyv		94Jo.A
$^{104}\text{Tc}(\beta^-)^{104}\text{Ru}$	ave. 2158	28			-0.1	1	73	^{104}Tc			average
	5620	70	5597	25	-0.3	-					78Su03
	5590	60			0.1	-			Bwg		87Gr18
	ave. 5600	50			-0.1	1	30	^{104}Tc			average
$^{104}\text{Rh}(\beta^-)^{104}\text{Pd}$	2440	30	2435.8	2.7	-0.1	U					55Bu.A
$^{104}\text{Ag}(\beta^+)^{104}\text{Pd}$	4276	15	4279	4	0.2	U					60Nu02 *
	4350	200			-0.4	U					72We.A
	4306	31			-0.9	U			Dlf		88Bo28 *
$^{104}\text{Pd}(\text{p},\text{n})^{104}\text{Ag}$	-5061	4				3					79De44
$^{104}\text{Cd}(\beta^+)^{104}\text{Ag}$	1587	60	1148	5	-7.3	B					70Mu17 *
	1403	26			-9.8	B			GSI		79Pi06 *
$^{104}\text{In}(\beta^+)^{104}\text{Cd}$	7100	200	7786	6	3.4	B					78Hu06 *
	7260	250			2.1	U			Brk		83Wo04 *
	7800	250			-0.1	U			Dlf		88Bo28
	7890	120			-0.9	U			Dlf		90Re08
	7880	100			-0.9	U			GSI		98Ka.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{104}\text{Sn}(\beta^+)^{104}\text{In}$	4550	300	4556	8	0.0	U					88Ba10 *
	4515	60			0.7	U			GSI		91Ke11
* $^{104}\text{Nb}-u$	$D_M=-77350(100) \mu\text{u}$ for mixture gs+m at 9.8(2.6) keV; $M-A=-72051(93) \text{ keV}$										Nub211 **
* $^{104}\text{Ag}-u$	$M-A=-85144(28) \text{ keV}$ for mixture gs+m at 6.90 keV										Nub211 **
* $^{104}\text{In}-^{85}\text{Rb}_{1.224}$	$D_M=26190.5(5.5) \mu\text{u}$ for mixture gs+m at 93.48 keV; $M-A=-76176.5(5.1) \text{ keV}$										Nub211 **
* $^{104}\text{Nb}-^{97}\text{Zr}_{1.072}$	Only long-lived state is measured										07Ri01 **
* $^{104}\text{In}-^{103}\text{In}$	From $^{103}\text{In}/^{104}\text{In}=0.99038900(222)$										AHW **
* $^{104}\text{Ru}(\text{d,t})^{103}\text{Ru}-^{148}\text{Gd}()^{147}\text{Gd}$	$Q=82(3)$ to $5/2^+$ level at 2.81 keV										Ens09a **
* $^{104}\text{Sb}(\text{p})^{103}\text{Sn}$	Below 550 keV; value and error estimated by evaluator										94Pa12 **
* $^{104}\text{Nb}^m(\beta^-)^{104}\text{Mo}$	Better use the difference of the two Q_{β^-} 's										GAU **
* $^{104}\text{Ag}(\beta^+)^{104}\text{Pd}$	$E_{\beta^+}=2705(15)$ from $^{104}\text{Ag}^m$ at 6.90 to 2^+ level at 555.81 keV										Ens079 **
* $^{104}\text{Ag}(\beta^+)^{104}\text{Pd}$	$E_{\beta^+}=2012(71)$ to 4^+ level at 1323.59 keV										Ens079 **
*	and $E_{\beta^+}=2729(24)$ from $^{104}\text{Ag}^m$ at 6.90 to 2^+ level at 555.81 keV										Ens079 **
*	plus systematic error 20 keV estimated by evaluator										GAU **
* $^{104}\text{Cd}(\beta^+)^{104}\text{Ag}$	$p^+=0.011(0.003)$ 0.0019(0.0005) respectively, to 1^+ level at 90.6 keV; recalculated E_{β^+}										Ens079 **
* $^{104}\text{In}(\beta^+)^{104}\text{Cd}$	$E_{\beta^+}=4600(200)$ 4750(250) respectively, to 4^+ level at 1492.1 keV										Ens079 **
* $^{104}\text{Sn}(\beta^+)^{104}\text{In}$	$p^+=0.71(0.07)$ to 1139.25 level										Ens079 **
$^{105}\text{Y}-u$	-55041	574	-54290#	430#	0.5	D			GT3	2.5	16Kn03 *
$^{105}\text{Rh}-u$	-94378	53	-94312.2	2.7	1.2	U			GS2	1.0	05Li24 *
$\text{C}_8 \text{H}_9-^{105}\text{Pd}$	165357	14	165345.8	1.2	-0.3	U			M16	2.5	63Da10
	165360	9			-1.1	U			R12	1.5	83De51
$\text{C}_7 \text{N H}_7-^{105}\text{Pd}$	152773	18	152769.7	1.2	-0.1	U			R12	1.5	83De51
$\text{C}_6^{13}\text{C N H}_6-^{105}\text{Pd}$	148309	26	148299.6	1.2	-0.2	U			R12	1.5	83De51
$\text{C}_7 \text{O H}_5-^{105}\text{Pd}$	128970	18	128960.3	1.2	-0.4	U			R12	1.5	83De51
$^{105}\text{Ag}-u$	-93534	31	-93474	5	1.9	U			GS2	1.0	05Li24 *
$^{105}\text{Cd}-^{96}\text{Mo}_{1.094}$	13748.5	5.4	13749.7	1.5	0.2	U			JY1	1.0	09El08
$^{105}\text{Cd}-^{85}\text{Rb}_{1.235}$	18403.4	1.5	18403.6	1.5	0.1	1	99	99 ^{105}Cd	MA8	1.0	09Br09
$^{105}\text{In}-^{85}\text{Rb}_{1.235}$	23442	11				2			SH1	1.0	07Ma92
$^{105}\text{In}-u$	-85536.8	33.3	-85498	11	1.2	U			GR1	1.0	20Ho03
$^{105}\text{Sn}-^{85}\text{Rb}_{1.235}$	30204.1	7.1	30208	4	0.6	1	36	36 ^{105}Sn	SH1	1.0	07Ma92
$^{105}\text{Sn}-^{87}\text{Rb}_{1.207}$	30890.0	5.6	30888	4	-0.4	1	58	58 ^{105}Sn	JY1	1.0	09El07
$^{105}\text{Zr}-^{97}\text{Zr}_{1.082}$	30359	13				2			JY1	1.0	06Ha03
$^{105}\text{Mo}-^{97}\text{Zr}_{1.082}$	13319.4	9.8	13319	10	-0.0	1	98	98 ^{105}Mo	JY1	1.0	06Ha03
$^{105}\text{Nb}-^{102}\text{Ru}_{1.029}$	23376.4	4.3				2			JY1	1.0	07Ha32
$^{105}\text{Pd}-^{104}\text{Pd}$	1049	35	1049.1	0.8	0.0	U			R12	1.5	83De51
$^{105}\text{In}-^{104}\text{In}$	-3618	144	-3712	13	-0.4	U			CR2	1.5	91Sh19 *
$^{105}\text{Te}(\alpha)^{101}\text{Sn}$	5079	50	5069	3	-0.2	U					06Se08 *
	5061.1	5.			1.6	o					06Li41 *
	5069.2	3.				4					10Da17 *
$^{104}\text{Ru}(\text{n},\gamma)^{105}\text{Ru}$	5909.9	0.5	5910.10	0.11	0.4	-					74Hr01
	5910.1	0.2			-0.0	-					78Gu14
	5910.11	0.14			-0.1	-			Bdn		06Fi.A
$^{104}\text{Ru}(\text{d,p})^{105}\text{Ru}$	3684	15	3685.53	0.11	0.1	U			ANL		71Fo01
	3684.5	1.0			1.0	U			Mun		76Ma49
$^{104}\text{Ru}(\text{n},\gamma)^{105}\text{Ru}$	ave.	5910.10	0.11	5910.10	0.11	0.0	1	100	69 ^{105}Ru		average
$^{104}\text{Pd}(\text{n},\gamma)^{105}\text{Pd}$	7094.1	0.7				2					70Bo29
$^{104}\text{Pd}(\text{d,p})^{105}\text{Pd}$	4867	20	4869.5	0.7	0.1	U			Pit		64Co11
$^{105}\text{Pd}(\text{d,t})^{104}\text{Pd}$	-851	30	-836.9	0.7	0.5	U			Pit		64Co11
$^{105}\text{Sb}(\text{p})^{104}\text{Sn}$	482.6	15.	323	22	-10.7	F			Bkp		94Ti03 *
$^{105}\text{Nb}(\beta^-)^{105}\text{Mo}$	6485	70	7415	10	13.3	B			Bwg		87Gr18
$^{105}\text{Mo}(\beta^-)^{105}\text{Tc}$	4950	45	4960	40	0.1	1	61	59 ^{105}Tc	Bwg		87Gr18
$^{105}\text{Tc}(\beta^-)^{105}\text{Ru}$	3640	55	3650	40	0.1	1	41	41 ^{105}Tc	Bwg		87Gr18
$^{105}\text{Ru}(\beta^-)^{105}\text{Rh}$	1916	4	1916.7	2.9	0.2	1	51	25 ^{105}Rh			67Se01
$^{105}\text{Rh}(\beta^-)^{105}\text{Pd}$	570	5	566.6	2.3	-0.7	-					51Du03
	560	5			1.3	-					56La24
	568	4			-0.3	-					64Ka23
	ave.	566.3	2.6		0.1	1	78	75 ^{105}Rh			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{105}\text{Ag}(\epsilon)^{105}\text{Pd}$	1347	25	1347	5	0.0	U					67Pi03
	1310	25			1.5	U					67Sc26 *
$^{105}\text{Cd}(\beta^+)^{105}\text{Ag}$	2738	5	2737	4	-0.2	-					53Jo20 *
	2600	200			0.7	U					72We.A
	2742	11			-0.5	-					86Bo28 *
ave.	2739	5			-0.4	1	92	91 ^{105}Ag			average
$^{105}\text{In}(\beta^+)^{105}\text{Cd}$	5140	200	4693	10	-2.2	U			Brk		83Wo04
	4849	13			-12.0	B					86Bo28 *
$^{105}\text{Sn}(\beta^+)^{105}\text{In}$	6230	80	6303	11	0.9	U			GSI		06Ka74
* $^{105}\text{Y}-\text{u}$	Trends from Mass Surface TMS suggest ^{105}Y 700 keV less bound										GAu **
* $^{105}\text{Rh}-\text{u}$	$M-A=-87847(32)$ keV for mixture gs+m at 129.742 keV										Nub211 **
* $^{105}\text{Ag}-\text{u}$	$M-A=-87113(28)$ keV for mixture gs+m at 25.468 keV										Nub211 **
* $^{105}\text{In}-^{104}\text{In}$	From $^{104}\text{In}/^{105}\text{In}=0.99050293(139)$										AHW **
* $^{105}\text{Te}(\alpha)^{101}\text{Sn}$	$E_\alpha=4720(50)$, $4703(5)$ to $7/2^+$ level at 171.7 keV (same group as next)										Ens07a **
* $^{105}\text{Te}(\alpha)^{101}\text{Sn}$	$E_\alpha=4711(3)$ to $7/2^+$ level at 171.7 keV; also $E_\alpha=4880(20)$ to ground state										Ens07a **
* $^{105}\text{Sb}(p)^{104}\text{Sn}$	F : expected 150 protons, no proton peak observed										05Li47 **
* $^{105}\text{Ag}(\epsilon)^{105}\text{Pd}$	$L/K=0.152(0.002) \rightarrow Q=222(12+\text{theor.error})$ to $3/2^-$ level at 1087.96 keV										Ens05b **
* $^{105}\text{Cd}(\beta^+)^{105}\text{Ag}$	$E_{\beta^+}=1691(5)$ to $^{105}\text{Ag}^m$ at 25.468 keV										Nub211 **
* $^{105}\text{Cd}(\beta^+)^{105}\text{Ag}$	$E_{\beta^+}=1695(11)$ to $^{105}\text{Ag}^m$ at 25.468 keV										Nub211 **
* $^{105}\text{In}(\beta^+)^{105}\text{Cd}$	$E_{\beta^+}=3696(13)$ to $7/2^+$ level at 131.11 keV										Ens05b **
$^{106}\text{Zr}-\text{u}$	-62674	322	-63070#	220#	-0.8	o			GT1	1.5	04Ma.A
	-62856	186			-0.5	D			GT3	2.5	16Kn03 *
$^{106}\text{Nb}-\text{u}$	-70843	258	-71071.5	1.5	-0.6	U			GT1	1.5	04Ma.A
$^{106}\text{Nb}-^{52}\text{Cr}_{2.038}$	50180.0	1.6	50179.9	1.5	-0.1	1	88	88 ^{106}Nb	CP2	1.0	18Or.A
$^{106}\text{Mo}-^{97}\text{Zr}_{1.093}$	15589.8	9.8				2			JY1	1.0	06Ha03
$\text{C}_8 \text{H}_{10}-^{106}\text{Pd}$	174764.0	4.3	174770.0	1.2	0.6	U			M16	2.5	63Da10
	174751	32			0.4	U			R12	1.5	83De51
	174766	8			0.3	U			R12	1.5	83De51
$\text{C}_7 \text{H}_9-^{106}\text{Pd}$	170285	32	170299.8	1.2	0.3	U			R12	1.5	83De51
	170298	30			0.0	U			R12	1.5	83De51
$\text{C}_7 \text{H}_8-^{106}\text{Pd}$	162186	18	162194.0	1.2	0.3	U			R12	1.5	83De51
$\text{C}_7 \text{O H}_6-^{106}\text{Pd}$	138378	20	138384.5	1.2	0.2	U			R12	1.5	83De51
$^{106}\text{Pd}-\text{u}$	-96495	30	-96519.7	1.2	-0.8	U			GS2	1.0	05Li24
	-96521.0	1.9			0.5	-			TG1	1.5	12Sm01
	-96524.9	4.7			0.7	-			TG1	1.5	12Sm01
ave.	-96521.5	2.6			0.7	1	20	20 ^{106}Pd			average
$^{106}\text{Ag}-\text{u}$	-93318	44	-93337	3	-0.4	U			GS2	1.0	05Li24 *
$\text{C}_8 \text{H}_{10}-^{106}\text{Cd}$	171789.3	2.7	171790.5	1.2	0.2	U			M16	2.5	63Da10
	171841	17			-2.0	U			R12	1.5	83De51
$\text{C}_7 \text{H}_8-^{106}\text{Cd}$	159210	15	159214.5	1.2	0.2	U			R12	1.5	83De51
$^{106}\text{Cd}-\text{u}$	-93540.6	1.7	-93540.2	1.2	0.2	-			TG1	1.5	12Sm01
	-93545.7	3.47			1.1	-			TG1	1.5	12Sm01
ave.	-93541.6	2.3			0.6	1	27	27 ^{106}Cd			average
$^{106}\text{Cd}-^{85}\text{Rb}_{1.247}$	16459.8	1.8	16458.0	1.2	-1.0	1	43	43 ^{106}Cd	MA8	1.0	09Br09
$^{106}\text{In}-\text{u}$	-86516	32	-86536	13	-0.6	U			GS2	1.0	05Li24 *
$^{106}\text{Sn}-^{85}\text{Rb}_{1.247}$	26959.4	8.7	26956	5	-0.4	1	39	39 ^{106}Sn	SH1	1.0	07Ma92
$^{106}\text{Sn}-^{87}\text{Rb}_{1.218}$	27578.0	7.6	27576	5	-0.3	1	52	52 ^{106}Sn	JY1	1.0	09El07
$^{106}\text{Sb}-^{87}\text{Rb}_{1.218}$	39256.1	8.0				2			JY1	1.0	09El07
$^{106}\text{Nb}-^{102}\text{Ru}_{1.039}$	28318.2	4.4	28318.9	1.6	0.2	1	13	12 ^{106}Nb	JY1	1.0	07Ha32
$^{106}\text{Tc}-^{102}\text{Ru}_{1.039}$	13780.7	4.7	13747	13	-7.2	F			JY1	1.0	07Ha20 *
$^{106}\text{Ru}-^{105}\text{Ru}_{1.010}$	511.8	9.1	505	6	-0.7	1	42	37 ^{106}Ru	JY1	1.0	07Ha20
$^{106}\text{Cd}-^{106}\text{Pd}$	2979.50	0.11	2979.50	0.11	0.0	1	100	70 ^{106}Pd	SH1	1.0	11Go23
	2979.08	0.60			0.5	U			TG1	1.5	12Sm01
$^{106}\text{Pd}-^{105}\text{Pd}$	-1608	25	-1599.2	0.3	0.2	U			R12	1.5	83De51

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{106}\text{Pd}-^{104}\text{Pd}$	-508	32	-550.1	0.8	-0.9	U			R12	1.5	83De51
$^{106}\text{Te}(\alpha)^{102}\text{Sn}$	4323.5	30.	4290	9	-1.1	U					81Sc17
	4290.2	9.				3					94Pa11
	4323.5	30.			-1.1	U					02Ma19
	4261.0	62.4			0.5	U					16Ca33
$^{106}\text{Cd}(^3\text{He}, ^6\text{He})^{103}\text{Cd}$	-9173	17	-9141.4	2.1	1.9	U			MSU		78Pa11 *
$^{104}\text{Ru}(\text{t}, \text{p})^{106}\text{Ru}$	5892	20	5888	5	-0.2	U			LAI		72Ca10
$^{104}\text{Pd}(^3\text{He}, \text{p})^{106}\text{Ag}$	5153	6	5189.5	2.9	6.1	F			Bld		75An07 *
$^{106}\text{Cd}(\text{p}, \text{t})^{104}\text{Cd}$	-10802	15	-10824.6	2.0	-1.5	U			MSU		82Cr01
	-10829	12			0.4	U			Pri		83De03
	-10819	12			-0.5	U			Ors		84Ro.A
$^{105}\text{Pd}(\text{n}, \gamma)^{106}\text{Pd}$	9562.8	1.1	9560.96	0.28	-1.7	U					70Bo29
	9560.5	0.4			1.1	-			BNn		87Fo20 *
	9561.4	0.4			-1.1	-			Bdn		06Fi.A
$^{105}\text{Pd}(\text{d}, \text{p})^{106}\text{Pd}$	7349	30	7336.39	0.28	-0.4	U			Pit		64Co11
$^{106}\text{Pd}(\text{d}, \text{t})^{105}\text{Pd}$	-3311	25	-3303.73	0.28	0.3	U			Pit		64Co11
$^{105}\text{Pd}(\text{n}, \gamma)^{106}\text{Pd}$ ave.	9560.95	0.28	9560.96	0.28	0.0	1	100	96 ^{105}Pd			average
$^{105}\text{Pd}(^3\text{He}, \text{d})^{106}\text{Ag}$	322	8	320.0	2.8	-0.3	1	12	12 ^{106}Ag	Bld		75An07
$^{106}\text{Cd}(\text{d}, \text{t})^{105}\text{Cd}$	-4661	50	-4612.4	1.8	1.0	U					73De16
$^{106}\text{Cd}(^3\text{He}, \alpha)^{105}\text{Cd}$	9728	25	9708.0	1.8	-0.8	U			Man		75Ch21
$^{106}\text{Mo}(\beta^-)^{106}\text{Tc}$	3510	45	3648	15	3.1	B			Bwg		87Gr18
	3520	17			7.5	C			Bwg		92Gr.A
	3520	17			7.5	C			Jyv		94Jo.A
$^{106}\text{Tc}(\beta^-)^{106}\text{Ru}$	6545	45	6547	11	0.0	o			Bwg		87Gr18
	6547	11				2			Bwg		92Gr.A
$^{106}\text{Ru}(\beta^-)^{106}\text{Rh}$	39.2	0.3	39.40	0.21	0.7	-					50Ag01
	39.6	0.3			-0.7	-					58Gr07
ave.	39.40	0.21			0.0	1	100	63 ^{106}Ru			average
$^{106}\text{Rh}(\beta^-)^{106}\text{Pd}$	3530	10	3545	5	1.5	-					52Al06
	3550	10			-0.5	-					58Gr07
	3550	20			-0.3	-					60Se05
ave.	3541	7			0.6	1	64	63 ^{106}Rh			average
$^{106}\text{Rh}^m(\beta^-)^{106}\text{Pd}$	3677	10				2					66De11 *
$^{106}\text{Ag}(\beta^+)^{106}\text{Pd}$	2980	20	2965.1	2.8	-0.7	U					53Be42
	3011	72			-0.6	U					86Bo28
$^{106}\text{Ag}(\epsilon)^{106}\text{Pd}$	2961	4			1.0	-					78Ge01 *
$^{106}\text{Pd}(\text{p}, \text{n})^{106}\text{Ag}$	-3754	13	-3747.5	2.8	0.5	U			Oak		64Jo11
	-3756	5			1.7	-					79De44
$^{106}\text{Ag}(\epsilon)^{106}\text{Pd}$ ave.	2966	3	2965.1	2.8	-0.3	1	81	81 ^{106}Ag			average
$^{106}\text{In}(\beta^+)^{106}\text{Cd}$	6516	30	6524	12	0.3	2					66Ca09 *
	6507	29			0.6	2					86Bo28 *
$^{106}\text{Cd}(\text{p}, \text{n})^{106}\text{In}$	-7312.9	15.	-7306	12	0.4	2			ANL		84Fi05 *
$^{106}\text{Sn}(\beta^+)^{106}\text{In}$	3195	60	3254	13	1.0	U			GSI		79Pi06 *
	3200	100			0.5	U					88Ba10
* $^{106}\text{Zr}-\text{u}$	Trends from Mass Surface TMS suggest ^{106}Zr 200 keV more bound										GAu **
* $^{106}\text{Ag}-\text{u}$	$M-A=-86880(32)$ keV for mixture gs+m at 89.66 keV										Nub211 **
* $^{106}\text{In}-\text{u}$	$M-A=-80575(29)$ keV for mixture gs+m at 28.6 keV										Nub211 **
* $^{106}\text{Tc}-^{102}\text{Ru}_{1,039}$	F : unidentified impurities in the trap										07Ha20 **
* $^{106}\text{Cd}(^3\text{He}, ^6\text{He})^{103}\text{Cd}$	First excited state at 187.89 keV yields $Q=-9146$ keV										GAu **
* $^{104}\text{Pd}(^3\text{He}, \text{p})^{106}\text{Ag}$	F : withdrawn by authors										AHW **
* $^{105}\text{Pd}(\text{n}, \gamma)^{106}\text{Pd}$	Calculated from 13 γ energies in 2 keV n-capture										AHW **
*	to levels in ^{106}Pd ; corrected for recoil										Ens086 **
* $^{106}\text{Rh}^m(\beta^-)^{106}\text{Pd}$	$E_{\beta^-}=920(10)$ to 5^+ level at 2757.06 keV										Ens086 **
* $^{106}\text{Ag}(\epsilon)^{106}\text{Pd}$	$L/K=0.203(0.003)$ gives $Q_{\beta^+}=99(4)$, recalculated Q										AHW **
*	from $^{106}\text{Ag}^m 6^+$ at 89.66 keV to 5^+ level at 2951.84 keV										Ens086 **
* $^{106}\text{In}(\beta^+)^{106}\text{Cd}$	$E_{\beta^+}=4890(30)$ from $^{106}\text{In}^m 2^+$ at 28.6 to 2^+ level at 632.64 keV										Ens086 **
* $^{106}\text{In}(\beta^+)^{106}\text{Cd}$	$E_{\beta^+}=2965(30)$ to 6^+ level at 2491.66 keV and 4908(29) from										Ens086 **
*	$^{106}\text{In}^m (2)^+$ at 28.6 to 2^+ level at 632.64 keV										Ens086 **
* $^{106}\text{Cd}(\text{p}, \text{n})^{106}\text{In}$	$T=7535(15)$ to 2^+ level at 151.1 keV										Ens086 **
* $^{106}\text{Sn}(\beta^+)^{106}\text{In}$	$p^+=0.253(0.021)$ to 1^+ level at 893.0 keV										Ens086 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{107}\text{Zr}-u$	-58379	482	-57990#	320#	0.3	D			GT3	2.5	16Kn03 *
$^{107}\text{Nb}-u$	-68326	279	-68410	9	-0.2	U			GT1	1.5	04Ma.A
$^{107}\text{Mo}-^{97}\text{Zr}_{1.103}$	20326.7	9.9				2			JY1	1.0	06Ha03
$^{107}\text{Pd}-u$	-95013	95	-94871.9	1.3	1.5	U			GS2	1.0	05Li24 *
$\text{C}_8\text{H}_{11}-^{107}\text{Ag}$	180986.4	3.1	180983.8	2.6	-0.3	1	11	^{107}Ag	M16	2.5	63Da10
	180994	17			-0.4	U			R12	1.5	83De51
$\text{C}_7\text{N H}_9-^{107}\text{Ag}$	168415	8	168407.8	2.6	-0.6	U			R12	1.5	83De51
$\text{C}_7\text{O H}_7-^{107}\text{Ag}$	144595	18	144598.3	2.6	0.1	U			R12	1.5	83De51
$\text{C}_6^{13}\text{C O H}_6-^{107}\text{Ag}$	140131	16	140128.1	2.6	-0.1	U			R12	1.5	83De51
$\text{C}_6\text{N O H}_5-^{107}\text{Ag}$	132025	16	132022.3	2.6	-0.1	U			R12	1.5	83De51
$^{107}\text{Cd}-u$	-93410	30	-93388.0	1.8	0.7	U			GS2	1.0	05Li24
	-93359	97			-0.3	U			GR1	1.0	19An10
$^{107}\text{Cd}-^{85}\text{Rb}_{1.259}$	17668.7	1.9	17668.8	1.8	0.0	1	88	^{107}Cd	MA8	1.0	09Br09
$^{107}\text{In}-u$	-89710	30	-89713	10	-0.1	-			GS2	1.0	05Li24
	-89730.0	29.0			0.6	-			GR1	1.0	20Ho03
ave.	-89720	21			0.4	1	25	^{107}In			average
$^{107}\text{Sn}-^{87}\text{Rb}_{1.230}$	27421.6	5.7				2			JY1	1.0	09El07
$^{107}\text{Sb}-^{87}\text{Rb}_{1.230}$	35866.2	5.8	35859	4	-1.3	1	59	^{107}Sb	JY1	1.0	09El07
$^{107}\text{Sb}-^{133}\text{Cs}_{8.805}$	251.8	9.7	262	4	1.0	1	21	^{107}Sb	SH1	1.0	07Ma92
$^{107}\text{Nb}-^{102}\text{Ru}_{1.049}$	31936.7	8.6				2			JY1	1.0	07Ha32
$^{107}\text{Tc}-^{105}\text{Ru}_{1.019}$	9465.8	8.9				2			JY1	1.0	07Ha20
$^{107}\text{Ru}-^{105}\text{Ru}_{1.019}$	3977.2	8.9				2			JY1	1.0	07Ha20
$^{107}\text{Sn}-^{106}\text{Sn}$	-1148	86	-1244	8	-0.7	U			CR2	1.5	92Sh.A *
$^{107}\text{Te}(\alpha)^{103}\text{Sn}$	3982.2	15.	4010	5	1.9	U					79Sc22
	4011.3	5.			-0.2	3					91He21
	4007	10			0.3	3			Ara		19Au02
	4001	17			0.6	U					20Ca01
$^{107}\text{Ag}(\text{p},\text{t})^{105}\text{Ag}$	-9015	15	-8997	5	1.2	1	11	^{105}Ag	Min		75Ku14 *
$^{106}\text{Pd}(\text{n},\gamma)^{107}\text{Pd}$	6536.4	0.5	6536.4	0.5	0.1	1	99	^{107}Pd	Bdn		06Fi.A
$^{107}\text{Ag}(\gamma,\text{n})^{106}\text{Ag}$	-9353	34	-9536	4	-5.4	B			Phi		60Ge01
$^{107}\text{Ag}(\text{p},\text{d})^{106}\text{Ag}$	-7305	11	-7311	4	-0.6	1	10	^{106}Ag	Bld		75An07
$^{107}\text{Mo}(\beta^-)^{107}\text{Tc}$	6160	60	6205	13	0.7	U			Bwg		89Gr23
$^{107}\text{Tc}(\beta^-)^{107}\text{Ru}$	4820	85	5113	12	3.4	B			Bwg		89Gr23
$^{107}\text{Ru}(\beta^-)^{107}\text{Rh}$	3140	300	3001	15	-0.5	U					62Pi02 *
	2900	135			0.7	U			Bwg		89Gr23
$^{107}\text{Rh}(\beta^-)^{107}\text{Pd}$	1510	40	1509	12	-0.0	U					62Pi02 *
$^{107}\text{Pd}(\beta^-)^{107}\text{Ag}$	33	3	34.0	2.3	0.3	1	60	^{107}Ag			49Pa.B
$^{107}\text{Cd}(\beta^+)^{107}\text{Ag}$	1417	4	1416.4	2.6	-0.2	1	41	^{107}Ag			62La10 *
$^{107}\text{In}(\beta^+)^{107}\text{Cd}$	3426	11	3424	10	-0.2	1	76	^{107}In			86Bo28 *
$^{107}\text{Zr}-u$	Trends from Mass Surface TMS suggest ^{107}Zr 360 keV less bound										GAu **
$^{107}\text{Pd}-u$	$M-A=-88397(62)$ keV for mixture gs+n at 214.6 keV										Nub211 **
$^{107}\text{Sn}-^{106}\text{Sn}$	From $^{107}\text{Sn}/^{106}\text{Sn}=1.00943053(81)$										AHW **
$^{107}\text{Ag}(\text{p},\text{t})^{105}\text{Ag}$	Recalibrated with (p,t) results on ^{104}Pd , ^{105}Pd , ^{106}Pd and ^{108}Pd										AHW **
$^{107}\text{Ru}(\beta^-)^{107}\text{Rh}$	$E_{\beta^-}=2100(300)$ to $(5/2^+, 7/2^+)$ level at 1041.950 keV										Ens085 **
$^{107}\text{Rh}(\beta^-)^{107}\text{Pd}$	$E_{\beta^-}=840(40)$ to $5/2^+$ level at 670.06 keV										Ens085 **
$^{107}\text{Cd}(\beta^+)^{107}\text{Ag}$	$E_{\beta^+}=302(4)$ to $^{107}\text{Ag}^m$ at 93.125 keV										Nub211 **
$^{107}\text{In}(\beta^+)^{107}\text{Cd}$	$E_{\beta^+}=2199(11)$ to $5/2^+$ level at 204.98 keV										Ens085 **
$^{108}\text{Nb}-u$	-64413	440	-63924	9	0.7	o			GT1	1.5	04Ma.A
	-63945	112			0.1	U			GT2	2.5	08Kn.A
$^{108}\text{Nb}-^{120}\text{Sn}_{900}$	24093.3	8.8				2			JY1	1.0	11Ha48
$^{108}\text{Mo}-^{97}\text{Zr}_{1.113}$	23144.8	9.9				2			JY1	1.0	06Ha03
$^{108}\text{Mo}-u$	-76039	215	-75952	10	0.3	U			GT1	1.5	04Ma.A
$^{108}\text{Rh}-^{120}\text{Sn}_{900}$	-3267	15				2			JY1	1.0	07Ha20
$^{108}\text{Rh}^m-^{120}\text{Sn}_{900}$	-3144	13				2			JY1	1.0	07Ha20

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_8 H_{12}-^{108}Pd$	190014	6	190008.6	1.2	-0.4	U			M16	2.5	63Da10
	190005	19			0.1	U			R12	1.5	83De51
$C_7 ^{13}C H_{11}-^{108}Pd$	185532	30	185538.4	1.2	0.1	U			R12	1.5	83De51
$C_7 N H_{10}-^{108}Pd$	177422	17	177432.5	1.2	0.4	U			R12	1.5	83De51
$C_6 ^{13}C N H_9-^{108}Pd$	172943	18	172962.3	1.2	0.7	U			R12	1.5	83De51
$C_7 O H_8-^{108}Pd$	153611	17	153623.1	1.2	0.5	U			R12	1.5	83De51
$C_6 N O H_6-^{108}Pd$	141031	16	141047.0	1.2	0.7	U			R12	1.5	83De51
$^{108}Pd-u$	-96109.6	1.3	-96108.2	1.2	0.7	-			TG1	1.5	12Sm01
	-96108.1	4.7			-0.0	-			TG1	1.5	12Sm01
ave.	-96109.5	1.9			0.7	1	40	$^{40}^{108}Pd$			average
$^{108}Ag-u$	-93973	50	-94049.8	2.6	-1.5	U			GS2	1.0	05Li24 *
$^{108}Ag-^{133}Cs_{812}$	-17218	84	-17276.7	2.6	-0.7	U			MA8	1.0	08Br.A *
$C_8 H_{12}-^{108}Cd$	189715.6	2.9	189716.8	1.2	0.2	U			M16	2.5	63Da10
	189695	16			0.9	U			R12	1.5	83De51
$C_7 N H_{10}-^{108}Cd$	177140	30	177140.7	1.2	0.0	U			R12	1.5	83De51
$C_6 ^{13}C N H_9-^{108}Cd$	172653	15	172670.5	1.2	0.8	U			R12	1.5	83De51
$C_6 N O H_6-^{108}Cd$	140746	15	140755.2	1.2	0.4	U			R12	1.5	83De51
$^{108}Cd-^{85}Rb_{1.271}$	16298.5	2.3	16298.8	1.2	0.1	1	27	$^{27}^{108}Cd$	MA8	1.0	09Br09
$^{108}Cd-u$	-95818.3	1.7	-95816.4	1.2	0.7	-			TG1	1.5	12Sm01
	-95817.1	4.87			0.1	-			TG1	1.5	12Sm01
ave.	-95818.2	2.4			0.7	1	25	$^{25}^{108}Cd$			average
$^{108}In-u$	-90277	31	-90306	9	-0.9	U			GS2	1.0	05Li24 *
$^{108}Sn-u$	-88102	32	-88106	6	-0.1	U			GS2	1.0	05Li24
$^{108}Sn-^{87}Rb_{1.241}$	24599.8	5.9	24601	6	0.2	1	96	$^{96}^{108}Sn$	JY1	1.0	09El07
$^{108}Sb-^{87}Rb_{1.241}$	34933.7	5.9				2			JY1	1.0	09El07
$^{108}Te-^{87}Rb_{1.241}$	42085.3	6.0	42087	6	0.4	1	94	$^{94}^{108}Te$	JY1	1.0	09El07
$^{108}Tc-^{105}Ru_{1.029}$	13423.4	9.0				2			JY1	1.0	07Ha20
$^{108}Ru-^{105}Ru_{1.029}$	5115.7	8.9				2			JY1	1.0	07Ha20
$^{108}Pd-^{108}Cd$	-292.05	0.59	-291.8	0.8	0.3	1	87	$^{46}^{108}Cd$	TG1	1.5	12Sm01
$^{108}Sn-^{107}Sn$	-3650	76	-3819	8	-1.5	U			CR2	1.5	92Sh.A *
$^{108}Pd-^{106}Pd$	425	40	411.5	1.7	-0.2	U			R12	1.5	83De51
$^{108}Cd-^{106}Cd$	-2232	32	-2276.2	1.7	-0.9	U			R12	1.5	83De51
$^{108}Te(\alpha)^{104}Sn$	3406.4	30.	3420	8	0.5	U					65Ma12
	3448.0	20.			-1.3	1	13	$^{7}^{104}Sn$			81Sc17
	3444.9	4.			-6.1	B					91He21
	3406.4	25.			0.6	U					91Pa05
$^{108}I(\alpha)^{104}Sb$	4034.7	25.	4099	5	1.1	F					91Pa05 *
	4099.1	5.			-0.1	5					94Pa12
	4097	10			0.2	5			Ara		19Au02
$^{108}Xe(\alpha)^{104}Te$	4569.6	208.0				5			Ara		18Au04
$^{108}Pd(d,^3He)^{107}Rh$	-4456	12				2			Grn		86Ka43
$^{108}Pd(d,t)^{107}Pd$	-2977	30	-2965.6	1.6	0.4	U			Pit		64Co11
$^{107}Ag(n,\gamma)^{108}Ag$	7269.6	0.6	7271.41	0.17	3.0	B			ILn		85Ma54 Z
	7271.41	0.17				2			Bdn		06Fi.A
$^{107}Ag(d,p)^{108}Ag$	5051	7	5046.84	0.17	-0.6	U			MIT		67Sp09
$^{108}I(p)^{107}Te$	597	13				4			Ara		19Au02
$^{108}Mo(\beta^-)^{108}Tc$	5135	60	5174	13	0.6	U			Bwg		92Gr.A
	5120	40			1.3	o					94Jo.A
	5100	60			1.2	U					95Jo02
$^{108}Tc(\beta^-)^{108}Ru$	7720	50	7739	12	0.4	U			Bwg		89Gr23
$^{108}Ru(\beta^-)^{108}Rh$	1315	100	1370	16	0.5	U					62Pi02 *
	1420	185			-0.3	U			Bwg		89Gr23
	1380	80			-0.1	o			Jyv		92Jo05
	1350	60			0.3	U			Jyv		94Jo.A
$^{108}Rh(\beta^-)^{108}Pd$	4500	600	4493	14	-0.0	U					62Pi02
	4505	105			-0.1	U			Bwg		89Gr23

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{108}\text{Rh}^m(\beta^-)^{108}\text{Pd}$	4434	50	4608	12	3.5	B					69Pi08 *
	4510	100			1.0	U					84Bh02 *
$^{108}\text{Ag}(\beta^+)^{108}\text{Pd}$	1902	25	1917.4	2.6	0.6	U					62Fr07
$^{108}\text{Pd}(\text{p,n})^{108}\text{Ag}$	-2675	100	-2699.8	2.6	-0.2	U			Oak		64Jo11
$^{108}\text{Ag}(\beta^-)^{108}\text{Cd}$	1650	40	1645.6	2.6	-0.1	U					60Wa10
$^{108}\text{In}(\beta^+)^{108}\text{Cd}$	5124	50	5133	9	0.2	U					62Ka23 *
	5125	14			0.5	-					86Bo28 *
$^{108}\text{Cd}(\text{p,n})^{108}\text{In}$	-5927	12	-5915	9	1.0	-			ANL		84Fi05 *
$^{108}\text{In}(\beta^+)^{108}\text{Cd}$	ave.	5136	5133	9	-0.4	1	89	89 ^{108}In			average
$^{108}\text{Sn}(\beta^+)^{108}\text{In}$	2078	25	2050	10	-1.1	1	15	11 ^{108}In	GSI		79Pi06 *
* $^{108}\text{Ag}-\text{u}$	$M-A=-87480(34)$ keV for mixture gs+m at 109.466 keV										Nub211 **
* $^{108}\text{Ag}-^{133}\text{Cs}_{812}$	$D_M=-17159(77)$ μu for mixture gs+m at 109.466 keV; $M-A=-87497(72)$ keV										Nub211 **
* $^{108}\text{In}-\text{u}$	$M-A=-84078(28)$ keV for mixture gs+m at 29.75 keV										Nub211 **
* $^{108}\text{Sn}-^{107}\text{Sn}$	From $^{107}\text{Sn}/^{108}\text{Sn}=0.99076701(70)$										AHW **
* $^{108}\text{I}(\alpha)^{104}\text{Sb}$	F : Same authors say: Consistent with new value, if recalibrated										94Pa12 **
* $^{108}\text{Ru}(\beta^-)^{108}\text{Rh}$	$E_{\beta^-}=1150(100)$ to 1^+ level at 164.98 keV										Ens08a **
* $^{108}\text{Rh}^m(\beta^-)^{108}\text{Pd}$	$E_{\beta^-}=1570(50)$ to $(4^+, 5^+, 6^+)$ level at 2863.70 keV										Ens08a **
* $^{108}\text{Rh}^m(\beta^-)^{108}\text{Pd}$	$E_{\beta^-}=1970(100)$ to 4^+ level at 2540.2 keV										Ens08a **
* $^{108}\text{In}(\beta^+)^{108}\text{Cd}$	$E_{\beta^+}=1290(80)$ to 6^+ level at 2807.81 keV, and $E_{\beta^+}=3500(50)$ from $^{108}\text{In}^m$										Ens08a **
*	at 29.75 to 2^+ level at 632.986 keV										Nub211 **
* $^{108}\text{In}(\beta^+)^{108}\text{Cd}$	$E_{\beta^+}=1887(28)$ to 4^+ level at 2239.35 keV, and $E_{\beta^+}=3494(14)$ from $^{108}\text{In}^m$										Ens08a **
*	at 29.75 to 2^+ level at 632.986 keV										Nub211 **
* $^{108}\text{Cd}(\text{p,n})^{108}\text{In}$	$Q(\text{not T, PrvCom Fi})=-6191(8), -6244(9)$, errors statistical only,										AHW **
*	to 3^+ levels at respectively, 198.36 and 266.06 keV										Ens08a **
* $^{108}\text{Sn}(\beta^+)^{108}\text{In}$	$p^+=35(6)\times 10^{-4}$ to 1^+ level at 698.85 keV										Ens08a **
$^{109}\text{Nb}-\text{u}$	-60784	376	-60860	460	-0.1	o			GT1	1.5	04Ma.A
	-60859	185				2			GT3	2.5	16Kn03
$^{109}\text{Mo}-^{97}\text{Zr}_{1.124}$	28515	12				2			JY1	1.0	06Ha03
$^{109}\text{Mo}-\text{u}$	-71552	247	-71562	12	-0.0	U			GT1	1.5	04Ma.A
$^{109}\text{Rh}-^{120}\text{Sn}_{908}$	-2463.6	7.2	-2450	4	1.8	1	37	36 ^{109}Rh	JY1	1.0	07Ha20
$\text{C}_8 \text{H}_{13}-^{109}\text{Ag}$	196972.1	3.8	196969.6	1.4	-0.3	U			M16	2.5	63Da10
	196972	6			-0.3	U			R12	1.5	83De51
$\text{C}_6^{13}\text{C O H}_8-^{109}\text{Ag}$	156110	16	156113.9	1.4	0.2	U			R12	1.5	83De51
$\text{C}_6 \text{N O H}_7-^{109}\text{Ag}$	148006	16	148008.1	1.4	0.1	U			R12	1.5	83De51
$^{109}\text{Ag}-^{133}\text{Cs}_{820}$	-17766	83	-17714.8	1.4	0.6	U			MA8	1.0	08Br.A *
$^{109}\text{Cd}-^{85}\text{Rb}_{1.282}$	18072.9	1.9	18072.3	1.6	-0.3	1	75	75 ^{109}Cd	MA8	1.0	09Br09
$^{109}\text{In}-\text{u}$	-92885.2	36.5	-92850	4	1.0	U			TR1	1.0	20Ho03
$^{109}\text{Sn}-\text{u}$	-88747	30	-88707	9	1.3	U			GS2	1.0	05Li24
$^{109}\text{Sb}-^{87}\text{Rb}_{1.253}$	31937.9	5.9	31938	6	0.0	1	92	92 ^{109}Sb	JY1	1.0	09El07
$^{109}\text{Sb}-^{133}\text{Cs}_{820}$	-4360	19	-4329	6	1.6	U			SH1	1.0	07Ma92
$^{109}\text{Te}-^{87}\text{Rb}_{1.253}$	41101.6	6.4	41101	5	-0.0	1	54	54 ^{109}Te	JY1	1.0	09El07
$^{109}\text{Te}-^{133}\text{Cs}_{820}$	4835.6	8.3	4834	5	-0.2	1	32	32 ^{109}Te	SH1	1.0	07Ma92
$^{109}\text{Tc}-^{105}\text{Ru}_{1.038}$	16014.3	10.0				2			JY1	1.0	07Ha20
$^{109}\text{Ru}-^{105}\text{Ru}_{1.038}$	9083.9	9.2				2			JY1	1.0	07Ha20
$^{109}\text{Ag}-^{107}\text{Ag}$	-335	28	-335.7	2.9	-0.0	U			R12	1.5	83De51
$^{109}\text{Te}(\alpha)^{105}\text{Sn}$	3197.6	30.	3198	6	0.0	U					65Ma12
	3197.6	15.			0.0	1	13	7 ^{109}Te			79Sc22
	3225.6	4.			-7.0	C					91He21
$^{109}\text{I}(\alpha)^{105}\text{Sb}$	3918.1	20.8				3			ORa		07Ma35
$^{109}\text{Xe}(\alpha)^{105}\text{Te}$	4217.0	7.3				5					06Li41 *
$^{109}\text{Ag}(\text{p,t})^{107}\text{Ag}$	-7995	15	-7973.6	2.7	1.4	U			Min		75Ku14 *
$^{108}\text{Pd}(\text{n},\gamma)^{109}\text{Pd}$	6153.8	0.3	6153.59	0.15	-0.7	-			ILn		80Ca02 Z
	6153.54	0.17			0.3	-			Bdn		06Fi.A
$^{108}\text{Pd}(\text{d,p})^{109}\text{Pd}$	3936	30	3929.02	0.15	-0.2	U			Pit		64Co11
$^{108}\text{Pd}(\text{n},\gamma)^{109}\text{Pd}$	ave.	6153.60	6153.59	0.15	-0.1	1	100	81 ^{109}Pd			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{109}\text{Ag}(\gamma, n)^{108}\text{Ag}$	-9196	26	-9184.0	2.7	0.5	U			Phi		60Ge01
$^{109}\text{Ag}(d, t)^{108}\text{Ag}$	-2947	30	-2926.7	2.7	0.7	U			Pit		64Co11
$^{108}\text{Cd}(^3\text{He}, d)^{109}\text{In}-^{110}\text{Cd}(^{111}\text{In})$	-806.5	2.6	-807.0	2.5	-0.2	1	91	70 ^{109}In			80Ta07
$^{109}\text{Te}(\varepsilon p)^{108}\text{Sn}$	7140	60	7066	7	-1.2	U					73Bo20
$^{109}\text{I}(p)^{108}\text{Te}$	819	5	820	4	0.2	o					84Fa04
	819.6	2.0			0.3	o					92He.A
	820.2	4.0			2						95Ho26
$^{109}\text{Tc}(\beta^-)^{109}\text{Ru}$	6315	70	6456	13	2.0	U			Bwg		89Gr23
$^{109}\text{Ru}(\beta^-)^{109}\text{Rh}$	4160	65	4261	10	1.6	U			Bwg		89Gr23
$^{109}\text{Rh}(\beta^-)^{109}\text{Pd}$	2577	50	2607	4	0.6	U					78Ka10 *
$^{109}\text{Pd}(\beta^-)^{109}\text{Ag}$	1116	2	1112.9	1.4	-1.5	1	49	30 ^{109}Ag			62Br15 *
$^{109}\text{Cd}(\varepsilon)^{109}\text{Ag}$	182	3	215.1	1.8	11.0	C					68Go.A *
	214	3			0.4	1	35	22 ^{109}Cd			Averag *
$^{109}\text{In}(\beta^+)^{109}\text{Cd}$	2015	8	2015	4	-0.0	-					62No06 *
	2030	15			-1.0	-					71Ba08 *
ave.	2018	7			-0.5	1	33	30 ^{109}In			average
$^{109}\text{Sn}(\beta^+)^{109}\text{In}$	4120	50	3859	9	-5.2	B					70Sh05
$^{109}\text{Sb}(\beta^+)^{109}\text{Sn}$	6380	16	6379	9	-0.1	1	30	22 ^{109}Sn			82Jo03 *
$^{109}\text{Ag}-^{133}\text{Cs}_{820}$	$D_M = -17719(78) \mu\text{u}$ for mixture gs+m at 88.0337 keV; $M - A = -88723(73) \text{ keV}$										Nub211 **
$^{109}\text{Xe}(\alpha)^{105}\text{Te}$	Also $E_\alpha = 3918(9) \text{ keV}$ to 150(13) level										06Li41 **
$^{109}\text{Ag}(p, t)^{107}\text{Ag}$	Recalibrated with (p, t) results on ^{104}Pd , ^{105}Pd , ^{106}Pd and ^{108}Pd										AHW **
$^{109}\text{Rh}(\beta^-)^{109}\text{Pd}$	$E_\beta = 2250(50)$ to $5/2^+$ level at 326.869 keV										Ens062 **
$^{109}\text{Pd}(\beta^-)^{109}\text{Ag}$	$E_\beta = 1028(2)$ to $^{109}\text{Ag}^m$ at 88.0337 keV										Nub211 **
$^{109}\text{Cd}(\varepsilon)^{109}\text{Ag}$	IBE=68(3) gives 94(3) to $^{109}\text{Ag}^m$ at 88.0337 keV										Nub211 **
$^{109}\text{Cd}(\varepsilon)^{109}\text{Ag}$	From average LM/K=0.2265(0.0026) $\rightarrow Q_{\beta^+} = 126(3)$; recalculated Q										AHW **
*	to $^{109}\text{Ag}^m$ at 88.0337 keV										Nub211 **
*	LMN/K=0.228(0.003)										65Le06 **
*	L/K=0.195(0.005) $\rightarrow$ LMN/K=0.258(0.006) $\rightarrow Q_{\beta^+} = 109(5)$ not used										65Le06 **
*	LMN/K=0.226(0.003)										70Go39 **
$^{109}\text{In}(\beta^+)^{109}\text{Cd}$	$E_{\beta^+} = 790(8) \text{ 805(15)}$ respectively, to $7/2^+$ level at 203.30 keV										Ens062 **
$^{109}\text{Sb}(\beta^+)^{109}\text{Sn}$	$E_{\beta^+} = 4416(21)$ to $3/2^+$ level at 925.6 keV, and other E_{β^+}										Ens062 **
$^{110}\text{Nb}-u$	-56157	360				2			GT3	2.5	16Kn03
$^{110}\text{Mo}-^{97}\text{Zr}_{1.134}$	31685	26				2			JY1	1.0	06Ha03
$^{110}\text{Mo}-u$	-69544	268	-69282	26	0.7	U			GT1	1.5	04Ma.A
$^{110}\text{Ru}-u$	-85897	78	-85961	10	-0.8	U			JY0	1.0	03Ko.A
$^{110}\text{Rh}-u$	-88835	130	-88920	19	-0.7	U			JY0	1.0	03Ko.A *
$^{110}\text{Rh}-^{120}\text{Sn}_{917}$	815	110	760	19	-0.5	U			JY1	1.0	07Ha20 *
$\text{C}_8 \text{H}_{14}-^{110}\text{Pd}$	204389	9	204377.6	0.7	-0.5	U			M16	2.5	63Da10
	204380	20			-0.1	U			R12	1.5	83De51
$\text{C}_7 \text{ }^{13}\text{C} \text{H}_{13}-^{110}\text{Pd}$	199913	20	199907.4	0.7	-0.2	U			R12	1.5	83De51
$\text{C}_6 \text{N} \text{O} \text{H}_8-^{110}\text{Pd}$	155418	17	155416.0	0.7	-0.1	U			R12	1.5	83De51
$\text{C}_5 \text{ }^{13}\text{C} \text{N} \text{O} \text{H}_7-^{110}\text{Pd}$	150946	17	150945.8	0.7	-0.0	U			R12	1.5	83De51
$\text{C}_6 \text{O}_2 \text{H}_6-^{110}\text{Pd}$	131609	18	131606.6	0.7	-0.1	U			R12	1.5	83De51
$^{110}\text{Pd}-u$	-94829.7	1.5	-94827.1	0.7	1.7	-			MA8	1.0	12Fi01
	-94829.5	1.7			0.9	-			TG1	1.5	12Sm01
	-94830.9	3.0			0.8	-			TG1	1.5	12Sm01
ave.	-94829.7	1.2			2.1	1	28	28 ^{110}Pd			average
$\text{C}_8 \text{H}_{14}-^{110}\text{Cd}$	206548.4	4.6	206543.0	0.4	-0.5	U			M16	2.5	63Da10
	206550	45			-0.1	U			R12	1.5	83De51
	206569	13			-1.3	U			R12	1.5	83De51
$\text{C}_7 \text{ }^{13}\text{C} \text{H}_{13}-^{110}\text{Cd}$	202093	14	202072.8	0.4	-1.0	U			R12	1.5	83De51
	202053	28			0.5	U			R12	1.5	83De51
$\text{C}_7 \text{O} \text{H}_{10}-^{110}\text{Cd}$	170156	16	170157.5	0.4	0.1	U			R12	1.5	83De51
$\text{C}_6 \text{N} \text{O} \text{H}_8-^{110}\text{Cd}$	157614	17	157581.4	0.4	-1.3	U			R12	1.5	83De51

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_5 \text{ }^{13}\text{C N O H}_7 - ^{110}\text{Cd}$	153131	17	153111.2	0.4	-0.8	U			R12	1.5	83De51
$C_6 \text{ O}_2 \text{ H}_6 - ^{110}\text{Cd}$	133801	18	133772.0	0.4	-1.1	U			R12	1.5	83De51
$C_9 \text{ H}_2 - ^{110}\text{Cd}$	112661	19	112642.6	0.4	-0.6	U			R12	1.5	83De51
$^{110}\text{Cd}-u$	-96993.6	1.5	-96992.5	0.4	0.7	-			MA8	1.0	12Fi01
	-96997.0	1.5			2.0	-			TG1	1.5	12Sm01
	-96992.2	2.4			-0.1	-			TG1	1.5	12Sm01
ave.	-96994.4	1.2			1.6	1	12	12 ^{110}Cd			average
$^{110}\text{In}-u$	-92898	36	-92829	12	1.9	U			GS2	1.0	05Li24 *
$^{110}\text{Sn}-u$	-92189	30	-92155	15	1.1	2			GS2	1.0	05Li24
$^{110}\text{Sb}-^{87}\text{Rb}_{1.264}$	31650.1	6.4				2			JY1	1.0	09El07
$^{110}\text{Te}-^{133}\text{Cs}_{8.827}$	643.8	7.7	649	7	0.7	1	84	84 ^{110}Te	SH1	1.0	07Ma92
$^{110}\text{Tc}-^{105}\text{Ru}_{1.048}$	20424.0	9.8				2			JY1	1.0	07Ha20
$^{110}\text{Ru}-^{105}\text{Ru}_{1.048}$	10719.5	9.3	10721	9	0.2	1	97	97 ^{110}Ru	JY1	1.0	07Ha20
$^{110}\text{Cd } ^{35}\text{Cl}-^{108}\text{Cd } ^{37}\text{Cl}$	1764	5	1774.0	1.3	0.5	U			H11	4.0	63Bi12
$^{110}\text{Pd}-^{110}\text{Cd}$	2166.24	0.69	2165.4	0.6	-1.2	-			MA8	1.0	12Fi01
	2166.2	1.3			-0.4	-			TG1	1.5	12Sm01
ave.	2166.2	0.7			-1.3	1	80	71 ^{110}Pd			average
$^{110}\text{Pd}-^{108}\text{Pd}$	1288	35	1281.1	1.3	-0.1	U			R12	1.5	83De51
$^{110}\text{Cd}-^{108}\text{Cd}$	-1219	34	-1176.1	1.3	0.8	U			R12	1.5	83De51
$^{110}\text{Te}(\alpha)^{106}\text{Sn}$	2723.2	15.6	2699	8	-1.6	1	25	16 ^{110}Te			81Sc17
$^{110}\text{I}(\alpha)^{106}\text{Sb}$	3574.2	10.	3580	60	0.1	3					81Sc17
	3586.7	5.			-0.1	3					91He21
$^{110}\text{Xe}(\alpha)^{106}\text{Te}$	3878.3	30.	3872	9	-0.2	4					81Sc17
	3886.6	15.			-0.9	4					92He.A
	3871.0	30.			0.0	4					02Ma19
	3857.5	19.7			0.7	4			Jya		07Sa36
	3860.7	20.8			0.6	4					16Ca33
$^{110}\text{Pd}(\text{p,t})^{108}\text{Pd}$	-6495	15	-6467.5	1.3	1.8	U			Min		75Ku14 *
$^{110}\text{Pd}(\text{d}, ^3\text{He})^{109}\text{Rh}$	-5134	5	-5127	4	1.4	1	65	64 ^{109}Rh	VUn		87Ka29
$^{110}\text{Pd}(\text{t}, \alpha)^{109}\text{Rh}$	9206	25	9193	4	-0.5	U			LAI		82Fi09
$^{109}\text{Ag}(\text{n}, \gamma)^{110}\text{Ag}$	6809.2	0.1	6809.19	0.10	-0.1	1	100	57 ^{109}Ag			81Bo.B
	6808.20	0.16			6.2	C			Bdn		06Fi.A
$^{109}\text{Ag}(\text{d,p})^{110}\text{Ag}$	4590	5	4584.63	0.10	-1.1	U			MIT		64Sp12
$^{110}\text{Cd}(\text{d,t})^{109}\text{Cd}$	-3664	30	-3657.7	1.6	0.2	U			Pit		64Ro17
$^{110}\text{Tc}(\beta^-)^{110}\text{Ru}$	6680	120	9038	13	19.7	C			Jyv		89Jo.A
	9021	55			0.3	U			Jyv		00Kr.A
$^{110}\text{Ru}(\beta^-)^{110}\text{Rh}$	2810	50	2756	19	-1.1	1	15	12 ^{110}Rh	Jyv		91Jo11 *
$^{110}\text{Rh}(\beta^-)^{110}\text{Pd}$	5500	500	5502	18	0.0	U					63Ka21
	5400	100			1.0	U					70Pi01 *
	5510	19			-0.4	1	88	88 ^{110}Rh	Bwg		00Kr.A
$^{110}\text{Ag}(\beta^-)^{110}\text{Cd}$	2891.4	3.0	2890.7	1.3	-0.2	-					63Da03 *
	2892.9	2.0			-1.1	-					67Mo12 *
ave.	2892.4	1.7			-1.1	1	59	57 ^{110}Ag			average
$^{110}\text{In}(\beta^+)^{110}\text{Cd}$	3928	20	3878	12	-2.5	2					51Mc11 *
	3868	20			0.5	2					53Bi44 *
	3838	20			2.0	2					62Ka08 *
$^{110}\text{Sb}(\beta^+)^{110}\text{Sn}$	8750	200	8392	15	-1.8	U					72Mi26 *
	9085	100			-6.9	B					72Si28 *
* $^{110}\text{Rh}-u$	$M-A=-82639(73)$ keV for mixture gs+m at 220#(150#) keV										Nub211 **
* $^{110}\text{Rh}-^{120}\text{Sn}_{917}$	$D_M=933.3(7.2)$ μu for mixture gs+m at 220#(150#) keV; $M-A=-82668.1(6.8)$ keV										Nub211 **
* $^{110}\text{In}-u$	$M-A=-86503(28)$ keV for mixture gs+m at 62.08 keV										Nub211 **
* $^{110}\text{Pd}(\text{p,t})^{108}\text{Pd}$	Recalibrated with (p,t) results on ^{104}Pd , ^{105}Pd , ^{106}Pd and ^{108}Pd										AHW **
* $^{110}\text{Ru}(\beta^-)^{110}\text{Rh}$	$E_{\beta^-}=2700(50)$ to 1^+ level at 112.19 keV										Ens126 **
* $^{110}\text{Rh}(\beta^-)^{110}\text{Pd}$	$E_{\beta^-}=2600(100)$ to (4^+) levels at 2790.64 and 2805.03 keV										Ens126 **
* $^{110}\text{Ag}(\beta^-)^{110}\text{Cd}$	$E_{\beta^-}=529(3)$ from $^{110}\text{Ag}^n$ at 117.59 to 6^+ level at 2479.9339 keV										Ens126 **
* $^{110}\text{Ag}(\beta^-)^{110}\text{Cd}$	$E_{\beta^-}=2891(4)$; and 531(2) from $^{110}\text{Ag}^n$ at 117.59 to 6^+ level at 2479.9339 keV										Ens126 **
* $^{110}\text{In}(\beta^+)^{110}\text{Cd}$	$E_{\beta^+}=2310(20)$ 2250(20) 2220(20) respectively, from $^{110}\text{In}^m$ at 62.08(0.04) keV										Nub211 **
*	to 2^+ level at 657.7623 keV										Ens126 **
* $^{110}\text{Sb}(\beta^+)^{110}\text{Sn}$	$E_{\beta^+}=6500(200)$ 6850(100) respectively, to 2^+ level at 1211.72; and other E_{β^+}										Ens126 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{111}\text{Mo}-u$	-64348	279	-64348	14	-0.0	U			GT1	1.5	04Ma.A
$^{111}\text{Ru}-u$	-82302	79	-82432	10	-1.7	U			JY0	1.0	03Ko.A
$^{111}\text{Rh}-u$	-88282	79	-88357	7	-0.9	U			JY0	1.0	03Ko.A
$^{111}\text{Rh}-^{120}\text{Sn}_{925}$	2105.8	7.3				2			JY1	1.0	07Ha20
$^{111}\text{Ag}-u$	-94741	51	-94703.2	1.6	0.7	U			GS2	1.0	05Li24 *
$\text{C}_8 \text{H}_{15}-^{111}\text{Cd}$	213184.4	3.9	213191.7	0.4	0.7	U			M16	2.5	63Da10
	213197	40			-0.1	U			R12	1.5	83De51
$\text{C}_7 \text{ } ^{13}\text{C} \text{H}_{14}-^{111}\text{Cd}$	208719	19	208721.5	0.4	0.1	U			R12	1.5	83De51
$\text{C}_7 \text{O} \text{H}_{11}-^{111}\text{Cd}$	176814	16	176806.2	0.4	-0.3	U			R12	1.5	83De51
$\text{C}_9 \text{H}_3-^{111}\text{Cd}$	119317	18	119291.3	0.4	-1.0	U			R12	1.5	83De51
$\text{C}_8 \text{N} \text{H}-^{111}\text{Cd}$	106723	17	106715.3	0.4	-0.3	U			R12	1.5	83De51
$^{111}\text{Cd}-u$	-95774	30	-95816.2	0.4	-1.4	U			GS2	1.0	05Li24 *
$^{111}\text{Sb}-u$	-86837	30	-86782	10	1.8	U			GS2	1.0	05Li24
$^{111}\text{Sb}-^{133}\text{Cs}_{835}$	-7834.2	9.5				2			SH1	1.0	07Ma92
$^{111}\text{Te}-^{133}\text{Cs}_{835}$	-51.8	6.9				2			SH1	1.0	07Ma92
$^{111}\text{I}-^{87}\text{Rb}_{1.276}$	46150.4	6.1	46155	5	0.7	1	70	70 ^{111}I	JY1	1.0	09EI07
$^{111}\text{I}-^{133}\text{Cs}_{835}$	9197	19	9217	5	1.0	U			SH1	1.0	07Ma92
$^{111}\text{Tc}-^{105}\text{Ru}_{1.057}$	23412	11				2			JY1	1.0	07Ha20
$^{111}\text{Ru}-^{105}\text{Ru}_{1.057}$	15080.6	10.0				2			JY1	1.0	07Ha20
$^{110}\text{Cd} \text{H}-^{111}\text{Cd}$	6638	18	6648.73	0.18	0.4	U			R12	1.5	83De51
$^{111}\text{Mo}-^{111}\text{Tc}$	9753.0	7.3				3			JY1	1.0	11Ha48 *
$^{111}\text{Cd}-^{110}\text{Cd}$	1180	11	1176.31	0.18	-0.1	U			M16	2.5	63Da10
	1208	34			-0.6	U			R12	1.5	83De51
$^{111}\text{Cd} \text{H}-^{110}\text{Cd}$	8994	35	9001.34	0.18	0.1	U			R12	1.5	83De51
$^{111}\text{I}(\alpha)^{107}\text{Sb}$	3270.1	10.	3275	5	0.4	-					79Sc22
	3293.0	10.			-1.8	-					92He.A
ave.	3281	7			-0.9	1	50	30 ^{111}I			average
$^{111}\text{Xe}(\alpha)^{107}\text{Te}$	3693.3	25.	3710	60	0.3	4					79Sc22
	3714.1	30.			-0.0	4					81Sc17
	3723.5	10.			-0.2	4					91He21
	3715.2	17.			-0.1	4					20Ca01
$^{110}\text{Pd}(\text{n},\gamma)^{111}\text{Pd}$	5726.3	0.4				2			Bdn		06Fi.A
$^{110}\text{Cd}(\text{n},\gamma)^{111}\text{Cd}$	6980	100	6975.60	0.17	-0.0	U					61Ja21
	6975.5	0.5			0.2	-					86Ba72
	6975.9	0.2			-1.5	-					90Ne.B
	6975.1	0.4			1.2	-			Bdn		06Fi.A
$^{111}\text{Cd}(\gamma,\text{n})^{110}\text{Cd}$	-6975	3	-6975.60	0.17	-0.2	U			McM		79Ba06
$^{110}\text{Cd}(\text{d},\text{p})^{111}\text{Cd}$	4740	30	4751.03	0.17	0.4	U			Pit		64Ro17
	4750.68	0.88			0.4	U			Rez		90Pi05 *
$^{111}\text{Cd}(\text{d},\text{t})^{110}\text{Cd}$	-745	30	-718.37	0.17	0.9	U			Pit		64Co11
$^{110}\text{Cd}(\text{n},\gamma)^{111}\text{Cd}$	ave.	6975.71	0.17	6975.60	0.17	-0.7	1	97	77 ^{110}Cd		average
$^{111}\text{Te}(\text{e}\text{p})^{110}\text{Sn}$	5070	70	4966	15	-1.5	U					68Ba53
$^{111}\text{Tc}(\beta^-)^{111}\text{Ru}$	7480	80	7761	14	3.5	B			Jyv		96Kl.A
	7449	80			3.9	B			Jyv		00Kr.A
$^{111}\text{Ru}(\beta^-)^{111}\text{Rh}$	5039	50	5519	12	9.6	C			Jyv		00Kr.A
$^{111}\text{Rh}(\beta^-)^{111}\text{Pd}$	3640	50	3682	7	0.8	U			Jyv		00Kr.A
	3650	33			1.0	U			Bwg		00Kr.A
$^{111}\text{Pd}(\beta^-)^{111}\text{Ag}$	2210	100	2229.6	1.6	0.2	U					52Mc34 *
	2190	50			0.8	U					57Kn.A *
	2160	100			0.7	U					60Pr07 *
$^{111}\text{Ag}(\beta^-)^{111}\text{Cd}$	1028	3	1036.8	1.4	2.9	B					67Le06
	1035	2			0.9	2					71Na02
	1038.6	2.			-0.9	2					77Re12
$^{111}\text{Cd}(\text{p},\text{n})^{111}\text{In}$	-1635	20	-1643	3	-0.4	U			Oak		74Ki02
$^{111}\text{Sn}(\beta^+)^{111}\text{In}$	2530	30	2453	6	-2.6	U					51Mc11
$^{111}\text{Sb}(\beta^+)^{111}\text{Sn}$	4470	50	5102	10	12.6	B					72Si28 *
* $^{111}\text{Ag}-u$	$M-A=-88221(44)$ keV for mixture gs+m at 59.82 keV										Nub211 **
* $^{111}\text{Cd}-u$	$M-A=-88817(28)$ keV for $^{111}\text{Cd}^m$ 11/2 ⁻ at 396.214 keV										Nub211 **
* $^{111}\text{Mo}-^{111}\text{Tc}$	Taken as low-spin isomer (see also ^{102}Y and ^{114}Tc doublets in same work)										GAu **
* $^{110}\text{Cd}(\text{d},\text{p})^{111}\text{Cd}$	Estimated systematic error 0.5 added to statistical error 0.73 keV										AHW **
* $^{111}\text{Pd}(\beta^-)^{111}\text{Ag}$	$Q_{\beta^-}=2150(100)$ 2130(50) 2100(100) respectively, to $^{111}\text{Ag}^m$ at 59.82 keV										Nub211 **
* $^{111}\text{Sb}(\beta^+)^{111}\text{Sn}$	$E_{\beta^+}=3290(50)$ to 5/2 ⁺ level at 154.48 keV										Ens095 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{112}\text{Tc}-^{102}\text{Ru}_{1.098}$	34976.0	5.9				2			JY1	1.0	07Ha20
$^{112}\text{Ru}-u$	-81033	78	-81193	10	-2.1	U			JY0	1.0	03Ko.A
$^{112}\text{Rh}-u$	-85506	119	-85590	50	-0.7	1	16	16 ^{112}Rh	JY0	1.0	03Ko.A *
$^{112}\text{Ag}-^{133}\text{Cs}_{.842}$	-13342.0	2.6				2			MA8	1.0	10Br02
$\text{C}_8 \text{H}_{16}-^{112}\text{Cd}$	222445.3	3.9	222436.61	0.27	-0.9	U			M16	2.5	63Da10
$\text{C}_7 \text{O} \text{H}_{12}-^{112}\text{Cd}$	186063	16	186051.11	0.27	-0.5	U			R12	1.5	83De51
$\text{C}_9 \text{H}_4-^{112}\text{Cd}$	128541	19	128536.23	0.27	-0.2	U			R12	1.5	83De51
	128550	10			-0.9	U			R12	1.5	83De51
$\text{C}_8 \text{N} \text{H}_2-^{112}\text{Cd}$	115979	14	115960.17	0.27	-0.9	U			R12	1.5	83De51
$^{112}\text{In}-u$	-94366	58	-94461	5	-1.6	U			GS2	1.0	05Li24 *
$\text{C}_8 \text{H}_{16}-^{112}\text{Sn}$	220384	9	220375.6	0.3	-0.4	U			M16	2.5	63Da10
	220385	8			-0.8	U			R13	1.5	83De51
$^{112}\text{Sn}-^{86}\text{Kr}_{1.302}$	21210.3	2.5	21209.9	0.3	-0.2	U			JY1	1.0	11Ha48
$^{112}\text{Sb}-u$	-87597	30	-87600	19	-0.1	2			GS2	1.0	05Li24
$^{112}\text{Te}-^{133}\text{Cs}_{.842}$	-3662.7	9.0				2			SH1	1.0	07Ma92
$^{112}\text{I}-^{133}\text{Cs}_{.842}$	7614	11				2			SH1	1.0	07Ma92
$^{112}\text{Rh}-^{120}\text{Sn}_{.933}$	5640	110	5650	50	0.1	1	19	19 ^{112}Rh	JY1	1.0	07Ha20 *
$^{112}\text{Pd}-^{120}\text{Sn}_{.933}$	-1423.7	7.4	-1424	7	-0.1	1	89	89 ^{112}Pd	JY1	1.0	07Ha20
$^{112}\text{Sn}-^{120}\text{Sn}_{.933}$	-3930.2	1.9	-3930.1	0.9	0.1	1	25	23 ^{120}Sn	JY1	1.0	11Ha48
$^{112}\text{Ru}-^{105}\text{Ru}_{1.067}$	17242.5	9.9				2			JY1	1.0	07Ha20
$^{112}\text{Cd}^{35}\text{Cl}-^{110}\text{Cd}^{37}\text{Cl}$	2701	2	2706.5	0.3	0.7	U			H11	4.0	63Bi12
$^{111}\text{Cd} \text{H}-^{112}\text{Cd}$	9255	20	9244.9	0.3	-0.3	U			R12	1.5	83De51
$^{112}\text{Sn}-^{112}\text{Cd}$	2061.01	0.17	2061.00	0.17	-0.1	1	99	97 ^{112}Sn	JY1	1.0	09Ra11
$^{112}\text{Cd}-^{110}\text{Cd} \text{H}$	-8060	40	-8068.6	0.3	-0.1	U			R12	1.5	83De51
$^{112}\text{Cd}-^{111}\text{Cd}$	-1419	11	-1419.9	0.3	-0.0	U			M16	2.5	63Da10
	-1410	42			-0.2	U			R12	1.5	83De51
$^{112}\text{Cd}-^{110}\text{Cd}$	-238	39	-243.6	0.3	-0.1	U			R12	1.5	83De51
$^{112}\text{Cd} \text{H}-^{111}\text{Cd}$	6402	35	6405.2	0.3	0.1	U			R12	1.5	83De51
$^{112}\text{I}(\alpha)^{108}\text{Sb}$	2986.9	31.1	2957	12	-0.5	U					81Sc17
$^{112}\text{Xe}(\alpha)^{108}\text{Te}$	3329.1	20.	3330	6	0.1	2					81Sc17
	3308.4	15.6			1.4	2					92He.A
	3335.4	7.			-0.7	2					94Pa11
$^{112}\text{Sn}(^3\text{He},^6\text{He})^{109}\text{Sn}$	-8686	9	-8686	8	0.0	1	78	78 ^{109}Sn	MSU		78Pa11
$^{110}\text{Pd}(\text{t},\text{p})^{112}\text{Pd}$	5659	20	5651	7	-0.4	1	11	11 ^{112}Pd	LAL		72Ca10
$^{112}\text{Cd}(^{14}\text{C},^{16}\text{O})^{110}\text{Pd}$	5543	29	5512.9	0.6	-1.0	U			LAL		84Co19
$^{110}\text{Cd}(\text{t},\text{p})^{112}\text{Cd}$	7888	20	7887.7	0.3	-0.0	U			Ald		67Hi01
$^{112}\text{Cd}(\text{p},\text{t})^{110}\text{Cd}$	-7891	5	-7887.7	0.3	0.7	U			Min		73Oo01
$^{112}\text{Sn}(\text{p},\text{t})^{110}\text{Sn}$	-10485	15	-10474	14	0.7	R			Roc		70Fi08
$^{111}\text{Cd}(\text{n},\gamma)^{112}\text{Cd}$	9460	50	9393.93	0.28	-1.3	U					61Ja21
	9394.3	0.3			-1.2	1	89	81 ^{111}Cd	ILn		97Dr03
$^{112}\text{Cd}(\gamma,\text{n})^{111}\text{Cd}$	-9403	5	-9393.93	0.28	1.8	U			McM		79Ba06
$^{111}\text{Cd}(\text{d},\text{p})^{112}\text{Cd}$	7183	30	7169.36	0.28	-0.5	U			Pit		64Co11
	7170	10			-0.1	U			Yal		67Ba15
	7171	5			-0.3	U			MIT		67Sp09
$^{112}\text{Cd}(\text{d},\text{t})^{111}\text{Cd}$	-3129	30	-3136.70	0.28	-0.3	U			Pit		64Ro17
$^{112}\text{Sn}(\text{d},^3\text{He})^{111}\text{In}$	-2050	50	-2058	3	-0.2	U			Sac		69Co03
$^{112}\text{Sn}(\text{p},\text{d})^{111}\text{Sn}$	-8574	15	-8563	5	0.7	2			Har		70Ca01
$^{112}\text{Sn}(\text{d},\text{t})^{111}\text{Sn}$	-4529.0	5.7	-4531	5	-0.3	2			SPa		75Be09
$^{112}\text{Cs}(\text{p})^{111}\text{Xe}$	814.3	7.	816	4	0.3	5					94Pa12
	817.3	5.			-0.2	5					12Wa10
$^{112}\text{Tc}(\beta^-)^{112}\text{Ru}$	6060	130	10372	11	33.2	C			Jyv		89Jo.A
	9484	100			8.9	C			Jyv		00Kr.A
$^{112}\text{Ru}(\beta^-)^{112}\text{Rh}$	4520	80	4100	50	-5.2	B			Jyv		91Jo11 *
$^{112}\text{Rh}(\beta^-)^{112}\text{Pd}$	6200	500	6590	40	0.8	U			Jyv		88Ay02
	6573	54			0.3	1	66	66 ^{112}Rh	Bwg		00Kr.A
$^{112}\text{Rh}^m(\beta^-)^{112}\text{Pd}$	6929	56				2			Bwg		00Kr.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{112}\text{Pd}(\beta^-)^{112}\text{Ag}$	299	20	263	7	-1.8	U					55Nu11 *
$^{112}\text{Ag}(\beta^-)^{112}\text{Cd}$	4057	20	3991.1	2.4	-3.3	C					57Je.A *
	3940	40			1.3	U					62In01 *
$^{112}\text{In}(\beta^+)^{112}\text{Cd}$	2582	20	2585	4	0.1	U					62Ru05
$^{112}\text{Cd}(\text{p,n})^{112}\text{In}$	-3399.3	20.	-3367	4	1.6	U			Oak		64Jo11
	-3376	6			1.5	1	50	50 ^{112}In	Tky		80Ad04
$^{112}\text{In}(\beta^-)^{112}\text{Sn}$	656	6	665	4	1.5	1	50	50 ^{112}In			53B144
$^{112}\text{Sb}(\beta^+)^{112}\text{Sn}$	7530	100	7056	18	-4.7	B					72Mi27 *
	7029	50			0.5	R					72Si28 *
	7062	26			-0.2	R					82Jo03 *
$^{112}\text{Sn}(\text{p,n})^{112}\text{Sb}$	-7995	55	-7838	18	2.8	U			VUn		76Ka19
* $^{112}\text{Rh}-\text{u}$	Average of 2 values; $M-A=-79479(36)$ keV for mixture gs+m at 340(70) keV										Nub211 **
* $^{112}\text{In}-\text{u}$	$M-A=-87823(30)$ keV for mixture gs+m at 156.592 keV										Nub211 **
* $^{112}\text{Rh}-^{120}\text{Sn}_{933}$	$D_M=5822.6(7.4)$ μu for mixture gs+m at 340(70) keV; $M-A=-79571.2(7.0)$ keV										Nub211 **
* $^{112}\text{Ru}(\beta^-)^{112}\text{Rh}$	$E_{\beta^-}=4190(80)$ to 1^+ level at 327.03 keV										Ens14c **
* $^{112}\text{Pd}(\beta^-)^{112}\text{Ag}$	$E_{\beta^-}=280(20)$ to (1^+) level at 18.5 keV										Ens14c **
* $^{112}\text{Ag}(\beta^-)^{112}\text{Cd}$	$E_{\beta^-}=3440(20)$ to 2^+ level at 617.518 keV										Ens14c **
* $^{112}\text{Ag}(\beta^-)^{112}\text{Cd}$	$E_{\beta^-}=3350(20)$ to 2^+ level at 617.518 keV; error increased by evaluator										Ens14c **
* $^{112}\text{Sb}(\beta^+)^{112}\text{Sn}$	$E_{\beta^+}=5200(100)$ 4750(50) 4783(26) respectively, to 2^+ level at 1256.69 keV										Ens14c **
$^{113}\text{Mo}-\text{u}$	-57317	337	-56520#	320#	0.9	D			GT3	2.5	16Kn03 *
$^{113}\text{Tc}-\text{u}$	-67633	268	-67431	4	0.5	o			GT1	1.5	04Ma.A
	-67502	106			0.3	U			GT2	2.5	08Kn.A
$^{113}\text{Tc}-^{129}\text{Xe}_{876}$	15981.0	3.6				2			JY1	1.0	11Ha48
$^{113}\text{Ru}-\text{u}$	-77032	94	-77150	40	-1.3	-			JY0	1.0	03Ko.A *
	-77240	110			0.3	o			GT2	2.5	08Kn.A
	-77295	140			0.4	-			GT2	2.5	08Su19
	ave. -77050	90			-1.1	1	20	20 ^{113}Ru			average
$^{113}\text{Rh}-\text{u}$	-84464	83	-84560	8	-1.2	U			JY0	1.0	03Ko.A
$\text{C}_9 \text{H}_5-^{113}\text{Cd}$	134721.1	3.9	134717.05	0.26	-0.4	U			M16	2.5	63Da10
	134727	19			-0.3	U			R12	1.5	83De51
	134728	5			-1.5	U			R12	1.5	83De51
$\text{C}_7 \text{O H}_{13}-^{113}\text{Cd}$	192250	16	192231.93	0.26	-0.8	U			R12	1.5	83De51
$\text{C}_6^{13}\text{C O H}_{12}-^{113}\text{Cd}$	187772	17	187761.73	0.26	-0.4	U			R12	1.5	83De51
$\text{C}_8 \text{N H}_3-^{113}\text{Cd}$	122161	19	122140.99	0.26	-0.7	U			R12	1.5	83De51
$^{113}\text{Cd}-\text{u}$	-95506	93	-95591.89	0.26	-0.9	U			GS2	1.0	05Li24 *
$\text{C}_9 \text{H}_5-^{113}\text{In}$	135015	9	135064.71	0.20	2.2	U			M16	2.5	63Da10
	135087	6			-2.5	U			R12	1.5	83De51
$\text{C}_8 \text{N H}_3-^{113}\text{In}$	122506	14	122488.65	0.20	-0.8	U			R12	1.5	83De51
$^{113}\text{In}-\text{u}$	-95969	126	-95939.55	0.20	0.2	U			GS2	1.0	05Li24 *
$^{113}\text{Sn}-\text{u}$	-94796	39	-94824.1	1.7	-0.7	U			GS2	1.0	05Li24 *
$^{113}\text{Sb}-\text{u}$	-90635	30	-90625	18	0.3	R			GS2	1.0	05Li24
$^{113}\text{Te}-\text{u}$	-84109	30				2			GS2	1.0	05Li24
$^{113}\text{I}-^{133}\text{Cs}_{850}$	4015.9	8.6				2			SH1	1.0	07Ma92
$^{113}\text{Xe}-^{133}\text{Cs}_{850}$	13585.5	8.1	13588	7	0.2	1	82	82 ^{113}Xe	SH1	1.0	07Ma92
$^{113}\text{Ru}-^{105}\text{Ru}_{1,076}$	22086	46	22110	40	0.6	1	80	80 ^{113}Ru	JY1	1.0	07Ha20 *
$^{113}\text{Rh}-^{120}\text{Sn}_{942}$	7565.4	7.6				2			JY1	1.0	07Ha20
$^{113}\text{Pd}-^{120}\text{Sn}_{942}$	2387.1	7.4				2			JY1	1.0	07Ha20
$^{113}\text{Cd }^{35}\text{Cl}-^{111}\text{Cd }^{37}\text{Cl}$	3174	2	3174.4	0.4	0.1	U			H11	4.0	63Bi12
$^{112}\text{Cd H}-^{113}\text{Cd}$	6164	20	6180.82	0.23	0.6	U			R12	1.5	83De51
$^{113}\text{Cd}-^{112}\text{Cd}_{1,009}$	2519.36	0.29	2519.33	0.23	-0.1	1	65	35 ^{112}Cd	MS1	1.0	16Ga33
$^{113}\text{In}-^{112}\text{Cd}_{1,009}$	2171.26	0.32	2171.68	0.26	1.3	1	65	48 ^{112}Cd	MS1	1.0	16Ga33
$^{113}\text{Cd}-^{111}\text{Cd H}$	-7623	42	-7600.7	0.4	0.4	U			R12	1.5	83De51
$^{113}\text{Cd}-^{112}\text{Cd}$	1642	11	1644.21	0.23	0.1	U			M16	2.5	63Da10
	1620	40			0.4	U			R12	1.5	83De51

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{113}\text{In}-^{112}\text{Cd}$	1297	45	1296.56	0.26	-0.0	U			R12	1.5	83De51
$^{113}\text{Cd}-^{110}\text{Cd}$ H	-6412	32	-6424.4	0.4	-0.3	U			R12	1.5	83De51
$^{113}\text{Cd}-^{111}\text{Cd}$	242	35	224.3	0.4	-0.3	U			R12	1.5	83De51
^{113}Cd H- ^{112}Cd	9467	35	9469.24	0.23	0.0	U			R12	1.5	83De51
$^{113}\text{I}(\alpha)^{109}\text{Sb}$	2706.0	41.5	2707	10	0.0	U					81Sc17
$^{113}\text{Xe}(\alpha)^{109}\text{Te}$	3094.8	15.	3087	8	-0.5	1	24	18 ^{113}Xe			79Sc22
$^{111}\text{Cd}(\text{t,p})^{113}\text{Cd}$	7456	20	7451.9	0.3	-0.2	U			Ald		67Hi01
$^{113}\text{Cd}(\text{p,t})^{111}\text{Cd}$	-7456	5	-7451.9	0.3	0.8	U			Min		73Oo01
$^{113}\text{In}(\text{p,t})^{111}\text{In}-^{115}\text{In}(\text{o})^{113}\text{In}$	-810	10	-806	3	0.4	1	12	12 ^{111}In	Roc		74Ma09
$^{113}\text{In}(\text{p,t})^{111}\text{In}-^{112}\text{Cd}(\text{o})^{110}\text{Cd}$	-746.3	4.1	-748	3	-0.5	1	69	69 ^{111}In	SPa		80Ta07
$^{112}\text{Cd}(\text{n},\gamma)^{113}\text{Cd}$	6550	100	6539.75	0.22	-0.1	U					61Ja21
	6542.0	0.2			-11.3	C					90Ne.A
$^{112}\text{Cd}(\text{d,p})^{113}\text{Cd}$	4318	30	4315.18	0.22	-0.1	U			Pit		64Ro17
	4315.56	0.64			-0.6	1	11	6 ^{112}Cd	Rez		90Pi05 *
$^{113}\text{Cd}(\text{d,t})^{112}\text{Cd}$	-254	30	-282.52	0.22	-1.0	U			Pit		64Co11
$^{113}\text{In}(\text{d,t})^{112}\text{In}$	-3180	25	-3191	4	-0.4	U			Pit		67Hj03
$^{112}\text{Sn}(\text{n},\gamma)^{113}\text{Sn}$	7741.9	2.3	7744.4	1.6	1.1	-			ORn		75Sl.A
$^{112}\text{Sn}(\text{d,p})^{113}\text{Sn}$	5504	25	5519.8	1.6	0.6	U			Pit		64Co11
	5518.2	3.2			0.5	-			SPa		75Be09
$^{112}\text{Sn}(\text{n},\gamma)^{113}\text{Sn}$	ave. 7742.2	1.9	7744.4	1.6	1.2	1	70	69 ^{113}Sn			average
$^{112}\text{Sn}(\text{He},\text{d})^{113}\text{Sb}$	-2400	40	-2443	17	-1.1	R			Sac		68Co22
$^{113}\text{Xe}(\varepsilon\text{p})^{112}\text{Te}$	7920	150	8075	11	1.0	o					82Pi05
	8300	150			-1.5	U					05Ja10
$^{113}\text{Cs}(\text{p})^{112}\text{Xe}$	967	4	972.8	2.2	1.5	o					84Fa04 *
	982.7	4.			-2.5	3					92He.A
	967.6	6.			0.9	3					94Pa12
	968.6	3.			1.4	3					95Ho26
$^{113}\text{Ru}(\beta^-)^{113}\text{Rh}$	6480	50	6900	40	8.4	C			Jyv		00Kr.A
$^{113}\text{Rh}(\beta^-)^{113}\text{Pd}$	5008	50	4824	10	-3.7	C			Jyv		00Kr.A
$^{113}\text{Pd}(\beta^-)^{113}\text{Ag}$	3360	150	3436	18	0.5	U					75Br.A
	3340	35			2.8	B			Stu		90Fo07
$^{113}\text{Ag}(\beta^-)^{113}\text{Cd}$	2010	20	2016	17	0.3	2					57Je.A
	2070	150			-0.4	U					70Ma47 *
	2031	30			-0.5	2			Stu		90Fo07 *
$^{113}\text{Cd}(\beta^-)^{113}\text{In}$	326.4	15.	323.84	0.27	-0.2	U					51Ca43 *
	316.4	30.			0.2	U					54De13 *
	320	10			0.4	U			CIT		88Mi13
	344.9	21.0			-1.0	U					07Be61
	322.2	0.9			1.8	U					09Da03
$^{113}\text{Sn}(\beta^+)^{113}\text{In}$	1034.6	5.0	1039.0	1.6	0.9	-					93Li10 *
$^{113}\text{In}(\text{p,n})^{113}\text{Sn}$	-1809	6	-1821.3	1.6	-2.1	-			Oak		73Ra13
$^{113}\text{Sn}(\beta^+)^{113}\text{In}$	ave. 1031	4	1039.0	1.6	2.0	1	17	17 ^{113}Sn			average
$^{113}\text{Sb}(\beta^+)^{113}\text{Sn}$	3934	30	3911	17	-0.8	2					61Se08 *
	3945	50			-0.7	2					69Ki16 *
$^{113}\text{Te}(\beta^+)^{113}\text{Sb}$	5520	300	6070	30	1.8	U					74Bu21
	5720	200			1.7	U					74Ch17
* $^{113}\text{Mo}-\text{u}$	Trends from Mass Surface TMS suggest ^{113}Mo 740 keV less bound										GAu **
* $^{113}\text{Ru}-\text{u}$	$M-A=-71689(77)$ keV for mixture gs+m at 131(33) keV										Nub211 **
* $^{113}\text{Ru}-\text{u}$	$M-A=-71882(93)$ keV for mixture gs+m at 131(33) keV										Nub211 **
* $^{113}\text{Ru}-\text{u}$	$M-A=-71931(120)$ keV for mixture gs+m at 131(33) keV										Nub211 **
* $^{113}\text{Cd}-\text{u}$	$M-A=-88832(41)$ keV for mixture gs+m at 263.54 keV										Nub211 **
* $^{113}\text{In}-\text{u}$	$M-A=-89199(30)$ keV for mixture gs+m at 391.699 keV										Nub211 **
* $^{113}\text{Sn}-\text{u}$	$M-A=-88263(29)$ keV for mixture gs+m at 77.389 keV										Nub211 **
* $^{113}\text{Ru}-^{105}\text{Ru}_{1,076}$	$D_M=22157(12)$ μu for mixture gs+m at 131(33) keV; $M-A=-71822(12)$ keV										Nub211 **
* $^{112}\text{Cd}(\text{d,p})^{113}\text{Cd}$	Estimated systematic error 0.5 added to statistical error 0.40										AHW **
* $^{113}\text{Cs}(\text{p})^{112}\text{Xe}$	E_p from another GSI work; superseded by 95Ho26										87Gi07 **
* $^{113}\text{Ag}(\beta^-)^{113}\text{Cd}$	$E_{\beta^-}=1530(150)$ from $^{113}\text{Ag}^m$ at 43.50 to 5/2 ⁺ level at 583.962 keV										Ens106 **
* $^{113}\text{Ag}(\beta^-)^{113}\text{Cd}$	$Q_{\beta^-}=2075(30)$ from $^{113}\text{Ag}^m$ at 43.50 keV										Nub211 **
* $^{113}\text{Cd}(\beta^-)^{113}\text{In}$	$Q_{\beta^-}=590(15)$ 580(30) respectively, from $^{113}\text{Cd}^m$ at 263.54 keV										Nub211 **
* $^{113}\text{Sn}(\beta^+)^{113}\text{In}$	$Q_{\beta^+}=642.9(5.0)$ to 1/2 ⁻ level at 391.699 keV										Ens106 **
* $^{113}\text{Sb}(\beta^+)^{113}\text{Sn}$	$E_{\beta^+}=2420(20)$ 2430(50) respectively, to 3/2 ⁺ level at 498.06 keV,										Ens106 **
*	plus 6% to 5/2 ⁺ at 409.83 keV										Ens106 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{114}\text{Tc}-u$	-62459	365	-62910	470	-0.8	o			GT1	1.5	04Ma.A
	-62910	186				2			GT3	2.5	16Kn03
$^{114}\text{Ru}-^{105}\text{Ru}_{1.086}$	24805	13	24803	5	-0.2	U			JY1	1.0	07Ha20
$^{114}\text{Ru}-u$	-75642	236	-75386	4	0.7	U			GT1	1.5	04Ma.A
$^{114}\text{Rh}-u$	-81193	120	-81280	80	-0.7	1	41	41 ^{114}Rh	JY0	1.0	03Ko.A *
$^{114}\text{Ag}-^{133}\text{Cs}_{.857}$	-10149.3	4.9				2			MA8	1.0	10Br02
$\text{C}_8 \text{H}_{18}-^{114}\text{Cd}$	237487.6	4.	237485.58	0.30	-0.2	U			M16	2.5	63Da10
$\text{C}_6 \text{O}_2 \text{H}_{10}-^{114}\text{Cd}$	164713	15	164714.56	0.30	0.1	U			R12	1.5	83De51
	164711	15			0.2	U			R12	1.5	83De51
$\text{C}_9 \text{H}_6-^{114}\text{Cd}$	143591	5	143585.19	0.30	-0.8	U			R12	1.5	83De51
	143586	8			-0.1	U			R12	1.5	83De51
$\text{C}_8 \text{ }^{13}\text{C} \text{H}_5-^{114}\text{Cd}$	139117	17	139115.00	0.30	-0.1	U			R12	1.5	83De51
$\text{C}_8 \text{N} \text{H}_4-^{114}\text{Cd}$	131017	12	131009.13	0.30	-0.4	U			R12	1.5	83De51
	131009	20			0.0	U			R12	1.5	83De51
$^{114}\text{Cd}-^{133}\text{Cs}_{.857}$	-15611.4	4.2	-15607.33	0.30	1.0	U			MA8	1.0	10Br02
$^{114}\text{In}-u$	-94986	68	-95083.6	0.3	-1.4	U			GS2	1.0	05Li24 *
$\text{C}_8 \text{H}_{18}-^{114}\text{Sn}$	238092	10	238070.44	0.03	-0.9	U			M16	2.5	63Da10
	238066	8			0.4	U			R13	1.5	83De51
$^{114}\text{Sb}-u$	-90731	30	-90711	21	0.7	-			GS2	1.0	05Li24
	-90711	50			0.0	-			GR1	1.0	19An10
	ave. -90726	26			0.6	1	68	68 ^{114}Sb			average
$^{114}\text{Te}-u$	-87911	30	-87912	26	-0.0	2			GS2	1.0	05Li24
	-87916	54			0.1	2			GR1	1.0	19An10
$^{114}\text{I}-u$	-77981.1	21.5				2			GR1	1.0	20Ay05
$^{114}\text{Xe}-^{133}\text{Cs}_{.857}$	9008	12				2			MA6	1.0	04Di18
$^{114}\text{Ru}-^{120}\text{Sn}_{.950}$	17522.0	3.7				2			JY1	1.0	11Ha48
$^{114}\text{Rh}-^{120}\text{Sn}_{.950}$	11570	100	11630	80	0.6	1	59	59 ^{114}Rh	JY1	1.0	07Ha20 *
$^{114}\text{Pd}-^{120}\text{Sn}_{.950}$	3277.0	7.4				2			JY1	1.0	07Ha20
$^{114}\text{Cd }^{35}\text{Cl}-^{112}\text{Cd }^{37}\text{Cl}$	3546	3	3551.22	0.28	0.4	U			H11	4.0	63Bi12
	3547	3			0.6	U			H20	2.5	66Ma05
	3548.5	1.0			1.1	U			H26	2.5	73Me28
$^{113}\text{Cd} \text{H}-^{114}\text{Cd}$	8859	18	8868.14	0.15	0.3	U			R12	1.5	83De51
$^{114}\text{Te}^m-^{114}\text{Ru}$	12651	13				3			JY1	1.0	11Ha48 *
$^{114}\text{Cd}-^{112}\text{Cd} \text{H}$	-7225	33	-7223.93	0.27	0.0	U			R12	1.5	83De51
$^{114}\text{Cd}-^{113}\text{Cd}$	-1040	11	-1043.11	0.15	-0.1	U			M16	2.5	63Da10
	-1032	33			-0.2	U			R12	1.5	83De51
$^{114}\text{Cd}-^{113}\text{In}$	-679	45	-695.5	0.3	-0.2	U			R12	1.5	83De51
$^{114}\text{Cd}-^{111}\text{Cd} \text{H}$	-8651	35	-8643.8	0.4	0.1	U			R12	1.5	83De51
$^{114}\text{Cd}-^{112}\text{Cd}$	587	33	601.10	0.27	0.3	U			R12	1.5	83De51
$^{114}\text{Cd} \text{H}-^{113}\text{Cd}$	6821	35	6781.93	0.15	-0.7	U			R12	1.5	83De51
$^{114}\text{Ba}(\gamma, ^{12}\text{C})^{102}\text{Sn}$	18110	780	19029	23	1.2	U					95Gu01
$^{114}\text{Cs}(\alpha)^{110}\text{I}$	3343.5	30.	3360	60	0.2	o			GSa		80Ro04
	3357.0	30.				4			GSa		81Sc17
$^{114}\text{Ba}(\alpha)^{110}\text{Xe}$	3534.2	40.	3592	19	1.4	5					02Ma19
	3606.8	20.			-0.7	5					16Ca33
$^{112}\text{Cd}(\text{t,p})^{114}\text{Cd}$	7105	20	7100.92	0.25	-0.2	U			Ald		67Hi01
$^{114}\text{Cd}(\text{p,t})^{112}\text{Cd}$	-7106	5	-7100.92	0.25	1.0	U			Min		73Oo01
$^{112}\text{Sn}(\text{t,p})^{114}\text{Sn}$	9579	15	9565.53	0.30	-0.9	U			Ald		69Bj01
$^{114}\text{Sn}(\text{p,t})^{112}\text{Sn}$	-9582	15	-9565.53	0.30	1.1	U			Roc		70Fi08
$^{113}\text{Cd}(\text{n},\gamma)^{114}\text{Cd}$	9042.76	0.20	9042.97	0.14	1.0	-			ILn		79Br25 Z
	9043.18	0.19			-1.1	-			Bdn		06Fi.A
$^{114}\text{Cd}(\gamma,\text{n})^{113}\text{Cd}$	-9050	4	-9042.97	0.14	1.8	U			McM		79Ba06
$^{113}\text{Cd}(\text{d,p})^{114}\text{Cd}$	6817	30	6818.40	0.14	0.0	U			Pit		64Co11
	6822	8			-0.5	U			MIT		67Co15
$^{114}\text{Cd}(\text{d,t})^{113}\text{Cd}$	-2801	30	-2785.74	0.14	0.5	U			Pit		64Ro17
$^{113}\text{Cd}(\text{n},\gamma)^{114}\text{Cd}$	ave. 9042.98	0.14	9042.97	0.14	-0.1	1	98	93 ^{114}Cd			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{113}\text{In}(\text{n},\gamma)^{114}\text{In}$	7274.0	1.2	7274.00	0.25	0.0	U					75Ra07 Z
	7273.83	0.27			0.6	1	88	82 ^{114}In	Bdn		06Fi.A
$^{113}\text{In}(\text{d},\text{p})^{114}\text{In}$	5082	25	5049.44	0.25	-1.3	U			Pit		67Hj03
$^{114}\text{Sn}(\text{d},^3\text{He})^{113}\text{In}$	-2980	50	-2988.11	0.19	-0.2	U			Sac		69Co03
$^{114}\text{Sn}(\text{p},\text{d})^{113}\text{Sn}$	-8101	15	-8078.4	1.6	1.5	U			Har		70Ca01
$^{114}\text{Sn}(\text{d},\text{t})^{113}\text{Sn}$	-4052	20	-4045.7	1.6	0.3	U			Pit		64Co11
	-4043.7	4.2			-0.5	1	14	14 ^{113}Sn	SPa		75Be09
$^{114}\text{Cs}(\text{e},\text{p})^{113}\text{I}$	8730	150	9140	90	2.8	B					82Pl05
$^{114}\text{Ru}(\beta^-)^{114}\text{Rh}$	6100	200	5490	70	-3.1	B			Jyv		92Jo05 *
	6120	200			-3.2	C			Jyv		94Jo.A
$^{114}\text{Rh}(\beta^-)^{114}\text{Pd}$	6500	500	7780	70	2.6	U			Jyv		88Ay02
	7392	53			7.3	C			Jyv		00Kr.A
$^{114}\text{Pd}(\beta^-)^{114}\text{Ag}$	1450	100	1440	8	-0.1	U					75Br.A
	1450	100			-0.1	o			Jyv		89Ay.A
	1450	100			-0.1	o			Jyv		89Ko22
	1414	30			0.9	U			Stu		90Fo07
	1451	25			-0.4	U			Jyv		94Jo.A
$^{114}\text{Ag}(\beta^-)^{114}\text{Cd}$	4850	150	5084	5	1.6	U					71Ro19
	4900	260			0.7	U					72Wa06
	5160	110			-0.7	o			Stu		84Lu02
	5018	35			1.9	U			Stu		90Fo07
$^{114}\text{In}(\beta^+)^{114}\text{Cd}$	1422	25	1445.1	0.4	0.9	U					56Gr35
	1417	20			1.4	U					57Dz64
$^{114}\text{In}(\beta^-)^{114}\text{Sn}$	1987	2	1989.9	0.3	1.5	U					61Da01
	1989	1			0.9	-					61Ni02
	1980	2			5.0	B					64An12
	1988.5	1.0			1.4	-					68Ze04
ave.	1988.8	0.7			1.7	1	18	18 ^{114}In			average
$^{114}\text{Sb}(\beta^+)^{114}\text{Sn}$	5690	100	6063	20	3.7	C					69Bu.A *
	6370	100			-3.1	B					72Mi27 *
$^{114}\text{Sn}(\text{p},\text{n})^{114}\text{Sb}$	-6875	35	-6845	20	0.8	1	32	32 ^{114}Sb	VUn		76Ka19
* $^{114}\text{Rh}-\text{u}$	Average of 2 values; $M-A=-75531(60)$ keV for mixture gs+m at 200#150 keV										Nub211 **
* $^{114}\text{In}-\text{u}$	$M-A=-88384(31)$ keV for mixture gs+m at 190.2682 keV										Nub211 **
* $^{114}\text{Rh}-^{120}\text{Sn}_{950}$	$D_M=11678.0(7.8)$ μu for mixture gs+m at 200#150 keV; $M-A=-75665.6(7.3)$ keV										Nub211 **
* $^{114}\text{Tc}^m-^{114}\text{Ru}$	Mixture of two isomeric states with stronger component of low-spin state										11Ri01 **
*	however, estimates from TMS suggest this is $^{114}\text{Tc}^m$										GAu **
* $^{114}\text{Ru}(\beta^-)^{114}\text{Rh}$	$E_{\beta^-}=5910(120)$ doublet to $(2)^+$ level at 127.0, 1^+ at 255.2 keV										Ens123 **
* $^{114}\text{Sb}(\beta^+)^{114}\text{Sn}$	$E_{\beta^+}=3365(50)$ to 2^+ at 1299.907, original error doubled see $^{114}\text{Sn}(\text{p},\text{n})$										Ens123 **
* $^{114}\text{Sb}(\beta^+)^{114}\text{Sn}$	$E_{\beta^+}=4050(100)$ to 2^+ at 1299.907 level, see $^{112}\text{Sb}(\beta^+)$										Ens123 **
$^{115}\text{Tc}-\text{u}$	-60462	339	-59900#	210#	0.7	D			GT3	2.5	16Kn03 *
$^{115}\text{Rh}-\text{u}$	-79664	85	-79688	8	-0.3	U			JY0	1.0	03Ko.A
$^{115}\text{Ag}-^{133}\text{Cs}_{865}$	-9439	24	-9448	20	-0.4	1	67	67 ^{115}Ag	MA8	1.0	10Br02 *
$\text{C}_9 \text{H}_7-^{115}\text{In}$	150910	8	150896.451	0.012	-0.7	U			M16	2.5	63Da10
	150932	16			-1.5	U			R12	1.5	83De51
$\text{C}_6 \text{O}_2 \text{H}_{11}-^{115}\text{In}$	172055	16	172025.817	0.012	-1.2	U			R12	1.5	83De51
$\text{C}_8 \text{N H}_5-^{115}\text{In}$	138355	13	138320.391	0.012	-1.8	U			R12	1.5	83De51
$^{115}\text{In}-\text{u}$	-96095	30	-96121.227	0.012	-0.9	U			GS2	1.0	05Li24
$\text{C}_9 \text{H}_7-^{115}\text{Sn}$	151411	8	151430.527	0.016	1.0	U			M16	2.5	63Da10
	151440	8			-0.8	U			R13	1.5	83De51
$^{115}\text{Sb}-\text{u}$	-93402	30	-93402	17	0.0	2			GS2	1.0	05Li24
$^{115}\text{Te}-\text{u}$	-88098	30				2			GS2	1.0	05Li24 *
$^{115}\text{I}-\text{u}$	-81952	31				2			GS2	1.0	05Li24
$^{115}\text{Xe}-^{133}\text{Cs}_{865}$	8078	13				2			MA6	1.0	04Di18
$^{115}\text{Ru}-^{120}\text{Sn}_{958}$	22723	27				2			JY1	1.0	07Ha20 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{115}\text{Rh}-^{120}\text{Sn}_{958}$	14001.6	7.8				2			JY1	1.0	07Ha20
$^{115}\text{Pd}-^{120}\text{Sn}_{958}$	7347	15	7349	15	0.2	1	94	94	^{115}Pd	1.0	07Ha20 *
$^{115}\text{Sn}-^{120}\text{Sn}_{958}$	-2963.9	2.0	-2965.4	0.9	-0.7	1	22	22	^{120}Sn	1.0	11Ha48
$^{113}\text{Cd}-^{115}\text{In}_{983}$	-1104.76	0.34	-1104.73	0.26	0.1	1	59	59	^{113}Cd	1.0	16Ga33
$^{113}\text{In}-^{115}\text{In}_{983}$	-1452.08	0.23	-1452.38	0.20	-1.3	1	77	77	^{113}In	1.0	16Ga33
$^{115}\text{In}-^{115}\text{Sn}$	534.0768	0.0104	534.077	0.010	-0.0	1	100	100	^{115}Sn	1.0	09Mo23
	534.28	0.18			-1.1	U			JY1	1.0	09Wi10
$^{115}\text{In}-^{114}\text{Cd}$	483	45	513.77	0.30	0.5	U			R12	1.5	83De51
$^{115}\text{Sn}-^{114}\text{Sn}$	573	11	564.566	0.027	-0.3	U			M16	2.5	63Da10
$^{115}\text{In}-^{113}\text{In}$	-200	28	-181.68	0.20	0.4	U			R12	1.5	83De51
$^{115}\text{In}-^{129}\text{Xe}$	-902.0845	0.0111	-902.085	0.011	-0.0	1	100	100	^{115}In	1.0	09Mo23
$^{115}\text{Sn}-^{129}\text{Xe}$	-1436.1613	0.0130	-1436.162	0.015	-0.0	o			FS1	1.0	09Mo23 *
$^{113}\text{Cd}(\text{t,p})^{115}\text{Cd}$	6702	20	6702.0	0.6	0.0	U			Ald		67Hi01
$^{114}\text{Cd}(\text{n},\gamma)^{115}\text{Cd}$	6160	100	6140.9	0.6	-0.2	U					61Ja21
$^{114}\text{Cd}(\text{d,p})^{115}\text{Cd}$	3923	30	3916.3	0.6	-0.2	U			Pit		64Ro17
	3929	20			-0.6	U			Oak		64Si18
	3916.30	0.59			-0.0	1	100	100	^{115}Cd		90Pi05 *
$^{114}\text{Cd}(\text{d},\text{He,d})^{115}\text{In}$	1320	15	1316.92	0.28	-0.2	U					70Th.A
$^{115}\text{In}(\gamma,\text{n})^{114}\text{In}$	-9025	29	-9037.9	0.3	-0.4	U			Phi		60Ge01
	-9039	5			0.2	U			McM		79Ba06
$^{115}\text{In}(\text{d,t})^{114}\text{In}$	-2789	30	-2780.6	0.3	0.3	U			Pit		64Co11
	-2766	25			-0.6	U			Pit		67Hj03
$^{114}\text{Sn}(\text{n},\gamma)^{115}\text{Sn}$	7545.5	2.0	7545.429	0.025	-0.0	U			ORn		78Ra16 Z
	7545.427	0.025			0.1	1	100	100	^{114}Sn		16Ur03
$^{114}\text{Sn}(\text{d,p})^{115}\text{Sn}$	5329	25	5320.862	0.025	-0.3	U			Pit		64Co11
	5320.6	3.4			0.1	U			SPa		75Be09
$^{115}\text{Sn}(\text{d,t})^{114}\text{Sn}$	-1304	30	-1288.198	0.025	0.5	U			Pit		64Co11
$^{115}\text{Xe}(\text{ep})^{114}\text{Te}$	6200	130	5944	27	-2.0	U					72Ho18
$^{115}\text{Ru}(\beta^-)^{115}\text{Rh}$	7780	100	8124	26	3.4	C			Jyv		00Kr.A
$^{115}\text{Rh}(\beta^-)^{115}\text{Pd}$	6000	500	6197	15	0.4	U			Jyv		88Ay01
	6566	50			-7.4	C			Jyv		00Kr.A
$^{115}\text{Pd}(\beta^-)^{115}\text{Ag}$	4584	50	4557	22	-0.5	1	19	12	^{115}Ag		90Fo07
$^{115}\text{Ag}(\beta^-)^{115}\text{Cd}$	3180	100	3102	18	-0.8	U					64Ba36 *
	3118	100			-0.2	U					78Ma18 *
	3091	40			0.3	1	21	21	^{115}Ag		90Fo07 *
$^{115}\text{Cd}(\beta^-)^{115}\text{In}$	1460	4	1451.9	0.7	-2.0	U					74Bo26 *
	1431	5			4.2	B					75Bo29 *
	1440	2			5.9	B					76Ra16 *
$^{115}\text{In}(\beta^-)^{115}\text{Sn}$	494	20	497.489	0.010	0.2	U					49Be53 *
	630	30			-4.4	B					50Ma76
	625	70			-1.8	U					61Be15
	494	30			0.1	U					62Se03 *
	480	30			0.6	U					62Wa15
	495	20			0.1	U					72Mu02
	482	15			1.0	U					78Pf01 *
$^{115}\text{Sb}(\beta^+)^{115}\text{Sn}$	3030	20	3030	16	0.0	R					61Se08 *
* $^{115}\text{Tc}-\text{u}$	Trends from Mass Surface TMS suggest ^{115}Tc 530 keV less bound										HWJ20a **
* $^{115}\text{Ag}-^{133}\text{Cs}_{865}$	$D_M=-9416.7(9.2) \mu\text{u}$ for ground state or $^{115}\text{Ag}^m$ at 41.16 keV; $M-A=-84952.9(8.6) \text{ keV}$										Nub211 **
* $^{115}\text{Te}-\text{u}$	$M-A=-82058(28) \text{ keV}$ for mixture gs+m at 10(6) keV										Nub211 **
* $^{115}\text{Ru}-^{120}\text{Sn}_{958}$	$D_M=22767.3(7.3) \mu\text{u}$ for mixture gs+m at 82(6) keV; $M-A=-66064.8(6.9) \text{ keV}$										Nub211 **
* $^{115}\text{Pd}-^{120}\text{Sn}_{958}$	$D_M=7348(15), 7442(15) \mu\text{u}$ for ground state, 89.21 isomer										Nub211 **
* $^{115}\text{Sn}-^{129}\text{Xe}$	Used are the equations for the $^{115}\text{In}-^{129}\text{Xe}$ and $^{115}\text{In}-^{115}\text{Sn}$ doublets										GAu **
* $^{114}\text{Cd}(\text{d,p})^{115}\text{Cd}$	Estimated systematic error 0.5 added to statistical error 0.32 keV										AHW **
* $^{115}\text{Ag}(\beta^-)^{115}\text{Cd}$	$E_{\beta^-}=2950(100)$ to $(3/2)^+$ level at 229.1 keV, and other E_{β^-}										Ens12a **
* $^{115}\text{Ag}(\beta^-)^{115}\text{Cd}$	$E_{\beta^-}=721(100)$ to $(23/2^-)$ level at 2397.2 keV, and other E_{β^-}										Ens12a **
* $^{115}\text{Ag}(\beta^-)^{115}\text{Cd}$	$Q_{\beta^-}=3132(40)$ from $^{115}\text{Ag}^m$ at 41.16 keV										Nub211 **
* $^{115}\text{Cd}(\beta^-)^{115}\text{In}$	$E_{\beta^-}=593(2), 636(2)$ to $1/2^+$ level at 864.139, $3/2^+$ at 828.588 keV										Ens12a **
* $^{115}\text{Cd}(\beta^-)^{115}\text{In}$	$E_{\beta^-}=320(5), 679(6)$ from $^{115}\text{Cd}^m$ 181.0 to $13/2^+$ 1290.592, $7/2^+$ 933.780										Ens12a **
* $^{115}\text{Cd}(\beta^-)^{115}\text{In}$	$Q_{\beta^-}=1621(2)$ from $^{115}\text{Cd}^m$ at 181.0 keV										Nub211 **
* $^{115}\text{In}(\beta^-)^{115}\text{Sn}$	$Q_{\beta^-}=830(20)$ from $^{115}\text{In}^m$ at 336.244 keV										Nub211 **
* $^{115}\text{In}(\beta^-)^{115}\text{Sn}$	$Q_{\beta^-}=830(30)$ from $^{115}\text{In}^m$ at 336.244 keV										Nub211 **
* $^{115}\text{In}(\beta^-)^{115}\text{Sn}$	Q_{β^-} is larger than first excitation energy 497.334(0.023) in ^{115}Sn										05Ca03 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{115}\text{Sb}(\beta^+)^{115}\text{Sn}$	$E_{\beta^+}=1510(20)$ to $3/2^+$ level at 497.334 keV										Ens12a **	
$^{116}\text{Ru}-^{129}\text{Xe}_{.899}$	16821.2	4.0				2			JY1	1.0	11Ha48	
$^{116}\text{Rh}-\text{u}$	-75936	140	-75940	80	-0.0	-			JY0	1.0	03Ko.A *	
	-75960	140			0.1	-			GT2	2.5	08Kn.A *	
ave.	-75940	130			0.0	1	37	37	^{116}Rh		average	
$^{116}\text{Ag}-^{133}\text{Cs}_{.872}$	-6167.3	3.5				2			MA8	1.0	10Br02	
$\text{C}_9 \text{H}_8-^{116}\text{Cd}$	157837.4	2.9	157837.02	0.17	-0.1	U			M16	2.5	63Da10	
	157851	5			-1.9	U			R12	1.5	83De51	
	157846	22			-0.3	U			R12	1.5	83De51	
$\text{C}_6 \text{O}_2 \text{H}_{12}-^{116}\text{Cd}$	178982	15	178966.39	0.17	-0.7	U			R12	1.5	83De51	
$\text{C}_8 \text{ }^{13}\text{C} \text{H}_7-^{116}\text{Cd}$	153376	8	153366.83	0.17	-0.8	U			R12	1.5	83De51	
	153382	22			-0.5	U			R12	1.5	83De51	
$\text{C}_8 \text{N} \text{H}_6-^{116}\text{Cd}$	145262	17	145260.96	0.17	-0.0	U			R12	1.5	83De51	
$\text{C}_9 \text{H}_8-^{116}\text{Sn}$	160861	8	160857.43	0.10	-0.2	U			M16	2.5	63Da10	
	160857	8			0.0	U			R13	1.5	83De51	
$^{116}\text{Sb}-\text{u}$	-93128	129	-93207	6	-0.6	U			GS2	1.0	05Li24 *	
$^{116}\text{Te}-\text{u}$	-91540	30	-91534	26	0.2	-			GS2	1.0	05Li24	
	-91539	55			0.1	-			GR1	1.0	19An10	
ave.	-91540	26			0.2	1	97	97	^{116}Te		average	
$^{116}\text{I}-\text{u}$	-83005	126	-83110	80	-0.9	1	41	41	^{116}I	1.0	20Ay05	
$^{116}\text{Xe}-^{133}\text{Cs}_{.872}$	4027	14	4027	14	-0.0	1	100	100	^{116}Xe	1.0	04Di18	
$^{116}\text{Rh}-^{120}\text{Sn}_{.967}$	18633	100	18630	80	-0.0	1	63	63	^{116}Rh	1.0	07Ha20 *	
$^{116}\text{Pd}-^{120}\text{Sn}_{.967}$	8868.0	7.6				2			JY1	1.0	07Ha20	
$^{116}\text{Cd }^{35}\text{Cl}-^{114}\text{Cd }^{37}\text{Cl}$	4353	3	4348.4	0.3	-0.4	U			H11	4.0	63Bi12	
	4344	2			0.9	U			H20	2.5	66Ma05	
	4348.7	1.2			-0.1	o			H26	2.5	73Me28	
	4347.46	0.44			0.8	1	10	7	^{114}Cd	2.5	10Mc04	
$^{116}\text{Cd}-^{116}\text{Sn}$	3020.42	0.14	3020.41	0.14	-0.1	1	99	98	^{116}Cd	1.0	11Ra24	
$^{116}\text{Cd}-^{114}\text{Cd} \text{H}$	-6452	32	-6426.8	0.3	0.5	U			R12	1.5	83De51	
$^{116}\text{Sn}-^{115}\text{Sn}$	-1602	11	-1601.87	0.10	0.0	U			M16	2.5	63Da10	
$^{116}\text{Cd}-^{113}\text{Cd} \text{H}$	-7458	32	-7469.9	0.3	-0.2	U			R12	1.5	83De51	
$^{116}\text{Cd}-^{114}\text{Cd}$	1370	32	1398.2	0.3	0.6	U			R12	1.5	83De51	
$^{116}\text{Cs}(\epsilon\alpha)^{112}\text{Te}$	12300	400	13100#	100#	2.0	D					77Bo28	
	12400	900			0.8	D					76Jo.A *	
$^{116}\text{Cd}(^{14}\text{C}, ^{16}\text{O})^{114}\text{Pd}$	2497	29	2535	7	1.3	U			LAl		84Co19	
$^{114}\text{Cd}(\text{t},\text{p})^{116}\text{Cd}$	6362	20	6358.4	0.3	-0.2	U			Ald		67Hi01	
$^{116}\text{Cd}(\text{p},\text{t})^{114}\text{Cd}$	-6363	5	-6358.4	0.3	0.9	U			Min		73Oo01	
$^{116}\text{Sn}(\text{p},\text{t})^{114}\text{Sn}$	-8619	15	-8627.08	0.10	-0.5	U			Roc		70Fi08	
$^{116}\text{Cd}(\gamma,\text{n})^{115}\text{Cd}$	-8702	4	-8699.3	0.7	0.7	U			McM		79Ba06	
$^{116}\text{Cd}(\text{d},\text{t})^{115}\text{Cd}$	-2458	30	-2442.1	0.7	0.5	U			Pit		64Ro17	
$^{115}\text{In}(\text{n},\gamma)^{116}\text{In}$	6783.8	1.2	6784.72	0.22	0.8	U					72Ra39 Z	
	6784.4	1.1			0.3	U					74Co35	
	6784.72	0.22				2			Bdn		06Fi.A	
$^{115}\text{In}(\text{d},\text{p})^{116}\text{In}$	4494	25	4560.15	0.22	2.6	U			Pit		64Co11	
	4580	30			-0.7	U			Pit		67Hj03	
$^{116}\text{Sn}(\text{d}, ^3\text{He})^{115}\text{In}$	-3740	50	-3785.12	0.10	-0.9	U			Sac		69Co03	
$^{115}\text{Sn}(\text{n},\gamma)^{116}\text{Sn}$	9562.2	1.5	9563.45	0.09	0.8	U					72Mc08	
	9563.5	0.5			-0.1	U					84Ga.B	
	9563.41	0.11			0.4	-			ORn		91Ra01 Z	
	9563.55	0.19			-0.5	-			Bdn		06Fi.A	
$^{115}\text{Sn}(\text{d},\text{p})^{116}\text{Sn}$	7358	30	7338.88	0.09	-0.6	U			Pit		64Co11	
$^{116}\text{Sn}(\text{p},\text{d})^{115}\text{Sn}$	-7344	15	-7338.88	0.09	0.3	U			Har		70Ca01	
$^{116}\text{Sn}(\text{d},\text{t})^{115}\text{Sn}$	-3309	20	-3306.22	0.09	0.1	U			Pit		64Co11	
	-3305.0	2.5			-0.5	U			SPa		75Be09	
$^{115}\text{Sn}(\text{n},\gamma)^{116}\text{Sn}$	ave.	9563.45	0.10	9563.45	0.09	0.1	1	99	99	^{116}Sn		average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{115}\text{Sn}(^3\text{He,d})^{116}\text{Sb}-^{120}\text{Sn}(^1)^{121}\text{Sb}$	-1722	10	-1713	5	0.9	1	30	25 ^{116}Sb	VUn		78Ka12
$^{116}\text{Cs}(\varepsilon\text{p})^{115}\text{I}$	6350	300	7010#	100#	2.2	U					78Da07 *
$^{116}\text{Rh}(\beta^-)^{116}\text{Pd}$	8000	500	9100	70	2.2	U			Jyv		88Ay02
$^{116}\text{Pd}(\beta^-)^{116}\text{Ag}$	2615	100	2712	8	1.0	U					75Br.A
	2620	100			0.9	o			Jyv		89Ay.A
	2607	30			3.5	B			Stu		90Fo07
	2620	100			0.9	U			Jyv		94Jo.A
$^{116}\text{Ag}(\beta^-)^{116}\text{Cd}$	6062	130	6170	3	0.8	o			Stu		82Al29 *
	5800	200			1.8	U					82Br10
	6194	50			-0.5	U			Stu		90Fo07 *
$^{116}\text{In}(\beta^-)^{116}\text{Sn}$	3290	60	3276.22	0.24	-0.2	U					54Bo39
$^{116}\text{Sb}(\beta^+)^{116}\text{Sn}$	4586	100	4704	5	1.2	U					61Fi05 *
	4606	50			2.0	U					68Ki07 *
$^{116}\text{Sn}(\text{p,n})^{116}\text{Sb}$	-5515	40	-5486	5	0.7	U			VUn		76Ka19
	-5483.2	6.			-0.5	1	75	75 ^{116}Sb	Oak		77Jo03
$^{116}\text{Sb}^n(\beta^+)^{116}\text{Sn}$	5090	40				2					60Je03 *
$^{116}\text{Te}(\beta^+)^{116}\text{Sb}$	1554	100	1558	25	0.0	U					61Fi05 *
$^{116}\text{I}(\beta^+)^{116}\text{Te}$	7760	130	7840	80	0.6	-					70Be.A
	7710	200			0.7	-					76Go02
ave.	7750	110			0.9	1	48	45 ^{116}I			average
$^{116}\text{Xe}(\beta^+)^{116}\text{I}$	4340	200	4370	80	0.2	1	14	14 ^{116}I			76Go02
* $^{116}\text{Rh}-\text{u}$	$M-A=-70634(96)$ keV for mixture gs+m at 200#150 keV										Nub211 **
* $^{116}\text{Rh}-\text{u}$	$M-A=-70662(93)$ keV for mixture gs+m at 200#150 keV										Nub211 **
* $^{116}\text{Sb}-\text{u}$	$M-A=-86553(34)$ keV for mixture gs+n at 390(40) keV										Nub211 **
* $^{116}\text{Rh}-^{120}\text{Sn}_{967}$	$D_M=18740.7(8.4)$ μu for mixture gs+m at 200#150 keV ; $M-A=-70635.5(7.9)$ keV										Nub211 **
* $^{116}\text{Cs}(\varepsilon\alpha)^{112}\text{Te}$	$Q=12500(900)$ from $^{116}\text{Cs}^m$ estimated at 100#60 keV										Nub211 **
* $^{116}\text{Cs}(\varepsilon\alpha)^{112}\text{Te}$	Trends from Mass Surface TMS suggest ^{116}Cs 780 keV less bound										GAu **
* $^{116}\text{Cs}(\varepsilon\text{p})^{115}\text{I}$	$Q=6450(300)$ from $^{116}\text{Cs}^m$ at estimated 100#60 keV										Nub211 **
* $^{116}\text{Ag}(\beta^-)^{116}\text{Cd}$	$Q_{\beta^-}=6110(130)$ from $^{116}\text{Ag}^m$ at 47.90 keV										Nub211 **
* $^{116}\text{Ag}(\beta^-)^{116}\text{Cd}$	$Q_{\beta^-}=6199(100)$; and 6241(50) from $^{116}\text{Ag}^m$ at 47.90 keV										Nub211 **
* $^{116}\text{Sb}(\beta^+)^{116}\text{Sn}$	$E_{\beta^+}=2270(100)$ 2290(50) respectively, to 2^+ level at 1293.560 keV										Ens104 **
* $^{116}\text{Sb}^n(\beta^+)^{116}\text{Sn}$	$E_{\beta^+}=1160(40)$ to 7^- level at 2908.85 keV										Ens104 **
* $^{116}\text{Te}(\beta^+)^{116}\text{Sb}$	$E_{\beta^+}=440(100)$ to 1^+ level at 93.99 keV										Ens104 **
$^{117}\text{Ru}-\text{u}$	-63897	419	-63870	470	0.1	o			GT1	1.5	04Ma.A
	-63865	186				2			GT3	2.5	16Kn03
$^{117}\text{Rh}-\text{u}$	-73903	408	-73964	10	-0.1	U			GT1	1.5	04Ma.A
$^{117}\text{Ag}-^{133}\text{Cs}_{880}$	-5029	16	-5024	15	0.3	1	83	83 ^{117}Ag	MA8	1.0	10Br02 *
$\text{C}_9 \text{H}_9-^{117}\text{Sn}$	167486	12	167471.3	0.5	-0.5	U			M16	2.5	63Da10
	167471	8			0.0	U			R13	1.5	83De51
$\text{C } ^{35}\text{Cl}_3-^{117}\text{Sn}$	3596	2	3604.0	0.5	1.0	U			H14	4.0	62Ba24
$^{117}\text{Te}-\text{u}$	-91318	30	-91354	14	-1.2	-			GS2	1.0	05Li24
	-91359	30			0.2	-			GS2	1.0	05Li24 *
	-91339	21			-0.7	1	46	46 ^{117}Te			average
$^{117}\text{I}-\text{u}$	-86350	30	-86354	27	-0.1	-			GS2	1.0	05Li24
	-86407	50			0.4	-			Gr1	2.5	19An10
	-86353	29			-0.0	1	88	88 ^{117}I			average
$^{117}\text{Xe}-\text{u}$	-79647	30	-79641	11	0.2	R			GS2	1.0	05Li24
$^{117}\text{Xe}-^{133}\text{Cs}_{880}$	3562	12	3561	11	-0.1	2			MA6	1.0	04Di18
$^{117}\text{Cs}-^{133}\text{Cs}_{880}$	11819	67				2			MA4	1.0	99Am05 *
$^{117}\text{Rh}-^{120}\text{Sn}_{975}$	21388.8	9.5				2			JY1	1.0	07Ha20
$^{117}\text{Pd}-^{120}\text{Sn}_{975}$	13309.4	7.9	13308	8	-0.2	1	96	96 ^{117}Pd	JY1	1.0	07Ha20
$^{117}\text{Sn}-^{116}\text{Sn}$	1219	11	1211.2	0.5	-0.3	U			M16	2.5	63Da10
$^{116}\text{Cd}(\text{d,p})^{117}\text{Cd}$	3538	30	3552.7	1.0	0.5	U			Pit		64Ro17
	3550	20			0.1	U			Oak		64Si18
	3552.66	1.0				2			Rez		90Pi05 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{116}\text{Sn}(n,\gamma)^{117}\text{Sn}$	6943.5	2.0	6943.1	0.5	-0.2	U					75Bh01 Z
	6943.3	1.5			-0.1	U			ORn		78Ra16 Z
	6942.9	0.5			0.4	-			Bdn		06Fi.A
$^{116}\text{Sn}(d,p)^{117}\text{Sn}$	4721.0	1.8	4718.5	0.5	-1.4	-			SPa		75Be09
$^{116}\text{Sn}(n,\gamma)^{117}\text{Sn}$	ave.	6943.1	0.5	6943.1	0.5	-0.0	1	97	^{117}Sn		average
$^{116}\text{Sn}(^3\text{He},d)^{117}\text{Sb}$	-1091	10	-1091	8	0.0	1	71	71	^{117}Sb	VUn	78Ka12 *
$^{117}\text{Xe}(\epsilon p)^{116}\text{Te}$	4100	200	3789	26	-1.6	U					72Ho18
$^{117}\text{Ba}(\epsilon p)^{116}\text{Xe}$	7900	300	8300	250	1.3	F					78Bo20 *
	8300	250				2			GSI		05Ja06
$^{117}\text{La}(p)^{116}\text{Ba}$	789.8	6.	820	3	5.0	B					01So02 *
	813.0	5.			1.4	o			Arp		01Ma69
	820.1	3.				3			Arp		11Li28
$^{117}\text{Pd}(\beta^-)^{117}\text{Ag}$	5735	32	5758	15	0.7	1	21	17	^{117}Ag	Jyv	00Kr.A
$^{117}\text{Ag}(\beta^-)^{117}\text{Cd}$	4160	50	4236	14	1.5	U			Stu		82Al29 *
$^{117}\text{Cd}(\beta^-)^{117}\text{In}$	2535	20	2525	5	-0.5	U					75Ta06 *
$^{117}\text{In}(\beta^-)^{117}\text{Sn}$	1456.6	5.	1455	5	-0.4	1	94	94	^{117}In		55Mc17 *
$^{117}\text{Sb}(\beta^+)^{117}\text{Sn}$	1751	40	1758	8	0.2	U					64Ba46 *
$^{117}\text{Sn}(p,n)^{117}\text{Sb}$	-2525	20	-2541	8	-0.8	1	18	18	^{117}Sb	Oak	71Ke21
$^{117}\text{Te}(\beta^+)^{117}\text{Sb}$	3552	20	3544	13	-0.4	-					62Kh05 *
	3492	30			1.7	-					67Be46 *
	ave.	3534	17		0.6	1	62	51	^{117}Te		average
$^{117}\text{I}(\beta^+)^{117}\text{Te}$	4680	100	4657	28	-0.2	-					69La33 *
	4610	110			0.4	-					70Be.A *
	ave.	4650	70		0.1	1	14	12	^{117}I		average
$^{117}\text{Xe}(\beta^+)^{117}\text{I}$	6270	300	6253	28	-0.1	U					85Le10 *
$^{117}\text{Cs}^x(\text{IT})^{117}\text{Cs}$	50	50				3					AHW
* $^{117}\text{Ag}-^{133}\text{Cs}_{880}$	$D_M = -5013.3(4.0) \mu\text{u}$ for ground state or $^{117}\text{Ag}^m$ at 28.6 keV; $M - A = -82172.3(3.7) \text{ keV}$										Nub211 **
* $^{117}\text{Te}-u$	$M - A = -84804(28) \text{ keV}$ for $^{117}\text{Te}^m 11/2^-$ at 296.1 keV										Nub211 **
* $^{117}\text{Cs}-^{133}\text{Cs}_{880}$	$D_M = 11900(21) \mu\text{u}$ for mixture gs+m at 150#80 keV; $M - A = -66418(19) \text{ keV}$										Nub211 **
* $^{116}\text{Cd}(d,p)^{117}\text{Cd}$	Estimated systematic error 0.5 added to statistical error 0.85 keV										AHW **
* $^{116}\text{Sn}(^3\text{He},d)^{117}\text{Sb}$	$Q - Q(^{120}\text{Sn}(^3\text{He},d)) = 1373(10, \text{Ka}), Q(120) = 282.1(2.0) \text{ keV}$										AHW **
* $^{117}\text{Ba}(\epsilon p)^{116}\text{Xe}$	F : disagrees with next and with TMS										GAu **
* $^{117}\text{La}(p)^{116}\text{Ba}$	Reports also an isomer $^{117}\text{La}^m E_p = 933(10) Q_p = 941.1 \text{ keV } T = 10(5) \text{ ms}$, not observed in reference using similar set-up and far greater statistics										01So02 **
* $^{117}\text{Ag}(\beta^-)^{117}\text{Cd}$	$Q_{\beta^-} = 4260(110)$; and $4170(50)$ from $^{117}\text{Ag}^m$ at 28.6 keV										11Li28 **
* $^{117}\text{Cd}(\beta^-)^{117}\text{In}$	$Q_{\beta^-} = 2220(20)$ to $^{117}\text{In}^m$ at 315.303 keV										Nub211 **
* $^{117}\text{In}(\beta^-)^{117}\text{Sn}$	$E_{\beta^-} = 740(10)$ to $7/2^+$ level at 711.54; and $1772(5), 1616(5)$ from $^{117}\text{In}^m$ at 315.303 to $1/2^+$ ground state, $3/2^+$ level at 158.56 keV										Ens111 **
* $^{117}\text{Sb}(\beta^+)^{117}\text{Sn}$	$E_{\beta^+} = 570(40)$ to $3/2^+$ level at 158.562 keV										Nub211 **
* $^{117}\text{Te}(\beta^+)^{117}\text{Sb}$	$E_{\beta^+} = 1810(20) 1750(30)$ respectively, to $1/2^+$ level at 719.7 keV										Ens111 **
* $^{117}\text{I}(\beta^+)^{117}\text{Te}$	$E_{\beta^+} = 3500(50), 3250(50)$ to $5/2^+$ level at 274.4, $(3/2)^+$ at 325.9 keV										Ens111 **
* $^{117}\text{I}(\beta^+)^{117}\text{Te}$	$Q_{\beta^+} = 4310(100)$ assumed to $5/2^+$ level at 274.4, $(3/2)^+$ at 325.9 keV										Ens111 **
* $^{117}\text{Xe}(\beta^+)^{117}\text{I}$	May be lower limit										AHW **
$^{118}\text{Ru}-u$	-61879	196	-61190#	220#	1.4	D			GT3	2.5	16Kn03 *
$^{118}\text{Rh}-u$	-69598	290	-69659	26	-0.1	U			GT1	1.5	04Ma.A
$^{118}\text{Pd}-^{129}\text{Xe}_{915}$	6193.6	4.3	6192.8	2.7	-0.2	1	39	39	^{118}Pd	JY1	1.0
$^{118}\text{Ag}-^{133}\text{Cs}_{887}$	-1540.4	2.7				2			MA8	1.0	10Br02 *
$\text{C}_9 \text{H}_{10}-^{118}\text{Sn}$	176645	7	176643.7	0.5	-0.1	U			M16	2.5	63Da10
	176637	8			0.6	U			R13	1.5	83De51
$^{118}\text{Te}-u$	-94162	30	-94140	20	0.7	R			GS2	1.0	05Li24
$^{118}\text{I}-u$	-86932	30	-86926	21	0.2	2			GS2	1.0	05Li24
	-86920	30			-0.2	2			GS2	1.0	05Li24 *
$^{118}\text{Xe}-u$	-83785	30	-83821	11	-1.2	R			GS2	1.0	05Li24
$^{118}\text{Xe}-^{133}\text{Cs}_{887}$	37	12	43	11	0.5	2			MA6	1.0	04Di18

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{118}\text{Cs}^x-^{133}\text{Cs}_{.887}$	10424	16	10429	13	0.3	o			MA1	1.0	90St25
	10429	13				2			MA1	1.0	99Am05
$^{118}\text{Rh}-^{120}\text{Sn}_{.983}$	26476	26				2			JY1	1.0	07Ha20
$^{118}\text{Pd}-^{120}\text{Sn}_{.983}$	15199.7	7.9	15202.2	2.6	0.3	—			JY1	1.0	07Ha20
	15202.1	3.6			0.0	—			JY1	1.0	11Ha48
ave.	15202	3			0.1	1	64	61 ^{118}Pd			average
$^{118}\text{Sn } ^{35}\text{Cl}-^{116}\text{Sn } ^{37}\text{Cl}$	2814	2	2813.9	0.5	—0.0	U			H15	4.0	62Ba23
$^{118}\text{Sn}-^{117}\text{Sn}$	—1338	11	—1347.41	0.14	—0.3	U			M16	2.5	63Da10
$^{117}\text{Cs}^x-^{118}\text{Cs}_{.496}^x$ $^{116}\text{Cs}_{.504}$	—1160	400	—1250#	100#	—0.1	U			P32	2.5	86Au02
$^{118}\text{Cs}(\varepsilon\alpha)^{114}\text{Te}$	10600	200	11055	28	2.3	U					77Bo28
	10750	200			1.5	U					78Da07 *
$^{116}\text{Cd}(\text{t,p})^{118}\text{Cd}$	5650	20				2			Ald		67Hi01
$^{116}\text{Sn}(\text{t,p})^{118}\text{Sn}$	7769	15	7787.7	0.5	1.2	U			Ald		68Bj02
$^{118}\text{Sn}(\text{p,t})^{116}\text{Sn}$	—7790	10	—7787.7	0.5	0.2	U			Roc		70Fl08
$^{118}\text{Sn}(\text{d},^3\text{He})^{117}\text{In}$	—4440	40	—4505	5	—1.6	U			Sac		69Co03
	—4481	15			—1.6	U			MSU		71We01
$^{117}\text{Sn}(\text{n},\gamma)^{118}\text{Sn}$	9326.5	2.	9326.42	0.13	—0.0	U					70Or.A
	9324.8	2.1			0.8	U			ORn		75Sl.A
	9326.42	0.13			—0.0	1	100	97 ^{118}Sn			02Bo11
	9327.9	1.1			—1.3	U			Bdn		06Fi.A
$^{117}\text{Sn}(\text{d,p})^{118}\text{Sn}$	7090	12	7101.85	0.13	1.0	U			Tal		64No06
$^{118}\text{Sn}(\text{p,d})^{117}\text{Sn}$	—7097	15	—7101.85	0.13	—0.3	U			Har		70Ca01
$^{118}\text{Cs}(\varepsilon\text{p})^{117}\text{I}$	4700	300	4740	29	0.1	U					78Da07
$^{118}\text{Pd}(\beta^-)^{118}\text{Ag}$	4100	200	4165	4	0.3	U			Jyv		89Ko22 *
$^{118}\text{Ag}(\beta^-)^{118}\text{Cd}$	7122	100	7148	20	0.3	U			Stu		82Al29 *
	7110	470			0.1	U			Stu		82Al29 *
	7155	76			—0.1	U					95Ap.A
$^{118}\text{In}(\beta^-)^{118}\text{Sn}$	4200	400	4425	8	0.6	U					61Gl02
	4200	300			0.7	U					64Ka10
	4310	100			1.1	U					87Ga.A
$^{118}\text{In}^m(\beta^-)^{118}\text{Sn}$	4270	100	4530#	50#	2.5	D					64Ka10 *
$^{118}\text{Sb}(\beta^+)^{118}\text{Sn}$	3610	50	3656.6	3.0	0.9	U					61Fi05
$^{118}\text{Sn}(\text{p,n})^{118}\text{Sb}$	—4477.7	5.7	—4439.0	3.0	6.8	F			Tkm		63Ok01 *
	—4439.0	3.				2			Oak		77Jo03
$^{118}\text{Sb}^n(\beta^+)^{118}\text{Sn}$	3907	5				2					61Bo13 *
$^{118}\text{I}(\beta^+)^{118}\text{Te}$	7080	150	6720	27	—2.4	U					68La18 *
	7068	100			—3.5	C					70Be.A *
$^{118}\text{Xe}(\beta^+)^{118}\text{I}$	3720	110	2892	22	—7.5	F					70Be.A *
$^{118}\text{Cs}(\beta^+)^{118}\text{Xe}$	9300	1000	9670	16	0.4	U					76Da.C
$^{118}\text{Cs}^x(\text{IT})^{118}\text{Cs}$	5	4				3					82Au01 *
* $^{118}\text{Ru}-\text{u}$	Trends from Mass Surface TMS suggest ^{118}Ru 640 keV less bound										GAu **
* $^{118}\text{Ag}-^{133}\text{Cs}_{.887}$	$D_M=-1403.5(2.7) \mu\text{u}$ for $^{118}\text{Ag}^n$ at 127.63(0.10) keV; $M-A=-79426.3(2.5) \text{ keV}$										Nub211 **
* $^{118}\text{I}-\text{u}$	$M-A=-80775(28) \text{ keV}$ for $^{118}\text{I}^m$ 7 [—] at 188.8 keV										Nub211 **
* $^{118}\text{Cs}(\varepsilon\alpha)^{114}\text{Te}$	As read from Fig. 2 (p.401)										GAu **
* $^{118}\text{Pd}(\beta^-)^{118}\text{Ag}$	Original value 4000(200) corrected for new branching ratios										93Ja03 **
* $^{118}\text{Ag}(\beta^-)^{118}\text{Cd}$	$E_{\beta^-}=4330(240), 3960(170), 3810(150)$ reinterpreted as feeding										95Ap.A **
*	(1) level at 2788.72, (1) at 3224.32, (2,3,4) at 3265.77 keV										Ens959 **
* $^{118}\text{Ag}(\beta^-)^{118}\text{Cd}$	$E_{\beta^-}=3990(720), 3910(630)$ reinterpreted as $^{118}\text{Ag}^n$ at 127.63(0.10) keV										95Ap.A **
	to (2,3,4) level at 3181.73, 3381.8 keV										Ens959 **
* $^{118}\text{In}^m(\beta^-)^{118}\text{Sn}$	$E_{\beta^-}=2000(100)$ to 4 ⁺ level at 2280.342 level, and other E_{β^-}										Ens959 **
* $^{118}\text{In}^m(\beta^-)^{118}\text{Sn}$	Trends from Mass Surface TMS suggest $^{118}\text{In}^m$ 255 keV less bound										GAu **
* $^{118}\text{Sn}(\text{p,n})^{118}\text{Sb}$	F : see note added in proof to reference										77Jo03 **
* $^{118}\text{Sb}^n(\beta^+)^{118}\text{Sn}$	$p^+=16(1)\times 10^{-4}$ to 7 [—] level at 2574.91 keV, recalculated Q										Ens959 **
* $^{118}\text{I}(\beta^+)^{118}\text{Te}$	$E_{\beta^+}=5450(150)$ to 2 ⁺ level at 605.70 keV										Ens959 **
* $^{118}\text{I}(\beta^+)^{118}\text{Te}$	$E_{\beta^+}=5440(100)$ to 2 ⁺ level at 605.70 keV										Ens959 **
* $^{118}\text{Xe}(\beta^+)^{118}\text{I}$	F : probably contaminated by isobars										GAu **
* $^{118}\text{Cs}^x(\text{IT})^{118}\text{Cs}$	Original 24(19) corrected for new estimated IT=100(60)#										Nub211 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{119}\text{Rh}-u$	-67698	268	-67443	10	0.6	U			GT1	1.5	04Ma.A	
$^{119}\text{Rh}-^{129}\text{Xe}_{.922}$	20349	10				2			JY1	1.0	11Ha48	
$^{119}\text{Pd}-u$	-76844	208	-76659	9	0.6	U			GT1	1.5	04Ma.A	
$^{119}\text{Ag}-^{133}\text{Cs}_{.895}$	188	16	191	16	0.2	1	97	97	^{119}Ag	MA8	1.0	10Br02 *
$\text{C}_9\text{H}_{11}-^{119}\text{Sn}$	182778	7	182764.1	0.8	-0.8	U			M16	2.5	63Da10	
	182762	8			0.2	U			R13	1.5	83De51	
$^{119}\text{Sn}-u$	-96726	39	-96688.7	0.8	1.0	U			GR1	1.0	19Mi18 *	
$^{119}\text{Sb}-u$	-96063	18	-96056	8	0.4	1	17	17	^{119}Sb	GR1	1.0	19Mi18
$^{119}\text{Te}-u$	-93731	190	-93594	8	0.7	U			GR1	1.0	19Mi18 *	
$^{119}\text{I}-u$	-89926	30	-89939	23	-0.4	2			GS2	1.0	05Li24	
	-89959	37			0.5	2			GR1	1.0	19An10	
$^{119}\text{Xe}-u$	-84601	30	-84589	11	0.4	1	14	14	^{119}Xe	GS2	1.0	05Li24
	-84613	61			0.4	U			GR1	1.0	19An10	
$^{119}\text{Xe}-^{133}\text{Cs}_{.895}$	33	12	31	11	-0.2	1	86	86	^{119}Xe	MA6	1.0	04Di18
$^{119}\text{Cs}-u$	-77532	57	-77623	15	-1.6	U			GS2	1.0	05Li24 *	
$^{119}\text{Cs}^x-^{133}\text{Cs}_{.895}$	7011	16	7015	9	0.2	o			MA1	1.0	90St25	
	7018	13			-0.2	2			MA1	1.0	99Am05	
	7012	13			0.2	2			MA4	1.0	99Am05	
$^{119}\text{Sn } ^{35}\text{Cl}-^{117}\text{Sn } ^{37}\text{Cl}$	3306	2	3307.4	0.6	0.2	U			H15	4.0	62Ba23	
$^{119}\text{Pd}-^{120}\text{Sn}_{.992}$	20356.2	8.8				2			JY1	1.0	07Ha20	
$^{119}\text{Sn}-^{118}\text{Sn}$	1709	12	1704.6	0.6	-0.1	U			M16	2.5	63Da10	
$^{119}\text{I}-^{118}\text{I}$	-2747	155	-3010	30	-1.1	U			CR2	1.5	92Sh.A *	
$^{119}\text{I}-^{117}\text{I}$	-3570	155	-3580	40	-0.1	U			CR2	1.5	92Sh.A *	
$^{118}\text{Cs}^x-^{119}\text{Cs}_{.661}^x\text{ }^{116}\text{Cs}_{.339}$	530	80	410#	40#	-0.6	U			P32	2.5	86Au02	
$^{118}\text{Cs}^x-^{119}\text{Cs}_{.496}^x\text{ }^{117}\text{Cs}_{.504}$	870	50	940	40	0.5	U			P22	2.5	82Au01	
	980	40			-0.4	U			P32	2.5	86Au02	
$^{119}\text{Sn}(t,\alpha)^{118}\text{In}-^{118}\text{Sn}(\gamma)^{117}\text{In}$	-127	6	-127	6	-0.0	1	100	100	^{118}In	McM		85Pi03
$^{118}\text{Sn}(n,\gamma)^{119}\text{Sn}$	6484.6	1.5	6483.5	0.5	-0.8	-			ORn			78Ra16
	6483.3	0.6			0.3	-			Bdn			06Fi.A
$^{118}\text{Sn}(d,p)^{119}\text{Sn}$	4238	12	4258.9	0.5	1.7	U			MIT			67Sp09
$^{118}\text{Sn}(n,\gamma)^{119}\text{Sn}$	ave. 6483.5	0.6	6483.5	0.5	-0.0	1	96	93	^{119}Sn			average
$^{118}\text{Sn}(^3\text{He},d)^{119}\text{Sb}$	-388	10	-382	7	0.6	1	49	49	^{119}Sb	VUn		78Ka12 *
$^{119}\text{Ba}(\epsilon p)^{118}\text{Xe}$	6200	200				3						78Bo20
$^{119}\text{Ag}(\beta^-)^{119}\text{Cd}$	5350	40	5330	40	-0.5	1	81	78	^{119}Cd	Stu		82Al29
$^{119}\text{Cd}(\beta^-)^{119}\text{In}$	3797	80	3720	40	-0.9	1	23	22	^{119}Cd	Stu		82Al29 *
$^{119}\text{In}(\beta^-)^{119}\text{Sn}$	2387	100	2366	7	-0.2	U						60Yu01 *
	2413	200			-0.2	U						61GI06 *
$^{119}\text{Sb}(\epsilon)^{119}\text{Sn}$	579	20	589	7	0.5	-						57OI05 *
$^{119}\text{Sn}(p,n)^{119}\text{Sb}$	-1369	15	-1372	7	-0.2	-			Oak			71Ke21
$^{119}\text{Sb}(\epsilon)^{119}\text{Sn}$	ave. 584	12	589	7	0.5	1	34	34	^{119}Sb			average
$^{119}\text{Te}(\beta^+)^{119}\text{Sb}$	2293	2				2						60Ko12 *
$^{119}\text{I}(\beta^+)^{119}\text{Te}$	3630	100	3405	23	-2.3	U						69La33 *
	3370	100			0.3	U						70Be.A
$^{119}\text{Xe}(\beta^+)^{119}\text{I}$	4990	120	4983	24	-0.1	U						70Be.A
$^{119}\text{Cs}(\beta^+)^{119}\text{Xe}$	6260	290	6489	17	0.8	U						83Pa.A *
$^{119}\text{Cs}^x(\text{IT})^{119}\text{Cs}$	16	11				3						82Au01 *
* $^{119}\text{Ag}-^{133}\text{Cs}_{.895}$	$D_M=198.4(5.7) \mu\text{u}$ for ground state or $^{119}\text{Ag}^m$ at 20#20 keV; $M-A=-78638.7(5.3) \text{ keV}$										Nub211 **	
* $^{119}\text{Sn}-u$	could be mixture of gs+iso, not corrected										WgM209**	
* $^{119}\text{Te}-u$	could be mixture of gs+iso, not corrected										WgM209**	
* $^{119}\text{Cs}-u$	$M-A=-72195(48) \text{ keV}$ for mixture gs+m at 50#30 keV										Nub211 **	
* $^{119}\text{I}-^{118}\text{I}$	From $^{118}\text{I}/^{119}\text{I}=0.99161584(117)$, originally $D_M=-3039(139) \mu\text{u}$, revised by										GAu **	
	authors: $-2849(139) \mu\text{u}$ for ^{118}I gs+m mixture at 188.8 keV										Nub211 **	
* $^{119}\text{I}-^{117}\text{I}$	From $^{117}\text{I}/^{119}\text{I}=0.98321059(130)$										GAu **	
* $^{118}\text{Sn}(^3\text{He},d)^{119}\text{Sb}$	$Q-Q(^{120}\text{Sn}(^3\text{He},d)^{121}\text{Sb})=-673(10)$, $Q(120)=285.1(2.1) \text{ keV}$										AHW **	
* $^{119}\text{Cd}(\beta^-)^{119}\text{In}$	$Q_{\beta^-}=3800(90)$; and $3940(80)$ from $^{119}\text{Cd}^m$ at 146.54 keV										Nub211 **	
* $^{119}\text{In}(\beta^-)^{119}\text{Sn}$	$E_{\beta^-}=1600(100)$ to $7/2^+$ level at 787.01 keV										Ens09a **	
* $^{119}\text{In}(\beta^-)^{119}\text{Sn}$	$E_{\beta^-}=2700(200)$ from $^{119}\text{In}^m$ at 311.37(0.03) to $3/2^+$ level at 23.871 keV										Ens09a **	
* $^{119}\text{Sb}(\epsilon)^{119}\text{Sn}$	IBE=526(20) to $3/2^+$ level at 23.871 keV										Ens09a **	
* $^{119}\text{Te}(\beta^+)^{119}\text{Sb}$	$E_{\beta^+}=627(2)$ to $1/2^+$ level at 644.03 keV										Ens09a **	
* $^{119}\text{I}(\beta^+)^{119}\text{Te}$	$E_{\beta^+}=2350(100)$ to $3/2^+$ level at 257.484 keV										Ens09a **	
* $^{119}\text{Cs}(\beta^+)^{119}\text{Xe}$	$E_{\beta^+}=4980(290)$ to $9/2^+$ level at 257.84 keV										Ens09a **	
* $^{119}\text{Cs}^x(\text{IT})^{119}\text{Cs}$	Original 33(22) corrected for new estimated IT=50(30)#										GAu **	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{120}\text{Pd}-^{129}\text{Xe}_{930}$	13107.0	4.4	13105.5	2.5	-0.3	1	31	31 ^{120}Pd	JY1	1.0	11Ha48
$^{120}\text{Ag}-^{133}\text{Cs}_{902}$	4067.1	4.8				2			MA8	1.0	10Br02
	4086	12	4067	5	-1.6	o			MA8	1.0	10Br02
$^{120}\text{Cd}-^{133}\text{Cs}_{902}$	-4849.6	4.0				2			MA8	1.0	10Br02
$\text{C}_9\text{H}_{12}-^{120}\text{Sn}$	191709	11	191697.8	1.0	-0.4	U			M16	2.5	63Da10
	191705	8			-0.6	U			R13	1.5	83De51
$^{13}\text{C}-^{35}\text{Cl}_2-^{37}\text{Cl}-^{120}\text{Sn}$	4758	3	4760.2	1.0	0.2	U			H14	4.0	62Ba24
$^{120}\text{Sb}-\text{u}$	-94796	76	-94920	8	-1.6	U			GS2	1.0	05Li24 *
$\text{C}_9\text{H}_{12}-^{120}\text{Te}$	189879	9	189834.6	1.9	-2.0	U			M16	2.5	63Da10
	189868	8			-2.8	U			R13	1.5	83De51
$^{120}\text{I}-\text{u}$	-90222	104	-89906	16	3.0	C			GS2	1.0	05Li24 *
$^{120}\text{Xe}-\text{u}$	-88231	30	-88216	13	0.5	R			GS2	1.0	05Li24
$^{120}\text{Xe}-^{133}\text{Cs}_{902}$	-2930	14	-2933	13	-0.2	2			MA6	1.0	04Di18
$^{120}\text{Cs}-\text{u}$	-79342	54	-79323	11	0.4	U			GS2	1.0	05Li24 *
$^{120}\text{Cs}^x-^{133}\text{Cs}_{902}$	5947	16	5965	10	1.1	o			MA1	1.0	90St25
	5956	12			0.7	2			MA1	1.0	99Am05
	5983	17			-1.1	2			MA4	1.0	99Am05
$^{120}\text{Sn}-^{35}\text{Cl}-^{118}\text{Sn}-^{37}\text{Cl}$	3546	2	3546.0	1.1	0.0	U			H15	4.0	62Ba23
$^{120}\text{Pd}-^{120}\text{Sn}$	22317.1	9.7	22349.2	2.4	3.3	B			JY1	1.0	07Ha20
	22348.6	2.8			0.2	1	72	69 ^{120}Pd	JY1	1.0	11Ha48
$^{120}\text{Te}-^{120}\text{Sn}$	1842.2	1.7	1863.2	1.9	12.4	B			CP1	1.0	09Sc19
	1839.7	1.7			13.8	B			CP1	1.0	09Sc19
$^{120}\text{Sn}-^{119}\text{Sn}$	-1113	11	-1108.7	1.2	0.2	U			M16	2.5	63Da10
$^{118}\text{Cs}^x-^{120}\text{Cs}_{328}^x-^{117}\text{Cs}_{672}^x$	460	120	480	60	0.1	U			P22	2.5	82Au01
$^{119}\text{Cs}^x-^{120}\text{Cs}_{661}^x-^{117}\text{Cs}_{339}^x$	-940	50	-928	29	0.1	U			P22	2.5	82Au01
$^{119}\text{Cs}^x-^{120}\text{Cs}_{496}^x-^{118}\text{Cs}_{504}^x$	-1220	30	-1167	11	0.7	U			P22	2.5	82Au01
	-1180	60			0.1	U			P32	2.5	86Au02
	-1200	30			0.4	U			P32	2.5	86Au02
	-1270	50			0.8	F			P32	2.5	86Au02 *
$^{120}\text{Cs}(\epsilon\alpha)-^{116}\text{Te}$	9200	300	8950	26	-0.8	U					76Jo.A
$^{118}\text{Sn}(\text{t,p})-^{120}\text{Sn}$	7107	15	7105.7	1.0	-0.1	U			Ald		68Bj02
$^{120}\text{Sn}(\text{p,t})-^{118}\text{Sn}$	-7109	10	-7105.7	1.0	0.3	U			Roc		70Fi08
$^{120}\text{Te}(\text{p,t})-^{118}\text{Te}$	-9343	24	-9332	18	0.4	2			Win		74De31 *
$^{120}\text{Sn}(\text{d},^3\text{He})-^{119}\text{In}$	-5160	40	-5195	7	-0.9	U			Sac		69Co03
	-5169	20			-1.3	1	13	13 ^{119}In	MSU		71We01
$^{120}\text{Sn}(\text{t},\alpha)-^{119}\text{In}-^{118}\text{Sn}()-^{117}\text{In}$	-692	6	-689	6	0.5	1	92	86 ^{119}In	McM		85Pi03
$^{119}\text{Sn}(\text{d,p})-^{120}\text{Sn}$	6890	12	6879.5	1.1	-0.9	U			Tal		64No06
$^{120}\text{Sn}(\text{p,d})-^{119}\text{Sn}$	-6889	15	-6879.5	1.1	0.6	U			Har		70Ca01
$^{120}\text{Sn}(\text{d,t})-^{119}\text{Sn}$	-2847.0	2.5	-2846.8	1.1	0.1	1	20	12 ^{120}Sn	SPa		75Be09
$^{120}\text{Pd}(\beta^-)-^{120}\text{Ag}$	5500	100	5372	5	-1.3	U			Jyv		94Jo.A
$^{120}\text{Ag}(\beta^-)-^{120}\text{Cd}$	8200	100	8306	6	1.1	U			Stu		82Al29
	8450	100			-1.4	U					95Ap.A
$^{120}\text{In}(\beta^-)-^{120}\text{Sn}$	5300	170	5370	40	0.4	U			Stu		78Al18
	5370	40				2					87Ga.A
$^{120}\text{In}^m(\beta^-)-^{120}\text{Sn}$	5280	200	5420#	50#	0.7	D					64Ka10 *
	5340	170			0.5	D			Stu		78Al18 *
$^{120}\text{Sb}(\beta^+)-^{120}\text{Sn}$	2720	20	2681	7	-2.0	U					50B192
	2770	30			-3.0	B					69Ki15
$^{120}\text{Sn}(\text{p,n})-^{120}\text{Sb}$	-3462.9	7.1				2			Tkm		63Ok01
$^{120}\text{I}(\beta^+)-^{120}\text{Te}$	5615	15				2					70Ga32 *
	5608	150	5615	15	0.0	U					68La18 *
$^{120}\text{Xe}(\beta^+)-^{120}\text{I}$	1960	40	1575	19	-9.6	B					74Mu10 *
$^{120}\text{Cs}(\beta^+)-^{120}\text{Xe}$	7300	500	8284	15	2.0	U					76Ba.A *
	7800	1000			0.5	U					76Da.C *
	7380	230			3.9	C					83Pa.A *
	8210	200			0.4	U			IRS		93Al03

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{120}\text{Cs}^x(\text{IT})^{120}\text{Cs}$	5	4				3					82Au01 *
$^{120}\text{Ba}(\beta^+)^{120}\text{Cs}$	5000	300				4					92Xu04
* $^{120}\text{Sb}-\text{u}$	$M-A=-88302(50)$ keV for mixture gs+m at 0#100 keV										Nub211 **
* $^{120}\text{I}-\text{u}$	$M-A=-83881(28)$ keV for mixture gs+n at 320(15) keV										Nub211 **
* $^{120}\text{Cs}-\text{u}$	$M-A=-73856(29)$ keV for mixture gs+m at 100#60 keV										Nub211 **
* $^{119}\text{Cs}^x_{-120}\text{Cs}^x_{496}$ ^{118}C	F : rejection based on line-shape analysis										86Au02 **
* $^{120}\text{Te}(\text{p,t})^{118}\text{Te}$	Original error 12; added systematic error 21 keV										GAu **
* $^{120}\text{In}^m(\beta^-)^{120}\text{Sn}$	$E_{\beta^-}=3100(200),2200(200)$ to 4^+ levels at 2194.299,3057.946 keV										Ens029 **
* $^{120}\text{In}^m(\beta^-)^{120}\text{Sn}$	$E_{\beta^-}=3100(170)$ to 4^+ level at 2194.299 keV, and other E_{β^-}										Ens029 **
* $^{120}\text{In}^m(\beta^-)^{120}\text{Sn}$	Trends from Mass Surface TMS suggest $^{120}\text{In}^m$ 105 keV less bound										GAu **
* $^{120}\text{I}(\beta^+)^{120}\text{Te}$	$E_{\beta^+}=4595(15), 4030(20)$ to ground state, 2^+ level at 560.438 keV										Ens029 **
* $^{120}\text{I}(\beta^+)^{120}\text{Te}$	$E_{\beta^+}=3130(150)$ from $^{120}\text{I}^m$ at 320(15) to 6^+ level at 1776.23 keV										Ens029 **
* $^{120}\text{Xe}(\beta^+)^{120}\text{I}$	$p^+=0.07(0.01)$ to 1^+ level at 25.1 keV, recalculated Q										Ens029 **
* $^{120}\text{Cs}(\beta^+)^{120}\text{Xe}$	$E_{\beta^+}=6000(500) 6500(1000) 6040(230)$, respectively, to 2^+ level at 322.61 keV										Ens029 **
* $^{120}\text{Cs}^x(\text{IT})^{120}\text{Cs}$	Original 24(19) corrected for new estimated IT=100(60)#										Nub211 **
$^{121}\text{Rh}-\text{u}$	-60387	266				2			GT3	2.5	16Kn03
$^{121}\text{Pd}-\text{u}$	-71820	311	-71050	4	1.7	U			GT1	1.5	04Ma.A
$^{121}\text{Ag}-^{133}\text{Cs}_{910}$	6164	13				2			MA8	1.0	10Br02 *
	6170	17	6164	13	-0.4	o			MA8	1.0	10Br02 *
$^{121}\text{Cd}-^{130}\text{Xe}_{931}$	2796.2	3.0	2796.5	2.1	0.1	2			JY1	1.0	12Ha25
	2796.7	2.9			-0.1	2			JY1	1.0	13Ka08 *
$\text{C}_9 \text{H}_{13}-^{121}\text{Sb}$	197910.5	3.7	197914.1	2.7	0.4	U			M16	2.5	63Da10
	197910	8			0.3	U			R13	1.5	83De51
$^{121}\text{Sb}-\text{C }^{35}\text{Cl }^{37}\text{Cl}_2$	3162	3	3153.5	2.7	-0.7	U			H14	4.0	62Ba24
$^{121}\text{Sb}-\text{u}$	-96180	30	-96188.6	2.7	-0.3	U			GS2	1.0	05Li24
$^{121}\text{I}-\text{u}$	-92609	30	-92589	5	0.7	U			GS2	1.0	05Li24
$^{121}\text{Xe}-\text{u}$	-88562	30	-88547	11	0.5	R			GS2	1.0	05Li24
$^{121}\text{Xe}-^{133}\text{Cs}_{910}$	-2495	13	-2508	11	-1.0	-			MA6	1.0	04Di18
	ave.	-2499	12		-0.7	1	85	85 ^{121}Xe			average
$^{121}\text{Cs}-^{133}\text{Cs}_{910}$	3247	25	3266	15	0.8	o			MA1	1.0	90St25 *
	3248	25			0.7	1	38	38 ^{121}Cs	MA1	1.0	99Am05 *
$^{121}\text{Cs}-\text{u}$	-82821	38	-82773	15	1.3	1	16	16 ^{121}Cs	GS2	1.0	05Li24 *
$^{121}\text{Pd}-^{129}\text{Xe}_{938}$	18265.9	3.6				2			JY1	1.0	11Ha48 *
$^{121}\text{Sb }^{35}\text{Cl}-^{119}\text{Sn }^{37}\text{Cl}$	3452	2	3450.2	2.8	-0.2	U			H14	4.0	62Ba24
$^{119}\text{Cs}^x_{-121}\text{Cs}^x_{328}$ $^{118}\text{Cs}^x_{672}$	-1080	30	-1047	13	0.4	U			P22	2.5	82Au01
$^{120}\text{Cs}^x_{-121}\text{Cs}^x_{661}$ $^{118}\text{Cs}^x_{339}$	280	30	240	15	-0.5	U			P22	2.5	82Au01
$^{120}\text{Cs}^x_{-121}\text{Cs}^x_{496}$ $^{119}\text{Cs}^x_{504}$	790	30	770	13	-0.3	U			P22	2.5	82Au01
	860	50			-0.7	U			P32	2.5	86Au02
	813	14			-1.2	U			P32	2.5	86Au02
$^{120}\text{Sn}(\text{n},\gamma)^{121}\text{Sn}$	6170.3	2.	6170.2	0.3	-0.0	U					76Ca24
	6170.5	0.7			-0.4	-					81Ba53
	6170.1	0.4			0.3	-			Bdn		06Fi.A
$^{120}\text{Sn}(\text{d,p})^{121}\text{Sn}$	3946.2	1.7	3945.6	0.3	-0.3	-			SPa		75Be09
$^{120}\text{Sn}(\text{n},\gamma)^{121}\text{Sn}$	ave.	6170.2	0.3	6170.2	0.3	-0.1	1	99	97 ^{121}Sn		average
$^{121}\text{Sb}(\gamma,\text{n})^{120}\text{Sb}$	-9310	60	-9253	8	0.9	U			Phi		60Ge01
	-9240	25			-0.5	U			McM		79Ba06
$^{120}\text{Te}(^3\text{He,d})^{121}\text{I}$	-1320.5	4.4	-1321	4	-0.1	1	99	99 ^{121}I	Hei		78Sz09
$^{121}\text{Ba}(\epsilon\text{p})^{120}\text{Xe}$	4200	300	4140	140	-0.2	R					78Bo20
$^{121}\text{Pr}(\text{p})^{120}\text{Ce}$	837	50	890	10	1.1	F					90Bo39 *
	889.6	10.				3			Arp		05Ro19 *
$^{121}\text{Ag}(\beta^-)^{121}\text{Cd}$	6400	120	6671	12	2.3	U			Stu		82Al29
$^{121}\text{Cd}(\beta^-)^{121}\text{In}$	4780	80	4761	27	-0.2	U			Stu		82Al29 *
$^{121}\text{In}(\beta^-)^{121}\text{Sn}$	3426	200	3362	27	-0.3	U					60Yu01 *
	3406	50			-0.9	R			Stu		78Al18 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{121}\text{Sn}(\beta^-)^{121}\text{Sb}$	383	5	402.5	2.5	3.9	B					49Du15
	383.4	3.			6.4	B					68Sn01 *
$^{121}\text{Te}(\beta^+)^{121}\text{Sb}$	1080	30	1056	26	-0.8	1	74	74 ^{121}Te			75Me23 *
$^{121}\text{I}(\beta^+)^{121}\text{Te}$	2364	50	2297	26	-1.3	1	27	26 ^{121}Te			53Fi.A *
	2384	100			-0.9	U					65Bu03 *
$^{121}\text{Xe}(\beta^+)^{121}\text{I}$	3790	100	3765	11	-0.3	U					60Mo.A
	4160	140			-2.8	U					70Be.A
$^{121}\text{Cs}(\beta^+)^{121}\text{Xe}$	5650	490	5379	14	-0.6	U					75We23
	5400	20			-1.1	-					81So06
	5210	220			0.8	U					83Pa.A *
	5300	100			0.8	U			IRS		93Al03 *
	5400	40			-0.5	-			JAE		96Os04 *
ave.	5400	18			-1.2	1	61	46 ^{121}Cs			average
$^{121}\text{Cs}^x(\text{IT})^{121}\text{Cs}$	46	8				2					GAu
$^{121}\text{Ba}(\beta^+)^{121}\text{Cs}$	6340	160	6360	140	0.1	2			JAE		96Os04
* $^{121}\text{Ag}-^{133}\text{Cs}_{910}$	$D_M=6175.1(5.0) \mu\text{u}$ for ground state or $^{121}\text{Ag}^m$ at 20#20 keV; $M-A=-74392.5(4.7) \text{ keV}$										Nub211 **
* $^{121}\text{Ag}-^{133}\text{Cs}_{910}$	$D_M=6180(12) \mu\text{u}$ for ground state or $^{121}\text{Ag}^m$ at 20#20 keV; $M-A=-74388(11) \text{ keV}$										Nub211 **
* $^{121}\text{Cd}-^{130}\text{Xe}_{931}$	$D_M=3027.4(2.9) \mu\text{u}$ for $^{121}\text{Cd}^m$ at 214.86(0.15) keV; $M-A=-80858.7(2.7) \text{ keV}$										Nub211 **
* $^{121}\text{Cs}-^{133}\text{Cs}_{910}$	$D_M=3284(16) \mu\text{u}$ for mixture gs+m at 68.5 keV										Nub211 **
* $^{121}\text{Cs}-^{133}\text{Cs}_{910}$	$D_M=3285(13) \mu\text{u}$ for mixture gs+m at 68.5 keV; $M-A=-77084(12) \text{ keV}$										Nub211 **
* $^{121}\text{Cs}-\text{u}$	$M-A=-77113(29) \text{ keV}$ for mixture gs+m at 68.5 keV										Nub211 **
* $^{121}\text{Pd}-^{129}\text{Xe}_{938}$	Taken as low-spin isomer (see also ^{102}Y and ^{114}Tc doublets in same paper)										GAu **
* $^{121}\text{Pr}(\text{p})^{120}\text{Ce}$	F : misassigned according to reference										05Ro19 **
* $^{121}\text{Pr}(\text{p})^{120}\text{Ce}$	$E_p=882(10)$; in publication $Q_p=900(10) \text{ keV}$										WgM10c**
* $^{121}\text{Pr}(\text{p})^{120}\text{Ce}$	Trends from Mass Surface TMS suggest ^{121}Pr 100 keV less bound										GAu **
* $^{121}\text{Cd}(\beta^-)^{121}\text{In}$	$Q_{\beta^-}=4890(150)$; and $4960(80)$ from $^{121}\text{Cd}^m$ at 214.86 keV										Nub211 **
* $^{121}\text{In}(\beta^-)^{121}\text{Sn}$	$E_{\beta^-}=3700(200)$ from $^{121}\text{In}^m$ at 313.68(0.07) to ground state and $1/2^+$ level at 60.34 keV										Ens106 **
* $^{121}\text{In}(\beta^-)^{121}\text{Sn}$	$E_{\beta^-}=2480(50)$ to $7/2^+$ level at 925.59 keV										Ens106 **
* $^{121}\text{Sn}(\beta^-)^{121}\text{Sb}$	$E_{\beta^-}=383(3)$; and $354(5)$ from $^{121}\text{Sn}^m$ at 6.31 to $7/2^+$ level at 37.1298 keV										Ens106 **
* $^{121}\text{Te}(\beta^+)^{121}\text{Sb}$	$p^+=0.024(0.011)$ gives $Q_{\beta^+}=315(30)$, recalculated Q_{β^+}										AHW **
*	from $^{121}\text{Te}^m$ at 293.974 to $7/2^+$ level at 37.1298 keV										Ens106 **
* $^{121}\text{I}(\beta^+)^{121}\text{Te}$	$E_{\beta^+}=1130(50) \text{ 1150(100)}$ respectively, to $3/2^+$ level at 212.191 keV										Ens106 **
* $^{121}\text{Cs}(\beta^+)^{121}\text{Xe}$	$E_{\beta^+}=3730(220)$ to $7/2^+$ level at 459.59 keV										Ens106 **
* $^{121}\text{Cs}(\beta^+)^{121}\text{Xe}$	$Q_{\beta^+}=5370(100) \text{ 5470(40)}$ respectively, from $^{121}\text{Cs}^m$ at 68.5 keV										Ens106 **
$^{122}\text{Pd}-\text{u}$	-69308	397	-69368	21	-0.1	U			GT1	1.5	04Ma.A
$^{122}\text{Ag}-\text{u}$	-76280	110	-76340	40	-0.2	o			GT2	2.5	08Kn.A *
	-76340	130			0.0	U			GT2	2.5	08Su19 *
$^{122}\text{Ag}-^{133}\text{Cs}_{917}$	10365	41				2			MA8	1.0	10Br02 *
$^{122}\text{Cd}-^{133}\text{Cs}_{917}$	155.1	4.7	159.6	2.5	1.0	1	28	28 ^{122}Cd	MA8	1.0	10Br02
$\text{C}_8 \text{H}_{12} \text{N}-^{122}\text{Sn}$	193541	8	193528.9	2.6	-0.6	U			M16	2.5	63Da10
	193558	8			-2.4	U			R13	1.5	83De51
$\text{C}_8 \text{H}_{12} \text{N}-^{122}\text{Te}$	193925	9	193929.7	1.5	0.2	U			M16	2.5	63Da10
	193926	8			0.3	U			R13	1.5	83De51
$^{122}\text{Xe}-\text{u}$	-91637	30	-91632	12	0.2	R			GS2	1.0	05Li24
$^{122}\text{Xe}-^{133}\text{Cs}_{917}$	-4931	13	-4932	12	-0.1	2			MA6	1.0	04Di18
$^{122}\text{Cs}-^{133}\text{Cs}_{917}$	2812	48	2810	40	-0.1	o			MA1	1.0	90St25 *
	2805	48			0.1	1	57	57 ^{122}Cs	MA1	1.0	99Am05 *
$^{122}\text{Cs}-\text{u}$	-83887	55	-83890	40	-0.1	1	43	43 ^{122}Cs	GS2	1.0	05Li24 *
$^{122}\text{Cs}^n-^{133}\text{Cs}_{917}$	2952	16	2959	10	0.4	o			MA1	1.0	90St25
	2961	12			-0.2	2			MA1	1.0	99Am05
	2955	17			0.2	2			MA4	1.0	99Am05
$^{122}\text{Ba}-\text{u}$	-80096	30				2			GS2	1.0	05Li24
$^{122}\text{Cd}-^{130}\text{Xe}_{938}$	3969.0	2.9	3967.3	2.5	-0.6	1	72	72 ^{122}Cd	JY1	1.0	12Ha25
$^{122}\text{Pd}-^{129}\text{Xe}_{946}$	20709	21				2			JY1	1.0	11Ha48

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{122}\text{Sn } ^{35}\text{Cl}-^{120}\text{Sn } ^{37}\text{Cl}$	4196	2	4193.1	2.5	-0.4	U			H15	4.0	62Ba23
$^{119}\text{Cs}^x-^{122}\text{Cs}_{244}^x$	-1600	80	-1511	15	0.4	U			P32	2.5	86Au02
$^{120}\text{Cs}^x-^{122}\text{Cs}_{492}^x$	-724	27	-694	20	0.4	U			P32	2.5	86Au02
$^{120}\text{Cs}^x-^{122}\text{Cs}_{328}^x$	350	50	321	16	-0.2	U			P22	2.5	82Au01
$^{119}\text{Cs}_{672}^x$	360	17			-0.9	U			P32	2.5	86Au02
$^{121}\text{Cs}^x-^{122}\text{Cs}_{496}^x$	-1100	40	-1066	24	0.3	U			P32	2.5	86Au02
$^{120}\text{Cs}_{504}^x$	-1169	15			2.7	U			P32	2.5	86Au02
$^{122}\text{Sn(p,t)}^{120}\text{Sn}$	-6504	15	-6503.1	2.3	0.1	U			Roc		70Fi08
$^{122}\text{Te(p,t)}^{120}\text{Te}$	-8560	24	-8612.0	1.1	-2.2	U			Win		74De31 *
$^{122}\text{Te(p,t)}^{120}\text{Te}-^{132}\text{Ba}()^{130}\text{Ba}$	227.0	0.2	227.01	0.20	0.0	1	100	98 ^{120}Te			08Su14
$^{122}\text{Te(p,t)}^{120}\text{Te}-^{144}\text{Sm}()^{142}\text{Sm}$	2032.6	0.4	2032.6	0.4	-0.1	1	100	97 ^{142}Sm			09Bu.A
$^{122}\text{Sn(d,}^3\text{He)}^{121}\text{In}$	-5910	50	-5901	27	0.2	2			Sac		69Co03
	-5861	43			-0.9	2			MSU		71We01
$^{122}\text{Sn(p,d)}^{121}\text{Sn}$	-6587	15	-6590.1	2.3	-0.2	U			Har		70Ca01
$^{122}\text{Sn(d,t)}^{121}\text{Sn}$	-2558.8	3.0	-2557.4	2.3	0.5	1	61	58 ^{122}Sn	SPa		75Be09
$^{121}\text{Sb(n,}\gamma)^{122}\text{Sb}$	6806.4	0.3	6806.37	0.13	-0.1	-					72Sh.A Z
	6806.36	0.15			0.0	-			Bdn		06Fi.A
ave.	6806.37	0.13			-0.0	1	100	96 ^{121}Sb			average
$^{122}\text{In}(\beta^-)^{122}\text{Sn}$	6440	200	6370	50	-0.4	U					71Ta07 *
	6510	230			-0.6	U			Stu		78Al18
$^{122}\text{Sn(t,}^3\text{He)}^{122}\text{In}$	-6350	50				2			LAl		78Aj01
$^{122}\text{In}^n(\beta^-)^{122}\text{Sn}$	6736	200	6660	130	-0.4	2					71Ta07 *
	6590	180			0.4	2			Stu		78Al18
$^{122}\text{Sb}(\beta^+)^{122}\text{Sn}$	1587	25	1606	3	0.7	U					58Pe17
$^{122}\text{Sb}(\beta^-)^{122}\text{Te}$	1970	5	1979.1	2.1	1.8	-					55Fa33
	1980	3			-0.3	-					68Hs02
ave.	1977.4	2.6			0.7	1	68	68 ^{122}Sb			average
$^{122}\text{I}(\beta^+)^{122}\text{Te}$	4140	40	4234	5	2.3	U					54Ma75
	4140	40			2.3	U					60Mo.A
	4234	5				2					77Re.A
$^{122}\text{Cs}(\beta^+)^{122}\text{Xe}$	7150	700	7210	40	0.1	U					75We23 *
	7050	180			0.9	U					83Pa.A *
	7000	150			1.4	U			IRS		93Al03
	7080	50			2.6	B			JAE		96Os04
$^{122}\text{Cs}^n(\beta^+)^{122}\text{Xe}$	6950	250	7350	14	1.6	U					83Pa.A *
	7300	150			0.3	U			IRS		93Al03
$^{122}\text{Cs}^x(\text{IT})^{122}\text{Cs}$	14	7				2					82Au01 *
* $^{122}\text{Ag}-\text{u}$	$M-A=-71014(94)$ keV for mixture gs+m at 80#(50#) keV										Nub211 **
* $^{122}\text{Ag}-\text{u}$	$M-A=-71065(120)$ keV for mixture gs+m at 80#(50#) keV										Nub211 **
* $^{122}\text{Ag}-^{133}\text{Cs}_{917}$	$D_M=10408(18)$ μu for mixture gs+m at 80#(50#) keV; $M-A=-71066(17)$ keV										Nub211 **
* $^{122}\text{Cs}-^{133}\text{Cs}_{917}$	$D_M=2887(12)$ μu for mixture gs+n at 140(30) keV										Nub211 **
* $^{122}\text{Cs}-^{133}\text{Cs}_{917}$	$D_M=2880(12)$ μu for mixture gs+n at 140(30) keV; $M-A=-78078(12)$ keV										Nub211 **
* $^{122}\text{Cs}-\text{u}$	$M-A=-78070(28)$ keV for mixture gs+m at 140(30) keV										Nub211 **
* $^{122}\text{Te(p,t)}^{120}\text{Te}$	Original error 12; added systematic error 21 keV										GAu **
* $^{122}\text{In}(\beta^-)^{122}\text{Sn}$	$E_{\beta^-}=5300(200)$ to 2^+ level at 1140.51 keV										Ens074 **
* $^{122}\text{In}^n(\beta^-)^{122}\text{Sn}$	$E_{\beta^-}=4400(200)$ to 4^+ level at 2331.09 keV										Ens074 **
* $^{122}\text{Cs}(\beta^+)^{122}\text{Xe}$	$E_{\beta^+}=5800(700)$ 5690(180) respectively, to 2^+ level at 331.28 keV										Ens074 **
* $^{122}\text{Cs}^n(\beta^+)^{122}\text{Xe}$	$E_{\beta^+}=3710(250)$ to 8^+ level at 2217.69 keV										Ens074 **
* $^{122}\text{Cs}^x(\text{IT})^{122}\text{Cs}$	Original was 45(33); revised using $^{122}\text{Cs}^n=140(30)$ keV										Nub211 **
$^{123}\text{Pd}-\text{u}$	-64423	290	-64870	850	-1.0	o			GT1	1.5	04Ma.A
	-64874	339				2			GT3	2.5	16Kn03
$^{123}\text{Ag}-\text{u}$	-74729	215	-74680	40	0.1	o			GT1	1.5	04Ma.A
	-74479	130			-0.6	U			GT2	2.5	08Su19 *
$^{123}\text{Ag}-^{133}\text{Cs}_{925}$	12700	120	12770	40	0.6	U			MA8	1.0	10Br02
	12772	35				2			MA8	1.0	10Br02 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{123}\text{Cd}-^{133}\text{Cs}_{925}$	4491	52	4349.4	2.9	-2.7	U			MA8	1.0	10Br02 *
$\text{C}_8 \text{H}_{13} \text{N}-^{123}\text{Sb}$	200580.0	3.3	200584.1	1.5	0.5	U			M16	2.5	63Da10
	200615	8			-2.6	U			R13	1.5	83De51
$\text{C}_8 \text{H}_{13} \text{N}-^{123}\text{Te}$	200538	16	200528.4	1.5	-0.2	U			M16	2.5	63Da10
	200515	8			1.1	U			R13	1.5	83De51
$^{123}\text{Te}-\text{u}$	-95615	83	-95729.0	1.5	-1.4	U			GS2	1.0	05Li24 *
$^{123}\text{I}-\text{u}$	-94444	30	-94410	4	1.1	U			GS2	1.0	05Li24
$^{123}\text{Xe}-^{133}\text{Cs}_{925}$	-4048	13	-4061	10	-1.0	1	62	62 ^{123}Xe	MA6	1.0	04Di18
$^{123}\text{Cs}-\text{u}$	-87007	57	-87004	13	0.1	U			GS2	1.0	05Li24 *
$^{123}\text{Cs}-^{133}\text{Cs}_{925}$	447	17	453	13	0.4	o			MA1	1.0	90St25
	453	13				2			MA1	1.0	99Am05
$^{123}\text{Ba}-^{133}\text{Cs}_{925}$	6238	13				2			MA5	1.0	00Be42
$^{123}\text{Ba}-\text{u}$	-81327	30	-81219	13	3.6	C			GS2	1.0	05Li24
$^{123}\text{Cd}-^{130}\text{Xe}_{946}$	8172.5	2.9	8172.6	2.9	0.0	1	100	100 ^{123}Cd	JY1	1.0	12Ha25
$^{123}\text{Cd}^m-^{130}\text{Xe}_{946}$	8326.5	3.3				2			JY1	1.0	13Ka08
$^{123}\text{Sb } ^{35}\text{Cl}-^{121}\text{Sb } ^{37}\text{Cl}$	3343	2	3354.1	2.3	1.4	U			H14	4.0	62Ba24
$^{123}\text{Te}-^{123}\text{Sb}$	55.729	0.071	55.73	0.07	0.0	1	100	98 ^{123}Sb	SH2	1.0	16Fi07
$^{119}\text{Cs}^x-^{123}\text{Cs}_{193}^{x_{118}\text{Cs}_{807}}$	-1480	60	-1447	13	0.2	U			P32	2.5	86Au02
$^{123}\text{Te}(\text{n},\alpha)^{120}\text{Sn}$	7564	30	7573.2	1.4	0.3	U			ILL		75Em04
$^{121}\text{Sb}(\text{t,p})^{123}\text{Sb}$	7295	20	7284.6	2.1	-0.5	U			Ald		67Hi01
$^{122}\text{Sn}(\text{n},\gamma)^{123}\text{Sn}$	5948	3	5946.0	1.2	-0.7	-					75Bh01
	5945.8	1.5			0.2	-					77Ca09
$^{122}\text{Sn}(\text{d,p})^{123}\text{Sn}$	3726	12	3721.5	1.2	-0.4	U			Tal		64Ne10
	3716	11			0.5	U					72Ca33
	3721.8	2.6			-0.1	-			SPa		75Be09
$^{122}\text{Sn}(\text{n},\gamma)^{123}\text{Sn}$	ave. 5946.3	1.2	5946.0	1.2	-0.2	1	94	52 ^{123}Sn			average
$^{123}\text{Sb}(\gamma,\text{n})^{122}\text{Sb}$	-8980	50	-8960.0	2.1	0.4	U			Phi		60Ge01
	-8966	4			1.5	1	28	28 ^{122}Sb	McM		79Ba06
$^{122}\text{Te}(\text{n},\gamma)^{123}\text{Te}$	6937	5	6929.01	0.08	-1.6	U					68Ch.A
	6929.1	0.5			-0.2	U					91Ho08
	6928.97	0.09			0.5	-					00Bo24
	6929.16	0.17			-0.9	-			Bdn		06Fi.A
$^{122}\text{Te}(\text{d,p})^{123}\text{Te}$	4706	6	4704.45	0.08	-0.3	U			MIT		75Li22
$^{122}\text{Te}(\text{n},\gamma)^{123}\text{Te}$	ave. 6929.01	0.08	6929.01	0.08	0.0	1	100	99 ^{122}Te			average
$^{122}\text{Te}(^3\text{He,d})^{123}\text{I}$	-574.2	3.5	-575	3	-0.3	1	97	96 ^{123}I	Hei		78Sz04
$^{123}\text{Cd}(\beta^-)^{123}\text{In}$	6033	35	6015	20	-0.5	1	32	32 ^{123}In	Stu		87Sp09 *
$^{123}\text{In}(\beta^-)^{123}\text{Sn}$	4400	30	4386	20	-0.5	1	44	43 ^{123}In	Stu		87Sp09 *
$^{123}\text{Sn}(\beta^-)^{123}\text{Sb}$	1395	10	1408.2	2.4	1.3	-					49Du15 *
	1420	10			-1.2	-					50Ke11
	1399	20			0.5	U					66Au04 *
	ave. 1408	7			0.1	1	12	10 ^{123}Sn			average
$^{123}\text{I}(\beta^+)^{123}\text{Te}$	1260	7	1228	3	-4.5	C					86Ag.A
$^{123}\text{Xe}(\beta^+)^{123}\text{I}$	2720	100	2694	10	-0.3	U					54Ma75
	2676	15			1.2	1	42	38 ^{123}Xe			60Mo.A *
$^{123}\text{Cs}(\beta^+)^{123}\text{Xe}$	4110	310	4205	15	0.3	U					75We23 *
	4000	140			1.5	U					81So06 *
	4050	180			0.9	U					83Pa.A *
	4200	100			0.0	U			IRS		93Al03
	4110	30			3.2	B			JAE		96Os04
$^{123}\text{Cs}^x(\text{IT})^{123}\text{Cs}$	7	4				3					82Au01 *
$^{123}\text{Ba}(\beta^+)^{123}\text{Cs}$	5330	100	5389	17	0.6	U			JAE		96Os04
* $^{123}\text{Ag}-\text{u}$	Isomer should be expected, but $^{123}\text{Ag}^m$ at 20#(20#) keV, correction negligible										GAu **
* $^{123}\text{Ag}-^{133}\text{Cs}_{925}$	$D_M=12805(30) \mu\text{u}$ for mixture gs+m at 59.5(0.5) keV; $M-A=-69538(28)$										Nub211 **
* $^{123}\text{Cd}-^{133}\text{Cs}_{925}$	$D_M=4568(26) \mu\text{u}$ for mixture gs+m at 143(4) keV; $M-A=-77210(25)$										Nub211 **
* $^{123}\text{Te}-\text{u}$	$M-A=-88941(30) \text{keV}$ for mixture gs+m at 247.47 keV										Nub211 **
* $^{123}\text{Cs}-\text{u}$	$M-A=-80968(28) \text{keV}$ for mixture gs+m at 156.27 keV										Nub211 **
* $^{123}\text{Cd}(\beta^-)^{123}\text{In}$	$Q=3590(51) 3464(41) 3547(36)$ from ground state to 2393 2529 2541 levels										89Hu03 **
* $^{123}\text{In}(\beta^-)^{123}\text{Sn}$	$Q_{\beta^-}=4410(31);$ and 4645(72) from $^{123}\text{In}^m$ at 327.21 keV										Nub211 **
* $^{123}\text{Sn}(\beta^-)^{123}\text{Sb}$	$E_{\beta^-}=1260(10)$ from $^{123}\text{Sn}^m$ at 24.6 to 5/2 ⁺ level at 160.33 keV										Ens04a **
* $^{123}\text{Sn}(\beta^-)^{123}\text{Sb}$	$E_{\beta^-}=310(20)$ to 9/2 ⁺ level at 1088.64 keV										Ens04a **
* $^{123}\text{Xe}(\beta^+)^{123}\text{I}$	$E_{\beta^+}=1505(15)$ to 1/2 ⁺ level at 148.92 keV										Ens04a **
* $^{123}\text{Cs}(\beta^+)^{123}\text{Xe}$	$E_{\beta^+}=2990(310)$ to 3/2 ⁺ level at 97.30 keV										Ens04a **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{123}\text{Cs}(\beta^+)^{123}\text{Xe}$	$E_{\beta^+}=2370(140)$ to $(1/2,3/2)^+$ level at 596.65 keV, and other E_{β^+}										Ens04a **
$^{123}\text{Cs}(\beta^+)^{123}\text{Xe}$	$E_{\beta^+}=2930(180)$ to $3/2^+$ level at 97.30 keV										Ens04a **
$^{123}\text{Cs}^x(\text{IT})^{123}\text{Cs}$	Based on $^{123}\text{Cs}^m(\text{IT})=156.27$ and isomeric ratio $R<0.1$										Nub211 **
$^{124}\text{Pd}-\text{u}$	-64617	399	-62700#	320#	1.9	D			GT3	2.5	16Kn03 *
$^{124}\text{Ag}-^{133}\text{Cs}_{932}$	17018	270				2			MA8	1.0	10Br02 *
$^{124}\text{Cd}-^{133}\text{Cs}_{932}$	5781	10	5778.5	2.8	-0.2	-			MA8	1.0	10Br02
	5786.1	5.7			-1.3	-			MA8	1.0	20Ma09
ave.	5785	5			-1.3	1	32	32 ^{124}Cd			average
$\text{C}_7\ ^{13}\text{C}\ \text{H}_{13}\ \text{N}-^{124}\text{Sn}$	202886	8	202874.6	1.4	-0.6	U			M16	2.5	63Da10
	202891	8			-1.4	U			R13	1.5	83De51
$^{124}\text{Sn}-^{13}\text{C}\ ^{37}\text{Cl}_3$	4210.47	0.71	4217.1	1.4	2.3	U			H39	4.0	84Ha20
$^{124}\text{Sn}-^{133}\text{Cs}_{932}$	-6598	21	-6601.6	1.4	-0.2	U			MA8	1.0	05Si34
$^{124}\text{Te}-^{13}\text{C}\ ^{37}\text{Cl}_3$	1754.63	1.26	1755.8	1.5	0.2	U			H39	4.0	84Ha20
$^{124}\text{Te}-^{54}\text{Fe}\ ^{35}\text{Cl}_2$	25501.65	2.56	25504.8	1.5	0.3	U			H39	4.0	84Ha20
$\text{C}_7\ ^{13}\text{C}\ \text{H}_{13}\ \text{N}-^{124}\text{Te}$	205336	13	205335.9	1.5	-0.0	U			M16	2.5	63Da10
	205325	8			0.9	U			R13	1.5	83De51
$^{124}\text{I}-\text{u}$	-93786	30	-93789.7	2.5	-0.1	U			GS2	1.0	05Li24
$^{124}\text{Xe}-^{13}\text{C}\ ^{37}\text{Cl}_3$	4831.15	1.58	4822.6	1.5	-1.3	U			H39	4.0	84Ha20
$^{124}\text{Xe}-^{54}\text{Fe}\ ^{35}\text{Cl}_2$	28575.78	0.99	28571.6	1.5	-1.1	U			H39	4.0	84Ha20
$^{124}\text{Xe}-^{133}\text{Cs}_{932}$	-5986	13	-5996.1	1.5	-0.8	U			MA6	1.0	04Di18
$^{124}\text{Cs}-^{133}\text{Cs}_{932}$	361	16	366	10	0.3	o			MA1	1.0	90St25
	370	13			-0.3	2			MA1	1.0	99Am05
	361	15			0.3	2			MA8	1.0	05Gu37
$^{124}\text{Cs}-\text{u}$	-87696	30	-87753	10	-1.9	U			GS2	1.0	05Li24
	-87693	30			-2.0	U			GS2	1.0	05Li24 *
	-87708	42			-1.1	U			GR1	1.0	19An10 *
$^{124}\text{Ba}-^{133}\text{Cs}_{932}$	3212	15	3212	13	0.0	2			MA1	1.0	99Am05
$^{124}\text{Ba}-\text{u}$	-84905	30	-84906	13	-0.0	R			GS2	1.0	05Li24
$^{124}\text{La}-\text{u}$	-75464	71	-75430	60	0.5	2			GS2	1.0	05Li24 *
$^{124}\text{Cd}-^{130}\text{Xe}_{954}$	9708.9	3.4	9711.9	2.8	0.9	1	68	68 ^{124}Cd	JY1	1.0	12Ha25
$^{124}\text{Sn}-^{129}\text{Xe}_{961}$	-3214.3	2.1	-3214.8	1.4	-0.2	1	45	45 ^{124}Sn	JY1	1.0	11Ha48
$^{124}\text{Sn}-^{120}\text{Sn}_{1,033}$	6305.1	2.1	6304.4	1.4	-0.3	1	46	34 ^{124}Sn	JY1	1.0	11Ha48
$^{124}\text{Sn}\ ^{35}\text{Cl}-^{122}\text{Sn}\ ^{37}\text{Cl}$	4784	2	4784.2	2.6	0.0	U			H15	4.0	62Ba23
$^{124}\text{Te}\ ^{35}\text{Cl}-^{122}\text{Te}\ ^{37}\text{Cl}$	2728	2	2723.75	0.15	-0.5	U			H16	4.0	63Ba47
$^{124}\text{Sn}-^{124}\text{Te}$	2458.51	0.89	2461.3	0.4	0.8	U			H39	4.0	84Ha20
	2461.26	0.42			0.0	1	99	83 ^{124}Te	SH1	1.0	12Ne10
$^{124}\text{Xe}-^{124}\text{Te}$	3076.00	1.78	3066.83	0.14	-1.3	U			H39	4.0	84Ha20
	3066.83	0.14			0.0	1	100	99 ^{124}Xe	SH1	1.0	12Ne10
$^{124}\text{Sn}-^{122}\text{Sn}$	1838	22	1834.1	2.6	-0.1	U			M16	2.5	63Da10
$^{120}\text{Cs}^x-^{124}\text{Cs}^x_{194}\ ^{119}\text{Cs}^x_{807}$	310	30	306	12	-0.1	U			P22	2.5	82Au01
$^{121}\text{Cs}^x-^{124}\text{Cs}^x_{244}\ ^{120}\text{Cs}^x_{756}$	-1360	30	-1263	19	1.3	U			P22	2.5	82Au01
$^{123}\text{Cs}^x-^{124}\text{Cs}^x_{744}\ ^{120}\text{Cs}^x_{256}$	-1390	30	-1329	21	0.8	U			P22	2.5	82Au01
$^{124}\text{Sn}(\text{d},^6\text{Li})^{120}\text{Cd}$	-5216	24	-5225	4	-0.4	U					79Ja21
$^{124}\text{Sn}(^3\text{He},^7\text{Be})^{120}\text{Cd}$	-5098	30	-5112	4	-0.5	U			MSU		76St11
$^{124}\text{Sn}(^{18}\text{O},^{20}\text{Ne})^{122}\text{Cd}$	-1266	39	-1360.0	2.6	-2.4	U					97Gu32 *
$^{122}\text{Sn}(\text{t,p})^{124}\text{Sn}$	5931	15	5952.4	2.4	1.4	U			Roc		70FI05
$^{124}\text{Sn}(\text{p,t})^{122}\text{Sn}$	-5956	10	-5952.4	2.4	0.4	U					64Al29
$^{124}\text{Sn}(\text{d},^3\text{He})^{123}\text{In}$	-6610	50	-6598	20	0.2	-			Sac		69Co03
	-6572	66			-0.4	-			MSU		71We01
ave.	-6600	40			-0.0	1	25	25 ^{123}In			average
$^{124}\text{Sn}(\text{p,d})^{123}\text{Sn}$	-6279	15	-6263.5	2.4	1.0	U			Har		70Ca01
$^{124}\text{Sn}(\text{d,t})^{123}\text{Sn}$	-2260	35	-2230.9	2.4	0.8	U			Pit		64Co11
	-2233.4	3.7			0.7	1	42	37 ^{123}Sn	SPa		75Be09
$^{123}\text{Sb}(\text{n},\gamma)^{124}\text{Sb}$	6467.55	0.10	6467.50	0.06	-0.5	2					73Sh.A Z
	6467.40	0.10			1.0	2					81Su.A Z
	6467.58	0.14			-0.6	2			Bdn		06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{123}\text{Te}(n,\gamma)^{124}\text{Te}$	9425	2	9424.48	0.09	-0.3	U					69Bu05
	9423.7	1.5			0.5	U					70Or.A
	9424.05	0.30			1.4	-			Ltn		95Ge06
	9423.89	0.20			3.0	C			Bdn		06Fi.A
	9424.53	0.10			-0.5	-					06Vo09
	ave. 9424.48	0.09			0.0	1	100	97 ^{123}Te			average
$^{124}\text{Cd}(\beta^-)^{124}\text{In}$	4166	39	4170	30	0.1	1	61	61 ^{124}In	Stu		87Sp09
$^{124}\text{In}(\beta^-)^{124}\text{Sn}$	7180	50	7360	30	3.7	B			Stu		78Al18
	7360	49			0.1	1	39	39 ^{124}In	Stu		87Sp09
$^{124}\text{Sn}(t,^3\text{He})^{124}\text{In}$	-7590	50	-7350	30	4.9	B			LAl		78Aj01
$^{124}\text{In}^m(\beta^-)^{124}\text{Sn}$	7370	210	7340	50	-0.1	o			Stu		78Al18
	7341	51				2			Stu		87Sp09
$^{124}\text{Sb}(\beta^-)^{124}\text{Te}$	2907.7	5.	2905.07	0.13	-0.5	U					65Hs02
	2903.7	4.			0.3	U					66Ca10
	2904.7	2.			0.2	U					69Na05
$^{124}\text{I}(\beta^+)^{124}\text{Te}$	3157	4	3159.6	1.9	0.6	2					71Bo01
	3160.3	2.1			-0.3	2					92Wo03
$^{124}\text{Cs}(\beta^+)^{124}\text{Xe}$	5920	460	5926	9	0.0	U					75We23
	5900	90			0.3	U			IRS		93Al03
	5910	30			0.5	U			JAE		96Os04
$^{124}\text{Cs}^x(\text{IT})^{124}\text{Cs}$	30	20				3					AHW
$^{124}\text{La}(\beta^+)^{124}\text{Ba}$	8930	110	8830	60	-0.9	R			JAE		98Ko66
* $^{124}\text{Pd}-u$	Trends from Mass Surface TMS suggest ^{124}Pd 1790 keV less bound										GAu
* $^{124}\text{Ag}-^{133}\text{Cs}_{932}$	$D_M=17050(270)$ μu for mixture gs+m at 50#(50#); $M-A=-66200(250)$ keV										Nub211
* $^{124}\text{Cs}-u$	$M-A=-81223(28)$ keV for $^{124}\text{Cs}^m$ at 462.63 keV										Nub211
* $^{124}\text{La}-u$	$M-A=-70244(32)$ keV for mixture gs+m at 100#100 keV										Nub211
* $^{124}\text{Sn}(^{18}\text{O},^{20}\text{Ne})^{122}\text{Cd}$	Original $Q=-1250(39)$ calibrated with $^{120}\text{Cd}=-83973(19)$ keV										GAu
* $^{124}\text{Sb}(\beta^-)^{124}\text{Te}$	$E_{\beta^-}=2305(5)$ 2301(4) 2302(2) respectively, to 2^+ level at 602.7271 keV										Ens087
* $^{124}\text{I}(\beta^+)^{124}\text{Te}$	Original error increased, see $^{84}\text{Rb}(\beta^+)$										AHW
* $^{124}\text{Cs}^x(\text{IT})^{124}\text{Cs}$	Based on $^{124}\text{Cs}^m(\text{IT})=462.63$ keV										Nub211
* $^{124}\text{Cs}^x(\text{IT})^{124}\text{Cs}$	Isomeric ratio assumed <0.1 as for ^{118}Cs , ^{120}Cs , ^{122}Cs										AHW
$^{125}\text{Ag}-u$	-68954	429	-69270	470	-0.5	o			GT1	1.5	04Ma.A
	-69265	186				2			GT3	2.5	16Kn03
$^{125}\text{Cd}-u$	-78770	120	-78742	3	0.1	o			GT2	2.5	08Kn.A
	-78780	140			0.1	U			GT2	2.5	08Su19
$\text{C}_7\text{H}_6\text{ }^{35}\text{Cl}-^{125}\text{Te}$	111363	6	111371.7	1.5	0.6	U			M16	2.5	63Da10
	111368	8			0.3	U			R13	1.5	83De51
$^{125}\text{I}-u$	-95374	30	-95369.4	1.5	0.2	U			GS2	1.0	05Li24
$^{125}\text{Cs}-u$	-90280	30	-90274	8	0.2	U			GS2	1.0	05Li24
	-90221	45			-1.2	U			GR1	1.0	19An10
$^{125}\text{Ba}-u$	-85569	30	-85528	12	1.4	R			GS2	1.0	05Li24
$^{125}\text{La}-u$	-79191	30	-79184	28	0.2	1	87	87 ^{125}La	GS2	1.0	05Li24
$^{125}\text{Cd}-^{133}\text{Cs}_{940}$	10133	29	10133	3	-0.0	U			TT1	1.0	17La16
$^{125}\text{Cd}^m-^{133}\text{Cs}_{940}$	10337	11	10333	3	-0.4	U			TT1	1.0	17La16
$^{125}\text{In}-^{133}\text{Cs}_{940}$	2549.0	1.9				2			TT1	1.0	18Ba08
$^{125}\text{In}^m-^{133}\text{Cs}_{940}$	2927	13				2			TT1	1.0	18Ba08
$^{125}\text{Cs}-^{133}\text{Cs}_{940}$	-1392	17	-1399	8	-0.4	o			MA1	1.0	90St25
	-1382	14			-1.2	-			MA1	1.0	99Am05
	-1386	14			-0.9	-			MA4	1.0	99Am05
	ave. -1384	10			-1.5	1	70	70 ^{125}Cs			average
$^{125}\text{Ba}-^{133}\text{Cs}_{940}$	3356	13	3347	12	-0.7	-			MA5	1.0	00Be42
	ave. 3348	12			-0.1	1	98	98 ^{125}Ba			average
$^{125}\text{Cd}-^{130}\text{Xe}_{962}$	14081.6	3.1				2			JY1	1.0	12Ha25

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{125}\text{Cd}^m - ^{130}\text{Xe}_{962}$	14281.6	3.4				2			JY1	1.0	13Ka08
$^{125}\text{Te} - ^{35}\text{Cl} - ^{123}\text{Te} - ^{37}\text{Cl}$	3090	2	3110.28	0.13	2.5	U			H16	4.0	63Ba47
$^{122}\text{Cs}^x - ^{125}\text{Cs}_{244} - ^{121}\text{Cs}_{756}^x$	715	23	640	40	-1.3	U			P32	2.5	86Au02
$^{123}\text{Sb}(t,p)^{125}\text{Sb}$	6696	20	6693.0	2.1	-0.1	U			Ald		67Hi01
$^{124}\text{Sn}(n,\gamma)^{125}\text{Sn}$	5733.1	1.5	5733.50	0.20	0.3	U					77Ca09
	5733.1	0.6			0.7	U					81Ba53
	5733.5	0.2				2					11To04
$^{124}\text{Sn}(d,p)^{125}\text{Sn}$	3530	30	3508.93	0.20	-0.7	U			Pit		64Co11
	3506	12			0.2	U			Tal		64Ne10
	3515	11			-0.6	U					72Ca33
	3509.4	3.6			-0.1	U			SPa		75Be09
$^{124}\text{Te}(n,\gamma)^{125}\text{Te}$	6569.0	1.0	6568.97	0.03	-0.0	U					71Gr.A
	6568.97	0.03			0.0	1	100	^{86}Te	Prn		99Ho01
	6569.39	0.19			-2.2	U			Bdn		06Fi.A
$^{125}\text{Te}(\gamma,n)^{124}\text{Te}$	-6560	60	-6568.97	0.03	-0.1	U			Phi		60Ge01
$^{124}\text{Te}(d,p)^{125}\text{Te}$	4344	8	4344.40	0.03	0.1	U			MIT		69Gr24
$^{124}\text{Te}(^3\text{He},d)^{125}\text{I}$	115.1	3.0	107.38	0.07	-2.6	U			Hei		78Sz04
$^{124}\text{Te}(\alpha,t)^{125}\text{I}$	-14203	7	-14213.01	0.07	-1.4	U			Hei		78Sz04
$^{124}\text{Xe}(n,\gamma)^{125}\text{Xe}$	7603.3	0.4	7603.3	0.4	-0.1	1	100	^{99}Xe			82Ka.A
$^{125}\text{Cd}(\beta^-)^{125}\text{In}$	7122	62	7064	3	-0.9	U			Stu		87Sp09 *
$^{125}\text{Cd}^m(\beta^-)^{125}\text{In}$	7172	35	7251	4	2.2	B			Stu		87Sp09 *
$^{125}\text{In}(\beta^-)^{125}\text{Sn}$	5418	30	5481.3	2.2	2.1	U			Stu		87Sp09 *
$^{125}\text{Sn}(\beta^-)^{125}\text{Sb}$	2330	10	2361.4	2.2	3.1	B					50Ha58
	2370	20			-0.4	U					50Ke11
	2335	40			0.7	U					64De02 *
$^{125}\text{Sb}(\beta^-)^{125}\text{Te}$	767.7	3.	766.7	2.1	-0.3	2					64Ma30 *
	765.7	3.			0.3	2					66Ma49 *
$^{125}\text{I}(\epsilon)^{125}\text{Te}$	184	7	185.77	0.06	0.3	U					64Le05 *
	185	8			0.1	U					66Sm05 *
	177.2	2.			4.3	C					68Go.A *
	186.1	0.3			-1.1	U					86Bo46
	179.3	2.0			3.2	B					90Li14 *
	185.77	0.06				2					94Hi04
$^{125}\text{Xe}(\epsilon)^{125}\text{I}$	1735	40	1636.7	0.4	-2.5	U					69Lu09 *
$^{125}\text{Cs}(\beta^+)^{125}\text{Xe}$	3072	20	3110	8	1.9	-					54Ma54
	3082	20			1.4	-					75We23
	3100	100			0.1	U			IRS		93Al03
ave.	3077	14			2.3	1	30	^{30}Cs			average
$^{125}\text{Ba}(\beta^+)^{125}\text{Cs}$	4560	250	4421	13	-0.6	U					68Da09 *
	4380	50			0.8	U			JAE		96Os04
$^{125}\text{La}(\beta^+)^{125}\text{Ba}$	5950	70	5909	28	-0.6	1	16	^{13}La	JAE		98Ko66
* $^{125}\text{Ag}-u$	Trends from Mass Surface TMS suggest ^{125}Ag 160 keV less bound										GAu **
* $^{125}\text{Cd}-u$	$M-A=-73274(93)$ keV for mixture gs+m at 186(4) keV										Nub211 **
* $^{125}\text{Cd}-u$	$M-A=-73287(120)$ keV for mixture gs+m at 186(4) keV										Nub211 **
* $^{125}\text{Cd}(\beta^-)^{125}\text{In}$	$E_{\beta^-}=4625(62)$ to $(1/2^+, 3/2^+)$ level at 2497.43 keV										Ens112 **
* $^{125}\text{Cd}^m(\beta^-)^{125}\text{In}$	$E_{\beta^-}=5009(109)$, 4581(126), 4533(39) to 2101.40, 2640.29, 2641.26 levels										Ens112 **
* $^{125}\text{In}(\beta^-)^{125}\text{Sn}$	$Q_{\beta^-}=5443(31)$; and 5730(43) from $^{125}\text{In}^m$ at 360.12 keV										Nub211 **
* $^{125}\text{Sn}(\beta^-)^{125}\text{Sb}$	$E_{\beta^-}=2030(40)$ from $^{125}\text{Sn}^m$ at 27.50(0.14) to $5/2^+$ level at 332.06 keV										Ens112 **
* $^{125}\text{Sb}(\beta^-)^{125}\text{Te}$	$E_{\beta^-}=623(3)$ 621(3) respectively, to $1/2^-$ level at 144.775 keV										Ens112 **
* $^{125}\text{I}(\epsilon)^{125}\text{Te}$	LMK=0.254(0.003) 0.253(0.005) IBE=110(2) 150.6(0.3) respectively, all to $3/2^+$ level at 35.4925 keV. $Q(\text{LMK})$ recalculated, error mainly theory										AHW **
*	IBE=112.0(2.0)(1s)+31.8 to $3/2^+$ level at 35.4925 keV										Ens112 **
* $^{125}\text{I}(\epsilon)^{125}\text{Te}$	IBE=112.0(2.0)(1s)+31.8 to $3/2^+$ level at 35.4925 keV										Ens112 **
* $^{125}\text{Xe}(\epsilon)^{125}\text{I}$	$E_{\beta^+}=470(40)$ to $3/2^+$ at 188.416 and $1/2^+$ at 243.382 keV, ratio 1:2										Ens112 **
* $^{125}\text{Ba}(\beta^+)^{125}\text{Cs}$	$E_{\beta^+}=3450(250)$ to $(5/2^+)$ level at 84.82 level										Ens112 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{126}\text{Ag}-u$	-65926	329	-65190#	220#	0.9	D			GT3	2.5	16Kn03 *	
$\text{C}_{10}\text{H}_6-^{126}\text{Te}$	143623	9	143638.0	1.5	0.7	U			M16	2.5	63Da10	
	143640	8			-0.2	U			R13	1.5	83De51	
$^{126}\text{Xe}-u$	-95647	30	-95702.578	0.006	-1.9	U			GS2	1.0	05Li24	
$^{126}\text{Cs}-u$	-90543	49	-90554	11	-0.2	U			GR1	1.0	19An10	
$^{126}\text{Ba}-u$	-88745	30	-88750	13	-0.2	R			GS2	1.0	05Li24	
$^{126}\text{La}-u$	-80503	232	-80490	100	0.1	2			GS2	1.0	05Li24 *	
$^{126}\text{Ce}-u$	-76029	30				2			GS2	1.0	05Li24	
$^{126}\text{Xe}-^{134}\text{Xe}_{.940}$	-6772.8	2.9	-6772.026	0.004	0.3	o			MA8	1.0	05He.A	
	-6773.2	3.8			0.3	1	0	0	^{126}Xe	MA8	1.0	06He29
$^{126}\text{Cd}-^{133}\text{Cs}_{.947}$	11966.5	4.5	11967.3	2.5	0.2	-			MA8	1.0	10Br02	
	11956	15			0.8	U			MA8	1.0	10Br02	
	11962	11			0.5	U			TT1	1.0	17La16	
	11973.7	6.7			-1.0	-			MA8	1.0	20Ma09	
	ave.	11969	4			-0.4	1	44	44	^{126}Cd	average	
$^{126}\text{In}-^{133}\text{Cs}_{.947}$	6005.2	4.5				2			TT1	1.0	18Ba08	
$^{126}\text{In}^m-^{133}\text{Cs}_{.947}$	6101.7	5.5				2			TT1	1.0	18Ba08	
$^{126}\text{Cs}-^{133}\text{Cs}_{.947}$	-1027	17	-1017	11	0.6	o			MA1	1.0	90St25	
	-1011	13			-0.5	1	73	73	^{126}Cs	MA1	1.0	99Am05
$^{126}\text{Ba}-^{133}\text{Cs}_{.947}$	786	15	787	13	0.1	2			MA1	1.0	99Am05	
$^{126}\text{Cd}-^{130}\text{Xe}_{.969}$	15928.6	3.3	15929.7	2.5	0.3	1	56	56	^{126}Cd	JY1	1.0	12Ha25
$^{126}\text{Te}-^{35}\text{Cl}-^{124}\text{Te}-^{37}\text{Cl}$	3432	2	3443.92	0.11	1.5	U			H16	4.0	63Ba47	
	3441.28	1.54			1.1	U			H43	1.5	90Dy04	
$^{126}\text{Xe}-^{128}\text{Xe}_{.984}$	-776.8366	0.0025	-776.8366	0.0025	-0.0	1	100	100	^{126}Xe	Hep	1.0	20Ri04
$^{123}\text{Cs}^x-^{126}\text{Cs}_{.390}$	-1160	30	-1136	17	0.3	U			P22	2.5	82Au01	
$^{124}\text{Cs}^x-^{126}\text{Cs}_{.590}$	-340	30	-351	24	-0.1	U			P22	2.5	82Au01	
$^{124}\text{Cs}^x-^{126}\text{Cs}_{.492}$	-570	30	-520	29	0.7	U			P22	2.5	82Au01	
$^{124}\text{Cs}^x-^{126}\text{Cs}_{.328}$	390	30	413	24	0.3	U			P22	2.5	82Au01	
$^{125}\text{Cs}-^{126}\text{Cs}_{.496}$	-1130	30	-1069	14	0.8	U			P22	2.5	82Au01	
$^{124}\text{Sn}(t,p)^{126}\text{Sn}$	5445	15	5444	11	-0.0	2			Ald		69Bj01	
	5444	15			0.0	2			Roc		70Fl05	
$^{125}\text{Te}(n,\gamma)^{126}\text{Te}$	9113.7	0.4	9113.69	0.08	-0.0	U					77Ko.A	
	9113.69	0.08			0.0	1	100	86	^{126}Te		03Vo03	
$^{126}\text{Te}(\gamma,n)^{125}\text{Te}$	-8840	120	-9113.69	0.08	-2.3	U			Phi		60Ge01	
$^{125}\text{Te}(d,p)^{126}\text{Te}$	6892	6	6889.13	0.08	-0.5	U			MIT		71Gr01	
$^{126}\text{Cd}(\beta^-)^{126}\text{In}$	5486	36	5554	5	1.9	U			Stu		87Sp09	
$^{126}\text{In}(\beta^-)^{126}\text{Sn}$	8207	39	8206	11	-0.0	U			Stu		87Sp09	
$^{126}\text{In}^m(\beta^-)^{126}\text{Sn}$	8309	51	8296	12	-0.3	U			Stu		87Sp09	
$^{126}\text{Sn}(\beta^-)^{126}\text{Sb}$	378	30				3					71Or04 *	
$^{126}\text{Sb}(\beta^-)^{126}\text{Te}$	3667	150	3670	30	0.0	U					71Or04 *	
$^{126}\text{I}(\beta^+)^{126}\text{Te}$	2151	5	2154	4	0.5	1	54	52	^{126}I		59Ha27	
$^{126}\text{I}(\beta^-)^{126}\text{Xe}$	1258	5	1236	4	-4.4	B					55Ko14 *	
$^{126}\text{Cs}(\beta^+)^{126}\text{Xe}$	4670	140	4796	10	0.9	U					75We23 *	
	4810	100			-0.1	U					76Pa11 *	
	4830	40			-0.9	U			JAE		92Os07	
	4730	100			0.7	U			IRS		93Al03	
	4780	20			0.8	1	27	27	^{126}Cs	JAE	96Os04	
	7700	100	7700	90	-0.0	R			JAE		98Ko66	
	7910	400				3			JAE		98Ko66	
* $^{126}\text{Ag}-u$	Trends from Mass Surface TMS suggest ^{126}Ag 690 keV less bound										GAu	**
* $^{126}\text{La}-u$	$M-A=-74883(28)$ keV for mixture gs+m at 210(410) keV										Nub211	**
* $^{126}\text{Sn}(\beta^-)^{126}\text{Sb}$	$E_{\beta^-}=250(30)$ to 2^+ level at 127.9 keV										Ens031	**
* $^{126}\text{Sb}(\beta^-)^{126}\text{Te}$	$E_{\beta^-}=1900(150)$ from mixture ground state and $^{126}\text{Sb}^m$ at 17.7 to 6^+ level at 1776.19 keV										Ens031	**
* $^{126}\text{I}(\beta^-)^{126}\text{Xe}$	$E_{\beta^-}=865(5)$ to 2^+ level at 388.631 keV, and other E_{β^-}										Ens031	**
* $^{126}\text{Cs}(\beta^+)^{126}\text{Xe}$	$E_{\beta^+}=3260(140)$ 3400(100) respectively, to 2^+ level at 388.631 keV										Ens031	**

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_{10}H_7-^{127}I$	150297	6	150303	4	0.4	U			M16	2.5	63Da10
	150305.3	3.4			-0.3	1	21	21 ^{127}I	M16	2.5	63Da10
	150322	8			-1.6	U			R13	1.5	83De51
$^{127}Cs-u$	-92571	30	-92582	6	-0.4	U			GS2	1.0	05Li24
$^{127}Ba-u$	-88923	39	-88909	12	0.4	R			GS2	1.0	05Li24 *
$^{127}La-u$	-83640	30	-83625	28	0.5	1	87	87 ^{127}La	GS2	1.0	05Li24 *
$^{127}Ce-u$	-77273	31				2			GS2	1.0	05Li24 *
$^{127}Cd-^{133}Cs_{955}$	16494.3	8.0	16497	7	0.3	2			TT1	1.0	17La16
	16502	12			-0.4	2			IS1	1.0	20Ma09
	16798.7	8.0	16803	5	0.5	2			TT1	1.0	17La16
$^{127}Cd^m-^{133}Cs_{955}$	16805.2	5.8			-0.4	2			IS1	1.0	20Ma09
	7763	12	7759	11	-0.3	1	80	80 ^{127}In	TT1	1.0	18Ba08
	8182	16				2			TT1	1.0	18Ba08
$^{127}In^m-^{133}Cs_{955}$	9585	52	9650	40	1.3	1	58	58 $^{127}In^n$	TT1	1.0	18Ba08
$^{127}In^n-^{133}Cs_{955}$	-7237	12	-7244	10	-0.5	1	68	68 ^{127}Sn	MA8	1.0	08Dw01 *
$^{127}Sn^{34}S-^{133}Cs_{1,211}$	-2303	17	-2289	6	0.8	o			MA1	1.0	90St25
$^{127}Cs-^{133}Cs_{955}$	-2287	13			-0.2	-			MA1	1.0	99Am05
	-2293.3	7.7			0.5	-			MA8	1.0	05Gu37
	ave. -2292	7			0.4	1	82	82 ^{127}Cs	average		
$^{127}Ba-^{133}Cs_{955}$	1389	13	1385	12	-0.3	-			MA5	1.0	00Be42
	ave. 1387	12			-0.2	1	98	98 ^{127}Ba	average		
	20598	89	20475	7	-1.4	U			JY1	1.0	12Ha25 *
$^{125}Cs-^{127}Cs_{591}$	-1098	18	-1088	16	0.2	U			P32	2.5	86Au02
$^{126}Te(n,\gamma)^{127}Te$	6289	3	6287.65	0.18	-0.5	U					72Mu.A
	6287.8	0.4			-0.4	-			Bdn		06Fi.A
	6287.6	0.2			0.2	-			Prn		05Ho15
$^{126}Te(d,p)^{127}Te$	4044	8	4063.08	0.18	2.4	U			MIT		68Gr16
$^{126}Te(n,\gamma)^{127}Te$	ave. 6287.64	0.18	6287.65	0.18	0.0	1	100	98 ^{127}Te			average
$^{127}I(\gamma,n)^{126}I$	-9135	22	-9144.0	2.7	-0.4	U			Phi		60Ge01
	-9145	3			0.3	1	83	48 ^{126}I	MMn		86Ts04
	8468	63	8424	11	-0.7	U			Stu		87Sp09 *
$^{127}Cd^m(\beta^-)^{127}In$	6514	31	6590	12	2.4	o			Stu		87Sp09 *
$^{127}In(\beta^-)^{127}Sn$	6579	20			0.5	1	36	20 ^{127}In	Stu		04Ga24 *
$^{127}In^n(\beta^-)^{127}Sn$	8442	56	8360	40	-1.6	1	44	42 $^{127}In^n$	Stu		04Ga24
$^{127}Sn(\beta^-)^{127}Sb$	3201	24	3229	10	1.2	1	18	14 ^{127}Sn	Stu		77Lu06 *
$^{127}Sb(\beta^-)^{127}Te$	1581	5	1582	5	0.2	1	96	96 ^{127}Sb			67Ra13 *
$^{127}Te(\beta^-)^{127}I$	683	10	703	4	2.0	-					55Da37
	695	10			0.8	-					56Kn20
	ave. 689	7			1.9	1	25	24 ^{127}I			average
$^{127}Xe(e)^{127}I$	663.3	2.2	662.3	2.0	-0.4	-					68Sc14
$^{127}I(^3He,t)^{127}Xe$	-676	6	-680.9	2.0	-0.8	-			Pri		89Ch01
$^{127}Xe(e)^{127}I$	ave. 662.6	2.1	662.3	2.0	-0.1	1	98	91 ^{127}Xe			average
$^{127}Cs(\beta^+)^{127}Xe$	2115	25	2081	6	-1.4	-					54Ma54 *
	2076	20			0.2	-					67Sp08 *
	2089	20			-0.4	-					75We23 *
$^{127}Ba(\beta^+)^{127}Cs$	ave. 2090	12			-0.8	1	27	18 ^{127}Cs			average
	3450	100	3422	13	-0.3	U					76Be11 *
	5010	70	4922	28	-1.3	1	16	13 ^{127}La	IAE		98Ko66
$^{127}La(\beta^+)^{127}Ba$											
$^{127}Ba-u$	$M-A=-82791(28)$ keV for mixture gs+m at 80.32 keV										Nub211 **
$^{127}La-u$	$M-A=-77903(28)$ keV for mixture gs+m at 14.2 keV										Nub211 **
$^{127}Ce-u$	$M-A=-71976(29)$ keV for mixture gs+m at 7.3 keV										Nub211 **
$^{127}Sn^{34}S-^{133}Cs_{1,211}$	$D_M=-7234.3(11.6)$ μ u for mixture gs+m at 5.07(0.06) keV										Nub211 **
$^{127}Cd-^{130}Xe_{977}$	$D_M=20741(14)$ μ u for mixture of gs+m at 283 keV										WgM204**
$^{127}Cd^m(\beta^-)^{127}In$	Also $E_{\beta^-}=7910(200)$ to $^{127}In^m$ at 394(18) keV										Nub211 **
$^{127}Cd^m(\beta^-)^{127}In$	Re-assigned to $^{127}Cd^m$ by the evaluator										GAu **
$^{127}In(\beta^-)^{127}Sn$	Also $E_{\beta^-}=6976(64)$ from $^{127}In^m$ at 394(18) keV										Nub211 **
$^{127}In(\beta^-)^{127}Sn$	Also $E_{\beta^-}=6999(63)$ from $^{127}In^m$ at 394(18) keV										Nub211 **
$^{127}Sn(\beta^-)^{127}Sb$	$Q_{\beta^-}=3206(24)$ from $^{127}Sn^m$ at 5.07 keV										Nub211 **
$^{127}Sb(\beta^-)^{127}Te$	$E_{\beta^-}=1493(5)$ to $11/2^-$ level at 88.23 keV, and other E_{β^-}										Ens118 **
$^{127}Cs(\beta^+)^{127}Xe$	$E_{\beta^+}=1063(10)$, 685(25) to mixture gs+124.751 and $1/2^+$ level at 411.965 keV										Ens118 **
$^{127}Cs(\beta^+)^{127}Xe$	$E_{\beta^+}=1068(20)$, 910(30), 650(30) to ground state, $3/2^+$ at 124.751, $1/2^+$ at 411.965										Ens118 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{127}\text{Cs}(\beta^+)^{127}\text{Xe}$	$E_{\beta^+}=1040(20), 650(20)$ to mixture gs+124.751 and $1/2^+$ level at 411.965 keV										Ens118	**
$^{127}\text{Ba}(\beta^+)^{127}\text{Cs}$	$E_{\beta^+}=2230(100)$ to $3/2^+$ level at 180.92 keV, and other E_{β^+}										Ens118	**
$^{128}\text{Sn}-\text{u}$	-89512	25	-89492	19	0.8	1	58	58	^{128}Sn	GS3	1.0	12Ch19
$\text{C}_{10}\text{H}_8-^{128}\text{Te}$	158112	9	158139.0	0.8	1.2	U				M16	2.5	63Da10
	158141.2	7.			-0.1	U				C3	2.5	70Ke05
	158151	8			-1.0	U				R13	1.5	83De51
$\text{C}_{10}\text{H}_8-^{128}\text{Xe}$	159068.2	4.2	159069.502	0.006	0.1	U				M16	2.5	63Da10
	159069.7	0.7			-0.1	1	0	0	^{128}Xe	C3	2.5	70Ke05
$^{128}\text{Cs}-\text{u}$	-92181	30	-92252	6	-2.4	U				GS2	1.0	05Li24
$^{128}\text{Ba}-\text{u}$	-91663	30	-91647.6	1.7	0.5	U				GS2	1.0	05Li24
$^{128}\text{La}-\text{u}$	-84436	69	-84410	60	0.4	2				GS2	1.0	05Li24
$^{128}\text{Ce}-\text{u}$	-81089	30				2				GS2	1.0	05Li24
$^{128}\text{Pr}-\text{u}$	-71209	32				2				GS2	1.0	05Li24
$^{128}\text{Cd}-^{133}\text{Cs}_{962}$	18759	11	18772	7	1.2	-				MA8	1.0	10Br02
	18787	15			-1.0	-				MA8	1.0	20Ma09
	ave.	18769			0.4	1	61	61	^{128}Cd			average
$^{128}\text{In}-^{133}\text{Cs}_{962}$	11329	11	11308.9	1.4	-1.8	U				TT1	1.0	18Ba08
$^{128}\text{In}^n-^{133}\text{Cs}_{962}$	11610.9	9.8	11615.0	2.1	0.4	U				TT1	1.0	18Ba08
$^{128}\text{Sn}-^{34}\text{S}-^{133}\text{Cs}_{1,218}$	-6396	14	-6466	19	-5.0	F				MA8	1.0	08Dw01
$^{128}\text{Cs}-^{133}\text{Cs}_{962}$	-1306	17	-1296	6	0.6	o				MA1	1.0	90St25
	-1293	13			-0.3	1	20	20	^{128}Cs	MA1	1.0	99Am05
$^{128}\text{Ba}-^{133}\text{Cs}_{962}$	-720	13	-692.3	1.7	2.1	1	2	2	^{128}Ba	MA1	1.0	99Am05
$^{128}\text{Cd}-^{130}\text{Xe}_{985}$	22865	11	22860	7	-0.4	1	39	39	^{128}Cd	JY1	1.0	12Ha25
$^{128}\text{Te}-^{35}\text{Cl}-^{126}\text{Te}-^{37}\text{Cl}$	4106	2	4099.2	1.6	-0.8	U				H16	4.0	63Ba47
	4102.3	1.8			-0.7	1	13	10	^{126}Te	C3	2.5	70Ke05
$^{128}\text{Xe}-^{129}\text{Xe}_{992}$	-2011.8587	0.0015	-2011.8587	0.0015	0.0	1	100	100	^{128}Xe	Hep	1.0	20Ri04
$^{128}\text{In}-^{128}\text{Te}$	15892.4	1.2				2				JY1	1.0	20Ne06
$^{128}\text{In}^n-^{128}\text{Te}$	16198.5	2.0				2				JY1	1.0	20Ne06
$^{128}\text{In}^p-^{128}\text{Te}$	17822.2	1.3				2				JY1	1.0	20Ne06
$^{128}\text{Te}-^{128}\text{Xe}$	931.26	1.20	930.5	0.8	-0.4	-				H43	1.5	90Dy04
	929.6	1.4			0.6	-				CP1	1.0	09Sc19
	ave.	930.2			0.2	1	47	47	^{128}Te			average
$^{128}\text{Xe}-^{126}\text{Xe}$	-774	45	-766.6687	0.0025	0.1	U				M16	2.5	63Da10
$^{126}\text{Cs}-^{128}\text{Cs}_{656}$	$^{122}\text{Cs}_{344}^x$	30	-1102	16	0.4	U				P22	2.5	82Au01
$^{124}\text{Cs}^x-^{128}\text{Cs}_{323}$	$^{122}\text{Cs}_{678}^x$	30	-980	30	1.2	U				P22	2.5	82Au01
$^{126}\text{Cs}-^{128}\text{Cs}_{591}$	$^{123}\text{Cs}_{410}^x$	30	-340	12	0.1	U				P22	2.5	82Au01
$^{124}\text{Cs}^x-^{128}\text{Cs}_{194}$	$^{123}\text{Cs}_{807}^x$	50	356	24	-0.1	U				P22	2.5	82Au01
$^{125}\text{Cs}-^{128}\text{Cs}_{244}$	$^{124}\text{Cs}_{756}^x$	30	-1349	18	1.2	U				P22	2.5	82Au01
$^{126}\text{Cs}-^{128}\text{Cs}_{492}$	$^{124}\text{Cs}_{508}^x$	30	-563	15	0.6	U				P22	2.5	82Au01
$^{127}\text{Cs}-^{128}\text{Cs}_{661}$	$^{125}\text{Cs}_{339}$	16	-933	7	0.8	U				P32	2.5	86Au02
$^{127}\text{Cs}-^{128}\text{Cs}_{496}$	$^{126}\text{Cs}_{504}$	30	-1105	8	0.7	U				P22	2.5	82Au01
$^{128}\text{Te}(\gamma, n)^{127}\text{Te}$	-8410	120	-8784.6	1.5	-3.1	B				Phi		60Ge01
$^{127}\text{I}(n, \gamma)^{128}\text{I}$	6825.7	0.5	6826.13	0.05	0.9	U						71Sc07
	6826.12	0.05			0.2	-				MMn		90Is03
	6826.22	0.14			-0.6	-				Bdn		06Fi.A
	ave.	6826.13			0.0	1	100	87	^{128}I			average
$^{128}\text{Cd}(\beta^-)^{128}\text{In}$	7070	290	6952	7	-0.4	U				Stu		87Sp09
$^{128}\text{In}(\beta^-)^{128}\text{Sn}$	9280	180	9171	18	-0.6	U				Stu		78Al18
	8984	37			5.1	F				Stu		87Sp09
	8950	103			2.1	F				Gsn		90St13
$^{128}\text{In}^n(\beta^-)^{128}\text{Sn}$	9390	220	9456	18	0.3	o				Stu		78Al18
	9306	30			5.0	F				Stu		87Sp09
	9230	90			2.5	U				Gsn		90St13
$^{128}\text{Sn}(\beta^-)^{128}\text{Sb}^m$	1265	30	1258	12	-0.2	-						76Nu01
	1290	40			-0.8	-				Stu		77Lu06
	1260	15			-0.1	-				Gsn		90St13
	ave.	1264			-0.4	1	87	45	$^{128}\text{Sb}^m$			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{128}\text{Sb}^m(\text{IT})^{128}\text{Sb}$	10	6				2					AHW *
$^{128}\text{Sb}(\beta^-)^{128}\text{Te}$	4640	100	4364	19	-2.8	U					71Ki15 *
$^{128}\text{Sb}^m(\beta^-)^{128}\text{Te}$	4391	40	4374	18	-0.4	-			Stu		77Lu06 *
	4395	30			-0.7	-			Gsn		90St13 *
ave.	4394	24			-0.8	1	55	55 $^{128}\text{Sb}^m$			average
$^{128}\text{I}(\beta^+)^{128}\text{Te}$	1277	13	1256	4	-1.6	U					61La16
$^{128}\text{I}(\beta^-)^{128}\text{Xe}$	2116	10	2123	4	0.7	1	13	13 ^{128}I			56Be18 *
$^{128}\text{Cs}(\beta^+)^{128}\text{Xe}$	3855	90	3929	5	0.8	U					75We23 *
	3928	6			0.1	1	80	80 ^{128}Cs			76Cr.B
	3907	40			0.5	o			IRS		83Al06
	3930	100			-0.0	U			IRS		93Al03
$^{128}\text{La}(\beta^+)^{128}\text{Ba}$	6650	400	6740	50	0.2	U					66Li04 *
	6820	100			-0.8	R			JAE		98Ko66 *
* $^{128}\text{La}-u$	$M-A=-78601(28)$ keV for mixture gs+m at 100#100 keV										Nub211 **
* $^{128}\text{Sn}^{34}\text{S}-^{133}\text{Cs}_{1,218}$	F : authors say "possible contamination, measurement abandoned"										GAu **
* $^{128}\text{In}(\beta^-)^{128}\text{Sn}$	$E_{\beta^-}=4980(180)$ to $(2)^+$ level at 4297.70 keV										Ens15a **
* $^{128}\text{In}(\beta^-)^{128}\text{Sn}$	$E_{\beta^-}=5464(37)$ to $(2)^+$ at 3519.86; others 6986(170), 7857(109) to 2104.07, 1168.82; different equipment/method than in previous; low E_{β^-} not seen										Ens15a **
* $^{128}\text{In}(\beta^-)^{128}\text{Sn}$	$E_{\beta^-}=4650(120)$, 5440(200) to 4297, 3520 levels and others										FGK126 **
* $^{128}\text{In}(\beta^-)^{128}\text{Sn}$	F : above 2 items conflict with 1st one and with trends in Ex of 8^- isomer										GAu **
* $^{128}\text{In}^n(\beta^-)^{128}\text{Sn}$	$E_{\beta^-}=5430(220)$ to 3958 level										FGK126 **
* $^{128}\text{In}^n(\beta^-)^{128}\text{Sn}$	$E_{\beta^-}=5239(40)$, 5350(44) to 4066, 3958 level										FGK126 **
* $^{128}\text{In}^n(\beta^-)^{128}\text{Sn}$	$E_{\beta^-}=5160(170)$, 5250(130) to 4066, 3958 levels										FGK126 **
* $^{128}\text{Sn}(\beta^-)^{128}\text{Sb}^m$	$E_{\beta^-}=630(30)$ 655(40) 625(15), to 1^+ level 635.2 above $^{128}\text{Sb}^m$ at 10(7) keV										Ens15a **
* $^{128}\text{Sb}^m(\text{IT})^{128}\text{Sb}$	From 3.6% IT for M3 transition										Ens15a **
* $^{128}\text{Sb}(\beta^-)^{128}\text{Te}$	$E_{\beta^-}=2300(100)$ to $(7)^-$ level at 2337.68 keV										Ens15a **
* $^{128}\text{Sb}^m(\beta^-)^{128}\text{Te}$	$E_{\beta^-}=2580(40)$ 2585(30) respectively, to 6^+ level at 1811.13 keV										Ens15a **
* $^{128}\text{I}(\beta^-)^{128}\text{Xe}$	$E_{\beta^-}=2120(10)$ and $E_{\beta^-}=1665(15)$ to ground state and 2^+ level at 442.911 keV										Ens15a **
* $^{128}\text{Cs}(\beta^+)^{128}\text{Xe}$	$E_{\beta^+}=2390(90)$ to 2^+ level at 442.911 keV										Ens15a **
* $^{128}\text{La}(\beta^+)^{128}\text{Ba}$	$E_{\beta^+}=3200(400)$ 3370(100) respectively, to $(4^-, 5^+)$ level at 2425.45 keV										Ens15a **
$^{129}\text{Cd}-u$	-67789	186	-67764	6	0.1	U			GT3	2.5	16Kn02
$^{129}\text{Sn}-u$	-86521	31	-86518	19	0.1	1	36	36 ^{129}Sn	MA8	1.0	05Si34 *
$^{129}\text{Xe}-^{120}\text{Sn}_{1,075}$	9913.3	2.4	9913.1	1.1	-0.1	1	20	20 ^{129}Sn	JY1	1.0	11Ha48
$\text{C}_{10}\text{H}_9-^{129}\text{Xe}$	165643.6	3.6	165644.430	0.005	0.1	U			M16	2.5	63Da10
$^{129}\text{Xe}-u$	-95228.7	5.4	-95219.143	0.005	0.7	U			ACC	2.5	90Me08
$^{129}\text{Xe}-\text{C}_2^{35}\text{Cl}_3$	-1777.98	0.68	-1777.23	0.11	0.7	U			H47	1.5	94Hy01
$^{129}\text{Xe}_2-^{86}\text{Kr}_3$	77729.8547	0.0250	77729.841	0.013	-0.6	1	28	16 ^{86}Kr	FS1	1.0	05Sh38 *
$^{129}\text{La}-u$	-87300	30	-87304	23	-0.1	1	58	58 ^{129}La	GS2	1.0	05Li24
$^{129}\text{Ce}-u$	-81898	30				2			GS2	1.0	05Li24
$^{129}\text{Pr}-u$	-74905	32				2			GS2	1.0	05Li24 *
$^{129}\text{Xe}-^{134}\text{Xe}_{963}$	-4114.7	3.8	-4112.6310	0.0030	0.5	o			MA8	1.0	05He.A
	-4119.3	5.1			1.3	U			MA8	1.0	06He29
$^{129}\text{Cd}-^{133}\text{Cs}_{970}$	23831	107	23947	6	1.1	U			MA8	1.0	15At03 *
	23947.2	5.7				2			IS1	1.0	20Ma09
$^{129}\text{Cd}^m-^{133}\text{Cs}_{970}$	24315.5	6.0				2			IS1	1.0	20Ma09
$^{129}\text{In}-^{133}\text{Cs}_{970}$	13518.6	6.7	13520.1	2.1	0.2	1	10	10 ^{129}In	TT1	1.0	18Ba08
$^{129}\text{In}^m-^{133}\text{Cs}_{970}$	13995	15	14004.0	2.1	0.6	U			TT1	1.0	18Ba08
$^{129}\text{Cs}-^{133}\text{Cs}_{970}$	-2234	18	-2222	5	0.6	o			MA1	1.0	90St25
	-2216	14			-0.5	1	12	12 ^{129}Cs	MA1	1.0	99Am05
$^{129}\text{In}-^{130}\text{Xe}_{992}$	17523.9	2.9	17527.3	2.1	1.2	1	53	53 ^{129}In	JY1	1.0	12Ha25
$^{129}\text{In}^m-^{130}\text{Xe}_{992}$	18016.5	3.5	18011.1	2.1	-1.5	1	37	37 $^{129}\text{In}^m$	JY1	1.0	13Ka08
$\text{C}_{10}\text{H}_{10}-^{129}\text{Xe}$	173469.4660	0.0147	173469.462	0.005	-0.3	1	14	14 ^{129}Xe	FS1	1.0	09Re03
$^{129}\text{Xe}-^{128}\text{Xe}$	1247	12	1250.1040	0.0015	0.1	U			M16	2.5	63Da10
$\text{C}_3\text{O}_6-^{129}\text{Xe}$	64706.8420	0.0255	64706.858	0.005	0.6	o			FS1	1.0	05Sh38 *
	64706.8516	0.0181			0.4	1	9	8 ^{129}Xe	FS1	1.0	09Re03

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{129}\text{Xe}_2-^{84}\text{Kr}_3$	75068.5115	0.0405	75068.534	0.014	0.5	1	11	7 ^{84}Kr	FS1	1.0	05Sh38 *
$^{128}\text{Cs}-^{129}\text{Cs}_{.661}$ $^{126}\text{Cs}_{.339}$	510	30	500	7	-0.1	U			P22	2.5	82Au01
$^{128}\text{Te}(\text{n},\gamma)^{129}\text{Te}$	6085	3	6082.41	0.08	-0.9	U					72Mu.A
	6082.42	0.09			-0.1	-			Prn		03Wi02
	6082.36	0.19			0.3	-			Bdn		06Fi.A
$^{128}\text{Te}(\text{d},\text{p})^{129}\text{Te}$	3857	10	3857.84	0.08	0.1	U			MIT		67Mo22
$^{128}\text{Te}(\text{n},\gamma)^{129}\text{Te}$	ave. 6082.41	0.08	6082.41	0.08	-0.0	1	100	99 ^{129}Te			average
$^{129}\text{Nd}(\text{ep})^{128}\text{Ce}$	5300	300	5870#	200#	1.9	D					78Bo.A *
$^{129}\text{In}^{\text{m}}(\text{IT})^{129}\text{In}$	450.72	0.16	450.73	0.16	0.1	1	100	63 $^{129}\text{In}^{\text{m}}$			FGK20c *
$^{129}\text{In}(\beta^-)^{129}\text{Sn}$	7655	32	7756	17	3.1	B			Stu		87Sp09
	7780	26			-0.9	1	44	44 ^{129}Sn	Stu		04Ga24 *
$^{129}\text{In}^{\text{m}}(\beta^-)^{129}\text{Sn}$	8033	66	8206	17	2.6	B			Stu		87Sp09
	8149	38			1.5	1	21	20 ^{129}Sn	Stu		04Ga24
$^{129}\text{In}^{\text{p}}(\beta^-)^{129}\text{Sn}$	9410	50				2			Stu		04Ga24
$^{129}\text{Sn}(\beta^-)^{129}\text{Sb}$	3996	120	4039	27	0.4	U			Stu		77Lu06 *
$^{129}\text{Sb}(\beta^-)^{129}\text{Te}$	2345	30	2375	21	1.0	2					70Oh05 *
$^{129}\text{Te}(\beta^-)^{129}\text{I}$	1453	28	1502	3	1.8	U					56Gr10 *
	1485	10			1.7	U					64De10 *
	1503	4			-0.2	1	61	60 ^{129}I			68Go34 *
$^{129}\text{I}(\beta^-)^{129}\text{Xe}$	190	5	189	3	-0.2	1	40	40 ^{129}I			54De17 *
$^{129}\text{Cs}(\beta^+)^{129}\text{Xe}$	1197	5	1197	5	0.0	1	83	83 ^{129}Cs			76Ma35
$^{129}\text{Ba}(\beta^+)^{129}\text{Cs}$	2446	15	2438	11	-0.5	1	50	45 ^{129}Ba			61Ar05 *
$^{129}\text{La}(\beta^+)^{129}\text{Ba}$	3720	50	3737	22	0.3	-					79Br05 *
	3740	40			-0.1	-			JAE		98Ko66
	ave. 3730	30			0.2	1	48	42 ^{129}La			average
$^{129}\text{Ce}(\beta^+)^{129}\text{La}$	5600	200	5040	40	-2.8	U			IRS		93Al03
* $^{129}\text{Sn}-\text{u}$	$M-A=-80576(27)$ keV for mixture gs+m at 35.15 keV										Nub211 **
* $^{129}\text{Xe}_2-^{86}\text{Kr}_3$	Corrected in reference of same group										09Re03 **
* $^{129}\text{Pr}-\text{u}$	Isomer at 382.57 with estimated $T=26\mu\text{s}$ not considered										Nub211 **
* $^{129}\text{Cd}-^{133}\text{Cs}_{.970}$	$D_M=24016(18)\mu\text{u}$ for mixture of gs+m at 343(8) keV; $M-A=-63058(17)$										WgM204**
* $\text{C}_3\text{O}_6-^{129}\text{Xe}$	Corrected in reference of same group										09Re03 **
* $^{129}\text{Xe}_2-^{84}\text{Kr}_3$	Corrected in reference of same group										09Re03 **
* $^{129}\text{Nd}(\text{ep})^{128}\text{Ce}$	Trends from Mass Surface TMS suggest ^{129}Nd 570 keV less bound										GAu **
* $^{129}\text{In}^{\text{m}}(\text{IT})^{129}\text{In}$	from a least-squares fit to the level scheme in $^{2015}\text{Ta}_{13}$										FGK20c **
* $^{129}\text{In}(\beta^-)^{129}\text{Sn}$	$E_{\beta^-}=7780(26)$, $9410(50)$ from ground state, $1688.0(0.5)$ levels										03Ge04 **
* $^{129}\text{Sn}(\beta^-)^{129}\text{Sb}$	$E_{\beta^-}=3350(120)$ to $(5/2^+)$ level at 645.14 keV										Ens148 **
* $^{129}\text{Sb}(\beta^-)^{129}\text{Te}$	$E_{\beta^-}=1800(30)$ to $5/2^+$ level at 544.585 keV, and other E_{β^-}										Ens148 **
* $^{129}\text{Te}(\beta^-)^{129}\text{I}$	$E_{\beta^-}=1453(5)$ to $5/2^+$ level at 27.793 keV and $1530(5)$ from $^{129}\text{Te}^{\text{m}}$										Ens148 **
*	at 105.51 to ground state (Birge=8.0: arithmetic average used)										Nub211 **
* $^{129}\text{Te}(\beta^-)^{129}\text{I}$	$E_{\beta^-}=1452(10)$ to $5/2^+$ level at 27.793 keV and $1595(10)$ from $^{129}\text{Te}^{\text{m}}$ at 105.51										Ens148 **
* $^{129}\text{Te}(\beta^-)^{129}\text{I}$	$E_{\beta^-}=1476(4)$ to $5/2^+$ level at 27.793 keV and $1607(7)$ from $^{129}\text{Te}^{\text{m}}$ at 105.51										Ens148 **
* $^{129}\text{I}(\beta^-)^{129}\text{Xe}$	$E_{\beta^-}=150(5)$ to $3/2^+$ level at 39.5774 keV										Ens148 **
* $^{129}\text{Ba}(\beta^+)^{129}\text{Cs}$	$E_{\beta^+}=1425(15)$; and $1243(35)$, $975(60)$ from $^{129}\text{Ba}^{\text{m}}$ at 8.42(0.06) keV										Nub211 **
*	to $7/2^+$ level at 188.91, $(9/2)^+$ at 426.47 keV										Ens148 **
* $^{129}\text{La}(\beta^+)^{129}\text{Ba}$	$E_{\beta^+}=2420(50)$ to $1/2^+$ level at 278.57 keV, and other E_{β^+}										Ens148 **
$^{130}\text{Cd}-\text{u}$	-66700	441	-65612	24	1.0	U			GT3	2.5	16Kn02
$\text{C}_9\text{H}_8\text{N}-^{130}\text{Te}$	159446	10	159451.514	0.012	0.2	U			M16	2.5	63Da10
$^{13}\text{C}\text{C}_8\text{NH}_7-^{130}\text{Te}$	154990.6	7.	154981.318	0.012	-0.5	U			C3	2.5	70Ke05
$\text{C}_9\text{H}_8\text{N}-^{130}\text{Te}$	159449	8	159451.514	0.012	0.2	U			R13	1.5	83De51
$\text{C}_{10}\text{H}_{10}-^{130}\text{Xe}$	174743.6	4.2	174740.972	0.010	-0.3	U			M16	2.5	63Da10
$^{13}\text{C}\text{C}_8\text{NH}_7-^{130}\text{Xe}$	157695.4	0.7	157694.716	0.010	-0.4	U			C3	2.5	70Ke05
$^{130}\text{Xe}-\text{C}^{13}\text{C}^{35}\text{Cl}_3$	-6407.63	1.21	-6403.57	0.11	2.2	U			H47	1.5	94Hy01
$^{130}\text{Cs}-\text{u}$	-93181	60	-93291	9	-1.8	U			GS2	1.0	05Li24 *
$\text{C}_{10}\text{H}_{10}-^{130}\text{Ba}$	171926	68	171924.3	0.3	-0.0	U			R07	1.5	68De17

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{130}\text{Ba}-^{85}\text{Rb}_{1.529}$	41195.8	3.4	41199.5	0.3	1.1	1	1	1	^{130}Ba	MA8	1.0	05Gu37
$^{130}\text{La}-\text{u}$	-87635	30	-87631	28	0.1	2			GS2	1.0	05Li24	
$^{130}\text{Ce}-\text{u}$	-85264	30				2			GS2	1.0	05Li24	
$^{130}\text{Pr}-\text{u}$	-76410	69				2			GS2	1.0	05Li24	
$^{130}\text{Nd}-\text{u}$	-71494	30				2			GS2	1.0	05Li24	
$^{130}\text{Xe}-^{134}\text{Xe}_{.970}$	-4726.6	5.6	-4721.890	0.009	0.8	o			MA8	1.0	05He.A	
	-4724.8	7.0			0.4	U			MA8	1.0	06He29	
$^{130}\text{Cd}-^{133}\text{Cs}_{.977}$	26761	24				2			MA8	1.0	15At03	
$^{130}\text{In}-^{133}\text{Cs}_{.977}$	17437	93	17325.7	1.9	-1.2	U			CP1	1.0	13Va12	
	17338	30			-0.4	U			TT1	1.0	18Ba08	
	17325.7	2.0			-0.0	1	92	92 ^{130}In	CP2	1.0	18Or.A	
$^{130}\text{In}^m-^{133}\text{Cs}_{.977}$	17399.0	2.4	17397.1	2.1	-0.8	1	79	79 $^{130}\text{In}^m$	CP2	1.0	18Or.A	
$^{130}\text{In}^n-^{133}\text{Cs}_{.977}$	17757	31	17739.4	2.2	-0.6	U			TT1	1.0	18Ba08	
	17739.3	3.6			0.0	-			CP2	1.0	18Or.A	
	17737.6	4.8			0.4	-			JY2	1.0	20Ne06	
	ave. 17738.7	2.9			0.3	1	60	60 $^{130}\text{In}^n$			average	
$^{130}\text{Sn}-^{133}\text{Cs}_{.977}$	6346	17	6348.0	2.0	0.1	U			MA8	1.0	05Si34	
	6344	12			0.3	U			MA8	1.0	05Si34	
	6349.5	3.9			-0.4	1	27	27 ^{130}Sn	CP1	1.0	13Va12	
$^{130}\text{Xe}-^{133}\text{Cs}_{.977}$	-4114	13	-4117.217	0.011	-0.2	U			MA6	1.0	04Di18	
$^{130}\text{Cs}-^{133}\text{Cs}_{.977}$	-928	17	-917	9	0.6	o			MA1	1.0	90St25	
	-916	13			-0.1	1	48	48 ^{130}Cs	MA1	1.0	99Am05	
$^{130}\text{Nd } ^{19}\text{F}-^{133}\text{Cs}_{1.120}$	32902	130	32800	30	-0.8	U			MA5	1.0	00Be42	
$^{130}\text{Te } ^{35}\text{Cl}-^{128}\text{Te } ^{37}\text{Cl}$	4715	2	4711.6	0.8	-0.4	o			H16	4.0	63Ba47	
	4711.7	1.8			-0.0	U			C3	2.5	70Ke05	
	4711.57	0.72			0.1	1	49	49 ^{128}Te	H43	1.5	90Dy04	
$^{130}\text{Xe}-^{129}\text{Xe}_{1.008}$	-509.78	0.34	-509.756	0.009	0.1	U			CP1	1.0	09Sc19	
	-509.96	0.26			0.8	U			SH2	1.0	13El01	
	-509.02	0.94			-0.8	U			SH1	1.0	13El01	
$^{130}\text{In}-^{130}\text{In}^n$	-413.8	6.5	-413.7	2.8	0.0	1	18	11 $^{130}\text{In}^n$	JY2	1.0	20Ne06	
$^{130}\text{In}^m-^{130}\text{In}^n$	-351.8	6.0	-342.3	2.9	1.6	1	23	12 $^{130}\text{In}^n$	JY2	1.0	20Ne06	
$^{130}\text{In}^m-^{130}\text{Te}$	18797.9	6.9	18800.9	2.1	0.4	1	10	10 $^{130}\text{In}^m$	JY1	1.0	20Ne06	
$^{130}\text{In}^n-^{130}\text{Te}$	19138.3	5.3	19143.2	2.2	0.9	1	18	18 $^{130}\text{In}^n$	JY1	1.0	20Ne06	
$^{130}\text{Sn}-^{130}\text{Xe}$	10463.9	3.6	10465.2	2.0	0.4	-			JY1	1.0	12Ha25	
	10465.4	3.1			-0.1	-			JY1	1.0	13Ka08	
	ave. 10464.8	2.3			0.2	1	73	73 ^{130}Sn			average	
$^{130}\text{Te}-^{130}\text{Xe}$	2706.2	7.	2713.399	0.012	0.4	U			C3	2.5	70Ke05	
	2712.98	3.02			0.1	U			H43	1.5	90Dy04	
	2713.416	0.034			-0.5	-			FS1	1.0	09Re07	
	2713.402	0.026			-0.1	-			FS1	1.0	09Re07	
	2713.402	0.014			-0.2	o			FS1	1.0	09Re07	
	2712.86	0.34			1.6	U			CP1	1.0	09Sc19	
	2712.82	0.25			2.3	U			JY1	1.0	11Ra24	
	2713.44	0.14			-0.3	U			SH1	1.0	12Ne10	
	ave. 2713.407	0.021			-0.4	1	35	22 ^{130}Te			average	
$^{130}\text{Ba}-^{130}\text{Xe}$	2816.71	0.31	2816.7	0.3	-0.2	1	99	99 ^{130}Ba	SH1	1.0	12Ne10	
$^{130}\text{Te}-^{129}\text{Xe}$	1441.885	0.012	1441.888	0.011	0.2	1	78	78 ^{130}Te	FS1	1.0	09Re07	
$^{130}\text{Xe}-^{129}\text{Xe}$	-1277	12	-1271.511	0.009	0.2	U			M16	2.5	63Da10	
	-1271.517	0.012			0.5	1	51	50 ^{130}Xe	FS1	1.0	09Re07	
$^{129}\text{Cs}-^{130}\text{Cs}_{.794}^x \text{ } ^{125}\text{Cs}_{.206}$	-1270	40	-1200	14	0.7	U			P22	2.5	82Au01	
$^{130}\text{Ba}(\text{p,t})^{128}\text{Ba}$	-9482	32	-9548.5	1.5	-2.1	U			Win		74De31	
$^{130}\text{Ba}(\text{p,t})^{128}\text{Ba}-^{144}\text{Sm}(\gamma)^{142}\text{Sm}$	1095.9	1.0	1096.1	1.0	0.2	1	99	98 ^{128}Ba			09Pa25	
$^{130}\text{Te}(\text{d},^3\text{He})^{129}\text{Sb}$	-4550	30	-4519	21	1.0	R			Oak		68Au04	
$^{129}\text{I}(\text{n},\gamma)^{130}\text{I}$	6500.33	0.04				2			ILn		89Sa11	
$^{129}\text{Xe}(\text{n},\gamma)^{130}\text{Xe}$	9255.3	1.0	9255.723	0.008	0.4	U					71Gr28	
	9256.1	0.8			-0.5	U					74Ge05	
	9255.57	0.30			0.5	U			Bdn		06Fi.A	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{129}\text{Xe}(^3\text{He,d})^{130}\text{Cs}$	5	20	−1	8	−0.3	1	17	17 ^{130}Cs	ChR		81Ha08
$^{130}\text{Ba(d,t)}^{129}\text{Ba}$	−4001	15	−4010	11	−0.6	1	49	49 ^{129}Ba	Tal		74Gr22
$^{130}\text{Eu(p)}^{129}\text{Sm}$	1028.0	15.0	1530#	200#	33.3	D			Arp		04Da04 *
$^{130}\text{Cd}(\beta^-)^{130}\text{In}$	8350	160	8789	22	2.7	B			Bwg		03Di06 *
$^{130}\text{In}(\beta^-)^{130}\text{Sn}$	10249	38	10225.7	2.6	−0.6	U			Stu		87Sp09
	9880	90			3.8	B			Gsn		90St13
$^{130}\text{In}^m(\beta^-)^{130}\text{Sn}$	10300	37	10292.2	2.7	−0.2	U			Stu		87Sp09
	10170	170			0.7	U			Gsn		90St13
$^{130}\text{In}^n(\beta^-)^{130}\text{Sn}$	10650	49	10611.1	2.8	−0.8	U			Stu		87Sp09
	9880	200			3.7	B			Gsn		90St13
$^{130}\text{Sn}(\beta^-)^{130}\text{Sb}$	2195	35	2153	14	−1.2	−			Stu		77Lu06 *
	2080	40			1.8	−					77Nu01 *
	2149	18			0.2	−			Gsn		90St13 *
ave.	2148	15			0.4	1	90	90 ^{130}Sb			average
$^{130}\text{Sb}(\beta^-)^{130}\text{Te}$	5046	100	5067	14	0.2	U					71Ki15 *
	5015	100			0.5	U			Stu		77Lu06 *
	4990	70			1.1	U			Gsn		90St13 *
	5015	45			1.2	1	10	10 ^{130}Sb	Stu		95Me16 *
$^{130}\text{I}(\beta^-)^{130}\text{Xe}$	2983	10	2944	3	−3.9	B					65Da01 *
	2964	50			−0.4	U					70Qa03 *
$^{130}\text{Cs}(\beta^+)^{130}\text{Xe}$	2992	20	2981	8	−0.6	−					52Sm41
	2972	20			0.4	−					75We23
ave.	2982	14			−0.1	1	35	35 ^{130}Cs			average
$^{130}\text{Cs}^x(\text{IT})^{130}\text{Cs}$	27	15				2					AHW *
$^{130}\text{Cs}(\beta^-)^{130}\text{Ba}$	442	50	357	8	−1.7	U					52Sm41 *
$^{130}\text{La}(\beta^+)^{130}\text{Ba}$	5660	70	5629	26	−0.4	R			JAE		98Ko66
* $^{130}\text{Cs-u}$	$M-A=-86716(30)$ keV for mixture gs+m at 163.25 keV										Nub211 **
* $^{130}\text{Pr-u}$	$M-A=-71125(29)$ keV for mixture gs+m at 100#100 keV										Nub211 **
* $^{130}\text{In}-^{133}\text{Cs}_{977}$	$D_M=17599(21)$ μu for mixture gs+m+n at 66.5(2.7) and 385.4(2.6) keV;										Nub211 **
* $^{130}\text{In}-^{133}\text{Cs}_{977}$	$M-A=-69652(20)$ keV										Nub211 **
* $^{130}\text{In}-^{133}\text{Cs}_{977}$	$D_M=17373(20)$ μu for mixture gs+m at 66.5(2.7) keV; $M-A=-69862(20)$ keV										Nub211 **
* $^{130}\text{Sn}-^{133}\text{Cs}_{977}$	$D_M=8434(12)$ μu for $^{130}\text{Sn}^m$ at 1946.88 keV; $M-A=-78189(11)$ keV										Nub211 **
* $^{130}\text{Nd}-^{19}\text{F}-^{133}\text{Cs}_{1,120}$	Tentative result, low statistics										00Be42 **
* $^{130}\text{Xe}-^{129}\text{Xe}_{1,008}$	Respectively for PI-ICR and Ramsey ToF-ICR techniques										13El01 **
* $^{130}\text{Sn}-^{130}\text{Xe}$	$D_M=12555.5(3.1)$ μu for $^{130}\text{Sn}^m$ at 1946.88 keV; $M-A=-78185.1(2.9)$ keV										Nub211 **
* $^{130}\text{Te}-^{130}\text{Xe}$	First item 1 ion; second item 2 ions - considered independent										GAu **
* $^{130}\text{Te}-^{130}\text{Xe}$	Combination of $^{130}\text{Xe}-^{129}\text{Xe}$ and $^{130}\text{Te}-^{129}\text{Xe}$										GAu **
* $^{130}\text{Ba(p,t)}^{128}\text{Ba}$	Not resolved peak. Original uncertainty 16 increased to 24 keV and added systematic error 21 keV										GAu **
* $^{130}\text{Eu(p)}^{129}\text{Sm}$	Trends from Mass Surface TMS suggest ^{130}Eu 500 keV less bound										GAu **
* $^{130}\text{Cd}(\beta^-)^{130}\text{In}$	$E_{\beta^-}=6224(+165-157)$ to 1^+ level at 2120.2 keV										Ens086 **
* $^{130}\text{Sn}(\beta^-)^{130}\text{Sb}$	$E_{\beta^-}=1490(90)$, 1150(35) to 1^+ levels at 702.32, 1047.67 keV										Ens017 **
* $^{130}\text{Sn}(\beta^-)^{130}\text{Sb}$	$E_{\beta^-}=1280(80)$, 1060(40) to 1^+ levels at 702.32, 1047.67 keV										Ens017 **
* $^{130}\text{Sn}(\beta^-)^{130}\text{Sb}$	$E_{\beta^-}=1415(30)$, 1112(18) to 1^+ levels at 702.32, 1047.67 keV										Ens017 **
*	and a 3σ conflicting 3955(50) from $^{130}\text{Sn}^m$ at 1946.88 keV										Nub211 **
* $^{130}\text{Sb}(\beta^-)^{130}\text{Te}$	$E_{\beta^-}=2900(100)$ to $^{130}\text{Te}^m$ at 2146.41 keV										Nub211 **
* $^{130}\text{Sb}(\beta^-)^{130}\text{Te}$	$Q=5020(100)$ from $^{130}\text{Sb}^m$ at 4.80 keV										Nub211 **
* $^{130}\text{Sb}(\beta^-)^{130}\text{Te}$	Also 4960(25) from $^{130}\text{Sb}^m$ at 4.80, in disagreement										Nub211 **
* $^{130}\text{Sb}(\beta^-)^{130}\text{Te}$	Derived from given average=5008(38) with 90St13=4990(70) keV										GAu **
* $^{130}\text{I}(\beta^-)^{130}\text{Xe}$	$E_{\beta^-}=1702(10)$ 1042(10) 618(10) to 4^+ 1204.614, 6^+ 1944.140, 5^+ 2362.073										Ens017 **
* $^{130}\text{I}(\beta^-)^{130}\text{Xe}$	$E_{\beta^-}=2480(50)$, 1850(80) from $^{130}\text{I}^m$ at 39.9525 to 2^+ levels at 536.068, and 1122.112 keV										GAu **
*											Ens017 **
* $^{130}\text{Cs}^x(\text{IT})^{130}\text{Cs}$	Combining isomer ratio of reference										82Au01 **
*	with $^{130}\text{Cs}^m(\text{IT})=163.25$ keV										Nub211 **
* $^{130}\text{Cs}(\beta^-)^{130}\text{Ba}$	Value given without associated error										AHW **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{131}\text{Cd}-u$	-59671	1023	-59272	21	0.2	U			GT3	2.5	16Kn02
	-59280	110			0.1	U			MR1	1.0	15At03
	-59300	26			1.1	1	63	63 ^{131}Cd	MR1	1.0	20Ma09
$^{131}\text{Sn}-u$	-82958	26	-82947	4	0.4	U			MA8	1.0	05Si34 *
	-82950	130			0.0	U			GT2	2.5	08Su19 *
	-88170	530	-88010.7	2.2	0.1	U			GT2	2.5	08Kn.A *
$\text{C}_{10}\text{H}_{11}-^{131}\text{Xe}$	180991.6	3.0	180991.223	0.005	-0.1	U			M16	2.5	63Da10
$^{131}\text{Xe}-u$	-94925.5	5.7	-94915.872	0.005	0.7	U			ACC	2.5	90Me08
$^{131}\text{Xe}-\text{C}_2\text{ }^{35}\text{Cl}_2\text{ }^{37}\text{Cl}$	1472.65	0.80	1476.17	0.09	2.9	B			H47	1.5	94Hy01
$^{131}\text{Ba}-u$	-92955	66	-93053.7	0.4	-1.5	U			GS2	1.0	05Li24 *
$^{131}\text{La}-u$	-89930	30				2			GS2	1.0	05Li24
$^{131}\text{Ce}-u$	-85579	36	-85570	40	0.2	1	96	96 ^{131}Ce	GS2	1.0	05Li24 *
$^{131}\text{Pr}-u$	-79741	56	-79770	50	-0.4	1	81	81 ^{131}Pr	GS2	1.0	05Li24 *
$^{131}\text{Nd}-u$	-72753	30	-72752	30	0.0	1	97	97 ^{131}Nd	GS2	1.0	05Li24
$^{131}\text{Cd}-^{133}\text{Cs}_{985}$	33905	34	33858	21	-1.4	1	37	37 ^{131}Cd	MA8	1.0	20Ma09
$^{131}\text{In}-^{133}\text{Cs}_{985}$	20262	37	20102.7	2.4	-4.3	C			CP1	1.0	13Va12 *
	20104.1	4.1			-0.4	1	33	33 ^{131}In	CP2	1.0	18Or.A
$^{131}\text{In}^m-^{133}\text{Cs}_{985}$	20507.9	2.8	20506.3	2.6	-0.6	1	88	88 $^{131}\text{In}^m$	CP2	1.0	18Or.A
$^{131}\text{Sn }^{34}\text{S}-^{133}\text{Cs}_{1.241}$	2253	11	2254	4	0.1	-			MA8	1.0	08Dw01 *
$^{131}\text{Sn}-^{133}\text{Cs}_{985}$	10188.2	4.7	10183	4	-1.1	-			CP1	1.0	13Va12
$^{131}\text{Sn }^{34}\text{S}-^{133}\text{Cs}_{1.241}$	ave. 2259	4	2254	4	-1.0	1	81	81 ^{131}Sn			average
$^{131}\text{Sb}-^{133}\text{Cs}_{985}$	5114	11	5119.2	2.2	0.5	U			CP1	1.0	13Va12
$^{131}\text{Xe}-^{129}\text{Xe}_{1.016}$	1826.7855	0.0096	1826.7800	0.0014	-0.6	U			FS1	1.0	13Ho22
	1826.7797	0.0015			0.2	1	89	48 ^{129}Xe	Hep	1.0	20Ri04
$^{131}\text{Cs}-^{133}\text{Cs}_{985}$	-1429	18	-1401.72	0.19	1.5	o			MA1	1.0	90St25
	-1419	14			1.2	o			MA1	1.0	99Am05
	-1401.72	0.19				2			IS1	1.0	19Ka48 *
$^{131}\text{Ba}-^{133}\text{Cs}_{985}$	72	14	76.1	0.4	0.3	U			MA5	1.0	00Be42
$^{131}\text{In}-^{130}\text{Xe}_{1.008}$	24234.7	2.9	24235.4	2.4	0.2	1	67	67 ^{131}In	JY1	1.0	12Ha25
$^{131}\text{In}^m-^{130}\text{Xe}_{1.008}$	24626.8	7.7	24639.0	2.6	1.6	1	12	12 $^{131}\text{In}^m$	JY1	1.0	13Ka08
$^{131}\text{Sn}-^{132}\text{Xe}_{992}$	12134	21	12131	4	-0.1	U			JY1	1.0	12Ha25 *
$^{131}\text{Sb}-^{130}\text{Xe}_{1.008}$	9250.7	2.3	9251.9	2.2	0.5	1	95	95 ^{131}Sb	JY1	1.0	12Ha25
$^{131}\text{Xe}-^{132}\text{Xe}_{992}$	162.283	0.030	162.2837	0.0017	0.0	U			SH1	1.0	14Ne15
	162.292	0.014			-0.6	U			FS1	1.0	13Ho22
	162.2842	0.0018			-0.3	1	84	58 ^{131}Xe	Hep	1.0	20Ri04
$^{131}\text{Xe}-^{130}\text{Xe}$	1574	11	1574.781	0.009	0.0	U			M16	2.5	63Da10
$^{128}\text{Cs}-^{131}\text{Cs}_{391}\text{ }^{126}\text{Cs}_{610}$	-100	30	-48	8	0.7	U			P22	2.5	82Au01
$^{128}\text{Cs}-^{131}\text{Cs}_{244}\text{ }^{127}\text{Cs}_{756}$	783	21	751	7	-0.6	F			P33	2.5	86Au02 *
$^{129}\text{Cs}-^{131}\text{Cs}_{328}\text{ }^{128}\text{Cs}_{672}$	-1030	30	-871	6	2.1	U			P22	2.5	82Au01
$^{130}\text{Te}(n,\gamma)^{131}\text{Te}$	5929.7	0.5	5929.38	0.06	-0.6	U					77Ko.A
	5929.5	0.4			-0.3	U					80Ho29 Z
	5929.38	0.06				2			Prn		03To08
	5930.16	0.19			-4.1	C			Bdn		06Fi.A
$^{130}\text{Te}(d,p)^{131}\text{Te}$	3703	6	3704.81	0.06	0.3	U			MIT		67Gr21
$^{130}\text{Ba}(n,\gamma)^{131}\text{Ba}$	7493.5	0.3				2					82Ka.A
$^{130}\text{Ba}(d,p)^{131}\text{Ba}$	5269	15	5268.90	0.30	-0.0	U			ANL		70Vo04
$^{131}\text{Nd}(\epsilon p)^{130}\text{Ce}$	4600	400	4370	40	-0.6	U					78Bo.A
$^{131}\text{Eu}(p)^{130}\text{Sm}$	957.4	8.	947	5	-1.3	3					98Da03
	939.2	7.			1.1	3					99So17 *
$^{131}\text{In}(\beta^-)^{131}\text{Sn}$	8820	200	9240	4	2.1	U					80De35
	8930	150			2.1	o			Stu		84Fo19
	9184	33			1.7	o			Stu		88Fo05
	9165	30			2.5	o			Stu		95Me16
	9174	22			3.0	B			Stu		99Fo01
	9222	18			1.0	U			Stu		04Fo06
$^{131}\text{In}^m(\beta^-)^{131}\text{Sn}$	9230	220	9616	4	1.8	o			Stu		84Fo19
	9547	46			1.5	o			Stu		88Fo05
	9480	70			1.9	o			Stu		95Me16
	9524	26			3.5	B			Stu		04Fo06

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{131}\text{In}(\beta^-)^{131}\text{Sn}$	13000	500	12990	90	-0.0	o			Stu		84Fo19
	13450	163			-2.8	B			Stu		88Fo05
	13230	80			-3.0	B			Stu		95Me16
	12986	86				2			Stu		04Fo06
$^{131}\text{Sn}(\beta^-)^{131}\text{Sb}$	4582	120	4717	4	1.1	o			Bwg		79Ke02 *
	4640	20			3.8	B			Stu		84Fo19 *
	4610	110			1.0	U			Bwg		87Gr.A *
	4689	14			2.0	o			Stu		95Me16
	4688	14			2.1	o			Stu		99Fo01
	4701	8			2.0	1	25	19 ^{131}Sn	Stu		04Fo06 *
$^{131}\text{Sb}(\beta^-)^{131}\text{Te}$	3190	70	3229.6	2.1	0.6	o			Stu		77Lu06
	3217	20			0.6	o			Stu		95Me16
	3200	26			1.1	U			Stu		99Fo01
$^{131}\text{Te}(\beta^-)^{131}\text{I}$	2275	10	2231.7	0.6	-4.3	B					61Be20 *
	2278	15			-3.1	B					65De22 *
$^{131}\text{I}(\beta^-)^{131}\text{Xe}$	971.0	0.7	970.8	0.6	-0.2	2					51Ve05 *
	970.4	1.2			0.4	2					52Ro16 *
$^{131}\text{Cs}(\epsilon)^{131}\text{Xe}$	355	10	358.00	0.18	0.3	U					54Sa22
	355	10			0.3	U					56Ho66
	360	15			-0.1	U					57Mi63
$^{131}\text{Ba}(\beta^+)^{131}\text{Cs}$	1370	16	1376.6	0.5	0.4	U					76Ge14 *
	1371	12			0.5	U					78Va04 *
$^{131}\text{La}(\beta^+)^{131}\text{Ba}$	2960	100	2910	28	-0.5	U					60Cr01
$^{131}\text{Ce}(\beta^+)^{131}\text{La}$	4020	400	4060	40	0.1	U					66No05 *
$^{131}\text{Pr}(\beta^+)^{131}\text{Ce}$	5250	150	5410	60	1.1	1	14	9 ^{131}Pr	IRS		93Al03
$^{131}\text{Nd}(\beta^+)^{131}\text{Pr}$	6560	150	6530	50	-0.2	1	13	9 ^{131}Pr	IRS		93Al03
* $^{131}\text{Sn}-u$	$M-A=-77242(15)$ keV for mixture gs+m at 65.1 keV										Nub211 **
*	next $^{131}\text{Sn } ^{34}\text{S}-^{133}\text{Cs}_{1,241}$ also from 1.4 GeV p on UC target										GAu **
* $^{131}\text{Sn}-u$	$M-A=-77238(120)$ keV for mixture gs+m at 65.1 keV										Nub211 **
* $^{131}\text{Sb}-u$	$M-A=-81291(96)$ keV for mixture gs+m at 1676.06 keV										Nub211 **
* $^{131}\text{Ba}-u$	$M-A=-86494(30)$ keV for mixture gs+m at 187.995 keV										Nub211 **
* $^{131}\text{Ce}-u$	$M-A=-79685(28)$ keV for mixture gs+m at 63.09 keV										Nub211 **
* $^{131}\text{Pr}-u$	$M-A=-74202(28)$ keV for mixture gs+m at 152.4 keV										Nub211 **
* $^{131}\text{In}-^{133}\text{Cs}_{985}$	Freq. ratio mistyped, derived from mass excess in col.3 Table 1 of paper										GAu **
* $^{131}\text{Sn } ^{34}\text{S}-^{133}\text{Cs}_{1,241}$	$D_M=2300.3(3.6)$ μu for mixture ^{131}Sn gs+m at 65.1 keV with $R=0.65(0.15)$										Nub211 **
* $^{131}\text{Cs}-^{133}\text{Cs}_{985}$	freq. ratio from three methods, PI-ICR (dominant), ToF-ICR, Ramsey										19Ka48 **
* $^{131}\text{Sn}-^{132}\text{Xe}_{992}$	$D_M=12168.7(3.4)$ μu for mixture gs+m at 65.1 keV; $M-A=-77229.6(3.2)$ keV										Nub211 **
*	identical to result given in reference										13Ka08 **
* $^{128}\text{Cs}-^{131}\text{Cs}_{244}$ ^{127}Cs .	F : rejection based on line-shape analysis										86Au02 **
* $^{131}\text{Eu}(p)^{130}\text{Sm}$	Trends from Mass Surface TMS suggest ^{131}Eu 240 keV less bound										GAu **
* $^{131}\text{Sn}(\beta^-)^{131}\text{Sb}$	$E_{\beta^-}=3870(120), 2620(180)$ from $^{131}\text{Sn}^m$ at 65.1 keV										Nub211 **
*	to $(5/2^+)$ level at 798.494, $(13/2^-)$ at 1980.39 keV										Ens06c **
* $^{131}\text{Sn}(\beta^-)^{131}\text{Sb}$	$Q_{\beta^-}=4638(20)$; and 4796(80) from $^{131}\text{Sn}^m$ at 65.1 keV										Nub211 **
* $^{131}\text{Sn}(\beta^-)^{131}\text{Sb}$	$Q_{\beta^-}=4600(110)$; and 4680(120) from $^{131}\text{Sn}^m$ at 65.1 keV										Nub211 **
* $^{131}\text{Sn}(\beta^-)^{131}\text{Sb}$	$Q_{\beta^-}=4698(11)$; and 4767(8) from $^{131}\text{Sn}^m$ at 65.1 keV										Nub211 **
* $^{131}\text{Te}(\beta^-)^{131}\text{I}$	$Q_{\beta^-}=2457(10)$ 2460(15) respectively, from $^{131}\text{Te}^m$ at 182.258 keV										Nub211 **
* $^{131}\text{I}(\beta^-)^{131}\text{Xe}$	$E_{\beta^-}=606.5(0.7)$ 605.9(1.2) respectively, to $5/2^+$ level at 364.490 keV										Ens06c **
* $^{131}\text{Ba}(\beta^+)^{131}\text{Cs}$	$p^+=22(11)\times 10^{-6}$ to $3/2^+$ level at 216.086, recalculated p^+ and Q										Ens06c **
* $^{131}\text{Ba}(\beta^+)^{131}\text{Cs}$	$L/K=0.165(0.001)$ to $3/2^+$ level at 1047.682 keV										Ens06c **
* $^{131}\text{Ce}(\beta^+)^{131}\text{La}$	$E_{\beta^+}=2800(400)$ to $7/2^+$ level at 195.68 keV										Ens06c **
$^{132}\text{Cd}-u$	-54213	77	-54180	60	0.5	2			MR1	1.0	20Ma09
	-54092	118			-0.7	2			MR1	1.0	20Ma09
$^{132}\text{Sb}-u$	-85820	140	-85492.0	2.6	0.9	U			GT2	2.5	08Su19 *
$\text{C}_{10}\text{H}_{12}-^{132}\text{Xe}$	189740.8	3.3	189745.299	0.005	0.5	U			M16	2.5	63Da10

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{132}\text{Xe}-u$	-95856.2	4.0	-95844.917	0.005	1.1	U			ACC	2.5	90Me08
$^{132}\text{Xe}-\text{C } ^{13}\text{C } ^{35}\text{Cl}_2 \text{ } ^{37}\text{Cl}$	-2803.73	1.40	-2807.71	0.09	-1.9	U			H47	1.5	94Hy01
$^{132}\text{Xe}-\text{C}_3 \text{O}_6$	-65332.6117	0.0248	-65332.632	0.005	-0.8	o			FS1	1.0	05Sh38 *
	-65332.6238	0.0140			-0.6	1	15	14 ^{132}Xe	FS1	1.0	09Re03
$\text{C}_{10} \text{H}_{12}-^{132}\text{Ba}$	188863	70	188839.2	1.1	-0.2	U			R07	1.5	68De17
	188821	88			0.1	U			R07	1.5	68De17
$^{132}\text{La}-u$	-89874	67	-89880	40	-0.1	1	34	34 ^{132}La	GS2	1.0	05Li24 *
$^{132}\text{Ce}-u$	-88542	30	-88534	22	0.3	1	53	53 ^{132}Ce	GS2	1.0	05Li24
$^{132}\text{Ce O}-^{142}\text{Sm}_{1.042}$	-5258	32	-5267	22	-0.3	1	47	47 ^{132}Ce	MA7	1.0	01Bo59 *
$^{132}\text{Pr}-u$	-80760	31				2			GS2	1.0	05Li24 *
$^{132}\text{Nd}-u$	-76690	30	-76679	26	0.4	2			GS2	1.0	05Li24
$^{132}\text{Xe}-^{129}\text{Xe}_{1.023}$	1564.20	0.32	1564.2665	0.0020	0.2	U			CP1	1.0	09Sc19
	1565.4	1.0			-1.1	U			CP1	1.0	09Sc19
$^{132}\text{Sb}-^{130}\text{Xe}_{1.015}$	12445.7	2.9	12446.0	2.6	0.1	1	83	83 ^{132}Sb	JY1	1.0	12Ha25
$^{132}\text{Te}-^{130}\text{Xe}_{1.015}$	6482.9	4.3	6485	4	0.4	1	76	76 ^{132}Te	JY1	1.0	12Ha25
$^{132}\text{Sn}-^{133}\text{Cs}_{.992}$	11621	19	11615.6	2.1	-0.3	U			MA8	1.0	05Si34
	11613.1	2.9			0.8	-			CP1	1.0	13Va12
$^{132}\text{Sn } ^{34}\text{S}-^{133}\text{Cs}_{1.248}$	3686.3	7.7	3686.9	2.1	0.1	-			MA8	1.0	08Dw01
$^{132}\text{Sn}-^{133}\text{Cs}_{.992}$	ave. 11613.3	2.7	11615.6	2.1	0.8	1	61	61 ^{132}Sn			average
$^{132}\text{Sb}-^{133}\text{Cs}_{.992}$	8301.3	6.5	8299.7	2.6	-0.3	1	17	17 ^{132}Sb	CP1	1.0	13Va12
$^{132}\text{Cs}-^{133}\text{Cs}_{.992}$	223	18	229.4	1.1	0.4	o			MA1	1.0	90St25
	232	14			-0.2	U			MA1	1.0	99Am05
	246.9	5.9			-3.0	C			MA8	1.0	09Bo.A
	230.8	1.3			-1.1	1	73	73 ^{132}Cs	MA8	1.0	17At01
$^{132}\text{Nd}-^{133}\text{Cs}_{.992}$	17147	52	17113	26	-0.7	R			MA5	1.0	00Be42
$^{132}\text{Sn}-^{132}\text{Xe}$	13672.3	3.4	13668.8	2.1	-1.0	1	39	39 ^{132}Sn	JY1	1.0	12Ha25
$^{132}\text{Xe}-^{131}\text{Xe}$	-930	11	-929.0446	0.0017	0.0	U			M16	2.5	63Da10
$^{132}\text{Xe}-\text{C}_{10} \text{H}_{10}$	-174095.2367	0.0095	-174095.236	0.005	0.1	1	33	33 ^{132}Xe	FS1	1.0	09Re03
$^{132}\text{Xe}-^{130}\text{Xe}$	645.724	0.014	645.737	0.009	0.9	1	38	37 ^{130}Xe	FS1	1.0	09Re07
$^{132}\text{Ba}-^{130}\text{Ba}$	-1241	4	-1264.8	1.2	-2.4	U			M17	2.5	66Be10
	-1253	68			-0.1	U			R07	1.5	68De17
$^{132}\text{Xe}-^{129}\text{Xe}$	-625.7755	0.0156	-625.7740	0.0020	0.1	o			FS1	1.0	05Sh38 *
	-625.7703	0.0119			-0.3	-			FS1	1.0	09Re03 *
	-625.7732	0.0125			-0.1	-			FS1	1.0	09Re03 *
	-625.771	0.013			-0.2	-			FS1	1.0	09Re07
	-625.7771	0.0083			0.4	-			FS1	1.0	10Mo30
	ave. -625.774	0.005			-0.0	1	14	7 ^{132}Xe			average
$^{132}\text{Xe}_2-^{84}\text{Kr}_3$	73816.9775	0.0594	73816.986	0.014	0.1	U			FS1	1.0	05Sh38 *
$^{132}\text{Xe}_2-^{86}\text{Kr}_3$	76478.3099	0.0412	76478.293	0.013	-0.4	1	11	6 ^{86}Kr	FS1	1.0	05Sh38 *
$^{14}\text{N}_{10}-^{132}\text{Xe}$	126584.9632	0.0168	126584.959	0.006	-0.2	1	11	10 ^{132}Xe	FS1	1.0	09Re03
$^{131}\text{Cs}-^{132}\text{Cs}_{.794} \text{ } ^{127}\text{Cs}_{.206}$	-1118	16	-1090.9	1.4	0.7	F			P33	2.5	86Au02 *
$^{131}\text{Cs}-^{132}\text{Cs}_{.744} \text{ } ^{128}\text{Cs}_{.256}$	-1200	30	-1215.4	1.6	-0.2	U			P22	2.5	82Au01
$^{130}\text{Cs}^x-^{132}\text{Cs}_{.492} \text{ } ^{128}\text{Cs}_{.508}$	-210	40	-340	17	-1.3	U			P22	2.5	82Au01
$^{131}\text{Xe}(n,\gamma)^{132}\text{Xe}$	8936.3	1.0	8936.7176	0.0016	0.4	U					71Ge05
	8935	2			0.9	U					71Gr28
	8936.65	0.22			0.3	U			Bdn		06Fi.A
$^{132}\text{In}(\beta^-)^{132}\text{Sn}$	13600	400	14140	60	1.3	U					86Bj01
	14135	60				2			Stu		95Me16
$^{132}\text{Sn}(\beta^-)^{132}\text{Sb}$	3080	40	3089	3	0.2	o			Stu		77Al09
	3103	10			-1.4	o			Stu		95Me16
	3115	10			-2.6	U			Stu		99Fo01
$^{132}\text{Sb}(\beta^-)^{132}\text{Te}$	5530	70	5553	4	0.3	o			Stu		77Al09
	5486	24			2.8	U			Stu		95Me16
	5491	20			3.1	B			Stu		99Fo01
$^{132}\text{Te}(\beta^-)^{132}\text{I}$	493	4	515	3	5.6	B					65Iv01 *
	517	4			-0.4	1	76	52 ^{132}I	Stu		99Fo01 *
$^{132}\text{I}(\beta^-)^{132}\text{Xe}$	3596	15	3575	4	-1.4	-					61De17 *
	3558	15			1.2	-					65Jo13 *
	3580	7			-0.6	-			Stu		99Fo01
	ave. 3579	6			-0.6	1	48	48 ^{132}I			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{132}\text{I}^m(\beta^-)^{132}\text{Xe}$	3685	10				2					74Di03 *
$^{132}\text{Cs}(\beta^+)^{132}\text{Xe}$	2090	25	2126.3	1.0	1.5	U					63Ta05 *
	2127.7	6.			-0.2	U					87De33 *
$^{132}\text{La}(\beta^+)^{132}\text{Ba}$	4820	100	4710	40	-1.1	-					60Wa03
	4680	50			0.6	-					67Fr02
ave.	4710	40			0.1	1	66	66 ^{132}La			average
$*^{132}\text{Sb}-\text{u}$	$M-A=-79870(124)$ keV for mixture gs+m at 150#(50#) keV										Nub211 **
$*^{132}\text{Xe}-\text{C}_3 \text{ O}_6$	Corrected in reference of same group										09Re03 **
$*^{132}\text{La}-\text{u}$	$M-A=-83623(30)$ keV for mixture gs+m at 188.20 keV										Nub211 **
$*^{132}\text{Ce O}-^{142}\text{Sm}_{1.042}$	Original error (22 keV) increased by 23 for BaF contamination in trap										GAu **
$*^{132}\text{Pr}-\text{u}$	$M-A=-75213(28)$ keV for mixture gs+m at 30#30 keV										Nub211 **
$*^{132}\text{Xe}-^{129}\text{Xe}$	Corrected in reference of same group										09Re03 **
$*^{132}\text{Xe}-^{129}\text{Xe}$	First item 5^+ ions; second item 3^+ ions - considered to be independent										09Re03 **
$*^{132}\text{Xe}_2-^{84}\text{Kr}_3$	Corrected in reference of same group										09Re03 **
$*^{132}\text{Xe}_2-^{86}\text{Kr}_3$	Corrected in reference of same group										09Re03 **
$*^{131}\text{Cs}-^{132}\text{Cs}_{.794} \text{ }^{127}\text{Cs.}$	F : Rejection based on line-shape analysis										86Au02 **
$*^{132}\text{Te}(\beta^-)^{132}\text{I}$	$E_{\beta^-}=215(4) \text{ } 239(4)$ respectively, to 1^+ level at 277.86 keV										Ens054 **
$*^{132}\text{I}(\beta^-)^{132}\text{Xe}$	$E_{\beta^-}=2156(15) \text{ } 2118(15)$ respectively, to 4^+ level at 1440.323 keV										Ens054 **
$*^{132}\text{I}^m(\beta^-)^{132}\text{Xe}$	$E_{\beta^-}=1465(10)$ to 7^- level at 2214.01 level, and other E_{β^-}										Ens054 **
$*^{132}\text{Cs}(\beta^+)^{132}\text{Xe}$	$E_{\beta^+}=400(25)$ to 2^+ level at 667.715 keV										Ens054 **
$*^{132}\text{Cs}(\beta^+)^{132}\text{Xe}$	$p^+=0.0042(0.0001)$ gives $E_{\beta^+}=438(6)$ recalculated Q										AHW **
*	to 2^+ level at 667.715 keV										Ens054 **
$^{133}\text{Sb}-\text{u}$	-84766	100	-84728	3	0.2	o			GT2	2.5	08Kn.A
	-84795	129			0.2	U			GT2	2.5	08Su19
	-84702	25			-1.0	U			GS3	1.0	12Ch19
$^{133}\text{Te}-\text{u}$	-89031	43	-89036.7	2.2	-0.1	U			GR1	1.0	19An10
$^{133}\text{I}-\text{u}$	-92166	16	-92172	6	-0.4	1	16	16 ^{133}I	GR1	1.0	19An10
$\text{C}_{10} \text{ H}_{13}-^{133}\text{Cs}$	196266	64	196273.456	0.009	0.1	U			R07	1.5	68De17
	196279	25			-0.1	U			R07	1.5	68De17
$\text{C}_9 \text{ }^{13}\text{C H}_{12}-^{133}\text{Cs}$	191796	34	191803.260	0.009	0.1	U			R07	1.5	68De17
$\text{C}_8 \text{ O N H}_7-^{133}\text{Cs}$	147321	25	147311.888	0.009	-0.2	U			R07	1.5	68De17
$\text{C}_7 \text{ }^{13}\text{C N O H}_6-^{133}\text{Cs}$	142835	31	142841.692	0.009	0.1	U			R07	1.5	68De17
$^{133}\text{Cs}-^{85}\text{Rb}_{1.565}$	43500	13	43501.027	0.011	0.1	U			MA5	1.0	00Be42
	43499.3	1.6			1.1	U			MA8	1.0	07Ke09
	43500.9	6.7			0.0	U			MA8	1.0	02Ke.A
	43500.1	6.7			0.1	U			MA8	1.0	09Na.A
	43470	47			0.7	U			MA9	1.0	09Na.A
	43501.2	1.7			-0.1	U			MA8	1.0	11He10
$^{133}\text{Cs}-\text{u}$	-94548.41	0.41	-94548.041	0.009	0.9	U			ST2	1.0	99Ca46 *
$^{133}\text{La}-\text{u}$	-91810	120	-91780	30	0.2	U			GS1	1.0	00Ra23
	-91782	30				2			GS2	1.0	05Li24
$^{133}\text{Ce}-\text{u}$	-88471	32	-88480	18	-0.3	2			GS2	1.0	05Li24 *
$^{133}\text{Ce O}-^{142}\text{Sm}_{1.049}$	-4618	21	-4620	18	-0.1	R			MA7	1.0	01Bo59 *
$^{133}\text{Pr}-\text{u}$	-83663	30	-83669	13	-0.2	R			GS2	1.0	05Li24
$^{133}\text{Nd}-\text{u}$	-77652	50				2			GS2	1.0	05Li24 *
$^{133}\text{Pm}-\text{u}$	-70218	54				2			GS2	1.0	05Li24 *
$^{133}\text{Sb}-^{136}\text{Xe}_{.978}$	6022	10	6016	3	-0.6	1	11	11 ^{133}Sb	CP1	1.0	12Va02
$^{133}\text{Sb}-^{130}\text{Xe}_{1.023}$	13984.7	4.0	13982	3	-0.7	1	70	70 ^{133}Sb	JY1	1.0	12Ha25
$^{133}\text{Te}-^{130}\text{Xe}_{1.023}$	9672.1	2.3	9673.3	2.2	0.5	1	93	93 ^{133}Te	JY1	1.0	12Ha25
$^{133}\text{Sn}-^{134}\text{Xe}_{.993}$	17856.5	2.4	17858.5	2.0	0.8	1	73	73 ^{133}Sn	JY1	1.0	12Ha25
$^{133}\text{Sn }^{34}\text{S}-^{133}\text{Cs}_{1.256}$	10562	25	10533.1	2.0	-1.2	U			MA8	1.0	08Dw01
$^{133}\text{Sn}-^{133}\text{Cs}$	18467.0	3.9	18461.8	2.0	-1.3	1	27	27 ^{133}Sn	CP1	1.0	13Va12
$^{133}\text{Sb}-^{133}\text{Cs}$	9818	13	9820	3	0.2	U			CP1	1.0	13Va12
$^{133}\text{Te}-^{133}\text{Cs}$	5551.4	6.9	5511.4	2.2	-5.8	B			CP1	1.0	13Va12

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{133}\text{I}-^{133}\text{Cs}$	2375.4	6.9	2376	6	0.2	1	84	84 ^{133}I	CP1	1.0	13Va12
$^{133}\text{Pr}-^{133}\text{Cs}$	10877	15	10879	13	0.1	2			MA5	1.0	00Be42
$^{133}\text{Cs}-\text{C}_3\text{O}_6$	-64035.786	0.026	-64035.757	0.009	1.1	1	11	11 ^{133}Cs	MI2	1.0	99Br47
$^{133}\text{Cs}-\text{C}_{10}\text{H}_{12}$	-188448.445	0.057	-188448.424	0.009	0.4	U			MI2	1.0	99Br47
$^{133}\text{Cs}-^{132}\text{Xe}$	1296.8803	0.0103	1296.875	0.007	-0.5	1	47	45 ^{133}Cs	FS1	1.0	10Mo30
$^{133}\text{Cs}-^{129}\text{Xe}$	671.1007	0.0103	671.101	0.007	0.0	1	47	45 ^{133}Cs	FS1	1.0	10Mo30
$^{133}\text{Cs}(\gamma, n)^{132}\text{Cs}$	-8988	33	-8989.6	1.0	-0.0	U			Phi		60Ge01
	-8986	2			-1.8	1	27	27 ^{132}Cs	MMn		85Ts02
$^{132}\text{Ba}(n, \gamma)^{133}\text{Ba}$	7189.91	0.36	7189.9	0.4	0.0	1	100	98 ^{132}Ba	MMn		90Is07 Z
$^{132}\text{Ba}(d, p)^{133}\text{Ba}$	4977	15	4965.4	0.4	-0.8	U			ANL		70Vo04
$^{133}\text{Sn}(\beta^-)^{133}\text{Sb}$	7830	70	8050	4	3.1	B			Stu		83Bl16 *
	8013	50			0.7	o			Stu		92Sp.A *
	7990	25			2.4	U			Stu		95Me16
$^{133}\text{Sb}(\beta^-)^{133}\text{Te}$	3966	50	4014	4	1.0	o			Stu		70Ru.A *
	4003	10			1.1	o			Stu		95Me16
	4002	7			1.7	1	25	18 ^{133}Sb	Stu		99Fo01
$^{133}\text{Te}(\beta^-)^{133}\text{I}$	2960	100	2920	6	-0.4	U					68Mc09
	2876	100			0.4	U					68Pa03 *
	3392	100			-4.7	C			Stu		70Ru.A *
	2890	15			2.0	o			Stu		95Me16
	2942	24			-0.9	U			Stu		99Fo01
$^{133}\text{I}(\beta^-)^{133}\text{Xe}$	1800	50	1786	6	-0.3	U					59Ho97 *
	1760	30			0.9	U					66Ei01 *
	1757	4			7.3	B			Stu		99Fo01
$^{133}\text{Xe}(\beta^-)^{133}\text{Cs}$	428.0	4.	427.4	2.4	-0.2	2					52Be55 *
	427.0	3.			0.1	2					61Er04 *
	424	11			0.3	U			Stu		99Fo01
$^{133}\text{Ba}(\epsilon)^{133}\text{Cs}$	517.3	1.0	517.4	1.0	0.1	1	98	98 ^{133}Ba			67Sc10 *
	498	5			3.9	F					68Mc06 *
	486	2			15.7	F					69Bo49 *
	521	5			-0.7	U					69To14 *
$^{133}\text{La}(\beta^+)^{133}\text{Ba}$	2230	200	2059	28	-0.9	U					50Na09 *
$^{133}\text{Nd}(\beta^+)^{133}\text{Pr}$	4800	200	5610	50	4.0	B					95Br24
* $^{133}\text{Cs}-\text{u}$	As revised in reference. Original: -94548.20(0.28) μu										02Bf02 **
* $^{133}\text{Ce}-\text{u}$	$M-A=-82392(28)$ keV for mixture gs+m at 37.2 keV										Nub211 **
* $^{133}\text{Ce O}-^{142}\text{Sm}_{1.049}$	$D_M=-4599(16)$ μu for mixture gs+m at 37.2 keV; $M-A=-87150(16)$ keV										Nub211 **
* $^{133}\text{Nd}-\text{u}$	$M-A=-72268(28)$ keV for mixture gs+m at 127.97 keV										Nub211 **
* $^{133}\text{Pm}-\text{u}$	$M-A=-65342(33)$ keV for mixture gs+m at 129.7 keV										Nub211 **
* $^{133}\text{Sn}(\beta^-)^{133}\text{Sb}$	$E_{\beta^-}=6870(70)$ to $5/2^+$ level at 962.30 keV										Ens114 **
* $^{133}\text{Sn}(\beta^-)^{133}\text{Sb}$	Private communication to reference										92Ch09 **
* $^{133}\text{Sb}(\beta^-)^{133}\text{Te}$	$E_{\beta^-}=1210(50)$ to $(5/2^+)$ level at 2755.51 keV; re-evaluated										Ens114 **
* $^{133}\text{Te}(\beta^-)^{133}\text{I}$	$Q_{\beta^-}=3210(100)$ from $^{133}\text{Te}^m$ at 334.26 keV										Nub211 **
*	reported as belonging to ground state, reinterpreted										AHW **
* $^{133}\text{Te}(\beta^-)^{133}\text{I}$	$E_{\beta^-}=850(100)$ to $(3/2^+, 5/2^+)$ level at 2541.74 keV										Ens114 **
* $^{133}\text{I}(\beta^-)^{133}\text{Xe}$	$E_{\beta^-}=1270(50)$ 1230(30) respectively, to $5/2^+$ level at 529.872 keV										Ens114 **
* $^{133}\text{Xe}(\beta^-)^{133}\text{Cs}$	$E_{\beta^-}=347(4)$ 346(3) respectively, to $5/2^+$ level at 80.9979 keV										Ens114 **
* $^{133}\text{Ba}(\epsilon)^{133}\text{Cs}$	From $L/K=0.371(0.007)$ to $1/2^+$ level at 437.0113; recalculated Q										Ens114 **
*	and $L/K=0.221(0.005)$ to $3/2^+$ level at 383.8491 keV: $Q=521(5)$ keV										Ens114 **
* $^{133}\text{Ba}(\epsilon)^{133}\text{Cs}$	F : badly resolved L-peak										Ens114 **
* $^{133}\text{Ba}(\epsilon)^{133}\text{Cs}$	$L/K=0.67(0.15)$ $LM/K=1.11(0.05)$ 0.45(0.04) respectively, to $1/2^+$ 437.0113 keV										Ens114 **
* $^{133}\text{La}(\beta^+)^{133}\text{Ba}$	$E_{\beta^+}=1200(200)$ to $3/2^+$ level at 12.327 keV										Ens114 **
$^{134}\text{Sb}-\text{u}$	-79351	131	-79463	3	-0.9	U			GR1	1.0	19An10
$^{134}\text{Te}-\text{u}$	-88844	130	-88603.6	2.9	0.7	U			GT2	2.5	08Su19
	-88614	44			0.2	U			GR1	1.0	19An10

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{134}\text{I}-\text{u}$	-90244	58	-90224	5	0.3	U			GR1	1.0	19An10
$\text{C}_{10}\text{H}_{14}-^{134}\text{Xe}$	204155.5	3.2	204157.416	0.006	0.2	U			M16	2.5	63Da10
$^{134}\text{Xe}-\text{u}$	-94634.4	5.4	-94606.970	0.006	2.0	U			ACC	2.5	90Me08
$^{134}\text{Xe}-\text{C } ^{13}\text{C } ^{35}\text{Cl } ^{37}\text{Cl}_2$	1381.76	0.60	1380.35	0.12	-1.6	U			H47	1.5	94Hy01
$\text{C}_{10}\text{H}_{14}-^{134}\text{Ba}$	205025	20	205042.20	0.27	0.3	U			M17	2.5	66Be10
	205010	46			0.5	U			R07	1.5	68De17
$\text{C}_{11}\text{H}_2-^{134}\text{Ba}$	111125	48	111141.81	0.27	0.2	U			R07	1.5	68De17
$\text{C}_8\text{N O H}_8-^{134}\text{Ba}$	156063	78	156080.63	0.27	0.2	U			R07	1.5	68De17
$\text{C}_{12}\text{H}_6-^{134}\text{Ba O}$	147531	64	147527.32	0.27	-0.0	U			R07	1.5	68De17
$^{134}\text{La}-\text{u}$	-91456	34	-91486	21	-0.9	2			GS2	1.0	05Li24
$^{134}\text{Ce}-\text{u}$	-91190	130	-91072	22	0.9	U			GS1	1.0	00Ra23
	-91056	30			-0.5	2			GS2	1.0	05Li24
$^{134}\text{Ce O}-^{142}\text{Sm}_{1.056}$	-6631	32	-6618	22	0.4	R			MA7	1.0	01Bo59 *
$^{134}\text{Pr}-\text{u}$	-84285	37	-84303	22	-0.5	2			GS2	1.0	05Li24 *
$^{134}\text{Nd}-\text{u}$	-81234	30	-81210	13	0.8	R			GS2	1.0	05Li24
$^{134}\text{Pm}-\text{u}$	-71674	45				2			GS2	1.0	05Li24 *
$^{134}\text{Sb}-^{130}\text{Xe}_{1.031}$	20016.8	2.2	20019	3	1.1	o			JY1	1.0	12Ha25
	20019.2	3.3				2			JY1	1.0	13Ka08 *
$^{134}\text{Te}-^{130}\text{Xe}_{1.031}$	10877.1	3.5	10878.2	2.9	0.3	1	71	71 ^{134}Te	JY1	1.0	12Ha25
$^{134}\text{Sb}-^{136}\text{Xe}_{.985}$	11906	36	11931	3	0.7	U			CP1	1.0	12Va02 *
$^{134}\text{Te}-^{136}\text{Xe}_{.985}$	2791.4	6.5	2790.1	2.9	-0.2	1	21	21 ^{134}Te	CP1	1.0	12Va02
$^{134}\text{Xe}-^{132}\text{Xe}_{1.015}$	2675.6193	0.0084	2675.6191	0.0028	-0.0	U			FS1	1.0	13Ho22
	2675.6191	0.0028			0.0	1	100	100 ^{134}Xe	Hep	1.0	20Ri04
$^{134}\text{Sn } ^{34}\text{S}-^{133}\text{Cs}_{1.263}$	16080	160	15962	3	-0.7	U			MA8	1.0	08Dw01
$^{134}\text{Sn}-^{133}\text{Cs}_{1.008}$	23974	17	23985	3	0.6	U			CP1	1.0	13Va12
$^{134}\text{Sb}-^{133}\text{Cs}_{1.008}$	15850	11	15842	3	-0.7	U			CP1	1.0	13Va12
$^{134}\text{I}-^{133}\text{Cs}_{1.008}$	5083.0	6.8	5080	5	-0.4	1	59	59 ^{134}I	CP1	1.0	13Va12
$^{134}\text{Cs}-^{133}\text{Cs}_{1.008}$	2025	18	2022.929	0.015	-0.1	o			MA1	1.0	90St25
	2033	14			-0.7	U			MA1	1.0	99Am05
$^{134}\text{Pr}-^{133}\text{Cs}_{1.008}$	10992	27	11001	22	0.3	R			MA5	1.0	00Be42 *
$^{134}\text{Nd}-^{133}\text{Cs}_{1.008}$	14100	14	14095	13	-0.4	2			MA5	1.0	00Be42
$^{134}\text{Sn}-^{134}\text{Xe}$	23287.4	3.4				2			JY1	1.0	12Ha25
$^{134}\text{Ba}-\text{C}_{10}\text{H}_{13}$	-197229	20	-197217.16	0.27	0.2	U			M17	2.5	66Be10
$^{134}\text{Ba}-^{132}\text{Ba}$	-553	4	-553.0	1.2	0.0	U			M17	2.5	66Be10
	-550	121			-0.0	U			R07	1.5	68De17
$^{131}\text{Cs}-^{134}\text{Cs}_{.244} \text{ } ^{130}\text{Cs}_{.756}$	-1313	50	-1178	13	1.1	U			P22	2.5	82Au01 *
$^{133}\text{Cs}(n,\gamma)^{134}\text{Cs}$	6891.540	0.017	6891.540	0.014	0.0	-			MMn		84Ke11 Z
	6891.540	0.027			0.0	-			ILn		87Bo24 Z
	6891.39	0.14			1.1	U			Bdn		06Fi.A
ave.	6891.540	0.014			0.0	1	100	100 ^{134}Cs			average
$^{134}\text{Sn}(\beta^-)^{134}\text{Sb}$	7370	90	7585	4	2.4	U			Stu		95Me16
$^{134}\text{Sb}(\beta^-)^{134}\text{Te}$	8306	210	8515	4	1.0	o			Stu		72Ke21 *
	7960	240			2.3	o			Stu		77Lu06 *
	8310	80			2.6	U			Bwg		79Ke02 *
	8390	45			2.8	U			Stu		95Me16
$^{134}\text{Te}(\beta^-)^{134}\text{I}$	1560	90	1510	5	-0.6	U			Stu		77Lu06 *
	1550	30			-1.3	U			Stu		95Me16
	1513	7			-0.5	1	50	41 ^{134}I	Stu		99Fo01
$^{134}\text{I}(\beta^-)^{134}\text{Xe}$	4170	60	4082	5	-1.5	U					61Jo08 *
	4175	15			-6.2	B			Stu		95Me16
	4052	8			3.8	B			Stu		99Fo01
$^{134}\text{Cs}(\beta^-)^{134}\text{Ba}$	2058.6	0.4	2058.84	0.25	0.6	1	39	39 ^{134}Ba			68Hs01 *
$^{134}\text{La}(\beta^+)^{134}\text{Ba}$	3772	50	3731	20	-0.8	R					65Bi12
	3692	30			1.3	R					73Al20
$^{134}\text{Ce}(\epsilon)^{134}\text{La}$	500	200	386	29	-0.6	U					76Gr09 *
$^{134}\text{Pr}(\beta^+)^{134}\text{Ce}$	6190	90	6305	29	1.3	U			Dbn		95Ve08 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{134}\text{Nd}(\beta^+)^{134}\text{Pr}$	2770	150	2882	24	0.7	U					77Ko.B
$^{134}\text{Pm}(\beta^+)^{134}\text{Nd}$	9170	200	8880	40	-1.4	U			Dbn		95Ve08 *
$^{134}\text{Ce O}-^{142}\text{Sm}_{1.056}$	Original error (22 keV) increased by 23 for BaF contamination in trap										GAu **
$^{134}\text{Pr}-\text{u}$	$M-A=-78477(28)$ keV for mixture gs+m at 67.7(0.4) keV										Nub211 **
$^{134}\text{Pm}-\text{u}$	$M-A=-66739(30)$ keV for mixture gs+m at 50#(50#) keV										Nub211 **
$^{134}\text{Sb}-^{130}\text{Xe}_{1.031}$	$D_M=20318.7(3.1)$ μu for $^{134}\text{Sb}^m$ at 279(1) keV; $M-A=-73740.0(2.9)$ keV										Nub211 **
$^{134}\text{Sb}-^{136}\text{Xe}_{.985}$	$D_M=12206(36)$ μu for $^{134}\text{Sb}^m$ at 279(1) keV; $M-A=-73762(33)$ keV assuming high spin is favored, as for $^{136}\text{I}^m$										Nub211 **
$^{134}\text{Pr}-^{133}\text{Cs}_{1.008}$	Most certainly gs. Mixture with isomer not completely excluded										GAu **
$^{134}\text{Pr}-^{133}\text{Cs}_{1.008}$	$D_M=11029(16)$ μu for mixture gs+m at 67.7(0.4) keV; $M-A=-78503(15)$ keV										00Be42 **
$^{131}\text{Cs}-^{134}\text{Cs}_{.244}$ $^{130}\text{Cs}^x$	$D_M=-1330(50)$ keV for mixture gs+m at 138.7441 keV										Nub211 **
$^{134}\text{Sb}(\beta^-)^{134}\text{Te}$	$E_{\beta^-}=8400(300)$, and 6800(300) from $^{134}\text{Sb}^m$ at 279(1) to $^{134}\text{Te}^m$ at 1691.34										Nub211 **
$^{134}\text{Sb}(\beta^-)^{134}\text{Te}$	$E_{\beta^-}=5840(240)$ from $^{134}\text{Sb}^m$ at 279(1) to $(6)^+$ level at 2397.70 keV										Ens04a **
$^{134}\text{Sb}(\beta^-)^{134}\text{Te}$	$E_{\beta^-}=8420(120)$, and 6710(210), 6980(210), 6070(150) from $^{134}\text{Sb}^m$ at 279(1) to 6^+ levels at 1691.34, 1691.34, 2397.70 keV										Nub211 **
$^{134}\text{Te}(\beta^-)^{134}\text{I}$	$E_{\beta^-}=730(110)$ 610(160) to 1^+ levels at 846.688 923.432 keV										Ens04a **
$^{134}\text{I}(\beta^-)^{134}\text{Xe}$	$E_{\beta^-}=2410(60)$ 1680(60) 1250(60) to 4^+ levels at 1731.17 2588.46 2867.38 keV										Ens04a **
$^{134}\text{Cs}(\beta^-)^{134}\text{Ba}$	$E_{\beta^-}=658.0(0.4)$ to 4^+ level at 1400.590 keV										Ens04a **
$^{134}\text{Ce}(\epsilon)^{134}\text{La}$	LK=0.798(0.04); also $Q_{\beta^+} > 375$ keV										76Gr09 **
$^{134}\text{Pr}(\beta^+)^{134}\text{Ce}$	$E_{\beta^+}=4120(90)$ to 4^+ level at 1048.68 keV										Ens04a **
$^{134}\text{Pm}(\beta^+)^{134}\text{Nd}$	$E_{\beta^+}=7360(200)$ to 4^+ level at 788.92 keV										Ens04a **
$^{135}\text{Sb}-\text{u}$	-74932	103	-74815.6	2.8	0.5	o			GT2	2.5	08Kn.A
	-74943	130			0.4	U			GT2	2.5	08Su19
$^{135}\text{Te}-\text{u}$	-83643	106	-83445.3	1.8	0.7	U			GT2	2.5	08Kn.A
	-83441	132			-0.0	U			GT2	2.5	08Su19
$\text{C}_8 \text{ N O H}_9-^{135}\text{Ba}$	162731	48	162725.46	0.26	-0.1	U			R07	1.5	68De17
$\text{C}_{11} \text{ H}_3-^{135}\text{Ba}$	117822	77	117786.65	0.26	-0.3	U			R07	1.5	68De17
$\text{C}_{12} \text{ H}_7-^{135}\text{Ba O}$	154160	46	154172.16	0.26	0.2	U			R07	1.5	68De17
$^{135}\text{Ce}-\text{u}$	-90779	30	-90839	11	-2.0	1	13	^{135}Ce	GS2	1.0	05Li24 *
$^{135}\text{Pr}-\text{u}$	-86897	30	-86888	13	0.3	R			GS2	1.0	05Li24
$^{135}\text{Nd}-\text{u}$	-81800	130	-81819	21	-0.1	o			GS1	1.0	00Ra23
	-81811	36			-0.2	R			GS2	1.0	05Li24 *
$^{135}\text{Pm}-\text{u}$	-75215	89				2			GS2	1.0	05Li24 *
$^{135}\text{Sm}-\text{u}$	-67480	166				2			GS2	1.0	05Li24 *
$^{135}\text{Sn}-^{130}\text{Xe}_{1.038}$	35065.9	3.3				2			JY1	1.0	12Ha25
$^{135}\text{Sb}-^{130}\text{Xe}_{1.038}$	25342.4	3.1	25341.7	2.8	-0.2	1	84	^{135}Sb	JY1	1.0	12Ha25
$^{135}\text{Te}-^{130}\text{Xe}_{1.038}$	16713.0	2.9	16712.0	1.8	-0.3	1	41	^{135}Te	JY1	1.0	12Ha25
$^{135}\text{Sn}-^{133}\text{Cs}_{1.015}$	30927	37	30875	3	-1.4	U			CP1	1.0	13Va12
$^{135}\text{Sb}-^{133}\text{Cs}_{1.015}$	21146.8	7.0	21150.6	2.8	0.5	1	16	^{135}Sb	CP1	1.0	13Va12
$^{135}\text{Te}-^{133}\text{Cs}_{1.015}$	12520.3	2.4	12521.0	1.8	0.3	1	59	^{135}Te	CP1	1.0	13Va12
$^{135}\text{I}-^{133}\text{Cs}_{1.015}$	6026.6	2.3	6025.6	2.2	-0.4	1	92	^{135}I	CP1	1.0	13Va12
$^{135}\text{Cs}-^{133}\text{Cs}_{1.015}$	1958	18	1943.2	0.4	-0.8	o			MA1	1.0	90St25
	1957	14			-1.0	U			MA1	1.0	99Am05
$^{135}\text{Pr}-^{133}\text{Cs}_{1.015}$	9080	14	9078	13	-0.1	2			MA5	1.0	00Be42
$^{135}\text{Nd}-^{133}\text{Cs}_{1.015}$	14144	25	14148	21	0.1	2			MA5	1.0	00Be42 *
$^{135}\text{Te}-^{136}\text{Xe}_{.993}$	8686	10	8690.7	1.8	0.5	U			CP1	1.0	12Va02
$^{135}\text{I}-^{136}\text{Xe}_{.993}$	2186.3	8.3	2195.4	2.2	1.1	U			CP1	1.0	12Va02
$^{135}\text{Cs}-^{135}\text{Ba}$	288.43	0.32	288.5	0.3	0.1	1	92	^{135}Cs	JY2	1.0	20De20
$^{135}\text{Ba}-\text{C}_{10} \text{ H}_{14}$	-203860	20	-203862.00	0.26	-0.0	U			M17	2.5	66Be10
$^{135}\text{Ba}-^{134}\text{Ba}$	1177	2	1180.20	0.11	0.6	U			M17	2.5	66Be10
	1161	70			0.2	U			R07	1.5	68De17
	1168	78			0.1	U			R07	1.5	68De17
$^{134}\text{Cs}(\text{n},\gamma)^{135}\text{Cs}$	8762	1	8762.1	0.4	0.1	1	13	^{135}Cs	ILn		92Ul.A
$^{134}\text{Ba}(\text{n},\gamma)^{135}\text{Ba}$	6973.2	0.4	6971.97	0.10	-3.1	C			BNn		77Ko.A
	6972.17	0.18			-1.1	-			MMn		90Is07 Z
	6971.84	0.17			0.8	-			Ltn		93Bo01 Z
	6973.24	0.22			-5.8	B			BNn		93Ch21
	6971.87	0.18			0.6	-			Bdn		06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{134}\text{Ba}(\text{d,p})^{135}\text{Ba}$		4746	15	4747.40	0.10	0.1	U		ANL		70Vo04
$^{134}\text{Ba}(\text{n},\gamma)^{135}\text{Ba}$	ave.	6971.96	0.10	6971.97	0.10	0.2	1	96	61 ^{134}Ba		average
$^{135}\text{Tb}(\text{p})^{134}\text{Gd}$		1188	7				3		Arp		04Wo07 *
$^{135}\text{Sb}(\beta^-)^{135}\text{Te}$		8120	50	8038	3	-1.6	U		Stu		89Ho08
$^{135}\text{Te}(\beta^-)^{135}\text{I}$		5950	240	6050.4	2.7	0.4	o		Stu		77Lu06
		5950	100			1.0	o		Bwg		79Ke02 *
		5970	200			0.4	U				85Sa15 *
		5960	100			0.9	U		Bwg		87Gr.A
		5888	13			12.5	B		Stu		07Fo02
$^{135}\text{I}(\beta^-)^{135}\text{Xe}$		2780	80	2634	4	-1.8	U				70Ma19
		2590	50			0.9	U		Stu		76Lu04 *
		2627	6			1.2	1	41	33 ^{135}Xe		99Fo01
$^{135}\text{Xe}(\beta^-)^{135}\text{Cs}$		1155	10	1169	4	1.4	-				52Be55 *
		1167	5			0.3	-		Stu		99Fo01 *
	ave.	1165	4			0.9	1	67	67 ^{135}Xe		average
$^{135}\text{Cs}(\beta^-)^{135}\text{Ba}$		205	5	268.70	0.29	12.7	B				53Li01
$^{135}\text{La}(\beta^+)^{135}\text{Ba}$		1200	10	1207	9	0.7	1	89	89 ^{135}La		71Ba18 *
$^{135}\text{Ce}(\beta^+)^{135}\text{La}$		2027	5	2027	5	0.0	-				76Ga.A *
		2016	13			0.9	-				81Sa09 *
	ave.	2026	5			0.3	1	98	87 ^{135}Ce		average
$^{135}\text{Pr}(\beta^+)^{135}\text{Ce}$		3720	150	3680	16	-0.3	U				54Ha68 *
$^{135}\text{Pm}^m(\beta^+)^{135}\text{Nd}$		6040	150	6390#	50#	2.3	D		Dbn		95Ve08 *
* $^{135}\text{Ce}-\text{u}$	$M-A=-84114(28)$ keV for $^{135}\text{Ce}^m$ at 445.81 keV										Nub211 **
* $^{135}\text{Nd}-\text{u}$	$M-A=-76174(28)$ keV for mixture gs+m at 64.95 keV										Nub211 **
* $^{135}\text{Pm}-\text{u}$	$M-A=-69952(28)$ keV for mixture gs+m at 220#90 keV										Nub211 **
* $^{135}\text{Sm}-\text{u}$	$M-A=-62857(38)$ keV for mixture gs+m at 0#300 keV										Nub211 **
* $^{135}\text{Nd}-^{133}\text{Cs}_{1.015}$	$D_M=14179(14)$ μu for gs+m mixture at 64.95 keV; $M-A=-76185(13)$ keV										Nub211 **
* $^{135}\text{Tb}(\text{p})^{134}\text{Gd}$	Trends from Mass Surface TMS suggest ^{135}Tb 300 keV less bound										GAu **
* $^{135}\text{Te}(\beta^-)^{135}\text{I}$	$E_{\beta^-}=5120(120)$ to $5/2^-$ level at 870.52 keV and other E_{β^-}										Ens083 **
* $^{135}\text{Te}(\beta^-)^{135}\text{I}$	$E_{\beta^-}=5370(100)$ to $5/2^+$ level at 603.68 keV										Ens083 **
* $^{135}\text{I}(\beta^-)^{135}\text{Xe}$	$E_{\beta^-}=1320(50)$ to $5/2^+$ level at 1260.416 keV, and other E_{β^-}										Ens083 **
* $^{135}\text{Xe}(\beta^-)^{135}\text{Cs}$	$E_{\beta^-}=905(10)$ 917(5) respectively, to $5/2^+$ level at 249.767 keV										Ens083 **
* $^{135}\text{La}(\beta^+)^{135}\text{Ba}$	$p^+=7(1)\times 10^{-5}$, from reanalysis										AHW **
* $^{135}\text{La}(\beta^+)^{135}\text{Ba}$	But 65Mo05 says $p^+<0.002\%$ (2×10^{-5}) or $E_{\beta^+}<125$ keV										65Mo05 **
* $^{135}\text{Ce}(\beta^+)^{135}\text{La}$	$E_{\beta^+}=705(5)$ 694(13) respectively, to $1/2^+$ level at 300.052 keV										Ens083 **
* $^{135}\text{Pr}(\beta^+)^{135}\text{Ce}$	$E_{\beta^+}=2500(100)$ to levels ($5/2^+$) 296.11 and $3/2^+$ 82.67 keV, roughly equal										Ens083 **
* $^{135}\text{Pm}^m(\beta^+)^{135}\text{Nd}$	$E_{\beta^+}=4920(150)$ to mixture ground state and ($11/2^-$) level at 198.5 keV										Ens083 **
* $^{135}\text{Pm}^m(\beta^+)^{135}\text{Nd}$	Trends from Mass Surface TMS suggest $^{135}\text{Pm}^m$ 350 keV less bound										GAu **
$^{136}\text{Te}-\text{u}$		-79945	25	-79898.8	2.4	1.8	U				12Ch19
$^{136}\text{Xe}-^{120}\text{Sn}_{1.133}$		18019.8	3.1	18019.0	1.1	-0.3	U				JY1 1.0 11Ha48
$\text{C}_{10}\text{H}_{16}-^{136}\text{Xe}$		217982.	3.9	217986.036	0.007	0.4	U				M16 2.5 63Da10
$^{136}\text{Xe}-\text{u}$		-92793.6	9.0	-92785.526	0.007	0.4	U				ACC 2.5 90Me08
$^{136}\text{Xe}-^{28}\text{Si}_4\text{D}_{12}$		-169712.9828	0.0162	-169712.998	0.007	-0.9	1	20	19 ^{136}Xe		FS1 1.0 07Re03
$\text{C}_{11}\text{H}_4-^{136}\text{Ba}$		126737	56	126724.33	0.26	-0.2	U				R07 1.5 68De17
$\text{C}_8\text{N O H}_{10}-^{136}\text{Ba}$		171635	56	171663.14	0.26	0.3	U				R07 1.5 68De17
$\text{C}_{12}\text{H}_8-^{136}\text{Ba O}$		163094	40	163109.84	0.26	0.3	U				R07 1.5 68De17
$^{136}\text{La}-\text{u}$		-92394	88	-92370	60	0.3	2		GS2	1.0	05Li24 *
$\text{C}_{10}\text{H}_{16}-^{136}\text{Ce}$		218128	50	218071.3	0.3	-0.3	U				R05 4.0 65De13
$\text{C}_{12}\text{H}_8-^{136}\text{Ce O}$		160563	36	160556.4	0.3	-0.0	U				R05 4.0 65De13
$^{136}\text{Nd}-\text{u}$		-85044	30	-85024	13	0.7	R				GS2 1.0 05Li24
$^{136}\text{Pm}-\text{u}$		-76378	79	-76400	70	-0.3	2		GS2	1.0	05Li24 *
$^{136}\text{Sm}-\text{u}$		-71768	30	-71724	13	1.5	R				GS2 1.0 05Li24
$^{136}\text{Sb}-^{130}\text{Xe}_{1.046}$		31675.1	6.8	31678	6	0.5	1	85	85 ^{136}Sb		JY1 1.0 12Ha25
$^{136}\text{Te}-^{130}\text{Xe}_{1.046}$		21029.9	3.1	21030.4	2.4	0.2	1	62	62 ^{136}Te		JY1 1.0 12Ha25

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference		
$^{136}\text{Sb}-^{133}\text{Cs}_{1.023}$	27489	16	27472	6	-1.1	1	15	15	^{136}Sb	CP1	1.0	13Va12	
$^{136}\text{Te}-^{133}\text{Cs}_{1.023}$	16827.7	6.8	16823.8	2.4	-0.6	1	13	13	^{136}Te	CP1	1.0	13Va12	
$^{136}\text{Xe}-^{133}\text{Cs}_{1.023}$	3936.5	1.9	3937.121	0.011	0.3	U			MA8	1.0	09Ne11		
$^{136}\text{Cs}-^{133}\text{Cs}_{1.023}$	4007	18	4034.1	2.0	1.5	o			MA1	1.0	90St25		
	4021	14			0.9	U			MA1	1.0	99Am05		
$^{136}\text{Pr}-^{133}\text{Cs}_{1.023}$	9418	15	9400	12	-1.2	1	67	67	^{136}Pr	MA5	1.0	00Be42	
$^{136}\text{Nd}-^{133}\text{Cs}_{1.023}$	11703	14	11699	13	-0.3	2			MA5	1.0	00Be42		
$^{136}\text{Pm}^m-^{133}\text{Cs}_{1.023}$	20429	100				2			MA5	1.0	00Be42	*	
$^{136}\text{Sm}-^{133}\text{Cs}_{1.023}$	25009	15	24998	13	-0.7	2			MA5	1.0	00Be42		
$^{136}\text{Xe}-^{134}\text{Xe}_{1.015}$	3245.8	3.8	3240.547	0.009	-1.4	o			MA8	1.0	05He.A		
	3244.3	4.0			-0.9	U			MA8	1.0	06He29		
$^{136}\text{Te}-^{136}\text{Xe}$	12887.9	5.0	12886.7	2.4	-0.2	1	24	24	^{136}Te	CP1	1.0	12Va02	
$^{136}\text{I}^m-^{136}\text{Xe}$	7611.2	4.9				2			CP1	1.0	12Va02	*	
$^{136}\text{Xe}-^{136}\text{Ba}$	2639.6	0.6	2638.67	0.26	-0.6	-			H49	2.5	10Mc04		
	2638.62	0.52			0.1	-			JY1	1.0	11Ko03		
ave.	2638.7	0.5			-0.1	1	29	29	^{136}Ba			average	
$^{136}\text{Ce}-^{136}\text{Ba}$	2553.46	0.29	2553.46	0.23	-0.0	-			JY1	1.0	11Ko03		
	2553.42	0.37			0.1	-			SH1	1.0	12Ne10		
ave.	2553.44	0.23			0.0	1	100	100	^{136}Ce			average	
$^{136}\text{Xe}-^{13}\text{C}_3\text{O}_6$	-72337.7553	0.0109	-72337.747	0.007	0.7	-			FS1	1.0	07Re03		
	-72337.7464	0.0109			-0.1	-			FS1	1.0	07Re03		
ave.	-72337.751	0.008			0.4	1	82	81	^{136}Xe			average	
$^{136}\text{Ba}-^{135}\text{Ba}$	-1115	3	-1112.65	0.04	0.3	U			M17	2.5	66Be10		
	-1119	50			0.1	U			R07	1.5	68De17		
	-1074	50			-0.5	U			R07	1.5	68De17		
$^{136}\text{Ba}-^{134}\text{Ba}$	67	5	67.55	0.11	0.0	U			M17	2.5	66Be10		
	69	128			-0.0	U			R07	1.5	68De17		
	72	78			-0.0	U			R07	1.5	68De17		
$\text{N}_{10}-^{136}\text{Xe}$	123525.5778	0.0235	123525.568	0.007	-0.4	U			FS1	1.0	07Re03		
$^{136}\text{Te}(\beta^-n)^{135}\text{I}$	1285	50	1283	3	-0.0	U						84Kr.B	
$^{136}\text{Xe}(d,^3\text{He})^{135}\text{I}$	-4438	30	-4445.5	2.1	-0.2	U			Oak			71Wi04	
$^{136}\text{Xe}(d,t)^{135}\text{Xe}$	-1723	40	-1830	4	-2.7	U			Oak			68Mo21	
$^{135}\text{Ba}(n,\gamma)^{136}\text{Ba}$	9106.4	0.8	9107.74	0.04	1.7	U						69Ge07	
	9107.74	0.04			0.1	-			MMn			90Is07	
	9107.73	0.19			0.1	-			Bdn			06Fi.A	
$^{135}\text{Ba}(d,p)^{136}\text{Ba}$	6886	15	6883.18	0.04	-0.2	U			ANL			70Vo04	
$^{135}\text{Ba}(n,\gamma)^{136}\text{Ba}$	9107.74	0.04	9107.74	0.04	0.1	1	99	59	^{135}Ba			average	
$^{136}\text{Te}(\beta^-)^{136}\text{I}$	5100	150	5120	14	0.1	U						77Sc21	
	5095	100			0.2	U			Bwg			87Gr.A	
	5086	20			1.7	1	50	50	^{136}I	Stu		07Fo02	
$^{136}\text{I}(\beta^-)^{136}\text{Xe}$	6960	100	6884	14	-0.8	U						59Jo37	*
	6690	150			1.3	U			Stu			76Lu04	*
	6925	70			-0.6	U			Bwg			87Gr.A	
	6850	20			1.7	1	50	50	^{136}I	Stu		07Fo02	
$^{136}\text{I}^m(\beta^-)^{136}\text{Xe}$	7100	230	7090	5	-0.0	o			Stu			76Lu04	*
	7705	120			-5.1	C			Bwg			87Gr.A	
	7051	12			3.2	B			Stu			07Fo02	
$^{136}\text{Cs}(\beta^-)^{136}\text{Ba}$	2548.1	2.0	2548.2	1.9	0.1	2						54OI05	*
	2549	5			-0.2	2						65Re07	*
$^{136}\text{La}(\beta^+)^{136}\text{Ba}$	2870	70	2850	50	-0.3	R						59Gi50	
$^{136}\text{Pr}(\beta^+)^{136}\text{Ce}$	5084	50	5168	11	1.7	U						68Zh04	*
	5114	75			0.7	U						71Ke07	*
	5134	20			1.7	1	33	33	^{136}Pr	IRS		83Al.B	*
$^{136}\text{Nd}(\beta^+)^{136}\text{Pr}$	2501	50	2141	16	-7.2	B						68Zh04	*
	2211	25			-2.8	B						75Br16	*
$^{136}\text{Pm}(\beta^+)^{136}\text{Nd}$	7850	200	8030	70	0.9	R			IRS			83Al06	*
$^{136}\text{La}-u$	$M-A=-85935(32)$ keV for mixture gs+m at 259.3 keV										Nub211	**	
$^{136}\text{Pm}-u$	$M-A=-71091(28)$ keV for mixture gs+m at 110(120) keV										Nub211	**	
$^{136}\text{Pm}^m-^{133}\text{Cs}_{1.023}$	Slightly contaminated by ground state, original error (20) increased										00Be42	**	
$^{136}\text{I}^m-^{136}\text{Xe}$	High spin isomer is preferred (see also ^{134}Sb)										GAu	**	
$^{136}\text{I}(\beta^-)^{136}\text{Xe}$	$E_{\beta^-}=7000(100)$, 5610(150), 4280(150) to ground state, 2^+ levels 1313.027, 2634.16 keV										Ens026	**	
$^{136}\text{I}(\beta^-)^{136}\text{Xe}$	$E_{\beta^-}=5370(400)$, 4700(320), 3920(220) to 2^+ levels 1313.027, 2289.53, 2634.16										Ens026	**	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
* $^{136}\text{I}^m(\beta^-)^{136}\text{Xe}$	E_{β^-} =5170(400) 4670(270) to $^{136}\text{Xe}^m$ 6 ⁺ at 1891.703 and 2444.39 level										Ens026	**
* $^{136}\text{Cs}(\beta^-)^{136}\text{Ba}$	E_{β^-} =341(2) 342(5) respectively, to 6 ⁺ level at 2207.077 keV										Ens026	**
* $^{136}\text{Pr}(\beta^+)^{136}\text{Ce}$	E_{β^+} =2970(50) 3000(75) 3020(20) respectively, to 2 ⁺ level at 1092.09 keV										Ens026	**
* $^{136}\text{Nd}(\beta^+)^{136}\text{Pr}$	E_{β^+} =1330(50) to 1 ⁺ level at 149.11 keV										Ens026	**
* $^{136}\text{Nd}(\beta^+)^{136}\text{Pr}$	K/β^+ =13.2(0.5) to 1 ⁺ level at 149.11 keV										Ens026	**
* $^{136}\text{Pm}(\beta^+)^{136}\text{Nd}$	E_{β^-} =4732(70) probably from high spin isomer going to several										AHW	**
*	high spin levels around 2100 keV										AHW	**
$^{137}\text{Sb}-u$	-64445	215	-64480	60	-0.1	o			GT1	1.5	04Ma.A	
	-65068	186			1.3	U			GT3	2.5	16Kn03	
	-74528	101	-74400.6	2.3	0.5	o			GT2	2.5	08Kn.A	
$^{137}\text{Te}-u$	-74386	129			-0.0	U			GT2	2.5	08Su19	
	-82145	130	-81972	9	0.5	U			GT2	2.5	08Su19	
$\text{C}_{11} \text{H}_5-^{137}\text{Ba}$	133366	24	133297.95	0.27	-1.9	U			R07	1.5	68De17	
$\text{C}_7 \text{ } ^{13}\text{C} \text{ N O H}_{10}-^{137}\text{Ba}$	173792	73	173766.57	0.27	-0.2	U			R07	1.5	68De17	
$\text{C}_{12} \text{H}_9-^{137}\text{Ba O}$	169692	39	169683.46	0.27	-0.1	U			R07	1.5	68De17	
$^{137}\text{La}-u$	-93556	30	-93549.6	1.8	0.2	U			GS2	1.0	05Li24	
$^{137}\text{Ce}-u$	-92101	85	-92237.6	0.4	-1.6	U			GS2	1.0	05Li24	
$^{137}\text{Nd}-u$	-85438	30	-85437	13	0.0	1	18	18	^{137}Nd	GS2	1.0	05Li24
$^{137}\text{Pm}-u$	-79608	62	-79520	14	1.4	U			GS2	1.0	05Li24	
$^{137}\text{Sm}-u$	-72982	51	-72990	30	-0.2	1	36	36	^{137}Sm	GS2	1.0	05Li24
$^{137}\text{Te}-^{130}\text{Xe}_{1.054}$	27300.0	2.7	27300.5	2.3	0.2	1	70	70	^{137}Te	JY1	1.0	12Ha25
$^{137}\text{Sb}-^{133}\text{Cs}_{1.030}$	32907	56				2			CP1	1.0	13Va12	
$^{137}\text{Te}-^{133}\text{Cs}_{1.030}$	22985.0	4.1	22983.8	2.3	-0.3	1	30	30	^{137}Te	CP1	1.0	13Va12
$^{137}\text{Xe}-^{133}\text{Cs}_{1.030}$	8943.6	2.0	8942.25	0.11	-0.7	U			MA8	1.0	09Ne11	
$^{137}\text{Cs}-^{133}\text{Cs}_{1.030}$	4452	19	4473.8	0.3	1.1	o			MA1	1.0	90St25	
	4470	14			0.3	U			MA1	1.0	99Am05	
$^{137}\text{Pr}-^{133}\text{Cs}_{1.030}$	8095	15	8064	9	-2.1	1	34	34	^{137}Pr	MA5	1.0	00Be42
$^{137}\text{Nd}-^{133}\text{Cs}_{1.030}$	11947	14	11948	13	0.0	1	81	81	^{137}Nd	MA5	1.0	00Be42
$^{137}\text{Pm}-^{133}\text{Cs}_{1.030}$	17864	14				2			MA5	1.0	00Be42	
$^{137}\text{Sm}-^{133}\text{Cs}_{1.030}$	24393	42	24390	30	-0.0	1	53	53	^{137}Sm	MA5	1.0	00Be42
$^{137}\text{Eu}-^{133}\text{Cs}_{1.030}$	32815.2	4.7				2			MA8	1.0	13Wo05	
$^{137}\text{Ba}-^{135}\text{Ba}$	3089.1	0.6	3088.88	0.11	-0.1	U			H49	2.5	10Mc04	
$^{137}\text{Te}-^{136}\text{Xe}_{1.007}$	19057	18	19034.4	2.3	-1.3	U			CP1	1.0	12Va02	
$^{137}\text{I}-^{136}\text{Xe}_{1.007}$	11463.2	9.0				2			CP1	1.0	12Va02	
$^{137}\text{Xe}-^{136}\text{Xe}_{1.007}$	5004	11	4992.79	0.11	-1.0	U			CP1	1.0	12Va02	
$^{137}\text{Ba}-^{136}\text{Ba}$	1249	3	1251.41	0.07	0.3	U			M17	2.5	66Be10	
	1222	50			0.4	U			R07	1.5	68De17	
	1227	44			0.4	U			R07	1.5	68De17	
$^{137}\text{Ba}-^{135}\text{Ba}$	143	3	138.76	0.08	-0.6	U			M17	2.5	66Be10	
	69	63			0.7	U			R07	1.5	68De17	
	106	46			0.5	U			R07	1.5	68De17	
$^{137}\text{I}(\beta^-n)^{136}\text{Xe}$	1850	30	2002	8	5.1	C					84Kr.B	
$^{136}\text{Xe}(n,\gamma)^{137}\text{Xe}$	4025.5	0.5	4025.56	0.10	0.1	U					77Fo02	
	4025.8	0.3			-0.8	2					77Pr07	
	4025.53	0.11			0.3	2			Bdn		06Fi.A	
$^{136}\text{Xe}(d,p)^{137}\text{Xe}$	1637	20	1801.00	0.10	8.2	F			Oak		68Mo21	
$^{136}\text{Xe}(^3\text{He},d)^{137}\text{Cs}$	1918	12	1912.1	0.3	-0.5	U			ChR		81Ha08	
$^{136}\text{Ba}(n,\gamma)^{137}\text{Ba}$	6891	5	6905.64	0.07	2.9	U					69Gr31	
	6905.54	0.10			1.0	-			MMn		90Is07	
	6905.70	0.12			-0.5	-			Ltn		95Bo03	
	6905.74	0.16			-0.6	-			Bdn		06Fi.A	
$^{137}\text{Ba}(\gamma,n)^{136}\text{Ba}$	-6949	38	-6905.64	0.07	1.1	U			Phi		60Ge01	
$^{136}\text{Ba}(d,p)^{137}\text{Ba}$	4680	15	4681.07	0.07	0.1	U			ANL		70Vo04	
$^{136}\text{Ba}(n,\gamma)^{137}\text{Ba}$	ave.	6905.63	0.07	6905.64	0.07	0.1	1	98	67	^{137}Ba	average	
$^{136}\text{Ce}(n,\gamma)^{137}\text{Ce}$	7481.3	0.4	7481.53	0.16	0.6	-					81Ko.A	
	7481.58	0.17			-0.3	-			Bdn		06Fi.A	
	ave.	7481.54	0.16		-0.0	1	100	100	^{137}Ce		average	
											Z	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{137}\text{Te}(\beta^-)^{137}\text{I}$	7030	300	7053	9	0.1	U					85Sa15
	6925	130			1.0	U			Bwg		87Gr.A
$^{137}\text{I}(\beta^-)^{137}\text{Xe}$	5880	60	6027	8	2.5	U			Bwg		87Gr.A
$^{137}\text{Xe}(\beta^-)^{137}\text{Cs}$	4140	70	4162.4	0.3	0.3	U					64On03
	4150	100			0.1	U					68Ho22
$^{137}\text{Cs}(\beta^-)^{137}\text{Ba}$	1173.29	0.84	1175.63	0.17	2.8	U					68Wo02 *
	1175.55	0.26			0.3	2					78Ch22 *
	1175.69	0.23			-0.3	2					83Be18 *
	1222.1	1.6				2					81Ar.A *
$^{137}\text{Ce}(\beta^+)^{137}\text{La}$	2702	10	2717	8	1.5	1	66	66 ^{137}Pr			73Bu17
$^{137}\text{Pr}(\beta^+)^{137}\text{Ce}$	3497	40	3618	14	3.0	B					73Bu18 *
$^{137}\text{Nd}(\beta^+)^{137}\text{Pr}$	3690	54			-1.3	U					85Af.A *
$^{137}\text{Pm}^m(\beta^+)^{137}\text{Nd}$	5690	130	5670	40	-0.2	-			IRS		83Al06 *
	5650	60			0.3	-			Dbn		95Ve08 *
ave.	5660	50			0.2	1	67	65 $^{137}\text{Pm}^m$			average
$^{137}\text{Sm}(\beta^+)^{137}\text{Pm}^m$	5900	70	5920	50	0.3	1	45	35 $^{137}\text{Pm}^m$	Dbn		95Ve08
* $^{137}\text{Ce}-u$	$M-A=-85665(29)$ keV for mixture gs+m at 254.29 keV										Nub211 **
* $^{137}\text{Pm}-u$	$M-A=-74079(28)$ keV for mixture gs+m at 150(50) keV										Nub211 **
* $^{137}\text{Sm}-u$	$M-A=-67932(28)$ keV for mixture gs+m at 100#50 keV										Nub211 **
* $^{137}\text{Sm}-^{133}\text{Cs}_{1.030}$	Might be a mixture of ground state and isomer say authors										00Be42 **
*	$D_M=24447(14)$ μ u for mixture gs+m at 100#50 keV; $M-A=-67941(13)$ keV										Nub211 **
* $^{136}\text{Xe}(d,p)^{137}\text{Xe}$	F : error severely underestimated and value low, see excitation energies										AHW **
* $^{137}\text{Cs}(\beta^-)^{137}\text{Ba}$	$E_{\beta^-}=511.63(0.84)$ 513.89(0.26) 514.03(0.23) to $^{137}\text{Ba}^m$ at 661.659 keV										Nub211 **
* $^{137}\text{Ce}(\beta^+)^{137}\text{La}$	$E_{\beta^+}=189.5(1.6)$ to $5/2^+$ level at 10.59 keV										Ens079 **
* $^{137}\text{Nd}(\beta^+)^{137}\text{Pr}$	$E_{\beta^+}=2400(40)$ $E_{\beta^+}=2592(54)$ respectively, to $3/2^+$ level at 75.5 keV										Ens079 **
* $^{137}\text{Pm}^m(\beta^+)^{137}\text{Nd}$	$E_{\beta^+}=4132(+150-115)$ 4110(60) respectively, to $11/2^-$ $^{137}\text{Nd}^m$ at 519.43 keV										Nub211 **
$^{138}\text{Sb}-u$	-58208	457	-58670#	320#	-0.4	D			GT3	2.5	16Kn03 *
$^{138}\text{Te}-u$	-70940	247	-70528	4	1.1	o			GT1	1.5	04Ma.A
	-70583	106			0.2	o			GT2	2.5	08Kn.A
	-70591	131			0.2	U			GT2	2.5	08Su19
$\text{C}_{10} \text{H}_{18}-^{138}\text{Ba}$	235609	20	235603.52	0.27	-0.1	U			M17	2.5	66Be10
$\text{C}_{11} \text{H}_7-^{138}\text{Ba H}$	141649	51	141703.13	0.27	0.7	U			R07	1.5	68De17
$\text{C}_{11} \text{H}_6-^{138}\text{Ba}$	141701	30	141703.13	0.27	0.0	U			R07	1.5	68De17
$\text{C}_{12} \text{H}_{10}-^{138}\text{Ba O}$	178106	15	178088.64	0.27	-0.8	U			R07	1.5	68De17
	178105	49			-0.2	U			R07	1.5	68De17
$\text{C}_{11} ^{13}\text{C H}_9-^{138}\text{Ba O}$	173612	37	173618.44	0.27	0.1	U			R07	1.5	68De17
$\text{C}_{10} \text{H}_{18}-^{138}\text{Ce}$	234799	60	234856.4	0.5	0.2	U			R05	4.0	65De13
$\text{C}_{12} \text{H}_{10}-^{138}\text{Ce O}$	177382	46	177341.5	0.5	-0.2	U			R05	4.0	65De13
$\text{C}_9 ^{13}\text{C H}_{17}-^{138}\text{Ce}$	230358	60	230386.2	0.5	0.1	U			R05	4.0	65De13
$^{138}\text{Pr}^m-u$	-88896	30	-88867	17	1.0	1	34	34 $^{138}\text{Pr}^m$	GS2	1.0	05Li24
$^{138}\text{Nd}-u$	-88060	130	-88049	12	0.1	o			GS1	1.0	00Ra23
	-88060	30			0.4	R			GS2	1.0	05Li24
$^{138}\text{Pm}-u$	-80226	140	-80424	12	-1.4	o			GS1	1.0	00Ra23
	-80437	30			0.4	1	17	17 ^{138}Pm	GS2	1.0	05Li24
$^{138}\text{Sm}-u$	-76766	30	-76756	13	0.3	R			GS2	1.0	05Li24
$^{138}\text{Eu}-u$	-66291	30				2			GS2	1.0	05Li24
$^{138}\text{Te}-^{130}\text{Xe}_{1.062}$	31945.3	4.7	31946	4	0.0	1	75	75 ^{138}Te	JY1	1.0	12Ha25
$^{138}\text{Te}-^{133}\text{Cs}_{1.038}$	27614.0	8.1	27613	4	-0.1	1	25	25 ^{138}Te	CP1	1.0	13Va12
$^{138}\text{Xe}-^{133}\text{Cs}_{1.038}$	12284.1	3.5	12287	3	0.9	1	74	74 ^{138}Xe	MA8	1.0	09Ne11
$^{138}\text{Cs}-^{133}\text{Cs}_{1.038}$	9158	14	9158	10	-0.0	1	49	49 ^{138}Cs	MA1	1.0	99Am05
$^{138}\text{Ba}-^{133}\text{Cs}_{1.038}$	3388	14	3387.92	0.27	-0.0	U			MA1	1.0	99Am05
$^{138}\text{Nd}-^{133}\text{Cs}_{1.038}$	10093	14	10092	12	-0.1	-			MA5	1.0	00Be42
ave.	10091	13			0.1	1	96	96 ^{138}Nd			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{138}\text{Pm}-^{133}\text{Cs}_{1.038}$	17721	14	17717	12	-0.3	1	79	^{138}Pm	MA5	1.0	00Be42
$^{138}\text{Sm}-^{133}\text{Cs}_{1.038}$	21387	14	21385	13	-0.2	2			MA5	1.0	00Be42
$^{138}\text{I}-^{136}\text{Xe}_{1.015}$	16903.7	6.4				2			CP1	1.0	12Va02
$^{138}\text{Xe}-^{136}\text{Xe}_{1.015}$	8332.2	5.9	8324	3	-1.5	1	26	^{138}Xe	CP1	1.0	12Va02
$^{138}\text{Ba}-^{136}\text{Xe}_{1.015}$	-575.98	0.48	-575.63	0.27	0.7	1	31	^{138}Ba	MS1	1.0	19Sa36
$^{138}\text{Ba } ^{35}\text{Cl}-^{136}\text{Ba } ^{37}\text{Cl}$	3621.1	0.6	3621.38	0.11	0.2	U			H49	2.5	10Mc04
$^{138}\text{Ce}-^{136}\text{Xe}_{1.015}$	170.9	1.7	171.5	0.5	0.3	1	10	^{138}Ce	MS1	1.0	19Sa36
$^{138}\text{La}-^{138}\text{Ba}$	1877.29	0.40	1877.0	0.4	-0.8	1	82	^{138}La	MS1	1.0	19Sa36
$^{138}\text{Ce}-^{138}\text{Ba}$	746.12	0.77	747.1	0.5	1.3	1	38	^{138}Ce	MS1	1.0	19Sa36
$^{138}\text{La}-^{138}\text{Ce}$	1129.36	0.51	1129.9	0.4	1.0	1	72	^{138}Ce	MS1	1.0	19Sa36
$^{138}\text{Ba}-^{137}\text{Ba}$	-582	2	-580.15	0.04	0.4	U			M17	2.5	66Be10
	-480	27			-2.5	U			R07	1.5	68De17
	-553	40			-0.5	U			R07	1.5	68De17
$^{138}\text{Ba}-^{136}\text{Ba}$	676	3	671.26	0.08	-0.6	U			M17	2.5	66Be10
	658	98			0.1	U			R07	1.5	68De17
	628	43			0.7	U			R07	1.5	68De17
$^{138}\text{Ce}-^{136}\text{Ce}$	-1040	47	-1135.1	0.5	-0.5	U			R05	4.0	65De13
	-1158	20			0.5	U			M17	2.5	66Be10
$^{138}\text{Ba H}-^{137}\text{Ba}$	7399	88	7244.88	0.04	-1.2	U			R07	1.5	68De17
	7280	43			-0.5	U			R07	1.5	68De17
$^{137}\text{Ba}(n,\gamma)^{138}\text{Ba}$	8611.3	0.8	8611.72	0.04	0.5	U					68Ma35
	8611.72	0.04			0.1	1	99	^{138}Ba	MMn		90Is07 Z
	8611.5	0.15			1.5	U			Ltn		95Bo05
	8611.63	0.18			0.5	U			Bdn		06Fi.A
$^{137}\text{Ba}(d,p)^{138}\text{Ba}$	6398	15	6387.16	0.04	-0.7	U			ANL		70Vo04
$^{138}\text{I}(\beta^-)^{138}\text{Xe}$	7820	70	7992	7	2.5	U			Bwg		87Gr.A
$^{138}\text{Xe}(\beta^-)^{138}\text{Cs}$	2700	50	2915	10	4.3	B					72Mo33 *
	2830	80			1.1	U			Trs		78Wo15
$^{138}\text{Cs}^x(\text{IT})^{138}\text{Cs}$	40	23				2					82Au01 *
$^{138}\text{Cs}(\beta^-)^{138}\text{Ba}$	5350	80	5375	9	0.3	U			Trs		78Wo15
	5388	25			-0.5	-			Gsn		81De25
	5370	15			0.3	-			McG		84He.A
ave.	5375	13			0.0	1	51	^{138}Cs			average
$^{138}\text{La}(\epsilon)^{138}\text{Ba}$	1620	15	1748.4	0.3	8.6	B					56Tu17 *
$^{138}\text{La}(\beta^-)^{138}\text{Ce}$	994	10	1052.5	0.4	5.8	B					57G120 *
	1159	40			-2.7	U					70El.A *
	1052.7	4.3			-0.1	U					16Qu01 *
$^{138}\text{Pr}(\beta^+)^{138}\text{Ce}$	4437	10				2					71Af05
$^{138}\text{Pr}^m(\beta^+)^{138}\text{Ce}$	4801	20	4787	16	-0.7	1	66	$^{138}\text{Pr}^m$			64Fu08 *
$^{138}\text{Nd}(\beta^+)^{138}\text{Pr}$	2020	100	1112	15	-9.1	C					61Bo.B
$^{138}\text{Pm}(\beta^+)^{138}\text{Nd}$	7090	100	7103	16	0.1	U			IRS		83Al06
	7080	60			0.4	1	7	^{138}Nd	Dbn		95Ve08
	7000	250			0.4	U					81De38 *
* $^{138}\text{Sb}-u$	Trends from Mass Surface TMS suggest ^{138}Sb 430 keV more bound										GAu **
* $^{138}\text{Xe}(\beta^-)^{138}\text{Cs}$	$E_{\beta^-}=2460(50), 2270(50)$ to $(1^- 2^-)$ level at 258.400, 1^- at 412.260 keV										Ens035 **
* $^{138}\text{Cs}^x(\text{IT})^{138}\text{Cs}$	Based on $^{138}\text{Cs}^m(\text{IT})=79.9$ keV										Nub211 **
* $^{138}\text{La}(\epsilon)^{138}\text{Ba}$	$L/K=1.40(0.25)$ to 2^+ level at 1435.816 keV										Ens035 **
* $^{138}\text{La}(\beta^-)^{138}\text{Ce}$	$E_{\beta^-}=205(10) 370(40) 264.0(4.3)$ respectively, to 2^+ level at 788.744 keV										Ens035 **
* $^{138}\text{Pr}^m(\beta^+)^{138}\text{Ce}$	$E_{\beta^+}=1650(20)$ to 7^- level at 2129.17 keV										Ens035 **
* $^{138}\text{Pm}(\beta^+)^{138}\text{Nd}$	$E_{\beta^+}=3900(200)$ to 5^- level at 1990.4, 6^+ at 2134.3 and (5^-) at 2221.8keV										Ens035 **
$^{139}\text{Te}-u$	-64541	333	-64633	4	-0.2	U			GT1	1.5	04Ma.A
$^{139}\text{Te}-^{130}\text{Xe}_{1.069}$	38515.7	3.8				2			JY1	1.0	12Ha25
$^{139}\text{I}-u$	-73838	102	-73507	4	1.3	U			GT2	2.5	08Kn.A
	-73567	130			0.2	U			GT2	2.5	08Su19

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_6^{13}C O_3 H_6-^{139}La$	128474	41	128686.0	0.7	1.3	U			R05	4.0	65De13
$C_7 O_3 H_7-^{139}La$	133063	32	133156.2	0.7	0.7	U			R05	4.0	65De13
$C_6 N O_3 H_5-^{139}La$	120496	21	120580.1	0.7	1.0	U			R05	4.0	65De13
$C_{12} H_{11}-^{139}La O$	184568	66	184797.8	0.7	0.9	U			R05	4.0	65De13
$C_{11}^{13}C H_{10}-^{139}La O$	180100	58	180327.6	0.7	1.0	U			R05	4.0	65De13
$^{139}Nd-u$	-87840	79	-88049	30	-2.6	U			GS2	1.0	05Li24 *
$^{139}Sm-u$	-77704	30	-77703	12	0.0	R			GS2	1.0	05Li24
	-77711	30			0.3	R			GS2	1.0	05Li24 *
$^{139}Eu-u$	-70215	30	-70208	14	0.2	R			GS2	1.0	05Li24
$^{139}Te-^{133}Cs_{1.045}$	34185	17	34170	4	-0.9	U			CP1	1.0	13Va12
$^{139}I-^{133}Cs_{1.045}$	25296.1	4.3				2			CP1	1.0	13Va12
$^{139}Xe-^{133}Cs_{1.045}$	17594.9	2.3				2			MA8	1.0	09Ne11 *
$^{139}Cs-^{133}Cs_{1.045}$	12163	14	12167	3	0.3	U			MA1	1.0	99Am05
$^{139}Ba-^{133}Cs_{1.045}$	7649	14	7643.86	0.27	-0.4	U			MA1	1.0	99Am05
$^{139}Pm-^{133}Cs_{1.045}$	15604	15	15602	15	-0.1	1	95	95 ^{139}Pm	MA5	1.0	00Be42
$^{139}Sm-^{133}Cs_{1.045}$	21101	14	21099	12	-0.1	2			MA5	1.0	00Be42
$^{139}Eu-^{133}Cs_{1.045}$	28597	16	28595	14	-0.1	2			MA5	1.0	00Be42
$^{139}I-^{136}Xe_{1.022}$	21333	31	21320	4	-0.4	U			CP1	1.0	12Va02
$^{139}Xe-^{136}Xe_{1.022}$	13618	12	13619.0	2.3	0.1	U			CP1	1.0	12Va02
$^{139}La-^{136}Xe_{1.022}$	1189.96	0.67	1189.7	0.7	-0.3	1	94	94 ^{139}La	MS1	1.0	19Sa39
$^{139}La-^{138}La$	-622	132	-761.1	0.8	-0.3	U			R05	4.0	65De13
$^{139}La-^{138}Ce$	485	74	368.7	0.8	-0.4	U			R05	4.0	65De13
$^{133}Cs-^{139}Cs_{239}^{131}Cs_{761}$	-1774	24	-1773.1	0.8	0.0	F			P33	2.5	86Au02 *
$^{138}Cs^x-^{139}Cs_{496}^{137}Cs_{504}$	770	40	800	25	0.3	U			P23	2.5	82Au01
$^{138}Ba(n,\gamma)^{139}Ba$	4723.4	0.7	4723.43	0.04	0.0	U					69Mo13
	4723.4	0.3			0.1	U					80Ba.A
	4723.43	0.04				2			MMn		90Is07 Z
	4723.20	0.14			1.6	U			Bdn		06Fi.A
$^{138}Ba(d,p)^{139}Ba$	2495	10	2498.86	0.04	0.4	U			MIT		64Sp12
	2496	15			0.2	U			Hei		67Wi08
	2493	10			0.6	U			ANL		70Vo04
$^{139}La(\gamma,n)^{138}La$	-8775	25	-8780.3	0.7	-0.2	U			Phi		60Ge01
$^{138}La(d,p)^{139}La$	6553	3	6555.7	0.7	0.9	U			Tal		71Du02
$^{139}La(d,t)^{138}La$	-2522	5	-2523.1	0.7	-0.2	U			Tal		72La20
$^{139}I(\beta^-)^{139}Xe$	6815	100	7174	5	3.6	C			Bwg		87Gr.A
	6806	23			16.0	B			Bwg		92Gr06
$^{139}Xe(\beta^-)^{139}Cs$	5020	60	5057	4	0.6	U			Trs		78Wo15
	5062	22			-0.2	U			Bwg		92Gr06
$^{139}Cs(\beta^-)^{139}Ba$	4290	70	4213	3	-1.1	U			Trs		78Wo15
	4190	25			0.9	o			Gsn		80Bl.A
	4213	5			-0.0	o			Gsn		81De25
	4214	4			-0.3	3			McG		84He.A
	4211	5			0.4	3			Gsn		92Pr04
$^{139}Ba(\beta^-)^{139}La$	2307	5	2308.5	0.7	0.3	U					75Fl07 *
	2336	25			-1.1	U			Gsn		81De25
	2316	4			-1.9	U			McG		84He.A
$^{139}Ce(\beta^+)^{139}La$	264.6	2.0	264.6	2.0	0.0	1	100	100 ^{139}Ce			96Hi14 *
$^{139}Ce(\epsilon)^{139}La$	278	7			-1.9	U					Averag *
$^{139}Pr(\beta^+)^{139}Ce$	2129	3	2129.1	3.0	0.0	1	100	100 ^{139}Pr			81Ar.A
$^{139}Nd(\beta^+)^{139}Pr$	2787	50	2812	28	0.5	1	31	30 ^{139}Nd			75Vy02 *
$^{139}Pm(\beta^+)^{139}Nd$	4450	100	4516	26	0.7	-					77De06 *
	4540	40			-0.6	-			IRS		83Al06
	4470	50			0.9	-			Dbn		95Ve08
	ave.	4507	30		0.3	1	75	70 ^{139}Nd			average
$^{139}Sm(\beta^+)^{139}Pm$	5430	150	5121	17	-2.1	U					82De06 *
	5510	150			-2.6	U			IRS		83Al06 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{139}\text{Eu}(\beta^+)^{139}\text{Sm}$	6080	50	6982	17	18.0	C			Dbn		95Ve08 *
* $^{139}\text{Nd}-\text{u}$	$M-A=-81707(30)$ keV for mixture gs+m at 231.15 keV										Nub211 **
* $^{139}\text{Sm}-\text{u}$	$M-A=-71930(28)$ keV for $^{139}\text{Sm}^m$ at 457.38 keV										Nub211 **
* $^{139}\text{Xe}-^{133}\text{Cs}_{1.045}$	Typo in original paper, ratio should read 1.045 245 4357(175)										GAu **
* $^{133}\text{Cs}-^{139}\text{Cs}_{239} \quad ^{131}\text{Cs}$	F : rejection based on line-shape analysis										86Au02 **
* $^{139}\text{Ba}(\beta^-)^{139}\text{La}$	$E_{\beta^-}=2141(5)$ to $5/2^+$ level at 165.8576 keV										Ens014 **
* $^{139}\text{Ce}(\beta^+)^{139}\text{La}$	$Q_{\beta^+}=98.7(1.7\text{stat})(1.0\text{syst})$ to $5/2^+$ level at 165.8576 keV										Ens014 **
* $^{139}\text{Ce}(\varepsilon)^{139}\text{La}$	Average pK=0.73(0.01) to $5/2^+$ level at 165.8576 keV in 10 references:										Ens014 **
*	pK=0.76 (0.04)										54Pr31 **
*	pK=0.73 (0.01)										56Ke23 **
*	pK=0.68 (0.02)										67Ma07 **
*	pK=0.75 (0.01)										68Ad08 **
*	pK=0.69 (0.02)										68Va08 **
*	pK=0.716(0.02)										72Ca07 **
*	pK=0.78 (0.02)										72Sc08 **
*	pK=0.726(0.010)										75Ha43 **
*	pK=0.801(0.034)										75Pl06 **
*	pK=0.705(0.020)										76Ha36 **
* $^{139}\text{Nd}(\beta^+)^{139}\text{Pr}$	$E_{\beta^+}=1770(50)$; and $1170(50)$ from $^{139}\text{Nd}^m$ at 231.15 to $1/2^-$ level at 821.98										Ens014 **
* $^{139}\text{Pm}(\beta^+)^{139}\text{Nd}$	$E_{\beta^+}=3020(120)$, $2990(100)$ to $3/2^+$ levels at 463.10, 402.77 keV										Ens014 **
* $^{139}\text{Sm}(\beta^+)^{139}\text{Pm}$	$E_{\beta^+}=4100(150)$ to $(1/2, 3/2)^+$ level at 306.69 keV										Ens014 **
* $^{139}\text{Sm}(\beta^+)^{139}\text{Pm}$	$E_{\beta^+}=4735(+180-130)$ from $^{139}\text{Sm}^m$ at 457.40 to $^{139}\text{Pm}^m$ at 188.7 keV										Nub211 **
* $^{139}\text{Eu}(\beta^+)^{139}\text{Sm}$	$E_{\beta^+}=4600(50)$ to $^{139}\text{Sm}^m$ at 457.38 keV										Nub211 **
$^{140}\text{Te}-\text{u}$	-60827	225	-60513	15	0.9	U			GT1	1.5	04Ma.A
$^{140}\text{Te}-^{130}\text{Xe}_{1.077}$	43419	30	43407	15	-0.4	1	26	26 ^{140}Te	JY1	1.0	12Ha25
$^{140}\text{I}-\text{u}$	-68181	193	-68284	13	-0.4	o			GT1	1.5	04Ma.A
	-68463	102			0.7	o			GT2	2.5	08Kn.A
	-68273	130			-0.0	o			GT2	2.5	08Su19
	-68202	186			-0.2	U			GT2	2.5	16Kn03
$^{140}\text{Xe}-\text{u}$	-78449	103	-78354.2	2.5	0.4	o			GT2	2.5	08Kn.A
	-78229	130			-0.4	U			GT2	2.5	08Su19
$\text{C}_{11} \text{H}_8-^{140}\text{Ce}$	157116	29	157151.8	1.4	0.3	U			R05	4.0	65De13
$\text{C}_{10} \text{ } ^{13}\text{C} \text{H}_7-^{140}\text{Ce}$	152553	17	152681.6	1.4	1.9	U			R05	4.0	65De13
$\text{C}_{10} \text{N} \text{H}_6-^{140}\text{Ce}$	144599	35	144575.8	1.4	-0.2	U			R05	4.0	65De13
$\text{C}_{10} \text{N}_2 \text{H}_8-^{140}\text{Ce} \text{O}$	168207	48	168385.2	1.4	0.9	U			R05	4.0	65De13
$^{140}\text{Nd}-\text{u}$	-90448	30	-90454	4	-0.2	U			GS2	1.0	05Li24
$^{140}\text{Pm}^m-\text{u}$	-83532	30	-83503	14	1.0	1	22	22 $^{140}\text{Pm}^m$	GS2	1.0	05Li24
$^{140}\text{Sm}-\text{u}$	-81018	30	-81005	13	0.4	R			GS2	1.0	05Li24
$^{140}\text{Gd}-\text{u}$	-66326	30				2			GS2	1.0	05Li24
$^{140}\text{Te}-^{133}\text{Cs}_{1.053}$	38822	67	39046	15	3.3	B			CP1	1.0	13Va12
	39042	18			0.2	1	74	74 ^{140}Te	CP2	1.0	18Or.A
$^{140}\text{I}-^{133}\text{Cs}_{1.053}$	31275	13				2			CP1	1.0	13Va12
$^{140}\text{Xe}-^{133}\text{Cs}_{1.053}$	21204.9	2.5				2			MA8	1.0	09Ne11
$^{140}\text{Cs}-^{133}\text{Cs}_{1.053}$	16837	14	16843	9	0.4	-			MA1	1.0	99Am05
	16857	14			-1.0	-			MA4	1.0	99Am05
ave.	16847	10			-0.4	1	79	79 ^{140}Cs			average
$^{140}\text{Ba}-^{133}\text{Cs}_{1.053}$	10150	14	10167	8	1.2	1	37	37 ^{140}Ba	MA1	1.0	99Am05
$^{140}\text{Ce} \text{O}-^{133}\text{Cs}_{1.173}$	11270.3	2.5	11267.9	1.4	-1.0	1	32	32 ^{140}Ce	MA8	1.0	19Hu15
$^{140}\text{Nd} \text{O}-^{133}\text{Cs}_{1.173}$	15365.6	3.5				2			MA8	1.0	19Hu15
$^{140}\text{Pm}^m-^{133}\text{Cs}_{1.053}$	16064	16	16056	14	-0.5	1	78	78 $^{140}\text{Pm}^m$	MA5	1.0	00Be42
$^{140}\text{Sm}-^{133}\text{Cs}_{1.053}$	18557	15	18554	13	-0.2	2			MA5	1.0	00Be42
$^{140}\text{Xe}-^{136}\text{Xe}_{1.029}$	17134	11	17122.1	2.5	-1.1	U			CP1	1.0	12Va02
$\text{C}_{11} \text{H}_9-^{140}\text{Ce}$	164956	40	164976.9	1.4	0.2	U			M17	2.5	66Be10
$^{140}\text{Ce}-^{139}\text{La}$	-1029	80	-914.5	1.4	0.4	U			R05	4.0	65De13

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_{11} H_{10} - {}^{140}\text{Ce}$	172765	40	172801.9	1.4	0.4	U			M17	2.5	66Be10
${}^{140}\text{Ce} - {}^{138}\text{Ce}$	-497	83	-545.7	1.5	-0.1	U			R05	4.0	65De13
	-543	8			-0.1	U			M17	2.5	66Be10
${}^{139}\text{Cs} - {}^{140}\text{Cs}_{.883} \quad {}^{131}\text{Cs}_{.118}$	-2280	40	-2276	8	0.0	U			P23	2.5	82Au01
${}^{139}\text{Cs} - {}^{140}\text{Cs}_{.869} \quad {}^{132}\text{Cs}_{.132}$	-2210	40	-2241	8	-0.3	U			P23	2.5	82Au01
${}^{138}\text{Ce}(t,p) {}^{140}\text{Ce}$	8184	15	8169.2	1.4	-1.0	U			LAl		72Mu09
${}^{140}\text{Ce}(p,t) {}^{138}\text{Ce}$	-8167	20	-8169.2	1.4	-0.1	U			Brk		77Sh06
${}^{139}\text{La}(n,\gamma) {}^{140}\text{La}$	5161.1	1.0	5160.997	0.019	-0.1	U					70Ju04
	5160	1			1.0	U					72Fu10
	5160.97	0.05			0.5	-			MMn		90Is09
	5161.00	0.10			-0.0	o			Bdn		06Fi.A
	5161.001	0.021			-0.2	-			Bdn		19Hu07
${}^{139}\text{La}(d,p) {}^{140}\text{La}$	2938	3	2936.430	0.019	-0.5	U			Tal		67Ke02
${}^{139}\text{La}(n,\gamma) {}^{140}\text{La}$	ave. 5160.996	0.019	5160.997	0.019	0.0	1	100	94 ${}^{140}\text{La}$			average
${}^{140}\text{Ho}(p) {}^{139}\text{Dy}$	1093.9	10.			3						99Ry04
${}^{140}\text{Xe}(\beta^-) {}^{140}\text{Cs}$	4060	60	4063	9	0.1	U			Trs		78Wo15
${}^{140}\text{Cs}(\beta^-) {}^{140}\text{Ba}$	6100	100	6218	10	1.2	U			Trs		78Wo15
	6235	25			-0.7	o			Gsn		80Bl.A
	6220	15			-0.1	o			Gsn		81De25
	6212	20			0.3	-			Gsn		92Pr04
	6199	25			0.8	-			Ida		93Gr17
	ave. 6207	16			0.7	1	40	21 ${}^{140}\text{Cs}$			average
${}^{140}\text{Ba}(\beta^-) {}^{140}\text{La}$	1060	20	1044	8	-0.8	-					49Be36
	1050	20			-0.3	-					59Bo61
	1055	30			-0.4	-					65Bu07
	ave. 1055	13			-0.8	1	38	38 ${}^{140}\text{Ba}$			average
${}^{140}\text{La}(\beta^-) {}^{140}\text{Ce}$	3760.2	2.0	3762.2	1.3	1.0	1	45	39 ${}^{140}\text{Ce}$			72Na04
${}^{140}\text{Pr}(\beta^+) {}^{140}\text{Ce}$	3388	6			2						68Ab17
${}^{140}\text{Nd}(\epsilon) {}^{140}\text{Pr}$	160	60	429	7	4.5	B					72Ba91
${}^{140}\text{Pm}(\beta^+) {}^{140}\text{Nd}$	6080	100	6045	24	-0.3	U					75Ke09
	6090	40			-1.1	3			IRS		83Al06
	6020	30			0.8	3			Dbn		95Ve08
${}^{140}\text{Pm}^m(\beta^+) {}^{140}\text{Nd}$	6484	70	6474	14	-0.1	U					75Ke09
${}^{140}\text{Sm}(\epsilon) {}^{140}\text{Pm}$	3400	300	2756	27	-2.1	U					87De04
${}^{140}\text{Eu}(\beta^+) {}^{140}\text{Sm}$	8400	400	8470	50	0.2	U			LBL		91Fi03
	8470	50			3				Dbn		95Ve08
${}^{140}\text{Gd}(\beta^+) {}^{140}\text{Eu}$	4800	400	5200	60	1.0	U			LBL		91Fi03
${}^{140}\text{Tb}(\beta^+) {}^{140}\text{Gd}$	11300	800			3				LBL		91Fi03
$* {}^{140}\text{Ho}(p) {}^{139}\text{Dy}$	Trends from Mass Surface TMS suggest ${}^{140}\text{Ho}$ 300 keV less bound										GAu
$* {}^{140}\text{Ba}(\beta^-) {}^{140}\text{La}$	$E_{\beta^-} = 1022(20), 480(20)$ to 2^- level at 29.9641, 0^- at 581.07 keV										Ens077
$* {}^{140}\text{Ba}(\beta^-) {}^{140}\text{La}$	$E_{\beta^-} = 1020(20), 830(50), 590(50)$ to 2^- level at 29.9641, 2^- at 162.6591, and 1^- at 467.653 keV										GAu
$* {}^{140}\text{Ba}(\beta^-) {}^{140}\text{La}$	$E_{\beta^-} = 1030(30), 1020(30)$ to 2^- level at 29.9641, 1^- at 43.844 keV										Ens077
$* {}^{140}\text{La}(\beta^-) {}^{140}\text{Ce}$	$E_{\beta^-} = 2164(2)$ to 2^+ level at 1596.237 keV										Ens077
$* {}^{140}\text{Pm}^m(\beta^+) {}^{140}\text{Nd}$	$E_{\beta^+} = 3240(70)$ to 7^- level at 2221.4 keV										Ens077
$* {}^{140}\text{Eu}(\beta^+) {}^{140}\text{Sm}$	From p^+ . May be lower limit										91Fi03
$* {}^{140}\text{Tb}(\beta^+) {}^{140}\text{Gd}$	Lower limit										91Fi03
${}^{141}\text{I}-u$	-64316	419	-64334	17	-0.0	o			GT1	1.5	04Ma.A
	-64549	120			0.7	o			GT2	2.5	08Kn.A
	-64736	137			1.2	o			GT2	2.5	08Su19
	-64445	186			0.2	U			GT3	2.5	16Kn03
${}^{141}\text{Xe}-u$	-73092	126	-73213	3	-0.4	o			GT2	2.5	08Kn.A
	-73560	136			1.0	U			GT2	2.5	08Su19
${}^{141}\text{Ba}-u$	-85603.5	7.5	-85596	6	1.0	1	58	58 ${}^{141}\text{Ba}$	CP1	1.0	06Sa56

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_{11} H_9 - ^{141}Pr$	162852	41	162765.7	1.6	-0.5	U			R05	4.0	65De13
$C_{10} N H_7 - ^{141}Pr$	150229	37	150189.6	1.6	-0.3	U			R05	4.0	65De13
$C_9 ^{13}C N H_6 - ^{141}Pr$	145722	65	145719.4	1.6	-0.0	U			R05	4.0	65De13
$^{141}Pr-u$	-92374	30	-92340.4	1.6	1.1	U			GS2	1.0	05Li24
$^{141}Nd-u$	-90401	30	-90383	3	0.6	U			GS2	1.0	05Li24
	-90365	30			-0.6	U			GS2	1.0	05Li24 *
$^{141}Sm-u$	-81496	62	-81518	9	-0.4	U			GS2	1.0	05Li24 *
$^{141}Eu-u$	-75048	42	-75068	14	-0.5	U			GS2	1.0	05Li24 *
$^{141}Gd-u$	-67881	30	-67874	21	0.2	2			GS2	1.0	05Li24
	-67867	30			-0.2	2			GS2	1.0	05Li24 *
$^{141}Tb-u$	-58552	113				2			GS2	1.0	05Li24 *
$^{141}I-^{133}Cs_{1.060}$	35887	17				2			CP1	1.0	13Va12
$^{141}Xe-^{133}Cs_{1.060}$	27008.1	3.1				2			MA8	1.0	09Ne11
$^{141}Cs-^{133}Cs_{1.060}$	20269	16	20266	10	-0.2	1	38	38 ^{141}Cs	MA4	1.0	99Am05
$^{141}Ba-^{133}Cs_{1.060}$	14625	15	14625	6	-0.0	-			MA1	1.0	99Am05
	14631	16			-0.4	-			MA4	1.0	99Am05
ave.	14628	11			-0.3	1	27	27 ^{141}Ba			average
$^{141}Pm-^{133}Cs_{1.060}$	13776	15				2			MA5	1.0	00Be42
$^{141}Sm-^{133}Cs_{1.060}$	18692	14	18702	9	0.7	1	43	43 ^{141}Sm	MA5	1.0	00Be42 *
$^{141}Eu-^{133}Cs_{1.060}$	25164	15	25153	14	-0.8	1	82	82 ^{141}Eu	MA5	1.0	00Be42 *
$^{141}Xe-^{136}Xe_{1.037}$	23003	10	23006	3	0.3	U			CP1	1.0	12Va02
$^{141}Cs-^{136}Xe_{1.037}$	16277	22	16264	10	-0.6	1	20	20 ^{141}Cs	CP1	1.0	12Va02
$^{139}Cs-^{141}Cs_{789} ^{131}Cs_{212}$	-3190	40	-3271	8	-0.8	U			P23	2.5	82Au01
$^{140}Cs-^{141}Cs_{894} ^{131}Cs_{107}$	-970	40	-1045	11	-0.8	U			P23	2.5	82Au01
$^{139}Cs-^{141}Cs_{767} ^{132}Cs_{234}$	-3210	40	-3183	8	0.3	U			P23	2.5	82Au01
$^{141}Cs(\beta^-n)^{140}Ba$	735	30	719	12	-0.5	1	15	9 ^{141}Cs			84Kr.B
$^{140}Ce(n,\gamma)^{141}Ce$	5428.6	0.6	5428.15	0.10	-0.8	U			BNn		70Ge03 Z
	5428.01	0.20			0.7	-			Ptn		80Ba.A Z
	5428.19	0.12			-0.3	-			Bdn		06Fi.A
$^{140}Ce(d,p)^{141}Ce$	3210	10	3203.58	0.10	-0.6	U			MIT		64Sp12
	3202	15			0.1	U			Hei		67Wi08
$^{140}Ce(n,\gamma)^{141}Ce$	ave. 5428.14	0.10	5428.15	0.10	0.1	1	100	75 ^{141}Ce			average
$^{141}Pr(\gamma,n)^{140}Pr$	-9361	23	-9400	6	-1.7	U			Phi		60Ge01
$^{141}Ho(p)^{140}Dy$	1177.4	8.	1177	7	-0.1	3					98Da03
	1172.9	20.			0.2	3					99Ry04 *
$^{141}Xe(\beta^-)^{141}Cs$	6150	90	6280	10	1.4	U			Trs		78Wo15
$^{141}Cs(\beta^-)^{141}Ba$	5200	80	5255	10	0.7	U			Trs		78Wo15
	5264	15			-0.6	o			Gsn		80Bl.A *
	5252	15			0.2	o			Gsn		81De25
	5242	15			0.9	1	41	33 ^{141}Cs	Gsn		92Pr04
$^{141}Ba(\beta^-)^{141}La$	3010	60	3197	7	3.1	B			Trs		78Wo15
	3208	35			-0.3	U			Gsn		81De25
	3217	20			-1.0	1	11	7 ^{141}Ba	McG		84He.A
$^{141}La(\beta^-)^{141}Ce$	2430	30	2501	4	2.4	U					51Du19
	2502	4			-0.2	1	96	96 ^{141}La	McG		84He.A
$^{141}Ce(\beta^-)^{141}Pr$	584	3	583.5	1.2	-0.2	-					50Fr58 *
	585	4			-0.4	-					52Ko27 *
	576.4	2.0			3.5	B					55Jo02 *
	581.4	2.0			1.0	-					68Be06 *
	582.2	2.6			0.5	-					79Ha09 *
ave.	582.5	1.3			0.7	1	79	54 ^{141}Pr			average
$^{141}Nd(\beta^+)^{141}Pr$	1816	8	1823.0	2.8	0.9	2					73Bu21
	1824	3			-0.3	2					76Ga.A *
$^{141}Pm(\beta^+)^{141}Nd$	3730	60	3669	14	-1.0	U					70Ch29 *
	3640	70			0.4	U					75Ke09
$^{141}Sm(\beta^+)^{141}Pm$	4580	50	4589	16	0.2	U					77Ke03 *
	4463	60			2.1	U			IRS		83Al06 *
	4524	80			0.8	U			IRS		93Al03 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{141}\text{Eu}(\beta^+)^{141}\text{Sm}$	6030	100	6008	14	-0.2	U					77De25 *
	5950	40			1.5	-			IRS		83Al06 *
	6035	60			-0.4	U					85Af.A
	5550	100			4.6	B			IRS		93Al03
	5980	40			0.7	-			Dbn		95Ve08 *
ave.	5965	28			1.5	1	26	18 ^{141}Eu			average
* $^{141}\text{Nd}-\text{u}$	$M-A=-83418(28)$ keV for $^{141}\text{Nd}^m$ at 756.51 keV										Nub211 **
* $^{141}\text{Sm}-\text{u}$	$M-A=-75825(28)$ keV for mixture gs+m at 175.9 keV										Nub211 **
* $^{141}\text{Eu}-\text{u}$	$M-A=-69858(28)$ keV for mixture gs+m at 96.45 keV										Nub211 **
* $^{141}\text{Gd}-\text{u}$	$M-A=-62840(28)$ keV for $^{141}\text{Gd}^m$ at 377.76 keV										Nub211 **
* $^{141}\text{Tb}-\text{u}$	$M-A=-54541(34)$ keV for mixture gs+m at 0#200 keV										Nub211 **
* $^{141}\text{Sm}-^{133}\text{Cs}_{1.060}$	$D_M=18694(14)$ and $D_M=18878(14)$ from $^{141}\text{Sm}^m$ at 175.9 keV										Nub211 **
* $^{141}\text{Eu}-^{133}\text{Cs}_{1.060}$	Slight (< 10%) isomeric contamination cannot be excluded										00Be42 **
* $^{141}\text{Ho}(p)^{140}\text{Dy}$	$E_p=1230(20)$ from $^{141}\text{Ho}^m$ at 66(2) keV										Nub211 **
* $^{141}\text{Cs}(\beta^-)^{141}\text{Ba}$	$E_{\beta^-}=5215(15)$ to $(5/2)^-$ level at 48.53 keV										Ens141 **
* $^{141}\text{Ce}(\beta^-)^{141}\text{Pr}$	$E_{\beta^-}=442(3)$ 444(4) 432(2) 436(2) 436.7(2.6) respectively, to $7/2^+$ level at 145.4434										Ens141 **
* $^{141}\text{Nd}(\beta^+)^{141}\text{Pr}$	Was erroneously quoted 77Ga.A in the 1993 tables										GAu **
* $^{141}\text{Pm}(\beta^+)^{141}\text{Nd}$	Original error 40 increased due to lack of information on calibration										GAu **
* $^{141}\text{Sm}(\beta^+)^{141}\text{Pm}$	$E_{\beta^+}=3180(50)$, $3100(50)$ to $3/2^+$ level at 403.82, $(1/2)^+$ at 438.69 keV										Ens141 **
*	and $E_{\beta^+}=1670(70)$, $1600(70)$ from $^{141}\text{Sm}^m$										77Ke03 **
*	at 175.9 to $11/2^-$ at 2091.71, $(9/2, 11/2, 13/2)^-$ at 2119.05										Ens141 **
* $^{141}\text{Sm}(\beta^+)^{141}\text{Pm}$	$E_{\beta^+}=3020(60)$ 32% to $3/2^+$ level at 403.82, 31% to $(1/2)^+$ 438.29 keV										Ens141 **
* $^{141}\text{Sm}(\beta^+)^{141}\text{Pm}$	$Q_{\beta^+}=4700(80)$ from $^{141}\text{Sm}^m$ at 175.9 keV										Nub211 **
* $^{141}\text{Eu}(\beta^+)^{141}\text{Sm}$	$E_{\beta^+}=4620(110)$ to $(5/2)^+$ level at 395.56 keV, and other E_{β^+} (not given)										Ens141 **
* $^{141}\text{Eu}(\beta^+)^{141}\text{Sm}$	$E_{\beta^+}=4925(40)$ to $3/2^+$ level at 1.58 keV										Ens141 **
* $^{141}\text{Eu}(\beta^+)^{141}\text{Sm}$	$E_{\beta^+}=4960(40)$ to $3/2^+$ level at 1.58 keV										Ens141 **
$^{142}\text{I}-\text{u}$	-58798	268	-58833	5	-0.1	U			GT1	1.5	04Ma.A
$^{142}\text{I}-^{133}\text{Cs}_{1.068}$	42143.9	5.3				2			CP2	1.0	20Or02
$^{142}\text{Xe}-\text{u}$	-70247	111	-70026.9	2.9	0.8	U			GT2	2.5	08Kn.A
$^{142}\text{Xe}-^{133}\text{Cs}_{1.068}$	30950.4	2.9				2			MA8	1.0	09Ne11
$^{142}\text{Cs}-^{133}\text{Cs}_{1.068}$	25270	16	25277	8	0.4	-			MA4	1.0	99Am05
	25304	23			-1.2	-			CP1	1.0	13Va12
ave.	25281	13			-0.3	1	33	33 ^{142}Cs			average
$^{142}\text{Ba}-^{133}\text{Cs}_{1.068}$	17410	15	17410	6	0.0	-			MA1	1.0	99Am05
	17420	16			-0.6	-			MA4	1.0	99Am05
ave.	17415	11			-0.4	1	34	34 ^{142}Ba			average
$^{142}\text{Ba}-\text{u}$	-83576.8	9.1	-83567	6	1.1	1	49	49 ^{142}Ba	CP1	1.0	06Sa56
$\text{C}_{11} \text{H}_{10}-^{142}\text{Ce}$	169111	15	169000.1	2.6	-1.8	U			R05	4.0	65De13
	168955	40			0.5	U			M17	2.5	66Be10
	168955	40			0.5	U			M17	2.5	66Be10
$\text{C}_{10} \text{ } ^{13}\text{C} \text{H}_9-^{142}\text{Ce}$	164528	82	164529.9	2.6	0.0	U			R05	4.0	65De13
$\text{C}_{10} \text{N} \text{H}_8-^{142}\text{Ce}$	156558	42	156424.1	2.6	-0.8	U			R05	4.0	65De13
$\text{C}_{11} \text{H}_{10}-^{142}\text{Nd}$	170509	36	170521.5	1.3	0.1	U			R05	4.0	65De13
$\text{C}_{10} \text{N} \text{H}_8-^{142}\text{Nd}$	157870	43	157945.4	1.3	0.4	U			R05	4.0	65De13
$\text{C}_{10} \text{O} \text{H}_6-^{142}\text{Nd}$	134076	36	134136.0	1.3	0.4	U			R05	4.0	65De13
$\text{C}_{10} \text{ } ^{13}\text{C} \text{H}_9-^{142}\text{Nd}$	166021	32	166051.3	1.3	0.2	U			R05	4.0	65De13 *
$^{142}\text{Pm}-\text{u}$	-87136	30	-87109	25	0.9	-			GS2	1.0	05Li24
ave.	-87124	27			0.6	1	89	89 ^{142}Pm			average
$^{142}\text{Sm}-^{133}\text{Cs}_{1.068}$	16173	14	16186.7	2.0	1.0	U			MA5	1.0	00Be42
$^{142}\text{Eu}^m-^{133}\text{Cs}_{1.068}$	24909	15	24910	13	0.1	2			MA5	1.0	00Be42
$^{142}\text{Eu}^m-\text{u}$	-76063	30	-76067	13	-0.1	R			GS2	1.0	05Li24
$^{142}\text{Gd}-\text{u}$	-71884	30				2			GS2	1.0	05Li24

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{142}\text{Cs}-^{136}\text{Xe}_{1.044}$	21171	11	21168	8	-0.3	1	48	48 ^{142}Cs	CP1	1.0	12Va02
$^{142}\text{Ce}-\text{C}_{11}\text{H}_9$	-161176	40	-161175.1	2.6	0.0	U			M17	2.5	66Be10
$^{142}\text{Nd}-\text{C}_{11}\text{H}_9$	-162665	30	-162696.5	1.3	-0.4	U			M17	2.5	66Be10
$^{142}\text{Ce}-^{140}\text{Ce}$	3818	3	3801.8	2.7	-2.2	U			M17	2.5	66Be10
$^{142}\text{Ce}-^{138}\text{Ce}$	3644	35	3256.0	2.7	-2.8	B			R05	4.0	65De13
$^{139}\text{Cs}-^{142}\text{Cs}_{.685}$ $^{132}\text{Cs}_{.316}$	-4840	40	-4858	6	-0.2	U			P23	2.5	82Au01
$^{140}\text{Cs}-^{142}\text{Cs}_{.789}$ $^{132}\text{Cs}_{.212}$	-2950	40	-2937	10	0.1	U			P23	2.5	82Au01
$^{141}\text{Cs}-^{142}\text{Cs}_{.794}$ $^{137}\text{Cs}_{.206}$	-580	40	-660	11	-0.8	U			P23	2.5	82Au01
$^{138}\text{Cs}^x-^{142}\text{Cs}_{.194}$ $^{137}\text{Cs}_{.806}$	550	40	589	25	0.4	U			P23	2.5	82Au01
$^{140}\text{Cs}-^{142}\text{Cs}_{.329}$ $^{139}\text{Cs}_{.671}$	260	40	300	9	0.4	U			P23	2.5	82Au01
$^{141}\text{Cs}-^{142}\text{Cs}_{.662}$ $^{139}\text{Cs}_{.338}$	-410	40	-520	10	-1.1	U			P23	2.5	82Au01
$^{141}\text{Cs}-^{142}\text{Cs}_{.496}$ $^{140}\text{Cs}_{.504}$	-640	40	-669	11	-0.3	U			P23	2.5	82Au01
	-663	19			-0.1	U			P33	2.5	86Au02
$^{142}\text{Ce}(\alpha)^{138}\text{Ba}$	1545	200	1304.0	2.5	-1.2	U					57Ri43
$^{140}\text{Ce}(\text{t,p})^{142}\text{Ce}$	4112	5	4119.5	2.5	1.5	1	24	21 ^{142}Ce	LAI		72Mu09
$^{142}\text{Ce}(\text{p,t})^{140}\text{Ce}$	-4170	20	-4119.5	2.5	2.5	U			Osa		70Ya05
$^{142}\text{Nd}(\text{p,t})^{140}\text{Nd}$	-9150	20	-9354	3	-10.2	B			Osa		71Ya10 *
$^{142}\text{Ce}(\gamma,\text{n})^{141}\text{Ce}$	-7240	70	-7173.2	2.5	1.0	U			Phi		60Ge01
$^{142}\text{Ce}(\text{d,t})^{141}\text{Ce}$	-909	15	-915.9	2.5	-0.5	U			Mtr		72Le17
$^{141}\text{Pr}(\text{n},\gamma)^{142}\text{Pr}$	5843.14	0.10	5843.15	0.08	0.1	-			MMn		81Ke11 Z
	5843.16	0.12			-0.1	-			Bdn		06Fi.A
$^{141}\text{Pr}(\text{d,p})^{142}\text{Pr}$	3626	10	3618.59	0.08	-0.7	U			MIT		64Sp12
$^{141}\text{Pr}(\text{n},\gamma)^{142}\text{Pr}$	ave.	5843.15	0.08	5843.15	0.08	0.0	1	100	54 ^{142}Pr		average
$^{142}\text{Xe}(\beta^-)^{142}\text{Cs}$	5040	100	5285	8	2.4	U			Trs		78Wo15
$^{142}\text{Cs}(\beta^-)^{142}\text{Ba}$	7230	70	7328	8	1.4	U			Trs		78Wo15
	7329	20			-0.1	o			Gsn		81De25
	7280	40			1.2	U			Bwg		87Gr.A
	7315	15			0.8	1	31	19 ^{142}Cs	Gsn		92Pr04
$^{142}\text{Ba}(\beta^-)^{142}\text{La}$	2200	25	2182	8	-0.7	1	11	6 ^{142}La			83Ch39
	2216	5			-6.9	C			McG		84He.A
$^{142}\text{La}(\beta^-)^{142}\text{Ce}$	4517	25	4509	6	-0.3	U					65Pr03
	4510	6			-0.2	1	95	94 ^{142}La	McG		84He.A
$^{142}\text{Pr}(\beta^-)^{142}\text{Nd}$	2164	2	2163.7	1.4	-0.2	-					66Be12
	2158	3			1.9	-					75Ra09
	ave.	2162.2	1.7		0.9	1	67	46 ^{142}Pr			average
$^{142}\text{Pm}(\beta^+)^{142}\text{Nd}$	4800	80	4809	24	0.1	R					60Ma.A
	4880	80			-0.9	R			IRS		83Al06
	4880	160			-0.4	U			LBL		91Fi03
$^{142}\text{Sm}(\beta^+)^{142}\text{Pm}$	2050	70	2160	24	1.6	1	11	11 ^{142}Pm			60Ma.A
	2100	400			0.1	U			LBL		91Fi03
$^{142}\text{Eu}(\beta^+)^{142}\text{Sm}$	8000	300	7673	30	-1.1	U					75Ke08
	7400	100			2.7	U					82Gr.A
	7000	300			2.2	U			LBL		91Fi03
	7673	30			2				Dbn		94Po26
$^{142}\text{Eu}^m(\beta^+)^{142}\text{Sm}$	8150	100	8126	13	-0.2	U					75Ke08 *
	8174	50			-1.0	U			IRS		83Al06 *
	7480	100			6.5	B			IRS		93Al03 *
	8150	60			-0.4	U			Dbn		94Po26 *
$^{142}\text{Gd}(\beta^+)^{142}\text{Eu}$	4200	300	4350	40	0.5	U			LBL		91Fi03
$^{142}\text{Tb}(\beta^+)^{142}\text{Gd}$	10400	700				3			LBL		91Fi03
$^{142}\text{Dy}(\beta^+)^{142}\text{Tb}$	7100	200	6440#	200#	-3.3	D			LBL		91Fi03 *
* $\text{C}_{10}^{13}\text{C H}_9-^{142}\text{Nd}$	Original 1002055(32) is certainly a typo; rebuilt from M=141.907760(36)u										WgM127**
* $^{142}\text{Nd}(\text{p,t})^{140}\text{Nd}$	Disagrees strongly with $^{140}\text{Nd}-\text{u}$										AHW **
* $^{142}\text{Eu}^m(\beta^+)^{142}\text{Sm}$	$E_{\beta^+}=4760(100)$ 4782(50) respectively, to 7^- level at 2372.1 keV										Ens118 **
* $^{142}\text{Eu}^m(\beta^+)^{142}\text{Sm}$	Measured half-life 73.4(0.5) s corresponds to $^{142}\text{Eu}^m$										GAu **
* $^{142}\text{Eu}^m(\beta^+)^{142}\text{Sm}$	$E_{\beta^+}=4756(60)$ to 7^- level at 2372.1 keV										Ens118 **
* $^{142}\text{Dy}(\beta^+)^{142}\text{Tb}$	Trends from Mass Surface TMS suggest ^{142}Dy 660 keV more bound										GAu **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{143}\text{I}-u$	-53849	495	-54530#	220#	-0.5	D			GT3	2.5	16Kn03 *
$^{143}\text{Xe}-u$	-64649	290	-64630	5	0.0	o			GT1	1.5	04Ma.A
	-64858	108			0.8	o			GT2	2.5	08Kn.A
	-64684	133			0.2	U			GT2	2.5	08Su19
$^{143}\text{Xe}-^{133}\text{Cs}_{1.075}$	37008.7	5.0				2			MA8	1.0	09Ne11
$^{143}\text{Cs}-u$	-72771	117	-72653	8	0.4	U			GT2	2.5	08Kn.A
$^{143}\text{Cs}-^{133}\text{Cs}_{1.075}$	28985.6	8.5	28986	8	0.1	1	91	91 ^{143}Cs	CP1	1.0	13Va12
$^{143}\text{Ba}-^{133}\text{Cs}_{1.075}$	22268	16	22264	7	-0.2	1	21	21 ^{143}Ba	MA1	1.0	99Am05
$^{143}\text{Ba}-u$	-79375.0	8.5	-79375	7	0.0	1	73	73 ^{143}Ba	CP1	1.0	06Sa56
$^{143}\text{La}-u$	-83918.1	8.7	-83921	8	-0.3	1	82	82 ^{143}La	CP1	1.0	06Sa56
$\text{C}_{10}\text{ N H}_9-^{143}\text{Nd}$	163719	31	163679.5	1.3	-0.3	U			R05	4.0	65De13
$\text{C}_{10}\text{ O H}_7-^{143}\text{Nd}$	139814	42	139870.0	1.3	0.3	U			R05	4.0	65De13
$^{143}\text{Pm}-^{133}\text{Cs}_{1.075}$	12567	15	12577	3	0.7	U			MA5	1.0	00Be42
$^{143}\text{Sm}-^{133}\text{Cs}_{1.075}$	16268	15	16274.0	3.0	0.4	U			MA5	1.0	00Be42
$^{143}\text{Sm}-u$	-85347	30	-85365.2	3.0	-0.6	U			GS2	1.0	05Li24 *
$^{143}\text{Eu}-^{133}\text{Cs}_{1.075}$	21947	14	21938	12	-0.7	2			MA5	1.0	00Be42
$^{143}\text{Eu}-u$	-79706	30	-79701	12	0.2	R			GS2	1.0	05Li24
$^{143}\text{Gd}-u$	-73012	56	-73250	220	-4.2	C			GS2	1.0	05Li24 *
$^{143}\text{Tb}-u$	-64879	64	-64860	60	0.3	U			GS2	1.0	05Li24 *
$^{143}\text{Tb}-^{85}\text{Rb}_{1.682}$	83507	55				2			SH1	1.0	07Ra37 *
$^{143}\text{Dy}-^{85}\text{Rb}_{1.682}$	92364	14				2			SH1	1.0	07Ra37 *
$^{143}\text{Nd}-^{35}\text{Cl}-^{141}\text{Pr}-^{37}\text{Cl}$	5116	4	5110.3	1.5	-0.6	U			H21	2.5	70Ma05
$^{143}\text{Nd}-\text{C}_{11}\text{ H}_{10}$	-168422	30	-168430.5	1.3	-0.1	U			M17	2.5	66Be10
$^{143}\text{Nd}-^{142}\text{Nd}$	2322	46	2090.99	0.07	-1.3	U			R05	4.0	65De13
	2084	2			1.4	U			M17	2.5	66Be10
$^{143}\text{Nd}-\text{C}_{11}\text{ H}_9$	-160594	30	-160605.5	1.3	-0.2	U			M17	2.5	66Be10
$^{141}\text{Cs}-^{143}\text{Cs}_{.493}-^{139}\text{Cs}_{.507}$	-230	40	-198	10	0.3	U			P23	2.5	82Au01
	-115	22			-1.5	U			P33	2.5	86Au02
$^{142}\text{Cs}-^{143}\text{Cs}_{.497}-^{141}\text{Cs}_{.504}$	647	15	657	9	0.3	U			P33	2.5	86Au02
$^{143}\text{Nd}(n,\alpha)^{140}\text{Ce}$	9699	15	9718.3	1.5	1.3	U			ILL		75Em.A
$^{143}\text{Nd}(p,t)^{141}\text{Nd}$	-7450	20	-7472	3	-1.1	U			Osa		71Ya10
$^{142}\text{Ce}(n,\gamma)^{143}\text{Ce}$	5145.9	0.5	5144.80	0.09	-2.2	U					76Ge02
	5144.78	0.15			0.1	-			Ptn		80Ba.A Z
	5144.81	0.12			-0.1	-			Bdn		06Fi.A
$^{142}\text{Ce}(d,p)^{143}\text{Ce}$	2945	15	2920.23	0.09	-1.7	U			Mtr		72Le17
$^{142}\text{Ce}(n,\gamma)^{143}\text{Ce}$	ave. 5144.80	0.09	5144.80	0.09	0.0	1	100	78 ^{142}Ce			average
$^{142}\text{Nd}(n,\gamma)^{143}\text{Nd}$	6123.62	0.08	6123.57	0.07	-0.6	-			MMn		82Is05 Z
	6123.41	0.14			1.2	-			Bdn		06Fi.A
$^{142}\text{Nd}(d,p)^{143}\text{Nd}$	3916	15	3899.01	0.07	-1.1	U			Kop		67Ch16
	3902	15			-0.2	U			Tal		67Ne04
	3902	15			-0.2	U			Hei		67Wi08
$^{142}\text{Nd}(n,\gamma)^{143}\text{Nd}$	ave. 6123.57	0.07	6123.57	0.07	0.1	1	100	77 ^{142}Nd			average
$^{142}\text{Nd}(^3\text{He},d)^{143}\text{Pm}$	-1099	25	-1193.9	2.7	-3.8	B			Oak		71Wi04
	-1195	5			0.2	1	29	29 ^{143}Pm	McM		80St10 *
$^{143}\text{Cs}(\beta^-)^{143}\text{Ba}$	6250	90	6262	10	0.1	o			Gsn		81De25 *
	6240	70			0.3	U			Bwg		87Gr.A
	6270	25			-0.3	1	15	9 ^{143}Cs	Gsn		92Pr04
$^{143}\text{Ba}(\beta^-)^{143}\text{La}$	4240	50	4234	10	-0.1	U					79Sc11
	4259	40			-0.6	U			Gsn		81De25
	4210	70			0.3	U			Bwg		87Gr.A
$^{143}\text{La}(\beta^-)^{143}\text{Ce}$	3425	17	3435	8	0.6	1	20	18 ^{143}La			84Is09 *
$^{143}\text{Ce}(\beta^-)^{143}\text{Pr}$	1460.6	2.	1461.8	1.9	0.6	1	87	77 ^{143}Ce			77Ra18 *
$^{143}\text{Pr}(\beta^-)^{143}\text{Nd}$	932	2	934.1	1.4	1.1	-					49Fe18
	935	2			-0.4	-					76Ra33
	ave. 933.5	1.4			0.4	1	93	90 ^{143}Pr			average
$^{143}\text{Pm}(\beta^+)^{143}\text{Nd}$	1000	70	1041.6	2.7	0.6	U					67Va01 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{143}\text{Sm}(\beta^+)^{143}\text{Pm}$	3492	30	3444	4	-1.6	U					66Be21
	3437	30			0.2	U			IRS		83Al06
	3500	60			-0.9	U			IRS		93Al03
	3461	40			-0.4	U			Dbn		94Po26
$^{143}\text{Eu}(\beta^+)^{143}\text{Sm}$	5100	50	5276	11	3.5	B					74Ch21
	5160	60			1.9	o			IRS		83Ve.A
	5240	70			0.5	o			IRS		83Al06
	5250	80			0.3	U			IRS		93Al03
	5236	30			1.3	R			Dbn		94Po26
$^{143}\text{Gd}(\beta^+)^{143}\text{Eu}$	6010	200				3			IRS		93Al03 *
* $^{143}\text{I}-\text{u}$	Trends from Mass Surface TMS suggest ^{143}I 630 keV more bound										GAu **
* $^{143}\text{Sm}-\text{u}$	$M-A=-78746(28)$ keV for $^{143}\text{Sm}^m$ at 753.99 keV										Nub211 **
* $^{143}\text{Gd}-\text{u}$	$M-A=-67934(28)$ keV for mixture gs+m at 152.6 keV										Nub211 **
* $^{143}\text{Tb}-\text{u}$	$M-A=-60434(32)$ keV for mixture gs+m at 0#100 keV										Nub211 **
*	outweighed by next item before correcting for isomeric mixture										GAu **
* $^{143}\text{Tb}-^{85}\text{Rb}_{1.682}$	$M-A=-60419.5(7.8)$ keV for mixture gs+m at 0#100 keV										Nub211 **
* $^{143}\text{Dy}-^{85}\text{Rb}_{1.682}$	$D_M=92354(17)$ and $D_M=92705(14)$ for $^{143}\text{Dy}^m$ at 310.7 keV										Nub211 **
* $^{142}\text{Nd}(^3\text{He,d})^{143}\text{Pm}$	Based on $^{146}\text{Nd}(^3\text{He,d})^{147}\text{Pm}$ $Q=-87.6(0.9)$ keV										AHW **
* $^{143}\text{Cs}(\beta^-)^{143}\text{Ba}$	$E_{\beta^-}=6070(50)$ and $5847(100)$ to $3/2^-$ level at 228.83 keV										Ens123 **
* $^{143}\text{La}(\beta^-)^{143}\text{Ce}$	$E_{\beta^-}=3419(17)$ 64% to $3/2^-$ ground state, 29% to $7/2^-$ level at 18.9 keV										Ens123 **
* $^{143}\text{Ce}(\beta^-)^{143}\text{Pr}$	$E_{\beta^-}=1110(2)$ to $3/2^+$ level at 350.622 keV										Ens123 **
* $^{143}\text{Pm}(\beta^+)^{143}\text{Nd}$	$pK=0.806(0.023)$ to $3/2^-$ level at 742.05 keV, and $p^+ < 1 \times 10^{-6}$										Ens123 **
* $^{143}\text{Gd}(\beta^+)^{143}\text{Eu}$	$Q_{\beta^+}=6160(200)$ from $^{143}\text{Gd}^m$ at 152.6 keV										Nub211 **
$^{144}\text{Xe}-^{133}\text{Cs}_{1.083}$	41340.6	5.7				2			MA8	1.0	09Ne11
$^{144}\text{Cs}-^{133}\text{Cs}_{1.083}$	34488	33	34471	22	-0.5	1	43	43 ^{144}Cs	CP1	1.0	13Va12
$^{144}\text{Ba}-^{133}\text{Cs}_{1.083}$	25347	15	25350	8	0.2	1	26	26 ^{144}Ba	MA1	1.0	99Am05
$^{144}\text{Ba}-\text{u}$	-77045.3	9.1	-77045	8	0.0	1	71	71 ^{144}Ba	CP1	1.0	06Sa56
$^{144}\text{La}-\text{u}$	-80337.1	19.3	-80354	14	-0.9	2			CP1	1.0	06Sa56
	-80373	20			0.9	2			GS3	1.0	12Ch19
$\text{C}_{10} \text{O H}_8-^{144}\text{Nd}$	147408	28	147422.1	1.3	0.1	U			R05	4.0	65De13
	147384	29			0.3	U			R05	4.0	65De13
$\text{C}_9 \text{ }^{13}\text{C N H}_9-^{144}\text{Nd}$	166777	28	166761.3	1.3	-0.1	U			R05	4.0	65De13
$\text{C}_{10} \text{H}_8 \text{O}-^{144}\text{Sm}$	145450	50	145508.6	1.6	0.3	U			R04	4.0	64De15
$\text{C}_9 \text{ }^{13}\text{C H}_9 \text{N}-^{144}\text{Sm}$	164955	46	164847.8	1.6	-0.6	U			R04	4.0	64De15
$^{144}\text{Eu}-^{133}\text{Cs}_{1.083}$	21223	17	21215	12	-0.5	1	46	46 ^{144}Eu	MA5	1.0	00Be42
$^{144}\text{Eu}-\text{u}$	-81117	30	-81181	12	-2.1	1	15	15 ^{144}Eu	GS2	1.0	05Li24
$^{144}\text{Gd}-\text{u}$	-77037	30				2			GS2	1.0	05Li24
$^{144}\text{Tb}-\text{u}$	-66955	30				2			GS2	1.0	05Li24 *
$^{144}\text{Dy}-\text{u}$	-60746	33	-60730	8	0.5	U			GS2	1.0	05Li24
$^{144}\text{Dy}-^{85}\text{Rb}_{1.694}$	88697.7	7.7				2			SH1	1.0	07Ra37
$^{144}\text{Ho}-^{85}\text{Rb}_{1.694}$	101537.9	9.1				2			SH1	1.0	07Ra37
$^{144}\text{Nd } ^{35}\text{Cl}-^{142}\text{Nd } ^{37}\text{Cl}$	5329	3	5314.09	0.12	-1.2	U			H12	4.0	64Ba15
	5308	3			0.8	U			H21	2.5	70Ma05
$^{144}\text{Sm}-^{144}\text{Nd}$	1951	3	1913.5	0.9	-3.1	B			H19	4.0	64Mc11
	1911.9	1.1			0.6	-			H25	2.5	72Ba08
	1913.68	0.94			-0.2	-			SH1	1.0	11Go23
ave.	1913.5	0.9			-0.0	1	92	86 ^{144}Sm			average
$^{144}\text{Nd}-^{143}\text{Nd}$	269	25	272.98	0.06	0.0	U			R05	4.0	65De13
	273	3			-0.0	U			M17	2.5	66Be10
$^{144}\text{Nd}-^{142}\text{Nd}$	2366	3	2363.97	0.09	-0.3	U			M17	2.5	66Be10
$^{142}\text{Cs}-^{144}\text{Cs}_{.592} \text{ } ^{139}\text{Cs}_{.409}$	-60	40	-51	14	0.1	U			P23	2.5	82Au01
$^{143}\text{Cs}-^{144}\text{Cs}_{.745} \text{ } ^{140}\text{Cs}_{.255}$	-920	50	-891	17	0.2	U			P23	2.5	82Au01
$^{142}\text{Cs}-^{144}\text{Cs}_{.329} \text{ } ^{141}\text{Cs}_{.671}$	290	40	276	11	-0.1	U			P23	2.5	82Au01
$^{143}\text{Cs}-^{144}\text{Cs}_{.662} \text{ } ^{141}\text{Cs}_{.338}$	-651	21	-617	16	0.7	U			P33	2.5	86Au02

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{143}\text{Cs}-^{144}\text{Cs}_{.497} \ ^{142}\text{Cs}_{.504}$	-790	50	-690	13	0.8	U			P23	2.5	82Au01
$^{144}\text{Nd}(\alpha)^{140}\text{Ce}$	1882.4	30.	1901.3	1.5	0.6	U					61Ma05
	1882.4	20.			0.9	U					65Is01
$^{144}\text{Sm}(^3\text{He}, ^6\text{He})^{141}\text{Sm}$	-8693	12	-8693	9	0.0	1	51	50	^{141}Sm	MSU	78Pa11
$^{142}\text{Ce}(\text{t}, \text{p})^{144}\text{Ce}$	3582	15	3560	3	-1.5	U			LAI		72Mu09
$^{142}\text{Nd}(\text{t}, \text{p})^{144}\text{Nd}$	5450	30	5458.81	0.09	0.3	U			Ald		72Ch11
$^{144}\text{Nd}(\text{p}, \text{t})^{142}\text{Nd}$	-5470	20	-5458.81	0.09	0.6	U			Osa		71Ya10
$^{144}\text{Sm}(\text{p}, \text{t})^{142}\text{Sm}$	-10649	15	-10644.5	1.2	0.3	U			Ham		73Oe02
$^{143}\text{Nd}(\text{n}, \gamma)^{144}\text{Nd}$	7817.11	0.07	7817.04	0.05	-1.1	-			MMn		82Is05 Z
	7816.93	0.08			1.3	-			ILn		91Ro.A Z
	7816.94	0.23			0.4	U			Bdn		06Fi.A
$^{144}\text{Nd}(\text{d}, \text{t})^{143}\text{Nd}$	-1555	15	-1559.81	0.05	-0.3	U			Ors		73Ga01
$^{143}\text{Nd}(\text{n}, \gamma)^{144}\text{Nd}$	ave. 7817.03	0.05	7817.04	0.05	0.1	1	100	61	^{143}Nd		average
$^{143}\text{Nd}(^3\text{He}, \text{d})^{144}\text{Pm}$	-804	5	-790.7	2.6	2.7	B			McM		80St10 *
$^{143}\text{Nd}(^3\text{He}, \text{d})^{144}\text{Pm}-^{142}\text{Nd}()^{143}\text{Pm}$	402.7	1.6	403.2	1.5	0.3	1	91	49	^{143}Pm		75Ma04
$^{144}\text{Sm}(\text{t}, \alpha)^{143}\text{Pm}$	13542	25	13519.9	2.7	-0.9	U			Ald		68Ha13
$^{144}\text{Sm}(\text{d}, \text{t})^{143}\text{Sm}$	-4262	10	-4262.6	2.3	-0.1	U					72Ja28
$^{144}\text{Sm}(\text{p}, \text{d})^{143}\text{Sm}-^{148}\text{Gd}()^{147}\text{Gd}$	-1536	2	-1536.0	2.0	0.0	1	100	100	^{143}Sm		86Ru04
$^{144}\text{Tm}(\text{p})^{143}\text{Er}$	1712.0	16.				3			ORp		05Gr32 *
$^{144}\text{Cs}(\beta^-)^{144}\text{Ba}$	8451	30	8496	20	1.5	o			Gsn		81De25
	8560	80			-0.8	-			Bwg		87Gr.A
	8462	35			1.0	-			Gsn		92Pr04
	ave. 8480	30			0.6	1	41	38	^{144}Cs		average
$^{144}\text{Ba}(\beta^-)^{144}\text{La}$	3055	70	3083	15	0.4	U			Bwg		87Gr.A
$^{144}\text{La}(\beta^-)^{144}\text{Ce}$	4300	100	5582	13	12.8	B					79Ik07
	5435	90			1.6	U			Bwg		87Gr.A
	5540	100			0.4	o			Kur		02Sh.B
	5540	100			0.4	U			Kur		02Sh16
$^{144}\text{Ce}(\beta^-)^{144}\text{Pr}$	315.6	1.5	318.6	0.8	2.0	3					66Da04
	320	1			-1.4	3					76Ra33
$^{144}\text{Pr}(\beta^-)^{144}\text{Nd}$	2996	3	2997.4	2.4	0.5	2					59Po77
	3000	4			-0.6	2					66Da04
$^{144}\text{Eu}(\beta^+)^{144}\text{Sm}$	6330	30	6346	11	0.5	-			IRS		83Al06
	6400	80			-0.7	U			IRS		93Al03
	6287	30			2.0	-			Dbn		94Po26
$^{144}\text{Sm}(\text{p}, \text{n})^{144}\text{Eu}$	-7110.0	30.	-7129	11	-0.6	-					65Me12
$^{144}\text{Eu}(\beta^+)^{144}\text{Sm}$	ave. 6315	17	6346	11	1.8	1	39	39	^{144}Eu		average
$^{144}\text{Gd}(\beta^+)^{144}\text{Eu}$	4300	400	3860	30	-1.1	U					70Ar04
* $^{144}\text{Tb}-\text{u}$	$M-A=-61971(28)$ keV for $^{144}\text{Tb}^m$ at 396.9 keV										Nub211 **
* $^{143}\text{Nd}(^3\text{He}, \text{d})^{144}\text{Pm}$	Based on $^{146}\text{Nd}(^3\text{He}, \text{d})^{147}\text{Pm}$ $Q=-87.6(0.9)$ keV										AHW **
* $^{144}\text{Tm}(\text{p})^{143}\text{Er}$	Trends from Mass Surface TMS suggest ^{144}Tm 160 keV less bound										GAU **
$^{145}\text{Xe}-^{133}\text{Cs}_{1.090}$	47777	12				2			MA8	1.0	09Ne11
$^{145}\text{Cs}-^{133}\text{Cs}_{1.090}$	38588	12	38586	10	-0.1	-			MA8	1.0	08We02
	38583	17			0.2	-			CP1	1.0	13Va12
	ave. 38586	10			-0.0	1	99	99	^{145}Cs		average
$^{145}\text{Ba}-\text{u}$	-72481.6	9.1				2			CP1	1.0	06Sa56
$^{145}\text{La}-\text{u}$	-78188.8	13.3	-78192	13	-0.2	1	98	98	^{145}La		06Sa56
$^{145}\text{Ce}-\text{u}$	-82771.8	92.2	-82730	40	0.4	1	16	16	^{145}Ce		06Sa56
$\text{C}_{10} \text{O H}_9-^{145}\text{Nd}$	152641	55	152760.8	1.4	0.5	U			R05	4.0	65De13
	152653	30			0.9	U			R05	4.0	65De13
$\text{C}_9 \ ^{13}\text{C O H}_8-^{145}\text{Nd}$	148231	31	148290.6	1.4	0.5	U			R05	4.0	65De13
$^{145}\text{Pm}-\text{u}$	-87255	30	-87244	3	0.4	U			GS2	1.0	05Li24
$^{145}\text{Sm}-\text{u}$	-86535	30	-86582.8	1.6	-1.6	U			GS2	1.0	05Li24
$^{145}\text{Eu}-^{133}\text{Cs}_{1.090}$	19338	17	19330	3	-0.5	U			MA5	1.0	00Be42
$^{145}\text{Gd}-\text{u}$	-78287	30	-78290	21	-0.1	-			GS2	1.0	05Li24
	-78294	30			0.1	-			GS2	1.0	05Li24 *
	ave. -78291	21			0.0	1	100	100	^{145}Gd		average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{145}\text{Tb}-\text{u}$	-71180	290	-71280	120	-0.4	1	17	17 ^{145}Tb	GS2	1.0	05Li24	*
$^{145}\text{Dy}-\text{u}$	-62575	49	-62526	7	1.0	U			GS2	1.0	05Li24	*
$^{145}\text{Dy}-^{85}\text{Rb}_{1.706}$	87960.7	7.0				2			SH1	1.0	07Ra37	*
$^{145}\text{Ho}-^{85}\text{Rb}_{1.706}$	97754.1	8.0				2			SH1	1.0	07Ra37	
$^{145}\text{Nd } ^{35}\text{Cl}_2-^{141}\text{Pr } ^{37}\text{Cl}_2$	10828	7	10819.8	1.5	-0.5	U			H21	2.5	70Ma05	
$^{145}\text{Nd } ^{35}\text{Cl}-^{143}\text{Nd } ^{37}\text{Cl}$	5744	5	5709.46	0.26	-1.7	U			H12	4.0	64Ba15	
	5703	4			0.6	U			H21	2.5	70Ma05	
$^{145}\text{Nd}-^{144}\text{Nd}$	2582	21	2486.35	0.25	-1.1	U			R05	4.0	65De13	
	2480	2			1.3	U			M17	2.5	66Be10	
$^{145}\text{Nd}-^{143}\text{Nd}$	2862	40	2759.34	0.25	-0.6	U			R05	4.0	65De13	
	2751	3			1.1	U			M17	2.5	66Be10	
$^{142}\text{Cs}-^{145}\text{Cs}_{.490} \text{ } ^{139}\text{Cs}_{.511}$	240	50	150	9	-0.7	U			P23	2.5	82Au01	
$^{144}\text{Cs}-^{145}\text{Cs}_{.828} \text{ } ^{139}\text{Cs}_{.173}$	450	50	415	21	-0.3	U			P23	2.5	82Au01	
$^{143}\text{Cs}-^{145}\text{Cs}_{.592} \text{ } ^{140}\text{Cs}_{.409}$	-700	80	-610	10	0.5	U			P23	2.5	82Au01	
$^{143}\text{Cs}-^{145}\text{Cs}_{.493} \text{ } ^{141}\text{Cs}_{.507}$	-310	40	-309	10	0.0	U			P23	2.5	82Au01	
$^{144}\text{Cs}-^{145}\text{Cs}_{.662} \text{ } ^{142}\text{Cs}_{.338}$	320	18	319	21	-0.0	1	21	20 ^{144}Cs	P33	2.5	86Au02	
$^{144}\text{Cs}-^{145}\text{Cs}_{.497} \text{ } ^{143}\text{Cs}_{.503}$	600	40	616	21	0.2	U			P23	2.5	82Au01	
$^{145}\text{Pm}(\alpha) ^{141}\text{Pr}$	2303.6	40.	2322.1	2.9	0.5	U					62Nu01	
$^{145}\text{Nd}(\text{n},\alpha) ^{142}\text{Ce}$	8706	30	8747.3	2.2	1.4	U			ILL		75Em04	
$^{145}\text{Nd}(\text{p},\text{t}) ^{143}\text{Nd}$	-5100	20	-5090.53	0.24	0.5	U			Osa		71Ya10	
$^{144}\text{Nd}(\text{n},\gamma) ^{145}\text{Nd}$	5755.3	0.7	5755.29	0.23	-0.0	-					75Na.A	
	5756.9	2.0			-0.8	U					77Mc09	
	5755.26	0.25			0.1	-			Bdn		06Fi.A	
$^{144}\text{Nd}(\text{d},\text{p}) ^{145}\text{Nd}$	3521	15	3530.73	0.23	0.6	U			Hei		67Wi08	
	3538	15			-0.5	U			Ors		73Ga01	
$^{144}\text{Nd}(\text{n},\gamma) ^{145}\text{Nd}$	ave. 5755.26	0.24	5755.29	0.23	0.1	1	96	84 ^{145}Nd			average	
$^{144}\text{Nd}(^3\text{He},\text{d}) ^{145}\text{Pm}$	-680	5	-685.0	2.5	-1.0	1	26	25 ^{145}Pm	McM		80St10	*
$^{144}\text{Nd}(^3\text{He},\text{d}) ^{145}\text{Pm}-^{143}\text{Nd}() ^{144}\text{Pm}$	105.2	1.6	105.7	1.5	0.3	1	91	57 ^{144}Pm			75Ma04	
$^{144}\text{Sm}(\text{n},\gamma) ^{145}\text{Sm}$	6757.1	0.3	6757.10	0.30	-0.0	1	99	92 ^{145}Sm			79Wa22	
$^{144}\text{Sm}(\text{d},\text{p}) ^{145}\text{Sm}$	4533	12	4532.53	0.30	-0.0	U			Tal		65Ke09	
	4547	15			-1.0	U			Kop		67Ch16	
$^{144}\text{Sm}(^3\text{He},\text{d}) ^{145}\text{Eu}$	-2184	4	-2178.6	2.7	1.3	-			Mun		82Sc25	
	-2174	4			-1.2	-					84Ru.A	
	ave. -2179.0	2.8			0.1	1	92	91 ^{145}Eu			average	
$^{145}\text{Dy}(\varepsilon\text{p}) ^{144}\text{Gd}$	6000	500	6228	29	0.5	U					83La.A	*
$^{145}\text{Tm}(\text{p}) ^{144}\text{Er}$	1740.1	10.	1736	7	-0.4	3			ORp		98Ba13	
	1732.1	10.			0.4	3			Arp		07Se06	
$^{145}\text{Cs}(\beta^-) ^{145}\text{Ba}$	7358	70	7462	12	1.5	U			Gsn		81De25	
	7930	75			-6.2	C			Bwg		87Gr.A	
	7865	50			-8.1	B			Gsn		92Pr04	
$^{145}\text{Ba}(\beta^-) ^{145}\text{La}$	4925	80	5319	15	4.9	C			Bwg		87Gr.A	
$^{145}\text{La}(\beta^-) ^{145}\text{Ce}$	4110	80	4230	40	1.5	1	19	18 ^{145}Ce	Bwg		87Gr.A	
$^{145}\text{Ce}(\beta^-) ^{145}\text{Pr}$	2490	100	2560	30	0.7	-					67Ho19	*
	2600	100			-0.4	-					80Ya07	*
	2530	50			0.6	-			Bwg		87Gr.A	
	ave. 2530	40			0.6	1	68	67 ^{145}Ce			average	
$^{145}\text{Pr}(\beta^-) ^{145}\text{Nd}$	1805	10	1806	7	0.1	1	50	50 ^{145}Pr			59Dr.A	
$^{145}\text{Pm}(\varepsilon) ^{145}\text{Nd}$	143	15	164.5	2.5	1.4	U					59Br65	*
	150	5			2.9	B					74To04	*
$^{145}\text{Sm}(\varepsilon) ^{145}\text{Pm}$	607	6	616.1	2.5	1.5	-					71My01	*
	622	5			-1.2	-					83Vo10	*
	ave. 616	4			0.1	1	44	41 ^{145}Pm			average	
$^{145}\text{Eu}(\beta^+) ^{145}\text{Sm}$	2710	15	2659.9	2.7	-3.3	B					68Ad04	*
	2647	12			1.1	U					83Sc28	*
$^{145}\text{Gd}(\beta^+) ^{145}\text{Eu}$	5070	60	5065	20	-0.1	U					79Fi07	
	5090	90			-0.3	o			IRS		83Ve.A	*
	5070	80			-0.1	U			IRS		85Al13	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{145}\text{Gd}(\epsilon)^{145}\text{Eu}$	5000	70	5065	20	0.9	U					77Ho18
$^{145}\text{Tb}(\beta^+)^{145}\text{Gd}$	6700	200	6530	110	-0.9	-					86Ve.A *
	6400	150			0.8	-			IRS		93Al03
ave.	6510	120			0.2	1	84	83	^{145}Tb		average
$^{145}\text{Dy}(\beta^+)^{145}\text{Tb}^m$	7300	200				3			IRS		93Al03
* $^{145}\text{Gd}-\text{u}$	$M-A=-72181(28)$ keV for $^{145}\text{Gd}^m$ at 749.1 keV										Nub211 **
* $^{145}\text{Tb}-\text{u}$	$M-A=-65881(28)$ keV for mixture gs+m at 850(230) keV										Nub211 **
* $^{145}\text{Dy}-\text{u}$	$M-A=-58230(30)$ keV for mixture gs+m at 118.2 keV										Nub211 **
* $^{145}\text{Dy}-^{85}\text{Rb}_{1.706}$	$D_M=88054.7(6.8)$ μu for mixture gs+m at 118.2 keV with ratio R=0.741(13)										Nub211 **
* $^{144}\text{Nd}(\beta^+\text{He,d})^{145}\text{Pm}$	Based on $^{146}\text{Nd}(\beta^+\text{He,d})^{147}\text{Pm}$ $Q=-87.6(0.9)$ keV										AHW **
* $^{145}\text{Dy}(\text{ep})^{144}\text{Gd}$	As read from graph										AHW **
* $^{145}\text{Ce}(\beta^-)^{145}\text{Pr}$	$E_{\beta^-}=1700(100)$ 1810(100) respectively, to $(3/2)^-$ level at 786.91; and other E_{β^-}										Ens092 **
* $^{145}\text{Pm}(\epsilon)^{145}\text{Nd}$	LM/K=0.85(0.03) to $3/2^-$ level at 67.167 keV										Ens092 **
* $^{145}\text{Pm}(\epsilon)^{145}\text{Nd}$	pK=0.554(0.025) to $5/2^-$ level at 72.486 keV, and other pK										Ens092 **
* $^{145}\text{Sm}(\epsilon)^{145}\text{Pm}$	pK=0.27(0.03) 0.35(0.025) respectively, to $3/2^+$ level at 492.31 keV										Ens092 **
* $^{145}\text{Eu}(\beta^+)^{145}\text{Sm}$	$E_{\beta^+}=794(15)$ to $3/2^+$ level at 893.788 keV										Ens092 **
* $^{145}\text{Eu}(\beta^+)^{145}\text{Sm}$	pK=0.72(0.02) to $(5/2^-, 7/2^-)$ level at 2508.31 and 9^- at 2513.37 levels										Ens092 **
* $^{145}\text{Gd}(\beta^+)^{145}\text{Eu}$	$E_{\beta^+}=2310(90)$ to $3/2^+$ level at 1758.03 keV, and other E_{β^+}										Ens092 **
* $^{145}\text{Tb}(\beta^+)^{145}\text{Gd}$	$E_{\beta^+}=3300(200)$ to $(9/2^-)$ level at 2382.3(0.2) keV										Ens092 **
$^{146}\text{Xe}-^{133}\text{Cs}_{1.098}$	52332	26				2			MA8	1.0	09Ne11
$^{146}\text{Cs}-^{133}\text{Cs}_{1.098}$	44437.4	3.3	44436	3	-0.5	2			MA8	1.0	17At01
	44421.8	9.2			1.5	2			CP1	1.0	13Va12
$^{146}\text{Ba}-\text{u}$	-69618	112	-69636.8	1.9	-0.1	o			GT2	2.5	08Kn.A
	-69963	141			0.9	U			GT2	2.5	08Su19
	-69717.5	23.7			3.4	B			CP1	1.0	06Sa56
	-69636.8	1.9				2			CP2	1.0	18Or.A *
$^{146}\text{La}-\text{u}$	-74259	54	-74312.0	1.8	-1.0	U			CP1	1.0	06Sa56 *
	-74311.9	1.8			-0.0	1	100	100	^{146}La	1.0	20Or02 *
$^{146}\text{La}^m-\text{u}$	-74160.1	1.8				2			CP2	1.0	20Or02 *
$^{146}\text{Ce}-\text{u}$	-81191.8	20.8	-81188	16	0.2	-			CP1	1.0	06Sa56
	-81171	40			-0.4	-			GS3	1.0	12Ch19
ave.	-81187	18			-0.0	1	73	73	^{146}Ce		average
$\text{C}_{12}\text{H}_2-^{146}\text{Nd}$	102453	31	102527.6	1.4	0.6	U			R05	4.0	65De13
$\text{C}_{10}\text{O H}_{10}-^{146}\text{Nd}$	160017	27	160042.5	1.4	0.2	U			R05	4.0	65De13
	159971	50			0.4	U			R05	4.0	65De13
$\text{C}_9\text{ }^{13}\text{C O H}_9-^{146}\text{Nd}$	155525	35	155572.3	1.4	0.3	U			R05	4.0	65De13
$^{146}\text{Pm}-\text{u}$	-85289	30	-85298	5	-0.3	U			GS2	1.0	05Li24
$^{146}\text{Eu}-^{133}\text{Cs}_{1.098}$	21029	15	21025	6	-0.3	1	18	18	^{146}Eu	1.0	00Be42
$^{146}\text{Tb}-\text{u}$	-72464	77	-72750	50	-3.7	C			GS2	1.0	05Li24 *
$^{146}\text{Dy}-\text{u}$	-67150	30	-67155	7	-0.2	U			GS2	1.0	05Li24
$^{146}\text{Dy}-^{85}\text{Rb}_{1.718}$	84390.0	7.2	84390	7	-0.0	1	100	100	^{146}Dy	1.0	07Ra37
$^{146}\text{Ho}-^{133}\text{Cs}_{1.098}$	48797	10	48807	7	1.0	1	50	50	^{146}Ho	1.0	07Ra37
$^{146}\text{Ho}-^{85}\text{Rb}_{1.718}$	96549	10	96539	7	-1.0	1	50	50	^{146}Ho	1.0	07Ra37
$^{146}\text{Er}-^{85}\text{Rb}_{1.718}$	103960.4	9.2	103964	7	0.3	1	61	61	^{146}Er	1.0	07Ra37
$^{146}\text{Nd }^{35}\text{Cl}-^{144}\text{Nd }^{37}\text{Cl}$	6003	3	5979.78	0.27	-1.9	U			H12	4.0	64Ba15
	5966	4			1.4	U			H21	2.5	70Ma05
	5982.8	1.1			-1.1	U			H25	2.5	72Ba08
$^{146}\text{Nd}-^{145}\text{Nd}$	526	33	543.31	0.09	0.1	U			R05	4.0	65De13
	536	2			1.5	U			M17	2.5	66Be10
$^{146}\text{Nd}-^{144}\text{Nd}$	3147	36	3029.66	0.26	-0.8	U			R05	4.0	65De13
	3026	3			0.5	U			M17	2.5	66Be10
$^{145}\text{Cs}-^{146}\text{Cs}_{.828} \text{ }^{140}\text{Cs}_{.173}$	-580	80	-928	9	-1.7	U			P23	2.5	82Au01
$^{144}\text{Cs}-^{146}\text{Cs}_{.329} \text{ }^{143}\text{Cs}_{.671}$	320	50	336	21	0.1	U			P23	2.5	82Au01
$^{145}\text{Cs}-^{146}\text{Cs}_{.662} \text{ }^{143}\text{Cs}_{.338}$	-440	30	-565	10	-1.7	U			P33	2.5	86Au02

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{145}\text{Cs}-^{146}\text{Cs}_{.497} \ ^{144}\text{Cs}_{.503}$	-730	30	-740	13	-0.1	U			P33	2.5	86Au02
$^{146}\text{Sm}(\alpha)^{142}\text{Nd}$	2529.5	20.	2528.8	2.8	-0.0	U					64Nu02
	2622.0	30.			-3.1	B					66Fr11
	2524.2	4.			1.1	1	47	46 ^{146}Sm			87Me08 Z
$^{144}\text{Nd}(\text{t,p})^{146}\text{Nd}$	4834	30	4838.73	0.25	0.2	U			Ald		72Ch11
$^{144}\text{Sm}(\text{t,p})^{146}\text{Sm}$	6681	25	6691.6	2.8	0.4	U			Ald		66Bj01
$^{144}\text{Sm}(\text{}^3\text{He,p})^{146}\text{Eu}$	2797	12	2794	6	-0.2	1	25	24 ^{146}Eu			84Ru.A
$^{144}\text{Sm}(\text{}^3\text{He,n})^{146}\text{Gd}$	977	30	980	4	0.1	U			Bld		79Al07
$^{144}\text{Sm}(\text{}^{12}\text{C}, \text{}^{10}\text{Be})^{146}\text{Gd}$	-18476	25	-18487	4	-0.5	U			MSU		80Pa07
$^{146}\text{Nd}(\text{d}, \text{}^3\text{He})^{145}\text{Pr}$	-3095	10	-3095	7	-0.0	1	50	50 ^{145}Pr	KVI		79Sa.A
$^{145}\text{Nd}(\text{n}, \gamma)^{146}\text{Nd}$	7565.28	0.10	7565.23	0.09	-0.5	-			MMn		82Is05 Z
	7565.05	0.18			1.0	-			Bdn		06Fi.A
ave.	7565.23	0.09			0.0	1	99	84 ^{146}Nd			average
$^{146}\text{Sm}(\text{}^3\text{He}, \alpha)^{145}\text{Sm}$	12161	5	12161.3	2.9	0.1	1	33	30 ^{146}Sm			86Ru04 *
$^{146}\text{Tm}(\text{p})^{145}\text{Er}$	895.2	8.	896	6	0.1	o			ORp		03Gi10
	896.2	8.			-0.1	3			Arp		05Ro40 *
	895.2	8.			0.1	3			ORp		06Ta08 *
$^{146}\text{Tm}^m(\text{p})^{145}\text{Er}$	1197.3	5.	1199.3	1.0	0.4	U			Dap		93Li18
	1198.3	10.			0.1	o			ORp		01Ry01
	1200.3	8.			-0.1	U			Arp		05Ro40
	1199.3	1.				3			ORp		06Ta08
$^{146}\text{Tm}^m(\text{p})^{145}\text{Er}^m$	994.5	4.				4			ORp		06Ta08
$^{146}\text{Tm}^m(\text{p})^{145}\text{Er}^m$	1126.8	5.	1127.8	1.0	0.2	U			Dap		93Li18
	1127.8	10.			-0.0	o			ORp		01Ry01
	1129.8	8.			-0.3	U			Arp		05Ro40
	1127.8	1.				5			ORp		06Ta08
$^{146}\text{Cs}(\beta^-)^{146}\text{Ba}$	9300	900	9556	3	0.3	o			Gsn		81De25
	9310	60			4.1	B			Bwg		87Gr.A
	9375	50			3.6	B			Gsn		92Pr04
$^{146}\text{Ba}(\beta^-)^{146}\text{La}$	4280	100	4354.9	2.4	0.7	U			Gsn		81De25 *
	4030	50			6.5	B			Bwg		87Gr.A
$^{146}\text{La}(\beta^-)^{146}\text{Ce}$	6175	100	6405	15	2.3	B			Gsn		81De25 *
	6380	30			0.8	1	24	24 ^{146}Ce	Trs		82Br23 *
	6620	70			-3.1	B			Bwg		87Gr.A
	6580	80			-2.2	U					01Ko07 *
$^{146}\text{Ce}(\beta^-)^{146}\text{Pr}$	1100	80	1050	30	-0.7	-					54Be10 *
	1050	100			-0.0	-					67Ho19 *
	951	50			1.9	-					80Ya07 *
	1065	100			-0.2	-					81Eb01 *
ave.	1010	40			1.0	1	80	76 ^{146}Pr			average
$^{146}\text{Pr}(\beta^-)^{146}\text{Nd}$	4150	200	4250	30	0.5	U					54Be10 *
	4250	200			0.0	U					65Ra02 *
	4080	100			1.7	-					68Da13 *
	4140	100			1.1	-					78Ik03 *
ave.	4110	70			2.0	1	24	24 ^{146}Pr			average
$^{146}\text{Pm}(\beta^-)^{146}\text{Sm}$	1542	3				2					74Sc06 *
$^{146}\text{Eu}(\beta^+)^{146}\text{Sm}$	3871	10	3879	6	0.8	-					62Fu16 *
	3871	20			0.4	-					64Ta11 *
	3896	20			-0.9	-					88Sa06 *
ave.	3875	8			0.4	1	52	46 ^{146}Eu	Got		average
$^{146}\text{Gd}(\beta^+)^{146}\text{Eu}$	1757	30	1032	7	-24.2	B					70Ag01 *
	1300	200			-1.3	U					81Ka07 *
$^{146}\text{Tb}(\beta^+)^{146}\text{Gd}$	8240	150	8320	40	0.5	o			IRS		83Al06
	7910	150			2.7	U			IRS		93Al03 *
	8310	50			0.2	1	80	80 ^{146}Tb	Dbn		94Po26
$^{146}\text{Dy}(\beta^+)^{146}\text{Tb}$	5160	100	5210	50	0.5	1	20	20 ^{146}Tb	IRS		93Al03
* $^{146}\text{Ba}-\text{u}$	Represents frequency ratio $^{146}\text{Ba}^{++}/(\text{C}_6\text{H}_6)^+ = 0.934887856(12)$										HWJ208 **
* $^{146}\text{La}-\text{u}$	$D_M = -74182.5(30.6) \mu\text{eV}$ for mixture gs+m at 141.5(2.4) keV;										Nub211 **
* $^{146}\text{La}-\text{u}$	$M - A = -69100.6(28.5) \text{ keV}$										Nub211 **
* $^{146}\text{La}-\text{u}$	Represents frequency ratio $^{146}\text{La}^{++}/(\text{C}_6\text{H}_6)^+ = 0.934857905(12)$										HWJ206 **
* $^{146}\text{La}^m-\text{u}$	Represents frequency ratio $^{146}\text{La}^{m++}/(\text{C}_6\text{H}_6)^+ = 0.934858878(12)$										HWJ206 **
* $^{146}\text{Tb}-\text{u}$	$M - A = -67424(28) \text{ keV}$ for mixture gs+m at 150#100 keV										Nub211 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
* $^{146}\text{Sm}(^3\text{He},\alpha)^{145}\text{Sm}$	$Q-Q(^{148}\text{Gd}(^3\text{He},\alpha))=-567(5)\text{ keV}$										AHW	**
* $^{146}\text{Tm}(p)^{145}\text{Er}$	Trends from Mass Surface TMS suggest ^{146}Tm 300 keV less bound										GAu	**
* $^{146}\text{Ba}(\beta^-)^{146}\text{La}$	$E_{\beta^-}=3910(100)$ to 1^+ level at 372.4 keV, and other E_{β^-}										Ens97c	**
* $^{146}\text{La}(\beta^-)^{146}\text{Ce}$	$E_{\beta^-}=5919(100)$ 6120(30) respectively, to 2^+ level at 258.46 keV, and other E_{β^-}										Ens97c	**
* $^{146}\text{La}(\beta^-)^{146}\text{Ce}$	$E_{\beta^-}=6580(100)$ and 6320(80) to ground state and 2^+ level at 258.46 keV										01Ko07	**
* $^{146}\text{Ce}(\beta^-)^{146}\text{Pr}$	$E_{\beta^-}=750(80)$ 700(100) 600(50) 715(100) respectively, to 1^+ at 351.78 keV										Ens97c	**
* $^{146}\text{Pr}(\beta^-)^{146}\text{Nd}$	$E_{\beta^-}=3700(200)$ 3800(200) respectively, to 2^+ level at 453.77 keV										Ens97c	**
* $^{146}\text{Pr}(\beta^-)^{146}\text{Nd}$	$E_{\beta^-}=4100(200)$, 3600(100), 2100(100) to ground state, 2^+ 453.77, 2^+ 1978.45 levels										Ens97c	**
* $^{146}\text{Pr}(\beta^-)^{146}\text{Nd}$	$E_{\beta^-}=4150(150)$, 3700(100), 2160(100) to ground state, 2^+ 453.77, 2^+ 1978.45 levels										Ens97c	**
* $^{146}\text{Pm}(\beta^-)^{146}\text{Sm}$	$E_{\beta^-}=795(3)$ to 2^+ level at 747.2 keV										Ens97c	**
* $^{146}\text{Eu}(\beta^+)^{146}\text{Sm}$	$E_{\beta^+}=2107(11)$ 2100(20) respectively, to 2^+ level at 747.2 keV, and other E_{β^+}										Ens97c	**
* $^{146}\text{Eu}(\beta^+)^{146}\text{Sm}$	e/β^+ to 2045.8 level										Ens97c	**
* $^{146}\text{Gd}(\beta^+)^{146}\text{Eu}$	$E_{\beta^+}=350(30)$ to 1^- level at 384.79 keV										Ens97c	**
* $^{146}\text{Gd}(\beta^+)^{146}\text{Eu}$	pK to 690.7 level, $p^+<1\times10^{-4}$ to 384.8 level, see $^{150}\text{Dy}(\beta^+)$										Ens97c	**
* $^{146}\text{Tb}(\beta^+)^{146}\text{Gd}$	Reported half-life 24.1(0.5)s corresponds to $^{146}\text{Tb}^m$										GAu	**
*	$Q_{\beta^-}=8060(100)\text{ keV}$ from $^{146}\text{Tb}^m$ at estimated 150#100 keV										Nub211	**
$^{147}\text{Cs}-^{133}\text{Cs}_{1.105}$	48640	64	48737	9	1.5	U			MA8	1.0	08We02	
	48737.1	9.0				2			MA8	1.0	17At01	
$^{147}\text{Ba}-\text{u}$	-64696.1	21.2				2			CP1	1.0	06Sa56	
$^{147}\text{La}-\text{u}$	-71582.2	11.5				2			CP1	1.0	06Sa56	
$^{147}\text{Ce}-\text{u}$	-77309.2	9.6	-77310	9	-0.1	1	92	92 ^{147}Ce	CP1	1.0	06Sa56	
$\text{C}_8\text{H}_5\text{N O}_2-^{147}\text{Sm}$	117197	40	117124.0	1.4	-0.5	U			R04	4.0	64De15	
$\text{C}_9\text{H}_7\text{O}_2-^{147}\text{Sm}$	129703	17	129700.1	1.4	-0.0	U			R04	4.0	64De15	
$^{147}\text{Eu}-^{133}\text{Cs}_{1.105}$	21215	16	21228.0	2.8	0.8	U			MA5	1.0	00Be42	
$^{147}\text{Tb}-\text{u}$	-75934	34	-75945	9	-0.3	U			GS2	1.0	05Li24 *	
$^{147}\text{Tb}-^{133}\text{Cs}_{1.105}$	28533	12	28530	9	-0.2	1	52	52 ^{147}Tb	SH1	1.0	07Ra37 *	
$^{147}\text{Dy}-\text{u}$	-68909	30	-68917	10	-0.3	U			GS2	1.0	05Li24	
	-68908	30			-0.3	U			GS2	1.0	05Li24 *	
$^{147}\text{Dy}-^{133}\text{Cs}_{1.105}$	35558.3	9.5				2			SH1	1.0	07Ra37 *	
$^{147}\text{Ho}-\text{u}$	-59944	30	-59858	5	2.9	B			GS2	1.0	05Li24	
$^{147}\text{Ho}-^{133}\text{Cs}_{1.105}$	44613.7	7.8	44618	5	0.5	1	47	47 ^{147}Ho	SH1	1.0	07Ra37	
$^{147}\text{Ho}-^{85}\text{Rb}_{1.729}$	92661.6	7.4	92658	5	-0.5	1	53	53 ^{147}Ho	SH1	1.0	07Ra37	
$^{147}\text{Er}-^{133}\text{Cs}_{1.105}$	54452	42	54440	40	-0.3	o			SH1	1.0	07Ra37 *	
$^{147}\text{Er}-^{85}\text{Rb}_{1.729}$	102480	41				2			SH1	1.0	07Ra37 *	
$^{147}\text{Tm}-^{85}\text{Rb}_{1.729}$	113900	11	113895	7	-0.4	1	45	45 ^{147}Tm	SH1	1.0	07Ra37	
$^{147}\text{Eu}-^{142}\text{Sm}_{1.035}$	4516	17	4510.7	2.9	-0.3	U			MA7	1.0	01Bo59	
$^{147}\text{Sm}\text{ }^{35}\text{Cl}-^{145}\text{Nd}\text{ }^{37}\text{Cl}$	5305	4	5275.4	0.6	-1.9	U			H12	4.0	64Ba15	
	5264	4			1.1	U			H21	2.5	70Ma05	
$^{145}\text{Cs}-^{147}\text{Cs}_{.705}\text{ }^{140}\text{Cs}_{.296}$	-170	170	-644	11	-1.1	U			P23	2.5	82Au01	
$^{144}\text{Cs}-^{147}\text{Cs}_{.490}\text{ }^{141}\text{Cs}_{.511}$	80	250	227	21	0.2	U			P23	2.5	82Au01	
$^{145}\text{Cs}-^{147}\text{Cs}_{.493}\text{ }^{143}\text{Cs}_{.507}$	-87	22	-146	11	-1.1	U			P33	2.5	86Au02	
$^{147}\text{Sm}(\alpha)^{143}\text{Nd}$	2292.5	10.	2311.3	0.5	1.9	U					62Si14 Z	
	2296.7	5.			2.9	U					66Ma05 Z	
	2300.8	5.			2.1	U					70Gu14 Z	
	2310.5	0.5			1.7	o					16Ca43 *	
$^{147}\text{Eu}(\alpha)^{143}\text{Pm}$	2990.6	10.	2991	3	0.1	U					62Si14 Z	
	2981.5	20.			0.5	U					64To04 Z	
	2987.2	5.			0.8	1	37	22 ^{143}Pm	DbA		67Go32 Z	
$^{147}\text{Sm}(n,\alpha)^{144}\text{Nd}$	10114	8	10128.4	0.5	1.8	U			ILL		74Em01	
$^{144}\text{Sm}(^{12}\text{C},^9\text{Be})^{147}\text{Gd}$	-17832	30	-17957.1	1.2	-4.2	B			MSU		80Pa07	
	-17921	25			-1.4	U			Ors		85Be24	
$^{144}\text{Sm}(^{14}\text{N},^{11}\text{Be})^{147}\text{Tb}$	-28280	50	-28537	8	-5.1	B			Hei		85Gy01	
$^{147}\text{Sm}(p,t)^{145}\text{Sm}$	-6287	8	-6275.5	1.0	1.4	U			Min		72De47	
$^{146}\text{Nd}(n,\gamma)^{147}\text{Nd}$	5292.19	0.15	5292.20	0.09	0.0	-			ILn		75Ro16 Z	
	5292.19	0.11			0.1	-			Bdn		06Fi.A	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item		Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{146}\text{Nd}(\text{d,p})^{147}\text{Nd}$		3070	15	3067.63	0.09	-0.2	U			Hei		67Wi08
$^{146}\text{Nd}(\text{n},\gamma)^{147}\text{Nd}$	ave.	5292.19	0.09	5292.20	0.09	0.1	1	99	85 ^{147}Nd			average
$^{147}\text{Sm}(\text{d,t})^{146}\text{Sm}$		-98	10	-83.8	2.8	1.4	U			McM		75Si03
$^{147}\text{Tb}(\text{p})^{146}\text{Gd}$		-1945	18	-1946	9	-0.0	1	23	19 ^{147}Tb			87Sc.A
$^{147}\text{Tm}(\text{p})^{146}\text{Er}$		1062.2	6.	1059	3	-0.6	o					82Ki03
		1058.2	3.3			0.1	1	94	55 ^{147}Tm	Dap		93Se04 *
		1052.3	10.			0.6	U					93To02
		1067.3	15.			-0.6	U			ORp		03Gi10
$^{147}\text{Tm}^m(\text{p})^{146}\text{Er}$		1124.7	6.	1120	3	-0.7	2					84Ho.A
		1118.5	3.9			0.5	2			Dap		93Se04
$^{147}\text{Ba}(\beta^-)^{147}\text{La}$		5750	50	6414	22	13.3	C			Bwg		87Gr.A
$^{147}\text{La}(\beta^-)^{147}\text{Ce}$		4945	55	5336	14	7.1	C			Bwg		87Gr.A
		5150	40			4.6	B			Kur		95Ik03
		5370	100			-0.3	o			Kur		02Sh.B
		5366	40			-0.8	U			Kur		09Ha.B
$^{147}\text{Ce}(\beta^-)^{147}\text{Pr}$		3290	40	3430	16	3.5	C			Bwg		87Gr.A
		3426	20			0.2	1	60	52 ^{147}Pr	Kur		95Ik03
		3380	100			0.5	U			Kur		02Sh.B
$^{147}\text{Pr}(\beta^-)^{147}\text{Nd}$		2700	200	2703	16	0.0	U					64Ho03
		2790	100			-0.9	U					81Ya06 *
		2711	28			-0.3	-			Kur		95Ik03
	ave.	2697	23			0.2	1	48	48 ^{147}Pr			average
$^{147}\text{Nd}(\beta^-)^{147}\text{Pm}$		894.6	1.0	895.2	0.6	0.6	1	32	18 ^{147}Pm			67Ca18 *
$^{147}\text{Pm}(\beta^-)^{147}\text{Sm}$		223.2	0.5	224.06	0.29	1.7	-					50La04
		224.3	1.3			-0.2	-					58Ha32
		224.5	0.4			-1.1	-					66Hs01
	ave.	224.0	0.3			0.2	1	94	82 ^{147}Pm			average
$^{147}\text{Eu}(\beta^+)^{147}\text{Sm}$		1767	10	1721.4	2.3	-4.6	B					67Ad03
		1723	3			-0.5	1	58	57 ^{147}Eu			80Bu04
		1702	13			1.5	U					84Sc18 *
		1692	18			1.6	U					84Sc18
$^{147}\text{Gd}(\beta^+)^{147}\text{Eu}$		2185	5	2187.7	2.5	0.5	1	26	19 ^{147}Eu			80Vy01 *
		2199	17			-0.7	U					84Sc18 *
$^{147}\text{Tb}(\beta^+)^{147}\text{Gd}$		4700	90	4614	8	-1.0	U					83Ve06 *
		4490	60			2.1	U			Got		85Ti01
		4560	50			1.1	U					Averag *
		4609	15			0.4	1	29	29 ^{147}Tb	GSI		91Ke11 *
		4509	60			1.8	U			IRS		93Al03 *
		4560	80			0.7	U					97Wa04
$^{147}\text{Dy}(\beta^+)^{147}\text{Tb}$		6334	60	6547	12	3.5	B			IRS		83Al06 *
		6480	100			0.7	U			IRS		83Al18 *
		6334	60			3.5	C					85Af.A *
		6480	100			0.7	U			IRS		85Al08 *
* $^{147}\text{Tb}-\text{u}$	$M-A=-70707(28)$ keV for mixture gs+m at 50.6 keV											Nub211 **
* $^{147}\text{Tb}-^{133}\text{Cs}_{1.105}$	$D_M=28574(12)$ μu for mixture gs+m at 50.6 keV with ratio $R=0.741(13)$											Nub211 **
* $^{147}\text{Dy}-\text{u}$	$M-A=-63437(28)$ keV for $^{147}\text{Dy}^m$ at 750.5 keV											Nub211 **
* $^{147}\text{Dy}-^{133}\text{Cs}_{1.105}$	$D_M=35567(14)$ and $D_M=36358.4(9.5)$ for $^{147}\text{Dy}^m$ at 750.5 keV											Nub211 **
* $^{147}\text{Er}-^{133}\text{Cs}_{1.105}$	$D_M=54531(11)$ μu for mixture gs+m at 100#50 keV with ratio $R=0.741(13)$											Nub211 **
*	error due to excitation energy, use only next item											GAu **
* $^{147}\text{Er}-^{85}\text{Rb}_{1.729}$	$D_M=102559.5(8.3)$ μu for mixture gs+m at 100#50 keV with $R=0.741(13)$											Nub211 **
* $^{147}\text{Sm}(\alpha)^{143}\text{Nd}$	Used as calibrant, without the origin											16Ca43 **
* $^{147}\text{Tm}(\text{p})^{146}\text{Er}$	Q_p from $E_p=1051.0(3.3)$, no screening correction should be applied											GAu **
* $^{147}\text{Pr}(\beta^-)^{147}\text{Nd}$	$E_{\beta^-}=2760(100)$ to 49.93, 1450(100) to 1310.7 and 1350.5 keV											Ens092 **
* $^{147}\text{Nd}(\beta^-)^{147}\text{Pm}$	$E_{\beta^-}=803.5(1.0)$ to $5/2^+$ level at 91.1047 keV											Ens092 **
* $^{147}\text{Eu}(\beta^+)^{147}\text{Sm}$	$p^+=2.9(0.3)\times 10^{-3}$ to $3/2^-$ level at 197.284 keV											Ens092 **
* $^{147}\text{Eu}(\beta^+)^{147}\text{Sm}$	$pK=0.724(0.026)$ to $(3/2^+, 5/2^+)$ level at 1548.634 keV											Ens092 **
* $^{147}\text{Gd}(\beta^+)^{147}\text{Eu}$	$E_{\beta^+}=933(5)$ to $7/2^+$ level at 229.323 keV											Ens092 **
* $^{147}\text{Gd}(\beta^+)^{147}\text{Eu}$	$pK=0.694(0.016)$ to 2073 level, recalculated by AHW											84Sc18 **
* $^{147}\text{Tb}(\beta^+)^{147}\text{Gd}$	$E_{\beta^+}=2460(80)$ to $3/2^+$ level at 1152.56 and $1/2^+$ at 1292.3 keV, reinterpretd											Ens092 **
* $^{147}\text{Tb}(\beta^+)^{147}\text{Gd}$	Average $KLM/\beta^+=2.03(0.15) \rightarrow E_{\beta^+}=2190(50)$ from $^{147}\text{Tb}^m$ at 50.6(0.9) to $9/2^-$											Ens092 **
*	level at 1397.00 from 3 references (no side-feeding correction applied):											AHW **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
*	$p^+ = 0.32(0.07)$ gives $KLM/\beta^+ = 2.2(0.8)$										84Ha.B **
*	$KLM/\beta^+ = 2.17(0.30)$										85Al08 **
*	$\beta^+/K = 0.59(0.05)$ gives $KLM/\beta^+ = 1.99(0.17)$										85Sc09 **
* $^{147}\text{Tb}(\beta^+)^{147}\text{Gd}$	$Q_{\beta^+} = 4660(15)$ 4560(60) respectively, from $^{147}\text{Tb}^m$ at 50.6(0.9) keV										Nub211 **
* $^{147}\text{Dy}(\beta^+)^{147}\text{Tb}$	$E_{\beta^+} = 6012(60)$ from $^{147}\text{Dy}^m$ at 750.5 to $^{147}\text{Tb}^m$ at 50.6(0.9) keV										Nub211 **
* $^{147}\text{Dy}(\beta^+)^{147}\text{Tb}$	$Q_{\beta^+} = 7180(100)$ from $^{147}\text{Dy}^m$ at 750.5 to $^{147}\text{Tb}^m$ at 50.6(0.9) keV										Nub211 **
* $^{147}\text{Dy}(\beta^+)^{147}\text{Tb}$	$E_{\beta^+} = 6012(60)$ from $^{147}\text{Dy}^m$ at 750.5 to $^{147}\text{Tb}^m$ at 50.6(0.9) keV										Nub211 **
* $^{147}\text{Dy}(\beta^+)^{147}\text{Tb}$	$Q_{\beta^+} = 7180(100)$ from $^{147}\text{Dy}^m$ at 750.5 to $^{147}\text{Tb}^m$ at 50.6(0.9) keV										Nub211 **
$^{148}\text{Cs} - ^{133}\text{Cs}_{1.113}$	54871	14				2			MA8	1.0	17At01
$^{148}\text{Ba} - \text{u}$	-61777.0	1.6				2			CP2	1.0	18Or.A *
$^{148}\text{La} - \text{u}$	-67320.6	20.9				2			CP1	1.0	06Sa56
$^{148}\text{Ce} - \text{u}$	-75578.2	13.0	-75576	12	0.2	1	85	85 ^{148}Ce	CP1	1.0	06Sa56
$^{148}\text{Pr} - \text{u}$	-77781	41	-77870	16	-2.2	B			CP1	1.0	06Sa56 *
$\text{C}_{12} \text{H}_4 - ^{148}\text{Nd}$	114261	34	114401.1	2.2	1.0	U			R05	4.0	65De13
$\text{C}_9 \text{N O H}_{10} - ^{148}\text{Nd}$	159186	59	159339.9	2.2	0.7	U			R05	4.0	65De13
$\text{C}_9 \text{H}_8 \text{O}_2 - ^{148}\text{Sm}$	137540	26	137600.3	1.3	0.6	U			R04	4.0	64De15
$\text{C}_9 \text{H}_{10} \text{N O} - ^{148}\text{Sm}$	161275	31	161409.7	1.3	1.1	U			R04	4.0	64De15
$\text{C}_8 ^{13}\text{C H}_7 \text{O}_2 - ^{148}\text{Sm}$	133030	60	133130.1	1.3	0.4	U			R04	4.0	64De15
$^{148}\text{Eu} - ^{133}\text{Cs}_{1.113}$	23315	15	23323	11	0.6	1	51	51 ^{148}Eu	MA5	1.0	00Be42
$^{148}\text{Tb} - \text{u}$	-75692	41	-75725	13	-0.8	U			GS2	1.0	05Li24 *
$^{148}\text{Dy} - ^{133}\text{Cs}_{1.113}$	32394	16	32382	9	-0.8	-			MA5	1.0	00Be42
	32380	14			0.1	-			SH1	1.0	07Ra37
	ave.	32386	11		-0.4	1	79	79 ^{148}Dy			average
$^{148}\text{Ho} - \text{u}$	-62201	100	-62260	90	-0.6	U			GS2	1.0	05Li24 *
$^{148}\text{Ho} - ^{85}\text{Rb}_{1.741}$	91318	90				2			SH1	1.0	07Ra37 *
$^{148}\text{Er} - ^{133}\text{Cs}_{1.113}$	49967	11				2			SH1	1.0	07Ra37
$^{148}\text{Tm} - ^{133}\text{Cs}_{1.113}$	63616	11				2			SH1	1.0	07Ra37
$^{148}\text{Eu} - ^{142}\text{Sm}_{1.042}$	6451	17	6443	11	-0.5	1	40	39 ^{148}Eu	MA7	1.0	01Bo59
$^{148}\text{Nd} ^{35}\text{Cl}_2 - ^{144}\text{Nd} ^{37}\text{Cl}_2$	12690	9	12706.5	1.8	0.7	U			H21	2.5	70Ma05
	12703.6	2.1			0.5	1	12	11 ^{148}Nd	H25	2.5	72Ba08
$^{148}\text{Sm} ^{35}\text{Cl}_2 - ^{144}\text{Sm} ^{37}\text{Cl}_2$	8710	10	8723.2	1.0	0.3	U			H12	4.0	64Ba15
	8721.4	2.6			0.3	U			H25	2.5	72Ba08
$^{148}\text{Nd} ^{35}\text{Cl} - ^{146}\text{Nd} ^{37}\text{Cl}$	6740	5	6726.7	1.8	-0.7	U			H12	4.0	64Ba15
	6721	4			0.6	U			H21	2.5	70Ma05
	6723.8	2.7			0.4	U			H25	2.5	72Ba08
	6725.7	0.9			0.4	1	62	61 ^{148}Nd	H26	2.5	73Me28
$^{148}\text{Sm} ^{35}\text{Cl} - ^{146}\text{Nd} ^{37}\text{Cl}$	4656	3	4656.9	0.5	0.1	U			H12	4.0	64Ba15
$^{148}\text{Sm} - ^{147}\text{Sm}$	110	44	-75.17	0.29	-1.1	U			R04	4.0	64De15
$^{148}\text{Nd} - ^{146}\text{Nd}$	3866	50	3776.6	1.8	-0.4	U			R05	4.0	65De13
	3773	3			0.5	U			M17	2.5	66Be10
$^{145}\text{Cs} - ^{148}\text{Cs}_{.392} ^{143}\text{Cs}_{.608}$	-370	90	-519	11	-0.7	U			P33	2.5	86Au02
$^{148}\text{Sm}(\alpha) ^{144}\text{Nd}$	2014.6	20.	1987.0	0.4	-1.4	U					70Gu14
	1987.3	0.5			-0.5	1	79	44 ^{144}Nd			16Ca43 *
$^{148}\text{Eu}(\alpha) ^{144}\text{Pm}$	2703.2	30.	2694	10	-0.3	1	11	10 ^{148}Eu			64To04
$^{148}\text{Gd}(\alpha) ^{144}\text{Sm}$	3271.29	0.03	3271.29	0.03	-0.0	1	100	97 ^{148}Gd			73Go29
$^{146}\text{Nd}(\text{t,p}) ^{148}\text{Nd}$	4139	30	4143.0	1.7	0.1	U			Ald		72Ch11
$^{148}\text{Sm}(\text{p,t}) ^{146}\text{Sm}$	-6011	8	-6000.5	2.8	1.3	1	13	12 ^{146}Sm	Min		72De47
	-6018	15			1.2	U			Ham		74Oe03
$^{148}\text{Gd}(\text{p,t}) ^{146}\text{Gd}$	-7844	14	-7844	4	-0.0	U			LAl		83FI05
$^{148}\text{Gd}(\text{p,t}) ^{146}\text{Gd} - ^{65}\text{Cu}() ^{63}\text{Cu}$	1500	4	1500	4	0.1	1	90	89 ^{146}Gd	Liv		86Ma40
$^{148}\text{Nd}(\text{d}, ^3\text{He}) ^{147}\text{Pr}$	-3726	40	-3759	16	-0.8	R			KVI		79Sa.A
$^{148}\text{Nd}(\text{d,t}) ^{147}\text{Nd}$	-1072	4	-1075.4	1.7	-0.8	1	17	17 ^{148}Nd	McM		77St22
$^{147}\text{Sm}(\text{n}, \gamma) ^{148}\text{Sm}$	8139.8	1.2	8141.34	0.27	1.3	U					69Re04 Z
	8141.1	1.5			0.2	U					70Bu19 Z
	8141.8	0.8			-0.6	-					71Gr37 Z
	8141.3	0.3			0.1	-			Bdn		06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{147}\text{Sm}(\text{d,p})^{148}\text{Sm}$	5920	10	5916.77	0.27	-0.3	U			Tal		64Ke03
$^{148}\text{Sm}(\text{d,t})^{147}\text{Sm}$	-1890	15	-1884.11	0.27	0.4	U			Kop		67Ve04
$^{147}\text{Sm}(\text{n},\gamma)^{148}\text{Sm}$	ave. 8141.36	0.28	8141.34	0.27	-0.1	1	93	72 ^{147}Sm			average
$^{148}\text{Gd}(\text{p,d})^{147}\text{Gd}$	-6755	5	-6759.2	1.2	-0.8	U					86Ru04
$^{148}\text{Gd}(\text{p,d})^{147}\text{Gd}-^{148}\text{Sm}(\text{)}^{147}\text{Sm}$	-842	2	-842.5	1.2	-0.2	-					86Ru04
$^{148}\text{Gd}(\text{d,t})^{147}\text{Gd}-^{148}\text{Sm}(\text{)}^{147}\text{Sm}$	-843	2			0.3	-					86Ru04
$^{148}\text{Gd}(\text{d},\alpha)^{147}\text{Gd}-^{148}\text{Sm}(\text{)}^{147}\text{Sm}$	-842	3			-0.2	-					86Ru04
$^{148}\text{Gd}(\text{p,d})^{147}\text{Gd}-^{148}\text{Sm}(\text{)}^{147}\text{S}$	ave. -842.4	1.3			-0.0	1	89	86 ^{147}Gd			average
$^{148}\text{Ba}(\beta^-)^{148}\text{La}$	5115	60	5164	20	0.8	U			Bwg		90Gr10
$^{148}\text{La}(\beta^-)^{148}\text{Ce}$	7310	140	7690	22	2.7	U			Trs		82Br23 *
	7255	55			7.9	B			Bwg		90Gr10
	7650	100			0.4	U			Kur		02Sh.B
	7732	70			-0.6	U			Kur		09Ha.B
$^{148}\text{Ce}(\beta^-)^{148}\text{Pr}$	2060	75	2137	13	1.0	U			Bwg		87Gr.A
	2140	14			-0.2	1	81	66 ^{148}Pr	Kur		95Ik03
$^{148}\text{Pr}(\beta^-)^{148}\text{Nd}$	4800	200	4873	15	0.4	U					79Ik06
	4965	100			-0.9	U			Bwg		87Gr.A
	4890	50			-0.3	-					88Ka14
	4880	30			-0.2	-			Kur		95Ik03
	4930	100			-0.6	U			Kur		02Sh.B
	ave. 4883	26			-0.4	1	34	34 ^{148}Pr			average
$^{148}\text{Pm}(\beta^-)^{148}\text{Sm}$	2480	15	2470	6	-0.7	R					62Sc04 *
	2475	30			-0.2	U					63Ba31 *
$^{148}\text{Eu}(\beta^+)^{148}\text{Sm}$	3122	30	3039	10	-2.8	U					63Ba32 *
	3150	30			-3.7	B					70Ag01 *
$^{148}\text{Tb}(\beta^+)^{148}\text{Gd}$	5630	80	5732	13	1.3	F					76Cr.B *
	5835	70			-1.5	U					83Ve06 *
	5710	100			0.2	U			Got		85Sc09 *
	5390	100			3.4	B			Got		85Ti01 *
	5760	80			-0.3	U			IRS		93Al03 *
	5752	40			-0.5	1	10	10 ^{148}Tb	GSI		95Ke05 *
$^{148}\text{Dy}(\beta^+)^{148}\text{Tb}$	2660	60	2678	10	0.3	U					81Sc21 *
	2805	60			-2.1	U					81Sp03 *
	2700	60			-0.4	U			IRS		82Al.A *
	2700	60			-0.4	U					82Ve.A *
	2722	60			-0.7	U					83Ve06 *
	2835	95			-1.7	U					84Ha.B *
	2740	60			-1.0	U			Got		85Sc09 *
	2682	10			-0.4	1	92	86 ^{148}Tb	GSI		95Ke05 *
$^{148}\text{Ho}^m(\beta^+)^{148}\text{Dy}$	9400	250	10120#	130#	2.9	B			IRS		93Al03
* $^{148}\text{Ba}-\text{u}$	Represents frequency ratio $^{148}\text{Ba}^{++}/(\text{C}_6\text{H}_6)^+ = 0.947751100(10)$										HWJ208 **
* $^{148}\text{Pr}-\text{u}$	$D_M = -77739.3(30.6) \mu\text{u}$ for mixture gs+m at 76.80; $M - A = -72413.7(28.5) \text{ keV}$										Nub211 **
* $^{148}\text{Tb}-\text{u}$	$M - A = -70462(28) \text{ keV}$ for mixture gs+m at 90.1 keV										Nub211 **
* $^{148}\text{Ho}-\text{u}$	$M - A = -57815(30) \text{ keV}$ for mixture gs+m at 250#100 keV										Nub211 **
*	outweighed by next item before correcting for isomeric mixture										GAu **
* $^{148}\text{Ho}-^{85}\text{Rb}_{1.741}$	$D_M = 91517.5(9.5) \mu\text{u}$ for mixture gs+m at 250#100 keV with $R=0.74(15)$										Nub211 **
* $^{148}\text{Sm}(\alpha)^{144}\text{Nd}$	Using $\text{Qa}(^{147}\text{Sm}) = 2310.5(0.5)$ as calibrant										16Ca43 **
* $^{148}\text{La}(\beta^-)^{148}\text{Ce}$	$E_{\beta^-} = 5862(100)$ supposed to go to levels around $E = 1450(100) \text{ keV}$										90Gr10 **
* $^{148}\text{Pm}(\beta^-)^{148}\text{Sm}$	$E_{\beta^-} = 2460(20) 1020(15)$ to ground state, 1^- level at 1465.137 keV										Ens144 **
* $^{148}\text{Pm}(\beta^-)^{148}\text{Sm}$	$E_{\beta^-} = 2480(30) 1930(30) 1020(30)$ to ground state, 2^+ at 550.255, 1^- at 1465.137 keV										Ens144 **
*	and $E_{\beta^-} = 400(30)$ from $^{148}\text{Pm}^m$ at 137.9 to 6^+ level at 2194.061 keV										Nub211 **
* $^{148}\text{Eu}(\beta^+)^{148}\text{Sm}$	$E_{\beta^+} = 920(30)$ to 1180.261 keV 4^+ level										Ens144 **
* $^{148}\text{Eu}(\beta^+)^{148}\text{Sm}$	$E_{\beta^+} = 540(30)$ to 1594.247 keV 5^- level, and other E_{β^+}										Ens144 **
* $^{148}\text{Tb}(\beta^+)^{148}\text{Gd}$	$E_{\beta^+} = 4610(80)$ assumed to ground state										76Cr.B **
*	F: since ^{148}Tb ground state 2^- , transition to ^{148}Gd ground state weak										AHW **
* $^{148}\text{Tb}(\beta^+)^{148}\text{Gd}$	$E_{\beta^+} = 2210(70)$ from $^{148}\text{Tb}^m$ at 90.1 to 8^+ level at 2693.35 keV and										Nub211 **
*	$E_{\beta^+} = 4560(80)$ mainly to 2^+ at 784.433 keV. Conflicting, not used										Ens144 **
* $^{148}\text{Tb}(\beta^+)^{148}\text{Gd}$	$p^+ = 0.271(0.10) \rightarrow E_{\beta^+} = 1920(30)$ from $^{148}\text{Tb}^m$ at 90.1 to 8^+ at 2693.35 keV										Ens144 **
*	but assuming 5(5)% side-feeding; see reference										90Sa32 **
* $^{148}\text{Tb}(\beta^+)^{148}\text{Gd}$	$KL/\beta^+ = 1.54(0.09)$ to 1863.42 level; yields $Q_{\beta^+} = 5295(45) \text{ keV}$										85Ti01 **
*	but assuming 7(7)% side-feeding; see 1990Sa32										AHW **
* $^{148}\text{Tb}(\beta^+)^{148}\text{Gd}$	$Q_{\beta^+} = 5700(80)$; and 5910(80) from $^{148}\text{Tb}^m$ at 90.1 keV										Nub211 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
* $^{148}\text{Tb}(\beta^+)^{148}\text{Gd}$	$Q_{\beta^+}=5750(40)$; and $5846(50)$ from $^{148}\text{Tb}^m$ at 90.1 keV										Nub211 **
* $^{148}\text{Dy}(\beta^+)^{148}\text{Tb}$	$p^+=0.069(0.014)$ to 1^+ level at 620.24 keV, recalculated Q										Ens144 **
* $^{148}\text{Dy}(\beta^+)^{148}\text{Tb}$	$E_{\beta^+}=1040(60)$, $1120(60)$ to 1^+ level at 620.24 keV										Ens144 **
* $^{148}\text{Dy}(\beta^+)^{148}\text{Tb}$	$p^+=0.055(0.015)$ to 1^+ level at 620.24 keV										Ens144 **
* $^{148}\text{Dy}(\beta^+)^{148}\text{Tb}$	$\beta^+/K=0.045(0.005)$ to 1^+ level at 620.24 keV, gives $Q_{\beta^+}=2680(30)$ keV										Ens144 **
* $^{148}\text{Dy}(\beta^+)^{148}\text{Tb}$	GSI average of $E_{\beta^+}=1043(10)$ and $1036(10)$ of reference										91Ke11 **
*	to 1^+ level at 620.24 keV										Ens144 **
$^{149}\text{Ba}-u$	-57027	188	-56716.0	2.7	0.7	U			GT3	2.5	16Kn03
	-56716.0	2.7				2			CP2	1.0	18Or.A *
$^{149}\text{Ce}-u$	-71573.1	11.0				2			CP1	1.0	06Sa56
$^{149}\text{Pr}-u$	-76263.9	10.6				2			CP1	1.0	06Sa56
$\text{C}_8^{13}\text{C H}_8 \text{O}_2-^{149}\text{Sm}$	138597	29	138593.1	1.2	-0.0	U			R04	4.0	64De15
$\text{C}_9 \text{H}_{11} \text{N O}-^{149}\text{Sm}$	166820	33	166872.8	1.2	0.4	U			R04	4.0	64De15
$\text{C}_8^{13}\text{C H}_{10} \text{N O}-^{149}\text{Sm}$	162408	46	162402.6	1.2	-0.0	U			R04	4.0	64De15
$^{149}\text{Eu}-^{133}\text{Cs}_{1.120}$	23849	17	23831	4	-1.1	U			MA5	1.0	00Be42
$^{149}\text{Tb}-u$	-76730	32	-76746	4	-0.5	U			GS2	1.0	05Li24 *
$^{149}\text{Dy}-^{133}\text{Cs}_{1.120}$	33278	109	33221	10	-0.5	U			MA5	1.0	00Be42
$^{149}\text{Dy}-u$	-72698	30	-72672	10	0.9	U			GS2	1.0	05Li24 *
$^{149}\text{Ho}-u$	-66179	34	-66180	13	-0.0	1	14	14 ^{149}Ho	GS2	1.0	05Li24 *
$^{149}\text{Er}-u$	-57694	30				2			GS2	1.0	05Li24 *
$^{149}\text{Eu}-^{142}\text{Sm}_{1.049}$	6909	18	6882	4	-1.5	U			MA7	1.0	01Bo59
$^{149}\text{Dy}-^{142}\text{Sm}_{1.049}$	16249	16	16273	10	1.5	1	38	37 ^{149}Dy	MA7	1.0	01Bo59
$^{149}\text{Sm}^{35}\text{Cl}-^{147}\text{Sm}^{37}\text{Cl}$	5257	4	5236.9	1.0	-1.3	U			H12	4.0	64Ba15
	5231	3			0.8	U			H21	2.5	70Ma05
	5239.8	0.8			-1.4	1	24	15 ^{147}Sm	M21	2.5	75Ka25
$^{149}\text{Sm}-^{148}\text{Sm}$	2282	31	2362.0	1.0	0.6	U			R04	4.0	64De15
$^{149}\text{Sm}-^{147}\text{Sm}$	2320	60	2286.8	1.0	-0.1	U			R04	4.0	64De15
$^{149}\text{Gd}(\alpha)^{145}\text{Sm}$	3102.3	10.	3099	3	-0.3	-					65Ma51 Z
	3096.1	10.3			0.3	-			ORa		66Wi12 Z
	3099.1	5.			0.0	-			DbA		67Go32 Z
ave.	3099	4			0.0	1	56	53 ^{149}Gd			average
$^{149}\text{Tb}(\alpha)^{145}\text{Eu}$	4074.4	3.	4078.0	2.2	1.2	-			DbA		67Go32 Z
	4073.8	7.			0.6	U			ORa		74To07 *
	4074.6	10.			0.3	U					81Ho.A Z
	4081.8	5.			-0.7	-			Bka		82Bo04 Z
	4082.8	4.			-1.2	-			Daa		96Pa01
ave.	4078.2	2.2			-0.1	1	95	86 ^{149}Tb			average
$^{149}\text{Sm}(n,\alpha)^{146}\text{Nd}$	9429	4	9436.4	1.0	1.9	U			McM		67Oa01
	9421	15			1.0	U			ILL		75Em.A
$^{149}\text{Sm}(p,t)^{147}\text{Sm}$	-5532	8	-5530.7	0.9	0.2	U			Min		72De47
	-5532	7			0.2	U			McM		73Ga04
$^{148}\text{Nd}(n,\gamma)^{149}\text{Nd}$	5038.76	0.10	5038.79	0.07	0.3	2			ILn		76Pi04 Z
	5038.82	0.11			-0.3	2			Bdn		06Fi.A
$^{148}\text{Nd}(^3\text{He},d)^{149}\text{Pm}$	455	5	451.8	2.5	-0.6	1	24	13 ^{149}Pm	McM		80St10 *
$^{149}\text{Sm}(d,^3\text{He})^{148}\text{Pm}$	-2064	6	-2066	6	-0.3	2					88No02
$^{148}\text{Sm}(n,\gamma)^{149}\text{Sm}$	5872.5	1.8	5871.1	0.9	-0.8	1	24	15 ^{148}Sm			70Sm.A
	5850.8	0.6			33.9	C					82Ba15
$^{149}\text{Sm}(\gamma,n)^{148}\text{Sm}$	-5890	160	-5871.1	0.9	0.1	U			Phi		60Ge01
$^{148}\text{Sm}(d,p)^{149}\text{Sm}$	3656	15	3646.6	0.9	-0.6	U			Kop		67Ve04
$^{149}\text{Er}(\epsilon p)^{148}\text{Dy}$	5758	900	6829	29	1.2	U					83La.A *
	7080	470			-0.5	U			LBL		89Fi01
$^{149}\text{La}(\beta^-)^{149}\text{Ce}$	6450	200				3			Kur		02Sh.B
$^{149}\text{Ce}(\beta^-)^{149}\text{Pr}$	4190	75	4369	14	2.4	U			Bwg		87Gr.A
	4380	60			-0.2	U			Kur		95Ik03
	4310	100			0.6	U			Kur		02Sh.B

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{149}\text{Pr}(\beta^-)^{149}\text{Nd}$	3000	200	3336	10	1.7	U					67Va14
	3390	90			-0.6	U			Kur		95Ik03
$^{149}\text{Nd}(\beta^-)^{149}\text{Pm}$	1669	10	1688.9	2.5	2.0	U					64Go08 *
$^{149}\text{Pm}(\beta^-)^{149}\text{Sm}$	1072	2	1071.5	1.9	-0.3	1	88	87 ^{149}Pm			60Ar05
	1062	2			4.7	B					78Re01
$^{149}\text{Eu}(\epsilon)^{149}\text{Sm}$	680	10	695	4	1.5	1	14	14 ^{149}Eu			85Ad.A
$^{149}\text{Gd}(\epsilon)^{149}\text{Eu}$	1308	6	1314	4	1.0	1	48	30 ^{149}Eu	Got		84Sc.B
$^{149}\text{Tb}(\beta^+)^{149}\text{Gd}$	3575	50	3639	4	1.3	U			Got		85Sc09 *
	3635	10			0.4	1	19	11 ^{149}Tb	GSI		91Ke06 *
$^{149}\text{Dy}(\beta^+)^{149}\text{Tb}$	3930	150	3795	9	-0.9	U					84Al36 *
	3925	65			-2.0	U			Got		90Sa32 *
	3797	13			-0.2	1	49	46 ^{149}Dy	GSI		91Ke11 *
	3950	100			-1.6	U			IRS		93Al03
$^{149}\text{Ho}(\beta^+)^{149}\text{Dy}$	6043	50	6048	13	0.1	-			IRS		83Al06 *
	5330	100			7.2	C					84Ha.B
	6000	90			0.5	U			IRS		93Al03
	6009	20			2.0	-			GSI		91Ke11 *
ave.	6014	19			1.9	1	47	32 ^{149}Ho			average
$^{149}\text{Er}(\epsilon)^{149}\text{Ho}$	8610	650	7900	30	-1.1	U			LBL		89Fi01 *
* $^{149}\text{Ba}-\text{u}$	Represents frequency ratio $^{149}\text{Ba}^{++}/(\text{C}_6\text{H}_6)^+ = 0.954189968(17)$										HWJ208 **
* $^{149}\text{Tb}-\text{u}$	$M-A=-71456(28)$ keV for mixture gs+m at 35.78 keV										Nub211 **
* $^{149}\text{Dy}-\text{u}$	$M-A=-65057(28)$ keV for $^{149}\text{Dy}^m$ at 2661.1 keV										Nub211 **
* $^{149}\text{Ho}-\text{u}$	$M-A=-61621(28)$ keV for mixture gs+m at 48.80 keV										Nub211 **
* $^{149}\text{Er}-\text{u}$	$M-A=-53000(28)$ keV for $^{149}\text{Er}^m$ at 741.8 keV										Nub211 **
* $^{149}\text{Tb}(\alpha)^{145}\text{Eu}$	$E_\alpha=3999(7)$ from $^{149}\text{Tb}^m$ at 35.78 keV										Nub211 **
* $^{148}\text{Nd}({}^3\text{He},\text{d})^{149}\text{Pm}$	Based on $^{146}\text{Nd}({}^3\text{He},\text{d})^{147}\text{Pm}$ $Q=-87.6(0.9)$ keV										AHW **
* $^{149}\text{Er}(\epsilon\text{p})^{148}\text{Dy}$	As read from graph; $Q=6500$ from $^{149}\text{Er}^m$ at 741.8 keV										Nub211 **
* $^{149}\text{Nd}(\beta^-)^{149}\text{Pm}$	$E_{\beta^-}=1555(10)$ to $5/2^+$ level at 114.312 level										Ens046 **
* $^{149}\text{Tb}(\beta^+)^{149}\text{Gd}$	$\beta^+/\text{K}=0.31(0.03)$ from $^{149}\text{Tb}^m$ at 35.78 to $9/2^-$ level at 795.82 keV										Ens046 **
* $^{149}\text{Tb}(\beta^+)^{149}\text{Gd}$	$E_{\beta^+}=1853(10)$ from $^{149}\text{Tb}^m$ at 35.78 to $9/2^-$ level at 795.82 keV										Ens046 **
* $^{149}\text{Dy}(\beta^+)^{149}\text{Tb}$	$E_{\beta^+}=1030(150)$ to 1728.31–1876.96 levels										Ens046 **
* $^{149}\text{Dy}(\beta^+)^{149}\text{Tb}$	KL/ $\beta^+=29.4(10.6)$, $14.5(3.7)$, $17.6(4.9)$ to 1883, 1735, 1728 levels										90Sa32 **
* $^{149}\text{Dy}(\beta^+)^{149}\text{Tb}$	Original $Q=3812(10)$ from $E_{\beta^+}=1965(10)$ to 825.16 level corrected to $E_{\beta^+}=1950(13)$ for background subtraction										GAu **
* $^{149}\text{Ho}(\beta^+)^{149}\text{Dy}$	$E_{\beta^+}=3930(50)$ to $9/2^-$ level at 1090.76 keV										Ens046 **
* $^{149}\text{Ho}(\beta^+)^{149}\text{Dy}$	$E_{\beta^+}=3896(20)$ to $9/2^-$ level at 1090.76 keV										Ens046 **
* $^{149}\text{Er}(\epsilon)^{149}\text{Ho}$	KLM/ $\beta^+=0.68(0.34)$ from $^{149}\text{Er}^m$ at 741.8 to 4699.7 level										Ens046 **
$^{150}\text{Ba}-\text{u}$	-55309	371	-53559	6	1.9	U			GT3	2.5	16Kn03
	-53558.9	6.1				2			CP2	1.0	18Or.A *
$^{150}\text{La}-\text{u}$	-60258	187	-60452.5	2.7	-0.4	U			GT3	2.5	16Kn03
	-60452.5	2.7				2			CP2	1.0	18Or.A *
$^{150}\text{Ce}-\text{u}$	-69618.6	13.1	-69616	13	0.2	1	92	92 ^{150}Ce	CP1	1.0	06Sa56
$^{150}\text{Pr}-\text{u}$	-73322.9	10.6	-73324	10	-0.1	1	83	83 ^{150}Pr	CP1	1.0	06Sa56
$\text{C}_{12}\text{H}_6-^{150}\text{Nd}$	126194	43	126048.9	1.2	-0.8	U			R05	4.0	65De13
$\text{C}_8\text{}^{13}\text{C}\text{}^{13}\text{N}\text{}^{13}\text{O}\text{}^{13}\text{H}_{11}-^{150}\text{Nd}$	166439	34	166517.5	1.2	0.6	U			R05	4.0	65De13
$\text{C}_9\text{}^{13}\text{C}\text{}^{13}\text{N}\text{}^{13}\text{O}\text{}^{13}\text{H}_{12}-^{150}\text{Nd}$	170931	46	170987.7	1.2	0.3	U			R05	4.0	65De13
$\text{C}_{12}\text{H}_6-^{150}\text{Sm}$	129810	140	129668.2	1.2	-0.3	U			R04	4.0	64De15
$\text{C}_8\text{}^{13}\text{C}\text{}^{13}\text{H}_{11}\text{}^{13}\text{N}\text{}^{13}\text{O}-^{150}\text{Sm}$	170029	25	170136.8	1.2	1.1	U			R04	4.0	64De15
$\text{C}_9\text{}^{13}\text{C}\text{}^{13}\text{H}_{12}\text{}^{13}\text{N}\text{}^{13}\text{O}-^{150}\text{Sm}$	174612	47	174607.0	1.2	-0.0	U			R04	4.0	64De15
$^{150}\text{Tb}^m-\text{u}$	-75850	30	-75840	28	0.3	1	89	89 $^{150}\text{Tb}^m$	GS2	1.0	05Li24
$^{150}\text{Ho}-^{133}\text{Cs}_{1,128}$	40150	29	40149	15	-0.1	-			MA5	1.0	00Be42
ave.	40132	21			0.8	1	53	53 ^{150}Ho			average
$^{150}\text{Ho}-\text{u}$	-66504	40	-66502	15	0.1	U			GS2	1.0	05Li24 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{150}\text{Er}-u$	-62060	30	-62084	18	-0.8	1	38	38	^{150}Er	GS2	1.0	05Li24
$^{150}\text{Nd } ^{35}\text{Cl}_2-^{146}\text{Nd } ^{37}\text{Cl}_2$	13654	9	13679.1	1.1	1.1	U			H21	2.5	70Ma05	
	13672.5	1.8			1.5	U			H25	2.5	72Ba08	
$^{150}\text{Nd } ^{35}\text{Cl}-^{148}\text{Nd } ^{37}\text{Cl}$	7006	4	6952.4	2.0	-3.3	B			H12	4.0	64Ba15	
	6939	4			1.3	U			H21	2.5	70Ma05	
$^{150}\text{Sm } ^{35}\text{Cl}-^{148}\text{Sm } ^{37}\text{Cl}$	5452	8	5402.9	0.9	-1.5	U			H12	4.0	64Ba15	
	5400	4			0.3	U			H21	2.5	70Ma05	
$^{150}\text{Nd}-^{150}\text{Sm}$	5404.8	0.6			-1.3	1	40	28	^{148}Sm	M21	2.5	75Ka25
	3633	4	3619.33	0.21	-0.9	U			H19	4.0	64Mc11	
	3617.0	1.2			0.8	U			H25	2.5	72Ba08	
	3619.33	0.21			-0.0	1	100	100	^{150}Nd	JY1	1.0	10Ko28
$^{150}\text{Sm}-^{149}\text{Sm}$	149	30	90.8	0.4	-0.5	U			R04	4.0	64De15	
$^{150}\text{Nd}-^{148}\text{Nd}$	3860	46	4002.3	2.0	0.8	U			R05	4.0	65De13	
	3988	3			1.9	U			M17	2.5	66Be10	
$^{150}\text{Sm}-^{148}\text{Sm}$	2430	50	2452.8	0.9	0.1	U			R04	4.0	64De15	
$^{150}\text{Nd}-^{146}\text{Nd}$	7719	67	7778.9	1.1	0.2	U			R05	4.0	65De13	
$^{150}\text{Gd}(\alpha)^{146}\text{Sm}$	2804.9	10.	2807	6	0.2	-						62Si14
	2792.6	18.			0.8	-						65Og01
	ave.	2802	9		0.6	1	45	39	^{150}Gd			average
	$^{150}\text{Tb}(\alpha)^{146}\text{Eu}$	3585.5	5.	3587	5	0.3	1	92	80	^{150}Tb	DbA	
$^{150}\text{Dy}(\alpha)^{146}\text{Gd}$	4345.8	5.	4351.3	1.5	1.1	-				DbA		67Go32 Z
	4349.5	5.			0.3	-				GSa		79Ho10 Z
	4351.3	3.			0.0	-				Bka		82Bo04 *
	4352.5	2.1			-0.6	-				Ora		82De11 Z
ave.	4351.3	1.5			-0.0	1	99	92	^{150}Dy			average
$^{148}\text{Nd}(\text{t,p})^{150}\text{Nd}$	3935	30	3932.7	1.9	-0.1	U				Ald		72Ch11
$^{148}\text{Sm}(\text{t,p})^{150}\text{Sm}$	5372	25	5376.1	0.9	0.2	U				Ald		66Bj01
$^{150}\text{Sm}(\text{p,t})^{148}\text{Sm}$	-5379	8	-5376.1	0.9	0.4	U				Min		72De47
	-5378	15			0.1	U				Ham		74Oe03
$^{150}\text{Nd}(\text{d},^3\text{He})^{149}\text{Pr}$	-4501	10	-4436	10	6.5	C				KVI		79Sa.A
$^{150}\text{Nd}(\text{d,t})^{149}\text{Nd}$	-1122	10	-1118.5	1.9	0.3	U				McM		73Bu02
$^{149}\text{Sm}(\text{n},\gamma)^{150}\text{Sm}$	7984.9	0.6	7986.8	0.4	3.1	B						69Re04 Z
	7986.7	1.5			0.0	-						70Bu19 Z
	7986.7	0.4			0.1	-				Bdn		06Fi.A
$^{149}\text{Sm}(\text{d,p})^{150}\text{Sm}$	5764	4	5762.2	0.4	-0.5	U				Tal		64Ke03
$^{150}\text{Sm}(\text{d,t})^{149}\text{Sm}$	-1738	15	-1729.5	0.4	0.6	U				Kop		67Ve04
$^{149}\text{Sm}(\text{n},\gamma)^{150}\text{Sm}$	ave.	7986.7	0.4	7986.8	0.4	0.1	1	95	82	^{149}Sm		average
	$^{150}\text{Lu}(\text{p})^{149}\text{Yb}$	1269.6	4.	1269.6	2.3	-0.0	3					84Ho.A
$^{150}\text{Lu}^m(\text{p})^{149}\text{Yb}$		1269.6	4.		-0.0	3				Dap		93Se04
		1269.6	4.		-0.0	3				ORp		03Gi10
		1303.8	15.	1291	5	-0.8	o			ORp		00Gi01
		1285.6	8.			0.7	3			ORp		03Gi10
		1294.7	6.			-0.5	3			Arp		03Ro21
$^{150}\text{Ce}(\beta^-)^{150}\text{Pr}$	3010	90	3454	14	4.9	C				Bwg		87Gr.A
$^{150}\text{Pr}(\beta^-)^{150}\text{Nd}$	3480	40			-0.7	1	13	8	^{150}Ce	Kur		95Ik03
	5690	80	5379	9	-3.9	C				Bwg		87Gr.A
$^{150}\text{Pr}(\beta^-)^{150}\text{Nd}$	5386	26			-0.3	1	12	12	^{150}Pr	Kur		95Ik03
	5290	100			0.9	U				Kur		02Sh.B
	$^{150}\text{Pm}(\beta^-)^{150}\text{Sm}$	3454	20				2					77Ho09
$^{150}\text{Eu}(\beta^+)^{150}\text{Sm}$	2222	25	2259	6	1.5	U						65Gu03 *
$^{150}\text{Eu}(\beta^-)^{150}\text{Gd}$	978	10	972	4	-0.6	-						63Yo07 *
	968	4			0.9	-						65Gu03 *
ave.	969	4			0.6	1	91	53	^{150}Eu			average
$^{150}\text{Tb}(\beta^+)^{150}\text{Gd}$	4720	40	4658	8	-1.5	U						68Wi21
	4670	15			-0.8	1	31	20	^{150}Tb			76Cr.B
	4760	50			-2.0	U						77Ha31 *
	4620	60			0.6	U						83Ve06

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{150}\text{Tb}^m(\beta^+)^{150}\text{Gd}$	5040	100	5119	27	0.8	U			IRS		93Al03
$^{150}\text{Dy}(\beta^+)^{150}\text{Tb}$	1760	40	1796	8	0.9	U					81Ka07 *
$^{150}\text{Ho}(\beta^+)^{150}\text{Dy}$	6980	150	7364	14	2.6	U					84Al36 *
	6560	100			8.0	B			IRS		93Al03
$^{150}\text{Ho}(\epsilon)^{150}\text{Dy}$	7400	200			-0.2	U					98Ag.A
	7372	27			-0.3	1	29	27 ^{150}Ho			00Ca.A
	7444	126			-0.6	U					01Ro35
$^{150}\text{Ho}^m(\beta^+)^{150}\text{Dy}$	7360	50			2				IRS		83Al06 *
	6575	75	7360	50	10.5	C					84Ha.B *
	6625	120			6.1	B			Got		85Sc09 *
	6900	130			3.5	B			Got		90Sa32 *
	7060	80			3.7	C			IRS		93Al03
$^{150}\text{Er}(\beta^+)^{150}\text{Ho}$	4010	80	4115	14	1.3	o					82No08 *
	4105	75			0.1	U					84Ha.B *
	4108	15			0.4	1	82	62 ^{150}Er	GSI		91Ke11 *
* $^{150}\text{Ba}-\text{u}$	Represent frequency ratio $^{150}\text{Ba}^{++}/(\text{C}_6\text{H}_6)^+ = 0.960616639(39)$										HWJ208 **
* $^{150}\text{La}-\text{u}$	Represents frequency ratio $^{150}\text{La}^{++}/(\text{C}_6\text{H}_6)^+ = 0.960572476(17)$										HWJ208 **
* $^{150}\text{Ho}-\text{u}$	$M-A = -61948(28)$ keV for mixture gs+m at $-0(50)$ keV										Nub211 **
* $^{150}\text{Dy}(\alpha)^{146}\text{Gd}$	Recalibrated as in reference										91Ry01 **
* $^{150}\text{Eu}(\beta^+)^{150}\text{Sm}$	$E_{\beta^+} = 1242(25)$ from $^{150}\text{Eu}^m$ at 41.7 keV										Nub211 **
* $^{150}\text{Eu}(\beta^-)^{150}\text{Gd}$	$Q_{\beta^-} = 1020(10)$ 1010(4) respectively, from $^{150}\text{Eu}^m$ at 41.7 keV										Nub211 **
* $^{150}\text{Tb}(\beta^+)^{150}\text{Gd}$	$E_{\beta^+} = 1655(80)$ to 2^+ at 2091.623, and other E_{β^+} . Orig. error 35 increased										Ens136 **
* $^{150}\text{Dy}(\beta^+)^{150}\text{Tb}$	$p^+ = 2(10) \times 10^{-3}$ to 1^+ 397.2 level combined with lower limit through $^{146}\text{Gd}(\epsilon)$										Ens136 **
* $^{150}\text{Ho}(\beta^+)^{150}\text{Dy}$	$E_{\beta^+} = 4550(150)$ to 1395.0 and 1456.8 levels										82No08 **
* $^{150}\text{Ho}^m(\beta^+)^{150}\text{Dy}$	$E_{\beta^+} = 3940(50)$ to 8^+ level at 2401.6 keV										Ens136 **
* $^{150}\text{Ho}^m(\beta^+)^{150}\text{Dy}$	$p^+ = 0.56(0.02)$ 0.58(0.07) respectively to 8^+ level at 2401.6 keV										Ens136 **
* $^{150}\text{Ho}^m(\beta^+)^{150}\text{Dy}$	$Q_{\beta^+} = 6819(+117-100)$ from p^+ to 2401.6 level; could be raised 140										90Sa32 **
*	if 4% feeding of higher Dy levels										90Sa32 **
* $^{150}\text{Er}(\beta^+)^{150}\text{Ho}$	$p^+ = 0.36(0.04)$ 0.39(0.04) to 1^+ level at 476.16 keV										Ens136 **
* $^{150}\text{Er}(\beta^+)^{150}\text{Ho}$	$E_{\beta^+} = 2610(15)$ to 1^+ level at 476.16 keV										Ens136 **
$^{151}\text{La}-\text{u}$	-58734	397	-57230	470	2.5	C			GT1	1.5	04Ma.A
	-57231	187				2			GT3	2.5	16Kn03
$^{151}\text{Ce}-\text{u}$	-65727.8	19.0				2			CP1	1.0	06Sa56
$^{151}\text{Pr}-\text{u}$	-71697.5	14.3	-71691	13	0.5	1	76	76 ^{151}Pr	CP1	1.0	06Sa56
$\text{C}_{12} \text{H}_7-^{151}\text{Eu}$	134920	37	134918.6	1.3	-0.0	U			R04	4.0	64De15
$\text{C}_{10} \text{H}_{15} \text{O}-^{151}\text{Eu}$	192490	70	192433.5	1.3	-0.2	U			R04	4.0	64De15
$^{151}\text{Eu}-^{85}\text{Rb}_{1.776}$	76520	15	76518.0	1.3	-0.1	U			MA5	1.0	00Be42
$^{151}\text{Tb}-\text{u}$	-76866	43	-76891	4	-0.6	U			GS2	1.0	05Li24 *
$^{151}\text{Dy}-\text{u}$	-73809	30	-73809	3	0.0	U			GS2	1.0	05Li24
$^{151}\text{Ho}-\text{u}$	-68323	33	-68302	9	0.6	U			GS2	1.0	05Li24 *
$^{151}\text{Er}-\text{u}$	-62528	30	-62551	18	-0.8	2			GS2	1.0	05Li24
	-62540	30			-0.4	2			GS2	1.0	05Li24 *
$^{151}\text{Eu}^{35}\text{Cl}-^{149}\text{Sm}^{37}\text{Cl}$	5620.3	2.6	5615.5	0.7	-0.7	U			H25	2.5	72Ba08
$^{151}\text{Eu}-^{150}\text{Sm}$	2800	60	2574.6	0.6	-0.9	U			R04	4.0	64De15
$^{151}\text{Eu}(\alpha)^{147}\text{Pm}$	1960	30	1964.0	1.1	0.1	U					07Be48
	1948.9	8.6			1.8	U					14Ca13
$^{151}\text{Gd}(\alpha)^{147}\text{Sm}$	2670.8	30.	2652.2	2.9	-0.6	U					65Si06
$^{151}\text{Tb}(\alpha)^{147}\text{Eu}$	3499.6	5.	3496	4	-0.7	1	58	48 ^{151}Tb	DbA		67Go32
$^{151}\text{Dy}(\alpha)^{147}\text{Gd}$	4175.5	5.	4179.6	2.6	0.8	2			DbA		67Go32 Z
	4181.1	3.			-0.5	2			Bka		82Bo04 Z
$^{151}\text{Ho}(\alpha)^{147}\text{Tb}$	4696.3	5.	4695.0	1.8	-0.3	2			GSa		79Ho10 *
	4695.8	3.			-0.3	2			Bka		82Bo04 *
	4693.8	3.			0.4	2			Ora		82De11 *
	4694.9	5.			0.0	2			Daa		96Pa01 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{151}\text{Eu}(n,\alpha)^{148}\text{Pm}$	7870	20	7859	6	-0.5	U			ILL		74Em01
$^{151}\text{Sm}(p,t)^{149}\text{Sm}$	-5100	4	-5101.4	0.4	-0.4	U			McM		73Ga04
$^{151}\text{Eu}(p,t)^{149}\text{Eu}$	-5872	5	-5873	4	-0.1	1	57	56	^{149}Eu		75Ta12
$^{150}\text{Nd}(n,\gamma)^{151}\text{Nd}$	5334.55	0.2	5334.55	0.10	0.0	-			ILn		76Pi13 Z
	5334.55	0.11			0.0	-			Bdn		06Fi.A
$^{150}\text{Nd}(d,p)^{151}\text{Nd}$	3084	15	3109.98	0.10	1.7	U			Tal		67Ne08
$^{150}\text{Nd}(n,\gamma)^{151}\text{Nd}$	ave. 5334.55	0.10	5334.55	0.10	0.0	1	100	100	^{151}Nd		average
$^{150}\text{Nd}(^3\text{He},d)^{151}\text{Pm}$	1503	5	1502	4	-0.2	1	80	80	^{151}Pm		80St10 *
$^{150}\text{Sm}(n,\gamma)^{151}\text{Sm}$	5596.5	1.8	5596.46	0.11	-0.0	U			McM		70Sm.A *
	5596.	1.5			0.3	U					71Gr22
	5596.42	0.20			0.2	-			ILn		86Va08 Z
	5596.44	0.13			0.2	-			Bdn		06Fi.A
$^{150}\text{Sm}(d,p)^{151}\text{Sm}$	3369	16	3371.90	0.11	0.2	U			Tal		65Ke09
$^{150}\text{Sm}(n,\gamma)^{151}\text{Sm}$	ave. 5596.43	0.11	5596.46	0.11	0.3	1	100	66	^{150}Sm		average
$^{151}\text{Eu}(\gamma,n)^{150}\text{Eu}$	-8040	110	-7932	6	1.0	U			Phi		60Ge01
$^{151}\text{Eu}(p,d)^{150}\text{Eu}$	-5721	9	-5707	6	1.5	1	47	47	^{150}Eu		82So.B
$^{151}\text{Yb}(ep)^{150}\text{Er}$	9000	300				2			ORa		86To12 *
$^{151}\text{Lu}(p)^{150}\text{Yb}$	1239.2	3.	1241.0	1.8	0.6	o			Dap		82Ho04
	1241.0	2.8			-0.0	3			Dap		93Se04
	1241.3	3.0			-0.1	3			ORp		03Gi10
	1240.3	4.0			0.2	3			Man		15Ta12
$^{151}\text{Lu}^m(p)^{150}\text{Yb}$	1318.8	10.	1298	3	-2.1	B			Dap		98Ba.B
	1293.6	4.			1.0	3			Man		15Ta12
	1303.7	5.			-1.2	3			Jyp		17Wa18
$^{151}\text{Ce}(\beta^-)^{151}\text{Pr}$	5270	100	5555	21	2.8	U			Kur		02Sh.B
$^{151}\text{Pr}(\beta^-)^{151}\text{Nd}$	4170	75	4164	12	-0.1	U			Bwg		90Gr10
	4136	40			0.7	-			Ida		93Gr17 *
	4210	30			-1.5	-			Kur		95Ik03
	ave. 4183	24			-0.8	1	24	24	^{151}Pr		average
$^{151}\text{Nd}(\beta^-)^{151}\text{Pm}$	2510	50	2443	4	-1.3	U					73Se12 *
	2480	50			-0.7	U			Kur		95Ik03
$^{151}\text{Pm}(\beta^-)^{151}\text{Sm}$	1195	10	1190	4	-0.5	1	20	20	^{151}Pm		64Be10 *
$^{151}\text{Sm}(\beta^-)^{151}\text{Eu}$	75.9	0.6	76.6	0.5	1.2	1	80	58	^{151}Eu		59Ac28
$^{151}\text{Gd}(\epsilon)^{151}\text{Eu}$	463	3	464.2	2.8	0.4	1	86	85	^{151}Gd		83Vo10 *
$^{151}\text{Tb}(\beta^+)^{151}\text{Gd}$	2562	5	2565	4	0.7	-					77Cr05 *
	2566	12			-0.1	-					84Sc18 *
	ave. 2563	5			0.6	1	66	52	^{151}Tb		average
$^{151}\text{Ho}(\beta^+)^{151}\text{Dy}$	5080	50	5130	9	1.0	U			IRS		83Al06 *
	5100	80			0.4	U			IRS		93Al03
$^{151}\text{Er}(\beta^+)^{151}\text{Ho}$	5130	110	5356	18	2.1	U					98Fo06
$^{151}\text{Tm}(\beta^+)^{151}\text{Er}$	6025	145	7495	25	10.1	C					84Ha.B *
	7074	50			8.4	F			GSI		91Ke11 *
$^{151}\text{Tm}^m(\text{IT})^{151}\text{Tm}$	96.4	7.0	93	6	-0.6	o					97Da07 *
$^{151}\text{Lu}^m(\text{IT})^{151}\text{Lu}$	77	5	57	4	-4.1	B			Dap		99Bi14
* $^{151}\text{Tb}-u$	$M-A=-71551(28)$ keV for mixture gs+m at 99.53 keV										Nub211 **
* $^{151}\text{Ho}-u$	$M-A=-63622(28)$ keV for mixture gs+m at 41.0 keV										Nub211 **
* $^{151}\text{Er}-u$	$M-A=-55670(28)$ keV for $^{151}\text{Er}^m$ at 2586.0 keV										Nub211 **
* $^{151}\text{Ho}(\alpha)^{147}\text{Tb}$	$E=4523.8(5,Z)$ to $^{147}\text{Tb}^m$ at 50.6(0.9); 4610.8(5,Z) from $^{151}\text{Ho}^m$ 41.0(0.2)										Nub211 **
* $^{151}\text{Ho}(\alpha)^{147}\text{Tb}$	$E=4521.5(3,Z)$ to $^{147}\text{Tb}^m$ at 50.6(0.9); 4611.5(3,Z) from $^{151}\text{Ho}^m$ 41.0(0.2)										Nub211 **
* $^{151}\text{Ho}(\alpha)^{147}\text{Tb}$	$E=4521.2(3,Z)$ to $^{147}\text{Tb}^m$ at 50.6(0.9); 4607.2(4,Z) from $^{151}\text{Ho}^m$ 41.0(0.2)										Nub211 **
* $^{151}\text{Ho}(\alpha)^{147}\text{Tb}$	$E_{\alpha}=4521(5,Z)$ to $^{147}\text{Tb}^m$ at 50.6(0.9)										Nub211 **
* $^{150}\text{Nd}(^3\text{He},d)^{151}\text{Pm}$	Based on $^{146}\text{Nd}(^3\text{He},d)^{147}\text{Pm}$ $Q=-87.6(0.9)$ keV										AHW **
* $^{150}\text{Sm}(n,\gamma)^{151}\text{Sm}$	$E(\gamma)=5591.7(1.8)$ to $3/2^-$ level at 4.821 keV										Ens091 **
* $^{151}\text{Yb}(ep)^{150}\text{Er}$	E_p estimated 7300(300) to levels around 1700 keV										GAu **
*	“Statistical p’s originate from $11/2^-$ isomer.”										86To12 **
* $^{151}\text{Pr}(\beta^-)^{151}\text{Nd}$	Two highest $Q_{\beta^-}=4135(50), 4137(40)$ keV										AHW **
* $^{151}\text{Nd}(\beta^-)^{151}\text{Pm}$	$E_{\beta^-}=2260(90) 2100(90) 1210(50)$ to $3/2^+$ level at 255.692, $1/2^+$ at 426.451, and $5/2^+$ at 1297.682 keV										Ens091 **
*											Ens091 **
* $^{151}\text{Pm}(\beta^-)^{151}\text{Sm}$	$E_{\beta^-}=1190(10)$ to ground state and $3/2^-$ level at 4.821 keV, and other E_{β^-}										Ens091 **
* $^{151}\text{Gd}(\epsilon)^{151}\text{Eu}$	$pK=0.652(0.007)$ to $(5/2^-, 7/2^-)$ level at 353.64 keV										Ens091 **
* $^{151}\text{Tb}(\beta^+)^{151}\text{Gd}$	$E_{\beta^+}=700(5) p^+ = 104(5) \times 10^{-4}$ respectively, to $1/2^-$ level at 839.320 keV, and other E_{β^+}										Ens091 **
* $^{151}\text{Ho}(\beta^+)^{151}\text{Dy}$	$E_{\beta^+}=3530(50)$ to $9/2^-$ level at 527.40 keV										Ens091 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$*^{151}\text{Tm}(\beta^+)^{151}\text{Er}$	$p^+ = 0.71(0.02)$ to $9/2^-$ level at 801.52 keV										Ens091	**
$*^{151}\text{Tm}(\beta^+)^{151}\text{Er}$	F : lower limit: positrons escape from detector; $E_{\beta^+} = 5250(50)$ to 801.6 level										91Ke11	**
$*^{151}\text{Tm}^m(\text{IT})^{151}\text{Tm}$	Only α -decay energies are used										GAu	**
$^{152}\text{Pr}-u$	-68447.1	19.9				2			CP1	1.0	06Sa56	
$\text{C}_{12} \text{H}_8-^{152}\text{Sm}$	142764	32	142861.6	1.1	0.8	U			R04	4.0	64De15	
	142867.0	5.0			-0.4	U			M22	2.5	75Ka25	
$^{152}\text{Eu}-u$	-78347	50	-78249.0	1.3	2.0	U			GS2	1.0	05Li24	
$\text{C}_{12} \text{H}_8-^{152}\text{Gd}$	142870	50	142801.8	1.1	-0.3	U			R04	4.0	64De15	
$^{152}\text{Gd O}-\text{C}_{14}$	-85290.7	3.5	-85287.0	1.1	0.7	U			TG1	1.5	11Ke03	
$^{152}\text{Tb}-u$	-76212	159	-75920	40	1.8	U			GS2	1.0	05Li24	
$^{152}\text{Dy}-u$	-75278	30	-75275	5	0.1	U			GS2	1.0	05Li24	
$^{152}\text{Ho}-u$	-68247	58	-68282	13	-0.6	U			GS2	1.0	05Li24	
$^{152}\text{Er}-u$	-64962	30	-64950	9	0.4	U			GS2	1.0	05Li24	
$^{152}\text{Tm}-u$	-55524	58	-55520	60	0.0	1	100	100 ^{152}Tm	GS2	1.0	05Li24	
$^{152}\text{Sm } ^{35}\text{Cl}_2-^{148}\text{Sm } ^{37}\text{Cl}_2$	10802	10	10809.7	1.1	0.3	U			H21	2.5	70Ma05	
	10810.8	2.0			-0.2	U			H25	2.5	72Ba08	
	10807.9	1.4			0.5	U			M21	2.5	75Ka25	
$^{152}\text{Sm } ^{35}\text{Cl}-^{150}\text{Sm } ^{37}\text{Cl}$	5429	4	5406.8	0.7	-1.4	U			H12	4.0	64Ba15	
	5396	4			1.1	o			H21	2.5	70Ma05	
	5402.7	0.8			2.0	1	11	8 ^{150}Sm	M21	2.5	75Ka25	
$^{152}\text{Gd}-^{152}\text{Sm}$	59.80	0.19	59.77	0.19	-0.2	1	98	78 ^{152}Sm	SH1	1.0	11El02	
$^{152}\text{Sm}-^{151}\text{Eu}$	95	42	-118.0	0.7	-1.3	U			R04	4.0	64De15	
$^{152}\text{Sm}-^{150}\text{Sm}$	2563	31	2456.7	0.7	-0.9	U			R04	4.0	64De15	
$^{152}\text{Gd}(\alpha)^{148}\text{Sm}$	2197.9	30.	2203.8	1.0	0.2	U					61Ma05	
$^{152}\text{Dy}(\alpha)^{148}\text{Gd}$	3728.0	8.	3727	4	-0.2	2					65Ma51	
	3726.0	5.			0.1	2			DbA		67Go32	
$^{152}\text{Ho}(\alpha)^{148}\text{Tb}$	4506.9	3.	4507.4	1.3	0.2	-			Bka		82Bo04	
	4508.0	2.			-0.3	-			Ora		82De11	
	4505.8	3.			0.5	-					82To14	
	4507.9	3.			-0.2	-					87St.A	
ave.	4507.3	1.3			0.0	1	100	95 ^{152}Ho			average	
$^{152}\text{Er}(\alpha)^{148}\text{Dy}$	4935.2	5.	4934.3	1.6	-0.2	-			GSa		79Ho10	
	4934.6	3.			-0.1	-			Bka		82Bo04	
	4934.3	2.			-0.0	-			Ora		82De11	
ave.	4934.5	1.6			-0.1	1	100	85 ^{152}Er			average	
$^{150}\text{Nd}(\text{t,p})^{152}\text{Nd}$	4125	30	4131	24	0.2	1	66	66 ^{152}Nd	Ald		72Ch11	
$^{150}\text{Sm}(\text{t,p})^{152}\text{Sm}$	5376	25	5372.5	0.6	-0.1	U			Ald		66Bj01	
$^{152}\text{Sm}(\text{p,t})^{150}\text{Sm}$	-5378	8	-5372.5	0.6	0.7	U			Min		72De47	
	-5376	4			0.9	U			McM		73Ga04	
	-5379	15			0.4	U			Ham		74Oe03	
$^{151}\text{Sm}(\text{n},\gamma)^{152}\text{Sm}$	8257.6	0.8	8257.8	0.6	0.3	1	57	44 ^{151}Sm			71Gr22	
$^{151}\text{Sm}(\text{p},\gamma)^{152}\text{Eu}$	5604	4	5601.0	0.5	-0.8	U					75Jo.A	
$^{151}\text{Eu}(\text{n},\gamma)^{152}\text{Eu}$	6306.70	0.10	6306.72	0.10	0.2	1	99	59 ^{152}Eu	ILn		85Vo15	
	6307.11	0.14			-2.8	C			Bdn		06Fi.A	
$^{152}\text{Gd}(\text{d,t})^{151}\text{Gd}$	-2338	10	-2332.5	2.9	0.6	U			Kop		67Tj01	
$^{152}\text{Pr}(\beta^-)^{152}\text{Nd}$	6350	120	6390	30	0.3	U			Kur		95Ik03	
$^{152}\text{Nd}(\beta^-)^{152}\text{Pm}$	1088	27	1105	19	0.6	-					93Sh23	
	1120	30			-0.5	-			Kur		95Ik03	
ave.	1102	20			0.1	1	85	51 ^{152}Pm			average	
$^{152}\text{Pm}(\beta^-)^{152}\text{Sm}$	3600	200	3509	26	-0.5	U					71Da19	
	3520	150			-0.1	U					72Wa04	
	3400	200			0.5	U					75Wi08	
	3500	100			0.1	-					77Ya07	
	3500	40			0.2	-			Kur		95Ik03	
ave.	3500	40			0.2	1	49	49 ^{152}Pm			average	
$^{152}\text{Pm}^m(\beta^-)^{152}\text{Sm}$	3603	100	3650	80	0.5	2					71Da19	
	3753	150			-0.7	2					72Wa04	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{152}\text{Eu}(\beta^+)^{152}\text{Sm}$	1871	5	1874.5	0.7	0.7	U					58A199 *
	1866	5			1.7	U					62Lo10 *
	1870.8	2.			1.8	—					72Sv02 *
	1872.8	1.5			1.1	—					77Mi.A *
ave.	1872.1	1.2			2.0	1	33	28	^{152}Eu		average
$^{152}\text{Eu}(\beta^-)^{152}\text{Gd}$	1809	10	1818.8	0.7	1.0	U					58A199 *
	1827	7			-1.2	U					60La04 *
	1836	20			-0.9	U					60Sc14 *
	1806	4			3.2	B					69An18 *
$^{152}\text{Tb}(\beta^+)^{152}\text{Gd}$	3990	40				2					76Cr.B *
$^{152}\text{Ho}(\beta^+)^{152}\text{Dy}$	6690	100	6513	13	-1.8	U			IRS		83A106 *
	6270	140			1.7	U					Averag *
	6225	90			3.2	B			IRS		93A103 *
$^{152}\text{Tm}(\beta^+)^{152}\text{Er}$	8820	240	8780	50	-0.2	U					16Na02
$^{152}\text{Tm}^m(\beta^+)^{152}\text{Er}$	6850	110	8680	240	16.6	C					84Ha.B *
	8680	240				2					16Na02
$^{152}\text{Yb}(\beta^+)^{152}\text{Tm}$	5065	165	5450	140	2.3	C					84Ha.B *
	5465	195			-0.1	—			Got		90Sa.A
	5434	200			0.1	—			GSI		04Na.A *
ave.	5450	140			0.0	1	100	100	^{152}Yb		average
* $^{152}\text{Eu}-u$	$M-A=-72915(35)$ keV for mixture gs+m+r at 45.5998 and 147.86 keV										Nub211 **
* $^{152}\text{Tb}-u$	$M-A=-70740(29)$ keV for mixture gs+n at 501.74 keV										Nub211 **
* $^{152}\text{Ho}-u$	$M-A=-63492(28)$ keV for mixture gs+m at 160(3) keV										Nub211 **
* $^{152}\text{Ho}(\alpha)^{148}\text{Tb}$	$E_\alpha=4389.1(3,Z)$; and $4455.1(3,Z)$ from $^{152}\text{Ho}^m$ to $^{148}\text{Tb}^m$										82Bo04 **
*	combined with $^{152}\text{Ho}^m(\text{IT})$ - $^{148}\text{Tb}^m(\text{IT})=160(1)$ - $90.1(0.3)$ keV										87St.A **
* $^{152}\text{Pm}^m(\beta^-)^{152}\text{Sm}$	$E_{\beta^-}=1800(100)$ 1950(150) respectively, to 5^- level at 1803.94 keV										Ens13b **
* $^{152}\text{Eu}(\beta^+)^{152}\text{Sm}$	$E_{\beta^+}=895(5)$ 890(5) respectively, from $^{152}\text{Eu}^m$ at 45.5998 keV										Nub211 **
* $^{152}\text{Eu}(\beta^+)^{152}\text{Sm}$	$E_{\beta^+}=727(2)$ 729(1.5) respectively, to 2^+ level at 121.7818 keV										Ens13b **
* $^{152}\text{Eu}(\beta^-)^{152}\text{Gd}$	$Q_{\beta^-}=1855(10)$ from $^{152}\text{Eu}^m$ at 45.5998 keV										Nub211 **
* $^{152}\text{Eu}(\beta^-)^{152}\text{Gd}$	$E_{\beta^-}=1483(7)$ to 2^+ level at 344.2790 keV										Ens13b **
* $^{152}\text{Eu}(\beta^-)^{152}\text{Gd}$	$E_{\beta^-}=1840(30)$ 1490(20) 1072(20) to ground state, 2^+ at 344.2790, 4^+ at 755.3961 keV										Ens13b **
* $^{152}\text{Eu}(\beta^-)^{152}\text{Gd}$	$Q_{\beta^-}=1852(4)$ from $^{152}\text{Eu}^m$ at 45.5998 keV										Nub211 **
* $^{152}\text{Tb}(\beta^+)^{152}\text{Gd}$	$E_{\beta^+}=2830(15)$ 8(4)% to ground state, 5.2(1)% to 2^+ level at 344.2790 keV										Ens13b **
* $^{152}\text{Ho}(\beta^+)^{152}\text{Dy}$	$E_{\beta^+}=3390(100)$ from $^{152}\text{Ho}^m$ at 160(1) to 8^+ level at 2437.42 keV										Ens13b **
* $^{152}\text{Ho}(\beta^+)^{152}\text{Dy}$	From adopted KLM/ $\beta^+=0.97(0.13)$										AHW **
*	from $^{152}\text{Ho}^m$ at 160(1) to 8^+ level at 2437.42 keV										Ens13b **
*	after extra 3(2)% side-feeding correction; see reference										90Sa32 **
*	$p^+=0.52(0.04)/.967$ gives KLM/ $\beta^+=0.86(0.14)$										85Sc09 **
*	KLM/ $\beta^+=1.12(0.10)$ after 0.967(0.008) side-feeding correction										90Sa32 **
* $^{152}\text{Ho}(\beta^+)^{152}\text{Dy}$	$Q_{\beta^+}=6270(90)$; and $6330(100)$ from $^{152}\text{Ho}^m$ at 160(3) keV										Nub211 **
* $^{152}\text{Tm}^m(\beta^+)^{152}\text{Er}$	$p^+=0.64(0.02)$ to 8^+ level at 2183.3 keV										Ens13b **
* $^{152}\text{Yb}(\beta^+)^{152}\text{Tm}$	$p^+=0.57(0.03)$ (1^+) level at 482.32 keV										Ens13b **
* $^{152}\text{Yb}(\beta^+)^{152}\text{Tm}$	As reported in reference										11Es03 **
$^{153}\text{Pr}-u$	-66110.5	15.3	-66096	13	0.9	—			CP1	1.0	06Sa56
	-66065	40			-0.8	—			CP1	1.0	12Va02 *
ave.	-66105	14			0.6	1	80	80	^{153}Pr		average
$^{153}\text{Pr}-^{80}\text{Kr}_{1.913}$	93906	40	93873	13	-0.8	1	10	10	^{153}Pr	1.0	12Va02
$^{153}\text{Pr}-^{86}\text{Kr}_{1.779}$	92958	40	92927	13	-0.8	1	10	10	^{153}Pr	1.0	12Va02
$^{153}\text{Nd}-u$	-72283.3	5.2	-72282.1	2.9	0.2	1	32	32	^{153}Nd	1.0	12Va02 *
$^{153}\text{Nd}-^{80}\text{Kr}_{1.913}$	87687.9	4.7	87687	3	-0.2	1	42	36	^{153}Nd	1.0	12Va02
$^{153}\text{Nd}-^{86}\text{Kr}_{1.779}$	86740.7	5.3	86741.6	2.9	0.2	1	31	31	^{153}Nd	1.0	12Va02
$^{153}\text{Pm}-u$	-75833	23	-75844	10	-0.5	1	18	18	^{153}Pm	1.0	12Va02 *
$^{153}\text{Pm}-^{80}\text{Kr}_{1.913}$	84139	23	84125	10	-0.6	1	18	18	^{153}Pm	1.0	12Va02
$^{153}\text{Pm}-^{86}\text{Kr}_{1.779}$	83192	23	83180	10	-0.5	1	18	18	^{153}Pm	1.0	12Va02

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_{12} H_9 - ^{153}Eu$	149103	18	149188.5	1.3	1.2	U			R04	4.0	64De15
$C_{11} ^{13}C H_8 - ^{153}Eu$	144606	30	144718.3	1.3	0.9	U			R04	4.0	64De15
$C_9 ^{13}C H_{16} O - ^{153}Eu$	201934	38	202233.2	1.3	2.0	U			R04	4.0	64De15
$^{153}Eu - ^{85}Rb_{1.800}$	80021	16	80015.3	1.3	-0.4	U			MA5	1.0	00Be42
$^{153}Ho - u$	-69814	37	-69793	5	0.6	U			GS2	1.0	05Li24 *
$^{153}Er - u$	-64942	30	-64914	10	0.9	U			GS2	1.0	05Li24
$^{153}Eu ^{35}Cl - ^{151}Eu ^{37}Cl$	4334	4	4330.30	0.18	-0.4	U			H21	2.5	70Ma05 *
$^{138}La O - ^{153}Eu$	-19266	123	-19198.1	1.3	0.1	U			R05	4.0	65De13
$^{153}Eu O - C_{14}$	-83849.6	5.8	-83848.6	1.3	0.1	U			TG1	1.5	11Ke03
$^{153}Eu - ^{152}Sm$	1544	42	1498.1	0.7	-0.3	U			R04	4.0	64De15
$^{153}Eu - ^{151}Eu$	1567	33	1380.18	0.17	-1.4	U			R04	4.0	64De15
$^{153}Dy(\alpha) ^{149}Gd$	3560.0	8.	3559	4	-0.1	-					65Ma51 Z
	3554.9	5.			0.8	-					67Go32 Z
	ave.	3556			0.6	1	69	48 ^{153}Dy	DbA		average
$^{153}Ho(\alpha) ^{149}Tb$	4052.3	5.	4052	4	-0.1	2					68Go.C *
	4051.0	5.			0.1	2			ORa		71To01 *
$^{153}Er(\alpha) ^{149}Dy$	4799.8	10.	4802.4	1.4	0.3	U					81Ho.A
	4804.5	3.			-0.7	-			Bka		82Bo04 Z
	4802.0	2.			0.2	-			Ora		82De11 Z
	4802.8	3.			-0.1	-					87Sc.A Z
	4799.7	4.			0.6	-			Daa		96Pa01
	ave.	4802.4			-0.0	1	100	98 ^{153}Er			average
$^{153}Tm(\alpha) ^{149}Ho$	5252.3	5.	5248.3	1.5	-0.8	U			GSa		79Ho10 *
	5246.1	3.			0.7	-			Bka		82Bo04 *
	5249.2	2.			-0.4	-			Ora		82De11 *
	5247.7	3.			0.2	o					87Sc.A *
	5247.7	3.			0.2	-					88Sc.A
	5249.5	5.			-0.2	U			Daa		96Pa01
	ave.	5248.1			0.1	1	100	53 ^{149}Ho			average
$^{153}Gd(n, \alpha) ^{150}Sm$	9790	30	9814.8	0.6	0.8	U			ILL		81Wa31
$^{153}Eu(p, v) ^{151}Eu$	-6374	5	-6375.21	0.16	-0.2	U			Min		75Ta12
$^{152}Sm(n, \gamma) ^{153}Sm$	5867.1	0.4	5868.40	0.13	3.2	B					69Re04 Z
	5868.4	0.3			-0.0	2					71Be41 Z
	5868.4	0.7			-0.0	U					82Ba15 Z
	5868.40	0.15			-0.0	2			Bdn		06Fi.A
$^{152}Sm(d, p) ^{153}Sm$	3645	12	3643.83	0.13	-0.1	U			Tal		65Ke09
$^{152}Eu(n, \gamma) ^{153}Eu$	8550.28	0.12	8550.28	0.12	0.0	1	100	86 ^{153}Eu	ILn		85Vo15 Z
$^{153}Eu(\gamma, n) ^{152}Eu$	-8650	130	-8550.28	0.12	0.8	U			Phi		60Ge01
$^{152}Gd(n, \gamma) ^{153}Gd$	6247.27	0.35	6246.96	0.13	-0.9	-			ILn		85Vo15 Z
	6246.89	0.14			0.5	-			ILn		93Sp.A
	6247.48	0.21			-2.5	B			Bdn		06Fi.A
$^{152}Gd(d, p) ^{153}Gd$	4015	10	4022.39	0.13	0.7	U			Kop		67Tj01
$^{152}Gd(n, \gamma) ^{153}Gd$	ave.	6246.94	6246.96	0.13	0.1	1	99	80 ^{152}Gd			average
$^{152}Gd(^3He, d) ^{153}Tb$	-1634	30	-1598	4	1.2	U			McM		76St10
$^{153}Pr(\beta^-) ^{153}Nd$	5720	100	5762	12	0.4	U			Kur		02Sh.B
$^{153}Nd(\beta^-) ^{153}Pm$	3336	25	3318	9	-0.7	1	14	13 ^{153}Pm	Ida		93Gr17
	3260	100			0.6	U			Kur		02Sh.B
$^{153}Pm(\beta^-) ^{153}Sm$	1777	50	1912	9	2.7	U					62Ko10 *
	1863	15			3.3	B			Ida		93Gr17
$^{153}Sm(\beta^-) ^{153}Eu$	810	10	807.4	0.7	-0.3	U					54Gr19
	795	10			1.2	U					54Le08
	820	10			-1.3	U					55Ma62
	825	10			-1.8	U					56Du31
	792	10			1.5	U					57Jo24
$^{153}Tb(\beta^+) ^{153}Gd$	1573	5	1569	4	-0.7	1	59	59 ^{153}Tb			78Cr02 *
$^{153}Dy(\beta^+) ^{153}Tb$	2171	2	2170.4	1.9	-0.3	1	93	52 ^{153}Dy			78Gr13 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{153}\text{Ho}(\beta^+)^{153}\text{Dy}$	4153	50	4131	6	-0.4	o			IRS		83Al06 *
	4160	60			-0.5	U			IRS		93Al03
$^{153}\text{Lu}^m(\text{IT})^{153}\text{Lu}$	80	5				4					97Ir01 *
* $^{153}\text{Pr}-\text{u}$	Represents frequency ratio $^{153}\text{Pr}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.99430250(26)$										WgM124**
* $^{153}\text{Nd}-\text{u}$	Represents frequency ratio $^{153}\text{Pr}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.994342.931(34)$										WgM124**
* $^{153}\text{Pm}-\text{u}$	Represents frequency ratio $^{153}\text{Pm}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.99436601(15)$										WgM124**
* $^{153}\text{Ho}-\text{u}$	$M-A=-64997(28)$ keV for mixture gs+m at 68.7 keV										Nub211 **
* $^{153}\text{Eu } ^{35}\text{Cl}-^{151}\text{Eu } ^{37}\text{Cl}$	Increased by 5 for systematic difference of H21 with later data										AHW **
* $^{153}\text{Ho}(\alpha)^{149}\text{Tb}$	$E_\alpha=4013.1(5,Z)$ from $^{153}\text{Ho}^m$ at 68.7 keV										Nub211 **
* $^{153}\text{Ho}(\alpha)^{149}\text{Tb}$	$E_\alpha=3910(5)$ to $^{149}\text{Tb}^m$ at 35.78 keV										Nub211 **
* $^{153}\text{Tm}(\alpha)^{149}\text{Ho}$	$E_\alpha=5114.2(5,Z)$ 5108.2(3,Z) 5111.2(2,Z) respectively, contain a 8%										AHW **
*	$^{153}\text{Tm}(\alpha)^{149}\text{Ho}-^{153}\text{Tm}^m(\alpha)^{149}\text{Ho}^m=48.80(0.20)-43.2(0.2)=$										Nub211 **
*	$=5.6(0.3)$ keV lower $^{153}\text{Tm}^m(\alpha)^{149}\text{Ho}^m$ branch, corrected thus +0.5keV										GAu **
* $^{153}\text{Tm}(\alpha)^{149}\text{Ho}$	$E_\alpha=5110.6(3,Z)$; and 5103.6(4,Z) for lower $^{153}\text{Tm}^m(\alpha)$ branch										87Sc.A **
* $^{153}\text{Pm}(\beta^-)^{153}\text{Sm}$	$E_{\beta^-}=1650(50)$ to $3/2^-$ level at 127.298 keV, and other E_{β^-}										Ens062 **
* $^{153}\text{Tb}(\beta^+)^{153}\text{Gd}$	$E_{\beta^+}=339(5)$ to $3/2^+$ level at 212.0078 keV										Ens062 **
* $^{153}\text{Dy}(\beta^+)^{153}\text{Tb}$	$E_{\beta^+}=886(2)$ to $9/2^-$ level at 262.831 keV										Ens062 **
* $^{153}\text{Ho}(\beta^+)^{153}\text{Dy}$	$E_{\beta^+}=2835(50)$ to $9/2^-$ level at 295.84 keV										Ens062 **
* $^{153}\text{Lu}^m(\text{IT})^{153}\text{Lu}$	Trends from Mass Surface TMS suggest ^{153}Lu 130 keV less bound										GAu **
$^{154}\text{Ce}-\text{u}$	-56404	619	-56060#	220#	0.2	D			GT3	2.5	16Kn03 *
$^{154}\text{Nd}-^{133}\text{Cs}_{1.158}$	39061	27	39084.0	1.1	0.9	U			JY1	1.0	20Vi04
$^{154}\text{Nd}-^{82}\text{Kr}_{1.878}$	92079.8	1.1				2			CP2	1.0	18Or02
$\text{C}_{12} \text{H}_{10}-^{154}\text{Sm}$	155830	29	156034.6	1.4	1.8	U			R04	4.0	64De15
	156035.7	4.0			-0.1	U			M22	2.5	75Ka25
$\text{C}_{12} \text{H}_{10}-^{154}\text{Gd}$	157149	40	157377.3	1.1	1.4	U			R04	4.0	64De15
$\text{C}_{11} ^{13}\text{C} \text{H}_9-^{154}\text{Gd}$	152550	110	152907.1	1.1	0.8	U			R04	4.0	64De15
$\text{C}_{10} ^{13}\text{C}_2 \text{H}_8-^{154}\text{Gd}$	148030	90	148437.0	1.1	1.1	U			R04	4.0	64De15
$\text{C}_{10} \text{H}_6 \text{N}_2-^{154}\text{Gd}$	131980	240	132225.2	1.1	0.3	U			R04	4.0	64De15
$^{154}\text{Gd}-^{138}\text{La} \text{O}$	19005	80	18834.3	1.2	-0.5	U			R05	4.0	65De13
$^{154}\text{Tb}-\text{u}$	-75420	120	-75320	50	0.9	R			GS2	1.0	05Li24 *
$^{154}\text{Dy}-^{133}\text{Cs}_{1.158}$	33903	19	33916	8	0.7	1	18	18 ^{154}Dy	MA5	1.0	00Be42 *
$^{154}\text{Ho}-\text{u}$	-69350	83	-69393	9	-0.5	U			GS2	1.0	05Li24 *
$^{154}\text{Tm}-\text{u}$	-58479	49	-58430	15	1.0	U			GS2	1.0	05Li24 *
$^{154}\text{Sm } ^{35}\text{Cl}_2-^{150}\text{Sm } ^{37}\text{Cl}_2$	10832.9	5.2	10834.0	1.1	0.1	U			M21	2.5	75Ka25
$^{154}\text{Sm } ^{35}\text{Cl}-^{152}\text{Sm } ^{37}\text{Cl}$	5480	4	5427.2	0.9	-3.3	B			H12	4.0	64Ba15
	5417	4			1.0	U			H21	2.5	70Ma05
	5427.2	0.4			0.0	1	80	78 ^{154}Sm	M21	2.5	75Ka25
$^{154}\text{Gd } ^{35}\text{Cl}-^{152}\text{Gd } ^{37}\text{Cl}$	4019.5	2.	4024.68	0.23	1.0	U			H25	2.5	72Ba08
	4016	30			0.1	U			H12	4.0	64Ba15
$^{154}\text{Sm}-^{154}\text{Gd}$	1338.2	3.8	1342.8	0.9	0.5	U			H25	2.5	72Ba08
	1342.8	0.8			-0.0	1	21	21 ^{154}Sm	M21	2.5	75Ka25
$^{154}\text{Sm}-\text{C}_{12} \text{H}_9$	-148211.0	8.0	-148209.5	1.4	0.1	U			M21	2.5	75Ka25
$^{139}\text{La} \text{O}-^{154}\text{Gd}$	-19616	55	-19595.4	1.2	0.1	U			R05	4.0	65De13
$^{154}\text{Sm}-^{153}\text{Eu}$	1082	42	979.0	1.2	-0.6	U			R04	4.0	64De15
$^{154}\text{Sm}-^{152}\text{Sm}$	2664	43	2477.1	0.9	-1.1	U			R04	4.0	64De15
$^{154}\text{Gd}-^{152}\text{Gd}$	1400	50	1074.56	0.22	-1.6	U			R04	4.0	64De15
$^{154}\text{Gd} \text{O}-\text{C}_{15}$	-84207.4	5.9	-84212.4	1.1	-0.6	U			TG1	1.5	09Ke.A
	-84206.6	4.3			-0.9	U			TG1	1.5	11Ke03
$^{154}\text{Dy}(\alpha)^{150}\text{Gd}$	2946.4	5.	2945	5	-0.2	1	93	82 ^{154}Dy	DbA		67Go32 Z
$^{154}\text{Ho}(\alpha)^{150}\text{Tb}$	4041.3	5.	4041	4	0.0	2					68Go.C Z
	4041.7	5.			-0.0	2			ORa		74Sc19 Z
$^{154}\text{Ho}^m(\alpha)^{150}\text{Tb}^m$	3819.2	10.	3823	5	0.4	-			ORa		71To01 Z
	3824.0	5.			-0.1	-			ORa		74Sc19 Z
	ave.	3823	5		0.1	1	100	89 $^{154}\text{Ho}^m$			average
$^{154}\text{Er}(\alpha)^{150}\text{Dy}$	4280.5	5.	4279.7	2.6	-0.2	-					68Go.C Z
	4279.5	3.			0.1	-			Bka		82Bo04 Z
	ave.	4279.8	2.6		-0.0	1	98	92 ^{154}Er			average
$^{154}\text{Tm}(\alpha)^{150}\text{Ho}$	5096.9	5.1	5093.8	2.6	-0.6	2			GSa		79Ho10 Z
	5092.7	3.			0.4	2			Bka		82Bo04
$^{154}\text{Tm}^m(\alpha)^{150}\text{Ho}^m$	5174.8	5.	5171.8	1.6	-0.6	3			GSa		79Ho10 Z

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{154}\text{Tm}^m(\alpha)^{150}\text{Ho}^m$	5170.8	3.	5171.8	1.6	0.3	3			Bka		82Bo04 Z
	5171.8	2.1			0.0	3			Ora		82De11 Z
$^{154}\text{Yb}(\alpha)^{150}\text{Er}$	5473.4	5.	5474.3	1.7	0.2	—			GSa		79Ho10 Z
	5474.7	2.			−0.2	—			Ora		82De11 Z
	5473.4	4.			0.2	—			Daa		96Pa01
ave.	5474.3	1.7			0.0	1	100	100	^{154}Yb		average
$^{152}\text{Sm}(\text{t},\text{p})^{154}\text{Sm}$	5361	25	5353.4	0.8	−0.3	U			Ald		66Bj01
$^{154}\text{Sm}(\text{p},\text{t})^{152}\text{Sm}$	−5357	8	−5353.4	0.8	0.4	U			Min		72De47
	−5353	15			−0.0	U			Ham		74Oe03
$^{154}\text{Gd}(\text{p},\text{t})^{152}\text{Gd}$	−6660	5	−6659.89	0.21	0.0	U			Min		73Oo01
$^{154}\text{Sm}(\text{d},^3\text{He})^{153}\text{Pm}$	−3623	25	−3603	9	0.8	—					76Su.B
$^{154}\text{Sm}(\text{t},\alpha)^{153}\text{Pm}$	10748	20	10717	9	−1.5	—			LAI		78Bu18
$^{154}\text{Sm}(\text{d},^3\text{He})^{153}\text{Pm}$	ave.	−3592	16	−3603	9	−0.7	1	34	33	^{153}Pm	average
$^{153}\text{Eu}(\text{n},\gamma)^{154}\text{Eu}$	6442.2	0.3	6442.22	0.24	0.1	—			ILn		87Ba52 Z
	6442.2	0.4			0.0	—			Bdn		06Fi.A
ave.	6442.20	0.24			0.1	1	99	85	^{154}Eu		average
$^{153}\text{Gd}(\text{n},\gamma)^{154}\text{Gd}$	8895.25	0.30	8894.73	0.17	−1.7	—			ILn		85Vo15 Z
	8894.47	0.20			1.3	—			ILn		93Sp.A Z
$^{154}\text{Gd}(\text{d},\text{t})^{153}\text{Gd}$	−2642	10	−2637.50	0.17	0.4	U			Kop		67Tj01
$^{153}\text{Gd}(\text{n},\gamma)^{154}\text{Gd}$	ave.	8894.71	0.17	8894.73	0.17	0.1	1	98	80	^{153}Gd	average
$^{154}\text{Pr}(\beta^-)^{154}\text{Nd}$	7720	100				3			Kur		02Sh.B *
$^{154}\text{Nd}(\beta^-)^{154}\text{Pm}$	2687	25				3			Ida		93Gr17
$^{154}\text{Pm}(\beta^-)^{154}\text{Sm}$	3900	200	4189	25	1.4	U					71Da28
	4396	180			−1.2	U					72Ta13
	3880	200			1.5	U					74Ya07 *
$^{154}\text{Pm}^m(\beta^-)^{154}\text{Sm}$	3900	200	3960	40	0.3	U					71Da28 *
	4190	170			−1.3	U					72Ta13 *
	3940	50			0.5	2					73Pr05 *
	3940	200			0.1	U					74Ya07 *
	4056	100			−0.9	2			Ida		93Gr17
$^{154}\text{Eu}(\beta^-)^{154}\text{Gd}$	1978	5	1968.0	0.8	−2.0	U					60La04 *
	1967	2			0.5	1	14	12	^{154}Eu		77Ra08 *
	1975	3			−2.3	U					81Bu.A *
$^{154}\text{Tb}(\beta^+)^{154}\text{Gd}$	3562	50	3550	50	−0.2	2					70Ag03 *
$^{154}\text{Ho}(\beta^+)^{154}\text{Dy}$	5700	80	5755	10	0.7	U			IRS		83Al06 *
	5750	80			0.1	U			IRS		93Al03
$^{154}\text{Ho}^m(\beta^+)^{154}\text{Dy}$	5994	100	5997	28	0.0	o			IRS		83Al.A *
	6070	80			−0.9	1	12	11	$^{154}\text{Ho}^m$		93Al03
$^{154}\text{Tm}^m(\beta^+)^{154}\text{Er}$	8234	150	8250	50	0.1	U			Dbn		94Po26 *
$^{154}\text{Lu}(\beta^+)^{154}\text{Yb}$	7556	450	10270#	200#	6.0	C					84Ha.B *
$^{154}\text{Lu}^m(\text{IT})^{154}\text{Lu}$	58.7	9.3	62	12	0.4	o			Ara		97Da07 *
* $^{154}\text{Ce}-\text{u}$	Trends from Mass Surface TMS suggest ^{154}Ce 320 keV less bound										GAu **
* $^{154}\text{Tb}-\text{u}$	$M-A=70142(43)$ keV for mixture gs+m+n at 130#50 and 200#150 keV										Nub211 **
* $^{154}\text{Dy}-^{133}\text{Cs}_{1,158}$	No contamination observed, but contamination by ^{154}Tb cannot be excluded										00Be42 **
* $^{154}\text{Ho}-\text{u}$	$M-A=64478(28)$ keV for mixture gs+m at 243(28) keV										Nub211 **
* $^{154}\text{Tm}-\text{u}$	$M-A=54438(32)$ keV for mixture gs+m at 70(50) keV										Nub211 **
* $^{154}\text{Pr}(\beta^-)^{154}\text{Nd}$	$E_{\beta^-}=7490(100)$ to 4^+ level at 233.2 keV										96To05 **
* $^{154}\text{Pm}(\beta^-)^{154}\text{Sm}$	$E_{\beta^-}=2410(180)$ to 3^- level at 1986.59 keV										Ens09a *W
* $^{154}\text{Pm}(\beta^-)^{154}\text{Sm}$	$E_{\beta^-}=2400(200)$, 1850(200) to 2^+ levels at 1440.04, 2069.07 keV										Ens09a **
* $^{154}\text{Pm}^m(\beta^-)^{154}\text{Sm}$	$E_{\beta^-}=3270$, 3090, 2810 (all 170) to 921.345 1^- , 1099.26 0^+ , 1475.81 1^-										Ens09a **
* $^{154}\text{Pm}^m(\beta^-)^{154}\text{Sm}$	$E_{\beta^-}=2810(170)$ to 1^- level at 1475.81 keV, and other E_{β^-}										Ens09a **
* $^{154}\text{Pm}^m(\beta^-)^{154}\text{Sm}$	$E_{\beta^-}=3950(50)$ 3010(80) to ground state, 1^- level at 921.345 keV										Ens09a **
* $^{154}\text{Pm}^m(\beta^-)^{154}\text{Sm}$	$E_{\beta^-}=3000(200)$, 1900(200), 1800(200) to 1^- level at 921.345, 2^+ at 2069.07, and $(1,2^+)$ at 2139.82 keV										Ens09a **
* $^{154}\text{Eu}(\beta^-)^{154}\text{Gd}$	$E_{\beta^-}=1855(5)$ 1844(2) respectively, to 2^+ level at 123.0709 keV										Ens09a **
* $^{154}\text{Eu}(\beta^-)^{154}\text{Gd}$	$E_{\beta^-}=257(3)$ to 2^- level at 1719.5593 keV, and other E_{β^-}										Ens09a **
* $^{154}\text{Tb}(\beta^+)^{154}\text{Gd}$	$E_{\beta^+}=2540(50)$ 1860(50) to ground state and 0^+ level at 680.6673 keV										Ens09a **
* $^{154}\text{Ho}(\beta^+)^{154}\text{Dy}$	$E_{\beta^+}=4340(80)$ to 2^+ level at 334.34 keV										Ens09a **
* $^{154}\text{Ho}^m(\beta^+)^{154}\text{Dy}$	$E_{\beta^+}=2500(100)$ to 7^+ level at 2472.40 keV										Ens09a **
* $^{154}\text{Tm}^m(\beta^+)^{154}\text{Er}$	$E_{\beta^+}=4882(150)$ to 8^+ level 2329.5 keV										Ens09a **
* $^{154}\text{Lu}(\beta^+)^{154}\text{Yb}$	$p^+=0.75(0.05)$ $Q=5710(450)$ from $^{154}\text{Lu}^m$ at 200#150 to 8^+ level at 2046.2 keV										Ens09a **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
* $^{154}\text{Lu}^m(\text{IT})^{154}\text{Lu}$	Use only their Q_α 's										GAu	**
$^{155}\text{Pr}-\text{u}$	-59492	31	-59491	18	0.0	1	35	35 ^{155}Pr	CP1	1.0	12Va02	*
$^{155}\text{Pr}-^{80}\text{Kr}_{1.938}$	102571	33	102569	18	-0.1	1	31	31 ^{155}Pr	CP1	1.0	12Va02	
$^{155}\text{Pr}-^{86}\text{Kr}_{1.802}$	101588	32	101589	18	0.0	1	33	33 ^{155}Pr	CP1	1.0	12Va02	
$^{155}\text{Nd}-\text{u}$	-66866	17	-66864	10	0.1	1	33	33 ^{155}Nd	CP1	1.0	12Va02	*
$^{155}\text{Nd}-^{80}\text{Kr}_{1.938}$	95197	17	95195	10	-0.1	1	34	33 ^{155}Nd	CP1	1.0	12Va02	
$^{155}\text{Nd}-^{86}\text{Kr}_{1.802}$	94215	17	94215	10	0.0	1	33	33 ^{155}Nd	CP1	1.0	12Va02	
$^{155}\text{Pm}-\text{u}$	-71863.8	8.8	-71863	5	0.1	1	33	33 ^{155}Pm	CP1	1.0	12Va02	*
$^{155}\text{Pm}-^{80}\text{Kr}_{1.938}$	90197.8	8.6	90196	5	-0.2	1	36	34 ^{155}Pm	CP1	1.0	12Va02	
$^{155}\text{Pm}-^{86}\text{Kr}_{1.802}$	89216.0	8.8	89217	5	0.1	1	33	33 ^{155}Pm	CP1	1.0	12Va02	
$^{155}\text{Sm}-\text{u}$	-75357	24	-75353.4	1.4	0.2	U			CP1	1.0	12Va02	*
$^{155}\text{Sm}-^{80}\text{Kr}_{1.938}$	86704	24	86706.2	2.0	0.1	U			CP1	1.0	12Va02	
$^{155}\text{Sm}-^{86}\text{Kr}_{1.802}$	85722	24	85726.3	1.4	0.2	U			CP1	1.0	12Va02	
$\text{C}_{12} \text{H}_{11}-^{155}\text{Gd}$	163530	70	163446.0	1.1	-0.3	U			R04	4.0	64De15	
$\text{C}_{11}^{13}\text{C} \text{H}_{10}-^{155}\text{Gd}$	158921	42	158975.8	1.1	0.3	U			R04	4.0	64De15	
$\text{C}_{10}^{13}\text{C}_2 \text{H}_9-^{155}\text{Gd}$	154450	140	154505.6	1.1	0.1	U			R04	4.0	64De15	
$\text{C}_{10} \text{H}_7 \text{N}_2-^{155}\text{Gd}$	138213	38	138293.9	1.1	0.5	U			R04	4.0	64De15	
$^{155}\text{Gd}-^{139}\text{La} \text{O}$	21252	32	21351.8	1.2	0.8	U			R05	4.0	65De13	
$^{155}\text{Tb}-\text{u}$	-76431	30	-76490	11	-2.0	U			GS2	1.0	05Li24	
$^{155}\text{Dy}-\text{u}$	-74227	30	-74242	10	-0.5	U			GS2	1.0	05Li24	
$^{155}\text{Ho}-\text{u}$	-70867	30	-70897	19	-1.0	1	39	39 ^{155}Ho	GS2	1.0	05Li24	
$^{155}\text{Er}-\text{u}$	-66785	30	-66784	7	0.0	U			GS2	1.0	05Li24	
$^{155}\text{Tm}-\text{u}$	-60814	33	-60790	11	0.7	U			GS2	1.0	05Li24	*
$^{155}\text{Gd}^{35}\text{Cl}_3-^{149}\text{Sm}^{37}\text{Cl}_3$	14282.4	6.3	14288.5	0.8	0.4	U			M21	2.5	75Ka25	
$^{155}\text{Gd}^{35}\text{Cl}-^{153}\text{Eu}^{37}\text{Cl}$	4345.4	2.4	4342.7	0.8	-0.5	U			H25	2.5	72Ba08	
$^{155}\text{Gd}-^{138}\text{La} \text{O}$	20558	49	20590.7	1.1	0.2	U			R05	4.0	65De13	
$^{155}\text{Gd}-^{154}\text{Gd}$	1480	60	1756.38	0.20	1.2	U			R04	4.0	64De15	
$^{155}\text{Gd} \text{O}-\text{C}_{15}$	-82452.8	5.0	-82456.0	1.1	-0.4	o			TG1	1.5	09Ke.A	
	-82452.2	2.6			-1.0	1	7	7 ^{155}Gd	TG1	1.5	11Ke03	
$^{155}\text{Er}(\alpha)^{151}\text{Dy}$	4118.3	5.				3			ORa		74To07	Z
$^{155}\text{Tm}(\alpha)^{151}\text{Ho}$	4578.3	10.3	4572	5	-0.6	3			ORa		71To01	*
	4568.1	10.			0.4	3			ORa		71To01	*
	4570.1	8.			0.2	3					92Ha10	*
$^{155}\text{Yb}(\alpha)^{151}\text{Er}$	5344.1	5.	5338.8	2.1	-1.1	3			GSa		79Ho10	
	5336.6	5.			0.4	3			Bka		82Bo04	Z
	5344.2	5.			-1.1	3					87Ka.A	
	5331.8	4.			1.7	3			ORa		91To08	
	5340.1	4.			-0.3	3			Daa		96Pa01	
$^{155}\text{Lu}(\alpha)^{151}\text{Tm}$	5796.9	5.	5801.6	2.2	0.9	-					89Ho12	*
	5797.9	5.			0.7	-			ORa		91To08	
	5805.1	5.			-0.7	-			Daa		96Pa01	
	5811.2	5.			-1.9	-			Ara		97Da07	
	5802.8	2.6			-0.4	1	76	76 ^{151}Tm			average	
$^{155}\text{Lu}^m(\alpha)^{151}\text{Tm}^m$	5723.0	10.	5730.6	2.8	0.7	3					89Ho12	
	5727.1	5.			0.7	3			ORa		91To08	
	5732.2	5.			-0.3	3			Daa		96Pa01	
	5734.2	5.			-0.7	3			Ara		97Da07	
$^{155}\text{Lu}^n(\alpha)^{151}\text{Tm}$	7574.9	15.	7581.9	2.6	0.1	U					89Ho12	*
	7586.2	5.			-0.8	o			Daa		96Pa01	*
	7579.0	4.1			0.7	1	39	24 ^{151}Tm			18Pa37	
$^{155}\text{Gd}(\text{n},\alpha)^{152}\text{Sm}$	8331	6	8339.1	0.3	1.3	U			McM		69Be17	
$^{155}\text{Gd}(\text{p},\text{t})^{153}\text{Gd}$	-6850	7	-6848.19	0.25	0.3	U			McM		73Lo08	
	-6853	5			1.0	U			Min		73Oo01	
$^{154}\text{Sm}(\text{n},\gamma)^{155}\text{Sm}$	5806.8	0.6	5806.96	0.27	0.3	2					82Ba15	Z
	5807.0	0.3			-0.1	2			ILn		82Sc03	Z

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{154}\text{Sm}(\text{d,p})^{155}\text{Sm}$	3584	12	3582.39	0.27	-0.1	U			Tal		65Ke09
$^{154}\text{Eu}(\text{n},\gamma)^{155}\text{Eu}$	8151.3	0.4	8151.3	0.4	-0.0	1	100	98 ^{155}Eu	ILn		86Pr03
$^{154}\text{Gd}(\text{n},\gamma)^{155}\text{Gd}$	6435.11	0.30	6435.26	0.18	0.5	-			ILn		86Sc25 Z
	6435.29	0.23			-0.1	-			Bdn		06Fi.A
$^{154}\text{Gd}(\text{d,p})^{155}\text{Gd}$	4217	10	4210.69	0.18	-0.6	U			Kop		67Tj01
$^{155}\text{Gd}(\text{d,t})^{154}\text{Gd}$	-190	10	-178.03	0.18	1.2	U			Kop		67Tj01
$^{154}\text{Gd}(\text{n},\gamma)^{155}\text{Gd}$	ave. 6435.22	0.18	6435.26	0.18	0.2	1	99	80 ^{154}Gd			average
$^{155}\text{Ta}(\text{p})^{154}\text{Hf}$	1776	10	1453	15	-32.3	B			Arp		99Uu01 *
	1453	15				3			Jya		07Pa27
$^{155}\text{Nd}(\beta^-)^{155}\text{Pm}$	4222	150	4656	10	2.9	U			Ida		93Gr17
$^{155}\text{Pm}(\beta^-)^{155}\text{Sm}$	3224	30	3251	5	0.9	U			Ida		93Gr17
$^{155}\text{Sm}(\beta^-)^{155}\text{Eu}$	1634	15	1627.1	1.2	-0.5	U					63Kr04 *
	1624	15			0.2	U					65Fu13 *
	1607	25			0.8	U			Ida		93Gr17
$^{155}\text{Eu}(\beta^-)^{155}\text{Gd}$	252	5	252.0	0.9	-0.0	U					54Le08
	245	5			1.4	U					58Gl56
	245	5			1.4	U					59Am16
$^{155}\text{Dy}(\beta^+)^{155}\text{Tb}$	2099	6	2094.5	1.9	-0.8	2					63Pe13 *
	2094	2			0.2	2					80Bu04 *
$^{155}\text{Ho}(\beta^+)^{155}\text{Dy}$	3102	20	3116	17	0.7	1	69	61 ^{155}Ho			72To07 *
$^{155}\text{Lu}^m(\text{IT})^{155}\text{Lu}$	23.0	6.2	21	4	-0.2	2					96Pa01
	19.9	6.2			0.3	2					97Da07
$^{155}\text{Lu}^n(\text{IT})^{155}\text{Lu}$	1781	2	1780.3	1.8	-0.3	1	85	85 $^{155}\text{Lu}^n$			96Pa01
* $^{155}\text{Pr}-\text{u}$	Represents frequency ratio $^{155}\text{Pr}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.98142559(20)$										WgM124**
* $^{155}\text{Nd}-\text{u}$	Represents frequency ratio $^{155}\text{Nd}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.98147230(11)$										WgM124**
* $^{155}\text{Pm}-\text{u}$	Represents frequency ratio $^{155}\text{Pm}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.981503965(56)$										WgM124**
* $^{155}\text{Sm}-\text{u}$	Represents frequency ratio $^{155}\text{Sm}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.98152610(15)$										WgM124**
* $^{155}\text{Tm}-\text{u}$	$M-A=-56627(28)$ keV for mixture gs+m at 41(6) keV										Nub211 **
* $^{155}\text{Tm}(\alpha)^{151}\text{Ho}$	First assigned to $^{156}\text{Tm}^m$ but belonging to ^{155}Tm ground state										94To10 **
* $^{155}\text{Tm}(\alpha)^{151}\text{Ho}$	Doublet from ground state and isomer, less than 5 keV apart										90Po13 **
* $^{155}\text{Lu}(\alpha)^{151}\text{Tm}$	Original value $E=5656(6)$ ($Q=5806.1$) recalibrated										79Ho10 **
* $^{155}\text{Lu}^n(\alpha)^{151}\text{Tm}$	Original value $E=7408(10)$ recalibrated										81Ho.A **
* $^{155}\text{Lu}^n(\alpha)^{151}\text{Tm}$	Replaced by authors' value for $^{155}\text{Lu}^n(\text{IT})$										AHW **
* $^{155}\text{Ta}(\text{p})^{154}\text{Hf}$	$E_p=1765(10)$ for $(11/2^-)$ state; ground state may be $1/2^+$, slightly lower										99Uu01 **
*	1776 keV proton not observed in coincidence with feeding α										07Pa27 **
* $^{155}\text{Sm}(\beta^-)^{155}\text{Eu}$	$E_{\beta^-}=1530(15)$ $E_{\beta^-}=1520(15)$ respectively, to $5/2^-$ level at 104.334 keV										Ens051 **
* $^{155}\text{Dy}(\beta^+)^{155}\text{Tb}$	$E_{\beta^+}=850(6)$ $845(2)$ respectively, to $5/2^-$ level at 226.918 keV, and other E_{β^+}										Ens051 **
* $^{155}\text{Ho}(\beta^+)^{155}\text{Dy}$	$E_{\beta^+}=1840(20)$ to $3/2^+$ level at 240.196 keV										Ens051 **
$^{156}\text{Pr}-\text{u}$	-55233.1	1.1				2			CP2	1.0	18Or.A *
$^{156}\text{Nd}-^{136}\text{Xe}_{1.147}$	41786.8	2.6	41795.4	1.4	3.3	B			JY1	1.0	18Vi02
$^{156}\text{Nd}-^{82}\text{Kr}_{1.902}$	99929.2	1.4				2			CP2	1.0	18Or02
$^{156}\text{Pm}-\text{u}$	-68883.4	6.9	-68885.9	1.3	-0.4	-			CP1	1.0	12Va02 *
	-68887.8	2.5			0.7	-			CP2	1.0	20Or03 *
	ave. -68887.3	2.4			0.6	1	29	29 ^{156}Pm			average
$^{156}\text{Pm}-^{80}\text{Kr}_{1.950}$	94181.7	6.4	94177.1	1.9	-0.7	1	9	5 ^{80}Kr	CP1	1.0	12Va02
$^{156}\text{Pm}-^{86}\text{Kr}_{1.814}$	93269.1	6.8	93266.4	1.3	-0.4	1	4	4 ^{156}Pm	CP1	1.0	12Va02
$^{156}\text{Pm}^m-\text{u}$	-68724.4	1.6	-68724.6	1.3	-0.1	1	63	63 $^{156}\text{Pm}^m$	CP2	1.0	20Or03 *
$\text{C}_{12}\text{H}_{12}-^{156}\text{Gd}$	171923	44	171770.3	1.1	-0.9	U			R04	4.0	64De15
$\text{C}_{11}\text{H}_{11}-^{156}\text{Gd}$	167384	43	167300.1	1.1	-0.5	U			R04	4.0	64De15
$\text{C}_{10}\text{H}_{10}-^{156}\text{Gd}$	162810	60	162829.9	1.1	0.1	U			R04	4.0	64De15
$\text{C}_{10}\text{H}_8\text{N}_2-^{156}\text{Gd}$	146661	38	146618.1	1.1	-0.3	U			R04	4.0	64De15
$^{156}\text{Tb}-\text{u}$	-75181	52	-75246	4	-1.2	U			GS2	1.0	05Li24 *
$\text{C}_{10}\text{H}_8\text{N}_2-^{156}\text{Dy}$	145130	100	144464.7	1.1	-1.7	U			R04	4.0	64De15
$^{156}\text{Dy}-^{133}\text{Cs}_{1.173}$	35195.1	4.5	35188.4	1.1	-1.5	U			MA8	1.0	19Hu15

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{156}\text{Ho}-\text{u}$	-70130	120	-70360	40	-1.9	o			GS1	1.0	00Ra23 *
$^{156}\text{Ho}''-\text{u}$	-70107	30				2			GS2	1.0	05Li24 *
$^{156}\text{Er}-\text{u}$	-68907	30	-68934	26	-0.9	1	78	78 ^{156}Er	GS2	1.0	05Li24
$^{156}\text{Tm}-\text{u}$	-61044	30	-61014	15	1.0	U			GS2	1.0	05Li24
$^{156}\text{Yb}-\text{u}$	-57202	30	-57183	10	0.6	U			GS2	1.0	05Li24
$^{156}\text{Gd } ^{35}\text{Cl}-^{154}\text{Gd } ^{37}\text{Cl}$	4199	5	4207.27	0.22	0.4	U			H12	4.0	64Ba15
	4206	10			0.1	U			H21	2.5	70Ma05
	4204.8	1.4			0.7	U			H25	2.5	72Ba08
	4203.0	1.0			1.7	U			M21	2.5	75Ka25
$^{156}\text{Dy}-^{156}\text{Gd}$	2153.47	0.11	2153.47	0.11	0.0	1	100	99 ^{156}Dy	SH1	1.0	11El05
$^{156}\text{Gd}-^{139}\text{La O}$	20618	71	20852.6	1.2	0.8	U			R05	4.0	65De13
$^{156}\text{Gd}-^{155}\text{Gd}$	-584	33	-499.24	0.07	0.6	U			R04	4.0	64De15
$^{156}\text{Gd O}-\text{C}_{15}$	-82946.5	5.8	-82955.3	1.1	-1.0	o			TG1	1.5	09Ke.A
	-82945.6	3.6			-1.8	U			TG1	1.5	11Ke03
$^{156}\text{Er}(\alpha)^{152}\text{Dy}$	3109.9	70.	3481	25	5.3	C					95Ka.A
$^{156}\text{Tm}(\alpha)^{152}\text{Ho}$	4341.6	10.	4345	7	0.4	-			ORa		71To10
	4345.6	10.			-0.0	-					81Ga36
	ave. 4344	7			0.2	1	98	94 ^{156}Tm			average
$^{156}\text{Tm}^m(\alpha)^{152}\text{Ho}$	4737.5	10.	*			F			ORa		71To01 *
$^{156}\text{Yb}(\alpha)^{152}\text{Er}$	4813.6	10.	4810	4	-0.4	-					77Ha48
	4809.6	10.			0.0	-			GSa		79Ho10
	4810.6	4.			-0.2	-			Daa		96Pa01
	ave. 4811	4			-0.3	1	98	83 ^{156}Yb			average
$^{156}\text{Lu}(\alpha)^{152}\text{Tm}$	5593.7	10.	5596	3	0.2	U			GSa		79Ho10
	5592.7	5.			0.6	2			Db		92Po14
	5597.9	4.			-0.5	2			Daa		96Pa01
$^{156}\text{Lu}^m(\alpha)^{152}\text{Tm}^m$	5713.7	5.	5710.4	2.2	-0.6	3			GSa		79Ho10 Z
	5709.7	5.			0.1	3			Db		92Po14
	5709.7	8.			0.1	3					92Ha10
	5711.7	4.			-0.3	3			Daa		96Pa01
	5707.6	4.1			0.7	3			Jya		19Pa27
$^{156}\text{Hf}(\alpha)^{152}\text{Yb}$	6033.0	10.	6026	3	-0.7	-			GSa		79Ho10
	6027.9	4.			-0.5	-			Daa		96Pa01
	ave. 6029	4			-0.7	1	65	65 ^{156}Hf			average
$^{156}\text{Hf}^m(\alpha)^{152}\text{Yb}$	8009.8	15.	7985	3	-1.7	U			GSa		81Ho.A
	7987.2	4.			-0.6	o			Daa		96Pa01 *
	7980.0	5.1			0.9	1	37	37 $^{156}\text{Hf}^m$			18Pa37
$^{154}\text{Sm}(\text{t,p})^{156}\text{Sm}$	4556	25	4566	9	0.4	1	12	11 ^{156}Sm	Ald		66Bj01
$^{154}\text{Eu}(\text{t,p})^{156}\text{Eu}$	6003	4	6005	3	0.6	1	71	70 ^{156}Eu	LAl		84La06 *
$^{154}\text{Gd}(\text{t,p})^{156}\text{Gd}$	6495.1	3.6	6489.81	0.19	-1.5	U			McM		89Lo07
$^{156}\text{Gd}(\text{p,t})^{154}\text{Gd}$	-6490	7	-6489.81	0.19	0.0	U			McM		73Lo08
	-6490	5			0.0	U			Min		73Oo01
$^{155}\text{Gd}(\text{n},\gamma)^{156}\text{Gd}$	8536.8	0.5	8536.35	0.07	-0.9	U			ILn		82Ba28
	8536.39	0.07			-0.5	-			MMn		82Is05 Z
	8536.04	0.19			1.6	-			Bdn		06Fi.A
$^{155}\text{Gd}(\text{d,p})^{156}\text{Gd}$	6319	10	6311.79	0.07	-0.7	U			Kop		67Tj01
$^{156}\text{Gd}(\text{d,t})^{155}\text{Gd}$	-2287	10	-2279.12	0.07	0.8	U			Kop		67Tj01
$^{155}\text{Gd}(\text{n},\gamma)^{156}\text{Gd}$	ave. 8536.35	0.07	8536.35	0.07	0.1	1	100	70 ^{155}Gd			average
$^{155}\text{Gd}(\alpha,\text{t})^{156}\text{Tb}-^{158}\text{Gd}()^{159}\text{Tb}$	-821.9	3.6	-822	4	-0.0	1	100	100 ^{156}Tb	McM		75Bu02
$^{156}\text{Dy}(\text{d,t})^{155}\text{Dy}$	-3184	10	-3188	10	-0.4	1	92	92 ^{155}Dy	Kop		70Gr46
$^{156}\text{Ta}(\text{p})^{155}\text{Hf}$	1028.6	13.	1020	4	-0.7	o			Dap		92Pa05
	1013.6	5.			1.2	o			Dap		96Pa01
	1017.9	5.			0.4	3			Dap		11Da12
$^{156}\text{Ta}^m(\text{p})^{155}\text{Hf}$	1110.2	12.	1114	7	0.3	3			Dap		93Li34
	1115.2	8.			-0.2	3			Dap		96Pa01
$^{156}\text{Nd}(\beta^-)^{156}\text{Pm}$	3690	200	3964.7	1.8	1.4	U			Kur		02Sh.B

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{156}\text{Pm}^m(\text{IT})^{156}\text{Pm}$	150.3	0.1	150.30	0.10	0.0	1	100	63	^{156}Pm		07Sh05
$^{156}\text{Pm}(\beta^-)^{156}\text{Sm}$	5155	35	5194	9	1.1	U			Stu		90He11
	5110	100			0.8	U			Kur		02Sh.B
$^{156}\text{Sm}(\beta^-)^{156}\text{Eu}$	721	10	722	8	0.1	—					63Gu04 *
	721	15			0.1	—					65Wi08 *
	ave. 721	8			0.1	1	90	89	^{156}Sm		average
$^{156}\text{Eu}(\beta^-)^{156}\text{Gd}$	2430	10	2452	3	2.2	—					62Ew01
	2460	10			−0.8	—					63Th02
	2450	15			0.2	—					64Pe17
	2478	20			−1.3	U					67Va23
	ave. 2446	6			1.0	1	28	28	^{156}Eu		average
$^{156}\text{Tb}(\beta^+)^{156}\text{Gd}$	3570	50	2444	4	−22.5	B					70Ag02 *
$^{156}\text{Ho}(\beta^+)^{156}\text{Dy}$	4400	400	4990	40	1.5	U					76Gr20 *
	5050	90			−0.7	o					02Iz01
	5050	60			−1.0	2					04Iz02 *
	4950	50			0.8	2					96By.A
$^{156}\text{Er}(\beta^+)^{156}\text{Ho}$	1670	70	1330	50	−4.9	B					82Vy06 *
$^{156}\text{Tm}(\beta^+)^{156}\text{Er}$	7458	50	7377	27	−1.6	1	29	22	^{156}Er	Dbn	94Po26 *
	7390	100			−0.1	U					95Ga.A
$^{156}\text{Hf}^m(\text{IT})^{156}\text{Hf}$	1959	1	1958.8	1.0	−0.2	1	98	63	$^{156}\text{Hf}^m$		96Pa01
* $^{156}\text{Pr}-u$	Represent frequency ratio $^{156}\text{Pr}^{++}/(\text{C}_6\text{H}_6)^+ = 0.9990445852(69)$										HWJ208 **
* $^{156}\text{Pm}-u$	Represents frequency ratio $^{156}\text{Pm}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.975190689(43)$										WgM124**
* $^{156}\text{Pm}-u$	Represents frequency ratio $^{156}\text{Pm}^{++}/(\text{C}_6\text{H}_6)^+ = 0.998957107(16)$										HWJ208 **
* $^{156}\text{Pm}^m-u$	Represents frequency ratio $^{156}\text{Pm}^{m++}/(\text{C}_6\text{H}_6)^+ = 0.998958154(10)$										HWJ208 **
* $^{156}\text{Tb}-u$	$M-A = -69968(32)$ keV for mixture gs+m+n at 88.4(0.2) and 100#50 keV										Nub211 **
* $^{156}\text{Ho}-u$	$M-A = -65230(100)$ keV for mixture gs+m+n at 52.37 and 230(50) keV										Nub211 **
* $^{156}\text{Ho}^n-u$	Assuming high spin isomer is favored										GAu **
* $^{156}\text{Tm}^m(\alpha)^{152}\text{Ho}$	F : originally $E_\alpha = 4460(10)$ to $^{152}\text{Ho}^m$ at 160(1), reassigned to ^{155}Tm										94To10 **
* $^{156}\text{Hf}^m(\alpha)^{152}\text{Yb}$	Replaced by authors' value for $^{156}\text{Hf}^m(\text{IT})$										AHW **
* $^{154}\text{Eu}(\text{t.p})^{156}\text{Eu}$	$Q = 5569(4)$ to 3^- level at 434.23 keV										91Ba06 **
* $^{156}\text{Sm}(\beta^-)^{156}\text{Eu}$	$E_{\beta^-} = 430(10)$ 430(15) respectively, to 1^+ level at 291.3037 keV										Ens12a **
* $^{156}\text{Tb}(\beta^+)^{156}\text{Gd}$	$E_{\beta^+} = 2640(50)$ from $^{156}\text{Tb}^m$ at 88.4 to ground state										Nub211 **
* $^{156}\text{Ho}(\beta^+)^{156}\text{Dy}$	$E_{\beta^+} = 1800(50)$ to levels around 1600										Ens12a **
* $^{156}\text{Ho}(\beta^+)^{156}\text{Dy}$	Original error 20 is for statistics only, increased by evaluator										GAu **
* $^{156}\text{Er}(\beta^+)^{156}\text{Ho}$	$p^+ = 0.0036(0.0017)$ to 2^- level at 82.23 keV, reanalyzed										Ens12a **
* $^{156}\text{Tm}(\beta^+)^{156}\text{Er}$	$E_{\beta^+} = 6091(50)$ to 2^+ level at 344.53 keV										Ens12a **
$^{157}\text{Pr}-u$	−51996.9	3.4				2			CP2	1.0	18Or.A *
$^{157}\text{Nd}-u$	−60614	47	−60648.9	2.3	−0.7	U			CP1	1.0	12Va02 *
	−60649.1	2.3			0.1	1	100	100	^{157}Nd	CP2	1.0 18Or.A *
$^{157}\text{Nd}-^{80}\text{Kr}_{1.963}$	103537	46	103501.2	2.7	−0.8	1	0	0	^{157}Nd	CP1	1.0 12Va02
$^{157}\text{Nd}-^{86}\text{Kr}_{1.826}$	102610	46	102576.1	2.3	−0.7	1	0	0	^{157}Nd	CP1	1.0 12Va02
$^{157}\text{Pm}-u$	−66880	13	−66879	8	0.1	1	33	33	^{157}Pm	CP1	1.0 12Va02 *
$^{157}\text{Pm}-^{80}\text{Kr}_{1.963}$	97273	13	97271	8	−0.1	1	34	33	^{157}Pm	CP1	1.0 12Va02
$^{157}\text{Pm}-^{86}\text{Kr}_{1.826}$	96346	13	96346	8	0.0	1	33	33	^{157}Pm	CP1	1.0 12Va02
$^{157}\text{Sm}-u$	−71582.2	8.3	−71581	5	0.1	1	33	33	^{157}Sm	CP1	1.0 12Va02 *
$^{157}\text{Sm}-^{80}\text{Kr}_{1.963}$	92570.0	8.0	92569	5	−0.2	1	36	34	^{157}Sm	CP1	1.0 12Va02
$^{157}\text{Sm}-^{86}\text{Kr}_{1.826}$	91643.0	8.3	91644	5	0.1	1	33	33	^{157}Sm	CP1	1.0 12Va02
$\text{C}_{10}\text{H}_9\text{N}_2-^{157}\text{Gd}$	152720	60	152605.9	1.0	−0.5	U			R04	4.0	64De15
$\text{C}_9^{13}\text{CH}_8\text{N}_2-^{157}\text{Gd}$	148170	70	148135.7	1.0	−0.1	U			R04	4.0	64De15
$\text{C}_{10}\text{H}_5\text{O}_2-^{157}\text{Gd}$	105080	60	104987.0	1.0	−0.4	U			R04	4.0	64De15
$^{157}\text{Ho}-u$	−71724	30	−71748	25	−0.8	1	71	71	^{157}Ho	GS2	1.0 05Li24
$^{157}\text{Er}-u$	−68084	30	−68077	28	0.2	1	90	90	^{157}Er	GS2	1.0 05Li24
$^{157}\text{Tm}-u$	−63027	30				2			GS2	1.0	05Li24
$^{157}\text{Yb}-u$	−57389	30	−57349	12	1.3	U			GS2	1.0	05Li24

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{157}\text{Lu}-u$	-49842	31	-49855	13	-0.4	1	17	17 ^{157}Lu	GS2	1.0	05Li24 *
$^{157}\text{Gd } ^{35}\text{Cl}-^{155}\text{Gd } ^{37}\text{Cl}$	4318	4	4288.19	0.19	-1.9	U			H12	4.0	64Ba15
	4287	3			0.2	U			H21	2.5	70Ma05
	4289.0	0.7			-0.5	U			M21	2.5	75Ka25
	4288.83	0.66			-0.4	U			H41	2.5	85Dy04
	1860	60	1837.30	0.16	-0.1	U			R04	4.0	64De15
$^{157}\text{Gd}-^{156}\text{Gd}$	-81114.2	5.4	-81118.0	1.0	-0.5	o			TG1	1.5	09Ke.A
$^{157}\text{Gd O}-\text{C}_{15}$	-81113.6	3.3			-0.9	U			TG1	1.5	11Ke03
$^{157}\text{Yb}(\alpha)^{153}\text{Er}$	4622.0	7.	4622	6	-0.0	-					77Ha48
	4623.0	10.			-0.1	-			GSa		79Ho10
	ave. 4622	6			-0.1	1	99	97 ^{157}Yb			average
$^{157}\text{Lu}(\alpha)^{153}\text{Tm}$	5097.2	5.	5107.9	2.9	2.1	o			DbA		91Le15 *
	5096.2	20.			0.6	U			Bka		91To09 *
	5111.5	5.			-0.7	o			DbA		92Po14 *
$^{157}\text{Lu}^m(\alpha)^{153}\text{Tm}$	5128.9	10.	5128.8	2.0	-0.0	U			IRa		79Al16 Z
	5131.8	5.			-0.6	-			GSa		79Ho10 Z
	5133.7	5.			-0.9	-			ORa		83To01 Z
	5128.9	5.			-0.0	o			DbA		91Le15
	5118.7	5.			2.0	-			Bka		91To09
	5125.8	6.			0.5	-					92Ha10
	5132.0	5.			-0.6	-			DbA		92Po14
	5127.9	4.			0.2	-			Daa		96Pa01
	ave. 5128.4	2.1			0.2	1	99	54 ^{153}Tm			average
	5869.4	10.	5880	3	1.0	3					73Ea01 Z
$^{157}\text{Hf}(\alpha)^{153}\text{Yb}$	5884.1	5.			-0.8	3			GSa		79Ho10 Z
	5879.1	4.			0.2	3			Daa		96Pa01
$^{157}\text{Ta}(\alpha)^{153}\text{Lu}^m$	6277.2	4.	6275	8	-0.6	o			Ara		97Ir01 *
$^{157}\text{Ta}^m(\alpha)^{153}\text{Lu}$	6381.9	10.	6377	4	-0.5	3			GSa		79Ho10
	6375.8	4.			0.2	3			Daa		96Pa01 *
$^{157}\text{Ta}^n(\alpha)^{153}\text{Lu}$	7946.9	8.	7948	8	0.0	o			Daa		96Pa01 *
$^{155}\text{Gd}(\text{t,p})^{157}\text{Gd}$	6417.8	2.9	6414.44	0.16	-1.2	U			McM		89Lo07
$^{157}\text{Gd}(\text{p,t})^{155}\text{Gd}$	-6414	7	-6414.44	0.16	-0.1	U			McM		73Lo08
	-6417	5			0.5	U			Min		73Oo01
$^{156}\text{Gd}(\text{n},\gamma)^{157}\text{Gd}$	6359.6	0.8	6359.88	0.15	0.4	U					70Bo29
	6360	1			-0.1	U					71Gr42
	6359.80	0.15			0.5	o			ILn		87Sp.A Z
	6359.86	0.15			0.1	1	99	74 ^{156}Gd	ILn		03Bo25
$^{157}\text{Gd}(\gamma,\text{n})^{156}\text{Gd}$	-6350	80	-6359.88	0.15	-0.1	U			Phi		60Ge01
$^{156}\text{Gd}(\text{d,p})^{157}\text{Gd}$	4136	10	4135.31	0.15	-0.1	U			Kop		67Tj01
$^{157}\text{Gd}(\text{d,t})^{156}\text{Gd}$	-112	10	-102.65	0.15	0.9	U			Kop		67Tj01
$^{156}\text{Gd}(\alpha,\text{t})^{157}\text{Tb}-^{158}\text{Gd}()^{159}\text{Tb}$	-616.2	2.0	-614.1	0.8	1.1	1	16	10 ^{159}Tb	McM		75Bu02
$^{156}\text{Dy}(\text{d,p})^{157}\text{Dy}$	4748	10	4742	5	-0.6	-			Tal		68Be.A
	4753	10			-1.1	-			Kop		70Gr46
	ave. 4751	7			-1.2	1	52	52 ^{157}Dy			average
$^{157}\text{Ta}(\text{p})^{156}\text{Hf}$	925.0	17.	935	10	0.6	o			Dap		96Pa01
	933.0	7.			0.2	o			Ara		97Ir01 *
$^{157}\text{Pm}(\beta^-)^{157}\text{Sm}$	4360	100	4381	8	0.2	U			Kur		02Sh.B
$^{157}\text{Sm}(\beta^-)^{157}\text{Eu}$	2700	200	2781	6	0.4	U					73Ka23 *
	2734	50			0.9	U			Ida		93Gr17
$^{157}\text{Eu}(\beta^-)^{157}\text{Gd}$	1350	20	1365	4	0.7	U					64Sh21 *
	1370	20			-0.3	U					66Fu05 *
$^{157}\text{Tb}(\epsilon)^{157}\text{Gd}$	62.4	0.6	60.05	0.30	-3.9	B					67Na08 *
	62.2	0.6			-3.6	B					83Be42 *
	60.0	0.3			0.2	1	98	94 ^{157}Tb			92Ra18
$^{157}\text{Ho}(\beta^+)^{157}\text{Dy}$	2540	50	2592	24	1.0	1	23	22 ^{157}Ho			72To05 *
$^{157}\text{Er}(\beta^+)^{157}\text{Ho}$	3470	80	3420	30	-0.6	1	18	10 ^{157}Er			75Al.A
	3805	100			-3.9	B			Dbn		94Po26 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{157}\text{Tm}(\beta^+)^{157}\text{Er}$	4480	100	4700	40	2.2	B			IRS		93Al03
	4482	100			2.2	B			Dbn		94Po26
$^{157}\text{Yb}(\beta^+)^{157}\text{Tm}$	5074	100	5289	30	2.2	U			Dbn		94Po26
$^{157}\text{Lu}^m(\text{IT})^{157}\text{Lu}$	32	2	20.9	2.0	-5.5	B			Dbn		91Le15
	21	2			-0.0	1	100	83 ^{157}Lu	Dbn		92Po14 *
$^{157}\text{Ta}^m(\text{IT})^{157}\text{Ta}$	22	5				3					97Ir01
$^{157}\text{Ta}^n(\text{IT})^{157}\text{Ta}^m$	1571	7				3			Daa		96Pa01
* $^{157}\text{Pr}-u$	Represents frequency ratio $^{157}\text{Pr}^{++}/(\text{C}_6\text{H}_6)^+ = 1.005471763(22)$										HWJ208 **
* $^{157}\text{Nd}-u$	Represents frequency ratio $^{157}\text{Nd}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.96892546(29)$										WgM124**
* $^{157}\text{Nd}-u$	Represents frequency ratio $^{157}\text{Nd}^{++}/(\text{C}_6\text{H}_6)^+ = 1.005416333(17)$										HWJ208 **
* $^{157}\text{Pm}-u$	Represents frequency ratio $^{157}\text{Pm}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.968964141(81)$										WgM124**
* $^{157}\text{Sm}-u$	Represents frequency ratio $^{157}\text{Sm}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.968993178(51)$										WgM124**
* $^{157}\text{Lu}-u$	$M-A = -46417(28)$ keV for mixture gs+m at 20.9(2.0) keV										Nub211 **
* $^{157}\text{Lu}(\alpha)^{153}\text{Tm}$	$E_\alpha = 4925(5)$ to $^{153}\text{Tm}^m$ at 43.2(0.2) keV										Nub211 **
* $^{157}\text{Lu}(\alpha)^{153}\text{Tm}$	$E_\alpha = 4924(20)$ to $^{153}\text{Tm}^m$ at 43.2(0.2) keV										Nub211 **
* $^{157}\text{Lu}(\alpha)^{153}\text{Tm}$	$E_\alpha = 4939(5)$ to $^{153}\text{Tm}^m$ at 43.2(0.2) keV; replaced by $^{157}\text{Lu}^m(\text{IT})$										Nub211 **
* $^{157}\text{Ta}(\alpha)^{153}\text{Lu}^m$	Replaced by $^{153}\text{Lu}^m(\text{IT})$										AHW **
* $^{157}\text{Ta}^m(\alpha)^{153}\text{Lu}$	Reassigned										97Ir01 **
* $^{157}\text{Ta}^n(\alpha)^{153}\text{Lu}$	Replaced by authors' value for $^{157}\text{Ta}^n(\text{IT})$										AHW **
* $^{157}\text{Ta}(\text{p})^{156}\text{Hf}$	Use instead $^{157}\text{Ta}^m(\text{IT})$										AHW **
* $^{157}\text{Sm}(\beta^-)^{157}\text{Eu}$	$E_{\beta^-} = 2400(200)$ to $5/2^-$ level at 197.863 and $3/2^+$ at 394.334 keV										Ens162 **
* $^{157}\text{Eu}(\beta^-)^{157}\text{Gd}$	$E_{\beta^-} = 870(30)$ 910(20) respectively, to $3/2^+$ level at 474.630 keV, and other E_{β^-}										Ens162 **
* $^{157}\text{Tb}(\epsilon)^{157}\text{Gd}$	LK=2.65(0.20); original value 66(6) recalculated										92Ha03 **
* $^{157}\text{Tb}(\epsilon)^{157}\text{Gd}$	LK=2.69(0.20); original value 62.9(0.7) recalculated										85Vo09 **
* $^{157}\text{Ho}(\beta^+)^{157}\text{Dy}$	$E_{\beta^+} = 1180(50)$ to $5/2^-$ level at 341.118 keV										Ens162 **
* $^{157}\text{Er}(\beta^+)^{157}\text{Ho}$	$E_{\beta^+} = 2525(100)$ to ground state yielding 3547(100), rather 24% to $(3/2^+)$										94Po26 **
*	level at 174.55 keV, 15% to $5/2^-$ at 391.32 keV $\rightarrow$ +258 keV										Ens162 **
* $^{157}\text{Lu}^m(\text{IT})^{157}\text{Lu}$	Derived from $^{157}\text{Lu}^m(\alpha)-^{157}\text{Lu}(\alpha)$ difference										92Po14 **
$^{158}\text{Nd}-^{136}\text{Xe}_{1.162}$	49956	39	50022.4	1.4	1.7	U			JY1	1.0	18Vi02
$^{158}\text{Nd}-^{84}\text{Kr}_{1.881}$	108678.4	1.4				2			CP2	1.0	18Or02
$^{158}\text{Pm}-u$	-63436	25	-63453.1	1.0	-0.7	U			CP1	1.0	12Va02 *
	-63453.7	1.1			0.6	1	75	75 ^{158}Pm	CP2	1.0	20Or03 *
$^{158}\text{Pm}-^{80}\text{Kr}_{1.975}$	101720	25	101700.5	1.8	-0.8	U			CP1	1.0	12Va02
$^{158}\text{Pm}-^{86}\text{Kr}_{1.837}$	100773	25	100755.2	1.0	-0.7	U			CP1	1.0	12Va02
$^{158}\text{Sm}-u$	-70049.2	9.5	-70051	5	-0.2	1	29	29 ^{158}Sm	CP1	1.0	12Va02 *
$^{158}\text{Sm}-^{80}\text{Kr}_{1.975}$	95106.5	9.1	95103	5	-0.4	1	33	31 ^{158}Sm	CP1	1.0	12Va02
$^{158}\text{Sm}-^{86}\text{Kr}_{1.837}$	94159.3	9.4	94158	5	-0.2	1	30	30 ^{158}Sm	CP1	1.0	12Va02
$^{158}\text{Eu}-u$	-72208	25	-72217.8	2.2	-0.4	U			CP1	1.0	12Va02 *
	-72218.2	2.2			0.2	1	98	98 ^{158}Eu	CP2	1.0	20Or03 *
$^{158}\text{Eu}-^{80}\text{Kr}_{1.975}$	92949	25	92935.8	2.6	-0.5	U			CP1	1.0	12Va02
$^{158}\text{Eu}-^{86}\text{Kr}_{1.837}$	92001	25	91990.5	2.2	-0.4	U			CP1	1.0	12Va02
$\text{C}_{10}\text{H}_6\text{O}_2-^{158}\text{Gd}$	112444	33	112668.2	1.0	1.7	U			R04	4.0	64De15
$\text{C}_{10}\text{H}_6\text{O}_2-^{158}\text{Dy}$	112870	100	112364.6	2.5	-1.3	U			R04	4.0	64De15
$^{158}\text{Ho}-u$	-71101	67	-71055	29	0.7	R			GS2	1.0	05Li24 *
$^{158}\text{Er}-u$	-70220	110	-70107	27	1.0	U			GS1	1.0	00Ra23
	-70107	30			0.0	1	81	81 ^{158}Er	GS2	1.0	05Li24
$^{158}\text{Tm}-u$	-63080	110	-63020	27	0.5	U			GS1	1.0	00Ra23
	-63020	30			-0.0	1	81	81 ^{158}Tm	GS2	1.0	05Li24
$^{158}\text{Yb}-^{142}\text{Sm}_{1.113}$	34252	22	34243	9	-0.4	1	15	15 ^{158}Yb	MA7	1.0	01Bo59
$^{158}\text{Lu}-u$	-50720	30	-50684	16	1.2	R			GS2	1.0	05Li24
$^{158}\text{Gd}-^{35}\text{Cl}-^{156}\text{Gd}-^{37}\text{Cl}$	4956	4	4931.20	0.19	-1.5	U			H12	4.0	64Ba15
	4929	3			0.3	U			H21	2.5	70Ma05
	4926.2	1.4			1.4	U			H25	2.5	72Ba08
	4930.8	0.7			0.2	U			M21	2.5	75Ka25
	4930.13	1.36			0.3	U			H41	2.5	85Dy04

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{158}\text{Dy } ^{35}\text{Cl}-^{156}\text{Dy } ^{37}\text{Cl}$	3081.4	3.3	3081.3	2.6	-0.0	U			H25	2.5	72Ba08
$^{158}\text{Pm}-^{158}\text{Gd}$	12436.8	1.4	12435.7	1.1	-0.8	1	60	35	^{158}Gd	JY1	1.0 18Vi02
$^{158}\text{Gd}-^{157}\text{Gd}$	392	48	143.78	0.07	-1.3	U			R04	4.0	64De15
$^{158}\text{Gd O}-\text{C}_{15}$	-80968.3	5.4	-80974.2	1.0	-0.7	o			TG1	1.5	09Ke.A
	-80967.8	3.2			-1.3	U			TG1	1.5	11Ke03
$^{158}\text{Gd O}-\text{C}_{14}$	-80964.7	8.2			-0.8	U			TG1	1.5	11Ke03
$^{158}\text{Yb}(\alpha)^{154}\text{Er}$	4174.9	10.	4170	7	-0.4	-					77Ha48
	4164.6	12.			0.5	-					92Ha10
	ave.	8			-0.0	1	80	71	^{158}Yb		average
$^{158}\text{Lu}(\alpha)^{154}\text{Tm}$	4792.2	10.	4790	5	-0.2	3			IRa		79A116 Z
	4789.5	5.			0.1	3			ORa		83To01 Z
$^{158}\text{Hf}(\alpha)^{154}\text{Yb}$	5406.0	5.	5404.8	2.7	-0.2	-			GSa		79Ho10 Z
	5401.4	5.			0.7	-			ORa		83To01 Z
	5406.1	4.			-0.3	-			Daa		96Pa01
	ave.	2.7			0.0	1	100	100	^{158}Hf		average
$^{158}\text{Ta}(\alpha)^{154}\text{Lu}$	6124.4	8.	6124	4	-0.1	9			Daa		96Pa01
	6123.3	5.			0.1	9			Ara		97Da07
$^{158}\text{Ta}^m(\alpha)^{154}\text{Lu}^m$	6208.5	6.	6202.8	2.3	-0.9	10			GSa		79Ho10
	6203.4	4.			-0.1	10			Daa		96Pa01
	6205.4	5.			-0.5	10			Ara		97Da07
	6198.2	4.1			1.1	10			Jya		19Pa27
$^{158}\text{Ta}^n(\alpha)^{154}\text{Lu}^m$	8869.0	11.3				11			Jya		14Ca03
$^{158}\text{W}(\alpha)^{154}\text{Hf}$	6600.4	30.	6612.5	2.6	0.4	U			GSa		81Ho10 *
	6609.7	30.			0.1	U			Daa		96Pa01
	6612.7	3.			-0.1	3			Ara		00Ma95
	6611.7	5.1			0.1	3			Jya		19Hi06
$^{158}\text{W}^m(\alpha)^{154}\text{Hf}$	8495.5	30.	8502	7	0.2	U			GSa		89Ho12
	8506.8	24.			-0.2	U			Daa		96Pa01
	8501.6	7.				3			Ara		00Ma95
$^{158}\text{Gd}(\text{p},\text{t})^{156}\text{Gd}$	-5818	5	-5815.48	0.16	0.5	U			Min		73Oo01
$^{158}\text{Dy}(\text{p},\text{t})^{156}\text{Dy}$	-7535	15	-7538.6	2.4	-0.2	U			Pri		77Ko04
$^{158}\text{Gd}(\text{t},\alpha)^{157}\text{Eu}-^{156}\text{Gd}()^{155}\text{Eu}$	-512	5	-514	4	-0.4	1	69	67	^{157}Eu	LAl	79Bu05
$^{157}\text{Gd}(\text{n},\gamma)^{158}\text{Gd}$	7937.39	0.07	7937.39	0.06	0.0	-			MMn		82Is05 Z
	7937.39	0.17			0.0	-			Bdn		06Fi.A
$^{157}\text{Gd}(\text{d},\text{p})^{158}\text{Gd}$	5724	10	5712.83	0.06	-1.1	U			Kop		67Tj01
	5706	5			1.4	U			Tal		71Sh04
$^{158}\text{Gd}(\text{d},\text{t})^{157}\text{Gd}$	-1688	10	-1680.16	0.06	0.8	U			Kop		67Tj01
$^{157}\text{Gd}(\text{n},\gamma)^{158}\text{Gd}$	ave.	0.06	7937.39	0.06	0.0	1	100	63	^{157}Gd		average
$^{158}\text{Gd}(\text{d},\text{t})^{157}\text{Gd}-^{159}\text{Tb}()^{158}\text{Tb}$	195.0	1.5	195.6	0.6	0.4	1	18	18	^{158}Tb	McM	84Bu14
$^{157}\text{Gd}(\alpha,\text{t})^{158}\text{Tb}-^{158}\text{Gd}()^{159}\text{Tb}$	-198.3	1.0	-195.6	0.6	2.7	o			McM		75Bu02
	-196.6	1.0			1.0	1	41	40	^{158}Tb	McM	84Bu14
$^{158}\text{Tb}(\text{p},\text{d})^{157}\text{Tb}$	-4560.3	4.2	-4553.8	1.0	1.6	U			Pri		85Al02 *
$^{158}\text{Dy}(\text{d},\text{t})^{157}\text{Dy}$	-2804	10	-2797	5	0.7	-			Tal		68Be.A
	-2804	10			0.7	-			Kop		70Gr46
	ave.	7			1.1	1	53	47	^{157}Dy		average
$^{158}\text{Pm}(\beta^-)^{158}\text{Sm}$	6120	100	6146	5	0.3	o			Kur		02Sh.A
	6085	80			0.8	o			Kur		07Ha57
	6060	80			1.1	U			Kur		10Ha38
$^{158}\text{Sm}(\beta^-)^{158}\text{Eu}$	1999	15	2019	5	1.3	1	12	10	^{158}Sm	Ida	93Gr17
$^{158}\text{Eu}(\beta^-)^{158}\text{Gd}$	3550	120	3419.5	2.3	-1.1	U					65Sc19 *
	3440	100			-0.2	U					66Da06 *
$^{158}\text{Tb}(\epsilon)^{158}\text{Gd}$	1237.542	0.018	1219.1	1.0	*****	F					83Ra25 *
	1220	13			-0.1	U					87Br33
	1222.1	3.			-1.0	U					85Vo13 *
$^{158}\text{Tb}(\beta^-)^{158}\text{Dy}$	952	10	936.3	2.5	-1.6	U					68Sc04 *
	933	6			0.5	1	17	14	^{158}Dy		85Vo03 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{158}\text{Ho}(\beta^+)^{158}\text{Dy}$	4350	100	4220	27	-1.3	U					61Bo24 *
	4230	30			-0.3	2					68Ab14 *
$^{158}\text{Er}(\beta^+)^{158}\text{Ho}$	1710	40	880	40	-20.7	F					82Vy06 *
$^{158}\text{Tm}(\beta^+)^{158}\text{Er}$	6530	100	6600	30	0.7	-			IRS		93Ai03
	6624	60			-0.4	-			Dbn		94Po26 *
ave.	6600	50			0.0	1	37	19 ^{158}Er			average
$^{158}\text{Lu}(\epsilon)^{158}\text{Yb}$	8960	200	8797	17	-0.8	U					95Ga.A
* $^{158}\text{Pm}-u$	Represents frequency ratio $^{158}\text{Pm}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.96280782(15)$										WgM124**
* $^{158}\text{Pm}-u$	Represents frequency ratio $^{158}\text{Pm}^{++}/(\text{C}_6\text{H}_6)^+ = 1.0118048112(69)$										HWJ208 **
* $^{158}\text{Sm}-u$	Represents frequency ratio $^{158}\text{Sm}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.962848141(58)$										WgM124**
* $^{158}\text{Eu}-u$	Represents frequency ratio $^{158}\text{Eu}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.96286130(15)$										WgM124**
* $^{158}\text{Eu}-u$	Represents frequency ratio $^{158}\text{Eu}^{++}/(\text{C}_6\text{H}_6)^+ = 1.011748662(14)$										HWJ208 **
* $^{158}\text{Ho}-u$	$M-A = -66148(29)$ keV for mixture gs+m+n at 67.2 and 180#70 keV										Nub211 **
* $^{158}\text{W}(\alpha)^{154}\text{Hf}$	Original value $E = 6450(30)$ ($Q = 6617.8$) recalibrated to $E = 6433(30)$ keV										89Ho12 **
* $^{158}\text{Tb}(\text{p,d})^{157}\text{Tb}$	$Q - Q(^{158}\text{Gd}(\text{p,d})) = 1152.5(4.2)$ keV										AHW **
* $^{158}\text{Eu}(\beta^-)^{158}\text{Gd}$	$E_{\beta^-} = 2520(120)$ 2430(100) respectively, to 2^- level at 1023.6974 keV										Ens043 **
*	and 3^- level at 1041.6376 keV, and other E_{β^-}										Ens043 **
* $^{158}\text{Tb}(\epsilon)^{158}\text{Gd}$	$\text{pK} = 0.00009(2)$ to 2^+ level at 1187.143, recalculated Q										Ens043 **
*	$F : \text{pK} < 0.00002$										87Br33 **
* $^{158}\text{Tb}(\epsilon)^{158}\text{Gd}$	$\text{pL} = 0.689(0.01)$ to 2^+ level at 1187.143 keV, recalculated Q										Ens043 **
* $^{158}\text{Tb}(\beta^-)^{158}\text{Dy}$	$E_{\beta^-} = 853(10)$ 834(6) respectively, to 2^+ level at 98.9180 keV										Ens043 **
* $^{158}\text{Ho}(\beta^+)^{158}\text{Dy}$	$E_{\beta^+} = 780(80)$ to 2436–2605 levels; originally assigned to $^{158}\text{Er}(\beta^+)$;										Ens043 **
*	reinterpreted by evaluator										AHW **
* $^{158}\text{Ho}(\beta^+)^{158}\text{Dy}$	$E_{\beta^+} = 2890(20)$, 700(60) to 317.139–637.712 and 2436.52–2605.96 levels, and										Ens043 **
*	$E_{\beta^+} = 1300(30)$, 1850(25) keV from $^{158}\text{Ho}^m$ at 67.2 to 1920.43–1940.75										Nub211 **
*	and 1441.75 levels; $E_{\beta^+} = 700(60)$ was originally assigned to $^{158}\text{Er}(\beta^+)$;										68Ab14 **
*	reinterpreted by evaluator										AHW **
* $^{158}\text{Er}(\beta^+)^{158}\text{Ho}$	$\text{p}^+ = 0.3(0.1)$ from annih. γ coinc. to 146.90 level										96Go06 **
*	$F : Q < 1550$ from upper limit on p^+										75Bu.A **
* $^{158}\text{Tm}(\beta^+)^{158}\text{Er}$	$E_{\beta^+} = 5410(60)$ to 2^+ level at 192.15 keV										Ens07a **
$^{159}\text{Nd}-^{133}\text{Cs}_{1.195}$	59604	32				2			CP2	1.0	18Or02
$^{159}\text{Pm}-u$	-60715	18	-60714	11	0.1	1	36	36 ^{159}Pm	CP1	1.0	12Va02 *
$^{159}\text{Pm}-^{80}\text{Kr}_{1.988}$	105529	19	105527	11	-0.1	1	32	32 ^{159}Pm	CP1	1.0	12Va02
$^{159}\text{Pm}-^{86}\text{Kr}_{1.849}$	104567	19	104567	11	0.0	1	32	32 ^{159}Pm	CP1	1.0	12Va02
$^{159}\text{Sm}-u$	-66784	11	-66783	6	0.1	1	34	34 ^{159}Sm	CP1	1.0	12Va02 *
$^{159}\text{Sm}-^{80}\text{Kr}_{1.988}$	99459	11	99458	6	-0.1	1	34	33 ^{159}Sm	CP1	1.0	12Va02
$^{159}\text{Sm}-^{86}\text{Kr}_{1.849}$	98498	11	98498	6	0.0	1	34	34 ^{159}Sm	CP1	1.0	12Va02
$^{159}\text{Eu}-u$	-70899	10	-70900	5	-0.1	1	22	22 ^{159}Eu	CP1	1.0	12Va02 *
$^{159}\text{Eu}-^{80}\text{Kr}_{1.988}$	95344	10	95340	5	-0.4	1	23	21 ^{159}Eu	CP1	1.0	12Va02
$^{159}\text{Eu}-^{86}\text{Kr}_{1.849}$	94382	10	94380	5	-0.2	1	22	22 ^{159}Eu	CP1	1.0	12Va02
$\text{C}_9 \text{ }^{13}\text{C} \text{ H}_6 \text{ O}_2 - ^{159}\text{Tb}$	114840	50	114780.6	1.2	-0.3	U			R04	4.0	64De15
$\text{C}_{10} \text{ H}_7 \text{ O}_2 - ^{159}\text{Tb}$	119238	25	119250.8	1.2	0.1	U			R04	4.0	64De15
$^{159}\text{Dy}-u$	-74285	30	-74254.1	1.5	1.0	U			GS2	1.0	05Li24
$^{159}\text{Ho}-u$	-72365	71	-72281	3	1.2	U			GS2	1.0	05Li24 *
$^{159}\text{Er}-u$	-69290	30	-69309	4	-0.6	U			GS2	1.0	05Li24
$^{159}\text{Tm}-u$	-65025	30				2			GS2	1.0	05Li24
$^{159}\text{Yb}-^{142}\text{Sm}_{1.120}$	35035	24	35026	19	-0.4	2			MA7	1.0	01Bo59
$^{159}\text{Yb}-u$	-59960	30	-59940	19	0.7	R			GS2	1.0	05Li24
$^{159}\text{Lu}-u$	-53420	61	-53360	40	0.9	2			GS2	1.0	05Li24 *
$^{159}\text{Hf}-u$	-46044	32	-46004	18	1.2	R			GS2	1.0	05Li24
$^{159}\text{Tb} \text{ }^{35}\text{Cl}_2 - ^{155}\text{Gd} \text{ }^{37}\text{Cl}_2$	8625.64	1.03	8624.6	0.8	-0.4	1	10	7 ^{159}Tb	H41	2.5	85Dy04
$^{159}\text{Tb} \text{ }^{35}\text{Cl} - ^{157}\text{Gd} \text{ }^{37}\text{Cl}$	4333.3	1.2	4336.4	0.8	1.0	U			H25	2.5	72Ba08
	4337.01	0.61			-0.4	1	28	20 ^{159}Tb	H41	2.5	85Dy04
$^{159}\text{Lu}(\alpha)^{155}\text{Tm}$	4534.3	10.	4490	40	-0.8	R			IRa		80Ai14
	4531.3	10.			-0.8	R					92Ha10

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{159}\text{Hf}(\alpha)^{155}\text{Yb}$	5221.2	10.	5225.1	2.7	0.4	U					73Ea01 Z
	5226.2	5.			-0.2	4			GSa		79Ho10 Z
	5223.0	5.			0.4	4			ORa		83To01 Z
	5219.6	6.			0.9	4					92Ha10
	5229.8	5.			-0.9	4			Daa		96Pa01
$^{159}\text{Ta}(\alpha)^{155}\text{Lu}^m$	5658.6	5.	5660	7	0.2	o			Daa		96Pa01 *
	5661.7	5.			-0.4	o			Ara		97Da07 *
$^{159}\text{Ta}^m(\alpha)^{155}\text{Lu}$	5745.8	6.	5745	3	-0.2	-			GSa		79Ho10
	5743.8	5.			0.2	-			Daa		96Pa01
	5744.8	5.			-0.0	-			Ara		97Da07
	ave. 5745	3			0.0	1	100	100	^{155}Lu		average
$^{159}\text{W}(\alpha)^{155}\text{Hf}$	6444.5	6.	6451	4	1.0	3			GSa		81Ho10 *
	6440.3	5.1			2.1	o			Daa		92Pa05
	6454.7	5.			-0.7	3			Daa		96Pa01
	6452.6	8.2			-0.2	3			Jya		19Hi06
$^{159}\text{Re}^m(\alpha)^{155}\text{Ta}$	6951.2	26.7	6968	23	0.6	R			Daa		07Pa27
$^{157}\text{Gd}(\text{t,p})^{159}\text{Gd}$	5398.9	2.3	5398.80	0.11	-0.0	U			McM		89Lo07
$^{158}\text{Gd}(\text{n},\gamma)^{159}\text{Gd}$	5942	1	5943.21	0.08	1.2	U					71Gr42
	5943.07	0.15			0.9	-			ILn		87Sp.A Z
	5943.1	0.2			0.5	-			Dbn		03Gr13
	5943.32	0.12			-1.0	-			BNn		03Gr27
	3717	10	3718.64	0.08	0.2	U			Kop		67Tj01
$^{158}\text{Gd}(\text{d,p})^{159}\text{Gd}$	ave. 5943.20	0.08	5943.21	0.08	0.1	1	100	93	^{159}Gd		average
$^{158}\text{Gd}(\text{n},\gamma)^{159}\text{Gd}$	-13686.6	10.	-13682.3	0.7	0.4	U			McM		75Bu02
$^{158}\text{Gd}(\alpha,\text{t})^{159}\text{Tb}$	-85.7	2.2	-87.8	1.0	-0.9	1	21	10	^{159}Tb	McM	84Bu14
$^{158}\text{Gd}(\alpha,\text{t})^{159}\text{Tb}-^{164}\text{Dy}()^{165}\text{Ho}$	-8141	39	-8133.0	0.6	0.2	U			Phi		60Ge01
$^{159}\text{Tb}(\gamma,\text{n})^{158}\text{Tb}$	-1870	15	-1875.8	0.6	-0.4	U			Tal		70Jo22
$^{159}\text{Tb}(\text{d,t})^{158}\text{Tb}$	-474.3	1.0	-474.9	0.6	-0.6	1	41	40	^{158}Tb	McM	84Bu14
$^{159}\text{Tb}(\text{d,t})^{158}\text{Tb}-^{164}\text{Dy}()^{163}\text{Dy}$	4608	10	4606.8	2.6	-0.1	U			Tal		68Be.A
$^{158}\text{Dy}(\text{d,p})^{159}\text{Dy}$	4600	10			0.7	U			Kop		70Gr46
$^{159}\text{Re}^m(\text{p})^{158}\text{W}$	1816.4	20.	1809	17	-0.4	4			Dap		06Jo10
$^{159}\text{Pm}(\beta^-)^{159}\text{Sm}$	5460	140	5653	12	1.4	o			Kur		07Ha57
	5430	140			1.6	U			Kur		10Ha38
$^{159}\text{Sm}(\beta^-)^{159}\text{Eu}$	3840	100	3836	7	-0.0	o			Kur		02Sh.A
	3805	65			0.5	o			Kur		07Ha57
	3800	65			0.5	U			Kur		10Ha38
$^{159}\text{Eu}(\beta^-)^{159}\text{Gd}$	2600	50	2518	4	-1.6	U					65Iw01 *
$^{159}\text{Gd}(\beta^-)^{159}\text{Tb}$	969.0	1.5	970.7	0.7	1.1	1	25	18	^{159}Tb		77Bo.A
$^{159}\text{Dy}(\epsilon)^{159}\text{Tb}$	365.9	1.3	365.4	1.2	-0.4	1	79	65	^{159}Dy		68My.A *
$^{159}\text{Ho}(\beta^+)^{159}\text{Dy}$	1837.6	6.	1837.6	2.7	-0.0	2					79Ad08 *
	1837.6	3.			-0.0	2					82Vy02 *
$^{159}\text{Er}(\beta^+)^{159}\text{Ho}$	2768.5	2.0				3					84Ka.A *
	2810	100	2768.5	2.0	-0.4	U			IRS		93Al03
$^{159}\text{Tm}(\beta^+)^{159}\text{Er}$	3400	300	3991	28	2.0	U					75St07
	3850	100			1.4	U			IRS		93Al03
	3670	100			3.2	B			Dbn		94Po26
$^{159}\text{Yb}(\beta^+)^{159}\text{Tm}$	5050	200	4740	30	-1.6	U			IRS		93Al03
	4554	150			1.2	U			Dbn		94Po26 *
$^{159}\text{Lu}(\beta^+)^{159}\text{Yb}$	5850	150	6120	40	1.8	U			IRS		93Al03
	5803	150			2.1	U			Dbn		94Po26
$^{159}\text{Ta}^m(\text{IT})^{159}\text{Ta}$	63.7	5.2				2			Ara		97Da07
* $^{159}\text{Pm}-\text{u}$	Represents frequency ratio $^{159}\text{Pm}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.95673359(11)$										WgM124**
* $^{159}\text{Sm}-\text{u}$	Represents frequency ratio $^{159}\text{Sm}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.956770122(65)$										WgM124**
* $^{159}\text{Eu}-\text{u}$	Represents frequency ratio $^{159}\text{Eu}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.956794898(63)$										WgM124**
* $^{159}\text{Ho}-\text{u}$	$M-A=67304(28)$ keV for mixture gs+m at 205.91 keV										Nub211 **
* $^{159}\text{Lu}-\text{u}$	$M-A=49710(28)$ keV for mixture gs+m at 100#80 keV										Nub211 **
* $^{159}\text{Ta}(\alpha)^{155}\text{Lu}^m$	Replaced by $^{155}\text{Lu}^m(\text{IT})$										AHW **
* $^{159}\text{W}(\alpha)^{155}\text{Hf}$	Original value $E_\alpha=6299(6)$ recalibrated to $E_\alpha=6282(6)$ keV										89Ho12 **
* $^{159}\text{Eu}(\beta^-)^{159}\text{Gd}$	$E_{\beta^-}=2350(50)$ to $7/2^-$ level at 227.412 level, and other E_{β^-}										Ens121 **
* $^{159}\text{Dy}(\epsilon)^{159}\text{Tb}$	From intensity of feeding $5/2^-$ level at 363.5449 keV										Ens121 **
* $^{159}\text{Ho}(\beta^+)^{159}\text{Dy}$	$E_{\beta^+}=506(6)$ 506(3) respectively, to $5/2^-$ level at 309.593 keV										Ens121 **
* $^{159}\text{Er}(\beta^+)^{159}\text{Ho}$	$E_{\beta^+}=1122(3)$ to $13/2^+$ level at 624.5 keV, and other E_{β^+}										Ens121 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$*^{159}\text{Yb}(\beta^+)^{159}\text{Tm}$	$E_{\beta^+}=3366(150)$ to $7/2^-$ level at 166.17 keV										Ens121 **	
$^{160}\text{Nd}-^{84}\text{Kr}_{1.905}$	118436	50				2			CP2	1.0	18Or02	
$^{160}\text{Pm}-^{136}\text{Xe}_{1.176}$	52377	18	52331.1	2.2	-2.6	B			JY1	1.0	18Vi02 *	
$^{160}\text{Pm}-^{84}\text{Kr}_{1.905}$	111812.1	2.2				2			CP2	1.0	20Or03	
$^{160}\text{Pm}^m-^{84}\text{Kr}_{1.905}$	112017	12				2			CP2	1.0	20Or03	
$^{160}\text{Sm}-u$	-64666	11	-64663.0	2.1	0.3	U			CP1	1.0	12Va02 *	
$^{160}\text{Sm}-^{80}\text{Kr}_{2.000}$	102581	11	102581.2	2.6	0.0	U			CP1	1.0	12Va02	
$^{160}\text{Sm}-^{86}\text{Kr}_{1.860}$	101599	11	101601.3	2.1	0.2	U			CP1	1.0	12Va02	
$^{160}\text{Sm}-^{82}\text{Kr}_{1.951}$	104135.3	2.1				2			CP2	1.0	18Or.A	
$^{160}\text{Eu}-u$	-68150	17	-68163.0	1.0	-0.8	U			CP1	1.0	12Va02 *	
$^{160}\text{Eu}-^{80}\text{Kr}_{2.000}$	99096	18	99081.1	1.8	-0.8	U			CP1	1.0	12Va02	
$^{160}\text{Eu}-^{86}\text{Kr}_{1.860}$	98115	18	98101.2	1.0	-0.8	U			CP1	1.0	12Va02	
$^{160}\text{Eu}-^{84}\text{Kr}_{1.905}$	100433.81	0.97				2			CP2	1.0	18Ha19	
$^{160}\text{Eu}^m-^{84}\text{Kr}_{1.905}$	100533.70	0.86				2			CP2	1.0	18Ha19	
$\text{C}_{12}\text{H}_{16}-^{160}\text{Gd}$	198150	50	198139.3	1.2	-0.1	U			R04	4.0	64De15	
$\text{C}_{12}\text{H}_{16}-^{160}\text{Dy}$	200050	70	199996.9	0.8	-0.2	U			R04	4.0	64De15	
$^{160}\text{Er}-u$	-70916	30	-70923	26	-0.2	-			GS2	1.0	05Li24	
ave.	-70914	27			-0.3	1	95	95	^{160}Er		average	
$^{160}\text{Tm}-u$	-64770	120	-64740	40	0.3	U			GS1	1.0	00Ra23 *	
	-64752	37			0.4	1	90	90	^{160}Tm	GS2	1.0	05Li24 *
$^{160}\text{Yb}-^{142}\text{Sm}_{1.127}$	33120	20	33118	6	-0.1	U			MA7	1.0	01Bo59	
$^{160}\text{Yb}-u$	-62440	120	-62441	6	-0.0	U			GS1	1.0	00Ra23	
	-62438	30			-0.1	U			GS2	1.0	05Li24	
$^{160}\text{Yb}-^{133}\text{Cs}_{1.203}$	51300.5	5.9				2			MA8	1.0	19Hu15	
$^{160}\text{Lu}-u$	-53967	61				2			GS2	1.0	05Li24 *	
$^{160}\text{Hf}-u$	-49334	30	-49317	10	0.6	U			GS2	1.0	05Li24	
$^{160}\text{Gd}\text{ }^{35}\text{Cl}_2-^{156}\text{Gd}\text{ }^{37}\text{Cl}_2$	10831.70	1.27	10831.3	0.8	-0.1	U			H41	2.5	85Dy04	
$^{160}\text{Gd}\text{ }^{35}\text{Cl}-^{158}\text{Gd}\text{ }^{37}\text{Cl}$	5890	5	5900.1	0.8	0.5	U			H12	4.0	64Ba15	
	5899	3			0.1	U			H21	2.5	70Ma05	
	5900.0	0.5			0.1	-			M21	2.5	75Ka25	
	5899.88	0.96			0.1	-			H41	2.5	85Dy04	
ave.	5900.0	1.1			0.1	1	50	39	^{160}Gd		average	
$^{160}\text{Dy}\text{ }^{35}\text{Cl}-^{158}\text{Dy}\text{ }^{37}\text{Cl}$	3731.8	2.3	3738.9	2.4	1.2	1	18	17	^{158}Dy	H25	2.5	72Ba08
$^{160}\text{Gd}-^{160}\text{Dy}$	1854.5	0.8	1857.6	1.2	1.6	1	35	29	^{160}Gd	H25	2.5	72Ba08
$^{160}\text{Gd}\text{ O}-\text{C}_{15}$	-78020.1	5.8	-78024.2	1.2	-0.5	o			TG1	1.5	09Ke.A	
	-78019.9	3.6			-0.8	U			TG1	1.5	11Ke03	
$^{160}\text{Hf}(\alpha)^{156}\text{Yb}$	4892.2	10.	4901.9	2.6	0.9	-					73Ea01 Z	
	4905.0	5.			-0.6	-			GSa		79Ho10 Z	
	4904.0	5.			-0.4	-			ORa		83To01 Z	
	4901.8	6.			0.0	-					92Ha10	
	4902.8	10.			-0.1	-					95Hi12	
	4900.8	6.			0.2	-			Daa		96Pa01	
ave.	4902.4	2.6			-0.2	1	99	82	^{160}Hf		average	
$^{160}\text{Ta}(\alpha)^{156}\text{Lu}$	5449.5	5.	5451	5	0.3	3			Daa		96Pa01	
	5456.6	10.			-0.6	3			Jya		09Ha42	
$^{160}\text{Ta}^m(\alpha)^{156}\text{Lu}^m$	5550.9	5.	5548.5	3.0	-0.5	4			GSa		79Ho10 Z	
	5538.7	6.			1.6	4					92Ha10	
	5552.1	5.			-0.7	4			Daa		96Pa01	
	5551.0	10.			-0.3	4			Jya		09Ha42	
$^{160}\text{W}(\alpha)^{156}\text{Hf}$	6072.1	10.	6066	5	-0.6	-			GSa		79Ho10	
	6063.9	5.			0.3	-			Daa		96Pa01	
ave.	6066	5			-0.0	1	100	100	^{160}W		average	
$^{160}\text{Re}(\alpha)^{156}\text{Ta}$	6704.9	16.	6698	4	-0.4	o			Daa		92Pa05	
	6711.1	16.			-0.8	o			Daa		96Pa01	
	6697.7	4.				4			Daa		11Da12	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{158}\text{Gd}(\text{t,p})^{160}\text{Gd}$	4912.0	2.2	4912.9	0.7	0.4	U			McM		89Lo07
$^{160}\text{Gd}(\text{p,t})^{158}\text{Gd}$	−4919	5	−4912.9	0.7	1.2	U			Min		73Oo01
$^{160}\text{Dy}(\text{p,t})^{158}\text{Dy}$	−6924	5	−6926.1	2.3	−0.4	−			Min		73Oo01
	−6925.1	3.4			−0.3	−			McM		88Bu08 *
ave.	−6924.8	2.8			−0.5	1	64	63 ^{158}Dy			average
$^{160}\text{Gd}(\text{t},\alpha)^{159}\text{Eu} - ^{158}\text{Gd}()^{157}\text{Eu}$	−666	5	−668	4	−0.4	1	69	36 ^{159}Eu	LAI		79Bu05
$^{160}\text{Gd}(\text{d,t})^{159}\text{Gd}$	−1200	10	−1194.3	0.7	0.6	U			Kop		67Tj01
$^{159}\text{Tb}(\text{n},\gamma)^{160}\text{Tb}$	6375.45	0.3	6375.21	0.13	−0.8	−					74Ke01 Z
	6375.13	0.15			0.5	−			Bdn		06Fi.A
$^{159}\text{Tb}(\text{d,p})^{160}\text{Tb}$	4165	20	4150.64	0.13	−0.7	U			MIT		64Sp12
	4153	5			−0.5	U			Tal		67St14
$^{159}\text{Tb}(\text{n},\gamma)^{160}\text{Tb}$	ave. 6375.19	0.13	6375.21	0.13	0.1	1	99	93 ^{160}Tb			average
$^{160}\text{Dy}(\text{d,t})^{159}\text{Dy}$	−2339	10	−2319.3	1.4	2.0	U			Tal		68Be.A
	−2323	10			0.4	U			Kop		70Gr46
$^{160}\text{Re}(\text{p})^{159}\text{W}$	1269.1	6.	1267	7	−0.4	o			Dap		92Pa05
	1271	9			−0.5	o			Dap		96Pa01 *
	1272.2	6.			−0.9	R			Dap		11Da12
$^{160}\text{Eu}(\beta^-)^{160}\text{Gd}$	3900	300	4448.6	1.4	1.8	U					73Da05
	4200	200			1.2	U					73Mo18
	4705	60			−4.3	B			Kur		07Ha57
	4695	60			−4.1	C			Kur		10Ha38
	4480	35			−0.9	U			Kur		14Ha38
$^{160}\text{Tb}(\beta^-)^{160}\text{Dy}$	1838	10	1836.0	1.1	−0.2	U					57Na03 *
	1827	10			0.9	U					59Gr93 *
	1825	10			1.1	U					63Wu01 *
$^{160}\text{Ho}(\beta^+)^{160}\text{Dy}$	3290	15				2					66Av03 *
$^{160}\text{Er}(\epsilon)^{160}\text{Ho}$	420	150	318	29	−0.7	U					82Vy06 *
$^{160}\text{Tm}(\beta^+)^{160}\text{Er}$	5600	300	5760	40	0.5	U					75St12 *
	5890	100			−1.3	1	15	10 ^{160}Tm	IRS		93Al03
$^{160}\text{Lu}(\beta^+)^{160}\text{Yb}$	7210	240	7890	60	2.8	U					83Ge08
	7340	100			5.5	C			IRS		83Vi.A
	7300	100			5.9	B			IRS		93Al03
* $^{160}\text{Pm} - ^{136}\text{Xe}_{1,176}$	Could be a mixture of ground state and isomer										HWJ208 **
* $^{160}\text{Sm} - \text{u}$	Represents frequency ratio $^{160}\text{Sm}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.950775181(67)$										WgM124**
* $^{160}\text{Eu} - \text{u}$	Represents frequency ratio $^{160}\text{Eu}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.95079589(10)$										WgM124**
* $^{160}\text{Tm} - \text{u}$	$M - A = -60300(110)$ keV for mixture gs+m at 67(14) keV										Nub211 **
* $^{160}\text{Tm} - \text{u}$	$M - A = -60283(28)$ keV for mixture gs+m at 67(12) keV										Nub211 **
* $^{160}\text{Lu} - \text{u}$	$M - A = -50270(28)$ keV for mixture gs+m at 0#100 keV										Nub211 **
* $^{160}\text{Dy}(\text{p,t})^{158}\text{Dy}$	$Q - Q(^{164}\text{Dy}(\text{p,t})) = -1477.9(3.4)$, see $^{164}\text{Dy}(\text{p,t})$										AHW **
* $^{160}\text{Re}(\text{p})^{159}\text{W}$	Corrected : Ame2003 assumed $E_p = 1271(9)$ thus $Q_p = 1279.1(9.)$ keV										WgM105**
* $^{160}\text{Tb}(\beta^-)^{160}\text{Dy}$	$E_{\beta^-} = 870(10)$ 858(10) 868(10) respectively, to 8^+ level at 966.85 keV, and other E_{β^-}										Ens059 **
* $^{160}\text{Ho}(\beta^+)^{160}\text{Dy}$	$E_{\beta^+} = 570(15)$ to 4^+ level at 1694.37 keV; and 1045(15) from										Ens059 **
*	$^{160}\text{Ho}^m$ at 59.98 to 1^- level at 1285.602 and 3^- at 1286.711 keV										Nub211 **
* $^{160}\text{Er}(\epsilon)^{160}\text{Ho}$	$pK = 0.795(0.2)$ to 1^+ level at 67.11 keV										Ens059 **
* $^{160}\text{Tm}(\beta^+)^{160}\text{Er}$	$E_{\beta^+} = 3700(300)$ to 854.4–1007.95 levels, reassigned by evaluator										Ens059 **
$^{161}\text{Pm} - ^{136}\text{Xe}_{1,184}$	56065	64	56088	10	0.4	U			JY1	1.0	20Vi04
$^{161}\text{Pm} - ^{84}\text{Kr}_{1,917}$	115888.7	9.7				2			CP2	1.0	18Or.A
$^{161}\text{Sm} - \text{u}$	−60841	13	−60840	7	0.1	1	32	32 ^{161}Sm	CP1	1.0	12Va02 *
$^{161}\text{Sm} - ^{80}\text{Kr}_{2,013}$	107493	12	107491	7	−0.1	1	38	37 ^{161}Sm	CP1	1.0	12Va02
$^{161}\text{Sm} - ^{86}\text{Kr}_{1,872}$	106496	13	106497	7	0.1	1	32	32 ^{161}Sm	CP1	1.0	12Va02
$^{161}\text{Eu} - \text{u}$	−66336	19	−66336	11	−0.0	1	35	35 ^{161}Eu	CP1	1.0	12Va02 *
$^{161}\text{Eu} - ^{80}\text{Kr}_{2,013}$	101996	19	101995	11	−0.0	1	35	34 ^{161}Eu	CP1	1.0	12Va02
$^{161}\text{Eu} - ^{86}\text{Kr}_{1,872}$	101000	20	101001	11	0.0	1	31	31 ^{161}Eu	CP1	1.0	12Va02
$\text{C}_{13}\text{H}_5 - ^{161}\text{Dy}$	112246	25	112185.7	0.7	−0.6	U			R04	4.0	64De15

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{161}\text{Tm}-u$	-66451	30				2			GS2	1.0	05Li24 *
$^{161}\text{Yb}-^{142}\text{Sm}_{1.134}$	34071	19	34065	16	-0.3	2			MA7	1.0	01Bo59
$^{161}\text{Yb}-u$	-62120	110	-62088	16	0.3	U			GS1	1.0	00Ra23
	-62107	30			0.6	R			GS2	1.0	05Li24
$^{161}\text{Lu}-u$	-56428	30				2			GS2	1.0	05Li24
$^{161}\text{Hf}-u$	-49733	30	-49722	25	0.4	1	70	70	^{161}Hf	1.0	05Li24
$^{161}\text{Dy } ^{35}\text{Cl}-^{159}\text{Tb } ^{37}\text{Cl}$	4535.0	1.0	4535.8	1.2	0.3	1	22	18	^{159}Tb	2.5	72Ba08
$^{161}\text{Hf}(\alpha)^{157}\text{Yb}$	4717.0	10.	4679	25	-0.7	-					73Ea01 Z
	4725.2	10.			-0.9	-					82Sc15 Z
	4724.2	5.			-0.9	-			ORa		83To01 Z
	4716.4	7.			-0.7	-					92Ha10
	4721.5	10.			-0.8	-					95Hi12
	ave.	4721			-0.8	1	21	18	^{161}Hf		average
$^{161}\text{Ta}^m(\alpha)^{157}\text{Lu}^m$	5278.9	5.	5276	6	-0.5	F			GSa		79Ho10 *
	5280.4	5.			-0.8	F					92Ha10
	5271.2	7.			0.8	F			Daa		96Pa01 *
	5282.5	7.			-0.9	F			Jya		05Sc22 *
	5273.2	6.			0.5	1	94	56	$^{161}\text{Ta}^m$		12Th13
$^{161}\text{W}(\alpha)^{157}\text{Hf}$	5923.4	5.	5923	4	-0.1	4			GSa		79Ho10 Z
	5922.4	5.			0.1	4			Daa		96Pa01
$^{161}\text{Re}^m(\alpha)^{157}\text{Ta}^m$	6439.3	10.	6430	4	-0.9	2			GSa		79Ho10
	6425.0	6.			0.8	2			Daa		96Pa01
	6432.1	7.			-0.3	2			Ara		97Ir01
$^{161}\text{Os}(\alpha)^{157}\text{W}$	7065.9	12.	7069	11	0.2	3					10Bi03
	7077.2	20.5			-0.4	3			Jya		19Hi06
$^{161}\text{Os}(\alpha)^{157}\text{W}^p$	6748.0	30.				4					10Bi03
$^{161}\text{Dy}(\text{p},\text{t})^{159}\text{Dy}$	-6546	5	-6549.1	1.4	-0.6	-			Min		73Oo01
	-6547.9	2.5			-0.5	-			McM		88Bu08 *
	ave.	-6547.5			-0.7	1	39	35	^{159}Dy		average
$^{160}\text{Gd}(\text{n},\gamma)^{161}\text{Gd}$	5635.4	1.0				2					71Gr42
$^{160}\text{Gd}(\text{d},\text{p})^{161}\text{Gd}$	3411	10	3410.8	1.0	-0.0	U			Kop		67Tj01
$^{160}\text{Gd}(\alpha,\text{t})^{161}\text{Tb}-^{158}\text{Gd}()^{159}\text{Tb}$	678.0	1.0	677.1	0.7	-0.9	1	55	31	^{160}Gd		75Bu02
$^{160}\text{Tb}(\text{n},\gamma)^{161}\text{Tb}$	7696.3	0.6	7696.6	0.5	0.5	1	84	77	^{161}Tb		75He.C
$^{160}\text{Dy}(\text{n},\gamma)^{161}\text{Dy}$	6451.5	2.	6454.39	0.08	1.4	U					77Be03
	6454.40	0.09			-0.1	-			ILn		86Sc16 Z
	6454.34	0.14			0.3	-			Bdn		06Fi.A
$^{160}\text{Dy}(\text{d},\text{p})^{161}\text{Dy}$	4231	10	4229.82	0.08	-0.1	U			Tal		68Be.A
	4237	10			-0.7	U			Kop		70Gr46
$^{161}\text{Dy}(\text{d},\text{t})^{160}\text{Dy}$	-205	10	-197.16	0.08	0.8	U			Kop		70Gr46
$^{160}\text{Dy}(\text{n},\gamma)^{161}\text{Dy}$	6454.38	0.08	6454.39	0.08	0.1	1	100	93	^{160}Dy		average
$^{160}\text{Dy}(^3\text{He},\text{d})^{161}\text{Ho}-^{164}\text{Dy}()^{165}\text{Ho}$	-1406.5	2.0	-1406.5	2.0	-0.0	1	100	100	^{161}Ho		75Bu02
$^{161}\text{Re}(\text{p})^{160}\text{W}$	1199.5	6.	1197	5	-0.4	1	79	79	^{161}Re		97Ir01
$^{161}\text{Re}^m(\text{p})^{160}\text{W}$	1323.3	7.	1321	5	-0.3	o			Ara		97Ir01 *
$^{161}\text{Sm}(\beta^-)^{161}\text{Eu}$	5065	130	5120	12	0.4	o			Kur		07Ha57
	5050	130			0.5	U			Kur		10Ha38
$^{161}\text{Eu}(\beta^-)^{161}\text{Gd}$	3705	60	3715	11	0.2	o			Kur		07Ha57
	3705	60			0.2	U			Kur		10Ha38
	3722	35			-0.2	U			Kur		14Ha38
$^{161}\text{Gd}(\beta^-)^{161}\text{Tb}$	1977	30	1955.6	1.4	-0.7	U					66Zy02 *
$^{161}\text{Tb}(\beta^-)^{161}\text{Dy}$	584	6	593.7	1.2	1.6	U					63Ko08
	590	10			0.4	U					64Fu11
$^{161}\text{Er}(\beta^+)^{161}\text{Ho}$	2050	40	1995	9	-1.4	U					65Gr35 *
	1980	18			0.8	R					84Ka.A *
$^{161}\text{Tm}(\beta^+)^{161}\text{Er}$	3100	200	3303	29	1.0	U					75Ad08 *
	3180	100			1.2	U			IRS		93Al03
$^{161}\text{Yb}(\beta^+)^{161}\text{Tm}$	3850	250	4060	30	0.9	U					81Ad02
	3585	200			2.4	U			Dbn		94Po26

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{161}\text{Lu}(\beta^+)^{161}\text{Yb}$	5300	100	5270	30	-0.3	o			IRS		83Vi.A	
	5300	100			-0.3	U			IRS		93Ai03	
	5255	150			0.1	U			Dbn		94Po26 *	
$^{161}\text{Re}^m(\text{IT})^{161}\text{Re}$	123.8	1.3	123.7	1.3	-0.1	1	99	78	$^{161}\text{Re}^m$		97Ir01	
* $^{161}\text{Sm}-\text{u}$	Represents frequency ratio $^{161}\text{Sm}^{++}/(\text{C}_{12}\text{H}_4)^+=0.944844874(74)$										WgM124**	
* $^{161}\text{Eu}-\text{u}$	Represents frequency ratio $^{161}\text{Eu}^{++}/(\text{C}_{12}\text{H}_4)^+=0.94487714(11)$										WgM124**	
* $^{161}\text{Tm}-\text{u}$	$M-A=-61895(28)$ keV for mixture gs+m at 7.51 keV										Nub211 **	
* $^{161}\text{Ta}^m(\alpha)^{157}\text{Lu}^m$	F : above 4 items are not unambiguously assigned										WgM151**	
* $^{161}\text{Dy}(\text{p,t})^{159}\text{Dy}$	$Q-Q(^{164}\text{Dy}(\text{p,t}))=-1100.7(2.5)$ keV										AHW **	
* $^{161}\text{Re}^m(\text{p})^{160}\text{W}$	Replaced by author's result for $^{161}\text{Re}^m(\text{IT})^{161}\text{Re}$										AHW **	
* $^{161}\text{Gd}(\beta^-)^{161}\text{Tb}$	$E_{\beta^-}=1560(30)$ mainly to $7/2^-$ level at 417.228 keV										Ens11b **	
* $^{161}\text{Er}(\beta^+)^{161}\text{Ho}$	$E_{\beta^+}=820(40)$ 748(18) respectively, to $1/2^+$ level at 211.15 keV										Ens11b **	
* $^{161}\text{Tm}(\beta^+)^{161}\text{Er}$	$E_{\beta^+}=1800(100)$ to several levels around $7/2^-$ one at 266.44 keV										Ens11b **	
* $^{161}\text{Lu}(\beta^+)^{161}\text{Yb}$	$E_{\beta^+}=3866(150)$ to 367.28 level										Ens11b **	
$^{162}\text{Sm}-^{136}\text{Xe}_{1.191}$	52127.2	5.3	52129	4	0.4	1	51	51	^{162}Sm	JY1	1.0	18Vi02
$^{162}\text{Sm}-^{84}\text{Kr}_{1.929}$	112344.7	5.4	112343	4	-0.4	1	49	49	^{162}Sm	CP2	1.0	18Or02
$^{162}\text{Eu}-^{136}\text{Xe}_{1.191}$	47449	50	47465.9	1.4	0.3	U				JY1	1.0	18Vi02 *
$^{162}\text{Eu}-^{84}\text{Kr}_{1.929}$	107678.2	1.6	107679.2	1.4	0.6	1	78	78	^{162}Eu	CP2	1.0	18Ha19
$^{162}\text{Eu}-^{133}\text{Cs}_{1.218}$	52124.0	6.0	52117.8	1.4	-1.0	1	6	6	^{162}Eu	JY1	1.0	20Vi04
$^{162}\text{Eu}^m-^{84}\text{Kr}_{1.929}$	107850.1	2.0	107848.8	1.4	-0.7	1	48	48	$^{162}\text{Eu}^m$	CP2	1.0	18Ha19
$^{162}\text{Eu}^m-^{133}\text{Cs}_{1.218}$	52286.4	2.4	52287.4	1.4	0.4	-				JY2	1.0	20Vi04
	52292.4	8.1			-0.6	-				JY1	1.0	20Vi04
ave.	52286.9	2.3			0.2	1	36	36	$^{162}\text{Eu}^m$			average
$^{162}\text{Tb}-\text{u}$	-70724.6	2.2				2				CP2	1.0	20Or03 *
$^{162}\text{Tb}^m-\text{u}$	-70418.1	2.7				2				CP2	1.0	20Or03 *
$\text{C}_{13}\text{H}_6-^{162}\text{Dy}$	120115	19	120145.7	0.7	0.4	U				R04	4.0	64De15
$\text{C}_{12}\text{H}_4\text{N}-^{162}\text{Er}$	105590	70	105586.8	0.8	-0.0	U				R04	4.0	64De15
$\text{C}_{13}\text{H}_6-^{162}\text{Er}$	118430	170	118162.9	0.8	-0.4	U				R04	4.0	64De15
$^{162}\text{Tm}-\text{u}$	-65942	55	-65999	28	-1.0	R				GS2	1.0	05Li24 *
$^{162}\text{Yb}-^{142}\text{Sm}_{1.141}$	32524	19	32525	16	0.1	2				MA7	1.0	01Bo59
$^{162}\text{Yb}-\text{u}$	-64210	110	-64221	16	-0.1	U				GS1	1.0	00Ra23
	-64223	30			0.1	R				GS2	1.0	05Li24
$^{162}\text{Lu}-\text{u}$	-56758	234	-56720	80	0.2	o				GS1	1.0	00Ra23 *
	-56781	190			0.3	2				GS2	1.0	05Li24 *
$^{162}\text{Hf}-\text{u}$	-52756	30	-52784	10	-0.9	U				GS2	1.0	05Li24
$^{162}\text{Er }^{35}\text{Cl}_2-^{158}\text{Gd }^{37}\text{Cl}_2$	10577.5	2.7	10576.3	1.1	-0.2	U				H25	2.5	72Ba08
$^{162}\text{Dy }^{35}\text{Cl}-^{160}\text{Dy }^{37}\text{Cl}$	4555	6	4551.05	0.12	-0.2	U				H12	4.0	64Ba15
	4552.1	1.1			-0.4	U				H25	2.5	72Ba08
$^{162}\text{Er }^{35}\text{Cl}-^{160}\text{Gd }^{37}\text{Cl}$	4674.6	1.9	4676.2	1.2	0.3	U				H25	2.5	72Ba08
$^{162}\text{Eu}^m-^{162}\text{Eu}$	167.4	3.1	169.6	1.8	0.7	1	33	17	^{162}Eu	JY2	1.0	20Vi04
$^{162}\text{Er}-^{162}\text{Dy}$	1982.79	0.32	1982.8	0.3	0.0	1	100	100	^{162}Er	SH1	1.0	11El04
$^{161}\text{Dy }^{37}\text{Cl}-^{162}\text{Dy }^{35}\text{Cl}$	-3080	70	-2815.20	0.09	0.9	U				R04	4.0	64De15
$^{162}\text{Dy}-^{161}\text{Dy}$	150	70	-134.92	0.06	-1.0	U				R04	4.0	64De15
	78	23			-2.3	U				R04	4.0	64De15
	22	40			-1.0	U				R04	4.0	64De15
$^{162}\text{Hf}(\alpha)^{158}\text{Yb}$	4417.2	10.	4416	5	-0.1	-						82Sc15
	4420.4	10.3			-0.4	-				ORa		83To01
	4414.2	9.			0.2	-						92Ha10
	4416.3	10.3			0.0	-						95Hi12
ave.	4417	5			-0.1	1	95	81	^{162}Hf			average
$^{162}\text{Ta}(\alpha)^{158}\text{Lu}$	5003.8	10.	5010	60	0.0	4						86Ru05
	5007.9	5.			-0.0	4						92Ha10
$^{162}\text{W}(\alpha)^{158}\text{Hf}$	5669.9	10.	5678.3	2.4	0.8	U						73Ea01 Z
	5668.0	10.			1.0	U				ORa		75To05 Z
	5677.5	5.			0.2	-				GSa		81Ho10 Z
	5674.5	4.			0.9	-				Ora		82De11 Z
	5681.6	5.			-0.6	-				Daa		96Pa01
	5681.5	5.1			-0.6	-				Jya		15Li24
ave.	5678.3	2.4			0.0	1	100	100	^{162}W			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{162}\text{Re}(\alpha)^{158}\text{Ta}$	6240.3	5.				8			Ara		97Da07
$^{162}\text{Re}^m(\alpha)^{158}\text{Ta}^m$	6274.2	6.	6274	3	-0.0	9			GSa		79Ho10
	6278.3	6.			-0.7	9			Daa		96Pa01
	6271.1	5.			0.6	9			Ara		97Da07
	6256	16			1.1	U			Jya		16Ca15 *
$^{162}\text{Os}(\alpha)^{158}\text{W}$	6778.8	30.	6767.8	2.9	-0.4	U			GSa		89Ho12 *
	6785.8	10.			-1.8	U			ORa		96Bi07
	6767.4	3.			0.1	4			Ara		00Ma95
	6781.7	13.			-1.1	U			Jya		04Jo12
	6770.5	8.2			-0.3	4			Jya		19Hi06
$^{160}\text{Gd}(\text{t,p})^{162}\text{Gd}$	3999.5	3.8				2			McM		89Lo07
$^{160}\text{Dy}(\text{t,p})^{162}\text{Dy}$	6169.5	1.9	6169.58	0.10	0.0	U			McM		88Bu08 *
$^{162}\text{Dy}(\text{p,t})^{160}\text{Dy}$	-6168	5	-6169.58	0.10	-0.3	U			Min		73Oo01
	-6169.7	2.1			0.1	U			McM		88Bu08 *
$^{162}\text{Er}(\text{p,t})^{160}\text{Er}$	-7944	55	-7931	24	0.2	R			Win		74De31 *
$^{161}\text{Dy}(\text{n},\gamma)^{162}\text{Dy}$	8196.99	0.06	8196.99	0.06	0.1	1	100	85	^{161}Dy		82Is05 Z
	8193	3			1.3	U			Bdn		06Fi.A
$^{161}\text{Dy}(\text{d,p})^{162}\text{Dy}$	5969	10	5972.43	0.06	0.3	U			Tal		67Ba34
	5981	10			-0.9	U			Kop		70Gr46
$^{162}\text{Dy}(\text{d,t})^{161}\text{Dy}$	-1944	10	-1939.76	0.06	0.4	U			Kop		70Gr46
	-1943	10			0.3	U			Tal		77Be03
$^{161}\text{Dy}(\text{}^3\text{He,d})^{162}\text{Ho}-^{164}\text{Dy}(\text{}^1\text{O})^{165}\text{Ho}$	-945.3	3.0	-945	3	-0.0	1	100	100	^{162}Ho		75Bu02
$^{162}\text{Er}(\text{d,t})^{161}\text{Er}$	-2952	10	-2947	9	0.5	2			Kop		69Tj01
$^{162}\text{Eu}(\beta^-)^{162}\text{Gd}$	5575	60	5558	4	-0.3	o			Kur		07Ha57
	5585	60			-0.5	o			Kur		10Ha38
	5577	35			-0.5	U			Kur		14Ha38
$^{162}\text{Gd}(\beta^-)^{162}\text{Tb}$	1442	100	1599	4	1.6	U					70Ch02 *
$^{162}\text{Tb}(\beta^-)^{162}\text{Dy}$	2448	100	2301.6	2.2	-1.5	U					66Fu08 *
	2523	50			-4.4	B					66Sc24 *
	2528	80			-2.8	U					77Ka08 *
$^{162}\text{Ho}(\beta^+)^{162}\text{Dy}$	2220	50	2141	3	-1.6	U					69Ak01
$^{162}\text{Tm}(\beta^+)^{162}\text{Er}$	4840	50	4857	26	0.3	2					63Ab02
	4705	70			2.2	2					74De47 *
	4900	100			-0.4	2			IRS		93Al03
	4892	50			-0.7	2			Dbn		94Po26 *
$^{162}\text{Lu}(\beta^+)^{162}\text{Yb}$	6740	270	6990	80	0.9	U					83Ge08
	6990	120			-0.0	o			IRS		83Vi.A
	6960	100			0.3	R			IRS		93Al03
	7111	150			-0.8	R			Dbn		94Po26 *
$*^{162}\text{Eu}-^{136}\text{Xe}_{1,191}$	$D_M=47536.0(3.8) \mu\text{u}$ for mixture gs+m at 158.2(1.7) keV; $M-A=-58657.6(3.5) \text{ keV}$										Nub211 **
$*^{162}\text{Tb}-\text{u}$	Represents frequency ratio $^{162}\text{Tb}^{++}/(\text{C}_6\text{H}_6)^+=1.037384012(14)$										HWJ208 **
$*^{162}\text{Tb}^m-\text{u}$	Represents frequency ratio $^{162}\text{Tbm}^{++}/(\text{C}_6\text{H}_6)^+=1.037385975(17)$										HWJ208 **
$*^{162}\text{Tm}-\text{u}$	$M-A=-61359(28) \text{ keV}$ for mixture gs+m at 130(40) keV										Nub211 **
$*^{162}\text{Lu}-\text{u}$	$M-A=-52730(130) \text{ keV}$ for mixture gs+m+n at 120#200 and 300#200 keV										Nub211 **
$*^{162}\text{Lu}-\text{u}$	$M-A=-52751(28) \text{ keV}$ for mixture gs+m+n at 120#200 and 300#200 keV										Nub211 **
$*^{162}\text{Re}^m(\alpha)^{158}\text{Ta}^m$	$E_\alpha=6037(16) \text{ keV}$ to 10^+ level 66 keV above the $9^+ ^{158}\text{Ta}^m$										16Ca15 **
$*^{162}\text{Os}(\alpha)^{158}\text{W}$	Original value $E=6640(20) (Q=6808.4) \text{ recalibrated}$										88Ho.B **
$*^{160}\text{Dy}(\text{t,p})^{162}\text{Dy}$	$Q-Q(^{162}\text{Dy}(\text{t,p}))=722.3(1.9) \text{ keV}$										AHW **
$*^{162}\text{Dy}(\text{p,t})^{160}\text{Dy}$	$Q-Q(^{164}\text{Dy}(\text{p,t}))=-722.5(2.1) \text{ keV}$										AHW **
$*^{162}\text{Er}(\text{p,t})^{160}\text{Er}$	Not resolved peak. Original uncertainty 28 increased to 51 keV and added systematic error 21 keV										GAu **
$*^{162}\text{Gd}(\beta^-)^{162}\text{Tb}$	$E_{\beta^-}=1000(100) \text{ to } 1^+ \text{ level at } 442.11 \text{ keV}$										Ens162 **
$*^{162}\text{Tb}(\beta^-)^{162}\text{Dy}$	$E_{\beta^-}=1300(100) 1375(50) 1380(80) \text{ respectively, to } 2^- \text{ level at } 1148.232 \text{ keV}$										Ens078 **
$*^{162}\text{Tm}(\beta^+)^{162}\text{Er}$	$E_{\beta^+}=2110(70) \text{ to } 2^- \text{ level at } 1572.84 \text{ keV}$										Ens078 **
$*^{162}\text{Tm}(\beta^+)^{162}\text{Er}$	$E_{\beta^+}=3768(50) \text{ to } 2^+ \text{ level at } 102.04 \text{ keV}$										Ens078 **
$*^{162}\text{Lu}(\beta^+)^{162}\text{Yb}$	$E_{\beta^+}=6006(150) \text{ to ground state and } 2^+ \text{ level at } 166.8, \text{ unknown intensity ratio}$										Ens078 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{163}\text{Sm}-^{84}\text{Kr}_{1.940}$	117373.5	7.9				2			CP2	1.0	18Or02
$^{163}\text{Sm}-^{136}\text{Xe}_{1.199}$	56980	39	56929	8	-1.3	U			JY1	1.0	20Vi04
$^{163}\text{Eu}-^{133}\text{Cs}_{1.226}$	55179.4	4.0	55181.4	1.0	0.5	U			JY1	1.0	20Vi04
$^{163}\text{Eu}-^{82}\text{Kr}_{1.988}$	111264.98	0.97				2			CP2	1.0	18Or.A
$^{163}\text{Gd}-u$	-65824	16	-65903.4	0.9	-5.0	U			CP1	1.0	12Va02 *
	-65849.5	3.7			-14.6	F			JY1	1.0	18Vi02 *
$^{163}\text{Gd}-^{80}\text{Kr}_{2.038}$	104600	16	104518.4	1.7	-5.1	F			CP1	1.0	12Va02 *
$^{163}\text{Gd}-^{86}\text{Kr}_{1.895}$	103569	15	103489.5	0.9	-5.3	B			CP1	1.0	12Va02
$^{163}\text{Gd}-^{136}\text{Xe}_{1.199}$	45353	11	45346.5	0.9	-0.6	U			JY1	1.0	20Vi04
$^{163}\text{Gd}-^{82}\text{Kr}_{1.988}$	106095.6	1.2	106096.1	0.9	0.4	1	51	51 ^{163}Gd	CP2	1.0	20Or02
$^{163}\text{Gd}^m-^{136}\text{Xe}_{1.199}$	45526	14	45494.9	0.9	-2.2	U			JY1	1.0	20Vi04
$^{163}\text{Gd}^m-^{82}\text{Kr}_{1.988}$	106245.0	1.2	106244.5	0.9	-0.4	1	51	51 $^{163}\text{Gd}^m$	CP2	1.0	20Or02
$\text{C}_{13}\text{H}_7-^{163}\text{Dy}$	125906	36	126038.0	0.7	0.9	U			R04	4.0	64De15
$^{163}\text{Tm}-u$	-67327	30	-67342	6	-0.5	U			GS2	1.0	05Li24
$^{163}\text{Yb}-^{142}\text{Sm}_{1.148}$	33686	19	33685	16	-0.1	2			MA7	1.0	01Bo59
$^{163}\text{Yb}-u$	-63663	30	-63655	16	0.3	R			GS2	1.0	05Li24
$^{163}\text{Lu}-u$	-58730	110	-58820	30	-0.8	U			GS1	1.0	00Ra23
	-58821	30				2			GS2	1.0	05Li24
$^{163}\text{Hf}-u$	-52911	30	-52893	28	0.6	1	85	85 ^{163}Hf	GS2	1.0	05Li24
$^{163}\text{Ta}-u$	-45855	54	-45660	40	3.6	C			GS2	1.0	05Li24 *
$^{163}\text{Dy}-^{35}\text{Cl}-^{161}\text{Dy}-^{37}\text{Cl}$	5200	60	4747.92	0.11	-1.9	U			R04	4.0	64De15
	4746	3			0.3	U			H23	2.5	70Wh01
	4744.7	1.2			1.1	U			H25	2.5	72Ba08
$^{163}\text{Eu}-^{163}\text{Dy}$	10693.5	3.7	10528.3	1.2	-44.7	F			JY1	1.0	18Vi02 *
$^{163}\text{Ho O}-^{163}\text{Dy O}$	2.72	0.77	3.039	0.024	0.3	U			TG1	1.5	15Sc13
$^{163}\text{Ho}-^{163}\text{Dy}$	3.042	0.037	3.039	0.024	-0.1	1	42	22 ^{163}Ho	SH2	1.0	15El03
$^{162}\text{Dy}-^{37}\text{Cl}-^{163}\text{Dy}-^{35}\text{Cl}$	-5069	42	-4882.83	0.08	1.1	U			R04	4.0	64De15
$^{163}\text{Dy}-^{162}\text{Dy}$	2164	35	1932.71	0.05	-1.7	U			R04	4.0	64De15
	1985	38			-0.3	U			R04	4.0	64De15
	2174	40			-1.5	U			R04	4.0	64De15
$^{163}\text{Dy O}-\text{C}_{15}$	-76349.06	0.86	-76348.2	0.7	0.7	1	33	33 ^{163}Dy	TG1	1.5	15Sc13
$^{163}\text{Ho O}-\text{C}_{15}$	-76346.61	0.97	-76345.1	0.7	1.0	1	26	26 ^{163}Ho	TG1	1.5	15Sc13
$^{163}\text{Ta}(\alpha)^{159}\text{Lu}$	4741.5	15.	4749	5	0.5	3					83Sc18 *
	4746.7	10.			0.2	3					86Ru05
	4751.8	7.			-0.4	3					92Ha10
$^{163}\text{W}(\alpha)^{159}\text{Hf}$	5520.3	5.	5520	60	-0.0	5					73Ea01 Z
	5518.1	5.			0.0	5			GSa		79Ho10 Z
	5519.9	3.			-0.0	5			Ora		82De11 Z
	5525.9	10.3			-0.1	U					84Sc06 *
	5518.7	6.			0.0	5			Daa		96Pa01
$^{163}\text{Re}(\alpha)^{159}\text{Ta}$	6017.9	5.	6012	8	-1.2	o			Ara		97Da07 *
$^{163}\text{Re}^m(\alpha)^{159}\text{Ta}^m$	6067.2	6.	6068	3	0.2	-			GSa		79Ho10
	6067.2	7.			0.1	-			Daa		96Pa01
	6069.2	5.			-0.2	-			Ara		97Da07
	6068	3			-0.0	1	100	100 $^{159}\text{Ta}^m$			average
$^{163}\text{Os}(\alpha)^{159}\text{W}$	6674.1	30.	6673	7	-0.0	4			GSa		81Ho10
	6678.2	10.			-0.5	4			ORa		96Bi07
	6676.2	19.			-0.2	4			Daa		96Pa01
	6674.1	30.			-0.0	o			Jya		13Dr06 *
	6662.8	12.3			0.8	4			Jya		19Hi06
$^{161}\text{Dy}(\text{t,p})^{163}\text{Dy}$	5986.3	1.5	5986.20	0.08	-0.1	U			McM		88Bu08 *
$^{163}\text{Dy}(\text{p,t})^{161}\text{Dy}$	-5985	5	-5986.20	0.08	-0.2	U			Min		73Oo01
	-5987.1	2.2			0.4	U			McM		88Bu08 *
$^{162}\text{Dy}(\text{n},\gamma)^{163}\text{Dy}$	6270.98	0.06	6271.01	0.05	0.4	-			MMn		82Is05 Z
	6271.00	0.09			0.1	-			ILn		89Sc31 Z
	6271.14	0.13			-1.0	-			Bdn		06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{163}\text{Dy}(\gamma, n)^{162}\text{Dy}$	−6320	110	−6271.01	0.05	0.4	U			Phi		60Ge01
$^{162}\text{Dy}(d, p)^{163}\text{Dy}$	4049	5	4046.44	0.05	−0.5	U			Tal		67Sc05
	4045	10			0.1	U			Kop		70Gr46
$^{163}\text{Dy}(d, t)^{162}\text{Dy}$	−14	5	−13.78	0.05	0.0	U					67Ba34
	−27	10			1.3	U			Kop		70Gr46
$^{162}\text{Dy}(n, \gamma)^{163}\text{Dy}$	ave. 6271.01	0.05	6271.01	0.05	0.0	1	100	100	^{162}Dy		average
$^{162}\text{Dy}(^3\text{He}, d)^{163}\text{Ho} - ^{164}\text{Dy}(^3\text{He}, d)^{165}\text{Ho}$	−734.3	1.0	−733.5	0.7	0.8	1	56	35	^{165}Ho	McM	75Bu02
$^{162}\text{Er}(d, p)^{163}\text{Er}$	4682	10	4680	5	−0.2	1	21	21	^{163}Er	Kop	69Tj01
$^{163}\text{Eu}(\beta^-)^{163}\text{Gd}$	4828	70	4814.8	1.2	−0.2	o			Kur		07Ha57 *
	4813	70			0.0	o			Kur		10Ha38 *
	4829	65			−0.2	U			Kur		14Ha38
$^{163}\text{Gd}^m(\text{IT})^{163}\text{Gd}$	137.8	1.0	138.22	0.20	0.4	U					14Ha38 *
	138.2	0.2			0.1	1	98	49	^{163}Gd		20Za04
$^{163}\text{Gd}(\beta^-)^{163}\text{Tb}$	3170	70	3207	4	0.5	o			Kur		07Ha57
	3150	70			0.8	o			Kur		10Ha38
	3187	40			0.5	U			Kur		14Ha38
$^{163}\text{Tb}(\beta^-)^{163}\text{Dy}$	1684	50	1785	4	2.0	U					66Fu08 *
	1721	100			0.6	U					71Ka22 *
$^{163}\text{Ho}(\epsilon)^{163}\text{Dy}$	2.83	0.05	2.831	0.022	0.0	−					82An19 *
	2.65	0.20			0.9	U					83Ba32
	2.84	0.10			−0.1	U					84La.A *
	2.56	0.05			5.4	B					85Ha12 *
	2.60	0.03			7.7	B					86Ya17
	2.561	0.020			13.5	B					92Ha15
	2.54	0.03			9.7	C					93Bo.A *
	2.71	0.10			1.2	U					94Ya07
	2.800	0.050			0.6	−					97Ga12
	2.858	0.051			−0.5	−					17Ra15
	ave. 2.829	0.029			0.1	1	58	31	^{163}Ho		average
$^{163}\text{Er}(\beta^+)^{163}\text{Ho}$	1210	6	1211	5	0.1	1	58	58	^{163}Er		63Pe16
$^{163}\text{Tm}(\beta^+)^{163}\text{Er}$	2439	3				2					82Vy07 *
	2360	100	2439	3	0.8	U			IRS		93Al03
$^{163}\text{Yb}(\beta^+)^{163}\text{Tm}$	3370	100	3435	16	0.6	U					75Ad09 *
$^{163}\text{Lu}(\beta^+)^{163}\text{Yb}$	4860	170	4500	30	−2.1	U					83Ge08
	4600	200			−0.5	o			IRS		83Vi.A
	4600	200			−0.5	U			IRS		93Al03
$^{163}\text{Re}^m(\text{IT})^{163}\text{Re}$	115.1	4.0	120	5	1.2	o			Ara		97Da07 *
* $^{163}\text{Gd}-u$	Represents frequency ratio $^{163}\text{Gd}^{++}/(\text{C}_{12}\text{H}_4)^+ = 0.933275822(91)$										WgM124**
* $^{163}\text{Gd}-u$	$M - A = -61200.6(3.4)$ for $^{163}\text{Gd}^m$ at 137.8(1.0)										HWJ188 **
* $^{163}\text{Gd}-u$	reference ion ^{163}Dy wrongly identified, most likely $^{146}\text{LaOH}$ or mixture										20Vi04 **
* $^{163}\text{Gd}-^{80}\text{Kr}_{2.038}$	Probably mixture state										HWJ205 **
* $^{163}\text{Gd}-^{80}\text{Kr}_{2.038}$	Probably mixture state										HWJ205 **
* $^{163}\text{Ta}-u$	$M - A = -42644(28)$ keV for mixture gs+m at 138#(18#) keV										Nub211 **
* $^{163}\text{Eu}-^{163}\text{Dy}$	reference ion ^{163}Dy wrongly identified, most likely $^{146}\text{LaOH}$ or mixture										20Vi04 **
* $^{163}\text{Ta}(\alpha)^{159}\text{Lu}$	Original assignment to 13 s ^{164}Ta changed to ^{163}Ta										86Ru05 **
* $^{163}\text{W}(\alpha)^{159}\text{Hf}$	Originally assigned to ^{166}Re , re-assigned in reference										92Me10 **
*	original $E_\alpha = 5372$ recalibrated using their $^{168}\text{Os} - ^{170}\text{Os}$ results										GAu **
* $^{163}\text{Re}(\alpha)^{159}\text{Ta}$	Replaced by author's value for $^{159}\text{Ta}^m(\text{IT})$										AHW **
* $^{163}\text{Os}(\alpha)^{159}\text{W}$	Error not given, estimated by evaluator										GAu **
* $^{161}\text{Dy}(t, p)^{163}\text{Dy}$	$Q - Q(^{162}\text{Dy}(t, p)) = 539.1(1.5)$ keV										AHW **
* $^{163}\text{Dy}(p, t)^{161}\text{Dy}$	$Q - Q(^{164}\text{Dy}(p, t)) = -539.9(2.2)$ keV										AHW **
* $^{163}\text{Eu}(\beta^-)^{163}\text{Gd}$	$E_{\beta^-} = 4690(70)$ to $^{163}\text{Gd}^m$ at 137.8 keV										14Ha38 **
* $^{163}\text{Eu}(\beta^-)^{163}\text{Gd}$	$E_{\beta^-} = 4675(70)$ to $^{163}\text{Gd}^m$ at 137.8 keV										14Ha38 **
* $^{163}\text{Tb}(\beta^-)^{163}\text{Dy}$	$E_{\beta^-} = 800(50)$ to $1/2^+$ level at 884.2943 keV										Ens105 **
* $^{163}\text{Tb}(\beta^-)^{163}\text{Dy}$	$E_{\beta^-} = 1300(100)$ to $3/2^-$ level at 421.8439 keV										Ens105 **
* $^{163}\text{Ho}(\epsilon)^{163}\text{Dy}$	Original 2.58(0.10) from partial $T = 40(12) \times 10^{+3}$ y, re-evaluated by authors										84La.A **
* $^{163}\text{Ho}(\epsilon)^{163}\text{Dy}$	Original value 2.82(+0.11−0.08)										84La.A **
* $^{163}\text{Ho}(\epsilon)^{163}\text{Dy}$	Original value 2.60(0.03) corrected to 2.561(0.020) for dynamic effects										87Sp02 **
*	error 0.020 is statistical only										87Sp02 **
* $^{163}\text{Ho}(\epsilon)^{163}\text{Dy}$	Original $2616 < Q < 2694$ eV 68% CL for charge $66^+ Q_{\beta^+}$,										92Ju01 **
*	corrected to $2511 < Q_{\beta^+} < 2572$ eV 68% CL										93Bo.A **
* $^{163}\text{Tm}(\beta^+)^{163}\text{Er}$	$E_{\beta^+} = 884(3)$ to $1/2^+$ level at 540.56 keV, and other E_{β^+}										Ens105 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
* $^{163}\text{Yb}(\beta^+)^{163}\text{Tm}$	$E_{\beta^+}=1400(100)$ to $5/2^-$ level at 947.29 keV										Ens105 **
* $^{163}\text{Re}^m(\text{IT})^{163}\text{Re}$	Redundant with $^{167}\text{Ir}(\alpha)^{163}\text{Re}$ in same paper										GAu **
$^{164}\text{Sm}-^{84}\text{Kr}_{1.952}$	121306.5	4.4				2			CP2	1.0	18Or02
$^{164}\text{Eu}-^{136}\text{Xe}_{1.206}$	54753.5	3.9	54752.3	2.2	-0.3	1	32	32	^{164}Eu JY1	1.0	20Vi04
$^{164}\text{Eu}-^{84}\text{Kr}_{1.952}$	115608.8	2.7	115609.4	2.2	0.2	1	68	68	^{164}Eu CP2	1.0	20Or03
$^{164}\text{Gd}-^{84}\text{Kr}_{1.952}$	108672.5	1.2	108672.6	1.1	0.1	1	80	80	^{164}Gd CP2	1.0	18Or.A
$^{164}\text{Tb}-^{84}\text{Kr}_{1.952}$	106084.0	2.0				2			CP2	1.0	20Or03
$^{164}\text{Tb}^m-^{84}\text{Kr}_{1.952}$	106240	13				2			CP2	1.0	20Or03
$\text{C}_{13}\text{H}_8-^{164}\text{Dy}$	133320	38	133419.4	0.7	0.7	U			R04	4.0	64De15
$\text{C}_{12}\text{H}_7-^{164}\text{Dy}$	128920	34	128949.2	0.7	0.2	U			R04	4.0	64De15
$\text{C}_{12}\text{H}_6\text{N}-^{164}\text{Er}$	120876	39	120816.5	0.8	-0.4	U			R04	4.0	64De15
$^{164}\text{Tm}-u$	-66446	31	-66462	27	-0.5	1	75	75	^{164}Tm GS2	1.0	05Li24 *
$^{164}\text{Yb}-^{142}\text{Sm}_{1.155}$	32429	19	32434	16	0.3	2			MA7	1.0	01Bo59
$^{164}\text{Yb}-u$	-65690	104	-65499	16	1.8	U			GS1	1.0	00Ra23
	-65493	30			-0.2	R			GS2	1.0	05Li24
$^{164}\text{Lu}-u$	-58750	110	-58660	30	0.8	U			GS1	1.0	00Ra23
	-58661	30				2			GS2	1.0	05Li24
$^{164}\text{Hf}-u$	-55620	110	-55629	17	-0.1	U			GS1	1.0	00Ra23
	-55596	30			-1.1	1	32	32	^{164}Hf GS2	1.0	05Li24
$^{164}\text{Ta}-u$	-46466	30				2			GS2	1.0	05Li24
$^{164}\text{Gd}-^{171}\text{Yb}_{.959}$	-3025.2	2.4	-3025.7	1.1	-0.2	1	20	20	^{164}Gd JY1	1.0	18Vi02
$^{164}\text{Tb}-^{171}\text{Yb}_{.959}$	-5597.6	3.6	-5614.4	2.0	-4.7	B			JY1	1.0	18Vi02 *
$^{164}\text{Dy}-^{35}\text{Cl}-^{162}\text{Dy}-^{37}\text{Cl}$	5347	5	5326.43	0.11	-1.0	U			H12	4.0	64Ba15
	5589	19			-3.5	B			R04	4.0	64De15
	5321	3			0.7	U			H23	2.5	70Wh01
	5326.5	0.9			-0.0	U			H25	2.5	72Ba08
$^{164}\text{Er}-^{35}\text{Cl}-^{162}\text{Er}-^{37}\text{Cl}$	3373.3	1.3	3370.6	0.4	-0.8	U			H25	2.5	72Ba08
$^{164}\text{Er}-^{164}\text{Dy}$	26.92	0.12	26.92	0.12	-0.0	1	100	100	^{164}Er SH1	1.0	11El08
$^{164}\text{Dy}-^{35}\text{Cl}-^{161}\text{Dy}-^{37}\text{Cl}$	5610	48	5191.51	0.13	-2.2	U			R04	4.0	64De15
$^{163}\text{Dy}-^{37}\text{Cl}-^{164}\text{Dy}-^{35}\text{Cl}$	-3360	50	-3393.72	0.10	-0.2	U			R04	4.0	64De15
$^{164}\text{Dy}-^{163}\text{Dy}$	392	48	443.60	0.07	0.3	U			R04	4.0	64De15
	540	25			-1.0	U			R04	4.0	64De15
	446	28			-0.0	U			R04	4.0	64De15
$^{164}\text{Er}-^{162}\text{Er}$	556	48	420.4	0.4	-0.7	U			R04	4.0	64De15
$^{164}\text{W}(\alpha)^{160}\text{Hf}$	5281.7	5.	5278.3	2.0	-0.7	-					73Ea01 Z
	5274.7	5.			0.7	-			ORa		75To05 Z
	5268.7	10.			1.0	U					78Sc26 *
	5279.0	5.			-0.1	-			GSa		79Ho10
	5279.2	3.			-0.3	-			Ora		82De11 Z
	5283.0	8.			-0.6	U					84Sc06 *
	5277.0	6.			0.2	-			Daa		96Pa01
$^{164}\text{Re}(\alpha)^{160}\text{Ta}$	ave. 5278.6	2.0			-0.2	1	99	81	^{164}W		average
	5922.7	10.	5926	5	0.4	4			GSa		79Ho10
	5928.9	7.			-0.4	4			Daa		96Pa01
	5924.7	10.			0.2	4			Jya		09Ha42
$^{164}\text{Re}^m(\alpha)^{160}\text{Ta}^m$	5763.8	10.				5			Jya		09Ha42
$^{164}\text{Os}(\alpha)^{160}\text{W}$	6478.3	20.	6479	5	0.1	U			GSa		81Ho10
	6473.2	10.			0.6	-			ORa		96Bi07
	6479.4	7.			0.0	-			Daa		96Pa01
	ave. 6477	6			0.4	1	80	80	^{164}Os		average
$^{164}\text{Ir}^m(\alpha)^{160}\text{Re}^m$	7052.3	10.2				6			Jya		14Dr02
$^{162}\text{Dy}(\text{t,p})^{164}\text{Dy}$	5447.3	1.9	5447.32	0.08	0.0	U			McM		88Bu08 *
$^{164}\text{Dy}(\text{p,t})^{162}\text{Dy}$	-5450	5	-5447.32	0.08	0.5	U			Min		73Oo01
$^{164}\text{Er}(\text{p,t})^{162}\text{Er}$	-7262	10	-7269.2	0.3	-0.7	U			Min		73Oo01

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{164}\text{Dy}(t,\alpha)^{163}\text{Tb}$	11153	4				2			McM		92Ga15 *
$^{163}\text{Dy}(n,\gamma)^{164}\text{Dy}$	7658.11	0.07	7658.11	0.07	-0.0	1	99	73 ^{164}Dy	MMn		82Is05 Z
	7658.90	0.06			-13.2	C					99Fo.A
	7655.0	0.9			3.5	C			Bdn		06Fi.A
$^{163}\text{Dy}(d,p)^{164}\text{Dy}$	5434	5	5433.54	0.07	-0.1	U			Tal		64Sh06
	5441	10			-0.7	U			Kop		70Gr46
$^{164}\text{Dy}(d,t)^{163}\text{Dy}$	-1407	10	-1400.88	0.07	0.6	U			Kop		70Gr46
	-1407	10			0.6	U			Kop		70Gr46
$^{163}\text{Dy}(^3\text{He},d)^{164}\text{Ho}-^{164}\text{Dy}(\gamma)^{165}\text{Ho}$	-331.6	1.4	-330.7	1.1	0.6	1	67	67 ^{164}Ho	McM		75Bu02 *
$^{164}\text{Er}(d,t)^{163}\text{Er}$	-2593	10	-2589	5	0.4	1	21	21 ^{163}Er	Kop		69Tj01
$^{164}\text{Ir}^m(p)^{163}\text{Os}$	1828	8	1824	6	-0.5	o			Jyp		01Ke05
	1818	14			0.4	5			Arp		02Ma61
	1825	6			-0.2	5			Jyp		14Dr02
$^{164}\text{Eu}(\beta^-)^{164}\text{Gd}$	6430	70	6461.5	2.3	0.5	o			Kur		07Ha57
	6440	70			0.3	o			Kur		10Ha38
	6393	50			1.4	U			Kur		14Ha38
$^{164}\text{Tb}(\beta^-)^{164}\text{Dy}$	3890	100	3862.7	2.0	-0.3	U					71Gu18 *
$^{164}\text{Ho}(\beta^-)^{164}\text{Er}$	990	30	962.1	1.4	-0.9	U					54Br96
	965	20			-0.1	U					66Se07
$^{164}\text{Tm}(\beta^+)^{164}\text{Er}$	3985	20	4034	25	2.4	B					67Vr04 *
	3989	50			0.9	1	25	25 ^{164}Tm	IRS		94Po26 *
$^{164}\text{Lu}(\beta^+)^{164}\text{Yb}$	6390	140	6370	30	-0.1	U					83Ge08
	6250	90			1.3	o			IRS		83Vi.A
	6290	90			0.9	U			IRS		93Al03 *
	6255	120			1.0	U			Dbn		94Po26 *
* $^{164}\text{Tm}-u$	$M-A=-61884(28)$ keV for mixture gs+m at 20(12) keV										Nub211 **
* $^{164}\text{Tb}-^{171}\text{Yb}_{959}$	Could be a mixture of ground state and iso.										HWJ208 **
* $^{164}\text{W}(\alpha)^{160}\text{Hf}$	Originally assigned to ^{168}Re										AHW **
* $^{164}\text{W}(\alpha)^{160}\text{Hf}$	Originally assigned to ^{167}Re , re-assigned in reference										92Me10 **
*	original $E_\alpha=5136$ recalibrated using their $^{168}\text{Os}-^{170}\text{Os}$ results										GAU **
* $^{162}\text{Dy}(t,p)^{164}\text{Dy}$	$Q-Q(^{160}\text{Dy}(t,p))=-722.3(1.9)$ keV, see $^{162}\text{Dy}(p,t)$										88Bu08 **
* $^{164}\text{Dy}(t,\alpha)^{163}\text{Tb}$	$Q-Q(^{162}\text{Dy}(t,\alpha))=-123(4)+54-584=-653(4)$ keV										AHW **
* $^{163}\text{Dy}(^3\text{He},d)^{164}\text{Ho}-^{164}\text{Dy}(\gamma)^{165}\text{H}$	See erratum										75Bu02 **
* $^{164}\text{Tb}(\beta^-)^{164}\text{Dy}$	$E_{\beta^-}=1700(100)$ to 4^+ level at 2194.44 and 4^+ at 2205.63 keV, and other E_{β^-}										Ens017 **
* $^{164}\text{Tm}(\beta^+)^{164}\text{Er}$	$E_{\beta^+}=2940(20)$ 29 to ground state 10 to 2^+ level at 91.38 keV										Ens017 **
* $^{164}\text{Tm}(\beta^+)^{164}\text{Er}$	$E_{\beta^+}=2944(50)$ 29 to ground state 10 to 2^+ level at 91.38 keV										Ens017 **
* $^{164}\text{Lu}(\beta^+)^{164}\text{Yb}$	$Q_{\beta^+}=6250(90)$ partly to 2^+ level at 123.31 keV										Ens017 **
* $^{164}\text{Lu}(\beta^+)^{164}\text{Yb}$	$E_{\beta^+}=5191(120)$ partly to 2^+ level at 123.31 keV										Ens017 **
$^{165}\text{Eu}-^{136}\text{Xe}_{1.213}$	58091.2	6.5	58089	6	-0.4	1	74	74 ^{165}Eu	JY1	1.0	20Vi04
$^{165}\text{Eu}-^{84}\text{Kr}_{1.964}$	119352	11	119359	6	0.6	1	26	26 ^{165}Eu	CP2	1.0	18Or.A
$^{165}\text{Gd}-^{84}\text{Kr}_{1.964}$	113134.9	1.5	113135.5	1.4	0.4	1	87	87 ^{165}Gd	CP2	1.0	18Or.A
$^{165}\text{Tb}-^{136}\text{Xe}_{1.213}$	47497.3	4.2	47504.0	1.7	1.6	1	16	16 ^{165}Tb	JY1	1.0	20Vi04
$^{165}\text{Tb}-^{84}\text{Kr}_{1.964}$	108774.9	1.8	108773.7	1.7	-0.7	1	84	84 ^{165}Tb	CP2	1.0	18Or.A
$\text{C}_{13}\text{H}_9-^{165}\text{Ho}$	140043	29	140096.2	0.8	0.5	U			R04	4.0	64De15
$\text{C}_{12}\text{H}_7\text{N}-^{165}\text{Ho}$	127537	28	127520.1	0.8	-0.2	U			R04	4.0	64De15
$\text{C}_{11}\text{C}_6\text{H}_6\text{N}-^{165}\text{Ho}$	122970	50	123049.9	0.8	0.4	U			R04	4.0	64De15
$^{165}\text{Tm}-^{142}\text{Sm}_{1.162}$	30970	20	30968.5	2.9	-0.1	U			MA7	1.0	01Bo59
$^{165}\text{Yb}-u$	-64721	30	-64730	28	-0.3	1	90	90 ^{165}Yb	GS2	1.0	05Li24
$^{165}\text{Lu}-u$	-60602	30	-60593	28	0.3	1	90	90 ^{165}Lu	GS2	1.0	05Li24
$^{165}\text{Hf}-u$	-55360	140	-55430	30	-0.5	U			GS1	1.0	00Ra23
	-55433	30				2			GS2	1.0	05Li24
$^{165}\text{Ta}-u$	-49191	30	-49220	15	-1.0	1	24	24 ^{165}Ta	GS2	1.0	05Li24
$^{165}\text{W}-u$	-41720	30	-41719	28	0.0	1	85	85 ^{165}W	GS2	1.0	05Li24
$^{165}\text{Gd O}-^{171}\text{Yb}_{1.058}$	1597.3	3.9	1593.0	1.4	-1.1	1	13	13 ^{165}Gd	JY1	1.0	18Vi02

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{165}\text{Ho } ^{35}\text{Cl}-^{163}\text{Dy } ^{37}\text{Cl}$	4539	4	4542.0	0.8	0.3	U			H23	2.5	70Wh01
$^{165}\text{W}(\alpha)^{161}\text{Hf}$	5031.0	5.	5030	30	-0.0	-			ORa		75To05 Z
	5034.2	10.			-0.1	-					84Sc06 *
	ave. 5030	40			-0.0	1	27	15 ^{165}W			average
$^{165}\text{Re}(\alpha)^{161}\text{Ta}$	5629.6	6.	5694	6	10.8	B			Jya		05Sc22
	5694.3	6.1				6			Jya		12Th13
$^{165}\text{Re}^m(\alpha)^{161}\text{Ta}^m$	5631.7	10.	5660.9	2.8	2.9	U					78Sc26 *
	5643.0	10.			1.8	U			GSa		81Ho10
	5664.5	4.			-0.9	-			Ora		82De11 *
	5655.4	5.			1.1	-			Daa		96Pa01 *
	5657.4	5.			0.7	o			Jya		05Sc22
	5657.4	6.			0.6	-			Jya		12Th13
	ave. 5660.2	2.8			0.2	1	99	55 $^{165}\text{Re}^m$			average
$^{165}\text{Os}(\alpha)^{161}\text{W}$	6354.3	20.	6335	6	-0.9	5			Ora		78Ca11
	6317.4	10.			1.8	5			GSa		81Ho10
	6342.1	7.			-0.9	5			Daa		96Pa01
	6342.1	30.			-0.2	U			Jya		13Dr06 *
$^{165}\text{Ir}^m(\alpha)^{161}\text{Re}^m$	6882.1	7.	6879	6	-0.4	1	70	48 $^{165}\text{Ir}^m$	Ara		97Da07
$^{165}\text{Pt}(\alpha)^{161}\text{Os}$	7453.0	14.3				4			Jya		19Hi06
$^{163}\text{Dy}(\text{t,p})^{165}\text{Dy}$	4890.6	2.9	4892.27	0.09	0.6	U			McM		88Bu08 *
$^{164}\text{Dy}(\text{n},\gamma)^{165}\text{Dy}$	5716.36	0.20	5715.96	0.05	-2.0	U			ILn		79Br25 Z
	5715.96	0.06			-0.0	2			MMn		82Is05 Z
	5715.70	0.30			0.9	U			ILn		90Ka21 Z
	5715.95	0.12			0.1	2			Bdn		06Fi.A
$^{164}\text{Dy}(\text{d,p})^{165}\text{Dy}$	3488	5	3491.39	0.05	0.7	U			Tal		64Sh13
	3496	10			-0.5	U			Kop		70Gr46
$^{164}\text{Dy}(^3\text{He,d})^{165}\text{Ho}$	717.3	10.	725.9	0.7	0.9	U			McM		75Bu02
$^{165}\text{Ho}(\gamma,\text{n})^{164}\text{Ho}$	-8160	80	-7988.8	1.1	2.1	U			Phi		60Ge01
	-7987	2			-0.9	1	33	33 ^{164}Ho	MMn		85Ts01
$^{165}\text{Ho}(\text{d,t})^{164}\text{Ho}$	-1730	15	-1731.6	1.1	-0.1	U			Tal		70Jo11
$^{164}\text{Er}(\text{n},\gamma)^{165}\text{Er}$	6650.1	0.6	6650.1	0.6	-0.0	1	96	96 ^{165}Er			70Bo29 Z
$^{164}\text{Er}(\text{d,p})^{165}\text{Er}$	4431	10	4425.5	0.6	-0.5	U			Kop		69Tj01
$^{164}\text{Er}(\alpha,\text{t})^{165}\text{Tm}-^{168}\text{Er}()^{169}\text{Tm}$	-1298.0	2.0	-1298.0	1.5	0.0	1	55	48 ^{165}Tm	McM		75Bu02
$^{165}\text{Ir}^m(\text{p})^{164}\text{Os}$	1717.5	7.	1721	6	0.4	1	72	52 $^{165}\text{Ir}^m$	Ara		97Da07
$^{165}\text{Eu}(\beta^-)^{165}\text{Gd}$	5800	120	5797	5	-0.0	o			Kur		07Ha57
	5800	120			-0.0	o			Kur		10Ha38
	5729	65			1.0	U			Kur		14Ha38
$^{165}\text{Gd}(\beta^-)^{165}\text{Tb}$	4113	65	4063.1	2.0	-0.8	U			Kur		14Ha38
$^{165}\text{Dy}(\beta^-)^{165}\text{Ho}$	1305	20	1285.7	0.8	-1.0	U					59Bo52
	1285	10			0.1	U					63Pe11
$^{165}\text{Er}(\epsilon)^{165}\text{Ho}$	370	10	376.7	1.0	0.7	U					63Ry01
	371	6			0.9	U					63Zy01
$^{165}\text{Tm}(\beta^+)^{165}\text{Er}$	1591.3	2.0	1591.3	1.5	0.0	1	55	52 ^{165}Tm			82Vy03 *
$^{165}\text{Yb}(\beta^+)^{165}\text{Tm}$	2762	20	2635	27	-6.4	B					67Pa04 *
$^{165}\text{Lu}(\beta^+)^{165}\text{Yb}$	4250	140	3850	40	-2.8	B					83Ge08
	3920	80			-0.8	o			IRS		83Vi.A
	3920	80			-0.8	1	20	10 ^{165}Yb	IRS		93Al03
* $^{165}\text{W}(\alpha)^{161}\text{Hf}$	Originally assigned to ^{168}Re , re-assigned in reference										92Me10 **
	original $E_\alpha=4894$ recalibrated using their $^{168}\text{Os}-^{170}\text{Os}$ results										GAu **
* $^{165}\text{Re}^m(\alpha)^{161}\text{Ta}^m$	Originally assigned to ^{166}Re										AHW **
* $^{165}\text{Re}^m(\alpha)^{161}\text{Ta}^m$	Originally assigned to ^{166}Re										AHW **
* $^{165}\text{Re}^m(\alpha)^{161}\text{Ta}^m$	Due to a high spin isomer										99Po09 **
* $^{165}\text{Os}(\alpha)^{161}\text{W}$	Error not given, estimated by evaluator										GAu **
* $^{163}\text{Dy}(\text{t,p})^{165}\text{Dy}$	$Q-Q(^{162}\text{Dy}(\text{t,p}))=-556.6(2.9)$ keV										AHW **
* $^{165}\text{Tm}(\beta^+)^{165}\text{Er}$	$E_{\beta^+}=272(2)$ to $1/2^-$ level at 297.371 keV										Ens066 **
* $^{165}\text{Yb}(\beta^+)^{165}\text{Tm}$	$E_{\beta^+}=1580(20)$ to $7/2^-$ level at 160.47 keV										Ens066 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{166}\text{Gd}-^{136}\text{Xe}_{1.221}$	54904.1	3.9	54921.5	1.7	4.5	B			JY1	1.0	18Vi02
$^{166}\text{Gd}-^{84}\text{Kr}_{1.976}$	116510.9	1.7				2			CP2	1.0	18Or.A
$^{166}\text{Tb}-^{136}\text{Xe}_{1.221}$	51232.1	4.1	51230.9	1.6	-0.3	1	15	15	^{166}Tb	1.0	20Vi04
$^{166}\text{Tb}-^{84}\text{Kr}_{1.976}$	112820.0	1.7	112820.2	1.6	0.1	1	85	85	^{166}Tb	1.0	20Or03
$\text{C}_{12}\text{H}_8\text{N}-^{166}\text{Er}$	135376	29	135373.2	0.4	-0.0	U			R04	4.0	64De15
	135420	60			-0.2	U			R04	4.0	64De15
$\text{C}_{13}\text{H}_{10}-^{166}\text{Er}$	147740	60	147949.3	0.4	0.9	U			R04	4.0	64De15
$^{166}\text{Lu}-\text{u}$	-60157	108	-60140	30	0.1	U			GS1	1.0	00Ra23 *
	-60141	32				2			GS2	1.0	05Li24 *
$^{166}\text{Hf}-\text{u}$	-57860	110	-57820	30	0.4	U			GS1	1.0	00Ra23
	-57820	30				2			GS2	1.0	05Li24
$^{166}\text{Ta}-\text{u}$	-49488	30				2			GS2	1.0	05Li24
$^{166}\text{W}-\text{u}$	-44957	30	-44968	10	-0.4	1	11	11	^{166}W	1.0	05Li24
$^{166}\text{Er}-^{35}\text{Cl}-^{164}\text{Er}-^{37}\text{Cl}$	4040.9	1.4	4043.4	0.8	0.7	U			H25	2.5	72Ba08
$^{166}\text{Er}-^{164}\text{Er}$	1214	46	1093.3	0.8	-0.7	U			R04	4.0	64De15
	1110	80			-0.1	U			R04	4.0	64De15
$^{166}\text{W}(\alpha)^{162}\text{Hf}$	4856.0	5.	4856	4	-0.0	-			ORa		75To05 Z
	4855.0	10.			0.1	-			GSa		79Ho10 Z
	4858.3	8.2			-0.3	-					89Hi04
$^{166}\text{Re}(\alpha)^{162}\text{Ta}$	ave. 4856	4			-0.1	1	97	78	^{166}W		average
	5461.8	10.	5520	60	1.1	5					78Sc26 *
	5574.5	3.			-1.1	5			Ora		82De11 *
	5637.0	13.			-2.3	B			Bea		92Me10 *
	5669.9	10.			-3.0	B			Daa		96Pa01 *
$^{166}\text{Os}(\alpha)^{162}\text{W}$	6148.5	20.	6143	3	-0.3	U					77Ca23
	6129.0	6.			2.2	-			GSa		81Ho10
	6148.5	6.			-0.9	-			Daa		96Pa01
	6148.4	5.1			-1.1	-			Jya		15Li24
$^{166}\text{Ir}(\alpha)^{162}\text{Re}$	ave. 6143	3			-0.0	1	100	100	^{166}Os		average
	6702.8	20.	6722	6	1.0	U			GSa		81Ho10
	6724.3	6.			-0.3	7			Ara		97Da07
	6713.1	13.			0.7	7			Jya		04Ke06 *
$^{166}\text{Ir}^m(\alpha)^{162}\text{Re}^m$	6718.2	11.	6719	4	0.0	8			Daa		96Pa01 *
	6723.3	5.			-0.9	8			Ara		97Da07
	6706.9	8.2			1.4	8			Jya		04Ke06
$^{166}\text{Pt}(\alpha)^{162}\text{Os}$	7285.9	15.	7292	7	0.4	5			ORa		96Bi07
	7294.1	8.2			-0.2	5			Jya		19Hi06
$^{164}\text{Dy}(\text{t,p})^{166}\text{Dy}$	4276.4	4.4	4277.7	0.4	0.3	U			McM		88Bu08 *
$^{166}\text{Er}(\text{p,t})^{164}\text{Er}$	-6641	5	-6642.4	0.7	-0.3	U			Min		73Oo01
$^{165}\text{Dy}(\text{n},\gamma)^{166}\text{Dy}$	7043.5	0.4				3					83Ke.A
$^{165}\text{Ho}(\text{n},\gamma)^{166}\text{Ho}$	6243.69	0.06	6243.640	0.020	-0.8	U			MMn		82Is05 Z
	6243.64	0.02			-0.0	1	100	51	^{165}Ho		84Ke15 Z
	6243.68	0.13			-0.3	U			Bdn		06Fi.A
$^{165}\text{Ho}(\text{d,p})^{166}\text{Ho}$	4025	7	4019.073	0.020	-0.8	U			Tal		65St06
$^{166}\text{Er}(\text{d,t})^{165}\text{Er}$	-2218	10	-2216.9	1.0	0.1	U			Kop		69Tj01
$^{166}\text{Ir}(\text{p})^{165}\text{Os}$	1152.0	8.0				6			Ara		97Da07
$^{166}\text{Ir}^m(\text{p})^{165}\text{Os}$	1324.1	8.	1323	10	-0.1	o			Ara		97Da07 *
$^{166}\text{Eu}(\beta^-)^{166}\text{Gd}$	7322	300	7620#	100#	1.0	D			Kur		14Ha38
$^{166}\text{Tb}(\beta^-)^{166}\text{Dy}$	4830	100	4775.7	1.7	-0.5	o			Kur		02Sh.A
	4695	70			1.2	o			Kur		07Ha57
	4700	70			1.1	U			Kur		10Ha38
$^{166}\text{Dy}(\beta^-)^{166}\text{Ho}$	483	5	485.9	0.9	0.6	U					60He09 *
$^{166}\text{Ho}(\beta^-)^{166}\text{Er}$	1859	3	1853.8	0.8	-1.7	-					63Fu17
	1857	3			-1.1	-					66Da04
	1854.7	1.5			-0.6	-					74Gr41
	1851.6	2.0			1.1	-					83Ra.A
$^{166}\text{Tm}(\beta^+)^{166}\text{Er}$	ave. 1854.7	1.0			-0.8	1	56	51	^{166}Ho		average
	3043	20	3038	12	-0.3	2					61Gr33 *
	3031	20			0.3	2					61Zy02 *
	3039	20			-0.1	2					63Pr13 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{166}\text{Yb}(\varepsilon)^{166}\text{Tm}$	280	40	293	14	0.3	U					Averag *	
$^{166}\text{Lu}(\beta^+)^{166}\text{Yb}$	5480	160	5570	30	0.6	U					74De09 *	
$^{166}\text{Ir}^m(\text{IT})^{166}\text{Ir}$	171.5	6.1				7			Ara		97Da07	
* $^{166}\text{Lu}-\text{u}$	$M-A=-56010(100)$ keV for mixture gs+m+n at 34.37 and 43.0 keV										Nub211 **	
* $^{166}\text{Lu}-\text{u}$	$M-A=-55995(28)$ keV for mixture gs+m+n at 34.37 and 43.0 keV										Nub211 **	
* $^{166}\text{Re}(\alpha)^{162}\text{Ta}$	Originally assigned to ^{167}Re										AHW **	
* $^{166}\text{Re}(\alpha)^{162}\text{Ta}$	Assignment uncertain, no other obvious attribution										AHW **	
* $^{166}\text{Re}(\alpha)^{162}\text{Ta}$	Originally assigned to ^{167}Re										AHW **	
* $^{166}\text{Re}(\alpha)^{162}\text{Ta}$	Assignment tentative, may be ^{165}Re										92Me10 **	
* $^{166}\text{Re}(\alpha)^{162}\text{Ta}$	Correlated to a ^{170}Ir 6003 line; assignment uncertain										GAu **	
* $^{166}\text{Ir}(\alpha)^{162}\text{Re}$	All Q_α of reference increased by 7 keV for calibration error										04Ke06 **	
* $^{166}\text{Ir}^m(\alpha)^{162}\text{Re}^m$	Correlated with $E_\alpha=6123$ of $^{162}\text{Re}^m$										96Pa01 **	
* $^{164}\text{Dy}(\text{t,p})^{166}\text{Dy}$	$Q-Q(^{162}\text{Dy}(\text{t,p}))=-1170.8(4.4)$ keV										AHW **	
* $^{166}\text{Ir}^m(\text{p})^{165}\text{Os}$	Replaced by author's value for $^{166}\text{Ir}^m(\text{IT})^{166}\text{Ir}$										97Da07 **	
* $^{166}\text{Eu}(\beta^-)^{166}\text{Gd}$	Trends from Mass Surface TMS suggest ^{166}Eu 300 keV less bound										GAu **	
* $^{166}\text{Dy}(\beta^-)^{166}\text{Ho}$	$E_{\beta^-}=402(5)$ to 1^- level at 82.47 keV, and other E_{β^-}										Ens084 **	
* $^{166}\text{Tm}(\beta^+)^{166}\text{Er}$	$E_{\beta^+}=1940(20)$ 1928(20) 1936(20) respectively, to 2^+ level at 80.5776 keV										Ens084 **	
* $^{166}\text{Yb}(\varepsilon)^{166}\text{Tm}$	Average $\text{pK}=0.712(0.038)$ to 1^+ level at 82.298 keV from 2 references:										Ens084 **	
*	$\text{pK}=0.74(0.05)$ to 82.298 level										63Ja06 **	
*	$\text{pK}=0.675(0.059)$ to 82.298 level										73De22 **	
* $^{166}\text{Lu}(\beta^+)^{166}\text{Yb}$	$E_{\beta^+}=2225(160)$ to $(6^-, 7^-)$ level at 2233.36 keV										Ens084 **	
$^{167}\text{Gd}-^{136}\text{Xe}_{1.228}$	59422	13	59431	6	0.7	1	19	19	^{167}Gd	JY1	1.0	20Vi04
$^{167}\text{Gd}-^{84}\text{Kr}_{1.988}$	121434.5	6.2	121433	6	-0.3	1	81	81	^{167}Gd	CP2	1.0	18Or.A
$^{167}\text{Tb}-^{136}\text{Xe}_{1.228}$	53947.0	4.1	53947.7	2.1	0.2	1	26	26	^{167}Tb	JY1	1.0	20Vi04
$^{167}\text{Tb}-^{84}\text{Kr}_{1.988}$	115949.8	2.4	115949.6	2.1	-0.1	1	74	74	^{167}Tb	CP2	1.0	18Or.A
$^{167}\text{Dy}-^{82}\text{Kr}_{2.037}$	111921.3	4.3				2				CP2	1.0	18Or.A
$\text{C}_{13}\text{H}_{11}-^{167}\text{Er}$	153840	130	154019.2	0.3	0.3	U				R04	4.0	64De15
	154040.4	6.2			-0.9	U				M23	4.0	79Ha32
$\text{C}_{12}\text{H}_9\text{N}-^{167}\text{Er}$	141480	27	141443.1	0.3	-0.3	U				R04	4.0	64De15
	141520	50			-0.4	U				R04	4.0	64De15
$^{167}\text{Lu}-\text{u}$	-61757	40				2				GS2	1.0	05Li24 *
$^{167}\text{Hf}-\text{u}$	-57490	110	-57400	30	0.8	U				GS1	1.0	00Ra23
	-57400	30				2				GS2	1.0	05Li24
$^{167}\text{Ta}-\text{u}$	-51870	120	-51910	30	-0.3	U				GS1	1.0	00Ra23
	-51907	30				2				GS2	1.0	05Li24
$^{167}\text{W}-\text{u}$	-45175	30	-45189	20	-0.5	R				GS2	1.0	05Li24
$^{167}\text{Er}-^{35}\text{Cl}-^{165}\text{Ho}-^{37}\text{Cl}$	4666	3	4677.2	0.8	1.5	U				H23	2.5	70Wh01
	4679.5	1.2			-0.8	U				H25	2.5	72Ba08
$^{167}\text{Er}-^{166}\text{Er}$	1722	31	1755.13	0.19	0.3	U				R04	4.0	64De15
$^{167}\text{W}(\alpha)^{163}\text{Hf}$	4661.9	20.	4751	30	1.7	-						89Me02
	4671.1	13.			1.6	-						91Me05
ave.	4670	40			1.4	1	23	15	^{163}Hf			average
$^{167}\text{Re}(\alpha)^{163}\text{Ta}^m$	5138.3	12.				12			Bea			92Me10
$^{167}\text{Re}^m(\alpha)^{163}\text{Ta}$	5408.8	3.	5407.0	2.9	-0.6	4			Ora			82De11 *
	5397.5	10.			0.9	4			ChR			84Sc06 *
	5392.4	12.			1.2	4			Bea			92Me10
$^{167}\text{Os}(\alpha)^{163}\text{W}$	5983.6	5.	5980	60	0.0	6			GSa			81Ho10 Z
	5978.7	2.			0.1	6			Ora			82De11 Z
	5996.9	5.			-0.2	6			Daa			96Pa01
	5979.5	5.			0.1	6			Bka			02Ro17
$^{167}\text{Ir}(\alpha)^{163}\text{Re}$	6507.1	5.	6504.9	2.6	-0.4	2			Ara			97Da07
	6504.0	3.			0.3	2			Jya			05Sc22
$^{167}\text{Ir}^m(\alpha)^{163}\text{Re}^m$	6543.0	10.	6561	3	1.7	-			GSa			81Ho10
	6567.6	11.			-0.6	-			Daa			96Pa01
	6567.6	5.			-1.4	-			Ara			97Da07
	6551.2	7.2			1.3	-			Jya			04Ke06
	6561.5	6.			-0.1	-			Jya			05Sc22
ave.	6561	3			0.0	1	100	100	$^{163}\text{Re}^m$			average
$^{167}\text{Pt}(\alpha)^{163}\text{Os}$	7159.8	10.	7160	60	-0.0	5			ORa			96Bi07
	7150.6	10.			0.1	o			Jya			04Ke06

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{167}\text{Pt}(\alpha)^{163}\text{Os}$	7156.4	8.2	7160	60	0.0	5			Jya		19Hi06
$^{167}\text{Er}(\text{p,t})^{165}\text{Er}$	-6427	6	-6428.7	0.9	-0.3	U			Min		73Oo01
	-6430	5			0.3	U					75St08
$^{166}\text{Er}(\text{n},\gamma)^{167}\text{Er}$	6436.35	0.50	6436.43	0.18	0.2	-					70Bo29 Z
	6436.51	0.40			-0.2	-					70Mi01 Z
	6436.46	0.22			-0.1	-			Bdn		06Fi.A
$^{167}\text{Er}(\gamma,\text{n})^{166}\text{Er}$	-6560	80	-6436.43	0.18	1.5	U			Phi		60Ge01
$^{166}\text{Er}(\text{d,p})^{167}\text{Er}$	4209	10	4211.86	0.18	0.3	U			Tal		68Ha10
	4214	10			-0.2	U			Kop		69Tj01
$^{167}\text{Er}(\text{d,t})^{166}\text{Er}$	-189	12	-179.20	0.18	0.8	U			Kop		69Bu01
$^{166}\text{Er}(\text{n},\gamma)^{167}\text{Er}$	ave.	6436.46	0.18	6436.43	0.18	-0.1	1	99	^{166}Er		average
$^{166}\text{Er}(\alpha,\text{t})^{167}\text{Tm}-^{168}\text{Er}(\gamma)^{169}\text{Tm}$	-666.5	1.0	-666.5	1.0	0.0	1	99	^{167}Tm	McM		75Bu02
$^{167}\text{Ir}(\text{p})^{166}\text{Os}$	1070.5	6.	1070	4	-0.1	-					97Da07
	1068.5	6.			0.3	-			Jyp		05Sc22 *
	ave.	1070	4		0.1	1	77	^{167}Ir			average
$^{167}\text{Ir}^m(\text{p})^{166}\text{Os}$	1245.5	7.	1246	4	0.0	o					97Da07 *
$^{167}\text{Dy}(\beta^-)^{167}\text{Ho}$	2350	60	2368	7	0.3	U					77Tu01 *
$^{167}\text{Ho}(\beta^-)^{167}\text{Er}$	970	20	1010	5	2.0	U					68Fu07
$^{167}\text{Yb}(\beta^+)^{167}\text{Tm}$	1954	4	1953	4	-0.2	1	90	^{167}Yb			77Kr.A *
$^{167}\text{Lu}(\beta^+)^{167}\text{Yb}$	3130	100	3060	40	-0.7	U					64Ag.A *
$^{167}\text{W}(\beta^+)^{167}\text{Ta}$	5620	270	6260	30	2.4	U			Got		89Me02
$^{167}\text{Ir}^m(\text{IT})^{167}\text{Ir}$	175.3	2.2	175.5	2.1	0.1	1	94	$^{167}\text{Ir}^m$	Ara		97Da07
$^{167}\text{Lu}-\text{u}$	$M-A=-57501(28)$ keV for mixture gs+m at 50#40 keV										Nub211 **
$^{167}\text{Re}^m(\alpha)^{163}\text{Ta}$	Original assignment to ^{168}Re changed in reference										92Me10 **
$^{167}\text{Re}^m(\alpha)^{163}\text{Ta}$	Original assignment to $^{168}\text{Re}^m$ changed in reference										92Me10 **
*	original $E_\alpha=5250$ recalibrated using their $^{168}\text{Os}-^{170}\text{Os}$ results										GAu **
$^{167}\text{Ir}(\text{p})^{166}\text{Os}$	$E_p=1062(6)$; also $E_p=1248(7)$ from $^{167}\text{Ir}^m$										05Sc22 **
$^{167}\text{Ir}^m(\text{p})^{166}\text{Os}$	Replaced by author's value for $^{167}\text{Ir}^m(\text{IT})^{167}\text{Ir}$										97Da07 **
$^{167}\text{Dy}(\beta^-)^{167}\text{Ho}$	$E_{\beta^-}=1780(60)$ to $3/2^-$ level at 569.69 keV										Ens008 **
$^{167}\text{Yb}(\beta^+)^{167}\text{Tm}$	$E_{\beta^+}=639(4)$ to $7/2^-$ level at 292.820 keV										Ens008 **
$^{167}\text{Lu}(\beta^+)^{167}\text{Yb}$	$E_{\beta^+}=2060(100)$ to $5/2^+$ level at 29.658, $7/2^-$ at 78.671 keV										Ens008 **
$^{168}\text{Tb}-^{136}\text{Xe}_{1,235}$	57927.2	4.5				2			JY1	1.0	20Vi04
$\text{C}_{13}\text{H}_{12}-^{168}\text{Er}$	161543.3	5.1	161522.10	0.28	-1.0	U			M23	4.0	79Ha32
$\text{C}_{12}\text{H}_{10}\text{N}-^{168}\text{Er}$	148884	44	148946.04	0.28	0.4	U			R04	4.0	64De15
$\text{C}_{11}\text{C}_2\text{H}_9\text{N}-^{168}\text{Er}$	144524	29	144475.84	0.28	-0.4	U			R04	4.0	64De15
$\text{C}_{12}\text{H}_{10}\text{N}-^{168}\text{Yb}$	147010	100	147433.03	0.10	1.1	U			R04	4.0	64De15
$^{168}\text{Lu}-\text{u}$	-61253	55	-61270	40	-0.3	1	55	^{168}Lu	GS2	1.0	05Li24 *
$^{168}\text{Lu}^m-^{133}\text{Cs}_{1,263}$	58320.7	6.2				2			MA8	1.0	19Hu15
$^{168}\text{Hf}-\text{u}$	-59560	104	-59430	30	1.2	U			GS1	1.0	00Ra23
	-59432	30				2			GS2	1.0	05Li24
$^{168}\text{Ta}-\text{u}$	-52020	110	-51950	30	0.6	U			GS1	1.0	00Ra23
	-51953	30				2			GS2	1.0	05Li24
$^{168}\text{W}-\text{u}$	-48181	30	-48195	14	-0.5	1	23	^{168}W	GS2	1.0	05Li24
$^{168}\text{Yb}^{35}\text{Cl}_2-^{164}\text{Dy}^{37}\text{Cl}_2$	10612.8	8.7	10610.7	0.8	-0.1	U			H27	2.5	74Ba90
$^{168}\text{Er}^{35}\text{Cl}-^{166}\text{Er}^{37}\text{Cl}$	5037	50	5027.34	0.24	-0.1	U			R08	1.5	69De19
	5026	3			0.2	U			H23	2.5	70Wh01
	5028.9	1.5			-0.4	U			H25	2.5	72Ba08
$^{168}\text{Yb}-^{170}\text{Yb}_{988}$	-1658.726	0.100	-1658.74	0.10	-0.1	1	99	^{168}Yb	JY2	1.0	20Ne.1
$^{168}\text{Yb}-^{168}\text{Er}$	1512.91	0.27	1513.02	0.26	0.4	1	96	^{168}Er	SH1	1.0	11El04
$^{168}\text{Er}-^{167}\text{Er}$	284	31	322.09	0.13	0.3	U			R04	4.0	64De15
	320.9	4.3			0.1	U			M24	2.5	79Ha32
$^{168}\text{W}(\alpha)^{164}\text{Hf}$	4506.5	12.	4501	11	-0.5	1	87	^{164}Hf			91Me05
$^{168}\text{Re}(\alpha)^{164}\text{Ta}$	5063	13				3			Bea		92Me10 *
$^{168}\text{Os}(\alpha)^{164}\text{W}$	5819.0	3.	5815.6	2.7	-1.1	-			Ora		82De11 Z
	5800.4	8.			1.9	-					84Sc06
	5812.7	8.			0.4	-					95Hi02
	ave.	5816.3	2.7		-0.2	1	99	^{168}Os			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{168}\text{Ir}(\alpha)^{164}\text{Re}$	6410.9	5.	6381	9	-5.9	B			Ora		82De11
	6379.2	15.			0.1	5			Daa		96Pa01
	6382.2	10.			-0.1	5			Jya		09Ha42
$^{168}\text{Ir}^m(\alpha)^{164}\text{Re}^m$	6477.5	8.	6476	6	-0.1	6			Daa		96Pa01
	6474.4	10.			0.2	6			Jya		09Ha42 *
$^{168}\text{Pt}(\alpha)^{164}\text{Os}$	6990.8	20.	6990	3	-0.1	U			GSa		81Ho10
	6998.9	10.			-0.9	U			ORa		96Bi07
	6986.7	8.			0.4	o			Jya		04Ke06
	6989.7	3.1				2			Jya		09Go16
$^{168}\text{Er}(\text{p,t})^{166}\text{Er}$	-5723	6	-5725.93	0.21	-0.5	U			Min		73Oo01
$^{168}\text{Yb}(\text{p,t})^{166}\text{Yb}$	-7647	7				2			Min		73Oo01
$^{167}\text{Er}(\text{n},\gamma)^{168}\text{Er}$	7771.43	0.40	7771.29	0.12	-0.3	-					70Mi01 Z
	7771.05	0.20			1.2	-			ILn		79Br25 Z
	7771.6	1.0			-0.3	U			ORn		84Ka22
	7771.0	0.5			0.6	U					85Va.A
	7771.45	0.16			-1.0	-			Bdn		06Fi.A
$^{167}\text{Er}(\text{d,p})^{168}\text{Er}$	5541	6	5546.73	0.12	1.0	U			Tal		67Ha25
$^{168}\text{Er}(\text{d,t})^{167}\text{Er}$	-1523	10	-1514.06	0.12	0.9	U			Kop		69Tj01
$^{167}\text{Er}(\text{n},\gamma)^{168}\text{Er}$	ave. 7771.31	0.12	7771.29	0.12	-0.1	1	99	96 ^{167}Er			average
$^{167}\text{Er}(\alpha,\text{t})^{168}\text{Tm}-^{168}\text{Er}()^{169}\text{Tm}$	-262.3	1.5	-262.3	1.5	0.0	1	100	100 ^{168}Tm	McM		75Bu02
$^{168}\text{Yb}(\text{d,t})^{167}\text{Yb}$	-2797	12	-2804	4	-0.6	1	11	11 ^{167}Yb	Kop		66Bu16
$^{168}\text{Ho}(\beta^-)^{168}\text{Er}$	2740	100	2930	30	1.9	U					73Ka07 *
	2930	30				2					90Ch37
$^{168}\text{Lu}(\beta^+)^{168}\text{Yb}$	4475	80	4510	40	0.4	o					70Ch28 *
	4475	80			0.4	-					72Ch44 *
	4500	80			0.1	-			IRS		83Vi.A
	ave. 4490	60			0.3	1	45	45 ^{168}Lu			average
$^{168}\text{Lu}^m(\beta^+)^{168}\text{Yb}$	4696	100	4672	6	-0.2	U					72Ch44 *
* $^{168}\text{Lu}-\text{u}$	$M-A=-56922(28)$ keV for mixture gs+m at 130(40) keV										Nub211 **
* $^{168}\text{Re}(\alpha)^{164}\text{Ta}$	$E_\alpha=4833(13)$ to level at 111.5 keV										Ens089 **
* $^{168}\text{Ir}^m(\alpha)^{164}\text{Re}^m$	$E_\alpha=6320(10), 6260(10)$ to ground state and level at 69 keV										09Ha42 **
* $^{168}\text{Ho}(\beta^-)^{168}\text{Er}$	$E_{\beta^-}=1900(100)$ to 2^+ level at 821.17 and 3^+ at 895.79 keV										Ens108 **
* $^{168}\text{Lu}(\beta^+)^{168}\text{Yb}$	$E_{\beta^+}=1230(80)$ to 2222.37 level										Ens108 **
* $^{168}\text{Lu}^m(\beta^+)^{168}\text{Yb}$	$E_{\beta^+}=1470(100)$ to 4^+ level at 2203.84 keV										Ens108 **
$\text{C}_{12}\text{H}_{11}\text{N}-^{169}\text{Tm}$	154920	60	154930.4	0.8	0.0	U			R04	4.0	64De15
$^{169}\text{Lu}-\text{u}$	-62362	31	-62354	3	0.3	U			GS2	1.0	05Li24 *
$^{169}\text{Hf}-\text{u}$	-58741	30				2			GS2	1.0	05Li24
$^{169}\text{Ta}-\text{u}$	-53960	110	-53990	30	-0.3	U			GS1	1.0	00Ra23
	-53989	30				2			GS2	1.0	05Li24
$^{169}\text{W}-\text{u}$	-48195	30	-48221	17	-0.9	1	31	31 ^{169}W	GS2	1.0	05Li24
$^{169}\text{Re}-\text{u}$	-41203	63	-41234	12	-0.5	U			GS2	1.0	05Li24 *
$^{169}\text{Tm}\ ^{35}\text{Cl}_2-^{165}\text{Ho}\ ^{37}\text{Cl}_2$	9793.0	1.1	9790.1	1.1	-1.1	1	16	8 ^{165}Ho	H25	2.5	72Ba08
$^{169}\text{Tm}\ ^{35}\text{Cl}-^{167}\text{Er}\ ^{37}\text{Cl}$	5107	3	5112.9	0.8	0.8	U			H23	2.5	70Wh01
	5113.2	1.1			-0.1	1	9	8 ^{169}Tm	H25	2.5	72Ba08
$^{169}\text{Re}(\alpha)^{165}\text{Ta}^m$	4989.3	12.				5			Bea		92Me10 *
$^{169}\text{Re}^m(\alpha)^{165}\text{Ta}$	5189.1	3.	5189	3	-0.1	1	99	76 ^{165}Ta	Ora		82De11
	5191.1	10.			-0.2	U			ChR		84Sc06 *
	5184.0	10.			0.5	U			Bea		92Me10
$^{169}\text{Os}(\alpha)^{165}\text{W}$	5717.6	4.	5713	3	-1.0	2			Ora		82De11
	5699.2	8.			1.7	2					84Sc06 *
	5713	8			0.1	2					95Hi02 *
	5711.5	8.			0.2	2			Daa		96Pa01
$^{169}\text{Ir}(\alpha)^{165}\text{Re}$	6150.8	8.	6141	4	-1.2	5			Ara		99Po09
	6138.5	4.			0.6	5			Jya		05Sc22
	6165	14			-1.7	U			Jya		12Th13

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{169}\text{Ir}^m(\alpha)^{165}\text{Re}^m$	6276.0	3.	6266.5	2.9	-3.2	B			Ora		82De11 Z
	6258.4	10.			0.8	U			GSa		84Sc.A
	6267.6	9.			-0.1	-			Daa		96Pa01
	6254.3	5.			2.4	B			Ara		99Po09
	6265.6	3.			0.3	-			Jya		05Sc22
	6268	14			-0.1	U			Jya		12Th13
	ave. 6265.8	2.9			0.2	1	99	54 $^{169}\text{Ir}^m$			average
$^{169}\text{Pt}(\alpha)^{165}\text{Os}$	6840.2	15.	6858	5	1.2	U			GSa		81Ho10
	6860.7	23.			-0.1	U			Daa		96Pa01
	6853.5	8.2			0.5	o			Jya		04Ke06
	6857.6	5.1				6			Jya		09Go16
$^{168}\text{Er}(n,\gamma)^{169}\text{Er}$	6002.5	0.7	6003.25	0.15	1.1	U					70Bo29 Z
	6003.5	0.3			-0.8	2					70Mu15 Z
	6003.16	0.18			0.5	2			Bdn		06Fi.A
$^{168}\text{Er}(d,p)^{169}\text{Er}$	3773	12	3778.68	0.15	0.5	U			Tal		68Ha10
	3781	10			-0.2	U			Kop		69Tj01
$^{168}\text{Er}(\alpha,t)^{169}\text{Tm}$	-14244.8	10.	-14239.5	0.8	0.5	U			McM		75Bu02
$^{169}\text{Tm}(\gamma,n)^{168}\text{Tm}$	-8110	50	-8033.6	1.5	1.5	U			Phi		60Ge01
$^{169}\text{Tm}(d,t)^{168}\text{Tm}$	-1775	6	-1776.4	1.5	-0.2	U			Pit		73Ko06
$^{168}\text{Yb}(n,\gamma)^{169}\text{Yb}$	6866.8	0.4	6866.98	0.15	0.4	2					68Mi08 Z
	6867.2	0.4			-0.6	2					68Sh12 Z
	6866.97	0.18			0.0	2			Bdn		06Fi.A
$^{168}\text{Yb}(d,p)^{169}\text{Yb}$	4636	12	4642.41	0.15	0.5	U			Kop		66Bu16
$^{169}\text{Dy}(\beta^-)^{169}\text{Ho}$	3200	300				3			LBL		90Ch34
$^{169}\text{Ho}(\beta^-)^{169}\text{Er}$	2070	100	2125	20	0.6	U					63Mi17 *
$^{169}\text{Er}(\beta^-)^{169}\text{Tm}$	343.8	3.	353.5	0.8	3.2	B					56Bi30 *
	347.8	5.			1.1	U					65Du02 *
$^{169}\text{Yb}(e)^{169}\text{Tm}$	913	12	899.1	0.8	-1.2	U					86Ad07 *
	900	100			-0.0	U					87Sa53 *
$^{169}\text{Lu}(\beta^+)^{169}\text{Yb}$	2293	3				3					77Bo31
$^{169}\text{Hf}(\beta^+)^{169}\text{Lu}$	3365	200	3366	28	0.0	U					69Ar23 *
	3250	90			1.3	U					73Me09 *
* $^{169}\text{Lu}-u$	$M-A=-58075(28)$ keV for mixture gs+m at 29.0 keV										Nub211 **
* $^{169}\text{Re}-u$	$M-A=-38293(29)$ keV for mixture gs+m at 175(13) keV										Nub211 **
* $^{169}\text{Re}(\alpha)^{165}\text{Ta}^m$	$E_\alpha=4871(12)$, and a stronger $E_\alpha=4700(12)$										92Me10 **
* $^{169}\text{Re}^m(\alpha)^{165}\text{Ta}$	Original $E_\alpha=5050$ recalibrated using their $^{168}\text{Os}-^{170}\text{Os}$ results										GAu **
* $^{169}\text{Os}(\alpha)^{165}\text{W}$	Used to recalibrate other results in same reference										GAu **
* $^{169}\text{Os}(\alpha)^{165}\text{W}$	$E_\alpha=5578(8)$, $5536(10)$ to ground state, $(3/2^-)$ level at 43 keV										Ens066 **
* $^{169}\text{Ho}(\beta^-)^{169}\text{Er}$	$E_{\beta^-}=1200(100)$ to $5/2^-$ level at 853.0 and $7/2^-$ at 941.04 keV										Ens089 **
* $^{169}\text{Er}(\beta^-)^{169}\text{Tm}$	$E_{\beta^-}=340(2)$ 344(4) respectively, 55% to ground state, 45% to $3/2^+$ level at 8.41 keV										Ens089 **
* $^{169}\text{Yb}(e)^{169}\text{Tm}$	From decay rates to $(5/2)^+$ level at 781.796, $(7/2^+)$ 878.35 of same band										Ens089 **
* $^{169}\text{Yb}(e)^{169}\text{Tm}$	$pK=0.812(0.029)$ to $9/2^-$ level at 472.88 keV										Ens089 **
* $^{169}\text{Hf}(\beta^+)^{169}\text{Lu}$	$E_{\beta^+}=1850(200)$ to $7/2^-$ level at 492.88 keV										Ens089 **
* $^{169}\text{Hf}(\beta^+)^{169}\text{Lu}$	$K/\beta^+=5.2(1.0)$ to $7/2^-$ level at 492.88 keV										Ens089 **
$\text{C}_{12} \text{H}_{12} \text{N}-^{170}\text{Er}$	161210	70	161502.5	1.5	1.0	U			R04	4.0	64De15
$\text{C}_{12} \text{H}_{12} \text{N}-^{170}\text{Yb}$	161831	43	162207.144	0.011	2.2	U			R04	4.0	64De15
$\text{C}_{11} \text{H}_8 \text{O N}-^{170}\text{Yb}$	125370	150	125821.636	0.011	0.8	U			R04	4.0	64De15
$\text{C}_{11}^{13} \text{C H}_{11} \text{N}-^{170}\text{Yb}$	157320	210	157736.948	0.011	0.5	U			R04	4.0	64De15
$^{170}\text{Yb}-^{129}\text{Xe}_{1.318}$	60266.078	0.012	60266.069	0.009	-0.8	1	55	54 ^{170}Yb	FS1	1.0	12Ra34
$^{170}\text{Yb}-^{132}\text{Xe}_{1.288}$	58215.483	0.013	58215.494	0.009	0.8	1	47	46 ^{170}Yb	FS1	1.0	12Ra34
$^{170}\text{Lu}-u$	-61529	42	-61521	18	0.2	R			GS2	1.0	05Li24 *
$^{170}\text{Hf}-u$	-60400	104	-60390	30	0.1	U			GS1	1.0	00Ra23
	-60391	30				2			GS2	1.0	05Li24
$^{170}\text{Ta}-u$	-53810	104	-53830	30	-0.1	U			GS1	1.0	00Ra23
	-53825	30				2			GS2	1.0	05Li24

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{170}\text{W}-u$	-50710	110	-50769	14	-0.5	U			GS1	1.0	00Ra23
	-50755	30			-0.5	1	22	22 ^{170}W	GS2	1.0	05Li24
$^{170}\text{Re}-u$	-41782	30	-41765	12	0.6	-			GS2	1.0	05Li24
	-41780	28			0.5	1	20	20 ^{170}Re			average
$^{170}\text{Os}-u$	-36454	31	-36421	10	1.1	1	11	11 ^{170}Os	GS2	1.0	05Li24
$^{170}\text{Er } ^{35}\text{Cl}-^{168}\text{Er } ^{37}\text{Cl}$	6073	31	6043.8	1.5	-0.6	U			R08	1.5	69De19
	6040	3			0.5	U			H23	2.5	70Wh01
	6046.9	1.8			-0.7	1	11	11 ^{170}Er	H25	2.5	72Ba08
$^{170}\text{Yb } ^{35}\text{Cl}-^{168}\text{Yb } ^{37}\text{Cl}$	3806.0	7.6	3826.07	0.12	1.1	U			H27	2.5	74Ba90
$^{170}\text{Er}-^{168}\text{Er}$	3450	70	3093.7	1.5	-1.3	U			R04	4.0	64De15
$^{170}\text{Yb}-^{168}\text{Yb}$	910	200	875.95	0.10	-0.0	U			R04	4.0	64De15
$^{170}\text{Os}(\alpha)^{166}\text{W}$	5533.5	10.	5536.9	2.7	0.3	-			ORa		72To06 Z
	5541.8	4.1			-1.2	-			Ora		82De11 Z
	5523.2	8.			1.7	-					84Sc06 *
	5533.4	8.			0.4	-					95Hi02
	5537.5	5.			-0.1	-			Bka		02Ro17
	5537.1	2.7			-0.1	1	99	89 ^{170}Os			average
$^{170}\text{Ir}(\alpha)^{166}\text{Re}^p$	5955.4	10.				7			Bka		02Ro17
$^{170}\text{Ir}^m(\alpha)^{166}\text{Re}$	6175.4	10.	6272	10	9.7	B					78Sc26 *
	6172.7	5.			19.9	B			Ora		82De11 *
	6147.9	10.			12.4	B			Daa		96Pa01 *
	6229.9	11.			3.9	B			Daa		96Pa01 *
	6272.4	10.				6			Jya		07Ha45 *
$^{170}\text{Pt}(\alpha)^{166}\text{Os}$	6703.0	8.	6707	3	0.5	-			GSa		81Ho10
	6705.0	10.			0.2	-					82En03
	6708.1	6.			-0.1	-			ORa		96Bi07
	6711.2	11.			-0.3	-			Jya		97Uu01
	6723.5	14.			-1.1	-			Bka		01Ro.B
	6707.1	7.			0.0	-			Jya		04Ke06
	6708	3			-0.1	1	84	84 ^{170}Pt			average
$^{170}\text{Au}(\alpha)^{166}\text{Ir}$	7174.1	11.	7177	15	0.3	o			Jya		02Ke.C
	7170.0	12.			0.6	U			Jya		04Ke06
$^{170}\text{Au}^m(\alpha)^{166}\text{Ir}^m$	7277.5	6.	7285	12	0.2	o			Jya		02Ke.C
	7226.3	15.			1.1	U			Ara		02Ma61
	7278.5	9.			0.1	U			Jya		04Ke06
$^{170}\text{Hg}(\alpha)^{166}\text{Pt}$	7773.2	30.7				6			Jya		19Hi06
$^{170}\text{Er}(p,\alpha)^{167}\text{Ho}$	7036	5				2			NDm		83Ta.A
$^{170}\text{Er}(^{18}\text{O}, ^{20}\text{Ne})^{168}\text{Dy}$	4710	140				2					98Lu08
$^{170}\text{Er}(p,t)^{168}\text{Er}$	-4785	5	-4779.1	1.4	1.2	U			Min		73Oo01
$^{170}\text{Yb}(p,t)^{168}\text{Yb}$	-6861	6	-6844.90	0.09	2.7	B			Min		73Oo01
$^{170}\text{Er}(d, ^3\text{He})^{169}\text{Ho}$	-3107	20				2					76Su.A
$^{170}\text{Er}(d,t)^{169}\text{Er}$	-1010	10	-1000.4	1.4	1.0	U			Kop		69Tj01
$^{169}\text{Tm}(n,\gamma)^{170}\text{Tm}$	6595.	2.5	6591.98	0.17	-1.2	U					66Sh03
	6592.1	1.5			-0.1	U					70Or.A
	6591.7	0.9			0.3	U			BNn		96Ho12 Z
	6591.95	0.17			0.2	1	99	66 ^{169}Tm	Bdn		06Fi.A
$^{169}\text{Tm}(d,p)^{170}\text{Tm}$	4420	20	4367.41	0.17	-2.6	U			CIT		66Ry01
	4369	15			-0.1	U			Tal		66Sh03
$^{170}\text{Yb}(d,t)^{169}\text{Yb}$	-2211	12	-2202.49	0.18	0.7	U			Kop		66Bu16
$^{170}\text{Au}(p)^{169}\text{Pt}$	1473.8	15.	1472	12	-0.1	o			Jyp		02Ke.C
	1471.7	12.				7			Jyp		04Ke06
$^{170}\text{Au}^m(p)^{169}\text{Pt}$	1749.5	8.	1751	5	0.2	o			Jyp		02Ke.C
	1745.4	10.			0.6	7			Arp		02Ma61
	1753.5	6.			-0.4	7			Jyp		04Ke06
$^{170}\text{Ho}(\beta^-)^{170}\text{Er}$	3870	50				2					78Tu04

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{170}\text{Ho}^m(\beta^-)^{170}\text{Er}$	3970	60				2					78Tu04
$^{170}\text{Tm}(\beta^-)^{170}\text{Yb}$	970	2	968.6	0.7	-0.7	-					54Po26
	967.3	1.			1.3	-					69Va17 *
ave.	967.8	0.9			0.9	1	67	67	^{170}Tm		average
$^{170}\text{Lu}(\beta^+)^{170}\text{Yb}$	3467	20	3458	17	-0.5	2					60Dz02
	3410	50			1.0	2					65Ha30
* $^{170}\text{Lu}-u$	$M-A=-57267(29)$ keV for mixture gs+m at 92.91 keV										Nub211 **
* $^{170}\text{Os}(\alpha)^{166}\text{W}$	Used to recalibrate other results in same reference										GAu **
* $^{170}\text{Ir}^m(\alpha)^{166}\text{Re}$	$E_\alpha=6029.8(10,Z)$ 6027.2(5,Z) 6003(10) most probably to low levels in ^{166}Re										GAu **
* $^{170}\text{Ir}^m(\alpha)^{166}\text{Re}$	Correlated with ^{166}Re $E_\alpha=5533$ keV										96Pa01 **
* $^{170}\text{Ir}^m(\alpha)^{166}\text{Re}$	$E_\alpha=5951(10)$ to level at 175, 6007(10) to 122, 6053(10) to 75 keV										07Ha45 **
* $^{170}\text{Tm}(\beta^-)^{170}\text{Yb}$	$E_{\beta^-}=883(1)$ to 2^+ level at 84.25468 keV										Ens02b **
$\text{C}_{11}^{13}\text{C H}_{12}\text{ N}-^{171}\text{Yb}$	164140	80	163997.707	0.014	-0.4	U					R04 4.0 64De15
$\text{C}_{10}\text{ H}_7\text{ O N}_2-^{171}\text{Yb}$	119640	270	119506.336	0.014	-0.1	U					R04 4.0 64De15
$^{171}\text{Yb}-^{129}\text{Xe}_{1.326}$	62592.096	0.012	62592.096	0.012	-0.0	1	100	100	^{171}Yb		FS1 1.0 12Ra34
$^{171}\text{Lu}-u$	-62132	41	-62081.4	2.0	1.2	U					GS2 1.0 05Li24 *
$^{171}\text{Hf}-u$	-59570	104	-59510	30	0.6	U					GS1 1.0 00Ra23 *
	-59508	31				2					GS2 1.0 05Li24 *
$^{171}\text{Ta}-u$	-55550	104	-55520	30	0.2	U					GS1 1.0 00Ra23
	-55524	30				2					GS2 1.0 05Li24
$^{171}\text{W}-u$	-50650	110	-50550	30	0.9	U					GS1 1.0 00Ra23
	-50549	30				2					GS2 1.0 05Li24
$^{171}\text{Re}-u$	-44284	30				2					GS2 1.0 05Li24
$^{171}\text{Os}-u$	-36796	30	-36820	20	-0.8	-					GS2 1.0 05Li24
ave.	-36801	21			-0.9	1	85	85	^{171}Os		average
$^{171}\text{Yb }^{35}\text{Cl}_2-^{167}\text{Er }^{37}\text{Cl}_2$	10178.0	1.7	10175.6	0.3	-0.6	U					H27 2.5 74Ba90
$^{171}\text{Yb }^{35}\text{Cl}-^{169}\text{Tm }^{37}\text{Cl}$	5055	3	5062.7	0.8	1.0	U					H23 2.5 70Wh01
	5061.9	1.7			0.2	U					H27 2.5 74Ba90
$^{171}\text{Yb}-^{170}\text{Yb}$	1220	60	1564.273	0.015	1.4	U					R04 4.0 64De15
$^{171}\text{Os}(\alpha)^{167}\text{W}$	5365.8	10.	5371	4	0.5	-					ORa 72To06
	5365.8	10.			0.5	-					78Sc26
	5393.4	15.			-1.5	-					ISa 79Ha10
	5367.9	8.			0.4	-					95Hi02 *
	5374.0	9.			-0.3	-					Daa 96Pa01
ave.	5371	4			0.1	1	100	93	^{167}W		average
$^{171}\text{Ir}(\alpha)^{167}\text{Re}^m$	5854.2	10.	5866	5	1.1	5					Bka 02Ro17 *
	5865.4	8.			0.0	5					Ara 11Ko.B *
	5871.6	7.2			-0.8	5					Anv 13An10
$^{171}\text{Ir}^m(\alpha)^{167}\text{Re}$	6159.2	3.	6161.1	2.3	0.6	11					Ora 82De11 *
	6159	5			0.4	11					92Sc16 *
	6180	11			-1.7	11					Daa 96Pa01 *
	6159.2	8.			0.2	11					Anv 10An01 *
	6172.4	8.			-1.4	11					Ara 11Ko.B *
$^{171}\text{Pt}(\alpha)^{167}\text{Os}$	6608.1	4.	6607	3	-0.2	7					Ora 81De22 Z
	6606.8	5.			0.1	7					GSa 81Ho10 Z
	6604.8	11.			0.2	7					Jya 97Uu01
	6600.6	15.			0.5	U					Anv 10An01
$^{171}\text{Au}^m(\alpha)^{167}\text{Ir}^m$	7163.9	6.	7164	4	0.1	-					Ara 97Da07
	7162.9	8.			0.2	-					Jya 04Ke06
ave.	7164	5			0.2	1	69	39	$^{171}\text{Au}^m$		average
$^{171}\text{Hg}(\alpha)^{167}\text{Pt}$	7667.7	15.				6					Jya 04Ke06
$^{171}\text{Yb}(p,t)^{169}\text{Yb}$	-6599	5	-6592.13	0.18	1.4	U					Min 73Oo01
$^{170}\text{Er}(n,\gamma)^{171}\text{Er}$	5681.5	0.5	5681.6	0.3	0.3	-					71Al01
	5681.6	0.5			0.1	-					Bdn 06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{170}\text{Er}(\text{d,p})^{171}\text{Er}$	3450	10	3457.1	0.3	0.7	U			Tal		68Ha10
	3458	10			-0.1	U			Kop		69Tj01
$^{170}\text{Er}(\text{n},\gamma)^{171}\text{Er}$	ave. 5681.6	0.4	5681.6	0.3	0.2	1	98	72	^{171}Er		average
$^{170}\text{Er}(\alpha,\text{t})^{171}\text{Tm}-^{168}\text{Er}(\gamma)^{169}\text{Tm}$	817.9	1.0	817.3	0.9	-0.6	1	80	63	^{170}Er		75Bu02
$^{170}\text{Yb}(\text{n},\gamma)^{171}\text{Yb}$	6614.3	0.6	6614.207	0.014	-0.2	U			McM		72Wa10
	6616.6	0.4			-6.0	C			Bdn		06Fi.A
$^{170}\text{Yb}(\text{d,p})^{171}\text{Yb}$	4390	12	4389.641	0.014	-0.0	U			Kop		66Bu16
$^{171}\text{Yb}(\text{d,t})^{170}\text{Yb}$	-359	12	-356.977	0.014	0.2	U			Kop		66Bu16
$^{170}\text{Yb}(\alpha,\text{t})^{171}\text{Lu}-^{174}\text{Yb}(\gamma)^{175}\text{Lu}$	-1156.2	2.0	-1156.6	1.7	-0.2	1	73	61	^{171}Lu		75Bu02
$^{171}\text{Au}(\text{p})^{170}\text{Pt}$	1452.6	17.	1448	10	-0.3	2			Arp		99Po09
	1445.6	12.			0.2	2			Jyp		04Ke06
$^{171}\text{Au}^{\text{m}}(\text{p})^{170}\text{Pt}$	1702.1	6.	1702	4	0.1	-					97Da07
	1704.1	6.			-0.3	-			Jyp		04Ke06
	ave. 1703	4			-0.1	1	77	61	$^{171}\text{Au}^{\text{m}}$		average
$^{171}\text{Ho}(\beta^-)^{171}\text{Er}$	3200	600				2			LBL		90Ch34
$^{171}\text{Er}(\beta^-)^{171}\text{Tm}$	1490	2	1492.4	1.1	1.2	1	29	28	^{171}Er		61Ar15
$^{171}\text{Tm}(\beta^-)^{171}\text{Yb}$	96.5	1.0	96.5	1.0	0.0	1	94	94	^{171}Tm		57Sm73
$^{171}\text{Lu}(\beta^+)^{171}\text{Yb}$	1479.3	3.	1478.4	1.9	-0.3	1	39	39	^{171}Lu		77Bo32
$^{171}\text{Re}(\beta^+)^{171}\text{W}$	5670	200	5840	40	0.8	U			Got		87Ru05
$^{171}\text{Au}^{\text{m}}(\text{IT})^{171}\text{Au}$	250	16	255	10	0.3	o					99Po09
$^{171}\text{Lu}-\text{u}$	$M-A=-57840(33)$ keV for mixture gs+m at 71.13 keV										Nub211
$^{171}\text{Hf}-\text{u}$	$M-A=-55480(100)$ keV for mixture gs+m at 21.93 keV										Nub211
$^{171}\text{Hf}-\text{u}$	$M-A=-55420(28)$ keV for mixture gs+m at 21.93 keV										Nub211
$^{171}\text{Os}(\alpha)^{167}\text{W}$	$E_{\alpha}=5241(8)$, 5166(8) to ground state and level at 79 keV										95Hi02
$^{171}\text{Ir}(\alpha)^{167}\text{Re}^{\text{m}}$	Correlated with $E_{\alpha}=6412$ of ^{175}Au										02Ro17
$^{171}\text{Ir}(\alpha)^{167}\text{Re}^{\text{m}}$	Correlated with $E_{\alpha}=6430(8)$ of ^{175}Au and 6556(8) of ^{179}Tl										11Ko.B
$^{171}\text{Ir}^{\text{m}}(\alpha)^{167}\text{Re}$	$E_{\alpha}=5925.2(3,Z)$ 5925(5) 5945(11) 5925(8) respectively, to $(11/2^-)$ level at 92 keV										92Sc16
*	$E_{\alpha}=5920$ correlated with ^{175}Au $E_{\alpha}=6438$ keV										02Ro17
$^{171}\text{Ir}^{\text{m}}(\alpha)^{167}\text{Re}$	$E_{\alpha}=5938(8)$ to 92 level; correlated with $E_{\alpha}=6431(8)$ of $^{175}\text{Au}^{\text{m}}$										11Ko.B
*	and 7194(8) of $^{179}\text{Tl}^{\text{m}}$										11Ko.B
$^{171}\text{Er}(\beta^-)^{171}\text{Tm}$	$E_{\beta^-}=1065(2)$ to $7/2^-$ level at 424.95 keV										Ens029
$^{171}\text{Lu}(\beta^+)^{171}\text{Yb}$	$E_{\beta^+}=362(3)$ to $7/2^+$ level at 95.28 keV										Ens029
$^{171}\text{Au}^{\text{m}}(\text{IT})^{171}\text{Au}$	Redundant; use only their Q_p										GAu
$\text{C}_{10}\text{H}_6\text{O}_2\text{N}-^{172}\text{Yb}$	103560	60	103466.780	0.015	-0.4	U			R04	4.0	64De15
$^{172}\text{Yb}-^{132}\text{Xe}_{1.303}$	61272.578	0.013	61272.578	0.013	0.0	1	100	100	^{172}Yb		FS1 1.0 12Ra34
$^{172}\text{Hf}-\text{u}$	-60555	30	-60550	26	0.2	2			GS2	1.0	05Li24
$^{172}\text{Ta}-\text{u}$	-55105	30				2			GS2	1.0	05Li24
$^{172}\text{W}-\text{u}$	-52770	110	-52710	30	0.6	U			GS1	1.0	00Ra23
	-52708	30				2			GS2	1.0	05Li24
$^{172}\text{Re}-\text{u}$	-44760	220	-44620	40	0.6	U			GS1	1.0	00Ra23
	-44647	53			0.4	1	52	52	^{172}Re		GS2 1.0 05Li24
$^{172}\text{Yb }^{35}\text{Cl}_2-^{168}\text{Er }^{37}\text{Cl}_2$	9906.7	1.7	9908.6	0.3	0.5	U			H27	2.5	74Ba90
$^{172}\text{Yb }^{35}\text{Cl}-^{170}\text{Yb }^{37}\text{Cl}$	4568.5	2.0	4569.53	0.07	0.2	U			H27	2.5	74Ba90
$^{172}\text{Yb}-^{170}\text{Yb}_{1.012}$	2402.251	0.085	2402.202	0.016	-0.6	U			JY2	1.0	20Ne.1
$^{172}\text{Yb}-^{171}\text{Yb}$	-50	230	55.139	0.018	0.1	U			R04	4.0	64De15
$^{172}\text{Os}(\alpha)^{168}\text{W}$	5226.8	10.	5224	7	-0.2	-					71Bo06
	5227.8	10.			-0.3	-			Daa		96Pa01
	ave. 5227	7			-0.4	1	93	59	^{168}W		average
$^{172}\text{Ir}(\alpha)^{168}\text{Re}$	5990.6	10.				4					92Sc16
$^{172}\text{Ir}^{\text{m}}(\alpha)^{168}\text{Re}$	6129.3	3.	6129.2	2.6	-0.0	4			Ora		82De11
	6161	20			-1.6	F			GSa		84Sc.A
	6129.1	5.			0.0	4					92Sc16
	6123.0	12.			0.5	U			Daa		96Pa01
$^{172}\text{Pt}(\alpha)^{168}\text{Os}$	6464.8	4.	6463	4	-0.3	1	97	77	^{172}Pt		81De22
	6474.8	15.			-0.8	U			Anv		09An20

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{172}\text{Au}(\alpha)^{168}\text{Ir}$	6923.3	10.2				6			Jya		09Ha42
$^{172}\text{Au}^m(\alpha)^{168}\text{Ir}^m$	7023.6	10.	7034	6	1.0	7					93Se09
	7042.1	9.			-0.9	7			Daa		96Pa01
	7033.8	10.			0.0	7			Jya		09Ha42 *
$^{172}\text{Hg}(\alpha)^{168}\text{Pt}$	7525.3	12.	7524	6	-0.1	3					99Se14
	7536.5	16.			-0.8	o			Jya		04Ke06
	7523.3	7.2			0.1	3			Jya		09Sa27
$^{170}\text{Er}(\text{t,p})^{172}\text{Er}$	4034	4	4036	4	0.5	1	89	87	^{172}Er		80Sh14
$^{172}\text{Yb}(\text{p,t})^{170}\text{Yb}$	-6161	5	-6152.367	0.015	1.7	U			Min		73Oo01
$^{171}\text{Yb}(\text{n},\gamma)^{172}\text{Yb}$	8020.3	0.7	8019.956	0.017	-0.5	U					71Al14 Z
	8020.1	0.5			-0.3	U					75Gr32
	8019.67	0.35			0.8	U			ILn		85Ge02 Z
	8019.27	0.17			4.0	C			Bdn		06Fi.A
$^{171}\text{Yb}(\text{d,p})^{172}\text{Yb}$	5797	12	5795.390	0.017	-0.1	U			Kop		66Bu16
	5789	5			1.3	U			Tal		66Sh14
$^{172}\text{Yb}(\text{d,t})^{171}\text{Yb}$	-1772	12	-1762.726	0.017	0.8	U			Kop		66Bu16
$^{171}\text{Yb}(\alpha,\text{t})^{172}\text{Lu}$	-792	34	-775.2	2.3	0.5	U			Roc		76El11
$^{171}\text{Yb}(\alpha,\text{t})^{172}\text{Lu}-^{174}\text{Yb}(\alpha)^{175}\text{Lu}$	-791.9	2.0	-791.9	2.0	-0.0	1	100	100	^{172}Lu	McM	75Bu02
$^{172}\text{Er}(\beta^-)^{172}\text{Tm}$	888	5	891	5	0.6	1	83	70	^{172}Tm		62Gu03 *
$^{172}\text{Tm}(\beta^-)^{172}\text{Yb}$	1870	10	1882	5	1.2	1	30	30	^{172}Tm		66Ha15
$^{172}\text{Hf}(\epsilon)^{172}\text{Lu}$	350	50	334	25	-0.3	R					79To18
$^{172}\text{Ta}(\beta^+)^{172}\text{Hf}$	4920	180	5070	40	0.8	U					73Ca10 *
$^{172}\text{W}(\beta^+)^{172}\text{Ta}$	3210	100	2230	40	-9.8	C					74Ca.A *
* $^{172}\text{Re}-\text{u}$	$M-A=-41640(200)$ keV for mixture gs+m at 110#50 keV										Nub211 **
* $^{172}\text{Re}-\text{u}$	$M-A=-41533(28)$ keV for mixture gs+m at 110#50 keV										Nub211 **
* $^{172}\text{Ir}(\alpha)^{168}\text{Re}$	$E_\alpha=5510(10)$ to 89.7+123.2+136.3 level										92Sc16 **
* $^{172}\text{Ir}^m(\alpha)^{168}\text{Re}$	$E_\alpha=5736$ followed by XK(Re), 128 M2 and 161 M3 γ 's										84Sc.A **
* $^{172}\text{Ir}^m(\alpha)^{168}\text{Re}$	F : first assigned to $^{173}\text{Ir}(\alpha)$; seen in neither nuclide in reference										92Sc16 **
* $^{172}\text{Ir}^m(\alpha)^{168}\text{Re}$	$E_\alpha=5828.2(3,Z)$ 5828(5) 5822(12) respectively, to (8^+) level at 162.1 keV										Ens108 **
* $^{172}\text{Au}^m(\alpha)^{168}\text{Ir}^m$	$E_\alpha=6870(10)$ 6800(10) to ground state and 70 keV level										09Ha42 **
* $^{172}\text{Er}(\beta^-)^{172}\text{Tm}$	$E_{\beta^-}=278(5)$ to 1^+ level at 610.062 keV										Ens15c **
* $^{172}\text{Ta}(\beta^+)^{172}\text{Hf}$	$E_{\beta^+}=2480(180)$ to 4^- level at 1418.55 keV										Ens959 **
* $^{172}\text{W}(\beta^+)^{172}\text{Ta}$	$E_{\beta^+}=1600(100)$ in coinc. with 459.2 keV γ from 586.3 level										Ens15c **
$\text{C}_{14} \text{H}_5-^{173}\text{Yb}$	101030	70	100908.948	0.012	-0.4	U			R04	4.0	64De15
$\text{C}_{10} \text{H}_7 \text{O}_2 \text{N}-^{173}\text{Yb}$	109810	60	109462.254	0.012	-1.4	U			R04	4.0	64De15
$^{173}\text{Yb}-^{129}\text{Xe}_{1.341}$	65905.075	0.013	65905.078	0.010	0.3	1	58	57	^{173}Yb	FS1	1.0 12Ra34
$^{173}\text{Yb}-^{132}\text{Xe}_{1.311}$	63868.901	0.015	63868.897	0.010	-0.3	1	44	43	^{173}Yb	FS1	1.0 12Ra34
$^{173}\text{Hf}-\text{u}$	-59487	30				2			GS2	1.0	05Li24
$^{173}\text{Ta}-\text{u}$	-56270	104	-56250	30	0.2	U			GS1	1.0	00Ra23
	-56250	30				2			GS2	1.0	05Li24
$^{173}\text{W}-\text{u}$	-52340	104	-52310	30	0.3	U			GS1	1.0	00Ra23
	-52311	30				2			GS2	1.0	05Li24
$^{173}\text{Re}-\text{u}$	-46910	110	-46760	30	1.4	U			GS1	1.0	00Ra23
	-46757	30				2			GS2	1.0	05Li24
$^{173}\text{Os}-\text{u}$	-40169	30	-40192	16	-0.8	1	29	29	^{173}Os	GS2	1.0 05Li24
$^{173}\text{Ir}-\text{u}$	-32449	100	-32495	11	-0.5	U			GS2	1.0	05Li24 *
$^{173}\text{Yb } ^{35}\text{Cl}_2-^{169}\text{Tm } ^{37}\text{Cl}_2$	9898.3	1.2	9897.5	0.8	-0.3	U			H27	2.5	74Ba90
$^{173}\text{Yb } ^{35}\text{Cl}-^{171}\text{Yb } ^{37}\text{Cl}$	4827	4	4834.82	0.07	0.8	U			H23	2.5	70Wh01
	4835.3	1.6			-0.1	U			H27	2.5	74Ba90
$^{173}\text{Yb}-^{172}\text{Yb}$	1970	120	1829.558	0.016	-0.3	U			R04	4.0	64De15
$^{173}\text{Os}(\alpha)^{169}\text{W}$	5057.2	10.	5055	6	-0.2	-					71Bo06
	5055.2	7.			-0.1	-			GSa		84Sc.A
	ave. 5056	6			-0.2	1	97	69	^{169}W		average
$^{173}\text{Ir}(\alpha)^{169}\text{Re}^m$	5544.4	10.	5541	10	-0.4	1	90	77	$^{169}\text{Re}^m$		92Sc16
$^{173}\text{Ir}^m(\alpha)^{169}\text{Re}$	5930.4	5.	5941.8	2.5	2.3	4					67Si02 *
	5947.1	4.			-1.3	4			Ora		82De11 *
	5937	10			0.5	4			GSa		84Sc.A *
	5944.8	5.			-0.6	4					92Sc16 *
	5951.9	13.			-0.8	4			Daa		96Pa01 *
	5927.3	20.			0.7	U			Ara		01Ko.B

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{173}\text{Pt}(\alpha)^{169}\text{Os}$	6359.3	8.2	6360	60	0.0	3			ISa	79Ha10	Z
	6352.3	3.			0.2	3			Ora	81De22	Z
	6382.9	10.			-0.4	o			GSa	84Sc.A	
	6372.6	9.			-0.2	3			Daa	96Pa01	
	6387.9	15.			-0.5	U			Anv	09An20	
$^{173}\text{Au}(\alpha)^{169}\text{Ir}$	6830.2	6.	6836	5	1.0	4			Ara	99Po09	
	6847.6	8.			-1.4	4			Ara	01Ko44	
	6846.6	14.			-0.7	U			Ara	12Th13	
$^{173}\text{Au}^m(\alpha)^{169}\text{Ir}^m$	6896.8	10.	6897	3	0.0	-			GSa	84Sc.A	
	6909.1	9.			-1.3	-			Daa	96Pa01	
	6891.6	4.			1.3	-			Ara	99Po09	
	6900.8	6.			-0.6	-			Ara	01Ko44	
	6899	15			-0.1	U			Jya	12Th13	
	ave.	6896	3		0.3	1	98	52	$^{173}\text{Au}^m$	average	
$^{173}\text{Hg}(\alpha)^{169}\text{Pt}$	7382.0	11.3	7378	4	-0.4	7				99Se14	
	7362.5	15.4			1.0	7			Jya	04Ke06	
	7378.9	5.			-0.2	7				12Od01	
$^{173}\text{Yb}(\text{p,t})^{171}\text{Yb}$	-5913	5	-5905.256	0.014	1.5	U			Min	73Oo01	
$^{172}\text{Yb}(\text{n},\gamma)^{173}\text{Yb}$	6367.3	0.4	6367.096	0.015	-0.5	U				71Al01	Z
	6367.2	0.6			-0.2	U			Bdn	06Fi.A	
$^{173}\text{Yb}(\gamma,\text{n})^{172}\text{Yb}$	-6500	80	-6367.096	0.015	1.7	U			Phi	60Ge01	
$^{172}\text{Yb}(\text{d,p})^{173}\text{Yb}$	4145	12	4142.530	0.015	-0.2	U			Kop	66Bu16	
$^{173}\text{Yb}(\text{d,t})^{172}\text{Yb}$	-114	12	-109.866	0.015	0.3	U			Kop	66Bu16	
$^{172}\text{Yb}(\alpha,\text{t})^{173}\text{Lu}-^{174}\text{Yb}()^{175}\text{Lu}$	-595.6	1.0	-595.6	1.0	-0.0	1	100	100	^{173}Lu	75Bu02	
$^{173}\text{Tm}(\beta^-)^{173}\text{Yb}$	1260	50	1295	4	0.7	U				63Ku22	
	1320	40			-0.6	U				63Or01	
$^{173}\text{Lu}(\varepsilon)^{173}\text{Yb}$	675	20	670.2	1.6	-0.2	U				73Ko13	
$^{173}\text{Ta}(\beta^+)^{173}\text{Hf}$	3670	200	3020	40	-3.3	B				73Re03	
$^{173}\text{W}(\beta^+)^{173}\text{Ta}$	4000	300	3670	40	-1.1	U				80Vi.A	
$^{173}\text{Ir}-\text{u}$	$M-A=-30113(70)$ keV for mixture gs+m at 226(9) keV										Nub211 **
$^{173}\text{Ir}^m(\alpha)^{169}\text{Re}$	$E_\alpha=5660.0(5,Z)$ 5676.2(4,Z) 5666(10) 5674(5) 5681(13) respectively,										AHW **
*	to (11/2 ⁻) level at 136.40 keV										Ens155 **
$\text{C}_{14}\text{H}_6-^{174}\text{Yb}$	108308	38	108082.646	0.012	-1.5	U			R04	4.0	64De15
$^{174}\text{Yb}-^{129}\text{Xe}_{1.349}$	67318.179	0.011	67318.167	0.009	-1.1	1	71	70	^{174}Yb	FS1	1.0 12Ra34
$^{174}\text{Yb}-^{132}\text{Xe}_{1.318}$	65191.113	0.017	65191.142	0.009	1.7	1	31	30	^{174}Yb	FS1	1.0 12Ra34
$^{174}\text{Ta}-\text{u}$	-55546	30				2			GS2	1.0	05Li24
$^{174}\text{W}-\text{u}$	-53940	104	-53920	30	0.2	U			GS1	1.0	00Ra23
	-53921	30				2			GS2	1.0	05Li24
$^{174}\text{Re}-\text{u}$	-46930	104	-46890	30	0.4	U			GS1	1.0	00Ra23
	-46885	30				2			GS2	1.0	05Li24
$^{174}\text{Os}-\text{u}$	-42880	110	-42937	11	-0.5	U			GS1	1.0	00Ra23
	-42919	30			-0.6	1	13	13	^{174}Os	GS2	1.0 05Li24
$^{174}\text{Ir}-\text{u}$	-33090	55	-33050	12	0.7	R			GS2	1.0	05Li24 *
$^{174}\text{Yb }^{35}\text{Cl}-^{172}\text{Yb }^{37}\text{Cl}$	5420	4	5431.01	0.07	1.1	U			H23	2.5	70Wh01
	5430.3	1.1			0.3	U			H27	2.5	74Ba90
$^{174}\text{Yb}-^{173}\text{Yb}$	700	50	651.334	0.013	-0.2	U			R04	4.0	64De15
$^{174}\text{Hf}(\alpha)^{170}\text{Yb}$	2558.9	30.	2494.4	2.3	-2.1	U					61Ma05
$^{174}\text{Os}(\alpha)^{170}\text{W}$	4872.2	10.	4871	10	-0.2	1	90	78	^{170}W		71Bo06
$^{174}\text{Ir}(\alpha)^{170}\text{Re}^m$	5624.1	10.	5620	5	-0.4	4					92Sc16 *
	5619.4	5.1			0.2	4			Jya		19Mo.B *
$^{174}\text{Ir}^m(\alpha)^{170}\text{Re}$	5817.6	6.	5817	4	-0.0	-					67Si02 *
	5816.4	5.			0.2	-					92Sc16 *
	5819.2	13.3			-0.1	-			Jya		19Mo.B *
	ave.	5817	4		0.1	1	98	80	^{170}Re	average	
$^{174}\text{Pt}(\alpha)^{170}\text{Os}$	6176.3	10.	6183	3	0.7	2			ISa	79Ha10	Z
	6185.7	5.			-0.5	2			Ora	81De22	Z
	6182.4	5.1			0.2	2			Ara	04Go38	
$^{174}\text{Au}(\alpha)^{170}\text{Ir}$	6700.3	10.	6699	7	-0.1	7			GSa	84Sc.A	
	6698.3	10.			0.1	7			Daa	96Pa01	*

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{174}\text{Au}^m(\alpha)^{170}\text{Ir}^m$	6683.9	20.	6784	8	5.0	B			GSa		83Sc24 *	
	6778	10			0.6	7			GSa		84Sc.A *	
	6793.5	13.			-0.7	7			Daa		96Pa01	
$^{174}\text{Hg}(\alpha)^{170}\text{Pt}$	7235.6	11.	7233	6	-0.2	2			Jya		97Uu01	
	7232.5	8.2			0.1	2					99Se14	
	7231.5	14.3			0.1	2			Bka		01Ro.B	
$^{174}\text{Yb}(\text{p,t})^{172}\text{Yb}$	-5359	5	-5349.904	0.015	1.8	U			Min		73Oo01	
$^{173}\text{Yb}(\text{n},\gamma)^{174}\text{Yb}$	7464.63	0.06	7464.605	0.013	-0.4	U			MMn		82Is05 Z	
	7464.58	0.35			0.1	U			ILn		87Ge01 Z	
	7465.5	0.4			-2.2	U			Bdn		06Fi.A	
$^{173}\text{Yb}(\text{d,p})^{174}\text{Yb}$	5239	12	5240.038	0.013	0.1	U			Kop		66Bu16	
	5229	5			2.2	U			Tal		66Sh14	
$^{174}\text{Yb}(\text{d,t})^{173}\text{Yb}$	-1218	12	-1207.374	0.013	0.9	U			Kop		66Bu16	
$^{173}\text{Yb}(\alpha,\text{t})^{174}\text{Lu}-^{174}\text{Yb}()^{175}\text{Lu}$	-202.1	1.0	-202.1	1.0	-0.0	1	100	100 ^{174}Lu	McM		75Bu02	
$^{174}\text{Tm}(\beta^-)^{174}\text{Yb}$	3080	100	3080	40	0.0	2					64Ka16 *	
	3080	50			0.0	2					67Gu12 *	
$^{174}\text{Lu}(\beta^+)^{174}\text{Yb}$	1402	5	1374.2	1.6	-5.6	B					68K108	
$^{174}\text{Lu}(\epsilon)^{174}\text{Yb}$	1370	7			0.6	U					68Li01 *	
$^{174}\text{Ta}(\beta^+)^{174}\text{Hf}$	3845	80	4104	28	3.2	B					71Ch26 *	
* $^{174}\text{Ir}-\text{u}$	$M-A=-30761(36)$ keV for mixture gs+m at 124(16) keV										Nub211	**
* $^{174}\text{Ir}(\alpha)^{170}\text{Re}^m$	$E_\alpha=5275(10)$ to (3^+) level at 224.7 keV above $^{170}\text{Re}^m$										Ens02b	**
* $^{174}\text{Ir}(\alpha)^{170}\text{Re}^m$	$E_\alpha=5271(5)$ to 224.2(0.3) keV level above $^{170}\text{Re}^m$										19Mo.B	**
* $^{174}\text{Ir}^m(\alpha)^{170}\text{Re}$	$E_\alpha=5478(6)$ to (7^+) level at 210.32 keV										Ens02b	**
* $^{174}\text{Ir}^m(\alpha)^{170}\text{Re}$	$E_\alpha=5478(5)$ 5316(10) to (7^+) at 210.32 and 370.1 level										Ens02b	**
* $^{174}\text{Ir}^m(\alpha)^{170}\text{Re}$	$E_\alpha=5480(13)$ to 210.1(0.1) keV level above ^{170}Re										19Mo.B	**
* $^{174}\text{Au}(\alpha)^{170}\text{Ir}$	$E_\alpha=6538$ correlated with ^{170}Ir $E_\alpha=5817$ keV										02Ro17	**
*	and only this with ^{178}Tl α 's										02Ro17	**
* $^{174}\text{Au}^m(\alpha)^{170}\text{Ir}^m$	$E_\alpha=6530(20)$ to level above 76 keV										84Sc.A	**
* $^{174}\text{Au}^m(\alpha)^{170}\text{Ir}^m$	$E_\alpha=6626, 6470, 6435$ to $^{170}\text{Ir}^m$ and levels above $^{170}\text{Ir}^m$ (9^+) at 152.5,										Ens082	**
*	$(7^-, 8^-, 9^-)$ at 190.56; last two E_α originally assigned to ^{175}Au										01Ko.B	**
* $^{174}\text{Tm}(\beta^-)^{174}\text{Yb}$	$E_{\beta^-}=1200(100)$ 1200(50) respectively, to 5^- level at 1884.674 keV, and other E_{β^-}										Ens998	**
* $^{174}\text{Lu}(\epsilon)^{174}\text{Yb}$	No K capture to 2^- level at 1318.361 keV $\rightarrow Q < 1380$; and L capture										Ens998	**
*	of $^{174}\text{Lu}^m$ at 170.83 to $^{174}\text{Yb}^m$ at 1518.148 keV $\rightarrow Q_{gs} > 1357$ keV										Nub211	**
* $^{174}\text{Ta}(\beta^+)^{174}\text{Hf}$	$E_{\beta^+}=2525(80)$ to 4^+ level at 297.38 keV										Ens04a	**
$^{175}\text{Lu } ^{37}\text{Cl}-^{142}\text{Nd } ^{35}\text{Cl}_2$	61249.5	2.5	61245.6	1.7	-0.6	U			H31	2.5	77So02	
$\text{C}_{14} \text{H}_7-^{175}\text{Lu}$	114121	37	113998.0	1.3	-0.8	U			R04	4.0	64De15	
$\text{C}_{13} ^{13}\text{C} \text{H}_6-^{175}\text{Lu}$	109763	36	109527.8	1.3	-1.6	U			R04	4.0	64De15	
$^{175}\text{Ta}-\text{u}$	-56350	120	-56260	30	0.7	U			GS1	1.0	00Ra23	
	-56263	30				2			GS2	1.0	05Li24	
$^{175}\text{W}-\text{u}$	-53290	104	-53280	30	0.1	U			GS1	1.0	00Ra23	
	-53283	30				2			GS2	1.0	05Li24	
$^{175}\text{Re}-\text{u}$	-48630	104	-48620	30	0.1	U			GS1	1.0	00Ra23	
	-48619	30				2			GS2	1.0	05Li24	
$^{175}\text{Os}-\text{u}$	-43120	110	-43055	13	0.6	U			GS1	1.0	00Ra23	
	-43024	30			-1.0	1	18	18 ^{175}Os	GS2	1.0	05Li24	
$^{175}\text{Ir}-\text{u}$	-34353	1288	-35850	13	-0.5	U				2.5	91Br17	
	-35828	30			-0.7	1	20	20 ^{175}Ir	GS2	1.0	05Li24	
$^{175}\text{Lu } ^{35}\text{Cl}-^{173}\text{Yb } ^{37}\text{Cl}$	5503	4	5511.1	1.3	0.8	U			H23	2.5	70Wh01	
	5507.3	1.4			1.1	1	14	14 ^{175}Lu	H27	2.5	74Ba90	
$^{175}\text{Lu} \text{O}-\text{C}_{16}$	-64316.3	4.5	-64308.2	1.3	1.2	U			TG1	1.5	11Ke03	
$^{175}\text{Ir}(\alpha)^{171}\text{Re}$	5709.0	5.	5430	30	-55.6	B					67Si02 *	
	5709.2	5.			-55.7	B					92Sc16 *	
$^{175}\text{Pt}(\alpha)^{171}\text{Os}$	6179	5	6164	4	-3.1	C			ISa		79Ha10 *	
	6178.1	3.			-4.8	B			Ora		82De11 *	
	6164	4			-0.1	1	100	92 ^{175}Pt	Jya		14Pe02 *	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{175}\text{Au}(\alpha)^{171}\text{Ir}$	6562.3	15.	6583.4	2.7	1.4	U			Bka		02Ro17 *
	6580.7	8.2			0.3	6			Ara		11Ko.B *
	6583.7	4.			-0.1	6			Anv		13An10
	6583.7	4.			-0.1	6			Jya		17Ba46
$^{175}\text{Au}^m(\alpha)^{171}\text{Ir}^m$	6590.9	10.	6583.5	2.5	-0.7	10			Ora		75Ca06
	6775.8	10.			-19.2	F			GSa		84Sc.A *
	6588.8	9.			-0.6	10			Daa		96Pa01
	6579.6	6.			0.6	10			Ara		01Ko44
	6582.7	5.			0.1	10			Anv		10An01
	6581.7	8.2			0.2	10			Ara		11Ko.B *
	6583.7	4.			-0.1	10			Jya		17Ba46
$^{175}\text{Hg}(\alpha)^{171}\text{Pt}$	7020.7	20.	7072	5	2.5	o			GSa		83Sc24
	7039.2	20.			1.7	U			GSa		84Sc.A
	7071.0	24.			0.1	o			Daa		96Pa01
	7058.7	11.			1.2	8			Jya		97Uu01
	7075	5			-0.5	8			Daa		09Od01
	7082.1	20.			-0.5	U			Anv		10An01
$^{174}\text{Yb}(n,\gamma)^{175}\text{Yb}$	5822.35	0.07	5822.35	0.07	0.1	1	100	100 ^{175}Yb	MMn		82Is05 Z
	5822.5	0.4			-0.4	U			Bdn		06Fi.A
$^{174}\text{Yb}(d,p)^{175}\text{Yb}$	3595	12	3597.79	0.07	0.2	U			Kop		66Bu16
$^{174}\text{Yb}(\alpha,t)^{175}\text{Lu}$	-14303	10	-14303.7	1.2	-0.1	U			McM		75Bu02
$^{175}\text{Lu}(\gamma,n)^{174}\text{Lu}$	-7880	80	-7666.7	1.0	2.7	U			Phi		60Ge01
$^{175}\text{Lu}(d,t)^{174}\text{Lu}$	-1400	10	-1409.5	1.0	-0.9	U			Tal		70Jo08
$^{174}\text{Hf}(n,\gamma)^{175}\text{Hf}$	6708.4	0.5	6708.5	0.4	0.2	-					71Al01 Z
	6708.8	0.6			-0.5	-			Bdn		06Fi.A
ave.	6708.6	0.4			-0.2	1	100	86 ^{175}Hf			average
$^{175}\text{Tm}(\beta^-)^{175}\text{Yb}$	2385	50				2					66Wi04 *
$^{175}\text{Yb}(\beta^-)^{175}\text{Lu}$	466	3	470.1	1.2	1.4	-					55De18
	468	5			0.4	-					55Mi90
	471	3			-0.3	-					56Co13
	467	3			1.0	-					62Ba32
ave.	468.0	1.6			1.3	1	54	54 ^{175}Lu			average
$^{175}\text{Hf}(\epsilon)^{175}\text{Lu}$	628	9	683.9	2.0	6.2	B					68Ja11 *
	650	20			1.7	U					69Jo16 *
	630	3			18.0	B					88Si22 *
* $^{175}\text{Ir}(\alpha)^{171}\text{Re}$	$E_\alpha=5392.8(5,Z)$ to 189.8 level										95Hi02 **
* $^{175}\text{Ir}(\alpha)^{171}\text{Re}$	$E_\alpha=5393(5)$ to 189.8 level										95Hi02 **
* $^{175}\text{Pt}(\alpha)^{171}\text{Os}$	$E_\alpha=6037(10)$, 5963.0(5,Z) to ground state, 76.4(0.5) level										84Sc.A **
* $^{175}\text{Pt}(\alpha)^{171}\text{Os}$	$E_\alpha=5959.2(3,Z)$ to 76.4(0.5) level										84Sc.A **
* $^{175}\text{Pt}(\alpha)^{171}\text{Os}$	$E_\alpha=6021(4)$, 5948(4) to ground state, 76.7(0.3) level										14Pe02 **
* $^{175}\text{Au}(\alpha)^{171}\text{Ir}$	Analysis by AHW of data in Fig. 3 of reference										02Ro17 **
* $^{175}\text{Au}(\alpha)^{171}\text{Ir}$	Correlated with $E_\alpha=6556(8)$ of ^{179}Tl and 5728(8) of ^{171}Ir										11Ko.B **
* $^{175}\text{Au}^m(\alpha)^{171}\text{Ir}^m$	$E_\alpha=6435(10)$ and 6470(20) to 190.0 and 152.7 levels										84Sc.A **
* $^{175}\text{Au}^m(\alpha)^{171}\text{Ir}^m$	F : reassigned to ^{174}Au !										01Ko.B **
* $^{175}\text{Au}^m(\alpha)^{171}\text{Ir}^m$	Correlated with $E_\alpha=7194(8)$ of $^{179}\text{Tl}^m$ and 5958(8) of $^{171}\text{Ir}^m$										11Ko.B **
*	different method and different detectors as compared to 01Ko44										11Ko.B **
* $^{175}\text{Tm}(\beta^-)^{175}\text{Yb}$	$E_{\beta^-}=1870(50)$ to $1/2^-$ level at 514.866 keV										Ens04a **
* $^{175}\text{Hf}(\epsilon)^{175}\text{Lu}$	pK=0.712(0.008) 0.740(0.015) 0.714(0.002) respectively,										AHW **
*	to $7/2^+$ level at 432.74 keV, and other capture ratios, recalculated										Ens04a **
$\text{C}_{14}\text{H}_8-^{176}\text{Yb}$	119980	46	120025.549	0.016	0.2	U			R04	4.0	64De15
$\text{C}_{13}\text{H}_6\text{N}-^{176}\text{Yb}$	107190	110	107449.489	0.016	0.6	U			R04	4.0	64De15
$^{176}\text{Yb}-^{129}\text{Xe}_{1.364}$	72453.619	0.016	72453.613	0.014	-0.4	1	74	74 ^{176}Yb	FS1	1.0	12Ra34
$^{176}\text{Yb}-^{132}\text{Xe}_{1.333}$	70335.958	0.027	70335.975	0.014	0.6	1	27	26 ^{176}Yb	FS1	1.0	12Ra34
$^{176}\text{Lu}\text{ }^{37}\text{Cl}-^{143}\text{Nd}\text{ }^{35}\text{Cl}_2$	61067.2	1.4	61069.1	1.7	0.5	1	25	13 ^{143}Nd	H31	2.5	77So02

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_{14}H_8-^{176}Lu$	119962	49	119908.5	1.3	-0.3	U			R04	4.0	64De15
$^{176}LuO-C_{16}$	-62394.1	7.6	-62393.7	1.3	0.0	U			TG1	1.5	11Ke03
$^{176}HfO-C_{16}$	-63668.5	9.8	-63675.6	1.6	-0.5	U			TG1	1.5	11Ke03
$^{176}Ta-u$	-55143	33				2			GS2	1.0	05Li24
$^{176}W-u$	-54420	104	-54370	30	0.5	U			GS1	1.0	00Ra23
	-54366	30				2			GS2	1.0	05Li24
$^{176}Re-u$	-48380	110	-48380	30	0.0	U			GS1	1.0	00Ra23
	-48377	30				2			GS2	1.0	05Li24
$^{176}Os-u$	-45150	110	-45230	12	-0.7	U			GS1	1.0	00Ra23
	-45194	30			-1.2	1	15	15 ^{176}Os	GS2	1.0	05Li24
$^{176}Ir-u$	-36328	30	-36374	9	-1.5	1	8	8 ^{176}Ir	GS2	1.0	05Li24
$^{176}Yb^{35}Cl_2-^{172}Yb^{37}Cl_2$	12088.9	2.4	12088.29	0.14	-0.1	U			H27	2.5	74Ba90
$^{176}Yb^{35}Cl-^{174}Yb^{37}Cl$	6652	3	6657.28	0.07	0.7	U			H23	2.5	70Wh01
	6656.3	1.4			0.3	U			H27	2.5	74Ba90
$^{176}Hf^{35}Cl-^{174}Hf^{37}Cl$	4106	16	4311.5	1.9	5.1	B			H24	2.5	73Ba40
	4314.21	0.86			-1.2	1	76	74 ^{174}Hf	H37	2.5	77Sh12
$^{176}Lu-^{175}Lu$	1980	60	1914.50	0.16	-0.3	U			R04	4.0	64De15
$^{176}Yb-^{174}Yb$	4000	50	3707.161	0.017	-1.5	U			R04	4.0	64De15
$^{176}Ir(\alpha)^{172}Re$	5237.3	8.	5260	40	0.4	1	49	48 ^{172}Re			67Si02
$^{176}Pt(\alpha)^{172}Os$	5890.1	5.	5884.8	2.0	-1.0	-			ISa		79Ha10 Z
	5881.4	4.			0.8	-			Bka		82Bo04 Z
	5887.3	3.			-0.8	-			Ora		82De11 Z
	5874.8	8.			1.2	-			Daa		96Pa01
	5881.9	7.			0.4	-			ISa		17An16
	ave.	5885.0	2.0		-0.1	1	99	66 ^{172}Os			average
$^{176}Au(\alpha)^{172}Ir$	6574.2	10.	6433	7	-14.1	B			Ora		75Ca06 *
	6541.5	10.			-10.8	C			GSa		84Sc.A *
	6433.4	7.				5			Anv		14An10 *
$^{176}Au^m(\alpha)^{172}Ir^m$	6436.6	10.	6433	4	-0.3	5			Ora		75Ca06 *
	6428.4	10.			0.5	5			GSa		84Sc.A *
	6433.4	6.			0.0	5			Ara		01Ko44 *
	6434.4	7.			-0.1	5			Anv		14An10 *
$^{176}Hg(\alpha)^{172}Pt$	6907.2	20.	6897	6	-0.5	o			GSa		83Sc24
	6924.7	10.			-2.8	C			GSa		84Sc.A
	6907.3	20.			-0.5	U			Daa		96Pa01
	6897.0	6.			0.0	-			Ara		99Po09
	6917.5	15.			-1.3	-			Anv		09An20
	ave.	6900	6		-0.5	1	95	72 ^{176}Hg			average
$^{176}Yb(p,\alpha)^{173}Tm$	7628.8	4.4				2			NDm		78Ta10
$^{176}Yb(p,t)^{174}Yb$	-4216	5	-4207.642	0.015	1.7	U			Min		73Oo01
$^{176}Hf(p,t)^{174}Hf$	-6397	5	-6392.7	1.7	0.9	1	12	12 ^{174}Hf	Min		73Oo01
$^{176}Yb(d,t)^{175}Yb$	-621	12	-609.85	0.07	0.9	U			Kop		66Bu16
$^{175}Lu(n,\gamma)^{176}Lu$	6293.2	1.2	6287.97	0.15	-4.4	B					70Wa20
	6287.96	0.15			0.1	1	100	79 ^{176}Lu	ILn		91KI02 Z
	6289.78	0.24			-7.5	C			Bdn		06Fi.A
$^{175}Lu(d,p)^{176}Lu$	4070	8	4063.41	0.15	-0.8	U			Tal		67St14
$^{176}Lu(d,t)^{175}Lu$	-25	15	-30.74	0.15	-0.4	U			Tal		71Mi01
$^{176}Hf(d,t)^{175}Hf$	-1925	8	-1908.8	1.8	2.0	U			Tal		73Za08
$^{176}Tl(p)^{175}Hg$	1265.2	18.				9			Jyp		04Ke06
$^{176}Tm(\beta^-)^{176}Yb$	4120	100				2					67Gu11 *
$^{176}Lu(\beta^-)^{176}Hf$	1162	25	1194.1	0.9	1.3	U					69Pr11 *
	1194.1	1.0			-0.0	1	76	75 ^{176}Hf			73Va11 *
$^{176}Ta(\beta^+)^{176}Hf$	3100	90	3210	30	1.2	U					71Be10 *
$*^{176}Au(\alpha)^{172}Ir$	$E_\alpha=6260(10)$ coinc. with $E(\gamma)=168.4(0.5)$ keV										75Ca06 **
$*^{176}Au(\alpha)^{172}Ir$	$E_\alpha=6228(10)$ to $168.4(0.5)$ γ										84Sc.A **
$*^{176}Au(\alpha)^{172}Ir$	$E_\alpha=6260$ correlated with ^{172}Ir $E_\alpha=5510$ keV										02Ro17 **
$*^{176}Au(\alpha)^{172}Ir$	and $E_\alpha=6157(20), 6138(15), 6054(20), 5798(20)$ to $126.3, 151.5, 236.6, 500.0$										14An10 **
$*^{176}Au^m(\alpha)^{172}Ir^m$	$E_\alpha=6286$ correlated with $^{172}Ir^m$ $E_\alpha=5828$ keV										02Ro17 **
$*^{176}Au^m(\alpha)^{172}Ir^m$	$E_\alpha=6119+E(\gamma)=175.1$ was misassigned to ^{177}Au										01Ko44 **
$*^{176}Au^m(\alpha)^{172}Ir^m$	$E_\alpha=6115(6)$ coinc. with 175.1 γ of reference										84Sc.A **
$*^{176}Au^m(\alpha)^{172}Ir^m$	$E_\alpha=6287(7), 6117(7), 6082(7)$ to $0, 175.2, 211.6$ above $^{172}Ir^m$										14An10 **
$*^{176}Tm(\beta^-)^{176}Yb$	$E_{\beta^-}=2000(100), 1150(100)$ to $(3^+, 4^+)$ level at $2053.34, (3^+, 4^+, 5^+) 3052.2$										Ens062 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
* $^{176}\text{Lu}(\beta^-)^{176}\text{Hf}$	$E_{\beta^-}=565(25)$ to 6^+ level at 596.82 keV										Ens062 **
* $^{176}\text{Lu}(\beta^-)^{176}\text{Hf}$	$Q_{\beta^-}=1317(1)$ from $^{176}\text{Lu}^m$ at 122.845 keV										Nub211 **
* $^{176}\text{Ta}(\beta^+)^{176}\text{Hf}$	KLM/ $\beta^+=119(50)$ to 2^- level at 1247.70 keV, 1^+ level at 2994 keV										Ens062 **
$^{177}\text{Ta}-u$	-55559	30	-55518	4	1.4	U			GS2	1.0	05Li24
$^{177}\text{W}-u$	-53420	110	-53360	30	0.6	U			GS1	1.0	00Ra23
	-53357	30				2			GS2	1.0	05Li24
$^{177}\text{Re}-u$	-49620	104	-49670	30	-0.5	U			GS1	1.0	00Ra23
	-49672	30				2			GS2	1.0	05Li24
$^{177}\text{Os}-u$	-45020	104	-45042	16	-0.2	U			GS1	1.0	00Ra23
	-45012	30			-1.0	R			GS2	1.0	05Li24
$^{177}\text{Ir}-u$	-38810	110	-38699	21	1.0	U			GS1	1.0	00Ra23
	-38699	30			0.0	2			GS2	1.0	05Li24
$^{177}\text{Pt}-u$	-31545	30	-31530	16	0.5	1	29	29 ^{177}Pt	GS2	1.0	05Li24
$^{177}\text{Hf O}-\text{C}_{16}$	-61845.2	7.2	-61855.2	1.5	-0.9	U			TG1	1.5	11Ke03
$^{177}\text{Ir}(\alpha)^{173}\text{Re}$	5127.1	10.	5080	30	-0.9	F					67Si02 *
$^{177}\text{Pt}(\alpha)^{173}\text{Os}$	5654.6	6.	5642.9	2.7	-1.9	-			ISa		79Ha10 Z
	5640.5	3.1			0.8	-			Bka		82Bo04 Z
	ave.	5643.3	2.7		-0.2	1	99	55 ^{177}Pt			average
$^{177}\text{Au}(\alpha)^{173}\text{Ir}$	6292.5	10.	6298	4	0.5	-			Daa		75Ca06
	6292.5	20.			0.3	U			GSa		84Sc.A
	6296.5	10.			0.1	-			Daa		96Pa01
	6298.6	6.			-0.1	-			Ara		01Ko44
	6303.7	7.			-0.8	o			Anv		09An14
	6301.6	7.2			-0.5	-			Anv		18Cu04
	ave.	6298	4		-0.1	1	98	88 ^{173}Ir			average
$^{177}\text{Au}^m(\alpha)^{173}\text{Ir}^m$	6251.5	10.	6262	4	1.0	3			Ora		75Ca06
	6260.8	10.			0.1	3			GSa		84Sc.A *
	6259.7	9.			0.2	3			Daa		96Pa01 *
	6263.8	6.			-0.3	3			Ara		01Ko44
	6265.8	7.			-0.6	3			Anv		09An14
$^{177}\text{Hg}(\alpha)^{173}\text{Pt}$	6732.4	8.	6740	60	0.1	4			ISa		79Ha10
	6747.8	10.			-0.2	4					91Ko.A
	6729.4	9.2			0.1	4			Daa		96Pa01
	6734.5	15.			0.0	4			Anv		09An20
$^{177}\text{Tl}(\alpha)^{173}\text{Au}$	7067.0	7.				3			Ara		99Po09
$^{177}\text{Tl}^m(\alpha)^{173}\text{Au}^m$	7660.4	13.	7660	9	-0.0	-			Ara		99Po09
	7645.1	13.			1.1	-			Jya		04Ke06
	ave.	7653	9		0.8	1	86	48 $^{173}\text{Au}^m$			average
$^{177}\text{Hf}(p,t)^{175}\text{Hf}$	-6071	5	-6059.8	2.0	2.2	1	16	14 ^{175}Hf	Min		73Oo01
$^{176}\text{Yb}(n,\gamma)^{177}\text{Yb}$	5565.1	1.0	5566.40	0.22	1.3	U					72Al19 Z
	5566.40	0.22				2			Bdn		06Fi.A
$^{176}\text{Yb}(d,p)^{177}\text{Yb}$	3340	16	3341.83	0.22	0.1	U			Tal		63Ve09
	3337	12			0.4	U			Kop		66Bu16
$^{176}\text{Yb}(\alpha,t)^{177}\text{Lu}-^{174}\text{Yb}()^{175}\text{Lu}$	674.1	1.0	671.42	0.22	-2.7	U			McM		75Bu02
$^{176}\text{Lu}(n,\gamma)^{177}\text{Lu}$	7071.2	0.4	7072.89	0.16	4.2	B					71Ma45 Z
	7073.1	0.4			-0.5	-					72Mi16 Z
	7072.85	0.17			0.2	-			Bdn		06Fi.A
$^{176}\text{Lu}(d,p)^{177}\text{Lu}$	4843	10	4848.32	0.16	0.5	U			Tal		71Mi01
$^{176}\text{Lu}(n,\gamma)^{177}\text{Lu}$	ave.	7072.89	0.16	7072.89	0.16	-0.0	1	99	92 ^{177}Lu		average
$^{176}\text{Hf}(n,\gamma)^{177}\text{Hf}$	6385.8	0.8	6375.6	1.0	-12.7	C			Bdn		06Fi.A
$^{177}\text{Hf}(\gamma,n)^{176}\text{Hf}$	-6400	30	-6375.6	1.0	0.8	U			Phi		60Ge01
$^{176}\text{Hf}(d,p)^{177}\text{Hf}$	4150	7	4151.1	1.0	0.2	U			Tal		68Ri07
$^{177}\text{Hf}(d,t)^{176}\text{Hf}$	-127	11	-118.4	1.0	0.8	U			Tal		72Za04
$^{177}\text{Tl}(p)^{176}\text{Hg}$	1162.6	20.	1156	19	-0.4	o			Arp		99Po09 *
$^{177}\text{Tl}^m(p)^{176}\text{Hg}$	1969.2	10.	1963	7	-0.7	-			Arp		99Po09
	1965.2	12.			-0.2	-			Jyp		04Ke06
	ave.	1968	8		-0.7	1	90	62 $^{177}\text{Tl}^m$			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{177}\text{Yb}(\beta^-)^{177}\text{Lu}$	1400	20	1397.5	1.2	-0.1	U					64Jo03
$^{177}\text{Lu}(\beta^-)^{177}\text{Hf}$	497	2	496.8	0.8	-0.1	-					55Ma12
	496.4	1.0			0.4	-					62El02 *
ave.	496.5	0.9			0.4	1	78	70 ^{177}Hf			average
$^{177}\text{Ta}(\beta^+)^{177}\text{Hf}$	1166	3				2					61We11
$^{177}\text{Au}^m(\text{IT})^{177}\text{Au}$	210	30	190	7	-0.7	o					01Ko44 *
$^{177}\text{Tl}^m(\text{IT})^{177}\text{Tl}$	807	18				2					99Po09
$^{177}\text{Ir}(\alpha)^{173}\text{Re}$	F : final state uncertain: possibly to $5/2^-$ level at 214.7 keV										95Hi02 **
$^{177}\text{Au}^m(\alpha)^{173}\text{Ir}^m$	Followed by a $175.1(0.5) \gamma$										84Sc.A **
*	γ associated with $E_\alpha=6116$ keV from ^{176}Au										01Ko44 **
*	yet $E_\alpha=6118$ correlated with $E_\alpha=5672$ of $^{173}\text{Ir}^m$										02Ro17 **
$^{177}\text{Au}^m(\alpha)^{173}\text{Ir}^m$	E_α correlated with ^{173}Ir $E_\alpha=5681(13)$; also with ^{181}Tl $E_\alpha=6180$ keV										96Pa01 **
*	evaluator doubts about correctness of latter remark										AHW **
$^{177}\text{Tl}(\text{p})^{176}\text{Hg}$	Replaced by $^{177}\text{Tl}^m(\text{IT})$										AHW **
$^{177}\text{Lu}(\beta^-)^{177}\text{Hf}$	$E_{\beta^-}=384(2)$, $175(1)$ to 112.95 , 321.32 levels										Ens035 **
$^{177}\text{Au}^m(\text{IT})^{177}\text{Au}$	Authors say $157.9+x$, x estimated by evaluator										AHW **
*	x is better known from $^{181}\text{Tl}^m(\text{IT})^{181}\text{Tl}$ combined with Q_α										09An14 **
$^{178}\text{Yb}-^{85}\text{Rb}_{2.094}$	131396.8	9.4	131382	7	-1.6	1	57	57 ^{178}Yb	MA8	1.0	19Hu15
$^{178}\text{W}-\text{u}$	-54152	30	-54114	16	1.3	U			GS2	1.0	05Li24
$^{178}\text{Re}-\text{u}$	-48800	110	-49010	30	-1.9	U			GS1	1.0	00Ra23
	-49011	30				2			GS2	1.0	05Li24
$^{178}\text{Os}-\text{u}$	-46790	104	-46747	15	0.4	U			GS1	1.0	00Ra23
	-46710	30			-1.2	1	24	24 ^{178}Os	GS2	1.0	05Li24
$^{178}\text{Ir}-\text{u}$	-38950	110	-38921	20	0.3	U			GS1	1.0	00Ra23
	-38888	30			-1.1	1	45	45 ^{178}Ir	GS2	1.0	05Li24
$^{178}\text{Pt}-\text{u}$	-34783	1181	-34351	11	0.1	U				2.5	91Br17
	-34300	110			-0.5	U			GS1	1.0	00Ra23
	-34333	30			-0.6	1	13	13 ^{178}Pt	GS2	1.0	05Li24
$^{178}\text{Au}-^{133}\text{Cs}_{1.338}$	102562	11				2			MA8	1.0	20Cu04
$^{178}\text{Au}^n-^{133}\text{Cs}_{1.338}$	102764	11	102762	10	-0.2	1	87	87 $^{178}\text{Au}^n$	MA8	1.0	20Cu04
$^{178}\text{Hf }^{35}\text{Cl}-^{176}\text{Hf }^{37}\text{Cl}$	5236	5	5248.6	1.1	1.0	U			H23	2.5	70Wh01
	5239.5	1.3			2.8	U			H27	2.5	74Ba90
$^{178}\text{Hf O}-\text{C}_{16}$	-61364.8	7.9	-61377.1	1.5	-1.0	U			TG1	1.5	11Ke03
$^{178}\text{Pt}-^{175}\text{Ir}$	-472	1052	1500	17	0.7	U				2.5	91Br17
$^{178}\text{Pt}(\alpha)^{174}\text{Os}$	5583.3	5.	5573.0	2.2	-2.0	-			ISa		79Ha10 Z
	5569.9	3.			1.0	-			Bka		82Bo04 Z
	5568.4	13.			0.4	U			Lvn		94Wa23
	5572.4	4.			0.1	-			Ara		00Ko16 *
ave.	5573.2	2.2			-0.1	1	99	75 ^{174}Os			average
$^{178}\text{Au}(\alpha)^{174}\text{Ir}$	6056.4	10.	6058	5	0.2	3					68Si01 *
	6058.4	5.1			-0.1	3			ISa		20Cu04
$^{178}\text{Au}^n(\alpha)^{174}\text{Ir}^m$	6117.7	20.	6120	6	0.1	-			GSa		86Ke03
	6119.0	7.0			0.1	-			ISa		20Cu04 *
ave.	6119	7			0.1	1	95	82 $^{174}\text{Ir}^m$			average
$^{178}\text{Hg}(\alpha)^{174}\text{Pt}$	6578.1	6.	6577.3	3.0	-0.1	3			ISa		79Ha10
	6576.1	9.			0.1	3			Daa		96Pa01
	6577.1	4.			0.1	3			Ara		00Ko48
	6578.1	8.			-0.1	3			Anv		09An14
$^{178}\text{Tl}(\alpha)^{174}\text{Au}$	7017.0	5.	7020	10	0.6	o			Bka		02Ro17 *
	7020.0	10.				8			Bka		13Li49 *
$^{178}\text{Pb}(\alpha)^{174}\text{Hg}$	7790.4	14.	7789	13	-0.1	3			Bka		01Ro.B
	7785.2	30.7			0.1	3			Jya		16Ba60
$^{178}\text{Pt}(\text{p},\alpha)^{175}\text{Ir}$	4420	980	6261	16	1.9	U					91Br17
$^{176}\text{Yb}(\text{t,p})^{178}\text{Yb}$	3865	10	3847	7	-1.8	1	43	43 ^{178}Yb	Phi		82Zu02

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{176}\text{Lu}(\text{t,p})^{178}\text{Lu}^m$	4482	5	4492.6	2.9	2.1	1	34	34 $^{178}\text{Lu}^m$	LAl		81Gi01
$^{178}\text{Hf}(\text{p,t})^{176}\text{Hf}$	-5531	5	-5519.8	1.0	2.2	U			Min		73Oo01
$^{177}\text{Hf}(\text{n},\gamma)^{178}\text{Hf}$	7625	1	7625.94	0.18	0.9	U					69Fa01
	7624.4	1.5			1.0	U					77St10
	7626.2	0.3			-0.9	-			ILn		86Ha22
	7625.80	0.22			0.6	-			Bdn		06Fi.A
$^{178}\text{Hf}(\text{d,t})^{177}\text{Hf}$	-1364	9	-1368.71	0.18	-0.5	U			Tal		68Ri07
$^{177}\text{Hf}(\text{n},\gamma)^{178}\text{Hf}$	ave. 7625.94	0.18	7625.94	0.18	-0.0	1	99	71 ^{178}Hf			average
$^{178}\text{Yb}(\beta^-)^{178}\text{Lu}$	641	30	661	7	0.7	U					73Or03
$^{178}\text{Lu}^m(\text{IT})^{178}\text{Lu}$	120	3	123.8	2.6	1.3	1	76	66 $^{178}\text{Lu}^m$	McM		93Bu02
$^{178}\text{Lu}(\beta^-)^{178}\text{Hf}$	2046	50	2097.5	2.1	1.0	U					73Or03
	2117	30			-0.7	U					75Ka15
$^{178}\text{Ta}^m(\beta^+)^{178}\text{Hf}$	1937	15				2					61Ga05
$^{178}\text{W}(\epsilon)^{178}\text{Ta}^m$	91.3	2.				3					67Ni02
$^{178}\text{Re}(\beta^+)^{178}\text{W}$	4660	180	4750	30	0.5	U					70Go20
$^{178}\text{Pt}(\alpha)^{174}\text{Os}$	Also $E_\alpha=5289(8)$ keV to 2^+ 158.601 level (not used)										GAu
$^{178}\text{Au}^n(\alpha)^{174}\text{Ir}^m$	$E_\alpha=5977(10), 5925(7), 5571(7), 5521(7)$ to n, 56.8, 421.4, 472.1										FGK209
$^{178}\text{Ti}(\alpha)^{174}\text{Au}$	And a stronger $E_\alpha=6704$; both correlated with ^{174}Au $E_\alpha=6538$ keV										02Ro17
$^{178}\text{Ti}(\alpha)^{174}\text{Au}$	Also 6693(10) and 6595(10) to 173.0, 273.0 levels										13Li49
$^{178}\text{Yb}(\beta^-)^{178}\text{Lu}$	$E_{\beta^-}=250(30)$ to 1^+ level at 390.8 keV										Ens097
$^{178}\text{Lu}(\beta^-)^{178}\text{Hf}$	$E_{\beta^-}=2000(50)$ to ground state and 50% to 2^+ level at 93.18 keV										Ens097
$^{178}\text{Lu}(\beta^-)^{178}\text{Hf}$	$E_{\beta^-}=2050(50)$ to ground state and 50% to 2^+ level at 93.18 keV and										Ens097
	$E_{\beta^-}=770(30)$ from $^{178}\text{Lu}^m$ at 123.8(2.6) to 8^- level 1479.025 keV										Nub211
$^{178}\text{Ta}^m(\beta^+)^{178}\text{Hf}$	$E_{\beta^+}=890(10)$ to ground state and 2^+ level at 93.18 keV, ratio 2.7 to 1										Ens097
$^{178}\text{Re}(\beta^+)^{178}\text{W}$	$E_{\beta^+}=3300(180)$ to 4^+ level at 342.74 keV										Ens097
$^{179}\text{Lu}-^{85}\text{Rb}_{2.106}$	132997	21	133104	6	5.1	C			MA8	1.0	13Ro.A
$\text{C}_{14}\text{H}_{11}-^{179}\text{Hf}$	140260.3	1.8	140249.6	1.5	-1.5	U			M23	4.0	79Ha32
$^{179}\text{W}-\text{u}$	-52964	76	-52921	16	0.6	U			GS2	1.0	05Li24
$^{179}\text{Re}-\text{u}$	-50010	30	-50010	26	-0.0	1	78	78 ^{179}Re	GS2	1.0	05Li24
$^{179}\text{Os}-\text{u}$	-46220	104	-46184	17	0.3	U			GS1	1.0	00Ra23
	-46176	30			-0.3	1	31	31 ^{179}Os	GS2	1.0	05Li24
$^{179}\text{Ir}-\text{u}$	-40910	104	-40882	10	0.3	U			GS1	1.0	00Ra23
	-40852	30			-1.0	1	12	12 ^{179}Ir	GS2	1.0	05Li24
$^{179}\text{Pt}-\text{u}$	-34710	110	-34641	9	0.6	U			GS1	1.0	00Ra23
	-34625	30			-0.5	U			GS2	1.0	05Li24
$^{179}\text{Au}-\text{u}$	-26811	31	-26826	13	-0.5	1	16	16 ^{179}Au	GS2	1.0	05Li24
$^{179}\text{Hg}-^{208}\text{Pb}_{.861}$	1900	34	1920	30	0.7	1	79	79 ^{179}Hg	MA6	1.0	01Sc41
$^{179}\text{Hf }^{35}\text{Cl}-^{177}\text{Hf }^{37}\text{Cl}$	5539	3	5545.64	0.22	0.9	U			H23	2.5	70Wh01
	5544.4	0.7			0.7	U			H27	2.5	74Ba90
$^{179}\text{Hf O}-\text{C}_{16}$	-59261.8	6.5	-59259.7	1.5	0.2	U			TG1	1.5	11Ke03
$^{179}\text{Pt}(\alpha)^{175}\text{Os}$	5370	10	5412	9	4.2	F					66Si08
	5416	10			-0.4	1	89	82 ^{175}Os	ISa		79Ha10
	5382	3			10.1	F			Bka		82Bo04
$^{179}\text{Au}(\alpha)^{175}\text{Ir}$	5981.8	5.	5981	5	-0.1	1	97	80 ^{175}Ir			68Si01
	5986.9	15.			-0.4	U			Jya		04Ra28
$^{179}\text{Hg}(\alpha)^{175}\text{Pt}$	6431.0	5.	6350	30	-1.6	-			ISa		79Ha10
	6418.7	9.			-1.3	-			Daa		96Pa01
	6430.0	4.			-1.6	-			Ara		02Ko09
	ave. 6427	29			-1.3	1	29	21 ^{179}Hg			average
$^{179}\text{Ti}(\alpha)^{175}\text{Au}$	6710.2	20.	6709.1	2.6	-0.1	U			GSa		83Sc24
	6718.4	18.			-0.5	U			Daa		96Pa01
	6719.4	10.			-1.0	7			Ara		98To14
	6706.1	8.			0.4	7			Ara		11Ko.B
	6710.2	4.			-0.3	7			Anv		13An10
	6707.2	4.1			0.5	7			Jya		17Ba46

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{179}\text{Tl}^m(\alpha)^{175}\text{Au}^m$	7364.5	20.	7369.7	3.0	0.3	U			GSa		83Sc24
	7366.0	20.			0.2	U			Daa		96Pa01
	7378.1	10.			-0.8	o			Ara		98To14
	7372.0	5.1			-0.4	9			Anv		10An01
	7358.7	8.2			1.3	9			Ara		11Ko.B
	7371.0	4.1			-0.3	9			Jya		17Ba46
$^{179}\text{Pb}(\alpha)^{175}\text{Hg}$	7598.3	20.	7596	5	-0.1	9			Anv		10An01 *
	7596	5			0.0	9			Jya		17Ba46 *
$^{179}\text{Hf}(\text{p,t})^{177}\text{Hf}$	-5249	5	-5243.13	0.19	1.2	U			Min		73Oo01
$^{179}\text{Hf}(\text{t},\alpha)^{178}\text{Lu}-^{178}\text{Hf}(\gamma)^{177}\text{Lu}$	-72	2	-73.7	1.9	-0.8	1	89	89 ^{178}Lu	McM		93Bu02
$^{178}\text{Hf}(\text{n},\gamma)^{179}\text{Hf}$	6099.02	0.10	6098.99	0.08	-0.3	-			ILn		89Ri03 Z
	6098.95	0.12			0.3	-			Bdn		06Fi.A
$^{179}\text{Hf}(\gamma,\text{n})^{178}\text{Hf}$	-6000	70	-6098.99	0.08	-1.4	U			Phi		60Ge01
$^{178}\text{Hf}(\text{d,p})^{179}\text{Hf}$	3877	14	3874.42	0.08	-0.2	U			Tal		63Ve09
$^{178}\text{Hf}(\text{n},\gamma)^{179}\text{Hf}$	ave.	6098.99	6098.99	0.08	-0.0	1	100	71 ^{179}Hf			average
$^{179}\text{Lu}(\beta^-)^{179}\text{Hf}$	1350	50	1404	5	1.1	U					61Ku10
	1380	70			0.3	U					63St06
$^{179}\text{Ta}(\epsilon)^{179}\text{Hf}$	129	16	105.6	0.4	-1.5	U					61Jo15 *
	105.61	0.41			-0.1	1	99	93 ^{179}Ta			01Hi06
$^{179}\text{Re}(\beta^+)^{179}\text{W}$	2710	50	2711	27	0.0	1	29	22 ^{179}Re			75Me20 *
$^{179}\text{W}-\text{u}$	$M-A=-49225(29)$ keV for mixture gs+m at 221.91 keV										Nub211 **
$^{179}\text{Pt}(\alpha)^{175}\text{Os}$	F : part of double line (with ^{180}Pt)										AHW **
$^{179}\text{Pt}(\alpha)^{175}\text{Os}$	$E_\alpha=5150(10)$ 5195(10) 5161(3) respectively, to $1/2^-$ level at 102.3 keV, recalibrated										Ens092 **
$^{179}\text{Au}(\alpha)^{175}\text{Ir}$	$E_\alpha=5853(15)$, 5810(15) to ground state, 49 keV level										04Ra28 **
$^{179}\text{Pb}(\alpha)^{175}\text{Hg}$	$E_\alpha=7350(20)$ to 80 keV level										10An01 **
$^{179}\text{Pb}(\alpha)^{175}\text{Hg}$	$E_\alpha=7348(5)$ to 80 keV level										17Ba46 **
$^{179}\text{Ta}(\epsilon)^{179}\text{Hf}$	As corrected in reference										76He03 **
$^{179}\text{Re}(\beta^+)^{179}\text{W}$	$E_{\beta^+}=950(50)$ to $3/2^+$ level at 720.18 and $5/2^+$ at 773.65 keV										Ens092 **
$\text{C}_{14}\text{H}_{12}-^{180}\text{Hf}$	147356.6	4.8	147340.8	1.5	-0.8	U			M23	4.0	79Ha32
$^{180}\text{W}-\text{u}$	-53299	30	-53286.7	1.5	0.4	U			GS2	1.0	05Li24
$^{180}\text{Re}-\text{u}$	-49209	30	-49208	23	0.0	2			GS2	1.0	05Li24
$^{180}\text{Os}-\text{u}$	-47650	104	-47618	17	0.3	U			GS1	1.0	00Ra23
	-47626	30			0.3	1	32	32 ^{180}Os	GS2	1.0	05Li24
$^{180}\text{Ir}-\text{u}$	-40800	104	-40771	23	0.3	U			GS1	1.0	00Ra23
	-40765	30			-0.2	2			GS2	1.0	05Li24
$^{180}\text{Pt}-\text{u}$	-36900	104	-36962	11	-0.6	U			GS1	1.0	00Ra23
	-36918	30			-1.5	1	13	13 ^{180}Pt	GS2	1.0	05Li24
$^{180}\text{Au}-\text{u}$	-27496	30	-27510	5	-0.5	U			GS2	1.0	05Li24
$^{180}\text{Au}-^{133}\text{Cs}_{1.353}$	100411.5	5.3	100413	5	0.3	1	93	93 ^{180}Au	MA8	1.0	17Ma29
$^{180}\text{Hg}-^{208}\text{Pb}_{.865}$	-1569	22	-1544	14	1.1	1	38	38 ^{180}Hg	MA6	1.0	01Sc41
$^{180}\text{Hf}^{35}\text{Cl}_2-^{176}\text{Hf}^{37}\text{Cl}_2$	11036.1	3.0	11050.0	1.1	1.9	U			H27	2.5	74Ba90
$^{180}\text{Hf}^{35}\text{Cl}-^{178}\text{Hf}^{37}\text{Cl}$	5797	3	5801.34	0.19	0.6	U			H23	2.5	70Wh01
	5798.4	0.7			1.7	U			H27	2.5	74Ba90
$^{180}\text{W}-^{180}\text{Hf}$	153.73	0.30	153.77	0.30	0.1	1	98	82 ^{180}W	SH1	1.0	12Dr01
$^{180}\text{Hf}-^{179}\text{Hf}$	730.8	4.7	733.83	0.16	0.3	U			M24	2.5	79Ha32
$^{180}\text{Hf O}-\text{C}_{16}$	-58524.5	6.5	-58525.8	1.5	-0.1	U			TG1	1.5	11Ke03
$^{180}\text{W}(\alpha)^{176}\text{Hf}$	2516.4	1.6	2515.3	1.0	-0.7	1	41	23 ^{176}Hf			04Co26
$^{180}\text{Pt}(\alpha)^{176}\text{Os}$	5257.1	10.	5276	5	1.9	F					66Si08 *
	5279	3			-0.9	F			Bka		82Bo04 *
	5277.5	5.1			-0.2	1	97	85 ^{176}Os			20Cu02
$^{180}\text{Au}(\alpha)^{176}\text{Ir}$	5814.4	10.2	5831	7	1.7	F			GSa		86Ke03 *
	5857	30			-0.9	F			Lvn		93Wa03 *
	5833.6	7.			-0.3	1	93	90 ^{176}Ir	ISa		20Ha24 *
$^{180}\text{Hg}(\alpha)^{176}\text{Pt}$	6258.3	5.	6258.5	2.3	0.0	-			ISa		79Ha10 Z
	6259.5	5.			-0.2	-			Lvn		93Wa03 *
	6258.3	4.			0.0	-			Ara		00Ko48
	6259.3	5.			-0.2	-			Anv		03An27
	6258.3	7.			0.0	-			ISa		17An16
	ave.	6258.8	2.3		-0.1	1	99	66 ^{176}Pt			average
$^{180}\text{Tl}(\alpha)^{176}\text{Au}$	6709.4	10.	6710	60	-0.1	6			Ara		98To14 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{180}\text{Tl}(\alpha)^{176}\text{Au}$	6701.9	7.	6710	60	0.1	6			Isa		17An16
$^{180}\text{Pb}(\alpha)^{176}\text{Hg}$	7394.6	40.	7419	5	0.6	U			ORa		96To08
	7415.1	15.			0.2	2			Ara		99To11
	7419.2	10.			-0.1	2			Anv		09An20
	7419.2	7.			-0.1	2			Jya		10Ra12
$^{180}\text{Hf}(\text{p,t})^{178}\text{Hf}$	-5011	5	-5004.95	0.17	1.2	U			Min		73Oo01
$^{180}\text{Hf}(\text{t},\alpha)^{179}\text{Lu}-^{178}\text{Hf}(\gamma)^{177}\text{Lu}$	-669	5	-669	5	0.0	1	100	100 ^{179}Lu	McM		92Bu12
$^{179}\text{Hf}(\text{n},\gamma)^{180}\text{Hf}$	7387.3	0.4	7387.76	0.15	1.1	-					74Bu22 Z
	7387.8	0.6			-0.1	-					90Bo52 Z
	7387.85	0.17			-0.5	-			Bdn		06Fi.A
$^{180}\text{Hf}(\gamma,\text{n})^{179}\text{Hf}$	-7470	110	-7387.76	0.15	0.7	U			Phi		60Ge01
$^{179}\text{Hf}(\text{d,p})^{180}\text{Hf}$	5167	7	5163.19	0.15	-0.5	U			Tal		72Za04
$^{180}\text{Hf}(\text{d,t})^{179}\text{Hf}$	-1112	4	-1130.53	0.15	-4.6	B			Tal		68Ri07
$^{179}\text{Hf}(\text{n},\gamma)^{180}\text{Hf}$	ave. 7387.77	0.15	7387.76	0.15	-0.1	1	99	83 ^{180}Hf			average
$^{180}\text{W}(\text{d,t})^{179}\text{W}$	-2155	15	-2155	15	-0.0	1	94	94 ^{179}W	Kop		72Ca01
$^{180}\text{Lu}(\beta^-)^{180}\text{Hf}$	3148	100	3100	70	-0.5	2					71Gu02 *
	3058	100			0.4	2					71Sw01 *
$^{180}\text{Ta}(\beta^-)^{180}\text{W}$	705	15	702.6	2.4	-0.2	U					51Br87
	712	15			-0.6	U					62Ga07
$^{180}\text{Re}(\beta^+)^{180}\text{W}$	3830	60	3799	21	-0.5	R					67Go22 *
	3790	40			0.2	R					67Ho12 *
* $^{180}\text{Pt}(\alpha)^{176}\text{Os}$	F : part of double line (with ^{179}Pt); $E_\alpha=5140(10)$ keV										AHW **
* $^{180}\text{Pt}(\alpha)^{176}\text{Os}$	F : part of double line (with ^{179}Pt)										AHW **
* $^{180}\text{Au}(\alpha)^{176}\text{Ir}$	$E_\alpha=5685(10)$ to $40(30)$ level, former treatment not supported by 20Ha24										93Wa03 **
* $^{180}\text{Au}(\alpha)^{176}\text{Ir}$	$E_\alpha=5647(10,Z)$ to $80(30)$ level, not supported by 20Ha24										93Wa03 **
* $^{180}\text{Au}(\alpha)^{176}\text{Ir}$	$E_\alpha=5639(7)$ to $46.0(0.9)$ keV level										20Ha24 **
* $^{180}\text{Hg}(\alpha)^{176}\text{Pt}$	$E_\alpha=6120$ 5862 5689(5) to ground state, 2^+ level at 264.0, 0^+ at 443 keV										Ens062 **
* $^{180}\text{Hg}(\alpha)^{176}\text{Pt}$	supersedes reference										93Wa.A **
* $^{180}\text{Tl}(\alpha)^{176}\text{Au}$	Highest E_α ; not necessarily ground state to ground state										98To14 **
* $^{180}\text{Lu}(\beta^-)^{180}\text{Hf}$	$E_{\beta^-}=1540(100)$ 1450(100) respectively, to 4^+ level at 1607.67 keV										Ens156 **
* $^{180}\text{Re}(\beta^+)^{180}\text{W}$	$E_{\beta^+}=1800(60)$ 1760(40) respectively, to 2^- level 1006.381 keV										Ens156 **
$^{181}\text{Lu}-\text{u}$	-48092	135				2			GS3	1.0	13Sh30
$^{181}\text{Ta O}-^{202}\text{Tl}_{975}$	-29891.4	2.5	-29893.0	1.9	-0.6	1	60	37 ^{202}Tl	MA8	1.0	17We09
$^{181}\text{Re}-\text{u}$	-49915	30	-49938	13	-0.8	R			GS2	1.0	05Li24
$^{181}\text{Os}-\text{u}$	-46670	110	-46753	27	-0.8	U			GS1	1.0	00Ra23 *
	-46756	34			0.1	1	64	64 ^{181}Os	GS2	1.0	05Li24 *
$^{181}\text{Ir}-\text{u}$	-42330	104	-42365	6	-0.3	U			GS1	1.0	00Ra23
	-42372	30			0.2	U			GS2	1.0	05Li24
$^{181}\text{Pt}-\text{u}$	-36880	104	-36910	15	-0.3	U			GS1	1.0	00Ra23
	-36900	30			-0.3	-			GS2	1.0	05Li24
	ave. -36890	21			-0.9	1	48	48 ^{181}Pt			average
$^{181}\text{Au}-\text{u}$	-30030	110	-29921	21	1.0	U			GS1	1.0	00Ra23
	-29920	30			-0.0	R			GS2	1.0	05Li24
$^{181}\text{Hg}-^{208}\text{Pb}_{870}$	-1929	40	-1868	17	1.5	1	17	17 ^{181}Hg	MA6	1.0	01Sc41
$^{181}\text{Tl}-^{133}\text{Cs}_{1361}$	114936	11	114940	10	0.4	1	79	79 ^{181}Tl	MA8	1.0	08We02
$^{181}\text{Ta }^{35}\text{Cl}-^{179}\text{Hf }^{37}\text{Cl}$	5128.6	2.1	5122.9	2.1	-1.1	1	15	9 ^{181}Ta	H35	2.5	80Sh06
$^{181}\text{Ta }^{17}\text{O }^{35}\text{Cl}-^{180}\text{Ta}^m \text{O }^{37}\text{Cl}$	7572	21	7617.32	0.21	0.9	U			H35	2.5	80Sh06
$^{181}\text{Pt}(\alpha)^{177}\text{Os}$	5133.7	20.	5150	5	0.8	U					66Si08
	5150.1	5.				2			ORa		95Bi01
$^{181}\text{Au}(\alpha)^{177}\text{Ir}$	5750.1	5.	5751.4	2.9	0.2	3					68Si01 Z
	5751.9	5.			-0.1	3			ISa		79Ha10 Z
	5735	4			4.1	F			IRa		92Sa03 *
	5752	5			-0.1	3			ORa		95Bi01 *
$^{181}\text{Hg}(\alpha)^{177}\text{Pt}$	6288	5	6284	4	-0.7	-			ISa		79Ha10 *
	6283	10			0.1	-			GSa		86Ke03 *
	6269.3	13.			1.2	-			Daa		96Pa01 *
	ave. 6285	4			-0.2	1	99	83 ^{181}Hg			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{181}\text{Tl}(\alpha)^{177}\text{Au}$	6319.9	20.	6322	4	0.1	U					92Bo.D
	6326.1	10.			-0.4	-			Ara		98To14
	6320.9	7.			0.2	-			Anv		09An14
	6323.0	7.2			-0.1	-			Anv		18Cu04
	ave. 6323	5			-0.2	1	98	^{89}Au			average
$^{181}\text{Tl}(\alpha)^{177}\text{Au}^m$	6120.3	20.	6132	5	0.6	2			GSa		84Sc.A *
	6132.6	10.			-0.1	2			Ara		98To14 *
	6133.1	6.4			-0.2	2			Anv		09An14 *
$^{181}\text{Pb}(\alpha)^{177}\text{Hg}$	7374.3	10.	7240	7	-13.4	F			GSa		86Ke03 *
	7203.5	15.			2.4	5			ORa		89To01
	7224.9	20.			0.7	o			Ara		96To01 *
	7250.7	10.			-1.0	5			Ara		05Ca.A *
	7252.0	15.			-0.8	5			Anv		09An20 *
$^{181}\text{Ta}(\text{p,t})^{179}\text{Ta}$	-5738	5	-5742.4	2.0	-0.9	1	15	^{81}Ta	Min		73Oo01
$^{180}\text{Hf}(\text{n},\gamma)^{181}\text{Hf}$	5695.2	0.6	5694.80	0.07	-0.7	U					71Al22
	5694.80	0.07				2			Prn		02Bo41
	5695.58	0.20			-3.9	C			Bdn		06Fi.A
$^{180}\text{Hf}(\text{d,p})^{181}\text{Hf}$	3440	25	3470.23	0.07	1.2	U			Sac		66Ga06
	3475	10			-0.5	U			Tal		68Ri07
$^{181}\text{Ta}(\gamma,\text{n})^{180}\text{Ta}$	-7713	25	-7576.8	1.3	5.4	B			Phi		60Ge01 *
	-7852	26			10.6	B			Phi		60Ge01
	-7580	5			0.6	U			McM		79Ba06
	-7579	2			1.1	2			McM		81Co17
$^{181}\text{Ta}(\text{d,t})^{180}\text{Ta}$	-1317.7	1.8	-1319.5	1.3	-1.0	2			NDm		79Ta.B
$^{180}\text{Ta}^m(\text{n},\gamma)^{181}\text{Ta}$	7651.8	0.5	7652.08	0.19	0.6	2			MMn		81Co17 Z
	7652.13	0.20			-0.2	2			ILn		84Fo.A Z
$^{180}\text{W}(\text{n},\gamma)^{181}\text{W}$	6669.02	0.16				2			Bdn		15Hu07
$^{180}\text{W}(\text{d,p})^{181}\text{W}$	4468	15	4444.45	0.16	-1.6	U			Kop		72Ca01
$^{181}\text{Hg}(\text{ep})^{180}\text{Pt}$	6150	200	6480	18	1.6	F					72Ho19 *
$^{181}\text{Hf}(\beta^-)^{181}\text{Ta}$	1023	8	1036.1	1.9	1.6	U					52Fa14 *
	1020	5			3.2	B					53Ba81 *
$^{181}\text{W}(\text{e})^{181}\text{Ta}$	184	12	205.1	1.9	1.8	U					66Ra03
	190	6			2.5	U					83Se17
$^{181}\text{Os}(\beta^+)^{181}\text{Re}$	2990	200	2967	28	-0.1	U					67Go25 *
$^{181}\text{Hg}^m(\text{IT})^{181}\text{Hg}$	212	50				2					09An17 *
* $^{181}\text{Os}-\text{u}$	$M-A=-43450(100)$ keV for mixture gs+m at 49.20 keV										Nub211 **
* $^{181}\text{Os}-\text{u}$	$M-A=-43529(28)$ keV for mixture gs+m at 49.20 keV										Nub211 **
* $^{181}\text{Au}(\alpha)^{177}\text{Ir}$	$E_\alpha=5609(8)$, $5462(4)$ to ground state and $(3/2^-)$ level at 148.00 keV										Ens035 **
*	F : all lines in ^{181}Au and ^{183}Au shifted by 16–20keV										GAu **
* $^{181}\text{Au}(\alpha)^{177}\text{Ir}$	$E_\alpha=5626(5)$ to gs; favored $5479(5)$ to $(3/2^-)$ level at 148.00 keV										Ens035 **
* $^{181}\text{Hg}(\alpha)^{177}\text{Pt}$	$E_\alpha=6147.0(10,Z)$, $6005.0(5,Z)$ to ground state and $1/2^-$ isomer at 147.5 keV										Nub211 **
* $^{181}\text{Hg}(\alpha)^{177}\text{Pt}$	$E_\alpha=6136.6(10,Z)$, $6005.6(10,Z)$ to ground state and $1/2^-$ isomer at 147.5 keV										Nub211 **
* $^{181}\text{Hg}(\alpha)^{177}\text{Pt}$	$E_\alpha=5986(13)$ to $1/2^-$ isomer at 147.5 keV										Nub211 **
* $^{181}\text{Tl}(\alpha)^{177}\text{Au}^m$	$E_\alpha=6566(20)$ $Q_\alpha=6956.2$ from $^{181}\text{Tl}^m$ at 835.9 to 241.5 above $^{177}\text{Au}^m$										Nub211 **
* $^{181}\text{Tl}(\alpha)^{177}\text{Au}^m$	$E_\alpha=6578(10)$ $Q_\alpha=6968.5$ from $^{181}\text{Tl}^m$ at 835.9 to 241.5 above $^{177}\text{Au}^m$										Nub211 **
* $^{181}\text{Tl}(\alpha)^{177}\text{Au}^m$	$E_\alpha=6818(15)$, $6578(7)$ from $^{181}\text{Tl}^m$ to $^{177}\text{Au}^m$ and level 241.5 above $^{177}\text{Au}^m$										09An14 **
* $^{181}\text{Pb}(\alpha)^{177}\text{Hg}$	F : This α -line not found in same reaction										96To01 **
* $^{181}\text{Pb}(\alpha)^{177}\text{Hg}$	Seen in correlation with ^{177}Hg $E_\alpha=6580$ keV										96To01 **
* $^{181}\text{Pb}(\alpha)^{177}\text{Hg}$	$E_\alpha=7015(10)$ to level at 77.2 keV										09An20 **
* $^{181}\text{Pb}(\alpha)^{177}\text{Hg}$	$E_\alpha=7016(15)$ to level at 77.2 keV										09An20 **
* $^{181}\text{Ta}(\gamma,\text{n})^{180}\text{Ta}$	$Q=-7640(25)$ to $^{180}\text{Ta}^m$ at 75.3(1.4) keV										Nub211 **
* $^{181}\text{Hg}(\text{ep})^{180}\text{Pt}$	F : retracted by authors in PrvCom										AHW **
* $^{181}\text{Hf}(\beta^-)^{181}\text{Ta}$	$E_{\beta^-}=408(8)$ $405(5)$ respectively, to $^{181}\text{Ta}^n$ at 615.19 keV										Nub211 **
* $^{181}\text{Os}(\beta^+)^{181}\text{Re}$	$E_{\beta^+}=1750(200)$ from $^{181}\text{Os}^m$ at 49.20 to $^{181}\text{Re}^m$ at 262.91 keV										Nub211 **
* $^{181}\text{Hg}^m(\text{IT})^{181}\text{Hg}$	From cascade x+90.3+71.4, with x estimated 50#										09An17 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{182}\text{Re}-u$	-48311	65	-48790	110	-7.3	C			GS2	1.0	03Li.A *
$^{182}\text{Os}-u$	-47883	30	-47890	23	-0.2	1	61	61 ^{182}Os	GS2	1.0	05Li24
$^{182}\text{Ir}-u$	-41942	30	-41924	23	0.6	1	56	56 ^{182}Ir	GS2	1.0	05Li24
$^{182}\text{Pt}-u$	-38870	104	-38828	14	0.4	U			GS1	1.0	00Ra23
	-38860	30			1.1	1	22	22 ^{182}Pt	GS2	1.0	05Li24
$^{182}\text{Au}-u$	-30420	110	-30386	20	0.3	U			GS1	1.0	00Ra23
	-30412	30			0.9	1	45	45 ^{182}Au	GS2	1.0	05Li24
$^{182}\text{Hg}-u$	-25297	30	-25311	11	-0.5	1	12	12 ^{182}Hg	GS2	1.0	05Li24
$^{182}\text{Hg}-^{208}\text{Pb}_{.875}$	-4893	19	-4881	11	0.6	-			MA6	1.0	01Sc41
	-4898	21			0.8	-			MA6	1.0	01Sc41
ave.	-4895	14			1.0	1	56	55 ^{182}Hg			average
$^{182}\text{Pt}(\alpha)^{178}\text{Os}$	4928.5	30.	4951	5	0.7	U					63Ga08
	4948.9	20.			0.1	U					66Si08
	4952.0	5.			-0.2	1	97	76 ^{178}Os	ORa		95Bi01
$^{182}\text{Au}(\alpha)^{178}\text{Ir}$	5529	10	5525	4	-0.4	-			ISa		79Ha10 *
	5525.5	5.			-0.0	-			ORa		95Bi01 *
ave.	5526	4			-0.2	1	99	55 ^{178}Ir			average
$^{182}\text{Hg}(\alpha)^{178}\text{Pt}$	5998.1	5.	5996	5	-0.5	-			ISa		79Ha10 Z
	5999.1	10.2			-0.3	o			Lvn		93Wa.A
	5989.9	13.3			0.4	-			Lvn		94Wa23
ave.	5997	5			-0.3	1	95	62 ^{178}Pt			average
$^{182}\text{Tl}(\alpha)^{178}\text{Au}$	6550.2	10.	6551	6	0.1	F			GSa		86Ke03 *
	6593.1	15.			-2.8	B					04Ra28 *
	6550.9	6.2				3			ISa		16Va01 *
$^{182}\text{Tl}(\alpha)^{178}\text{Au}^q$	6186.2	20.				4					92Bo.D
$^{182}\text{Pb}(\alpha)^{178}\text{Hg}$	7076.8	10.	7066	6	-1.1	4			GSa		86Ke03
	7074.8	15.			-0.6	4			ORa		87To09
	7050.2	10.			1.5	4			Ara		99To11
	7066.6	10.			-0.1	4			Jya		00Je09
$^{180}\text{Hf}(t,p)^{182}\text{Hf}$	3931	6				2			McM		83Bu03
$^{180}\text{W}(t,p)^{182}\text{W}$	6265	5	6270.7	1.6	1.1	U			LAI		76Ca10 *
$^{182}\text{W}(p,t)^{180}\text{W}$	-6261	10	-6270.7	1.6	-1.0	U			Min		73Oo01
$^{181}\text{Ta}(n,\gamma)^{182}\text{Ta}$	6063.0	0.4	6062.94	0.11	-0.2	-					71He13 Z
	6063.1	0.5			-0.3	-					77Si15 Z
	6063.1	0.5			-0.3	-			MMn		81Co17 Z
	6062.95	0.2			-0.1	-			ILn		83Fo.B
	6062.89	0.14			0.3	-			Bdn		06Fi.A
$^{181}\text{Ta}(d,p)^{182}\text{Ta}$	3832	8	3838.37	0.11	0.8	U			MIT		64Er02
$^{181}\text{Ta}(n,\gamma)^{182}\text{Ta}$	6062.93	0.11	6062.94	0.11	0.1	1	100	68 ^{182}Ta			average
$^{182}\text{W}(d,t)^{181}\text{W}$	-1809	10	-1826.3	1.6	-1.7	U			Kop		72Ca01
$^{182}\text{Hf}(\beta^-)^{182}\text{Ta}$	431	50	381	6	-1.0	U					74Wa14 *
$^{182}\text{Ta}(\beta^-)^{182}\text{W}$	1809	5	1815.5	1.5	1.3	-					64Da15 *
	1813	3			0.8	-					67Ba01 *
ave.	1811.9	2.6			1.4	1	35	32 ^{182}Ta			average
$^{182}\text{Re}^m(\beta^+)^{182}\text{W}$	2860	20				2					63Ba37 *
$^{182}\text{Re}^m(\text{IT})^{182}\text{Re}$	60	100				3					63Ba37
$^{182}\text{Os}(\varepsilon)^{182}\text{Re}^m$	848	15	777	30	-4.7	B					70Ak02 *
$^{182}\text{Ir}(\beta^+)^{182}\text{Os}$	5700	200	5560	30	-0.7	U					72We.A
$^{182}\text{Pt}(\beta^+)^{182}\text{Ir}$	2900	200	2883	25	-0.1	U					72We.A
$^{182}\text{Au}(\beta^+)^{182}\text{Pt}$	6850	200	7864	23	5.1	C					72We.A
$^{182}\text{Hg}(\beta^+)^{182}\text{Au}$	4950	200	4727	21	-1.1	U					72We.A
* $^{182}\text{Re}-u$	$M-A=-44972(29)$ keV for mixture gs+m at 60(100) keV										Nub211 **
* $^{182}\text{Au}(\alpha)^{178}\text{Ir}$	$E_\alpha=5353(10)$ to 2^+ level at 54.4(0.5) keV										Ens097 **
* $^{182}\text{Au}(\alpha)^{178}\text{Ir}$	$E_\alpha=5403(5)$, 5352(5) to ground state, 54.4 level										95Bi01 **
* $^{182}\text{Tl}(\alpha)^{178}\text{Au}$	F : identification from excitation function assuming 100% α decay										WgM118**
* $^{182}\text{Tl}(\alpha)^{178}\text{Au}$	$E_\alpha=6403(15)$ in coincidence with 46 keV γ										04Ra28 **
* $^{182}\text{Tl}(\alpha)^{178}\text{Au}$	$E_\alpha=6165(6)$ in coincidence with 247.2(0.5) keV γ										16Va01 **
* $^{180}\text{W}(t,p)^{182}\text{W}$	$Q-Q(^{170}\text{Y}(t,p))=112(5,\text{Ca})$, $Q(170)=-6153(4)$ keV										AHW **
* $^{182}\text{Hf}(\beta^-)^{182}\text{Ta}$	$E_{\beta^-}=970(70)$ 480(50) from $^{182}\text{Hf}^m$ to 651.215 (4) ⁻ , 1115.96 (7) ⁻ levels										Ens15c **
* $^{182}\text{Ta}(\beta^-)^{182}\text{W}$	$E_{\beta^-}=520(5)$ to 2^- level at 1289.1498 keV										Ens15c **
* $^{182}\text{Ta}(\beta^-)^{182}\text{W}$	$E_{\beta^-}=1713(3)$ to 2^+ level at 100.10598 keV										Ens15c **
* $^{182}\text{Re}^m(\beta^+)^{182}\text{W}$	$E_{\beta^+}=1740(20)$, 550(20) to 2^+ level at 100.10598, 2^- at 1289.1498 keV										Ens15c **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{182}\text{Os}(\epsilon)^{182}\text{Re}^m$	pK=0.47(0.07) to 1^+ level at 726.97 keV above $^{182}\text{Re}^m$, recalculated Q										Ens15c	**
$^{183}\text{Lu}-u$	-42637	86				2			GS3	1.0	13Sh30	
$^{183}\text{W O}-\text{C}_2 \text{ } ^{35}\text{Cl}_5$	100858.0	2.7	100875.6	0.8	2.6	U			H29	2.5	77Sh04	
	100873.6	0.8			1.0	1	16	16 ^{183}W	H48	2.5	03Ba49	
$^{183}\text{Re}-u$	-49151	30	-49179	9	-0.9	U			GS2	1.0	05Li24	
$^{183}\text{Os}-u$	-46879	61	-46870	50	0.1	1	77	77 ^{183}Os	GS2	1.0	05Li24	
$^{183}\text{Ir}-u$	-43160	104	-43159	26	0.0	U			GS1	1.0	00Ra23	
	-43145	30			-0.5	1	78	78 ^{183}Ir	GS2	1.0	05Li24	
$^{183}\text{Pt}-u$	-38440	107	-38404	15	0.3	U			GS1	1.0	00Ra23	
	-38400	32			-0.1	1	23	23 ^{183}Pt	GS2	1.0	05Li24	
$^{183}\text{Au}-u$	-32440	104	-32412	10	0.3	U			GS1	1.0	00Ra23	
	-32371	30			-1.4	1	11	11 ^{183}Au	GS2	1.0	05Li24	
$^{183}\text{Hg}-u$	-25537	35	-25555	8	-0.5	U			GS2	1.0	05Li24	
$^{183}\text{Hg}-^{208}\text{Pb}_{.880}$	-5009	19	-5009	8	-0.0	-			MA6	1.0	01Sc41	
	-5002	19			-0.4	-			MA6	1.0	01Sc41	
ave.	-5006	13			-0.3	1	32	32 ^{183}Hg			average	
$^{183}\text{Tl}-^{133}\text{Cs}_{1.376}$	112286	11	112291	10	0.5	1	83	83 ^{183}Tl	MA8	1.0	08We02	
$^{183}\text{W O}_2-^{178}\text{Hf } ^{37}\text{Cl}$	30455.7	5.0	30442.8	1.7	-1.0	U			H35	2.5	80Sh06	
$^{183}\text{W O}_2-^{180}\text{W } ^{35}\text{Cl}$	24421	9	24487.7	1.7	3.0	B			H24	2.5	73Ba40	
	24509	6			-1.4	U			H28	2.5	77Sh04	
$^{183}\text{W } ^{35}\text{Cl}-^{181}\text{Ta } ^{37}\text{Cl}$	5177.2	1.2	5176.0	1.6	-0.4	1	30	27 ^{181}Ta	H35	2.5	80Sh06	
$^{183}\text{W O}_2 \text{ } ^{37}\text{Cl}-^{182}\text{W } ^{35}\text{Cl}_2$	20045.6	1.8	20045.20	0.11	-0.1	U			H28	2.5	77Sh04	
$^{183}\text{Pt}(\alpha)^{179}\text{Os}$	4846.1	30.	4822	9	-0.8	U					63Gr08	
	4835.9	20.0			-0.7	-					66Si08	
	4819.6	10.2			0.2	-			ORa		95Bi01	
ave.	4823	9			-0.1	1	93	69 ^{179}Os			average	
$^{183}\text{Au}(\alpha)^{179}\text{Ir}$	5462.6	5.	5465.3	2.9	0.5	-					68Si01	
	5465.5	5.			-0.0	-			Bka		82Bo04	
	5449.3	10.			1.6	F					84Br.A	
	5468.8	5.			-0.7	-			ORa		95Bi01	
ave.	5465.6	3.0			-0.1	1	99	88 ^{179}Ir			average	
$^{183}\text{Hg}(\alpha)^{179}\text{Pt}$	6043.4	6.	6039	4	-0.8	-			ORa		76To06	
	6036.2	5.			0.5	-			ISa		79Ha10	
ave.	6039	4			-0.1	1	98	93 ^{179}Pt			average	
$^{183}\text{Tl}^m(\alpha)^{179}\text{Au}$	6593.4	30.	6605	9	0.4	U			GSa		80Sc09	
	6600.6	30.			0.2	U			Jya		04Ra28	
	6609.5	10.			-0.4	1	84	67 ^{179}Au	Anv		11Ve01	
$^{183}\text{Pb}(\alpha)^{179}\text{Hg}$	6928	7				2			Anv		02Je09	
$^{183}\text{Pb}^m(\alpha)^{179}\text{Hg}$	6950.1	25.	7022	4	2.9	B			GSa		80Sc09	
	7029	20			-0.3	U			GSa		84Sc.A	
	7026.9	10.			-0.5	2			GSa		86Ke03	
	6868.4	10.			15.4	B			ORa		87To09	
	7034	10			-1.2	2			ORa		89To01	
	7018	5			0.8	2			Anv		02Je09	
$^{183}\text{W}(\text{p,t})^{181}\text{W}$	-5810	10	-5792.6	1.6	1.7	U			Min		73Oo01	
$^{182}\text{Ta}(\text{n},\gamma)^{183}\text{Ta}$	6934.18	0.20				2			ILn		83Fo.B	
$^{182}\text{W}(\text{n},\gamma)^{183}\text{W}$	6191.6	2.0	6190.84	0.04	-0.4	U					67Sp03	
	6190.1	1.5			0.5	U					70Or.A	
	6190.76	0.12			0.6	-			Ltn		97Pr02	
	6190.89	0.13			-0.4	o			Bdn		06Fi.A	
	6190.81	0.06			0.4	-			ILn		11Bo09	
	6190.88	0.06			-0.7	-			Bdn		14Hu02	
$^{183}\text{W}(\gamma,\text{n})^{182}\text{W}$	-6290	50	-6190.84	0.04	2.0	U			Phi		60Ge01	
$^{182}\text{W}(\text{d,p})^{183}\text{W}$	3967	5	3966.27	0.04	-0.1	U			ANL		65Er03	
	3979	10			-1.3	U			Kop		72Ca01	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{183}\text{W}(\text{d,t})^{182}\text{W}$		57	15	66.39	0.04	0.6	U				72Ca01	
$^{182}\text{W}(\text{n},\gamma)^{183}\text{W}$	ave.	6190.84	0.04	6190.84	0.04	0.0	1	100	100 ^{182}W		average	
$^{182}\text{W}(^3\text{He,d})^{183}\text{Re}$		-610	40	-641	8	-0.8	U				71Lu01	
$^{183}\text{Hg}(\text{ep})^{182}\text{Pt}$		5000	200	5075	15	0.4	F				72Ho19	
$^{183}\text{Hf}(\beta^-)^{183}\text{Ta}$		2010	30				3				67Mo13 *	
$^{183}\text{Ta}(\beta^-)^{183}\text{W}$		1068	10	1072.1	1.5	0.4	U				55Mu19 *	
$^{183}\text{Re}(\epsilon)^{183}\text{W}$		556	8				2				69Ku03 *	
$^{183}\text{Ir}(\beta^+)^{183}\text{Os}$		3450	100	3460	50	0.1	1	28	23 ^{183}Os		70Be.A *	
$^{183}\text{Tl}^m(\text{IT})^{183}\text{Tl}$		628.7	0.5	628.7	0.5	0.0	1	100	83 $^{183}\text{Tl}^m$		11Ve.A	
* $^{183}\text{Os}-\text{u}$	$M-A=-43582(28)$ keV for mixture gs+m at 170.73 keV										Nub211 **	
* $^{183}\text{Pt}-\text{u}$	$M-A=-35752(28)$ keV for mixture gs+m at 34.74 keV										Nub211 **	
* $^{183}\text{Hg}-\text{u}$	Existence of isomeric state under discussion (see Nubase); not corrected										Nub211 **	
* $^{183}\text{Au}(\alpha)^{179}\text{Ir}$	F : all lines in ^{181}Au and ^{183}Au shifted by 16-20keV										GAu **	
* $^{183}\text{Tl}^m(\alpha)^{179}\text{Au}$	$E_\alpha=6449(15)$, error increased since partially summed with electrons										GAu **	
* $^{183}\text{Tl}^m(\alpha)^{179}\text{Au}$	$E_\alpha=6456(15)$, error increased since partially summed with electrons										GAu **	
* $^{183}\text{Tl}^m(\alpha)^{179}\text{Au}$	$E_\alpha=6400(10)$, 6378(10) summed with e^- , in coinc. with 62.4, 89.5 γ										GAu **	
* $^{183}\text{Pb}(\alpha)^{179}\text{Hg}$	$E_\alpha=6775(7)$, 6570(10) to ground state, 217 level										02Je09 **	
* $^{183}\text{Pb}^m(\alpha)^{179}\text{Hg}$	$E_\alpha=6868(20)$, 6715(20) to ground state, 171.4 isomer										02Je09 **	
*	original assignment to ^{182}Pb changed										AHW **	
* $^{183}\text{Pb}^m(\alpha)^{179}\text{Hg}$	$E_\alpha=6874(15)$, 6712(10) to ground state, 171.4 isomer; and an 6784(15) line										Nub211 **	
* $^{183}\text{Pb}^m(\alpha)^{179}\text{Hg}$	$E_\alpha=6860(11)$, 6698(5) to ground state, 171.4 isomer										Nub211 **	
* $^{183}\text{Hg}(\text{ep})^{182}\text{Pt}$	F : retracted by authors in PrvCom										AHW **	
* $^{183}\text{Hf}(\beta^-)^{183}\text{Ta}$	$E_{\beta^-}=1540(30)$ to $(5/2^+)$ level at 459.062 keV, and other E_{β^-}										Ens164 **	
* $^{183}\text{Ta}(\beta^-)^{183}\text{W}$	$E_{\beta^-}=615(10)$ to $7/2^-$ level at 453.0695 keV										Ens164 **	
* $^{183}\text{Re}(\epsilon)^{183}\text{W}$	$pK=0.40(0.07)$ to $7/2^-$ level at 453.0695 keV										Ens164 **	
* $^{183}\text{Ir}(\beta^+)^{183}\text{Os}$	$Q_{\beta^+}=3190(100)$ mainly to $3/2^-$ level at 258.34 keV										Ens164 **	
$^{184}\text{W}-\text{u}$		-49066.76	0.99	-49066.8	0.8	-0.0	1	28	28 ^{184}W	TG1	1.5	12Sm07
$^{184}\text{Os}-\text{u}$		-47504.3	1.2	-47507.1	0.9	-1.5	1	24	24 ^{184}Os	TG1	1.5	12Sm07
$^{184}\text{Ir}-\text{u}$		-42460	110	-42520	30	-0.6	U			GS1	1.0	00Ra23
		-42524	30				2			GS2	1.0	05Li24
$^{184}\text{Pt}-\text{u}$		-40120	104	-40078	16	0.4	U			GS1	1.0	00Ra23
		-40068	30			-0.3	1	28	28 ^{184}Pt	GS2	1.0	05Li24
$^{184}\text{Au}-\text{u}$		-32540	104	-32548	24	-0.1	U			GS1	1.0	00Ra23 *
		-32557	37			0.2	R			GS2	1.0	05Li24 *
$^{184}\text{Hg}-\text{u}$		-28230	110	-28282	10	-0.5	U			GS1	1.0	00Ra23
		-28296	30			0.5	-			GS2	1.0	05Li24
	ave.	-28301	21			0.9	1	23	23 ^{184}Hg			average
$^{184}\text{Hg}-^{204}\text{Pb}_{.902}$		-3986	20	-3968	10	0.9	1	26	26 ^{184}Hg	MA6	1.0	01Sc41
$^{184}\text{Hg}-^{208}\text{Pb}_{.885}$		-7620	19	-7619	10	0.0	1	29	29 ^{184}Hg	MA6	1.0	01Sc41
$^{184}\text{Tl}-\text{u}$		-18115	112	-18125	11	-0.1	U			GS2	1.0	05Li24 *
$^{184}\text{Tl}-^{133}\text{Cs}_{1.383}$		112645.4	23.2	112635	11	-0.5	1	21	21 ^{184}Tl	MA8	1.0	14Bo26 *
$^{184}\text{W O}_2-^{181}\text{Ta }^{35}\text{Cl}$		23917.5	2.8	23911.2	1.6	-0.9	U			H35	2.5	80Sh06
$^{184}\text{W }^{35}\text{Cl}-^{182}\text{W }^{37}\text{Cl}$		5675	3	5677.66	0.16	0.4	U			H22	2.5	70Mc03
		5676.3	2.2			0.2	U			H28	2.5	77Sh04
$^{184}\text{W O}_2-^{37}\text{Cl}-^{183}\text{W }^{35}\text{Cl}_2$		18734.7	3.0	18735.19	0.17	0.1	U			H28	2.5	77Sh04
$^{184}\text{Os}-^{184}\text{W}$		1560.59	0.70	1559.7	0.7	-0.8	1	46	31 ^{184}Os	TG1	1.5	12Sm07
$^{184}\text{Pt}(\alpha)^{180}\text{Os}$		4579.8	20.	4599	8	0.9	-					63Gr08
		4600.2	20.			-0.1	-					66Si08
		4602.2	10.			-0.3	-			ORa		95Bi01
	ave.	4598	8			0.1	1	94	68 ^{180}Os			average
$^{184}\text{Au}(\alpha)^{180}\text{Ir}$		5218.6	15.	5234	5	1.0	U			ISa		70Ha18 *
		5233.9	5.				3			ORa		95Bi01 *
$^{184}\text{Hg}(\alpha)^{180}\text{Pt}$		5658.2	15.	5660	4	0.1	-			ISa		70Ha18
		5662.3	5.1			-0.4	-			ORa		76To06
		5658.2	10.2			0.2	o			Lvn		93Wa.A
		5662.3	10.2			-0.2	-			Lvn		93Wa03 Z
	ave.	5662	4			-0.4	1	96	75 ^{180}Pt			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{184}\text{Tl}(\alpha)^{180}\text{Au}$	6299.4	5.	6317	9	0.4	U			ORa		76To06 Z
	6292.9	10.			0.5	U			GSa		80Sc09 Z
$^{184}\text{Pb}(\alpha)^{180}\text{Hg}$	6315.2	10.2			0.2	1	83	79 ^{184}Tl	ISa		16Va01 *
	6765.4	10.	6774	3	0.8	—					80Du02
	6779.6	10.			−0.6	—			GSa		80Sc09
	6773.6	10.			0.0	—			GSa		84Sc.A
	6781.7	10.2			−0.8	—			ORa		87To09
	6773.6	6.			0.1	—			Jya		98Co27
	6772.5	10.			0.1	—			Ara		99To11
	6773.6	6.			0.1	—			Anv		04An07
	ave. 6774	3			0.0	1	99	70 ^{184}Pb			average
$^{184}\text{Bi}(\alpha)^{180}\text{Tl}$	8024.8	50.	8220#	100#	2.8	D			Anv		03An27 *
$^{182}\text{W}(\text{t,p})^{184}\text{W}$	5127	7	5120.15	0.14	−1.0	U			LAI		76Ca10
$^{184}\text{W}(\text{p,t})^{182}\text{W}$	−5124	5	−5120.15	0.14	0.8	U			Min		73Oo01
$^{183}\text{W}(\text{n},\gamma)^{184}\text{W}$	7411.2	0.5	7411.11	0.13	−0.2	U					74Gr11 Z
	7411.8	0.3			−2.3	B					75Bu01 Z
	7411.15	0.16			−0.3	o			Bdn		06Fi.A
	7411.11	0.13			−0.0	1	99	73 ^{183}W	Bdn		14Hu02
$^{183}\text{W}(\text{d,p})^{184}\text{W}$	5187	15	5186.54	0.13	−0.0	U			Kop		72Ca01
$^{184}\text{W}(\text{d,t})^{183}\text{W}$	−1154	10	−1153.88	0.13	0.0	U			Kop		72Ca01
$^{184}\text{Hf}(\beta^-)^{184}\text{Ta}$	1340	30				3					73Wa18 *
$^{184}\text{Ta}(\beta^-)^{184}\text{W}$	2866	26				2					73Ya02 *
$^{184}\text{Ir}(\beta^+)^{184}\text{Os}$	5100	250	4642	28	−1.8	U					70Be.A *
	4300	100			3.4	B					73Ho09 *
	4285	70			5.1	B					89Po09 *
$^{184}\text{Au}(\beta^+)^{184}\text{Pt}$	6380	50	7014	27	12.7	C					84Da.A *
$^{184}\text{Hg}(\beta^+)^{184}\text{Au}$	3760	30	3974	24	7.1	C					84Da.A
* $^{184}\text{Au}-\text{u}$	$M-A=-30280(100)$ keV for mixture gs+m at 68.46 keV										Nub211 **
* $^{184}\text{Au}-\text{u}$	$M-A=-30292(28)$ keV for mixture gs+m at 68.46 keV										Nub211 **
* $^{184}\text{Tl}-\text{u}$	$M-A=-16899(102)$ keV for mixture gs+m at −50(30) keV										Nub211 **
* $^{184}\text{Tl}-^{133}\text{Cs}_{1,383}$	$D_M=112618.6(6.2)$ μu for mixture gs+m at −50(30) keV; $M-A=-16898.5(5.8)$ keV										Nub211 **
* $^{184}\text{Au}(\alpha)^{180}\text{Ir}$	$E_\alpha=5172(15)$ from $^{184}\text{Au}^m$ at 68.6(0.1) keV										94Ib01 **
*	transition to ground state in ^{180}Ir										95Bi01 **
* $^{184}\text{Au}(\alpha)^{180}\text{Ir}$	$E_\alpha=5187(5)$ from $^{184}\text{Au}^m$ at 68.6(0.1) keV										94Ib01 **
* $^{184}\text{Tl}(\alpha)^{180}\text{Au}$	$E_\alpha=6161(10)$ to level at 17.1(3) keV										16Va01 **
* $^{184}\text{Bi}(\alpha)^{180}\text{Tl}$	Trends from Mass Surface TMS suggest ^{184}Bi 200 keV less bound										GAu **
* $^{184}\text{Hf}(\beta^-)^{184}\text{Ta}$	$E_{\beta^-}=1100(30)$ to 1^- level at 228.4 keV, and other E_{β^-}										Ens102 **
* $^{184}\text{Ta}(\beta^-)^{184}\text{W}$	$E_{\beta^-}=1165(26)$ to 6^+ level at 1746.03 keV, and other E_{β^-}										Ens102 **
* $^{184}\text{Ir}(\beta^+)^{184}\text{Os}$	$Q_{\beta^+}=4720(250)$ to 4^+ level at 383.68 keV										Ens102 **
* $^{184}\text{Ir}(\beta^+)^{184}\text{Os}$	$E_{\beta^+}=2900(100)$ to 4^+ level at 383.68 keV										Ens102 **
* $^{184}\text{Ir}(\beta^+)^{184}\text{Os}$	$E_{\beta^+}=2320(70)$ to 2^+ level at 942.86 keV										Ens102 **
* $^{184}\text{Au}(\beta^+)^{184}\text{Pt}$	$Q_{\beta^+}=6450(50)$ from $^{184}\text{Au}^m$ at 68.6(0.1) keV										94Ib01 **
$^{185}\text{Hf}-\text{u}$	−41138	69				2			GS3	1.0	13Sh30
$^{185}\text{Os}-\text{u}$	−46037	31	−45954.0	0.9	2.7	F			GS2	1.0	03Li.A *
$^{185}\text{Ir}-\text{u}$	−43340	110	−43300	30	0.3	U			GS1	1.0	00Ra23
	−43302	30				2			GS2	1.0	05Li24
$^{185}\text{Pt}-\text{u}$	−39334	112	−39386	28	−0.5	U			GS1	1.0	00Ra23 *
	−39381	44			−0.1	1	40	40 ^{185}Pt	GS2	1.0	05Li24 *
$^{185}\text{Au}-\text{u}$	−34190	100	−34201.1	2.8	−0.1	o			GS1	1.0	00Ra23 *
	−34197	43			−0.1	U			GS2	1.0	05Li24 *
$^{185}\text{Au}-^{133}\text{Cs}_{1,391}$	97315.2	2.8				2			MA8	1.0	17Ma29
$^{185}\text{Hg}-\text{u}$	−28070	107	−28109	15	−0.4	U			GS1	1.0	00Ra23
	−28088	44			−0.5	1	11	11 ^{185}Hg	GS2	1.0	05Li24 *
$^{185}\text{Hg}-^{208}\text{Pb}_{889}$	−7373	29	−7353	15	0.7	1	26	25 ^{185}Hg	MA6	1.0	01Sc41 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{185}\text{Tl}-u$	-21354	145	-21211	22	1.0	U			GS2	1.0	05Li24 *
$^{185}\text{Re } ^{16}\text{O}_2 - ^{182}\text{W } ^{35}\text{Cl}$	25731	6	25729.2	0.7	-0.1	U			H22	2.5	70Mc03
$^{185}\text{Re } ^{35}\text{Cl} - ^{183}\text{W } ^{37}\text{Cl}$	5695	3	5684.0	0.7	-1.5	U			H22	2.5	70Mc03
	5678.7	1.0			2.1	U			H28	2.5	77Sh04
$^{185}\text{Re}(\alpha, ^8\text{He})^{181}\text{Re}$	-26480	14	-26486	13	-0.5	2			INS		90Ka19
$^{185}\text{Pt}(\alpha)^{181}\text{Os}$	4436.6	10.2	4437	10	0.0	1	96	60 ^{185}Pt	ORa		91Bi04 *
$^{185}\text{Au}(\alpha)^{181}\text{Ir}$	5180.2	5.	5180	5	-0.0	3					68Si01 Z
	5182.9	15.			-0.2	U			ISa		70Ha18 Z
	5179	10			0.1	3			ORa		91Bi04 *
$^{185}\text{Hg}(\alpha)^{181}\text{Pt}$	5777	15	5773	4	-0.3	-			ISa		70Ha18 *
	5775	5			-0.4	-			ORa		76To06 *
	5761	15			0.8	-					76Gr.A *
	5774	5			-0.2	1	97	52 ^{181}Pt			average
$^{185}\text{Tl}^m(\alpha)^{181}\text{Au}$	6112.6	7.	6143	5	4.4	C					75Co.A Z
	6143.3	5.				4			ORa		76To06 *
	6145.6	15.			-0.2	U			GSa		80Sc09 Z
$^{185}\text{Pb}(\alpha)^{181}\text{Hg}$	6693	15	6695	5	0.1	U			GSa		80Sc09 *
	6555.0	15.			2.7	B			ORa		87To09
	6695	5				2			Anv		02An15 *
$^{185}\text{Pb}^m(\alpha)^{181}\text{Hg}^m$	6622.9	20.	6550	5	-3.7	B			Ora		75Ca06
	6679.7	20.			-6.5	B			GSa		80Sc09
	6549.8	5.				3			Anv		02An15
$^{185}\text{Bi}^m(\alpha)^{181}\text{Tl}$	8258.9	30.	8218	12	-1.3	-			Ara		01Po05 *
	8207.8	15.3			0.7	-			Anv		04An07
	8218	14			-0.0	1	76	64 $^{185}\text{Bi}^m$			average
$^{184}\text{W}(n,\gamma)^{185}\text{W}$	5753.7	0.3	5753.74	0.05	0.1	U			BNn		87Br05 Z
	5754.62	0.24			-3.7	C			Bdn		06Fi.A
	5753.74	0.05			-0.1	1	100	84 ^{185}W	Bdn		14Hu02
$^{184}\text{W}(d,p)^{185}\text{W}$	3524	5	3529.17	0.05	1.0	U			ANL		65Er03
	3533	10			-0.4	U			Kop		72Ca01
$^{184}\text{W}(^3\text{He},d)^{185}\text{Re}$	-98	40	-90.9	0.7	0.2	U			Roc		71Lu01
$^{185}\text{Re}(d,t)^{184}\text{Re} - ^{187}\text{Re}()^{186}\text{Re}$	-310	4	-310	4	-0.0	1	100	100 ^{184}Re	Roc		76El12
$^{184}\text{Os}(n,\gamma)^{185}\text{Os}$	6625.4	0.9	6624.66	0.27	-0.8	U					74Pr15
	6624.52	0.28			0.5	1	95	51 ^{185}Os	Bdn		06Fi.A
$^{185}\text{Bi}^m(p)^{184}\text{Pb}$	1669	50	1607	13	-1.2	o					95Da.A *
	1606.8	16.			0.0	1	67	36 $^{185}\text{Bi}^m$	Ara		01Po05 *
	1568.6	50.			0.8	o					03An27 *
	1591.7	5.			3.0	F					04An07 *
$^{185}\text{Ta}(\beta^-)^{185}\text{W}$	2013	20	1994	14	-1.0	2					69Ku07 *
$^{185}\text{W}(\beta^-)^{185}\text{Re}$	432.6	1.0	431.2	0.7	-1.4	1	44	28 ^{185}Re			67Wi19
$^{185}\text{Os}(\varepsilon)^{185}\text{Re}$	1012.7	1.0	1013.1	0.4	0.4	-					67Sc15 *
	1012.8	0.5			0.7	-					70Sc06 *
	1012.8	0.4			0.8	1	88	49 ^{185}Os			average
$^{185}\text{Au}(\beta^+)^{185}\text{Pt}$	4707	40	4830	26	3.1	F					86Da.A *
$^{185}\text{Tl}^m(\text{IT})^{185}\text{Tl}$	454.8	1.5				5					Ens061
* $^{185}\text{Os}-u$	F : contaminated by isomeric state AND by other nuclides										03Li.B **
* $^{185}\text{Pt}-u$	$M-A=-36590(100)$ keV for mixture gs+m at 103.41 keV										Nub211 **
* $^{185}\text{Pt}-u$	$M-A=-36631(28)$ keV for mixture gs+m at 103.41 keV										Nub211 **
* $^{185}\text{Au}-u$	$M-A=-31820(90)$ keV for mixture gs+m at 50#50 keV										Nub211 **
* $^{185}\text{Au}-u$	$M-A=-31829(28)$ keV for mixture gs+m at 50#50 keV										Nub211 **
* $^{185}\text{Hg}-u$	$M-A=-26112(28)$ keV for mixture gs+m at 103.7 keV										Nub211 **
* $^{185}\text{Hg}-^{208}\text{Pb}_{889}$	Original error (17keV) increased by 20 due to isomer+ground state lines in trap										01Sc41 **
* $^{185}\text{Tl}-u$	$M-A=-19664(31)$ keV for mixture gs+m at 454.8(1.5) keV										Nub211 **
* $^{185}\text{Pt}(\alpha)^{181}\text{Os}$	$E_\alpha=4444(10)$ assumed from $(1/2^-)$ isomer at 103.41 keV										Nub211 **
* $^{185}\text{Au}(\alpha)^{181}\text{Ir}$	$E_\alpha=5069(10)$, 4826(10) to ground state, 243.3 level										91Bi04 **
*	unhindered $E_\alpha=5069(10)$ to ground state or very low level; from coinc.										95Bi01 **
* $^{185}\text{Hg}(\alpha)^{181}\text{Pt}$	$E_\alpha=5653.4(15,Z)$, 5576.4(15,Z) to ground state, $3/2^-$ level at 79.41 keV										Ens061 **
*	and $E_\alpha=5376.4(15,Z)$ from $^{185}\text{Hg}^m$ at 103.8 to $13/2^+$ level at 380.92 keV										Ens061 **
* $^{185}\text{Hg}(\alpha)^{181}\text{Pt}$	$E_\alpha=5653(5)$, 5569(5) to ground state, $3/2^-$ level at 79.41 keV										Ens061 **
*	and 5371(10) from $^{185}\text{Hg}^m$ at 103.8 to $13/2^+$ level at 380.92 keV										Ens061 **
* $^{185}\text{Hg}(\alpha)^{181}\text{Pt}$	$E_\alpha=5365(15)$ from $^{185}\text{Hg}^m$ at 103.8 to $13/2^+$ level at 380.92 keV										Ens061 **
* $^{185}\text{Tl}^m(\alpha)^{181}\text{Au}$	$E_\alpha=6010.2(5,Z)$; also an $E_\alpha=5975.2(5,Z)$, 4 times stronger branch										76To06 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
* $^{185}\text{Tl}^m(\alpha)^{181}\text{Au}$	$E_\alpha=6012.5(15,Z)$; also an $E_\alpha=5970.5(15,Z)$, 4 times stronger branch										80Sc09	**
* $^{185}\text{Pb}(\alpha)^{181}\text{Hg}$	$E_\alpha=6485(15)$ to 64 level										02An15	**
* $^{185}\text{Pb}(\alpha)^{181}\text{Hg}$	$E_\alpha=6486(5),6288(5)$ to 64, 269 levels										02An15	**
* $^{185}\text{Bi}^m(\alpha)^{181}\text{Tl}$	$E_\alpha=8030$, by same authors, from only one event										96Da06	**
* $^{185}\text{Bi}^m(p)^{184}\text{Pb}$	Read from graph										AHW	**
* $^{185}\text{Bi}^m(p)^{184}\text{Pb}$	Average by authors of $E_p=1618(11)$, and $1585(9)$ in reference										96Da06	**
* $^{185}\text{Bi}^m(p)^{184}\text{Pb}$	As read from graph										GAu	**
* $^{185}\text{Bi}^m(p)^{184}\text{Pb}$	F : rejected by authors: no dedicated calibration with known proton activity										04An07	**
* $^{185}\text{Ta}(\beta^-)^{185}\text{W}$	$E_{\beta^-}=1770(20)$ to $7/2^-$ level at 243.62 keV										Ens061	**
* $^{185}\text{Os}(\varepsilon)^{185}\text{Re}$	L/K=0.600(0.006) to $3/2^+$ level at 874.81 and $1/2^+$ at 880.33 keV										Ens061	**
* $^{185}\text{Os}(\varepsilon)^{185}\text{Re}$	pK=0.109(0.005) to $3/2^+$ level at 931.06 keV, and other pK, recalculated										Ens061	**
* $^{185}\text{Au}(\beta^+)^{185}\text{Pt}$	F : insufficient information										GAu	**
$^{186}\text{Hf}-u$	-39103	55				2			GS3	1.0	13Sh30	
$^{186}\text{W O}-\text{C }^{13}\text{C }^{35}\text{Cl}_4 \text{ }^{37}\text{Cl}$	104592.7	3.2	104611.6	1.3	2.4	U			H29	2.5	77Sh04	
$^{186}\text{Ir}-u$	-42063	30	-42053	18	0.3	2			GS2	1.0	05Li24	
$^{186}\text{Pt}-u$	-40656	30	-40649	23	0.2	1	61	61 ^{186}Pt	GS2	1.0	05Li24	
$^{186}\text{Au}-u$	-34029	30	-34047	23	-0.6	1	56	56 ^{186}Au	GS2	1.0	05Li24	
$^{186}\text{Hg}-u$	-30660	104	-30638	13	0.2	U			GS1	1.0	00Ra23	
	-30630	30			-0.3	1	17	17 ^{186}Hg	GS2	1.0	05Li24	
$^{186}\text{Hg}-^{204}\text{Pb}_{912}$	-6065	20	-6054	13	0.6	-			MA6	1.0	01Sc41	
	ave.	-6058	17		0.2	1	56	56 ^{186}Hg			average	
$^{186}\text{Tl}-u$	-21650	220	-21345	22	1.4	o			GS1	1.0	00Ra23	
	-21513	105			1.6	U			GS2	1.0	05Li24	
$^{186}\text{Tl}-^{133}\text{Cs}_{1,398}$	110831	24	110833	22	0.1	o			MA8	1.0	08We02	
	110829	24			0.2	1	86	86 ^{186}Tl	MA8	1.0	14Bo26	
$^{186}\text{Tl}^n-^{133}\text{Cs}_{1,398}$	111254	34				2			MA8	1.0	14Bo26	
$^{186}\text{W O}_2-^{183}\text{W }^{35}\text{Cl}$	25122	5	25117.3	1.3	-0.4	U			H28	2.5	77Sh04	
$^{186}\text{W }^{35}\text{Cl}-^{184}\text{W }^{37}\text{Cl}$	6374	3	6382.1	1.2	1.1	U			H22	2.5	70Mc03	
	6382.0	1.4			0.0	1	13	11 ^{186}W	H28	2.5	77Sh04	
$^{186}\text{Os}(\alpha)^{182}\text{W}$	2820.6	50.	2821.2	0.9	0.0	U					75Vi01	
$^{186}\text{Pt}(\alpha)^{182}\text{Os}$	4323.2	20.	4320	18	-0.2	1	79	39 ^{182}Os			63Gr08	
$^{186}\text{Au}(\alpha)^{182}\text{Ir}$	4907	15	4912	14	0.3	1	87	44 ^{182}Ir	ORa		90Ak04	
$^{186}\text{Hg}(\alpha)^{182}\text{Pt}$	5206.2	15.	5204	10	-0.1	-			ISa		70Ha18	
	5204.2	15.			0.0	-					96Ri12	
	ave.	5205	11		-0.1	1	83	57 ^{182}Pt			average	
$^{186}\text{Tl}(\alpha)^{182}\text{Au}$	6016.1	52.1	5996	26	-0.4	1	25	14 ^{186}Tl	ISa		20St11	
$^{186}\text{Tl}^m(\alpha)^{182}\text{Au}^m$	5891.9	7.	5892	5	-0.0	4					75Co.A	
	5891.9	7.			-0.0	4			ORa		77Ij01	
$^{186}\text{Pb}(\alpha)^{182}\text{Hg}$	6458.2	20.	6471	5	0.6	2			Ora		74Le02	
	6470.1	10.			0.1	2			GSa		80Sc09	
	6474.7	10.			-0.4	2			ORa		84To09	
	6474.5	10.2			-0.3	2			Lvn		93Wa.A	
	6476.5	15.			-0.4	2			ORa		97Ba25	
	6459.2	15.			0.8	2			Anv		97An09	
$^{186}\text{Bi}(\alpha)^{182}\text{Tl}$	7760	20	7757	12	-0.2	4			Ara		97Ba21	
	7755	15			0.1	4			Anv		03An27	
$^{186}\text{Bi}^m(\alpha)^{182}\text{Tl}^p$	7349.3	25.	7423	5	2.9	U			GSa		84Sc.A	
	7420.9	20.			0.1	U			Ara		97Ba21	
	7422.9	5.				5			Anv		03An27	
$^{186}\text{Po}(\alpha)^{182}\text{Pb}$	8493	30	8501	14	0.3	5					05Hu.A	
	8503.2	15.3			-0.1	5					13An13	
$^{186}\text{W}(p,t)^{184}\text{W}$	-4474	5	-4464.0	1.2	2.0	U			Min		73Oo01	
$^{186}\text{W}(p,t)^{184}\text{W}-^{184}\text{W}(t)^{182}\text{W}$	660.1	1.6	656.2	1.2	-2.5	o					09Le03	
	657.0	1.8			-0.5	1	42	35 ^{186}W			09Le.A	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{186}\text{W}(\text{t},\alpha)^{185}\text{Ta}$	11430	20	11411	14	-1.0	R			LAI		80Lo10
$^{186}\text{W}(\gamma,\text{n})^{185}\text{W}$	-7120	60	-7192.0	1.2	-1.2	U			Phi		60Ge01
$^{186}\text{W}(\text{d},\text{t})^{185}\text{W}$	-939	10	-934.8	1.2	0.4	U			Kop		72Ca01
$^{185}\text{Re}(\text{n},\gamma)^{186}\text{Re}$	6179.8	0.8	6179.59	0.05	-0.3	U			Tal		69La11 Z
	6178.6	1.5			0.7	U					70Or.A
	6179.34	0.18			1.4	o			Bdn		06Fi.A
	6179.59	0.05			0.0	1	100	72 ^{186}Re	Bdn		16Ma35
$^{185}\text{Re}(\text{d},\text{p})^{186}\text{Re}$	3939	25	3955.02	0.05	0.6	U			Tal		69La11
$^{186}\text{Ta}(\beta^-)^{186}\text{W}$	3901	60				2					69Mo16 *
$^{186}\text{Re}(\beta^-)^{186}\text{Os}$	1064	2	1072.7	0.8	4.4	B					56Jo05
	1071.5	1.3			0.9	-					56Po28
	1076	3			-1.1	-					64Ma36
	1064	3			2.9	B					68An11
	ave.	1072.2			0.4	1	49	28 ^{186}Re	average		
$^{186}\text{Ir}(\beta^+)^{186}\text{Os}$	3760	200	3828	17	0.3	U					62Bo22 *
	3831	20			-0.2	R					63Em02
$^{186}\text{Au}(\beta^+)^{186}\text{Pt}$	5950	200	6150	30	1.0	U					72We.A
$^{186}\text{Hg}(\beta^+)^{186}\text{Au}$	3250	200	3176	24	-0.4	U					72We.A
$^{186}\text{Tl}^{\text{m}}(\text{IT})^{186}\text{Tl}^{\text{m}}$	373.9	0.5	374.16	0.09	0.5	o			Lvn		91Va04
	374.0	0.2			0.8	3					Ens036
	374.2	0.1			-0.4	3					20St11
* $^{186}\text{Ir}-\text{u}$	$M-A=-39181(28)$ keV for mixture gs+m at 0.8 keV										Nub211 **
* $^{186}\text{Tl}-\text{u}$	$M-A=-20030(180)$ keV for mixture gs+m+n at 20(40) and 390(40) keV										Nub211 **
* $^{186}\text{Tl}-\text{u}$	$M-A=-19900(29)$ keV for mixture gs+m at 20(40) keV,										Nub211 **
*	n state can be resolved										Nub211 **
* $^{186}\text{Tl}-^{133}\text{Cs}_{1.398}$	$D_M=110842.1(9.2)$ μu for mixture gs+m at 20(40) keV; $M-A=-19874.4(8.6)$ keV										Nub211 **
* $^{186}\text{Tl}-^{133}\text{Cs}_{1.398}$	$D_M=110840.4(8.6)$ μu for mixture gs+m at 20(40) keV; $M-A=-19876.0(8.0)$ keV										Nub211 **
* $^{186}\text{Au}(\alpha)^{182}\text{Ir}$	$E_\alpha=4653(15)$ to 3^- level at 152.3 keV										95Sa42 **
* $^{186}\text{Tl}(\alpha)^{182}\text{Au}$	$E_\alpha=5760(51)$ to 129.4(0.1) keV										20St11 **
* $^{186}\text{Bi}(\alpha)^{182}\text{Tl}$	$E_\alpha=7158(20)$ followed by $E(\gamma)=444$ keV										03An27 **
* $^{186}\text{Bi}(\alpha)^{182}\text{Tl}$	$E_\alpha=7152(15)$, 7085(15) followed by $E(\gamma)=444$, 520 keV										03An27 **
* $^{186}\text{Ta}(\beta^-)^{186}\text{W}$	$E_{\beta^-}=2240(60)$ to $(2^-, 3^-)$ level at 1661.3817 keV										Ens036 **
* $^{186}\text{Ir}(\beta^+)^{186}\text{Os}$	$E_{\beta^+}=2600(200)$ assumed to 2^+ level at 137.159 keV, also other E_{β^+}										Ens036 **
* $^{186}\text{Ir}(\beta^+)^{186}\text{Os}$	$E_{\beta^+}=1940(20)$ to 6^+ level at 868.94 keV										Ens036 **
$^{187}\text{Ta}-\text{u}$	-39609	60				2			GS3	1.0	13Sh30
$^{187}\text{Ir}-\text{u}$	-42458	30				2			GS2	1.0	05Li24
$^{187}\text{Pt}-\text{u}$	-39500	110	-39383	26	1.1	U			GS1	1.0	00Ra23
	-39413	30			1.0	1	74	74 ^{187}Pt	GS2	1.0	05Li24
$^{187}\text{Au}-\text{u}$	-35470	114	-35458	24	0.1	U			GS1	1.0	00Ra23 *
	-35441	30			-0.6	1	65	65 ^{187}Au	GS2	1.0	05Li24
$^{187}\text{Hg}-\text{u}$	-30190	110	-30186	14	0.0	U			GS1	1.0	00Ra23 *
	-30155	36			-0.9	1	15	15 ^{187}Hg	GS2	1.0	05Li24 *
$^{187}\text{Hg}-^{208}\text{Pb}_{.899}$	-9210	20	-9197	14	0.7	1	48	48 ^{187}Hg	MA6	1.0	01Sc41
$^{187}\text{Hg}^{\text{m}}-^{208}\text{Pb}_{.899}$	-9152	19	-9134	16	0.9	o			MA6	1.0	01Sc41 *
$^{187}\text{Tl}-\text{u}$	-24120	107	-24095	9	0.2	U			GS1	1.0	00Ra23
	-23928	109			-1.5	U			GS2	1.0	05Li24 *
$^{187}\text{Tl}^{\text{m}}-^{133}\text{Cs}_{1.406}$	109151	24	109198	8	1.9	F			MA8	1.0	08We02 *
$^{187}\text{Pb}-\text{u}$	-16076	45	-16089	5	-0.3	U			GS2	1.0	05Li24 *
$^{187}\text{Pb}-^{133}\text{Cs}_{1.406}$	116843.5	5.9	116845	5	0.3	1	86	86 ^{187}Pb	MA8	1.0	05We11 *
$^{187}\text{Pb}^{\text{m}}-^{133}\text{Cs}_{1.406}$	116871.6	5.6	116866	12	-1.0	o			MA8	1.0	05We11 *
$^{187}\text{Re O}_2-^{184}\text{W }^{35}\text{Cl}$	25797.4	3.5	25795.6	0.9	-0.2	U			H28	2.5	77Sh04
$^{187}\text{Re }^{35}\text{Cl}-^{185}\text{Re }^{37}\text{Cl}$	5737	3	5744.0	0.9	0.9	U			H22	2.5	70Mc03
	5744.2	1.2			-0.1	1	10	6 ^{185}Re	H28	2.5	77Sh04
$^{187}\text{Re}-^{187}\text{Os}$	2.676	0.036	2.6481	0.0017	-0.8	U			SH1	1.0	14Ne15

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{187}\text{Au}(\alpha)^{183}\text{Ir}$	4792.7	20.	4750	30	-0.8	1	31	17 ^{183}Ir			68Si01 *
$^{187}\text{Hg}(\alpha)^{183}\text{Pt}$	5229.9	20.	5230	14	-0.0	1	49	29 ^{183}Pt	ISa		70Ha18 *
$^{187}\text{Hg}^m(\alpha)^{183}\text{Pt}$	5293.4	20.	5288	14	-0.3	1	52	27 $^{187}\text{Hg}^m$	ISa		70Ha18 *
$^{187}\text{Tl}^m(\alpha)^{183}\text{Au}$	5643	20	5656	6	0.6	-			ORa		76To06 *
	5661.5	10.			-0.6	-			GSa		80Sc09 *
	5645.1	12.			0.9	o			Lvn		85Co06 *
	5661.5	10.			-0.6	-			Lvn		91Wa21 *
ave.	5659	7			-0.6	1	91	77 ^{183}Au			average
$^{187}\text{Pb}(\alpha)^{183}\text{Hg}$	6393.0	10.	6393	6	-0.0	-			Ora		75Ca06 *
	6395.0	19.			-0.1	o			GSa		80Sc09 *
	6398.4	10.			-0.6	-			GSa		81Mi12 *
ave.	6396	7			-0.4	1	77	63 ^{183}Hg			average
$^{187}\text{Pb}^m(\alpha)^{183}\text{Hg}^m$	6213.1	20.	6208	7	-0.2	o			Ora		74Le02
	6213.1	10.			-0.5	2			Ora		75Ca06
	6223.3	10.			-1.5	o			GSa		80Sc09
	6206.0	10.2			0.2	2			GSa		81Mi12
	6202.9	15.			0.4	2			Anv		99An36
$^{187}\text{Bi}(\alpha)^{183}\text{Tl}^m$	7139.0	10.	7150	4	1.1	2			GSa		84Sc.A
	7153.3	8.			-0.4	2			ORa		99Ba45 *
	7158.4	10.			-0.8	o			Anv		03An27
	7147.2	8.			0.4	2			Jya		03Ke08 *
	7153.3	10.			-0.3	o			Anv		04An07
	7153.3	5.			-0.6	2			Anv		06An11 *
$^{187}\text{Bi}^m(\alpha)^{183}\text{Tl}$	7749.1	10.	7887	7	13.8	F			GSa		84Sc.A *
	7890.1	15.			-0.2	2			ORa		99Ba45
	7882.9	11.			0.4	2			Jya		03Ke08
	7890.1	10.			-0.3	2			Anv		06An11
$^{187}\text{Po}(\alpha)^{183}\text{Pb}$	7978.9	15.				3			Anv		06An11 *
$^{187}\text{Po}^m(\alpha)^{183}\text{Pb}^m$	7889.1	20.				3			Anv		06An11
$^{186}\text{W}(\text{n},\gamma)^{187}\text{W}$	5466.3	0.3	5466.76	0.04	1.5	U			BNn		87Br05 Z
	5467.22	0.3			-1.5	U			Ltn		92Be17 *
	5466.59	0.12			1.4	o			Bdn		06Fi.A
	5466.83	0.05			-1.4	-			Prn		08Bo26
	5466.62	0.07			2.0	-			Bdn		14Hu02
$^{186}\text{W}(\text{d},\text{p})^{187}\text{W}$	3236	5	3242.19	0.04	1.2	U			ANL		65Er03
	3240	10			0.2	U			Kop		72Ca01
ave.	5466.76	0.04	5466.76	0.04	0.0	1	100	55 ^{186}W			average
$^{186}\text{W}(\text{n},\gamma)^{187}\text{W}$	530	40	503.4	1.1	-0.7	U			Roc		71Lu01
$^{186}\text{W}(\text{d},\text{p})^{187}\text{Re}$	-7180	80	-7360.5	0.9	-2.3	U			Phi		60Ge01
$^{187}\text{Re}(\gamma,\text{n})^{186}\text{Re}$	-1055	25	-1103.3	0.9	-1.9	U			Tal		69La11
$^{187}\text{Re}(\text{d},\text{t})^{186}\text{Re}$	6291.1	1.0	6290.3	0.5	-0.8	-					74Pr15 Z
$^{186}\text{Os}(\text{n},\gamma)^{187}\text{Os}$	6289.4	0.8			1.1	-			Bdn		06Fi.A
ave.	6290.1	0.6			0.4	1	70	39 ^{186}Os			average
$^{187}\text{W}(\beta^-)^{187}\text{Re}$	1314	2	1312.5	1.1	-0.7	-					69Na03
	1310	2			1.3	-					70He14
ave.	1312.0	1.4			0.4	1	63	55 ^{187}W			average
$^{187}\text{Re}(\beta^-)^{187}\text{Os}$	2.62	0.09	2.4667	0.0016	-1.7	U					65Br12
	2.64	0.05			-3.5	B					67Hu05
	2.667	0.020			-10.0	B					92Co23
	2.70	0.09			-2.6	U					93As02
	2.460	0.011			0.6	U					99Al20
	2.470	0.004			-0.8	-					01Ga01 *
	2.4661	0.0017			0.4	-					03Ar36
ave.	2.4667	0.0016			0.0	1	100	89 ^{187}Re			average
$^{187}\text{Ir}(\beta^+)^{187}\text{Os}$	1550	200	1670	28	0.6	U					71Ma24 *
$^{187}\text{Os}(\text{d},\text{p})^{187}\text{Ir}$	-1521	6	-1688	28	-27.9	B			INS		90Ka27

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{187}\text{Au}(\beta^+)^{187}\text{Pt}$	3600	40	3657	27	1.4	1	47	26 ^{187}Pt			83Gn01 *
$^{187}\text{Hg}^m(\text{IT})^{187}\text{Hg}$	54	21	58	14	0.2	1	46	28 $^{187}\text{Hg}^m$	MA6		01Sc41 *
$^{187}\text{Tl}^m(\text{IT})^{187}\text{Tl}$	330	5	334	3	0.7	1	45	31 ^{187}Tl			77Sc03
$^{187}\text{Pb}^m(\text{IT})^{187}\text{Pb}$	33	13	19	10	-1.1	1	61	61 $^{187}\text{Pb}^m$	MA8		05We11
$^{187}\text{Au}-\text{u}$	$M-A=-32980(100)$ keV for mixture gs+m at 120.33 keV										Nub211 **
$^{187}\text{Hg}-\text{u}$	$M-A=-28090(100)$ keV for mixture gs+m at 58(14) keV										Nub211 **
$^{187}\text{Hg}-\text{u}$	$M-A=-28060(28)$ keV for mixture gs+m at 58(14) keV										Nub211 **
$^{187}\text{Hg}^m-^{208}\text{Pb}_{.899}$	Use instead their difference between ground state and $^{187}\text{Hg}^m$ lines										GAu **
$^{187}\text{Tl}-\text{u}$	$M-A=-22121(28)$ keV for mixture gs+m at 334(3) keV										Nub211 **
$^{187}\text{Tl}^m-^{133}\text{Cs}_{1.406}$	F : contamination from ground state not resolved										08We02 **
$^{187}\text{Pb}-\text{u}$	$M-A=-14965(41)$ keV for mixture gs+m at 19(10) keV										Nub211 **
$^{187}\text{Pb}-^{133}\text{Cs}_{1.406}$	$D_M=116851.3(4.3)$ μu for mixture gs+m at 19(10) keV with $R=0.62(0.02)$;										Nub211 **
	$M-A=-14981.5(4.0)$ keV										GAu **
$^{187}\text{Pb}^m-^{133}\text{Cs}_{1.406}$	$D_M=116869.5(5.5)$ for mixture gs+m at 19(10) $R=8.7(0.7)$; $M-A=-14964.5(5.1)$ keV										Nub211 **
	used are only the equations for the ^{187}Pb doublet and $^{187}\text{Pb}^m(\text{IT})^{187}\text{Pb}$										GAu **
$^{187}\text{Au}(\alpha)^{183}\text{Ir}$	Assignment uncertain										Ens095 **
$^{187}\text{Hg}(\alpha)^{183}\text{Pt}$	$E_\alpha=5035(20)$ to $3/2^-$ level at 84.73 keV										Ens164 **
$^{187}\text{Hg}^m(\alpha)^{183}\text{Pt}$	$E_\alpha=4870(20)$ to $(13/2^+)$ level at 316.9 level										Ens164 **
$^{187}\text{Tl}^m(\alpha)^{183}\text{Au}$	$E_\alpha=5510(20)$ 5528(10) 5512(12) 5528(10) respectively, to $(9/2^-)$ at 12.4(0.4)keV										Ens164 **
$^{187}\text{Pb}(\alpha)^{183}\text{Hg}$	$E_\alpha=6190(10)$ to $3/2^-$ level at 67.16 keV										Ens164 **
$^{187}\text{Pb}(\alpha)^{183}\text{Hg}$	$E_\alpha=6194(10)$ 5993(10) to $3/2^-$ levels at 67.16 and 275.33 keV										Ens164 **
$^{187}\text{Bi}(\alpha)^{183}\text{Tl}^m$	Also $E_\alpha=7612(15)$ keV to ground state										99Ba45 **
$^{187}\text{Bi}(\alpha)^{183}\text{Tl}^m$	Also $E_\alpha=7605(16)$ keV to ground state										03Ke08 **
$^{187}\text{Bi}(\alpha)^{183}\text{Tl}^m$	Also $E_\alpha=7612(5)$, 7342(15) keV to ground state, 273(1) keV										06An11 **
$^{187}\text{Bi}^m(\alpha)^{183}\text{Tl}$	F : for $T=700$ μs instead of Nubase=370(20) μs										Nub211 **
$^{187}\text{Po}(\alpha)^{183}\text{Pb}$	$E_\alpha=7528(15)$ to 286(1) keV level; also 1 event $E_\alpha=7796(15)$ to ground state										06An11 **
$^{186}\text{W}(\text{n},\gamma)^{187}\text{W}$	Only statistical error 0.04 keV given; Z recalibrated										GAu **
$^{187}\text{Re}(\beta^-)^{187}\text{Os}$	supersedes 2.481(0.006) of same group										98Ga50 **
$^{187}\text{Ir}(\beta^+)^{187}\text{Os}$	$p^+ < 0.15(0.05)$, resulting $Q_i 1550$ keV										Ens095 **
$^{187}\text{Au}(\beta^+)^{187}\text{Pt}$	$K/\beta^+=31.6(2.8)$ to $1/2^+$ level at 1341.07 keV, recalculated										Ens095 **
$^{187}\text{Hg}^m(\text{IT})^{187}\text{Hg}$	Original error (7 keV) increased by 20 due to isomer+ground state lines in trap										01Sc41 **
$^{188}\text{Ta}-\text{u}$	-36084	59	-36400#	220#	-5.4	D			GS3	1.0	13Sh30 *
$^{188}\text{Au}-\text{u}$	-34750	104	-34752.0	2.9	-0.0	U			GS1	1.0	00Ra23
	-34674	30			-2.6	B			GS2	1.0	05Li24
$^{188}\text{Au}-^{133}\text{Cs}_{1.414}$	98938.9	2.9				2			MA8	1.0	17Ma29
$^{188}\text{Hg}-\text{u}$	-32500	104	-32419	7	0.8	U			GS1	1.0	00Ra23
	-32428	30			0.3	1	6	6 ^{188}Hg	GS2	1.0	05Li24
$^{188}\text{Hg}-^{208}\text{Pb}_{.904}$	-11330	20	-11313	7	0.9	-			MA6	1.0	01Sc41
ave.	-11317	17			0.2	1	19	19 ^{188}Hg			average
$^{188}\text{Tl}-\text{u}$	-23827	110	-23980	30	-1.4	U			GS1	1.0	00Ra23 *
	-23994	38			0.4	2			GS2	1.0	05Li24 *
$^{188}\text{Pb}-\text{u}$	-19070	110	-19121	11	-0.5	U			GS1	1.0	00Ra23
	-19144	30			0.8	R			GS2	1.0	05Li24
$^{188}\text{Os } ^{35}\text{Cl}-^{186}\text{W } ^{37}\text{Cl}$	4426	3	4422.3	1.2	-0.5	U			H22	2.5	70Mc03
$^{188}\text{Pt}(\alpha)^{184}\text{Os}$	4015.7	10.	4007	5	-0.9	-					63Gr08
	4000.3	10.			0.6	-			ORa		78El11
	3990.1	15.			1.1	-			ISa		79Ha10
ave.	4005	7			0.3	1	65	65 ^{188}Pt			average
$^{188}\text{Hg}(\alpha)^{184}\text{Pt}$	4710.4	20.	4709	15	-0.1	1	52	47 ^{184}Pt	ISa		79Ha10
$^{188}\text{Pb}(\alpha)^{184}\text{Hg}$	6110.3	10.	6109	3	-0.1	2			Ora		74Le02 Z
	6109.2	10.			-0.0	2			Ora		77De32 Z
	6120.5	15.			-0.8	2			GSa		80Sc09 Z
	6110.5	5.			-0.3	2			ORa		81To02 Z
	6110.3	10.3			-0.1	o			Lvn		93Wa.A
	6109.3	10.			-0.0	2			Lvn		93Wa03 Z
	6100.0	8.			1.1	2			Jya		03Ke04

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{188}\text{Bi}(\alpha)^{184}\text{Tl}$	7274.5	25.	7264	5	-0.4	o			GSa		80Sc09 *
	7279.7	10.			-1.6	2			GSa		84Sc.A *
	7255.2	7.			1.2	2			Lvn		97Wa05 *
	7259.3	5.			0.9	o			Anv		03An26 *
	7264.8	10.			-0.1	2			Anv		06An04 *
$^{188}\text{Bi}^n(\alpha)^{184}\text{Tl}^m$	7462.9	5.				5			Anv		03An26 *
$^{188}\text{Bi}^n(\alpha)^{184}\text{Tl}^n$	6968.5	20.	6965	5	-0.2	o			GSa		80Sc09
	6968.5	10.			-0.4	5			GSa		84Sc.A
	6963.5	6.			0.2	5			Lvn		97Wa05
	6961.3	5.			0.6	o			Anv		03An26
	6963.5	5.			0.1	5			Anv		06An04
$^{188}\text{Po}(\alpha)^{184}\text{Pb}$	8087.4	25.	8082	15	-0.2	o			Anv		99An52
	8080.2	15.			0.1	o			Anv		01Va.B
	8082.3	15.				2			Anv		03Va16
$^{188}\text{Os}(\text{p,t})^{186}\text{Os}$	-5802	5	-5798.1	0.5	0.8	U			Min		73Oo01
	-5803	4			1.2	U			McM		75Th04
$^{187}\text{Re}(\text{n},\gamma)^{188}\text{Re}$	5871.77	0.3	5871.65	0.04	-0.4	U					72Sh13 Z
	5871.75	0.13			-0.8	U			Bdn		06Fi.A
	5871.65	0.04				2			Prn		10Ba48
$^{188}\text{Os}(\text{t},\alpha)^{187}\text{Re}$	12604	10	12604.14	0.15	0.0	U			McM		76Hi08
$^{187}\text{Os}(\text{n},\gamma)^{188}\text{Os}$	7989.6	0.3	7989.60	0.15	0.0	-					83Fe06 Z
	7989.58	0.17			0.1	-			Bdn		06Fi.A
ave.	7989.58	0.15			0.1	1	98	58 ^{187}Os			average
$^{188}\text{W}(\beta^-)^{188}\text{Re}$	349	3				3					64Bu10
$^{188}\text{Re}(\beta^-)^{188}\text{Os}$	2116	2	2120.42	0.15	2.2	U					56Jo05
	2111	3			3.1	B					68An11
$^{188}\text{Ir}(\beta^+)^{188}\text{Os}$	2833	10	2792	9	-4.1	B					62Wa20 *
	2781	20			0.6	-					69Ya02 *
	2827	30			-1.2	-					70Ag03 *
ave.	2795	17			-0.2	1	32	32 ^{188}Ir			average
$^{188}\text{Pt}(\epsilon)^{188}\text{Ir}$	525	10	524	9	-0.1	1	75	68 ^{188}Ir	ORa		78El11 *
$^{188}\text{Au}(\beta^+)^{188}\text{Pt}$	5520	30	5450	6	-2.3	U					84Da.A
$^{188}\text{Hg}(\beta^+)^{188}\text{Au}$	2040	20	2173	7	6.6	B					84Da.A
* $^{188}\text{Ta}-\text{u}$	Trends from Mass Surface TMS suggest ^{188}Ta 300 keV more bound										GAu **
* $^{188}\text{Tl}-\text{u}$	$M-A=-22180(100)$ keV for mixture gs+m at 30(40) keV										GAu **
* $^{188}\text{Tl}-\text{u}$	$M-A=-22335(28)$ keV for mixture gs+m at 30(40) keV										GAu **
* $^{188}\text{Bi}(\alpha)^{184}\text{Tl}$	$E_\alpha=7005(25)$ 7010(10) 6987(6) respectively, to (3^+) level at 117.5(0.5) keV										Ens102 **
*	$E_\alpha=7029(7)$ 3 times weaker exists too, possible mixture in older results										97Wa05 **
* $^{188}\text{Bi}(\alpha)^{184}\text{Tl}$	$E_\alpha=7106(5)$, 6992(5), 6889(10) to ground state, 117.5, 216 levels										03An26 **
* $^{188}\text{Bi}(\alpha)^{184}\text{Tl}$	$E_\alpha=6995(10)$ to 117.5 level										06An04 **
* $^{188}\text{Bi}^n(\alpha)^{184}\text{Tl}^m$	$E_\alpha=7302(5)$, 7232(10), 6995(15) to ground state, 70.5, 320 levels										03An26 **
* $^{188}\text{Ir}(\beta^+)^{188}\text{Os}$	$E_{\beta^+}=1656(10)$ 1605(20) 1650(30) respectively, to 2^+ level at 155.021 keV										Ens024 **
* $^{188}\text{Pt}(\epsilon)^{188}\text{Ir}$	$pL=0.67(0.05)$ to 1^+ level at 478.17 keV										Ens024 **
$^{189}\text{W}-\text{u}$	-38237	43	-38440#	220#	-4.8	B			GS3	1.0	13Sh30
$\text{C}_{14}\text{H}_{21}-^{189}\text{Os}$	206188.3	6.2	206179.7	0.7	-0.3	U			M23	4.0	79Ha32
$^{189}\text{Au}-\text{u}$	-36080	140	-36052	22	0.2	U			GS1	1.0	00Ra23 *
	-36045	31			-0.2	2			GS2	1.0	05Li24
	-36058	30			0.2	2			GS2	1.0	05Li24 *
$^{189}\text{Hg}-\text{u}$	-31788	111	-31810	30	-0.2	U			GS1	1.0	00Ra23 *
	-31791	42			-0.3	1	65	65 ^{189}Hg	GS2	1.0	05Li24 *
$^{189}\text{Hg}^m-^{208}\text{Pb}_{909}$	-10501	20	-10498	19	0.2	1	92	92 $^{189}\text{Hg}^m$	MA6	1.0	01Sc41
$^{189}\text{Tl}-\text{u}$	-26497	139	-26426	9	0.5	U			GS1	1.0	00Ra23 *
	-26313	93			-1.2	U			GS2	1.0	05Li24 *
$^{189}\text{Pb}-\text{u}$	-19206	97	-19156	15	0.5	U			GS1	1.0	00Ra23 *
	-19193	34			1.1	1	20	20 ^{189}Pb	GS2	1.0	05Li24 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{189}\text{Os } ^{35}\text{Cl}-^{187}\text{Re } ^{37}\text{Cl}$	5341	3	5343.9	0.5	0.4	U			H22	2.5	70Mc03
$^{189}\text{Pb}(\alpha)^{185}\text{Hg}$	5954.2	10.	5915	4	-3.9	B			Ora		72Ga27 *
	5943.9	10.			-2.9	B			Ora		74Le02 *
	5915	10			-0.0	-					05Fr.A *
	5914.8	5.4			-0.0	-			ISa		13Sa43 *
	ave. 5915	5			-0.0	1	82	67 ^{189}Pb			average
$^{189}\text{Pb}^m(\alpha)^{185}\text{Hg}$	5958	10	5955	5	-0.3	1	28	25 $^{189}\text{Pb}^m$			05Fr.A *
	5955	5			0.0	o			ISa		13Sa43 *
$^{189}\text{Bi}(\alpha)^{185}\text{Tl}$	7269.4	10.	7268.2	2.7	-0.1	6			Ora		74Le02 *
	7274.5	10.			-0.6	6			GSa		84Sc.A *
	7271.2	5.			-0.6	6			Lvn		85Co06 *
	7271.8	15.			-0.2	U			Anv		97An09 *
	7268.1	6.			0.0	6			Lvn		97Wa05 *
	7271.5	5.			-0.7	o			Jya		02Hu14 *
	7264.2	4.5			0.9	6			Jya		03Ke08 *
$^{189}\text{Bi}^m(\alpha)^{185}\text{Tl}$	7362.1	20.	7452	4	1.7	U			GSa		84Sc.A *
	7499.0	30.			-1.6	U			Dbb		93An19 *
	7458.2	40.			-0.2	U			ORa		95Ba75 *
	7458.2	15.			-0.4	6			Anv		97An09 *
	7450.0	6.			0.4	6			Lvn		97Wa05 *
	7453.1	6.			-0.2	6			Jya		03Ke08 *
$^{189}\text{Po}(\alpha)^{185}\text{Pb}$	7699.4	15.	7694	15	-0.3	o			Anv		99An52 *
	7694.3	15.				3			Anv		05Va04 *
$^{189}\text{Os}(\text{p,t})^{187}\text{Os}$	-5431	5	-5428.6	0.5	0.5	U			Min		73Oo01 *
	-5432	4			0.8	U			McM		75Th04 *
$^{188}\text{Os}(\text{n},\gamma)^{189}\text{Os}$	5920.8	2.	5920.8	0.4	0.0	U					76Be50 *
	5920.6	0.5			0.4	1	80	59 ^{188}Os	ILn		92Br17 *
	5922.0	0.4			-3.0	C			Bdn		06Fi.A *
$^{188}\text{Os}(\text{d,p})^{189}\text{Os}$	3689	10	3696.3	0.4	0.7	U			Kop		75Mo29 *
$^{189}\text{Os}(\text{d,t})^{188}\text{Os}$	335	15	336.4	0.4	0.1	U			Tal		75Th06 *
$^{189}\text{W}(\beta^-)^{189}\text{Re}$	2500	200	2170#	200#	-1.6	D					65Ka07 *
$^{189}\text{Re}(\beta^-)^{189}\text{Os}$	1000	20	1008	8	0.4	R					63Cr06 *
	1015	20			-0.4	R					65Bi06 *
$^{189}\text{Pt}(\beta^+)^{189}\text{Ir}$	1950	20	1980	14	1.5	1	46	30 ^{189}Ir			71Pl08 *
$^{189}\text{Au}(\beta^+)^{189}\text{Pt}$	3160	300	2887	22	-0.9	U					75Un.A *
$^{189}\text{Hg}(\beta^+)^{189}\text{Au}$	4200	200	3960	40	-1.2	U					75Un.A *
$^{189}\text{Hg}^m(\text{IT})^{189}\text{Hg}$	100	50	80	30	-0.4	1	43	35 ^{189}Hg	MA6		01Sc41 *
$^{189}\text{Tl}^m(\beta^+)^{189}\text{Hg}$	5460	200	5300	30	-0.8	U					75Un.A *
$^{189}\text{Pb}^m(\text{IT})^{189}\text{Pb}$	40	4	40	4	0.1	1	88	75 $^{189}\text{Pb}^m$	ISa		13Sa43 *
$^{189}\text{Au}-\text{u}$	$M-A=-33490(100)$ keV for mixture gs+m at 247.25 keV										Nub211 **
$^{189}\text{Au}-\text{u}$	$M-A=-33341(28)$ keV for $^{189}\text{Au}^m$ at 247.25 keV										Nub211 **
$^{189}\text{Hg}-\text{u}$	$M-A=-29570(100)$ keV for mixture gs+m at 80(30) keV										Nub211 **
$^{189}\text{Hg}-\text{u}$	$M-A=-29573(28)$ keV for mixture gs+m at 80(30) keV										Nub211 **
$^{189}\text{Tl}-\text{u}$	$M-A=-24540(100)$ keV for mixture gs+m at 285(6) keV										Nub211 **
$^{189}\text{Tl}-\text{u}$	$M-A=-24369(28)$ keV for mixture gs+m at 285(6) keV										Nub211 **
$^{189}\text{Pb}-\text{u}$	$M-A=-17870(90)$ keV for mixture gs+m at 40(4) keV										Nub211 **
$^{189}\text{Pb}-\text{u}$	$M-A=-17858(29)$ keV for mixture gs+m at 40(4) keV										Nub211 **
$^{189}\text{Pb}(\alpha)^{185}\text{Hg}$	$E_\alpha=5730.1(10,Z)$ possibly from ground state, to $3/2^-$ level at 26.1 keV										Ens061 **
$^{189}\text{Pb}(\alpha)^{185}\text{Hg}$	$E_\alpha=5720(10)$ possibly from ground state, to $3/2^-$ level at 26.1 keV										Ens061 **
$^{189}\text{Pb}(\alpha)^{185}\text{Hg}$	$E_\alpha=5761$ to $3/2^-$ level at 26.1 and $E_\alpha=5623$ to 173.8 level										05Fr.A **
$^{189}\text{Pb}(\alpha)^{185}\text{Hg}$	$E_\alpha=5764(3)(5)$ to $3/2^-$ level at 26.02 and $E_\alpha=5619(2)(5)$ to 173.9 level										13Sa43 **
$^{189}\text{Pb}^m(\alpha)^{185}\text{Hg}$	$E_\alpha=5730$ to 103.8 level										05Fr.A **
$^{189}\text{Pb}^m(\alpha)^{185}\text{Hg}$	$E_\alpha=5727(2)(5)$ to 103.7(0.4) level										13Sa43 **
$^{189}\text{Pb}^m(\alpha)^{185}\text{Hg}$	use instead $^{189}\text{Pb}^m(\text{IT})^{189}\text{Pb}$ where systematic errors cancel										GAu **
$^{189}\text{Bi}(\alpha)^{185}\text{Tl}$	$E_\alpha=6670.1(10,Z)$ to $^{185}\text{Tl}^m$ at 454.8(1.5) keV										Nub211 **
$^{189}\text{Bi}(\alpha)^{185}\text{Tl}$	$E_\alpha=6675(10)$ to $^{185}\text{Tl}^m$ at 454.8(1.5) keV										Nub211 **
$^{189}\text{Bi}(\alpha)^{185}\text{Tl}$	$E_\alpha=7115.6(15,Z)$ and $6671.6(5,Z)$ to $^{185}\text{Tl}^m$ at 454.8(1.5) keV										Nub211 **
$^{189}\text{Bi}(\alpha)^{185}\text{Tl}$	$E_\alpha=7120(15)$, 6670(15) to ground state and $^{185}\text{Tl}^m$ at 454.8(1.5) keV										Nub211 **
$^{189}\text{Bi}(\alpha)^{185}\text{Tl}$	$E_\alpha=6674(5)$ to $^{185}\text{Tl}^m$ at 452.8(2.0) keV										77Sc03 **
$^{189}\text{Bi}(\alpha)^{185}\text{Tl}$	$E_\alpha=6667(4)$ to $^{185}\text{Tl}^m$ at 452.8(2.0) keV										77Sc03 **
	and also $E_\alpha=7114(6)$ to ground state										03Ke08 **
$^{189}\text{Bi}^m(\alpha)^{185}\text{Tl}$	Only one event; not seen in reference										93An19 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		ν_i	Dg	Signf.	Main infl.	Lab	F	Reference
* $^{189}\text{Po}(\alpha)^{185}\text{Pb}$	$E_\alpha=7264(15)$ to $278(1)$ level										99An52 **
* $^{189}\text{Po}(\alpha)^{185}\text{Pb}$	$E_\alpha=7259(15)$ to $278(1)$ level										05Va04 **
* $^{189}\text{W}(\beta^-)^{189}\text{Re}$	Trends from Mass Surface TMS suggest ^{189}W 330 keV more bound										GAu **
* $^{189}\text{Pt}(\beta^+)^{189}\text{Ir}$	$E_{\beta^+}=885(10)$ to ground state, $1/2^+$ level at 94.34 and $3/2^+$ at 176.53 keV										Ens039 **
* $^{189}\text{Tl}^m(\beta^+)^{189}\text{Hg}$	$E_{\beta^+}=4140(200)$ to several levels around 300 keV										75Un.A **
$^{190}\text{W}-u$	-36917	44	-36900	40	0.5	1	75	75 ^{190}W	GS3	1.0	13Sh30
$^{190}\text{Ir}-u$	-39479	188	-39456.6	1.5	0.1	U			GS1	1.0	97Sc.B *
$^{190}\text{Au}-u$	-35213	106	-35248	4	-0.3	U			GS2	1.0	05Li24 *
$^{190}\text{Au}-^{133}\text{Cs}_{1.429}$	99860.9	3.7				2			MA8	1.0	17Ma29
$^{190}\text{Hg}-u$	-33670	107	-33678	17	-0.1	U			GS1	1.0	00Ra23
$^{190}\text{Hg}-^{208}\text{Pb}_{.913}$	-12361	20	-12361	17	-0.0	1	73	73 ^{190}Hg	MA6	1.0	01Sc41
$^{190}\text{Tl}^m-u$	-26055	107	-26083	6	-0.3	o			GS1	1.0	00Ra23 *
	-26048	30			-1.2	U			GS2	1.0	05Li24 *
$^{190}\text{Tl}^m-^{133}\text{Cs}_{1.429}$	109033.5	6.9	109026	6	-1.1	1	64	64 $^{190}\text{Tl}^m$	MA8	1.0	14Bo26
$^{190}\text{Pb}-u$	-21940	104	-21918	13	0.2	U			GS1	1.0	00Ra23
	-21905	30			-0.4	R			GS2	1.0	05Li24
$^{190}\text{Bi}^m-^{133}\text{Cs}_{1.429}$	123800	27	123870	30	2.4	F			MA8	1.0	08We02 *
$^{186}\text{Os}-^{190}\text{Pt}_{.979}$	-6953.13	0.86	-6953.3	0.6	-0.2	1	53	39 ^{186}Os	MS1	1.0	16Ei01
$^{190}\text{Os }^{35}\text{Cl}-^{188}\text{Os }^{37}\text{Cl}$	5557	3	5558.3	0.5	0.2	U			H22	2.5	70Mc03
$^{190}\text{Os}-^{190}\text{Pt}$	-1504.31	0.59	-1504.4	0.5	-0.1	1	62	33 ^{190}Pt	MS1	1.0	16Ei01
$^{190}\text{Os}-\text{C}_{14}\text{H}_{21}$	-205897.8	5.8	-205880.2	0.7	0.8	U			M23	4.0	79Ha32
$^{190}\text{Os}-^{189}\text{Os}$	285.2	5.2	299.49	0.20	1.1	U			M24	2.5	79Ha32
$^{190}\text{Pt}(\alpha)^{186}\text{Os}$	3238.3	20.	3268.6	0.6	1.5	U					61Pe23
	3248.5	20.			1.0	U					63Gr08
$^{190}\text{Pb}(\alpha)^{186}\text{Hg}$	5699.8	10.	5698	5	-0.2	2			Ora		74Le02 Z
	5697.0	5.			0.1	2			ORa		81Ei03 Z
$^{190}\text{Bi}(\alpha)^{186}\text{Tl}$	6862.2	5.	6862	3	-0.0	2			Lvn		91Va04 *
	6863.3	5.			-0.2	2			Anv		03An26 *
	6860.3	6.			0.3	2			Jya		13Ny01 *
$^{190}\text{Bi}^m(\alpha)^{186}\text{Tl}^m$	6967.9	5.	6966.4	2.8	-0.3	4			Lvn		91Va04 *
	6969.1	5.			-0.5	4			Anv		03An26 *
	6963	10			0.3	4			Ora		74Le02 *
	6963.1	5.			0.7	4			Jya		13Ny01 *
$^{190}\text{Bi}^m(\alpha)^{186}\text{Tl}^n$	6589.0	10.	6592.3	2.8	0.3	U			Ora		74Le02
$^{190}\text{Po}(\alpha)^{186}\text{Pb}$	7643.2	20.	7693	7	2.5	C			GSa		88Qu.A
	7651.4	40.			1.0	U			ORa		96Ba35
	7691.2	10.			0.2	3			ORa		97Ba25
	7695.3	10.			-0.2	3			Anv		00An14 *
$^{190}\text{Os}(\text{p,t})^{188}\text{Os}$	-5234	5	-5231.4	0.5	0.5	U			Min		73Oo01
	-5237	4			1.4	U			McM		75Th04
$^{190}\text{Pt}(\text{p,t})^{188}\text{Pt}$	-7150	10	-7146	5	0.4	1	28	28 ^{188}Pt	Ors		78Ve10
$^{190}\text{Os}(\text{t},\alpha)^{189}\text{Re}$	11796	10	11796	8	0.0	2			McM		76Hi08
$^{189}\text{Os}(\text{n},\gamma)^{190}\text{Os}$	7791.8	1.0	7792.34	0.19	0.5	U			BNn		79Ca02 Z
	7792.31	0.19			0.2	1	97	79 ^{189}Os	Bdn		06Fi.A
$^{190}\text{Os}(\text{d,t})^{189}\text{Os}$	-1541	10	-1535.11	0.19	0.6	U			Kop		75Mo29
	-1530	4			-1.3	U			Tal		76Be50
$^{190}\text{Pt}(\text{p,d})^{189}\text{Pt}$	-6693	11	-6684	10	0.8	1	84	84 ^{189}Pt	Ors		80Ka19
$^{190}\text{W}(\beta^-)^{190}\text{Re}$	1270	70	1210	40	-0.8	1	26	25 ^{190}W			76Ha39 *
$^{190}\text{Re}(\beta^-)^{190}\text{Os}$	3090	300	3125	5	0.1	U					55At21 *
	3190	300			-0.2	U					69Ha44 *
	3146	200			-0.1	U					64Fl02 *
$^{190}\text{Ir}(\beta^+)^{190}\text{Os}$	2000	200	1954.2	1.2	-0.2	U					60Ka14 *
$^{190}\text{Au}(\beta^+)^{190}\text{Pt}$	4442	15	4473	4	2.1	U					73Jo11
	4380	55			1.7	U					74Di.A
	4380	200			0.5	U					75Un.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{190}\text{Hg}(\beta^+)^{190}\text{Au}$	2105	80	1463	16	-8.0	C					74Di.A
$^{190}\text{Tl}(\beta^+)^{190}\text{Hg}$	7000	400	7004	17	0.0	U					75Un.A *
$^{190}\text{Tl}^m(\beta^+)^{190}\text{Hg}$	6975	300	7074	17	0.3	U					76Bi09 *
$^{190}\text{Bi}(\beta^+)^{190}\text{Pb}$	8700	500	9821	24	2.2	F					76Bi09 *
$^{190}\text{Ir}-u$	$M-A=-36761(175)$ keV for mixture gs+m at 26.1 keV										Nub211 **
$^{190}\text{Au}-u$	$M-A=-32701(28)$ keV for mixture gs+m at 200#150 keV										Nub211 **
$^{190}\text{Tl}^m-u$	Assumed by evaluator to be the 7^+ excited isomer										GAu **
$^{190}\text{Bi}^m-^{133}\text{Cs}_{1.429}$	F : contamination due to ground state not resolved										08We02 **
$^{190}\text{Bi}(\alpha)^{186}\text{Tl}$	$E_\alpha=6716(5)$, $6507(5)$, $6431(5)$ to ground state, 215.2, 293.7 levels										91Va04 **
$^{190}\text{Bi}(\alpha)^{186}\text{Tl}$	$E_\alpha=6431(5)$ to 293.7 level										03An26 **
$^{190}\text{Bi}(\alpha)^{186}\text{Tl}$	$E_\alpha=6428(6)$ to 293.7 level										13Ny01 **
$^{190}\text{Bi}^m(\alpha)^{186}\text{Tl}^m$	$E_\alpha=6819(5)$, $6734(5)$, $6456(5)$ to levels 0, 89.5, 373.9 above $^{186}\text{Tl}^m$										91Va04 **
$^{190}\text{Bi}^m(\alpha)^{186}\text{Tl}^m$	$E_\alpha=6456(5)$ to 374.0 level above $^{186}\text{Tl}^m$										03An26 **
$^{190}\text{Bi}^m(\alpha)^{186}\text{Tl}^m$	$Q_\alpha=6589(10)$ to 374.0 level above $^{186}\text{Tl}^m$										Nub211 **
$^{190}\text{Bi}^m(\alpha)^{186}\text{Tl}^m$	$E_\alpha=6450(5)$ to 374.0 level above $^{186}\text{Tl}^m$										Nub211 **
$^{190}\text{Po}(\alpha)^{186}\text{Pb}$	$E_\alpha=7545(15)$ same dataset as in reference 00An14										97An09 **
$^{190}\text{W}(\beta^-)^{190}\text{Re}$	$E_\beta=950(70)$ to 1^+ level at 319.7 keV										Ens036 **
$^{190}\text{Re}(\beta^-)^{190}\text{Os}$	$E_\beta=1700(300)$ $1800(300)$ respectively, to 3^- level at 1387.00 keV										Ens036 **
$^{190}\text{Re}(\beta^-)^{190}\text{Os}$	$E_\beta=1600(200)$ from isomer at 204(10) to several levels around 1750 keV										Nub211 **
$^{190}\text{Ir}(\beta^+)^{190}\text{Os}$	$p^+=6(1)\times 10^{-5}$ to 4^+ levels at 1163.19 and 955.37 keV, level at 1872.15 keV fed										Ens036 **
$^{190}\text{Tl}(\beta^+)^{190}\text{Hg}$	$E_\beta=5700(400)$ to ground state and 2^+ level at 416.32 keV										Ens036 **
$^{190}\text{Tl}^m(\beta^+)^{190}\text{Hg}$	$E_\beta=4180(300)$ to 6^+ level at 1772.94 keV										Ens036 **
$^{190}\text{Bi}(\beta^+)^{190}\text{Pb}$	F : $E_\beta=5700(300)$ to a level around 2000 at least										AHW **
$^{191}\text{W}-u$	-33469	45				2			GS3	1.0	13Sh30
$^{191}\text{Au}-u$	-36180	88	-36284	5	-1.2	U			GS2	1.0	05Li24 *
$^{191}\text{Au}-^{133}\text{Cs}_{1.436}$	99487.3	5.3	99487	5	0.0	1	100	100 ^{191}Au	MA8	1.0	13Kr15
$^{191}\text{Hg}-u$	-32811	51	-32842	24	-0.6	1	22	22 ^{191}Hg	GS2	1.0	05Li24 *
$^{191}\text{Hg}-^{208}\text{Pb}_{918}$	-11414	29	-11408	24	0.2	1	68	68 ^{191}Hg	MA6	1.0	01Sc41 *
$^{191}\text{Tl}-u$	-28340	130	-28216	8	1.0	U			GS1	1.0	00Ra23 *
	-28234	30			0.6	U			GS2	1.0	05Li24
	-28192	31			-0.8	U			GS2	1.0	05Li24 *
$^{191}\text{Pb}-u$	-21770	110	-21784	7	-0.1	U			GS1	1.0	00Ra23 *
	-21744	35			-1.1	1	4	4 ^{191}Pb	GS2	1.0	05Li24 *
$^{191}\text{Bi}-^{133}\text{Cs}_{1.436}$	121552.1	8.6	121558	8	0.7	1	87	87 ^{191}Bi	MA8	1.0	08We02
$^{191}\text{Pb}^m(\alpha)^{187}\text{Hg}^m$	5403.4	20.	5402	15	-0.1	1	51	44 $^{187}\text{Hg}^m$	Ora		74Le02
$^{191}\text{Bi}(\alpha)^{187}\text{Tl}$	6780.8	5.	6780	3	-0.1	-			Lvn		85Co06 Z
	6785.3	10.2			-0.5	-			ORa		98Bi.A
	6782.3	10.2			-0.2	-			Anv		99An36
	6783.3	7.2			-0.4	-			Jya		13Ny01
ave.	6782	4			-0.5	1	71	69 ^{187}Tl			average
$^{191}\text{Bi}(\alpha)^{187}\text{Tl}^m$	6440.0	5.	6446.6	2.5	1.3	-					67Tr06 Z
	6455.0	10.			-0.8	U			Ora		74Le02 Z
	6445.9	5.			0.2	-			Lvn		85Co06 Z
	6447	10			-0.0	U			ORa		98Bi.A
	6458.5	20.			-0.6	U			Rla		99Ta20
	6445	10			0.2	U			Anv		99An36
	6443.2	3.			1.1	o			Jya		03Ke04
	6445.2	10.			0.1	U			Jya		13Uu01
	6450.3	4.			-0.9	-			Jya		13Ny01
ave.	6446.2	2.7			0.2	1	83	72 $^{187}\text{Tl}^m$			average
$^{191}\text{Bi}^m(\alpha)^{187}\text{Tl}$	7022.8	5.	7023	3	-0.0	2			Lvn		85Co06 Z
	7023.4	10.			-0.1	U			ORa		98Bi.A
	7016.2	20.			0.3	U			Rla		99Ta20
	7017.2	3.			1.7	o			Jya		03Ke04
	7031.0	15.			-0.6	U			Jya		13Uu01 *
	7022.4	4.			0.0	2			Jya		13Ny01

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{191}\text{Po}(\alpha)^{187}\text{Pb}$	7470.8	20.	7493	5	1.1	F			GSa		93Qu03 *
	7487.1	15.			0.4	U			ORa		97Ba25
	7491.2	5.			0.4	1	94	94 ^{191}Po	Anv		02An19 *
$^{191}\text{Po}(\alpha)^{187}\text{Pb}^m$	7493.2	15.	7474	10	-1.2	1	45	39 $^{187}\text{Pb}^m$	Anv		02An19
$^{191}\text{Po}^m(\alpha)^{187}\text{Pb}^m$	7535	5				2			Anv		02An19 *
$^{191}\text{At}(\alpha)^{187}\text{Bi}^m$	7713.9	11.				3			Jya		03Ke08
$^{191}\text{At}^m(\alpha)^{187}\text{Bi}$	7880.4	15.				3			Jya		03Ke08
$^{191}\text{Ir}(\text{p},\text{t})^{189}\text{Ir}$	-5903	15	-5920	13	-1.1	1	70	70 ^{189}Ir	McM		78Lo07
$^{190}\text{Os}(\text{n},\gamma)^{191}\text{Os}$	5758.2	2.	5758.73	0.11	0.3	U					77Be15
	5759.1	1.5			-0.2	U					77Ca19
	5758.67	0.16			0.4	-			ILn		91Bo35 Z
	5758.81	0.15			-0.5	-			Bdn		06Fi.A
ave.	5758.74	0.11			-0.1	1	100	99 ^{191}Os			average
$^{190}\text{Os}(\alpha,\text{t})^{191}\text{Ir}$	-14569	15	-14523.9	1.1	3.0	B			McM		71Pr13
$^{191}\text{Ir}(\text{d},\text{t})^{190}\text{Ir}$	-1769.3	0.4				2					95Ga04 *
$^{191}\text{Os}(\beta^-)^{191}\text{Ir}$	313.3	3.	313.6	1.1	0.1	-					48Sa18 *
	314.3	2.			-0.4	-					51Ko17 *
	316.3	3.			-0.9	-					58Na15 *
	314.3	3.			-0.2	-					60Fe03 *
	318.3	3.			-1.6	-					63Pl01 *
ave.	315.1	1.2			-1.3	1	90	90 ^{191}Ir			average
$^{191}\text{Pt}(\epsilon)^{191}\text{Ir}$	1000	15	1010	4	0.7	U					70Sc20 *
$^{191}\text{Au}(\beta^+)^{191}\text{Pt}$	1830	50	1900	6	1.4	U					76Vi.A *
$^{191}\text{Hg}(\beta^+)^{191}\text{Au}$	3430	200	3206	23	-1.1	U					75Un.A *
	3180	70			0.4	1	11	10 ^{191}Hg			76Vi.A *
$^{191}\text{Tl}^m(\beta^+)^{191}\text{Hg}$	5178	200	4606	23	-2.9	C					75Un.A *
* $^{191}\text{Au}-\text{u}$	$M-A=-33568(28)$ keV for mixture gs+m at 266.2 keV										Nub211 **
* $^{191}\text{Hg}-\text{u}$	$M-A=-30499(28)$ keV for mixture gs+m at 128(22) keV										Nub211 **
* $^{191}\text{Hg}-^{208}\text{Pb}_{918}$	Original error (19keV) increased by 20 due to isomer+ground state lines in trap										01Sc41 **
* $^{191}\text{Tl}-\text{u}$	$M-A=-26250(90)$ keV for mixture gs+m at 297(7) keV										Nub211 **
* $^{191}\text{Tl}-\text{u}$	$M-A=-25964(28)$ keV for $^{191}\text{Tl}^m$ at 297(7) keV										Nub211 **
* $^{191}\text{Pb}-\text{u}$	Possible isomeric contamination										00Ra23 **
* $^{191}\text{Pb}-\text{u}$	$M-A=-20226(28)$ keV for mixture gs+m at 57(10) keV										Nub211 **
* $^{191}\text{Bi}^m(\alpha)^{187}\text{Tl}$	average 6882(20) 6885(15)										13Uu01 **
* $^{191}\text{Po}(\alpha)^{187}\text{Pb}$	F : probably mainly $^{189}\text{Bi}^m$										97Ba25 **
* $^{191}\text{Po}(\alpha)^{187}\text{Pb}$	$E_\alpha=7334(10)$, 6960(15) to ground state, 375(1) superseded by 02An19										99An10 **
* $^{191}\text{Po}^m(\alpha)^{187}\text{Pb}^m$	$E_\alpha=7376(5)$, 6888(5) to $^{187}\text{Pb}^m$ and 494(1) above										02An19 **
* $^{191}\text{Po}^m(\alpha)^{187}\text{Pb}^m$	$E_\alpha=7378(10)$, 6888(15) superseded by 02An19										99An10 **
* $^{191}\text{Ir}(\text{d},\text{t})^{190}\text{Ir}$	Feeds ground state										96Ga30 **
* $^{191}\text{Os}(\beta^-)^{191}\text{Ir}$	$E_{\beta^-}=142(3)$ 143(2) 145(3) 143(3) 147(3) respectively, to $11/2^-$ level at 171.29 keV										Ens07a **
* $^{191}\text{Pt}(\epsilon)^{191}\text{Ir}$	$pL=0.73(0.12)$ to $(1/2^+, 3/2, 5/2^+)$ at 935.46 keV, no K capture										Ens07a **
* $^{191}\text{Au}(\beta^+)^{191}\text{Pt}$	$E_{\beta^+}=850(30)$ to ground state and $(5/2^-, 7/2^-)$ level at 9.547 keV; also $E_{\beta^+}=470(60)$ to $(3/2^-, 5/2^-)$ level at 277.88 and $5/2^-$ level at 293.458 keV										Ens07a **
*	Reassigned by evaluator to mainly ground state, partly $3/2^+$ 207.9 level										Ens07a **
* $^{191}\text{Hg}(\beta^+)^{191}\text{Au}$	$E_{\beta^+}=3820(200)$ to level $(5/2^-)$ at 336.32 keV										Ens07a **
* $^{191}\text{Tl}^m(\beta^+)^{191}\text{Hg}$											
$^{192}\text{Re}-\text{u}$	-33912	76				2			GS3	1.0	13Sh30
$^{192}\text{Hg}-\text{u}$	-34440	104	-34366	17	0.7	U			GS1	1.0	00Ra23
	-34342	30			-0.8	R			GS2	1.0	05Li24
$^{192}\text{Hg}-^{208}\text{Pb}_{923}$	-12826	20	-12816	17	0.5	2			MA6	1.0	01Sc41
$^{192}\text{Tl}-\text{u}$	-27830	123	-27780	30	0.4	U			GS1	1.0	00Ra23 *
	-27775	34				2			GS2	1.0	05Li24
$^{192}\text{Pb}-\text{u}$	-24280	104	-24210	6	0.7	U			GS1	1.0	00Ra23
	-24185	30			-0.8	R			GS2	1.0	05Li24
$^{192}\text{Bi}-\text{u}$	-14783	128	-14530	30	2.0	U			GS1	1.0	00Ra23 *
	-14489	60			-0.7	R			GS2	1.0	05Li24 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{192}\text{Bi}^m \rightarrow ^{133}\text{Cs}_{1.444}$	122143.5	9.6				2			MA8	1.0	08We02
$^{192}\text{Os} \text{ O}_2 \rightarrow ^{189}\text{Os} \text{ }^{35}\text{Cl}$	24301	6	24309.4	2.4	0.6	U			H22	2.5	70Mc03
$^{192}\text{Os} \text{ }^{35}\text{Cl} \rightarrow ^{190}\text{Os} \text{ }^{37}\text{Cl}$	5984	3	5983.4	2.4	-0.1	U			H22	2.5	70Mc03
$^{192}\text{Pb}(\alpha)^{188}\text{Hg}$	5221.0	5.	5222	5	0.1	1	88	69 ^{188}Hg	ORa		79To06 Z
$^{192}\text{Bi}(\alpha)^{188}\text{Tl}$	6376.0	5.	6377	4	0.2	3			Lvn		91Va04 *
	6377.9	5.			-0.2	3			Jya		13Ny01 *
$^{192}\text{Bi}^m(\alpha)^{188}\text{Tl}^m$	6481.4	5.	6485	3	0.7	3					67Tr06 *
	6491.6	10.			-0.7	3			Ora		74Le02 *
	6483.3	5.			0.3	3			Lvn		91Va04 *
	6494	8			-1.1	3			Jya		03Ke04 *
$^{192}\text{Po}(\alpha)^{188}\text{Pb}$	7322.8	20.	7320	3	-0.2	U					81Le23
	7319.8	7.			-0.0	3			Lvn		93Wa04
	7364.6	35.			-1.3	U			Rla		95Mo14
	7349.4	30.			-1.0	U			Rla		97Pu01
	7319.8	11.			-0.0	o			Jya		01Ke06
	7318.8	8.			0.1	3			Jya		03Ke04
	7319.8	4.			-0.0	3			Anv		03Va16
$^{192}\text{At}(\alpha)^{188}\text{Bi}$	7695.6	25.				3			Anv		06An04
$^{192}\text{At}(\alpha)^{188}\text{Bi}^m$	7629.3	15.				4			Anv		06An04 *
$^{192}\text{At}^m(\alpha)^{188}\text{Bi}$	7695.6	25.				3			Anv		06An04
$^{192}\text{At}^m(\alpha)^{188}\text{Bi}^n$	7542.4	15.				4			Anv		06An04 *
$^{192}\text{Os}(\text{p,t})^{190}\text{Os}$	-4835	5	-4835.3	2.2	-0.1	-			Min		73Oo01
	-4837	4			0.4	-			McM		75Th04
	ave.	-4836	3		0.3	1	51	51 ^{192}Os			average
$^{192}\text{Pt}(\text{p,t})^{190}\text{Pt}$	-6629	7	-6642.9	2.5	-2.0	1	13	13 ^{192}Pt	Ors		80Ka19
$^{192}\text{Os}(\text{t},\alpha)^{191}\text{Re}$	10993	10				2			McM		76Hi08
$^{192}\text{Os}(\text{d,t})^{191}\text{Os}$	-1265	15	-1301.2	2.2	-2.4	U			Tal		77Be15
$^{191}\text{Ir}(\text{n},\gamma)^{192}\text{Ir}$	6197.7	0.3	6198.12	0.11	1.4	o			ILn		87Ke.A
	6198.1	0.2			0.1	-			ILn		91Ke10
	6198.14	0.13			-0.1	-			Bdn		06Fi.A
	ave.	6198.13	0.11		-0.1	1	100	91 ^{192}Ir			average
$^{192}\text{Pt}(\text{p,d})^{191}\text{Pt}$	-6448	6	-6436.9	2.9	1.8	1	23	26 ^{191}Pt	Ors		80Ka19
$^{192}\text{Pt}(\text{p,d})^{191}\text{Pt} \rightarrow ^{194}\text{Pt} \rightarrow ^{193}\text{Pt}$	-307	3	-309.8	2.7	-0.9	1	81	74 ^{191}Pt	Ors		78Be09
$^{192}\text{Ir}(\beta^+)^{192}\text{Os}$	1468	10	1046.7	2.4	-42.1	B					60An04 *
$^{192}\text{Ir}(\beta^-)^{192}\text{Pt}$	1456.7	4.	1452.9	2.3	-1.0	-					65Jo04 *
	1453.3	3.			-0.1	-					77Ra17 *
	ave.	1454.5	2.4		-0.7	1	90	87 ^{192}Pt			average
$^{192}\text{Au}(\beta^+)^{192}\text{Pt}$	3514	20	3516	16	0.1	2					66Ny01
	3520	25			-0.1	2					74Di.A
$^{192}\text{Hg}(\beta^+)^{192}\text{Au}$	1745	30	761	22	-32.8	F					74Di.A *
$^{192}\text{Tl}(\beta^+)^{192}\text{Hg}$	6380	200	6140	40	-1.2	U					75Un.A *
* $^{192}\text{Tl} \rightarrow \text{u}$	$M - A = -25830(100)$ keV for mixture gs+m at 196(7) keV										Nub211 **
* $^{192}\text{Bi} \rightarrow \text{u}$	$M - A = -13700(110)$ keV for mixture gs+m at 140(30) keV										Nub211 **
* $^{192}\text{Bi} \rightarrow \text{u}$	$M - A = -13426(31)$ keV for mixture gs+m at 140(30) keV										Nub211 **
* $^{192}\text{Bi}(\alpha)^{188}\text{Tl}$	$E_\alpha = 6245(5), 6060(5)$ to ground state, 184.6 level										91Va04 **
* $^{192}\text{Bi}(\alpha)^{188}\text{Tl}$	$E_\alpha = 6064(5)$ to 184.6 level										Ens024 **
* $^{192}\text{Bi}^m(\alpha)^{188}\text{Tl}^m$	$E_\alpha = 6050(5)$ to (10^-) level, 302.4 above $^{188}\text{Tl}^m$										91Va04 **
* $^{192}\text{Bi}^m(\alpha)^{188}\text{Tl}^m$	$E_\alpha = 6060(10)$ to (10^-) level, 302.4 above $^{188}\text{Tl}^m$										91Va04 **
* $^{192}\text{Bi}^m(\alpha)^{188}\text{Tl}^m$	$E_\alpha = 6348(5), 6253(5), 6081(10), 6052(5)$ to $^{188}\text{Tl}^m$ and to levels 103.2, 268.8, 302.4 above $^{188}\text{Tl}^m$										91Va04 **
* $^{192}\text{Bi}^m(\alpha)^{188}\text{Tl}^m$	$E_\alpha = 6062(5)$ to level 302.4 above $^{188}\text{Tl}^m$										03Ke04 **
* $^{192}\text{At}(\alpha)^{188}\text{Bi}^m$	Also $E_\alpha = 7435(15)$ keV followed by 36 keV γ										06An04 **
* $^{192}\text{At}^m(\alpha)^{188}\text{Bi}^n$	Also $E_\alpha = 7224(15), 7195(15)$ keV followed by 165 and 188 keV γ										06An04 **
* $^{192}\text{Ir}(\beta^+)^{192}\text{Os}$	$E_{\beta^+} = 240(10)$ to 2^+ level at 205.7944 keV										Ens129 **
* $^{192}\text{Ir}(\beta^-)^{192}\text{Pt}$	$E_{\beta^-} = 672(4), 666(2)$ respectively, to 4^+ level at 784.5759, and other E_{β^-}										Ens129 **
* $^{192}\text{Hg}(\beta^+)^{192}\text{Au}$	F : most probably due to backscattering of 2.5 MeV Au positons										AHW **
* $^{192}\text{Tl}(\beta^+)^{192}\text{Hg}$	$E_{\beta^+} = 4940(200)$ to 2^+ level at 422.79 keV										Ens129 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{193}\text{Re}-u$	-32455	42				2			GS3	1.0	13Sh30
$^{193}\text{Au}-u$	-35736	96	-35862	9	-1.3	U			GS2	1.0	05Li24 *
$^{193}\text{Hg}-u$	-33288	53	-33347	17	-1.1	U			GS2	1.0	05Li24 *
$^{193}\text{Hg}-^{208}\text{Pb}_{928}$	-11673	29	-11680	17	-0.2	1	33	33 ^{193}Hg	MA6	1.0	01Sc41 *
$^{193}\text{Tl}-u$	-29690	158	-29498	7	1.2	U			GS1	1.0	00Ra23 *
	-29329	120			-1.4	U			GS2	1.0	05Li24 *
$^{193}\text{Tl}-^{133}\text{Cs}_{1.451}$	107691.2	7.2				2			MA8	1.0	14Bo26 *
$^{193}\text{Pb}-u$	-23840	110	-23864	11	-0.2	o			GS1	1.0	00Ra23 *
	-23826	43			-0.9	-			GS2	1.0	05Li24 *
	ave. -23820	40			-1.1	1	8	8 ^{193}Pb			average
$^{193}\text{Bi}-u$	-16980	110	-17053	8	-0.7	U			GS1	1.0	00Ra23
	-17025	30			-0.9	R			GS2	1.0	05Li24
$^{193}\text{Bi}-^{133}\text{Cs}_{1.451}$	120147	11	120136	8	-1.0	-			MA8	1.0	08We02
	ave. 120149	10			-1.2	1	62	62 ^{193}Bi			average
$^{193}\text{Bi}(\alpha)^{189}\text{Tl}$	6304.5	5.	6307	5	0.4	1	92	70 ^{189}Tl	Lvn		85Co06 Z
$^{193}\text{Bi}(\alpha)^{189}\text{Tl}^m$	6017.8	5.	6021	3	0.7	2					67Tr06 Z
	6024.6	10.			-0.3	2			Ora		74Le02 Z
	6023.7	5.			-0.5	2			Lvn		85Co06 Z
$^{193}\text{Bi}^m(\alpha)^{189}\text{Tl}$	6617.4	10.	6611	4	-0.6	-			Ora		74Le02
	6611.9	5.			-0.1	-			Lvn		85Co06 Z
	6618.4	14.			-0.5	U			Jya		05Uu02
	ave. 6613	5			-0.4	1	94	64 $^{193}\text{Bi}^m$			average
$^{193}\text{Po}(\alpha)^{189}\text{Pb}$	7128.1	20.	7094	4	-1.7	U					67Si09
	7087.1	20.			0.3	U			Ora		77De32
	7096.4	5.			-0.5	2			Lvn		93Wa04
	7093.3	30.			0.0	U			RIa		95Mo14
	7089.2	6.			0.7	2			Jya		96En02
	7096.4	10.			-0.3	2			Anv		02Va13
$^{193}\text{Po}^m(\alpha)^{189}\text{Pb}^m$	7143.3	10.	7154	3	1.0	2			Ora		77De32
	7148.4	20.			0.3	U					81Le23
	7152.5	5.			0.2	2			Lvn		93Wa04
	7139.2	30.			0.5	U			RIa		95Mo14
	7159.7	6.			-1.0	2			Jya		96En02
	7152.5	10.			0.1	2			Anv		02Va13
$^{193}\text{At}(\alpha)^{189}\text{Bi}^m$	7388.5	5.				7			Jya		03Ke08
$^{193}\text{At}^m(\alpha)^{189}\text{Bi}$	7556.9	20.	7580	5	1.1	o			Jya		95Le15
	7490	6			15.1	C			Jya		98En.A
	7580.4	5.				7			Jya		03Ke08 *
$^{193}\text{At}^n(\alpha)^{189}\text{Bi}$	7614.3	5.				7			Jya		03Ke08 *
$^{193}\text{Rn}(\alpha)^{189}\text{Po}$	8040.0	12.				4			Anv		06An36 *
$^{193}\text{Ir}(p,t)^{191}\text{Ir}$	-5490	15	-5488.32	0.23	0.1	U			McM		78Lo07
$^{192}\text{Os}(n,\gamma)^{193}\text{Os}$	5583.5	2.	5583.42	0.20	-0.0	U					78Be22
	5583.40	0.20			0.1	1	100	81 ^{193}Os			79Wa04
	5584.01	0.16			-3.7	C			Bdn		06Fi.A
$^{192}\text{Os}(\alpha,t)^{193}\text{Ir}$	-13923	15	-13870.9	2.4	3.5	B			McM		71Pr13
$^{193}\text{Ir}(t,\alpha)^{192}\text{Os}-^{191}\text{Ir}^{190}\text{Os}$	-661	4	-653.0	2.2	2.0	1	31	31 ^{192}Os	LAI		82La22
$^{192}\text{Ir}(n,\gamma)^{193}\text{Ir}$	7772.0	0.2	7771.99	0.20	-0.0	1	100	94 ^{193}Ir			85Co.B Z
$^{193}\text{Ir}(\gamma,n)^{192}\text{Ir}$	-7790	50	-7771.99	0.20	0.4	U			Phi		60Ge01
$^{192}\text{Pt}(n,\gamma)^{193}\text{Pt}$	6247	3	6262.5	2.3	5.2	B					68Sa13
$^{193}\text{Os}(\beta^-)^{193}\text{Ir}$	1132	5	1141.9	2.4	2.0	1	23	19 ^{193}Os			58Na15
$^{193}\text{Pt}(\epsilon)^{193}\text{Ir}$	56.6	0.3	56.63	0.30	0.1	1	100	96 ^{193}Pt			83Jo04
$^{193}\text{Au}(\beta^+)^{193}\text{Pt}$	1355	20	1075	9	-14.0	B					76Di15 *
$^{193}\text{Hg}(\beta^+)^{193}\text{Au}$	2341	30	2343	14	0.1	-					58Br88 *
	2340	20			0.1	-					76Di15 *
	ave. 2340	17			0.1	1	75	67 ^{193}Hg			average
* $^{193}\text{Au}-u$	$M-A=-33143(29)$ keV for mixture gs+m at 290.19 keV										Nub211 **
* $^{193}\text{Hg}-u$	$M-A=-30937(28)$ keV for mixture gs+m at 140.76 keV										Nub211 **
* $^{193}\text{Hg}-^{208}\text{Pb}_{928}$	Original error (18keV) increased by 20 due to isomer+ground state lines in trap										01Sc41 **
* $^{193}\text{Tl}-u$	$M-A=-27470(100)$ keV for mixture gs+m at 372(4) keV										Nub211 **
* $^{193}\text{Tl}-u$	$M-A=-27134(28)$ keV for mixture gs+m at 372(4) keV										Nub211 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
* $^{193}\text{Tl}-^{133}\text{Cs}_{1.451}$	$D_M=108091.5(5.6) \mu\text{u}$ for $^{193}\text{Tl}^m$ at 372(4) keV; $M-A=-27104.3(5.2)$ keV										Nub211 **
* $^{193}\text{Pb}-u$	$M-A=-22160(100)$ keV for mixture gs+m at 94(13) keV										Nub211 **
* $^{193}\text{Pb}-u$	$M-A=-22147(28)$ keV for mixture gs+m at 94(13) keV										Nub211 **
* $^{193}\text{At}^m(\alpha)^{189}\text{Bi}$	$E_\alpha=7423(5), 7325(5)$ to ground state and 99.6(5) level										03Ke08 **
* $^{193}\text{At}^n(\alpha)^{189}\text{Bi}$	$E_\alpha=7106(5)$ to 357.6(0.5) $^{189}\text{Bi}^n$ level										03Ke08 **
* $^{193}\text{Rn}(\alpha)^{189}\text{Po}$	$E_\alpha=7875(20), 7685(15)$ to ground state and 194 level										06An36 **
* $^{193}\text{Au}(\beta^+)^{193}\text{Pt}$	$E_{\beta^+}=153(15)$ to $3/2^-$ level at 187.81 keV, and other E_{β^+}										Ens062 **
* $^{193}\text{Hg}(\beta^+)^{193}\text{Au}$	$E_{\beta^-}=1170(30)$ from $^{193}\text{Hg}^m$ at 140.76 to $11/2^-$ level at 290.19 keV										Ens062 **
* $^{193}\text{Hg}(\beta^+)^{193}\text{Au}$	$E_{\beta^+}=1287(15)$ reinterpreted by AHW as going to ground state and $1/2^+$ level at 38.22keV										Ens062 **
$^{194}\text{Pt}-u$	-37315.72	0.67	-37316.5	0.5	-1.2	1	63	63 ^{194}Pt	MS1	1.0	16Ei01
$^{194}\text{Au}-u$	-34768	114	-34580.9	2.3	1.6	U			GS2	1.0	05Li24 *
$^{194}\text{Hg}-u$	-34527	30	-34551	3	-0.8	U			GS2	1.0	05Li24
$^{194}\text{Hg}-^{133}\text{Cs}_{1.459}$	103394.7	3.1				2			MA8	1.0	10E111
$^{194}\text{Hg}-^{208}\text{Pb}_{.933}$	-12766	19	-12767	3	-0.1	U			MA6	1.0	01Sc41
$^{194}\text{Tl}-u$	-28803	135	-28919	15	-0.9	o			GS1	1.0	00Ra23 *
	-28778	87			-1.6	U			GS2	1.0	05Li24 *
$^{194}\text{Tl}-^{133}\text{Cs}_{1.459}$	109027	15				2			MA8	1.0	14Bo26
$^{194}\text{Tl}^m-^{133}\text{Cs}_{1.459}$	109306.4	4.1	109306	4	0.0	o			MA8	1.0	14Bo26
	109306.4	4.1				2			MA8	1.0	13St25
$^{194}\text{Pb}-u$	-25980	104	-25988	19	-0.1	U			GS1	1.0	00Ra23
$^{194}\text{Bi}-u$	-17160	120	-17201	6	-0.3	o			GS1	1.0	00Ra23 *
	-17165	59			-0.6	U			GS2	1.0	05Li24 *
$^{194}\text{Bi}^m-^{133}\text{Cs}_{1.459}$	120900	54				2			MA8	1.0	08We02 *
$^{190}\text{Os}-^{194}\text{Pt}_{.979}$	-5022.45	0.68	-5021.7	0.5	1.1	1	57	52 ^{190}Os	MS1	1.0	16Ei01
$^{190}\text{Pt}-^{194}\text{Pt}_{.979}$	-3516.95	0.68	-3517.3	0.5	-0.5	1	58	53 ^{190}Pt	MS1	1.0	16Ei01
$^{194}\text{Pt}-^{197}\text{Au}_{.985}$	-4396.4	3.2	-4388.1	0.6	2.6	U			CP1	1.0	05Sh52
$^{194}\text{Au}-^{197}\text{Au}_{.985}$	-1652.5	2.2				2			MA8	1.0	10E111
$^{194}\text{Pb}(\alpha)^{190}\text{Hg}$	4737.9	20.	4738	17	-0.0	1	67	40 ^{194}Pb	ORa		87El09
$^{194}\text{Bi}(\alpha)^{190}\text{Tl}$	5918.3	5.				3			Lvn		91Va04 *
$^{194}\text{Bi}^n(\alpha)^{190}\text{Tl}^m$	6015.7	5.	6012	4	-0.8	1	78	42 $^{194}\text{Bi}^n$	Lvn		91Va04 *
$^{194}\text{Po}(\alpha)^{190}\text{Pb}$	6991.5	10.	6987	3	-0.4	3					67Si09 Z
	6990.9	7.			-0.5	3					67Tr06 Z
	6984.4	5.			0.5	3			Ora		77De32 Z
	6990.0	5.			-0.6	o			Lvn		85Va03 Z
	6986.3	6.			0.1	3			Lvn		93Wa04
	6993.4	4.			-1.6	o			Jya		96En02
	6987.3	14.			-0.0	3			Jya		05Uu02
$^{194}\text{At}(\alpha)^{190}\text{Bi}$	7412.5	20.	7454	11	2.1	o			Jya		95Le15 *
	7462.5	15.			-0.5	3			Anv		09An11 *
	7446.5	15.			0.5	3			Jya		13Ny01 *
$^{194}\text{At}^m(\alpha)^{190}\text{Bi}^m$	7362.1	20.	7309	5	-2.6	o					80Ya.A
	7351.9	20.			-2.1	U					84Ya.A
	7341.7	20.			-1.6	o			Jya		95Le15
	7329.4	15.3			-1.3	5			Anv		09An11
	7306.9	5.1			0.4	5			Jya		13Ny01
$^{194}\text{Rn}(\alpha)^{190}\text{Po}$	7862.5	10.				4			Anv		06An36
$^{193}\text{Ir}(n,\gamma)^{194}\text{Ir}$	6067.0	0.4	6066.79	0.11	-0.5	2					82Ra.A
	6066.9	0.2			-0.6	2					98Ba85
	6066.71	0.14			0.6	2			Bdn		06Fi.A
$^{194}\text{Pt}(t,\alpha)^{193}\text{Ir}$	12286	20	12301.1	1.3	0.8	U			Tal		78Ya07
$^{194}\text{Pt}(d,t)^{193}\text{Pt}$	-2126	20	-2094.5	1.3	1.6	U			Pit		64Co11
$^{194}\text{Pt}(p,d)^{193}\text{Pt}-^{196}\text{Pt}()^{195}\text{Pt}$	-445	3	-429.8	1.3	5.1	B			Ors		78Be09
$^{194}\text{Os}(\beta^-)^{194}\text{Ir}$	96.6	2.				3					64Wi07 *
$^{194}\text{Ir}(\beta^-)^{194}\text{Pt}$	2254	4	2228.3	1.3	-6.4	B					76Ra33

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{194}\text{Ir}^n(\beta^-)^{194}\text{Pt}$	2600	70				2					68Su02 *
$^{194}\text{Pt}(d,\alpha)^{192}\text{Ir}-^{192}\text{Os}(\gamma)^{190}\text{Re}$	375	4	375	4	-0.0	1	100	100 ^{190}Re			20Gr08
$^{194}\text{Au}(\beta^+)^{194}\text{Pt}$	2465	20	2548.2	2.1	4.2	B					56Th11
	2509	15			2.6	U					60Ba17
	2485	30			2.1	U					70Ag03 *
$^{194}\text{Hg}(\epsilon)^{194}\text{Au}$	40	20	28	4	-0.6	U					81Ho18
* $^{194}\text{Au}-u$	$M-A=-32192(29)$ keV for mixture gs+m+n at 107.4 and 475.8 keV										Nub211 **
* $^{194}\text{Tl}-u$	$M-A=-26700(100)$ keV for mixture gs+m at 260(14) keV										Nub211 **
* $^{194}\text{Tl}-u$	$M-A=-26677(28)$ keV for mixture gs+m at 260(14) keV										Nub211 **
* $^{194}\text{Bi}-u$	$M-A=-15870(100)$ keV for mixture gs+m+n at 150(50) and 163(4) keV										Nub211 **
* $^{194}\text{Bi}-u$	$M-A=-15885(28)$ keV for mixture gs+m+n at 150(50) and 163(4) keV										Nub211 **
* $^{194}\text{Bi}^m-^{133}\text{Cs}_{1.459}$	Original error $16\mu\text{u}$ increased to include possible 3^+ and 10^- contam.										08We02 **
* $^{194}\text{Bi}(\alpha)^{190}\text{Tl}$	$E_\alpha=5799(5)$, $5645(5)$ to ground state, 151.3 level										91Va04 **
* $^{194}\text{Bi}^m(\alpha)^{190}\text{Tl}^m$	$E_\alpha=5892(5)$, $5781(5)$ to levels 0, 112.2 above $^{190}\text{Tl}^m$										91Va04 **
* $^{194}\text{At}(\alpha)^{190}\text{Bi}$	$E_\alpha=7140(20)$ to 121(15)										09An11 **
* $^{194}\text{At}(\alpha)^{190}\text{Bi}$	$E_\alpha=7190(15)$ to 121(15); further E_α : 7310(15), 7266(15), 7145(15) keV										09An11 **
* $^{194}\text{At}(\alpha)^{190}\text{Bi}$	$E_\alpha=7174(8)$ to 121(15)										13Ny01 **
* $^{194}\text{Os}(\beta^-)^{194}\text{Ir}$	$E_{\beta^-}=54.5(2.0)$ to 0^- level at 43.119 keV, and other E_{β^-}										Ens066 **
* $^{194}\text{Ir}^n(\beta^-)^{194}\text{Pt}$	$E_{\beta^-}<250$ to 10^+ level at 2438.41 keV										Ens066 **
* $^{194}\text{Au}(\beta^+)^{194}\text{Pt}$	$E_{\beta^+}=1230(30)$ to 2^+ level at 328.464 keV, and other E^+										Ens066 **
$^{195}\text{Os}-u$	-31682	60				2					GS3 1.0 13Sh30
$^{195}\text{Hg}-u$	-33283	62	-33294	25	-0.2	U					GS2 1.0 05Li24 *
$^{195}\text{Hg}-^{208}\text{Pb}_{.938}$	-11381	28	-11394	25	-0.5	1	79	79 ^{195}Hg	MA6	1.0	01Sc41 *
$^{195}\text{Tl}-u$	-30320	200	-30226	12	0.5	U					GS1 1.0 00Ra23 *
	-30209	40			-0.4	-					GS2 1.0 05Li24 *
	-30264	33			1.2	-					GS2 1.0 05Li24 *
	ave. -30242	25			0.6	1	22	22 ^{195}Tl			average
$^{195}\text{Tl}-^{133}\text{Cs}_{1.466}$	108375	27	108381	12	0.2	-			MA8	1.0	14Bo26
	108472	79			-1.1	-			MA8	1.0	14Bo26 *
	ave. 108385	26			-0.1	1	22	22 ^{195}Tl			average
$^{195}\text{Pb}-u$	-25423	150	-25484	5	-0.4	o					GS1 1.0 00Ra23 *
	-25461	70			-0.3	-					GS2 1.0 05Li24 *
	ave. -25457	25			-1.1	1	5	5 ^{195}Pb			average
$^{195}\text{Bi}-u$	-19320	100	-19351	6	-0.3	U					GS1 1.0 00Ra23
	-19537	128			1.5	U					GS2 1.0 05Li24 *
$^{195}\text{Bi}-^{133}\text{Cs}_{1.466}$	119258.2	6.0	119256	6	-0.3	1	89	89 ^{195}Bi	MA8	1.0	08We02
$^{195}\text{Po}-^{133}\text{Cs}_{1.466}$	126671.8	6.6	126673	6	0.2	1	97	97 ^{195}Po	MA8	1.0	17Al34
$^{195}\text{Po}^m-^{133}\text{Cs}_{1.466}$	126832.4	7.8	126833	8	0.0	1	94	94 $^{195}\text{Po}^m$	MA8	1.0	17Al34
$^{195}\text{Pt}-^{197}\text{Au}_{.990}$	-2119.9	3.2	-2110.1	0.6	3.1	B			CP1	1.0	05Sh52
$^{195}\text{Bi}(\alpha)^{191}\text{Tl}$	5832.5	5.				2			Lvn		85Co06 Z
$^{195}\text{Bi}(\alpha)^{191}\text{Tl}^m$	5542.9	10.	5535	5	-0.8	2			Ora		74Le02 Z
	5533.3	5.			0.4	2			Lvn		85Co06 Z
$^{195}\text{Bi}^m(\alpha)^{191}\text{Tl}$	6228.1	5.	6232	3	0.7	3					67Tr06 Z
	6238.4	10.			-0.6	3			Ora		74Le02 Z
	6233.7	5.			-0.4	3			Lvn		85Co06 Z
$^{195}\text{Po}(\alpha)^{191}\text{Pb}$	6763.1	8.	6749.7	2.8	-1.6	-					67Si09 Z
	6747.4	5.			0.5	-					67Tr06 Z
	6744.6	5.			1.0	-					93Wa04
	6752.8	14.			-0.2	o			Jya		96Le09
	6744.6	10.			0.5	-			Anv		02Va13
	6755.9	6.			-1.0	-			Jya		05Uu02
	ave. 6749.9	2.8			-0.1	1	99	96 ^{191}Pb			average
$^{195}\text{Po}^m(\alpha)^{191}\text{Pb}^m$	6850.8	10.	6840.6	2.8	-1.0	-					67Si09
	6839.4	5.			0.2	-					67Tr06 Z
	6839.6	5.			0.2	-			Lvn		93Wa04
	6852.8	10.			-1.2	o			Jya		96Le09
	6839.6	10.			0.1	-			Anv		02Va13
	6840.6	6.			0.0	-			Jya		05Uu02

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{195}\text{Po}^m(\alpha)^{191}\text{Pb}^m$	ave.	6840.6	2.9	6840.6	2.8	-0.0	1	99	93 $^{191}\text{Pb}^m$		average
$^{195}\text{At}(\alpha)^{191}\text{Bi}^m$		7095.8	20.	7102	4	0.3	U		Jya		95Le15
		7105	20			-0.2	U		RIa		99Ta20
		7098.9	3.			1.0	o		Jya		03Ke04 *
		7113.2	10.			-1.1	o		Jya		13Uu01
		7101.9	4.				3		Jya		13Ny01
$^{195}\text{At}^m(\alpha)^{191}\text{Bi}$		7340.9	30.	7373	4	1.1	o		Jya		83Le.A *
		7371.5	30.			0.1	o		Jya		95Le.A *
		7403	30			-1.0	U		RIa		99Ta20 *
		7372.5	4.0			0.2	o		Jya		03Ke04 *
		7373.5	4.0				2		Jya		13Ny01 *
$^{195}\text{Rn}(\alpha)^{191}\text{Po}$		7694.1	11.				2		Jya		01Ke06
$^{195}\text{Rn}^m(\alpha)^{191}\text{Po}^m$		7713.5	11.				3		Jya		01Ke06
$^{194}\text{Ir}(n,\gamma)^{195}\text{Ir}$		7231.92	0.11	7231.86	0.06	-0.5	o		ILn		87Ci.A
		7231.86	0.06				3		ILn		87Co08 Z
$^{194}\text{Pt}(n,\gamma)^{195}\text{Pt}$		6105.06	0.12	6105.09	0.12	0.3	1	99	72 ^{195}Pt		81Ho.B Z
		6109.17	0.13			-31.3	F		Bdn		06Fi.A
$^{195}\text{Pt}(\gamma,n)^{194}\text{Pt}$		-6205	44	-6105.09	0.12	2.3	U		Phi		60Ge01
$^{194}\text{Pt}(d,p)^{195}\text{Pt}$		3908	20	3880.53	0.12	-1.4	U		Pit		64Co11
$^{195}\text{Pt}(d,t)^{194}\text{Pt}$		140	20	152.14	0.12	0.6	U		Pit		64Co11
$^{195}\text{Os}(\beta^-)^{195}\text{Ir}$		2000	500	2180	60	0.4	U				57Ba08
$^{195}\text{Ir}(\beta^-)^{195}\text{Pt}$		1116	20	1101.6	1.3	-0.7	U				73Ja10 *
$^{195}\text{Au}(\epsilon)^{195}\text{Pt}$		226.8	1.0	226.8	1.0	0.0	1	100	100 ^{195}Au		Averag *
$^{195}\text{Hg}(\beta^+)^{195}\text{Au}$		1510	50	1554	23	0.9	1	21	21 ^{195}Hg		71Fr03 *
$^{195}\text{Tl}(\beta^+)^{195}\text{Hg}$		3000	300	2858	26	-0.5	U				78Go15 *
$^{195}\text{Pb}^m(\text{IT})^{195}\text{Pb}$		202.9	0.7	202.9	0.7	-0.0	1	100	95 ^{195}Pb	Oak	91Gr12
$^{195}\text{Bi}(\beta^+)^{195}\text{Pb}$		4850	550	5712	7	1.6	U		Oak		91Gr12
$^{195}\text{Hg}-u$	$M-A=-30914(28)$ keV for mixture gs+m at 176.07 keV										Nub211 **
$^{195}\text{Hg}-^{208}\text{Pb}_{938}$	Corrected 40(20) keV for isomeric mixture $R=0.3(0.2)$ $E=176.07$ keV										Nub211 **
$^{195}\text{Tl}-u$	$M-A=-28000(100)$ keV for mixture gs+m at 482.63 keV										Nub211 **
$^{195}\text{Tl}-u$	$M-A=-27708(31)$ keV for $^{195}\text{Tl}^m$ at 482.63 keV										Nub211 **
$^{195}\text{Tl}-^{133}\text{Cs}_{1.466}$	$D_M=108990(79)$ μu for $^{195}\text{Tl}^m$ at 482.63 keV; $M-A=-27589(73)$ keV										Nub211 **
$^{195}\text{Pb}-u$	$M-A=-23580(100)$ keV for mixture gs+m at 202.9 keV										Nub211 **
$^{195}\text{Pb}-u$	$M-A=-23615(28)$ keV for mixture gs+m at 202.9 keV										Nub211 **
$^{195}\text{Bi}-u$	$M-A=-17999(28)$ keV for mixture gs+m at 399(6) keV										Nub211 **
$^{195}\text{At}(\alpha)^{191}\text{Bi}^m$	Correlated with $E_\alpha=6313$ of $^{191}\text{Bi}^m$										03Ke04 **
$^{195}\text{At}^m(\alpha)^{191}\text{Bi}$	$E_\alpha=7190(30)$ to 148.7(0.5) level										03Ke04 **
*	correlated with α of 12 s ^{191}Bi ground state										95Le15 **
$^{195}\text{At}^m(\alpha)^{191}\text{Bi}$	$E_\alpha=7105(30)$ to 148.7(0.5) level										03Ke04 **
$^{195}\text{At}^m(\alpha)^{191}\text{Bi}$	$E_\alpha=7221(4)$ and 7075(4) to 148.7(0.5) level										03Ke04 **
$^{195}\text{At}^m(\alpha)^{191}\text{Bi}$	$E_\alpha=7222(4)$ and 7076(5) to 148.7(0.5) level										13Ny01 **
$^{195}\text{Ir}(\beta^-)^{195}\text{Pt}$	$E_{\beta^-}=980(30)$ to $3/2^-$ level at 98.880 keV and $5/2^-$ level at 129.772 keV,										Ens148 **
*	and $E_{\beta^-}=410(20)$ from $^{195}\text{Ir}^m$ at 100(5) to $9/2^-$ at 814.50, and other E_{β^-}										Ens148 **
$^{195}\text{Au}(\epsilon)^{195}\text{Pt}$	Average $pK=0.179(0.006)$ to $5/2^-$ level at 129.772 from the following references:										Ens148 **
*	$pK=0.195(0.015)$ to 129.78 level										65De20 **
*	$pK=0.166(0.020)$ to 129.78 level										68Ja11 **
*	$pK=0.160(0.017)$ to 129.78 level										73Go05 **
*	$pK=0.183(0.009)$ to 129.78 level										80Sa11 **
*	$pK=0.176(0.012)$ to 129.78 level										82Be.A **
$^{195}\text{Hg}(\beta^+)^{195}\text{Au}$	Assuming 511 γ is annihilation of β^+ to ground state and $1/2^+$ level at 61.434										Ens148 **
$^{195}\text{Tl}(\beta^+)^{195}\text{Hg}$	$K/\beta^+=6(1)$ to ground state and $3/2^-$ level at 37.083 keV										Ens148 **
$^{196}\text{Hg}-u$	-34161.8	16.3	-34167	3	-0.3	U			MA8	1.0	20Ku19 *
$^{196}\text{Hg}-^{208}\text{Pb}_{942}$	-12178	20	-12173	3	0.3	U			MA6	1.0	01Sc41
$^{196}\text{Tl}-u$	-29188	126	-29519	13	-2.6	U			GS2	1.0	05Li24 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{196}\text{Tl}-^{133}\text{Cs}_{1.474}$	109845	13				2			MA8	1.0	08We02 *
$^{196}\text{Pb}-^{208}\text{Pb}_{942}$	-5228	22	-5219	8	0.4	-			MA6	1.0	01Sc41
	ave.	-5231			0.7	1	21	21 ^{196}Pb			average
$^{196}\text{Pb}-\text{u}$	-27200	104	-27212	8	-0.1	U			GS1	1.0	00Ra23
	-27232	30			0.7	R			GS2	1.0	05Li24
$^{196}\text{Bi}-\text{u}$	-19309	137	-19333	26	-0.2	o			GS1	1.0	00Ra23 *
	-19325	30			-0.3	2			GS2	1.0	05Li24
	-19361	54			0.5	2			MA8	1.0	08We02 *
$^{196}\text{Po}-^{133}\text{Cs}_{1.474}$	124905.4	6.3	124905	6	-0.1	1	84	84 ^{196}Po	MA8	1.0	17Al34
$^{196}\text{Pt}-^{197}\text{Au}_{995}$	-1781.1	3.0	-1782.6	0.6	-0.5	U			CP1	1.0	05Sh52
$^{196}\text{Bi}(\alpha)^{192}\text{Tl}^p$	5260.6	5.				3			Lvn		91Va04
$^{196}\text{Po}(\alpha)^{192}\text{Pb}$	6662.3	8.2	6658.2	2.4	-0.5	-					67Si09 Z
	6653.7	5.			0.9	-					67Tr06 Z
	6658.4	8.			-0.0	-					71Ho01 Z
	6656.7	5.			0.3	o			Lvn		85Va03 Z
	6656.7	5.			0.3	-			Lvn		93Wa04
	6653.1	18.			0.3	o			Ara		95Le04
	6657.1	10.			0.1	o			Jya		96Le09
	6654.0	5.0			0.8	-			Ara		96Ta18 *
	6669.4	6.			-1.8	-			Jya		05Uu02
	6658.2	25.			-0.0	U			Anv		10He25
$^{196}\text{At}(\alpha)^{192}\text{Bi}$	ave.	6658.1			0.1	1	97	81 ^{192}Pb			average
	7202.3	7.	7196.4	2.7	-0.8	4					67Tr06
	7187.0	25.			0.4	U			Jya		95Le15
	7200.2	30.			-0.1	U			RIa		95Mo14
	7191.0	7.			0.8	o			Jya		96En01
	7195.1	5.			0.3	o			Jya		00Sm06
	7202.3	12.			-0.5	o			Anv		05De01
	7195.1	12.			0.1	o			Jya		13Uu01 *
	7194.1	5.			0.5	4			Jya		13Ny01
	7192.1	5.			0.9	4			Anv		14Ka23
	7200.2	5.			-0.7	4			Anv		16Tr07
$^{196}\text{At}^m(\alpha)^{192}\text{Bi}^m$	7023.6	15.				3			Jya		96En01 *
$^{196}\text{Rn}(\alpha)^{192}\text{Po}$	7583.1	35.	7617	9	0.9	o			RIa		95Mo14
	7648.4	30.			-1.1	U			RIa		97Pu01
	7616.7	9.				4			Jya		01Ke06
$^{196}\text{Pt}(\text{t},\alpha)^{195}\text{Ir}$	11565	20	11572.7	1.3	0.4	U			Tal		78Ya07
	11545	20			1.4	U			LAl		81Fl.A
$^{195}\text{Pt}(\text{n},\gamma)^{196}\text{Pt}$	7921.96	0.20	7921.98	0.13	0.1	-			ILn		81Ho.B Z
	7921.92	0.17			0.3	-			Bdn		06Fi.A
$^{196}\text{Pt}(\gamma,\text{n})^{195}\text{Pt}$	-8290	140	-7921.98	0.13	2.6	U			Phi		60Ge01
$^{195}\text{Pt}(\text{d},\text{p})^{196}\text{Pt}$	5712	25	5697.41	0.13	-0.6	U			Pit		64Co11
$^{196}\text{Pt}(\text{d},\text{t})^{195}\text{Pt}$	-1686	20	-1664.75	0.13	1.1	U			Pit		64Co11
$^{195}\text{Pt}(\text{n},\gamma)^{196}\text{Pt}$	ave.	7921.94	0.13	7921.98	0.13	0.3	1	99	71 ^{196}Pt		average
$^{196}\text{Os}(\beta^-)^{196}\text{Ir}$	900	40	1160	60	6.5	B					77Ha32 *
$^{196}\text{Ir}(\beta^-)^{196}\text{Pt}$	3150	60	3210	40	1.0	2					66Vo05 *
	3250	50			-0.8	2					67Mo10
$^{196}\text{Ir}^m(\beta^-)^{196}\text{Pt}$	3418	20				2					65Bi04 *
	3630	100	3418	20	-2.1	U					68Ja06 *
$^{196}\text{Au}(\beta^+)^{196}\text{Pt}$	1498	7	1505.8	3.0	1.1	1	18	18 ^{196}Au			63Ik01 *
$^{196}\text{Au}(\epsilon)^{196}\text{Pt}$	1490	10			1.6	U					62Wa16 *
$^{196}\text{Au}(\beta^-)^{196}\text{Hg}$	685	4	687	3	0.6	1	61	31 ^{196}Au			62Li03 *
* $^{196}\text{Hg}-\text{u}$	from frequency ratio $^{196}\text{Hg}^+ / ^{39}\text{K}^+ = 2.514744076(209)$										20Ku19 **
* $^{196}\text{Tl}-\text{u}$	$M-A=-26991(28)$ keV for mixture gs+m at 394.2 keV										Nub211 **
* $^{196}\text{Tl}-^{133}\text{Cs}_{1.474}$	$Q=110268(13)$ μu $M-A=-27103(12)$ keV for $^{196}\text{Tl}^m$ at 394.2 keV										Nub211 **
* $^{196}\text{Bi}-\text{u}$	$M-A=-17850(100)$ keV for mixture gs+n at 272(3) keV										Nub211 **
* $^{196}\text{Bi}-\text{u}$	$Q=120182(15)$ μu for $^{196}\text{Bi}^m-^{133}\text{Cs}_{1.474}$, $M-A(^{196}\text{Bi}^m)=-17868(14)$ keV at 167(3) keV; error increased to include possible 3^+ and 10^- contam.										08We02 **
* $^{196}\text{Po}(\alpha)^{192}\text{Pb}$	Including systematic uncertainty 5 keV										96Ta18 **
* $^{196}\text{At}(\alpha)^{192}\text{Bi}$	Same group, but much less events than in 00Sm06										WgM151**
* $^{196}\text{At}^m(\alpha)^{192}\text{Bi}^m$	Correlated with $E_\alpha=7550$ of $^{200}\text{Fr}(\alpha)$										96En01 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
* $^{196}\text{Os}(\beta^-)^{196}\text{Ir}$	$E_{\beta^-}=435(20)$ to $(0,1)^+$ levels at 407.88, 522.37 keV										Ens076	**
* $^{196}\text{Ir}(\beta^-)^{196}\text{Pt}$	Original value 3170(60) recalibrated using ^{62}Cu										AHW	**
* $^{196}\text{Ir}^m(\beta^-)^{196}\text{Pt}$	$E_{\beta^-}=950(20)$ to $(10^-,11^-)$ level at 2468.0 keV										Ens076	**
* $^{196}\text{Ir}^m(\beta^-)^{196}\text{Pt}$	$E_{\beta^-}=1160(100)$ to $(10^-,11^-)$ level at 2468.0 keV										Ens076	**
* $^{196}\text{Au}(\beta^+)^{196}\text{Pt}$	$KL/\beta^+=2.0(0.4)\times 10^{+6}$ to 2^+ level at 355.68 keV, recalculated										Ens076	**
* $^{196}\text{Au}(\varepsilon)^{196}\text{Pt}$	$pL=0.64(0.06)$ to 3^- level at 1447.043 keV										Ens076	**
* $^{196}\text{Au}(\beta^-)^{196}\text{Hg}$	$E_{\beta^-}=259(4)$ to 2^+ level at 425.98 keV										Ens076	**
$^{197}\text{Hg}-u$	-32766	30	-32786	3	-0.7	U			GS2	1.0	05Li24	
	-32765	30			-0.7	U			GS2	1.0	05Li24	
$^{197}\text{Hg}-^{208}\text{Pb}_{.947}$	-10664	30	-10676	4	-0.4	U			MA6	1.0	01Sc41	
$^{197}\text{Tl}-u$	-30450	30	-30440	15	0.3	R			GS2	1.0	05Li24	
$^{197}\text{Pb}-u$	-26520	110	-26565	5	-0.4	U			GS1	1.0	00Ra23	
	-26609	30			1.5	U			GS2	1.0	05Li24	
	-26543	30			-0.7	U			GS2	1.0	05Li24	
$^{197}\text{Pb}^m-^{133}\text{Cs}_{1.481}$	113799.6	6.0	113803	5	0.6	1	74	74 $^{197}\text{Pb}^m$	MA8	1.0	08We02	
$^{197}\text{Bi}-^{208}\text{Pb}_{.947}$	982	22	975	9	-0.3	R			MA6	1.0	01Sc41	
$^{197}\text{Bi}-u$	-21381	192	-21135	9	1.3	U			GS1	1.0	00Ra23	
	-21187	31			1.7	U			GS2	1.0	05Li24	
$^{197}\text{Bi}-^{133}\text{Cs}_{1.481}$	118870	26	118891	9	0.8	R			MA8	1.0	08We02	
$^{197}\text{Po}-u$	-14420	130	-14378	11	0.3	o			GS1	1.0	00Ra23	
	-14288	71			-1.3	R			GS2	1.0	05Li24	
$^{197}\text{Po}-^{133}\text{Cs}_{1.481}$	125644	11	125648	11	0.3	1	93	93 ^{197}Po	MA8	1.0	17Al34	
$^{197}\text{Po}^m-^{133}\text{Cs}_{1.481}$	125858.0	7.0				2			MA8	1.0	17Al34	
$^{197}\text{At}-^{133}\text{Cs}_{1.481}$	133186	20	133203	9	0.8	1	18	18 ^{197}At	MA8	1.0	17Ma29	
$^{197}\text{At}^m-^{133}\text{Cs}_{1.481}$	133234	15	133251	10	1.1	1	42	42 $^{197}\text{At}^m$	MA8	1.0	17Ma29	
$^{197}\text{Au}-\text{C}_{16}$	-33432.5	7.3	-33429.9	0.6	0.2	o			TG1	1.5	09Ke.A	
	-33432.9	5.4			0.4	U			TG1	1.5	10Ke09	
$^{197}\text{Au}(\alpha,^8\text{He})^{193}\text{Au}$	-26919	9	-26920	9	-0.1	1	93	93 ^{193}Au			89Ka04	
$^{197}\text{Bi}^m(\alpha)^{193}\text{Tl}$	5890.8	10.	5898	5	0.7	o			Ora		72Ga27	
	5889.7	10.			0.8	3			Ora		74Le02	
	5899.6	5.			-0.4	3			Lvn		85Co06	
$^{197}\text{Po}(\alpha)^{193}\text{Pb}$	6420.7	10.	6411	3	-0.9	-					67Si09	
	6410.1	5.			0.2	-					67Tr06	
	6409.4	9.			0.2	-					71Ho01	
	6411.4	5.0			-0.0	-			Ara		96Ta18	
	ave. 6412	3			-0.1	1	99	92 ^{193}Pb			average	
$^{197}\text{Po}^m(\alpha)^{193}\text{Pb}^m$	6510.1	5.	6514.7	2.1	0.9	3					67Tr06	
	6511.4	9.			0.4	U					71Ho01	
	6518.0	3.			-1.1	3			Bka		82Bo04	
	6512.4	5.0			0.4	3			Ara		96Ta18	
	6517.6	10.			-0.3	o			Anv		02Va13	
	6516.6	30.			-0.1	U			Anv		10He25	
	6513.0	4.6			0.4	3			Tex		12Fo09	
$^{197}\text{At}(\alpha)^{193}\text{Bi}$	7103.0	5.	7104	3	0.3	-					67Tr06	
	7100.5	5.			0.8	o			Jya		96En01	
	7104.5	5.			-0.0	o			Jya		99Sm07	
	7103.5	6.			0.1	-			Jya		05Uu02	
	7107.5	5.			-0.6	-			Anv		14Ka23	
	ave. 7105	3			-0.1	1	98	82 ^{197}At			average	
$^{197}\text{At}^m(\alpha)^{193}\text{Bi}^m$	6846.2	10.	6844	4	-0.2	o			Lvn		86Co12	
	6846.2	5.			-0.4	-			Jya		99Sm07	
	6845.2	9.			-0.1	-			Jya		05Uu02	
	6837.0	16.			0.5	U			Anv		14Ka23	
	ave. 6846	4			-0.4	1	94	58 $^{197}\text{At}^m$			average	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{197}\text{Rn}(\alpha)^{193}\text{Po}$	7410.8	20.	7411	7	-0.0	o			RIa		95No.A
	7411.8	30.			-0.0	U			RIa		95Mo14
	7410.8	7.				3			Jya		96En02
$^{197}\text{Rn}^m(\alpha)^{193}\text{Po}^m$	7523.1	30.	7509	6	-0.5	U			RIa		95Mo14
	7508.7	7.			0.1	3			Jya		96En02
	7510.7	14.			-0.1	3			Jya		05Uu02
$^{197}\text{Fr}(\alpha)^{193}\text{At}^m$	7888.4	15.				8			Anv		13Ka16
$^{196}\text{Pt}(n,\gamma)^{197}\text{Pt}$	5846.4	0.4	5846.55	0.26	0.4	-					78Ya07 Z
	5846.0	0.9			0.6	-			ILn		81Ho.B Z
	5846.6	0.5			-0.1	-			BNn		83Ca04 Z
	5846.0	0.7			0.8	-			Bdn		06Fi.A
$^{196}\text{Pt}(d,p)^{197}\text{Pt}$	3627	20	3621.99	0.26	-0.3	U			Pit		64Co11
	3606	20			0.8	U			Tal		78Ya07
$^{196}\text{Pt}(n,\gamma)^{197}\text{Pt}$	ave.	5846.36	0.27	5846.55	0.26	0.7	1	94	65 ^{197}Pt		average
$^{197}\text{Au}(\gamma,n)^{196}\text{Au}$	-8057	22	-8072.4	2.9	-0.7	U			Phi		60Ge01
	-8080	5			1.5	-			McM		79Ba06
	-8072	7			-0.1	-					79Be.A
$^{197}\text{Au}(d,t)^{196}\text{Au}$	-1820	30	-1815.1	2.9	0.2	U			Pit		64Co11
$^{197}\text{Au}(\gamma,n)^{196}\text{Au}$	ave.	-8077	4	-8072.4	2.9	1.2	1	52	52 ^{196}Au		average
$^{196}\text{Hg}(n,\gamma)^{197}\text{Hg}$	6785.3	1.5	6785.6	1.5	0.2	1	97	84 ^{197}Hg	BNn		78Zg.A Z
$^{197}\text{Ir}(\beta^-)^{197}\text{Pt}$	2000	200	2156	20	0.8	U					61Ho10
$^{197}\text{Pt}(\beta^-)^{197}\text{Au}$	719.0	0.6	720.0	0.5	1.6	1	70	36 ^{197}Au			71Pr03
$^{197}\text{Hg}(\epsilon)^{197}\text{Au}$	415	20	600	3	9.2	B					65De20 *
	610	100			-0.1	U					92Da14 *
$^{197}\text{Tl}(\beta+)^{197}\text{Hg}$	2220	100	2186	14	-0.3	U					61Ju05
$^{197}\text{Pb}^m(\text{IT})^{197}\text{Pb}$	319.31	0.11	319.31	0.11	-0.0	1	100	74 ^{197}Pb			Ens053
* $^{197}\text{Hg}-u$	$M-A=-30221(28)$ keV for $^{197}\text{Hg}^m$ at 298.93 keV										Nub211 **
* $^{197}\text{Pb}-u$	$M-A=-24405(28)$ keV for $^{197}\text{Pb}^m$ at 319.31 keV										Nub211 **
* $^{197}\text{Bi}-u$	$M-A=-19650(90)$ keV for mixture gs+m at 533(12) keV										Nub211 **
* $^{197}\text{Bi}-^{133}\text{Cs}_{1.481}$	$Q=118887(12)$ μu $M-A=-19690(11)$ keV corrected by $-16(22)$ keV due to possible contamination from $^{197}\text{Bi}^m$										08We02 **
* $^{197}\text{Po}-u$	$M-A=-13330(110)$ keV for mixture gs+m at 198(12) keV										08We02 **
* $^{197}\text{Po}-u$	$M-A=-13210(32)$ keV for mixture gs+m at 198(12) keV										Nub211 **
* $^{197}\text{Po}(\alpha)^{193}\text{Pb}$	Also $E_\alpha=6283(5)$ keV from uncorrelated decays										96Ta18 **
* $^{197}\text{Po}^m(\alpha)^{193}\text{Pb}^m$	Also $E_\alpha=6381(5)$ keV from uncorrelated decays										96Ta18 **
* $^{197}\text{Hg}(\epsilon)^{197}\text{Au}$	$pK=0.54(0.06)$ to $3/2^+$ level at 268.788 keV										Ens053 **
* $^{197}\text{Hg}(\epsilon)^{197}\text{Au}$	$pK=0.746(0.033)$ to 268.75 level $\rightarrow Q=574(+139-62)$ keV										Ens053 **
$^{198}\text{Hg}-^{161}\text{Dy } ^{37}\text{Cl}$	74130	60	73927.2	0.9	-0.8	U			R04	4.0	64De15
$^{198}\text{Hg}-^{163}\text{Dy } ^{35}\text{Cl}$	68979	37	69179.3	0.9	1.4	U			R04	4.0	64De15
$^{198}\text{Hg}-u$	-33231.6	0.6	-33230.8	0.5	1.3	1	67	67 ^{198}Hg	ST2	1.0	02Bf02
$^{198}\text{Tl}-^{133}\text{Cs}_{1.489}$	111228.7	8.1				2			MA8	1.0	14Bo26 *
$^{198}\text{Pb}-^{208}\text{Pb}_{.952}$	-5748	23	-5757	9	-0.4	-			MA6	1.0	01Sc41
	ave.	-5739	18		-1.0	1	26	26 ^{198}Pb			average
$^{198}\text{Pb}-u$	-27990	104	-27985	9	0.1	U			GS1	1.0	00Ra23
	-27951	30			-1.1	R			GS2	1.0	05Li24
$^{198}\text{Bi}-u$	-21070	160	-20799	30	1.7	o			GS1	1.0	00Ra23 *
	-20794	30			-0.2	1	97	97 ^{198}Bi	GS2	1.0	05Li24
$^{198}\text{Bi}^n-u$	-20222	30				2			GS2	1.0	05Li24
$^{198}\text{Po}-^{208}\text{Pb}_{.952}$	5616	24	5616	19	0.0	1	61	61 ^{198}Po	MA6	1.0	01Sc41
$^{198}\text{Po}-u$	-16600	104	-16611	19	-0.1	U			GS1	1.0	00Ra23
$^{198}\text{At}-^{133}\text{Cs}_{1.489}$	133570.0	7.3	133580	5	1.4	o			MA8	1.0	13Ma.A
	133573.7	6.3			1.0	1	70	70 ^{198}At	MA8	1.0	13St25
$^{198}\text{At}^m-^{133}\text{Cs}_{1.489}$	133898	39	133866	6	-0.8	U			MA8	1.0	13St25

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{198}\text{Hg}-^{35}\text{Cl}-^{196}\text{Hg}-^{37}\text{Cl}$	3885.91	1.66	3886	3	-0.0	1	57	57 ^{196}Hg	H33	2.5	80Ko25
$^{198}\text{Pt}-^{197}\text{Au}_{1.005}$	1494.7	3.0	1493.8	2.2	-0.3	1	54	53 ^{198}Pt	CP1	1.0	05Sh52
$^{198}\text{Po}(\alpha)^{194}\text{Pb}$	6312.8	5.	6309.7	1.4	-0.6	U					67Si09 Z
	6305.7	5.			0.8	U					67Tr06 Z
	6301.2	8.			1.1	U					71Ho01 Z
	6311.1	3.			-0.5	-			Bka		82Bo04 Z
	6307.7	5.			0.4	U			Lvn		93Wa04
	6309.7	5.0			-0.0	U			Ara		96Ta18 *
	6309.3	1.7			0.2	-			Tex		12Fo09
$^{198}\text{At}(\alpha)^{194}\text{Bi}$	ave. 6309.7	1.4			-0.0	1	100	60 ^{194}Pb			average
	6887.5	5.	6889.4	1.9	0.4	2					67Tr06 Z
	6904.9	7.			-2.2	U			Ora		75Ba.B Z
	6889.4	15.			-0.0	U					80Ew03 Z
	6893.3	3.5			-1.1	2			Lvn		92Hu04 *
	6892.5	4.			-0.8	o			Jya		96En01
	6887.4	6.			0.3	2			Jya		05Uu02
	6888.4	3.6			0.3	2			Tex		12Fo09
	6886.4	5.			0.6	2			Anv		14Ka23
	6888.9	8.2			0.1	2			ISa		19Gh11 *
$^{198}\text{At}^m(\alpha)^{194}\text{Bi}^n$	6990.0	5.	6992.8	2.3	0.6	-					67Tr06 Z
	6997.5	10.			-0.5	-					80Ew03 Z
	6997.6	4.			-1.2	-			Lvn		92Hu04
	6996.6	4.			-0.9	o			Jya		96En01
	6991.5	6.			0.2	-			Jya		05Uu02
	6990.5	5.			0.5	-			Anv		14Ka23
	6996.9	8.2			-0.5	-			ISa		19Gh11 *
$^{198}\text{Rn}(\alpha)^{194}\text{Po}$	ave. 6993.7	2.3			-0.4	1	95	58 $^{194}\text{Bi}^n$			average
	7344.7	10.	7349	4	0.5	4					84Ca32
	7353.8	5.			-0.9	4			Lvn		95Bi17
	7344.7	6.			0.8	4			Jya		96En02
$^{198}\text{Fr}(\alpha)^{194}\text{At}$	7869.2	20.4				4			Anv		13Ka16 *
$^{198}\text{Fr}^m(\alpha)^{194}\text{At}^m$	7889.6	20.4				6			Anv		13Ka16
$^{198}\text{Pt}(^{14}\text{C}, ^{16}\text{O})^{196}\text{Os}$	6130	40				2			BNL		83Bo29
$^{198}\text{Pt}(t, \alpha)^{197}\text{Ir}$	10885	20				2			LAL		83Ci01
$^{198}\text{Pt}(p, d)^{197}\text{Pt}$	-5332	3	-5331.0	2.1	0.3	1	47	47 ^{198}Pt	Ors		78Be09 *
$^{198}\text{Pt}(d, t)^{197}\text{Pt}$	-1305	20	-1298.3	2.1	0.3	U			Pit		64Co11
	-1311	20			0.6	U			Tal		78Ya07
$^{197}\text{Au}(n, \gamma)^{198}\text{Au}$	6512.35	0.11	6512.36	0.09	0.1	-			ILn		79Br26 Z
	6512.32	0.16			0.2	-			Bdn		06Fi.A
$^{197}\text{Au}(d, p)^{198}\text{Au}$	4282	30	4287.79	0.09	0.2	U			Pit		64Co11
	4298	5			-2.0	U			MIT		67Sp09
$^{197}\text{Au}(n, \gamma)^{198}\text{Au}$	ave. 6512.34	0.09	6512.36	0.09	0.2	1	99	63 ^{197}Au			average
$^{198}\text{Au}(\beta^-)^{198}\text{Hg}$	1372.3	0.7	1373.5	0.5	1.7	-					65Ke04 *
	1372.8	1.2			0.6	-					65Pa08 *
	ave. 1372.4	0.6			1.8	1	66	44 ^{198}Au			average
$^{198}\text{Tl}(\beta^+)^{198}\text{Hg}$	3460	80	3426	8	-0.4	U					61Gu02
$^{198}\text{Bi}^n(\text{IT})^{198}\text{Bi}^m$	248.5	0.5				3			Lvn		92Hu04
$^{198}\text{At}^m(\text{IT})^{198}\text{At}$	265	3	266.6	2.7	0.5	1	84	57 $^{198}\text{At}^m$	ISa		19Gh11 *
$^{198}\text{Tl}-^{133}\text{Cs}_{1.489}$	$D_M=111812.3(8.1) \mu\text{u}$ for $^{198}\text{Tl}^m$ at 543.6(0.4) keV; $M-A=-26985.1(7.5) \text{ keV}$										Nub211 **
$^{198}\text{Bi}-\text{u}$	$M-A=-19350(100) \text{ keV}$ for mixture gs+m+n at 290(40) and 540(40) keV										Nub211 **
$^{198}\text{Po}(\alpha)^{194}\text{Pb}$	Also $E_\alpha=6182(5) \text{ keV}$ from uncorrelated decays										96Ta18 **
$^{198}\text{At}(\alpha)^{194}\text{Bi}$	$E_\alpha=6755(4), 6539(10), 6360(10) \text{ to ground state, 218, 396 levels}$										92Hu04 **
$^{198}\text{At}(\alpha)^{194}\text{Bi}$	$E_\alpha=6358(8) \text{ keV to } 399.7(0.2) \text{ keV; Also } E_\alpha=6275(8), 6361(9),$										19Gh11 **
$^{198}\text{At}(\alpha)^{194}\text{Bi}$	$6535(8) \text{ to } 485.6(0.7), 382.4(0.1), 218.2(0.1) \text{ keV}$										19Gh11 **
$^{198}\text{At}^m(\alpha)^{194}\text{Bi}^n$	$E_\alpha=6753(8) \text{ keV to level at } 104.5(0.2) \text{ keV; Also } E_\alpha=6322(12),$										19Gh11 **
$^{198}\text{At}^m(\alpha)^{194}\text{Bi}^n$	$6338(8) \text{ keV to levels at } 538.3(0.4), 525.4(0.2) \text{ keV}$										19Gh11 **
$^{198}\text{Fr}(\alpha)^{194}\text{At}$	Evaluator's interpretation of Fig. 3d, no 210 keV γ										GAu **
$^{198}\text{Pt}(p, d)^{197}\text{Pt}$	$Q-Q(^{196}\text{Pt}(p, d))=365(3) \text{ keV}$										AHW **
$^{198}\text{Au}(\beta^-)^{198}\text{Hg}$	$E_{\beta^-}=960.5(0.7) 961.0(1.2) \text{ respectively, to } 2^+ \text{ level at } 411.80251 \text{ keV}$										Ens163 **
$^{198}\text{At}^m(\text{IT})^{198}\text{At}$	$E(\gamma)=511(3)+130.0(0.1)+150.8(0.1) - E(\gamma)=527.3(0.2)$										19Gh11 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{199}\text{Hg}-\text{C}_2^{35}\text{Cl}_5$	124023.43	0.53	124017.5	0.6	-4.5	B			H34	2.5	80Ko25
	124017.21	0.37			0.3	1	38	35 ^{199}Hg	H48	2.5	03Ba49
$^{199}\text{Hg}-^{183}\text{W O}$	23144.4	0.9	23142.0	0.9	-1.1	1	16	11 ^{183}W	H48	2.5	03Ba49
$^{199}\text{Hg}-^{162}\text{Dy }^{37}\text{Cl}$	75661	41	75573.9	0.9	-0.5	U			R04	4.0	64De15
$^{199}\text{Hg}-^{164}\text{Dy }^{35}\text{Cl}$	70087	31	70247.5	0.9	1.3	U			R04	4.0	64De15
$^{199}\text{Hg}-^{164}\text{Er }^{35}\text{Cl}$	70310	80	70220.6	0.9	-0.3	U			R04	4.0	64De15
$^{199}\text{Tl}-\text{u}$	-30123	30				2			GS2	1.0	05Li24
$^{199}\text{Pb}-\text{u}$	-27028	137	-27087	7	-0.4	U			GS2	1.0	05Li24 *
$^{199}\text{Bi}-\text{u}$	-22328	31	-22327	11	0.0	-			GS2	1.0	05Li24
	-22263	30			-2.1	-			GS2	1.0	05Li24 *
ave.	-22294	22			-1.5	1	28	28 ^{199}Bi			average
$^{199}\text{Po}-\text{u}$	-16249	144	-16360	6	-0.8	U			GS1	1.0	00Ra23 *
	-16327	38			-0.9	R			GS2	1.0	05Li24
	-16339	38			-0.5	R			GS2	1.0	05Li24 *
$^{199}\text{Po}^m-^{133}\text{Cs}_{1.496}$	125416.5	5.3	125419	5	0.5	1	93	93 $^{199}\text{Po}^m$	MA8	1.0	17Ai34
$^{199}\text{Bi}^m(\alpha)^{195}\text{Tl}$	5598.7	6.	5600	6	0.1	1	93	56 ^{195}Tl			66Ma51
$^{199}\text{Po}(\alpha)^{195}\text{Pb}$	6074.1	2.	6074.3	1.9	0.1	2			Db		68Go.B Z
	6075.3	5.0			-0.2	2			Ara		96Ta18
$^{199}\text{Po}^m(\alpha)^{195}\text{Pb}^m$	6190.7	5.	6183.1	1.7	-1.5	-					67Si09 Z
	6177.6	5.1			1.1	-					67Tr06 Z
	6182.3	3.1			0.3	-			Db		68Go.B Z
	6183.5	3.			-0.1	-			Bka		82Bo04 Z
	6183.5	5.0			-0.1	-			Ara		96Ta18 *
	6173.3	3.6			2.7	B			Tex		12Fo09
ave.	6183.3	1.7			-0.1	1	99	95 $^{195}\text{Pb}^m$			average
$^{199}\text{At}(\alpha)^{195}\text{Bi}$	6775.1	5.	6777.3	1.2	0.4	-					67Tr06 Z
	6781.3	3.			-1.3	-			Ora		75Ba.B Z
	6775.4	5.0			0.4	-			Ara		96Ta18
	6779.4	6.			-0.4	U			Jya		05Uu02
	6776.8	1.5			0.3	-			Tex		12Fo09
ave.	6777.4	1.2			-0.1	1	100	89 ^{199}At			average
$^{199}\text{Rn}(\alpha)^{195}\text{Po}$	7133.7	15.	7132	4	-0.1	2					80Di07
	7132.7	10.			-0.1	2					82Hi14
	7138.8	10.			-0.7	2					84Ca32
	7112.2	15.			1.3	o			Jya		96Le09
	7132.6	6.			-0.1	2			Jya		05Uu02
	7121.4	10.2			1.0	2			Anv		14Ka23
$^{199}\text{Rn}^m(\alpha)^{195}\text{Po}^m$	7205.1	15.	7203	4	-0.1	2					80Di07
	7205.1	10.			-0.2	2					82Hi14
	7204.1	10.			-0.1	2					84Ca32
	7205.1	15.			-0.1	o			Jya		96Le09
	7205.1	6.			-0.3	2			Jya		05Uu02
	7194.9	10.2			0.8	2			Anv		14Ka23
$^{199}\text{Fr}(\alpha)^{195}\text{At}$	7812.3	40.	7817	10	0.1	U					99Ta20
	7821.5	11.			-0.4	4			Anv		13Ka16
	7801.1	20.			0.8	4			Jya		13Uu01
$^{199}\text{Fr}^m(\alpha)^{195}\text{At}^m$	7833.7	6.	7833	6	-0.2	3			Anv		13Ka16
	7825.6	15.3			0.5	3			Jya		13Uu01
$^{199}\text{Fr}^m(\alpha)^{195}\text{At}^p$	7968.4	20.				4			Jya		13Uu01
$^{199}\text{Hg}(\text{p,t})^{197}\text{Hg}$	-6734	29	-6667	3	2.3	U			Pri		81Ko13
	-6658	8			-1.1	1	16	16 ^{197}Hg	Ors		82Be21
$^{198}\text{Pt}(^{18}\text{O}, ^{17}\text{F})^{199}\text{Ir}$	-8240	41				2					95Zh10
$^{198}\text{Pt}(\text{n}, \gamma)^{199}\text{Pt}$	5602	3	5556.0	0.5	-15.3	B					68Sa13
	5556.0	0.5				2			BNn		83Ca04 Z
$^{198}\text{Pt}(\text{d,p})^{199}\text{Pt}$	3347	20	3331.4	0.5	-0.8	U			Pit		64Co11
$^{198}\text{Au}(\text{n}, \gamma)^{199}\text{Au}$	7584.27	0.15	7584.28	0.06	0.1	o			ILn		79Br26 Z
	7584.28	0.06			-0.0	1	100	80 ^{199}Au	ILn		91Ma65

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{198}\text{Hg}(\text{n},\gamma)^{199}\text{Hg}$	6665.2	0.5	6663.1	0.6	-4.3	B			CRn		75Lo03
$^{199}\text{Hg}(\gamma,\text{n})^{198}\text{Hg}$	-6590	90	-6663.1	0.6	-0.8	U			Phi		60Ge01
$^{199}\text{Pt}(\beta^-)^{199}\text{Au}$	1690	50	1705.1	2.1	0.3	U					64Jo09
$^{199}\text{Au}(\beta^-)^{199}\text{Hg}$	453.0	1.0	452.3	0.6	-0.7	1	38	20 ^{199}Au			68Be06
$^{199}\text{Tl}(\beta^+)^{199}\text{Hg}$	1420	150	1487	28	0.4	U					75Ma05 *
$^{199}\text{Pb}(\beta^+)^{199}\text{Tl}$	2870	110	2828	29	-0.4	U					70Do.A *
$^{199}\text{Bi}^m(\text{IT})^{199}\text{Bi}$	667	5	667	3	-0.1	-					80Br23
	667	5			-0.1	-					85St02
ave.	667	4			-0.1	1	98	64 $^{199}\text{Bi}^m$			average
$^{199}\text{At}^m(\text{IT})^{199}\text{At}$	255	20	244.0	1.0	-0.6	o			Jya		13Ja06 *
	244	1				2			Jya		14Au03
* $^{199}\text{Pb}-\text{u}$	$M-A=-24961(28)$ keV for mixture gs+m at 429.5(2.7) keV										Nub211 **
* $^{199}\text{Bi}-\text{u}$	$M-A=-20071(28)$ keV for $^{199}\text{Bi}^m$ at 667(3) keV										Nub211 **
* $^{199}\text{Po}-\text{u}$	$M-A=-14980(100)$ keV for mixture gs+m at 311.7(2.7) keV										Nub211 **
* $^{199}\text{Po}-\text{u}$	$M-A=-14909(35)$ keV for $^{199}\text{Po}^m$ at 311.7(2.7) keV										Nub211 **
* $^{199}\text{Po}^m(\alpha)^{195}\text{Pb}^m$	Also $E_\alpha=6059(5)$ keV from uncorrelated decays										96Ta18 **
* $^{199}\text{Tl}(\beta^+)^{199}\text{Hg}$	KL+<500(100) giving $Q < 1620$, ($1/2^-$, $3/2^-$) level at 1221.17 fed. Reanalyzed										Ens073 **
* $^{199}\text{Pb}(\beta^+)^{199}\text{Tl}$	$p^+ = 0.04(0.01)$ to $3/2^+$ level at 366.89 keV, recalculated										Ens073 **
* $^{199}\text{At}^m(\text{IT})^{199}\text{At}$	Combining lines 110(20) 160(20) 240(30) as read from Fig. 8a										GAu **
$^{200}\text{Au}-\text{u}$	-29237	34	-29243	29	-0.2	1	71	71 ^{200}Au	GS3	1.0	08Ch.A
$^{200}\text{Au}^m-\text{u}$	-28135	33	-28163	28	-0.8	1	73	73 $^{200}\text{Au}^m$	GS3	1.0	08Ch.A
$^{200}\text{Hg}-^{13}\text{C }^{35}\text{Cl}_5$	120707.97	1.22	120708.6	0.6	0.2	U			H34	2.5	80Ko25
$^{200}\text{Hg}-^{165}\text{Ho }^{35}\text{Cl}$	69116	33	69145.1	1.0	0.2	U			R04	4.0	64De15
$^{200}\text{Hg}-^{163}\text{Dy }^{37}\text{Cl}$	73527	42	73687.1	0.9	1.0	U			R04	4.0	64De15
$^{200}\text{Pb}-\text{u}$	-28179	30	-28181	11	-0.1	R			GS2	1.0	05Li24
$^{200}\text{Bi}-\text{u}$	-21888	57	-21869	24	0.3	R			GS2	1.0	05Li24 *
$^{200}\text{Po}-\text{u}$	-18170	104	-18188	8	-0.2	U			GS1	1.0	00Ra23
	-18204	30			0.5	U			GS2	1.0	05Li24
$^{200}\text{Hg}-^{208}\text{Pb}_{962}$	-9205	28	-9212.3	1.2	-0.3	U			MA6	1.0	01Sc41
$^{200}\text{Hg }^{35}\text{Cl}-^{198}\text{Hg }^{37}\text{Cl}$	4525	2	4507.9	0.6	-2.1	U			H17	4.0	64Mc07
	4508.80	0.48			-0.8	1	28	17 ^{200}Hg	H33	2.5	80Ko25
$^{200}\text{Po}(\alpha)^{196}\text{Pb}$	5979.8	5.	5981.6	1.8	0.4	-					67Si09 Z
	5980.0	3.			0.5	-					67Tr06 Z
	5983.4	3.			-0.6	-					70Ra14 Z
	5981.8	5.0			-0.0	-			Ara		96Ta18 *
ave.	5981.5	1.9			0.1	1	99	79 ^{196}Pb			average
$^{200}\text{At}(\alpha)^{196}\text{Bi}$	6594.9	5.	6596.2	1.3	0.3	3					67Tr06 Z
	6596.9	2.			-0.4	3			Ora		75Ba.B Z
	6593.1	5.			0.6	o			Lvn		87Va09
	6596.1	2.			0.0	3			Lvn		92Hu04
	6593.1	5.0			0.6	3			Ara		96Ta18
	6599.1	6.			-0.5	U			Jya		05Uu02
$^{200}\text{At}^m(\alpha)^{196}\text{Bi}$	6708.3	5.	6709.1	2.6	0.2	3			Ora		75Ba.B Z
	6705.4	5.			0.7	o			Lvn		87Va09
	6709.5	3.			-0.1	3			Lvn		92Hu04
$^{200}\text{At}^m(\alpha)^{196}\text{Bi}^m$	6542.8	5.	6542.7	1.3	-0.0	4					67Tr06 Z
	6542.9	2.			-0.1	4			Ora		75Ba.B Z
	6540.0	5.			0.5	o			Lvn		87Va09
	6542.1	2.			0.3	4			Lvn		92Hu04
	6545.1	5.0			-0.5	4			Ara		96Ta18
	6544.1	6.			-0.2	U			Jya		05Uu02
$^{200}\text{At}^m(\alpha)^{196}\text{Bi}^n$	6439.5	5.	6437.5	2.0	-0.4	4					67Tr06 *
	6438.5	5.			-0.2	4			Ora		75Ba.B *
	6433.8	5.			0.7	o			Lvn		87Va09 *
	6439.2	3.			-0.6	4			Lvn		92Hu04 *
	6430.5	5.0			1.4	4			Ara		96Ta18 *
	6436.7	6.			0.1	4			Jya		05Uu02 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{200}\text{Rn}(\alpha)^{196}\text{Po}$	7020.6	10.	7043.4	2.1	2.3	U					67Va.A
	7050.3	8.			-0.9	U					71Ho01
	7040.1	10.			0.3	U			Lvn		84Co.A
	7043.5	2.5			-0.1	2			Lvn		93Wa04
	7042.1	12.			0.1	o			Ara		95Le04
	7039.0	10.			0.4	o			Jya		96Le09
	7042.1	5.1			0.2	2			Ara		96Ta18
	7044.1	6.			-0.1	2			Jya		05Uu02
	7055.4	30.			-0.4	U			Anv		10He25
	7653.4	30.	7622	4	-1.1	U			RIa		95Mo14
$^{200}\text{Fr}(\alpha)^{196}\text{At}$	7620.7	9.			0.1	5			Jya		96En01
	7625.8	12.			-0.3	o			Anv		05De01
	7620.7	15.			0.1	o			Jya		13Uu01 *
	7622.7	5.			-0.2	5			Anv		14Ka23
	7618.6	12.2			0.3	5			ISa		19Gh11
	7704.4	15.				4			Jya		96En01 *
$^{200}\text{Fr}^m(\alpha)^{196}\text{At}^m$	4356	20				2					81Ci01
$^{198}\text{Pt}(t,p)^{200}\text{Pt}$	8029.1	0.3	8028.52	0.11	-1.9	-			BNn		67Sc30 Z
$^{199}\text{Hg}(n,\gamma)^{200}\text{Hg}$	8029.6	0.5			-2.2	U			CRn		75Lo03 Z
	8028.51	0.18			0.1	-			ILn		79Br25 Z
	8028.37	0.17			0.9	-			Bdn		06Fi.A
	ave.	8028.53	0.11		-0.1	1	98	64 ^{200}Hg			average
$^{200}\text{Au}(\beta^-)^{200}\text{Hg}$	2273	100	2263	27	-0.1	-					59Ro53 *
	2200	100			0.6	-					60Gi01
	2260	70			0.0	-					72He36 *
	ave.	2250	50		0.3	1	29	29 ^{200}Au			average
$^{200}\text{Au}^m(\beta^-)^{200}\text{Hg}$	3202	50	3270	26	1.4	1	27	27 $^{200}\text{Au}^m$			72Cu07 *
$^{200}\text{Tl}(\beta^+)^{200}\text{Hg}$	2450	10	2456	6	0.6	2					57He43 *
	2459	7			-0.4	2					62Va10 *
$^{200}\text{At}^n(1T)^{200}\text{At}^m$	230.9	0.2				4			Lvn		92Hu04
* $^{200}\text{Bi}-u$	$M-A=-20338(28)$ keV for mixture gs+m at 100#70 keV										Nub211 **
* $^{200}\text{Po}(\alpha)^{196}\text{Pb}$	Also $E_\alpha=5863(5)$ keV from uncorrelated decays										96Ta18 **
* $^{200}\text{At}^m(\alpha)^{196}\text{Bi}^n$	$E_\alpha=6536.7(5,Z)$ from $^{200}\text{At}^n$ 230.9 above $^{200}\text{At}^m$										92Hu04 **
* $^{200}\text{At}^m(\alpha)^{196}\text{Bi}^n$	$E_\alpha=6535.8(5,Z)$ from $^{200}\text{At}^n$ 230.9 above $^{200}\text{At}^m$										92Hu04 **
* $^{200}\text{At}^m(\alpha)^{196}\text{Bi}^n$	$E_\alpha=6301(5); 6535(5)$ from $^{200}\text{At}^n$ 230.9 above $^{200}\text{At}^m$										92Hu04 **
* $^{200}\text{At}^m(\alpha)^{196}\text{Bi}^n$	$E_\alpha=6306(5); 6538(3)$ from $^{200}\text{At}^n$ 230.9 above $^{200}\text{At}^m$										92Hu04 **
* $^{200}\text{At}^m(\alpha)^{196}\text{Bi}^n$	$E_\alpha=6528(5)$ from $^{200}\text{At}^n$ 230.9 above $^{200}\text{At}^m$										92Hu04 **
* $^{200}\text{At}^m(\alpha)^{196}\text{Bi}^n$	$E_\alpha=6534(6)$ from $^{200}\text{At}^n$ 230.9 above $^{200}\text{At}^m$										92Hu04 **
* $^{200}\text{Fr}(\alpha)^{196}\text{At}$	Same group, but much less events than preceding item										WgM151**
* $^{200}\text{Fr}^m(\alpha)^{196}\text{At}^m$	Correlated with $^{196}\text{At}^m$ $E_\alpha=6880(15)$; two events only										96En01 **
* $^{200}\text{Au}(\beta^-)^{200}\text{Hg}$	$E_{\beta^-}=2250(200)$ to ground state, and $700(100)$ to levels 1^+ at 1570.275,										Ens077 **
*	2^+ at 1573.663, 2^+ at 1593.423 keV										Ens077 **
* $^{200}\text{Au}(\beta^-)^{200}\text{Hg}$	$E_{\beta^-}=2260(100), 670(70)$ to ground state, 2^+ level at 1593.423 keV										Ens077 **
* $^{200}\text{Au}^m(\beta^-)^{200}\text{Hg}$	$E_{\beta^-}=560(50)$ to 11^- level at 2641.54 keV										Ens077 **
* $^{200}\text{Tl}(\beta^+)^{200}\text{Hg}$	$E_{\beta^+}=1052(10) 1069(7)$ respectively, to 2^+ level at 367.943 keV, and other E_{β^+}										Ens077 **
$^{201}\text{Hg}-^{185}\text{Re O}$	22440	5	22430.1	1.1	-0.8	U			H48	2.5	03Ba49
$^{201}\text{Hg}-\text{C}_2^{35}\text{Cl}_4^{37}\text{Cl}$	128995.43	0.61	128989.7	0.8	-3.8	B			H34	2.5	80Ko25
$^{201}\text{Hg}-^{164}\text{Dy}^{37}\text{Cl}$	75086	42	75219.7	1.1	0.8	U			R04	4.0	64De15
$^{201}\text{Hg}-^{166}\text{Er}^{35}\text{Cl}$	71186	35	71149.3	0.8	-0.3	U			R04	4.0	64De15
$^{201}\text{Pb}-u$	-27418	198	-27130	15	1.5	U			GS2	1.0	05Li24 *
$^{201}\text{Bi}-u$	-22935	30	-23005	13	-2.3	R			GS2	1.0	05Li24
	-22995	30			-0.3	R			GS2	1.0	05Li24 *
$^{201}\text{Po}-u$	-17760	190	-17736	5	0.1	U			GS1	1.0	00Ra23 *
	-17649	30			-2.9	U			GS2	1.0	05Li24

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{201}\text{Po}^m\text{--u}$	−17305	30	−17281	5	0.8	U			GS2	1.0	05Li24
$^{201}\text{At}\text{--u}$	−11573	31	−11583	9	−0.3	U			GS2	1.0	05Li24
$^{201}\text{Hg } ^{35}\text{Cl}\text{--}^{199}\text{Hg } ^{37}\text{Cl}$	4981	2	4972.2	0.6	−1.1	U			H17	4.0	64Mc07
	4972.65	0.37			−0.5	1	47	^{201}Hg	H33	2.5	80Ko25
	4971.8	1.0			0.2	U			H48	2.5	03Ba49
$^{201}\text{Bi}(\alpha)^{197}\text{Tl}$	4500.3	6.				4					66Ma51 *
$^{201}\text{Po}(\alpha)^{197}\text{Pb}$	5793.9	5.	5799.3	1.7	1.1	−					67Tr06 Z
	5799.4	2.			−0.1	−			DbA		68Go.B Z
	5800.4	4.			−0.3	−					70Ra14 Z
	ave. 5799.0	1.7			0.2	1	98	^{201}Po			average
$^{201}\text{Po}^m(\alpha)^{197}\text{Pb}^m$	5899.0	5.1	5903.8	1.7	0.9	2					67Tr06 Z
	5904.5	2.0			−0.4	2			DbA		68Go.B Z
	5903.9	4.1			−0.0	2					70Ra14 Z
$^{201}\text{At}(\alpha)^{197}\text{Bi}$	6470.7	3.	6472.8	1.6	0.7	4					67Tr06 Z
	6476.2	5.			−0.7	U					74Ho27 Z
	6474.0	2.			−0.6	4			Ora		75Ba.B Z
	6471.0	5.0			0.4	U			Ara		96Ta18
	6472.0	4.			0.2	4			Anv		05De01
$^{201}\text{Rn}(\alpha)^{197}\text{Po}$	6862.8	8.	6860.7	2.3	−0.3	U					67Va.A
	6858.8	8.			0.2	U					71Ho01
	6866.9	20.			−0.3	U			GSa		87He10
	6860.5	2.5			0.1	2			Lvn		93Wa04
	6863.8	7.			−0.4	o			Ara		95Le04
	6861.8	5.0			−0.2	2			Ara		96Ta18
$^{201}\text{Rn}^m(\alpha)^{197}\text{Po}^m$	6906.8	5.	6909.5	2.1	0.5	3					67Va17 Z
	6909.0	8.			0.1	U					71Ho01 Z
	6907.7	20.			0.1	U			GSa		87He10
	6909.9	2.5			−0.1	3			Lvn		93Wa04
	6915.9	7.			−0.9	o			Ara		95Le04
	6910.7	5.0			−0.3	3			Ara		96Ta18
	6925.1	30.			−0.5	U			Anv		10He25
$^{201}\text{Fr}(\alpha)^{197}\text{At}$	7538.0	15.	7519	4	−1.3	U					80Ew03
	7510.8	7.			1.1	o			Jya		96En01
	7529.1	7.			−1.4	o			Anv		05De01
	7519.0	8.			−0.0	2			Jya		05Uu02
	7519.0	5.			−0.0	2			Anv		14Ka23
$^{201}\text{Fr}^m(\alpha)^{197}\text{At}^m$	7605.7	8.	7603	5	−0.3	2			Jya		05Uu02
	7596.4	8.			0.8	2			Anv		14Ka23
	7608.7	9.2			−0.6	2			Jya		20Au01
$^{201}\text{Ra}(\alpha)^{197}\text{Rn}$	8001.5	12.				4			Anv		14Ka23
$^{201}\text{Ra}^m(\alpha)^{197}\text{Rn}^m$	8065.8	20.				4			Jya		05Uu02
$^{201}\text{Hg}(\gamma, n)^{200}\text{Hg}$	−6210	70	−6230.6	0.6	−0.3	U			Phi		60Ge01
$^{201}\text{Pt}(\beta^-)^{201}\text{Au}$	2660	50				2					63Go06
$^{201}\text{Au}(\beta^-)^{201}\text{Hg}$	1270	100	1262	3	−0.1	U					72Pa24
$^{201}\text{Tl}(\epsilon)^{201}\text{Hg}$	470	70	482	14	0.2	U					60Gu05 *
$^{201}\text{Pb}(\beta^+)^{201}\text{Tl}$	1900	40	1910	19	0.2	1	21	^{201}Tl			79Do09 *
$^{201}\text{Pb}\text{--u}$	$M - A = -25225(28)$ keV for mixture gs+m at 629.1 keV										Nub211 **
$^{201}\text{Bi}\text{--u}$	$M - A = -20573(28)$ keV for $^{201}\text{Bi}^m$ at 846.35 keV										Nub211 **
$^{201}\text{Po}\text{--u}$	$M - A = -16330(100)$ keV for mixture gs+m at 423.8(2.4) keV										Nub211 **
$^{201}\text{Bi}(\alpha)^{197}\text{Tl}$	$E_\alpha = 5240(6)$ from $^{201}\text{Bi}^m$ at 846.35 keV										Nub211 **
$^{201}\text{Tl}(\epsilon)^{201}\text{Hg}$	$pK = 0.70(0.04)$ to $1/2^-$ level at 167.47 keV, recalculated										Ens073 **
$^{201}\text{Pb}(\beta^+)^{201}\text{Tl}$	$p^+ = 10(2) \times 10^{-3}$ to $3/2^+$ level at 331.16 keV										Ens073 **
$^{202}\text{Pt}\text{--u}$	−24425	34	−24361	27	1.9	o			GS3	1.0	08Ch.A
	−24361	27				2			GS3	1.0	12Ch19
$^{202}\text{Au}\text{--u}$	−26202	34	−26144	25	1.7	o			GS3	1.0	08Ch.A
	−26144	25				2			GS3	1.0	12Ch19

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{202}\text{Hg}-^{13}\text{C } ^{35}\text{Cl}_4 ^{37}\text{Cl}$	125976.01	1.32	125975.4	0.8	-0.2	U			H34	2.5	80Ko25
$\text{C}_{16}\text{H}_{10}-^{202}\text{Hg}$	107663	40	107606.7	0.8	-0.9	U			R08	1.5	69De19
$\text{C}_{15}^{13}\text{C H}_9-^{202}\text{Hg}$	103102	60	103136.5	0.8	0.4	U			R08	1.5	69De19
$^{202}\text{Hg}-^{167}\text{Er } ^{35}\text{Cl}$	69740	60	69734.7	0.8	-0.0	U			R04	4.0	64De15
$^{202}\text{Hg}-^{165}\text{Ho } ^{37}\text{Cl}$	74470	50	74411.9	1.1	-0.3	U			R04	4.0	64De15
$^{202}\text{Pb}-\text{u}$	-27823	30	-27848	4	-0.8	U			GS2	1.0	05Li24 *
$^{202}\text{Pb}-^{133}\text{Cs}_{1.519}$	115773.4	3.6	115770	4	-0.9	o			MA8	1.0	10Bo.A
	115769.2	4.4			0.2	1	86	86 ^{202}Pb	MA8	1.0	14Bo26
$^{202}\text{Bi}-\text{u}$	-22282	30	-22277	15	0.2	1	25	25 ^{202}Bi	GS2	1.0	05Li24
$^{202}\text{Po}-\text{u}$	-19270	104	-19261	9	0.1	U			GS1	1.0	00Ra23
	-19243	30			-0.6	U			GS2	1.0	05Li24
$^{202}\text{Hg } ^{35}\text{Cl}_2-^{198}\text{Hg } ^{37}\text{Cl}_2$	9774.87	1.06	9774.7	0.8	-0.1	U			H33	2.5	80Ko25
$^{202}\text{Hg } ^{35}\text{Cl}-^{200}\text{Hg } ^{37}\text{Cl}$	5271	3	5266.8	0.6	-0.4	U			H17	4.0	64Mc07
	5266.76	0.43			0.0	1	35	28 ^{202}Hg	H33	2.5	80Ko25
$^{202}\text{Tl}-^{203}\text{Tl}_{.995}$	-372.4	2.1	-373.5	1.8	-0.5	1	73	63 ^{202}Tl	MA8	1.0	17We09
$^{202}\text{Po}(\alpha)^{198}\text{Pb}$	5700.9	2.	5701.0	1.7	0.0	-			Db		68Go.B Z
	5701.6	3.			-0.2	-					70Ra14 Z
	5701.1	1.7			-0.1	1	99	74 ^{198}Pb			average
$^{202}\text{At}(\alpha)^{198}\text{Bi}$	6355.8	3.	6353.8	1.3	-0.6	-					63Ho18 Z
	6351.7	3.			0.7	-					67Tr06 Z
	6353.2	5.			0.1	-					74Ho27 Z
	6353.9	2.			-0.0	-			Ora		75Ba.B Z
	6354	5			-0.0	-			Lvn		92Hu04 *
	6355.0	6.0			-0.2	-			Ara		96Ta18
	6353.8	1.3			-0.0	1	100	97 ^{202}At			average
$^{202}\text{At}^m(\alpha)^{198}\text{Bi}^m$	6259.9	2.	6259.0	1.3	-0.5	4					63Ho18 Z
	6256.8	3.			0.7	4					67Tr06 Z
	6257.2	5.			0.4	U					74Ho27 Z
	6259.0	2.			-0.0	4			Ora		75Ba.B *
	6260.0	5.			-0.2	U			Lvn		92Hu04 *
	6257.1	6.0			0.3	U			Ara		96Ta18
$^{202}\text{Rn}(\alpha)^{198}\text{Po}$	6771.0	3.	6773.8	1.8	0.9	2					67Va17 Z
	6772.3	10.			0.2	U			GSa		87He10
	6775.3	2.5			-0.6	2			Lvn		93Wa04
	6773.4	7.			0.1	o			Ara		95Le04
	6775.4	5.0			-0.3	2			Ara		96Ta18
$^{202}\text{Fr}(\alpha)^{198}\text{At}$	7397.7	15.	7385	4	-0.8	2					80Ew03 *
	7382.5	11.			0.3	2			Lvn		92Hu04 *
	7389.6	6.			-0.7	o			Jya		96En01 *
	7387.6	8.			-0.3	2			Jya		05Uu02
	7384.5	5.1			0.2	2			Anv		14Ka23
	7383.3	8.2			0.3	2			ISa		19Gh11 *
$^{202}\text{Fr}^m(\alpha)^{198}\text{At}$	7643.7	8.2	7643	4	-0.1	1	27	24 $^{202}\text{Fr}^m$	ISa		19Gh11 *
$^{202}\text{Fr}^m(\alpha)^{198}\text{At}^m$	7382.5	11.	7376	4	-0.6	-			Lvn		92Hu04 *
	7388.6	6.			-2.0	o			Jya		96En01
	7381.5	8.			-0.7	-			Jya		05Uu02
	7372.2	5.			0.8	-			Anv		14Ka23
	7387.3	8.2			-1.4	U			ISa		19Gh11 *
	7376	4			0.1	1	82	76 $^{202}\text{Fr}^m$			average
$^{202}\text{Ra}(\alpha)^{198}\text{Rn}$	8019.1	60.	7880	7	-2.3	o			Jya		96Le09
	7896.7	20.			-0.8	5			Jya		05Uu02
	7878.3	7.1			0.3	5			Anv		14Ka23
$^{202}\text{Hg}(\text{t},\alpha)^{201}\text{Au}$	11567	15	11580	3	0.9	U			LAL		81Fl05
$^{202}\text{Hg}(\text{d},^3\text{He})^{201}\text{Au}-^{206}\text{Pb}()^{205}\text{Tl}$	-979.9	3.1	-980	3	-0.0	1	100	100 ^{201}Au			94Gr07
$^{201}\text{Hg}(\text{n},\gamma)^{202}\text{Hg}$	7754.9	0.5	7754.10	0.20	-1.6	-			BNn		75Br02 Z
	7756.4	0.5			-4.6	B			CRn		75Lo03 Z
	7753.93	0.22			0.8	-			Bdn		06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{202}\text{Hg}(\gamma, n)^{201}\text{Hg}$		−7600	130	−7754.10	0.20	−1.2	U				60Ge01
$^{201}\text{Hg}(n, \gamma)^{202}\text{Hg}$	ave.	7754.09	0.20	7754.10	0.20	0.1	1	97	59 ^{201}Hg		average
$^{202}\text{Au}(\beta^-)^{202}\text{Hg}$		3500	300	2992	23	−1.7	U				67Wa23
		2700	300			1.0	U				72Bu05
$^{202}\text{Tl}(\epsilon)^{202}\text{Hg}$		1245	25	1364.9	1.8	4.8	B				66Le06 *
$^{202}\text{Pb}(\epsilon)^{202}\text{Tl}$		55	20	40	4	−0.8	U				54Hu61
$^{202}\text{At}^n(\text{IT})^{202}\text{At}^m$		391.7	0.2				5		Lvn		92Hu04
$*^{202}\text{Pb}-u$	$M-A=-23747(28)$ keV for $^{202}\text{Pb}^m$ at 2169.85 keV										Nub211 **
$*^{202}\text{At}(\alpha)^{198}\text{Bi}$	$E_\alpha=6228(5)$, 6070(10), 5929(10) to ground state, 164, 303 levels										92Hu04 **
$*^{202}\text{At}^m(\alpha)^{198}\text{Bi}^m$	Assignment to $^{202}\text{At}^m$ in reference; Z recalibrated										92Hu04 **
$*^{202}\text{At}^m(\alpha)^{198}\text{Bi}^m$	$E_\alpha=6135(5)$; and 6277(5) from $^{202}\text{At}^n(\alpha)^{198}\text{Bi}^n$, with										92Hu04 **
*	$^{202}\text{At}^n(\text{IT})^{202}\text{At}^m=391.7(0.5)$ and $^{198}\text{Bi}^n(\text{IT})^{198}\text{Bi}^m=248.5(0.5)$ keV										Nub211 **
$*^{202}\text{Fr}(\alpha)^{198}\text{At}$	$E_\alpha=7251(10)$ has a doublet structure										92Hu04 **
$*^{202}\text{Fr}(\alpha)^{198}\text{At}$	$E_\alpha=7237(8)$, is a doublet										92Hu04 **
$*^{202}\text{Fr}(\alpha)^{198}\text{At}$	^{202}Fr E_α 's in correlation with At daughters										96En01 **
$*^{202}\text{Fr}(\alpha)^{198}\text{At}$	$E_\alpha=7105(8)$ keV to level at 130.0(0.1) keV; Also $E_\alpha=7086(8)$ keV										19Gh11 **
$*^{202}\text{Fr}(\alpha)^{198}\text{At}$	to level at 154.2(0.3) keV										19Gh11 **
$*^{202}\text{Fr}^m(\alpha)^{198}\text{At}$	$E_\alpha=7217(8)$ to level at 280.8(0.1) keV										19Gh11 **
$*^{202}\text{Fr}^m(\alpha)^{198}\text{At}^m$	$E_\alpha=7237(8)$, is a doublet										92Hu04 **
$*^{202}\text{Fr}^m(\alpha)^{198}\text{At}^m$	$E_\alpha=6724(8)$ keV to level at 527.3(0.2) keV above $^{198}\text{At}^m$ and at										19Gh11 **
$*^{202}\text{Fr}^m(\alpha)^{198}\text{At}^m$	792(3) keV above ^{198}At ; not independent										19Gh11 **
$*^{202}\text{Tl}(\epsilon)^{202}\text{Hg}$	pK=0.305(0.020) to 2^+ level at 959.94 keV										Ens083 **
$\text{C}_{16} \text{H}_{11}-^{203}\text{Tl}$		113735	43	113731.3	1.3	−0.1	U		R08	1.5	69De19
$\text{C}_{15}^{13}\text{C} \text{H}_{10}-^{203}\text{Tl}$		109216	95	109261.1	1.3	0.3	U		R08	1.5	69De19
$\text{C}_{14} \text{N}_2 \text{H}_7-^{203}\text{Tl}$		88540	48	88579.1	1.3	0.5	U		R08	1.5	69De19
$^{203}\text{Tl}-^{166}\text{Er}^{37}\text{Cl}$		76190	48	76140.5	1.3	−0.7	U		R08	1.5	69De19
$^{203}\text{Tl}-^{168}\text{Er}^{35}\text{Cl}$		71069	36	71113.1	1.3	0.8	U		R08	1.5	69De19
$^{203}\text{Pb}-u$		−26594	30	−26609	7	−0.5	U		GS2	1.0	05Li24
$^{203}\text{Po}-u$		−18581	30	−18584	5	−0.1	U		GS2	1.0	05Li24
$^{203}\text{Po}-^{133}\text{Cs}_{1.526}$		125697.0	6.0	125696	5	−0.1	1	69	69 ^{203}Po	MA8	1.0 17Al34
$^{203}\text{At}-u$		−13042	30	−13057	11	−0.5	1	14	14 ^{203}At	GS2	1.0 05Li24
$^{203}\text{Fr}-^{133}\text{Cs}_{1.526}$		145205	17	145221	7	1.0	1	15	15 ^{203}Fr	MA8	1.0 08We02
$^{203}\text{At}-^{208}\text{Pb}_{.976}$		9690	25	9731	11	1.6	1	21	21 ^{203}At	MA6	1.0 01Sc41
$^{203}\text{Rn}^m-^{208}\text{Pb}_{.976}$		16579	30	16538	5	−1.4	1	3	3 $^{203}\text{Rn}^m$	SH1	1.0 13Dr04
$^{203}\text{Tl}^{35}\text{Cl}-^{201}\text{Hg}^{37}\text{Cl}$		4997	3	4991.2	1.2	−0.5	U			H17	4.0 64Mc07
		4995.23	1.49			−1.1	1	10	9 ^{203}Tl	H36	2.5 85De40
$^{202}\text{Hg} \text{H}-^{203}\text{Tl}$		6154	34	6124.5	1.2	−0.6	U			R08	1.5 69De19
$^{203}\text{Tl}-^{167}\text{Er}^{35}\text{Cl}$		71436	36	71435.2	1.3	−0.0	U			R08	1.5 69De19
$^{167}\text{Er}^{37}\text{Cl}-^{203}\text{Tl}$		−74404	33	−74385.3	1.3	0.4	U			R08	1.5 69De19
$^{169}\text{Tm}^{35}\text{Cl}-^{203}\text{Tl}$		−69257	29	−69272.4	1.5	−0.4	U			R08	1.5 69De19
$^{203}\text{Tl}-^{202}\text{Hg}$		1722	20	1700.5	1.2	−0.7	U			R08	1.5 69De19
$^{203}\text{Tl}-^{201}\text{Hg}$		1999	29	2041.0	1.2	1.0	U			R08	1.5 69De19
$^{203}\text{Po}(\alpha)^{199}\text{Pb}$		5496	5				2		Db		68Go.B *
$^{203}\text{At}(\alpha)^{199}\text{Bi}$		6210.3	1.	6210.1	0.8	−0.2	−				63Ho18 Z
		6208.7	3.			0.5	−				67Tr06 Z
		6209.4	2.			0.3	−		Db		68Go.B Z
		6211.7	3.			−0.5	−		Ora		75Ba.B
		6210.6	5.0			−0.1	U		Ara		96Ta18
	ave.	6210.1	0.8			−0.0	1	100	62 ^{203}At		average
$^{203}\text{Rn}(\alpha)^{199}\text{Po}$		6628.6	5.	6629.9	2.1	0.3	3				67Va17 Z
		6630.2	2.5			−0.1	3		Lvn		93Wa04
		6630	10			−0.0	U		Jya		95Uu01
		6629.8	5.0			−0.0	3		Ara		96Ta18
$^{203}\text{Rn}^m(\alpha)^{199}\text{Po}^m$		6679.5	3.	6680.6	1.6	0.3	−				67Va17 Z
		6681.9	10.			−0.1	U		GSa		87He10
		6680.9	2.5			−0.1	−		Lvn		93Wa04
		6683.9	7.			−0.5	o		Ara		95Le04
		6679.8	3.			0.2	−		Jya		96Le09
		6682.9	5.0			−0.5	−		Ara		96Ta18
	ave.	6680.5	1.6			0.1	1	100	97 $^{203}\text{Rn}^m$		average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{203}\text{Fr}(\alpha)^{199}\text{At}$	7275.6	5.	7275	4	-0.1	-					67Va20 Z	
	7281.7	10.			-0.7	-					80Ew03 Z	
	7263.4	25.			0.5	o			Jya		94Le05	
	7273.6	6.			0.2	-			Jya		05Uu02	
	ave. 7276	4			-0.2	1	95	85 ^{203}Fr			average	
$^{203}\text{Fr}^m(\alpha)^{199}\text{At}^m$	7391.9	5.				3			Jya		13Ja06	
$^{203}\text{Ra}(\alpha)^{199}\text{Rn}$	7729.6	20.	7736	6	0.3	o			Jya		96Le09	
	7741.8	8.			-0.7	3			Jya		05Uu02	
	7727.6	10.			0.9	3			Anv		14Ka23	
$^{203}\text{Ra}^m(\alpha)^{199}\text{Rn}^m$	7768.4	20.	7763	6	-0.3	o			Jya		96Le09	
	7765.3	8.			-0.3	3			Jya		05Uu02	
	7760.2	8.2			0.3	3			Anv		14Ka23	
$^{203}\text{Tl}(\text{p},\text{t})^{201}\text{Tl}$	-6240	15	-6241	14	-0.1	1	89	89 ^{201}Tl	Yal		71Ki01	
$^{202}\text{Hg}(\text{d},\text{p})^{203}\text{Hg}-^{204}\text{Hg}(\text{)}^{205}\text{Hg}$	325	5	326	4	0.3	1	52	48 ^{205}Hg	Pit		72Mo12	
$^{203}\text{Tl}(\text{p},\text{d})^{202}\text{Tl}$	-5630	20	-5627.6	1.7	0.1	U			Yal		71Ki01	
$^{203}\text{Au}(\beta^-)^{203}\text{Hg}$	2040	60	2126	3	1.4	U					94We02	
$^{203}\text{Hg}(\beta^-)^{203}\text{Tl}$	489.2	2.	492.1	1.2	1.5	-					54Th17 *	
	493.2	2.			-0.5	-					55Ma40 *	
	493.2	3.			-0.4	-					58Ni28 *	
	ave. 491.6	1.3			0.4	1	92	85 ^{203}Hg			average	
	940	50	975	6	0.7	U					55Ha.A *	
$^{203}\text{Pb}(\epsilon)^{203}\text{Tl}$	980	20			-0.3	1	10	10 ^{203}Pb			65Le07 *	
$^{203}\text{Bi}(\beta^+)^{203}\text{Pb}$	3260	50	3262	14	0.0	U					58No30 *	
$^{203}\text{At}(\beta^+)^{203}\text{Po}$	5060	200	5148	12	0.4	U					87Se04	
$*^{203}\text{Po}(\alpha)^{199}\text{Pb}$	$E_\alpha=5383.8(3,\text{Z})$ to 4(4) level (this is level x<9.3 in Ensdf)										Ens073	**
$*^{203}\text{Hg}(\beta^-)^{203}\text{Tl}$	$E_{\beta^-}=210(2) 214(2) 214(3)$ respectively, to $3/2^+$ level at 279.1958 keV										Ens057	**
$*^{203}\text{Pb}(\epsilon)^{203}\text{Tl}$	$pK=0.36(0.07) 0.71(0.01)$ respectively, to $5/2^+$ level at 680.5164 keV										Ens057	**
$*^{203}\text{Bi}(\beta^+)^{203}\text{Pb}$	$E_{\beta^+}=1350(50), 740(50)$ to levels around 840, 1550 keV										Ens057	**
$^{204}\text{Hg}-\text{C }^{13}\text{C }^{35}\text{Cl}_3 \text{ }^{37}\text{Cl}_2$	131776.05	1.25	131776.0	0.6	-0.0	U			H34	2.5	80Ko25	
$^{204}\text{Hg}-^{169}\text{Tm }^{35}\text{Cl}$	70420	100	70422.4	1.0	0.0	U			R04	4.0	64De15	
$^{204}\text{Hg}-^{167}\text{Er }^{37}\text{Cl}$	75430	60	75535.3	0.6	0.4	U			R04	4.0	64De15	
$^{204}\text{Hg}-\text{u}$	-26505.8	0.6	-26506.0	0.5	-0.3	1	79	79 ^{204}Hg	ST2	1.0	02Bf02	
$^{204}\text{Po}-\text{u}$	-19689	30	-19690	11	-0.0	R			GS2	1.0	05Li24	
	-19426	213			-1.2	U			RI1	1.0	17Sc02 *	
	-12748	30	-12749	24	-0.0	-			GS2	1.0	05Li24	
	-13099	208			1.7	U			RI1	1.0	17Sc02 *	
	ave. -12752	27			0.1	1	84	84 ^{204}At			average	
$^{204}\text{Rn}-^{133}\text{Cs}_{1.534}$	136394	60	136480	8	1.4	U			SH1	1.0	13Dr04	
$^{204}\text{Rn}-\text{u}$	-8660	86	-8556	8	1.2	U			RI1	1.0	17Sc02	
$^{204}\text{Fr}-\text{u}$	595	115	652	26	0.5	U			RI1	1.0	17Sc02 *	
$^{204}\text{Hg }^{35}\text{Cl}_2-^{200}\text{Hg }^{37}\text{Cl}_2$	11066.85	0.55	11067.3	0.7	0.4	1	24	12 ^{200}Hg	H33	2.5	80Ko25	
$^{204}\text{Pb}-^{208}\text{Pb}_{981}$	-4047	21	-4052.11	0.18	-0.2	U			MA6	1.0	01Sc41	
$^{204}\text{Rn}-^{208}\text{Pb}_{981}$	14358.1	9.4	14348	8	-1.1	-			SH1	1.0	13Dr04	
	14303	25			1.8	-			SH1	1.0	13Dr04	
	ave. 14351	9			-0.4	1	81	81 ^{204}Rn			average	
	5807	2	5800.6	0.8	-0.8	U			H17	4.0	64Mc07	
$^{204}\text{Hg }^{35}\text{Cl}-^{202}\text{Hg }^{37}\text{Cl}$	5800.67	0.53			-0.1	1	35	26 ^{202}Hg	H33	2.5	80Ko25	
	1161	25	1149.9	1.3	-0.3	U			R08	1.5	69De19	
$^{204}\text{Pb}(\alpha)^{200}\text{Hg}$	2650	100	1968.5	1.1	-6.8	B					58Ri23	
$^{204}\text{Pb}(\alpha, ^8\text{He})^{200}\text{Pb}$	-28043	13	-28044	10	-0.1	-			INS		90Ka10	
$^{204}\text{Po}(\alpha)^{200}\text{Pb}$	ave. -28044	11			-0.0	1	84	84 ^{200}Pb			average	
	5484.6	1.5	5484.9	1.4	0.2	-			DbA		69Go23 *	
	5486.3	3.			-0.5	-					70Ra14 Z	
$^{204}\text{At}(\alpha)^{200}\text{Bi}$	ave. 5484.9	1.4			0.0	1	100	84 ^{204}Po			average	
	6069.9	3.	6070.4	1.2	0.2	2					63Ho18 Z	
	6066.2	3.			1.4	2					67Tr06 Z	
	6071.3	3.			-0.3	2			Ora		75Ba.B	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{204}\text{At}(\alpha)^{200}\text{Bi}$	6071.3	2.0	6070.4	1.2	-0.4	2					79Sc01
	6072.0	3.			-0.5	2			Db		81Va27 Z
$^{204}\text{Rn}(\alpha)^{200}\text{Po}$	6544.3	3.	6546.7	1.8	0.8	-					67Va17 Z
	6547.5	2.5			-0.3	-			Lvn		93Wa04
	6537.4	7.			1.3	o			Ara		95Le04
	6548.6	5.0			-0.4	-			Ara		96Ta18
	ave. 6546.5	1.8			0.1	1	99	80	^{200}Po		average
$^{204}\text{Fr}(\alpha)^{200}\text{At}$	7170.4	5.	7170.3	2.4	-0.0	4					67Va20 Z
	7169.4	5.			0.2	4					74Ho27 Z
	7170.6	5.			-0.1	4			Lvn		92Hu04 *
	7179.0	6.			-1.4	o			Jya		94Le05
	7167.8	7.			0.3	4			Ara		95Le04
	7173.9	6.			-0.6	o			Jya		05Uu02
	7171.9	5.			-0.3	4			Jya		13Ja06
$^{204}\text{Fr}^m(\alpha)^{200}\text{At}$	7218.8	8.	7221	4	0.2	o			Lvn		92Hu04
$^{204}\text{Fr}^m(\alpha)^{200}\text{At}^m$	7108.2	5.	7107.6	2.0	-0.1	4					74Ho27 Z
	7105.5	3.			0.7	4			Bka		82Bo04 Z
	7108.4	5.			-0.2	4			Lvn		92Hu04 *
	7115.6	7.			-1.1	o			Jya		94Le05 *
	7114.7	7.			-1.0	4			Ara		95Le04
	7117.7	6.			-1.7	o			Jya		05Uu02 *
	7108.7	5.			-0.2	4			Jya		13Ja06
$^{204}\text{Fr}^n(\alpha)^{200}\text{At}^n$	7157.6	6.1	7152.8	2.1	-0.8	o			Jya		05Uu02
	7153.5	5.			-0.1	o			Jya		13Ja06
$^{204}\text{Ra}(\alpha)^{200}\text{Rn}$	7638.1	12.	7637	7	-0.1	3			Ara		95Le04
	7638.1	25.			-0.1	o			Jya		95Le15
	7634.0	10.			0.3	o			Jya		96Le09
	7636.1	8.			0.1	3			Jya		05Uu02
	7638.1	25.			-0.1	U			Anv		10He25
$^{204}\text{Pb}(\text{p,t})^{202}\text{Pb}$	-6835	10	-6830	4	0.5	1	15	14	^{202}Pb	Yal	71Ki01
$^{204}\text{Hg}(\text{t},\alpha)^{203}\text{Au}$	10962	15	10978	3	1.1	U			LAl		81Fi05
$^{204}\text{Hg}(\text{d},^3\text{He})^{203}\text{Au}-^{206}\text{Pb}(\gamma)^{205}\text{Tl}$	-1582.0	3.0	-1582.0	3.0	0.0	1	100	100	^{203}Au		94Gr07
$^{204}\text{Hg}(\text{d,t})^{203}\text{Hg}$	-1242	5	-1235.0	1.6	1.4	1	11	10	^{203}Hg	Ald	70An14
$^{203}\text{Tl}(\text{n},\gamma)^{204}\text{Tl}$	6656.0	0.3	6656.08	0.29	0.3	1	94	70	^{203}Tl	MMn	74Co21 Z
	6654.88	0.14			8.6	C			Bdn		06Fi.A
$^{204}\text{Pb}(\text{p,d})^{203}\text{Pb}$	-6165	10	-6170	6	-0.5	-			Yal		71Ki01
$^{204}\text{Pb}(\text{d,t})^{203}\text{Pb}$	-2160	20	-2137	6	1.1	-			Ald		67Bj01
$^{204}\text{Pb}(\text{p,d})^{203}\text{Pb}$	ave. -6171	9	-6170	6	0.1	1	52	52	^{203}Pb		average
$^{204}\text{Au}(\beta^-)^{204}\text{Hg}$	4500	300	4300#	200#	-0.7	F					67Wa23 *
$^{204}\text{Tl}(\epsilon)^{204}\text{Hg}$	314	20	344.1	1.2	1.5	U					64Ch17
	332	20			0.6	U					66Kl02
	385	20			-2.0	U					73La17
$^{204}\text{Tl}(\beta^-)^{204}\text{Pb}$	764.24	0.31	763.75	0.18	-1.6	-					67Pa08
	763.47	0.22			1.3	-					68Wo02
	ave. 763.73	0.18			0.1	1	97	73	^{204}Tl		average
$^{204}\text{At}(\beta^+)^{204}\text{Po}$	6220	160	6466	25	1.5	U					86Ve.B *
$^{204}\text{Fr}(\text{IT})^{204}\text{Fr}^m$	276.1	0.5				5					95Bi.A
* $^{204}\text{Po}-\text{u}$	A systematic error 75 keV is added to all data in the same paper.										HWJ182 **
*	freq ratio rounded, not used										HWJ182 **
* $^{204}\text{At}-\text{u}$	$M-A=-11908(94)$ keV for mixture gs+m at 587.3(0.2) keV										Nub211 **
* $^{204}\text{Fr}-\text{u}$	$M-A=679(79)$ keV for mixture at gs+m+n 50(4) 326(4) keV										Nub211 **
* $^{204}\text{Po}(\alpha)^{200}\text{Pb}$	Printing error in reference: ^{204}Po not ^{206}Po ; Z recalibrated										AHW **
* $^{204}\text{Fr}(\alpha)^{200}\text{At}$	$E_\alpha=7031(5)$, 6916(8) to ground state, 113 level										92Hu04 **
* $^{204}\text{Fr}^m(\alpha)^{200}\text{At}^m$	$E_\alpha=6969(5)$; and 7013(5) from $^{204}\text{Fr}^n$ 276.1 above $^{204}\text{Fr}^m$ to $^{200}\text{At}^n$										95Bi.A **
*	230.9 above $^{200}\text{At}^m$										92Hu04 **
* $^{204}\text{Fr}^m(\alpha)^{200}\text{At}^m$	$E_\alpha=7020(7)$ from $^{204}\text{Fr}^n$ 276.1 above Fr^m to $^{200}\text{At}^n$ 230.9 above $^{200}\text{At}^m$										95Bi.A **
* $^{204}\text{Fr}^m(\alpha)^{200}\text{At}^m$	$E_\alpha=6976(6)$; and 7017(6) from $^{204}\text{Fr}^n$ 276.1 above $^{204}\text{Fr}^m$ to $^{200}\text{At}^n$										GAu **
* $^{204}\text{Au}(\beta^-)^{204}\text{Hg}$	F : reported 4 s activity does not exist										Ens87a **
* $^{204}\text{At}(\beta^+)^{204}\text{Po}$	$E_{\beta^+}=2950(160)$ to 8^+ level at 2248.17 keV										Ens102 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$C_{16}H_{13}-^{205}Tl$	127345	29	127298.1	1.3	-1.1	U			R08	1.5	69De19
$C_{14}N_2H_9-^{205}Tl$	102091	36	102146.0	1.3	1.0	U			R08	1.5	69De19
$^{205}Tl-^{168}Er-^{37}Cl$	76198	44	76146.5	1.4	-0.8	U			R08	1.5	69De19
$^{205}Tl-^{170}Er-^{35}Cl$	70034	23	70102.7	2.0	2.0	U			R08	1.5	69De19
$^{205}Tl-^{133}Cs_{1.541}$	120129	11	120125.9	1.3	-0.3	U			MA8	1.0	08We02
$^{205}Bi-u$	-22559	30	-22615	5	-1.9	U			GS2	1.0	05Li24
	-22358	616			-0.4	U			RI1	1.0	17Sc02
$^{205}Po-u$	-18773	30	-18810	11	-1.2	-			GS2	1.0	05Li24
	-18811	538			0.0	U			RI1	1.0	17Sc02
ave.	-18790	25			-0.8	1	19	19 ^{205}Po			average
$^{205}At-u$	-13841	111	-13939	13	-0.9	U			RI1	1.0	17Sc02
$^{205}Rn-^{133}Cs_{1.541}$	137456	50	137422	5	-0.7	U			SH1	1.0	13Dr04
	137458	29			-1.2	U			SH1	1.0	13Dr04
$^{205}Rn-u$	-8494	222	-8277	5	1.0	U			RI1	1.0	17Sc02
$^{205}Fr-^{133}Cs_{1.541}$	144293.8	9.7	144292	8	-0.1	2			MA8	1.0	08We02
$^{205}Rn-^{208}Pb_{.986}$	14748	11	14744	6	-0.3	-			SH1	1.0	13Dr04
	14772	20			-1.4	-			SH1	1.0	13Dr04
ave.	14754	10			-1.0	1	33	32 ^{205}Rn			average
$^{205}Tl-^{35}Cl-^{203}Tl-^{37}Cl$	5040	4	5033.3	0.6	-0.4	U			H17	4.0	64Mc07
	5031.43	1.07			0.7	-			H36	2.5	85De40
	5032.88	1.01			0.3	-			H42	1.5	93Si05
ave.	5032.5	1.3			0.6	1	19	15 ^{205}Tl			average
$^{205}Tl-^{167}Er-^{37}Cl$	76426	47	76468.6	1.4	0.6	U			R08	1.5	69De19
$^{205}Tl-^{169}Tm-^{35}Cl$	71355	25	71355.7	1.5	0.0	U			R08	1.5	69De19
$^{169}Tm-^{37}Cl-^{205}Tl$	-74316	32	-74305.8	1.5	0.2	U			R08	1.5	69De19
$^{205}Tl-^{204}Hg$	938	27	933.3	1.4	-0.1	U			R08	1.5	69De19
$^{205}Tl-^{203}Tl$	2092	20	2083.2	0.6	-0.3	U			R08	1.5	69De19
$^{205}Po(\alpha)^{201}Pb$	5324.1	10.	5325	10	0.1	1	95	90 ^{201}Pb			67Ti04
$^{205}At(\alpha)^{201}Bi$	6016.3	4.	6019.6	1.7	0.8	3					63Ho18 Z
	6020.5	2.			-0.5	3			DbA		68Go.B Z
	6018.9	5.			0.1	3					74Ho27 Z
$^{205}Rn(\alpha)^{201}Po$	6386.6	3.	6386.5	1.8	-0.0	-					67Va17 Z
	6386.6	6.			-0.0	-					71Ho01 Z
	6385.7	2.5			0.3	-			Lvn		93Wa04
ave.	6386.1	1.9			0.2	1	97	68 ^{205}Rn			average
$^{205}Fr(\alpha)^{201}At$	7056.5	5.	7054.7	2.4	-0.3	3					67Va20 Z
	7052.2	5.			0.5	3					74Ho27 Z
	7057.3	5.			-0.5	3			ORa		81Ri04 Z
	7052.9	7.			0.3	3			Ara		95Le04
	7053.9	5.			0.2	3			Anv		05De01
$^{205}Ra(\alpha)^{201}Rn$	7506.7	20.	7486	20	-0.4	F			GSa		87He10 *
	7496.6	25.			-0.4	o			Jya		95Le15
	7486.4	20.				3			Jya		96Le09
$^{205}Ra^m(\alpha)^{201}Rn^m$	7501.7	10.	7505	9	0.3	4			Ara		95Le04
	7522.1	25.			-0.7	o			Jya		95Le15
	7517.0	20.			-0.6	4			Jya		96Le09
	7526.1	30.			-0.7	U			Anv		10He25
$^{205}Ac(\alpha)^{201}Fr$	8093.2	30.6				3			Lza		14Zh03
$^{204}Hg(d,p)^{205}Hg$	3443	5	3444	4	0.3	1	52	52 ^{205}Hg	Ald		70An14
$^{205}Tl(\gamma,n)^{204}Tl$	-7515	29	-7546.1	0.5	-1.1	U			Phi		60Ge01
	-7548	3			0.6	U			McM		79Ba06
$^{205}Tl(d,t)^{204}Tl$	-1288.7	0.6	-1288.8	0.5	-0.2	1	64	60 ^{205}Tl	Mun		90Li40
$^{204}Pb(n,\gamma)^{205}Pb$	6731.53	0.15	6731.67	0.11	0.9	-			ILn		83Hu13 Z
	6731.80	0.16			-0.8	-			Bdn		06Fi.A
$^{204}Pb(d,p)^{205}Pb$	4516	20	4507.10	0.11	-0.4	U			Ald		67Bj01
$^{204}Pb(n,\gamma)^{205}Pb$	ave.	6731.66	6731.67	0.11	0.1	1	99	70 ^{204}Pb			average

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{205}\text{Hg}(\beta^-)^{205}\text{Tl}$	1620	200	1533	4	-0.4	U					40Kr08
	1750	200			-1.1	U					51Ly10
$^{205}\text{Pb}(\epsilon)^{205}\text{Tl}$	41.4	1.1	50.6	0.5	8.4	B					78Pe08
$^{205}\text{Bi}(\beta^+)^{205}\text{Pb}$	2701.4	10.	2705	5	0.3	-					62Bo25 *
	2715.4	10.			-1.1	-					62Pe08 *
ave.	2708	7			-0.5	1	46	45 ^{205}Bi			average
$^{205}\text{Po}(\beta^+)^{205}\text{Bi}$	3390	150	3544	11	1.0	U					69Ho37 *
$^{205}\text{Po}-\text{u}$	$M-A=-16791(268)$ keV for mixture at gs+p 1461.2(0.2) keV										Nub211 **
$^{205}\text{Rn}-\text{u}$	$M-A=-7584(82)$ keV for mixture at gs+m 657.1(0.5) keV										Nub211 **
$^{205}\text{Ra}(\alpha)^{201}\text{Rn}$	F : possibly mixed with $^{205}\text{Ra}^m(\alpha)^{201}\text{Rn}^m$										87He10 **
$^{205}\text{Bi}(\beta^+)^{205}\text{Pb}$	$E_{\beta^+}=976(10)$ 990(10) respectively, to $7/2^-$ level at 703.427 keV										Ens044 **
$^{205}\text{Po}(\beta^+)^{205}\text{Bi}$	$p^+=3(1)\times 10^{-3}$ to $7/2^-$ level at 849.84 and $7/2^-$ at 1001.22 keV										Ens044 **
$^{206}\text{Bi}-\text{u}$	-21429	30	-21501	8	-2.4	U			GS2	1.0	05Li24
$^{206}\text{Po}-\text{u}$	-19471	30	-19526	4	-1.8	U			GS2	1.0	05Li24
$^{206}\text{At}-\text{u}$	-13305	30	-13354	15	-1.6	1	23	23 ^{206}At	GS2	1.0	05Li24
	-13416	2309			0.0	U			RI1	1.0	17Sc02
$^{206}\text{Rn}-\text{u}$	-9195	649	-9805	9	-0.9	U			RI1	1.0	17Sc02
$^{206}\text{Rn}-^{133}\text{Cs}_{1.549}$	136641	15	136650	9	0.6	1	38	38 ^{206}Rn	SH1	1.0	13Dr04
$^{206}\text{Fr}-\text{u}$	-1520	220	-1339	30	0.8	U			RI1	1.0	17Sc02 *
$^{206}\text{Pb }^{35}\text{Cl}_2-^{202}\text{Hg }^{37}\text{Cl}_2$	9722.09	0.57	9721.8	1.1	-0.2	1	62	54 ^{206}Pb	H36	2.5	85De40
$^{206}\text{Pb }^{35}\text{Cl}-^{204}\text{Hg }^{37}\text{Cl}$	3929	4	3921.3	1.3	-0.5	U			H17	4.0	64Mc07
$^{206}\text{Pb }^{35}\text{Cl}-^{204}\text{Pb }^{37}\text{Cl}$	4378	3	4371.82	0.15	-0.5	U			H17	4.0	64Mc07
	4370.72	1.17			0.4	U			H36	2.5	85De40
	4371.29	0.81			0.4	U			H42	1.5	93Si05
$^{206}\text{Rn}-^{208}\text{Pb}_{990}$	13307	15	13310	9	0.2	1	38	37 ^{206}Rn	SH1	1.0	13Dr04
$^{206}\text{Po}(\alpha)^{202}\text{Pb}$	5327.4	4.	5327.0	1.3	-0.1	2					67Ti04 Z
	5327.4	1.5			-0.3	2			Db		69Go23 *
	5325.1	3.			0.6	2					70Ra14 Z
$^{206}\text{At}(\alpha)^{202}\text{Bi}$	5884.4	3.6	5887	5	0.6	o			Db		68Go.B *
	5886.4	5.			0.0	1	98	75 ^{202}Bi	Db		81Va27 *
$^{206}\text{Rn}(\alpha)^{202}\text{Po}$	6381.8	3.	6383.7	1.6	0.6	-					67Va17 Z
	6384.6	3.			-0.3	-			Db		71Go35 Z
	6384.8	2.5			-0.4	-			Lvn		93Wa04
ave.	6383.9	1.6			-0.1	1	99	75 ^{202}Po			average
$^{206}\text{Fr}(\alpha)^{202}\text{At}$	6925.9	7.	6923	3	-0.4	-					67Va20 *
	6918.9	7.			0.6	-					74Ho27 *
	6924.0	7.			-0.1	-			ORa		81Ri04 *
	6924.8	7.			-0.2	-			Lvn		92Hu04 *
ave.	6923	4			-0.0	1	100	97 ^{206}Fr			average
$^{206}\text{Fr}^n(\alpha)^{202}\text{At}^n$	7068.8	5.	7068	4	-0.2	6			ORa		81Ri04 Z
	7067.1	5.			0.2	6			Lvn		92Hu04 *
$^{206}\text{Ra}(\alpha)^{202}\text{Rn}$	7416.3	5.	7415	4	-0.2	3					67Va22 Z
	7414.3	10.			0.1	3			GSa		87He10
	7412.2	10.			0.3	o			Jya		95Le15
	7406	15			0.6	o			Jya		95Uu01
	7412.2	10.			0.3	3			Jya		96Le09
$^{206}\text{Ac}(\alpha)^{202}\text{Fr}$	7944.6	30.	7960	60	0.2	3			Jya		98Es02
	7972.0	30.			-0.2	3			Lza		14Zh03
$^{206}\text{Ac}^m(\alpha)^{202}\text{Fr}^m$	7903.8	30.				2			Jya		98Es02
$^{204}\text{Pb}(\text{t,p})^{206}\text{Pb}$	6322	20	6336.53	0.12	0.7	U			Ald		67Ha.A
$^{204}\text{Pb}(\alpha,\text{d})^{206}\text{Bi}$	-15798.	11.5	-15792	8	0.5	R			Pit		76Da20
$^{205}\text{Tl}(\text{n},\gamma)^{206}\text{Tl}$	6503.7	0.4	6503.8	0.4	0.3	1	93	84 ^{206}Tl	MMn		74Co21 Z
	6502.87	0.27			3.5	C			Bdn		06Fi.A
$^{205}\text{Tl}(\text{d,p})^{206}\text{Tl}$	4276	5	4279.2	0.4	0.6	U			ANL		65Er02

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{205}\text{Tl}(^3\text{He},d)^{206}\text{Pb}$	1761.7	1.4	1760.2	0.5	-1.1	1	13	12 ^{205}Tl	Mun		90Li40
$^{205}\text{Pb}(n,\gamma)^{206}\text{Pb}$	8086.66	0.06	8086.66	0.06	0.0	1	100	69 ^{205}Pb			96Ra16 Z
$^{206}\text{Pb}(\gamma,n)^{205}\text{Pb}$	-8090	70	-8086.66	0.06	0.0	U			Phi		60Ge01
	-8087	3			0.1	U			McM		79Ba06
$^{206}\text{Pb}(d,t)^{205}\text{Pb}$	-1830	100	-1829.43	0.06	0.0	U			MIT		53Ha66
	-1831.2	0.5			3.5	B			Mun		90Li40
$^{206}\text{Hg}(\beta^-)^{206}\text{Tl}$	1240	62	1308	20	1.1	U					68Wo09 *
$^{206}\text{Tl}(\beta^-)^{206}\text{Pb}$	1534	5	1532.2	0.6	-0.4	U					71Pe23
	1527	4			1.3	U					72Wi18
$^{206}\text{Bi}(\beta^+)^{206}\text{Pb}$	3683	33	3757	8	2.3	U					62Pe08 *
$^{206}\text{Bi}(\epsilon)^{206}\text{Pb}$	3753	10			0.4	2					74Go20 *
$^{206}\text{At}(\beta^+)^{206}\text{Po}$	5687	150	5749	14	0.4	U					77Li16 *
$^{206}\text{Fr}^n(\text{IT})^{206}\text{Fr}^m$	531	2				7			ORa		81Ri04
$^{206}\text{Fr}^x(\text{IT})^{206}\text{Fr}$	100	100				2					AHW *
$^{206}\text{Fr}-u$	$M-A=-1104(131)$ keV for mixture gs+m+n at 200(40) 730(40) keV										Nub211 **
$^{206}\text{Po}(\alpha)^{202}\text{Pb}$	Printing error in reference: ^{206}Po not ^{211}Po ; Z recalibrated										AHW **
$^{206}\text{At}(\alpha)^{202}\text{Bi}$	$E_\alpha=5702.8(2,Z)$ to $(5)^+$ level at 68(3) keV										Ens044 **
$^{206}\text{At}(\alpha)^{202}\text{Bi}$	$E_\alpha=5773.8(5,Z)$, 5702.8(5,Z) to ground state and $(5)^+$ level at 68(3) keV										Ens044 **
$^{206}\text{Fr}(\alpha)^{202}\text{At}$	$E_\alpha=6793.1(5,Z)$; correction -2 for being a doublet										AHW **
$^{206}\text{Fr}(\alpha)^{202}\text{At}$	$E_\alpha=6786.3(5,Z)$; correction -2 for being a doublet										AHW **
$^{206}\text{Fr}(\alpha)^{202}\text{At}$	$E_\alpha=6791.3(5,Z)$; correction -2 for being a doublet										AHW **
$^{206}\text{Fr}(\alpha)^{202}\text{At}$	$E_\alpha=6792(5)$; correction -2 for being a doublet										AHW **
$^{206}\text{Fr}^n(\alpha)^{202}\text{At}^n$	$E_\alpha=6930(5)$ and 6792(7) combined with $E(\gamma)$'s 531, 391.7 keV										92Hu04 **
$^{206}\text{Hg}(\beta^-)^{206}\text{Tl}$	$E_{\beta^-}=935(62)$ to 1^- level at 304.896 keV										Ens085 **
$^{206}\text{Bi}(\beta^+)^{206}\text{Pb}$	$E_{\beta^+}=977(33)$ to 4^+ level at 1683.99 keV										Ens085 **
$^{206}\text{Bi}(\epsilon)^{206}\text{Pb}$	LK=0.509(0.015) to 5^- level at 3562.92 keV, original error 22, recalculated										Ens085 **
$^{206}\text{At}(\beta^+)^{206}\text{Po}$	$E_{\beta^+}=3092(150)$ to 6^+ level at 1573.38 keV										Ens085 **
$^{206}\text{Fr}^x(\text{IT})^{206}\text{Fr}$	Assuming a 0.15(0.20)% isomeric mixture										AHW **
$^{207}\text{Hg}-u$	-17721	33	-17700	30	0.6	o			GS3	1.0	08Ch.A
	-17700	32				2			GS3	1.0	12Ch19
$^{207}\text{Rn}-^{133}\text{Cs}_{1.556}$	137794	28	137847	5	1.9	U			SH1	1.0	13Dr04
$^{207}\text{Fr}-^{133}\text{Cs}_{1.556}$	144062	20	144058	19	-0.2	1	89	89 ^{207}Fr	MA8	1.0	14Bo26
$^{207}\text{Pb }^{35}\text{Cl}-^{205}\text{Tl }^{37}\text{Cl}$	4413	4	4419.6	0.6	0.4	U			H17	4.0	64Mc07
	4415.60	2.40			0.7	U			H36	2.5	85De40
	4417.32	1.40			1.1	U			H42	1.5	93Si05
$^{206}\text{Pb H}-^{207}\text{Pb}$	6394.2	1.1	6393.42	0.10	-0.3	U			C4	2.5	71Ke02
$^{206}\text{Fr}^x-^{207}\text{Fr}_{.498} \text{ } ^{205}\text{Fr}_{.502}$	930	90	930	100	-0.0	U			P24	2.5	82Au01
$^{207}\text{Po}(\alpha)^{203}\text{Pb}$	5216.0	2.5	5215.9	2.5	-0.0	1	96	59 ^{207}Po	DBa		70Af.A
$^{207}\text{At}(\alpha)^{203}\text{Bi}$	5872.5	3.				3			DBa		69Go23 Z
$^{207}\text{Rn}(\alpha)^{203}\text{Po}$	6256.3	3.	6251.2	1.6	-1.7	-					67Va20 Z
	6247.3	3.			1.2	-			DBa		71Go35 Z
	6250.4	2.5			0.3	-			Lvn		93Wa04
	ave. 6251.2	1.6			-0.0	1	97	66 ^{207}Rn			average
$^{207}\text{Fr}(\alpha)^{203}\text{At}$	6907.8	5.	6889	20	-0.4	-					67Va20 Z
	6895.8	5.			-0.1	-					74Ho27 Z
	6900.9	5.			-0.2	-			ORa		81Ri04 Z
	ave. 6902	29			-0.2	1	12	9 ^{207}Fr			average
$^{207}\text{Ra}(\alpha)^{203}\text{Rn}$	7273.8	5.	7270	60	-0.0	4					67Va22 Z
	7268.7	10.			0.1	4			GSa		87He10
	7276.9	12.2			-0.1	4			Jya		95Uu01
$^{207}\text{Ra}^m(\alpha)^{203}\text{Rn}^m$	7464.5	10.2	7468	8	0.3	2			GSa		87He10
	7474.7	15.			-0.4	o			Jya		95Le15
	7475.7	15.			-0.5	2			Jya		96Le09
$^{207}\text{Ac}(\alpha)^{203}\text{Fr}$	7864.3	25.	7840	60	-0.3	o			Jya		94Le05
	7844.9	25.				2			Jya		98Es02

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{205}\text{Tl}(\text{t,p})^{207}\text{Tl}$	4880	15	4874	5	-0.4	1	13	13 ^{207}Tl	Ald		69Ha11
$^{207}\text{Pb}(\text{t},\alpha)^{206}\text{Tl}$	12321	25	12326.2	0.6	0.2	U			Ald		67Ha.A
$^{206}\text{Pb}(\text{n},\gamma)^{207}\text{Pb}$	6737.85	0.15	6737.78	0.10	-0.5	-			MMn		81Ke11 Z
	6737.72	0.18			0.3	-			ILn		83Hu13 Z
	6737.74	0.17			0.2	-			Bdn		06Fi.A
$^{207}\text{Pb}(\gamma,\text{n})^{206}\text{Pb}$	-6742	3	-6737.78	0.10	1.4	U			McM		79Ba06
$^{206}\text{Pb}(\text{d,p})^{207}\text{Pb}$	4480	30	4513.21	0.10	1.1	U			MIT		53Ha66
	4510	20			0.2	U					58Mc64
	4526	30			-0.4	U			Pit		64Co11
$^{206}\text{Pb}(\text{n},\gamma)^{207}\text{Pb}$	ave.	6737.78	0.10	6737.78	0.10	0.0	1	100	86 ^{207}Pb		average
$^{207}\text{Hg}(\beta^-)^{207}\text{Tl}$	4815	150	4550	30	-1.8	U					81Jo.B *
$^{207}\text{Tl}(\beta^-)^{207}\text{Pb}$	1431	8	1418	5	-1.7	1	46	45 ^{207}Tl			67Da10
$^{207}\text{Bi}(\epsilon)^{207}\text{Pb}$	2392	10	2397.4	2.1	0.5	U					Averag *
$^{207}\text{Po}(\beta^+)^{207}\text{Bi}$	2907	10	2909	7	0.2	1	44	41 ^{207}Po			58Ar56 *
$^{207}\text{Rn}(\beta^+)^{207}\text{At}$	4617	70	4593	13	-0.3	U					75Ze.A *
$^{*207}\text{Hg}(\beta^-)^{207}\text{Tl}$	$E_{\beta^-}=1800(150)$, 14%, 32%, 16%, 7% to $(7/2^-, 9/2)$ level at 2911.83 keV										81Jo.B **
*	$(9/2)^+$ at 2985.23 and 3104.43, $(7/2^-, 9/2, 11/2)$ at 3143.1 keV										Ens112 **
$^{*207}\text{Bi}(\epsilon)^{207}\text{Pb}$	Average $pL=0.61(0.05)$ to $7/2^-$ level at 2339.921 keV from two references:										Ens112 **
*	$pL=0.663(0.014)$										64De16 **
*	$pL=0.56(0.04)$; original error 0.08 is 2σ										82Ta18 **
$^{*207}\text{Po}(\beta^+)^{207}\text{Bi}$	$E_{\beta^+}=893(10)$ to $7/2^-$ level at 992.43 keV, and other E_{β^+}										Ens112 **
$^{*207}\text{Rn}(\beta^+)^{207}\text{At}$	$E_{\beta^+}=3250(70)$ to $7/2^-$ level at 344.55 keV										Ens112 **
$^{208}\text{Hg}-\text{u}$	-14241	33	-14240	30	0.0	o			GS3	1.0	08Ch.A
	-14241	33				2			GS3	1.0	09Ch08
$^{208}\text{Pb}-^{133}\text{Cs}_{1.564}$	124532.0	5.6	124525.1	1.2	-1.2	U			MA8	1.0	08We02
	124524.3	5.5			0.2	U			MA8	1.0	14Bo26
$^{208}\text{Po}-\text{u}$	-18710	31	-18754.0	1.8	-1.4	U			GS2	1.0	05Li24
$^{208}\text{Po}-^{133}\text{Cs}_{1.564}$	129125.3	6.0	129119.2	1.8	-1.0	1	9	9 ^{208}Po	MA8	1.0	17Al34
$^{208}\text{Fr}-^{133}\text{Cs}_{1.564}$	144984	20	145012	13	1.4	o			MA8	1.0	12Bo.A
	145030	16			-1.1	o			MA8	1.0	12Bo.A
	145012	13			0.0	1	93	93 ^{208}Fr	MA8	1.0	14Bo26
$^{208}\text{Pb } ^{35}\text{Cl}-^{206}\text{Pb } ^{37}\text{Cl}$	5136	2	5136.92	0.14	0.1	U			H17	4.0	64Mc07
	5136.23	1.08			0.3	U			H36	2.5	85De40
	5136.93	0.41			-0.0	U			H42	1.5	93Si05
$^{207}\text{Fr}-^{208}\text{Fr}_{.498} \text{ } ^{206}\text{Fr}_{.502}$	-890	60	-950	60	-0.4	U			P24	2.5	82Au01
$^{208}\text{Po}(\alpha)^{204}\text{Pb}$	5216.3	2.	5215.7	1.3	-0.3	-			DbA		69Go23 Z
	5214.0	3.			0.6	-					70Ra14 Z
	5215.1	2.			0.3	-					89Ma05
	ave.	5215.4	1.3		0.2	1	95	91 ^{208}Po			average
$^{208}\text{At}(\alpha)^{204}\text{Bi}$	5750.6	3.	5751.1	2.2	0.2	3			DbA		69Go23 Z
	5751.6	3.			-0.2	3			DbA		81Va27 Z
$^{208}\text{Rn}(\alpha)^{204}\text{Po}$	6269.3	4.	6260.7	1.7	-2.1	-					55Mo69 Z
	6260.0	3.			0.2	-			DbA		71Go35 Z
	6257.5	5.			0.6	-					74Ho27
	6258.7	2.5			0.8	-			Lvn		93Wa04
$^{208}\text{Fr}(\alpha)^{204}\text{At}$	ave.	6260.7	1.7		0.0	1	100	83 ^{208}Rn			average
	6778.3	5.	6785	25	0.1	-					67Va20 Z
	6767.7	5.			0.4	-					74Ho27 Z
	6767.7	5.			0.4	-			ORa		81Ri04 Z
$^{208}\text{Ra}(\alpha)^{204}\text{Rn}$	6739.8	51.0			0.6	-			GSa		15De22
	6772.4	18.4			0.2	-			Lza		19Zh23
	ave.	6768	24		0.3	1	20	16 ^{204}At			average
	7273.1	5.				2					67Va22 Z
$^{208}\text{Ac}(\alpha)^{204}\text{Fr}$	7720.8	15.	7730	60	0.1	5			Jya		94Le05
$^{208}\text{Ac}^m(\alpha)^{204}\text{Fr}^m$	7769.7	40.			-0.6	5			Jaa		96Ik01
	7707.5	21.4			0.4	5			Lza		14Ya19
	7892.1	20.	7899	14	0.3	6			Dbb		94An01
	7910.4	20.			-0.6	6			Jya		94Le05
	7871.7	50.			0.5	6			Jaa		96Ik01

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{208}\text{Th}(\alpha)^{204}\text{Ra}$	8202.0	30.				4			Anv		10He25
$^{206}\text{Pb}(\text{t,p})^{208}\text{Pb}$	5622	30	5623.85	0.11	0.1	U			Ald		67Ha.A
$^{207}\text{Pb}(\text{n},\gamma)^{208}\text{Pb}$	7367.95	0.15	7367.87	0.05	-0.5	-			MMn		81Ke11 Z
	7367.96	0.10			-0.9	-					81Su.A Z
	7367.81	0.11			0.5	-			ILn		83Hu13 Z
	7367.774	0.098			1.0	-					98Be19 Z
	7367.92	0.16			-0.3	-			Bdn		06Fi.A
$^{208}\text{Pb}(\gamma,\text{n})^{207}\text{Pb}$	-7370	3	-7367.87	0.05	0.7	U			McM		79Ba06
$^{208}\text{Pb}(\text{d,t})^{207}\text{Pb}$	-1114	25	-1110.64	0.05	0.1	U			Pit		64Co11
$^{207}\text{Pb}(\text{n},\gamma)^{208}\text{Pb}$	ave. 7367.87	0.05	7367.87	0.05	-0.0	1	100	87 ^{208}Pb			average
$^{208}\text{Tl}(\beta^-)^{208}\text{Pb}$	4989.7	7.	4998.4	1.7	1.2	U					48Ma29 *
	4997.7	10.			0.1	U					54El24 *
$^{208}\text{Bi}(\epsilon)^{208}\text{Pb}$	2810	4	2878.4	2.0	17.1	B					59Mi19 *
$^{208}\text{Tl}(\beta^-)^{208}\text{Pb}$	$E_{\beta^-}=1792(7)$ 1800(10) respectively, to 5^- level at 3197.711 keV										Ens077 **
$^{208}\text{Bi}(\epsilon)^{208}\text{Pb}$	$\text{pK}=0.24(0.01)$ to 3^- level at 2614.522 keV, recalculated										Ens077 **
$^{209}\text{Bi}-^{133}\text{Cs}_{1.571}$	128937.6	4.7	128933.6	1.5	-0.9	U			MA8	1.0	08We02
$^{209}\text{Fr}-^{226}\text{Ra}_{925}$	-27584	36	-27563	12	0.6	-			MA3	1.0	92Bo28
$^{209}\text{Bi}-^{35}\text{Cl}-^{207}\text{Pb}-^{37}\text{Cl}$	ave. -27550	16			-0.8	1	62	62 ^{209}Fr			average
	7444	3	7451.9	0.8	0.7	U			H17	4.0	64Mc07
	7454.13	1.51			-0.6	U			H36	2.5	85De40
$^{208}\text{Fr}-^{209}\text{Fr}_{498}$ $^{207}\text{Fr}_{502}$	720	60	649	16	-0.5	U			P24	2.5	82Au01
$^{209}\text{Bi}(\alpha)^{205}\text{Tl}$	3137.0	2.2	3137.3	0.8	0.1	1	12	10 ^{209}Bi			03De11
$^{209}\text{Po}(\alpha)^{205}\text{Pb}$	4974	5	4979.2	1.4	1.0	2					66Ha29 *
	4980.0	2.			-0.4	2			DbA		69Go23 *
	4979.3	2.			-0.0	2					89Ma05 *
$^{209}\text{At}(\alpha)^{205}\text{Bi}$	5757.2	2.	5756.9	2.0	-0.2	1	96	55 ^{205}Bi	DbA		69Go23 Z
$^{209}\text{Rn}(\alpha)^{205}\text{Po}$	6157.5	3.	6155.4	2.0	-0.7	-			DbA		71Go35 Z
	6154.2	2.5			0.5	-			Lvn		93Wa04
$^{209}\text{Fr}(\alpha)^{205}\text{At}$	ave. 6155.5	2.0			-0.1	1	99	75 ^{205}Po			average
	6777.7	5.	6777	4	-0.0	2					67Va20 Z
	6777.3	5.			0.0	2					74Ho27 Z
$^{209}\text{Ra}(\alpha)^{205}\text{Rn}$	7147.0	5.	7143.1	2.7	-0.8	2					67Va22 Z
	7141	5			0.4	2			GSa		03He06 *
	7142.0	4.			0.3	2					08Ha12
$^{209}\text{Ac}(\alpha)^{205}\text{Fr}$	7733.3	15.	7730	60	-0.1	3					68Va04
	7738.4	20.			-0.2	3			Dbb		94An01
	7729.2	15.			0.0	3			Jya		94Le05
	7728.2	40.			0.0	U			JAa		96Ik01
	7725.1	10.			0.1	3			GSa		00He17
	7723.0	23.			0.1	3			Lza		14Ya19
$^{209}\text{Th}^m(\alpha)^{205}\text{Ra}^m$	8238.0	50.	8273	23	0.7	5			JAa		96Ik01 *
	8281.8	25.			-0.3	5			Anv		10He25 *
$^{207}\text{Pb}(\text{t,p})^{209}\text{Pb}$	2814	12	2823.4	1.3	0.8	U			Ald		68Bj03
$^{209}\text{Bi}(\text{p,t})^{207}\text{Bi}$	-5864.8	2.0	-5864.9	2.0	-0.0	1	98	97 ^{207}Bi	MSU		76Be.B *
$^{208}\text{Pb}(\text{d,p})^{209}\text{Pb}$	1705	15	1712.8	1.3	0.5	U			MIT		64Sp12
	1700	10			1.3	U					67Mu16
	1718	4			-1.3	1	11	11 ^{209}Pb	Pit		72Ko03 *
	1715	10			-0.2	U			Yal		74Ko20
$^{209}\text{Bi}(\text{t},\alpha)^{208}\text{Pb}$	16003	25	16014.8	0.8	0.5	U			Ald		68Bj01
$^{209}\text{Bi}(\gamma,\text{n})^{208}\text{Bi}$	-7432	10	-7459.8	1.9	-2.8	U			Phi		60Ge01
	-7460	2			0.1	2			McM		79Ba06

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{209}\text{Bi}(\text{d,t})^{208}\text{Bi}$	-1216	30	-1202.5	1.9	0.4	U			Pit		64Co11
	-1201	5			-0.3	2			ANL		64Er06
$^{209}\text{Pb}(\beta^-)^{209}\text{Bi}$	644.6	1.2	644.0	1.1	-0.5	1	91	87 ^{209}Pb			72Be44
$^{209}\text{Rn}(\beta^+)^{209}\text{At}$	3928	40	3943	11	0.4	R					74Vy01 *
$^{209}\text{Po}(\alpha)^{205}\text{Pb}$	$E_\alpha=4876.8(5,Z)$ 4882.8(2,Z) respectively, to (20% ground state + 80% $^{205}\text{Pb}^m$ at 2.329 keV)										Ens044 **
$^{209}\text{Po}(\alpha)^{205}\text{Pb}$	$E_\alpha=4882.6(2.0)$ 4622(5) to (ground state + 80% $^{205}\text{Pb}^m$ at 2.329), $3/2^-$ at 262.8keV										Ens044 **
$^{209}\text{Ra}(\alpha)^{205}\text{Rn}$	$E_\alpha=7003(10)$ to ground state, 6625(5) to 387.0 level										03He06 **
$^{209}\text{Th}^m(\alpha)^{205}\text{Ra}^m$	the decay is from ^{209}Th isomer, following $\text{Ea}_1-\text{Ea}_2-\text{Ea}_3-\text{Ea}_4$ correlations										FGK141 **
$^{209}\text{Bi}(\text{p,t})^{207}\text{Bi}$	$Q-Q(^{208}\text{Pb}(\text{p,t}))=-241(2,\text{Be})$, $Q(\text{Pb})=-5623.82(0.20)$ keV										AHW **
$^{208}\text{Pb}(\text{d,p})^{209}\text{Pb}$	$Q-Q(^{209}\text{Bi}(\text{d,p}))=-662(4)$, $Q(\text{Bi})=2380.01(0.14)$ keV										AHW **
$^{209}\text{Rn}(\beta^+)^{209}\text{At}$	$E_{\beta^+}=2160(40)$ to $7/2^-$ level at 745.81 keV										Ens156 **
$^{210}\text{Fr}-^{226}\text{Ra}_{929}$	-27198	24	-27194	14	0.2	1	36	36 ^{210}Fr	MA3	1.0	92Bo28
$^{210}\text{Ra}-\text{u}$	441.3	35.5	475	10	1.0	U			RII	1.0	18Ro14
$^{210}\text{Ac}-\text{u}$	9260	169	9410	70	0.9	1	16	16 ^{210}Ac	RII	1.0	18Ro14
$^{209}\text{Fr}-^{210}\text{Fr}_{498}$ $^{208}\text{Fr}_{502}$	-770	50	-779	15	-0.1	U			P24	2.5	82Au01
$^{210}\text{Pb}(\alpha)^{206}\text{Hg}$	3792.4	20.				2					62Ka27
$^{210}\text{Bi}(\alpha)^{206}\text{Tl}$	5042.8	2.	5036.5	0.8	-3.2	B			Orm		60Wa14 *
	5037.3	1.1			-0.8	1	50	33 ^{210}Bi			76Tu.A *
$^{210}\text{Po}(\alpha)^{206}\text{Pb}$	5407.53	0.07	5407.53	0.07	0.0	1	100	98 ^{210}Po			73Go39 Z
	5407.7	2.			-0.1	U					89Ma05
$^{210}\text{At}(\alpha)^{206}\text{Bi}$	5630.9	1.5	5631.2	1.0	0.2	3			DbA		69Go23 *
	5631.4	1.3			-0.2	3			DbA		81Va27 *
$^{210}\text{Rn}(\alpha)^{206}\text{Po}$	6162.1	3.	6159.0	2.2	-1.0	3					55Mo69 Z
	6155.9	3.			1.0	3			DbA		71Go35 Z
$^{210}\text{Fr}(\alpha)^{206}\text{At}$	6699.9	5.	6671	5	-5.8	B					67Va20 *
	6672.3	5.			-0.3	1	96	54 ^{206}At	GSa		05Ku06
$^{210}\text{Ra}(\alpha)^{206}\text{Rn}$	7156.6	5.	7151	3	-1.1	2					67Va22 Z
	7147	5			0.8	2			GSa		03He06 *
	7146.4	5.			0.5	2			Jya		07Le14
$^{210}\text{Ac}(\alpha)^{206}\text{Fr}$	7607.2	8.	7590	60	-0.4	-					68Va04
	7607.2	10.			-0.4	-			GSa		00He17
	ave.	7610	40		-0.3	1	87	84 ^{210}Ac			average
$^{210}\text{Th}(\alpha)^{206}\text{Ra}$	8052.7	17.	8069	6	0.9	4			Jya		95Uu01
	7962.0	50.			2.1	F			JAA		96Ik01 *
	8071.0	6.			-0.3	4			Anv		10He25
$^{208}\text{Pb}(\text{t,p})^{210}\text{Pb}$	628	12	640.7	0.9	1.1	U			Ald		68Bj03
$^{209}\text{Bi}(\text{n},\gamma)^{210}\text{Bi}$	4604.5	0.3	4604.63	0.08	0.4	-					71Mo03
	4604.68	0.14			-0.3	-			MMn		83Ts01 Z
	4604.63	0.10			0.0	-			Bdn		06Fi.A
$^{209}\text{Bi}(\text{d,p})^{210}\text{Bi}$	2369	10	2380.07	0.08	1.1	U			MIT		64Sp12
$^{209}\text{Bi}(\text{n},\gamma)^{210}\text{Bi}$	ave.	4604.64	0.08	4604.63	0.08	-0.0	1	100	86 ^{209}Bi		average
$^{210}\text{Tl}(\beta^-)^{210}\text{Pb}$	5500	100	5481	12	-0.2	U					64We06 *
$^{210}\text{Pb}(\beta^-)^{210}\text{Bi}$	63.5	0.5	63.5	0.5	-0.0	1	100	97 ^{210}Pb			67Ha03 *
$^{210}\text{Bi}(\beta^-)^{210}\text{Po}$	1160.5	1.5	1161.2	0.8	0.4	-					62Da03
	1161.5	1.5			-0.2	-					67Hs01
	ave.	1161.0	1.1		0.1	1	52	50 ^{210}Bi			average
$^{210}\text{At}(\epsilon)^{210}\text{Po}$	3870	30	3981	8	3.7	B					63Sc15 *
$^{210}\text{Bi}(\alpha)^{206}\text{Tl}$	$E_\alpha=4685.3(2,Z)$, 4648.3(2,Z) to 2^- level at 265.832, 1^- at 304.896 keV										Ens085 **
$^{210}\text{Bi}(\alpha)^{206}\text{Tl}$	$E_\alpha=4946(1)$, 4909(1) from $^{210}\text{Bi}^m$ at 271.31 keV										Nub211 **
*	to 2^- level at 265.832, 1^- level at 304.896 keV										Ens085 **
$^{210}\text{At}(\alpha)^{206}\text{Bi}$	$E_\alpha=5523.8$, 5464.8, 5441.8(1.5,Z) to ground state, 4^+ at 59.897, 5^+ at 82.818										Ens085 **
$^{210}\text{At}(\alpha)^{206}\text{Bi}$	$E_\alpha=5524.1$, 5465.3, 5442.8(1.3,Z) to ground state, 4^+ at 59.897, 5^+ at 82.818										Ens085 **
$^{210}\text{Fr}(\alpha)^{206}\text{At}$	$E_\alpha=6572.0(5,Z)$ 6542(5,Z) to ground state and level at 31.05 keV										Ens085 **
$^{210}\text{Ra}(\alpha)^{206}\text{Rn}$	$E_\alpha=7003(10)$ to ground state, 6447(5) to 574.9 level										03He06 **
$^{210}\text{Th}(\alpha)^{206}\text{Ra}$	F : Low energy; may be escape										96Ik01 **
$^{210}\text{Tl}(\beta^-)^{210}\text{Pb}$	$E_{\beta^-}=1870(100)$ to 3625(19) level, and other E_{β^-}										Ens148 **
$^{210}\text{Pb}(\beta^-)^{210}\text{Bi}$	$E_{\beta^-}=17.0(0.5)$ to 0^- level at 46.5390 keV										Ens148 **
$^{210}\text{At}(\epsilon)^{210}\text{Po}$	pK=0.46(0.10) to $(6)^-$ level at 3727.34 keV										Ens148 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{211}\text{Tl}-u$	-6525	45				2			GS3	1.0	08Ch.A
$^{211}\text{Po}-u$	-13519	147	-13346.8	1.3	1.2	U			GR1	1.0	19An10
$^{211}\text{Fr}-^{133}\text{Cs}_{1.586}$	145517	15	145508	13	-0.6	1	74	74 ^{211}Fr	MA8	1.0	09Ko35
$^{211}\text{Fr}-u$	-4410	43	-4445	13	-0.8	U			GR1	1.0	19An10
$^{211}\text{Fr}-^{226}\text{Ra}_{.934}$	-28200	25	-28176	13	1.0	1	27	26 ^{211}Fr	MA3	1.0	92Bo28
$^{211}\text{Ra}-^{133}\text{Cs}_{1.586}$	150846.4	8.5	150846	5	-0.0	1	39	39 ^{211}Ra	MA8	1.0	09Ko35
$^{211}\text{Ra}-u$	879	27	893	5	0.5	1	4	4 ^{211}Ra	RII	1.0	18Ro14
$^{211}\text{Ac}-u$	7522	106	7670	60	1.4	1	30	30 ^{211}Ac	RII	1.0	18Ro14
$^{211}\text{Po}^m-^{211}\text{Po}$	1580	129	1570	5	-0.0	U				2.5	15Di03
$^{207}\text{Fr}-^{211}\text{Fr}_{.327}$ $^{205}\text{Fr}_{.673}$	-930	100	-614	19	1.3	U			P24	2.5	82Au01
$^{208}\text{Fr}-^{211}\text{Fr}_{.394}$ $^{206}\text{Fr}_{.606}$	-260	50	-340	60	-0.6	U			P24	2.5	82Au01
$^{210}\text{Fr}-^{211}\text{Fr}_{.498}$ $^{209}\text{Fr}_{.502}$	580	50	617	16	0.3	U			P24	2.5	82Au01
$^{211}\text{Bi}(\alpha)^{207}\text{Tl}$	6749.5	0.7	6750.4	0.5	1.2	-					61Ry02 Z
	6751.1	0.6			-1.2	-					71Gr17 Z
	ave.	6750.4			-0.1	1	100	57 ^{211}Bi			average
$^{211}\text{Po}(\alpha)^{207}\text{Pb}$	7594.5	0.5				2			Orm		62Wa18 Z
	7594.3	3.	7594.6	0.5	0.1	U			DbA		69Go23 Z
	7600.6	2.			-3.0	B					85La17 Z
	7586.0	15.3			0.6	U					15Di03
	7604.5	30.6			-0.3	U			GSa		18Mi11
$^{211}\text{Po}^m(\alpha)^{207}\text{Pb}$	9057.1	5.1				2			Bka		82Bo04
	9049.0	15.	9057	5	0.5	U					89Ku08 *
	9043.0	15.			0.9	U					15Di03 *
$^{211}\text{At}(\alpha)^{207}\text{Bi}$	5979.4	2.	5982.4	1.3	1.5	2			DbA		69Go23 Z
	5981.6	3.			0.3	2			Bka		82Bo04 *
	5985.9	2.			-1.7	2					85La17 Z
$^{211}\text{Rn}(\alpha)^{207}\text{Po}$	5967.9	2.	5965.5	1.4	-1.2	2					55Mo69 Z
	5963.1	2.			1.2	2			DbA		71Go35 Z
$^{211}\text{Fr}(\alpha)^{207}\text{At}$	6660.2	5.1	6662	3	0.4	2					67Va20 Z
	6663.5	4.			-0.3	2			GSa		05Ku06
$^{211}\text{Ra}(\alpha)^{207}\text{Rn}$	7045.3	5.	7041.7	2.9	-0.7	-					67Va22 Z
	7040	5			0.3	-			GSa		03He06 *
	7039.7	6.			0.3	-			Jya		07Le14
	7033.5	15.3			0.5	U			Lza		19Zh54
	ave.	7042			-0.1	1	91	57 ^{211}Ra			average
$^{211}\text{Ac}(\alpha)^{207}\text{Fr}$	7624.8	8.	7570	50	-1.1	-					68Va04
	7616.7	10.			-1.0	-			GSa		00He17
	ave.	7620			-0.9	1	73	70 ^{211}Ac			average
$^{211}\text{Th}(\alpha)^{207}\text{Ra}$	7942.9	14.	7940	60	-0.1	5			Jya		95Uu01
	7930.6	30.6			0.1	5			Lza		15Ya13
$^{211}\text{Pa}(\alpha)^{207}\text{Ac}$	8481.0	40.8				3			Jya		20Au04
$^{211}\text{Pb}(\beta^-)^{211}\text{Bi}$	1378	8	1366	5	-1.5	1	47	43 ^{211}Bi			65Co06
$^{*211}\text{Po}^m(\alpha)^{207}\text{Pb}$	$E_\alpha=7275(15)$ to $13/2^+$ level at 1633.356 keV										Ens112 **
$^{*211}\text{Po}^m(\alpha)^{207}\text{Pb}$	$E_\alpha=7269(15)$ to $13/2^+$ level at 1633.356 keV										Ens112 **
$^{*211}\text{At}(\alpha)^{207}\text{Bi}$	Recalibrated as in reference										91Ry01 **
$^{*211}\text{Ra}(\alpha)^{207}\text{Rn}$	Average of $E_\alpha=6907(5)$ and several branches to known levels										03He06 **
$^{212}\text{Bi}^n-u$	-7127	32				2			GS3	1.0	08Ch.A
$^{212}\text{At}-u$	-9234	92	-9262.7	2.6	-0.3	U			GR1	1.0	19An10
$^{212}\text{Rn}-u$	-9242	32	-9296	3	-1.7	U			GR1	1.0	19An10
$^{212}\text{Fr}-^{133}\text{Cs}_{1.594}$	146938	10	146935	9	-0.3	1	89	89 ^{212}Fr	MA8	1.0	09Ko35
$^{212}\text{Fr}-u$	-3790	30	-3775	9	0.5	U			GR1	1.0	19An10
$^{212}\text{Fr}-^{226}\text{Ra}_{.938}$	-27631	28	-27607	10	0.8	1	12	11 ^{212}Fr	MA3	1.0	92Bo28
$^{212}\text{Ra}-u$	-212.7	26.6	-213	11	-0.0	1	17	17 ^{212}Ra	RII	1.0	18Ro14
$^{212}\text{Ac}-u$	7840.7	25.4	7836	23	-0.2	1	86	86 ^{212}Ac	RII	1.0	18Ro14

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{209}\text{Fr} - ^{212}\text{Fr}_{.563} \quad ^{205}\text{Fr}_{.437}$	-1270	70	-1230	13	0.2	U			P24	2.5	82Au01
$^{206}\text{Fr}^x - ^{212}\text{Fr}_{.139} \quad ^{205}\text{Fr}_{.861}$	340	130	470	100	0.4	U			P24	2.5	82Au01
$^{207}\text{Fr} - ^{212}\text{Fr}_{.163} \quad ^{206}\text{Fr}_{.837}$	-1150	70	-1320	90	-0.9	U			P24	2.5	82Au01
$^{212}\text{Bi}(\alpha)^{208}\text{Tl}$	6207.12	0.04	6207.262	0.028	3.5	B			BIP		61Ry02 Z
	6207.09	0.08			2.1	o			BIP		69Gr28 *
	6207.262	0.028				2			BIP		72Go.A *
$^{212}\text{Bi}^m(\alpha)^{208}\text{Tl}$	6458.1	30.				3					78Ba44
$^{212}\text{Po}(\alpha)^{208}\text{Pb}$	8953.6	0.8	8954.19	0.11	0.7	U					61Ry02 Z
	8953.85	0.31			1.1	-					71De52 Z
	8953.3	0.6			1.5	U					71Gr17 Z
	8954.25	0.12			-0.5	-					74Hu15 Z
	ave. 8954.20	0.11			-0.0	1	100	91 ^{212}Po			average
$^{212}\text{Po}^m(\alpha)^{208}\text{Pb}$	11874.6	20.	11877	4	0.1	2					62Pe15
	11859.3	15.			1.2	o					75Fr.B
	11884.8	10.2			-0.7	2					76Fr.A
	11875.6	5.1			0.3	2			GSa		12Ho12
$^{212}\text{At}(\alpha)^{208}\text{Bi}$	7829.0	9.	7817.1	0.6	-1.3	U					70Re02 *
	7817.0	0.6				3					76Fr.A *
	7828.0	10.			-1.1	U					96Li37 *
$^{212}\text{At}^m(\alpha)^{208}\text{Bi}$	8049.3	10.	8040.0	0.6	-0.9	U					68Va18
	8054.3	9.			-1.6	U					70Re02 *
	8040.00	0.61				3					76Fr.A *
	8051.2	10.			-1.1	U					96Li37 *
$^{212}\text{Rn}(\alpha)^{208}\text{Po}$	6392.3	5.	6385.1	2.6	-1.4	2					55Mo69 Z
	6382.5	3.			0.9	2			DbA		71Go35 Z
$^{212}\text{Fr}(\alpha)^{208}\text{At}$	6531.3	3.	6529.0	1.6	-0.7	2					66Va.A Z
	6528.0	3.			0.3	2			DbA		81Va27
	6527.5	3.			0.5	2			Bka		82Bo04 *
	6529.5	4.			-0.1	2			GSa		05Ku06
$^{212}\text{Ra}(\alpha)^{208}\text{Rn}$	7030.0	5.	7031.7	1.7	0.3	-					67Va22 Z
	7034.0	5.			-0.4	-					74Ho27 Z
	7032.2	2.			-0.3	-			Bka		82Bo04 Z
	7028	5			0.7	-			GSa		03He06 *
	7040	24			-0.3	U			Lza		14Ya19
	ave. 7031.7	1.7			0.0	1	100	83 ^{212}Ra			average
$^{212}\text{Ac}(\alpha)^{208}\text{Fr}$	7521.2	8.	7540	24	0.4	-					68Va04
	7515.1	10.			0.5	-			GSa		00He17
	7514.0	18.			0.5	-			Lza		14Ya19
	7518.0	17.			0.4	-			Lza		19Zh23
	ave. 7517	26			0.4	1	18	14 ^{212}Ac			average
$^{212}\text{Th}(\alpha)^{208}\text{Ra}$	7952.3	10.	7958	5	0.6	3					80Ve01
	7959.5	5.			-0.3	3			Anv		10He25
	7980.8	20.4			-1.1	U			GSa		15De22
$^{212}\text{Pa}(\alpha)^{208}\text{Ac}$	8429.4	30.	8410	60	-0.3	6			JAa		97Mi03
	8406.9	20.			0.1	6			Lza		14Ya19 *
	8398.7	20.			0.2	6			Jva		20Au04
$^{210}\text{Pb}(\text{t,p})^{212}\text{Pb}$	515	25	481.3	2.1	-1.3	U					71El05
$^{212}\text{Pb}(\beta^-)^{212}\text{Bi}$	569.3	2.5	569.0	1.8	-0.1	-					48Ma30 *
	576.6	5.			-1.5	-					58Se71 *
	ave. 570.8	2.2			-0.8	1	67	34 ^{212}Bi			average
$^{212}\text{Bi}(\beta^-)^{212}\text{Po}$	2256	3	2251.5	1.7	-1.5	-					48Fe09
	2250.5	2.5			0.4	-					48Ma30
	ave. 2252.8	1.9			-0.7	1	75	66 ^{212}Bi			average
$^{*212}\text{Bi}(\alpha)^{208}\text{Tl}$	$E_\alpha=6089.86(0.08,Z)$, $6050.57(0.07,Z)$ to ground state, 4^+ level at 39.858 keV										Ens077 **
$^{*212}\text{Bi}(\alpha)^{208}\text{Tl}$	$E_\alpha=6089.883(0.037,Z)$, $6050.837(0.028,Z)$ to ground state, 4^+ level at 39.858 keV										72Go.A **
$^{*212}\text{At}(\alpha)^{208}\text{Bi}$	Original $E_\alpha=7679(8)$; calibration ^{211}Po 7448(1), now 7450.3(0.5) keV										AHW **
$^{*212}\text{At}(\alpha)^{208}\text{Bi}$	Original $E_\alpha=7669.0(0.2)$; calibration ^{211}Po 7450(2), now 7450.3(0.5)										05Ma.A **
$^{*212}\text{At}(\alpha)^{208}\text{Bi}$	$E_\alpha=7679(10)$ to ground state, $7618(10)$ to 63.3 level										96Li37 **
$^{*212}\text{At}(\alpha)^{208}\text{Bi}$	error estimated by the evaluators										GAU **
$^{*212}\text{At}^m(\alpha)^{208}\text{Bi}$	Original $E_\alpha=7900(8)$; calibration ^{211}Po 7448(1), now 7450.3(0.5) keV										GAU **
$^{*212}\text{At}^m(\alpha)^{208}\text{Bi}$	Original $E_\alpha=7887.7(0.2)$; calibration ^{211}Po 7450(2), now 7450.3(0.5)										GAU **
$^{*212}\text{At}^m(\alpha)^{208}\text{Bi}$	$E_\alpha=7897(10)$ to ground state, $7837(10)$ to 63.3 level										96Li37 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
*	error estimated by the evaluators										GAu	**
* $^{212}\text{Fr}(\alpha)^{208}\text{At}$	$E_\alpha=6341(3)$ (recalibrated as in reference) to 63.70 level										91Ry01	**
* $^{212}\text{Ra}(\alpha)^{208}\text{Rn}$	$E_\alpha=6898(5)$ to ground state, 6269(5) to 635.1 level										03He06	**
* $^{212}\text{Pa}(\alpha)^{208}\text{Ac}$	$E_\alpha=8250(20)$ keV in reference by combining with 97Mi03										14Ya19	**
* $^{212}\text{Pb}(\beta^-)^{212}\text{Bi}$	$E_{\beta^-}=330.7(2.5)$ 338(5) respectively to 0^- level at 238.632 keV										Ens053	**
$^{213}\text{Tl}-u$	1893	65	1915	29	0.3	o			GS3	1.0	10Ch19	
	1915	29				2			GS3	1.0	12Ch19	
	-6159	63	-6115	4	0.7	U			GR1	1.0	19An10	
	147649.1	7.4	147650	5	0.2	1	47	47 ^{213}Fr	MA8	1.0	09Ko35	
	-3823.9	13.2	-3816	5	0.6	1	15	15 ^{213}Fr	GR1	1.0	19An10	
	151833	12	151837	11	0.3	1	77	77 ^{213}Ra	SH1	1.0	13Dr04	
	6572.2	19.4	6593	13	1.1	1	42	42 ^{213}Ac	RII	1.0	18Ro14	
	-2540	330	-2108	24	0.5	U			P24	2.5	82Au01	
	-700	60	-850	80	-1.0	U			P24	2.5	82Au01	
	-670	60	-703	17	-0.2	U			P24	2.5	82Au01	
$^{213}\text{Po}(\alpha)^{209}\text{Pb}$	-980	60	-943	15	0.2	U			P24	2.5	82Au01	
	-830	60	-727	15	0.7	U			P24	2.5	82Au01	
	270	50	332	11	0.5	U			P24	2.5	82Au01	
	5982.6	6.	5988	3	0.9	2					64Gr11	
	5990.7	4.			-0.6	2			Gea		13Ma13	
	8537.1	5.	8536.1	2.6	-0.2	-					64Va20	
	8536.5	3.			-0.1	-			Bka		82Bo04	
	8530.5	10.2			0.6	U			GSt		18Sa45	
	ave. 8536.7	2.6			-0.2	1	95	93 ^{213}Po			average	
	9254.2	12.	9254	5	-0.0	2					70Bo13	
$^{213}\text{At}(\alpha)^{209}\text{Bi}$	9254.2	5.			-0.0	2			Lvn		87De.A	
	8245.1	8.	8245.2	2.9	-0.0	3					67Va20	
	8240.0	10.			0.5	U					70Va13	
	8242	10			0.3	U			GSa		00He17	
	8245.2	3.1			-0.0	3			Jya		01Ku07	
	8218.6	44.			0.6	U					05Li17	
	8204.4	30.6			1.3	U			GSa		18Mi11	
	8245.0	10.			0.0	U			GSa		19Mi08	
	6904.0	5.	6904.7	1.3	0.1	U					67Va20	
	6908.0	5.			-0.7	U					74Ho27	
$^{213}\text{Fr}(\alpha)^{209}\text{At}$	6904.6	2.			0.0	-			Bka		82Bo04	
	6904.9	1.7			-0.2	-			GSa		05Ku06	
	6880	20			1.2	U			GSa		15De22	
	6869.2	20.4			1.7	U			GSa		19Mi08	
	ave. 6904.8	1.3			-0.1	1	98	59 ^{209}At			average	
	6860.3	5.	6861.7	2.3	0.3	-					67Va22	
	6862.4	5.			-0.1	-					76Ra37	
	6862.2	3.			-0.2	-			GSa		06Ku26	
	ave. 6861.8	2.3			-0.1	1	99	76 ^{209}Rn			average	
	8630.3	5.1	8630	4	-0.1	2					76Ra37	
$^{213}\text{Ra}^m(\alpha)^{209}\text{Rn}$	8629.3	5.			0.1	2			GSa		06Ku26	
	7505.2	8.	7498	4	-0.9	-					68Va04	
	7497.0	10.			0.1	o			GSa		00He17	
	7497.0	5.			0.2	-			GSa		02He.A	
	ave. 7499	4			-0.3	1	97	58 ^{213}Ac			average	
	7837.4	10.	7837	7	-0.1	3					68Va18	
	7836.5	10.			0.0	3					80Ve01	
	8393.9	15.	8384	12	-0.6	4			GSa		00He17	
	8367.4	20.4			0.8	4			Jva		20Au04	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{213}\text{Bi}(\beta^-)^{213}\text{Po}$	1430	10	1422	5	-0.8	1	30	23 ^{213}Bi			68Va17 *
$^{207}\text{Fr}-^{213}\text{Fr}_{324}$ ^{204}Fr .	$D_M=-2470(330)$ keV for ^{204}Fr mixture gs+m+n at 50(4), 326(4) keV										Nub211 **
$^{213}\text{Rn}(\alpha)^{209}\text{Po}$	$E_\alpha=8088(10)$, 7550(15) to ground state, 540.3 level										00He17 **
$^{213}\text{Ra}(\alpha)^{209}\text{Rn}$	$E_\alpha=6730.7$, 6623.7, 6520.7(3,Z) to ground state, $1/2^-$ at 110.25, $3/2^-$ at 214.93										Ens156 **
$^{213}\text{Ra}(\alpha)^{209}\text{Rn}$	$E_\alpha=6731.9$, 6624.9, 6523.9(5,Z) to ground state, $1/2^-$ at 110.25, $3/2^-$ at 214.93										Ens156 **
$^{213}\text{Ra}(\alpha)^{209}\text{Rn}$	$E_\alpha=6733(3)$, 6625(3) to ground state and $1/2^-$ level at 110.25 keV										Ens156 **
$^{213}\text{Ra}^m(\alpha)^{209}\text{Rn}$	$E_\alpha=8467(5)$ 8358(10) to ground state and $1/2^-$ level at 110.25 keV										Ens156 **
$^{213}\text{Ac}(\alpha)^{209}\text{Fr}$	Original Q increased by 2 keV, as in reference										91Ry01 **
$^{213}\text{Th}(\alpha)^{209}\text{Ra}$	Original Q decreased by 0.5 keV, as in reference										91Ry01 **
$^{213}\text{Bi}(\beta^-)^{213}\text{Po}$	$E_{\beta^-}=1420(10)$ 1018(15) to ground state and $7/2^+$ level at 440.45 keV										Ens074 **
$^{214}\text{Ra}-^{133}\text{Cs}_{1.609}$	152235	22	152227	6	-0.3	U			MA8	1.0	08We02
$^{214}\text{Ra}-u$	63	38	100	6	1.0	U			RII	1.0	18Ro14
$^{214}\text{Ac}-u$	6866.8	30.9	6906	15	1.3	1	22	22 ^{214}Ac	RII	1.0	18Ro14
$^{214}\text{Bi}(\alpha)^{210}\text{Tl}$	5621.3	3.0				2					91Ry01 *
$^{214}\text{Po}(\alpha)^{210}\text{Pb}$	7833.54	0.06	7833.54	0.06	-0.0	1	100	97 ^{214}Po			71Gr17 Z
$^{214}\text{At}(\alpha)^{210}\text{Bi}$	8987.2	4.	8988	4	0.1	1	84	84 ^{214}At	Bka		82Bo04 Z
$^{214}\text{At}^m(\alpha)^{210}\text{Bi}$	9046.4	8.				2					82Ew01
$^{214}\text{At}^n(\alpha)^{210}\text{Bi}$	9220.8	5.	9220	4	-0.2	1	76	76 $^{214}\text{At}^n$			82Ew01 *
$^{214}\text{Rn}(\alpha)^{210}\text{Po}$	9212.6	20.	9208	9	-0.2	2					70To07
	9207.5	10.			0.1	2					70Va13
	9334.8	51.			-2.5	U			GSa		18Mi11
$^{214}\text{Fr}(\alpha)^{210}\text{At}$	8585.5	8.	8589	4	0.4	4					68Va18 *
	8590.9	5.			-0.4	4					70To18 *
	8583.8	10.			0.5	4			Dbb		89An.A
	8590.8	20.			-0.1	U			GSa		90Ni05
	8578.7	48.			0.2	U					05Li17
	8590.9	10.2			-0.2	4			GSa		19Mi08
$^{214}\text{Fr}^m(\alpha)^{210}\text{At}$	8711.7	8.	8710	3	-0.2	4					68Va04 Z
	8711.7	5.			-0.3	4					70To18 *
	8708.1	5.			0.4	4			GSa		05Ku06
$^{214}\text{Ra}(\alpha)^{210}\text{Rn}$	7271.7	5.	7272.6	2.6	0.2	4					67Va22 Z
	7275.6	5.			-0.6	4					74Ho27 Z
	7273.2	10.			-0.1	4			GSa		00He17 *
	7271.2	4.			0.3	4			GSa		06Ku26 *
$^{214}\text{Ra}(\alpha)^{210}\text{Rn}^m$	5563.9	30.				5			GSa		06Ku26 *
$^{214}\text{Ac}(\alpha)^{210}\text{Fr}$	7351.7	5.	7351.9	2.5	0.0	-					68Va04 Z
	7347.6	10.			0.4	-			Dbb		89An13
	7347.6	10.			0.4	o			GSa		00He17 *
	7349.6	5.			0.4	o			GSa		02He.A
	7352.7	3.			-0.3	-			GSa		04Ku24 *
	7360.6	17.3			-0.5	U			Lza		20Ma27
	7352.1	2.5			-0.1	1	99	78 ^{214}Ac			average
$^{214}\text{Th}(\alpha)^{210}\text{Ra}$	7828.6	10.	7827	5	-0.1	3					68Va18
	7823.5	10.			0.4	3					80Ve01
	7828.6	8.			-0.2	3			Jya		07Le14
	8270.9	15.				2			GSa		00He17
$^{214}\text{Pa}(\alpha)^{210}\text{Ac}$	1024	20	1018	11	-0.3	1	32	31 ^{214}Bi			52Be78 *
$^{214}\text{Pb}(\beta^-)^{214}\text{Bi}$	3260	30	3269	11	0.3	-					56Da06
$^{214}\text{Bi}(\beta^-)^{214}\text{Po}$	3275	15			-0.4	-					60Lu07
	ave.	3272	13		-0.2	1	69	69 ^{214}Bi			average
$^{214}\text{Bi}(\alpha)^{210}\text{Tl}$	$E_\alpha=5516(3)$ recommended in place of the following E_α :										91Ry01 **
*	$E_\alpha=5510.5(1.0)$ keV										34Le01 **
*	$E_\alpha=5515.8(3.0)$ keV										60Wa14 **
$^{214}\text{At}^n(\alpha)^{210}\text{Bi}$	$E_\alpha=8782(5)$ to 9^- level at 271.31 keV										Ens148 **
$^{214}\text{Fr}(\alpha)^{210}\text{At}$	$E_\alpha=8425.5$, 8352.5(8,Z) to ground state, $(4)^+$ level at 72.65 keV										Ens148 **
$^{214}\text{Fr}(\alpha)^{210}\text{At}$	$E_\alpha=8428.3$, 8360.3(5,Z) to ground state, $(4)^+$ level at 72.65 keV										Ens148 **
$^{214}\text{Fr}^m(\alpha)^{210}\text{At}$	$E_\alpha=8546.8$, 8477.8(5,Z) to ground state, $(4)^+$ level at 72.65 keV										Ens148 **
$^{214}\text{Ra}(\alpha)^{210}\text{Rn}$	$E_\alpha=7137(10)$, 6505(15) to ground state, 641.9 level										00He17 **
$^{214}\text{Ra}(\alpha)^{210}\text{Rn}$	Also $E_\alpha=8950(30)$ keV $Q_\alpha=9120.9$ keV from $^{214}\text{Ra}^n$ at 1865.2 keV										Nub211 **
$^{214}\text{Ra}(\alpha)^{210}\text{Rn}^m$	$E_\alpha=7290(30)$ $Q_\alpha=7429.1$ from $^{214}\text{Ra}^n$ at 1865.2 keV										Nub211 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
* $^{214}\text{Ac}(\alpha)^{210}\text{Fr}$	$E_\alpha=7210(10)$, 7080(15) to ground state, 138.6 level										00He17	**
* $^{214}\text{Ac}(\alpha)^{210}\text{Fr}$	Also $E_\alpha=7081(4)$ keV to 139.0(1) level										04Ku24	**
* $^{214}\text{Pb}(\beta^-)^{214}\text{Bi}$	$E_{\beta^-}=670(20)$ to $(0^-, 1^-)$ level at 351.9324 keV, and another branch										Ens092	**
$^{215}\text{Bi}-^{133}\text{Cs}_{1.617}$	154654	16	154633	6	-1.3	1	14	14 ^{215}Bi	MA8	1.0	08We02	
$^{215}\text{Po}(\alpha)^{211}\text{Pb}$	7526.45	0.8	7526.3	0.8	-0.2	1	99	96 ^{211}Pb			71Gr17 Z	
$^{215}\text{At}(\alpha)^{211}\text{Bi}$	8178.5	4.	8178	4	-0.2	2			Bka		82Bo04 Z	
	8172.3	10.2			0.5	2			GSt		18Sa45	
$^{215}\text{Rn}(\alpha)^{211}\text{Po}$	8834.7	20.	8839	6	0.2	3			ORa		69Ha32	
	8839.8	8.			-0.1	3					70Va13	
	8865.3	30.6			-0.9	3			GSa		18Mi11	
	8834.7	10.2			0.4	3			GSt		18Sa45	
$^{215}\text{Fr}(\alpha)^{211}\text{At}$	9543.0	15.	9540	7	-0.2	3					70Bo13	
	9532.7	10.			0.8	3					74No02	
	9546.9	10.			-0.6	3					84De16	
	9364.4	50.9			3.5	B			GSa		19Mi08 *	
$^{215}\text{Ra}(\alpha)^{211}\text{Rn}$	8862.7	5.	8862.4	2.3	-0.1	3					68Va18 Z	
	8865.5	5.			-0.6	3					70To18 Z	
	8865.3	10.			-0.3	U			GSa		00He17	
	8865.3	46.			-0.1	U					05Li17	
	8861.1	3.1			0.4	3			Lza		20Su02	
$^{215}\text{Ac}(\alpha)^{211}\text{Fr}$	7748.4	5.	7746	3	-0.5	2					68Va04 Z	
	7746	10			-0.0	o			GSa		00He17 *	
	7740.3	5.			1.1	o			GSa		02He.A	
	7744.4	4.			0.4	2			GSa		04Ku24	
$^{215}\text{Th}(\alpha)^{211}\text{Ra}$	7664.9	8.	7665	4	-0.0	2					68Va18	
	7667.0	10.			-0.2	o			GSa		89He03	
	7664	15			0.0	o			GSa		00He17 *	
	7665	5			-0.1	2			GSa		05Ku31 *	
	7662.8	10.			0.2	2			Jya		07Le14 *	
	7652.2	15.2			0.8	U			Lza		19Zh54	
$^{215}\text{Pa}(\alpha)^{211}\text{Ac}$	8238.6	15.	8240	60	-0.0	2					79Sc09	
	8244.7	15.			-0.2	o			GSa		00He17	
	8233.5	20.4			0.0	2			GSa		15De22	
$^{215}\text{U}(\alpha)^{211}\text{Th}$	8588.0	30.6				6			Lza		15Ya13	
* $^{215}\text{Ac}(\alpha)^{211}\text{Fr}$	$E_\alpha=7602(10)$ 7026(15) 6960(15) to ground state, 11/2 ⁻ at 583.28, 13/2 ⁻ at 652.62										Ens136	**
* $^{215}\text{Th}(\alpha)^{211}\text{Ra}$	$E_\alpha=7520(15)$, 7387(15), 7336(15) to ground state, 133.6, 192.4 levels										00He17	**
* $^{215}\text{Th}(\alpha)^{211}\text{Ra}$	$E_\alpha=7523(5)$, 7392(4), 7335(5), 7236(7) to ground state, 133.9, 194.5, 295.1 levels										05Ku31	**
* $^{215}\text{Th}(\alpha)^{211}\text{Ra}$	Also $E_\alpha=7399(20)$ keV to 133.9 level										07Le14	**
$^{216}\text{Bi}-^{133}\text{Cs}_{1.624}$	159852	12				2			MA8	1.0	08We02	
$^{216}\text{Po}(\alpha)^{212}\text{Pb}$	6906.44	0.5	6906.3	0.5	-0.2	1	98	67 ^{212}Pb			71Gr17 Z	
$^{216}\text{At}(\alpha)^{212}\text{Bi}$	7949.7	3.				2			Bka		82Bo04 Z	
$^{216}\text{At}^m(\alpha)^{212}\text{Bi}$	8110.5	10.				2					71Br13	
$^{216}\text{Rn}(\alpha)^{212}\text{Po}$	8199.2	10.	8198	6	-0.1	2					61Ru06	
	8201.2	10.			-0.3	2					70Va13	
	8192.0	10.2			0.6	2					71Br13	
	8202.2	20.4			-0.2	2			GSt		18Sa45	
$^{216}\text{Fr}(\alpha)^{212}\text{At}$	9175.3	12.	9174	3	-0.1	4					70Bo13	
	9174.1	5.			0.0	4					96Li37 *	
	9174.3	5.			0.0	4			GSa		07Ku30	
$^{216}\text{Fr}^m(\alpha)^{212}\text{At}^m$	9170.2	5.				4			GSa		07Ku30	
$^{216}\text{Ra}(\alpha)^{212}\text{Rn}$	9525.8	8.	9526	7	-0.0	3					73No09	
	9525.8	17.3			-0.0	3			Lza		17Su18	
$^{216}\text{Ac}(\alpha)^{212}\text{Fr}$	9243.3	8.	9240.8	2.9	-0.3	-					70To18 Z	
	9223.1	10.			1.8	U			GSa		00He17	
	9241.4	50.9			-0.0	U					05Li17	
	9240.5	3.0			0.1	-			Lza		17Hu08	
	9240.5	6.1			0.1	o			Lza		18Hu13	
ave.	9240.8	2.9			0.0	1	100	100 ^{216}Ac			average	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{216}\text{Ac}^m(\alpha)^{212}\text{Fr}$	9280.0	5.	9279	4	-0.2	2					70To18 Z
	9284	10			-0.5	o			GSa		00He17 *
	9278.2	5.			0.2	o			GSa		02He.A
	9277.2	7.			0.3	2			GSa		04Ku24 *
$^{216}\text{Th}(\alpha)^{212}\text{Ra}$	8070.7	8.	8072	4	0.2	2					68Va18
	8071	10			0.1	o			GSa		00He17 *
	8073	5			-0.1	2			GSa		05Ku31 *
	8069.7	44.			0.1	U					05Li17
	8070.7	23.			0.1	U			Lza		14Ya19
	8068.6	15.3			0.2	U			Lza		19Zh54
$^{216}\text{Th}^m(\alpha)^{212}\text{Ra}$	10099.4	20.	10114	7	0.7	2					83Hi08
	10107.4	40.			0.2	U			Dbb		93An07
	10120.8	15.			-0.5	2			GSa		00He17
	10117.5	10.			-0.4	2					05Ku31 *
	10105.5	15.3			0.5	2			Lza		19Zh54
$^{216}\text{Pa}(\alpha)^{212}\text{Ac}$	8013.7	20.	8099	11	4.3	B					79Sc09
	8110.5	50.			-0.2	U			JAa		98Ik01
	8097	15			0.2	2			GSa		00He17 *
	8102.3	17.3			-0.2	2			Lza		19Zh23
$^{216}\text{U}(\alpha)^{212}\text{Th}$	8542.5	30.6	8531	26	-0.4	4			Lza		15Ma37
	8497.6	50.9			0.6	4			GSa		15De22
$^{216}\text{U}^m(\alpha)^{212}\text{Th}$	10782.0	30.6				4			Lza		15Ma37
$^{216}\text{Fr}(\alpha)^{212}\text{At}$	$E_\alpha=9004(5)$; and $E_\alpha=8933(8)$ from 133.3 level to 205.6 keV										96Li37 **
$^{216}\text{Ac}^m(\alpha)^{212}\text{Fr}$	$E_\alpha=9110(10)$, 9026(15), 8586(15) to ground state, 82.4, 542.2 levels										00He17 **
$^{216}\text{Ac}^m(\alpha)^{212}\text{Fr}$	Also $E_\alpha=9029(7)$ keV to 82.6(1) level										04Ku24 **
$^{216}\text{Th}(\alpha)^{212}\text{Ra}$	$E_\alpha=7923(10)$, 7302(15) to ground state, 618.3 level										00He17 **
$^{216}\text{Th}(\alpha)^{212}\text{Ra}$	$E_\alpha=7923(5)$, 7304(4) to ground state, 629.3(1) level										05Ku31 **
$^{216}\text{Th}^m(\alpha)^{212}\text{Ra}$	$E_\alpha=9930(10)$, 9312(12) to ground state, 629.3(1) level										05Ku31 **
$^{216}\text{Pa}(\alpha)^{212}\text{Ac}$	$E_\alpha=7948(15)$, 7815(15) to ground state, 133.6 level										00He17 **
$^{217}\text{Bi}-u$	9420	32	9372	19	-1.5	o			GS3	1.0	08Ch.A
	9372	19				2			GS3	1.0	12Ch19
$^{217}\text{At}-u$	4759	145	4718	5	-0.3	U			GR1	1.0	19An10
$^{217}\text{Po}(\alpha)^{213}\text{Pb}$	6660.3	4.	6662.1	2.4	0.4	4			DbA		77Vy02 Z
	6660.0	4.			0.5	4			Orm		97Li23
	6666.1	4.1			-1.0	4			Anv		03Ku25
$^{217}\text{At}(\alpha)^{213}\text{Bi}$	7200.3	3.	7201.4	1.2	0.4	-					60Vo05 Z
	7200.3	2.			0.5	-			Orm		62Wa28 Z
	7204.6	5.			-0.6	-					64Va20 Z
	7193.1	5.			1.6	-			DbA		77Vy02 Z
	7204.0	2.			-1.3	-			Bka		82Bo04
	ave.	1.2			-0.1	1	99	^{213}Bi			average
$^{217}\text{Rn}(\alpha)^{213}\text{Po}$	7887.5	4.	7887.2	2.9	-0.1	2					61Ru06 Z
	7886.9	4.			0.1	2			Bka		82Bo04 Z
	7885.6	10.2			0.2	U			GSt		18Sa45
$^{217}\text{Fr}(\alpha)^{213}\text{At}$	8471.5	8.	8469	4	-0.3	3					70Bo13
	8468.4	5.			0.2	3			Lvn		87De.A
$^{217}\text{Ra}(\alpha)^{213}\text{Rn}$	9159.1	8.	9161	6	0.2	4					70To07
	9163.2	10.			-0.3	4					70Va13
	9159.2	40.8			0.0	U			GSa		19Mi08 *
	9157.1	26.5			0.1	4			Lza		19Ya04
$^{217}\text{Ac}(\alpha)^{213}\text{Fr}$	9831.6	10.				2					73No09
	9474.9	203.8	9832	10	1.8	U			GSa		19Mi08
$^{217}\text{Ac}^m(\alpha)^{213}\text{Fr}$	11843.8	17.				2					85De14
$^{217}\text{Th}(\alpha)^{213}\text{Ra}$	9424.1	10.	9435	4	1.1	2					68Va18
	9424.1	20.			0.6	U					73Ha32
	9421.1	15.			0.9	U					00Ni02
	9442	15			-0.4	o			GSa		00He17 *
	9435.6	5.			-0.1	2			GSa		02He29 *
	9443.5	9.			-0.9	2			GSa		05Ku31
	9424.1	47.			0.2	U					05Li17

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{217}\text{Th}(\alpha)^{213}\text{Ra}$	9431.0	15.3	9435	4	0.3	U			Lza		19Zh54
$^{217}\text{Pa}(\alpha)^{213}\text{Ac}$	8486.7	10.	8489	4	0.2	2					68Va18
	8489.8	15.			-0.1	U					79Sc09
	8486.7	50.			0.0	U			JAa		98Ik01
	8490.8	15.			-0.1	U			GSa		00He17
	8489.3	5.			-0.1	2			GSa		02He29 *
$^{217}\text{Pa}^m(\alpha)^{213}\text{Ac}$	10351	20	10349	5	-0.1	U					79Sc09
	10330.8	50.			0.4	U			JAa		98Ik01
	10346.1	15.			0.2	o			GSa		00He17
	10349.1	5.				2			GSa		02He29 *
$^{217}\text{U}(\alpha)^{213}\text{Th}^p$	8155.6	20.	8170	60	0.2	5					00Ma65
	8175.0	14.3			-0.2	5			Jya		05Le42 *
$^{217}\text{Th}(\alpha)^{213}\text{Ra}$	$E_\alpha=9268(15)$, $8731(15)$, $8459(15)$ to ground state, 546.35, 822.7 levels										00He17 **
$^{217}\text{Th}(\alpha)^{213}\text{Ra}$	$E_\alpha=9261(5)$, $8725(5)$, $8455(5)$ to ground state, 546.35, 822.7 levels										02He29 **
$^{217}\text{Pa}(\alpha)^{213}\text{Ac}$	$E_\alpha=8337(5)$, $7873(5)$, $7728(5)$, $7710(5)$ to ground state, 466.1, 612.5, 634.3 levels										02He29 **
$^{217}\text{Pa}^m(\alpha)^{213}\text{Ac}$	Average of 5 E_α 's to known levels										02He29 **
$^{217}\text{U}(\alpha)^{213}\text{Th}^p$	Only one event. Not reported in later publication 07Le14										WgM115**
$^{218}\text{Bi}-u$	14178	34	14188	29	0.3	o			GS3	1.0	08Ch.A
	14188	29				2			GS3	1.0	12Ch19
$^{218}\text{Rn}-u$	5463	58	5601.1	2.5	2.4	U			GR1	1.0	19An10
$^{218}\text{Po}(\alpha)^{214}\text{Pb}$	6114.76	0.09	6114.75	0.09	-0.0	1	100	99 ^{214}Pb			71Gr17 Z
$^{218}\text{At}(\alpha)^{214}\text{Bi}$	6874	3	6876.1	2.6	0.7	2			Orm		58Wa.A *
	6882.1	5.1			-1.2	2					19Cu02 *
$^{218}\text{Rn}(\alpha)^{214}\text{Po}$	7265.0	5.	7262.5	1.9	-0.5	-					56As38 Z
	7262.4	2.			0.0	-			Bka		82Bo04 Z
	ave.	7262.8	1.9		-0.2	1	96	93 ^{218}Rn			average
$^{218}\text{Fr}(\alpha)^{214}\text{At}$	8014.0	2.	8013.7	1.4	-0.2	2			Bka		82Bo04 Z
	8013.3	2.			0.2	2					99Sh03
$^{218}\text{Fr}^m(\alpha)^{214}\text{At}$	8099.9	5.	8101	3	0.2	-					82Ew01 Z
	8100.9	5.			-0.0	-					99Sh03
	ave.	8100	4		0.1	1	88	72 $^{218}\text{Fr}^m$			average
$^{218}\text{Fr}^m(\alpha)^{214}\text{At}^n$	7870.7	7.1	7869	5	-0.3	1	52	28 $^{218}\text{Fr}^m$			99Sh03
$^{218}\text{Ra}(\alpha)^{214}\text{Rn}$	8549.1	8.	8540	3	-1.1	3					70To07
	8541.0	10.			-0.1	3					70Va13
	8577.7	51.			-0.7	U			GSa		18Mi11
	8537.9	4.1			0.6	3			Jya		19Pa45
$^{218}\text{Ac}(\alpha)^{214}\text{Fr}$	9377.4	15.3	9380	60	0.1	5					70Bo13
	9391.3	15.			-0.1	5			Lza		17Su18
	9392.6	30.6			-0.1	5			GSa		19Mi08
	9376.3	26.			0.1	5			Lza		19Ya04
$^{218}\text{Th}(\alpha)^{214}\text{Ra}$	9861.3	20.4	9849	9	-0.6	5					73Ha32
	9846.1	10.			0.3	5					73No09
	9851.0	81.5			-0.0	U			GSa		15Kh09
$^{218}\text{Pa}(\alpha)^{214}\text{Ac}$	9794.1	20.	9791	12	-0.1	2					79Sc09 *
	9815	10			-2.4	F			GSa		00He17 *
	9790.0	14.2			0.1	2			Lza		20Zh01 *
	9781.8	17.3			0.6	o			Lza		20Ma27
$^{218}\text{Pa}^m(\alpha)^{214}\text{Ac}$	9872.4	15.0				2			Lza		20Zh01
$^{218}\text{U}(\alpha)^{214}\text{Th}$	8786.6	25.	8775	9	-0.5	4			Dbb		92An04
	8773.2	9.			0.2	4			Jya		07Le14
	8780.4	35.7			-0.2	U			Lza		18Ya01
$^{218}\text{U}^m(\alpha)^{214}\text{Th}$	10878.1	17.	10884	15	0.3	4			Jya		07Le14
	10901.4	30.6			-0.6	4			Lza		15Ma37
$^{218}\text{At}(\alpha)^{214}\text{Bi}$	$E_\alpha=6696.3(3.0, Z)$ to $(2)^-$ level at 53.2282 keV										Ens092 **
$^{218}\text{At}(\alpha)^{214}\text{Bi}$	$E_\alpha=6694(5)$ to 3^- level at 63 keV										19Cu02 **
$^{218}\text{At}(\alpha)^{214}\text{Bi}$	also $E_\alpha=6655(7)$ to 4^- level at 100 keV										19Cu02 **
$^{218}\text{Pa}(\alpha)^{214}\text{Ac}$	$E_\alpha=9614(20)$										00He17 **
$^{218}\text{Pa}(\alpha)^{214}\text{Ac}$	$E_\alpha=9544(10)$ to 91.8 level; F : from single α -spectra										00He17 **
$^{218}\text{Pa}(\alpha)^{214}\text{Ac}$	Also $E_\alpha=9524(16)$ to 91.8(0.8)										20Zh01 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{219}\text{Po}-u$	13601	32	13614	17	0.4	o			GS3	1.0	08Ch.A
	13614	17				2			GS3	1.0	12Ch19
$^{219}\text{At}-^{133}\text{Cs}_{1.647}$	166879.4	8.4	166881	3	0.2	1	17	^{219}At	MA8	1.0	17Ma29
$^{219}\text{Po}(\alpha)^{215}\text{Pb}$	5914.2	5.				3			ISa		15Fi07
$^{219}\text{At}(\alpha)^{215}\text{Bi}$	6390.9	50.	6342	5	-1.0	U					53Hy83
	6344.0	5.			-0.4	1	90	^{215}Bi	ISa		15Fi07
$^{219}\text{Rn}(\alpha)^{215}\text{Po}$	6946.21	0.3	6946.2	0.3	-0.1	1	100	^{215}Po			71Gr17 Z
$^{219}\text{Fr}(\alpha)^{215}\text{At}$	7448.7	2.0	7448.6	1.8	-0.1	3			Orm		68Ba73 Z
	7448.2	4.			0.1	3			Bka		82Bo04 Z
$^{219}\text{Ra}(\alpha)^{215}\text{Rn}$	8139.0	20.	8138	3	-0.1	U			ORa		69Ha32
	8128.7	10.			0.9	U					70Va13
	8128.7	20.			0.5	U			Dbb		89An13
	8138.0	3.				4					94Sh02
	8160	31			-0.7	U			GSa		18Mi11
	8128.8	10.2			0.9	U			GSt		18Sa45 *
$^{219}\text{Ac}(\alpha)^{215}\text{Fr}$	8826.5	10.				4					70Bo13
	9330.6	40.7	8830	50	-7.8	B			GSa		19Mi08 *
$^{219}\text{Th}(\alpha)^{215}\text{Ra}$	9514.1	20.	9510	60	-0.2	4					73Ha32
	9503.9	50.9			0.0	4			GSa		15Kh09
	9503.6	15.3			0.1	o			Lza		17Su18
	9503.9	20.			0.0	4			Lza		19Zh54
	9493.6	24.			0.2	4			Lza		19Ya04
	9512.0	16.3			-0.1	4			Lza		20Su02
$^{219}\text{Pa}(\alpha)^{215}\text{Ac}$	10084.6	50.	10130	70	0.6	3					87Fa.A
	10161.8	38.			-0.5	3			Lza		17Su18
$^{219}\text{U}(\alpha)^{215}\text{Th}$	9860.4	40.	9950	12	2.2	U			Dbb		93An07
	9956.2	18.			-0.4	3			Jya		07Le14
	9945.0	15.3			0.3	3			Lza		19Zh54
$^{219}\text{Np}(\alpha)^{215}\text{Pa}$	9167.8	203.7	9210	40	0.2	U			GSa		15De22 *
	9207.5	40.7				3			Lza		18Ya01
$^{219}\text{Ra}(\alpha)^{215}\text{Rn}$	also $E_\alpha=7.66(2)$ to 316-keV level										18Sa45 **
$^{219}\text{Ac}(\alpha)^{215}\text{Fr}$	$E_\alpha=8520(40)$ 9160(40)										19Mi08 **
$^{219}\text{Np}(\alpha)^{215}\text{Pa}$	$E_\alpha>9000$ keV										15De22 **
$^{220}\text{Po}-u$	16420	32	16386	19	-1.1	o			GS3	1.0	08Ch.A
	16386	19				2			GS3	1.0	12Ch19
$^{220}\text{At}-u$	15427	32	15433	15	0.2	o			GS3	1.0	08Ch.A
	15433	15				2			GS3	1.0	12Ch19
$^{220}\text{Rn}-^{133}\text{Cs}_{1.654}$	167777	11	167774.9	1.9	-0.2	U			MA8	1.0	09Ne03
$^{220}\text{Ra}-u$	11389	344	11028	8	-1.1	U			GR1	1.0	19An10
$^{210}\text{Fr}-^{220}\text{Fr}_{159}$ $^{208}\text{Fr}_{841}$	-2930	60	-2928	17	0.0	U			P24	2.5	82Au01
$^{211}\text{Fr}-^{220}\text{Fr}_{240}$ $^{208}\text{Fr}_{761}$	-4850	70	-4868	15	-0.1	U			P24	2.5	82Au01
$^{212}\text{Fr}-^{220}\text{Fr}_{321}$ $^{208}\text{Fr}_{679}$	-5450	60	-5392	12	0.4	U			P24	2.5	82Au01
$^{212}\text{Fr}-^{220}\text{Fr}_{263}$ $^{209}\text{Fr}_{738}$	-3730	60	-3745	12	-0.1	U			P24	2.5	82Au01
$^{213}\text{Fr}-^{220}\text{Fr}_{352}$ $^{209}\text{Fr}_{649}$	-5170	50	-5141	9	0.2	U			P24	2.5	82Au01
$^{212}\text{Fr}-^{220}\text{Fr}_{193}$ $^{210}\text{Fr}_{808}$	-3160	60	-3031	14	0.9	U			P24	2.5	82Au01
$^{220}\text{At}(\alpha)^{216}\text{Bi}^m$	6053.3	6.				3					89Bu09
$^{220}\text{Rn}(\alpha)^{216}\text{Po}$	6404.75	0.10	6404.74	0.10	-0.0	1	100	^{216}Po			71Gr17 Z
$^{220}\text{Fr}(\alpha)^{216}\text{At}$	6799.0	2.	6800.7	1.9	0.9	3			Orm		68Ba73 *
	6811.6	5.			-2.2	3					74Ho27 *
$^{220}\text{Ra}(\alpha)^{216}\text{Rn}$	7593.3	10.	7594	5	0.1	3					61Ru06
	7595.3	10.			-0.1	3					70Va13
	7598.4	20.4			-0.2	3			Dbb		90An19
	7587.2	10.			0.7	3			GSa		00He17
	7598.4	10.2			-0.4	3			GSt		18Sa45
$^{220}\text{Ac}(\alpha)^{216}\text{Fr}$	8347.1	10.	8348	4	0.1	5					70Bo13
	8348	5			-0.0	5					97Sh09 *
$^{220}\text{Th}(\alpha)^{216}\text{Ra}$	8953.1	20.	8973	11	1.0	4					73Ha32
	8981.6	13.2			-0.6	4			Jya		19Pa45

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{220}\text{Pa}(\alpha)^{216}\text{Ac}$	9829.1	50.	9704	11	-1.8	B					87Fa.A
	9696.7	16.3			0.4	2			Lza		17Hu08
	9718	20			-0.7	2			Lza		19Zh54
	9698.7	25.5			0.2	2			Lza		19Ya04
$^{220}\text{Pa}^m(\alpha)^{216}\text{Ac}$	9730.3	20.4	9730	20	-0.0	1	95	95 $^{220}\text{Pa}^m$	Lza		18Hu13
$^{220}\text{Pa}^n(\alpha)^{216}\text{Ac}$	9993.1	63.	10000	50	0.1	1	55	55 $^{220}\text{Pa}^n$	Lza		18Hu13
$^{220}\text{Np}(\alpha)^{216}\text{Pa}$	10226.3	18.3				3			Lza		19Zh23
$^{220}\text{Fr}(\alpha)^{216}\text{At}$	$E_\alpha=6675.2, 6631.0, 6570.2(2,Z)$ to ground state, $(2)^-$ at 44.59, $(0)^-$ at 105.89										Ens075 **
$^{220}\text{Fr}(\alpha)^{216}\text{At}$	$E_\alpha=6687.5, 6642.5, 6583.5(2,Z)$ to ground state, $(2)^-$ at 44.59, $(0)^-$ at 105.89										Ens075 **
$^{220}\text{Ac}(\alpha)^{216}\text{Fr}$	$E_\alpha=7792, 7855$ to levels at 409.3, 349.3 keV										Ens075 **
$^{221}\text{Po}-u$	21238	62	21228	21	-0.2	o			GS3	1.0	10Ch19
	21228	21				2			GS3	1.0	12Ch19
$^{221}\text{At}-u$	18028	32	18017	15	-0.3	o			GS3	1.0	08Ch.A
	18017	15				2			GS3	1.0	12Ch19
$^{221}\text{Fr}-^{226}\text{Ra}_{978}$	-10590	34	-10595	5	-0.2	U			MA3	1.0	92Bo28
$^{221}\text{Fr}-^{221}\text{Fr}_{159} \text{ } ^{209}\text{Fr}_{841}$	-3080	60	-3071	15	0.1	U			P24	2.5	82Au01
$^{221}\text{Rn}(\alpha)^{217}\text{Po}$	6161.8	5.8	6163	3	0.2	3			Db		77Vy02 *
	6163.5	5.4			-0.1	3			Orm		97Li23 *
	6163.5	5.4			-0.1	3			Orm		04Li28 *
$^{221}\text{Fr}(\alpha)^{217}\text{At}$	6457.3	2.0	6457.7	1.4	0.2	-			Orm		62Wa28 *
	6458.5	2.0			-0.4	-			Orm		68Le07 *
	ave.	6457.9	1.4		-0.1	1	99	78 ^{217}At			average
$^{221}\text{Ra}(\alpha)^{217}\text{Rn}$	6883.7	5.	6880.4	2.0	-0.7	3					61Ru06 *
	6881.3	3.			-0.3	3					95Ch74 *
	6878.3	3.			0.7	3					97Li12 *
	6884.8	10.2			-0.4	U			GS		18Sa45
$^{221}\text{Ac}(\alpha)^{217}\text{Fr}$	7786.2	10.	7790	60	0.1	4					70Bo13
	7782.1	5.			0.2	4			Lvn		87De.A
	7791.3	15.			0.0	4			Db		92An.A
	7811.4	30.6			-0.3	4			GS		18Mi11
$^{221}\text{Th}(\alpha)^{217}\text{Ra}$	8628.5	5.	8625	4	-0.6	5					70To07 Z
	8626.0	10.			-0.1	5					70Va13 Z
	8626.4	10.			-0.1	5			Db		90An19
	8614.2	10.			1.1	5			GS		00He17
	8596.9	66.			0.4	U					05Li17
	8667.2	30.6			-1.4	U			GS		18Mi11
	8381.9	41.			5.9	B			GS		19Mi08
	8617.2	19.			0.4	5			Lza		19Ya04 *
$^{221}\text{Pa}(\alpha)^{217}\text{Ac}$	9247.7	30.				3					89Mi17
	8250	200	9250	60	4.8	B			GS		19Mi08
$^{221}\text{U}(\alpha)^{217}\text{Th}$	9889.3	50.9				3			GS		15Kh09
$^{221}\text{Rn}(\alpha)^{217}\text{Po}$	$E_\alpha=6035.3(3,Z)$ to a tentative $(11/2^+)$ level at 15(5)keV above $(9/2^+)$ ground state										04Li28 **
$^{221}\text{Rn}(\alpha)^{217}\text{Po}$	$E_\alpha=6037(2)$ to a tentative $(11/2^+)$ level at 15(5) keV above $(9/2^+)$ ground state										04Li28 **
$^{221}\text{Rn}(\alpha)^{217}\text{Po}$	$E_\alpha=6037(2)$ to a tentative $(11/2^+)$ level at 15(5) keV above $(9/2^+)$ ground state										04Li28 **
$^{221}\text{Fr}(\alpha)^{217}\text{At}$	$E_\alpha=6341.1(2,Z), 6125.1(3,Z)$ to ground state, $5/2^-$ level at 218.12 keV										Ens039 **
$^{221}\text{Fr}(\alpha)^{217}\text{At}$	$E_\alpha=6341.3(2,Z), 6127.2(3,Z)$ to ground state, $5/2^-$ level at 218.12 keV										Ens039 **
$^{221}\text{Ra}(\alpha)^{217}\text{Rn}$	$E_\alpha=6761.2, 6668.2, 6613.2, 6591.2(5,Z)$ to ground state, levels at 88.9 149.18 174.3keV										Ens039 **
$^{221}\text{Ra}(\alpha)^{217}\text{Rn}$	$E_\alpha=6610(3,Z)$ to 149.2 level										97Li12 **
$^{221}\text{Ra}(\alpha)^{217}\text{Rn}$	$E_\alpha=6754, 6662, 6607(..)$ to ground state, 93.02, 149.2 level										97Li12 **
$^{222}\text{Po}-u$	24133	72	24140	40	0.1	o			GS3	1.0	10Ch19
	24140	43				2			GS3	1.0	12Ch19
$^{222}\text{At}-u$	22459	32	22494	17	1.1	o			GS3	1.0	08Ch.A
	22494	17				2			GS3	1.0	12Ch19

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{222}\text{Fr}-^{133}\text{Cs}_{1.669}$	175383.3	8.0				2			MA8	1.0	14Kr09
$^{222}\text{Fr}-^{226}\text{Ra}_{.982}$	-7410	25	-7368	8	1.7	U			MA3	1.0	92Bo28
$^{213}\text{Fr}-^{222}\text{Fr}_{.240}$ $^{210}\text{Fr}_{.761}$	-4810	60	-4941	11	-0.9	U			P24	2.5	82Au01
$^{213}\text{Fr}-^{222}\text{Fr}_{.096}$ $^{212}\text{Fr}_{.904}$	-1940	60	-1948	9	-0.1	U			P24	2.5	82Au01
$^{221}\text{Fr}-^{222}\text{Fr}_{.498}$ $^{220}\text{Fr}_{.502}$	-610	90	-643	6	-0.1	U			P34	2.5	86Au02
$^{222}\text{Rn}(\alpha)^{218}\text{Po}$	5590.39	0.3	5590.4	0.3	-0.0	1	100	99 ^{218}Po			71Gr17 Z
$^{222}\text{Ra}(\alpha)^{218}\text{Rn}$	6680.0	5.	6678	4	-0.4	1	69	62 ^{222}Ra			56As38 Z
$^{222}\text{Ac}(\alpha)^{218}\text{Fr}$	7137.5	2.				3			Bka		82Bo04 Z
$^{222}\text{Ac}^m(\alpha)^{218}\text{Fr}^m$	7128.6	22.4				2					72Es03 *
$^{222}\text{Th}(\alpha)^{218}\text{Ra}$	8127.7	10.	8132.6	2.9	0.5	U					70To07
	8130.7	8.			0.2	4					70Va13
	8126.7	15.			0.4	U			Dbb		92An.A
	8120.6	10.			1.2	U			GSa		00He17
	8116.4	48.			0.3	U					05Li17
	8132.8	3.			-0.1	4			Jya		16Pa28
	8157.3	30.6			-0.8	U			GSa		18Mi11
$^{222}\text{Pa}(\alpha)^{218}\text{Ac}$	8788.6	41.				6			GSa		19Mi08 *
$^{222}\text{Pa}(\alpha)^{218}\text{Ac}^m$	8697.0	30.	8736	13	1.3	7					70Bo13
	8745.5	15.			-0.6	7			GSa		95Ho.C *
$^{222}\text{U}(\alpha)^{218}\text{Th}$	9481.1	50.9				6			GSa		15Kh09
$^{222}\text{Np}(\alpha)^{218}\text{Pa}$	10200.2	33.7				3			Lza		20Ma27
* $^{222}\text{Ac}^m(\alpha)^{218}\text{Fr}^m$	$E_\alpha=7000(20)$ keV from $^{1972}\text{Es}_{03}$ to $^{218}\text{Fr}^m$ at 86(8) keV										FGK20c **
* $^{222}\text{Pa}(\alpha)^{218}\text{Ac}^m$	$E_\alpha=8210(15)$ to $^{218}\text{Ac}^n$ at 384.49 keV above $^{218}\text{Ac}^m$										Nub211 **
$^{223}\text{At}-u$	25172	32	25151	15	-0.7	o			GS3	1.0	08Ch.A
	25151	15				2			GS3	1.0	12Ch19
$^{223}\text{Rn}-^{133}\text{Cs}_{1.677}$	180453	11	180446	8	-0.6	1	58	58 ^{223}Rn	MA8	1.0	09Ne03
$^{223}\text{Rn}-u$	21899	32	21889	8	-0.3	o			GS3	1.0	08Ch.A
	21880	13			0.7	1	42	42 ^{223}Rn	GS3	1.0	12Ch19
$^{213}\text{Fr}-^{223}\text{Fr}_{.087}$ $^{212}\text{Fr}_{.913}$	-1900	60	-1943	9	-0.3	U			P24	2.5	82Au01
$^{222}\text{Fr}-^{223}\text{Fr}_{.498}$ $^{221}\text{Fr}_{.502}$	790	100	559	8	-0.9	U			P34	2.5	86Au02
$^{223}\text{Fr}(\alpha)^{219}\text{At}$	5431.6	80.	5561.4	2.8	1.6	U					55Ad10
	5562	3			-0.2	1	85	79 ^{219}At			01Li44
$^{223}\text{Ra}(\alpha)^{219}\text{Rn}$	5978.9	0.3	5978.99	0.21	0.3	-			Orm		62Wa18 *
	5979.1	0.3			-0.4	-			BIP		71Gr17 *
ave.	5979.00	0.21			-0.0	1	100	96 ^{219}Rn			average
$^{223}\text{Ac}(\alpha)^{219}\text{Fr}$	6783.2	1.0				4			Orm		69Le.A *
$^{223}\text{Th}(\alpha)^{219}\text{Ra}$	7602	23	7567	4	-1.5	U			ORa		69Ha32 *
	7589	14			-1.6	U					70Va13 *
	7570	25			-0.1	U					84Mi.A *
	7568	10			-0.1	5					87El02 *
	7567.4	10.			-0.1	5			Dbb		90An19 *
	7566.1	5.			0.1	5					92Li09 *
$^{223}\text{Pa}(\alpha)^{219}\text{Ac}$	8345.0	10.	8340	60	-0.0	5					70Bo13
	8339.9	10.2			0.1	o			Dbb		89An.A *
	8350.0	15.			-0.1	5			Dbb		90An19 *
	8339.9	15.			0.1	5			GSa		95Ho.C *
	8321.6	5.			0.4	5			Jya		99Ho28
	8370.3	41.			-0.4	5			GSa		19Mi08 *
$^{223}\text{U}(\alpha)^{219}\text{Th}$	8940.7	40.7	9158	17	3.4	F			Dbb		91An10 *
	9157.5	17.3				5			Lza		20Su02 *
$^{223}\text{Np}(\alpha)^{219}\text{Pa}$	9650.4	44.8				4			Lza		17Su18 *
$^{223}\text{Fr}(\beta^-)^{223}\text{Ra}$	1170	10	1149.1	0.8	-2.1	U					75We23 *
* $^{223}\text{Ra}(\alpha)^{219}\text{Rn}$	$E_\alpha=5747.0(0.4,Z)$, $5715.7(0.3,Z)$, $5606.7(0.3,Z)$ keV										62Wa18 **
*	to $11/2^+$ level at 126.77, $7/2^+$ at 158.64, $3/2^+$ at 269.48 keV										Ens01a **
* $^{223}\text{Ra}(\alpha)^{219}\text{Rn}$	$E_\alpha=5747.0(0.40,Z)$, $5716.23(0.29,Z)$, $5606.73(0.30,Z)$ keV										71Gr17 **
*	to $11/2^+$ level at 126.77, $7/2^+$ at 158.64, $3/2^+$ at 269.48 keV										Ens01a **
* $^{223}\text{Ac}(\alpha)^{219}\text{Fr}$	$E_\alpha=6661.6$, 6646.7 , $6563.7(1.0,Z)$ to ground state, $5/2^+$ at 15.0, $7/2^-$ at 98.58										Ens01a **
* $^{223}\text{Th}(\alpha)^{219}\text{Ra}$	$E_\alpha=7330(20)$ to mixture of excited states at 138(10) keV										GAu **
* $^{223}\text{Th}(\alpha)^{219}\text{Ra}$	$E_\alpha=7317(10)$ to mixture of excited states at 138(10) keV										GAu **
* $^{223}\text{Th}(\alpha)^{219}\text{Ra}$	$E_\alpha=7324(10)$ to 113.8, 7285(10) 55% to 140.0, 26% to 152.0 level										92Li09 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
* $^{223}\text{Th}(\alpha)^{219}\text{Ra}$	$E_\alpha=7290(10)$ 55% to 140.0, 26% to 152.0 level										92Li09 **
* $^{223}\text{Th}(\alpha)^{219}\text{Ra}$	$E_\alpha=7318(5), 7293(5), 7281(5)$ to 113.8, 140.0, 152.0 levels										92Li09 **
* $^{223}\text{U}(\alpha)^{219}\text{Th}$	Might be 8753 to 224 keV level										20Su02 **
* $^{223}\text{U}(\alpha)^{219}\text{Th}$	Also $E_\alpha=8753$ to 224 keV level										20Su02 **
* $^{223}\text{Np}(\alpha)^{219}\text{Pa}$	Extracted as difference of sum energy 19453(23) and $E_\alpha(^{219}\text{Pa})=9976(37)$										17Su18 **
* $^{223}\text{Fr}(\beta^-)^{223}\text{Ra}$	$E_{\beta^-}=1120(10)$ to $3/2^-$ level at 50.128 keV										Ens01a **
$^{224}\text{At}-\text{u}$	29744	63	29749	24	0.1	o			GS3	1.0	10Ch19
	29749	24				2			GS3	1.0	12Ch19
$^{224}\text{Rn}-^{133}\text{Cs}_{1.684}$	183304	16	183315	11	0.7	1	43	43 ^{224}Rn	MA8	1.0	09Ne03
$^{224}\text{Rn}-\text{u}$	24073	32	24096	11	0.7	o			GS3	1.0	08Ch.A
	24104	14			-0.6	1	57	57 ^{224}Rn	GS3	1.0	12Ch19
$^{224}\text{Fr}-\text{u}$	23399	32	23348	12	-1.6	o			GS3	1.0	08Ch.A
	23398	14			-3.6	B			GS3	1.0	12Ch19
$^{224}\text{Fr}-^{133}\text{Cs}_{1.684}$	182567	12				2			MA8	1.0	14Kr09
$^{224}\text{Ra}-^{133}\text{Cs}_{1.684}$	179430	30	179429.3	1.9	-0.0	U			MA8	1.0	14Bo26
$^{223}\text{Fr}-^{224}\text{Fr}_{.747}$ $^{220}\text{Fr}_{.253}$	-620	70	-769	9	-0.9	U			P34	2.5	86Au02
$^{222}\text{Fr}-^{224}\text{Fr}_{.496}$ $^{220}\text{Fr}_{.505}$	10	70	-260#	50#	-1.5	U			P24	2.5	82Au01
$^{223}\text{Fr}-^{224}\text{Fr}_{.747}$ $^{220}\text{Fr}_{.253}$	-410	70	-840#	80#	-2.5	B			P24	2.5	82Au01
$^{223}\text{Fr}-^{224}\text{Fr}_{.664}$ $^{221}\text{Fr}_{.336}$	780	110	-520	8	-4.7	F			P34	2.5	86Au02 *
$^{223}\text{Fr}-^{224}\text{Fr}_{.664}$ $^{221}\text{Fr}_{.336}$	-110	70	-590#	70#	-2.7	B			P24	2.5	82Au01
$^{224}\text{Ra}(\alpha)^{220}\text{Rn}$	5788.93	0.15	5788.92	0.15	-0.1	1	100	69 ^{220}Rn			71Gr17 Z
$^{224}\text{Ac}(\alpha)^{220}\text{Fr}$	6326.9	0.7				4			Orm		69Le.A *
$^{224}\text{Th}(\alpha)^{220}\text{Ra}$	7304.7	10.	7299	6	-0.6	4					61Ru06
	7304.7	10.			-0.6	4					70Va13
	7300.7	20.			-0.1	U			Dbb		89An13
	7286.4	10.			1.2	4			GSa		00He17
$^{224}\text{Pa}(\alpha)^{220}\text{Ac}$	7695.2	10.	7694	4	-0.2	6					70Bo13 *
	7692.6	10.			0.1	F			Dbb		90An19 *
	7680	15			0.9	U			GSa		95Ho.C
	7693.3	5.			0.1	6					96Li05 *
$^{224}\text{U}(\alpha)^{220}\text{Th}$	8624.3	15.	8628	7	0.3	5			Dbb		91An10
	8612.1	20.			0.8	5			ORa		92To02
	8631.9	8.1			-0.5	5			Orm		14Lo10 *
$^{224}\text{Np}(\alpha)^{220}\text{Pa}^m$	9303.5	20.4	9303	20	-0.0	1	95	91 ^{224}Np	Lza		18Hu13
$^{224}\text{Np}(\alpha)^{220}\text{Pa}^n$	9029.6	63.1	9030	50	0.1	1	55	45 $^{220}\text{Pa}^n$	Lza		18Hu13
$^{224}\text{Fr}(\beta^-)^{224}\text{Ra}$	2830	50	2923	11	1.9	U					75We23 *
* $^{223}\text{Fr}-^{224}\text{Fr}_{.664}$ ^{221}Fr .	F : rejection based on line-shape analysis										86Au02 **
* $^{224}\text{Ac}(\alpha)^{220}\text{Fr}$	$E_\alpha=6213.8, 6207.0, 6141.7, 6059.8(0.7.Z)$ keV										69Le.A **
*	to ground state, 3^+ at 6.92, 3^+ at 72.99, 2^- at 156.82 keV										Ens114 **
* $^{224}\text{Pa}(\alpha)^{220}\text{Ac}$	$E_\alpha=7490(10)$ to 5^- level at 68.71 keV										Ens114 **
* $^{224}\text{Pa}(\alpha)^{220}\text{Ac}$	F : intensities in contradiction with reference										96Li05 **
* $^{224}\text{Pa}(\alpha)^{220}\text{Ac}$	$E_\alpha=7488(5), 7375(5)$ to (5^-) level at 68.71 keV and 184.21 level										Ens114 **
* $^{224}\text{U}(\alpha)^{220}\text{Th}$	$E_\alpha=8479(8), 8095(11)$ to ground state, 386.7(1.8) 2^+ level										14Lo10 **
* $^{224}\text{Fr}(\beta^-)^{224}\text{Ra}$	$E_{\beta^-}=1780(50)$ to 1^- level at 1053.041 keV, and other E_{β^-}										Ens15c **
$^{225}\text{Rn}-^{133}\text{Cs}_{1.692}$	188484	23	188461	12	-1.0	1	27	27 ^{225}Rn	MA8	1.0	09Ne03
	28498	32	28486	12	-0.4	o			GS3	1.0	08Ch.A
$^{225}\text{Fr}-\text{u}$	28477	14			0.6	1	73	73 ^{225}Rn	GS3	1.0	12Ch19
	25574	14	25572	13	-0.1	1	84	84 ^{225}Fr	GS3	1.0	12Ch19
$^{221}\text{Fr}-^{225}\text{Fr}_{.655}$ $^{213}\text{Fr}_{.346}$	-1110	60	-1095	9	0.1	U			P24	2.5	82Au01
$^{224}\text{Fr}^x-^{225}\text{Fr}_{.747}$ $^{221}\text{Fr}_{.253}$	50	80	700#	100#	3.2	B			P24	2.5	82Au01
$^{224}\text{Fr}^x-^{225}\text{Fr}_{.498}$ $^{223}\text{Fr}_{.502}$	190	80	760#	100#	2.8	B			P24	2.5	82Au01
$^{225}\text{Ra}(\alpha)^{221}\text{Rn}$	5096.7	5.1				2					00Li37
$^{225}\text{Ac}(\alpha)^{221}\text{Fr}$	5936.1	2.	5935.1	1.4	-0.5	-			Orm		67Ba51 Z
	5934.5	2.			0.3	-					67Dz02 Z
	ave. 5935.3	1.4			-0.1	1	99	79 ^{221}Fr			average
$^{225}\text{Th}(\alpha)^{221}\text{Ra}$	6920.7	3.	6921.4	2.1	0.2	4					61Ru06 *
	6922.1	3.			-0.2	4					87Li.A *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{225}\text{Pa}(\alpha)^{221}\text{Ac}$	7381.5	20.	7400	60	0.4	U			ORa		68Ha14
	7376.4	10.			0.5	F					70Bo13 *
	7392.7	5.1			0.2	5			Lvn		87De.A
	7383.5	19.			0.3	5					00Sa52
	7432.1	30.5			-0.5	5			GSa		18Mi11
$^{225}\text{U}(\alpha)^{221}\text{Th}$	8012.7	20.	8007	6	-0.3	o			Dbb		89An13
	8022.9	20.			-0.8	6			GSa		89He13
	8021.9	15.			-1.0	6			ORa		92To02
	8012.7	20.4			-0.3	6			Dbb		94Ye08
	8010	10			-0.3	6			GSa		00He17 *
	7992.4	30.5			0.5	U			GSa		18Mi11
	7992.3	10.2			1.5	6			GSa		19Mi08
$^{225}\text{Np}(\alpha)^{221}\text{Pa}$	8786.5	20.	8820	70	0.6	4			Dbb		94Ye08
	8959.5	102.			-1.2	4			GSa		19Mi08
$^{225}\text{Fr}(\beta^-)^{225}\text{Ra}$	1820	30	1828	12	0.3	1	16	16 ^{225}Fr			75We23 *
$^{225}\text{Ra}(\beta^-)^{225}\text{Ac}$	360	10	356	5	-0.4	1	25	20 ^{225}Ac			55Ma.A *
	360	30			-0.1	U					55Pe24 *
* $^{225}\text{Th}(\alpha)^{221}\text{Ra}$	$E_\alpha=6800.2, 6746.2, 6503.2, 6480.2, 6443.2(3,Z)$ keV										61Ru06 **
*	to ground state, $7/2^+ 53.14, 7/2^+ 299.16, 3/2^+ 321.39, 5/2^+ 359.02$ levels										Ens075 **
* $^{225}\text{Th}(\alpha)^{221}\text{Ra}$	$E_\alpha=6799.3, 6745.3, 6504.3, 6483.3, 6447.3(3,Z)$ keV										87Li.A **
*	to ground state, $7/2^+ 53.14, 7/2^+ 299.16, 3/2^+ 321.39, 5/2^+ 359.02$ levels										Ens075 **
* $^{225}\text{Pa}(\alpha)^{221}\text{Ac}$	F : average of two branches										87De.A **
* $^{225}\text{U}(\alpha)^{221}\text{Th}$	$E_\alpha=7868(15), 7621(15)$ to ground state, 250.9 level										00He17 **
* $^{225}\text{Fr}(\beta^-)^{225}\text{Ra}$	$E_{\beta^-}=1640(10)$. 28% to $3/2^-$ level at 225.2										Ens095 **
* $^{225}\text{Ra}(\beta^-)^{225}\text{Ac}$	$E_{\beta^-}=320(10) 320(30)$ respectively, to $3/2^+$ level at 40.09 keV										Ens095 **
$^{226}\text{Rn}-^{133}\text{Cs}_{1.699}$	191490	17	191499	11	0.5	1	44	44 ^{226}Rn	MA8	1.0	09Ne03
$^{226}\text{Rn}-u$	30864	32	30861	11	-0.1	o			GS3	1.0	08Ch.A
	30868	15			-0.4	1	56	56 ^{226}Rn	GS3	1.0	12Ch19
$^{226}\text{Fr}-u$	29565	32	29545	7	-0.6	o			GS3	1.0	08Ch.A
	29566	13			-1.7	1	26	26 ^{226}Fr	GS3	1.0	12Ch19
$^{226}\text{Fr}-^{133}\text{Cs}_{1.699}$	190173.9	7.8	190182	7	1.0	1	74	74 ^{226}Fr	MA8	1.0	14Kr09
$^{133}\text{Cs}-^{226}\text{Ra}_{.588}$	-109487	9	-109488.1	1.2	-0.1	U			MA3	1.0	92Bo28
	-109499	13			0.8	U			MA4	1.0	99Am05
$^{223}\text{Fr}-^{226}\text{Fr}_{.493} \quad ^{220}\text{Fr}_{.507}$	-800	80	-1007	4	-1.0	U			P24	2.5	82Au01
$^{225}\text{Fr}-^{226}\text{Fr}_{.796} \quad ^{221}\text{Fr}_{.204}$	-570	100	-794	13	-0.9	U			P24	2.5	82Au01
$^{225}\text{Fr}-^{226}\text{Fr}_{.498} \quad ^{224}\text{Fr}_{.502}$	-260	90	-850#	50#	-2.6	B			P24	2.5	82Au01
$^{226}\text{Ra}(\alpha)^{222}\text{Rn}$	4870.70	0.25	4870.70	0.25	-0.0	1	100	99 ^{222}Rn			71Gr17 Z
$^{226}\text{Ac}(\alpha)^{222}\text{Fr}$	5496.1	5.	5506	8	0.2	U			DbA		75Va.A Z
$^{226}\text{Th}(\alpha)^{222}\text{Ra}$	6448.5	3.0	6452.5	1.0	1.3	U					56As38 *
	6454.8	3.6			-0.6	U			DbA		75Va.A *
	6452.6	1.0			-0.1	1	99	61 ^{226}Th	Gea		12Ma30
$^{226}\text{Pa}(\alpha)^{222}\text{Ac}$	6986.9	10.				4					64Mc21
$^{226}\text{U}(\alpha)^{222}\text{Th}$	7747.4	30.	7701	4	-1.6	U					73Vi10 *
	7706.6	15.			-0.4	5			Dbb		90An22
	7701.6	5.			-0.1	5			Jya		99Gr28
	7691.4	10.			0.9	o			GSa		00He17
	7696.5	10.			0.4	5			GSa		01Ca.B
	7696.4	20.4			0.2	5			RIa		16Ka13
	7706.6	30.5			-0.2	U			GSa		18Mi11
$^{226}\text{Np}(\alpha)^{222}\text{Pa}$	8189.2	20.4	8330	50	2.6	B			GSa		90Ni05
	8205.5	20.			2.3	B			Dbb		94Ye08
	8327.6	20.4				7			GSa		19Mi08
$^{226}\text{Ra}(\text{p,t})^{224}\text{Ra}$	-2816	15	-2819.1	1.9	-0.2	U			ANL		74Fr01
$^{226}\text{Ra}(\text{d,t})^{225}\text{Ra}$	-146	10	-139.6	2.9	0.6	U					83Ny01

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{226}\text{Fr}(\beta^-)^{226}\text{Ra}$	3804	330	3853	7	0.1	U					75We23 *
	3704	100			1.5	U					87Ve.A *
$^{226}\text{Ac}(\beta^-)^{226}\text{Th}$	1115	7	1112	5	-0.5	-					68Va17 *
ave.	1115	6			-0.5	1	52	39 ^{226}Th			average
$^{226}\text{Th}(\alpha)^{222}\text{Ra}$	$E_\alpha=6334.6(3,Z)$, 6224.6(3,Z) to ground state, 2^+ level at 111.12 keV										Ens11b **
$^{226}\text{Th}(\alpha)^{222}\text{Ra}$	$E_\alpha=6337.1(1.0,Z)$, 6233.6(1.0,Z) to ground state, 2^+ level at 111.12 keV										Ens11b **
$^{226}\text{U}(\alpha)^{222}\text{Th}$	$E_\alpha=7430(30)$ to 2^+ level at 183.3(0.3) keV										94Ye08 **
$^{226}\text{Fr}(\beta^-)^{226}\text{Ra}$	$E_{\beta^-}=3550(330)$ 3450(100) respectively, to 1^- level at 253.73 keV										Ens964 **
$^{226}\text{Ac}(\beta^-)^{226}\text{Th}$	$E_{\beta^-}=885(7)$ to 1^- level at 230.37 keV										Ens964 **
$^{227}\text{Rn}-^{133}\text{Cs}_{1.707}$	196686	19	196698	15	0.6	1	63	63 ^{227}Rn	MA8	1.0	09Ne03
$^{227}\text{Rn}-u$	35288	33	35304	15	0.5	o			GS3	1.0	08Ch.A
	35325	25			-0.8	1	37	37 ^{227}Rn	GS3	1.0	12Ch19
$^{227}\text{Fr}-u$	31868	32	31865	6	-0.1	o			GS3	1.0	08Ch.A
	31869	14			-0.3	1	20	20 ^{227}Fr	GS3	1.0	12Ch19
$^{227}\text{Fr}-^{133}\text{Cs}_{1.707}$	193258.0	7.1	193259	6	0.1	1	80	80 ^{227}Fr	MA8	1.0	14Kr09
$^{225}\text{Fr}-^{227}\text{Fr}_{708}$ $^{220}\text{Fr}_{292}$	-410	130	-547	13	-0.4	U			P24	2.5	82Au01
$^{224}\text{Fr}^x-^{227}\text{Fr}_{493}$ $^{221}\text{Fr}_{507}$	-220	80	480#	100#	3.5	B			P24	2.5	82Au01
$^{227}\text{Ac}(\alpha)^{223}\text{Fr}$	5043.0	2.0	5042.27	0.14	-0.4	U					66Ba19 Z
	5042.27	0.14			-0.0	1	100	94 ^{223}Fr			86Ry04 Z
$^{227}\text{Th}(\alpha)^{223}\text{Ra}$	6146.60	0.10	6146.60	0.10	-0.0	1	100	96 ^{223}Ra	BIP		71Gr17 *
$^{227}\text{Pa}(\alpha)^{223}\text{Ac}$	6581.5	3.	6580.4	2.1	-0.4	5					63Su.A *
	6579.3	3.			0.4	5					90Sh15 *
$^{227}\text{U}(\alpha)^{223}\text{Th}$	7230	30	7235	3	0.2	U			ORa		69Ha32 *
	7206	16			1.8	U					91Ho05
	7234.7	3.1				6			GSa		15Ka24
$^{227}\text{Np}(\alpha)^{223}\text{Pa}$	7818.0	10.	7816	14	-0.0	o			Dbb		90An19
	7815.0	20.			0.1	6			GSa		90Ni05
	7818.0	20.			-0.1	6			Dbb		94Ye08
$^{226}\text{Ra}(n,\gamma)^{227}\text{Ra}$	4561.43	0.27				2			ILn		81Vo03 Z
$^{227}\text{Fr}(\beta^-)^{227}\text{Ra}$	2476	100	2505	6	0.3	U					75We23 *
$^{227}\text{Ra}(\beta^-)^{227}\text{Ac}$	1345	20	1327.9	2.3	-0.9	U					53Bu63 *
	1335	15			-0.5	U					71Lo15 *
$^{227}\text{Ac}(\beta^-)^{227}\text{Th}$	45.5	1.0	44.8	0.8	-0.7	-					55Be20
	43.5	1.5			0.8	-					59No41
ave.	44.9	0.8			-0.2	1	99	96 ^{227}Th			average
$^{227}\text{Th}(\alpha)^{223}\text{Ra}$	$E_\alpha=6038.01(0.15,Z)$, 5977.72(0.10,Z), 5756.89(0.15,Z) keV										71Gr17 **
*	to ground state, $7/2^+$ at 61.424, $1/2^+$ at 286.182 keV										Ens01a **
$^{227}\text{Pa}(\alpha)^{223}\text{Ac}$	$E_\alpha=6465.8(3,Z)$, 6423.8(3,Z), 6415.8(3,Z), 6401.7(3,Z), 6356.7(3,Z)										63Su.A **
*	to ground state, $7/2^-$ at 42.4, $5/2^-$ at 50.7, $5/2^+$ at 64.62, $7/2^+$ at 110.06										Ens01a **
$^{227}\text{Pa}(\alpha)^{223}\text{Ac}$	$E_\alpha=6463$, 6421, 6355 keV (all errors 3 keV, estimated by evaluator)										90Sh15 **
*	to ground state, $7/2^-$ at 42.4, $5/2^-$ at 50.7, $7/2^+$ at 110.06 keV										Ens01a **
$^{227}\text{U}(\alpha)^{223}\text{Th}$	$E_\alpha=6860(30)$ to $3/2^+$ level at 247(1) keV										Ens01a **
$^{227}\text{Fr}(\beta^-)^{227}\text{Ra}$	$E_{\beta^-}=1800(100)$ to $1/2^-$ level at 675.863 keV										Ens162 **
$^{227}\text{Ra}(\beta^-)^{227}\text{Ac}$	$E_{\beta^-}=1310(20)$ 1300(15) respectively, to $3/2^+$ level at 27.369 and $5/2^+$ at 46.354										Ens162 **
$^{228}\text{Rn}-^{133}\text{Cs}_{1.714}$	199897	24	199891	19	-0.3	1	63	63 ^{228}Rn	MA8	1.0	09Ne03
$^{228}\text{Rn}-u$	37856	33	37835	19	-0.6	o			GS3	1.0	08Ch.A
	37825	31			0.3	1	37	37 ^{228}Rn	GS3	1.0	12Ch19
$^{228}\text{Fr}-u$	35833	34	35839	7	0.2	o			MA8	1.0	11Kr.A
	35852	32			-0.4	o			GS3	1.0	08Ch.A
	35821	16			1.2	1	20	20 ^{228}Fr	GS3	1.0	12Ch19
$^{228}\text{Fr}-^{133}\text{Cs}_{1.714}$	197899.5	8.1	197895	7	-0.6	1	80	80 ^{228}Fr	MA8	1.0	14Kr09

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{224}\text{Fr} \leftarrow ^{228}\text{Fr}_{.491} \rightarrow ^{220}\text{Fr}_{.509}$	−540	320	−390#	100#	0.2	U			P24	2.5	82Au01
$^{228}\text{Th}(\alpha)^{224}\text{Ra}$	5520.17	0.22	5520.15	0.22	−0.1	1	100	69 ^{224}Ra			71Gr17 Z
$^{228}\text{Pa}(\alpha)^{224}\text{Ac}$	6266.7	3.	6264.5	1.5	−0.7	5					58Hi.A *
	6264.7	3.			−0.1	5					93Sh07 *
	6263.5	2.			0.5	5					94Ah03 *
$^{228}\text{U}(\alpha)^{224}\text{Th}$	6803.6	10.	6800	9	−0.4	5					61Ru06
	6774.1	25.4			1.0	5			GSa		18Mi11
$^{228}\text{Np}(\alpha)^{224}\text{Pa}$	7308.5	36.	7540#	100#	3.7	D			JAA		03Ni10 *
$^{228}\text{Pu}(\alpha)^{224}\text{U}$	7949.7	20.	7940	18	−0.5	6			DBb		94An02
	7911.0	35.			0.8	6			JAA		03Ni10
$^{228}\text{Ra}(\beta^-)^{228}\text{Ac}$	46.7	2.	45.5	0.6	−0.6	3					61To10 *
	45.5	0.9			0.0	3					72He.A *
	45.3	1.0			0.2	3					95So11 *
$^{228}\text{Ac}(\beta^-)^{228}\text{Th}$	2240	20	2123.8	2.6	−5.8	B					53Ky19 *
	2158	20			−1.7	U					57Bj56 *
$^{228}\text{Pa}(\epsilon)^{228}\text{Th}$	2109	15	2153	4	2.9	B					73Ku09 *
$^{228}\text{Pa}(\alpha)^{224}\text{Ac}$	$E_\alpha=6119.2(3,Z), 6106.2(3,Z), 6079.2(3,Z)$ to 37.2, 51.9, 78.4 levels										93Sh07 **
$^{228}\text{Pa}(\alpha)^{224}\text{Ac}$	$E_\alpha=6118(3)$ to 37.2 level										93Sh07 **
$^{228}\text{Pa}(\alpha)^{224}\text{Ac}$	$E_\alpha=6117(2)$ to 37.1 level										94Ah03 **
$^{228}\text{Np}(\alpha)^{224}\text{Pa}$	Trends from Mass Surface TMS suggest ^{228}Np 230 keV less bound										GAu **
$^{228}\text{Ra}(\beta^-)^{228}\text{Ac}$	$E_{\beta^-}=40(2)$ 39(1) respectively, to 1^- level at 6.28 keV, and other E_{β^-}										Ens143 **
$^{228}\text{Ra}(\beta^-)^{228}\text{Ac}$	$E_{\beta^-}=39.0(1.0)$ to 1^+ level at 6.28 keV										Ens143 **
$^{228}\text{Ac}(\beta^-)^{228}\text{Th}$	$E_{\beta^-}=2180(20)$ to 2^+ level at 57.773 keV, and other E_{β^-}										Ens143 **
$^{228}\text{Ac}(\beta^-)^{228}\text{Th}$	$E_{\beta^-}=2100(20), 1760, 1180$ to 2^+ at 57.773, 3^- at 396.094, 2^+ at 968.984										Ens143 **
$^{228}\text{Pa}(\epsilon)^{228}\text{Th}$	$pK=0.33(0.08)$ to 3^+ level at 1944.904 keV, recalculated										Ens143 **
$^{229}\text{Rn} \leftarrow ^{133}\text{Cs}_{1.722}$	205069	14				2			MA8	1.0	09Ne03
$^{229}\text{Fr} \leftarrow ^{133}\text{Cs}_{1.722}$	201262	40	201103	5	−4.0	B			MA8	1.0	08We02 *
	201104.5	6.4			−0.2	1	70	70 ^{229}Fr	MA8	1.0	14Kr09
$^{229}\text{Fr} \rightarrow \text{u}$	38343	32	38291	5	−1.6	o			GS3	1.0	08Ch.A
	38298	15			−0.4	1	13	13 ^{229}Fr	GS3	1.0	12Ch19
$^{229}\text{Fr} \leftarrow ^{238}\text{U}_{.962}$	−10576	13	−10566	6	0.8	1	18	17 ^{229}Fr	MA8	1.0	14Kr09
$^{229}\text{Ra} \leftarrow ^{133}\text{Cs}_{1.722}$	197782	21	197768	17	−0.6	2			MA8	1.0	08We02
	197746	27			0.8	2			MA8	1.0	05He26
$^{229}\text{Ac} \rightarrow \text{u}$	32947	13				2			GS3	1.0	12Ch19
$^{229}\text{Th}(\alpha)^{225}\text{Ra}$	5167.4	1.2	5167.6	1.0	0.1	−			Kum		71Bb10 *
	5168.2	2.			−0.3	−					87He28 Z
	ave. 5167.6	1.0			−0.1	1	99	95 ^{225}Ra			average
$^{229}\text{Pa}(\alpha)^{225}\text{Ac}$	5835.6	5.	5835	4	−0.2	1	73	60 ^{225}Ac			63Su.A *
$^{229}\text{U}(\alpha)^{225}\text{Th}$	6475.5	3.				5					61Ru06 Z
$^{229}\text{Np}(\alpha)^{225}\text{Pa}$	7012.7	20.	7020	60	0.1	6			ORa		68Ha14
	7015.8	23.			0.1	6					00Sa52
	7032.8	30.5			−0.2	6			GSa		18Mi11
$^{229}\text{Pu}(\alpha)^{225}\text{U}$	7592.9	30.	7600	60	0.1	7			DBb		94An02
	7598.0	10.			0.0	o			GSa		01Ca.B
	7589.8	20.			0.2	7			GSa		10Kh06
	7613	31			−0.3	7			GSa		18Mi11
$^{229}\text{Am}(\alpha)^{225}\text{Np}$	8137.4	20.4				5			GSa		15De22 *
$^{229}\text{Ra}(\beta^-)^{229}\text{Ac}$	1760	40	1872	20	2.8	B					75We23
$^{229}\text{Ac}(\beta^-)^{229}\text{Th}$	1140	150	1104	12	−0.2	U					73Ch24
	1090	50			0.3	U					75We23
$^{229}\text{Fr} \leftarrow ^{133}\text{Cs}_{1.722}$	Could be influenced by ^{229}Rn contaminant										08We02 **
$^{229}\text{Th}(\alpha)^{225}\text{Ra}$	$E_\alpha=4978.3(1.2,Z), 4967.3(1.2,Z), 4845.1(1.2,Z)$ keV										71Gr17 **
*	to 100.60, 111.60, 236.25 levels										71Gr17 **
$^{229}\text{Th}(\alpha)^{225}\text{Ra}$	$E_\alpha=4979.3(2,Z), 4968.3(2,Z), 4845.1(2,Z)$ keV										87He28 **
*	to $9/2^+$ level at 100.50, $7/2^+$ at 111.60, $5/2^+$ at 236.25 keV										Ens095 **
*	calibrated with 71BaB2 value for 4845 level										AHW **
$^{229}\text{Pa}(\alpha)^{225}\text{Ac}$	$E_\alpha=5670.2, 5630.2, 5615.2, 5580.2, 5536.2$ (all 3,Z) keV to										63Su.A **
*	$5/2^+ 64.70, 7/2^+ 105.06, 5/2^- 120.80, 5/2^+ 155.65, 7/2^+ 199.85$										Ens095 **
$^{229}\text{Am}(\alpha)^{225}\text{Np}$	$E_\alpha=7990(20)$ and $E_\alpha=8000(20)$										15De22 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{230}\text{Fr}-^{133}\text{Cs}_{1.729}$	205878	32	205864	7	-0.4	o			MA8	1.0	05He26
	205860.1	7.5			0.6	1	88	88 ^{230}Fr	MA8	1.0	14Kr09
$^{230}\text{Fr}-\text{u}$	42401	34	42391	7	-0.3	o			GS3	1.0	08Ch.A
	42421	20			-1.5	1	12	12 ^{230}Fr	GS3	1.0	12Ch19
$^{230}\text{Ra}-^{133}\text{Cs}_{1.729}$	200530	13	200528	11	-0.1	2			MA8	1.0	08We02
	200524	21			0.2	2			MA8	1.0	05He26
$^{230}\text{Ac}-\text{u}$	36328	32	36327	17	-0.0	o			GS3	1.0	08Ch.A
	36327	17				2			GS3	1.0	12Ch19
$^{230}\text{Ra}-^{226}\text{Ra}_{1.018}$	11225	35	11189	11	-1.0	U			MA3	1.0	92Bo28
$^{230}\text{Th}(\alpha)^{226}\text{Ra}$	4770.1	1.5	4770.0	1.5	-0.1	1	98	98 ^{226}Ra	Orm		66Ba14 Z
$^{230}\text{Pa}(\alpha)^{226}\text{Ac}$	5439.5	0.7	5439.4	0.7	-0.1	1	99	87 ^{226}Ac	Orm		66Ba14 Z
$^{230}\text{U}(\alpha)^{226}\text{Th}$	5992.8	0.7	5992.5	0.5	-0.5	2			Orm		66Ba14 Z
	5992.1	0.7			0.5	2			Gea		12Ma30
$^{230}\text{Np}(\alpha)^{226}\text{Pa}$	6778.1	20.				5			ORa		68Ha14
$^{230}\text{Pu}(\alpha)^{226}\text{U}$	7175.0	15.	7178	9	0.2	6			Dbb		90An22
	7180.1	17.			-0.1	6			Jya		99Gr28
	7182.2	10.			-0.4	o			GSa		01Ca.B
	7185.1	20.4			-0.3	6			RIa		16Ka13
	7174.0	25.4			0.2	6			GSa		18Mi11
$^{230}\text{Th}(\text{p,t})^{228}\text{Th}$	-3550	15	-3569.2	1.1	-1.3	U			ANL		74Fr01
$^{230}\text{Th}(\text{p,t})^{228}\text{Th}-^{232}\text{Th}^{230}\text{Th}$	-493.5	1.0	-492.6	0.5	0.9	o					91Gr13
	-492.5	0.5			-0.2	1	98	69 ^{228}Th			94Le22
$^{230}\text{Th}(\text{p,t})^{228}\text{Th}-^{184}\text{W}^{182}\text{W}$	1564.0	1.6	1550.9	1.2	-8.2	B					09Le03
	1564.0	1.8			-7.3	C					09Le.A
$^{230}\text{Th}(\text{d,t})^{229}\text{Th}$	-541	6	-537.1	2.2	0.7	-					90Bu17
	-525	6			-2.0	-			ANL		67Er02 *
ave.	-533	4			-1.0	1	27	26 ^{229}Th			average
$^{230}\text{Ra}(\beta^-)^{230}\text{Ac}$	710	300	678	19	-0.1	U					80Gi04 *
$^{230}\text{Ac}(\beta^-)^{230}\text{Th}$	2700	100	2976	16	2.8	U					80Gi04 *
$^{230}\text{Pa}(\epsilon)^{230}\text{Th}$	1310.3	3.	1311.0	2.8	0.2	1	89	88 ^{230}Pa			70Lo02 *
$^{230}\text{Pa}(\beta^-)^{230}\text{U}$	561	15	559	5	-0.2	R					70Lo02
$^{230}\text{Th}(\text{d,t})^{229}\text{Th}$	$Q=-525(6)$ to $^{229}\text{Th}^m$ at 0.0035(0.0010) keV										94He08 **
$^{230}\text{Ra}(\beta^-)^{230}\text{Ac}$	$E_{\beta^-}=500(200)$ to level at 211.78 keV										Ens129 **
$^{230}\text{Ac}(\beta^-)^{230}\text{Th}$	$E_{\beta^-}=1400(100)$ to 0^+ level at 1297.14 keV										Ens129 **
$^{230}\text{Pa}(\epsilon)^{230}\text{Th}$	$pK=0.42(0.01)$ to 3^- level at 1127.789, recalculated										Ens129 **
$^{231}\text{Fr}-\text{u}$	45191	39	45175	8	-0.4	o			GS3	1.0	08Ch.A
	45158	27			0.6	U			GS3	1.0	12Ch19
$^{231}\text{Fr}-^{133}\text{Cs}_{1.737}$	209405.3	8.3				2			MA8	1.0	14Kr09
$^{231}\text{Ra}-^{133}\text{Cs}_{1.737}$	205267	21	205257	12	-0.5	1	34	34 ^{231}Ra	MA8	1.0	05He26
$^{231}\text{Ra}-\text{u}$	41052	32	41027	12	-0.8	o			GS3	1.0	08Ch.A
	41022	15			0.3	1	66	66 ^{231}Ra	GS3	1.0	12Ch19
$^{231}\text{Ac}-\text{u}$	38404	32	38393	14	-0.3	o			GS3	1.0	08Ch.A
	38393	14				2			GS3	1.0	12Ch19
$^{231}\text{Pa}(\alpha)^{227}\text{Ac}$	5150.2	1.5	5149.9	0.8	-0.2	o			Orm		66Ba14
	5146.9	1.0			3.0	B			Kum		68Ba25 *
	5150.7	1.5			-0.5	-			Orm		69Le.A *
	5149.8	1.0			0.1	-			Kum		76Ba68 *
ave.	5150.1	0.8			-0.2	1	98	91 ^{227}Ac			average
$^{231}\text{U}(\alpha)^{227}\text{Th}$	5551.3	50.	5576.3	1.7	0.4	U					53Cr.A
	5576.9	3.			-0.2	2					94Li12 *
	5576	2			0.1	2					97Mu08
$^{231}\text{Np}(\alpha)^{227}\text{Pa}$	6368.4	8.				6					73Ja06
$^{231}\text{Pu}(\alpha)^{227}\text{U}$	6838.6	20.				7					99La14
$^{231}\text{Pa}(\text{p,t})^{229}\text{Pa}$	-4133	3	-4133.4	2.8	-0.1	1	90	87 ^{229}Pa	Mun		91Gr13 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{230}\text{Th}(\text{n},\gamma)^{231}\text{Th}$	5118.00	0.20	5118.02	0.20	0.1	1	98	73 ^{231}Th	ILn		87Wh01 Z
$^{230}\text{Th}(\text{d},\text{p})^{231}\text{Th}$	2907	7	2893.45	0.20	-1.9	U			ANL		67Er02
$^{231}\text{Ac}(\beta^-)^{231}\text{Th}$	2100	100	1947	13	-1.5	U					60Ta19
$^{231}\text{Th}(\beta^-)^{231}\text{Pa}$	389.2	2.	391.5	1.5	1.1	1	53	47 ^{231}Pa			75Ho14 *
$^{231}\text{Pa}(\alpha)^{227}\text{Ac}$	$E_\alpha=5057.6(1.0,Z)$, $4985.9(1.0,Z)$, $4950.4(1.0,Z)$ to ground state, $7/2^-$ at 74.149 keV, and $9/2^+$ at 109.992 keV										Ens129 **
$^{231}\text{Pa}(\alpha)^{227}\text{Ac}$	$E_\alpha=5015.9(1.5,Z)$, $4735.9(1.5,Z)$ to $5/2^+$ at 46.354, $3/2^-$ at 330.040 keV										Ens129 **
$^{231}\text{Pa}(\alpha)^{227}\text{Ac}$	$E_\alpha=4736.2(1.0,Z)$ to 330.040 level										Ens129 **
$^{231}\text{U}(\alpha)^{227}\text{Th}$	$E_\alpha=5471(3)$, $5456(3)$, $5404(3)$ to 9.3, 24.4, 77.7 levels										94Li12 **
$^{231}\text{Pa}(\text{p},\text{t})^{229}\text{Pa}$	$Q=-4145(3)$ to 11.6 level										98Le15 **
$^{231}\text{Th}(\beta^-)^{231}\text{Pa}$	$E_{\beta^-}=305(2)$ to $5/2^+$ level at 84.2148 keV										Ens136 **
$^{232}\text{Fr}-^{133}\text{Cs}_{1.744}$	214353	15				2					
$^{232}\text{Ra}-^{133}\text{Cs}_{1.744}$	208368	13	208367	10	-0.1	1	57	57 ^{232}Ra	MA8	1.0	14Kr09
$^{232}\text{Ra}-\text{u}$	43518	32	43475	10	-1.3	o					
	43474	15			0.1	1	43	43 ^{232}Ra	GS3	1.0	08Ch.A
$^{232}\text{Ac}-\text{u}$	42052	32	42034	14	-0.6	o					
	42034	14				2					
$\text{C}_{18}\text{H}_{16}-^{232}\text{Th}$	87142.4	2.	87146.9	1.5	0.9	U					
$\text{C}_{24}\text{H}_{16}-^{232}\text{Th}$ ^{37}Cl ^{35}Cl	152393.4	1.8	152391.6	1.5	-0.4	1	12	11 ^{232}Th	M20	2.5	73Br06
$^{232}\text{Th}(\alpha)^{228}\text{Ra}$	4082.5	5.	4081.6	1.4	-0.2	U					57Ha08 Z
	4084.6	5.			-0.6	U					61Ko11 Z
	4083.5	5.			-0.4	U					62Ko12 Z
	4081.6	1.4				2					89Sa01 *
$^{232}\text{U}(\alpha)^{228}\text{Th}$	5413.63	0.09	5413.63	0.09	-0.0	1	100	99 ^{232}U	BIP		72Go33 *
$^{232}\text{Pu}(\alpha)^{228}\text{U}$	6716.0	10.				6					73Ja06
$^{232}\text{Th}(\text{p},\text{t})^{230}\text{Th}$	-3070	15	-3076.6	1.1	-0.4	U			ANL		74Fr01
$^{232}\text{Th}(\text{p},\text{t})^{230}\text{Th}-^{184}\text{W}()$ ^{182}W	2056.4	1.6	2043.5	1.1	-8.1	B					09Le03
	2056.5	1.8			-7.2	B					09Le.A
$^{232}\text{Th}(\text{d},\text{t})^{231}\text{Th}$	-174	6	-183.2	1.1	-1.5	U			ANL		67Er02
	-187	10			0.4	U			MIT		72Gr19
$^{232}\text{Ac}(\beta^-)^{232}\text{Th}$	3700	100	3708	13	0.1	U					90Be.B
$^{232}\text{Pa}(\beta^-)^{232}\text{U}$	1344	20	1337	7	-0.3	2					63Bj01 *
	1336	8			0.1	2					71Ka42 *
$^{232}\text{Th}(\alpha)^{228}\text{Ra}$	$E_\alpha=4012.3(1.4)$, $3947.2(2.0)$ to ground state, 2^+ level at 63.823 keV										Ens143 **
$^{232}\text{U}(\alpha)^{228}\text{Th}$	$E_\alpha=5320.12(0.14,Z)$, $5263.36(0.09,Z)$ to ground state, 2^+ level at 57.773 level										Ens143 **
$^{232}\text{Pa}(\beta^-)^{232}\text{U}$	$E_{\beta^-}=1295(20)$ to 2^+ level at 47.573 keV, and other E_{β^-}										Ens06a **
$^{232}\text{Pa}(\beta^-)^{232}\text{U}$	$E_{\beta^-}=314(8)$ to 2^- level at 1016.85 keV, and other E_{β^-}										Ens06a **
$^{233}\text{Fr}-^{133}\text{Cs}_{1.752}$	218166	21				2					
$^{233}\text{Ra}-\text{u}$	47602	32	47595	9	-0.2	o					
	47582	17			0.7	1	30	30 ^{233}Ra	GS3	1.0	08Ch.A
$^{233}\text{Ra}-^{133}\text{Cs}_{1.752}$	213248	11	213243	9	-0.5	1	70	70 ^{233}Ra	MA8	1.0	12Ch19
$^{233}\text{Ac}-\text{u}$	44363	32	44346	14	-0.5	o					
	44346	14				2					
$^{233}\text{U}(\alpha)^{229}\text{Th}$	4908.4	1.2	4908.7	1.2	0.2	1	93	70 ^{229}Th	Kum		68Ba25 Z
$^{233}\text{Np}(\alpha)^{229}\text{Pa}$	5626.7	50.9				2					50Ma14
$^{233}\text{Pu}(\alpha)^{229}\text{U}$	6416.3	20.				6					57Th10
$^{233}\text{Am}(\alpha)^{229}\text{Np}^p$	6898.6	17.3				8					00Sa52
$^{233}\text{Cm}(\alpha)^{229}\text{Pu}$	7468.5	10.	7470	50	0.1	o			GSa		01Ca.B
	7473.5	20.				8			GSa		10Kh06
$^{233}\text{Bk}(\alpha)^{229}\text{Am}^p$	7905.9	20.3				7			GSa		15De22
$^{232}\text{Th}(\text{n},\gamma)^{233}\text{Th}$	4786.69	0.25	4786.39	0.09	-1.2	-					74Ke13 Z
	4786.34	0.10			0.5	-			Bdn		06Fi.A

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{232}\text{Th}(\text{d,p})^{233}\text{Th}$	2555	10	2561.82	0.09	0.7	U			MIT		72Gr19
	2567	7			-0.7	U			ANL		72Vo08
$^{232}\text{Th}(\text{n},\gamma)^{233}\text{Th}$	ave.	4786.39	0.09	4786.39	0.09	-0.0	1	100	92 ^{233}Th		average
$^{233}\text{Th}(\beta^-)^{233}\text{Pa}$		1245	3	1242.2	1.1	-0.9	1	14	8 ^{233}Th		57Fr.A *
$^{233}\text{Pa}(\beta^-)^{233}\text{U}$		568	4	570.3	2.0	0.6	-				54Br37 *
		568	5			0.5	-				55On05 *
		566	5			0.9	-				63Bi03 *
	ave.	567.4	2.6			1.1	1	56	51 ^{233}U		average
$^{*233}\text{Th}(\beta^-)^{233}\text{Pa}$	PrvCom to reference										58St50 **
$^{*233}\text{Pa}(\beta^-)^{233}\text{U}$	$E_{\beta^-}=568(5)$, 256(4) to ground state, $3/2^+$ level at 311.904 keV										Ens057 **
$^{*233}\text{Pa}(\beta^-)^{233}\text{U}$	$E_{\beta^-}=568(5)$, 257(5) to ground state, $3/2^+$ level at 311.904 keV										Ens057 **
$^{*233}\text{Pa}(\beta^-)^{233}\text{U}$	$E_{\beta^-}=254(5)$ to $3/2^+$ level at 311.904 keV										Ens057 **
$^{234}\text{Ra}-\text{u}$	50358	33	50382	9	0.7	o			GS3	1.0	08Ch.A
	50342	33			1.2	U			GS3	1.0	12Ch19
$^{234}\text{Ra}-^{133}\text{Cs}_{1.759}$	216692.1	9.0				2			MA8	1.0	14Kr09
$^{234}\text{Ac}-\text{u}$	48137	32	48139	15	0.1	o			GS3	1.0	08Ch.A
	48139	15				2			GS3	1.0	12Ch19
$^{234}\text{U}(\alpha)^{230}\text{Th}$	4857.4	1.0	4857.5	0.7	0.1	-					55Go.A Z
	4860.4	2.			-1.4	-			Kum		67Ba43 Z
	ave.	4858.0	0.9		-0.5	1	55	39 ^{230}Th			average
$^{234}\text{Pu}(\alpha)^{230}\text{U}$	6310.1	5.				3					60Ho.A *
$^{234}\text{Am}(\alpha)^{230}\text{Np}$	6572.6	20.	6800#	150#	11.4	F					90Ha02 *
$^{234}\text{Cm}(\alpha)^{230}\text{Pu}$	7365.2	10.	7365	9	0.0	7			GSa		01Ca.B
	7366.1	20.3			-0.0	7			RIa		16Ka13
$^{234}\text{Bk}(\alpha)^{230}\text{Am}$	8087	50	8100	50	0.2	o			RIa		02Mo.B *
	8098.6	20.3				9			RIa		16Ka13
$^{232}\text{Th}(\text{t,p})^{234}\text{Th}$	2487	20	2494.6	2.6	0.4	U			LAI		69Br11
$^{234}\text{U}(\text{p,t})^{232}\text{U}$	-4099	15	-4125.3	1.6	-1.8	U			ANL		74Fr01
$^{234}\text{U}(\text{p,t})^{232}\text{U}-^{184}\text{W}(\text{t})^{182}\text{W}$	1007.6	1.6	994.8	1.6	-8.0	B					09Le03
	1007.6	1.8			-7.1	C					09Le.A
$^{233}\text{U}(\text{d,p})^{234}\text{U}$	4656	15	4620.9	2.0	-2.3	U			Kop		68Bj05
$^{234}\text{U}(\text{d,t})^{233}\text{U}$	-579	6	-588.2	2.0	-1.5	1	11	11 ^{233}U	ANL		67Er02
$^{234}\text{Th}(\beta^-)^{234}\text{Pa}^m$	192	2	195.1	1.0	1.5	3					55De40 *
	193	2			1.0	3					63Bj02 *
	198.	1.5			-1.9	3					73Go40 *
$^{234}\text{Pa}^m(\text{IT})^{234}\text{Pa}$	79	3				4					Nub211
$^{234}\text{Pa}(\beta^-)^{234}\text{U}$	2230	40	2194	4	-0.9	U					62Bj01
$^{234}\text{Pa}^m(\beta^-)^{234}\text{U}$	2290	20	2272.9	2.6	-0.9	U					63Bj02
$^{234}\text{Np}(\beta^+)^{234}\text{U}$	1812	10	1810	8	-0.2	2					67Ha04 *
	1805	15			0.3	2					67Wa09 *
$^{*234}\text{Pu}(\alpha)^{230}\text{U}$	With correction similar to reference										91Ry01 **
$^{*234}\text{Am}(\alpha)^{230}\text{Np}$	F : not believed to be measured in this work, replaced by estimate										GAu **
$^{*234}\text{Bk}(\alpha)^{230}\text{Am}$	$E_{\alpha}=7850(50)$ to 100 keV excited state										16Ka13 **
$^{*234}\text{Th}(\beta^-)^{234}\text{Pa}^m$	$E_{\beta^-}=100(2)$ 100(2) 104.0(1.5) respectively, to (1^-) 92.38 above $^{234}\text{Pa}^m$, and other E_{β^-}										Ens074 **
$^{*234}\text{Np}(\beta^+)^{234}\text{U}$	$E_{\beta^+}=790(10)$ pK=0.48(0.03) respectively, to 1^+ at 1570.69 and 1^+ at 1601.8 keV										Ens074 **
$^{235}\text{Ac}-\text{u}$	50872	32	50840	15	-1.0	o			GS3	1.0	08Ch.A
	50840	15				2			GS3	1.0	12Ch19
$^{235}\text{Th}-\text{u}$	47252	32	47255	14	0.1	o			GS3	1.0	08Ch.A
	47255	14				2			GS3	1.0	12Ch19
$^{235}\text{Pa}-\text{u}$	45421	32	45399	15	-0.7	o			GS3	1.0	08Ch.A
	45399	15				2			GS3	1.0	12Ch19
$^{235}\text{U}-^{206}\text{Pb}$ C ₂ H ₅	30341.0	10.	30337.7	1.6	-0.1	U			C4	2.5	71Ke02

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{235}\text{U}-\text{C}_{18}\text{H}_{18}$	-96932.8	3.8	-96922.5	1.2	1.1	U			M20	2.5	73Br06
$\text{C}_{18}\text{H}_{20}-^{235}\text{U}$	112584.2	4.8	112572.5	1.2	-1.0	U			M20	2.5	73Br06
$^{235}\text{U}(\alpha)^{231}\text{Th}$	4678	2	4678.1	0.7	0.0	-			Kum		60Ba44 *
	4681	3			-1.0	-					60Vo07 *
	4675.5	3.0			0.9	-					64Sc27 *
	4677	3			0.4	-					66Ga03 *
ave.	4677.9	1.3			0.1	1	28	21 ^{231}Th			average
$^{235}\text{Np}(\alpha)^{231}\text{Pa}$	5197.2	2.0	5193.8	1.5	-1.7	1	54	42 ^{231}Pa	Bka		73Br12 *
$^{235}\text{Pu}(\alpha)^{231}\text{U}$	5951.5	20.				3					57Th10
$^{235}\text{Am}(\alpha)^{231}\text{Np}$	6559	100	6576	13	0.2	o			JAa		99Sa.D *
	6576	15			-0.0	o			JAa		04Sa05 *
	6576	13				7			JAa		04As12 *
$^{235}\text{Cm}(\alpha)^{231}\text{Pu}$	7131.6	20.0	7280#	100#	7.4	D			GSa		20Kh10 *
$^{233}\text{U}(\text{t,p})^{235}\text{U}$	3668	10	3661.2	2.0	-0.7	U					67Ri.A *
$^{234}\text{U}(\text{n},\gamma)^{235}\text{U}$	5297.1	0.5	5297.50	0.23	0.8	-					72Ri08 Z
	5297.4	0.3			0.3	-					77Ko15 Z
$^{234}\text{U}(\text{d,p})^{235}\text{U}$	3075	7	3072.93	0.23	-0.3	U			ANL		70Br01
$^{235}\text{U}(\text{d,t})^{234}\text{U}$	935	15	959.73	0.23	1.6	U			Kop		68Bj05
$^{234}\text{U}(\text{n},\gamma)^{235}\text{U}$	ave.	5297.32	0.26	5297.50	0.23	0.7	1	81	63 ^{234}U		average
$^{235}\text{Th}(\beta^-)^{235}\text{Pa}$	1470	80	1729	19	3.2	B					89Yu01
$^{235}\text{Pa}(\beta^-)^{235}\text{U}$	1410	50	1370	14	-0.8	U					68Tr07
$^{235}\text{Np}(\epsilon)^{235}\text{U}$	123.5	2.	124.3	0.9	0.4	-					58Gi05
	123.6	1.			0.7	-					72Mc25
ave.	123.6	0.9			0.8	1	91	88 ^{235}Np			average
$^{235}\text{U}(\alpha)^{231}\text{Th}$	$E_\alpha=4596\ 4398\ 4372(\text{all } 2,Z) \text{ to ground state, } (7/2^-) \text{ at } 205.310, 9/2^- \text{ at } 236.906$										Ens136 **
$^{235}\text{U}(\alpha)^{231}\text{Th}$	$E_\alpha=4598.6(3,Z), 4402.6(3,Z) \text{ to ground state, } (7/2^-) \text{ at } 205.310 \text{ keV}$										Ens136 **
$^{235}\text{U}(\alpha)^{231}\text{Th}$	$E_\alpha=4595\ 4394\ 4364(\text{all } 3,Z) \text{ to ground state, } (7/2^-) \text{ at } 205.310, 9/2^- \text{ at } 236.906$										Ens136 **
$^{235}\text{U}(\alpha)^{231}\text{Th}$	$E_\alpha=4595.3, 4397.3, 4365.3(\text{all } 3,Z) \text{ to ground state, } (7/2^-) \text{ level at } 205.310 \text{ keV}$										Ens136 **
*	and $9/2^-$ at 236.906 keV										Ens136 **
$^{235}\text{Np}(\alpha)^{231}\text{Pa}$	$E_\alpha=5105.2(3), 5097.2(3), 5050.8(2,Z), 5024.8(2,Z), 4924.8(2,Z) \text{ to ground state, } 1/2^- \text{ at } 9.206, 7/2^- \text{ at } 58.5699, 5/2^+ \text{ at } 84.2148, 5/2^+ \text{ at } 183.4962$										AHW **
*	$E_\alpha=6440(100) \text{ to level below } 15 \text{ keV}$										Ens136 **
$^{235}\text{Am}(\alpha)^{231}\text{Np}$	$E_\alpha=6457(14) \text{ to level below } 15 \text{ keV}$										04As12 **
$^{235}\text{Am}(\alpha)^{231}\text{Np}$	$E_\alpha=6457(12) \text{ to level below } 15 \text{ keV}$										04As12 **
$^{235}\text{Cm}(\alpha)^{231}\text{Pu}$	Trends from Mass Surface TMS suggest ^{235}Cm 150 keV less bound										GAu **
$^{233}\text{U}(\text{t,p})^{235}\text{U}$	$Q=3335(10,\text{Ri}) \text{ to } 5/2^+ \text{ level at } 332.845 \text{ keV}$										Ens14b **
$^{236}\text{Ac}-\text{u}$	55037	73	54990	40	-0.7	o			GS3	1.0	10Ch19
	54988	41				2			GS3	1.0	12Ch19
$^{236}\text{Th}-\text{u}$	49665	32	49657	15	-0.2	o			GS3	1.0	08Ch.A
	49657	15				2			GS3	1.0	12Ch19
$^{236}\text{Pa}-\text{u}$	48666	32	48668	15	0.1	o			GS3	1.0	08Ch.A
	48668	15				2			GS3	1.0	12Ch19
$^{236}\text{U}(\alpha)^{232}\text{Th}$	4572.2	3.	4573.0	0.9	0.2	o					60Ko04 Z
	4570.0	3.1			1.0	-					61Ko11 Z
	4573.1	1.0			-0.2	-			Kum		78Ba.C
ave.	4572.8	1.0			0.1	1	82	83 ^{232}Th			average
$^{236}\text{Pu}(\alpha)^{232}\text{U}$	5867.15	0.08	5867.15	0.08	-0.0	1	100	99 ^{236}Pu			84Ry02 Z
$^{236}\text{Am}(\alpha)^{232}\text{Np}$	6256.2	40.				3			JAa		04Sa05
$^{236}\text{Cm}(\alpha)^{232}\text{Pu}$	7074.1	20.	7067	5	-0.4	U			GSa		10Kh06
	7066.9	5.				7			JAa		10As.A
$^{236}\text{Bk}(\alpha)^{232}\text{Am}$	7447	14	7700#	200#	4.8	D			Bka		20Po07 *
$^{236}\text{U}(\text{p,t})^{234}\text{U}$	-3330	15	-3361.2	0.3	-2.1	U			ANL		74Fr01
$^{235}\text{U}(\text{n},\gamma)^{236}\text{U}$	6545	2	6545.52	0.26	0.3	U					70Ka22
	6545.1	0.5			0.8	-					74Ju.B Z
	6545.4	0.5			0.2	-					75We.A Z

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{236}\text{U}(\text{d,t})^{235}\text{U}$		−281	6	−288.29	0.26	−1.2	U				70Br01
$^{235}\text{U}(\text{n},\gamma)^{236}\text{U}$	ave.	6545.3	0.4	6545.52	0.26	0.8	1	54	30 ^{235}U		average
$^{236}\text{Pa}(\beta^-)^{236}\text{U}$		3350	100	2889	14	−4.6	B				63Wo04
		2900	200			−0.1	U				68Tr07 *
$^{236}\text{Np}^m(\text{IT})^{236}\text{Np}$		60	50				3				Ens06a
$^{236}\text{Np}^m(\beta^-)^{236}\text{Pu}$		525	10	537	6	1.2	2				56Gr11 *
		544	8			−0.9	2				69Le05 *
* $^{236}\text{Bk}(\alpha)^{232}\text{Am}$	Trends from Mass Surface TMS suggest ^{236}Bk 250 keV less bound										GAu **
* $^{236}\text{Pa}(\beta^-)^{236}\text{U}$	E_{β^-} =2000(200) to 1^- level at 687.59 keV, and other E_{β^-} , reinterpreted										Ens06a **
* $^{236}\text{Np}^m(\beta^-)^{236}\text{Pu}$	E_{β^-} =518(10) 537(8) respectively, to ground state and 2^+ level at 44.63 keV										Ens06a **
$^{237}\text{Th}-\text{u}$		53690	32	53629	17	−1.9	o				08Ch.A
		53629	17				2				12Ch19
$^{237}\text{Pa}-\text{u}$		51038	32	51023	14	−0.5	o				08Ch.A
		51023	14				2				12Ch19
$^{237}\text{Np}(\alpha)^{233}\text{Pa}$		4959.9	3.	4957.3	0.7	−0.9	U				61Ba44 *
		4956.7	1.5			0.4	−				68Ba25 *
		4959.9	3.			−0.9	U				69Va06 *
		4956.9	0.9			0.4	−				00Si02 *
	ave.	4956.8	0.8			0.6	1	91	90 ^{233}Pa		average
$^{237}\text{Pu}(\alpha)^{233}\text{U}$		5753.3	20.	5747.6	2.3	−0.3	U				57Th10
		5747	5			0.1	1	20	15 ^{233}U		93Dm02
$^{237}\text{Am}(\alpha)^{233}\text{Np}^p$		6146.2	5.				4				75Ah05 Z
$^{237}\text{Cm}(\alpha)^{233}\text{Pu}$		6774.5	10.	6770	50	−0.1	o				02As08
		6770.4	10.				7				06As03
$^{237}\text{Cf}(\alpha)^{233}\text{Cm}$		8220	20				9				10Kh06
$^{235}\text{U}(\text{t,p})^{237}\text{U}$		3206	20	3189.5	0.5	−0.8	U				64Mi.A *
		3178	20			0.6	U				69Br11
$^{237}\text{Np}(\text{p,t})^{235}\text{Np}$		−3816	15	−3832.3	0.9	−1.1	U				74Fr01
$^{236}\text{U}(\text{n},\gamma)^{237}\text{U}$		5125.9	0.5	5125.8	0.5	−0.3	1	85	84 ^{237}U		79Vo05 Z
$^{236}\text{U}(\text{d,p})^{237}\text{U}$		2898	8	2901.2	0.5	0.4	U				67Er02
$^{237}\text{Pa}(\beta^-)^{237}\text{U}$		2250	100	2137	13	−1.1	U				74Ka05
$^{237}\text{U}(\beta^-)^{237}\text{Np}$		520	5	518.5	0.5	−0.3	U				53Wa05 *
		524	5			−1.1	U				56Ba39 *
		523	5			−0.9	U				57Ra04 *
$^{237}\text{Pu}(\epsilon)^{237}\text{Np}$		222	8	220.1	1.3	−0.2	U				58Ho02 *
		207	18			0.7	U				59Gi54 *
* $^{237}\text{Np}(\alpha)^{233}\text{Pa}$	E_{α} =4876.7 4774.2 4769.1(3,Z) to ground state, $7/2^+$ at 103.635, $9/2^+$ at 109.07 keV										Ens057 **
* $^{237}\text{Np}(\alpha)^{233}\text{Pa}$	E_{α} =4787.9(1.5,Z) to $5/2^+$ level at 86.48 keV										Ens057 **
* $^{237}\text{Np}(\alpha)^{233}\text{Pa}$	E_{α} =4791.0(3,Z), 4774.0(3,Z), 4770.0(3,Z) keV										69Va06 **
*	to $5/2^+$ at 86.468, $7/2^+$ at 103.635, $9/2^+$ at 109.07 keV										Ens057 **
* $^{237}\text{Np}(\alpha)^{233}\text{Pa}$	E_{α} =4788.0(0.9) keV to $5/2^+$ at 86.468 keV										00Si02 **
* $^{235}\text{U}(\text{t,p})^{237}\text{U}$	Q =2980(20) to $7/2^+$ level at 426.15 keV										Ens068 **
* $^{237}\text{U}(\beta^-)^{237}\text{Np}$	E_{β^-} =245(5), 249(5), 248(5) respectively, to 53% $3/2^-$ level at 267.556, and										Ens068 **
*	43% $1/2^-$ level at 281.356 keV										Ens068 **
* $^{237}\text{Pu}(\epsilon)^{237}\text{Np}$	LK=2.8(0.8) capture to $5/2^-$ level at 59.541 keV, recalculated										Ens068 **
* $^{237}\text{Pu}(\epsilon)^{237}\text{Np}$	pK=0.38(0.06) to ground state, $7/2^+$ level at 33.196, $5/2^-$ at 59.541 keV										Ens068 **
$^{238}\text{Pa}-\text{u}$		54648	32	54637	17	−0.3	o				08Ch.A
		54637	17				2				12Ch19
$^{238}\text{U}-^{206}\text{Pb}$ ^{32}S		104253.9	10.	104250.6	1.9	−0.1	U				71Ke02
$\text{C}_{18}\text{H}_{22}-^{238}\text{U}$		121366.0	2.4	121363.8	1.6	−0.4	U				73Br06
$\text{C}_{24}\text{H}_{20}-^{238}\text{U}$ $^{35}\text{Cl}_2$		168010.8	1.4	168008.3	1.6	−0.7	1	21	21 ^{238}U		73Br06
$^{238}\text{U}-^{235}\text{U}$		6858.6	10.	6858.8	1.3	0.0	U				71Ke02
$^{238}\text{U}(\alpha)^{234}\text{Th}$		4271.5	5.	4269.9	2.1	−0.3	2				57Ha08 Z
		4265.1	5.			0.9	2				60Vo07 Z
		4272.9	5.			−0.6	2				61Ko11 Z
		4269.9	3.			−0.0	2				14Po02
									Gea		

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{238}\text{Pu}(\alpha)^{234}\text{U}$	5593.20	0.2	5593.27	0.19	0.4	1	90	69 ^{238}Pu			71Gr17 Z
$^{238}\text{Am}(\alpha)^{234}\text{Np}$	6041.7	30.				3					72Ah04
$^{238}\text{Cm}(\alpha)^{234}\text{Pu}$	6611.5	50.	6670	10	1.2	U					48St.A *
	6632.0	50.			0.8	U					52Hi.A
	6672.3	10.			-0.2	o			JAA		02As08
	6670.3	10.				4			JAA		06As03
$^{238}\text{U}(\text{n},\alpha)^{235}\text{Th}$	8700	50	8936	13	4.7	B					81Wa11
$^{236}\text{U}(\text{t},\text{p})^{238}\text{U}$	2900	20	2797.7	1.2	-5.1	F			Ald		64Mi.A *
	2782	10			1.6	U			ANL		67Er02
	2780	20			0.9	U			LAI		69Br11
$^{238}\text{U}(\text{p},\text{t})^{236}\text{U}$	-2765	15	-2797.7	1.2	-2.2	U			ANL		74Fr01
$^{238}\text{U}(\text{p},\text{d})^{237}\text{U}$	-3951	20	-3929.2	1.3	1.1	U			Ald		64Mi.A
$^{238}\text{U}(\text{d},\text{t})^{237}\text{U}$	116	6	103.5	1.3	-2.1	U			ANL		67Er02
$^{237}\text{Np}(\text{n},\gamma)^{238}\text{Np}$	5488.32	0.20				2			BNn		79Io01 Z
$^{238}\text{Pu}(\text{d},\text{t})^{237}\text{Pu}$	-746	10	-742.6	1.3	0.3	U			Kop		73Gr26
$^{238}\text{Pa}(\beta^-)^{238}\text{U}$	3600	300	3586	16	-0.0	U					68Tr07 *
	3460	60			2.1	U					85Ba57 *
$^{238}\text{Np}(\beta^-)^{238}\text{Pu}$	1295	10	1291.4	0.5	-0.4	U					55Ra27 *
	1300	15			-0.6	U					56Ba95 *
$^{238}\text{Cm}(\alpha)^{234}\text{Pu}$	PrvCom to reference										58St50 **
$^{236}\text{U}(\text{t},\text{p})^{238}\text{U}$	F : authors not satisfied with target material										AHW **
$^{238}\text{Pa}(\beta^-)^{238}\text{U}$	$E_{\beta^-}=1700(300)$ to (3^-) level at 1992.2 keV, and other E_{β^-} , reinterpreted										Ens157 **
$^{238}\text{Pa}(\beta^-)^{238}\text{U}$	Reports result from thesis										82Gi.A **
$^{238}\text{Np}(\beta^-)^{238}\text{Pu}$	$E_{\beta^-}=270(10)$ 280(10) respectively, to 2^+ level at 1028.537 keV, and other E_{β^-}										Ens157 **
$^{239}\text{Pu}(\alpha)^{235}\text{U}$	5244.60	0.25	5244.52	0.21	-0.3	1	68	41 ^{235}U			79Ry.A *
$^{239}\text{Am}(\alpha)^{235}\text{Np}$	5924.6	2.0	5922.4	1.4	-1.1	2			Bka		71Go01 *
	5920.2	2.0			1.1	2					75Ah05 *
$^{239}\text{Cm}(\alpha)^{235}\text{Pu}$	6539.7	140.				4			JAA		02Sh.C *
$^{239}\text{Cf}(\alpha)^{235}\text{Cm}$	7760.1	25.	7760	60	0.1	9			GSa		81Mu12
	7766.2	8.			-0.1	9			GSa		20Kh10
$^{238}\text{U}(\text{n},\gamma)^{239}\text{U}$	4806.55	0.30	4806.38	0.17	-0.6	2			ANL		72Bo46 Z
	4806.30	0.21			0.4	2			ILn		79Br25 Z
$^{238}\text{U}(\text{d},\text{p})^{239}\text{U}$	2588	20	2581.82	0.17	-0.3	U			Ald		64Mi.A
	2579	7			0.4	U			Tal		66Sh16
	2585	6			-0.5	U			ANL		67Er02
$^{238}\text{Pu}(\text{n},\gamma)^{239}\text{Pu}$	5646.7	0.5	5646.2	0.3	-0.9	1	38	31 ^{238}Pu			75Ma.A Z
$^{238}\text{Pu}(\text{d},\text{p})^{239}\text{Pu}$	3432	10	3421.7	0.3	-1.0	U			Kop		73Gr26
$^{239}\text{Pu}(\text{d},\text{t})^{238}\text{Pu}$	604	10	611.0	0.3	0.7	U			ANL		73Fr01
$^{239}\text{U}(\beta^-)^{239}\text{Np}$	1290	20	1261.7	1.5	-1.4	U					64Bl11 *
$^{239}\text{Np}(\beta^-)^{239}\text{Pu}$	722.5	1.0	722.8	0.9	0.3	1	87	67 ^{239}Np			59Co63 *
$^{239}\text{Pu}(\alpha)^{235}\text{U}$	$E_{\alpha}=5156.59(0.25,Z)$ to $1/2^+$ level at 0.0760 keV										Ens14b **
$^{239}\text{Am}(\alpha)^{235}\text{Np}$	$E_{\alpha}=5824.6(4,Z)$ 5775.6(2,Z) 5733.6(2,Z); ground state, $(5/2)^-$ 49.10, $(7/2)^-$ 91.6										Ens14b **
$^{239}\text{Am}(\alpha)^{235}\text{Np}$	$E_{\alpha}=5772.7(2,Z)$ to $(5/2)^-$ level at 49.10 keV										Ens14b **
$^{239}\text{Cm}(\alpha)^{235}\text{Pu}$	Private communication to reference										08Qi03 **
$^{239}\text{U}(\beta^-)^{239}\text{Np}$	$E_{\beta^-}=1211(20)$ to $5/2^-$ level at 74.6640 keV, and other E_{β^-}										Ens14b **
$^{239}\text{Np}(\beta^-)^{239}\text{Pu}$	$E_{\beta^-}=437(1)$ to $5/2^+$ level at 285.460 keV, and other E_{β^-}										Ens14b **
$^{240}\text{Pu}(\alpha)^{236}\text{U}$	5255.88	0.15	5255.82	0.14	-0.4	1	90	77 ^{236}U			72Go33 Z
$^{240}\text{Am}(\alpha)^{236}\text{Np}^p$	5468.9	1.0				3					70Go42 Z
$^{240}\text{Cm}(\alpha)^{236}\text{Pu}$	6397.8	0.6	6397.8	0.6	-0.0	1	100	99 ^{240}Cm	Kum		71Bb10 *
$^{240}\text{Cf}(\alpha)^{236}\text{Cm}$	7718.9	10.	7711	4	-0.8	8					70Si19
	7713.8	20.			-0.1	U			GSa		10Kh06
	7709.7	4.1			0.3	8			JAA		10As.A
$^{240}\text{Es}(\alpha)^{236}\text{Bk}$	8330	30	8260	60	-1.2	4					17Ko02 *
	8258	77			0.0	4			Bka		20Po07
	8186.6	31.0			1.2	4			GSt		20Kh10

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{238}\text{U}(\text{t,p})^{240}\text{U}$	2242	20	2253.1	2.9	0.6	U			Ald		64Mi.A
	2253	20			0.0	U			LAI		69Br11
$^{240}\text{Pu}(\text{p,t})^{238}\text{Pu}$	-3692	15	-3698.7	0.4	-0.4	U			ANL		74Fr01
$^{239}\text{Pu}(\text{n},\gamma)^{240}\text{Pu}$	6534.1	1.0	6534.22	0.23	0.1	-					70Ch.A
	6534.3	0.4			-0.2	-					74Ju.B Z
	6534.2	0.4			0.0	-					75We.A Z
$^{239}\text{Pu}(\text{d,p})^{240}\text{Pu}$	4300	10	4309.65	0.23	1.0	U			ANL		73Fr01
$^{239}\text{Pu}(\text{n},\gamma)^{240}\text{Pu}$	ave.	6534.24	0.27	6534.22	0.23	-0.1	1	72	46 ^{239}Pu		average
$^{240}\text{U}(\beta^-)^{240}\text{Np}^m$	386	20	381	13	-0.2	1	43	42 $^{240}\text{Np}^m$			53Kn23 *
$^{240}\text{Np}^m(\text{IT})^{240}\text{Np}$	20	15	18	14	-0.1	1	83	68 ^{240}Np			81Hs02 *
$^{240}\text{Np}(\beta^-)^{240}\text{Pu}$	2199	30	2191	17	-0.3	1	32	32 ^{240}Np			51Or.A *
$^{240}\text{Np}^m(\beta^-)^{240}\text{Pu}$	2210	20	2209	13	-0.1	1	43	43 $^{240}\text{Np}^m$			59Bu20 *
$^{240}\text{Am}(\epsilon)^{240}\text{Pu}$	1395	35	1385	14	-0.3	R					72Ah07 *
$^{240}\text{Cm}(\alpha)^{236}\text{Pu}$	$E_\alpha=6290.5, 6247.7(0.6,Z)$ to ground state, 2^+ level at 44.63 keV										Ens06a **
$^{240}\text{Es}(\alpha)^{236}\text{Bk}$	Highest decay energy is assigned to ground state to ground state transition										HWJ202 **
$^{240}\text{U}(\beta^-)^{240}\text{Np}^m$	$E_{\beta^-}=360(20)$ to $^{240}\text{Np}^m$, and 1^+ level at 44.17 keV above										Ens08a **
$^{240}\text{Np}^m(\text{IT})^{240}\text{Np}$	From fraction IT=0.0012(0.0001)										AHW **
$^{240}\text{Np}(\beta^-)^{240}\text{Pu}$	$E_{\beta^-}=890(30)$ to 5^- level at 1308.74 keV										Ens08a **
$^{240}\text{Np}^m(\beta^-)^{240}\text{Pu}$	$E_{\beta^-}=2180(20)$ to ground state and 2^+ level at 42.824 keV, and other E_{β^-}										Ens08a **
$^{240}\text{Am}(\epsilon)^{240}\text{Pu}$	$pK=0.635(0.020)$ to 3^+ level at 1030.55 keV, recalculated										Ens08a **
$^{241}\text{Am O}-\text{C}_{22}$	51744.8	1.9	51742.0	1.2	-1.0	1	18	18 ^{241}Am	TG1	1.5	14Ei01
$^{241}\text{Pu}(\alpha)^{237}\text{U}$	5139.6	3.	5140.1	0.5	0.2	U					68Ah01 *
	5139.3	1.2			0.6	1	16	16 ^{237}U	Kum		68Ba25 *
$^{241}\text{Am}(\alpha)^{237}\text{Np}$	5637.81	0.12	5637.82	0.12	0.1	1	100	99 ^{237}Np			71Gr17 *
$^{241}\text{Cm}(\alpha)^{237}\text{Pu}$	6182.8	2.0	6185.2	0.6	1.2	U			Kum		67Ba42 *
	6185.2	0.6			-0.0	-			Kum		71Bb10 *
	6185.0	2.0			0.1	-					75Ah05 *
	ave.	6185.2	0.6		0.0	1	99	94 ^{237}Pu			average
$^{241}\text{Cf}(\alpha)^{237}\text{Cm}^p$	7459.0	5.	7455	3	-0.9	9					70Si19
	7451.9	4.1			0.7	9			JAA		10As.A
	7451.9	15.3			0.2	U			GSa		20Kh10
$^{241}\text{Es}(\alpha)^{237}\text{Bk}$	8064.1	30.	8259	17	6.5	C			GSa		85Hi.A *
	8250.2	20.			0.4	10			GSa		96Ni09
	8277.6	31.			-0.6	10			GSt		20Kh08
$^{239}\text{Pu}(\text{t,p})^{241}\text{Pu}$	3242	20	3293.94	0.23	2.6	U			LAI		69Br11
$^{240}\text{Pu}(\text{n},\gamma)^{241}\text{Pu}$	5241.3	0.7	5241.522	0.030	0.3	U					75Ma.A
	5241.52	0.03			0.1	1	100	61 ^{240}Pu	ILn		98Wh01 Z
$^{240}\text{Pu}(\text{d,p})^{241}\text{Pu}$	3018	6	3016.956	0.030	-0.2	U			ANL		67Er02
$^{241}\text{Am}(\text{d,t})^{240}\text{Am}$	-388	15	-390	14	-0.1	2			Kop		76Gr19
$^{241}\text{Np}(\beta^-)^{241}\text{Pu}$	1360	100	1360	100	1.1	B					59Va32
	1250	100									66Qa02 *
$^{241}\text{Pu}(\beta^-)^{241}\text{Am}$	20.8	0.2	20.78	0.17	-0.1	-					56Sh31
	20.7	0.3			0.3	-					99Dr13
	20.78	0.20			-0.0	o					99Ya.A
	21.6	0.5			-1.6	U					10Lo14 *
	ave.	20.77	0.17		0.1	1	99	81 ^{241}Am			average
$^{241}\text{Cm}(\epsilon)^{241}\text{Am}$	767.5	1.2	767.4	1.2	-0.1	1	95	94 ^{241}Cm			89Su.A *
$^{241}\text{Pu}(\alpha)^{237}\text{U}$	$E_\alpha=4896.6(3,Z), 4853.6(3,Z)$ to $5/2^+$ at 159.962, $11/2^+$ at 204.06 keV										Ens068 **
$^{241}\text{Pu}(\alpha)^{237}\text{U}$	$E_\alpha=4896.3(1.2,Z), 4853.3(1.2,Z)$ to $5/2^+$ at 159.962, $11/2^+$ at 204.06 keV										Ens068 **
$^{241}\text{Am}(\alpha)^{237}\text{Np}$	$E_\alpha=5485.56(0.12,Z), 5442.80(0.13,Z)$ to $5/2^-$ at 59.54, $7/2^-$ at 102.96										Ens068 **
$^{241}\text{Cm}(\alpha)^{237}\text{Pu}$	$E_\alpha=6080.6(2,Z), 5926.6(2,Z)$ to ground state, $3/2^+$ level at 155.456 keV										Ens068 **
$^{241}\text{Cm}(\alpha)^{237}\text{Pu}$	$E_\alpha=5939.0(0.6,Z), 5884.7(0.6,Z)$ to $1/2^+$ at 145.543, $5/2^+$ at 201.179 keV										Ens068 **
$^{241}\text{Cm}(\alpha)^{237}\text{Pu}$	$E_\alpha=5938.7(2,Z), 5884.7(2,Z)$ to $1/2^+$ at 145.543, $5/2^+$ at 201.179 keV										Ens068 **
$^{241}\text{Es}(\alpha)^{237}\text{Bk}$	C : new data from same group (next item) is more reliable										96Ni09 **
$^{241}\text{Np}(\beta^-)^{241}\text{Pu}$	Trends from Mass Surface TMS favors the higher $Q-\beta$										GAu **
$^{241}\text{Pu}(\beta^-)^{241}\text{Am}$	No quoted uncertainty, estimated by evaluator										GAu **
$^{241}\text{Cm}(\epsilon)^{241}\text{Am}$	$Q(\epsilon)=5.5(1.2)$ to $3/2^-$ level at 636.861 keV										Ens152 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{242}\text{Pu}(\alpha)^{238}\text{U}$	4987.3	2.0	4984.2	1.0	-1.5	-					53As.A *
	4989.5	3.0			-1.8	U					56Ko67 *
	4982.9	1.2			1.1	-					68Ba25 *
	ave. 4984.1	1.0			0.2	1	92	78 ^{238}U	Kum		average
$^{242}\text{Am}(\alpha)^{238}\text{Np}$	5587.5	0.8	5588.50	0.25	1.3	U					79Ba67 *
	5589.9	0.8			-1.7	U					90Ho02 *
$^{242}\text{Cm}(\alpha)^{238}\text{Pu}$	6215.63	0.08	6215.63	0.08	0.0	1	100	100 ^{242}Cm			71Gr17 Z
$^{242}\text{Cf}(\alpha)^{238}\text{Cm}$	7516.9	4.				5					70Si19 Z
$^{242}\text{Es}(\alpha)^{238}\text{Bk}$	8160.2	20.				8			GSa		10An08
$^{240}\text{Pu}(\text{t,p})^{242}\text{Pu}$	3043	20	3069.3	0.7	1.3	U			LAI		69Br11
$^{242}\text{Pu}(\text{p,t})^{240}\text{Pu}$	-3045	15	-3069.3	0.7	-1.6	U			ANL		74Fr01
$^{241}\text{Pu}(\text{n},\gamma)^{242}\text{Pu}$	6309.5	0.7	6309.6	0.7	0.1	1	95	81 ^{242}Pu			72Ma.A
$^{242}\text{Pu}(\text{d,t})^{241}\text{Pu}$	-49	7	-52.3	0.7	-0.5	U			ANL		67Er02
$^{241}\text{Am}(\text{n},\gamma)^{242}\text{Am}$	5541.5	1.5	5537.64	0.10	-2.6	U					75Ij.A
	5537.64	0.1				2			ILn		88Sa18 Z
$^{241}\text{Am}(\text{d,p})^{242}\text{Am}$	3308	15	3313.07	0.10	0.3	U			Kop		76Gr19
$^{242}\text{Np}(\beta^-)^{242}\text{Pu}$	2700	200				2					79Ha26
$^{242}\text{Am}(\beta^-)^{242}\text{Cm}$	651	5	664.3	0.4	2.7	U					50Ok52 *
	667	5			-0.5	U					55Ba.A
$^{242}\text{Pu}(\alpha)^{238}\text{U}$	$E_\alpha=4904.6, 4860.6(2,Z)$ to ground state, 2^+ level at 44.916 keV										Ens157 **
$^{242}\text{Pu}(\alpha)^{238}\text{U}$	$E_\alpha=4905.2(3,Z), 4863.2(3,Z)$ to ground state, 2^+ level at 44.916 keV										Ens157 **
$^{242}\text{Pu}(\alpha)^{238}\text{U}$	$E_\alpha=4900.4(1.2,Z), 4856.1(1.2,Z)$ to ground state, 2^+ level at 44.916 keV										Ens157 **
$^{242}\text{Am}(\alpha)^{238}\text{Np}$	$E_\alpha=5206.6(0.5,Z), 5141.4(0.5,Z)$ from $^{242}\text{Am}^m$ at 48.60 to 5^- at 342.439, and 6^- at 407.59 keV; error increased due to conflict with next item										Ens157 **
*											GAu **
$^{242}\text{Am}(\alpha)^{238}\text{Np}$	$E_\alpha=5208.3(0.8,Z), 5144.3(0.9,Z)$ from $^{242}\text{Am}^m$ to 5^- 6^- levels (see above)										Ens157 **
$^{242}\text{Am}(\beta^-)^{242}\text{Cm}$	$E_\beta=628(5)$ to ground state and 2^+ level at 42.13 keV										Ens026 **
$^{243}\text{Am O-C}_{22}$	56295.8	1.5	56294.5	1.5	-0.6	1	44	44 ^{243}Am	TG1	1.5	14Ei01
$^{243}\text{Am}(\alpha)^{239}\text{Np}$	5438.8	1.0	5439.1	0.9	0.3	1	87	54 ^{243}Am	Kum		68Ba25 *
$^{243}\text{Cm}(\alpha)^{239}\text{Pu}$	6165.4	3.0	6168.8	1.0	1.1	U					57As.A *
	6165.7	3.0			1.0	U					63Dz07 *
	6165.4	3.0			1.1	o			Kum		66Ba07 *
	6168.8	1.0				2					69Ba57 *
$^{243}\text{Bk}(\alpha)^{239}\text{Am}$	6874.4	4.				3			Bka		66Ah.A Z
$^{243}\text{Cf}(\alpha)^{239}\text{Cm}^p$	7178	10				6					67Fi04 *
$^{243}\text{Es}(\alpha)^{239}\text{Bk}$	8072.1	10.				9			RIa		89Ha27
$^{243}\text{Es}(\alpha)^{239}\text{Bk}^p$	8022.3	20.	8030.9	2.9	0.4	U					73Es02
	8031.4	3.			-0.2	10			RIa		89Ha27
	8027.3	20.			0.2	o			GSa		93Ho.A
	8025.4	10.			0.6	10			GSa		10An08
$^{243}\text{Fm}(\alpha)^{239}\text{Cf}$	8689.3	25.4	8690	50	-0.0	o			GSa		81Mu12
	8693.4	20.3			-0.1	o			GSa		08Kh10
	8689.2	8.1				10			GSa		20Kh10
$^{243}\text{Am}(\text{p,t})^{241}\text{Am}$	-3407	15	-3420.2	1.2	-0.9	U			ANL		74Fr01
$^{242}\text{Pu}(\text{n},\gamma)^{243}\text{Pu}$	5034.2	3.	5033.6	2.4	-0.2	1	63	59 ^{243}Pu			76Ca25
$^{242}\text{Pu}(\text{d,p})^{243}\text{Pu}$	2807	8	2809.1	2.4	0.3	U			ANL		67Er02
$^{243}\text{Am}(\text{d,t})^{242}\text{Am}$	-111	15	-107.1	1.2	0.3	U			Kop		76Gr19
$^{243}\text{Pu}(\beta^-)^{243}\text{Am}$	578	10	579.6	2.6	0.2	-					69Ho10
	580	10			-0.0	-					77Dr07
	ave. 579	7			0.1	1	14	11 ^{243}Pu			average
$^{243}\text{Am}(\alpha)^{239}\text{Np}$	$E_\alpha=5275.2(1.0,Z), 5233.3(1.0,Z)$ to $5/2^-$ at 74.6640, $7/2^-$ at 117.715										Ens14b **
$^{243}\text{Cm}(\alpha)^{239}\text{Pu}$	$E_\alpha=6063.7, 5989.7, 5782.7, 5738.7(3,Z)$ to ground state, $7/2^+$ level at 75.705, $5/2^+$ at 285.460, and $7/2^+$ at 330.124 keV										57As.A **
*											Ens14b **
$^{243}\text{Cm}(\alpha)^{239}\text{Pu}$	$E_\alpha=5990.5, 5783.5, 5738.5(3,Z)$ to $7/2^+$ level at 75.705, $5/2^+$ at 285.46, and $7/2^+$ at 330.124 keV										Ens14b **
*											Ens14b **
$^{243}\text{Cm}(\alpha)^{239}\text{Pu}$	$E_\alpha=6067.4, 5992.4(2,Z)$ to ground state, $7/2^+$ at 75.705 keV										Ens14b **
$^{243}\text{Cm}(\alpha)^{239}\text{Pu}$	$E_\alpha=5785.7(1.0,Z), 5742.8(1.0,Z)$ to $5/2^+$ at 285.46, $7/2^+$ at 330.124keV										Ens14b **
$^{243}\text{Cf}(\alpha)^{239}\text{Cm}^p$	Unhindered $E_\alpha=7060(10)$; there is a weaker $E_\alpha=7170(10)$ keV										AHW **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{244}\text{Pu O-C}_{22}$	59119.3	1.9	59119.0	2.5	-0.1	1	78	78 ^{244}Pu	TG1	1.5	14Ei01
$^{244}\text{Pu}(\alpha)^{240}\text{U}$	4665.6	1.0	4665.6	1.0	-0.0	1	100	99 ^{240}U			69Be06 Z
$^{244}\text{Cm}(\alpha)^{240}\text{Pu}$	5901.60	0.03				2			BIP		71Gr17 *
$^{244}\text{Bk}(\alpha)^{240}\text{Am}$	6778.8	4.				3					66Ah.B *
$^{244}\text{Cf}(\alpha)^{240}\text{Cm}$	7327.1	2.	7329.0	1.8	0.9	-					67Fi04 Z
	7336.4	4.			-1.8	-					67Si08 Z
	7330.4	20.			-0.1	U			GSa		08Kh10
	ave. 7329.0	1.8			-0.0	1	99	98 ^{244}Cf			average
$^{244}\text{Es}(\alpha)^{240}\text{Bk}^p$	7696.4	20.				4					73Es02
$^{244}\text{Md}(\alpha)^{240}\text{Es}$	8446.9	19.3	8950	80	9.3	B			Bka		20Po07 *
	8807.4	23.4			2.5	B			Bka		20Po07
	8947	61				5			GSt		20Kh08 *
$^{242}\text{Pu}(\text{t,p})^{244}\text{Pu}$	2576	20	2571.7	2.5	-0.2	U			LAI		69Br11
$^{244}\text{Pu}(\text{p,t})^{242}\text{Pu}$	-2560	15	-2571.7	2.5	-0.8	U			ANL		72Ma15
$^{244}\text{Pu}(\text{t},\alpha)^{243}\text{Np}^p$	12405	10				2					79Fi02
$^{244}\text{Pu}(\text{d,t})^{243}\text{Pu}$	234	5	237.4	2.9	0.7	1	34	19 ^{243}Pu	ANL		76Ca25
$^{243}\text{Am}(\text{n},\gamma)^{244}\text{Am}^m$	5277.90	0.07				2			ILn		84Vo07 Z
$^{244}\text{Cm}(\text{d,t})^{243}\text{Cm}$	-530	7	-544.2	1.0	-2.0	U			ANL		67Er02
$^{244}\text{Am}^m(\text{IT})^{244}\text{Am}$	85.0	1.0	89.3	1.6	4.3	F					84Ho02 *
$^{244}\text{Am}(\beta^-)^{244}\text{Cm}$	1427.3	1.0				3					62Va08 *
* $^{244}\text{Cm}(\alpha)^{240}\text{Pu}$	$E_\alpha=5804.77(0.05,\text{Z})$, $5762.65(0.03,\text{Z})$ to ground state, 2^+ level at 42.824 keV										Ens08a **
* $^{244}\text{Bk}(\alpha)^{240}\text{Am}$	$E_\alpha=6667.5(4,\text{Z})$, $6625.5(3,\text{Z})$ to ground state, 2^+ level at 42.824 keV										Ens08a **
* $^{244}\text{Md}(\alpha)^{240}\text{Es}$	Unweighted average of $E_\alpha=8860(30)$ at $E=231.5$ MeV and $E_\alpha=8740(30)$										20Kh08 **
* $^{244}\text{Md}(\alpha)^{240}\text{Es}$	at $E=239.8$ MeV										20Kh08 **
* $^{244}\text{Md}(\alpha)^{240}\text{Es}$	Trends from Mass Surface TMS favors the higher $Q-\alpha$										GAU **
* $^{244}\text{Am}^m(\text{IT})^{244}\text{Am}$	F : value in Fig. 1 only, no source no error										AHW **
* $^{244}\text{Am}(\beta^-)^{244}\text{Cm}$	$E_{\beta^-}=387(1)$ to 6^+ level at 1040.188 keV; also $E_{\beta^-}=1498(10)$ from										Ens036 **
*	$^{244}\text{Am}^m$ at 89.3(1.6) to ground state and 2^+ at 42.965 keV, not used										Nub211 **
$^{245}\text{Cm}(\alpha)^{241}\text{Pu}$	5621.9	0.5	5624.5	0.5	5.2	B			Kum		75Ba65 *
	5624.8	0.5			-0.6	1	96	68 ^{245}Cm	Ara		16Ko.B *
$^{245}\text{Bk}(\alpha)^{241}\text{Am}$	6454.7	4.	6454.5	1.4	-0.0	2					74Po08 *
	6454.5	1.5			0.0	2			Kum		75Ba25 *
$^{245}\text{Cf}(\alpha)^{241}\text{Cm}$	7257.5	2.0	7258.5	1.8	0.5	-					67Fi04 *
	7265	5			-1.3	-					96Ma72 *
	7260.8	11.			-0.2	U			GSa		04He28
	ave. 7258.5	1.9			-0.0	1	98	97 ^{245}Cf			average
$^{245}\text{Es}(\alpha)^{241}\text{Bk}$	7858.5	20.	7909	3	0.9	U					73Es01
	7884.0	20.			0.5	U			GSa		85He22
	7909.4	3.				4			RIa		89Ha27
$^{245}\text{Es}(\alpha)^{241}\text{Bk}^p$	7827.9	30.	7858.5	1.0	1.0	U					67Mi06
	7858.5	1.				4			RIa		89Ha27
$^{245}\text{Fm}(\alpha)^{241}\text{Cf}^p$	8285.5	20.	8290	7	0.2	11					67Nu01
	8290.6	7.			-0.1	11			GSa		20Kh10
$^{245}\text{Md}(\alpha)^{241}\text{Es}^p$	8779.0	20.	8780	60	-0.0	12			GSa		96Ni09 *
	8773.4	30.			0.1	12			GSt		20Kh10
$^{244}\text{Pu}(\text{d,p})^{245}\text{Pu}$	2469	15	2475	13	0.4	2			ANL		75Er.A *
$^{244}\text{Cm}(\text{d,p})^{245}\text{Cm}$	3297	7	3294.1	0.5	-0.4	U			ANL		67Er02
$^{245}\text{Pu}(\beta^-)^{245}\text{Am}$	1257	30	1278	14	0.7	R					68Da02 *
$^{245}\text{Am}(\beta^-)^{245}\text{Cm}$	905	5	895.9	1.5	-1.8	U					55Br02
* $^{245}\text{Cm}(\alpha)^{241}\text{Pu}$	$E_\alpha=5529.0$, 5488.5 , $5436.1(0.5,\text{Z})$, 5303.6 , $5234.4(1.2,\text{Z})$ keV										75Ba65 **
* $^{245}\text{Cm}(\alpha)^{241}\text{Pu}$	to ground state, $7/2^+$ 41.9722, $9/2^+$ 95.7795, $9/2^+$ 231.935, $11/2^+$ 301.172 levels										Ens15c **
*	$E_\alpha=5361.8(1.2,\text{Z})$ to $7/2^+$ 175.05 level in error in 75Ba65; excluded										FGK169 **
* $^{245}\text{Cm}(\alpha)^{241}\text{Pu}$	$E_\alpha=5532.7$, $5491.3(0.5)$, $5436.6(2.0)$, 5360.6 , 5305.2 , $5624.5(0.5)$ keV										16Ko.B **
*	to ground state, 41.97, 95.78, 175.05, 231.94, 301.17 levels										Ens15c **
* $^{245}\text{Bk}(\alpha)^{241}\text{Am}$	$E_\alpha=6349.0$, 6309.0 , 6146.0 , 5886.0 (all 4,Z)										91Ry01 **
*	to ground state, $7/2^-$ at 41.176, $5/2^+$ at 205.883, $3/2^-$ at 471.810 keV										Ens15c **
* $^{245}\text{Bk}(\alpha)^{241}\text{Am}$	$E_\alpha=6347.8$, 6307.8 , 6146.8 , 5885.8 recalibrated as in reference										91Ry01 **
*	to ground state, $7/2^-$ at 41.176, $5/2^+$ at 205.883, $3/2^-$ at 471.810 keV										Ens15c **
* $^{245}\text{Cf}(\alpha)^{241}\text{Cm}$	$E_\alpha=7136.8(2.0,\text{Z})$, $7083.8(2.0,\text{Z})$ to gs+5.6 and 56.1 level										96Ma72 **
* $^{245}\text{Cf}(\alpha)^{241}\text{Cm}$	$E_\alpha=7145(5)$, $7090(5)$ to gs+5.6 and 56.1 level										96Ma72 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
* ²⁴⁵ Md(α) ²⁴¹ Es ^p	Second E_α 8635(20) keV										96Ni09 **
* ²⁴⁵ Md(α) ²⁴¹ Es ^p	Interpreted as e ⁻ summing with E_α =8635(20) keV										21He.A **
* ²⁴⁴ Pu(d,p) ²⁴⁵ Pu	$Q=2252(15)$ to 217 level (estimated energy for 15/2 ⁻ level)										06Ma.A **
* ²⁴⁵ Pu(β^-) ²⁴⁵ Am	E_{β^-} =1210(40), 930(30) to (9/2 ⁺) level at 47.07, 7/2 ⁺ at 327.428 keV										Ens112 **
²⁴⁶ Es-u	72799	122	72810	100	0.1	1	63	63 ²⁴⁶ Es	RI1	1.0	18It04
²⁴⁶ Cm(α) ²⁴² Pu	5475.2	4.	5475.1	0.9	-0.0	U					63Dz07 Z
	5474.9	2.			0.1	-			Kum		66Ba07 *
	5475.2	1.			-0.1	-					84Sh31 *
ave.	5475.1	0.9			-0.0	1	99	98 ²⁴⁶ Cm			average
²⁴⁶ Cf(α) ²⁴² Cm	6871.0	1.0	6861.6	1.0	-9.4	B		99 ²⁴⁶ Cf	Kum		63Fr04 *
	6861.6	1.			0.0	1	100				77Ba69 *
²⁴⁶ Es(α) ²⁴² Bk ^p	7451.2	30.	7492	4	1.4	U					67Mi06
	7481.9	30.			0.3	U					73Es01
	7492.0	4.				2			RIa		89Ha27
²⁴⁶ Fm(α) ²⁴² Cf	8371.4	20.	8379	5	0.4	U					66Ak01
	8376.5	20.			0.1	U			Bka		67Nu01
	8386.7	20.			-0.4	o			GSa		96Ni09
	8378.4	10.			0.1	6			GSa		10An08
	8379.5	5.			-0.0	6			GSa		11Ve03
²⁴⁶ Md(α) ²⁴² Es	8884.7	20.	8890	40	0.1	o			GSa		96Ni09 *
	8888.8	40.				9			GSa		10An08
²⁴⁶ Md ^m (α) ²⁴² Es	8944.5	50.				9			GSa		10An08 *
²⁴⁴ Pu(t,p) ²⁴⁶ Pu	2085	20	2072	15	-0.6	1	56	55 ²⁴⁶ Pu	LAI		79Br19
²⁴⁶ Cm(d,t) ²⁴⁵ Cm	-196	6	-201.7	1.2	-1.0	U			ANL		67Er02
²⁴⁶ Pu(β^-) ²⁴⁶ Am ^m	374	10	371	9	-0.3	1	89	45 ²⁴⁶ Pu			56Ho23 *
²⁴⁶ Am ^m (β^-) ²⁴⁶ Cm	2300	100	2407	15	1.1	U					55En16 *
	2420	20			-0.6	1	56	56 ²⁴⁶ Am ^m			56Sm85 *
²⁴⁶ Bk(ϵ) ²⁴⁶ Cm	1350	60				2					89Sc26
* ²⁴⁶ Cm(α) ²⁴² Pu	E_α =5385.3(2,Z), 5342.3(2,Z) to ground state, 2 ⁺ level at 44.54 keV										Ens026 **
* ²⁴⁶ Cm(α) ²⁴² Pu	E_α =5385.6(1,Z), 5342.6(1,Z) to ground state, 2 ⁺ level at 44.54 keV										Ens026 **
* ²⁴⁶ Cf(α) ²⁴² Cm	E_α =6757.4(1.0,Z), 6718.4(0.7,Z) to ground state, 2 ⁺ level at 42.13 keV										Ens026 **
* ²⁴⁶ Cf(α) ²⁴² Cm	E_α =6750.0(1.0,Z), 6708.2(1.0,Z) to ground state, 2 ⁺ level at 42.13 keV										Ens026 **
* ²⁴⁶ Md(α) ²⁴² Es	Also a lower E_α =8530(30) keV										96Ni09 **
* ²⁴⁶ Md ^m (α) ²⁴² Es	E_α =8178(10) to level at 531+x; x estimated to be 100#50 keV										10An08 **
* ²⁴⁶ Pu(β^-) ²⁴⁶ Am ^m	E_{β^-} =150(10) to 1 ⁺ level at 223.74 keV above ²⁴⁶ Am ^m										Ens118 **
* ²⁴⁶ Am ^m (β^-) ²⁴⁶ Cm	E_{β^-} =1222(100) 1350(20) respectively, to 1 ⁻ level at 1078.845, 2 ⁻ level at 1104.854										Ens118 **
²⁴⁷ Cm(α) ²⁴³ Pu	5354.6	4.	5354	3	-0.2	1	71	60 ²⁴⁷ Cm			71Fi01 *
²⁴⁷ Bk(α) ²⁴³ Am	5889.6	5.				2					69Fr01 *
²⁴⁷ Cf(α) ²⁴³ Cm ^p	6399.6	5.				3					84Ah02 Z
²⁴⁷ Es(α) ²⁴³ Bk ^p	7450.7	30.	7443.8	1.0	-0.2	U					67Mi06
	7430.5	30.			0.4	U					73Es01
	7443.8	1.				5			RIa		89Ha27
²⁴⁷ Fm(α) ²⁴³ Cf	8060.8	50.	8258	10	3.9	B			DBa		67Fi15
	8213	18			2.5	o			GSa		89He03 *
	8287.3	20.			-1.5	o			GSa		04He28 *
	8268.1	10.			-1.0	o			GSa		06He27 *
²⁴⁷ Fm ^m (α) ²⁴³ Cf	8314.9	30.	8307	5	-0.3	U			DBa		67Fi15 *
	8260.0	30.			1.5	o			GSa		97He29 *
	8304.8	11.			0.2	o			GSa		04He28
	8306.8	5.				7			GSa		06He27
²⁴⁷ Md(α) ²⁴³ Es	8776.6	25.	8764	10	-0.5	o			GSa		81Mu12 *
	8772.5	20.			-0.4	o			GSa		93Ho.A *
	8770.5	10.			-0.6	o			GSa		05He27 *
	8764.4	10.				10			GSa		10An08 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{247}\text{Md}(\alpha)^{243}\text{Es}$	9027.9	40.				10			GSa		10An08 *
$^{246}\text{Cm}(\text{d,p})^{247}\text{Cm}$	2931	8	2931	4	-0.1	1	22	21 ^{247}Cm	ANL		67Er02
$^{247}\text{Cf}(\epsilon)^{247}\text{Bk}$	646	6	620	15	-4.4	C					56Ch.A *
$^{247}\text{Cm}(\alpha)^{243}\text{Pu}$	$E_\alpha=5267.3(4,Z)$ 5212.3(4,Z) 4870.3(4,Z) to ground state, $9/2^+$ 58.13, $9/2^-$ 402.6										Ens148 **
$^{247}\text{Bk}(\alpha)^{243}\text{Am}$	$E_\alpha=5794$, 5710, 5688(5,Z) to ground state, $5/2^+$ level at 84.00, $7/2^+$ at 109.22 keV										Ens148 **
$^{247}\text{Fm}(\alpha)^{243}\text{Cf}$	$E_\alpha=8060(15)$ summed with e^-										AHW **
$^{247}\text{Fm}(\alpha)^{243}\text{Cf}$	$E_\alpha=7840(20)$ to 318 level										04He28 **
$^{247}\text{Fm}(\alpha)^{243}\text{Cf}$	$E_\alpha=7824(10)$ to 315 level										06He27 **
$^{247}\text{Fm}^m(\alpha)^{243}\text{Cf}$	Only one event										97He29 **
$^{247}\text{Fm}^m(\alpha)^{243}\text{Cf}$	Not observed in later work on ^{251}No decay										01He35 **
$^{247}\text{Md}(\alpha)^{243}\text{Es}$	$E_\alpha=8428(25)$ to 209.6 level										10An08 **
$^{247}\text{Md}(\alpha)^{243}\text{Es}$	$E_\alpha=8424(20)$ to 209.6 level										10An08 **
$^{247}\text{Md}(\alpha)^{243}\text{Es}$	$E_\alpha=8422(10)$ to 209.6 level										10An08 **
$^{247}\text{Md}(\alpha)^{243}\text{Es}$	$E_\alpha=8616(20)$, 8416(10) to ground state, 209.6 level										10An08 **
$^{247}\text{Md}^m(\alpha)^{243}\text{Es}$	$E_\alpha=8783(40)$ to $1/2^-$ level at 100 keV										10An08 **
$^{247}\text{Cf}(\epsilon)^{247}\text{Bk}$	LMK=10(3) assuming first-forbidden to $(5/2^-)$ level at 447.80 keV, yields 646.8										Ens153 **
*	contradicts LMK=75 allowed to $(5/2)^+$ level at 334.92 keV, yields 550;										WgM10a**
*	both conflict with 613 from Shiptrap data to ^{255}No plus α chain										WgM10a**
$^{248}\text{Cm}(\alpha)^{244}\text{Pu}$	5161.81	0.25	5161.81	0.25	0.0	1	100	95 ^{248}Cm	Kum		77Ba69 Z
$^{248}\text{Cf}(\alpha)^{244}\text{Cm}$	6364.8	5.1	6361	5	-0.7	o					78Gr10
	6361.2	5.				3					84Ah02 *
$^{248}\text{Es}(\alpha)^{244}\text{Bk}$	7165.8	20.	7160#	50#	-0.3	F					84Li.A *
$^{248}\text{Es}(\alpha)^{244}\text{Bk}^p$	6982.8	15.	7020	5	2.5	U					56Ch67
	6982.8	10.			3.8	B					70Ah01
	7020.4	5.				5			RIa		89Ha27
$^{248}\text{Fm}(\alpha)^{244}\text{Cf}$	8009.4	30.	7995	8	-0.5	-					66Ak01
	7999.3	20.			-0.2	-					67Nu01
	8002.3	15.			-0.5	o			GSa		85He.A
	7992.2	11.			0.2	-			GSa		04He28
ave.	7995	9			-0.0	1	79	77 ^{248}Fm			average
$^{248}\text{Md}(\alpha)^{244}\text{Es}$	8497.3	30.				5					73Es01
$^{248}\text{Cm}(\text{p,t})^{246}\text{Cm}$	-2894	15	-2885.0	2.7	0.6	U			ANL		74Fr01
$^{248}\text{Cm}(\text{d,t})^{247}\text{Cm}$	49	8	46	4	-0.4	1	25	19 ^{247}Cm	ANL		67Er02
$^{248}\text{Bk}^m(\beta^-)^{248}\text{Cf}$	870	20				4					78Gr10
$^{248}\text{Cf}(\alpha)^{244}\text{Cm}$	$E_\alpha=6257.8(5,Z)$, 6216.8(5,Z) to ground state, 2^+ level at 42.965 keV										Ens036 **
$^{248}\text{Es}(\alpha)^{244}\text{Bk}$	F : this line is not observed in more recent works										AHW **
$^{249}\text{Md}-u$	82940	239	82860	180	-0.3	1	55	55 ^{249}Md	RII	1.0	18It04
$^{249}\text{Cf O}-\text{C}_{22}$	69760.0	1.4	69765.0	1.3	2.4	1	37	37 ^{249}Cf	TG1	1.5	14Ei01
$^{249}\text{Bk}(\alpha)^{245}\text{Am}$	5520.4	2.0	5521.0	1.4	0.3	3			Bka		66Ah.A *
	5526.1	1.0			-5.1	B			Kum		71Bb10 *
	5521.6	2.0			-0.3	3			Ara		13Ah03 *
$^{249}\text{Cf}(\alpha)^{245}\text{Cm}$	6296.0	0.7	6293.3	0.5	-3.9	B			Kum		71Bb10 *
	6293.6	0.5			-0.6	1	96	63 ^{249}Cf	Ara		15Ah03 *
$^{249}\text{Es}(\alpha)^{245}\text{Bk}^p$	6881.3	5.	6886.0	1.9	0.9	4					70Ah01 Z
	6886.8	2.			-0.4	4			RIa		89Ha27
$^{249}\text{Fm}(\alpha)^{245}\text{Cf}$	7718.3	20.	7709	6	-0.5	-					73Es01 *
	7705.1	23.			0.2	o			GSa		85He06 *
	7705.0	14.			0.3	-			GSa		04He28 *
	7710.2	8.1			-0.1	-			Orm		11Lo06 *
ave.	7710	7			-0.1	1	80	77 ^{249}Fm			average
$^{249}\text{Md}(\alpha)^{245}\text{Es}^p$	8161.3	20.	8158	9	-0.2	2					73Es01
	8157.3	20.			0.0	o			GSa		85He22
	8165	20			-0.3	o			GSa		01He35 *
	8157.3	10.			0.1	o			GSa		05He27
	8157.3	10.			0.1	2			GSa		09He20

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{249}\text{Md}^m(\alpha)^{245}\text{Es}^q$	8212.2	20.				3			GSa		01He35
$^{248}\text{Cm}(\text{n},\gamma)^{249}\text{Cm}$	4713.37	0.25				2			ILn		82Ho07 Z
$^{248}\text{Cm}(\text{d},\text{p})^{249}\text{Cm}$	2488	6	2488.80	0.25	0.1	U			ANL		67Er02
$^{249}\text{Cm}(\beta^-)^{249}\text{Bk}$	870	100	904.4	2.6	0.3	U					58Ea06 *
	885	15			1.3	U			ANB		05Ah03 *
$^{249}\text{Bk}(\beta^-)^{249}\text{Cf}$	125	2	123.6	0.4	-0.7	U					59Va02
	123	2			0.3	U					74G110
	123.6	0.4				2					14Ch47
$^{249}\text{Bk}(\alpha)^{245}\text{Am}$	$E_\alpha=5431.8, 5412.8, 5384.8(\text{all } 2, Z)$ to ground state, $7/2^+$ 19.20, $9/2^+$ 47.07 keV										Ens112 **
$^{249}\text{Bk}(\alpha)^{245}\text{Am}$	$E_\alpha=5437.1(1.0, Z)$ to gs. Energies of higher branches										71Bb10 **
*	rather different from reference, calibrated with same ground state α										75Ba27 **
$^{249}\text{Bk}(\alpha)^{245}\text{Am}$	$E_\alpha=5433(2), 5414(2), 5386(2)$ to ground state, $7/2^+$ 19.20, $9/2^+$ 47.07 keV										Ens112 **
$^{249}\text{Cf}(\alpha)^{245}\text{Cm}$	$E_\alpha=6193.8(0.7, Z), 5813.3(1.0, Z)$ to ground state, $9/2^-$ level at 388.181 keV										Ens112 **
$^{249}\text{Cf}(\alpha)^{245}\text{Cm}$	$E_\alpha=6192.4(0.5), 5810.5(0.5)$ to ground state, $9/2^-$ level at 388.181 keV										Ens112 **
$^{249}\text{Fm}(\alpha)^{245}\text{Cf}$	$E_\alpha=7540(20)$ to corresponding $7/2^+$ [624] level at 55(10) keV										04He28 **
$^{249}\text{Fm}(\alpha)^{245}\text{Cf}$	$E_\alpha=7527(23)$ to corresponding $7/2^+$ [624] level at 55(10) keV										04He28 **
$^{249}\text{Fm}(\alpha)^{245}\text{Cf}$	Also $E_\alpha=7530(10)$ keV to $7/2^+$ [624] level at 57(4) keV										11Lo06 **
$^{249}\text{Fm}(\alpha)^{245}\text{Cf}$	$E_\alpha=7530(7)$ to $7/2^+$ [624] level at 57(4) keV										11Lo06 **
$^{249}\text{Md}(\alpha)^{245}\text{Es}^p$	$E_\alpha=8022(20)$ partly summed with conversion electrons										01He35 **
$^{249}\text{Cm}(\beta^-)^{249}\text{Bk}$	$E_{\beta^-}=860(100)$ 876(15) respectively, to $3/2^-$ level at 8.777 keV										Ens118 **
$^{250}\text{Md}-\text{u}$	84177	155	84160	100	-0.1	1	40	^{250}Md	RII	1.0	18It04 *
$^{250}\text{Cf}(\alpha)^{246}\text{Cm}$	6129.1	0.6	6128.51	0.19	-1.0	2			Kum		71Bb10 *
	6128.44	0.2			0.3	2					86Ry04 Z
$^{250}\text{Fm}(\alpha)^{246}\text{Cf}$	7550.9	50.	7557	8	0.1	U					57Am47
	7540.9	30.5			0.5	-					66Ak01
	7561.1	30.			-0.1	-					73Es01
	7560.1	15.			-0.2	-			ORb		77Be36
	7556.1	35.6			0.0	o			GSa		81Mu06
	7555.0	12.			0.2	-			GSa		04He28
	7544.8	35.			0.3	U			Bka		06Fo02
ave.	7556	9			0.1	1	80	^{250}Fm			average
$^{250}\text{Md}(\alpha)^{246}\text{Es}$	8155	28	8155	28	0.0	1	98	^{250}Md	GSa		19Vo03 *
$^{250}\text{Md}(\alpha)^{246}\text{Es}^p$	7947.4	30.	7970	40	0.7	F					73Es01 *
	7964.7	20.			0.1	o			GSa		85He22
	7967.7	40.				2			GSa		08An16
$^{250}\text{Md}^m(\alpha)^{246}\text{Es}$	8278	28				2			GSa		19Vo03 *
$^{248}\text{Cm}(\text{t},\text{p})^{250}\text{Cm}$	2064	10				2					73Ba72
$^{250}\text{Bk}(\beta^-)^{250}\text{Cf}$	1760	15	1781.7	2.5	1.4	U					59Va02 *
$^{250}\text{Md}-\text{u}$	$M-A=78472(140)$ keV for mixture gs+m at 123(28) keV										WgM209**
$^{250}\text{Cf}(\alpha)^{246}\text{Cm}$	$E_\alpha=6030.6(0.6, Z), 5988.9(0.6, Z)$ to ground state, 2^+ level at 42.852 keV										Ens118 **
$^{250}\text{Md}(\alpha)^{246}\text{Es}$	$E_\alpha=7875(20)$ to 2-# level at 151.9(2.0), unc. as given in reference										19Vo03 **
$^{250}\text{Md}(\alpha)^{246}\text{Es}^p$	Lacking spectroscopy information										19Vo03 **
$^{250}\text{Md}^m(\alpha)^{246}\text{Es}$	$E_\alpha=7741(20)$ to 411.4, 7773(20) to 379.5 keV, unc. as given in reference										19Vo03 **
$^{250}\text{Bk}(\beta^-)^{250}\text{Cf}$	$E_{\beta^-}=725(15)$ to 2^+ level at 1031.852 keV and 3^+ level at 1071.37 keV										Ens01c **
$^{251}\text{Fm}-\text{u}$	81585	45	81545	15	-0.9	1	12	^{251}Fm	RII	1.0	18It04
$^{251}\text{Md}-\text{u}$	84837	69	84774	20	-0.9	U			RII	1.0	18It04 *
$^{251}\text{Cf}(\alpha)^{247}\text{Cm}$	6177.2	1.0	6177.0	0.9	-0.2	2			Kum		71Bb10 *
	6176	2			0.5	2					03Ah07 *
$^{251}\text{Es}(\alpha)^{247}\text{Bk}$	6593.5	5.	6597.1	1.0	0.7	o					70Ah01 *
	6597.8	3.			-0.2	o					79Ah03 *
	6597.1	1.0				3			Ara		19Ah04 *
$^{251}\text{Fm}(\alpha)^{247}\text{Cf}$	7425.1	2.0	7424.5	1.0	-0.3	o					73Ah02 *
	7424.5	1.0				2			Ara		19Ah04 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{251}\text{Md}(\alpha)^{247}\text{Es}$	7965.5	20.	7963	4	-0.1	U					73Es01 *
	7955.2	10.			0.8	4			GSa		05He27 *
	7965.5	5.			-0.4	4			Jya		06Ch52 *
$^{251}\text{No}(\alpha)^{247}\text{Fm}$	8739.5	20.	8752	4	0.6	U			Bka		67Gh01
	8732.4	15.			1.3	o			GSa		89He03
	8762.9	20.			-0.6	o			GSa		97He29
	8760.9	20.			-0.9	o			GSa		01He35
	8747.7	11.			0.4	o			GSa		04He28
	8751.8	4.				10			GSa		06He27
$^{251}\text{No}^m(\alpha)^{247}\text{Fm}^m$	8619.6	30.	8809	4	6.3	F			GSa		97He29 *
	8805.5	13.			0.2	o			GSa		04He28
	8808.6	4.				8			GSa		06He27
$^{251}\text{Cm}(\beta^-)^{251}\text{Bk}$	1420	20				4					78Lo13
$^{251}\text{Bk}(\beta^-)^{251}\text{Cf}$	1093	10				3					84Li05 *
$^{251}\text{No}^m(\text{IT})^{251}\text{No}$	106	6				9			GSa		06He27
$^{251}\text{Md}-\text{u}$	possible mixture of gs+p, not corrected										WgM209**
$^{251}\text{Cf}(\alpha)^{247}\text{Cm}$	$E_\alpha=5680.1(1.0, Z)$ to $1/2^+$ level at 404.90 keV										Ens153 **
$^{251}\text{Cf}(\alpha)^{247}\text{Cm}$	$E_\alpha=6078(2), 5679(2)$ to ground state, $1/2^+$ level at 404.90 keV, and others										Ens153 **
$^{251}\text{Es}(\alpha)^{247}\text{Bk}$	$E_\alpha=6488.5(5, Z), 6458.5(5, Z)$ to ground state, $(5/2^-)$ level at 29.88 keV										Ens153 **
$^{251}\text{Es}(\alpha)^{247}\text{Bk}$	$E_\alpha=6492.8(3, Z), 6462.8(3, Z)$ to ground state, $(5/2^-)$ level at 29.88 keV										Ens153 **
$^{251}\text{Es}(\alpha)^{247}\text{Bk}$	re-evaluated data of $^{1979}\text{Ah}_{03}$; $E_\alpha=6492.8(3, Z)$										19Ah04 **
$^{251}\text{Fm}(\alpha)^{247}\text{Cf}$	$E_\alpha=7305.7(3, Z), 6833.7(2, Z)$ to ground state and $(9/2^-)$ level at 480.40 keV										Ens153 **
$^{251}\text{Fm}(\alpha)^{247}\text{Cf}$	re-evaluated data from 73Ah02; $E_\alpha=7306.0(1.0), 6833.4(1.0)$ keV										19Ah04 **
$^{251}\text{Fm}(\alpha)^{247}\text{Cf}$	to ground state and $(9/2^-)$ level at 480.40 keV										19Ah04 **
$^{251}\text{Md}(\alpha)^{247}\text{Es}$	$E_\alpha=7550(20) 7540(10) 7550(1)$ respectively, to $7/2^-$ level at 293.7 keV										06Ch52 **
$^{251}\text{Md}(\alpha)^{247}\text{Es}$	Original error 1 keV in third reference increased for calibration										GAu **
$^{251}\text{No}^m(\alpha)^{247}\text{Fm}^m$	F : not observed in later work on ^{251}No decay										01He35 **
$^{251}\text{Bk}(\beta^-)^{251}\text{Cf}$	$E_{\beta^-}=915(10)$ to $3/2^+$ level at 177.602 keV										Ens139 **
$^{252}\text{Md}-\text{u}$	86385	98				2			RI1	1.0	18It04
$^{252}\text{No}-^{133}\text{Cs}_{1.895}$	268111	34	268135	10	0.7	o			SH1	1.0	10Dw01
	268133	18			0.1	1	31	31 ^{252}No	SH1	1.0	12Mi27
$^{252}\text{Cf}(\alpha)^{248}\text{Cm}$	6216.9	0.5	6216.95	0.04	0.1	U			Kum		71Bb10 Z
	6216.95	0.04				2					86Ry04 Z
$^{252}\text{Es}(\alpha)^{248}\text{Bk}$	6739.5	3.	6738.6	0.5	-0.3	o					73Fi06 *
	6738.6	0.5				4			Ara		19Ah04 *
$^{252}\text{Fm}(\alpha)^{248}\text{Cf}$	7152.7	2.	7153.7	1.0	0.5	o					84Ah02 *
	7153.7	1.0				4			Ara		19Ah04 *
$^{252}\text{No}(\alpha)^{248}\text{Fm}$	8545.9	20.	8549	5	0.1	U			Bka		67Gh01
	8545.9	30.			0.1	U			DbA		67Mi03
	8551.0	6.			-0.4	-					77Be09
	8542.8	15.			0.4	o			GSa		85He.A
	8538.7	13.			0.7	-			GSa		04He28
ave.	8549	6			-0.0	1	92	69 ^{252}No			average
$^{252}\text{Lr}(\alpha)^{248}\text{Md}$	9163.8	20.	9164	17	0.0	6			GSa		01He35
	9165.8	30.			-0.0	6			Bka		08Ne01 *
$^{252}\text{Es}(\epsilon)^{252}\text{Cf}$	1260	50				3					73Fi06 *
$^{252}\text{Es}(\alpha)^{248}\text{Bk}$	$E_\alpha=6632.1(3, Z), 6522.1(3, Z)$ to 0, 7^+ level at 70.65 keV above ^{248}Bk										Ens14b **
$^{252}\text{Es}(\alpha)^{248}\text{Bk}$	re-evaluated data of 73Fi06; $E_\alpha=6631.5(0.5), 6050.8(0.5)$ to ground state and										19Ah04 **
$^{252}\text{Es}(\alpha)^{248}\text{Bk}$	(5^-) level at 590 keV										19Ah04 **
$^{252}\text{Fm}(\alpha)^{248}\text{Cf}$	$E_\alpha=7038.9(2, Z), 6998.1(2, Z)$ to ground state, 2^+ level at 41.53 keV										Ens14b **
$^{252}\text{Fm}(\alpha)^{248}\text{Cf}$	re-evaluated data from 84Ah02										19Ah04 **
$^{252}\text{Lr}(\alpha)^{248}\text{Md}$	Other $E_\alpha=9610(20)$ unexplained, and 8990, 8820 keV										08Ne01 **
$^{252}\text{Es}(\epsilon)^{252}\text{Cf}$	$pK=0.45(0.10)$ to 3^+ level at 969.8 keV, recalculated for non-unique first										Ens061 **
*	forbidden or allowed transition; unique first forbidden would give 1440(100)										AHW **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{253}\text{No}-^{133}\text{Cs}_{1,902}$	270390	13	270393	7	0.2	1	33	^{253}No	SH1	1.0	10Dw01
$^{253}\text{Cf}(\alpha)^{249}\text{Cm}$	6127.3	5.	6126	4	-0.3	3					66Rg01 *
	6124.6	5.			0.3	3					68Be21 *
$^{253}\text{Es}(\alpha)^{249}\text{Bk}$	6739.24	0.05				3					71Gr17 Z
	6739.7	2.0	6739.24	0.05	-0.2	U			Ara		19Ah04
$^{253}\text{Fm}(\alpha)^{249}\text{Cf}$	7199	3	7197.9	1.0	-0.4	o					67Ah02 *
	7194.2	6.			0.6	U			Orm		11Lo06 *
	7197.9	1.0				2			Ara		19Ah04 *
$^{253}\text{Md}(\alpha)^{249}\text{Es}$	7567.5	15.	7573	8	0.4	o			GSa		05He27 *
	7574.0	10.			-0.1	5			Orm		11Lo06 *
	7571	15			0.1	5			GSa		12He09 *
$^{253}\text{No}(\alpha)^{249}\text{Fm}$	8419	20	8415	4	-0.2	U			Bka		67Gh01 *
	8419	30			-0.1	U			DbA		67Mi03 *
	8430	20			-0.8	o			GSa		85He.A *
	8420	10			-0.5	o			GSa		01He.A *
	8412.5	11.			0.2	o			GSa		04He28 *
	8411.5	5.			0.6	o			Orm		06Lo12 *
	8415.6	5.0			-0.2	-			Orm		11Lo06 *
	8412.4	11.			0.2	-			GSa		12He09 *
	ave.	8415	5		-0.1	1	90	^{253}No			average
$^{253}\text{Lr}(\alpha)^{249}\text{Md}$	8941.6	20.	8918	20	-1.2	o			GSa		85He22
	8935.6	10.			-1.8	o			GSa		01He35
	8927.4	15.			-0.6	o			GSa		09He20
	8918.3	20.			-0.0	1	100	^{253}Lr	GSa		10He11
$^{253}\text{Lr}^m(\alpha)^{249}\text{Md}^m$	8862.4	20.	8850	20	-0.6	o			GSa		85He22
	8862.4	10.			-1.2	o			GSa		01He35
	8859.4	15.			-0.6	o			GSa		09He20
	8850.2	20.				3			GSa		10He11
$^{253}\text{Cf}(\beta^-)^{253}\text{Es}$	270	50	291	4	0.4	U					59Gh.A
$^{253}\text{Md}^p(\text{IT})^{253}\text{Md}$	60	30				6					Ens139
$^{253}\text{Cf}(\alpha)^{249}\text{Cm}$	$E_\alpha=5981(5,Z)$ to $7/2^+$ level at 48.76 keV										Ens118 **
$^{253}\text{Cf}(\alpha)^{249}\text{Cm}$	$E_\alpha=5978.4(5,Z)$, $5920.4(5,Z)$ to $7/2^+$ at 48.76, $7/2^+$ at 110.173 keV										Ens118 **
$^{253}\text{Fm}(\alpha)^{249}\text{Cf}$	$E_\alpha=7083.2(4,Z)$, $6943.2(3,Z)$, $6846.2(3,Z)$, $6673.2(3,Z)$ keV										67Ah02 **
*	to ground state and levels $5/2^+$ at 144.98, $9/2^+$ at 243.13, $1/2^+$ at 416.8 keV										Ens118 **
$^{253}\text{Fm}(\alpha)^{249}\text{Cf}$	$E_\alpha=6670(6)$ to 416.8 level										11Lo06 **
$^{253}\text{Fm}(\alpha)^{249}\text{Cf}$	re-evaluated data from 67Ah02; $E_\alpha=7083.9(1.0)$, $6673.7(1.0)$ to ground state,										19Ah04 **
$^{253}\text{Fm}(\alpha)^{249}\text{Cf}$	$1/2^+$ at 416.8 keV										19Ah04 **
$^{253}\text{Md}(\alpha)^{249}\text{Es}$	$E_\alpha=7100(15)$ to $353.2(0.4)$ level										05He27 **
$^{253}\text{Md}(\alpha)^{249}\text{Es}$	$E_\alpha=7105(10)$ to $354.6(0.6)$ level										11Lo06 **
$^{253}\text{Md}(\alpha)^{249}\text{Es}$	$E_\alpha=7103(15)$ to $353.2(0.4)$ level										12He09 **
$^{253}\text{No}(\alpha)^{249}\text{Fm}$	$E_\alpha=8010(20)$ to 279.7 level										04He28 **
$^{253}\text{No}(\alpha)^{249}\text{Fm}$	$E_\alpha=8010(30)$ to 279.7 level										04He28 **
$^{253}\text{No}(\alpha)^{249}\text{Fm}$	$E_\alpha=8021(20)$ to 279.7 level										04He28 **
$^{253}\text{No}(\alpha)^{249}\text{Fm}$	$E_\alpha=8011(10)$ to 279.7 level										04He28 **
$^{253}\text{No}(\alpha)^{249}\text{Fm}$	$E_\alpha=8004(11)$ to 279.7(5) level										04He28 **
$^{253}\text{No}(\alpha)^{249}\text{Fm}$	$E_\alpha=8003(5)$ to 279.7(5) level; and 8280(10) to ground state										04He28 **
$^{253}\text{No}(\alpha)^{249}\text{Fm}$	$E_\alpha=8007(4)$ to 279.8(0.2) level; also $E_\alpha=7615(30)$ to 669(3) and										11Lo06 **
*	$E_\alpha=8080(10)$ to 209.5(0.5) levels										11Lo06 **
$^{253}\text{No}(\alpha)^{249}\text{Fm}$	$E_\alpha=8004(10)$ to 279.5(5) level										12He09 **
$^{254}\text{No}-^{133}\text{Cs}_{1,910}$	271552	15	271541	10	-0.7	o			SH1	1.0	10Dw01
	271544	16			-0.2	1	42	^{254}No	SH1	1.0	12Mi27
$^{254}\text{No}-u$	90902	49	90954	10	1.1	U			RII	1.0	18It04
$^{254}\text{Cf}(\alpha)^{250}\text{Cm}$	5926.9	5.				3					68Be21 Z
$^{254}\text{Es}(\alpha)^{250}\text{Bk}$	6615.7	1.5	6617.2	0.5	1.0	6			Kum		72Bb24 *
	6617.4	0.5			-0.3	6			Ara		19Ah04 *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{254}\text{Es}(\alpha)^{250}\text{Bk}^n$	6531.6	1.5				7			Kum		72Bb24
$^{254}\text{Es}^m(\alpha)^{250}\text{Bk}$	6699.9	2.0	6697.6	1.0	-1.2	o					73Ah04 *
	6697.6	1.0				5			Ara		19Ah04 *
$^{254}\text{Fm}(\alpha)^{250}\text{Cf}$	7306.8	5.	7307.3	1.0	0.1	U			Bka		64As01 Z
	7307.6	2.			-0.2	o					84Ah02 *
	7307.2	1.0				3			Ara		19Ah04 *
$^{254}\text{No}(\alpha)^{250}\text{Fm}$	8229.8	20.	8226	8	-0.2	o			Bka		67Gh01
	8240.0	30.			-0.5	U			Db		67Mi03
	8215.6	20.			0.5	o			GSa		85He22
	8177.0	30.			1.6	U			Bka		06Fo02
	8225.7	10.			0.0	-			GSa		10He10
	8222.7	20.3			0.2	-			RIa		15KaZX
	8225	9			0.1	1	78	58 ^{254}No			average
$^{254}\text{Lr}(\alpha)^{250}\text{Md}$	8804.7	20.	8822	8	0.9	o			GSa		85He22 *
	8804.7	20.			0.9	2			Lza		01Ga20 *
	8820.6	25.7			0.0	2			Bka		06Fo02 *
	8825.2	20.			-0.2	o			GSa		08An16 *
	8826.3	10.			-0.4	2			GSa		19Vo03 *
$^{254}\text{Es}^m(\beta^-)^{254}\text{Fm}$	1172	2				4					62Un01 *
$^{254}\text{Es}(\alpha)^{250}\text{Bk}$	re-evaluated from 08Ah02; $E_\alpha=6430.5(0.5)$ to 7^+ level at 83.88(18) keV										19Ah04 **
$^{254}\text{Es}^m(\alpha)^{250}\text{Bk}$	$E_\alpha=6558.9(2,Z)$, 6383.9(2,Z) to 4^+ at 35.59, 2^+ at 211.82 keV										Ens01c **
$^{254}\text{Es}^m(\alpha)^{250}\text{Bk}$	re-evaluated from 73Ah04; $E_\alpha=6383.5(1.0)$ to 2^+ at 211.8(0.2) keV										19Ah04 **
$^{254}\text{Fm}(\alpha)^{250}\text{Cf}$	$E_\alpha=7192.3(2,Z)$, 7150.3(2,Z) to ground state, 5^+ level at 42.721 keV										Ens01c **
$^{254}\text{Fm}(\alpha)^{250}\text{Cf}$	re-evaluated data from 84Ah02										19Ah04 **
$^{254}\text{Lr}(\alpha)^{250}\text{Md}$	$E_\alpha=8460(20)$ to 209.1 level										08An16 **
$^{254}\text{Lr}(\alpha)^{250}\text{Md}$	$E_\alpha=8460(20)$ to 209.1 level										08An16 **
$^{254}\text{Lr}(\alpha)^{250}\text{Md}$	$E_\alpha=8437(50)$ to 209.1 and $E_\alpha=8394(30)$ to 306.2 keV										08An16 **
$^{254}\text{Lr}(\alpha)^{250}\text{Md}$	$E_\alpha=8480(20)$ to 209.1 and $E_\alpha=8385(20)$ to 306.2 keV										08An16 **
$^{254}\text{Lr}(\alpha)^{250}\text{Md}$	$E_\alpha=8480(15)$ to 209.1 and $E_\alpha=8386(10)$ to 306.8 keV										19Vo03 **
$^{254}\text{Es}^m(\beta^-)^{254}\text{Fm}$	$E_{\beta^-}=1127(2)$ to 2^+ level at 45.000 keV										Ens05b **
$^{255}\text{No}-^{133}\text{Cs}_{1,917}$	274440	16	274445	15	0.3	1	89	89 ^{255}No	SH1	1.0	12Mi27
$^{255}\text{Lr}-^{133}\text{Cs}_{1,917}$	277811	19				2			SH1	1.0	12Mi27 *
$^{255}\text{Es}(\alpha)^{251}\text{Bk}$	6439.3	3.0	6436.3	1.3	-1.0	4					66Rg01 *
	6435.6	1.5			0.5	4			Kum		71Bb10 *
$^{255}\text{Fm}(\alpha)^{251}\text{Cf}$	7237.0	4.	7240.6	0.5	0.9	U					64As01 *
	7240.4	2.			0.1	o					75Ah01 *
	7240.6	0.5				3			Ara		19Ah04 *
$^{255}\text{Md}(\alpha)^{251}\text{Es}$	7901.8	5.	7905.6	1.7	0.8	4					70Fi12 *
	7910.7	5.			-1.0	4					71Ho16 *
	7905.4	4.			0.1	o			Ara		00Ah02 *
	7891.2	15.			1.0	U			GSa		05He27 *
	7905.4	2.0			0.1	4			Ara		19Ah04 *
$^{255}\text{No}(\alpha)^{251}\text{Fm}$	8451.1	6.	8428.2	3.0	-3.8	B			ORb		71Di03 *
	8426.4	20.			0.1	o			GSa		98Ho13 *
	8391.9	35.			1.0	U			RIa		04Mo40 *
	8449.3	20.			-1.1	U			Bka		04Fo08 *
	8426.4	10.			0.2	U			GSa		06He20 *
	8403.8	60.			0.4	U			Bka		06Gr24 *
	8428.4	3.			-0.1	1	100	88 ^{251}Fm	JAA		11As03 *
$^{255}\text{Lr}(\alpha)^{251}\text{Md}$	8563.6	18.	8556	7	-0.4	3			ORb		76Be.A *
	8534.1	30.			0.7	U			Lza		01Ga20 *
	8554.4	10.			0.1	3			Jya		06Ch52 *
	8554.4	10.			0.1	3			Orm		08Ha31 *
$^{255}\text{Lr}(\alpha)^{251}\text{Md}^p$	8503.6	18.	8503	4	-0.0	3			ORb		76Be.A *
	8442.7	50.			1.2	F			Bka		95Gh04 *
	8493.5	30.			0.3	U			Lza		01Ga20 *
	8498.6	5.			0.8	3			Jya		06Ch52 *
	8506.7	11.			-0.3	3			GSa		08An16 *
	8509.7	10.2			-0.7	3			Orm		08Ha31 *
	8509.7	10.2			-0.7	3			RIa		11As.A *

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value	Adjusted value	v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{255}\text{Lr}^m(\alpha)^{251}\text{Md}$	8592.0 5.	8597 4	0.9	4			Jya		06Ch52 *
	8602.2 11.		-0.5	4			GSa		08An16
	8603.1 10.2		-0.6	4			Orm		08Ha31 *
	8604.2 10.2		-0.7	4			RIa		11As.A *
$^{255}\text{Rf}(\alpha)^{251}\text{No}$	9042 20	9055 4	0.7	10			Bka		69Gh01 *
	9053 15		0.2	o			GSa		85He06 *
	9064 20		-0.4	o			GSa		97He29 *
	9062 10		-0.7	o			GSa		01He35 *
	9056 4		-0.1	10			GSa		06He27 *
$^{255}\text{Rf}^m(\text{IT})^{255}\text{Rf}$	135 20	150 22	0.7	R					15An05 *
$^{255}\text{Lr}-^{133}\text{Cs}_{1.917}$	$D_M=277822(17) \mu\text{u}$ for mixture 71% ground state + 29% $^{255}\text{Lr}^m$ at 41(8); $M-A=89958(16) \text{ keV}$								Nub211 **
$^{255}\text{Es}(\alpha)^{251}\text{Bk}$	$E_\alpha=6303(3,Z) 6299.3(1.5,Z)$ respectively, to $(7/2^+)$ level at 35.5 keV								Ens139 **
$^{255}\text{Fm}(\alpha)^{251}\text{Cf}$	$E_\alpha=7121.5, 7018.5(4,Z)$ to ground state, $7/2^+$ level at 106.309 keV								Ens139 **
$^{255}\text{Fm}(\alpha)^{251}\text{Cf}$	$E_\alpha=7126.8, 7021.8(2,Z)$ to ground state, $7/2^+$ level at 106.309 keV								Ens139 **
$^{255}\text{Fm}(\alpha)^{251}\text{Cf}$	re-evaluated data from 75Ah01; Also $E_\alpha=7022.0(0.5), 6591.3(0.5)$ to								19Ah04 **
*	$7/2^+$ level at 106.309 keV, $5/2^+$ level at 543.99 keV								19Ah04 **
$^{255}\text{Md}(\alpha)^{251}\text{Es}$	$E_\alpha=7323.5(5,Z) 7332.3(5,Z) 7327(4) 7313(15)$ respectively, to $7/2^-$ at 461.5 keV								Ens139 **
$^{255}\text{Md}(\alpha)^{251}\text{Es}$	re-evaluated data from 00Ah02; $E_\alpha=7327.0(2.0)$ to $7/2^-$ at 461.5 keV								19Ah04 **
$^{255}\text{No}(\alpha)^{251}\text{Fm}$	$E_\alpha=8312(9), 8121(6)$ to ground state and 199.9 level								06He20 **
$^{255}\text{No}(\alpha)^{251}\text{Fm}$	$E_\alpha=8296(20), 8092(20)$ to ground state and 199.9 level								06He20 **
$^{255}\text{No}(\alpha)^{251}\text{Fm}$	$E_\alpha=8060$ to 199.9 level; also $E_\alpha=7800 \text{ keV}$								04Mo40 **
$^{255}\text{No}(\alpha)^{251}\text{Fm}$	$E_\alpha=8341, 8092$ to ground state and 199.9 level; also $E_\alpha=7873 \text{ keV}$								06He20 **
$^{255}\text{No}(\alpha)^{251}\text{Fm}$	$E_\alpha=8290(20), 8095(10)$ to ground state and 199.9 level								06He20 **
$^{255}\text{No}(\alpha)^{251}\text{Fm}$	$E_\alpha=8150, 8000$ to 199.9 level								06He20 **
$^{255}\text{No}(\alpha)^{251}\text{Fm}$	$E_\alpha=8100(3)$ to $^{251}\text{Fm}^m$ at 200.09(0.11) keV								11As03 **
$^{255}\text{Lr}(\alpha)^{251}\text{Md}$	This is the faint α from long-lived isomer to the $7/2^-$ ground state								06Ch52 **
$^{255}\text{Lr}(\alpha)^{251}\text{Md}$	Line is mixed with ^{254}Lr 's α								01Ga20 **
$^{255}\text{Lr}(\alpha)^{251}\text{Md}$	As interpreted from Fig. 1								08Ha31 **
$^{255}\text{Lr}(\alpha)^{251}\text{Md}^p$	This is the most intense α from long-lived isomer to $1/2^-$								06Ch52 **
$^{255}\text{Lr}(\alpha)^{251}\text{Md}^p$	F : one event in a questionable ^{267}Ds decay chain								AHW **
$^{255}\text{Lr}(\alpha)^{251}\text{Md}^p$	No γ observed in coincidence								08An16 **
$^{255}\text{Lr}(\alpha)^{251}\text{Md}^p$	Original $E_\alpha=8371(10)$ corrected for recoil to $E_\alpha=8476(10)$								16Hu.A **
$^{255}\text{Lr}(\alpha)^{251}\text{Md}^p$	Uncertainty estimated by evaluator								GAu **
$^{255}\text{Lr}^m(\alpha)^{251}\text{Md}$	Original error 2 keV increased for calibration								GAu **
$^{255}\text{Lr}^m(\alpha)^{251}\text{Md}$	Original $E_\alpha=8463(10)$ corrected for recoil to $E_\alpha=8468(10)$								16Hu.A **
$^{255}\text{Lr}^m(\alpha)^{251}\text{Md}$	Uncertainty estimated by evaluator								GAu **
$^{255}\text{Rf}(\alpha)^{251}\text{No}$	$E_\alpha=8700(20)$ to 203 level								01He35 **
$^{255}\text{Rf}(\alpha)^{251}\text{No}$	$E_\alpha=8766(15), 8715(15)$ to 142, 203 levels,								01He35 **
$^{255}\text{Rf}(\alpha)^{251}\text{No}$	$E_\alpha=8905(20), 8739(20)$ to ground state, 203 level								01He35 **
$^{255}\text{Rf}(\alpha)^{251}\text{No}$	$E_\alpha=8722(10)$ to 203(3) level								01He35 **
$^{255}\text{Rf}(\alpha)^{251}\text{No}$	$E_\alpha=8716(4)$ to 203.6(0.2) level								06He27 **
$^{255}\text{Rf}^m(\text{IT})^{255}\text{Rf}$	From 105 keV EC from K shell, error estim. by eval.								WgM164**
$^{256}\text{Lr}-^{133}\text{Cs}_{1.925}$	280499 89			2			SH1	1.0	12Mi27
$^{256}\text{Fm}(\alpha)^{252}\text{Cf}$	7027.3 5.	7025.3 1.9	-0.4	3					68Ho13 Z
	7024.9 2.0		0.2	3			Ara		19Ah04
$^{256}\text{Md}^m(\alpha)^{252}\text{Es}$	7834.6 20.	7900 50	1.2	B					71Ho16 *
	7896.6 16.			4					93Mo18 *
	7798.0 8.		1.9	B					00Ah02
$^{256}\text{No}(\alpha)^{252}\text{Fm}$	8553.9 30.	8582 5	0.9	U					67FI05 *
	8553.9 20.		1.4	U			Bka		67Gh01 *
	8578.3 12.		0.3	5					81Be03
	8582.3 6.		-0.1	5					90Ho03
$^{256}\text{Lr}(\alpha)^{252}\text{Md}^p$	8787.6 20.	8771 11	-0.8	3					71Es01
	8761.1 25.		0.4	o			ORb		76Be.A
	8777.4 20.		-0.3	3			ORb		76Di.A
	8767.2 35.		0.1	3			RIa		04Mo26
	8749.9 20.		1.0	3			Bka		04Fo08

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{256}\text{Rf}(\alpha)^{252}\text{No}$	8952.1	23.	8926	15	-1.1	o			GSa		85He06	
	8929.8	20.			-0.2	o			GSa		97He29	
	8925.7	15.2				2			GSa		10St14	
$^{256}\text{Db}(\alpha)^{252}\text{Lr}$	9336.2	20.				7			Bka		08Ne01	
$^{256}\text{Db}(\alpha)^{252}\text{Lr}^p$	9157.4	20.	9169	14	0.6	8			GSa		01He35	
	9179.7	20.			-0.6	8			Bka		08Ne01	*
$^{*256}\text{Md}^m(\alpha)^{252}\text{Es}$	Also $E_\alpha=7210(5,Z)$ keV to 520(20) level										70Fi12	**
$^{*256}\text{Md}^m(\alpha)^{252}\text{Es}$	Very weak line; more precise E_α to excited levels										93Mo18	**
*	α summed with electrons										WgM129	**
$^{*256}\text{No}(\alpha)^{252}\text{Fm}$	Probably mixture of two branches										AHW	**
$^{*256}\text{Db}(\alpha)^{252}\text{Lr}^p$	5 events $E_\alpha=9030\ 9060\ 9020\ 9040\ 9030$ keV										08Ne01	**
$^{257}\text{Db}-u$	107422	260	107520	180	0.4	1	46	46 ^{257}Db	RI1	1.0	20Sc.1	*
$^{257}\text{Fm}(\alpha)^{253}\text{Cf}$	6862.7	2.	6863.7	0.9	0.5	4			Bka		67As02	*
	6864.4	2.			-0.4	o					82Ah01	*
	6863.9	1.0			-0.2	4			Ara		19Ah04	*
$^{257}\text{Md}(\alpha)^{253}\text{Es}$	7549.3	5.	7557.1	0.9	1.6	U					70Fi12	*
	7557.6	1.			-0.5	4					93Mo18	*
	7552.3	3.0			1.6	4			Ara		19Ah04	*
$^{257}\text{No}(\alpha)^{253}\text{Fm}$	8474.1	30.	8477	6	0.1	U					70Es02	*
	8480	30			-0.1	U			GSa		96Ho13	*
	8476.6	6.				3			JAA		05As05	*
$^{257}\text{Lr}(\alpha)^{253}\text{Md}^p$	9020.6	20.	9008	9	-0.6	7					71Es01	
	9001.3	12.			0.5	o			ORb		76Be.A	
	9001.3	12.			0.5	7			ORb		77Be36	
	9015.5	15.2			-0.5	o			GSa		97He29	
	9030.8	50.			-0.5	U			Lza		04Ga29	
	9010.4	15.			-0.2	7			GSa		10St14	
$^{257}\text{Rf}(\alpha)^{253}\text{No}$	9079.8	15.	9083	8	0.2	2			ORb		73Be33	*
	9083.7	15.			-0.1	o			GSa		85He06	
	9044.0	15.			2.6	B			GSa		97He29	
	9084.1	20.			-0.1	o			GSa		07St12	*
	9084.1	10.			-0.1	2			GSa		10St14	*
	9106.2	100.			-0.2	U			Ara		09Qi04	*
$^{257}\text{Rf}^m(\alpha)^{253}\text{No}$	9142.5	20.	9156	7	0.7	2			Bka		69Gh01	
	9158.8	15.			-0.2	o			ORb		73Be33	
	9155.8	8.			0.0	2			ORb		90Be.A	
	9163.9	15.			-0.5	2			GSa		97He29	
	9144.0	100.			0.1	U			Ara		09Qi04	*
$^{257}\text{Db}(\alpha)^{253}\text{Lr}$	9112.1	20.	9206	20	4.7	B			GSa		85He22	
	9218	10			-1.2	o			GSa		01He35	*
	9209.2	20.			-0.2	o			GSa		09He20	*
	9206.6	20.			-0.0	1	100	54 ^{257}Db	GSa		10He11	
$^{257}\text{Db}^m(\alpha)^{253}\text{Lr}^m$	9305.1	20.	9313	20	0.4	o			GSa		85He22	
	9308.2	10.			0.5	o			GSa		01He35	
	9300.0	20.3			0.7	o			GSa		09He20	
	9313.2	20.3				4			GSa		10He11	
$^{257}\text{Rf}^m(\text{IT})^{257}\text{Rf}^m$	1082	4				3					10Be16	
$^{*257}\text{Db}-u$	100063(231)(2) keV for mixture of ground state and isomer, not corrected										20Sc.1	**
$^{*257}\text{Fm}(\alpha)^{253}\text{Cf}$	$E_\alpha=6518.5(2,Z)$ to $(9/2^+)$ level at 241.01 keV										Ens139	**
$^{*257}\text{Fm}(\alpha)^{253}\text{Cf}$	$E_\alpha=6756.5(3,Z)$, 6520.5(2,Z) to ground state, $(9/2^+)$ level at 241.01 keV										Ens139	**
$^{*257}\text{Fm}(\alpha)^{253}\text{Cf}$	re-evaluated data from 82Ah01; $E_\alpha=6519.7(1.0)$ to $(9/2^+)$ level at										19Ah04	**
$^{*257}\text{Fm}(\alpha)^{253}\text{Cf}$	241.01 keV										19Ah04	**
$^{*257}\text{Md}(\alpha)^{253}\text{Es}$	$E_\alpha=7066(5,Z)$ to 371.4 level										93Mo18	**
$^{*257}\text{Md}(\alpha)^{253}\text{Es}$	$E_\alpha=7440(2)$, 7074(1) to ground state, 371.4 level										93Mo18	**
$^{*257}\text{Md}(\alpha)^{253}\text{Es}$	re-evaluated data from 70Fi12; $E_\alpha=7069.0(3.0)$ to 371.4 level										19Ah04	**
$^{*257}\text{No}(\alpha)^{253}\text{Fm}$	$E_\alpha=8320(30)$ to 22.3 keV										05As05	**
$^{*257}\text{No}(\alpha)^{253}\text{Fm}$	$E_\alpha=8340(20)$; one event only; may be summing with e^-										AHW	**
$^{*257}\text{No}(\alpha)^{253}\text{Fm}$	$E_\alpha=8222(6)$ to 124.1(1) level; also $E_\alpha=8323(7)$ to 22.3 keV										05As05	**
$^{*257}\text{Rf}(\alpha)^{253}\text{No}$	$E_\alpha=8778(15)$ to 166.7 level										07St12	**
$^{*257}\text{Rf}(\alpha)^{253}\text{No}$	$E_\alpha=8778(20)$ and 8495(20) to 166.7 and 455 level										07St12	**

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
* ²⁵⁷ Rf(α) ²⁵³ No	E_α =8778(10) to 166.7; and 8950(15) to ground state is sum with conversion e ⁻										10St14	**
* ²⁵⁷ Rf(α) ²⁵³ No	E_α =8800(100) to 166.7 level										09Qi04	**
* ²⁵⁷ Rf ^m (α) ²⁵³ No	E_α =9000(100) and 8950(100) to ground state and 54(14) level										09Qi04	**
* ²⁵⁷ Db(α) ²⁵³ Lr	E_α =9074(10) partly summed with conversion e ⁻										01He35	**
* ²⁵⁷ Db(α) ²⁵³ Lr	E_α =8965(20) coinc. E(γ)=102.2; E_α =9066(20)summed with conversion e-										09He20	**
²⁵⁸ Md(α) ²⁵⁴ Es	7266.8	5.	7271.3	1.9	0.9	7					70Fi12	*
	7272	2			-0.4	7					93Mo18	*
²⁵⁸ Lr(α) ²⁵⁴ Md	8870	50	8904	19	0.7	5			ORb		76Be.A	*
	8910	20			-0.3	5					88Gr30	*
²⁵⁸ Rf(α) ²⁵⁴ No	9192.8	30.5	9196	13	0.1	2					08Ga08	
	9196.8	14.2			-0.1	2			GSa		16He15	
	9197.8	15.2			-0.1	o			GSa		19Vo03	
²⁵⁸ Db(α) ²⁵⁴ Lr	9531.0	50.	9437	10	-1.3	o			GSa		97Ho14	
	9500.6	15.			-1.2	o			GSa		09He20	
	9436.9	10.				3			GSa		19Vo03	*
²⁵⁸ Db(α) ²⁵⁴ Lr ^m	9341.1	10.	9330	21	-0.2	o			GSa		09He20	
	9330	21				4			GSa		19Vo03	*
²⁵⁸ Db(α) ²⁵⁴ Lr ^p	9445.7	15.	*			F			GSa		85He22	*
	9446.8	20.	*			F			Lza		01Ga20	*
	9341.1	10.	*			F			GSa		09He20	*
²⁵⁸ Db ^m (α) ²⁵⁴ Lr	9489.5	10.				3			GSa		19Vo03	*
* ²⁵⁸ Md(α) ²⁵⁴ Es	E_α =6713(5) to 447.9 level										93Mo18	**
* ²⁵⁸ Md(α) ²⁵⁴ Es	E_α =6763(4), 6718(2) to 403.8, 447.9 levels										93Mo18	**
* ²⁵⁸ Lr(α) ²⁵⁴ Md	E_α =8648(10) in coincidence with X(L) not X(K) $\rightarrow$ E(γ)=90(50) keV										AHW	**
* ²⁵⁸ Lr(α) ²⁵⁴ Md	E_α =8752 observed as sum of α 's and conversion electrons										AHW	**
* ²⁵⁸ Lr(α) ²⁵⁴ Md	Mass assignment confirmed										92Gr02	**
* ²⁵⁸ Db(α) ²⁵⁴ Lr	E_α =9074(10) to 220.7 keV, E_α =9093(10) to 199.1 keV										19Vo03	**
* ²⁵⁸ Db(α) ²⁵⁴ Lr ^m	E_α =9185(15), unc. as given in reference										19Vo03	**
* ²⁵⁸ Db(α) ²⁵⁴ Lr ^p	Lacking spectroscopy information, might misidentified										19Vo03	**
* ²⁵⁸ Db ^m (α) ²⁵⁴ Lr	E_α =9154(15) 9144(10) to 199.1 keV, 9165(15) to 178 keV										19Vo03	**
²⁵⁹ No(α) ²⁵⁵ Fm	7849.2	15.	7854	5	0.3	U					73Si40	*
	7869.6	15.			-1.0	U					93Mo18	*
	7854	5				4			JAa		13As02	*
²⁵⁹ Lr(α) ²⁵⁵ Md ^p	8582.8	20.	8574	9	-0.4	6					71Es01	
	8571.6	10.			0.2	6					92Ha22	
	8577.7	29.			-0.1	U			Bka		92Kr01	
²⁵⁹ Rf(α) ²⁵⁵ No ^p	8999.2	20.	9030	11	1.5	o			Bka		69Gh01	
	9030	20			0.0	3					81Be03	*
	9034.7	20.			-0.2	3			GSa		98Ho13	
	9026.6	35.			0.1	3			RIa		04Mo40	
	9026.6	20.3			0.2	3			Bka		04Fo08	
	9017	60			0.2	U			Bka		06Gr24	
	8940.4	11.			8.2	F			GSa		10Ni14	*
²⁵⁹ Db(α) ²⁵⁵ Lr	8968.8	50.			1.2	U					12Zh04	
	9618.8	20.				3			Lza		01Ga20	
²⁵⁹ Sg(α) ²⁵⁵ Rf	9771.2	30.5	9765	8	-0.1	o			GSa		85Mu11	
	9807.7	23.			-0.8	U			Bka		09Fo02	*
	9784	50			-0.4	o			GSa		09He20	*
²⁵⁹ Sg ^m (α) ²⁵⁵ Rf	9765.0	8.1				11			GSa		15An05	
	9852.4	20.3				11			GSa		15An05	
²⁵⁹ Sg ^m (α) ²⁵⁵ Rf ^m	9700.0	8.1	9702	8	0.3	12			GSa		15An05	
* ²⁵⁹ No(α) ²⁵⁵ Fm	suffer summing effect; highest E_α seen 7685(10); extra unc. added										WgM147	**
* ²⁵⁹ No(α) ²⁵⁵ Fm	suffer summing effect; highest E_α seen 7689(4); extra unc. added										WgM147	**
* ²⁵⁹ No(α) ²⁵⁵ Fm	E_α =7505(5) to 231.1 level										13As02	**
* ²⁵⁹ Rf(α) ²⁵⁵ No ^p	E_α =8870(20); partly summed E_α =8770(20) and e ⁻										AHW	**
* ²⁵⁹ Rf(α) ²⁵⁵ No ^p	F : lifetime 107 ms is much shorter than T=2.63 s in Nubase										Nub211	**
* ²⁵⁹ Sg(α) ²⁵⁵ Rf	One event only, resolution 23 keV; also a wide group at lower 9593(46)keV										09Fo02	**
* ²⁵⁹ Sg(α) ²⁵⁵ Rf	One event with E_α 9050 in coincidence with 593 γ ; also groups at										09He20	**

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
*	E_α =9607(10), 9550(10) keV										09He20 **
$^{260}\text{Lr}(\alpha)^{256}\text{Md}^p$	8155.6	20.3				6					71Es01
$^{260}\text{Db}(\alpha)^{256}\text{Lr}$	9191.5	30.	9500#	40#	10.3	F			RIa		04Mo26 *
	9516.5	30.			-0.5	F			RIa		04Mo26 *
	9563.2	20.			-3.1	F			Bka		04Fo08 *
$^{260}\text{Db}(\alpha)^{256}\text{Lr}^p$	9283.1	20.	9271	13	-0.6	4			Bka		70Gh02
	9262.8	17.			0.5	4			ORb		77Be36
	9316.5	60.			-0.8	U			GSa		95Ho04 *
	9285.1	60.			-0.2	U			GSa		02Ho11 *
	9181.3	60.			1.5	U			Lza		04Ga29 *
	9310.4	60.			-0.7	U			RIa		04Mo26 *
$^{260}\text{Sg}(\alpha)^{256}\text{Rf}$	9923.0	30.	9901	10	-0.7	o			GSa		85Mu11
	9900.6	10.				3			GSa		09He20
$^{260}\text{Bh}(\alpha)^{256}\text{Db}$	10400.3	30.5				8					08Ne01 *
$^{260}\text{Db}(\alpha)^{256}\text{Lr}$	Highest energy event; other two E_α =8810 and 8500 keV										04Mo26 **
$^{260}\text{Db}(\alpha)^{256}\text{Lr}$	F : not observed in experiments with greater statistics										77Be36 **
$^{260}\text{Db}(\alpha)^{256}\text{Lr}^p$	Two events E_α =9200 and 9146; error estimated by evaluator										FGK126 **
$^{260}\text{Db}(\alpha)^{256}\text{Lr}^p$	Two events E_α =9156 and 9129; error estimated by evaluator										FGK126 **
$^{260}\text{Db}(\alpha)^{256}\text{Lr}^p$	Eight events out of 14, E_α =9170 9050 9340 9400 9010 9100 9140 9130 keV										04Mo26 **
$^{260}\text{Db}(\alpha)^{256}\text{Lr}^p$	Two longer-lived α -escapes are assigned to the daughter										FGK126 **
$^{260}\text{Bh}(\alpha)^{256}\text{Db}$	Other events E_α =10170, 10170 and 10190; 10080 and 10030										08Ne01 **
$^{261}\text{Rf}(\alpha)^{257}\text{No}$	8652.9	20.	8650	70	-0.1	o			GSa		96Ho13
	8652.9	20.			-0.1	o			GSa		02Ho11
	8659	52			-0.2	o			GSa		03Tu05 *
	8642.6	50.			0.1	4			GSa		08Dv02
	8652.9	50.			-0.1	4			RIIm		11Ha13 *
	8642.6	60.			0.0	4			RIIm		12Ha05 *
$^{261}\text{Rf}^m(\alpha)^{257}\text{No}^p$	8409.1	20.	8415	15	0.3	6			Bka		70Gh01
	8388.8	30.			0.9	o			GSa		98Tu01 *
	8429.4	30.5			-0.5	6			DbA		00La34
	8470.0	50.			-1.1	U					08Ga08 *
	8419.3	50.			-0.1	6			GSa		08Dv02
	8409.1	50.			0.1	o			RIIm		11Ha13
	8409.1	40.6			0.1	6			RIIm		13Mu08
$^{261}\text{Db}(\alpha)^{257}\text{Lr}^p$	9069.2	20.	9068	14	-0.1	9			Bka		71Gh01
	9069.2	40.			-0.0	9			Lza		04Ga29
	9066.2	20.			0.1	9			GSa		10St14
$^{261}\text{Sg}(\alpha)^{257}\text{Rf}$	9709.0	30.	9714	15	0.2	o			GSa		85Mu11
	9713.1	20.			0.0	F			GSa		95Ho03 *
	9770.0	20.3			-2.8	o			GSa		07St12
	9713.7	15.				3			GSa		10St14 *
$^{261}\text{Bh}(\alpha)^{257}\text{Db}$	10562.1	25.	10500	70	-1.1	o			GSa		89Mu09
	10507.3	75.			-0.1	o			Bka		06Fo02 *
	10492.1	75.			0.1	2			Bka		08Ne08 *
	10504.3	40.			-0.1	2			GSa		10He11 *
$^{261}\text{Rf}(\alpha)^{257}\text{No}$	Two events with E_α =8500(+70-30) keV										GAu **
$^{261}\text{Rf}(\alpha)^{257}\text{No}$	From direct production (fusion-evaporation)										11Ha13 **
$^{261}\text{Rf}(\alpha)^{257}\text{No}$	Decay chain of ^{265}Sg , observation is independent of previous item										12Ha05 **
$^{261}\text{Rf}^m(\alpha)^{257}\text{No}^p$	In addition 60% E_α =8380(30) keV										98Tu01 **
$^{261}\text{Rf}^m(\alpha)^{257}\text{No}^p$	Single event, decay time 103.2 s										08Ga08 **
$^{261}\text{Sg}(\alpha)^{257}\text{Rf}$	F : α 's to 157 level summed with conversion electron										FGK10a **
$^{261}\text{Sg}(\alpha)^{257}\text{Rf}$	E_α =9410(15) to 157(1) level										10St14 **
$^{261}\text{Bh}(\alpha)^{257}\text{Db}$	E_α =10346(75) one event; error estimated by evaluator										06Fo02 **
$^{261}\text{Bh}(\alpha)^{257}\text{Db}$	Highest E_α ; error estimated; others 10054, 10285, 10113, 10165, 9989										08Ne08 **
$^{261}\text{Bh}(\alpha)^{257}\text{Db}$	Average of 2 highest 10331, 10355 as read from graph; error estimated										10He11 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{262}\text{Db}(\alpha)^{258}\text{Lr}^p$	8794.5	20.	8806	11	0.6	7			Bka		71Gh01	
	8815.8	20.			-0.5	7				88Gr30		
	8804.7	20.			0.1	7		GSa		99Dr09		
	8875.8	20.			-1.4	o		RIa		09Mo12		
	8814.8	30.5			-0.3	7		RIa		14Ha04		
$^{262}\text{Sg}(\alpha)^{258}\text{Rf}$	9599.8	15.2				3		GSa		10Ac.A		
$^{262}\text{Bh}(\alpha)^{258}\text{Db}$	10216.2	25.	10319	15	4.1	F		GSa		89Mu09	*	
	10300.5	25.4			0.7	o		GSa	97Ho14			
	10231.4	25.4			3.5	B		Bka	06Fo02			
	10239.2	30.			2.7	B		Bka	08Ne08	*		
	10319.5	15.				4		GSa	09He20	*		
$^{262}\text{Bh}^m(\alpha)^{258}\text{Db}$	10531.1	25.4	10540	70	0.1	o		GSa		89Mu09		
	10605.2	25.4			-1.2	o		GSa	97Ho14			
	10508.7	76.2			0.3	o		Bka	06Fo02	*		
	10544.3	76.2			-0.1	4		Bka	08Ne08			
	10534.1	15.			0.0	4		GSa	09He20			
$*^{262}\text{Bh}(\alpha)^{258}\text{Db}$	F : not highest line, see reference										97Ho14	**
$*^{262}\text{Bh}(\alpha)^{258}\text{Db}$	$E_\alpha=10096, 10025, 10125$ keV										08Ne08	**
$*^{262}\text{Bh}(\alpha)^{258}\text{Db}$	$E_\alpha=10008(15)$ to 156.5 level										09He20	**
$*^{262}\text{Bh}^m(\alpha)^{258}\text{Db}$	Single event, error estimated by evaluator										GAu	**
$^{263}\text{Rf}(\alpha)^{259}\text{No}^p$	8022.2	40.6	8022	29	-0.0	6					93Gr.C	
	8022.2	40.6			-0.0	6				99Ga.A		
$^{263}\text{Db}(\alpha)^{259}\text{Lr}^p$	8484.3	27.				8		Bka		92Kr01		
$^{263}\text{Sg}(\alpha)^{259}\text{Rf}$	9393.1	40.	9400	60	0.3	o		Bka		74Gh04		
	9403.3	60.						Bka	06Gr24	*		
$^{263}\text{Sg}(\alpha)^{259}\text{Rf}^q$	9200.2	40.	9200	60	-0.1	o		Bka		74Gh04		
	9149.4	60.9			0.8	o		Bka	94Gr08			
$^{263}\text{Sg}^m(\alpha)^{259}\text{Rf}^p$	9198.1	60.9				5		Bka		06Gr24	*	
	9391.1	20.	9390	13	-0.1	7		GSa		98Ho13		
	9382.9	50.8			0.1	o		Bka	03Gi05			
	9393.1	35.			-0.1	7		RIa	04Mo40	*		
	9388.0	20.			0.1	7		Bka	04Fo08			
9198.1	11.	17.5			F		GSa	10Ni14	*			
$^{263}\text{Hs}(\alpha)^{259}\text{Sg}$	10733.5	60.				12		Bka		09Dr02		
$^{263}\text{Hs}^m(\alpha)^{259}\text{Sg}$	11058.5	60.				12		Bka		09Dr02	*	
$*^{263}\text{Sg}(\alpha)^{259}\text{Rf}$	Two events $E_\alpha=9290$ and 9230 keV										06Gr24	**
$*^{263}\text{Sg}(\alpha)^{259}\text{Rf}^q$	Four events $E_\alpha=9010, 9100, 9060$ and 9060 keV										06Gr24	**
$*^{263}\text{Sg}^m(\alpha)^{259}\text{Rf}^p$	Also lower $E_\alpha=9130, 9040, 9150$ keV										04Mo40	**
$*^{263}\text{Sg}^m(\alpha)^{259}\text{Rf}^p$	F : the α chain originating from ^{267}Hs is in conflict with other data										10Ni14	**
$*^{263}\text{Hs}^m(\alpha)^{259}\text{Sg}$	Assignment assumed by evaluator										GAu	**
$^{264}\text{Bh}(\alpha)^{260}\text{Db}^p$	9767.3	20.	9760	18	-0.4	6			GSa		95Ho04	*
	9636.0	60.			2.1	U		Lza	04Ga29	*		
	9737.8	35.5			0.6	6		RIa	04Mo26	*		
$^{264}\text{Hs}(\alpha)^{260}\text{Sg}$	10870	210	10591	20	-1.3	o		GSa		87Mu15	*	
	10590.8	20.				4		GSa	95Ho.B			
	10966.4	80.			-4.7	B		RIa	11Sa41	*		
	$*^{264}\text{Bh}(\alpha)^{260}\text{Db}^p$ Three more events in reference $E_\alpha=9365, 9514$ and 9113 keV										02Ho11	**
$*^{264}\text{Bh}(\alpha)^{260}\text{Db}^p$	Three more events $E_\alpha=9501, 9481, 9440$ keV										04Ga29	**
$*^{264}\text{Bh}(\alpha)^{260}\text{Db}^p$	Six events; also two $E_\alpha=9830$ keV										04Mo26	**
$*^{264}\text{Hs}(\alpha)^{260}\text{Sg}$	$Q_\alpha=11000(+100-300)$ derived from T(1/2), one event only										87Mu15	**
$*^{264}\text{Hs}(\alpha)^{260}\text{Sg}$	Also $E_\alpha=10610(40)$ keV										11Sa41	**

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference	
$^{265}\text{Sg}(\alpha)^{261}\text{Rf}^m$	8945.3	60.	8980	70	0.5	F			Db		94La22 *	
	8904.7	30.			1.3	F			GSa		96Ho13 *	
	8975.7	30.			0.1	o			GSa		98Tu01	
	9077.3	30.			-1.7	F			GSa		98Tu01 *	
	9036.6	50.8			-0.8	o			GSa		03Tu05	
	8985.9	50.			-0.1	6			GSa		08Du09	
	8975.7	50.8			0.1	6			RIm		12Ha05	
$^{265}\text{Sg}^m(\alpha)^{261}\text{Rf}^p$	8823.5	50.	8810	40	-0.2	o			GSa		03Tu05	
	8813.3	40.			0.0	o			GSa		06Dv01	
	8823.5	50.			-0.2	6			GSa		08Dv02	
	8843.8	40.			-0.7	o			RIa		08Mo09	
	8823.5	50.			-0.2	o			RIa		12Ha05	
	8792.9	81.2			0.3	6			RIa		13Su04	
	9381.9	50.				11			Lza		04Ga29	
$^{265}\text{Bh}(\alpha)^{261}\text{Db}^p$	10524.2	25.	10470	15	-2.1	o			GSa		87Mu15	
$^{265}\text{Hs}(\alpha)^{261}\text{Sg}$	10468.3	20.			0.1	o			GSa		95Ho03	
	10459.2	15.			0.7	o			GSa		99He11 *	
	10470.3	15.2				4			GSa		09He20 *	
$^{265}\text{Hs}^m(\alpha)^{261}\text{Sg}$	10712.0	20.	10700	15	-0.6	o			GSa		95Ho03	
	10734.3	15.2			-2.3	o			GSa		99He11 *	
	10699.8	15.				4			GSa		09He20	
$*^{265}\text{Sg}(\alpha)^{261}\text{Rf}^m$	F : average but probably due to several groups, see reference										98Tu01 **	
$*^{265}\text{Sg}(\alpha)^{261}\text{Rf}^m$	F : this event is not trusted, see reference from same group										02Ho11 **	
$*^{265}\text{Sg}(\alpha)^{261}\text{Rf}^m$	F : most probably not from ^{265}Sg										GAu **	
$*^{265}\text{Hs}(\alpha)^{261}\text{Sg}$	Also $E_\alpha=10426(15)$ keV										99He11 **	
$*^{265}\text{Hs}(\alpha)^{261}\text{Sg}$	Most intense line, also $E_\alpha=10282(15)$, 10428(15), 10573(15) keV										09He20 **	
$*^{265}\text{Hs}^m(\alpha)^{261}\text{Sg}$	Also $E_\alpha=10726(15)$ keV										99He11 **	
$^{266}\text{Sg}(\alpha)^{262}\text{Rf}$	8762.0	50.	8800#	100#	0.8	F			Db		94La22 *	
	8904.1	40.			-2.6	F			GSa		98Tu01 *	
	8853.4	50.			-1.1	F			GSa		02Tu05 *	
	$^{266}\text{Bh}(\alpha)^{262}\text{Db}^p$	9432	50	9380	30	-1.1	9			Bka		00Wi15
		9245.3	81.2			1.6	9			Lza		06Qi03
		9371.2	50.			0.1	9			RIa		09Mo12 *
	$^{266}\text{Hs}(\alpha)^{262}\text{Sg}$	10335.7	20.3	10346	16	0.5	4			GSa		01Ho06
		10360	24			-0.6	4			GSa		11Ac.A
	$^{266}\text{Hs}^m(\alpha)^{262}\text{Sg}^p$	10592.6	20.				7			GSa		11Ac.A
	$^{266}\text{Mt}(\alpha)^{262}\text{Bh}$	10995.7	25.				5			GSa		97Ho14
$^{266}\text{Mt}^m(\alpha)^{262}\text{Bh}^m$	11269.7	50.	11920	50	13.0	B			GSa		84Mu07 *	
	11168.1	30.			25.0	B			GSa		89Mu16	
	11918.6	50.				5			GSa		97Ho14 *	
$*^{266}\text{Sg}(\alpha)^{262}\text{Rf}$	Average of two groups										02Tu05 **	
$*^{266}\text{Sg}(\alpha)^{262}\text{Rf}$	F : no α decay from ^{266}Sg , all re-assigned to ^{265}Sg										08Dv02 **	
$*^{266}\text{Bh}(\alpha)^{262}\text{Db}^p$	Also $E_\alpha=9770(40)$, 9080(40) keV from ^{278}Nh decay chain										08Mo09 **	
$*^{266}\text{Mt}^m(\alpha)^{262}\text{Bh}^m$	One E_α only; may be ground state										AHW **	
$*^{266}\text{Mt}^m(\alpha)^{262}\text{Bh}^m$	One $E_\alpha=11739(50)$, one 11306; several smaller										AHW **	
$^{267}\text{Sg}(\alpha)^{263}\text{Rf}^p$	8325.0	50.				8			GSa		08Dv02	
$^{267}\text{Bh}(\alpha)^{263}\text{Db}^p$	8964.5	30.5	8970	26	0.2	10			Bka		00Wi15	
	8984.8	50.			-0.3	10			GSa		02Tu05	
$^{267}\text{Hs}(\alpha)^{263}\text{Sg}$	10015.3	60.	10038	13	0.4	5			Db		95La20 *	
	10032.6	20.			0.2	5			GSa		98Ho13	
	10035.7	50.			0.0	o			Bka		03Gi05	
	10069.1	35.			-0.9	5			RIa		04Mo40 *	
	10034.6	20.			0.1	5			Bka		04Fo08 *	
	10145.2	50.			-2.2	U			GSa		10Ni14 *	
$^{267}\text{Hs}(\alpha)^{263}\text{Sg}^m$	9978.8	20.	9987	13	0.4	6			GSa		98Ho13 *	
	9979.8	35.			0.2	6			RIa		04Mo40 *	
	9997.0	20.			-0.5	6			Bka		04Fo08 *	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{267}\text{Hs}^m(\alpha)^{263}\text{Sg}^p$	9979.8	20.				8			Bka		04Fo08
$^{267}\text{Ds}(\alpha)^{263}\text{Hs}$	11776.8	50.8				13			Bka		95Gh04 *
* $^{267}\text{Hs}(\alpha)^{263}\text{Sg}$	Selecting two events at $E_\alpha=9860, 9870$; and one $E_\alpha=9740(60)$ keV										95La20 **
* $^{267}\text{Hs}(\alpha)^{263}\text{Sg}$	Selecting four events at $E_\alpha=9970, 9890, 9900, 9910$ keV										04Mo40 **
* $^{267}\text{Hs}(\alpha)^{263}\text{Sg}$	Selecting two events at $E_\alpha=9880, 9888$ keV										04Fo08 **
* $^{267}\text{Hs}(\alpha)^{263}\text{Sg}$	Directly produced ^{267}Hs ; daughter and grand-daughter also conflicting										10Ni14 **
* $^{267}\text{Hs}(\alpha)^{263}\text{Sg}^m$	And one $E_\alpha=9749(20)$ at 13 ms										98Ho13 **
* $^{267}\text{Hs}(\alpha)^{263}\text{Sg}^m$	Selecting 7 events at $E_\alpha=9830, 9820, 9820, 9860, 9820, 9830, 9830$ keV										04Mo40 **
* $^{267}\text{Hs}(\alpha)^{263}\text{Sg}^m$	Selecting 2 events at $E_\alpha=9864, 9830$ keV										04Fo08 **
* $^{267}\text{Ds}(\alpha)^{263}\text{Hs}$	Maybe the upper isomer at about 250 keV excitation energy										AHW **
$^{268}\text{Hs}(\alpha)^{264}\text{Sg}$	9622.9	16.	9760#	100#	8.6	D			GSa		10Ni14 *
$^{268}\text{Mt}(\alpha)^{264}\text{Bh}^p$	10395.5	20.	10438	19	2.1	o			GSa		95Ho04 *
	10431.9	20.			0.3	8			GSa		02Ho11 *
	10476.7	50.			-0.8	8			RIa		04Mo26 *
	10268	20			8.5	B			Bka		04Fo08 *
* $^{268}\text{Hs}(\alpha)^{264}\text{Sg}$	Trends from Mass Surface TMS suggest ^{268}Hs 140 keV less bound										GAu **
* $^{268}\text{Mt}(\alpha)^{264}\text{Bh}^p$	Two events $E_\alpha=10221$ coinc. $E(\gamma)=93$ and 10259; event #3 $E_\alpha=10097$ keV could be decay of an isomer with lifetime=171 ms										95Ho04 **
*	Average of event 95Ho04 $E_\alpha=10259$ and present 10294 keV										02Ho11 **
* $^{268}\text{Mt}(\alpha)^{264}\text{Bh}^p$	Also $E_\alpha=10340$ keV										02Ho11 **
* $^{268}\text{Mt}(\alpha)^{264}\text{Bh}^p$	One event only										04Fo08 **
$^{269}\text{Sg}(\alpha)^{265}\text{Rf}$	8699.6	100.	8580	70	-1.1	7					10El06
	8628.5	60.9			-0.6	o			Db		15Ut02
	8536.9	40.6			0.6	7			Db		18Ut02
$^{269}\text{Hs}(\alpha)^{265}\text{Sg}^m$	9369.6	30.	9280	30	-3.0	B			GSa		96Ho13 *
	9349.3	30.			-2.3	o			GSa		02Ho11 *
	9278.2	50.8			0.0	o			GSa		03Tu05 *
	9207.1	30.5			2.4	o			GSa		06Dv01
	9268.0	50.8			0.2	7			GSa		08Dv02
	9349.2	71.1			-1.0	o			RIa		08Mo09
	9288.3	40.6			-0.2	7			RIa		13Su04
$^{269}\text{Ds}(\alpha)^{265}\text{Hs}^m$	11280.1	20.				5			GSa		95Ho03
* $^{269}\text{Hs}(\alpha)^{265}\text{Sg}^m$	Event number 2 only; first event rejected, see reference										02Ho11 **
* $^{269}\text{Hs}(\alpha)^{265}\text{Sg}^m$	Two events $E_\alpha=9230, 9180$ both following $300 \mu\text{s}$ ^{273}Ds										02Ho11 **
* $^{269}\text{Hs}(\alpha)^{265}\text{Sg}^m$	Three events $E_\alpha=9180, 9100, 8880$; latter probably due to energy loss										03Tu05 **
$^{270}\text{Db}(\alpha)^{266}\text{Lr}$	8019.0	30.5	8310#	200#	5.0	D			GSa		14Kh04 *
$^{270}\text{Bh}(\alpha)^{266}\text{Db}$	9064.5	81.2				10			Db		07Og02
$^{270}\text{Hs}(\alpha)^{266}\text{Sg}$	9324.3	52.8	9070	40	-4.8	F			GSa		03Tu05 *
	9024	52			1.5	o			GSa		06Dv01 *
	9013.8	50.			1.1	7			GSa		08Dv02 *
	9123.4	77.			-0.7	7			GSa		10Gr04 *
	9155.8	80.			-1.1	7			Db		13Og03
$^{270}\text{Mt}(\alpha)^{266}\text{Bh}$	10181.1	70.	10180#	100#	-0.0	o			RIa		08Mo09
	10414.5	71.			-2.7	D			RIa		12Mo25 *
$^{270}\text{Ds}(\alpha)^{266}\text{Hs}$	11196.2	50.8	11117	28	-1.6	o			GSa		01Ho06
	11117.0	28.4				5			GSa		11Ac.A
$^{270}\text{Ds}^m(\alpha)^{266}\text{Hs}$	12333	50	12510	50	2.5	B			GSa		01Ho06
	12508.6	20.				5			GSa		11Ac.A
$^{270}\text{Ds}^m(\alpha)^{266}\text{Hs}^m$	11318	50	11410	70	1.2	o			GSa		01Ho06
	11405.2	52.				6			GSa		11Ac.A
* $^{270}\text{Db}(\alpha)^{266}\text{Lr}$	Trends from Mass Surface TMS suggest ^{270}Db 290 keV less bound										GAu **
* $^{270}\text{Hs}(\alpha)^{266}\text{Sg}$	F : re-assigned to ^{269}Hs										06Dv01 **
* $^{270}\text{Hs}(\alpha)^{266}\text{Sg}$	4 events at 8850(1.62s), 8900(0.05s), 8920(0.5s), 8880(0.4s)										GAu **
* $^{270}\text{Hs}(\alpha)^{266}\text{Sg}$	2 events at 8760(0.27s), 8810(0.27s); $E_\alpha=8880(50)$ for both experiments										GAu **
* $^{270}\text{Hs}(\alpha)^{266}\text{Sg}$	Symmetrized from $E_\alpha=9020(+50-100)$; independent from previous item										GAu **
* $^{270}\text{Mt}(\alpha)^{266}\text{Bh}$	Trends from Mass Surface TMS suggest ^{270}Mt 230 keV more bound										GAu **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{271}\text{Sg}(\alpha)^{267}\text{Rf}^p$	8658	80	8670	100	0.1	o			Db		04Og12
	8668.2	81.2				10			Db		06Og05
$^{271}\text{Bh}(\alpha)^{267}\text{Db}$	9490.3	162.4	9420	90	-0.4	o			Db		12St.A
	9409.1	71.0			0.1	o			GSa		13Ru11
	9420	70				8			SRv		17Og01
$^{271}\text{Hs}(\alpha)^{267}\text{Sg}^p$	9439.6	50.7				10			GSa		08Dv02
$^{271}\text{Ds}(\alpha)^{267}\text{Hs}$	10869.8	20.	10870	18	0.0	6			GSa		98Ho13
	10870.8	35.			-0.0	6			RIa		04Mo40 *
$^{271}\text{Ds}^m(\alpha)^{267}\text{Hs}$	10937.8	20.				6			Bka		04Fo08 *
	10803.8	50.	10938	20	2.7	F					12Zh04 *
$^{271}\text{Ds}^m(\alpha)^{267}\text{Hs}^m$	10899.2	20.	10899	13	-0.0	7			GSa		98Ho13
	10880.8	50.			0.3	o			Bka		03Gi05
	10883.0	35.			0.4	7			RIa		04Mo40 *
	10903.3	20.			-0.2	7			Bka		04Fo08 *
$^{*271}\text{Ds}(\alpha)^{267}\text{Hs}$	Decay chain number 6 for the long-lived isomer, GAu interpretation										04Mo40 **
$^{*271}\text{Ds}^m(\alpha)^{267}\text{Hs}$	Decay chain number 6, GAu interpretation										04Fo08 **
$^{*271}\text{Ds}^m(\alpha)^{267}\text{Hs}$	F : α escaped ?										GAu **
$^{*271}\text{Ds}^m(\alpha)^{267}\text{Hs}^m$	GAu : average of decay chains number 2, 5, 10, 13 for short-lived isomer										04Mo40 **
$^{*271}\text{Ds}^m(\alpha)^{267}\text{Hs}^m$	Decay chains number 5 and 7, GAu interpretation										04Fo08 **
$^{272}\text{Bh}(\alpha)^{268}\text{Db}$	9303.0	20.				8			Bka		15Ga24 *
	9210	10	9300	50	9.3	B			SRv		17Og01 *
$^{272}\text{Bh}(\alpha)^{268}\text{Db}^p$	9154.9	60.	9160	60	0.0	o			Db		04Og03
	9144.7	60.9			0.2	o			Db		12Og02
	9126.4	30.4			0.5	9			Db		13Og01 *
	9187.3	30.4			-0.5	9			GSa		13Ru11 *
$^{272}\text{Rg}(\alpha)^{268}\text{Mt}$	10981.9	20.	11197	13	10.8	B			GSa		95Ho04 *
	11191.9	20.			0.3	9			GSa		02Ho11 *
	11184.7	35.			0.4	9			RIa		04Mo26 *
	11207.1	20.3			-0.5	9			Bka		04Fo08 *
$^{*272}\text{Bh}(\alpha)^{268}\text{Db}$	Average 9.16(2) 9.20(5) 9.18(5) 9.16(5)										15Ga24 **
$^{*272}\text{Bh}(\alpha)^{268}\text{Db}$	$E_\alpha=8.55-9.20$ MeV										17Og01 **
$^{*272}\text{Bh}(\alpha)^{268}\text{Db}^p$	Average E_α of 20 events; error increased										13Og01 **
$^{*272}\text{Bh}(\alpha)^{268}\text{Db}^p$	Average E_α of 8 events; error increased										13Ru11 **
$^{*272}\text{Rg}(\alpha)^{268}\text{Mt}$	B : one event only; E(K) in coincidence may explain disagreement										GAu **
$^{*272}\text{Rg}(\alpha)^{268}\text{Mt}$	Two events $E_\alpha=11008$ and 11046 keV										02Ho11 **
$^{*272}\text{Rg}(\alpha)^{268}\text{Mt}$	Also others up to $E_\alpha=11560$ keV										04Mo26 **
$^{*272}\text{Rg}(\alpha)^{268}\text{Mt}$	One event only										04Fo08 **
$^{273}\text{Hs}(\alpha)^{269}\text{Sg}$	9732.9	40.	9650	60	-1.3	o					10El06
	9671.9	40.6			-0.3	o			Db		15Ut02
	9670	40			-0.3	o			SRv		16Ho09
	9670	40			-0.3	o			SRv		17Og01
	9650	40				8			Db		18Ut02
$^{273}\text{Ds}(\alpha)^{269}\text{Hs}$	9875.0	20.	11370	50	27.7	F			GSa		96Ho13 *
	11248.1	30.			2.0	F			GSa		96Ho13 *
	11519.1	60.			-1.9	F			Db		96La12 *
	11366.9	20.3				8			GSa		02Ho11 *
	11311.0	70.			0.6	U			RIa		08Mo09
	11194.3	81.2			1.8	B			RIa		13Su04
$^{*273}\text{Ds}(\alpha)^{269}\text{Hs}$	F : this event is distrusted, see reference										02Ho11 **
$^{*273}\text{Ds}(\alpha)^{269}\text{Hs}$	F : event number 2, probably to excited state in ^{269}Hs										GAu **
$^{*273}\text{Ds}(\alpha)^{269}\text{Hs}$	F : this event is distrusted, see reference; average 4 others $E_\alpha=11720$ keV										02Ho11 **
$^{*273}\text{Ds}(\alpha)^{269}\text{Hs}$	And one $E_\alpha=11080$ keV										02Ho11 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{274}\text{Bh}(\alpha)^{270}\text{Db}$	8930.6	101.5	8940	60	0.1	o			Db		11Og04 *
	8890.0	50.7			0.7	o			Db		13Og04
	8971.2	30.5			-0.5	o			GSa		14Kh04
	8940	60				6			SRv		17Og01
	9904.8	101.5				12			Db		07Og02
$^{274}\text{Mt}(\alpha)^{270}\text{Bh}^p$	11477.9	70.				11			RIa		08Mo09 *
$^{274}\text{Rg}(\alpha)^{270}\text{Mt}$	All results from this work were first published in reference										10Og01 **
$^{274}\text{Bh}(\alpha)^{270}\text{Db}$	$E_\alpha=8.73-8.84$ MeV, unc. increased from 30 to 60 keV for "O" state										17Og01 **
$^{274}\text{Bh}(\alpha)^{270}\text{Db}$	Also one $E_\alpha=11150(70)$ keV										08Mo09 **
$^{275}\text{Hs}(\alpha)^{271}\text{Sg}$	9437.5	71.0	9450	50	0.1	o			Db		04Og12
	9437.5	60.9			0.2	o			Db		06Og05
	9450	15			-0.0	o			SRv		16Ho09
	9450	20				11			SRv		17Og01
	10482.8	90.	10480	50	-0.0	U			Db		04Og03
$^{275}\text{Mt}(\alpha)^{271}\text{Bh}$	10503.0	60.			-0.3	U			Db		12StA
	10482.7	10.1				9			GSa		13Ru11
$^{276}\text{Mt}(\alpha)^{272}\text{Bh}$	10253	65	10100	10	-2.4	o			Db		04Og03 *
	10212	84			-1.3	U			Db		12Og02 *
	10160	16			-3.8	B			Db		13Og01 *
	10102.3	10.1			-0.2	o			GSa		13Ru11 *
	10093.0	20.			0.3	o			Bka		15Ga24 *
$^{276}\text{Mt}(\alpha)^{272}\text{Bh}$	10100	10				9			SRv		17Og01
	$E_\alpha=9710(60)$ to 434 and 363 keV levels										13Ru11 **
	$E_\alpha=9670(80)$ to 434 and 363 keV levels, and one 9260(64) keV										13Ru11 **
	Average for 18 events $E_\alpha=9622(16)$ to 434 and 363 keV levels										13Ru11 **
	$E_\alpha=9530(10)$ to 434 keV level; also $E_\alpha=9600(10)$ to 362 keV level										13Ru11 **
$^{276}\text{Mt}(\alpha)^{272}\text{Bh}$	Average of 9 events $E_\alpha=9570-9630=9590$ to 362 keV level										15Ga24 **
$^{276}\text{Mt}(\alpha)^{272}\text{Bh}$	also one event $E_\alpha=9890(20)$ to 60 keV level										15Ga24 **
$^{276}\text{Mt}(\alpha)^{272}\text{Bh}$	$E_\alpha=8.52-10.01$ MeV										17Og01 **
$^{277}\text{Ds}(\alpha)^{273}\text{Hs}^p$	10725.2	40.	10700	60	-0.4	o					10El06
	10704.8	40.6			-0.1	o			Db		15Ut02
	10720	40			-0.3	o			SRv		16Ho09
	10710	40			-0.2	o			SRv		17Og01
	10700	40				10			Db		18Ut02
$^{277}\text{Cn}(\alpha)^{273}\text{Ds}$	11622.2	30.				9			GSa		96Ho13 *
	11821.0	30.	11620	60	-3.4	F			GSa		96Ho13 *
	11486.2	40.6			2.1	C			RIa		04MoZU
	11486.2	40.			2.1	B			RIa		08Mo09 *
	11232.4	81.2			4.1	B			RIa		13Su04
$^{277}\text{Cn}(\alpha)^{273}\text{Ds}$	And one $E_\alpha=11170(20)$ $Q_\alpha=11334(20.)$ keV										02Ho11 **
$^{277}\text{Cn}(\alpha)^{273}\text{Ds}$	F : this event is distrusted, see reference										02Ho11 **
$^{277}\text{Cn}(\alpha)^{273}\text{Ds}$	And one $E_\alpha=11090(70)$ keV										08Mo09 **
$^{278}\text{Mt}(\alpha)^{274}\text{Bh}$	9689.7	190.	9580	30	-0.6	U			Db		11Og04
	9527.3	71.			0.6	o			Db		12Og06
	9689.6	30.4			-1.9	o			Db		13Og04
	9588.2	30.			-0.1	o			GSa		14Kh04
	9580	30				7			SRv		17Og01 *
$^{278}\text{Rg}(\alpha)^{274}\text{Mt}$	10846.3	81.2				13			Db		07Og02

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{278}\text{Nh}(\alpha)^{274}\text{Rg}$	11850.8	40.	11990	80	2.2	o			RIa		08Mo09 *
	11992.8	61.				12			RIa		12Mo25
$^{*278}\text{Mt}(\alpha)^{274}\text{Bh}$	$E_\alpha=9.38-9.55$ MeV										17Og01 **
$^{*278}\text{Nh}(\alpha)^{274}\text{Rg}$	Also one $E_\alpha=11520(40)$ keV										08Mo09 **
$^{279}\text{Ds}(\alpha)^{275}\text{Hs}^p$	9841.3	60.9	9850	60	0.1	o			Db		04Og12
	9841.3	60.9			0.1	13			Db		06Og05
	9847	15			0.0	13			Db		06Og05
	9850	20			-0.1	13			Db		06Og05
$^{279}\text{Rg}(\alpha)^{275}\text{Mt}$	10521.1	162.3	10530	170	0.1	o			Db		04Og03
	10530	160				10			SRv		17Og01
$^{280}\text{Rg}(\alpha)^{276}\text{Mt}$	10128.6	60.	10149	10	0.3	o			Db		04Og03 *
	10128.6	60.			0.3	U			Db		12Og02 *
	10148.8	10.2			-0.0	o			GSa		13Ru11 *
	10088.0	15.2			4.0	B			Db		13Og01 *
	10148.8	10.2				10			Bka		15Ga24 *
$^{*280}\text{Rg}(\alpha)^{276}\text{Mt}$	$E_\alpha=9750(60)$ to 237 keV level										13Ru11 **
$^{*280}\text{Rg}(\alpha)^{276}\text{Mt}$	$E_\alpha=9750(60)$ to 237 keV level										13Ru11 **
$^{*280}\text{Rg}(\alpha)^{276}\text{Mt}$	$E_\alpha=9770(10)$ to 237 keV level										13Ru11 **
$^{*280}\text{Rg}(\alpha)^{276}\text{Mt}$	Average of 19 events $E_\alpha=9710(15)$ to 237 keV level										13Ru11 **
$^{*280}\text{Rg}(\alpha)^{276}\text{Mt}$	$E_\alpha=9770(10)$ to 237 keV level, combining results from 13Ru11										15Ga24 **
$^{*280}\text{Rg}(\alpha)^{276}\text{Mt}$	$E_\alpha=9.09-10.07$ MeV										17Og01 **
$^{281}\text{Ds}(\alpha)^{277}\text{Hs}^p$	8957.8	180.	8850	60	-0.6	F			Db		99Og10 *
	8825.9	100.			0.2	F			Db		04Mo15 *
	8856.3	30.4			-0.1	o			GSt		10Du06
	8853.3	25.			-0.0	o			GSt		11Ga19
	8853.3	15.			-0.0	o			GSa		12Ho12
	8853	30				5			SRv		16Ho09
	8850	30			0.1	o			SRv		17Og01
$^{281}\text{Ds}^m(\alpha)^{277}\text{Hs}^m$	9449.8	15.				5			GSa		12Ho12 *
$^{281}\text{Rg}(\alpha)^{277}\text{Mt}$	9454.8	40.6	9900#	400#	6.9	o			Db		13Og04
	9410	50			6.9	D			SRv		17Og01 *
$^{281}\text{Cn}(\alpha)^{277}\text{Ds}$	10459.2	40.	10430	60	-0.5	o					10El06
	10448.9	40.6			-0.3	o			Db		15U02
	10450	40			-0.3	o			SRv		16Ho09
	10450	40			-0.3	o			SRv		17Og01
	10430	40				11			Db		18Ut02
$^{*281}\text{Ds}(\alpha)^{277}\text{Hs}^p$	F : wrong α chain, see ^{285}Cn and ^{289}Fl										GAu **
$^{*281}\text{Ds}(\alpha)^{277}\text{Hs}^p$	F : non traceable information										GAu **
$^{*281}\text{Ds}^m(\alpha)^{277}\text{Hs}^m$	Assignment of $^{293}\text{Lv}^m-^{289}\text{Fl}^m-^{285}\text{Cn}^m-^{281}\text{Ds}^m-^{277}\text{Hs}^m$ chain is tentative										12Ho12 **
$^{*281}\text{Rg}(\alpha)^{277}\text{Mt}$	Trends from Mass Surface TMS suggest ^{281}Rg 490 keV less bound										GAu **
$^{282}\text{Mt}(\alpha)^{278}\text{Bh}$	8960	180	8660#	200#	-1.6	D			SRv		16Ho09 *
$^{282}\text{Rg}(\alpha)^{278}\text{Mt}^p$	9129.7	101.4	9160	30	0.3	o			Db		11Og04
	9139.8	50.7			0.3	o			Db		13Og04
	9180.4	30.4			-0.3	o			GSa		14Kh04 *
	9160	30				9			SRv		17Og01 *
$^{282}\text{Cn}(\alpha)^{278}\text{Ds}$	9667.4	100.	10150#	200#	4.3	F			Db		04Mo15 *
$^{282}\text{Nh}(\alpha)^{278}\text{Rg}$	10783.2	81.2				14			Db		07Og02
$^{*282}\text{Mt}(\alpha)^{278}\text{Bh}$	re-interpret results from 1999Og10										16Ho09 **
$^{*282}\text{Mt}(\alpha)^{278}\text{Bh}$	Trends from Mass Surface TMS suggest ^{282}Mt 300 keV more bound										GAu **
$^{*282}\text{Rg}(\alpha)^{278}\text{Mt}^p$	also observed $E_\alpha=8860(30)$ keV										14Kh04 **
$^{*282}\text{Rg}(\alpha)^{278}\text{Mt}^p$	$E_\alpha=8.86-9.05$ MeV										17Og01 **
$^{*282}\text{Cn}(\alpha)^{278}\text{Ds}$	F : non traceable information										GAu **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{283}\text{Cn}(\alpha)^{279}\text{Ds}^p$	9677	70	9660	50	-0.2	o			DbA		04Og07
	9677.0	60.9			-0.2	o			DbA		04Og12
	9677.0	60.9			-0.2	o			DbA		06Og05
	9606.0	60.9			0.7	o					07Ei02
	9656.7	15.2			0.0	o			GSa		07Ho18
	9788.6	100.			-1.2	U			Bka		09St21
	9658	15				15			SRv		16Ho09
	9660	20			-0.0	o			SRv		17Og01
	9585.5	51.			1.0	U			RIa		17Ka31
	10265.4	90.	10380	50	1.1	U			DbA		04Og03
$^{283}\text{Nh}(\alpha)^{279}\text{Rg}^p$	10376.9	10.1				12			GSa		13Ru11
$^{284}\text{Cn}(\alpha)^{280}\text{Ds}$	9301.3	50.	9670#	150#	7.4	F			DbA		01Og01 *
	9269.0	100.			4.0	F			DbA		04Mo15 *
	9869.2	192.7			-1.0	D			RIa		17Ka66 *
$^{284}\text{Nh}(\alpha)^{280}\text{Rg}$	10254.6	20.3	10280	40	0.5	11			GSa		13Ru11 *
	10305.3	20.			-0.5	11			Bka		15Ga24 *
	10120	50			3.2	B			SRv		17Og01 *
$^{284}\text{Cn}(\alpha)^{280}\text{Ds}$	F : no α observed in later work by same group; re-assigned to ^{285}Cn										04Og07 **
$^{284}\text{Cn}(\alpha)^{280}\text{Ds}$	F : non traceable information										GAu **
$^{284}\text{Cn}(\alpha)^{280}\text{Ds}$	Alternatively ^{284}Cn could decay by SF										17Ka66 *W
$^{284}\text{Cn}(\alpha)^{280}\text{Ds}$	Trends from Mass Surface TMS suggest ^{284}Cn 200 keV more bound										GAu **
$^{284}\text{Nh}(\alpha)^{280}\text{Rg}$	Highest $E_\alpha=10110(10)$, error increased										13Ru11 **
$^{284}\text{Nh}(\alpha)^{280}\text{Rg}$	Highest $E_\alpha=10160(20)$										15Ga24 **
$^{284}\text{Nh}(\alpha)^{280}\text{Rg}$	$E_\alpha=9.10-10.11$ MeV; unc. increased from 10 to 50 keV for "O" state										17Og01 **
$^{284}\text{Nh}(\alpha)^{280}\text{Rg}$	Trends from Mass Surface TMS disagrees with the Superheavy Review SRv										GAu **
$^{285}\text{Cn}(\alpha)^{281}\text{Ds}^p$	8793.7	50.	9320	60	7.4	F			DbA		99Og10 *
	8793.7	100.			4.7	F			DbA		04Mo15 *
	9290.6	60.9			0.4	o			DbA		04Og07
	9280.5	50.			0.5	o			DbA		07Og01
	9341.3	30.4			-0.4	o			GSt		10Du06
	9341.3	30.4			-0.4	o			GSt		11Ga19
	9314.9	15.			0.1	7			GSa		12Ho12
	9320	15			-0.0	7			SRv		16Ho09
	9320	20			-0.0	o			SRv		17Og01
	9320.8	30.4			-0.0	7			RIa		17Ka66
	9845.4	15.2				6			GSa		12Ho12 *
	9878.9	81.1	10010	40	1.4	B			DbA		11Og04 *
	10026.9	62.			-0.2	o			DbA		12Og02 *
	9767.3	183.			1.3	B			DbA		12Og06
$^{285}\text{Cn}^m(\alpha)^{281}\text{Ds}^m$	10031.0	162.			-0.1	o			DbA		13Og01
	9990.4	41.			0.3	o			DbA		13Og04
	10008.7	20.3			0.0	o			DbA		16Fo16 *
	10010	40				11			SRv		17Og01 *
	10558.4	50.7	10560	70	0.0	o			DbA		15U02
	10560	50			-0.0	o			SRv		16Ho09
	10560	50			-0.0	o			SRv		17Og01
	10560	50				12			DbA		18U02
$^{285}\text{Cn}(\alpha)^{281}\text{Ds}^p$	F : one event at 15.4 m, later work yields much shorter half-lives										GAu **
$^{285}\text{Cn}(\alpha)^{281}\text{Ds}^p$	F : non traceable information										GAu **
$^{285}\text{Cn}^m(\alpha)^{281}\text{Ds}^m$	Assignment of $^{293}\text{Lv}^m-^{289}\text{Fl}^m-^{285}\text{Cn}^m-^{281}\text{Ds}^m-^{277}\text{Hs}^m$ chain is tentative										12Ho12 **
$^{285}\text{Nh}(\alpha)^{281}\text{Rg}$	And $E_\alpha=9480(110)$ keV										11Og04 **
$^{285}\text{Nh}(\alpha)^{281}\text{Rg}$	Also $E_\alpha=9740(80)$ and $9480(110)$ keV										12Og02 **
$^{285}\text{Nh}(\alpha)^{281}\text{Rg}$	reanalyzed data of 11Og04, 12Og02, 12Og06, 13Og01 and 13Og04										16Fo16 **
*	unweighted average of the 4 highest energy events as selected by										GAu **
*	the evaluator: s06=9857 s11=9902 s12=9845 s13=9867 in Table 2										GAu **
$^{285}\text{Nh}(\alpha)^{281}\text{Rg}$	$E_\alpha = 9.47-10.18$ MeV										17Og01 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{286}\text{Rg}(\alpha)^{282}\text{Mt}$	8793	45	8630#	100#	-2.4	D			SRv		16Ho09 *
$^{286}\text{Nh}(\alpha)^{282}\text{Rg}$	9766.9	100.	9790	50	0.2	o			DbA		11Og04
	9817.5	111.6			-0.2	o			DbA		12Og06
	9432.1	304.3			1.2	F			GSa		14Kh04 *
	9790	50				10			SRv		17Og01 *
$^{286}\text{Fl}(\alpha)^{282}\text{Cn}$	10142.1	100.	10360	40	2.1	F			DbA		04Mo15 *
	10172.5	314.4			0.6	o			DbA		04Og07
	10345	60			0.2	o			DbA		04Og12
	10334.7	60.9			0.3	o			DbA		06Og05
	10375.4	40.			-0.5	o			Bka		09St21
	10456.5	100.			-1.0	o			Bka		10El06
	10355.1	40.6			-0.0	o			SRv		16Ho09
	10355.1	40.6				11			SRv		17Og01
* $^{286}\text{Rg}(\alpha)^{282}\text{Mt}$	Trends from Mass Surface TMS suggest ^{286}Rg 160 keV more bound										GAu **
* $^{286}\text{Nh}(\alpha)^{282}\text{Rg}$	Only partial energy										19Kh04 **
* $^{286}\text{Nh}(\alpha)^{282}\text{Rg}$	$E_\alpha = 9.61-9.75$ MeV										17Og01 **
* $^{286}\text{Fl}(\alpha)^{282}\text{Cn}$	F : non traceable information										GAu **
$^{287}\text{Fl}(\alpha)^{283}\text{Cn}$	10435.7	20.3	10170	50	-5.0	F			DbA		99Og07 *
	10182.2	70.			-0.2	o			DbA		04Og07
	10161.9	60.8			0.1	o			DbA		04Og12
	10161.9	60.8			0.1	o			DbA		06Og05
	10166.8	15.2				16			SRv		16Ho09
	10170	20			-0.1	o			SRv		17Og01
$^{287}\text{Mc}(\alpha)^{283}\text{Nh}$	10740	90	10760	70	0.2	o			DbA		04Og03
	10740	60			0.3	o			DbA		12StA
	10780.5	50.7			-0.3	o			GSa		13Ru11
	10760	50				13			SRv		17Og01
* $^{287}\text{Fl}(\alpha)^{283}\text{Cn}$	F : 2 evts at 1.32 s, 14.4 s, later work yields T=1.7 s										GAu **
$^{288}\text{Fl}(\alpha)^{284}\text{Cn}$	9968.8	50.	10076	12	2.2	F			DbA		01Og01 *
	9958.8	100.			1.2	F			DbA		04Mo15 *
	10090.3	80.			-0.2	o			DbA		04Og07
	10090.3	70.			-0.2	o			DbA		04Og12
	10080.2	60.8			-0.1	11			DbA		07Og01
	10090.3	30.			-0.5	o			GSt		10Du06
	10090.3	30.			-0.5	o			GSt		11Ga19
	10067.0	15.			0.6	o			GSa		12Ho12
	10074.1	15.2			0.2	11			SRv		16Ho09
	10070	30			0.2	o			SRv		17Og01
	10080.2	20.3			-0.2	11			RIa		17Ka66
$^{288}\text{Mc}(\alpha)^{284}\text{Nh}$	10607.6	60.8	10650	50	0.5	o			DbA		04Og03
	10627.8	60.			0.3	o			DbA		12Og02
	10727.2	82.1			-0.8	o			DbA		13Og01 *
	10698.8	10.1			-1.0	o			GSa		13Ru11 *
	10754.6	20.3			-1.9	o			Bka		15Ga24 *
	10650	50				12			SRv		17Og01 *
* $^{288}\text{Fl}(\alpha)^{284}\text{Cn}$	F : T=1800(+2100-600) ms, later work yields shorter half-lives										GAu **
*	re-assigned to ^{289}Fl										04Og07 **
* $^{288}\text{Fl}(\alpha)^{284}\text{Cn}$	F : non traceable information										GAu **
* $^{288}\text{Mc}(\alpha)^{284}\text{Nh}$	Highest $E_\alpha=10578(81)$; average 24 events would yield $E_\alpha=10480$ $Q_\alpha=10627.8$										13Og01 **
* $^{288}\text{Mc}(\alpha)^{284}\text{Nh}$	Average of 2 highest $E_\alpha=10560(10)$ $10540(10)$										13Ru11 **
* $^{288}\text{Mc}(\alpha)^{284}\text{Nh}$	Average of 2 highest $E_\alpha=10610(20)$ $10600(20)$										15Ga24 **
* $^{288}\text{Mc}(\alpha)^{284}\text{Nh}$	$E_\alpha=10.21-10.67$ MeV, $Q_\alpha=10.65(0.01)$ MeV in reference; unc. increased										17Og01 **

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference		
$^{289}\text{Fl}(\alpha)^{285}\text{Cn}$	9846.5	50.7	9950	70	1.5	F			DbA		99Og10 *		
	9846.5	101.4			1.0	F			DbA		04Mo15 *		
	9958.1	60.			-0.1	o			DbA		04Og07		
	9958.1	50.			-0.1	o			DbA		07Og01		
	10008.8	30.			-0.9	o			GSt		10Du06		
	10008.8	30.			-0.9	o			GSt		11Ga19		
	9956.0	15.2			-0.0	o			GSa		12Ho12		
	9974	15			-0.4	8			SRv		16Ho09		
	9980	20			-0.5	o			SRv		17Og01		
	9917.3	50.7			0.5	8		RIa		17Ka66			
$^{289}\text{Fl}^m(\alpha)^{285}\text{Cn}^m$	10169.9	15.				7		GSa		12Ho12 *			
$^{289}\text{Mc}(\alpha)^{285}\text{Nh}$	10455.0	90.	10490	50	0.3	o			DbA		11Og04		
	10522.8	62.			-0.4	o			DbA		12Og02 *		
	10404.2	71.			1.0	o			DbA		12Og06		
	10546.2	81.			-0.6	o			DbA		13Og01		
	10607.0	152.			-0.7	o			DbA		13Og04		
	10510.7	20.3			-0.4	o			DbA		16Fo16 *		
		10490			50		12			SRv		17Og01 *	
$^{289}\text{Fl}(\alpha)^{285}\text{Cn}$	F : one event at 30.4 s, later work yields much shorter half-lives										GAu	**	
$^{289}\text{Fl}(\alpha)^{285}\text{Cn}$	F : non traceable information										GAu	**	
$^{289}\text{Fl}^m(\alpha)^{285}\text{Cn}^m$	Assignment of $^{293}\text{Lv}^m-^{289}\text{Fl}^m-^{285}\text{Cn}^m-^{281}\text{Ds}^m-^{277}\text{Hs}^m$ chain is tentative										12Ho12	**	
$^{289}\text{Mc}(\alpha)^{285}\text{Nh}$	Also $E_\alpha=10310(90)$ keV										12Og02	**	
$^{289}\text{Mc}(\alpha)^{285}\text{Nh}$	reanalyzed data of 11Og04, 12Og02, 12Og06, 13Og01 and 13Og04										16Fo16	**	
*	unweighted average of the 3 highest energy events selected by										GAu	**	
*	the evaluator: s06=10370 s12=10364 s14=10362 in Table 2										GAu	**	
$^{289}\text{Mc}(\alpha)^{285}\text{Nh}$	$E_\alpha=10.15-10.54$ MeV										17Og01	**	
$^{290}\text{Nh}(\alpha)^{286}\text{Rg}$	9846	45	9380#	100#	-6.9	D			SRv		16Ho09 *		
	9856.2	30.4				3			RIa		17Ka66 *		
	10454.4	40.6			10410	40	-0.7	o			GSa		14Kh04
	10410	40						11			SRv		17Og01 *
	10920.9	100.			11000	60	0.7	F			DbA		04Mo15 *
	11002.0	81.1					-0.1	o			DbA		04Og07
	10991.8	81.1					0.1	o			DbA		06Og05
	10999.9	65.9					-0.0	o			DbA		12Og06
	11002.0	60.8					-0.1	12			SRv		16Ho09
	11002.0	71.0					-0.1	o			SRv		17Og01
11013.2	142.0	-0.1	o					DbA		18Br13			
	10952.3	178.5	0.2	12			DbA		20Vo07 *				
$^{290}\text{Nh}(\alpha)^{286}\text{Rg}$	Authors say assignment is tentative										16Ho09	**	
$^{290}\text{Nh}(\alpha)^{286}\text{Rg}$	Trends from Mass Surface TMS suggest ^{290}Nh 470 keV more bound										GAu	**	
$^{290}\text{Fl}(\alpha)^{286}\text{Cn}$	Alternatively it could be $^{289}\text{Fl}(\alpha)^{285}\text{Cn}$										17Ka66	**	
$^{290}\text{Mc}(\alpha)^{286}\text{Nh}$	$E_\alpha=9.78-10.31$ MeV										17Og01	**	
$^{290}\text{Lv}(\alpha)^{286}\text{Fl}$	F : non traceable information										GAu	**	
$^{290}\text{Lv}(\alpha)^{286}\text{Fl}$	Same event as in 18Br13										20Vo07	**	
$^{291}\text{Lv}(\alpha)^{287}\text{Fl}$	10890	70	10890	90	-0.0	o			DbA		04Og07		
	10890	70			-0.0	o			DbA		06Og05		
	10890	70			-0.0	o			SRv		16Ho09		
	10890	70				17			SRv		17Og01		
$^{292}\text{Lv}(\alpha)^{288}\text{Fl}$	10707.0	50.	10791	12	1.7	F			DbA		01Og01 *		
	10676.5	100.			1.1	F			DbA		04Mo15 *		
	10808.3	71.0			-0.2	U			DbA		04Og12		
	10772.8	15.2			1.2	o			GSa		12Ho12		
	10775.9	15.2			1.0	12			SRv		16Ho09		
	10777.9	20.3			0.7	o			SRv		17Og01		
	10818.4	20.3			-1.3	12			RIa		17Ka66		
$^{292}\text{Lv}(\alpha)^{288}\text{Fl}$	F : daughter and grand-daughter re-assigned to ^{289}Fl and ^{285}Cn										GAu	**	
$^{292}\text{Lv}(\alpha)^{288}\text{Fl}$	F : non traceable information										GAu	**	

Table I. Comparison of input data and adjusted values (continued, Explanation of Table on page 030002-30)

Item	Input value		Adjusted value		v_i	Dg	Signf.	Main infl.	Lab	F	Reference
$^{293}\text{Lv}(\alpha)^{289}\text{Fl}$	10676	60	10680	60	0.0	o			DbA		04Og07
	10686.1	60.8			-0.1	o			DbA		07Og01
	10679.0	15.2			-0.0	o			GSa		12Ho12
	10705	15			-0.5	9			SRv		16Ho09
	10710	20			-0.6	o			SRv		17Og01
	10635.2	40.6			0.7	9			RIa		17Ka66
$^{293}\text{Lv}^m(\alpha)^{289}\text{Fl}^m$	10647.6	15.2				8			GSa		12Ho12 *
$^{293}\text{Ts}(\alpha)^{289}\text{Mc}$	11182.9	81.1	11320	50	1.4	o			DbA		11Og04 *
	10949.7	71.			4.3	B			DbA		12Og06 *
	11253.9	91.2			0.6	o			DbA		13Og04 *
	11293.4	20.3			0.5	o			DbA		16Fo16 *
	11320	50				13			SRv		17Og01 *
$*^{293}\text{Lv}^m(\alpha)^{289}\text{Fl}^m$	Assignment of $^{293}\text{Lv}^m_{-289}\text{Fl}^m_{-285}\text{Cn}^m_{-281}\text{Ds}^m_{-277}\text{Hs}^m$ chain is tentative										12Ho12 **
$*^{293}\text{Ts}(\alpha)^{289}\text{Mc}$	reanalyzed data of 11Og04, 12Og02, 12Og06, 13Og01 and 13Og04										16Fo16 **
*	unweighted average of the 8 highest energy events selected by										GAu **
*	the evaluator: s02=11140 s03=11080 s07=11142 s08=11114 s12=11183										GAu **
*	s13=11203 s14=11059 s16=11190 in Table 2										GAu **
$*^{293}\text{Ts}(\alpha)^{289}\text{Mc}$	$E_\alpha=10.60-11.20$ MeV										17Og01 **
$^{294}\text{Ts}(\alpha)^{290}\text{Mc}$	11202.6	40.6	11180	40	-0.4	o			GSa		14Kh04
	11180	40				12			SRv		17Og01 *
$^{294}\text{Og}(\alpha)^{290}\text{Lv}$	11800.9	100.	11870	30	0.7	F			DbA		04Mo15 *
	11810.9	60.			0.9	o			DbA		04Og12
	11810.9	60.			0.9	o			DbA		06Og05
	11840.3	65.9			0.4	o			DbA		12Og06
	11821.1	60.8			0.8	o			SRv		16Ho09
	11821.1	60.8			0.8	13			SRv		17Og01
	11878.9	30.4			-0.4	o			DbA		18Br13
	11883.9	36.5			-0.5	13			DbA		20Vo07 *
$*^{294}\text{Ts}(\alpha)^{290}\text{Mc}$	$E_\alpha=10.81-11.07$ MeV										17Og01 **
$*^{294}\text{Og}(\alpha)^{290}\text{Lv}$	F : non traceable information										GAu **
$*^{294}\text{Og}(\alpha)^{290}\text{Lv}$	Same event as in 18Br13										20Vo07 **
$^{295}\text{Og}(\alpha)^{291}\text{Lv}$	11810.4	71.0	11700#	200#	-1.3	F			DbA		04Og05 *